ALLIGATION OR MIXTURE

NATIONALISED BANKS & IBPS PO/MT/SO

1. In a mixture of milk and water the proportion of water by weight was 75%. If in the 60 gms mixture 15 gms water was added, what would be the percentage of water in the new mixture ?
(1) 75% (2) 88%
(3) 90% (4) 100%
(5) None of these
(Indian Overseas Bank PO Exam. 05.04.2009)

SBI PO EXAMS

1. In 1kg mixture of sand and iron, 20% is iron. How much sand should be added, so that the proportion of iron becomes 5% ?
(1) 3 kg (2) 4 gms
(3) 5 gms (4) 6 kg
(5) None of these
(SBI Associate Banks PO Exam. 14.02.1999)

2. The wheat sold by a grocer contained 10% low quality wheat. What quantity of good quality wheat should be added to 150 kgs of wheat so that the percentage of low quality wheat becomes 5% ?
(1) 150 kgs (2) 135 kgs
(3) 50 kgs (4) 85 kgs
(5) None of these
(SBI Associate Banks PO Exam. 21.07.2002)

3. A grocer purchased 2 kg. of rice at the rate of Rs. 15 per kg. and 3 kg. of rice at the rate of Rs. 13 per kg. At what price per kg should he sell the mixture to earn $33\frac{1}{3}\%$ profit on the cost price ?
(1) Rs. 28.00 (2) Rs. 20.00
(3) Rs. 18.40 (4) Rs. 17.40
(5) None of these
(SBI PO Exam. 26.11.2006)

INSURANCE EXAMS

1. The wheat sold by a grocer contained 10% low quality wheat. What quantity of good quality wheat should be added to 150 kgs of wheat so that the percentage of low quality wheat becomes 5% ?
(1) 150 kgs (2) 135 kgs
(3) 50 kgs (4) 85 kgs
(5) None of these
(LIC Assistant Administrative Officer (AAO) Exam. 2006)

2. A man purchased 35 kg of rice at the rate of Rs. 9.50 per kg and 30 kg at the rate of Rs. 10.50 per kg. He mixed the two. Approximately, at what price (in Rupees) per kg should he sell the mixture to make 35 per cent profit in the transaction ?
(1) 12 (2) 12.50
(3) 13 (4) 13.50
(LIC Assistant Administrative Officer (AAO) Exam. 07.06.2009)

3. To m litres of a m% solution of acid, x litres of water is mixed to yield $(m-10)$% solution of acid. If $m > 25$, then x equals
(1) $\frac{10m}{m-10}$ (2) $\frac{5m}{m-10}$
(3) $\frac{2m}{m-10}$ (4) $\frac{m}{m-10}$
(New India Assurance AO Exam. 25.10.2009)

4. When one litre of water is added to a mixture of acid and water, the new mixture contains 20% acid. When one litre of acid is added to the new mixture, then the resulting mixture contains $33\frac{1}{3}\%$ acid. The percentage of acid in the original mixture was
(1) 20% (2) 22%
(3) 24% (4) 25%
(New India Insurance AAO Exam. 22.05.2011)

5. Two barrels contain a mixture of ethanol and gasoline. The content of the ethanol is 60% in the first barrel and 30% in the second barrel. In what ratio must the mixtures from the first and the second barrels be taken to form a mixture containing 50% ethanol ?
(1) 1 : 2 (2) 2 : 1
(3) 2 : 3 (4) 3 : 2
(General Insurance Corporation AAO Exam. 11.12.2011)

6. To x litres of an x% solution of acid, y litres of water is added to get $(x-10)$% solution of acid. If $x > 20$, then value of y is
(1) $\frac{x^2}{100}$ (2) $\frac{10x}{x-10}$

(3) $\frac{10x}{x+10}$ (4) $\frac{10x^2}{x-10}$

(Oriental Insurance Company AAO Exam. 08.04.2012)

7. A and B are two alloys of gold and copper prepared by mixing metals in the ratio 7 : 2 and 7 : 11 respectively. If equal quantities of the alloys are melted to form a third alloy C, the ratio of gold and copper in C will be :

(1) 5 : 7 (2) 5 : 9

(3) 7 : 5 (4) 9 : 5

(Oriental Insurance Company AAO Exam. 08.04.2012)

8. A container has 30 litres of water. If 3 litres of water is replaced by 3 litres of spirit and this operation is repeated twice, what will be the quantity of water in the new mixture ?

(1) 24 litres (2) 23 litres

(3) 24.3 litres (4) 23.3 litres

(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 12.05.2013)

SHORT ANSWERS

NATIONALISED BANKS & IBPS PO/MT/SO

1. (5)

SBI PO EXAMS

I. (1) **2.** (1)
3. (3)

INSURANCE EXAMS

1. (1) **2.** (4)
3. (1) **4**. (4)
5. (2) **6.** (2)
7. (3) **8.** (3)

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (5) In 60 gm of mixture, Quantity of water

$= 60 \times \frac{75}{100} = 45$gm

Quantity of milk = 15 gm
After mixing 15 gm of more water, Quantity of water in the new mixture = 45 + 15
= 60 gm
$\because$ Quantity of water in 75 gm of mixture = 60 gm
$\therefore$ 100 gm of of mixture will contain

$= \frac{60}{75} \times 100 = 80\%$ of water

SBI PO EXAMS

1. (1) Amount of iron in 1 kg mixture = 20% of 1000 gms

$= \frac{20 \times 1000}{100}$ gms

= 200 gms
$\therefore$ Amount of sand in mixture
= (1000 - 200) gms = 800 gms
Now, let the total mixture is x kg in which iron is 20%
$\therefore$ According to the question,
5% of x = 200 gm
$\Rightarrow$ 5% of x = 200

$\Rightarrow \frac{5 \times x}{100} = 200$

$\Rightarrow x = \frac{200 \times 100}{5}$ gms

$\Rightarrow x = \frac{2000}{5} = 4000$ gms

$\therefore$ Required answer = 4000 gms - 1000 gms = 3000 gms = 3 kg.

2. (1) Good quality content in 150 kgs of wheat = 90% of 150 = 135 kg.
In new mixture, low quality wheat is 5%, so good quality wheat 95%
$\therefore$ 5% of the new mixture = 15 kg,

$\therefore$ New mixture $= \frac{15 \times 100}{5} = 300$ kg

$\therefore$ Good quality of wheat added
= (300 - 150)kg. = 150 kg.

3. (3) Mixture : 2 kg of rice at Rs. 15/kg + 3 kg of rice at Rs. 13/kg
Total weight = 2 + 3 = 5 kg
Total cost price = (2 × 15) + (3 × 13)
= 30 + 39 = Rs. 69
Cost price per kg of the mixture

$= \frac{69}{5} =$ Rs.13.80

Selling price to get $33\frac{1}{3}\%$ profit

$= \frac{10 + 33\frac{1}{3}}{100} \times$ Rs 13.80

$= \frac{400}{3 \times 100} \times$ Rs 13.80

$= \frac{4}{3} \times$ Rs 13.80 = Rs 18.40

INSURANCE EXAMS

1. (1) Good quality content in 150 kgs of wheat = 90% of 150 = 135 kg.
In new mixture, low quality wheat is 5%, so good quality wheat 95%
$\therefore$ 5% of the new mixture =15 kg

$\therefore$ New mixture $= \frac{15 \times 100}{5} = 300$ kg

$\therefore$ Good quality of wheat added
= (300 - 150) kg. = 150 kg.

2. (4) CP of 65 kg of the mixture
= Rs. (35 × 9.50 + 30 × 10.50)
= Rs. (332.5 + 315) = Rs. 647.5

$\therefore$ Rate per kg of the mixture = Rs $\left(\frac{647.5}{65}\right)$

$\therefore$ Required rate $= \frac{647.5}{65} \times \frac{135}{100}$

$\approx$ Rs 13.50/kg

3. (i) Tricky approach

Acid = $\frac{m}{100}\times m = \frac{m^2}{100}$

New percentage = $\frac{m^2\times 100}{(m+x)\times 100}$

$\Rightarrow \frac{m^2}{m+x} = m - 10$

$\Rightarrow m^2 - 10m + mx - 10x = m^2$

$\Rightarrow x(m - 10) = 10m$

$\Rightarrow x = \frac{10m}{m-10}$

4. (4) If there be 1 litre of acid in 4 litres of mixture, then in case I.

Percentage of acid

$= \frac{1}{4+1} \times 100 = 20\%$

Case II,

Percentage of acid

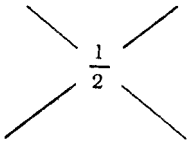
$= \frac{2}{6}\times 100 = \frac{100}{3} = 33\frac{1}{3}\%$

$\therefore$ Percentage of acid in original mixture = 25%

5. (2) By alligation,

Mixture-1 Mixture -II

Ethanol- $\frac{3}{5}$ Ethanol- $\frac{3}{10}$



$\frac{1}{2} - \frac{3}{10} = \frac{2}{10} = \frac{1}{5}$ $\frac{3}{5} - \frac{1}{2} = \frac{1}{10}$

$\therefore$ Required ratio

$= \frac{1}{5} : \frac{1}{10} = 2 : 1$

6. (2) In litres of solution,

Acid = $\frac{x^2}{100}$

Water = $x - \frac{x^2}{100}$

$\therefore \frac{\frac{x^2}{100}}{x+y} \times 100 = x - 10$

$\Rightarrow \frac{x^2}{x-y} = x - 10$

$\Rightarrow x^2 = y^2 - 10x - y(x - 10)$

$\Rightarrow y = \frac{-10x}{x-10} = \frac{10x}{x-10}$

because $x > 20$

7. (3) In 1 kg of alloy A,

Gold = $\frac{7}{18}$

Copper = $\frac{2}{9}$

In 1 kg of alloy B,

Gold = $\frac{7}{18}$ Copper = $\frac{11}{18}$

$\therefore$ Ratio of gold and copper in alloy C

$= \frac{7}{9}+\frac{7}{18} : \frac{2}{9}+\frac{11}{18} = 21 : 15 = 7 : 5$

8. (3) Suppose a container contains x units of liquid from which y units are taken out and replaced by water. After n operations, the quantity of pure liquid

$= x\left(1-\frac{y}{x}\right)^n$ units

Remaining water

$= 30\left(1-\frac{3}{30}\right)^2$

$= \frac{30\times 9\times 9}{100} = 24.3$ litres

MODEL EXERCISES

1. several litres of acid were drawn off a 54 L vessel full of acid and an equal amount of water added, Again, the same volume of the mixture was drawn off and replaced by water. As a result the vessel contained 24 L of pure acid. How much of the acid was drawn off initially ?
(1) 12 L (2) 16 L
(3) 18 L (4) 24 L
(5) None of these

2. 37.85% and 92% alcoholic solutions are mixed to get 35 L of an 89% alcoholic solution. How many litres of each solution are there in the new mixture ?
(1) 10 of the first and 25 of the second
(2) 20 of the first and 15 of the second
(3) 15 of the first and 20 of the second
(4) None of the above
(5) None of these

3. Three containers, A, B and C are having mixtures of milk and water in the ratio 1 : 5, 3 : 5 and 5 : 7 respectively. If the capacities of the containers are in the ratio 5 : 4 : 5, then find the ratio of the milk to the water, if the mixtures of all the three containers are mixed together.
(1) 51 : 115 (2) 52 : 115
(3) 53 : 115 (4) 54 : 115
(5) None of these

4. A chemist has 10 L of a solution that is 10% nitric acid by volume. He wants to dilute the solution to 40% strength by adding water. How many litres of water must be add ?
(1) 15 (2) 20
(3) 18 (4) 25
(5) None of these

5. 5 L of water is added to a certain quantity of pure milk costing Rs 3/L. If by selling the mixture at the same price as before, a profit of 20% is made, then what is the amount of pure milk in the mixture ?
(1) 20 L (2) 30 L
(3) 25 L (4) 35 L
(5) None of these

6. How many kg of sugar costing Rs 5.75 per kg should be mixed with 75 kg of cheaper sugar costing Rs 4.50 per kg so that the mixture is worth Rs 5.50 per kg ?
(1) 350 kg (2) 300 kg
(3) 250 kg (4) 325 kg
(5) None of these

7. The amount of water (in ml.) that should be added to reduce 9 mL lotion, containing 50% alcohol, to a lotion containing 30% alcohol is :
(1) 5 mL (2) 4 mL
(3) 3 mL (4) 6 mL
(5) None of these

8. A person has a chemical of Rs 25 per litre. In what ratio should water be mixed in that chemical, so that after selling the mixture at Rs 20 per litre he may get a profit of 25% ?
(1) 13 : 16 (2) 12 : 15
(3) 16 : 9 (4) 19 : 22
(5) None of these

9. 50 g of an alloy of gold and silver contains 80% gold (by weight). The quantity of gold, that is to be mixed up with this alloy, so that it may contain 95% gold, is
(1) 200 g (2) 50 g
(3) 150 g (4) 10 g
(5) None of these

10. A bottle contains 3/4 of milk and the rest water. How much of the mixture must be taken away and replaced by an equal quantity of water so that the mixture has half milk and half water ?
(1) 25% (2) $33\frac{1}{3}\%$
(3) 45% (4) 50%
(5) None of these

11. A mixture of 45 L of spirit and water contains 20% of water in it. How much water must be added to it make the water 25% in the new mixture ?
(1) 5 L (2) 3 L
(3) 4 L (4) 6 L
(5) None of these

12. In a mixture of 60 L, the ratio of milk and water is 2 : 1. If this ratio is to be 1 : 2, then the quantity of water to be further added is
(1) 30 L (2) 20 L
(3) 40 L (4) 60 L
(5) None of these

13. One test tube contains some acid and another test tube contains an equal quantity of water. To prepare a solution, 20 g of the acid is poured into the second test tube. Then, two-thirds of the so formed solution is poured from the second tube into the first. If the fluid in the first test tube is four times that in the second, what quantity of water was taken initially ?
(1) 80 g (2) 60 g
(3) 40 g (4) None of these
(5) All of these

14. One type of liquid contains 25% of milk, the other contains 30% of milk. A container is filled with 6 parts of the first liquid and 4 parts of the second liquid. The percentage of milk in the mixture is—
(1) 27% (2) 31%
(3) 29% (4) 33%
(5) None of these

15. A solution of sugar syrup has 15% sugar. Another solution has 5% sugar. How many litres of the second solution must be added to 20 L of the first solution to make a solution of 20% sugar.
(1) 10 L (2) 5 L
(3) 15 L (4) 20 L
(5) None of these

16. Mixture of milk and water has been kept in two separate containers. Ratio of milk to water in one of the containers is 5 : 1 and that in the other container is 7 : 2. In that ratio the mixture of these two containes should be added together, so that the quantity of milk in the new mixture may become 80% ?
(1) 2 : 3 (2) 3 : 2
(3) 4 : 5 (4) 3 : 4
(5) None of these

17. Vijay purchased two different kinds of alcohol. In the first mixture the ratio of alcohol to water is 3 : 4 and in the second mixture it is 5 : 6. If he mixes the two given mixtures and makes a third mixture of 18 litres in which the ratio of alcohol to water is 4 : 5, the quantity of the first mixture (whose ratio is 3 : 4) that is required to make 18 litres of the third kind of mixture is
(1) 6 (2) 7
(3) 8 (4) 9
(5) None of these

18. Two casks of 48 litres and 42 litres are filled with mixtures of milk and water; the proportions in the two casks being respectively, 13 : 7 and 18 : 17. If the contents of the two casks be mixed, and 20 litres of water be added to the whole, what will be the proportion of milk and water in the resulting mixture ?
(1) 5 : 12 (2) 7 : 13
(3) 12 : 13 (4) 8 : 15
(5) None of these

SHORT ANSWERS

1. (3) **2.** (4)
3. (3) **4.** (1)
5. (3) **6.** (2)
7. (4) **8.** (3)
9. (3) **10.** (2)
11. (2) **12.** (4)
13. (4) **14.** (1)
15. (4) **16.** (1)
17. (2) **18.** (3)

EXPLANATIONS

1. (3) According to question, if a container contains y units of liquid and x units of liquid is taken out. If this operation is repeated n times. The final quantity of the luquid in the container is

$$x\left(1-\frac{x}{y}\right)^n$$

$$\therefore 24 = 54\left(1-\frac{x}{54}\right)^2$$

$$\Rightarrow \left(1-\frac{x}{54}\right)^2 = \frac{24}{54}=\frac{4}{9}$$

$$\Rightarrow \left(1-\frac{x}{54}\right)^2 = \frac{2}{3}$$

$$\Rightarrow \frac{x}{54} = \frac{1}{3}$$

$\Rightarrow x = 18$ L

2. (4) Let x L of 37.85% of alcoholic solution and (35 - x)L of 92% alcoholic solution are required to get 35 L of 89% solution.
$\Rightarrow$ 37.85% of x + 92% of (35 - x)
= 89% of 35
$\Rightarrow x = 1.94$
$\therefore$ 1.94 L of 1st solution and 33.06 L of IInd solution are there in the new mixture.

3. (3) By question ratio of milk and water

$$=\left(\frac{1}{6}\times5+\frac{3}{8}\times4+\frac{5}{12}\times5\right) : \left(\frac{5}{6}\times5+\frac{5}{8}\times4+\frac{7}{12}\times5\right)$$

= 106 : 230 = 53 : 115

4. (1) According to question, quantity of nitric acid

$$= 10 \times \frac{1}{10} = 1 \text{ L}$$

Water = 10 - 1 = 9 L
Let x L of water be added.
Then, $(10 + x) \times \frac{4}{100} = 1$

$\Rightarrow x = 15$ L

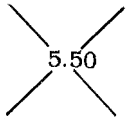
5. (3) Let the quantity of pure milk be x L.
If 5 L of water is added to it, then cost of $(5 + x)$L
= Rs $(3x + 5)$
$\therefore$ Rs Profit = Rs 15
Given, 20% of $3x$ =15

$$\Rightarrow \frac{3x}{5} = 15$$

$\Rightarrow 3x = 75$
$\Rightarrow x = 25$ L
$\therefore$ The amount of pure milk in the mixture was 25 L.

6. (2) Using Alligation Method,

Sugar I	Sugar II
5.75	4.50



5.50 - 4.50	5.75 - 5.50
1.00	0.25

i.e., 4 : 1
Hence, the required quantity of Sugar I

$$= \frac{75}{1} \times 4 = 300 \text{ kg}$$

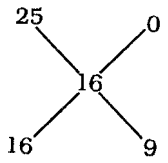
7. (4) Required quantity of water

$$=\frac{9\times(50-30)}{30}=\frac{9\times20}{30} = 6 \text{ mL}$$

8. (3) Selling price of mixture = Rs 20
Cost price of mixture

$$= \frac{100}{125} \times 20 = \text{Rs}16$$

By the rule of alligation,



So, required ratio = 16 : 9

9. (3) According to question, quantity of gold in alloy $= 50 \times \frac{80}{100} = 40$ g
Let x g of gold is mixed with this alloy.

$\frac{40+x}{50+x} \times 100 = 95$

$\Rightarrow \frac{40+x}{50+x} = \frac{95}{100}$

$\Rightarrow 4000 + 100x = 4750 + 95x$

$\Rightarrow x = \frac{750}{5} = 150$ g

10. (2) According to question,
Ratio of milk and water = 3 : 1

Water = $\frac{1}{4} \times 100 = 25$

Let x L of mixture is taken away, then

quantity of milk left = $\left(3 - \frac{3x}{4}\right)$

and water left = $\left(1 - \frac{x}{4}\right) + x$

Given, $3 - \frac{3x}{4} = 1 - \frac{x}{4} + x$

$\Rightarrow 3 - 1 = \frac{3x}{4} - \frac{x}{4} + x$

$\Rightarrow 2 = \frac{6x}{4}$

$\Rightarrow x = \frac{4}{3}$

Required percentage = $\frac{4}{3\times 4} \times 100 = 33\frac{1}{3}\%$

11. (2) According to the question,
Quantity of water in the mixture

$= \frac{20}{100} \times 45 = 9$ L

Let x L of water must be added to make the water 25% in the

$\therefore \frac{9+x}{45+x} \times 100 = 25$

$\Rightarrow \frac{9+x}{45+x} = \frac{25}{100}$

$\Rightarrow 36 + 4x = 45 + x$

$\Rightarrow 3x = 9$

$\Rightarrow x = 3$ L

12. (4) By question quantity of milk in mixture

$= \frac{2}{3} \times 60 = 40$ L

Quantity of water in mixture

$= \frac{1}{3} \times 60 = 20$ L

Let quantity of water to be further added be x.

Then, $\frac{40}{20+x} = \frac{1}{2}$

$\Rightarrow 20 + x = 80$

$\Rightarrow x = 60$ L

13. (4) Initially, let x g of water and Acid was taken. Initially 1st process
First test tube = $(x - 20)$ g
Second test tube = $(x + 20)$ g
2nd process

First test tube = $(x - 20) + (x + 20) \times \frac{2}{3}$

Second test tube = $(x + 20) \times \frac{1}{3}$

A/q,

$(x - 20) + \frac{2}{3}(x + 20) = 4 \times \frac{1}{3}(x + 20)$

$\Rightarrow x - 20 = \frac{2}{3}(x + 20)$

$\Rightarrow 3x - 60 = 2x - 40$

$\Rightarrow x = 100$ g

14. (1) According to question, milk in mixture

$6 \times \frac{25}{100} + 4 \times \frac{30}{100} = \frac{270}{100}$

For 10 parts = $\frac{270\times 10}{100} = 27\%$

15. (4) Let x L of second solution must be added.

Then, $\frac{15\times 20 + 5\times x}{20+x} = 10$

$\Rightarrow 300 + 5x = 200 + 10x$

$\Rightarrow x = \frac{100}{5} = 20$L

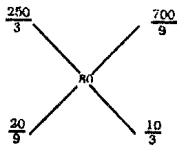
16. (1) Percentage of milk in first mixture

$= \frac{5}{6} \times 100 = \frac{250}{3}\%$

Percentage of milk in second mixture

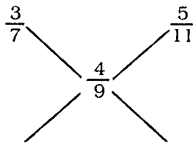
$= \frac{7}{9} \times 100 = \frac{700}{9}\%$

Using Alligation method,



So, required ratio = $\frac{20}{9} : \frac{10}{3} = 2 : 3$

17. (2) Using Alligation method,



$$\frac{5}{11} - \frac{4}{9} = \frac{1}{99} \qquad \frac{4}{9} - \frac{3}{7} = \frac{1}{63}$$

$\therefore$ Ratio = $\frac{1}{99} : \frac{1}{63} = \frac{1}{11} : \frac{1}{7}$

= 7:11

$\therefore$ Required quantity

= $\frac{7}{18} \times 18 = 7$ litres

18. (3) In mixture A of 48 litres,

Milk= $\frac{13}{20} \times 48 = \frac{156}{5}$ litres

Water = $\frac{7}{20} \times 48 = \frac{84}{5}$ litres

In mixture B of 42 litres,

Milk = $\frac{18}{35} \times 42 = \frac{108}{5}$ litres

Water = $\frac{17}{35} \times 42 = \frac{102}{5}$ litres

In the final mixture,

Milk = $\frac{156}{5} + \frac{108}{5} = \frac{264}{5}$ litres

Water = $\frac{84}{5} + \frac{102}{5} = \frac{186}{5}$ litres

$\therefore$ Required ratio

= $\frac{264}{5} : \frac{186}{5} + 20$

264 : 286 = 12 : 13

APPROXIMATE VALUE

NATIONALISED BANKS & IBPS PO/MT/SO

Directions (1-5): Find out the **approximate** value which should replace the question mark (?) in the following questions. (You are not expected to find out the exact value)

(Canara Bank PO Exam. 09.02.2003)

1. $95^{3.7} \div 95^{0.9989} = 95^{?}$

(1) 1.9 (2) 3
(3) 2.99 (4) 3.6
(5) 2.7

2. $\sqrt{10000} + \frac{3.001}{4.987}$ of 1891.992 = ?

(1) 2500 (2) 1230
(3) 1640 (4) 1525
(5) 2130

3. 0.0004 ÷ 0.0001 × 36.000009 = ?

(1) 0.10 (2) 1.45
(3) 145 (4) 14.5
(5) 1450

4. 137% of 12345 = ?

(1) 17000 (2) 15000
(3) 1500 (4) 14300
(5) 6300

5. 3739 + 164 × 27 = ?

(1) 102400 (2) 4000
(3) 8200 (4) 690
(5) 6300

Directions (6-10): What **approximate value** will come in place of the question mark (?) in the following equations ?

(Syndicate Bank PO Exam, 10.10.2004)

6. 447.75 ÷ 28 × 4.99 = ?

(1) 60 (2) 70
(3) 72 (4) 80
(5) 75

7. $(3.5)^2 \times 19.25 + ? = 275$

(1) 15 (2) 20
(3) 30 (4) 28
(5) 40

8. 85% of 225 + 32.91 × 5.01 = ?

(1) 340 (2) 355
(3) 375 (4) 345
(5) 370

9. $(15.96)^2$ + 75% of 285 = ?

(1) 435 (2) 485
(3) 440 (4) 420
(5) 470

10. 1679 ÷ 14.95 × 5.02 = ?

(1) 540 (2) 525
(3) 545 (4) 565
(5) 520

Directions (11-15): What **approximate** value should come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value).

(Union Bank of India PO Exam. 27.11.2005)

11. 63.9872 × 9449.8780 ÷ 243.0034 = $(?)^2$

(1) 2489 (2) 2500
(3) 50 (4) 45
(5) 150

12. 5237.897 - 6629.010 + 7153.999 - 2205.102 = ?

(1) 6340 (2) 4688
(3) 5240 (4) 3558
(5) 6290

13. 4985.0346 ÷ 215.987 - 3768.112 ÷ 206.868 = ?

(1) 8 (2) 5
(3)18 (4)11
(5) 15

14. $\sqrt{956240}$ = ?

(1) 979 (2) 864
(3) 1009 (4) 647
(5) 783

15. 459% of 849.947 + 266% of 6284.012 - 1486.002 = ?

(1) 20330 (2) 12640
(3)15000 (4) 22160
(5) 19130

Directions (16-20) : What **aproximate** value should come in place of the question-mark (?) in the followlg questions ? (You are not expected calculate the exact value).

(Corporation Bank PO Exam. 9.07.2006)

16. 6,23,898 × 99 = ? × 60,000

(1) 1000 (2) 1030
(3) 1050 (4) 1065
(5) 1010

17. $\frac{4}{5} \times \frac{3}{7} \div \frac{6}{7} \div \frac{5}{9}$ = ?

(1) $\frac{9}{17}$ (2) $\frac{20}{49}$
(3) $\frac{18}{25}$ (4) $\frac{1}{2}$
(5) $\frac{4}{7}$

18. $(399.98)^2$ = ?

(1) 160000 (2) 15999
(3) 1600 (4) 1599
(5) 16000

19. $\sqrt{624.9995} + (4.9989)^2 = ? \div \frac{1}{4.9900865}$

(1) 6 (2) 50
(3) 10 (4) 125
(5) 15

20. 989.001 + 1.00982 × 76.792 = ?
(1) 1000 (2) 1100
(3) 1065 (4) 110
(5) 100

Directions (21-25) : What **approximate** value should come in place of the question mark (?) in the following question ? (You are not expected to calculate the exact value).

(Bank Of Maharashtra PO Exam. 25.05.2008

21. $\frac{3}{7} \times \frac{4}{9} \times \frac{2}{5} \times 3719 = ?$
(1) 341 (2) 283
(3) 274 (4) 301
(5) 288

22. 0.008 + 6.009 ÷ $(0.7)^2$ = ?
(1) 21 (2) 6
(3) 12 (4) 8
(5) 18

23. $\left(\sqrt[3]{795657} \times 7\right) \div (3.8 \times 5.5) = ?$
(1) 48 (2) 22
(3) 43 (4) 26
(5) 31

24. 98 × 785 ÷ $(285)^2$ = ?
(1) 0.3 (2) 1.8
(3) 2.2 (4) 0.9
(5) 0.08

25. $\sqrt{749}$ × 0.56 + 14.38 = ?
(1) 30 (2) 35
(3) 42 (4) 25
(5) 45

Directions (26-30) : What **approximate** value should come in place question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Indian Overseas Bank PO Exam. 15.06.2008)

26. 459.008 + 3.0056 × 88.862 = ?
(1) 738 (2) 725
(3) 695 (4) 752
(5) 666

27. $(621.52)^2$ = ?
(1) 386300 (2) 379300
(3) 398300 (4) 365300
(5) 356300

28. 561204 × 58 = ? × 55555
(1) 606 (2) 646
(3) 556 (4) 716
(5) 586

29. (444% of 531) ÷ 972 = ?
(1) 4.5 (2) 0.5
(3) 2.5 (4) 8.5
(5) 6.5

30. (9321 + 5406 + 1001) ÷ (498 + 929 + 660) = ?
(1) 13.5 (2) 4.5
(3) 16.5 (4) 7.5
(5) 10.5

Directions (31-35) : What **approximate** value should come in place of question mark (?) in the folowing questions ?(You are not expected to calculate the exact value.)

(Andhra Bank Po Exam. 14.09.2008)

31. $(11.49)^4$ = ?
(1) 15544 (2) 16729
(3) 17430 (4) 18443
(5) 19031

32. (2198 - 1347 - 403) ÷ (159 - 113 - 27) = ?
(1) 15 (2) 24
(3) 37 (4) 49
(5) 53

33. (825 % of 330) ÷ 507 = ?
(1) 5 12) 11
(3) 17 (4) 23
(5) 27

34. $\sqrt{888888}$ × 1.486 = ?
(1) 1200 (2) 1000
(3) 1600 (4) 1400
(5) 800

35. 564.666 + 82.5091 × 44.581 - 34.111 = ?
(1) 28450 (2) 4000
(3) 1600 (4) 14225
(5) 4210

Directions (36-40) : What **approximate** value will come in place of the question mark (?) in the following questions ?

(Bank Of Baroda Specialist Officer Exam. 05.10.2008

36. (47% of 1442 - 36% of 1412) ÷ 63 = ?
(1) 4 (2) 5
(3) 3 (4) 6
(5) 1

37. $\left(\sqrt{7921} - \sqrt{2070.25}\right) \times \frac{1}{4} = ?$
(1) 11 (2) 14
(3) 15 (4) 9
(5) 13

38. (341789 + 265108) ÷(8936 - 3578) = ?
(1) 150 (2) 113
(3) 135 (4) 100
(5) 125

39. 29% of 725 = 60% of 315 + ?
(1) 28 (2) 30
(3) 15 (4) 18
(5) 21

40. 1595 ÷ 25 × 36.5 = ?
(1) 2459 (2) 2329
(3) 2359 (4) 2429
(5) 2349

Directions (41-45) : What **approximate** value should come in place of the question Mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Oriental Bank of Commerce PO Exam. 21.12.2008)

41. 63251 × 82 = ? × 42105
(1) 101 (2) 123
(3) 147 (4) 165
(5) 189

42. $\sqrt{84111}$ = ?
(1) 240 (2) 270
(3) 330 (4) 290
(5) 310

43. $(54.78)^2$ = ?
(1) 3000 (2) 3300
(3) 3500 (4) 3700
(5) 3900

44. (7171 + 3854 + 1195) ÷ (892 + 214 + 543) = ?
(1) 13 (2) 18
(3) 3 (4) 26
(5) 7

45. (562% of 816) + 1449 = ?
(1) 4145 (2) 5675
(3) 6035 (4) 7325
(5) 8885

Directions (46-50) : What **approximate** value should come in place of the question mark (?) in the following question ?

(PNB Agriculture Officer Exam. 04.01.2009)

46. 888888 ÷ 88 ÷ 8 = ?
(1) 80800 (2) 1047
(3) 1263 (4) 70600
(5) 1526

47. 193.999 + 228.008 + ? + 422.005 = 1168.01
(1) 226 (2) 484
(3) 168 (4) 196
(5) 324

48. 27.8 × 28.74 × 17.3 = ?
(1) 13822 (2) 12546
(3) 10228 (4) 15183
(5) 14995

49. $1\frac{5}{7} \times 6\frac{12}{13} \times 5\frac{8}{9}$ = ?
(1) 110 (2) 70
(3) 30 (4) 20
(5) 50

50. 16.8% of 222 × 12.1% of 923 = ?
(1) 3325 (2) 5085
(3) 2925 (4) 4165
(5) 6245

Directions (51-55) : What **approximate** value should come in place of question mark (?) in the following questions ?(You are not expected to calculate the exact value.)

(Canara Bank PO Exam. 15.03.2009)

51. (4576 + 3286 + 5639) ÷ (712 + 415 + 212) = ?
(1) 18 (2) 22
(3) 34 (4) 10
(5) 46

52. 675.456 + 12.492 × 55.671 = ?
(1) 971 (2) 1071
(3) 1171 (4) 1271
(5) 1371

53. $(447.22)^2$ = ?
(1) 200000 (2) 210000
(3) 220000 (4) 230000
(5) 240000

54. 4374562 × 64 = ? × 7777
(1) 360 (2) 3600
(3) 36000 (4) 360000
(5) 3600000

55. (872% of 659) ÷543 = ?
(1) 17 (2) 11
(3) 21 (4) 27
(5) 31

Directions (56-60) : What **approximate** value should come in place of the question mark (?) in the following questions ? **(Note**: You are not expected to calculate the exact value.)

(UCO Bank PO Exam. 22.03.2009)

56. $\sqrt[3]{1500}$ = ?
(1) 11 (2) 6
(3) 15 (4) 19
(5) 4

57. $1\frac{3}{5} \times 2\frac{1}{7} \times 7\frac{1}{3}$ = ?
(1) 17 (2) 13
(3) 9 (4) 29
(5) 25

58. 18.999 × 12.005 × 25.998 = ?
(1) 4860 (2) 6470
(3) 3320 (4) 5930
(5) 4590

59. 11.5% of 666 × 18.3% of 888 = ?
(1) 15608 (2) 12446
(3) 10520 (4) 18338
(5) 11542

60. $2898 \div 22 \div 2 = ?$
(1) 278 (2) 52
(3) 66 (4) 43
(5) 263

Directions (61-65) : What approximate value will come in place of the question mark (?) in the following questions ?

(United Bank of India PO Exam. 21.06.2009)

61. $175 \times 28 + 275 \times 27.98 = ?$
(1) 11800 (2) 12600
(3) 12800 (4) 11600
(5) 12200

62. $324.995 \times 15.98 \div 4.002 + 36.88 = ?$
(1) 1300 (2) 1230
(3) 1440 (4) 1380
(5) 1340

63. $1164 \times 128 \div 8.008 + 969.007 = ?$
(1) 18800 (2) 19000
(3) 19600 (4) 19200
(5) 18600

64. $\sqrt{624.98} + \sqrt{729.25} = ?$
(1) 58 (2) 56
(3) 52 (4) 63
(5) 61

65. 69.008% of 699.998 + 32.99% of 399.999 ?
(1) 615 (2) 645
(3) 675 (4) 715
(5) 725

Directions (66 - 70): What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate exact value).

(Andhra Bank PO Exam. 05.07.2009)

66. $7999.99 + 72 \times 49.99 = ?$
(1) 12000 (2) 12600
(3) 12500 (4) 11600
(5) 11000

67. $(25.01)^2 - (15.99)^2 = ?$
(1) 361 (2) 381
(3) 369 (4) 375
(5) 356

68. $380 \times 12.25 - 365 \div 15 = ?$
(1) 4500 (2) 4550
(3) 4800 (4) 4850
(5) 4630

69. 180% of 25501 + 50% of 28999 = ?
(1) 62400 (2) 64000
(3) 60400 (4) 64200
(5) 61600

70. $171.995 \times 14.995 \div 25 = ?$
(1) 105 (2) 115
(3) 110 (4) 125
(5) 120

Directions (71-75) : What **approximate** value will come in place of the question mark (?) in the following questions? (You are not expected to calculate exact value).

(PNB Specialist Officer's Exam. 16.08.2009)

71. $1580.05 \times 23.98 = ?$
(1) 36900 (2) 36800
(3) 37500 (4) 37900
(5) 37200

72. $77.077 \div 7.07 \times 6.08 = ?$
(1) 57 (2) 46
(3) 48 (4) 77
(5) 66

73. $(16.01)^2 - (8.99)^2 = ?$
(1) 175 (2) 180
(3) 170 (4) 165
(5) 185

74. 171% of 399 = ?
(1) 740 (2) 720
(3) 680 (4) 640
(5) 620

75. $\sqrt{224} \times \sqrt{785} = ?$
(1) 400 (2) 420
(3) 440 (4) 405
(5) 435

Directions (76-80) : What **approximate** value should come in place of question mark (?) in the following questions ? **(Note :** You are not expected to calculate the exact value.)

(Corporation Bank PO Exam. 22.11.2009)

76. $23.999 \times 9.004 \times 16.997 = ?$
(1) 3200 (2) 4100
(3) 2700 (4) 3700
(5) 4500

77. $5\frac{7}{9} \times 8\frac{4}{5} \times 9\frac{2}{3} = ?$
(1) 490 (2) 590
(3) 540 (4) 460
(5) 520

78. $5940 \div 28 \div 6 = ?$
(1) 40 (2) 35
(3) 46 (4) 52
(5) 27

79. 15.5% of 850 + 24.8% of 650 = ?
(1) 295 (2) 330
(3) 270 (4) 375
(5) 220

80. $\sqrt{2230} = ?$
(1) 54 (2) 59
(3) 41 (4) 37
(5) 47

Directions (81-85) : What **approximate** value should come in the place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

81. 15.5% of 323 - 20.8% of 198 = ?
(1) 12 (2) 5
(3) 15 (4) 3
(5) 90

82. 3058 ÷ 27 × 3 = ?
(1) 360 (2) 348
(3) 340 (4) 330
(5) 321

83. $(3.58)^2 \times (1.75)^2$ = ?
(1) 25 (2) 40
(3) 30 (4) 35
(5) 50

84. $\sqrt{5138} \div \sqrt{36}$ = ?
(1) 21 (2) 6
(3) 12 (4) 18
(5) 26

85. 37.5 × 34.9 ÷ 2.75 = ?
(1) 476 (2) 491
(3) 464 (4) 453
(5) 486

Directions (86-90) : What **aparoximate** value should come in place the question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Indian Bank PO Exam. 17.10.2010)

86. 18% of 609 + 27.5% of 450 = ?
(1) 220 (2) 233
(3) 267 (4) 248
(5) 274

87. 3942 ÷ 64 ÷ 3 = ?
(1) 29 (2) 32
(3) 21 (4) 17
(5) 11

88. $2\frac{3}{10} \times 4\frac{6}{7} \times 7\frac{1}{2}$ = ?
(1) 68 (2) 72
(3) 93 (4) 84
(5) 101

89. 12.564 × 22.009 × 17.932 = ?
(1) 4901 (2) 4895
(3) 4800 (4) 4959
(5) 4350

90. 16.978 + 27.007 + 36.984 - 12.969 - 9.003 = ?
(1) 72 (2) 42
(2) 60 (4) 51
(5) 65

Directions (91-95) : What **approximate** value should come in place of the question mark (?) in the following questions ? (Note : You are not expected to calculate the exact value)

(Bank Of India Banking Officer Exam. 24.01.2010)

91. 8399.999 ÷ 375.002 × 14.996 = ?
(1) 565 (2) 225
(3) 335 (4) 625
(5) 455

92. $\sqrt{?}$ = 37.0005
(1) 1150 (2) 1220
(3) 1570 (4) 1480
(5) 1370

93. 14.998% of 619.999 = ?
(1) 95 (2) 80
(3) 115 (4) 105
(5) 75

94. 11.003 × 19.998 × 9.010 = ?
(1) 1710 (2) 1680
(3) 1800 (4) 1980
(5) 1750

95. 1088.88 + 1800.08 + 1880.80 = ?
(1) 3950 (2) 4770
(3) 4620 (4) 5040
(5) 6810

Directions (96-100): What **approximate** value should come in place of the question mark (?) in each of the following questions ? (You are not expected to find out the exact value.)

(Allahabad Bank PO Exam. 21.02.2010)

96. 1548.45 + 3065.15 ÷ 15.058 = ?
(1) 1700 (2) 1650
(3) 1840 (4) 1750
(5) 1950

97. $6\frac{2}{5}$ of 248.65 = ? of 2398.59
(1) $\frac{2}{5}$ (2) $\frac{1}{4}$
(3) $\frac{1}{2}$ (4) $\frac{1}{3}$
(5) $\frac{2}{3}$

98. 39% of 695 = 10% of ?
(1) 2800 (2) 2400
(3) 3200 (4) 31000
(5) 2500

99. $6\sqrt{2}$ + 14.275 = ? of 196.35
(1) $\frac{1}{3}$ (2) $\frac{1}{4}$

(3) $\frac{1}{8}$ (4) $\frac{1}{5}$

(5) $\frac{1}{2}$

100. 1524.79 × 19.92 + 495.26 = ?
(1) 33,000 (2) 78,535
(3) 31,000 (4) 26,575
(5) 34,000

Directions (101-105): What **approximate** value should come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value).

(Corporation Bank PO Exam. 09.05.2010)

101. 57% of 394 - 2.5% of 996 = ?
(1) 215 (2) 175
(3) 200 (4) 180
(5) 205

102. 96.996 × 9.669 + 0.96 = ?
(1) 860 (2) 870
(3) 1020 (4) 940
(5) 1100

103. $\frac{3}{5} \times \frac{1125}{1228} \times 7 = ?$
(1) 7 (2) 12
(3) 9 (4) 4
(5) 15

104. $\left(\sqrt{339} \times 25\right) \div 30 = ?$
(1) 12 (2) 15
(3) 24 (4) 21
(5) 9

105. (638 + 9709 - 216) ÷ 26 = ?
(1) 275 (2) 365
(3) 420 (4) 300
(5) 390

Directions (106-110) : What **approximate** value should come in place of the question mark (?) in the following questions ? (You are not ex-pected to calculate the exact value.)

(Punjab & Sind Bank PO Exam. 16.05.2010)

106. $\sqrt{8938} \times (5.96)^2 = ?$
(1) 3050 (2) 3780
(3) 2340 (4) 3400
(5) 3950

107. 4734.96 - 3454.03 - 1612.86 = ? - 1611.43
(1) 1280 (2) 2290
(3) 1020 (4) 18150
(5) 1040

108. $\frac{323}{55} \times \frac{971}{251} \times \frac{56}{61} = ?$
(1) 27 (2) 9
(3) 4 (4) 16
(5) 21

109. 133.008 × 2.97 - 111.87 + 74.13 = ?
(1) 311 (2) 234
(3) 357 (4) 290
(5) 399

110. 32.1 × 2799 ÷ 549 + 120 = ?
(1) 220 (2) 284
(3) 375 (4) 505
(5) 190

Directions (111-115): What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value).

(Bank Of Baroda PO Exam. 30.05.2010)

111. 21.7% of 514.9 -43.44 = $\frac{?}{5.5}$
(1) 320 (2) 335
(3) 475 (4) 375
(5) 420

112. 1599 × 199 ÷ 49 - 1398 + 3877 = ?
(1) 9400 (2) 9000
(3) 8700 (4) 8400
(5) 9200

113. 4433.764 - 2211.993 - 1133.667 + 3377.442 = ?
(1) 4466 (2) 4377
(3) 3633 (4) 4144
(5) 3344

114. $(13.96)^2 - (15.03)^2 + (18.09)^2 - 32.65 = ?$
(1) 223 (2) 264
(3) 334 (4) 354
(5) 201

115. $7\frac{7}{12} \times 6\frac{7}{19} \div 9\frac{1}{3}$
(1) 9 (2) 11
(3) 2 (4) 5
(5) 13

Directions (116-120): What **approximate** value will come in place of question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Central Bank Of India PO Exam. 25.07.2010)

116. 9228.789 - 5021.832 + 1496.989 = ?
(1) 6500 (2) 6000
(3) 6300 (4) 5700
(5) 5100

117. 1002 ÷ 49 × 99 - 1299 = ?
(1) 700 (2) 600
(3) 900 (4) 250
(5) 400

118. 29.8% of 260 + 60.01% of 510 - 103.57 = ?

(1) 450 (2) 320
(3) 210 (4) 280
(5) 350

119. $(21.98)^2 - (25.02)^2 + (13.03)^2 = ?$
(1) 25 (2) 120
(3) 10 (4) 65
(5) 140

120. $\sqrt{2498} \times \sqrt{626} \div \sqrt{99} = ?$
(1) 110 (2) 90
(3) 200 (4) 160
(5) 125

Directions (121- 125) : What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Syndicate Bank PO Exam. 29.08.2010)

121. $\frac{150}{17} \times \frac{199}{13} \div \frac{16}{91} = ?$
(1) 650 (2) 700
(3) 770 (4) 820
(5) 850

122. 151.011 - 419.999 + 649.991 = ?
(1) 380 (2) 420
(3) 350 (4) 410
(5) 360

123. 1299 ÷ 19.99 × 25.01 + 400.01 = ?
(1) 2025 (2) 2300
(3) 1925 (4) 2200
(5) 1700

124. 30.06% of 499 + 39.99% of 799
(1) 420 (2) 380
(3) 440 (4) 470
(5) 510

125. $(14.99)^2 - (7.01)^2 + (4.99)^3 = ?$
(1) 250 (2) 200
(3) 150 (4) 300
(5) 350

Directions (126-130): What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Punjab National Bank Specialist Officer Exam, 24.10.2010)

126. $\frac{3.5 \times 1.35 \times 4.5}{0.5} = ?$
(1) 35 (2) 20
(3) 40 (4) 50
(5) 55

127. $(128.4 + 11.101 + 35.025) \div ? = 12$
(1) 8 (2) 10
(3) 18 (4) 14
(5) 20

128. $572 \div \sqrt{1755} \times 12 = ?$
(1) 150 (2) 170
(3) 155 (4) 165
(5) 175

129. $\sqrt{925} \div \sqrt[3]{350} = ?$
(1) 4 (2) 8
(3) 12 (4) 15
(5) 6

130. 12.36 × 4.26 + 13.38 = ?
(1) 72 (2) 66
(3) 58 (4) 52
(5) 80

Directions (131-135) : What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expectec to calculate the exact value.)

(Bank Of India PO Exam. 31.10.2010;

131. 465.84 + 764.86 - 211.99 = ?
(1) 1100 (2) 1080
(3) 1000 (4) 1020
(5) 1060

132. 149.9% of 149.9 + 149.9 = ?
(1) 375 (2) 400
(3) 350 (4) 425
(5) 450

133. 3001 × 749 ÷ 1001 - 1399 = ?
(1) 650 (2) 700
(3) 950 (4) 850
(5) 1000

134. $\sqrt{2642} - \sqrt{1156} + \sqrt{459} = ?$
(1) 50 (2) 90
(3) 40 (4) 20
(5) 30

135. $\frac{901}{29} \times \frac{91}{301} \div \frac{51}{599} = ?$
(1) 140 (2) 120
(3) 60 (4) 80
(5) 110

Directions (136-140): What **approximate** value should come in place of the question mark (?) in the following questions? **(Note** : you are not expected to calculate the exact value.)

(United Bank Of India PO Exam. 14.11.2010)

136. 18.505% of 550.010 = ?
(1) 135 (2) 85
(3) 100 (4) 120
(5) 90

137. 969.69 + 996.96 + 966.66 = ?
(1) 2560 (2) 2870
(3) 2930 (4) 2390
(5) 2900

138. $\sqrt{1599}$ = ?
(1) 40 (2) 45
(3) 35 (4) 30
(5) 50

139. 24.996 × 13.005 × 17.080 = ?
(1) 6225 (2) 5525
(3) 5405 (4) 5875
(5) 6025

140. 8599.999 ÷ 420.002 × 14.996 = ?
(1) 250 (2) 325
(3) 275 (4) 300
(5) 350

Directions (141-145): What **approximate** value should come in place the question mark (?) in the following questions? (You are not expected to calculate the exact value)

(PNB Management Trainee Exam. 28.11.2010)

141. 31.85 ÷ 3.90 × 15 = ?
(1) 120 (2) 90
(3) 80 (4) 140
(5) 160

142. 4.99 × 12.865 + 599 = ?
(1) 650 (2) 655
(3) 665 (4) 670
(5) 675

143. 21 + 63 ÷ 17 = ?
(1) 35 (2) 40
(3) 10 (4) 25
(5) 15

144. 1562 ÷ 24% of 356 = ?
(1) 24 (2) 18
(3) 12 (4) 28
(5) 8

145. 5986 ÷ 364 × $\sqrt{220}$ = ?
(1) 250 (2) 245
(3) 230 (4) 235
(5) 255

Directions (146 -150): What **approximate** value will come in place of Ihe question mark (?) in the following qestions ? (You are not expected to calculate the exact value.)

(Bank Of Maharashtra Exam. 19.12.2010)

146. 5682 ÷63 × 36 = ? × 19
(1) 170 (2) 190
(3) 210 (4) 240
(5) 140

147. $\frac{340}{33} \div \frac{43}{510} \times \frac{113}{93}$ = ?
(1) 150 (2) 120
(3) 210 (4) 240
(5) 170

148. $(31.33)^2 + (3.96)^3 - (12.02)^2$ = ?
(1) 800 (2) 900
(3) 950 (4) 980
(5) 1000

149. $\sqrt{3178} \times \sqrt{1330} \div \sqrt{360}$ = ?
(1) 130 (2) 110
(3) 140 (4) 160
(5) 90

150. 39% of 405 + 62% of 610 - 183.57 = ?
(1) 450 (2) 300
(3) 230 (4) 280
(5) 350

Directions (151-155): What **approximate** value should come in place of the question mark (?) in the following questions? **(Note** : you are not expected to calculate the exact value.)

(Oriental Bank Of Commerce PO Exam. 26.12.2010 (1st Sitting)

151. 20.06% of 599 + 10.01% of 901 = ?
(1) 150 (2) 210
(3) 250 (4) 280
(5) 300

152. $\frac{249}{15} \times \frac{299}{19} \div \frac{14}{99}$ = ?
(1) 1850 (2) 1700
(3) 1750 (4) 1900
(5) 2000

153. $(11.99)^2 - (8.01)^2 + (5.99)^3$ = ?
(1) 250 (2) 450
(3) 300 (4) 400
(5) 350

154. 1201 ÷ 14.99 × 19.91 + 400.01 = ?
(1) 1700 (2) 1850
(3) 1800 (4) 1950
(5) 2000

155. 251.01 - 429.99 + 549.99 = ?
(1) 370 (2) 420
(3) 340 (4) 410
(5) 320

Directions (156-160): What **approximate** value should come in place of the question mark (?) in the following questions ? **(Note** : You are not expected to calculate the exact value.)

(Indian Bank PO Exam. 02.01.2011)

156. 12959.998 ÷ 18.010 = ?
(1) 840 (2) 990
(3) 570 (4) 680
(5) 720

157. 40.005% of 439.998 + ?% of 655.011 = 228.5
(1) 8 (2) 17
(3) 12 (4) 20
(5) 5

158. 6894.986 + 5025.005 + 600.020 = ?

(1) 12170 (2) 13540
(3) 12950 (4) 11560
(5) 12520

159. $31.999 \times 12.001 \times 17.5001 = ?$
(1) 6600 (2) 6720
(3) 6480 (4) 6070
(5) 6270

160. $(10.998)^3 = ?$
(1) 1440 (2) 1730
(3) 1330 (4) 1640
(5) 1000

Directions (161-165): What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value).

(Union Bank Of India PO Exam. 09.01.2001)

161. $(41.33)^2 + (7.96)^2 - (22.02)^2 = ?$
(1) 1280 (2) 1440
(3) 1580 (4) 1540
(5) 1380

162. 41% of 601 - 250.17 = ? - 77% of 910
(1) 800 (2) 500
(3) 700 (4) 650
(5) 550

163. $52001 \div 61 \times 29 = ? \times 41$
(1) 700 (2) 600
(3) 500 (4) 550
(5) 680

164. $\frac{701}{52} \div \frac{11}{699} \times \frac{112}{107} = ?$
(1) 700 (2) 850
(3) 900 (4) 800
(5) 650

165. $\sqrt{6378} \times \sqrt{3330} \div \sqrt{360} = ?$
(1) 200 (2) 250
(3) 300 (4) 225
(5) 325

Directions (166-170): What **aproximate** value will come in place of the question mark (?) in the following questions? (You are not expected to calculate the exact value)

(Corporation Bank PO Exam. 16.01.2011)

166. $499.99 + 1999 \div 39.99 \times 50.01 = ?$
(1) 3200 (2) 2700
(3) 3000 (4) 2500
(5) 2400

167. $[(7.99)^2 - (13.001)^2 + (4.01)^3]^2 = ?$
(1) -1800 (2) 1450
(3) -1660 (4) 1660
(5) -1450

168. $\frac{601}{49} \times \frac{399}{81} \div \frac{29}{201} = ?$
(1) 520 (2) 360
(3) 460 (4) 500
(5) 420

169. $441.01 - 232.99 + 1649.99 = ? + 1225.92$
(1) 600 (2) 630
(3) 660 (4) 690
(5) 720

170. $(21.5\% \text{ of } 999)^{1/3} + (42\% \text{ of } 601)^{1/2} = ?$
(1) 18 (2) 22
(3) 26 (4) 30
(5) 33

Directions (171-175): What **approximate** value should come in place of the questions mark (?) in the following questions ? **(Note** : You are not expected to calculate the exact value.)

(Punjab & Sind Bank PO Exam. 23.01.2011)

171. $5554.999 \div 50.007 = ?$
(1) 110 (2) 150
(3) 200 (4) 50
(5) 125

172. $(18.001)^3 = ?$
(1) 5832 (2) 5500
(3) 6000 (4) 6480
(5) 5240

173. $23.001 \times 18.999 \times 7.998 = ?$
(1) 4200 (2) 3000
(3) 3500 (4) 4000
(5) 2500

174. $9999 \div 99 \div 9 = ?$
(1) 18 (2) 15
(3) 6 (4) 11
(5) 20

175. 22.005% of 449.999 = ?
(1) 85 (2) 100
(3) 125 (4) 75
(5) 150

Directions (176-180): What **approximate** value will come in place of the question mark (?) in the following questions? (You are not expected to calculate the exact value.)

(UCO Bank PO Exam. 30.01.2011)

176. 73.99 % of 1299 + 9.98% of 1899 = ?
(1) 1250 (2) 1230
(3) 1150 (4) 1180
(5) 1200

177. $5891 \div 14.99 + 589.01 - 111.99 = ?$
(1) 870 (2) 920
(3) 840 (4) 810
(5) 770

178. $(9.979)^3 - (23.99)^2 + (1.99)^5 = ?$
(1) 350 (2) 490
(3) 390 (4) 420
(5) 450

179. $\left(\frac{18}{4}\right)^2 \times \frac{455}{19} \div \frac{61}{799} = ?$

(1) 6320 (2) 6350
(3) 6400 (4) 6430
(5) 6490

180. 2439.97 - 1234.01 + 401.99 = ? + 989.99

(1) 620 (2) 650
(3) 680 (4) 700
(5) 600

Directions (181-185): What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Bank Of Baroda PO Exam. 13.03.2011)

181. $\left(\frac{24}{9}\right)^2 \times \frac{399}{39} \div \frac{41}{899} = ?$

(1) 1600 (2) 1650
(3) 1700 (4) 1550
(5) 1750

182. 67.99% of 1401 - 13.99% of 1299 = ?

(1) 700 (2) 720
(3) 770 (4) 800
(5) 740

183. 5466.97 - 3245.01 + 1122.99 = ? + 2309.99

(1) 1130 (2) 1000
(3) 1100 (4) 1030
(5) 1060

184. 5998 ÷ 9.98 + 670.99 - 139.99 = ?

(1) 1080 (2) 1280
(3) 1180 (4) 1130
(5) 1230

185. $-(4.99)^3 + (29.98)^2 - (3.01)^4 = ?$

(1) 550 (2) 590
(3) 620 (4) 650
(5) 690

Directions (186-170): What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(Allahabad Bank PO Exam. 17.04.2011)

186. $\sqrt{3100} \times \sqrt{567} \div \sqrt{250} = ? \div 8$

(1) 620 (2) 670
(3) 770 (4) 750
(5) 700

187. 89.988% of 699.9 + 50.002% of 999.99 - 170.015 = ?

(1) 990 (2) 900
(3) 920 (4) 960
(5) 860

188. $\frac{340}{20.002} \div \frac{29.997}{510} \times \frac{179.909}{59.919} = ?$

(1) 760 (2) 800
(3) 690 (4) 870
(5) 780

189. 6999 ÷ 70.005 × 94.998 = ? × 19.999

(1) 475 (2) 420
(3) 320 (4) 540
(5) 525

190. $(49.99)^2 - (8.9)^2 - (15.9)^2 = ?$

(1) 2165 (2) 2000
(3) 1965 (4) 1920
(5) 1885

Directions (191-195): What **approximate** value will come in place of the question mark (?) in the following questions? (You are not expected to calculate the exact value.)

(Indian Overseas Bank PO Exam. 22.05.2011)

191. 7441 ÷ 34 × 12 = ? × 9 + 110

(1) 420 (2) 280
(3) 590 (4) 350
(5) 220

192. $\frac{989}{34} \div \frac{65}{869} \times \frac{515}{207} = ?$

(1) 840 (2) 920
(3) 970 (4) 780
(5) 1000

193. $(32.13)^2 + (23.96)^2 - (17.11)^2 = ?$

(1) 1270 (2) 1420
(3) 1450 (4) 1360
(5) 1310

194. $\sqrt{5456} \times \sqrt{2120} \div \sqrt{460} = ?$

(1) 120 (2) 140
(3) 160 (4) 180
(5) 200

195. 67% of 801 - 231.17 = ? - 23% of 789

(1) 490 (2) 440
(3) 540 (4) 520
(5) 590

Directions (196-200): What **approximate** value should come in place of the question mark (?) in the following questions ?

(IBPS Bank PO/MT CWE 18.09.2011)

(Note : You are not expected to calculate the exact value.)

196. 39.897% of 4331 + 58.779% of 5003 = ?

(1) 4200 (2) 4600
(3) 4700 (4) 4800
(5) 5200

197. 43931.03 ÷ 2111.02 × 401.04 = ?

(1) 8900 (2) 6600
(3) 6400 (4) 8000
(5) 8300

198. $\sqrt{3178} \times \sqrt{1330} \div \sqrt{360} = ?$

(1) 130 (2) 110
(3) 140 (4) 160
(5) 90

199. $\sqrt[3]{4663} \div 349 = ? \div 21.003$
(1) 6700 (2) 8640
(3) 6680 (4) 9520
(5) 7680

200. $5682 \div 63 \times 36 = ? \times 19$
(1) 170 (2) 190
(3) 210 (4) 240
(5) 140

Directions (201-205): What **approximate** value should come in place of the quesiton mark (?) in the following questions ?

(IBPS Bank PO/MT CWE 17.06.2012)

(Note : You are not expected to calculate the exact value.)

201. $8787 \div 343 \times \sqrt{50} = ?$
(1) 250 (2) 140
(3) 180 (4) 100
(5) 280

202. $\sqrt[3]{54821} \times (303 \div 8) = (?)^2$
(1) 48 (2) 38
(3) 28 (4) 18
(5) 58

203. $\frac{5}{8}$ of 4011.33 + $\frac{7}{10}$ of 3411.22 = ?
(1) 4810 (2) 4980
(3) 4890 (4) 4930
(5) 4850

204. 23% of 6783 + 57% of 8431 = ?
(1) 6460 (2) 6420
(3) 6320 (4) 6630
(5) 6360

205. $335.01 \times 244.99 \div 55 = ?$
(1) 1490 (2) 1550
(3) 1420 (4) 1590
(5) 1400

206. 24% of 4568 ÷ 8% of 246 is **approximately** equal to
(1) 32 (2) 43
(3) 89 (4) 78
(4) 55

(IDBI Bank Officer Exam. 16.09.2012)

Directions (207-211): What approximate value should come in the following questions at the questions places.

(IBPS Specialist Officer CWE 17.03.2013)

(You are not required to calculate the exact value)

207. $(13.001)^3 = ?$
(1) 1900 (2) 2200
(3) 2000 (4) 1800
(5) 2100

208. $55.003 \times 54.998 + 5.001 = ?$
(1) 3500 (2) 3630
(3) 2540 (4) 3030
(5) 2750

209. 50.001% of 99.99 ÷ 49.999 = ?
(1) 1 (2) 0.1
(3) 0.01 (4) 0.02
(5) None of these

210. 999.0001 + 899.999 - 349.88 = ?
(1) 1549 (2) 1560
(3) 1449 (4) 1460
(5) None of these

211. $(2.0001)^3 \times (1.999)^{-2} \div (3.999)^{-4} = ?$
(1) 32 (2) 16
(3) 64 (4) 256
(5) 512

Directions (212-216): What **approximate** value should come in place of the question mark (?) in the following questions ?

(NOTE : You are not expected to calculate the exact value)

(IBPS RRBs Office Assistant CWE Exam. 09.09.2012)

212. $(10.97)^2 + (4.13)^3 \times 3.79 = ?$
(1) 428 (2) 376
(3) 197 (4) 204
(5) 302

213. 12.13% of 935.81 + 1498% of 25.85 = ?
(1) 500 (2) 550
(3) 478 (4) 341
(5) 596

214. $\sqrt[3]{65} \times 23.93 - 31.04 = ?$
(1) 98 (2) 65
(3) 102 (4) 35
(5) 79

215. $1624.12 \times 3.891 = ?$
(1) 6100 (2) 6900
(3) 6000 (4) 6400
(5) 6500

216. $3018.19 \div 2.87 - 841.02 = ?$
(1) 365 (2) 90
(3) 387 (4) 1000
(5) 200

Directions (217-226) : What approximate value will come in place of the question mark (?) in the following qestions ? (You are not required to find the exact value).

(Indian Overseas Bank PO Online Exam. 01.09.2013)

217. $2371 \div 6 + (43 \times 4.35) = ?$
(1) 582 (2) 590
(3) 600 (4) 570
(5) 595

218. $\sqrt[3]{3380} + \sqrt{1300} = ?$
(1) 56 (2) 51
(3) 53 (4) 54
(5) 55

219. $(4.989)^2 + (21.012)^3 + \sqrt{1090} = ?$
(1) 9219 (2) 9391
(3) 9319 (4) 9129
(5) None of these

220. $7020 \div 2.99 \times \frac{13}{29} = ?$
(1) 1040 (2) 1100
(3) 1060 (4) 1050
(5) None of these

221. 24.99% of 5001 - 65.01% of 2999 = ?
(1) 840 (2) 500
(3) 700 (4) -500
(5) -700

222. $(81)^{-\frac{1}{2}} - (64)^{-\frac{2}{3}} = ?$
(1) $\frac{3}{19}$ (2) $\frac{1}{16}$
(3) $\frac{7}{144}$ (4) $\frac{1}{9}$
(5) None of these

223. $331.8 \div 23.7 + (-21)^2 - 94 = (?)^2$
(1) 15 (2) 16
(3) 18 (4) 19
(5) 17

224. 34% of 576 + 18% of 842 = ?% of 400 + 83.4
(1) 75 (2) 72
(3) 62 (4) 65
(5) 66

225. $\frac{\sqrt{29241}}{\sqrt{361}} \times 5\frac{2}{9} = ?$
(1) 47 (2) 49
(3) 46 (4) 45
(5) 61

226. $3\frac{1}{4} + 6\frac{2}{7} + ? = 13\frac{3}{28}$
(1) $3\frac{2}{7}$ (2) $3\frac{4}{7}$
(3) $3\frac{3}{7}$ (4) $3\frac{5}{7}$
(5) $3\frac{6}{7}$

Directions (227-231) : What value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value)

(IBPS Bank PO/MT CWE-III 26.10.2013)

227. 21 + 3.9 × 2.9 + 8.99 = ?
(1) 42 (2) 46
(3) 44 (4) 34
(5) 36

228. 22.9889 + 0.002 ÷ ? = 23
(1) 23 (2) 1
(3) 23^2 (4) 24
(5) None of these

229. $\sqrt{1000000.000001} = ?$
(1) 1000 (2) 100
(3) 1000.001 (4) 10000
(5) 999

230. 134% of 3894 + 38.94 of 134 = ?
(1) 11452 (2) 10000
(3)10452 (4)1100
(5) None of these

231. $10^3 \times 100^3 + 999999999 = 10^? + 10^?$
(1) 6 (2) 9
(3) 7 (4) 10
(5) 12

Directions (232-236): What approximate value will come in place of the question mark (?) in the following questions ? (You are not required to calculate the exact value).

(Corporation Bank Specialist Officer (Marketing) Exam 22.12.2014)

232. $4 \times \frac{3}{13} \times 952 - 128\frac{5}{7} = ?$
(1) 823 (2) 840
(3) 835 (4) 839
(5) 845

233. $105.01\% \text{ of } 8451 - \frac{3}{7}\% \text{ of } 5006 + 9.999 = ?$
(1) 8879 (2) 8860
(3) 8850 (4) 8760
(5) None of these

234. $10^3 \times 100^3 + 999999999 = 10^? + 10^?$
(1) 6, 9 (2) 9, 9
(3) 6, 12 (4) 16, 9
(5) 6, 18

235. 21 + 3.7 × 2.9 = ?
(1) 74 (2) 70
(3) 27 (4) 32
(5) 44

236. 22.9782 + 9.002 - ? = 23.001
(1) 9 (2) 8
(3) 6 (4) 11
(5) 12

SBI PO EXAMS

1. The radius of the cylinder is half of its height and area of the inner part is 616sq. metres. **Approximately** how many litres of milk can it contains ?
(1) 1.4 (2) 1.5
(3) 1.9 (4) 1.7
(5) 2.2

(SBI Associate Banks PO Exam. 16.07.2000)

2. In the following equation what **approximate value** will come in place of the question mark (?) ?
85.147 + 34.912 × 6.2 + ? = 802.293
(1) 400 (2) 450
(3) 550 (4) 600
(5) 500

(SBI Banks PO Exam. 20.08.2000)

3. In the following equation what **approximate value** will come in place of the question mark (?)?
248.251 ÷ 12.62 × 20.52 = ?
(1) 400 (2) 450
(3) 600 (4) 350
(5) 375

(SBI Banks PO Exam. 20.08.2000)

4. In the following equation what **approximate value** will come in place of the question mark (?)
158.25 × 4.6 + 21% of 847 + ? = 950.93
(1) 35 (2) 40
(3) 25 (4) 50
(5) 45

(SBI Banks PO Exam. 20.08.2000)

5. What **approximate value** should come in place of the question mark (?) in the following equation ?
39.05 × 14.95 - 27.99 × 10.12 = (36 + ?) × 5
(1) 22 (2) 29
(3) 34 (4) 32
(5) 25

(SBI Banks PO Exam. 11.02.2001)

6. What **approximate value** should come in place of ? in the following equation ?
9876 ÷ 24.96 + 215.005 - ? = 309.99
(1) 395 (2) 295
(3) 300 (4)315
(5) 310

(SBI Associate Banks PO Exam. 21.07.2002)

Directions (7-9): What **approximate value** will come in place of the question mark (?) in the following equations ?

(SBI PO Exam. 09.01.2005)

7. 125% of 4875 + 88.005 × 14.995 = ?
(1) 7395 (2) 7490
(3) 7510 (4) 7375
(5) 7415

8. 127.001 × 7.998 + 6.05 × 4.001 = ?
(1) 1440 (2) 1400
(3) 1000 (4) 1040
(5) 1140

9. 1010 ÷ 36 + 187 × 20.05 = ?
(1) 3650 (2) 3770
(3) 3825 (4) 3800
(5) 3700

10. What approximate value will come in place of the question mark (?) in the following equation ?

$33\frac{1}{3}\%$ of 768.9 + 25% of 161.2 - 58.12 = ?

(1) 230 (2) 225
(3) 235 (4) 220
(5) 240

(SBI PO Exam. 26.11.2006)

Directions (11-15) : Find out the **approximate** value which should reolace the question mark (?) in the following questions. (You are not expected to find out the exact value)

(SBI Associate Banks PO Exam. 07.01.2007)

11. $95^{3.7} \div 95^{0.9989} = 95^{?}$
(1) 1.9 (2) 3
(3) 2.99 (4) 3.6
(5) 2.7

12. $\sqrt{10000} + \frac{3.001}{4.987}$ of 1891.992 = ?
(1) 2500 (2) 1230
(3) 1640 (4) 1525
(5)2130

13. 0.0004 ÷ 0.0001 × 36.000009 = ?
(1) 0.10 (2) 1.45
(3) 145 (4) 14.5
(5) 1450

14. 137% of 12345 = ?
(1) 17000 (2) 15000
(3) 1500 (4) 14300
(5) 900

15. 3739 + 164 × 27 = ?
(1) 105400 (2) 4000
(3) 8200 (4) 690
(5) 6300

16 What **approximate** value should come in place of the question mark(?) in the following question ?
6523 ÷ 544 × 1.2 = ?
(1) 21 (2) 33
(3) 14 (4) 8
(5) 28

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

17. If an amount of Rs. 74,336 is equally divided amongst 150 people, how much **approximate** amount would each person get ?
(1) Rs.522 (2) Rs.485
(3) Rs.496 (4) Rs.488
(5) Rs.510

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

Directions (18 - 23): What **approximate** value will come in place of the question mark (?) in the following questions ?

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

18. 2959.85 ÷ 16.001 - 34.99 = ?
(1) 160 (2) 150
(3) 140 (4) 180
(5) 170

19. (1702 ÷ 68) × 136.05 = ?
(1) 3500 (2) 3550
(3) 3450 (4) 3400
(5) 3525

20. 2950 ÷ 12.25 + 160 = ?
(1) 440 (2) 350
(3) 380 (4) 360
(5) 400

21. 25.05% of 2845 + 14.95 × 2400 = ?
(1) 36,700 (2) 36,500
(3) 35,800 (4) 35,600
(5) 36,200

22. (186 × 270.99) ÷ 40 = ?
(1) 1160 (2) 1200
(3) 1300 (4) 1180
(5) 1260

23. 424.99 × 23.95 ÷8.05 = ?
(1) 1300 (2) 1225
(3) 1325 (4) 1275
(5) 1375

Directions (24-28) : What **approximate** value will come in place of the question mark (?) in the following questions ?

(SBI Associate Banks PO Exam. 07.08.2011)

24. 32.05% of 259.99 = ?
(1) 92 (2) 88
(3) 78 (4) 90
(5) 83

25. $\frac{1}{8}$ of $\frac{2}{3}$ of $\frac{3}{5}$ of 1715 = ?
(1) 80 (2) 85
(3) 90 (4) 95
(5) 75

26. 25.05 × 123.95 + 388.999 × 15.001 = ?
(1) 900 (2) 8950
(3) 8935 (4) 8975
(5) 8995

27. 561 ÷ 35.05 × 19.99 = ?
(1) 320 (2) 330
(3) 315 (4) 325
(5) 335

28. $(15.01)^2 \times \sqrt{730} = ?$
(1) 6125 (2) 6225
(3) 6200 (4) 6075
(5) 6250

RBI GRADE-B OFFICER EXAMS

Directions (1-5): What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate exact value.)

(RBI Grade-B Officer Exam. 11.10.2009)

1. $(32.51)^2 - (17.45)^2 = ?$
(1) 780 (2) 850
(3) 680 (4) 820
(5) 750

2. 88.25% of 450 = ?% of 530
(1) 70 (2) 68
(3) 75 (4) 80
(5) 65

3. $\sqrt{898} \times (12.005)^2 + ? = 5000$
(1) 680 (2) 720
(3) 750 (4) 620
(5) 630

4. 3745 ÷ 24.05 × 17.98 = ?
(1) 2860 (2) 2800
(3) 2760 (4) 2720
(5) 2840

5. 117.95 × 8.017 × 4.98 = ?
(1) 4670 (2) 4780
(3) 4840 (4) 4720
(5) 4800

INSURANCE EXAMS

1. What **approximate** value will come in place of question mark (?) in the following equation ?
21.0091 - 6.085 + 13.24 = (35 + ?) × 2
(1) 6.5 (2) 10.5
(3) 15.5 (4) 20.5
(5) 24.5

(LIC Assistant Administrative Officer (AAO) Exam. 24.04.2005)

2. What approximate value should come in place of ? in the following equation ?
9876 ÷ 24.96 + 215.005 - ? = 309.99

(1) 395 (2) 295
(3) 300 (4) 315
(5) 310

(LIC Assistant Administrative Officer (AAO) Exam. 2006)

Directions (3 -7) : What **approximate** value should come in place of the question mark (?) in the following questions ? (Note : You are not expected to calculate the exact value)

(New India Assurance AO Exam. 25.10.2009)

3. 8537.986 - 2416.005 -221.996 = ?
(1) 6500 (2) 5900
(3) 4300 (4) 3900
(5) 5050

4. 1019.999 ÷ 60.007 = ?
(1) 11 (2) 23
(3) 17 (4) 27
(5) 13

5. 111111 ÷ 1111 ÷11 = ?
(1) 1180 (2) 15
(3) 1100 (4) 9
(5) 2

6. $\sqrt[3]{5000}$ = ?
(1) 15 (2) 9
(3) 29 (4) 32
(5) 17

7. 16.001 × 30.999 × 8.998 = ?
(1) 4450 (2) 4800
(3) 4100 (4) 3900
(5) 5000

Directions (8-12) : What **approximate** value will come in place of the question mark (?) in the following questions ? (You are not expected to calculate the exact value.)

(United India Insurance AO Exam. 27.03.2011)

8. 17001 ÷ 81 × 19 = ? × 29
(1) 100 (2) 110
(3) 140 (4) 170
(5) 130

9. $\frac{901}{51} \div \frac{21}{1201} \times \frac{101}{301} = ?$
(1) 320 (2) 350
(3) 400 (4) 410
(5) 430

10. $(9.99)^3 + (30.01)^2 - (17.01)^2 = ?$
(1) 1610 (2) 1630
(3) 1580 (4) 1540
(5) 1510

11. $\sqrt{3444} \times \sqrt{2121} \div \sqrt{1111} \times 10 = ?$
(1) 720 (2) 740
(3) 810 (4) 840
(5) 760

12. 79% of 801 - 259.99 = ? - 66% of 499
(1) 800 (2) 700
(3) 500 (4) 650
(5) 550

Directions (13-17) : What approximate value will come in place of the question mark (?) in the following questions ? (you are not required to find the exact value).

(LIC Assistant Administrative Officer (AAO) Exam. 12.05.2013)

13. $\frac{25}{9} \times \frac{16}{53} \times 91 = ?$
(1) 65 (2) 75
(3) 80 (4) 85
(5) None of these

14. $\frac{4}{9} \times 5671 - \frac{8}{15} \times 2524 = ?$
(1) 1200 (2) 1120
(3) 1100 (4) 1175
(5) None of these

15. 4568.6531 - 2431.3178 + 134.675 = ?
(1) 2272 (2) 2372
(3) 2172 (4) 2200
(5) None of these

16. 24.9% of 5679 + 44.9% of 4301 = ?
(1) 3455 (2) 3355
(3) 3255 (4) 3555
(5) None of these

17. $(6.99)^2 + (8.01)^2 - \sqrt{85}$ = ?
(1) 95 (2) 115
(3)110 (4) 104
(5) None of these

SHORT ANSWERS

NATIONALISED BANKS & IBPS PO/MT/SO

1. (5) **2.** (2) **3.** (3) **4.** (1) **5.** (3) **6.** (4) **7.** (3) **8.** (2) **9.** (5) **10.** (4) **11.** (3) **12.** (4) **13.** (2) **14.** (1) **15.** (5) **16.** (2) **17.** (3) **18.** (1) **19.** (3) **20.** (3) **21.** (2) **22.** (3) **23.** (5) **24.** (4) **25.** (1) **26.** (2) **27.** (1) **28.** (5) **29.** (3) **30.** (4) **31.** (3) **32.** (2) **33.** (1) **34.** (4) **35.** (5) **36.** (3) **37.** (1) **38.** (2) **39.** (5) **40.** (2) **41.** (2) **42.** (4) **43.** (1) **44.** (5) **45.** (3) **46.** (3) **47.** (5) **48.** (1) **49.** (2) **50.** (4) **51.** (4) **52.** (5) **53.** (1) **54.** (3) **55.** (2) **56.** (1) **57.** (5) **58.** (4) **59.** (2) **60.** (3) **61.** (2) **62.** (5) **63.** (3) **64.** (3) **65.** (1) **66.** (4) **67.** (3) **68.** (5) **69.** (3) **70.** (1) **71.** (4) **72.** (5) **73.** (1) **74.** (3) **75.** (2) **76.** (4) **77.** (1) **78.** (2) **79.** (1) **80.** (5) **81.** (5) **82.** (3) **83.** (2) **84.** (3) **85.** (1) **86.** (2) **87.** (3) **88.** (4) **89.** (4) **90.** (3) **91.** (3) **92.** (5) **93.** (1) **94.** (4) **95.** (2) **96.** (4) **97.** (5) **98.** (1) **99.** (3) **100.** (3) **101.** (3) **102.** (4) **103.** (4) **104.** (4) **105.** (5) **106.** (4) **107.** (1) **108.** (5) **109.** (3) **110.** (2) **111.** (4) **112.** (2) **113.** (1) **114.** (2) **115.** (4) **116.** (4) **117.** (1) **118.** (4) **119.** (1) **120.** (5) **121.** (3) **122.** (1) **123.** (1) **124.** (4) **125.** (4) **126.** (2) **127.** (4) **128.** (4) **129.** (1) **130.** (2) **131.** (4) **132.** (1) **133.** (4) **134.** (3) **135.** (5) **136.** (3) **137.** (3) **138.** (1) **139.** (2) **140.** (4) **141.** (1) **142.** (3) **143.** (4) **144.** (2) **145.** (2) **146.** (1) **147.** (1) **148.** (2) **149.** (2) **150.** (5) **151.** (2) **152.** (3) **153.** (3) **154.** (5) **155.** (1) **156.** (5) **157.** (1) **158.** (5) **159.** (2) **160.** (3) **161.** (1) **162.** (3) **163.** (2) **164.** (3) **165.** (4) **166.** (3) **167.** (4) **168.** (5) **169.** (2) **170.** (2) **171.** (1) **172.** (1) **173.** (3) **174.** (4) **175.** (2) **176.** (3) **177.** (1) **178.** (5) **179.** (1) **180.** (1) **181.** (1) **182.** (3) **183.** (4) **184.** (4) **185.** (5) **186.** (2) **187.** (4) **188.** (4) **189.** (1) **190.** (1) **191.** (2) **192.** (3) **193.** (5) **194.** (3) **195.** (1) **196.** (3) **197.** (5) **198.** (2) **199.** (5) **200.** (1) **201.** (3) **202.** (2) **203.** (3) **204.** [5] **205.** (1) **206.** (5) **207.** (2) **208.** (4) **209.** (1) **210.** (1) **211.** (5) **212.** (2) **213.** (1) **214.** (2)

215. (5) **216.** (5)
217. (1) **218.** (2)
219. (3) **220.** (4)
221. (5) **222.** (3)
223. (4) **224.** (5)
225. (1) **226.** (2)
227. (1) **228.** (2)
229. (1) **230.** (3)
231. (2) **232.** (1)
233. (2) **234.** (2)
235. (4) **236.** (1)

SBIPO EXAMS

1. (5) **2.** (5)
3. (1) **4.** (5)
5. (5) **6.** (3)
7. (5) **8.** (4)
9. (2) **10.** (5)
11. (5) **12.** (2)
13. (3) **14.** (1)
15. (3) **16.** (3)
17. (3) **18.** (2)
19. (4) **20.** (5)
21. (1) **22.** (5)
23. (4) **24.** (5)
25. (2) **26.** (2)
27. (1) **28.** (4)

RBI GRADE-B OFFICER EXAMS

1. (5) **2.** (3)
3. (1) **4.** (2)
5. (4)

INSURANCE EXAMS

1. (2) **2.** (3)
3. (2) **4.** (3)
5. (4) **6.** (5)
7. (1) **8.** (3)
9. (2) **10.** (1)
11. (3) **12.** (2)
13. (3) **14.** (4)
15. (1) **16.** (2)
17. (4)

EXPLANATIONS

NATIONALISED BANKS & IBPS PO/MT/SO

1. (5) $\Rightarrow 95^{?} = 95^{3.7} \div 95^{0.9989}$
$\Rightarrow 95^{?} = 95^{3.7-0.9989} = 95^{2.7011}$
$\Rightarrow ? \approx 2.7$

2. (2) $? \approx \sqrt{10000} + \frac{3}{5} \times 1892$
$= 100 + 1135.2$
$= 1235.2 \approx 1230$

3. (3) $? \approx \frac{0.0004}{0.0001} \times 36 = 4 \times 36$
$= 144 \approx 145$

4. (1) $? = 12345 \times \frac{137}{100}$
$= 16912.65 \approx 17000$

5. (3) $? = 3739 + 164 \times 27$
$= 3739 + 4428$
$= 8167 \approx 8200$

6. (4) Taking approximate integral values we have,
$? \approx 448 \div 28 \times 5$
$= \frac{448}{28} \times 5 = 80$

7. (3) $(3.5)^2 + 19.95 + ? = 275$
$\Rightarrow 12.25 \times 19.95 + ? = 275$
$\Rightarrow ? = 275 - 235.81$
$= 39.18 \approx 40$

8. (2) $? = 85\%$ of $225 + 32.91 \times 5.01$
$\approx 85\%$ of $225 + 33 \times 5$
$\approx \frac{85 \times 225}{100} + 33 \times 5$
$= 191.25 + 165$
$= 356.25 \approx 355$

9. (5) $? = (15.96)^2 + 75\%$ of 285
$\approx (16)^2 + \frac{75 \times 285}{100}$
$= 256 + 213.75$
$\approx 469.75 = 470$

10. (4) $? = 1679 \div 14.95 \times 5.02$
$\approx 1680 \div 15 \times 5$
$\approx \frac{1680}{15} \times 5 = 560 \approx 565$

11. (3) $(?)^2 = 63.9872 \times 9449.8780 \div 243.0034$
Taking approximate integral values,
$(?) = 64 \times 9450 \div 240$
$\approx \frac{64 \times 9450}{240} = 2520 \approx 2500$
$\therefore ? = \sqrt{2500} = 50$

12. (4) $? = 5237.897 - 6629.010 + 7153.999 - 2205.102$
$\approx 5238 - 6629 + 7154 - 2205$
$\approx (5238 + 7154) - (6629 + 2205)$
$\approx 12392 - 8834 = 3558$

13. (2) $? = 4985.0346 \div 215.987 - 3768.112 \div 206.868$
$\approx 4985 \div 216 - 3768 \div 207$
$= 23.078 - 18.202$

$= 4.876 \approx 5$

14. (1) $? \approx \sqrt{956240} \approx 977.8 \approx 979$

15. (5) ? = 459% of 849.947 + 266% of 6284.012 - 1486.002

$$\approx \frac{460 \times 850}{100} + \frac{260 \times 6280}{100} - 1486$$

$= 3910 + 16328 - 1486 = 18752$

This can be treated approximate to 19130.

Note : Here numbers have been approximated as multiples of 10. It makes simplification easy. The other point is the numbers in answer choices have larger gap. Hence, this approximation makes answer correct.

16. (2) $? = \frac{623898 \times 99}{60000} = 1029.43 \approx 1030$

17. (3) $? = \frac{4}{3} \times \frac{3}{7} \div \frac{6}{7} \div \frac{5}{9}$

$$= \frac{4}{5} \times \frac{3}{7} \times \frac{7}{6} \times \frac{9}{5} = \frac{18}{25}$$

18. (1) $(399.98)^2 = ?$

$\Rightarrow ? \approx (400)^2 = 160000$

19. (3) $\sqrt{624.9995} + (4.9989)^2$

Taking approximate values,

$\Rightarrow \sqrt{625} + (5)^2 \approx ? \div \frac{1}{5}$

$\Rightarrow 25 + 25 \approx ? \times 5$

$\Rightarrow ? = \frac{50}{5} = 10$

20. (3) $989.001 + 1.00982 \times 76.792 = ?$

$\Rightarrow ? \approx 989 + 1 \times 77$

$= 989 + 77 = 1066 \approx 1065$

21. (2) $? = \frac{3}{7} \times \frac{4}{9} \times \frac{2}{5} \times 3719 \approx 283$

22. (3) $? = 0.008 + 6.009 \div (0.7)^2$

$$= 0.008 + \frac{6.009}{0.7 \times 0.7}$$

$= 0.008 + 12.26 = 12.27 \approx 12$

23. (5) $? \approx (92 \times 7) \div (3.8 \times 5.5)$

$= 644 \div 20.9 = 30.81 \approx 31$

$[\sqrt[3]{95657} \approx 92]$

Hence, we can choose 31 as our answer.

24. $? = \frac{98 \times 785}{285 \times 285} = 0.94 \approx 0.9$

25. (1) $? = \sqrt{749} \times 0.56 + 14.38$

$\approx 27 \times 0.6 + 14.38$ [0.56 ≈ 0.6]

$\approx 16.2 + 14.38$

$\approx 30.58 \approx 30$

26. (2) $? \approx 459 + 3 \times 89$

[459.008 ≈ 459, 88.862 ≈ 89]

$\approx 459 + 267 = 726 \approx 725$

27. (1) $? = (621.52)^2$

$\approx 622 \times 622 = 386884 \approx 386300$

We have taken 622 > 621.52 here.

$\therefore$ Required answer = 386300.

28. (5) $561204 \times 58 = ? \times 55555$

$$\Rightarrow ? = \frac{561204 \times 58}{55555} = 585.90 \approx 586$$

29. (3) $? = \left(531 \times \frac{444}{100}\right) \div 972$

$\approx 2357.64 \div 972 = 2.42 \approx 2.5$

30. (4) $? = (9321 + 5406 + 1001) \div (498 + 929 + 660)$

$= 15728 \div 2087 = 7.53 \approx 7.5$

31. (3) $? = (11.49)^4 = 17429.30 \approx 17430$

32. (2) $? = \frac{2198 - 1347 - 403}{159 - 113 - 27}$

$$= \frac{448}{19} = 23.58 \approx 24$$

33. (1) $? = \left(330 \times \frac{825}{100}\right) \div 507$

$$= \frac{2722.50}{507} = 5.369 \approx 5$$

34. (4) $? = \sqrt{888888} \times 1.486$

$\approx 943 \times 1.5 = 1414.5$

$\therefore$ Nearest answer = 1400

35. (5) $? = 564.666 + 82.5091 \times 44.581 - 34.111$

$\approx 565 + 82.5 \times 45 - 34$

$= 565 + 3712.5 - 34$

$= 4243.5$

$\therefore$ Approximate answer = 4210

36. (3) $? = \left(1442 \times \frac{47}{100} - \frac{1412 \times 36}{100}\right) \div 63$

$= (677.74 - 508.32) \div 63$

$= = \frac{169.42}{63} = 2.689 \approx 3$

37. (1) $? = \left(\sqrt{7921} - \sqrt{2070.25}\right) \times \frac{1}{4}$

$= (89 - 45.5) \times \frac{1}{4}$

$= \frac{43.5}{4} = 10.875 \approx 11$

38. (2) ? = (341789 + 265108) ÷ (8936 - 3578)
= 606897 ÷ 5358 = 113.27 ≈ 113

39. (5) $\frac{725 \times 29}{100} = \frac{315 \times 60}{100} + ?$

$\Rightarrow 210.25 = 189 + ?$

$\Rightarrow ? = 210.25 - 189$

$= 21.25 \approx 21$

40. (2) ? = 1595 ÷ 25 × 36.5

$= \frac{1595}{25} \times 36.5 = 2328.7 \approx 2329$

41. (2) 63251 × 82 = ? × 42105

$\Rightarrow ? = \frac{63251 \times 82}{42105} = 123.182 \approx 123$

42. (4) $? = \sqrt{84111} \approx \sqrt{84100} = 290.01 \approx 290$

43. (1) $? = (54.78)^2 \approx (55)^2 = 3025$

$\therefore$ Approximate answer = 3000

44. (5) ? = (7171 + 3854 + 1195) ÷ (892 + 214 + 543)
= 12220 ÷ 1649 = 7.41 ≈ 7

45. (3) $? = \left(\frac{816 \times 562}{100}\right) + 1449$

= 4585.92 + 1449 = 6034.92 ≈ 6035

46. (3) $? = \frac{888888}{88 \times 8} = 1262.625 \approx 1263$

47. (5)193.999 + 228.008 + ? + 422.005
= 1168.01
≈ 194 + 228 + ? + 422 ≈ 1168
$\Rightarrow$ 844 + ? ≈ 1168
$\Rightarrow$? ≈ 1168 - 844 = 324

48. (1) ? = 27.8 × 28.74 × 17.3
= 13822.2156 ≈ 13822

49. $? = \frac{12}{7} \times \frac{90}{13} \times \frac{53}{9} = 69.89 \approx 70$

50. (4) $? \approx 222 \times \frac{17}{100} \times 923 \times \frac{12}{100}$

= 4180.08 ≈ 4165

51. (4) ? = (4576 + 3286 + 5639) ÷ (712 + 415 + 212) = 13501 ÷ 1339 = 10.08 ≈ 10

52. (5) ? = 675.456 + 12.492 × 55.671
≈ 675 + 12.5 × 56
= 675 + 700 ≈ 1375 ≈ 1371

53. (1) $? \approx (447)^2 = 199809 \approx 200000$

54. (3) $? = \frac{4374562 \times 64}{7777} = 35999.99 \approx 36000$

55. (2) $? = \frac{659 \times 872}{100} \div 543 = 10.58 \approx 11$

56. (1) $? = \sqrt[3]{1500} \approx 11.4$
Illustration:
11× 11 × 11
= 1331 ; 150 - 1331 = 169
12 × 12 × 12
= 1728 ; 1728 - 1500 = 228

57. (5) $? = \frac{8}{5} \times \frac{15}{7} \times \frac{22}{3} = 25.142 \approx 25$

58. (4) ? = 18.999 × 12.005 × 25.998
≈ 19 × 12 × 26 = 5928 ≈ 5930

59. (2) $? = 666 \times \frac{11.5}{100} \times 888 \times \frac{18.3}{100}$

= 12446.18 ≈ 12446

60. (3) $? = \frac{2898}{22 \times 2} = 65.863 \approx 66$

61. (2) ? = 175 × 28 + 275 × 27.98
≈ 175 × 28 + 275 × 28
= 28 (175 + 275)
= 28 × 450 = 12600

62. (5) $? \approx 325 \times 16 \div 4 + 37$

$= \frac{325 \times 16}{4} + 37$

= 1300 + 37 = 1337 ≈ 1340

63. (3) ? = 1164 × 128 ÷ 8.008 + 969.007

$\approx \frac{1164 \times 128}{8} + 969$

= 18624 + 969 = 19593 ≈ 19600

64. (3) $? = \sqrt{624.98} + \sqrt{729.25}$

$\approx \sqrt{625} + \sqrt{729}$

= 25 + 27 = 52

65. (1) $? \approx \frac{700 \times 69}{100} + \frac{400 \times 33}{100}$

= 483 + 132 = 615

66. (4) ? ≈ 8000 + 72 × 50
= 8000 + 3600 = 11600

67. (3)? $\approx (25)^2 - (16)^2$
= (25 + 16) (25 - 16) = 41 × 9 = 369

68. (5) $? = 380 \times 12.25 - \frac{365}{15}$

= 4655 - 24.33 = 4630.67 ≈ 4630

69. (3) $? \approx \frac{180 \times 25501}{100} + \frac{28999 \times 50}{100}$

= 45901.8 + 14499.5
= 60401.3 ≈ 60400
≈ 45900 + 14500 ≈ 60400

70. (1) $? \approx 172 \times 15 \div 25$

$= \frac{172 \times 15}{25} = 103.2 \approx 105$

71. (4) $? \approx 1580 \times 24 = 37920$
$\therefore$ Approximate answer = 37900

72. (5) $? \approx \frac{77}{7} \times 6 = 66$

73. (1) $? \approx (16)^2 - (9)^2$
$= (16 + 9)(16 - 9) = 25 \times 7 = 175$

74. (3) $? \approx \frac{170 \times 400}{100} = 680$

75. (2) $? \approx \sqrt{225} \times \sqrt{784} = 15 \times 28 = 420$

76. (4) $? = 23.999 \times 9.004 \times 16.997$
$\approx 24 \times 9 \times 17 = 3672 \approx 3700$

77. (1) $? = 5\frac{7}{9} \times 8\frac{4}{5} \times 9\frac{2}{3}$
$= 6 \times 9 \times 9 = 486 \approx 490$

78. (2) $? = 5940 \div 28 \div 6$
$= \frac{5940}{28 \times 6} = 35.35 \approx 35$

79. (1) $? = \frac{850 \times 15.5}{100} + \frac{650 \times 24.8}{100}$
$= 131.75 + 161.20$
$= 292.95 \approx 295$

80. (5) $47 \times 47 = 2209$
$\therefore \sqrt{2230} \approx 47$

81. (5) $? = 15.5 \times \frac{323}{100} + 198 \times \frac{20.8}{100}$
$= 50.06 + 41.18 = 91.24 \approx 90$

82. (3) $? = \frac{3058}{27} \times 3 \approx 340$

83. (2) $? = (3.55)^2 \times (1.75)^2$
$= 12.81 \times 3.06 = 39.23 \approx 40$

84. (3) $? \approx \frac{72}{6} = 12$

85. (1) $? = \frac{37.5 \times 34.9}{2.75} = 475.90 \approx 476$

86. (2) $? = \frac{18 \times 609}{100} + \frac{27.5 \times 450}{100}$
$= 109.62 + 123.75 = 233.37 \approx 233$

87. (3) $? = \frac{3942}{64 \times 3} = 20.53 \approx 21$

88. (4) $? = \frac{23}{10} \times \frac{34}{7} \times \frac{15}{2} = 83.785 \approx 84$

89. (4) $? \approx 12.6 \times 22 \times 18 = 4989.6 \approx 4959$

90. (3) $? \approx 17 + 27 + 37 - 13 - 9 \approx 59 \approx 60$

91. (3) $? \approx 8400 \div 375 \times 15$
[$\because 8399.999 \approx 8400, 14.996 \approx 15$]
$= \frac{8400 \times 15}{375} = 336 \approx 335$

92. (5) $? \approx 37 \times 37 = 1369 \approx 1370$

93. (1) $? \approx \frac{620 \times 15}{100} = 93 \approx 95$

94. (4) $? \approx 11 \times 20 \times 9 = 1980$

95. (2) $? = 1088.88 + 1800.08 + 1880.80$
$= 4769.76 \approx 4770$

96. (4) $? \approx 1548 + 3065 \times \frac{1}{15}$
$= 1548 + 204.33 = 1752.33 \approx 1750$

97. (5) $\approx 250 \times \frac{32}{5} \approx 2400 \times ?$

$\Rightarrow ? \approx \frac{1600}{2400} = \frac{2}{3}$

98. (1) $? = \frac{695 \times 39 \times 10}{100} = 2710.5 \approx 2800$

99. (3) $6 \times 1.414 + 14.275 = 196.35 \times ?$
$\Rightarrow 22.759 = 196.35 \times ?$

$\Rightarrow ? = \frac{22.759}{196.35} \approx \frac{1}{8}$

100. (3) $? \approx 1525 \times 20 + 495$
$= 30500 + 495 = 30995 \approx 31000$

101. (3) $? = \frac{394 \times 57}{100} - \frac{996 \times 2.5}{100}$
$= 224.58 - 24.9$
$= 199.68 \approx 200$

102. (4) $? \approx 97 \times 10 + 1 = 971 \approx 940$

103. (4) $? = \frac{3}{5} \times \frac{1125}{1228} \times 7 = 3.847 \approx 4$

104. (2) $? = \frac{\sqrt{339} \times 25}{30} = 15.34 \approx 15$

105. (5) $? = \frac{638 + 9709 - 216}{26} = 389.65 \approx 390$

106. (4) $? \approx 95 \times 6 \times 6 = 3420 \approx 3400$

107. (1) $4735 - 3454 - 1613$
$= ? - 1611$
$\therefore ? = -332 + 1611 = 1279 \approx 1280$

108. (5) $? = \frac{323}{53} \times \frac{971}{251} \times \frac{56}{61} = 20.856 \approx 21$

109. (3) $? \approx 133 \times 3 - 112 + 74 = 361 \approx 357$

110. (2) $? \approx 32 \times 2800 \div 550 + 120$
$= 282.9 \approx 284$

111. (4) $\frac{515 \times 22}{100} - 43 = \frac{?}{5.5}$

$\Rightarrow 113.3 - 43 = \frac{?}{5.5}$

$\therefore ? = 70.3 \times 5.5 = 386.65 \approx 375$

112. (2) $? \approx \frac{1600 \times 200}{50} - 1400 + 3900$

$= 6400 - 1400 + 3900 = 8900 \approx 9000$

113. (1) $? \approx 4434 - 2212 - 1134 + 3377$

$= 4465 \approx 4466$

114. (2) $? \approx (14)^2 - (15)^2 + (18)^2 - 33$

$= 196 - 225 + 324 - 33 = 262 \approx 264$

115. (4) $? = 8 \times 6 \div 9 = 5.33 \approx 5$

116. (4) $? \approx 9230 - 5022 + 1500 = 5708 \approx 5700$

Required answer = 5700

117. (1) $? \approx \frac{1000}{50} \times 100 - 1300$

$= 2000 - 1300 = 700$

118. (4) $? \approx \frac{260 \times 30}{100} + \frac{510 \times 60}{100} - 100$

$= 78 + 306 - 100 = 284 \approx 280$

119. (1) $? \approx (22)^2 - (25)^2 + (13)^2$

$= 484 - 625 + 169 = 28 \approx 25$

120. (5) $? \approx \sqrt{2500} \times \sqrt{625} \div \sqrt{100}$

$= 50 \times 25 \times \frac{1}{10} = 125$

121. (3) $? = \frac{150}{17} \times \frac{199}{13} \times \frac{91}{16}$

$= 768.19 \approx 770$

122. (1) $? \approx 151 - 420 + 650 = 382 \approx 380$

123. (1) $? \approx \frac{1300}{20} \times 25 + 400$

$= 1625 + 400 = 2025$

124. (4) $? \approx \frac{30 \times 500}{100} + \frac{40 \times 800}{100}$

$= 150 + 320 = 470$

125. (4) $? \approx 15^2 - 7^2 + 5^3$

$= 225 - 49 + 125 = 301 \approx 300$

126. (3) $? = \frac{3.5 \times 1.35 \times 4.5}{0.5} = 42.525 \approx 40$

127. (4) $\frac{175}{?} = 12 \Rightarrow ? = \frac{175}{12} = 14.58 \approx 14$

128. (4) $? \approx 572 \div 42 \times 12 = 163.42 \approx 165$

129. (1) $? = \sqrt{925} + \sqrt[3]{350}$

$\approx 30 \div 7 = 4.285 \approx 4$

130. (2) $? \approx 53 + 13 = 66$

131. (4) $? \approx 466 + 765 - 212 = 1019 \approx 1020$

132. (1) $? \approx \frac{150 \times 150}{100} + 150$

$= 225 + 150 = 375$

133. (4) $? \approx \frac{3000 \times 750}{1000} - 1400$

$= 2250 - 1400 = 850$

134. (3) $? \approx 51 - 34 + 21 = 38 \approx 40$

135. (5) $? \approx \frac{900}{30} \times \frac{90}{300} \times \frac{600}{50} = 108 \approx 110$

136. (3) $? \approx \frac{550 \times 19}{100} = \frac{10450}{100} = 104.50 \approx 100$

137. (3) $? \approx 970 + 997 + 967 = 2934 \approx 2930$

138. (1) $? = \sqrt{1599} \approx \sqrt{1600} = 40$

139. (2) $? \approx 25 \times 13 \times 17 = 5525$

140. (4) $? \approx \frac{8600}{420} \times 15 = 307 \approx 300$

141. (1) $? \approx 32 \div 4 \times 15$

$= \frac{32}{4} \times 15 = 120$

142. (3) $? \approx 5 \times 13 + 600 = 665$

143. (4) $? = 21 + \frac{63}{17} = 24.70 \approx 25$

144. (2) $? = 1562 \div \frac{356 \times 24}{100}$

$= 1562 \div 85.44 = 18.28 \approx 18$

145. (2) $? = 5986 \div 364 \times \sqrt{220}$

$\approx \frac{5986}{364} \times 15 = 246.67 \approx 245$

146. (1) $\frac{5282}{63} \times 36 = ? \times 19$

$\Rightarrow ? = \frac{5682 \times 36}{63 \times 19} = 170.88 \approx 170$

147. (1) $? = \frac{340}{33} \times \frac{510}{43} \times \frac{113}{93}$

$= 148.47 \approx 150$

148. (2) $? = (31)^2 + (4)^3 - (12)^2$

$= 961 + 64 - 144 = 881 \approx 900$

149. (2) $? = \frac{\sqrt{3178} \times \sqrt{1330}}{\sqrt{360}}$

$\approx \frac{56 \times 36}{19} = 106.105 = 110$

150. (5) $? = \frac{405\times39}{100} + \frac{610\times62}{100} - 183.57$

$\approx 158 + 378 - 184 = 352 \approx 350$

151. (2) $? \approx \frac{600\times20}{100} + \frac{900\times10}{100}$

$= 120 + 90 = 210$

152. (3) $? = \frac{249}{15} \times \frac{299}{19} \times \frac{99}{14}$

$\Rightarrow ? \approx \frac{250}{15} \times \frac{300}{20} \times \frac{98}{14} = 1750$

153. (3) $? \approx (12)^2 - (8)^2 + (6)^3$

$= 144 - 64 + 216 = 296 \approx 300$

154. (5) $? \approx 1200 \div 15 \times 20 + 400$

$= \frac{1200}{15} \times 20 + 400$

$= 1600 + 400 = 2000$

155. (1) $? \approx 251 - 430 + 550 = 371 \approx 370$

156. (5) $? = 12959.998 \div 18.010$

$\approx \frac{12960}{18} = 720$

157. (1) $\frac{440\times40}{100} + \frac{655\times?}{100} \approx 228$

$\Rightarrow 176 + \frac{655\times?}{100} \approx 228$

$\Rightarrow \frac{655\times?}{100} \approx 228 - 176 = 52$

$\Rightarrow ? = \frac{52\times100}{655} = 7.93 \approx 8$

158. (5) $? \approx 6895 + 5025 + 600 = 12520$

159. (2) $? \approx 32 \times 12 \times 17.5 = 6720$

160. (3) $? \approx (11)^3 = 11 \times 11 \times 11 = 1331 \approx 1330$

161. (1) $? \approx (41)^2 + (8)^2 - (22)^2$

$= 1681 + 64 - 484 = 1261 \approx 1280$

162. (3) $\frac{600\times40}{100} - 250 \approx ? - \frac{900\times80}{100}$

$\Rightarrow 240 - 250 \approx ? - 720$

$\Rightarrow ? = 720 + 240 - 250 = 710 \approx 700$

163. (2) $52000 \div 60 \times 30 \approx ? \times 40$

$\Rightarrow \frac{52000}{60} \times 30 \approx ? \times 40$

$\Rightarrow 26000 \approx ? \times 40$

$\therefore ? = \frac{26000}{40} = 650 \approx 600$

164. (3) $? = \frac{701}{52} \times \frac{699}{11} \times \frac{112}{107}$

$\approx \frac{700}{50} \times \frac{700}{11} \times \frac{110}{100} = 915.88 \approx 900$

165. (4) $? = \frac{\sqrt{6378}\times\sqrt{3330}}{\sqrt{360}}$

$\approx \frac{80\times58}{19} = 244.21 \approx 250$

166. (3) $? \approx 500 + 2000 \div 40 \times 50$

$= 500 + \frac{2000}{40} \times 50$

$= 500 + 2500 = 3000$

167. (4) $? \approx [8^2 - (13)^2 + 4^3]^2$

$= (64 - 169 + 64]^2$

$= (-41)^2 = 1681 \approx 1680$

168. (5) $? \approx \frac{600}{50} \times \frac{400}{80} \div \frac{30}{200}$

$= \frac{600}{50} \times \frac{400}{80} \times \frac{200}{30} = 400 \approx 420$

169. (2) $441 - 233 + 1650 \approx ? + 1226$

$\Rightarrow 1858 \approx ? + 1226$

$\Rightarrow ? = 1858 - 1226 = 632 \approx 630$

170. (2) $? \approx \left(\frac{1000\times21.5}{100}\right)^{\frac{1}{3}} + \left(\frac{600\times43}{100}\right)^{\frac{1}{3}}$

$= (215)^{1/3} + (258)^{1/2} \approx 6 + 16 = 22$

171. (1) $? \approx \frac{5555}{50} = 111.1 \approx 110$

172. (1) $? \approx (18)^3 = 5832$

173. (3) $? \approx 23 \times 19 \times 8 = 3496 \approx 3500$

174. (4) $? = \frac{9999}{99\times9} = 11.22 \approx 11$

175. (2) $? \approx \frac{450\times22}{100} = 99 \approx 100$

176. (3) $? \approx \frac{1300\times74}{100} + \frac{1900\times10}{100}$

$= 962 + 190 = 1152 \approx 1150$

177. (1) $? \approx 5900 \div 15 + 589 - 112$

$= 393 + 589 - 112 = 870$

178. (5) $? \approx (10)^3 - (24)^2 + (2)^5$

$= 1000 - 576 + 32 = 456 \approx 450$

179. (1) $? = \left(\frac{18}{4}\right)^2 \times \frac{455}{19} \times \frac{799}{61}$

$\approx (4.5)^2 \times 24 \times 13 = 6318 \approx 6320$

180. (1) $2440 - 1234 + 402 \approx ? + 990$
$\Rightarrow 1608 \approx ? + 990$
$\Rightarrow ? = 1608 - 990 = 618 \approx 620$

181. (1) $? \approx \left(\frac{8}{3}\right)^2 \times \frac{400}{40} \times \frac{900}{40} = 1600$

182. (3) $? \approx \frac{1400 \times 68}{100} - \frac{1300 \times 14}{100}$
$= 952 - 182 = 770$

183. (4) $5466.97 - 3245.01 + 1122.99$
$= ? + 2309.99$
$\Rightarrow 3344.95 = ? + 2309.99$
$\Rightarrow ? = 3344.95 - 2309.99$
$= 1034.96 \approx 1030$

184. (4) $? \approx 6000 \div 10 + 671 - 140$
$= 600 + 671 - 140 = 1131 \approx 1130$

185. (5) $? \approx -(5)^3 + (30)^2 - (3)^4$
$= -125 + 900 - 81 = 694 \approx 690$

186. (2) $\sqrt{3100} \times \sqrt{567} \div \sqrt{250}$
$\approx ? + 8 \Rightarrow 56 \times 24 \div 16 = ? \div 8$
$\Rightarrow \frac{56 \times 24}{16} \approx \frac{?}{8}$
$\Rightarrow 84 \approx \frac{?}{8}$
$\Rightarrow ? = 8 \times 84 = 672 \approx 670$

187. (4) $? \approx \frac{700 \times 90}{100} + \frac{1000 \times 50}{100} - 170$
$= 630 + 500 - 170 = 960$

188. (4) $? \approx \frac{340}{20} \div \frac{30}{510} \times \frac{180}{60}$
$= \frac{340}{20} \times \frac{510}{30} \times \frac{180}{60} = 867 \approx 870$

189. (1) $7000 \div 70 \times 95 \approx ? \times 20$
$? = \frac{7000 \times 95}{70 \times 20} = 475$

19O. (1) $? \approx (50)^2 - (9)^2 - (16)^2$
$= 2500 - 81 - 256$
$= 2163 \approx 2165$

191. (2) $\frac{7441}{34} \times 12 = ? \times 9 + 110$
$\Rightarrow 2626 \approx ? \times 9 + 110$
$\Rightarrow ? \times 9 \approx 2516$
$\Rightarrow ? \approx \frac{2516}{9} = 279.55 \approx 280$

192. (3) $? = \frac{989}{34} \times \frac{869}{65} \times \frac{515}{207} = 967.52 \approx 970$

193. (5) $? \approx (32)^2 + (24)^2 - (17)^2$
$= 1024 + 576 - 289 = 1311 \approx 1310$

194. (3) $? = \sqrt{5456} \times \sqrt{2120} \div \sqrt{460}$
$\approx 74 \times 46 \div 21 = 162.09 \approx 160$

195. (1) $\frac{800 \times 67}{100} - 231 \approx ? - \frac{800 \times 23}{100}$
$\Rightarrow 536 - 231 \approx ? - 184$
$\Rightarrow 305 \approx ? - 184$
$\therefore ? \approx 305 + 184 = 489 \approx 490$

196. (3) $? \approx 4330 \times \frac{40}{100} + 5000 \times \frac{59}{100}$
$= 1732 + 2950 = 4682 \approx 4700$

197. (5) $? \approx 44000 \div 2100 \times 400$
$= \frac{44000}{2100} \times 400 = 8380.95 \approx 8300$

198. (2) $? = \frac{\sqrt{3178} \times \sqrt{1330}}{\sqrt{360}}$
$\approx \frac{56 \times 36}{19} = 106.105 \approx 110$

199. (5) $\sqrt[3]{4663} + 349 = ? \div 21.003$
$\Rightarrow 17 + 349 \approx ? \div 21$
$\Rightarrow 366 \approx \frac{?}{21}$
$\Rightarrow ? \approx 366 \times 21 = 7686 \approx 7680$

200. (1) $\frac{5682}{63} \times 36 = ? \times 19$
$\Rightarrow ? = \frac{5682 \times 36}{63 \times 19} = 171.187 \approx 170$

201. (3) $8800 \div 340 \times \sqrt{50}$
$[\because 8787 \approx 8800; 343 \approx 340]$
$\Rightarrow ? \approx 25.9 \times 7 = 181.3 \approx 180$

202. (2) $?^2 = \sqrt[3]{54821} \times (303 \div 8)$
$\approx \sqrt[3]{55000} \times 38$
$\approx 38 \times 38$
$\therefore ? = 38$

203. (3) $? = \frac{4011.33 \times 5}{8} + \frac{3411.22 \times 7}{10}$
$\approx \frac{4000 \times 5}{8} + \frac{3400 \times 7}{10}$
$= 2500 + 2380 = 4880 \approx 4890$

204. (5) $? = \frac{6783 \times 23}{100} + \frac{8431 \times 57}{100}$

$= \frac{6800 \times 23}{100} + \frac{8400 \times 57}{100}$

$= 1564 + 4788 = 6352 \approx 6360$

205. (1) $? \approx 335 \times 245 \div 55 = 1492.27 \approx 1490$

206. (5) $\frac{4568 \times 24}{100} \times \frac{100}{246 \times 8} = 55.70 \approx 55$

207. (2) $? = (13.001)^3 \approx (13)^3 = 2197 \approx 2200$

208. (4) $? \approx 55 \times 55 + 5$

$= 3025 + 5 = 3030$

209. (1) $? \approx \frac{100 \times 50}{100} \div 50 = 1$

210. (1) $? \approx 999 + 900 - 350 = 1549$

211. (5) $? \approx 2^3 \times (2)^{-2} \div (4)^{-4}$

$= \frac{2}{(4)^{-2}} = 2 \times 2^8 = 2^9 = 512$

212. (2) $? = (10.97)^2 + (4.13)^3 \times 3.79$

$\approx (11)^2 + (4)^3 \times 4$

$= 121 + 256 = 377 \approx 376$

213. (1)$? \approx \frac{936 \times 12}{100} + \frac{26 \times 1500}{100}$

$= 112.32 + 390 = 502.32 \approx 500$

214. (2) $? = \sqrt[3]{65} \times 23.93 - 31.04$

$\approx \sqrt[3]{64} \times 24 - 31$

$= 4 \times 24 - 31 = 96 - 31 = 65$

215. (5) $? = 1624.12 \times 3.891$

$\approx 1624 \times 4 = 6496 \approx 6500$

216. (5) $? \approx 3018 \div 3 - 841$

$= 1006 - 841 = 209 \approx 200$

217. (1) $? \approx 395 + 187 = 582$

218. (2) $? = \sqrt[3]{3380} + \sqrt{1300}$

$\approx \sqrt[3]{3375} + \sqrt{1296}$

$= 5 + 36 = 51$

219. (3) $? \approx (5)^2 + (21)^3 + \sqrt{1089}$

$= 25 + 9261 + 33 = 9319$

220. (4) $? \approx \frac{7020}{3} \times \frac{13}{29} = 1048.96 \approx 1049$

221. (5) $? \approx \frac{5000 \times 25}{100} - \frac{3000 \times 65}{100}$

$= 1250 - 1950 = -700$

222. (3) $? = (81)^{-\frac{1}{2}} - (64)^{-\frac{2}{3}}$

$= \left(\frac{1}{81}\right)^{\frac{1}{2}} - \left(\frac{1}{64}\right)^{\frac{2}{3}} = \frac{1}{9} - \frac{1}{16} = \frac{16-9}{144} = \frac{7}{144}$

223. (4) $(?)^2 = 331.8 \div 23.7 + (-21)^2 - 94$

$= 14 + 441 - 94 = 361$

$\Rightarrow ? = \sqrt{361} = 19$

224. (5) $\frac{576 \times 34}{100} + \frac{842 \times 18}{100} = \frac{400 \times ?}{100} + 83.4$

$\Rightarrow 195.84 + 151.56 = 4 \times ? + 83.4$

$\Rightarrow 347.4 = 4 \times ? + 83.4$

$\Rightarrow 4 \times ? = 347.4 - 83.4 = 264$

$\Rightarrow ? = \frac{264}{4} = 66$

225. (1) $? = \frac{\sqrt{29241}}{\sqrt{361}} \times \frac{47}{9}$

$= \frac{171}{19} \times \frac{47}{9} = 47$

226. (2) $\Rightarrow \frac{13}{4} + \frac{44}{7} + ? = 13 + \frac{3}{28}$

$\Rightarrow \frac{91+176}{28} + ? = 13 + \frac{3}{28}$

$\Rightarrow \quad + ? = 13 + \frac{3}{28}$

$\Rightarrow \frac{264}{28} + ? = 13$

$\Rightarrow \frac{66}{7} + ? = 13$

$\Rightarrow ? = 13 - \frac{66}{7} = \frac{91-66}{7}$

$\Rightarrow ? = \frac{25}{7} = 3\frac{4}{7}$

227. (1) $? \approx 21 + 4 \times 3 + 9$

$= 21 + 12 + 9 = 42$

228. (2) $\frac{23}{?} \approx 23$

$\Rightarrow ? = \frac{23}{23} = 1$

229. (3) $? \approx \sqrt{100000} = 1000$

230. (3) $? \approx \frac{3900 \times 134}{100} + 134 \times 39$

$= 2 \times (39 \times 134) = 10452$

231. (2) $10^3 \times 10^6 + 10^9 = 10^? + 10^?$

$\Rightarrow 10^9 + 10^9 = 2 \times 10^?$

$\Rightarrow 2 \times 10^9 = 2 \times 10^?$

$\Rightarrow ? = 9$

232. (1) $? \approx \frac{4\times 3}{12} \times 952 - 129$

$= 952 - 129 = 823$

233. (2) $? \approx \frac{8450\times 105}{100} - 5006 \times \frac{3}{700} + 10$

$= 8872 - 21 + 10 = 8861$

234. (2) $10^3 \times 100^3 + 10^9 = 10^? + 10^?$

$\Rightarrow 10^9 + 10^9 = 10^? + 10^7$

$\Rightarrow ? = 9$

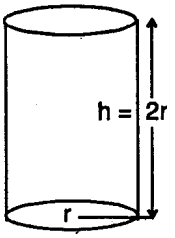
235. (4) $? = 21 + 3.7 \times 3$

$= 21 + 11.1 = 32.1 \approx 32$

236. (1) $23 + 9 - ? = 23 \Rightarrow ? = 9$

SBI PO EXAMS

1. (5)



Volume of cylinder = $\pi r^2 h$

$v = \pi r^2 h\ (2r) = 2\pi r^3$

Curve Surface area = $2\pi rh$

$= 2\pi r\ (2r) = 4\pi r^2$

$4\pi r^2 = 616$

$r^2 = 49$

$r = 7$

$v = 2\pi\ (7)^3 = 2156\ cm^3$

$= 2.156$ litres

≈ 2.2 litres

2. (5) $85.147 + 34.912 \times 6.2 + ? = 802.293$

or, $85.147 + 216.4544 + ? = 802.293$

or, $301.6014 + ? = 802.293$

$\therefore ? = 500.6916 \approx 500$

Approximate calculation:

$85.147 + 34.912 \times 6.2 + ? = 802.293$

Approximate equation can be written as

$85 + 35 \times 6 + ? = 802$

$85 + 210 + ? = 802$

$295 + ? = 802$

$? = 507 \approx 500$

3. (1) $248.251 \div 12.62 \times 20.52$

Its approximate value may be written as

$\approx 250 \div 12.5 \times 20.5$

$= 20 \times 20.5 = 410 \approx 400$

The answer choice nearest to 410 is 400.

4. (5) $158.25 \times 4.6 + 21\%$ of $847 + ? = 950.93$

or, $727.95 + 0.21 \times 847 + ? = 950.93$

or, $727.95 + 177.87 + ? = 950.93$

or, $905.82 + ? = 950.93$

$? = 45.11 = 45$ (approx.)

5. (5) $39.05 \times 14.95 - 27.99 \times 10.12$

$= (36 + ?) \times 5$

$583.7975 - 283.2588$

$= (36 + ?) \times 5$

$300.5387 = (36 + ?) \times 5$

$\frac{300.5387}{5} = 36 + ?$

$60.10774 = 36 + .?$

$60.10774 - 36 = ?$

$\therefore ? = 24.10774 \approx 24$ (approx.)

or 25 (approx.) as per the answer choice.

Approximate calculation :

$39.05 \times 14.95 - 27.99 \times 10.12 = (36 + ?) \times 5$

This can be approximated as

$39 \times 15 - 28 \times 10 = (36 + ?) \times 5$

$585 - 280 = (36 + ?) \times 5$

$305 = (36 + ?) \times 5$

$\frac{305}{5} = 36 + ?$

$61 = 36 + ?$

$? = 61 - 36$

$? = 25$ (approx.)

6. (3) $9876 \div 24.96 + 215.006 - ? = 309.99$

$\Rightarrow ? = 9876 \div 24.96 + 215.005 - 309.99$

Approximating the terms to the nearest values, we have

$? = 9900 \div 25 + 215 - 310$

$\Rightarrow \frac{9900}{25} + 215 - 310$

$= 396 + 215 - 310$

$= 301 \approx 300$

7. (5) $? = 125\%$ of $4875 + 88.005 \times 14.995$

$\approx 125\%$ of $4900 + 88 \times 15$

$= \frac{125\times 4900}{100} + 1320$

$= 6125 + 1320$

$= 7445 \approx 7415$

8. (4) $? = 127.001 \times 7.998 + 6.05 \times 4.001$

$\approx 127 \times 8 + 6 \times 4$

$= 1016 + 24 = 1040$

9. (2) $? = 1010 \div 36 + 187 \times 20.05$

$\approx 1008 \div 36 + 187 \times 20$

[1010 ≈ 1008 which is a multiple of 36]

$= 28 + 3740$

$= 3768 \approx 3770$

10. (5) $33\frac{1}{3}\%$ of $768.9 + 25\%$ of $161.2 - 58.12$

$= \frac{100}{3\times 100} \times 768.9 + \frac{25}{100} \times 161.2 - 58.12$

= 256.3 + 40.3 - 58.12
= 238.48 ≈ 240

11. (5) $\Rightarrow 95^{?} \approx 95^{3.7} \div 95^{1.0}$
$\Rightarrow 95^{?} = 95^{3.7-1} = 95^{2.7}$
$\Rightarrow ? = 2.7$

12 (2) $? \approx \sqrt{10000} + \frac{3}{5} \times 1892$
= 100 + 1135.2 = 1235.2 ≈ 1230

13. (3) $\approx \frac{0.0004}{0.0001} \times 36 = 4 \times 36$
= 144 ≈ 145

14. (1) ? ≈ 140% of 12300
$= \frac{140 \times 12300}{100}$
= 17220 ≈ 17000

15. (3) ? 3739 + 164 × 27
= 3739 + 4428 = 8167 ≈ 8200

16. (3) ? = 6523 ÷ 544 × 1.2
$= \frac{6523}{544} \times 1.2 = 14.4 \approx 14$

17. (3) Amount received by each person
$= \frac{74336}{150}$ = Rs 495.57 ≈ Rs 496

18. (2) ? = 2959.85 ÷ 16.001 - 34.99
≈ 2960 ÷ 16 - 35
$= \frac{2960}{16} - 35 = 185 - 35 = 150$

19. (4) ? = (1702 ÷ 68) × 136.05
$\approx \frac{1700}{68} \times 136 = 3400$

20. (5) $? = \frac{2950}{12.25} + 160 = 400.81 \approx 405$

21. (1) ? = 25.05 % 2845 + 14.95 × 2400
$\approx \frac{25}{100} \times 2845 + 15 \times 2400$
= 711.25 + 36000
= 36711.25 ≈ 36700

22. (5) $? \approx \frac{186 \times 271}{40} = 1260.15 \approx 1260$

23. (4) $? \approx \frac{425 \times 24}{8} = 1275$

24. (5) $? \approx \frac{260 \times 32}{100} = 83.2 \approx 83$

25. (2) $? = 1715 \times \frac{3}{5} \times \frac{2}{3} \times \frac{1}{8}$
$= 1715 \times \frac{3}{5} \times \frac{2}{3} \times \frac{1}{8} = 85.75 \approx 85$

26. (2) ? ≈ 25 × 124 + 390 × 15
= 3100 + 5850 = 8950

27. (1) ? ≈ 560 ÷ 35 × 20
$= \frac{560}{35} \times 20 = 320$

28. (4) $? \approx (15)^2 \times \sqrt{729}$
= 225 × 27 = 6075

RBI GRADE-B OFFICER EXAMS

1. (5) ? = (32.51 + 17.45) (32.51 - 17.45)
= 49.96 × 15.06 = 752.39 ≈ 750

2. (3) $\frac{450 \times 88}{100} \approx \frac{530 \times ?}{100}$

$\Rightarrow ? = \frac{450 \times 88}{530} = 74.71 \approx 75$

3. (1) $\sqrt{900} \times (12)^2 + ? \approx 5000$
(898 ≈ 900; 12.005 ≈ 12)
$\Rightarrow 30 \times 144 + ? \approx 5000$
$\Rightarrow ? \approx 5000 - 4320 = 680$

4. (2) $? \approx \frac{3745}{24} \times 18 = 2808.75 \approx 2800$

5. (4) ? ≈ 118 × 8 × 5 = 4720

INSURANCE EXAMS

1. (2) Let *x* will come in the place of question mark (?)
21.0091 - 6.085 + 13.24 = (3.5 + *x*) × 2
28.1641 = (3.5 + *x*) × 2

$3.5 + x = \frac{28.1641}{2} = 14.082$

$\Rightarrow x = 14.082 - 3.5 = 10.582 \approx 10.5$

2. (3) 9876 ÷ 24.96 + 215.006 - ? = 309.99
$\Rightarrow$? = 9876 ÷ 24.96 + 215.005 - ? = 309.99
Approximating the terms to the nearest values, we have
? = 9900 ÷ 25 + 215 - 310
$= \frac{9900}{25} + 215 - 310$
= 396 + 215 - 310
= 301 ≈ 300

3. (2) ? ≈ 8538 - 2416 - 222 = 5900
[8537.986 ≈ 8538]

4. (3) $? \approx \frac{1020}{60} = 17$

5. (4) $? = \frac{111111}{1111 \times 11} = 9.09 \approx 9$

6. (5) $\sqrt[3]{5000} \approx 17$

7. (1) $? \approx 16 \times 31 \times 9 = 4464 \approx 4450$

8. (3) $? \times 30 \approx 17000 \div 80 \times 20$

$\Rightarrow ? \times 30 = \frac{17000}{80} \times 20 = 4250$

$\Rightarrow ? = \frac{4250}{30} = 141.66 \approx 140$

9. (2) $? \approx \frac{900}{50} \times \frac{1200}{20} \times \frac{100}{300} = 360 \approx 350$

10. (1) $? \approx (10)^4 + (30)^2 - (17)^2$
$= 1000 + 900 - 289 = 1611 \approx 1610$

11. (3) $? \approx \frac{58 \times 46 \times 10}{33} = 808.48 \approx 810$

12. (2) $\frac{80 \times 800}{100} - 260 \approx ? - \frac{66 \times 500}{100}$

$\Rightarrow 640 - 260 = ? - 330$

$\Rightarrow ? = 640 - 260 + 330 = 710 \approx 700$

13. (3) $? \approx \frac{25}{9} \times \frac{16}{50} \times 90 = 80$

14. (4) $? \approx \frac{4}{9} \times 5670 - \frac{8}{15} \times 2520$

$= 2520 - 1344 = 1176 \approx 1175$

15. (1) $? = 4568.6531 + 134.675 - 2431.3178$
$= 2272.0103 \approx 2272$

16. (2) $\approx 5680 \times \frac{25}{100} + 4300 \times \frac{45}{100}$

$= 1420 + 1995 = 3355$

17. (2) $? \approx 7^2 + 8^2 - \sqrt{81}$

$= 49 + 64 - 9 = 104$

AVERAGE

NATIONALISED BANKS & IBPS PO/MT/SO

1. The average of 5 numbers is 306.4. The average of the first two numbers is 431 and the average of the last two numbers is 214.5. What is the third number ?
(1) 108
(2) 52
(3) 321
(4) Cannot be determined
(5) None of these

(Bank Of Maharashtra PO Exam. 25.05.2008)

2. Kamlesh bought 65 books for Rs. 1,050 from one shop and 50 books for Rs. 1,020 from another. What is the average price he paid per book ?
(1) Rs. 36.40 (2) Rs. 18.20
(3) Rs. 24 (4) Rs. 18
(5) None of these

(Bank Of Maharashtra PO Exam. 25.05.2008)

3. The sum of five numbers is 290. The average of the first two numbers is 48.5 and the average of last two numbers is 53.5. What is the third number ?
(1) 72 (2) 84
(3) 96 (4) 108
(5) None of these

(Indian Overseas Bank PO Exam. 05.04.2009)

4. The average weight of a group of 53 girls was calculated as 58 kgs. It was later discovered that the weight of one of the girls was read as 65 kgs., whereas her actual weight was 45 kgs. What is the actual average weight of the group of 53 girls ? (rounded off to two digits after decimal)
(1) 58.62 kgs. (2) 58.37 kgs.
(3) 57.37 kgs. (4) 57.62 kgs.
(5) None of these

(Indian Overseas Bank PO Exam. 05.04.2009)

5. The average marks in Science subject of a class of 20 students is 68. If the marks of two students were misread as 48 and 65 of the actual marks 72 and 61 respectively, then what would be the correct average ?
(1) 68.5 (2) 69
(3) 69.5 (4) 70
(5) 66

(Corporation Bank PO Exam. 22.11.2009)

6. The average age of the family of five members is 24. If the present age of youngest member is 8 years, then what was the average age of the family at the time of the birth of the youngest member ?
(1) 20 years (2) 16 years
(3) 12 years (4) 18 years
(5) 21 years

(Corporation Bank PO Exam. 22.11.2009)

7. The average of 5 numbers is 65. The average of the first two numbers is 81 and the average of the last two numbers is 38. What is the third number ?
(1) 63
(2) 87
(3) 99
(4) Cannot be determined
(5) None of these

(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

8. The total marks obtained by a student in Physics, Chemistry and Mathematics together is 120 more than the marks obtained by him in Chemistry. What are the average marks obtained by him in Physics and Mathematics together ?
(1) 60
(2) 120
(3) 40
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 21.02.2010)

9. The average age of 80 boys in a class is 15 years. The average age of a group of 15 boys in the class is 16 years and the average age of another 25 boys in the class is 14 years. What is the average age of the remaining boys in the class ?
(1) 15.25 years
(2) 14 years
(3) 14.75 years
(4) Cannot be determined
(5) None of these

(Corporation Bank PO Exam. 09.05.2010)

10. The average age of seven boys sitting in a row facing North is 26 years. If the average age of first three boys is 19 years and the average age of last three boys is 32 years, what is the age of the boy who is sitting in middle of the row ?
(1) 28 years (2) 29 years

(3) 24 years (4) 31 years
(5) None of these

(Punjab & Sind Bank PO Exam. 16.05.2010)

11. Ram's present age is three times his son's present age and two-fifth of his father's present age. The average of the present ages of all of them is 46 years. What is the difference between the Ram's son's present age and Ram's father's present age ?
(1) 68 years
(2) 88 years
(3) 58 years
(4) Cannot be determined
(5) None of these

(Bank Of Baroda PO Exam. 30.05.2010)

12. The average of five numbers is 56. If the average of first four numbers is 54, what is the value of the fifth number ?
(1) 68 (2) 72
(3) 56 (4) 64
(5) None of these

(Bank Of India PO Exam. 31.10.2010)

13. The average weight of 45 students in a class was calculated as 36 kg. It was later found that the weight of two students in the class was wrongly calculated. The actual weight of one of the boys in the class was 32 kg., but it was calculated as 34 kg. and the weight of another boy in the class was 45 kg. whereas it was calculated as 40 kg. What is the actual average weight of the 45 students in the class ?
(Rounded off to two-digits after decimal)
(1) 36.07 kg.
(2) 36.16 kg.
(3) 35.84 kg.
(4) Cannot be determined
(5) None of these

(PNB Management Trainee Exam. 28.11.2010)

14. Seema's present age is four times her son's present age and four-seventh of her father's present age. The average of the present ages of all three of them is 32 years. What is the difference between the Seema's son's present age and Seema's father's present age ?
(1) 44 years
(2) 48 years
(3) 46 years
(4) Cannot be determined
(5) None of these

(Bank Of Maharashtra Exam. 19.12.2010)

15. The sum of eight consecutive even numbers of set-A is 376. What is the sum of different set of five consecutive numbers whose lowest number is 15 more than the mean of set-A ?
(1) 296 (2) 320
(3) 324 (4) 284
(5) None of these

(Union Bank Of India PO Exam. 09.01.2001)

16. Average score of Rahul, Manish and Suresh is 63. Rahul's score is 15 less than Ajay and 10 more than Manish. If Ajay scored 30 marks more than the average score of Rahul, Manish and Suresh, what is the sum of Manish's and Suresh's scores ?
(1) 120
(2) 111
(3) 117
(4) Cannot be determined
(5) None of these

(Corporation Bank PO Exam. 16.01.2011)

17. The sum of 8 consecutive odd numbers is 656. Also average of four consecutive even numbers is 87. What is the sum of the smallest odd number and second largest even number ?
(1) 165
(2) 175
(3) 163
(4) Cannot be determined
(5) None of these

(Bank Of Baroda PO Exam. 13.03.2011)

18. The average marks in English subject of a class of 24 students is 56. If the marks of three students were misread as 44, 45 and 61 of the actual marks 48,59 and 67 respectively, then what would be the correct average ?
(1) 56 (2) 55
(3) 57.5 (4) 58.5
(5) None of these

(IBPS Bank PO/MT CWE Exam. 18.09.2011)

19. Amit's present age is three times Pratibha's present age and nine-thirteenth of his father's present age. The sum of the present age of all of them is 150 years. What is the difference between Pratibha's present age and Amit's father's present age ?
(1) 56 years
(2) 64 years
(3) 60 years
(4) Cannot be determined
(5) None of these

(IDBI Bank Officer Exam. 16.09.2012)

20. Farah was married 8 years ago. Today her age is $1\frac{2}{7}$ times to that at the time of marriage. At present her daughter's age is $\frac{1}{6}$th of her age. What was her daughter's age 3 years ago ?
(1) 6 years
(2) 7 years
(3) 3 years
(4) Cannot be determined
(5) None of these

(IBPS Specialist Officer CWE 17.03.2013)

21. The average of four consecutive even numbers P, Q, R and S respectively (in increasing order) is 51. What is the product of P and R ?
(1) 2592 (2) 2400
(3) 2600 (4) 2808
(5) None of these

(IBPS RRBs Office Assistant CWE Exam. 09.09.2012

22. The average height of 27 persons was recorded as 162 cm. If the height of Shreyas was deleted from the observation, the average height reduced by 1 cm. What was Shreyas' height ?
(1) 184 cm. (2) 226 cm.
(3) 179 cm. (4) 186 cm.
(5) None of these

(IBPS RRBs Office Assistant CWE Exam. 09.09.2013)

23. The average of the 9 consecutive positive integers is 63. The product of the largest and smallest integers is
(1) 3935 (2) 3953
(3) 3853 (4) 3835
(5) 3635

(Indian Overseas Bank PO Online Exam. 01.09.2013

24. The present age of Romila is one fourth of that of her father. After 6 years the father's age will be twice the age of Kapil. If Kapil celebrated fifth birth day 8 years ago, What is Romils's present age ?
(1) 7 years (2) 7.5 years
(3) 8 years (4) 8.5 years
(5) None of these

(IBPS Bank PO/MT CWE-III 26.10.2013)

25. The average height of 16 students is 142 cm. If the height of the teacher is included, the average height increases by 1 cm. The height of the teacher is
(1) 156 cm (2) 159 cm
(3) 158 cm (4) 157 cm
(5) 159.5 cm

(Corporation Bank Specialist Officer (Marketing Exam. 22.12.2014)

SBI PO EXAMS

1. If the ages of P and R are added to twice the age of Q, the total becomes 59; if the ages of Q and R are added to thrice age of P, the total becomes 68 and if the age of P is added to thrice the age of Q and thrice the age of R, the total becomes 108. What is the age of P ?
(1)15 years (2) 19 years
(3) 17 years (4) 12 years
(5) None of these

(SBI Associate Banks PO Exam. 14.02.1999)

2. The product of the ages of Harish and Seema is 240. If twice the age of Seema is more than Harish's age by 4 years, what is Seema's age in years ?
(1) 12 years (2) 20 years
(3) 10 years (4) 14 years
(5) Data inadequate

(SBI Associate Banks PO Exam. 14.02.1999)

3. Of the three numbers, the average of the first and the second is greater than the average of the second and the third by 15. What is the difference between the first and the third of the three numbers ?
(1) 15
(2) 45
(3) 60
(4) Data inadequate
(5) None of these

(SBI Banks PO Exam. 20.08.2000)

4. In Arun's opinion his weight is greater than 65 kg but less than 72 kg. His brother does not agree with Arun and he thinks that Arun's weight is greater than 60 kg but less than 70 kg. His mother's view is that his weight cannot be greater than 68 kg. If all of them are correct in their es-timation, what is the average of different probably weights of .Arun ?
(1) 69 kg (2) 67 kg
(3) 68 kg (4) Data inadequate
(5) None of these

(SBI Banks PO Exam. 20.08.2000)

5. Average weight of three boys P, T and R is $54\frac{1}{3}$ kgs while the average weight of three boys, T, F and G is 53 kgs. What is the average weight of P, T, R, F and H ?
(1) 53.8 kgs (2) 52.4 kgs
(3) 53.2 kgs (4) Data inadequate
(5) None of these

(SBI Associate Banks PO Exam. 21.07.2002)

6. Present age of Rahul is 5 years less than Ritu's present age. If 3 years ago Ritu's age was *x*, which of the following represents Rahul's present age ?
(1) *x* + 3 (2) *x* - 2
(3) *x* - 3 + 8 (4) *x* + 3 + 8
(5) None of these

(SBI PO Exam. 26.11.2006)

7. The average weight of a group of 75 girls was calculated as 47 kgs. It was later discovered that the weight of one of the girls was read as 45 kgs., whereas her actual weight was 25 kgs. What is the actual average weight of the group of 75 girls ? (Rounded off to two digits after decimal)
(1) 46.73 kgs
(2) 46.64 kgs
(3) 45.96 kgs
(4) Cannot be determined
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

8. The difference between the present ages of Arun and Deepak is 14 years. Seven years ago the ratio of their ages was 5:7 respectively. What is Deepak's present age ?
(1) 49 years (2) 42 years
(3) 63 years (4) 35 years
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

9. The average age of 80 girls was 20 years, the average age of 20 of them was 22 years and that of another 20 was 24 years. Find the average age of the remaining girls.
(1) 17 years (2) 19 years
(3) 21 years (4) 15 years
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

10. The average of the ages of Sumit, Krishna and Rishabh is 43 years and the average of the ages of Sumit, Rishabh and Rohit is 49 years. If Rohit is 54 years old, what is Krishna's age ?
(1) 45 years
(2) 24 years
(3) 36 years
(4) Cannot be determined
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

11. Father's age is 30 years more than the son's age. Ten years hence the father's age will become three times the son's age that time. What is son's present age in years ?
(1) 8
(2) 7
(3) 5
(4) Cannot be determined
(5) None of these

(SBI Associate Banks PO Exam. 07.08.2011)

INSURANCE EXAMS

1. The number of heat wave days in the Indian sub continent in 1995 was 29 and deaths due to heat wave were 406. What was the average of deaths per heat wave day in 1995 ?
(1) 12 (2) 13
(3) 14 (4) 15

(United India Insurance Co. AAO Exam. 21.04.2002)

2. If twelve years hence a man is going to be $1\frac{1}{3}$ his age twelve years ago, What is his present age ?
(1) 24 years (2) 72 years
(3) 96 years (4) 84 years

(United India Insurance Co. AAO Exam. 21.04.2002

3. Average weight of three boys P, T and R is $54\frac{1}{3}$ kgs while the average weight of three boys, T, F and G is 53 kgs. What is the average weight of P, T, R, F and H ?
(1) 53.8 kgs (2) 52.4 kgs
(3) 53.2 kgs (4) Data inadequate
(5) None of these

(LIC Assitant Administrative Officer (AAO) Exam. 2006)

4. The mean temperature of Monday to Wednesday was 37°C and of Tuesday to Thursday was 34° C. If the temperature on Thursday was 4/5th that of Monday, then what was the temperature on Thursday ?
(1) 36.5°C (2) 36°C

(3) 35.5°C (4) 34°C

(United India Insurance Co. (AAO) Exam. 11.03.2007)

5. Three yeas ago the average age of A and B was 18 years. While C joining them now, the average becomes 22 years. How old (in years) is C now ?
(1) 24 (2) 27
(3) 28 (4) 30

(United India Insurance Co. (AAO) Exam. 11.03.2007)

6. What fraction must be subtracted from the sum of $\frac{1}{4}$ and $\frac{1}{6}$ to have an average of $\frac{1}{12}$ of all the three fractions ?

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

(3) $\frac{1}{4}$ (6) $\frac{1}{6}$

(LIC Assistant Administrative Officer (AAO) Exam. 07.06.2009)

7. A team of 8 persons joins in a shooting competition. The best marksman scored 85 points. If he had scored 92 points, the average score for the team would have been 84. The number of points, the team scored was :
(1) 645 (2) 665
(3) 588 (4) 672

(LIC Assistant Administrative Officer (AAO) Exam. 07.06.2009)

8. A is as much younger to B as he is elder to C. If the sum of the ages of B and C is 48 years, what is the age of A in years ?
(1) 20 (2) 24
(3) 30 (4) 32

(LIC Assistant Administrative Officer (AAO) Exam. 07.06.2009)

9. The ages of Melwyn and Louis are in the ratio of 7 : 10 respectively. After 6 years the ratio of their ages will be 17 : 23. What is the difference in their ages ?
(1) 8 years (2) 4 years
(3) 12 years (4) 10 years
(5) None of these

(New India Assurance AO Exam. 25.10.2009)

10. The sum of seven consecutive even numbers of a set is 532. What is the average of first four consecutive even numbers of the same set ?
(1) 76 (2) 75
(3) 74 (4) 73
(5) None of these

(United India Insurance AO Exam. 27.03.2011)

11. Four numbers are written in a row. The average of first two numbers is 7, the average of the middle two terms is 2.3 and the average of the last two numbers is 8.4. The average of first number and the last number is
(1) 5.9
(2) 10.7
(3) 13.1
(4) cannot be determined

(New India Insurance AAO Exam. 22.05.2011)

12. The age of Mr. Bhagat in 2002 was 1 /90 of his birth year. What is his age in 2006 ?
(1) 30 (2) 28
(3) 26 (4) 22

(General Insurance Corporation AAO Exam. 11.12.2011)

13. The average of marks obtained in 120 students was 35. If the average of passed candidates was 39 and that of failed candidates is 15, the number of candidates who passed the examination is:
(1) 100 (2) 110
(3) 120 (4) 80

(General Insurance Corporation AAO Exam. 11.12.2011)

14. The mean of *n* observations is $\bar{x}_1$. If the first observation is increased by 1, second by 2, and so on, then their mean is $\bar{x}_2$. The value of $\bar{x}_2 - \bar{x}_1$ is

(1) *n* (2) $\frac{n}{2} + 1$

(3) $\frac{n(n+1)}{2}$ (4) $\frac{(n+1)}{2}$

(Oriental Insurance Company AAO Exam. 08.04.2012)

15. In a group of 5 friends, the sums of ages (in years) of each group of 4 of them are 124, 128, 130, 136 and 142. The age (in years) of the youngest of them is
(1) 18 (2) 21
(3) 23 (4) 27

(United India Insurance AAO Exam. 03.06.2012)

16. The mean of 15 different natural numbers is 13. The maximum value of the second largest of these numbers is

(1) 53 (2) 52
(3) 51 (4) 50

(United India Insurance AAO Exam. 03.06.2012

17. What will be the average of the following set of numbers ?
678, 193, 285, 494, 342, 762
(1) 395 (2) 359
(3) 495 (4) 459
(5) None of these

(LIC Apprentice Development Officer (ADO) Online Exam. 03.02.2013

18. The average weight of M,N and O is equal to 65 kg. The average weight of M and N is equal to 69.5 kg. The average weight of N and O is 66.5 kg. What is the weight of N ?
(1) 74 kg (2) 75 kg
(3) 77 kg (4) 80 kg
(5) None of these

(LIC Assistant Administrative Office (AAO) Exam. 12.05.2013

19. The average temperature of Gwalior, Agra and Lucknow is 37° C. The average temperature of Agra, Gwalior and Delhi is 38° C. If the temperature of Delhi is 39°C, find the ratio of temperature of Delhi to that of Lucknow ?
(1) 12 : 11
(2) 13 : 12
(3) 12 : 13
(4) Can't be determined

(NICL (GIC) AO (Finance Exam. 08.09.2013 (Paper-I)

20. The average of the two-digit numbers, which remain the same when the digits interchange ther positions is:
(1) 55 (2) 55.5
(3) 56 (4) 56.5

(NICL (GIC) AO (Finance Exam. 08.09.2013 (Paper-I)

21. The average of 8 numbers is A and the average of four of these is B. If the average of remainig four is C, then
(1) 2A = B + C (2) A = B + 2C
(3) A = 2B + C (4) A = 2A + 2B

(NICL (GIC) AO (Finance) Exam. 08.09.2013 (Paper-I)

22. Average of first five multiples of 5 is :
(1) 15 (2) 15.1
(3) 15.5 (4) 15.3

(NICL (GIC) AO (Finance) Exam. 08.09.2013 (Paper-I)

23. The sum of five numbers is 555. The average of first two numbers is 75 and the third number is 115. What is the average of last two-numbers ?
(1) 145 (2) 150
(3) 265 (4) 290

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

24. The mean temperature of Monday and Thursday was 36°C. If the temperature on Thursday was $\frac{4}{5}$th of that of Monday, then what was the temperature on Thursday ?
(1) 36.5°C (2) 32°C
(3) 35.5°C (4) 34°C

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (5) **2.** (4)
3. (5) **4.** (4)
5. (2) **6.** (2)
7. (2) **8.** (1)
9. (1) **10.** (2)
11. (5) **12.** (4)
13. (1) **14.** (2)
15. (2) **16.** (2)
17. (3) **18.** (5)
19. (3) **20.** (1)
21. (5) **22.** (5)
23. (2) **24.** (2)
25. (2)

SBI PO EXAMS

1. (4) **2.** (1)
3. (5) **4.** (4)
5. (4) **6.** (2)
7. (1) **8.** (5)
9. (1) **10.** (3)
11. (3)

INSURANCE EXAMS

1. (3) **2.** (4)
3. (4) **4.** (2)
5. (1) **6.** (4)
7. (2) **8.** (2)
9. (3) **10.** (4)
11. (3) **12.** (3)
13. (1) **14.** (4)
15. (2) **16.** (3)
17. (4) **18.** (3)
19. (2) **20.** (1)
21. (1) **22.** (1)
23. (1) **24.** (2)

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (5) According to the question,
Sum of five numbers = 5 × 306.4
= 1532
∴ Third number
= 1532 - 2 × 431 - 2 × 214.5
= 1532 - 862 - 429
= 241

2. (4) Required average

$$= \frac{1050+1020}{65+50}$$

$$= \frac{2070}{115} = \text{Rs. } 18$$

3. (5) Third number
= 290 - 2 × 48.5 - 2 × 53.5
= 290 - 97 - 107 = 86

4. (4) Actual weight of 53 girls
= 53 × 58 + 45 - 65
= 3074 + 45 - 65 = 3054 kg.
∴ Required actual average weight

$$= \frac{3054}{53} = 57.62 \text{ kg.}$$

Or,
Decrease in weight = 65-45
= 20 kg
∴ Required actual average weight

$$= \left(58 - \frac{20}{53}\right) \text{kg} = 57.62 \text{ kg.}$$

5. (2) Difference of marks
= 72 + 61 - 48 - 65 = 20
∴ Correct average marks

$$= 68 + \frac{20}{20} = 69$$

6. (2) Total age of 5 member family
= 24 × 5 = 120 years
Total age 8 years ago
= 120 - 5 × 8 = 80 years
∴ Required average age

$$= \frac{80}{5} = 16 \text{ years}$$

7. (2) The third number
= 5 × 65 - 2 × 81 - 2 × 38
= 325 - 162 - 76 = 87

8. (1) Required average $= \frac{120}{2} = 60$

9. (1) <u>Tricky Approach</u>
Let the required average age be x years.
∴ 80 × 15 = 15 × 16 + 25 × 14 + 40 × x
$\Rightarrow 40x$ =1200 - 240 - 350 = 610

$$\therefore x = \frac{610}{40} = 15.25 \text{ years}$$

10. (2) Age of the fourth boy
= (26 × 7 - 3 × 19 - 3 × 32) years
= (182 - 57 - 96) years
= 29 years

11. (5) <u>Tricky approach</u>
Present age of Ram's son = x years
∴ Ram's present age = 3x years

Ram's father's present age = $3x \times \frac{5}{2}$

$= \frac{15x}{2}$ years

$\therefore\ x + 3x + \frac{15x}{2} = 46 \times 3$

$\Rightarrow 23x = 46 \times 3 \times 2$

$\Rightarrow x = 12$

$\therefore$ Required difference

$= \frac{15x}{2} - x = \frac{13x}{2}$

$= \frac{13 \times 12}{2} = 78$ years

12. (4) Fifth number
$= 5 \times 56 - 4 \times 54$
$= 280 - 216 = 64$

13. (1) Actual weight of all the students
$= 36 \times 45 - 34 + 32 - 40 + 45$
$= 1620 + 3 = 1623$kg.
$\therefore$ Actual average weight

$= \frac{1623}{45} = 36.07$kg.

14. (2) Let Seema's son's present age = x years
$\therefore$ Seema's present age = $4x$ years
$\therefore$ Seema's husband's present age

$= 4x \times \frac{7}{4} = 7x$ years

$\therefore\ x + 4x + 7x = 32 \times 3$

$\Rightarrow 12x = 96 \qquad \Rightarrow x = \frac{96}{12} = 8$

$\therefore$ Required difference
$= 7x - x = 6x$
$= 6 \times 8 = 48$ years

15. (2) Mean of set A $= \frac{376}{8} = 47$

The lowest number of second set
$= 47 + 15 = 62$
$\therefore$ Required $= 62 + 63 + 64 + 65 + 66$
$= 320$

16. (2) Ajay's score $= 63 + 30 = 93$
$\therefore$ Rahul's score $= 93 - 15 = 78$
$\therefore$ Sum of Manish's and Suresh's scores
$= 3 \times 63 - 78 = 189 - 78 = 111$

17. (3) $x + x + 2 + x + 4 + x + 6 + x + 8 + x + 10 + x + 12 + x + 14 = 656$

$\Rightarrow\ 8x + 56 = 656$

$\Rightarrow\ 8x = 600$

$\Rightarrow\ x = \frac{600}{8} = 75$

and,

$x + x + 2 + x + 4 + x + 6 = 87 \times 4$

$\Rightarrow\ 4x + 12 = 348$

$\Rightarrow\ 4x = 336$

$\Rightarrow\ x = \frac{336}{4} = 84$

$\therefore$ Required sum $= 75 + (84 + 4) = 163$

18. (5) Difference $= 48 + 59 + 67 - 44 - 45 - 61$
$= 24$
$\therefore$ Correct average

$= 56 + \frac{24}{24} = 57$

19. (3) Pratibha's present age = x years
$\therefore$ Amit's present age = $3x$ years
$\therefore$ Father's present age

$= \frac{3x \times 13}{9} = \frac{13x}{3}$ years

$\therefore x + 3x + \frac{13x}{3} = 150$

$\Rightarrow 3x + 9x + 13x = 150 \times 3$

$\Rightarrow 25x = 150 \times 3$

$\Rightarrow = \frac{150 \times 3}{25} = 18$

$\therefore$ Required difference

$= \frac{13x}{3} - x = \frac{10x}{3}$

$= \frac{10}{3} \times 18 = 60$ years

20. (3) Let Farah's age 8 years ago be x years
Farah's present age = $(x + 8)$ years

$\therefore x + 8 = \frac{9x}{7} \qquad \Rightarrow 7x + 56 = 9x$

$\Rightarrow 2x = 56 \qquad \Rightarrow x = 28$

Farah's present age
$= 28 + 8 = 36$ years
Her daughter's age 3 years ago

$= 36 \times \frac{1}{6} - 3 = 3$ years

21. (5) $x + x + 2 + x + 4 + x + 6$
$= 4 \times 51$
$\Rightarrow 4x + 12 = 204$
$\Rightarrow 4x = 204 - 12 = 192$
$\Rightarrow x = 48 = P$
$\therefore R = 48 + 4 = 52$
$\therefore P \times R = 48 \times 52 = 2496$

22. (5) Height of Shreyas
$= 162 \times 27 - 161 \times 26 = 188$ cm

23. (2) Average of 9 consecutive positive integers = 63
$\therefore$ Fifth number = 63
$\therefore$ Smallest number = 59
Largest number = 67
$\therefore$ Required product
= 59 × 67 = 3953

24. (2) Kapil's present age = 12 years
After 6 years, Kapil's age = 18 years
$\therefore$ Father's present age = 18 × 2 - 6
= 30 years

$\therefore$ Romlla's present age = $\frac{1}{4}$ × 30 = 7.5 years

25. (2) Height of teacher = 143 × 17 - 142 × 16
= 159 cm

SBI PO EXAMS

1. (4) According to question,
P + R + 2Q = 59 ....(i)
Q + R + 3P = 68 .... (ii)
P + 3Q + 3R = 108 .....(iii)
Subtract equation (iii) from thrice the equation (ii), we get
3Q + 3R + 9P - P - 3Q - 3R = 204 - 108
$\Rightarrow$ 8P = 96

$\Rightarrow$ P = $\frac{96}{8}$ = 12 years

2. (1) Let the ages of Harish and Seema be x and y years respectively.
$\therefore$ According to the question,
$x \times y = 240$
and $2y = x + 4$
Now,
$(2y - 4) \times y = 240$
$\Rightarrow 2y^2 - 4y - 240 = 0$
$\Rightarrow y^2 - 2y - 120 = 0$
$\Rightarrow y^2 - 12y + 10y - 120 = 0$
$\Rightarrow y(y - 12) + 10(y - 12) = 0$
$\Rightarrow (y + 10)(y - 12) = 0$
$\Rightarrow y = -10, 12$
$\therefore$ age of Seema = 12 years

3. (5) Let the three numbers be x, y and z.

or, $\frac{x+y}{2} - \frac{y+z}{2} = 15$

or, $\frac{x+y-y-z}{2} = 15$

or, $x - z = 30$

4. (4) Arun's opinion : 65 < W < 72
Brother's opinion : 60 < W < 70
Mother's opinion : W not greater than 68
or W < 68

5. (4) We are to determine the average weight of P, T, R, F and H. Obviously, this can't be determined as we don't know tin weight of H.

6. (2) 3 years ago Ritu's age = x
$\therefore$ Ritu's present age = $x + 3$
Rahul's present age = Ritu present age - 5
= $x + 3 - 5 = x - 2$

7. (1) Total actual weight of all gin
= 47 × 75 - 45 + 25
= 3525- 20 = 3505 kg.
$\therefore$ Average weight

= $\frac{3505}{75}$ = 46.73 kg

8. (5) Seven years ago, let Arun's and Deepak's age be $5x$ and $7x$ years respectively.
$\therefore$ Arun's present age = $(5x + 7)$ years
Deepak' present age = $(7x + 7)$ years
According to the question,
$7x + 7 - 5x - 7 = 14$
$\Rightarrow 2x = 14$

$\Rightarrow x = \frac{14}{2} = 7$

$\therefore$ Deepak's present age = $7x + 7$
= 7 × 7 + 7 = 56 years

9. (1) Total age of remaining 40 girls
= (80 × 20 - 20 × 22 - 20 × 24) years
= (1600 - 440 - 480) years = 680 years

$\therefore$ Required average age = $\frac{680}{40}$ = 17 years

10. (3) (Sumit + Krishna + RishabhJ's age
= 43 × 3 = 129 years ....(i)
(Sumit + Rishabh + Rohit)'s age
= 3 × 49 = 147 years ....(ii)
By equation (i)-(ii), we have
Krishna's age - Rohit's age = 129 - 147
= -18
$\therefore$ Krishna's age = 54 - 18 = 36 years

11. (3) Let the son's present age be x years.
$\therefore$ Father's present age = $(x + 30)$ years
After 10 years,
$x + 40 = 3(x + 10)$
$\Rightarrow x + 40 = 3x + 30$
$\Rightarrow 2x = 10 \Rightarrow x = 5$ years

INSURANCE EXAMS

1. (3) Required answer = $\frac{406}{29}$ = 14

2. (4) Let present age of man be x years
$\therefore$ According to the question,

$(x + 12) = 1\frac{1}{3} \times (x - 12)$

or, $x + 12 = \frac{4}{3}(x - 12)$

or, $x + 12 = \frac{4x}{3} - 16$

or, $\frac{4x}{3} - x = 12 + 16$

or, $\frac{x}{3} = 28$

$\therefore x = 28 \times 3 = 84$ years

3. (4) We are to determine the average weight of P, T, R, F and H. Obviously, this can't be determined as we don't know the weight of H.

4. (2) M + T + W = 3 × 37 = 111°C
T + W + Th = 3 × 34 = 102° C
Subtracting equation (ii) from equation (i),
M - Th = (111 -102)°C

$\Rightarrow \frac{5}{4}$Th - Th = 9°C

$\Rightarrow \frac{\text{Th}}{4} = 9^{\circ}\text{C} \Rightarrow$ Th = 9 × 4 = 36°C

5. (1) Three years ago,
Sum of the ages of A and B = 18 × 2
= 36 years
Sum of the present ages of A and B = 36 + 6
= 42 years
Sum of the present ages of A, B and C
= 3 × 22 = 66 years
$\therefore$ Present age of C = 66 - 42 = 24 years

6. (4) Required fraction

$= \frac{1}{4} + \frac{1}{6} - 3 \times \frac{1}{12}$

$= \frac{1}{4} + \frac{1}{6} - \frac{1}{4} = \frac{1}{6}$

7. (2) The numbers of points term scored
= 8 × 84 - 92 + 85 = 672 - 92 + 85 = 665

8. (2) According to the question,
B - A = A - C
$\Rightarrow$ 2A = B + C = 48

$\Rightarrow$ A = $\frac{48}{2}$ = 24 years

9. (3) Let the present ages of Melwyn and Louis be 7x and 10x years respectively.
After 6 years

$\frac{7x+6}{10x+6} = \frac{17}{23}$

$\Rightarrow$ 170 x + 102 = 161 x + 138
$\Rightarrow$ 9x = 138 - 102 = 36

$\Rightarrow x = \frac{36}{9} = 4$

$\therefore$ Required difference = 10x - 7x = 3x
= 3 × 4 = 12 years

10. (4) Fourth even numbers = $\frac{532}{7} = 76$

First four even numbers are 70, 72, 74 and 76.
$\therefore$ Required average

$= \frac{70+72+74+76}{4}$

$= \frac{292}{4} = 73$

11. (3) Let the numbers be a, b, c and d respectively.
$a + b = 14$ ...(i)
$b + c = 4.6$ ...(ii)
$c + d = 16.8$ ...(iii)
By equations (i) - (ii) + (iii)
$a + d$ = 14 - 4.6 + 16.8 = 26.2

$\therefore \frac{a+d}{2} = \frac{26.2}{2} = 13.1$

12. (3) Let the birth year be x.

$\therefore \frac{1}{90}x = 2002 - x$

$\Rightarrow x = 2002 \times 90 - 90x$
$\Rightarrow 91x = 2002 \times 90$
$\Rightarrow x = 1980$
$\therefore$ Bhagat's age in 2006
= 2006 - 1980
= 26 years

13. (1) If the number of successful candidates be x, then number of unsuccessful candidates = 120 - x
$\therefore x \times 39 + (120 - x)\,15 = 120 \times 35$
$\Rightarrow 39x + 1800 - 15x = 4200$
$\Rightarrow 24x = 4200 - 1800 = 2400$
$\Rightarrow x = 100$

14. (4) New mean = $\bar{x}_2$

$= \frac{x_1 + x_2 + \ldots\ldots + x_x}{n} + \frac{1+2+3+\ldots.+n}{n}$

$= \bar{x}_1 + \frac{n(n+1)}{2n}$

$= \bar{x}_1 + \frac{n+1}{2}$

$\therefore \overline{x_2} - \overline{x_1} = \overline{x_1} + \frac{n+1}{2} - \overline{x_1}$

$= \frac{n+1}{2}$

15. (3) $a + b + c + d = 124$
$b + c + d + e = 128$
$c + d + e + a = 130$
$d + e + a + b = 136$
$e + a + b + c = 142$
On adding,
$4(a + b + c + d + e) = 660$
$\therefore a + b + c + d + e = 165$
$\therefore$ Age of the youngest of them
$= d = 165 - 142 = 23$ years

16. (3) Sum of 15 numbers $= 15 \times 13 = 195$
For maximum value, the other 13 numbers will be least.
$\therefore 1 + 2 + ...+ 13$

$= \frac{13\times14}{2} = 91$

$\therefore$ Sum of last two numbers
$= 195 - 91 = 104$
$\therefore$ Numbers = 51 and 53

17. (4) Sum of numbers
$= 678 + 193 + 285 + 494 + 342 + 762$
$= 2754$
$\therefore$ Required average

$= \frac{2754}{6} = 459$

18. (3) $M + N + O = 3 \times 65 = 195$ kg
$M + N = 2 \times 69.5 = 139$ kg.
$N + O = 2 \times 66.5 = 133$ kg.
$\therefore$ Weight of N = (139 + 133 - 195) kg.
= 77 kg.

19. (2) Gwalior + Agra + Lucknow
$= 37 \times 3 = 111°C$ ...(i)
Agra + Delhi + Gwalior
$= 38 \times 3 = 114°C$ ...(ii)
By equation (i) - (ii)
Lucknow - Delhi = 111 - 114 = - 3°C
$\therefore$ Lucknow = -3 + 39 = 36°C
$\therefore$ Required ratio
= 39 : 36 = 13 : 12

20. (1) Required average

$= \frac{11+22+33+...+99}{9}$

$= \frac{11(1+2+3+....+9)}{9}$

$= \frac{11\times45}{9} = 55$

21. (1) $8A = 4B + 4C$
$\Rightarrow 2A = B + C$

22. (1) Required average

$= \frac{5+10+15+20+25}{5}$

$= \frac{5\times(1+2+3+4+5)}{5} = 15$

23. (1) Sum of last two numbers
$= 555 - 2 \times 75 - 115$
$= 555 - 150 - 115 = 290$

$\therefore$ Required average $= \frac{290}{2} = 145$

24. (2) Monday + Thursday $= 2 \times 36 = 72°C$
If the temperature on Thursday be T°C, then

temperature on Monday $= \frac{5}{4}T$

$\therefore \frac{5}{4}T + T = 72$

$\Rightarrow \frac{9T}{4} = 72$

$\Rightarrow T = \frac{4\times32}{9} = 32°$

MODEL EXERCISES

1. A batsman makes a score of 98 runs in the 19th innings and thus increases his average by 4. What is his average after 19th innings ?
(1) 22 (2) 24
(3) 28 (4) 26
(5) None of these

2. The average weight of 45 students in a class is 52 kg. 5 of them whose average weight is 48 kg leave the class and other 5 students whose average weight is 54 kg join the class. What is the new average weight (in kg) of the class ?
(1) 52.6 (2) $52\frac{2}{3}$
(3) $52\frac{1}{3}$ (4) $52\frac{3}{4}$
(5) None of these

3. In three numbers, the first is twice the second and thrice the third. If the average of these three numbers is 44, then the first number is —
(1) 72 (2) 24
(3) 36 (4) 44
(5) None of these

4. The batting average of 40 innings of a cricket player is 50 runs. His highest score exceeds his lowest score by 172 runs. If these two innings are excluded the average of the remaining 38 innings is 48. His highest score was :
(1) 172 (2) 173
(3) 174 (4) 176
(5) None of these

5. Nine persons went to a hotel for taking their meals. Eight of them spent Rs 12 each over their meals and the ninth spent Rs 8 more than the average expending of all the nine. Total money spent by them was :
(1) Rs 104 (2) Rs 105
(3) Rs 116 (4) Rs 117
(5) None of these

6. A demographic survey of 100 families in which two parents were present revealed that the average age A, of the oldest child is 20 years less than $\frac{1}{2}$ the sum of the ages of the two parents. If F represents the age of one parents and M the age of the other parents, then which of the following is equivalent to A ?
(1) $\frac{F+M-20}{2}$ (2) $\frac{F+M}{2}+20$
(3) $\frac{F+M}{2}-20$ (4) $F + M - 10$
(5) None of these

7. The average of three numbers is 135. The largest number is 180 and the difference between the others is 25. The smallest number is —
(1) 130 (2) 125
(3) 120 (4) 100
(5) None of these

8. Of the three numbers, the first is twice the second and the second is twice the third. The average of these numbers is 21. Find the largest number out of these.
(1) 36 (2) 38
(3) 47 (4) 48
(5) None of these

9. There are 30 students in a class. The average age of the first 10 students is 12.5 years. The average age of the next 20 students is 13.1 years. The average age of the whole class is —
(1) 12.5 years (2) 12.7 years
(3) 12.8 years (4) 12.9 years
(5) None of these

10. The weight in kilograms of 10 students are 52, 45, 31, 35, 40, 55, 60, 38, 44, 36. If 44 is replaced by 46 and 40 is replacec by 35, then the new median will be —
(1) 42 (2) 40.5
(3) 40 (4) 41.5
(5) None of these

11. In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs ?
(1) 6.25 (2) 6.50
(3) 6.75 (4) 7
(5) None of these

12. The average (arithmetic mean) of x and y is 40. If $z = 10$, then what is the average of x, y and z ?
(1) $16\frac{2}{3}$ (2) 30
(3) 25 (4) $17\frac{1}{2}$
(5) None of these

13. The mean of 1, 7, 5, 3, 4 and 10 is m. The mean of 10, 12, 3, 5, 1, 7 and p is $m + 1$ and their median is q. Then, q is equal to —
(1) 4 (2) 5
(3) 10 (4) 12
(5) None of these

14. In a class, the average score of girls in an examination is 73 and that of boys is 71. The average score for the whole class is 71.8. The percentage of the girls in the class is —
(1) 60% (2) 40%
(3) 1.8% (4) 18%
(5) None of these

15. The average of 10 numbers is 40.2. Later it is found that two numbers have been wrongly added. The first is 18 greater than the actual number and the second number added is 13 instead of 31. Find the correct average.
(1) 40.2 (2) 40.4
(3) 40.6 (4) 40.8
(5) None of these

16. The average of 11 numbers is 10.9. If the average of the first six numbers is 10.5 and that of the last six numbers is 11.4, then the middle number is —
(1) 11.5 (2) 11.4
(3) 11.3 (4) 11.0
(5) None of these

17. The average temperature of the first four days of a week is 40.2°C and that of the last four days is 41.3°C. If the average temperature for the whole week is 40.6°C, then the temperature on the fourth day is —
(1) 40.8°C (2) 38.5°C
(3) 41.3°C (4) 41.8°C
(5) None of these

18. The average salary of all the workers in a workshop is Rs 8000. The average salary of 7 technicians is Rs 12000 and the average salary of the rest is Rs 6000. The total number of workers in the workshop is —
(1) 21 (2) 20
(3) 23 (4) 22
(5) None of these

19. A number of friends decided to go on a picnic and planned to spend Rs 96 on eatables. Four of them, did not turn up. As a consequence, the remaining ones had to contribute Rs 4 each extra. The number of those who attended the picnic was —
(1) 8 (2) 16
(3) 12 (4) 24
(5) None of these

20. Some students planned a picnic. The budget for food was Rs 500. But 5 of them failed to go and thus the cost of food for each member increased by Rs 5. How many students attend the picnic ?
(1) 15 (2) 25
(3) 20 (4) 30
(5) None of these

21. The average score of boys in an examination in a school is 71 and that of the girls is 73. The average score of the school is 71.8. The ratio of the number of boys to that of the girls that appeared in the examination is —
(1) 1 : 2 (2) 3 : 2
(3) 2 : 3 (4) 4 ; 2
(5) None of these

22. The mean monthly salary paid to 75 workers in a factory is Rs 5680. The mean salary of 25 of them is Rs 5400 and that of 30 others is Rs 5700. The mean salary of the remaining workers is —
(1) Rs 5000 (2) Rs 7000
(3) Rs 6000 (4) Rs 8000
(5) None of these

23. The average monthly expenditure of a family was Rs 2200 during the first 3 months; Rs 2250 during the next 4 months and Rs 3120 during the last 5 months of a year. If the total savings during the year were 1260, then the average monthly income was—
(1) Rs 2605 (2) Rs 2805
(3) Rs 2705 (4) Rs 2905
(5) None of these

24. A monthly return railway ticket costs 25% more than a single ticket. A week's extension can be had for the former by paying 5% of the monthly ticket's cost. If the money paid for the monthly ticket (with extension) is 84, the price of the single ticket is —
(1) Rs 64 (2) Rs 80
(3) Rs 48 (4) Rs 72
(5) None of these

25. The average monthly salary of employees, consisting of officers and workers of an organization is Rs 3000. The average salary of an officer is Rs 10000 while that of a worker is Rs 2000 per month. If there are total 400 employees in the organization, find the number of officers and workers separately.
(1) 50, 350 (2) 350, 450
(3) 50, 275 (4) 325, 350
(5) None of these

SHORT ANSWERS

1.	(4)	**2.**	(2)
3.	(1)	**4.**	(3)
5.	(4)	**6.**	(3)
7.	(4)	**8.**	(1)
9.	(4)	**10.**	(4)
11.	(1)	**12.**	(2)
13.	(2)	**14.**	(2)
15.	(1)	**16.**	(1)
17.	(4)	**18.**	(1)
19.	(1)	**20.**	(3)
21.	(2)	**22.**	(3)
23.	(3)	**24.**	(1)
28.	(1)		

EXPLANATIONS

1. (4) Let us suppose that the average of 18 innings be x.

Then, $\frac{18x+98}{19} = x + 4$

$\Rightarrow 19x + 76 = 18x + 98$

$\Rightarrow x = 22$

Thus, average after 19th innings
$= x + 4 = 22 + 4 = 26$

2. (2) According to the question,
Total weight of 45 students
$= 45 \times 52 = 2340$ kgs
Total weight of 5 students who leave
$= 5 \times 48 = 240$ kgs
Total weight of 5 students who join
$= 5 \times 54 = 270$ kgs
Hence, new total weight of 45 students
$= 2340 - 240 + 270 = 2370$

$\therefore$ New Average $= \frac{2370}{45} = 52\frac{2}{3}$ kg

3.(1) Let the three numbers be a, b and c.
$a = 2b$ and $3c$
$a = 3c$

$\therefore b = \frac{a}{2}$ and $c = \frac{a}{3}$

Given, $\frac{a+\frac{a}{2}+\frac{a}{3}}{3} = 44$

$\Rightarrow \frac{11a}{18} = 44$

$\Rightarrow a = 72$

4. (3) According to total of 40 innings
$= 40 \times 50 = 2000$
Total of 38 innings $= 38 \times 48 = 1824$
$\therefore$ Sum of highest and lowest score

$\Rightarrow x + y = 2000 - 1824$

$\Rightarrow x + y = 176$

and $x - y = 172$

Solving Eqs. (i) and (ii), we get

$x = 174$

5. (4) Let the ninth person spent Rs x.
Then, average of all the nine

$= \frac{12\times 8+x}{9} = \frac{96+x}{9}$

Given, $x = \frac{96+x}{9} + 8$

$9x = 96 + x + 72$

$\Rightarrow 8x = 168$

$\Rightarrow x = 21$

Hence, total money spent
$= 96 + 21 =$ Rs 117

6. (3) According to question

A= $\frac{F+M}{2}$ - 20

7. (4) According to question,

$\frac{(x-25)+(x)+180}{3} = 135$

$\therefore 2x + 155 = 405$

$= x = \frac{250}{2} = 125$

Hence, the smallest number
$= 125 - 25 = 100$

8. (1) Let the numbers be a, b and c. Then,

$a = 2b$ and $b = 2c$ or $c = \frac{b}{2}$

Then, $\frac{2b+b+\frac{b}{2}}{3} = 21$

$\Rightarrow \frac{7b}{6} = 21 \Rightarrow b = 18$

$\therefore$ Numbers are 36, 18 and 9
$\therefore$ The largest number is 36.

9. (4) Total age of 10 students = 125 years
Total age of 20 students = 262 years
$\therefore$ Average age of 30 students

$= \frac{125+262}{30} = 12.9$ years

10. (4) According to question, replacing the two number and arranging them in ascending order, we get 31, 35, 35, 36, 38, 45, 46, 52, 55, 60

$\therefore$ Median $= \frac{38+45}{2} = \frac{83}{2} = 41.5$

11. (1) Total runs in the first 10 overs
$= 10 \times 3.2 = 32$
Runs rate in the remaining 40 overs

$= \frac{282-32}{40} = \frac{250}{40} = 6.25$

12. (2) Total of x and $y = 40 \times 2 = 80$
$\therefore$ Average of x, y and z

$= \frac{80+10}{3} = \frac{90}{3} = 30$

13. (2) According to question,

$m = \frac{1+7+5+3+4+10}{6}$

$\Rightarrow m = 5$
and $(m + 1)$

$= \frac{10+12+3+5+1+7+p}{7}$

$\Rightarrow \frac{38+p}{7} = m + 1$

$\Rightarrow \frac{38+p}{7} = 5 + 1$

$\therefore p = 42 - 38 = 4$
$\therefore$ Numbers are 1, 3, 4, 5, 7, 10, 12.
$\therefore$ Median $q = 5$

14. (2) Let the number of boys be x and that of girls be y.
Then,
total score of boys $= 71x$
total score of girls $= 73y$

$\therefore \frac{71x+73y}{(x+y)} = 71.8$

$\Rightarrow 71x + 73y = 71.8x + 71.8y$

$0.8x = 1.2y \quad \frac{x}{y} = \frac{1.2}{0.8} = \frac{3}{2}$

$\therefore$ Percentage of girls in the class

$= \frac{2}{5} \times 100 = 40\%$

15. (1) Sum of 10 numbers = 402
Correct sum of 10 numbers
402 - 13 + 31 - 18 = 402

Hence, new average $= \frac{402}{10} = 40.2$

16. (1) Sum of 11 numbers $= 11 \times 10.9$
Sum of first 6 numbers $= 6 \times 10.5$
Sum of last 6 numbers $= 6 \times 11.4$
$\therefore$ The middle number
$= 6 \times 10.5 + 6 \times 11.4 - 11 \times 10.9$
$= 63 + 68.4 - 119.9$
$= 131.4 - 119.9 = 11.5$

17. (4) Temperature on the fourth day
$= 40.2 \times 4 + 41.3 \times 4 - 40.6 \times 7$
$= 160.8 + 165.2 - 284.2 = 41.8°C$

18. (1) Let total no. of workers be x.
Then, $8000x = 7 \times 12000 + (x - 7) \times 6000$
$\Rightarrow 8000x = 84000 + 6000x - 42000$

$\Rightarrow x = \frac{42000}{2000} = 21$

19. (1) Let there were x friends, then contribution of one friend $= \frac{96}{x}$

If four friends have left, then contribution of each friend

$= \frac{96}{x-4}$

$\therefore \frac{96}{x-4} - \frac{96}{x} = 4$

$\Rightarrow x = 12$
Hence, number of friends who attended the picnic $= 12 - 4 = 8$

20. (3) Let there were x students, then contribution of one student $= \frac{500}{x}$

Contribution of each students where 5 of them have left $= \frac{500}{(x-5)}$

Given, $\frac{500}{(x-5)} - \frac{500}{x} = 5$

using options, we find $x = 25$ satisfies the equation. Therefore, number of students who attended the picnic
$= (25 - 5) = 20$.

21. (2) Let the total number of boys and girls be B and G respectively.
$\therefore$ Total score of boys $= 71B$
Total score of girls $= 73G$
Total score of the class $= 71.8 (B+G)$
$\therefore 71 B + 73 G = 71.8 (B+G)$

$\Rightarrow 0.8B = 1.2G \Rightarrow \frac{B}{G} = \frac{1.2}{0.8} = \frac{3}{2}$

22. (3) Total salary of 75 workers $= 75 \times 5680$
$=$ Rs 426000
Total salary of 25 workers $= 25 \times 5400$
$=$ Rs 135000
Total salary of 30 workers $= 30 \times 5700$
$=$ Rs 171000
$\therefore$ Total salary of remaining 20 workers

= 426000 - (135000 + 171000) = 120000

$\therefore$ Mean salary of 20 workers = $\frac{120000}{20}$

= Rs 6000

23. (3) Total expenditure for the year
= [2200 × 3 + 2250 × 4 + 3120 × 5)
= 6600 + 9000 + 15600 = Rs 31200
Total saving = Rs 1260
Total income = expenses + savings
= 31200 + 1260 = Rs 32460

$\therefore$ Average income = $\frac{32460}{12}$ = Rs 2705

24. (1) Required price of the single ticket

= $\frac{84}{105} \times \frac{100}{1} \times \frac{100}{125}$

= 84 × $\frac{100}{105} \times \frac{100}{125}$ = Rs 64

25. (1) Since, there are 400 employes.
hence option (1) is correct.

COMPOUND INTEREST

NATIONALISED BANKS & IBPS PO/MT/SO

1. Suresh invested a sum of Rs. 15000 at 9 per cent per annum simple interest and Rs. 12000 at 8 per cent per annum compound interest for a period of 2 years. What amount of interest did Suresh earn in 2 years ?
(1) Rs. 4096.60 (2) Rs. 4696.80
(3) Rs. 4896.60 (4) Rs. 4698.60
(5) None of these

(Syndicate Bank PO Exam. 10.10.2004)

2. Satish invests Rs. 35,500 in a scheme which earns him simple interest at the rate of 15 p.c.p.a. for two years. At the end of two years he reinvests the principal amount plus interest earned in another scheme which earns him compound Interest at the rate of 20 p.c.p.a. What will be the total interest earned by Satish over the principal amount at the end of 5 years ?
(1) Rs. 30,956.80
(2) Rs. 35,017.20
(3) Rs. 43,597.80
(4) Rs. 44,247.20
(5) None of these

(Union Bank of India PO Exam. 27.11.2005

3. The simple interest accrued on an amount of Rs. 14,800 at the end of three years is Rs. 6,216. What would be the compound interest accrued on the same amount at the same rate in the same period ?
(1) Rs. 6986.1142
(2) Rs. 7042.2014
(3) Rs. 7126.8512
(4) Rs. 8321.4166
(5) None of these

(Corporation Bank PO Exam. 29.07.2006)

4. The simple interest accrued on an amount of Rs. 40,000 at the end of three years is Rs. 12,000. What would be the compound interest accrued on the same amount at the same rate in the same period ?
(1) Rs. 18,765 (2) Rs. 15,350
(3) Rs. 21,555 (4) Rs. 13,240
(5) None of these

(Andhra Bank PO Exam. 14.09.2008)

5. The compound interest accrued on an amount of Rs. 22,000 at the end of two years is Rs. 5,596.8. What would be the simple interest accrued on the same amount at the same rate in the same period ?
(1) Rs.5,420 (2) Rs.5,360
(3) Rs.5,280 (4) Rs.5,140
(5) None of these

(Oriental Bank of Commerce PO Exam. 21.12.2008)

6. The compound interest accrued on an amount of Rs. 25,500 at the end of three years is Rs. 8,440.5. What would be the simple interest accrued on the same amount at the same rate in the same period ?
(1) Rs. 4,650 (2) Rs. 5,650
(3) Rs. 6,650 (4) Rs. 7,650
(5) None of these

(Canara Bank PO Exam. 15.03.2009)

7. What would be the compound interest obtained on an amount of Rs. 20,000 at the rate of 15 p.c.p.a. after 4 years ?
(1) Rs. 14,980.125
(2) Rs. 19,680.125
(3) Rs. 16,780.125
(4) Rs. 18,980.125
(5) None of these

(Indian Overseas Bank PO Exam. 05.04.2009)

8. Mr. Duggal invested Rs. 20,000 with rate of interest at 20 p.c.p.a. The interest was compounded half yearly for first year and in the next year it was compounded yearly. What will be the total interest earned at the end of two years ?
(1) Rs. 8,800 (2) Rs. 9,040
(3) Rs. 8,040 (4) Rs. 9,800
(5) None of these

(United Bank of India PO Exam. 21.06.2009)

9. What will be the compound interest accrued on an amount. Rs. 10,000 at the rate of 20 p.c.p.a. in two years if the interest is compounded half yearly ?
(1) Rs. 4,400 (2) Rs. 4,600
(3) Rs. 4,641 (4) Rs. 4,680
(5) None of these

(Andhra Bank PO Exam. 05.07.2009

10. What would be the compound interest accrued on an amount of Rs. 7,400 at the rate of 13.5 p.c.p.a. at the end of two years ? (rounded off to two digits after decimal)
(1) Rs. 2,136.87
(2) Rs. 2,306.81
(3) Rs. 2,032.18
(4) Rs. 2,132.87
(5) None of these

(Indian Bank PO Exam. 17.10.2010)

11. What would be the compound interest accrued on an amout of Rs. 45,400 at the end of two years at the rate of 15 p.c.p.a ?
(1) Rs. 16411.5 (2) Rs. 14461.5
(3) Rs. 16461.5 (4) Rs. 14641.5
(5) None of these

(Bank Of India Banking Officer Exam. 24.01.2010)

12. Sonika invested an amount of Rs. 5,800 for 2 years. At what rate of compound interest will she get an amount of Rs. 594.5 at the end of two years ?
(1) 5 p.c.p.a. (2) 4 p.c.p.a.
(3) 6 p.c.p.a. (4) 8 p.c.p.a.
(5) None of these

(Corporation Bank PO Exam. 09.05.2010)

13. A man gets a simple interest of Rs. 1,000 on a certain principal at the rate of 5 p.c.p.a. in 4 years. What compound interest will the man get on twice the principal in two years at the same rate ?
(1) Rs. 1,050 (2) Rs. 1,005
(3) Rs. 11,025 (4) Rs. 10,125
(5) None of these

(Punjab & Sind Bank Exam. 16.05.2010)

14. What will be the **approximate** difference in the simple and compound interest accrued on an amount of Rs 2600 at the rate of 15 p.c. p.a. at the end of three years ?
(1) Rs 167 (2) Rs 194
(3) Rs 202 (4) Rs 172
(5) Rs 184

(United Bank Of India PO Exam. 14.11.2010)

15. Pamela invested an amount of Rs 35,000 for two years at the rate of 5 p.c.p.a. What amount of compound interest would she receive at the end of two years ?
(1) Rs 3587.50 (2) Rs 3500
(3) Rs 3580.50 (4) Rs 3565.50
(5) None of these

(PNB Management Trainee Exam. 28.11.2010)

16. The simple interest accrued on a sum of certain principal is Rs 2,000 in five years at the rate of 4 p.c.p.a. What would be the compound interest accrued on same principal at same rate in two years ?
(1) Rs 716 (2) Rs 724
(3) Rs 824 (4) Rs 816
(5) None of these

(Corporation Bank PO Exam. 16.01.2011)

17. The difference between the amount of compound interest and simple interest accured on an amount of Rs 26000 at the end of 3 years is Rs 2994.134. What is the rate of interest p.c.p.a. ?
(1) 22
(2) 17
(3) 19
(4) Cannot be determined
(5) None of these

(Punjab & Sind Bank PO Exam. 23.01.2011)

18. The simple interest accrued on a sum of certain principal is Rs 6500 in eight years at the rate of 13 per cent per year. What would be the compound interest accrued on that principal at the rate of 8 per cent per year in 2 years?
(1) Rs 1040 (2) Rs 1020
(3) Rs 1060 (4) Rs 1200
(5) None of these

(UCO Bank PO Exam. 30.01.2011)

19. The simple interest accrued on a sum of certain principal is Rs 7,200 in six years at the rate of 12 p.c.p.a. What would be the compound interest accrued on that principal at the rate of 5 p.c.p.a. in 2 years ?
(1) Rs 1,020 (2) Rs 1,055
(3) Rs 1,050 (4) Rs 1,025
(5) None of these

(Bank Of Baxoda PO Exam. 13.03.2011)

20. The simple interest accrued on an amount of Rs 22,500 at the end of four years is Rs 10,800. What would be the compound interest accrued on the same amount at the same rate at the end of two years ?
(1) Rs 14,908 (2) Rs 5,724
(3) Rs 26,234 (4) Rs 8,568
(5) None of these

(IBPS Bank PO/MT CWE 18.09.2011)

21. What is the difference between the simple and compound interest on Rs 7300 at the rate of 6 p.c.p.a. in 2 years ?
(1) Rs 29.37 (2) Rs 26.28
(3) Rs 31.41 (4) Rs 23.22
(5) Rs 21.34

(IBPS Bank PO/MT CWE 17.06.2012)

22. The compound interest earned on a sum in 3 years at 15% per annum compounded annually is Rs 6500.52. What is the sum ?
(1) Rs 12480 (2) Rs 10500
(3) Rs 14800 (4) Rs 13620
(5) None of these

(IBPS Specialist Officer CWE 17.03.2013)

23. What would be the compound interest obtained on an amount of Rs. 7,790 at the rate of 10 p.c.p.a. after two years ?
(1) Rs. 1532.60 (2) Rs. 1495.90
(3) Rs. 1653.50 (4) Rs. 1635.90
(5) None of these
(IBPS RRBs Office Assistant CWE Exam. 09.09.2012)

SBI PO EXAMS

1. The difference between the simple interest on a certain sum at the rate of 10% per annum for 2 years and compound interest which is compounded every 6 months is Rs. 124.05. What is the principal sum ?
(1) Rs. 10,000 (2) Rs. 6,000
(3) Rs. 12,000 (4) Rs. 8,000
(5) None of these
(SBI Banks PO Exam. 20.08.2000)

2. Sudharshan invested Rs. 15,000 at interest @ 10 p.c. p.a. for one year. If the interest is compounded every six months what amount will Sudharshan get at the end of the year ?
(1) Rs. 16,537.50
(2) Rs. 16, 500
(3) Rs. 16, 525.50
(4) Rs. 18,150
(5) None of these
(SBI Associate Banks PO Exam. 21.07.2002)

3. What will be the compound interest on a sum of Rs. 25,000 after three years at the rate of 12 per cent p.a. ?
(1) Rs. 10123.20
(2) Rs. 9000.30
(3) Rs. 10483.20
(4) Rs. 9720
(5) None of these
(SBI Banks PO Exam. 18.05.2003)

4. Mr. Rao invests a sum of Rs. 41,250 at the rate of 6 p.c.p.a. What approximate amount of compound interest will he obtain at the end of 3 years ?
(1) Rs.8,100 (2) Rs.7,425
(3) Rs.8,210 (4) Rs.7,879
(5) Rs.7,295
(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

5. The simple interest accrued on an amount of Rs. 20,000 at the end of three years is Rs. 7,200. What would be the compound interest accrued on the same amount at the same rate in the same period ?
(1) Rs. 8342.36
(2) Rs. 8098.56
(3) Rs. 8246.16
(4) Rs. 8112.86
(5) None of these
(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

6. The compound interest accrued on an amount at the end of three years at 15 p.c.p.a. is Rs. 6500.52. What is the amount ?
(1) Rs. 12480 (2) Rs. 10500
(3) Rs. 14800 (4) Rs. 13620
(5) None of these
(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

7. The compound interest earned by Suresh on a certain amount at the end of two years at the rate of 8 p.c.p.a was Rs. 1,414.4. What was the total amount that Suresh got back at the end of two years in the form of principal plus interest earned ?
(1) Rs.9,414.4 (2) Rs.9.914.4
(3) Rs.9,014.4 (4) Rs.8,914.4
(5) None of these
(SBI & Rural Business PO Exam. 18.04.2010)

RBI GRADE—B OFFICER EXAMS

1. The compound interest accrued on an amount at the end of two years at the rate of 12 p.c.p.a is Rs 2,862. What is the amount ?
(1) Rs 11,250 (2) Rs 12,200
(3) Rs 13,500 (4) Rs 10,000
(5) None of these
(RBI Grade-B Officer Exam.06.02.2011)

INSURANCE EXAMS

1. Rajan invested an amount of Rs. 8,000/- in a fixed deposit scheme for 2 years at compound interest rate 5 percent p.a. How much amount will Rajan get on maturity of the fixed deposit ?
(1) Rs. 8,620/- (2) Rs. 8,840/-
(3) Rs. 8,800/- (4) Rs. 8,600/-
(5) None of these
(LIC Assistant Administrative Officer (AAO) Exam. 24.04.2005)

2. Sudharshan invested Rs. 15,000 at interest @ 10 p.c. p.a. for one year. If the interest is compounded every six months what amount will Sudharshan get at the end of the year ?
(1) Rs. 16,537.50 (2) Rs. 16,500
(3) Rs. 16,525.50 (4) Rs. 18,150
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 2006)

3. A sum of money becomes Rs. 13,380 after 3 years and Rs. 20,070 after 6 years on compound interest. The sum (in Rupees) is :
(1) 8800 (2) 8890
(3) 8920 (4) 9040

(United India Insurance Co. (AAO) Exam. 11.03.2007)

4. The compound interest (in Rupees) on Rs. 5,600 for $1\frac{1}{2}$ years at 10% per annum, compounded annually, is
(1) 882.70 (2) 873.50
(3) 868 (4) 840

(LIC Assistant Administrative Officer (AAO) Exam, 07.06.2009)

5. The simple interest accrued on an amount of Rs. 27,500 at the end of three years is Rs. 10,230. What would be the **approximate** compound interest accrued on the same amount at the same rate in the same period ?
(1) Rs. 11,550 (2) Rs. 12,620
(3) Rs. 10,950 (4) Rs. 11,900
(5) Rs. 13,500

(New India Assurance AO Exam. 25.10.2009)

6. A sum of money lent out at compound interest increases in value by 50% in 5 years. A person wants to lend three different sums *x, y* and *z* for 10, 15 and 20 years respectively at the above rate in such a way that he gets back equal sums at the end of their respective periods. The ratio *x* : y : *z* is
(1) 6 : 9 : 4 (2) 9 : 4 : 6
(3) 9 : 6 : 4 (4) 6 : 4 : 9

(New India Assurance AO Exam. 25.10.2009)

7. The simple interest accrued on a sum of certain principal is Rs 6,400 in four years at the rate of 8 p.c.p.a. What would be the compound interest accrued on that principal at the rate of 2 p.c.p.a. in 2 years ?
(1) Rs 800 (2) Rs 808
(3) Rs 704 (4) Rs 700
(5) None of these

(United India Insurance AO Exam. 27.03.2011)

8. A sum of Rs. 91,000 is borrowed at 20% per annum compounded annually. If the amount is to be paid in three equal instalments,the annual instalment will be
(1) Rs. 43,200 (2) Rs. 42,800
(3) Rs. 42,500 (4) Rs. 42,300

(United India Insurance AAO Exam. 03.06.2012)

9. The compound interest accrued on an amount of Rs. 22,000 at the end of two years is Rs 5,596.8. What would be the simple interest accrued on the same amount at the same rate in the same period ?
(1) Rs.5,420 (2) Rs.5,360
(3) Rs.5,280 (4) Rs.5,140
(5) None of these

(United India Insurance AO Exam. 26.05.2013

10. The difference between the compound interest and simple interest for the amount Rs. 5,000 in 2 years is Rs. 32. The rate of interest is
(1) 5% (2) 8%
(3) 10% (4) 12%

(NICL (GIC) AO (Finance) Exam. 15.12.2013)

SHORT ANSWERS

NATIONALISED BANKS & IBPS PO/MT/SO

1. (2) **2.** (4)
3. (3) **4.** (4)
5. (3) **6.** (4)
7. (1) **8.** (2)
9. (3) **10.** (4)
11. (4) **12.** (1)
13. (5) **14.** (5)
15. (1) **16.** (4)
17. (3) **18.** (1)
19. (4) **20.** (2)
21. (2) **22.** (1)
23. (4)

SBI PO EXAMS

1. (4) **2.** (1)
3. (1) **4.** (4)
5. (2) **6.** (1)
7. (2)

RBI GRADE—B OFFICER EXAMS

1. (1)

INSURANCE EXAMS

1. (5) **2.** (1)
3. (3) **4.** (3)
5. (1) **6.** (3)
7. (2) **8.** (1)
9. (3) **10.** (2)

EXPLANATIONS

NATIONALISED BANKS & IBPS PO/MT/SO

1. (2) SI = $\frac{15000\times9\times2}{100}$ = Rs. 2700

CI = 12000 $\left[\left(1+\frac{8}{100}\right)^2-1\right]$

= 12000 $\left[\left(\frac{27}{25}\right)^2-1\right]$

= 12000 $\left[\frac{729-625}{625}\right]$

= 12000 × $\frac{104}{625}$ = Rs. 1996.8

∴ Total interest earned
= Rs. (2700 + 1996.8) = Rs. 4696.8

2. (4) Simple interest
= $\frac{35500\times15\times2}{100}$ = Rs. 10650

Principal for another investment
= 35500 + 10650 = Rs. 46150

∴ C.I. = Rs. 46150 $\left[\left(1+\frac{20}{100}\right)^3-1\right]$

= Rs. 46150 $\left[\left(\frac{6}{5}\right)^3-1\right]$

= Rs. 46150 $\left(\frac{216-125}{125}\right)$

= Rs. $\frac{46150\times91}{125}$

= Rs. 33597.20

Total interest earned
= Rs (10650 + 33597.20) = Rs. 44247.20

3. (3) Let the rate p.c. p.a. be *r*.

r = $\frac{6216\times100}{14800\times3}$ = 14

∴ C.I = P $\left[\left(1+\frac{r}{100}\right)^3-1\right]$

= Rs. 14800 $\left[\left(1+\frac{14}{100}\right)^3-1\right]$

= Rs. 14800 $\left[(1.14)^3-1\right]$

= Rs. 14800 (1.481544- 1)
= Rs. 14800 × 0.481544
= Rs. 7126.8512

4. (4) Rate = $\frac{\text{SI}\times100}{\text{Principal}\times\text{Time}}$

= $\frac{12000\times100}{40000\times3}$ = 10%

∴ CI = Principal $\left[\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}-1\right]$

= 40000 $\left[\left(1+\frac{10}{100}\right)^3-1\right]$

= 40000 [$(1.1)^3$ - 1]
= 40000 (1.331 - 1)
= 40000 × 0.331 = Rs. 13240

5. (3) Let the rate of interest be R% per annum.

$\therefore CI = P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$\Rightarrow 5596.8 = 22000\left[\left(1+\frac{R}{100}\right)^{2}-1\right]$

$\Rightarrow \frac{5596.8}{22000}=\left(1+\frac{R}{100}\right)^{2}-1$

$\Rightarrow \left(1+\frac{R}{100}\right)^{2} = 1+\frac{5596.8}{22000}$

$\Rightarrow \left(1+\frac{R}{100}\right)^{2} = \frac{22000+5596.8}{22000} = \frac{27596.8}{22000}$

$\Rightarrow \left(1+\frac{R}{100}\right)^{2} = \frac{275968}{220000} = \frac{12544}{10000}$

$\Rightarrow 1 + \frac{R}{100} = \sqrt{\frac{12544}{10000}} = \frac{112}{100}$

$\Rightarrow \frac{R}{100} = \frac{112}{100} - 1 = \frac{112-110}{100} = \frac{12}{100}$

$\Rightarrow R = 12\%$

$\therefore SI = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$

$= \frac{22000\times 2\times 12}{100} = \text{Rs } 5280$

6. (4) Amount
= Rs. (25500 + 8440.5)
= Rs. 33940.5

$A = P\left(1+\frac{R}{100}\right)^{T}$

$\therefore 33940.5 = 25500\left(1+\frac{R}{100}\right)^{3}$

$\Rightarrow \frac{33940.5}{25500} = \left(1+\frac{R}{100}\right)^{3}$

$\Rightarrow \frac{339405}{255000} = \left(1+\frac{R}{100}\right)^{3}$

$\Rightarrow \frac{1331}{1000} = \left(1+\frac{R}{100}\right)^{3}$

$\Rightarrow \left(\frac{11}{10}\right)^{3} = \left(1+\frac{R}{100}\right)^{3}$

$\Rightarrow 1 + \frac{R}{100} = \frac{11}{10}$

$\Rightarrow \frac{R}{100} = \frac{11}{10} - 1 = \frac{1}{10}$

$\therefore R = 100 \times \frac{1}{10} = 10\%$

$\therefore S.I = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$

$= \frac{25500\times 3\times 10}{100} = \text{Rs. } 7650$

7. (1) $CI = P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$= 20000\left[\left(1+\frac{15}{100}\right)^{4}-1\right]$

$= 20000\left[\left(1+\frac{3}{20}\right)^{4}-1\right]$

$= 20000\left[\left(\frac{23}{20}\right)^{4}-1\right]$

$= 20000\left[\frac{279841-160000}{160000}\right]$

$= 20000 \times \frac{119841}{160000} = \text{Rs } 14980.125$

8. (2) $\text{Amount} = \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}$

$= 20000\left(1+\frac{10}{100}\right)^{2}\left(1+\frac{20}{100}\right)$

(Rate of interest for the first year
= 10%, Time = 2 half years)

$= \text{Rs. } \left(20000\times\frac{11}{10}\times\frac{11}{10}\times\frac{6}{5}\right) = \text{Rs. } 29040$

$\therefore$ C.I. = Rs. (29040 - 20000) = Rs. 9040

9. (3) Interest is compounded half yearly.
$\therefore$ R = 20% p.a. = 10% / half year
T = 2 years = 4 half years

$\therefore C.I = P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$= 10000\left[\left(1+\frac{10}{100}\right)^{4}-1\right]$

$= 10000\left[\left(\frac{11}{10}\right)^{4}-1\right]$

$= 10000\left[\left(\frac{121}{100}+1\right)\left(\frac{121}{100}-1\right)\right]$

$= 10000 \times \frac{221}{100} \times \frac{21}{100} = \text{Rs.}4641$

10. (4) C.I. $= P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$= 7400\left[\left(1+\frac{13.5}{100}\right)^{2}-1\right]$

$= 7400\left[(1.135)^{2}-1\right]$

$= 7400 \times 0.288225 = \text{Rs. } 2132.87$

11. (4) C.I. $= P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$= 45400\left[\left(1+\frac{15}{100}\right)^{2}-1\right]$

$= 45400\left[(1.15)^{2}-1\right]$

$= 45400 \times 0.3225 = \text{Rs. } 14641.5$

12. (1) Amount $= P\left(1+\frac{R}{100}\right)^{T}$

$\Rightarrow 5800 + 594.5 = 5800\left(1+\frac{R}{100}\right)^{2}$

$\Rightarrow \frac{63945}{58000} = \left(1+\frac{R}{100}\right)^{2}$

$\Rightarrow \frac{441}{400} = \left(1+\frac{R}{100}\right)^{2}$

$\Rightarrow \left(\frac{21}{20}\right)^{2} = \left(1+\frac{R}{100}\right)^{2}$

$\Rightarrow 1 + \frac{R}{100} = \frac{21}{20}$

$\Rightarrow R = \frac{1}{20} \times 100$

$= 5\%$ per annum.

13. (5) Principal $= \frac{SI \times 100}{Time \times Rate}$

$= \frac{1000 \times 100}{4 \times 5} = \text{Rs. } 5000$

Case II

Principal = Rs. 10000

$\therefore CI = P\left[\left(1+\frac{Rate}{100}\right)^{Time}-1\right]$

$= 10000\left[\left(1+\frac{5}{100}\right)^{2}-1\right]$

$= 10000 \times \left[\left(\frac{21}{20}\right)^{2}-1\right]$

$= 10000 \times \frac{41}{400} = \text{Rs. } 1025$

14. (5) $A = P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$= 2600\left[\left(1+\frac{15}{100}\right)^{3}-1\right]$

$= 2600 \times \frac{4167}{8000} = \text{Rs. } 1354.275$

Simple interest $= \frac{P \times R \times T}{100}$

$= \frac{2600 \times 15 \times 3}{100} = \text{Rs. } 1170$

$\therefore$ Difference = Rs. (1354.275 - 1170)

= Rs. 184.275 $\approx$ Rs 184

15. (1) Compound interest

$= P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$= 35000\left[\left(1+\frac{5}{100}\right)^{2}-1\right]$

$= 35000 \times \frac{41}{400} = \text{Rs.}3587.5$

16. (4) Principal

$= \frac{SI \times 100}{Time \times Rate} = \frac{2000 \times 100}{5 \times 4} = \text{Rs } 10000$

$\therefore C.I = P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$= 10000\left[\left(1+\frac{4}{100}\right)^{2}-1\right]$

$= 10000\left[\left(\frac{26}{25}\right)^{2}-1\right]$

$= 10000\left(\frac{676}{625}-1\right)$

$= \frac{10000\times 51}{625}$ = Rs 816

17. (3) CI - SI for 3 years = $P\left(\frac{r}{100}\right)^2\left(\frac{300+r}{100}\right)$

$\Rightarrow$ 2994.134 = $2600\left(\frac{r}{100}\right)^2\left(\frac{300+r}{100}\right)$

$\Rightarrow \frac{2994.134}{26000} = \frac{(300+r)r^2}{1000000}$

$\Rightarrow \frac{2994134000}{26000} = 300r^2 + r^3$

$\Rightarrow 115159 = 300r^2 + r^3$

Option (3) r = 19% satisfies the equation.

18. (1) Principal = $\frac{\text{SI}\times 100}{\text{Time}\times\text{Rate}}$

$= \frac{6500\times 100}{8\times 13}$ = Rs. 6250

$\text{CI} = P\left[\left(1+\frac{R}{100}\right)^T-1\right]$

$= 6250\left[\left(1+\frac{8}{100}\right)^2-1\right]$

$= 6250\left[\left(\frac{27}{25}\right)^2-1\right]$

$= 6250\left(\frac{729-625}{625}\right)$

$= \frac{6250\times 104}{625}$ = Rs 1040

19. (4) Principal = $\frac{\text{S.I}\times 100}{\text{Time}\times\text{Rate}}$

$= \frac{7200\times 100}{6\times 12}$ = Rs 10000

$\therefore \text{C.I} = P\left[\left(1+\frac{R}{100}\right)^T-1\right]$

$= 10000\left[\left(1+\frac{5}{100}\right)^2-1\right]$

$= 10000\left[\left(\frac{21}{20}\right)^2-1\right]$

$= 10000\left(\frac{441}{400}-1\right)$

$= \frac{10000\times 41}{400}$ = Rs. 1025

20. (2) Rate = $\frac{\text{S.I}\times 100}{\text{Principal}\times\text{Time}}$

$= \frac{10800\times 100}{22500\times 4}$ = 12% per annum

$\therefore \text{CI} = P\left[\left(1+\frac{R}{100}\right)^T-1\right]$

$= 22500\left[\left(1+\frac{12}{100}\right)^2-1\right]$

$= 22500\left[\left(\frac{28}{25}\right)^2-1\right]$

$= 22500\left(\frac{784-625}{625}\right)$

$= \frac{22500\times 159}{625}$ = Rs. 5724

21. (2) Difference = $\frac{P\times R^2}{(100)^2}$

$= \frac{7300\times 36}{10000}$ = Rs. 26.28

22. (1) Let the principal be Rs. x, then

$x\left(1+\frac{15}{100}\right)^3 - x = \frac{650052}{100}$

$\Rightarrow x\times\frac{23}{20}\times\frac{23}{20}\times\frac{23}{20} - x = \frac{650052}{100}$

$\Rightarrow 12167x - 8000x = \frac{650052\times 8000}{100}$

$\Rightarrow x = \frac{650052\times 8000}{100\times 4167} = 156\times 80$

= Rs. 12480

23. (4) C.I. = $P\left[\left(1+\frac{R}{100}\right)^T-1\right]$

$= 7790\left[\left(1+\frac{10}{100}\right)^2-1\right]$

$= 7790\left[\left(\frac{121}{100}-1\right)\right]$

$= \frac{7790\times 21}{100}$ = Rs. 1635.90

SBI PO EXAMS

1. (4) Let the principal sum be Rs. P

$SI = \frac{PRT}{100}$

$CI = P\left[\left(1+\frac{R}{100}\right)^{nt}-1\right]$

R = 10%, T = 2 years, n = 2
because CI is compounded half

$SI = \frac{P\times 10\times 2}{100} = \frac{P}{5}$

$CI = P\left[\left(1+\frac{10}{100\times 2}\right)^{2\times 2}-1\right]$

$CI = P\left[\left(\frac{21}{20}\right)^4-1\right]$

$CI = P\left[\frac{194481}{160000}-1\right]$

$CI = \frac{34481}{16000}P$

$CI - SI = \frac{34481}{160000}P-\frac{P}{5}$

$= \frac{34481-32000}{160000}P$

$= \frac{2481}{160000}P$

$\therefore \frac{2481}{16000}P = 124.05$

$P = \frac{124.05\times 160000}{2481}$

= Rs. 8000

2. (1) Interest is compounded half yearly.
Hence rate of interest

$= \frac{10}{2}$ = 5%

No. of periods = 2
$\therefore$ Amount

$= 15000\left(1+\frac{5}{100}\right)^2$

$= 15000 \times \frac{21}{20}\times\frac{21}{20}$

= Rs. 16537.50

3. (1)

$CI. = 25000\left\{\left(1+\frac{12}{100}\right)^3-1\right\}$

$\left[\because C.I. = P\left\{\left(1+\frac{R}{100}\right)^T-1\right\}\right]$

$= 25000\ \{(1.12)^3 - 1\}$
$= 25000\ (1.404928 - 1)$
$= 25000 \times 0.404928$
= Rs. 10123.2

4. (4) $C.I. = P\left[\left(1+\frac{R}{100}\right)^T-1\right]$

$= 41250\left[\left(1+\frac{6}{100}\right)^3-1\right]$

$= 41250\left[\frac{148877-125000}{125000}\right]$

$= 41250 \times \frac{23877}{125000}$ = Rs 7879.41

$\approx$ Rs. 7879

5. (2) $Rate = \frac{S.I.\times 100}{Principal\times Time}$

$= \frac{7200\times 100}{20000\times 3}$ = 12% Per annum

$\therefore C.I. = P\left[\left(1+\frac{R}{100}\right)^T-1\right]$

$= 20000\left[\left(1+\frac{12}{100}\right)^2-1\right]$

$= 20000\ \left[(1.12)^3-1\right]$
$= 20000 \times (1.404928 - 1)$
= Rs. 8098.56

6. (1) Let the principal be Rs. P.
According to the question,

$$C.I. = P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$$

$$\Rightarrow 6500.52 = P\left[\left(1+\frac{15}{100}\right)^{3}-1\right]$$

$$\Rightarrow 6500.52 = P\,[1.520875 - 1]$$

$$\Rightarrow P = \frac{6500.52}{0.520875} = \text{Rs.}12480$$

7. (2) $CI = P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$

$$\Rightarrow 1414.4 = P\left[\left(1+\frac{8}{100}\right)^{2}-1\right]$$

$$\Rightarrow 1414.4 = P \times 0.1664$$

$$\Rightarrow P = \frac{1414.4}{0.1664} = \text{Rs. } 8500$$

$\therefore$ Amount = Rs. (8500 + 1414.4)
= Rs. 9914.40

RBI GRADE-B OFFICER EXAMS

1. (1) If the principal be Rs. P, then

$$C.I = P\left[\left(1+\frac{R}{100}\right)^{T}-1\right]$$

$$\Rightarrow 2862 = P\left[\left(1+\frac{12}{100}\right)^{2}-1\right]$$

$$\Rightarrow 2862 = P\left[\left(\frac{28}{25}\right)^{2}-1\right]$$

$$\Rightarrow 2862 = P \times \left(\frac{784-625}{625}\right)$$

$$\Rightarrow 2862 = P \times \frac{159}{625}$$

$$\therefore P = \frac{2862\times 625}{159} = \text{Rs. } 11250$$

INSURANCE EXAMS

1. (5) P = 8000
t = 2 years
r = 5%

$$A = P\left(1+\frac{r}{100}\right)^{t}$$

$$= 8000\left(1+\frac{5}{100}\right)^{2}$$

$$= 8000\left(\frac{105}{100}\right)^{2}$$

$$= 8000 \times \frac{105}{100}\times\frac{105}{100} = \text{Rs.}8820$$

2. (1) Interest is compounded half yearly.
Hence rate of interest

$$= \frac{10}{2} = 5\%$$

No. of periods = 2
$\therefore$ Amount

$$= 15000\left(1+\frac{5}{100}\right)^{2}$$

$$= 15000 \times \frac{21}{20}\times\frac{21}{20} = \text{Rs. } 16537.50$$

3. (3) Let the sum be Rs. x. Then,

$$x\left(1+\frac{R}{100}\right)^{3} = 13380 \quad \ldots(i)$$

and,

$$x\left(1+\frac{R}{100}\right)^{6} = 20070 \quad \ldots(ii)$$

Dividing equation (ii) by (i),

$$\left(1+\frac{R}{100}\right)^{3} = \frac{20070}{13380} = \frac{3}{2}$$

From equation (i),

$$x \times \frac{3}{2} = 13380$$

$$\Rightarrow x = 13380 \times \frac{2}{3} = 8920$$

Hence, the sum is Rs. 8920.

4. (3) Amount

$$= \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}$$

$$= \text{Rs. } \left[5600\left(1+\frac{10}{100}\right)^{1\frac{1}{2}}\right]$$

$$= \text{Rs. } \left[5600\left(1+\frac{10}{100}\right)\times\left(1+\frac{\frac{1}{2}\times 10}{100}\right)\right]$$

$$= \text{Rs.}\left(5600\times\frac{11}{10}\times\frac{21}{20}\right) = \text{Rs. } 6468$$

$\therefore$ Compound interest
= Rs. (6468 - 5600) = Rs. 868

5. (1) Rate = $\frac{\text{SI}\times 100}{\text{Principal}\times\text{Time}}$

$= \frac{10230\times 100}{27500\times 3} = 12.4\%$

$\therefore \text{C.I} = \text{P}\left[\left(1+\frac{\text{R}}{100}\right)^{\text{T}}-1\right]$

$= 27500\left[\left(1+\frac{12.4}{100}\right)^{3}-1\right]$

$\approx$ 27500 (1.42- 1)
= 27500 × 0.42
= Rs. 11550

6. (3) Tricky approach
According to the question,

$\left(\frac{3}{2}\right)^2 x = \left(\frac{3}{2}\right)^3 y = \left(\frac{3}{2}\right)^4 z = k$ let

$\Rightarrow x = \left(\frac{2}{3}\right)^2 k,\ y = \left(\frac{2}{3}\right)^3 k$

and $z = \left(\frac{2}{3}\right)^4 k$

$\therefore x : y : z = \left(\frac{2}{3}\right)^2 k : \left(\frac{2}{3}\right)^3 k : \left(\frac{2}{3}\right)^4 k$

$= 1 : \frac{2}{3} : \left(\frac{2}{3}\right)^2 = 9 : 6 : 4$

7. (2) $\text{P} = \frac{\text{S.I}\times 100}{\text{Rate}\times\text{Time}}$

$= \frac{6400\times 100}{8\times 4}$
= Rs. 20000

$\text{C.I} = \text{P}\left[\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}-1\right]$

$= 20000\left[\left(1+\frac{2}{100}\right)^{2}-1\right]$
= 20000 × 0.0404 = Rs. 808

8. (1) If each instalment be Rs. x, then.
91000 =

$\frac{x}{1+\frac{r}{100}} + \frac{x}{\left(1+\frac{r}{100}\right)^2} + \frac{x}{\left(1+\frac{r}{100}\right)^3}$

$\Rightarrow 91000 = x\left[\frac{5}{6}+\left(\frac{5}{6}\right)^2+\left(\frac{5}{6}\right)^3\right]$

$\Rightarrow 91000 = x\left(\frac{5}{6}+\frac{25}{36}+\frac{125}{216}\right)$

$\Rightarrow 91000 = x\left(\frac{180+150+125}{216}\right)$

$\Rightarrow 91000 = \frac{x\times 455}{216}$

$\therefore x = \frac{91000\times 216}{455}$ = Rs. 43200

9. (3) Let the rate of interest be R% per annum.

$\therefore \text{CI} = \text{P}\left[\left(1+\frac{\text{R}}{100}\right)^{\text{T}}-1\right]$

$\Rightarrow 5596.8 = 22000\left[\left(1+\frac{\text{R}}{100}\right)^{2}-1\right]$

$\Rightarrow \frac{5596.8}{22000} = \left(1+\frac{\text{R}}{100}\right)^2 - 1$

$\Rightarrow \left(1+\frac{\text{R}}{100}\right)^2 = 1 + \frac{5596.8}{22000}$

$\Rightarrow \left(1+\frac{\text{R}}{100}\right)^2 = \frac{22000+5596.8}{22000}$

$\Rightarrow \left(1+\frac{\text{R}}{100}\right)^2 = \frac{27596.8}{22000}$

$\Rightarrow 1 + \frac{\text{R}}{100} = \sqrt{\frac{12544}{10000}} = \frac{112}{100}$

$\Rightarrow \frac{\text{R}}{100} = \frac{112}{100} - 1 = \frac{112-110}{100} = \frac{12}{100}$

$\Rightarrow$ R = 12%

$\therefore \text{SI} = \frac{\text{Principal}\times\text{Time}\times\text{Rate}}{100}$

$= \frac{22000\times 2\times 12}{100}$ = Rs. 5280

10. (2) Difference

$\Rightarrow \text{Principal}\left(\frac{\text{Rate}^2}{10000}\right)$

$\Rightarrow 32 = 5000\left(\frac{r^2}{10000}\right)$

$\Rightarrow r^2 = 32 \times 2 = 64$

$\Rightarrow \text{r} = \sqrt{64} = 8\%$ per annum

MODEL EXERCISES

1. A sum of money becomes eight times in 3 years, if the rate is compounded annually. In how much time the same amount at the same compound interest rate will become sixteen times ?
(1) 6 years (2) 4 years
(3) 8 years (4) 5 years
(5) None of these

2. If the difference between the simple and the compound interests on same principal amount at 20% for 3 years is Rs 48, then the principal amount must be
(1) Rs 650 (2) Rs 600
(3) Rs 375 (4) Rs 400
(5) None of these

3. The difference between the simple interest and the compound interest compounded annually at the rate of 12% per annum on Rs 5000 for two years will be
(1) Rs 17.50 (2) Rs 36
(3) Rs 45 (4) Rs 72
(5) None of these

4. Sanjay borrowed a certain sum from Anil at a certain rate of simple interest for 2 yr. He lent this sum to Ram at the same rate of interest compounded annually for the same period. At the end of two years, he received Rs 4200 as compound interest but paid Rs 4000 only as simple interest, find the rate of interest.
(1) 15% (2) 20%
(3) 35% (4) 10%
(5) None of these

5. Under the Rural Housing Scheme, the Delhi Development Authority (DDA) allotted a house to Kamal Raj for Rs 126100. This payment is to be made in three equal annual instalments. If the money is reckoned at 5% per annum compound interest, then how much is to be paid by Kamal Raj in each instalment ?
(1) Rs 45205 (2) Rs 47405
(3) Rs 46305 (4) Rs 48505
(5) None of these

6. Ram takes a loan Rs 10000 and pays back Rs 13310 after 3 years. The compound interest rate per annum will be approximately.
(1) 8% (2) 9%
(3) 10% (4) 11%
(5) None of these

7. The difference between the compound interest and the simple interest on a certain sum of money for 2 years at 15% per annum is Rs 45. Find the sum.
(1) Rs 7000 (2) Rs 2750
(3) Rs 2000 (4) Rs 2250
(5) None of these

8. Two friends A and B jointly lent out Rs 81600 at 4% per annum compound interest. After 2 years, A gets the same amount as B gets after 3 years. The investment made by B was
(1) Rs 40000 (2) Rs 30000
(3) Rs 45000 (4) Rs 38000
(5) None of these

9. A sum of money is accumulating at compound interest at a certain rate of interest. If simple interest instead of compound were reckoned, the interest for the first two years would be diminished by Rs 20 and that for the first three years by Rs 61. Find the sum.
(1) Rs 7000 (2) Rs 8000
(3) Rs 7500 (4) Rs 6500
(5) None of these

10. Rs 5887 is divided between Shyam and Ram, such that Shyam's share at the end of 9 years is equal to Ram's share at the end of 11 years, compound annually at the rate of 5%. The share of Shyam is
(1) Rs 2088 (2) Rs 2000
(3) Rs 3087 (4) Rs 2200
(5) None of these

11. Amit borrowed a certain sum of money for 2 years at 8% per annum on simple interest and immediately lent it to Ravi but at compound interest and gained by Rs 16. What amount did Amit borrow ?
(1) Rs 1600 (2) Rs 2500
(3) Rs 24000 (4) Rs 1800
(5) None of these

12. A sum of Rs 30600 is divided between Anjali and Arun, who are respectively 18 and 19 years old, in such a way that if their shares are invested at 4% per annum compounded annually, they shall receive the same amount on reaching 21 years of age. What is the share of Anjali ?
(1) Rs 16000 (2) Rs 15000
(3) Rs 15600 (4) Rs 14600
(5) None of these

13. A man borrows Rs 4000 from a bank at $7\frac{1}{2}\%$ compound interest. At the end of every year, he pays Rs 1500 as part

repayment of loan and interest. How much does he still owe to the bank after three such instalments ?

(1) Rs 123.25 (2) Rs 125
(3) Rs 400 (4) Rs 469.18
(5) None of these

14. Sanju puts equal amounts of money one at 10% per annum compound interest payable half yearly and the second at a certain per cent per annum compound interest payable yearly. If he gets equal amounts after 3 years, what is the value of the second per cent ?

(1) $10\frac{1}{4}\%$ (2) 10%

(3) $9\frac{1}{2}\%$ (4) $8\frac{1}{4}\%$

(5) None of these

15. Ravi Shankar takes money from the employees cooperative society at lower rate of interest and saves in a scheme, which gives him a compound interest of 20%, compounded annually. Finally the least number of complete years after which his sum will be more than double.

(1) 4 years (2) 2 years
(3) 6 years (4) 8 years
(5) None of these

16. Equal sums of money are deposited in two different banks by M/s Enterprise, one at compound interest, compounded annually and the other at simple interest, both at 5% per annum. If after two years, the difference in the amounts comes to Rs 200, what are the amounts deposited with each bank ?

(1) Rs 72000 (2) Rs 64000
(3) Rs 80000 (4) Rs 8400
(5) None of these

17. A sum of money lent at compound interest for 2 years at 20% per annum would fetch Rs 482 more if the interest was payable half yearly than if it was payable annually. The sum is

(1) Rs 10,000 (2) Rs 20,000
(3) Rs 40,000 (4) Rs 50,000
(5) None of these

SHORTANSWERS

1. (2) **2.** (3)
3. (4) **4.** (4)
5. (3) **6.** (3)
7. (3) **8.** (1)
9. (2) **10.** (3)
11. (2) **12.** (2)
13. (1) **14.** (1)
15. (2) **16.** (3)
17. (2)

EXPLANATIONS

1. (2) Let the sum of money be Rs *x*.

Then. $8x = x\left(1+\frac{r}{100}\right)^3$

$8 = \left(1+\frac{r}{100}\right)^3$

$\Rightarrow 2^3 = \left(1+\frac{r}{100}\right)^3$

$\Rightarrow \left(1+\frac{r}{100}\right)^3 = 2^3$

$\Rightarrow 1+\frac{r}{100} = 2$

Again, let the sum will become 16 times in *n* years
Then,

$\Rightarrow 16x = x\left(1+\frac{r}{100}\right)^n$

$\Rightarrow 16 = 2^n \Rightarrow 2^4 = 2^n$

$n = 4$ years

2. (3) Let principal amount be Rs 100.

Then, S.I. $= \frac{100\times 20\times 3}{100}$ = Rs 60

and C.I. $= 100\left(1+\frac{20}{100}\right)^3 - 100$

$= 100\times\left(\frac{6}{5}\right)^3 - 100 = \frac{364}{5}$

$\therefore$ C.I. - S.I. $= \frac{364}{5} - 60 = \frac{64}{5}$

If differences is Rs $\frac{64}{5}$, Principal

= Rs 100
If difference is if 48,

Principal $= \frac{100\times 5}{64}\times 48$ = Rs 375

3. (4) CI - SI for 2 years $= P\left(\frac{r}{100}\right)^2$

$= 5000\left(\frac{12}{100}\right)^2$

$= 5000 \times \frac{3\times 3}{25\times 25}$ = Rs 72

4. (4) Let the money borrowed be Rs *x* and rate be *r*%.
and Time = 2 yr

$\therefore 4000 = \frac{x\times r\times 2}{100}$

$\Rightarrow rx = 200000$

and $x\left(1+\frac{r}{100}\right)^2 = x + 4200$

$\Rightarrow x + \frac{xr^2}{10000} + \frac{2xr}{100} = 4200 + x$

$\Rightarrow 20r + 4000 = 4200$

$\Rightarrow r = 10\%$

5. (3) Let the amount of each instalment be Rs *x*.
Then,

$126100 = \frac{x}{1+\frac{r}{100}} + \frac{x}{\left(1+\frac{r}{100}\right)^2} + \frac{x}{\left(1+\frac{r}{100}\right)^3}$

$\Rightarrow 126100 = x\left[\frac{20}{21}+\left(\frac{20}{21}\right)^2+\left(\frac{20}{21}\right)^3\right]$

$\Rightarrow 126100 = x\times\frac{20}{21}\left[1+\frac{20}{21}+\frac{400}{441}\right]$

$\Rightarrow 126100 = x\times\frac{20}{21}\left[\frac{441+420+400}{441}\right]$

$\Rightarrow 126100 = x\times\frac{20}{21}\times\frac{1261}{441}$

$\Rightarrow x = \frac{126100\times 21\times 441}{20\times 1261}$ = Rs 46305

6. (3) Let the rate of compound interest be *r*.

Then, $10000\times\left(1+\frac{r}{100}\right)^3 = 13310$

$\Rightarrow \left(1+\frac{r}{100}\right)^3 = \frac{1331}{1000}$

$\Rightarrow \left(1+\frac{r}{100}\right)^3 = \left(\frac{11}{10}\right)^3$

$\Rightarrow 1 + \frac{r}{100} = \frac{11}{10}$

$\Rightarrow \frac{r}{100} = \frac{1}{10}$

$\therefore r = 10\%$

7. (3) CI - SI for 2 years = $P\left(\frac{r}{100}\right)^2$

$\Rightarrow 45 = P\left(\frac{15}{100}\right)^2$

$\Rightarrow P = \frac{45\times100\times100}{15\times15}$ = Rs 2000

8. Let the investment made by B = x
Then, investment made by A = (81600 - x)

$\therefore (81600 - x)\left(1+\frac{4}{100}\right)^2 = x\left(1+\frac{4}{100}\right)^3$

$\Rightarrow 81600 - x = 1.04x$

$\Rightarrow x = \frac{81600}{2.04}$ = Rs. 40000

9. (2) Let the principal be P and rate of interest be r%. Then, principal (when difference between C.I. and SI is for 2 years) is given by

$P = \frac{20\times(100)^2}{r^2}$ .....(i)

and difference between CI and SI is for 3 years is given by

$P = \frac{61\times10^6}{r^2(300+r)}$ .....(ii)

From eqs. (i) and (ii),

$\frac{20\times10^4}{r^2} = \frac{61\times10^6}{r^2(300+r)}$

$\Rightarrow r = 305 - 300 = 5\%$

From Eq. (i), $P = \frac{20\times10^4}{25}$ = Rs 8000

10. (3) Let Shyam's share = x

Then, $x\left[1+\frac{5}{100}\right]^9 = (5887 - x)\left[1+\frac{5}{100}\right]^{11}$

$\Rightarrow \frac{x}{5887-x} = 1.1025$

$\Rightarrow x = 5887 \times 1.1025 - 1.1025x$

$\Rightarrow 2.1025x = 5887 \times 1.1025$

$\Rightarrow x$ = Rs 3087

11. (2) Let Amit borrow Rs x.
According to the question,

CI - SI for 2 years = $P\left(\frac{r}{100}\right)^2$

$\Rightarrow 16 = x\left(\frac{8}{100}\right)^2$

$\Rightarrow x = \frac{16\times10000}{64}$ = Rs 2500

12. (2) Let the share of Anjali be x.
$\therefore$ Then, the share of Arun is (30600 - x).

$x \times \left(1+\frac{4}{100}\right)^3 = (30600 - x)\left(1+\frac{4}{100}\right)^2$

$\Rightarrow x \times \frac{104}{100} = 30600 - x$

$\Rightarrow \frac{204}{100}x = 30600$

$\Rightarrow x = \frac{30600\times100}{204}$ = Rs 15000

13. (1) Amount remaining after

1 year = $4000\left(1+\frac{7.5}{100}\right) - 1500$ = Rs 2800

2 years = $2800\left(1+\frac{7.5}{100}\right) - 1500$ = Rs 1510

3 years = $1510\left(1+\frac{7.5}{100}\right) - 1500$ = Rs 123.25

14. (1) Let the value of second rate of interest be x% and equal amounts be P each.

$P \times \left(1+\frac{5}{100}\right)^6 = P \times \left(1+\frac{x}{100}\right)^3$

$\Rightarrow \left(1+\frac{5}{100}\right)^2 = \left(1+\frac{x}{100}\right)$

$\Rightarrow \frac{105\times105}{100\times100}=\frac{100+x}{100}$

$\Rightarrow 110.25 = 100 + x$

$\Rightarrow$ = 10.25%

$\Rightarrow x = 10\ \frac{25}{100} = 10\frac{1}{4}\%$

15. (2) Let the sum of money be P.

$\therefore P\left(1+\frac{20}{100}\right)^{T} \geq 2P$

$\Rightarrow \left(\frac{6}{5}\right)^{T} \geq 2$

$\Rightarrow (1.2)^{T} \geq 2$

Hence, least value of T is 2 yr.

16. (3) Let P be the amount deposit in two different banks.

In 1st bank,

$CI = P\left(1+\frac{5}{100}\right)^{2} - P$

$= P\left(\frac{21}{20}\right)^{2} - P$

$= \frac{441P}{400} - P = \frac{441P-400P}{400} = \frac{41P}{400}$

In IInd bank, we get

$SI = \frac{P\times R\times T}{100} = \frac{P\times5\times2}{100} = \frac{P}{10}$

$\therefore \frac{41P}{100} - \frac{P}{100} = 200$

$\Rightarrow$ P = Rs 80000

17. (2) Let the sum be Rs. x.

$x\left(1+\frac{10}{100}\right)^{4} - x\left(1+\frac{20}{100}\right)^{2} = 482$

$\Rightarrow x\left(\frac{11}{10}\right)^{4} - x\left(\frac{6}{5}\right)^{2} = 482$

$\Rightarrow x$ (1.4641 - 1.44) = 482

$\Rightarrow 0.0241x = 482$

$\Rightarrow x = \frac{482}{0.0241} = 20000$

DATA INTERPRETATION TABLE GRAPH

Directions (Q. 1-5): Following table shows the marks scored by seven students in six different subjects.

Subjects → / Full marks → / Students ↓	Hindi (80) Marks obtained	Eng (80) Marks obtained	Maths (100) Marks obtained	Phy (40) Marks obtained	Chem (40) Marks obtained	Bio (40) Marks obtained
P	44	65	87	36	30	24
Q	51	48	93	28	27	31
R	62	57	74	32	28	32
S	65	55	67	21	25	28
T	54	64	69	27	24	27
U	48	60	78	35	32	30
V	55	70	81	30	28	33

1. What is the percentage difference between the marks scored by student 'V in Hindi and student 'S' in Chemistry?
   (1) 5.75% (2) 6.25% (3) 6.75% (4) 7.25% (5) 7.5%
2. What is the average of marks obtained by all students in English? (Answer in approximate value)
   (1) 57 (2) 55 (3) 60 (4) 64 (5) 51
3. In how many subjects did student 'Q' get more than 65% marks?
   (1) nil (2) one (3) two (4) three (5) four
4. What is the difference between the percentage of marks obtained by student 'R' in Hindi and Physics together and the percentage of marks obtained by student 'Q' in English and Chemistry together?
   (1) 11.4% (2) 15.8% (3) 12.6% (4) 17.5% (5) 21%
5. What is the overall percentage of marks scored by student 'V in all subjects together? (Answer in approximate value)
   (1) 68% (2) 73% (3) 75% (4) 78% (5) 81%

Directions (Q. Nos. 6-10) ***Study the table carefully to answer the questions that follow:***
Candidates who appeared and passed in the test from four schools in six different years

Year	School							
	A		B		C		D	
	Appeared	Passed	Appeared	Passed	Appeared	Passed	Appeared	Passed
2004	124	78	445	354	454	343	546	345
2005	234	124	545	435	732	567	565	456
2006	456	235	664	454	693	456	235	112
2007	398	156	345	144	645	545	546	234
2008	546	346	584	354	354	258	656	564
2009	547	435	704	347	578	313	456	252

6. What was the total number of failed candidates from school-C in the year 2008 and the number of candidates who appeared in the exam from school-D in the year 2006?
   (1) 335 (2) 325 (3) 322 (4) 332 (5) None of these

7. In which year was the difference between the number of candidates who appeared and passed in the exam from school-B second lowest?
(1) 2004 (2) 2005 (3) 2006 (4) 2007 (5) 2008

8. What was the respective ratio between the number of candidates who appeared from school-C in the year 2006 and the number of candidates who passed in the exam from school-D in the year 2009?
(1) 11 : 4 (2) 11 : 5 (3) 5 : 11 (4) 9 : 11 (5) None of these

9. Number of candidates who passed in the exam from school B in the year 2005 was approximately what per cent of number of candidates who appeared from school-A in the year 2008?
(1) 76 (2) 87 (3) 90 (4) 84 (5) 80

10. What was the approximate percent increase in the number of candidates who passed in the exam from school-A in the year 2009 as compared to the previous year?
(1) 22 (2) 39 (3) 26 4) 30 (5) 34

Directions (Q. Nos. 11-15) Study the following table carefully to answer the questions that follow.

Amount earned (in lacs) by five persons in six different years

Year	Person				
	A	B	C	D	E
2005	2.24	4.33	5.64	3.73	1.69
2006	1.44	3.34	6.93	5.52	5.52
2007	4.63	2.79	7.52	5.68	4.28
2008	6.65	6.63	5.83	6.74	6.83
2009	5.34	4.5	5.94	8.42	5.53
2010	7.38	5.36	7.84	9.45	9.94

11. What was the average of the earning of Person-B in **the** year 2006, that of person C in the year 2008 and that of E in the year 2005 together?
(1) ` 3.62 lac (2) ` 2.64 lac (3) ` 3.64 lac (4) ` 10.86 lac (5) None of these

12. What was the respective ratio between the amount earned by Person-B in the year 2007 and Person-D in the year 2010?
(1) 32 : 107 (2) 31 : 105 (3) 29 : 107 (4) 32 : 105 (5) None of these

13. What is the **approximate** per cent increase in the amount earned by Person-D in the year 2010 as compared to the previous year?
(1) 7 (2) 21 (3) 18 (4) 15 (5) 12

14. Whose earning increased consistently from the year 2005 to the year 2010?
(1) A (2) B (3) C (4) D (5) E

15. Total amount earned by Person-A in the year 2006 and Person-C in the year 2010 together was **approximately** what per cent of the amount earned by Person-E in the year 2009?
(1) 151 (2) 155 (3) 168 (4) 174 (5) 162

Directions (Q. 16-20): Following table shows the number of candidates appeared and qualified in an entrance examination of six schools during the period of 2005-2010.

YEAR	2005		2006		2007		2008		2009		2010	
SCHOOL	A	Q	A	Q	A	Q	A	Q	A	Q	A	Q
S_1	840	275	625	215	910	525	825	480	890	480	595	390
S_2	935	355	740	320	885	440	745	360	815	450	615	320
S_3	715	310	780	410	765	410	550	240	720	410	810	425
S_4	720	400	575	240	775	350	470	225	590	250	925	540
S_5	685	275	645	300	810	370	630	310	680	280	780	450
S_6	760	280	530	225	925	480	690	345	650	375	725	375

A → Appeared, Q → Qualified

16. What is the difference between the total number of candidates who appeared and the total number of candidates who qualified in the year 2006 in all six schools together?
(1) 2175 (2) 2180 (3) 2185 (4) 2190 (5) 2195

17. For which of the following years the number of candidates who qualified as a percentage of those who appeared for School S_6 is maximum?
(1) 2005 (2) 2006 (3) 2007 (4) 2009 (5) 2010

18. What is the percentage of the total students who qualified with respect to the total students who appeared for School S_1, taking all six years together?
(1) 47.24% (2) 50.48% (3) 51.75% (4) 53% (5) 56.25%

19. Which of the following schools has the maximum percentage of students who qualified with respect to the number of candidates who appeared in the year 2009?
(1) S_1 (2) S_2 (3) S_3 (4) S_5 (5) S_6

20. What is the per cent rise in the number of candidates who qualified from 2009 to 2010 for School S_4?
(1) 46% (2) 96% (3) 112% (4) 116% (5) 216%

Directions (Q. 21-25): Following table shows the percentage marks scored by seven students in six different subjects. Maximum marks of each paper are 80.

Students	Percentage of Marks (out of 80)					
	P_1	P_2	P_3	P_4	P_5	P_6
A	58.75%	78.75%	81.25%	82.50%	77.50%	76.25%
B	63.75%	60%	65%	88.75%	83.75%	85%
C	68.75%	71.25%	58.75%	83.75%	55%	67.50%
D	52.50%	76.25%	63.75%	61.25%	58.75%	66.25%
E	85%	78.75%	70%	73.75%	67.50%	80%
F	87.50%	90%	77.50%	71.25%	73.75%	76.25%
G	81.25%	72.50%	87.50%	70%	81.25%	93.75%

21. What is the total marks scored by A in all six subjects?
(1) 357 (2) 361 (3) 363 (4) 364 (5) 365

22. What is the approximate average of marks obtained by all seven students in subject P_5? (Rounded off up to two digits)
(1) 55.24 (2) 56.85 (3) 57.54 (4) 58.48 (5) 59.62

23. The marks scored by E in paper P_1 is approximately what per cent of the marks scored by D in the same paper?
(1) 154.6% (2) 158.4% (3) 161.9% (4) 163.2% (5) 167.5%

24. What is the overall percentage of marks of student C?
(1) 67.5% (2) 68.5% (3) 69.5% (4) 70.5% (5) 71.5%

25. What is the average of the percentage of marks obtained by all students in papers P_2 and P_5 together?
(1) 79.82% (2) 77.42% (3) 75.04% (4) 74.43% (5) 73.21%

Directions (Q. 26-30): Following table shows the percentage of boys and difference between the number of boys and the number of girls among the students of six different schools who appeared in board examination in different years.

	1986		1987		1988		1989	
	% boys	Diff	% boys	Diff	% boys	Diff	% boys	Diff
A	70%	68	60%	35	75%	92	60%	43
B	40%	42	48%	9	45%	24	60%	45
C	44%	30	55%	12	60%	26	56%	12
D	44%	42	57%	42	55%	36	65%	96
E	75%	140	60%	68	70%	132	66%	112
F	44%	45	56%	48	65%	114	45%	42

26. What is the average of the number of boys who appeared from School E, taking all the four years together?
(1) 212 (2) 217 (3) 219 (4) 222 (5) 227

27. What is the total number of girls who appeared in the examination from all the six schools in the year 1987?
(1) 682 (2) 693 (3) 702 (4) 707 (5) None of these

28. What is the difference between the total number of students appearing from School B in the year 1987 and that in 1989?
(1) 17 (2) 29 (3) 35 (4) 46 (5) None of these

29. What is the ratio of the total number of boys appeared from School C in 1986 to the total number of girls appeared from School E in the year 1988?
(1) 5:4 (2) 8:7 (3) 9:8 (4) 10:9 (5) None of these

30. Total number of students appearing from School F in the year 1986 is what per cent of the total number of students appearing from School C in the year 1986?
(1) 66.66% (2) 90% (3) 120% (4) 150% (5) None of these

Directions (Q.31-35): Study the table carefully to answer the questions that follow :
Number of animals in grassland of four different countries in five different years

Year	Country											
	South Africa			China			Sri Lanka			England		
	Tiger	Lion	Bear	Tiger	Lion	Bear	Tiger	Lion	Bear	Tiger	Lion	Bear
1990	145	156	250	320	346	436	280	468	255	423	342	234
1995	134	165	354	445	256	542	354	354	343	368	136	345
2000	120	135	324	583	325	454	433	345	545	354	267	456
2005	110	184	285	466	475	322	343	324	546	562	235	567
2010	160	224	264	411	535	534	535	532	453	349	345	324

31. What is the average of the number of tigers in grassland of Sri Lanka over all the years together ?
(1) 386 (2) 389 (3) 369 (4) 276 (5) None of these

32. What was the difference between the total number of lions and bears in the grassland of England in the year 2005 and the number of tigers in the grassland of South Africa in the year 1995 ?
(1) 597 (2) 558 (3) 677 (4) 668 (5) None of these

33. Total number of animals together in grassland, of China in the year 1990 was approximately what percent of total number of bears in the grassland of Sri Lanka overall the years together ?
(1) 44% (2) 56% (3) 41% (4) 47% (5) 51%

34. If 35 percent of the total number of animals in the grassland of China in the year 2010 died due to an epidemic, how many animals remained in the grassland of China in the year 2010 ?
(1) 976 (2) 952 (3) 986 (4) 962 (5) None of these

35. What was three fourth of the total number of lions in the grassland of all the four countries in the year 2000 ?
(1) 848 (2) 868 (3) 804 (4) 824 (5) None of these

Directions (Q. 36-40): Study the following table carefully and answer the given questions.

Company	2009				2010			
	Total Production	$I_1 : I_2$	% Sold	Sold $I_1 : I_2$	Total Production	$I_1 : I_2$	% Sold	Sold $I_1 : I_2$
A	36	5 : 4	42%	3 : 4	48	9 : 7	65%	7 : 6
B	28	3 : 4	60%	8 : 7	40	5 : 3	56%	3 : 5
C	32	1 : 3	55%	5 : 6	36	1 : 2	50%	3 : 2
D	40	3 : 5	72%	5 : 4	50	2 : 3	48%	5 : 3
E	25	3 : 2	50%	2 : 3	30	3 : 2	40%	1 : 1
F	30	2 : 1	75%	8 : 7	45	4 : 5	80%	7 : 9

Total production is in lakhs and I_1 and I_2 are the two different models of the items.

36. What is the total number of items sold by all six companies in 2009?
(1) 107.48 lakh (2) 109.76 lakh (3) 113.32 lakh (4) 115.8 lakh (5) 160 lakh

37. What is the total number of I_1 items sold by Company D in year 2009 and 2010 together?
(1) 28.8 lakh (2) 30.6 lakh (3) 31 lakh (4) 32.4 lakh (5) 36 lakh

38. The percentage items sold by Company B in the year 2010 is what per cent of the percentage of items sold by CompanyEin2010?
(1) 48% (2) 96% (3) 120% (4) 140% (5) 71.42%

39. What is the total number of I_2 items which remained unsold in Company D in 2009 and 2010 together?
(1) 12.2 lakh (2) 21 lakh (3) 33.2 lakh (4) 36.4 lakh (5) None of these

40. I_1 items sold by Company A in the year 2010 is what percentage of I_1 items sold by Company E in the year 2009? (Approximate value)
(1) 336% (2) 240% (3) 180% (4) 112.5% (5) 29.76%

Directions (Q. 41-45): Following table shows the number of items (in thousand) produced by four different companies (A, B, C and D) and the ratio of sold to unsold items among them.

Company → / Year ↓	A		B		C		D	
	Total	S : US	Total	S : US	Total	S : US	Total	S : US
2006	45.5	4 : 3	64.8	5 : 3	42.14	4 : 3	50	3 : 2
2007	48.6	5 : 4	70.15	3 : 2	49.5	4 : 5	52.7	8 : 9
2008	40	2 : 3	77.11	5 : 6	51	9 : 8	56.4	1 : 1
2009	55	3 : 2	86.4	5 : 3	54	1 : 1	51	2 : 1
2010	64.4	3 : 4	85	8 : 9	66.22	6 : 5	60.5	2 : 3
2011	68	5 : 3	81.18	5 : 4	68.8	5 : 3	62.1	3 : 2

41. What is the number of items sold by Company A in all six years together? (Answer options are in thousand)
(1) 168.4 (2) 171.6 (3) 172.1 (4) 173.2 (5) None of these

42. What is the average number of items produced by Company D in all six years (Answer options are in thousand)
(1) 54.25 (2) 55.45 (3) 56.75 (4) 57.5 (5) None of these

43. The number of items sold by Company D in the year 2009 is what percentage of the number of items which remain unsold by Company D in the year 2006?
(1) 58.82% (2) 80% (3) 120% (4) 150% (5) 170%

44. The number of items which remain unsold by Company C in 2008 is what percentage more or less than the number of items which are sold by Company B in the year 2010?
(1) 16% (2) 24% (3) 32% (4) 40% (5) 48%

45. What is the difference between the total items sold and the total items that remain unsold by Company D in all six years together?
(1) 24220 (2) 25640 (3) 26380 (4) 27550 (5) None of these

Direction(Q.46-50): Following table shows the marks scored by six students in different subjects:

Student	Subject					
	Maths (150)	Hindi (120)	English (100)	Science (100)	Sanskrit (50)	GK (80)
A	84	66	73	61	24	52
B	75	90	82	54	38	60
C	96	48	65	62	40	44
D	128	75	62	76	34	68
E	108	78	78	70	39	48
F	142	84	48	81	42	38

46. What overall percentage did student B get in all subjects together?
(1) 62.5% (2) 64% (3) 66.5% (4) 67.5% (5) 72%

47. What is the ratio of the total marks obtained by A to that obtained by F?
(1) 4:5 (2) 5:6 (3) 5:7 (4) 3:5 (5) None of these

48. What is the average of marks obtained by all the students in Hindi?
(1) 73.5 (2) 74.5 (3) 75 (4) 76.5 (5) 77.5

49. What is the average percentage of marks obtained by all the students in Maths? (Answer in approximate value)
(1) 62% (2) 65% (3) 68% (4) 70% (5) 72%

50. The total marks obtained by E is what percentage more than the total marks obtained by A? (Answer in approximate value)
(1) 9% (2) 11% (3) 13% (4) 15% (5) 17%

Directions (Q. 51-55): Following table shows the total number of tyres produced by six companies (in lakh), percentage of tyres rejected and percentage of tyres sold by these companies during the year 2008 and 2009.

Year	2008			2009		
Company	Total	Rejected	Sold	Total	Rejected	Sold
A	12.8	3.80%	67.90%	16.4	4.10%	72%
B	13.2	5.70%	88%	15.2	3.40%	76.40%
C	16	2.40%	72.10%	18.8	3.60%	82.10%
D	12.4	9.20%	76.40%	16.2	4.80%	87.50%
E	17.5	4.10%	81.90%	20.5	5.20%	80.90%
F	8.6	4.70%	90.60%	12.2	4.40%	81%
G	14.8	3.60%	83.70%	17.5	3.90%	78.20%

51. What is the percentage rise in the production of Company C from year 2008 to 2009?
(1) 12.5% (2) 15% (3) 17.5% (4) 20% (5) 22.5%

52. What is the percentage rise in the sale of Company F from year 2008 to 2009?
(1) 20.2% (2) 22.4% (3) 24.6% (4) 26.8% (5) 29%

53. What is the total number of rejected tyres from all companies together in year 2008?
(1) 441810 (2) 441820 (3) 441830 (4) 441840 (5) 441850

54. Total number of tyres sold by all companies in year 2009 is what percentage of total tyres produced in that year?
(1) 72% (2) 75% (3) 80% (4) 84% (5) 96%

55. For which of the following companies the rise in production is maximum from year 2008 to 2009?
(1) A (2) B (3) C (4) E (5) G

Directions (Q. 56-60): Following table shows the number of students appeared and passed in Board exam from four schools A, B, C and D.

School								
Year	**A**		**B**		**C**		**D**	
	A	**P**	**A**	**P**	**A**	**P**	**A**	**P**
2000	782	360	612	310	720	410	1020	802
2001	804	472	608	324	728	480	1135	840
2002	720	448	636	298	680	390	1084	864
2003	750	360	655	305	695	396	1096	766
2004	824	504	640	346	712	424	1180	752
2005	850	496	600	315	740	464	1165	780

P = Passed, A = Appeared

56. What is the difference between the total students appeared and total students passed from School A in all the six years together?
(1) 2060 (2) 2070 (3) 2080 (4) 2090 (5) None of these

57. What is the average number of students passed from all the four schools in the year 2001?
(1) 528 (2) 529 (3) 530 (4) 531 (5) 532

58. For which of me following schools is the percentage of students passed among those who appeared for the exam the minimum in the year 2005?
(1) A (2) B (3) C (4) D (5) None of these

59. What is the percentage rise in the number of students who passed from School A in the year 2003 to that in 2004?
(1) 32% (2) 36% (3) 40% (4) 44% (5) None of these

60. The total number of students who passed from School C in all the six years together is what percentage of the total students who appeared from School C in all the six years together? (Answer in approximate value)
(1) 56% (2) 58% (3) 60% (4) 62% (5) 64%

Directions (Q. 61-65): Study the table carefully and answer the questions that follow. The table represents the percentage expenditure of the income of A, B, C, D, E and F on different items.

% Expenditure from Annual Income						
Person	**Food**	**Rent**	**Transport**	**Clothes**	**Entertainment**	**Misc**
A	21.8%	15.0%	18.4%	12.5%	13.3%	19.0%
B	17.2%	18.0%	22.6%	15.0%	11.4%	15.8%
C	24.0%	16.3%	14.8%	11.2%	7.8%	25.9%
D	18.0%	19.5%	15.5%	12.0%	16.4%	18.6%
E	20.2%	16.4%	17.5%	14.0%	8.6%	23.3%
F	23.6%	18.5%	16.0%	13.8%	11.0%	17.1%

61. If the annual incomes of B and C are ` 216000 and ` 264000 respectively, what is the difference between the amount spent by them on transport?
(1) ` 9248 (2) ` 9414 (3) ` 9518 (4) ` 9608 (5) ` 9744

62. If the amounts of money spent on food by C and D are ` 72000 and ` 86400 respectively, then the annual income of C is what percentage of the annual income of D?
(1) 47.5% (2) 60% (3) 62.5% (4) 120% (5) 160%

63. The percentage of amount of money spent by E on entertainment is what percentage of the amount of money spent by F on transport?
(1) 53.75% (2) 72.5% (3) 87.25% (4) 112.5% (5) 186%

64. If the annual income of C and D together is ` 420000,what is the sum of the amount spent by C on rent and that by D oh miscellaneous items?
(1) `144410 (2) `145260 (3) `146580 (4) `147850 (5) None of these

65. If the monthly incomes of A and D are `40000 and `36000 respectively, then the amount of money spent by A on rent is what percentage more than the amount spent by D on clothes?
(1) 32.62% (2) 34.24% (3) 36.54% (4) 38.88% (5) 40%

Directions (Q. 66-70): Following table shows the number of viewers of different channels and the ratio of male to female among them. Based on the data given in the table, answer the given questions.

City	STAR PLUS		ZEE TV		SONY TV		COLORS	
	Total	M : F	Total	M : F	Total	M : F	Total	M : F
A	1394	7 : 10	1173	2 : 1	1043	3 : 4	1155	1 : 2
B	1265	2 : 3	1547	8 : 9	1323	1 : 2	1179	5 : 4
C	1056	4 : 7	1305	3 : 2	1404	7 : 5	1200	2 : 3
D	1236	5 : 7	1488	7 : 9	1195	3 : 2	1089	6 : 5
E	1053	4 : 5	1335	8 : 7	1428	8 : 9	1469	6 : 7
F	1302	1 : 2	1199	5 : 6	1254	9 : 10	1215	8 : 7

66. What is the average number of female viewers of ZEE TV taking all six cities together?
(1) 621 (2) 631 (3) 641 (4) 651 (5) 661

67. The total number of female viewers of COLORS TV from City C is what percentage of the total number of female viewers of STAR PLUS from City A? (Answer in approximate value)
(1) 82% (2) 88% (3) 96% (4) 108% (5) 114%

68. The average number of male viewers of SONY TV from all cities together is what percentage of the total number of viewers of STAR PLUS TV from City D? (Answer in approximate value)
(1) 30% (2) 40% (3) 50% (4) 60% (5) 70%

69. The total number of male viewers of ZEE TV from City C is what percentage more or less than the total number of female viewers of SONY TV from City F?
(1) 12.4% (2) 15.2% (3) 17% (4) 18.6% (5) 19.8%

70. What is the difference between the total number of male viewers and female viewers of ZEE TV from all six cities?
(1) 351 (2) 352 (3) 353 (4) 354 (5) 355

Directions (Q. 71-75) : Following table shows the total number of students appeared from different cities, ratio of boys and girls among those appeared students, percentage of passed students and number of passed girls among them.

	Total Appeared	Apeared Boys : Girls	Pass %	Number of girls passed
S_1	7210	3 : 2	60%	1268
S_2	4800	9 : 7	66%	1146
S_3	5670	5 : 4	70%	1432
S_4	6400	11 : 5	68%	975
S_5	7200	11 : 7	57%	1224
S_6	7080	7 : 5	65%	1565

71. What is the average number of boys appeared in the examination from all six cities?
(1) 3851 (2) 3852 (3) 3853 (4) 3854 (5) 3855

72. The total number of girls passed from City S_4 is what percentage of the total number of girls appeared from City S_4?
(1) 43.25% (2) 48.75% (3) 52.5% (4) 55% (5) 62.5%

73. What is the total number of boys failed in the examination from all six cities together?
(1) 6175 (2) 6180 (3) 6185 (4) 6190 (5) 6195

74. The total number of girls passed in the examination is approximately what percentage of the total number of girls appeared in the examination, taking all cities together?
(1) 42% (2) 50% (3) 56% (4) 64% (5) 72%

75. The total number of boys passed from City S_2 is what percentage more than the total number of girls passed from that city?
(1) 70.2% (2) 76.5% (3) 78.4% (4) 80% (5) 82.8%

Directions (Q. 76-80) : The following table shows the price (Rs. per 100 kg) of different items during different years. Answer the questions based on this table.

	1990	1995	2000	2005	2010
Rice	800	1150	1680	2400	3500
Wheat	450	700	1200	1650	2100
Pulses	2000	2700	3650	4600	6400
Sugar	1500	2200	3000	3800	4500
Groundnut	1200	1700	2450	3500	4200
Oil	4200	5500	6400	8000	11000

76. What is the percentage rise in the price of rice from year 1990 to year 2000?
(1) 10% (2) 110% (3) 52.3% (4) 90% (5) None of these

77. The price of 3 kg wheat in the year 1995 is what percentage more than the price of 1 kg of groundnut in the year 1990?
(1) 60% (2) 75% (3) 42.85% (4) 25% (5) None of these

78. What is the average price of 10 kg pulses (in Rs) over the years 1990 to 2010?
(1) 387 (2) 391 (3) 395 (4) 378 (5) 38.7

79. The average price of sugar is what percentage of the highest price of sugar over this period?
(1) 40% (2) 66.66% (3) 72.5% (4) 80% (5) None of these

80. In which of the following years was the percentage increase in the price of oil the highest over its preceding year?
(1) 1990-1995 (2) 1995-2000 (3) 2000-2005 (4) 2005-2010 (5) None of these

Directions (Q. 81-85) : In the following table the percentages of population of different age groups for five cities are given. Answer the questions based on this table.

City	$0 < \text{Age} \leq 13$	$13 < \text{Age} \leq 19$	$19 < \text{Age} \leq 35$	$35 < \text{Age} \leq 60$	Age > 60
A	18%	12%	24%	30%	16%
B	16%	18%	22%	29%	15%
C	20%	20%	20%.	25%	15%
D	15%	18%	21%	26%	20%
E	18%	15%	25%	24%	18%

81. If the number of peopLe of City A which belongs to 19-35 age group is 15840 how many people are there in the age group above 60 years?
(1) 10560 (2) 12140 (3) 11840 (4) 9675 (5) None of these

82. If the population of City E in the age group (0-13) years is 8100, then the population of the age group (0-13) years is what percentage of the population of the age group (13-19) years?
(1) 60% (2) 75% (3) 80% (4) 90% (5) 120%

83. If the population of City C and City D in the age group above 60 years are equal to 12000 each, what is the sum of the total population of City C and City D?
(1) 1.2 lakh (2) 1.4 lakh (3) 1.6 lakh (4) 2.0 lakh (5) 2.4 lakh

84. If the population of City A and City B in the age group (19-35) years are 8640 and 10560 respectively, what is the ratio of the total population of A to that of B?
(1) 2 : 3 (2) 3 : 4 (3) 4 : 5 (4) 5 : 6 (5) None of these

85. If the total population of City B and City E are 48000 and 65000 respectively, then the population of City E in the age group (0-13) years is what percentage more or less than the population of City B in the same age group?
(1) 47.24% (2) 49.5% (3) 56% (4) 57.5% (5) None of these

Directions (Q. 86-90): The following table shows the proportion of students passed in different streams in graduation from different cities. It also shows the ratio of Males to Females among the students.

City	Arts : Science : Commerce	Arts M : F	Science M : F	Commerce M : F
A	2 : 4 : 5	31 : 14	23 : 27	11 : 7
B	7 : 2 : 4	37 : 33	43 : 32	29 : 21
C	1 : 4 : 2	34 : 16	57 : 43	31 : 29
D	5 : 7 : 4	17 : 13	51 : 33	23 : 17
E	4 : 3 : 8	23 : 17	41 : 34	57 : 23
F	2 : 4 : 3	47 : 28	11 : 7	16 : 11
G	3 : 5 : 4	29 : 21	27 : 24	53 : 47

86. If the total number of Males who passed in Commerce stream from City G is 1272, what is the total number of students who passed in Arts from City G?
(1) 1800 (2) 2100 (3) 2400 (4) 3000 (5) 7200

87. If the total number of Males who passed from City A in Arts is 1240, what is the difference between the total number of students who passed in Commerce and that in Science from City A?
(1) 300 (2) 500 (3) 700 (4) 900 (5) 1100

88. If the total number of students who passed in Commerce from City F is 2700, the total number of students who passed from City F is what percentage of the total number of Science students who passed from City F?
(1) 44.44% (2) 75% (3) 150% (4) 180% (5) 225%

89. If the number of Females who passed, in Arts from City C is 384, the total number of Males who passed in Commerce from City C is what percentage of the total number **of** students who passed from City **C?** (Approximate value)
(1) 14.76% (2) 18.24% (3) 27.8% (4) 32.5% (5) 36%

90. The number of Females who passed in Commerce from City F is what percentage more or less than the total number of Males who passed in Commerce from City F?
(1) 45.45% less (2) 45.45% more (3) 31.25% less (4) 31.25% more (5) Can't be determined

Directions (Q. 91-95) : Following table shows the marks obtained by six students in six different subjects.

Students ↓	**Subjects**					
	S_1 (Out of 80)	S_2 (Out of 80)	S_3 (Out of 60)	S_4 (Out of 60)	S_5 (Out of 100)	S_6 (Out of 120)
A	38	42	33	28	77	72
B	60	50	42	38	68	66
C	64	36	32	35	72	80
D	42	65	48	42	52	84
E	32	64	45	46	87	35
F	35	48	30	28	82	48

91. What is the overall percentage of marks Student A scored in all subjects together?
(1) 55% (2) 56% (3) 57% (4) 58% (5) 59%

92. What is the average marks scored in the Subject S_5?
(1) 71 (2) 72 (3) 73 (4) 74 (5) 75

93. What is the ratio of the total marks scored by Student B to the total marks scored by Student D?
(1) 16 : 17 (2) 26 : 27 (3) 36 : 37 (4) 46 : 47 (5) 56 : 57

94. If for getting first division, a student needs to score minimum 60% marks in aggregate, then how many students are there who didn't get first class?
(1) One (2) Two (3) Three (4) Four (5) Five

95. The marks scored by Student B and Student C together in subject S_1 is what percentage of the marks scored by A and D together in that subject?
(1) 64.5% (2) 96% (3) 120% (4) 145% (5) 155%

Directions (Q. 96-100) : Study the following table and answer the questions given below. The given table shows the total number of candidates appeared, passed and selected in a competitive examination in different states for the period 2006 to 2011.

State	A			B			C			D		
Year	A	P	S	A	P	S	A	P	S	A	P	S
2006	5600	780	80	7500	480	75	4800	800	80	7500	700	95
2007	4200	800	120	6400	600	72	5500	450	60	7200	540	84
2008	5500	840	72	5400	520	104	4500	540	66	6500	660	77
2009	7200	600	96	6000	540	112	5100	500	55	5400	720	78
2010	8500	800	64	5100	700	60	6800	650	52	6400	640	64
2011	8000	850	68	7000	720	75	6000	640	60	5000	500	58

96. What is the difference between the average number of students selected in State B and that in State D during the whole period?
(1) 6 (2) 7 (3) 8 (4) 9 (5) 10

97. In the year 2006, which state had the highest percentage candidates passed over the candidates appeared?
(1) A (2) B (3) C (4) D (5) None of these

98. The total number of students selected in State C is approximately what percentage of the total number of students selected in State A?
(1) 70% (2) 75% (3) 80% (4) 85% (5) 90%

99. In which of the following years is the percentage of selected candidates with respect to passed candidates the highest in State D?
(1) 2006 (2) 2007 (3) 2008 (4) 2009 (5) 2011

100. The total candidates passed in State A in the year 2006 is what percentage more than the total candidates passed in State C in the year 2009?
(1) 16% (2) 36% (3) 44.4% (4) 51% (5) 56%

Directions (Q. 101-105) : Study the table carefully to answer the questions that follow

Number of cars (in thousand) of two models (Basic and Premium) produced by five different companies in five different years

Company	A		B		C		D		E	
Year	Basic	Premium	Basic	Premium	Basic	Premium	Basic	Premium	Basic	Premium
2006	4.4	2.5	5.6	2.4	5.4	6.1	7.6	7.5	2.7	5.1
2007	4.9	7.2	9.4	7.2	7.5	8.3	8.4	4.9	4.2	5.5
2008	13.6	15.5	14.8	9.5	12.8	9.9	9.2	8.2	7.7	11.5
2009	6.6	13.9	11.8	11.4	16.6	18.2	10.6	10.4	7.2	12.8
2010	5.8	14.9	12.2	7.2	19.9	22.3	14.6	12.2	13.2	12.2

101. The number of cars of premium model produced by Company D in the year 2009 was approximately what per cent of the total number of cars (both models) produced by Company C in the year 2007?
(1) 70 (2) 51 (3) 56 (4) 61 (5) 66

102. What was the approximate percentage decrease in the number of cars of basic model produced by Company B in the year 2009 as compared to the previous year?
(1) 15 (2) 20 (3) 10 (4) 80 (5) 85

103. What was the average number of cars of premium model produced by Company A over all the years together?
(1) 9000 (2) 8000 (3) 6000 (4) 48000 (5) None of these

104. In which year was the difference between the basic model and the premium model of cars produced by Company E the second highest?
(1) 2010 (2) 2006 (3) 2007 (4) 2008 (5) 2009

105. In which company did the production of cars of premium model consistently increase from the year 2006 to the year 2010?
(1) Both C and E (2) Both C and D (3) C only (4) D only (5) E only

Directions (Q. 106-110) : The table given below is a score card of a test match between two teams T_1 and T_2.

T_1					T_2				
Player	**1st innings**		**2nd innings**		**Player**	**1st innings**		**2nd innings**	
	Run	**Ball**	**Run**	**Ball**		**Run**	**Ball**	**Run**	**Ball**
A_1	105	156	44	64	A_2	28	40	92	172
B_1	44	72	60	88	B_2	46	72	26	30
C_1	65	110	112	145	C_2	97	167	65	78
D_1	8	25	47	62	D_2	63	90	87	116
E_1	86	110	30	64	E_2	56	70	46	76
F_1	34	56	36	42	F_2	74	90	57	72
G_1	15	35	42	95	G_2	25	20	35	32
H_1	7	9	28	22	H_2	8	8	DNB	0
I_1	18	26	4	3	I_2	14	47	DNB	0
J_1	9	4	16	12	J_2	5	8	DNB	0
K_1	5	12	10	5	K_2	2	3	DNB	0

106. What is the average runs scored by the players of T_1 in the 1st innings?
(1) 35 (2) 36 (3) 37 (4) 38 (5) 40

107. The runs scored by players A_2, B_2 and C_2 in 1st innings is what percentage of the total runs scored by T_2 in 1st innings (approximate) ?
(1) 35 (2) 36 (3) 37 (4) 38 (5) 40

108. What is the ratio of runs scored by players G_1, H_1, I_1 and J_1 in 2nd innings to the runs scored by A_2, B_2, C_2 and D_2 in the 2nd innings?
(1) 1 : 3 (2) 2 : 3 (3) 3 : 4 (4) 4 : 5 (5) 3 : 5

109. What is the percentage rise/fall of runs scored by player G_1 from 1st innings to 2nd innings?
(1) 60% (2) 90% (3) 120% (4) 150% (5) 180%

110. The strike rate of player D_2 in the 2nd innings is how much more or less than the strike rate of E_2 in the 1st innings (strike rate is runs scored per 100 balls) ?
(1) 17.5% (2) 11.25% (3) 7.5% (4) 6.25% (5) 5%

Directions (Q. 111-115) : Study the table carefully to answer the questions that follow:

Number of Research Papers and Articles published by six different scholars (person) in five different journals

Journal	**Edutrack**		**Frontier**		**Educon**		**New Era**		**Eduforms**	
Person	**Research Papers**	**Articles**	**Research Papers**	**Articles**	**Research Papers**	**Articles**	**Research Papers**	**Articles**	**Research Papers**	**Articles**
Anand	27	45	17	48	42	38	8	12	22	11
Vijay	16	35	6	24	12	4	6	14	38	25
Naidu	26	39	12	32	22	18	2	24	57	35
Mohan	42	75	22	39	62	36	12	16	39	48
Neeta	48	32	28	30	54	49	32	24	44	32
Ronit	13	23	29	21	69	56	19	4	11	18

111. How much more is the approximate percentage of the number of Research papers that were published by Neeta in Educon as compared to the number of Research papers that were published by Vijay in Eduforms?
(1) 52 (2) 42 (3) 152 (4) 147 (5) 47

112. What is the difference between the total number of Research papers published by Anand, Vijay and Neeta together in Educon and the total number of Articles published by Mohan, Naidu and Ronit together in Edutrack?
(1) 33 (2) 27 (3) 32 (4) 29 (5) None of these

113. Who published the third highest number of Research papers and Articles together in Eduforms?
(1) Anand (2) Vijay (3) Neeta (4) Mohan (5) Naidu

114. What is the average number of Research papers published by all the six scholars together in Frontier?
(1) 14 (2) 16 (3) 17 (4) 15 (5) None of these

115. The total number of Research papers and Articles together published by Mohan in Edutrack is approximately what percentage of the total number of Articles published by all the six scholars together in New Era?
(1) 145 (2) 117 (3) 137 (4) 132 (5) 124

Directions (Q. 116-115) : Read the following table carefully and answer the following questions. The table shows the percentage of marks of students A, B, C, D, E and F got in different subjects—Maths, Physics, Chemistry, Biology, Hindi, English and Sanskrit—and each subject has different maximum marks.

Max Marks	Subject						
	Maths (200)	Physics (100)	Chemistry (100)	Biology (100)	Hindi (150)	English (150)	Sanskrit (80)
A	72%	77%	61%	67%	72%	78%	40%
B	44%	62%	78%	73%	60%	84%	55%
C	80%	68%	45%	56%	48%	64%	60%
D	66%	45%	65%	53%	46%	52%	30%
E	70%	55%	66%	63%	58%	38%	50%
F	63%	42%	48%	51%	66%	46%	75%

116. What is the percentage marks scored by Student B in all the subjects together?
(1) 62.2% (2) 63.75% (3) 64% (4) 67.5% (5) 57.5%

117. The marks scored by Student F in Hindi is what percentage of the marks scored by Student B in Maths?
(1) 112.5% (2) 88.88% (3) 78.5% (4) 117.5% (5) 120%

118. What is the average marks scored in English?
(1) 90 (2) 90.5 (3) 91 (4) 91.5 (5) 92

119. The total marks scored by Student A is what percentage more than the total marks scored by Student D? (Answer in approximate value)
(1) 18% (2) 24% (3) 30% (4) 32% (5) 36%

120. The percentage marks scored by Student B in Chemistry is what per cent of the percentage marks scored by C in Hindi?
(1) 122.5% (2) 132.5% (3) 142.5% (4) 152.5% (5) 162.5%

Directions (Q. 121-125) : Study the table below and answer the questions that follow:

Oil import from different countries over the years (in million tonnes)

Country	2007-08	2008-09	2009-10	2010-11	2011-12
Saudi Arabia	28.8	29.9	27.2	27.4	32.6
Iran	20.5	21.8	21.2	18.5	17.5
Iraq	15.8	14.4	15	17.2	24.6
Nigeria	11.6	10.5	13.2	15.9	14.2
Kuwait	13.9	14.8	11.8	11.5	17.8
Venezuela	7.2	7.6	7.3	10.3	9.6

121. What is the ratio of average of imports from Iraq to that from Venezuela for all the years?
(1) 14 : 29 (2) 29 : 14 (3) 39 : 23 (4) 23 : 39 (5) None of these

122. In which of the following years is the percentage increase/decrease in oil import from Nigeria the maximum?
(1) 2008-09 (2) 2010-11 (3) 2009-10 (4) 2011-12 (5) None of these

123. What is the approximate percentage of oil import from Iran in the year 2009-10 with respect to total oil import in all the years together?
(1) 20% (2) 23% (3) 21% (4) 25% (5) None of these

124. What is the approximate average of percentage increase or decrease in oil import from Kuwait over its previous year for the given period?
(1) 4% (2) 5% (3) 15% (4) 21% (5) None of these

125. Average oil import from all the countries in the year 2011-12 is approximately what percentage of that in the year 2009-10?
(1) 21.32% (2) 15.38% (3) 115.38% (4) 121.32% (5) None of these

Directions (Q. 126-130): Study the table and answer the questions that follow:

The first table shows the net sales of different organisations and YoY% change in their sales for the first quarter of FY 2012

Organisation	Net profit (in Rs. crore)	% change
Dutch Bank	7570	26.6
CLSA	6186	2.6
Morgan Stanley	7372	23
Motilal Oswal Security	599	24.1
HDFC Bank	609	26.1
Citi Bank	597	24.0

The second table shows the net profit and YoY% change in their profit for the first quarter of FY 2012.

Organisation	Net profit (in Rs. crore)	% change
Dutch Bank	546	-15.2
CLSA	502	-22
Morgan Stanley	623	-3
Motilal Oswal Security	377	20.4
HDFC Bank	359	14.6
Citi Bank	388	24.0

126. What was the approximate average (in ` crore) of net profits of Dutch Bank and CLSA in the first quarter of the previous year?
(1) 700 (2) 644 (3) 636 (4) 605 (5) None of these

127. What is approximate percentage of net sales of Dutch Bank with respect to the net sales of all the organisations in the first 'quarter of fiscal year 2012?
(1) 35% (2) 30% (3) 29% (4) 33% (5) None of these

128. Which of the following organisations has net profit to net sales ratio the maximum?
(1) CLSA (2) Morgan Stanley (3) Motilal Oswal
(4) HDFC Bank (5) Citi Bank

129. Which of the following banks has net profit to net sales ratio the least?
(1) Dutch Bank (2) CLSA (3) Morgan Stanley
(4) Motilal Oswal (5) HDFC Bank

130. What was the approximate average (in `) of net sales of HDFC and Citi Bank sales in the first quarter of the previous year?
(1) 482 crore (2) 473 crore (3) 462 crore
(4) 445 crore (5) Can't be determined

Directions (Q. 131-135) : The following table shows the population of six different cities, ratio of males to females among them, percentage of adult males and adult females (Population is given in lakh) :

City	Population (in lakh)	Males : Females	% Adult males	% Adult females
A	7.8	7 : 6	62%	65%
B	3.6	5 : 4	70%	72%
C	4.5	2 : 3	68%	64%
D	6.8	9 : 8	72%	70%
E	7.2	4 : 5	65%	72%
F	5.4	2 : 1	75%	64%

131. What is the difference between total adult males and total adult females in City A?
(1) 21500 (2) 22800 (3) 24200 (4) 26400 (5) 27500

132. What is the average number of adult males taking all six cities together?
(1) 1.98 lakh (2) 2.1 lakh (3) 2.42 lakh (4) 2.64 lakh (5) 3 lakh

133. The total number of minor females in City C is approximately what percentage more or less than the total number of minor males in City F?
(1) 8% (2) 10% (3) 12% (4) 15% (5) 16%

134. The total number of minor males in City E is approximately what percentage of the total number of adult males in City B?
(1) 60% (2) 75% (3) 80% (4) 96% (5) 120%

135. What is the difference between adult females and minor males in City C?
(1) 1.1141akh (2) 1.3261akh (3) 1.152 lakh (4) 1.6521akh (5) None of these

Directions (Q. 136-140) : The following table shows the percentage of marks scored by six students in six different subjects.

Students	Physics (80)	Chemistry (80)	Biology (80)	Hindi (100)	English (120)	Maths (150)
A	58.75%	55%	62.50%	67%	55%	84%
B	77.50%	60%	60%	72%	60%	72%
C	80%	71.25%	81.25%	65%	75%	66%
D	68.75%	78.75%	72.50%	55%	80%	60%
E	75%	70%	65%	48%	65%	78%
F	67.50%	87.50%	50%	75%	50%	70%

136. What is the total marks scored by Student D in all six subjects together?
(1) 411 (2) 413 (3) 415 (4) 417 (5) 419

137. What is the average marks scored by all students in Physics?
(1) 51 (2) 54 (3) 57 (4) 60 (5) 63

138. The marks scored by Student B in Maths is approximately what per cent of marks scored by Student E in Physics?
(1) 55.55% (2) 80% (3) 120% (4) 150% (5) 180%

139. What is the ratio of marks scored by Student B in English to marks scored by Student A in Maths?
(1) 3:5 (2) 4:7 (3) 5:9 (4) 3:4 (5) 4:5

140. The marks scored by Student F in Maths is approximately what per cent more or less than the marks scored by Student E in Chemistry?
(1) 75% (2) 77.5% (3) 82.5% (4) 85% (5) 87.5%

Directions (Q. 141-145) : The following table shows the population of six different cities, ratio of males to females among them, the percentage of literate males and the percentage of literate females. Answer the given questions based on this table.

City	Population (in lakh)	Males : Females	% Literate males	% Literate females
A	1.2	7 : 5	67%	57%
B	1.75	3 : 2	64%	60%
C	3.4	8 : 9	71%	53%
D	2.5	2 : 3	73%	61%
E	1.8	1 : 1	65%	65%
F	3.0	3 : 2	68%	56%

141. What is the total number of illiterate females in all six cities together?
(1) 2.769 lakh (2) 2.842 lakh (3) 2.888 lakh (4) 2.926 lakh (5) 2.964 lakh

142. The total number of illiterate females of City C is approximately what per cent of the total number of literate males of City F?
(1) 65% (2) 69% (3) 74% (4) 78% (5) 81%

143. What is the average number of literate females taking all six cities together?
(1) 62140 (2) 63580 (3) 63850 (4) 62410 (5) 64550

144. What is the ratio of illiterate males to literate females of City B?
(1) 3 : 5 (2) 4 : 9 (3) 9 : 10 (4) 3 : 10 (5) 5 : 8

145. What is the difference between total literate males of City A and B together and the total literate females of City C and D together?
(1) 64400 (2) 72800 (3) 84100 (4) 84400 (5) 9200

Directions (Q. 146-150): The following table shows the total number of students appeared in an entrance exam from six different schools in different years, and the ratio of passed to failed students among them. Answer the given questions based on this table.

School	2010		2011		2012	
	Total appeared	Pass : Fail	Total Appeared	Pass : Fail	Total appeared	Pass : Fail
A	646	11 : 8	754	7 : 6	672	3 : 5
B	847	4 : 7	845	8 : 5	952	9 : 8
C	810	8 : 7	792	7 : 4	637	4 : 3
D	876	7 : 5	828	11 : 7	988	7 : 12
E	870	3 : 2	726	7 : 4	715	8 : 5
F	986	17 : 12	867	12 : 5	924	8 : 13

146. What is the difference between the total number of passed students from School D in the year 2010 and the total number of failed students from School B in the year 2012?
(1) 56 (2) 60 (3) 63 (4) 68 (5) 72

147. What is the total number of failed students from School F in all three years together?
(1) 1145 (2) 1235 (3) 1325 (4) 1415 (5) 1505

148. What is the total number of passed students from all six schools in the year 2011?
(1) 2850 (2) 2940 (3) 2990 (4) 3010 (5) 3060

149. What is the average number of failed students from School C in all three years together?
(1) 311 (2) 312 (3) 313 (4) 314 (5) 315

150. The total number of passed students from School E in the year 2010 is approximately what percentage of the total number of failed students from School A in the year 2011?
(1) 66.66% (2) 80% (3) 112.5% (4) 125% (5) 150%

Directions (Q. 151-155): The following table shows the expenditure (in `crore) of three companies A, B and C and the percentage profit of these companies in different years.

Year	Company A		Company B		Company C	
	Expenditure	Profit	Expenditure	Profit	Expenditure	Profit
2007	17.8	16.20%	16.5	18.50%	26	20.50%
2007	19.6	24.50%	17.4	18%	27.5	30%
2009	21	19%	20.5	21.80%	24.3	28.40%
2010	20.4	34.80%	23	25%	22.5	22%
2011	21.5	30%	22.6	28%	25.4	21.50%
2012	23.2	31.50%	24.8	27.50%	29.75	20%

151. What is the income (in ?)of Company C in the year 2011 ?
(1) 25.461 crore (2) 19.312crore (3) 30.861 crore (4) 32.612 crore (5) None of these

152. What is the difference between the profits of Company A and Company B in the year 201.2?
(1) ` 42.4 1akh (2) ` 48.8 1akh (3) ` 51.4 lakh (4) ` 56.2 1akh (5) ` 57.5 1akh

153. The expenditure of Company A in the year 2007 and 2012 together is approximately what per cent of the expenditure of Company C in the year 2008 and 2010 together?
(1) 64% (2) 72% (3) 78% (4) 82% (5) 86%

154. The percentage profit of Company C in the year 2009 is approximately what per cent more or less than the percentage profit of Company A in the year 2007?
(1) 72% (2) 75% (3) 78% (4) 81% (5) 89%

155. The income of Company B in the year 2010 is approximately what per cent of the expenditure of Company A in the year 2009?
(1) 112% (2) 123% (3) 137% (4) 142% (5) 148%

Directions (Q. 156-160): Six companies A, B, C, D, E and F produce items which come in three models I_1 **I_2 and I_3. The following table shows the total items produced by these companies and the ratios of I_1, I_2 and I_3 among them.**

Company	Total items	$I_1 : I_2 : I_3$
A	80370	25 : 23 : 9
B	61050	19 : 15 : 21
C	77490	23 : 18 : 22
D	61880	21 : 23 : 24
E	73130	25 : 24 : 22
F	93160	3 : 5 : 9

156. What is the total number of items I_1 produced by Company A and B together?
(1) 51280 (2) 53410 (3) 54720 (4) 55860 (5) 56340

157. What is the difference between the total number of items I_1 and I_3 produced by Company E?
(1) 3090 (2) 3140 (3) 3270 (4) 3320 (5) 3450

158. The total number of items I_2 produced by Company A is approximately what per cent of the total number of items I_1 produced by it?
(1) 23% (2) 67.64% (3) 92% (4) 108.7% (5) None of these

159. The total number of items I_1 produced by Company D is approximately what per cent more/less than the total number of items I_1 produced by Company F?
(1) 13.5% (2) 16.25% (3) 17.75% (4) 19.5% (5) 24%

160. What is the total number of items I_2 produced by all six companies together?
(1) 142580 (2) 144270 (3) 146820 (4) 148360 (5) None of these

Directions (Q. 161-165): The following table shows the percentage of marks obtained by six students in five different subjects. Answer the following questions based on this table.

Students	Physics (Out of 75)	Chemistry (Out of 75)	Maths (Out of 200)	Hindi (Out of 50)	English (Out of 150)
A	84%	42%	67%	44%	74%
B	68%	64%	49%	74%	52%
C	72%	54%	58%	68%	64%
D	48%	82%	63%	48%	70%
E	70%	78%	71%	56%	78%
F	56%	66%	55%	76%	66%

161. What is the average marks scored by all the students in Physics?
(1) 49.75 (2) 52.25 (3) 54 (4) 57.5 (5) 47.5

162. What is the total marks scored by Student F in all the subjects together?
(1) 332 (2) 334.5 (3) 335 (4) 336.5 (5) 338.5

163. What is the overall percentage of marks scored by Student B? (Answer in approximate value.)
(1) 53% (2) 57% (3) 61% (4) 63% (5) 51%

164. The marks scored by Student C in Physics is approximately what per cent of the marks scored by him in English?
(1) 56% (2) 60% (3) 62% (4) 67% (5) 69%

165. What is the difference between the total marks obtained by Student D in Chemistry and English and that obtained by Student F in the same subject?
(1) 14.5 (2) 16 (3) 18 (4) 19.5 (5) 16.5

Directions (166-170) : Study the following table carefully to answer these questions.

Number of students enrolled in five colleges over the years

College → Year ↓	A	B	C	D	E
2007	550	430	600	420	300
2008	400	450	300	620	520
2009	1000	900	700	650	520
2010	850	450	720	650	420
2011	800	650	850	420	850

166. In the year 2009, 80% of the students enrolled in College A appeared in a competitive examination. Out of these, 60% students passed. How many students passed the examination?
(1) 320 (2) 455 (3) 535 (4) 480 (5) None of these

167. In 2008, from all the colleges together overall 70% of the students got enrolled for computer course. How many students got enrolled for the course?
(1) 1702 (2) 1593 (3) 1603 (4) 1105 (5) None of these

168. What is the ratio of the average number of students enrolled with all the colleges together during the year 2009 to that during 2010?
(1) 375 : 364 (2) 364 : 365 (3) 377 : 309 (4) 389 : 367 (5) None of these

169. The number of students enrolled in College A in the year 2009 is approximately what per cent more than the number of students enrolled in College B in the year 2011 ?
(1) 65% (2) 70% (3) 35% (4) 54% (5) None of these

170. In 2010, from all colleges together 10% of the students enrolled went abroad. How many students went abroad?
(1) 409 (2) 429 (3) 609 (4) 509 (5) 309

Directions (Q. 171-175) : Study the table carefully to answer the questions that follow:

The table shows the percentage of 25000 people who are involved in different professions, and the percentage of female and male professionals among them.

Professions	Percentage of people	Percentage of females	Percentage of males
Banking	20	40	-
Law	15	20	-
Teaching	30	-	40
Engineering	25	-	30
Medical	10	60	-

171. The total number of people in the Teaching profession is what percentage of the total number of people in the Medical profession?
(1) 175% (2) 225% (3) 325% (4) 140% (5) 300%

172. What is the ratio of the total number of males in the Medical and Banking professions together to the total number of females in the same profession together?
(1) 3:5 (2) 7:5 (3) 8:7 (4) 7:8 (5) None of these

173. The females in the Engineering profession are approximately what per cent of the males in the Banking profession?
(1) 135% (2) 125% (3) 146% (4) 153% (5) None of these

174. What is the ratio of the total number of males in the Banking and Medical professions together to the total number of females in the Law and Teaching professions together?
(1) 4:5 (2) 3:7 (3) 16:21 (4) 21:16 (5) 21:4

175. The total number of females in the Engineering profession is approximately what percentage more than the number of males in the Law profession?
(1) 46% (2) 51% (3) 37% (4) 54% (5) None of these

Directions (Q. 176-180) : Study the table carefully to answer the questions that follow:

Monthly Bill (in rupees) landline phone, electricity of laundry and mobile phone paid, by three different people in five months

Month	Monthly Bills											
	Landline Phone			Electricity			Laundry			Mobile Phone		
	Ravi	Dev	Manu	Ravi	Dev	Manu	Ravi	Dev	Manu	Ravi	Dev	Manu
March	234	190	113	145	245	315	93	323	65	144	234	345
April	124	234	321	270	220	135	151	134	35	164	221	325
May	156	432	211	86	150	98	232	442	132	143	532	332
June	87	123	124	124	150	116	213	324	184	245	134	125
July	221	104	156	235	103	131	143	532	143	324	432	543

176. What is the total amount of bill paid by Dev in the month of June for all the four commodities?
(1) ` 608 (2) ` 763 (3) ` 731 (4) ` 683 (5) ` 674

177. What is the average electricity bill paid by Manu over all the five months together?
(1) ` 183 (2) ` 149 (3) ` 159 (4) ` 178 (5) ` 164

178. What is the difference between the mobile phone bill paid by Ravi in the month of May and the laundry bill paid by Dev in the month of March?
(1) ` 180 (2) ` 176 (3) ` 190 (4) ` 167 (5) ` 196

179. In which months respectively did Manu pay the second highest mobile phone bill and the lowest electricity bill?
(1) April and June (2) April and May (3) March and June
(4) March and May (5) July and May

180. What is the ratio of the electricity bill paid by Manu in the month of April to the mobile phone bill paid by Ravi in the month of June?
(1) 27:49 (2) 27:65 (3) 34:49 (4) 135:184 (5) 13:24

Directions (Q. 181-185) : Study the following table carefully and answer the questions that follow:

Station	Arrival time	Departure time	Halt time (in minutes)	Distance travelled from origin (in km)	No. of passengers boarding the trainat each station
Dadar	Starting	12.05 am	___	0 km	437
Vasai Road	12.53 am	12.56 am	3 minutes	42 km	378
Surat	4.15 am	4.20 am	5 minutes	257 km	458
Vadodara	6.05 am	6.10 am	5 minutes	386 km	239
Anand Jn	6.43 am	6.45 am	2 minutes	422 km	290
Nadiad Jn	7.01 am	7.03 am	2 minutes	440 km	132
Ahmedabad	8.00 am	8.20 am	20 minutes	486 km	306
Bhuj	5.40 pm	Ending point	___	977 km	None

181. What is the distance travelled by the train from Surat to Nadiad Jn?
(1) 176km (2) 188 km (3) 183 km (4) 193 km (5) 159 km

182. How much time does the train take to reach Ahmedabad after departing from Anand Jn (including the halt time) ?
(1) 1 hr 59 min (2) 1 hr 17 min (3) 1 hr 47 min (4) 1 hr 45 min (5) 1 hr 15 min

183. What is the ratio of the number of passengers boarding from Vasai Road to that from Ahmedabad in the train ?
(1) 21:17 (2) 13:9 (3) 21:19 (4) 15:13 (5) 13:15

184. If the halt time (stopping time) of the train at Vadodara is decreased by 2 minutes and increased by 23 minutes at Ahmedabad, at what time will the train reach Bhuj?
(1) 6.10am (2) 6.01 pm (3) 6.05 am (4) 6.50 pm (5) 6.07 pm

185. The distance between which two stations is the second lowest?
(1) Nadiad Jn to Ahmedabad (2) Anand Jn to Nadiad Jn (3) Dadar to Vasai Road
(4) Anand Jn to Vadodara (5) Vasai Road to Surat

Directions (Q. 186-190) : Study the table carefully to answer the questions that follow.

Maximum and Minimum temperature (in degree Celsius) recorded on 1st day of each month of five different cities

Month	Temperature									
	Bhuj		Sydney		Ontario		Kabul		Beijing	
	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min
1st September	24	14	12	2	5	1	34	23	12	9
1st October	35	21	5	-1	15	6	37	30	9	3
1st November	19	8	11	3	4	0	45	36	15	1
1st December	9	2	-5	-9	-11	-7	31	23	2	-3
1st January	-4	-7	-11	-13	-14	-19	20	11	5	-13

186. What is the difference between the maximum temperature of Ontario on 1st November and the minimum temperature of Bhuj on 1st January?
(1) 3°C (2) 18°C (3) 15°C (4) 9°C (5) 11°C

187. In which month respectively is the maximum temperature of Kabul the second highest and the minimum temperature of Sydney the highest?
(1) 1st October and 1st January (2) 1st October and 1st November
(3) 1st December and 1st January (4) 1st September and 1st January
(5) 1st December and 1st September

188. In which month (on 1st day) is the difference between maximum temperature and minimum temperature of Bhuj the second highest?
(1) 1st September (2) 1st October (3) 1st November
(4) 1st December (5) 1st January

189. What is the average maximum temperature of Beijing over all the months together?
(1) 8.4°C (2) 9.6°C (3) 7.6°C (4) 9.2°C (5) 8.6°C

190. What is the ratio of the minimum temperature of Beijing on 1st September to the maximum temperature of Ontario on 1st October?
(1) 3:4 (2) 3:5 (3) 4:5 (4) 1:5 (5) 1:4

Directions (Q. 191-195): Study the following table carefully and answer the questions given.

Number of 5 types of cars (Swift, SX4, Ertiga, Zen, Echo) manufactured (in thousand) by Maruti over the years

Year	Types of Car				
	Swift	SX4	Ertiga	Zen	Echo
2007	250	200	128	140	115
2008	200	230	150	155	120
2009	230	225	142	160	135
2010	245	210	170	175	125
2011	260	135	180	185	130
2012	275	155	230	220	120

191. Which type of cars manufactured by Maruti during 2007 to 2012 is the maximum?
(1) Swift (2) Zen (3) Echo (4) Ertiga (5) SX4

192. What was the percentage increase in the production of Swift from 2007 to 2012?
(1) 10% (2) 12% (3) 16% (4) 22% (5) 8%

193. Which type of cars registered a continuous increase in the production over the years?
(1) Swift (2) Zen (3) SX4 (4) Ertiga (5) Echo

194. The production of Echo in the year 2011 was what per cent of the production of SX4 in the year 2010?
(1) 67.21% (2) 57.97% (3) 59% (4) 61.9% (5) 65.4%

195. What was the percentage increase in the production of Zen from 2008 to 2010?
(1) 7.8% (2) 10.8% (3) 12.9% (4) 13.5% (5) 14.2%

Directions (Q. 196-200): Study the following table carefully and answer the questions given below:

The table shows the number of people working in various departments of various organisations.

Department	Organisation				
	P	Q	R	S	T
Production	1050	1015	976	888	1004
IT	1017	960	786	1025	963
Accounts	1382	1384	1275	1300	1290
Legal	786	745	801	800	735
Finance	1542	1545	1550	1570	1580
Marketing	48	54	36	30	53

196. The total number of employees working in the Marketing Departments is approximately what per cent of the total number of employees working in the Production Departments of all the organisations together?
(1) 4.5% (2) 7% (3) 8.5% (4) 10% (5) 12%

197. What is the approximate difference between the average number of people working in the Accounts Departments and that in the Finance Departments of all the organisations together?
(1) 331 (2) 231 (3) 430 (4) 546 (5) 210

198. What is the ratio of the total number of employees working in Organisation P to the total number of, employees working in Organisation T?
(1) 45 : 233 (2) 225 : 233 (3) 125 : 233 (4) 233 : 225 (5) 625 : 233

199. What is the total number of employees working in all departments of all the organisations together?
(1) 28910 (2) 27690 (3) 28901 (4) 26960 (5) 28190

200. The number of people working in the IT Department of Organisation Q is approximately what per cent of the total number of employees working in Organisation Q?
(1) 27% (2) 15% (3) 17% (4) 12% (5) 29%

Directions(Q. 201-205): Study the following table carefully to answer the questions that follow.

Total number of students studying in various colleges over the years

Year	College				
	A	B	C	D	E
2007	860	890	780	900	840
2008	910	980	820	970	880
2009	930	1040	910	908	990
2010	990	1000	980	940	1000
2011	940	940	980	960	1050
2012	980	960	1020	920	1120

201. What is the ratio of the number of students studying in College A to the number of students studying in College E in the year 2012?
(1) 15 : 14 (2) 7 : 8 (3) 9 : 8 (4) 10 : 11 (5) None of these

202. What is the difference between the average number of students studying in College A over the given period and the average number of students studying in College C over the same period?
(1) 23 (2) 128 (3) 120 (4) 32 (5) 20

203. What is the difference between the total number of students studying in College B over the given period and the total number of students studying in College D over the same period?
(1) 218 (2) 35 (3) 32 (4) 212 (5) None of these

204. What is-the average number of students studying in College E over the given period?
(1) 928 (2) 930 (3) 933 (4) 941 (5) 980

205. The number of students studying in College C in the year 2010 is approximately what per cent of the total number of students studying in various colleges in that year?
(1) 20 (2) 23 (3) 17 (4) 25 (5) None of these

SHORT ANSWER

1. (2)	2. (3)	3. (5)	4. (2)	5. (4)	6. (5)	7. (1)	8. (1)
9. (5)	10. (3)	11. (1)	12. (2)	13. (5)	14. (4)	15. (3)	16. (3)
17. (4)	18. (2)	19. (5)	20. (4)	21. (4)	22. (2)	23. (3)	24. (1)
25. (5)	26. (3)	27. (1)	28. (5)	29. (4)	30. (4)	31. (2)	32. (4)
33. (5)	34. (4)	35. (3)	36. (3)	37. (3)	38. (4)	39. (3)	40. (1)
41. (3)	42. (2)	43. (5)	44. (4)	45. (1)	46. (3)	47. (5)	48. (1)
49. (4)	50. (5)	51. (3)	52. (4)	53. (1)	54. (3)	55. (2)	56. (4)
57. (2)	58. (2)	59. (3)	60. (3)	61. (5)	62. (3)	63. (1)	64. (3)
65. (4)	66. (3)	67. (2)	68. (3)	69. (4)	70. (5)	71. (1)	72. (2)
73. (5)	74. (2)	75. (2)	76. (2)	77. (2)	78. (1)	79. (2)	80. (4)
81. (1)	82. (5)	83. (2)	84. (2)	85. (5)	86. (1)	87. (4)	88. (5)
89. (1)	90. (3)	91. (4)	92. (3)	93. (3)	94. (2)	95. (5)	96. (2)
97. (3)	98. (2)	99. (2)	100. (5)	101. (5)	102. (2)	103. (5)	104. (5)
105. (3)	106. (2)	107. (3)	108. (1)	109. (5)	110. (4)	111. (2)	112. (4)
113. (3)	114. (5)	115. (4)	116. (2)	117. (1)	118. (2)	119. (3)	120. (5)
121. (2)	122. (3)	123. (3)	124. (4)	125. (4)	126. (2)	127. (4)	128. (5)
129. (1)	130. (1)	131. (4)	132. (2)	133. (1)	134. (3)	135. (3)	136. (4)
137. (3)	138. (5)	139. (2)	140. (5)	141. (1)	142. (2)	143. (3)	144. (3)
145. (2)	146. (3)	147. (2)	148. (4)	149. (3)	150. (5)	151. (3)	152. (2)
153. (4)	154. (2)	155. (3)	156. (5)	157. (1)	158. (3)	159. (2)	160. (2)
161. (1)	162. (5)	163. (2)	164. (1)	165. (3)	166. (4)	167. (3)	168. (3)
169. (4)	170. (5)	171. (5)	172. (3)	173. (3)	174. (3)	175. (1)	176. (3)
177. (3)	178. (1)	179. (4)	180. (1)	181. (3)	182. (5)	183. (1)	184. (2)
185. (3)	186. (5)	187. (1)	188. (3)	189. (5)	190. (2)	191. (1)	192. (1)
193. (2)	194. (4)	195. (3)	196. (1)	197. (2)	198. (4)	199. (5)	200. (3)
201. (2)	202. (5)	203. (4)	204. (5)	205. (1)			

DETAIL - EXPLANATIONS

1. 2; $V_{Hindi} = \frac{55}{80} \times 100 = 68.75\%$

$S_{che} = \frac{25}{40} \times 100 = 62.5\%$

$\therefore$ Difference = 68.75 - 62.5 = 6.25%

2. 3; $\text{Avg} = \frac{65+48+57+55+64+60+70}{7}$

$= \frac{419}{7} \approx 60$

3. 5; Hindi = $\frac{51}{80}$ × 100 = 63.75%,

Eng = $\frac{48}{80}$ × 100 = 60%,

Maths = $\frac{93}{100}$ × 100 = 93%,

Phy = $\frac{28}{40}$ × 100 = 70%,

Chem = $\frac{27}{40}$ × 100 = 67.5%,

Bio = $\frac{31}{40}$ × 100 = 77.5%

4. 2; % Marks of 'R' = $\frac{(62+32)}{80+40} \times 100$

$= \frac{9400}{120} = 78.33\%$

% marks of 'Q' = $\frac{(48+27)}{80+40} \times 100$

$= \frac{7500}{120} = 62.5\%$

$\therefore$ Diff = 78.33 - 62.5 = 15.83% = 15.8%

5. 4; Total_v = 55 + 70 + 81 + 30 + 28 + 33 = 297
Maximum marks = 80 + 80 + 100 + 40 + 40 + 40 = 380

$\therefore$ Reqd % = $\frac{297}{380}$ × 100 = 78.15 $\approx$ 78%

6. 5; Total failed in school C in 2008 = 354 - 258 = 96
Total appeared in school D in 2006 = 235
Total = 331

7. 1

8. 1; 11 : 4

9. 5; $\frac{435}{546} \times 100 = 80\%$

10. 3; $\frac{435-346}{346} \times 100 = 25.7\%$

11. 1; Avg = $\frac{3.34+5.83+1.69}{3} = \frac{10.86}{3}$ = 3.62 lac

12. 2; $\frac{2.79}{9.45} = \frac{31}{105} = 31:105$

13. 5; Reqd % $\frac{9.45-8.42}{8.42} \times 100 = 12.23\%$

14. 4

15. 3; Reqd % = $\frac{1.44+7.84}{5.53} \times 100$

$= \frac{9.28}{5.53} \times 100 = 167.82\%$

16. 3; Total students who appeared = 3895
Total student who qualified = 1710
$\therefore$ Diff = 3895 - 1710 = 2185

17. 4; 2005 = $\frac{280}{760} \times 100$ = 36.84%

2006 $\rightarrow \frac{225}{530} \times 100 = 42.45\%$

2007 $\rightarrow \frac{480 \times 100}{925} = 51.89\%$

2008 $\rightarrow \frac{345 \times 100}{690} = 50\%$

2009 $\rightarrow \frac{375 \times 100}{650} = 57.69\%$

2010 $\rightarrow \frac{375 \times 100}{725} = 51.72\%$

18. 2; Total qualified = 275 + 215 + 525 + 480 + 480 + 390 = 2365
Total appeared = 840 + 625 + 910 + 825 + 890 + 595 = 4685

$\therefore$ Reqd % = $\frac{2365}{4685} \times 100 = 50.48\%$

19. 5; $S_1 = \frac{480 \times 100}{890} = 53.93\%$,

$S_2 = \frac{450 \times 100}{815} = 55.21\%$

$S_3 = \frac{410 \times 100}{720} = 56.94\%$

$S_4 = \frac{250 \times 100}{590} = 42.37\%$

$S_5 = \frac{280 \times 100}{680} = 41.17\%$

$S_6 = \frac{375 \times 100}{650} = 57.69\%$

20. 4; $Q_{2009} = 250, \quad Q_{2010} = 540$

$\therefore$ % rise = $\frac{540 - 250}{250} \times 100 = \frac{290 \times 100}{250}$

$= 29 \times 4 = 116\%$

21. 4; Total = 58.75 × 0.80 + 78.75 × 0.80 + 81.25 × 0.80 + 82.5 × 0.80 + 77.5 × 0.80 + 76.25 × 0.80
= 47 + 63 + 65 + 66 + 62 + 61 = 364

22. 2; Total P_5 = 0.80 × (77.5 + 83.75 + 55 + 58.75 + 67.5 + 73.75 + 81.25)
= 0.80 × 497.5 = 398

$\therefore$ Avg = $\frac{398}{7} = 56.857 = 56.85$

23. 3; Score of E in $P_1 = 80 \times \frac{85}{100} = 68$

Score of D in $P_1 = 80 \times \frac{52.5}{100} = 42$

$\therefore$ Reqd% = $\frac{68}{42} \times 100 = 161.9\%$

24. 1; Total marks of C =

$\frac{80}{100}\{68.75 + 71.25 + 58.75 + 83.75 + 55 + 67.5\}$

$= 80 \times \frac{405}{100} = 324$

$\therefore$ Reqd percentage = $\frac{324}{480} \times 100 = 67.5\%$

25. 5; Avg of percentage of marks in P_2

$= \frac{78.75 + 60 + 71.25 + 76.25 + 78.75 + 90 + 72.5}{7}$

$= \frac{527.5}{7}$

Avg of percentage of marks in P_5 =

$\frac{77.5 + 83.75 + 55 + 58.75 + 67.5 + 73.75 + 81.25}{7}$

$= \frac{497.5}{7}$

$\therefore$ Avg = $\frac{527.5 + 497.5}{7 \times 2} = \frac{1025}{14} = 73.21$

26. 3; Avg = $\frac{210 + 204 + 231 + 231}{4}$

$= \frac{876}{4} = 219$

27. 1; Total number of girls = 70 + 117 + 54 + 129 + 136 + 176 = 682

28. 5; Diff = 225 - 225 = 0

29. 4; Let the total number of students be x.

$\therefore$ Boys = $\frac{44x}{100}$ and girls = $\frac{56x}{100}$

Diff = $\frac{12x}{100} = 30 \qquad \therefore x = \frac{3000}{12} = 250$

$\therefore$ Boys = $\frac{44}{100} \times 250 = 110$

Similarly,

Total students = $\frac{132 \times 100}{40} = 330$

Girls = $\frac{30 \times 330}{100} = 99$

$\therefore$ Ratio = $\frac{110}{99} = \frac{10}{9}$

30. 4; Students from $F_{1986} = 375$
Students from $C_{1986} = 250$

% = $\frac{375}{250} \times 100 = 150\%$

31. 2; $\frac{1945}{5} = 389$

32. 4

33. 5; $\frac{1102}{2142} \times 100 = 51.44\%$

34. 4; $1480 \times \frac{65}{100} = 962$

35. 3; $1072 \times \frac{3}{4} = 804$

36. 3

37. 3; $I_1(2009)_{sold} = 40 \times \frac{72}{100} \times \frac{5}{9} = 16$ lakh

$I_1(2010)_{sold} = 50 \times \frac{48}{100} \times \frac{5}{8} = 15$ lakh

$\therefore$ Total = 16 + 15 = 31 lakh

38. 4; % $Sale_B = 56\%$; % $Sale_E = 40\%$

$\therefore$ Reqd % = $\frac{56}{40} \times 100 = 140\%$

39. 3; Company D $I_{2\ \text{Produced-2009}}$

$= 40 \times \frac{5}{8} = 25$ lakh

Sold $I_2 = 40 \times \frac{72}{100} \times \frac{4}{9} = 12.8$ lakh

$\therefore$ $Unsold_{2009} = 25 - 12.8 = 12.2$ lakh,

$I_{2\ \text{Produced-2010}} = 50 \times \frac{3}{5} = 30$ lakh

Sold = $50 \times \frac{48}{100} \times \frac{3}{8} = 9$ lakh

$\therefore$ $I_{2\ \text{unsold-2010}} = 30 - 9 = 21$ lakh

$\therefore$ Total = 21 + 12.2 = 33.2 lakh

40. 1; $I_{1\ A\ \text{in}\ 2010} = 48 \times \frac{65}{100} \times \frac{7}{13} = 16.8$ lakh

$I_{1\ E\ \text{in}\ 2009} = 25 \times \frac{50}{100} \times \frac{2}{5} = 5$ lakh

Reqd % = $\frac{16.8}{5} \times 100 = 336\%$

41. 3: $Total_{sold} = 45.5 \times \frac{4}{7} + 48.6 \times \frac{5}{9} + 40 \times \frac{2}{5} +$

$55 \times \frac{3}{5} + 64.4 \times \frac{3}{7} + 68 \times \frac{5}{8}$

= 26 + 27 + 16 + 33 + 27.6 + 42.5

= 172.1 thousand

42. 2

43. 5; $Sold_{2009} = 34$ thousand

$Unsold_{2006} = 20$ thousand

Reqd% = $\frac{34}{20}$ xl00 = 170%

44. 4; Unsold $C_{2O08} = 51x\frac{8}{17} = 24$ thousand

Sold $B_{20l0} = 85x\frac{8}{17} = 40$ thousand

% less = $\frac{40-24}{40} \times 100 = \frac{1600}{40} = 40\%$

45. 1; Sold = 30 + 24.8 + 28.2 + 34 + 24.2 + 37.26 = 178.46 thousand

Unsold = 20 + 27.9 + 28.2 + 17 + 36.3 + 24.84 = 154.24 thousand

$\therefore$ Diff = 178.46 - 154.25 = 24.22 thousand

46. 3; Overall by B in all subjects

% = $\frac{399}{600} \times 100 = 66.5$

47. 5; Ratio = $\frac{360}{435} = \frac{24}{99}$ ie 24 : 29

48. 1; Average = $\frac{441}{6} = 73.5$

49. 4; Average marks = $\frac{633}{6} = 105.5$

$\therefore$ % average marks = $\frac{105.5}{150} \times 100 \approx 70.3\%$

50. 5;

$Total_A = 84 + 66 + 73 + 61 + 24 + 52 = 360$

$Total_E = 108 + 78 + 78 + 70 + 39 + 48 = 421$

$\therefore$ Reqd % = $\frac{421-360}{360} \times 100 = \frac{6100}{360} \approx 17\%$

51. 3; % rise = $\frac{18.8-16}{16} \times 100 = \frac{280}{16}$

= 17.5%

52. 4; $Sale_{2008} = 860000 \times \frac{90.6}{100} = 779160$

$Sale_{2009} = \frac{1120000 \times 81}{100} = 988200$

$\therefore$ Reqd % = $\frac{988200 - 779160}{779160} \times 100 = 26.8\%$

53. 1; Total rejected

$$= \frac{\{12.8 \times 3.8 + 13.2 \times 5.7 + 16 \times 2.4 + 12.4 \times 9.2 + 17.5 \times 4.1 + 8.6 \times 4.7 + 14.8 \times 3.6\}}{100}$$

$$= \frac{48.64 + 75.24 + 38.4 + 114.08 + 71.75 + 40.42 + 53.28}{100}$$

$$= \frac{441.81}{100} = 441810$$

54. 3; Reqd % = $\frac{9318210}{11680000} \times 100$

= 79.778 = 80%

55. 2; Percentage rise = A = 28.125%,

B = 15.15%, C = 17.5 %, D = 30.64%,

E = 17.14%, F = 41.86%, G = 18.2%
So Company B has maximum rise.

56. 4; Total appeared = 4730, Total passed = 2640,
$\therefore$ Difference = 4730 - 2640 = 2090

57. 2; $\text{Avg} = \frac{472+324+480+840}{4}$

$= \frac{2116}{4} = 529$

58. 2; $A = \frac{496}{850} \times 100 = 58.35\%$,

$B = \frac{315}{600} \times 100 = 52.5\%$

$C = \frac{464}{740} \times 100 = 62.7\%$

$D = \frac{780}{1165} \times 100 = 66.95\%$

59. 3; $A_{2003} = 360$, $A_{2004} = 504$

$\therefore$ % rise $= \frac{(504-360)}{360} \times 100 = 40\%$

60. 3; Total passed = 2564
Total appeared = 4275

$\therefore$ Reqd % $= \frac{2564}{4275} \times 100 \approx 60\%$

61. 5; $T_B = 216000 \times \frac{22.6}{100} = 48816$

$T_C = 264000 \times \frac{14.8}{100} = 39072$
Difference = 48816 – 39072 = 9744

62. 3; $\text{Income}_C = \frac{72000 \times 100}{24} = 300000$

$\text{Income}_D = \frac{86400 \times 100}{18} = 480000$

$\therefore$ Reqd % $= \frac{300000}{480000} \times 100 = 62.5\%$

63. 1; $E_{Ent} = 8.6\%$, $F_{trn} = 16\%$

$\therefore$ Reqd % $= \frac{8.6}{16} \times 100 = 53.75\%$

64. 3; Reqd amount = (16.3 + 18.6)% of 420000
= 146580

65. 4; $A_{Rent} = 40000 \times \frac{15}{100} = 6000$

$D_{Clothes} = 36000 \times \frac{12}{100} = 4320$

$\therefore$ Reqd % $= \frac{(6000-4320)}{4320} \times 100 = \frac{168000}{4320} =$ 38.88%

66. 3; Total females $= \frac{1173}{3} \times 1 + \frac{1547}{17} \times 9 + \frac{1305}{5} \times 2 +$

$\frac{1488}{16} \times 9 + \frac{1335}{15} \times 7 + \frac{1199}{11} \times 6$
= 391 + 819 + 522 + 837 + 623 + 654 = 3846

$\therefore$ Average $= \frac{3846}{6} = 641$

67. 2; $\therefore$ Reqd % $= \frac{720}{820} \times 100 = 87.8 \approx 88\%$

68. 3; Total $\text{Male}_{SONY} = 3690$
$\therefore$ Average = 615
$\Rightarrow \text{Star}_D = 1236$

$\therefore$ Reqd % $= \frac{615}{1236} \times 100 = 49.75 \approx 50\%$

69. 4; $\text{Male}_c = 783 \quad \Rightarrow \text{Female}_F = 660$

$\therefore$ Reqd% $= \frac{783-660}{660} \times 100 = 18.636\%$

70. 5; $\text{Male}_{ZEE} = 4201$
$\Rightarrow \text{Female}_{ZEE} = 3846$
$\therefore$ Difference = 4201 - 3846 = 355

71. 1; Total $= \frac{7210}{5} \times 3 + \frac{4800}{16} \times 9$

$+ \frac{5670}{9} \times 5 + \frac{6400}{16} \times 11 + \frac{7200}{18} \times 11 + \frac{7080}{12} \times 7$
= 4326 + 2700 + 3150 + 4400 + 4400 + 4130 = 23106

$\therefore$ Average $= \frac{23106}{6} = 3851$

72. 2; Appeared girls $= \frac{6400}{16} \times 5 = 2000$
Number of girls passed from $S_4 = 975$

$\therefore$ Reqd % $= \frac{975}{2000} \times 100 = 48.75\%$

73. 5; Total number of boys appeared from all cities together = 23106
Total number of boys passed from all cities together = 3058 + 2022 + 2537 + 3377 + 2880 + 3037= 16911
$\therefore$ Total number of boys failed from all cities = Number of boys appeared from all cities - number of boys passed from all cities
= 23106 - 16911 = 6195

74. 2; Girls appeared $= \frac{7210}{5} \times 2 + \frac{4800}{16} \times 7$

$+\frac{5670}{9}\times 4+\frac{5670}{9}\times 4+\frac{6400}{16}\times 5+\frac{7200}{18}\times 7+\frac{7080}{12}\times 5$

= 2884 + 2100 + 2520 + 2000 + 2800 + 2950 = 15254

Girls Passed = 1268 + 1146 + 1432 + 975 + 1224 + 1565 = 7610

$\therefore$ Reqd % = $\frac{7610}{15254}\times 100 = 49.88 \approx 50\%$

75. 2; Total number of students passed from City S_2

$= 4800 \times \frac{66}{100} = 3168$

Total number of girls passed from City S_2 = 1146

$\therefore$ Total numbers of boys passed from City S_2

= 3168 - 1146 = 2022

$\therefore$ Reqd % = $\frac{2022-1146}{1146}\times 100$

$= \frac{876\times 100}{1146} = 76.43\% \approx 76.5\%$

76. 2; % rise = $\frac{1680-800}{800}\times 100 = \frac{880}{800}\times 100\%$ = 110%

= 110%

77. 2; In year 1995 price of 3kg wheat

$= \frac{700}{100}\times 3 = 21$

Groundnut (1 kg) = $\frac{1200}{100} = 12$

Percentage difference

$= \frac{21-12}{12}\times 100 = \frac{9}{12}\times 100 = 75\%$

78. 1; Average

$= \frac{200+270+365+460+640}{5} = \frac{1935}{5} = 387$

79. 2; $\text{Average}_{\text{sugar}} = \frac{15000}{5\times 100} = 30$

Price of suger is highest in 2010.

So, price of suger_{2010} = 45

Percentage = $\frac{30}{45}\times 100 = 66.66\%$

80. 4;

$1990-1995 \rightarrow \frac{5500-4200}{4200}\times 100 = 30.95\%$

$1995-2000 \rightarrow \frac{6400-5500}{5500}\times 100 = 16.36\%$

$2000-2005 \rightarrow \frac{8000-6400}{6400}\times 100 = 25\%$

$2005-2010 \rightarrow \frac{11000-8000}{8000}\times 100 = 37.5\%$

81. 1; Let the total population of City A be 'x'.

$\therefore x\times\frac{24}{100} = 15840$

$\therefore x = \frac{1584000}{24} = 66000$

$\therefore$ 16% of 66000 = 10560

82. 5; 0 < Age $\leq$ 13 = 18%

and 13 < Age $\leq$ 19 =15%

$\therefore$ Reqd% = $\frac{18}{15}\times 100$= 120%

83. 2; Total population of City C

$= 12000\times\frac{100}{15} = 80000$

Total population of City ,D

$= 12000\times\frac{100}{20} = 60000$

$\therefore$ Sum = 1.4 lakh

84. 2; Total population of City A

$= \frac{8640\times 100}{24} = 36000$

Total population of City B

$= \frac{10560\times 100}{22} = 48000$

$\therefore$ Ratio = $\frac{36000}{48000} = \frac{3}{4} = 3:4$

85. 5; Population of City B in age group $_{(0-13)}$

$= 48000\times\frac{16}{100} = 7680$

Population of City - E in age group$_{(0-13)}$

$= 65000\times\frac{18}{100} = 11700$

$\therefore$ Reqd% = $\frac{(11700-7680)}{7680}\times 100$

$= \frac{4020}{7680}\times 100 = 52.34\%$

86. 1; For Commerce, M : F = 53 : 47

$\therefore$ Number of Females = $\frac{47\times 1272}{53} = 1128$

$\therefore$ Total students in Commerce = 1272 + 1128 = 2400

Arts : Science : Commerce = 3 : 5 : 4

$\therefore$ Number of students in Arts = $2400\times\frac{3}{4}$ = 1800

87. 4; For Arts stream, M : F = 31 : 14

$\therefore$ Females = $\frac{1240}{31}\times 14 = 560$

$\therefore$ Total number of students in Arts = 1800

$\therefore$ Arts : Science : Commerce..
2 : 4 : 5

$\therefore$ Science = $\frac{1800}{2} \times 4 = 3600$

$\therefore$ Commerce = 4500

$\therefore$ Difference = 4500 - 3600 = 900

88. 5; Ratio of the numbers of students passed in streams Arts, Science and Commerce in City F = 2 : 4 : 3

$\therefore$ Reqd % = $\frac{9}{4} \times 100 = 225\%$

89. 1; In Arts, M : F = 34 : 16

$\therefore$ Number of Males = $\frac{34}{16} \times 384 = 816$

$\therefore$ Total number of students in Arts stream in City C
= 816 + 384 = 1200
So, total number of students passed from, City C
= 7 × 1200 = 8400
$\therefore$ Total number of students in Commerce

= 8400 × $\frac{2}{7}$ = 2400

$\therefore$ Number of Males passed in Commerce

= 2400 × $\frac{31}{60}$ = 1240

$\therefore$ Reqd % = $\frac{1240}{8400} \times 100 = 14.76\%$

90. 3; M : F = 16 : 11

$\therefore$ Reqd % = $\frac{(16-11)}{16} \times 100 = \frac{500}{16}$ = 31.25% less

91. 4; $Total_A$ = 290, Total marks = 500

$\therefore$ Reqd% = $\frac{290}{500} \times 100 = 58\%$

92. 3; Average = $\frac{438}{6}$ = 73

93. 3; Ratio = $\frac{324}{333} = \frac{36}{37}$ = 36 : 37

94. 2; Only Student A and F didn't get 1st class.
A = 58% and F = 54.2%

95. 5; A + D = 38 + 42 = 80
B + C = 60 + 64 = 124

$\therefore$ Reqd % = $\frac{125}{80} \times 100 = 155\%$

96. 2; The total number of selected students in State B = 75 + 72 + 104 + 112 + 60 + 75 = 498

$\therefore$ Average = $\frac{498}{6}$ = 83

The total number of selected students in State D = 95 + 84 + 77 + 78 + 64 + 58 = 456

$\therefore$ Average = $\frac{456}{6}$ = 76

$\therefore$ Difference = 83 - 76 = 7

97. 3; Percentage of candidates passed in

State A = $\frac{780}{5600} \times$ 100 = 13.92%

Percentage of candidates passed in State B

$= \frac{480}{7500} \times 100 = 6.4\%$

Percentage of candidates passed in State C

$= \frac{800}{4800} \times 100 = 16.66\%$

Percentage of candidates passed in-State D

$= \frac{700}{7500} \times 100 = 9.33\%$

98. 2; Total number of students selected in State C
= 80 + 60 + 66 + *55*+ 52 + 60 = 373
Total number of students selected in State A
= 80 + 120 + 72 + 96 + 64 + 68 = 500

$\therefore$ Reqd % = $\frac{373}{500} \times 100 = 74.6\%$

99. 2; Percentage of selected candidates in State D

in 2006 → $\frac{95}{700} \times 100 = 13.57\%$

Percentage of selected candidates in State D

in 2007 → $\frac{84}{540} \times 100$ = 15.5%

Percentage of selected candidates in State D

in 2008 → $\frac{77}{660} \times 100 = 11.6\%$

Percentage of selected candidates in State D

in 2009 → $\frac{78}{720} \times 100$ = 10.83%

Percentage of selected candidates in State D

in 2010 → $\frac{64}{640} \times 100$ = 10%

Percentage of selected candidates in State D

in 2011 → $\frac{58}{500} \times 100$ = 11.6%

100. 5; Total candidates passed in State A in 2006 = 780
Total candidates passed in State C in 2009 = 500

$\therefore$ Reqd % = $\frac{(780-500)}{500} \times 100 = \frac{280}{5} = 56\%$

101. 5; Premium model of Company D in the year 2009 = 10.4 thousand
Production of both the models by Company C in the year 2007 = 7.5 + 8.3 = 15.8

Required percentage = $\frac{10.4}{15.8} \times 100 = 66\%$

102. 2; Basic model produced by Company B in the year 2009 = 11.8
Basic model produced by Company B in the year 2008 = 14.8

$\therefore$ decrease % = $\frac{14.8 - 11.8}{14.8} \times 100$

$= \frac{3}{14.8} \times 100 = \frac{30}{148} \times 100$

$= \frac{3000}{148} = 20.27 \approx 20\%$

103. 5; Average = $\frac{2.5 \times 7.2 + 15.5 + 13.9 + 14.9}{5}$

= 10.8 = 10.8 × 1000 = 10800

104. 5; Company E_{2006} = 5.1 - 2.7 = 2.4
Company E_{2007} = 5.5 - 4.2 = 1.3
Company E_{2008} = 11.5 - 7.7 = 3.8
Company E_{2009} = 12.8 - 7.2 = 5.6
Company E_{2010} = 13.2 - 12.2 = 1
In the year 2009 the difference is the maximum.

105. 3

106. 2; Average = $\frac{396}{11}$ = 36

107. 3; $A_2 + B_2 + C_2$ = 28 + 46 + 97 = 171
Total runs scored by T_2 in 1st innings = 418

$\therefore$ Reqd % = $\frac{171}{418} \times 100 = 40.9 \approx 41\%$

108. 1; $G_1 + H_1 + I_1 + J_1 = 90$
$A_2 + B_2 + C_2 + D_2 = 270$
$\therefore$ Ratio = 1 : 3

109. 5; % rise = $\frac{42 - 15}{15} \times 100 = \frac{2700}{15} = 180\%$

110. 4; Strike rate of D_2 = $\frac{87}{116} \times 100 = 75$

Strike rate of E_2 = $\frac{56}{70} \times 100 = 80$

% Difference = $\frac{80 - 75}{75} \times 100 = \frac{500}{75} = 6.25\%$

111. 2; Number of Research Papers published by Neeta in Educon = 54
Number of Research Papers published by Vijay in Eduforms = 38

$\therefore$ Reqd % = $\frac{54 - 38}{38} \times 100 = \frac{1600}{38}$

$= 42.10 \approx 42\%$

112. 4; Total number of Research Papers published by Anand, Vijay and Neeta together in Educon = 42 + 12 + 54 = 108
Total Number of Articles published by Mohan, Naidu and Ronit together in Edutrack
= 75 + 39 + 23 = 137
$\therefore$ Required difference = 137 - 108 = 29

113. 3; Research Papers and Articles together published by
Anand = 22+ 11 =33
Vijay = 38 + 25 = 63
Naidu = 57 + 35 = 92
Mohan = 39 + 48 = 87
Neeta = 44 + 32 =76
and Ronit = 11 + 18 = 29
Hence, third hightest published by Neeta.

114. 5; Average

$= \frac{17 + 6 + 12 + 22 + 28 + 29}{6} = \frac{114}{6}$ =19

115. 4; Total number of Reasearch Papers and Articles together published by Mohan in Edutrack = 42 + 75 = 117
Total Number of articles published by all six persons in New Era = 94

$\therefore$ Reqd % = $\frac{117}{94} \times 100 = 124\%$

116. 2; $(Total_B)$ = 200 × 0.44 + 62 + 78 + 73 + 150 × 0.6 + 150 × 0.84 + 80 × 0.55 = 88 + 62 + 78 + 73 + 90 + 126 + 44 = 561

$\therefore$ % marks = $\frac{561}{880} \times 100 = 63.75\%$

117. l; $F_{Hindi} = 150 \times \frac{66}{100} = 99$

$B_{Maths} = 200 \times \frac{44}{100} = 88$

$\therefore$ Reqd % = $\frac{99}{88} \times 100 = 112.5\%$

118. 2; Average marks = $\frac{150}{6}$ {0.78 + 0.84 + 0.64 + 0.52 + 0.38 + 0.46}
= 25 × 3.62 = 90.5

119. 3; Total marks scored by Student A
= 200 × 0.72 + 77 + 61 + 67 + 150 × 0.72 + 150 × 0.78 + 80 × 0.4
= 144 + 77 + 61 + 67 + 108 + 117 + 32 = 606
Total marks scored by Student D
= 200 × 0.66 + 45 + 65 + 53 + 150 × 0.46 +

$150 \times 0.52 + 80 \times 0.3$
$= 132 + 45 + 65 + 53 + 69 + 78 + 24 = 466$

$\therefore$ Reqd% = $\frac{606-466}{466} \times 100$

$= \frac{14000}{466} = 30.04 \approx 30\%$

120. 5; The percentage marks scored by Student B in Chemistry = 78%
The percentage marks scored by Student C in Hindi = 48%

$\therefore$ Reqd % = $\frac{78}{48} \times 100 = 162.5\%$

121. 2; Average oil import from Iraq = $\frac{87}{5} = 17.4$

Average oil import from Venezuela = $\frac{42}{5} = 8.4$

$\therefore$ Ratio = $\frac{17.4}{8.4}$ = 29 : 14

122. 3; 2009-10 oil import from Nigeria is max with respect to its previous year.

123. 3;

Reqd = $\frac{21.2}{20.5 + 21.8 + 21.2 + 18.5 + 17.5} \times 100$

$= \frac{21.2}{99.5} \times 100 = 21.38 \approx 21\%$

124. 4; $\frac{0.9}{13.4} \times 100 + \frac{3}{14.8} \times 100 + \frac{0.3}{11.8} \times 100$

$+ \frac{6.3}{11.5} \times 100 = 6.4 + 20.27 + 2.54 + 54.78$

$\frac{6.4 + 20.27 + 2.54 + 54.78}{4} = 21\%$

125. 4; Average in 2011-12 = 19.36 million tonnes
Average in 2009-10 = 15.95 million tonnes.

$\therefore$ Reqd % = $\frac{19.36}{15.95} \times 100 = 121.37 \approx 121\%$

126. 2; Net profit of Dutch Bank last year

$= \frac{546}{1 - 0.152}$

$= \frac{546}{0.848} = 643.86$ crore

Net profit of CLSA last year

$= \frac{502}{0.78} = 643.589$ crore

Average net profit = $\frac{643.86 + 647.584}{2}$

= 643.72 crore = 644 crore

127. 4; Total net sales of all the organisations
= 7570 + 6186 + 7372 + 599 + 609 + 597
= 22933
Net sales of Dutch Bank = 7570

Reqd % = $\frac{7570}{22931} \times 100 = 33.009\% \approx 33\%$

128. 5; Reqd ratio = $\frac{546}{7570} \rightarrow$ Dutch Bank

CLSA $\rightarrow \frac{502}{6186}$

Morgan $\rightarrow \frac{623}{7372}$

Motilal $\rightarrow \frac{377}{599}$

HDFC Bank $\rightarrow \frac{359}{609}$

Citi Bank $\rightarrow \frac{388}{597}$

Thus, ratio of Citi Bank is the maximum.

129. 1; Dutch Bank

130. 1; Net sales of HDFC Bank = $\frac{609}{1 + 0.261}$

$= \frac{609}{1.261} = 482.95$ crore

Net sale of Citi Bank = $\frac{597}{1.24} = 481.45$ crore

$\therefore$ Average = $\frac{482.95 + 481.45}{2}$

= 482.2 crore $\approx$ 482

131. 4; Adult males in City A = $7.8 \times \frac{7}{13} \times \frac{62}{100}$

= 2.604 lakh

Adult females in City A = $7.8 \times \frac{6}{13} \times \frac{65}{100}$

= 2.34 lakh
$\therefore$ Difference = 2.604 - 2.34 = 0.264 lakh
= 26400

132. 2; Total adult males = $7.8 \times \frac{7}{13} \times \frac{62}{100}$

$+ 3.6 \times \frac{5}{9} \times \frac{70}{100} + 4.5 \times \frac{2}{5} \times \frac{68}{100} + 6.8 \times \frac{9}{17} \times \frac{72}{100} +$

$7.2 \times \frac{4}{9} \times \frac{65}{100} + 5.4 \times \frac{2}{3} \times \frac{75}{100} = 2.604 + 1.4 +$

1.224 + 2.592 + 2.08 + 2.7 = 12.6 lakh

$\therefore$ Average = $\frac{12.6}{6} = 2.1$ lakh

133. 1; Minor females in City C

$= 4.5 \times \frac{3}{5} \times \frac{36}{100} = 0.972$ lakh

Minor males in City F

$= 5.4 \times \frac{2}{3} \times \frac{25}{100} = 0.90$ lakh

$\therefore$ Reqd % $= \frac{(0.972 - 0.90)}{0.9} \times 100$

$= \frac{0.072}{0.9} \times 100 = 8\%$

134. 3; Minor males in City E

$= 7.2 \times \frac{4}{9} \times \frac{35}{100} = 1.12$ lakh

Adult males in City B

$= 3.6 \times \frac{5}{9} \times \frac{70}{100} = 1.4$ lakh

$\therefore$ Reqd % $= \frac{1.12}{1.4} \times 100 = 80\%$

135. 3; Adult females in City C

$= 4.5 \times \frac{3}{5} \times \frac{64}{100} = 1.728$ lakh

Minor males in City C

$= 4.5 \times \frac{2}{5} \times \frac{32}{100} = 0.576$ lakh

$\therefore$ Difference = 1.728 - 0.576 = 1.152 lakh

136. 4; Total marks scored by Student D

$= \frac{1}{100}$ {68.75 × 80 + 78.75 × 80 + 72.5 × 80 + 55 × 100 + 80 × 120 + 60 × 150}

$= \frac{1}{100}$ {5500 + 6300 + 5800 + 5500 + 9600 + 9000} $= \frac{1}{100} \times 41700 = 417$

137. 3; $\therefore$ Average $= \frac{1}{6} \times \frac{80}{100}$ {58.75 + 77.5 + 80 + 68.75 + 75 + 67.5} $= \frac{8}{60} \times 427.5 = 57$

138. 5; Marks scored by Student B in Maths

$= 150 \times \frac{72}{100} = 108$

Marks scored by Student E in Physics

$= 80 \times \frac{75}{100} = 60$

$\therefore$ Reqd % $= \frac{108}{60} \times 100 = 180\%$

139. 2; Marks scored by Student B in English

$= 120 \times \frac{60}{100} = 72$

Marks scored by Student A in Maths

$= 150 \times \frac{84}{100} = 126$

$\therefore$ Ratio $= \frac{72}{126} = \frac{4}{7} = 4 : 7$

140. 5; Marks scored by Student F in Maths $= 150 \times \frac{70}{100} = 105$

Marks scored by Student E in Chemistry $= 80 \times \frac{70}{100} = 56$

$\therefore$ Reqd % $= \frac{(105 - 56)}{56} \times 100 = \frac{4900}{56} = 87.5\%$

141. 1; Total illiterate females

$= 1.2 \times \frac{5}{12} \times \frac{43}{100} + 1.75 \times \frac{2}{5} \times \frac{40}{100} + 3.4 \times \frac{9}{17} \times \frac{47}{100} + 2.5 \times \frac{3}{5} \times \frac{39}{100} + 1.8 \times \frac{1}{2} \times \frac{35}{100} + 3.0 \times \frac{2}{5} \times \frac{44}{100}$

= 0.215 + 0.28 + 0.846 + 0.585 + 0.315 + 0.528 = 2.769 lakh

142. 2; Literate males from City F

$= 3 \times \frac{3}{5} \times \frac{68}{100} = 1.224$

Literate females from City C

$= 3.4 \times \frac{9}{17} \times \frac{47}{100} = 0.846$

$\therefore$ Read % $= \frac{0.846 \times 100}{1.224} = 69.11 \approx 69\%$

143. 3; $\frac{1}{6} \times \frac{1}{100}$ {$1.2 \times \frac{5}{12} \times 57 + 1.75 \times \frac{2}{5} \times 60 + 3.4 \times \frac{9}{17} \times 53 + 2.5 \times \frac{3}{5} \times 61 + 1.8 \times \frac{1}{2} \times 65 + 3 \times \frac{2}{5} \times 56$}

$= \frac{1}{600}$ {28.5 + 42 + 95.4 + 91.5 + 58.5 + 67.2}

$= \frac{383.1}{600} = 0.6385$ lakh = 63850

144. 3; Literate males in City B $= 1.75 \times \frac{3}{5} \times \frac{36}{100}$

Literate females in City B = $1.75 \times \frac{2}{5} \times \frac{60}{100}$

$\therefore$ Ratio = $\frac{3 \times 36}{2 \times 60} = \frac{9}{10} = 9:10$

145. 2; Literate males

$= 1.2 \times \frac{7}{12} \times \frac{67}{100} + 1.75 \times \frac{3}{5} \times \frac{64}{100}$

= 0.469 + 0.672 = 1.141 lakh

Literate females

$= 3.4 \times \frac{9}{17} \times \frac{53}{100} + 2.5 \times \frac{3}{5} \times \frac{61}{100}$

= 0.954 + 0.915 = 1.869 lakh

$\therefore$ Difference = 1.869 - 1.141

= 0.728 lakh = 72800

146. 3; Passed students from School D in the year

2010 = $876 \times \frac{7}{12} = 511$

Failed students from School B in the year 2012

$= 952 \times \frac{8}{17} = 448$

$\therefore$ Difference = 511 - 448 = 63

147. 2; Total failed students from School F

$= 986 \times \frac{12}{29} + 867 \times \frac{5}{17} + 924 \times \frac{13}{21}$

= 408 + 255 + 572 = 1235

148. 4; Total passed students from all six schools in the year 2011 = $754 \times \frac{7}{13} + 845 \times \frac{8}{13}$ + $792 \times \frac{7}{11} + 828 \times \frac{11}{18} + 726 \times \frac{7}{11} + 867 \times \frac{12}{17}$

= 406 + 520 + 504 + 506 + 462 + 612 = 3010

149. 3;

Average= $\frac{1}{3}\{810 \times \frac{7}{15} + 792 \times \frac{4}{11} + 637 \times \frac{3}{7}\}$

$= \frac{1}{3}\{378 + 288 + 273\} = \frac{939}{3} = 313$

150. 5; Passed students from School E in the year

2010 = $870 \times \frac{3}{5}$ = 522

Failed students from School A in the year 2011

$= 754 \times \frac{6}{13} = 348$

$\therefore$ Reqd % = $\frac{522}{348} \times 100 = 150\%$

151. 3; Income of Company C_{2011}

= 25.4 + 25.4 × $\frac{21.5}{100}$

= 25.4 + 5.461 = 30.861 crore

152. 2; Profit of Company A in 2012

= 23.2 × $\frac{31.5}{100}$ = 7.308

Profit of Company B in 2012

= 24.8 × $\frac{27.5}{100}$ = 6.82

$\therefore$ Difference = 7.308 - 6.820 = 0.488 crore

153. 4; Expenditure of Company A

= 17.8 + 23.2 = 41 crore

Expenditure of Company C

= 27.5 + 22.5 = 50 crore

Reqd% = $\frac{41}{50}$ × 100 = 82%

154. 2; Percentage profit of Company C_{20O9} = 28.4%

Percentage profit of Company A_{2007} = 16.2%

$\therefore$ Reqd % = $\frac{28.4 - 16.2}{16.2} \times 100$

$= \frac{12.2 \times 100}{16.2} = 75.3\% \approx 75\%$

155. 3; Income of Company B_{2010} = 23 + 23 × $\frac{25}{100}$

= 28.75 crore

Expenditure of Company A_{2009} = 21 crore

$\therefore$ Reqd % = $\frac{28.75}{21}$ × 100 = 136.9 $\approx$ 137%

156. 5; Number of I_1 produced by A

$= \frac{80370}{57}$ × 25 = 35250

Number of I_1 produced by B

= $\frac{61050}{55}$ × 19 = 21090

$\therefore$ Total = 35250 + 21090 = 56340

157. 1; Difference

$$= \frac{73130}{(25+24+22)} \times (25-22) = \frac{73130 \times 3}{71}$$

$= 3090$

158. 3; Required % $= \frac{23}{25} \times 100 = 92\%$

159. 2; Number of I_1 produced by D

$= \frac{61880}{68} \times 21 = 19110$

Number of I_1 produced by F.

$= \frac{93160}{17} \times 3 = 16440$

Required % $= \frac{19110-16440}{16440} \times 100$

$= \frac{2670 \times 100}{16440} = 16.25\%$

160. 2; Total $= 80370 \times \frac{23}{57} + 61050 \times \frac{15}{55} + 77490 \times \frac{18}{63} + 61880 \times \frac{23}{68} + 73130 \times \frac{24}{71} + 93160 \times \frac{5}{17}$

$= 32430 + 16650 + 22140 + 20930 + 24720 + 27400 = 14420$

161. 1; Average marks of all students in Physics

$= \frac{1}{6}$ [75{0.84 + 0.68 + 0.72 + 0.48 + 0.70 + 0.56}] $= \frac{1}{6}$ [75 × 3.98] =

$\therefore$ Average $= \frac{298.5}{6} = 49.75$

162. 5; Total marks scored by Student F in all the subjects together = 75 × 0.56 + 75 × 0.66 + 200 × 0.55 + 50 × 0.76 + 150 × 0.66 = 42 + 49.5 + 110 + 38 + 99.

$= 338.5$

163. 2; Marks scored by Student B = 75 × 0.68 + 75 × 0.64 + 200 × 0.49 + 50 × 0.74 + 150 × 0.52 = 51 +48 + 98 + 37 + 78=312

$\therefore$ Reqd % $= \frac{312}{550} \times 100 = 56.27 \approx 57\%$

164. 1; Marks scored by Student C in Physics

$= 75 \times 0.72 = 54$

Marks scored by Student C in English

$= 150 \times 0.64 = 96$

$\therefore$ Reqd % $= \frac{54}{96} \times 100 = 56.25\% \approx 56\%$

165. 3; Total marks obtained by Student D = (75 × 0.82) + (150 × 0.70) = 61.5 + 105 = 166.5

Total marks obtained by Student F = (75 × 0.66) + (150 + 0.66) = 49.5 + 99 = 148.5

$\therefore$ Difference = 166.5 - 148.5 = 18

166. 4; Number of students enrolled in College A in the year 2009 = 1000

$\therefore$ Number of students passed

$= 1000 \times \frac{80}{100} \times \frac{60}{100} = 480$

167. 3; Reqd number of students

$= 2290 \times \frac{70}{100} = 1603$

168. 3; Average number of students enrolled in all colleges together in the year 2009

$= \frac{3770}{5} = 754$

Average number of students enrolled in all colleges together in the year 2010

$= \frac{3090}{5} = 618$

$\therefore$ Reqd ratio $= \frac{754}{618} = \frac{377}{309} = 377 : 309$

169. 4; Number of students enrolled in College A in the year 2009= 1000

Number of students enrolled in College B in the year 2011 = 650

$\therefore$ Reqd% $= \frac{350}{650} \times 100 = 53.84\% \approx 54$

170. 5; Total number of students in the year 2010 from all the colleges = 3090

$\therefore$ Reqd number of students = 10% of 3090 = 309

171. 5; Number of people in Teaching profession

$= \frac{30}{100} \times 25000 = 7500$

Number of people in Medical profession

$= \frac{10}{100} \times 25000 = 2500$

$\therefore$ Reqd% $= \frac{7500}{2500} \times 100 = 300\%$

172. 3; Total numbers of males in Banking and Medical professions

$= 25000 \times \frac{20}{100} \times \frac{60}{100} + 25000 \times \frac{10}{100} \times \frac{40}{100}$

= 3000 + 1000 = 4000

The total number of females in Medical and Banking profession = 10% of 60% of 25000 + 20% of 40% of 25000 = 1500 + 2000 = 3500

$\therefore$ Reqd ratio = $\frac{4000}{3500} = \frac{8}{7} = 8:7$

173. 3; Females in Engineering professions

$= 25000 \times \frac{25}{100} \times \frac{7}{100} = 625 \times 7 = 4375$

Males in Banking profession

$= 25000 \times \frac{25}{100} \times \frac{60}{100} = 3000$

Reqd% = $\frac{4375}{3000} \times 100 = 145.83 \approx 146\%$

174. 3; Number of males in Banking and Medical = 20% of 60% of 25000 + 10% of 40% of 25000 = 3000 + 1000 = 4000

Number of females in Law and Teaching

$= \frac{15}{100} \times \frac{20}{100} \times 25000 + 25000 \times \frac{30}{100} \times \frac{60}{100} = 5250$

$\therefore$ Reqd ratio = $\frac{4000}{5250} = \frac{16}{21} = 16:21$

175. 1; Number of females in Engineering profession = 25% of 70% of 25000 = 4375

Number of males in Law profession = 15% of 80% of 25000 = 3000

Reqd % = $\frac{4375 - 3000}{3000} \times 100$

$= \frac{1375}{3000} \times 100 = 45.83 \approx 46\%$

176. 3; Total amount of bill paid by Dev in the month of June for all commodities = 123 + 150 + 324 + 134 = ` 731

177. 3; Average = $\frac{315 + 135 + 98 + 116 + 131}{5}$

$= \frac{795}{5} =$ `159

178. 1; Reqd difference = 323 - 143 = ` 180

Alternate Method:

Mobile bill paid by Ravi in May = `143

Laundry bill paid by Dev in March = `323

$\therefore$ Difference = 323 - 143 = `180

179. 4; Manu paid second highest mobile bill in the month of March = `345

And Manu paid lowest electricity bill in the month of May.

180. 1; Reqd ratio = $\frac{135}{245}$ = 27 : 49

181. 3; Total distance from Surat to Nadiad Junction = 440 - 253 = 183 km

182. 5; Total time taken by the train from Anand Junction to Ahmedabad = 8:00 - 6:45 = 1hr 15 min

183. 1; Reqd ratio = $\frac{378}{306} = 21:17$

184. 2; Arrival time of the train at Bhuj = (5:40 + 0:23 - 0:2) = 6:01 pm

185. 3; We see in the graph that there is second lowest distance between Dadar and Vasai Road = 42 km

186. 5; Maximum temperature of Ontario on 1st November = 4°C

Minimum temperature of Bhuj on 1st January = -7°C

$\therefore$ Difference = 4 + 7 = 11°C

187. 1; There is second highest temperature of Kabul on 1st October = 37°C

The minimum temperature of Sydney is on 1st January (13°C).

188. 3; Diff of temp in Bhuj on 1st September → 24 - 14 = 10°C

Diff of temp in Bhuj on 1st October → 35 - 21 = 14°C

Diff of temp in Bhuj on 1st November → 19 - 8 = 11°C

Diff of temp in of Bhuj on 1 st December → 9 - 2 = 7°C

Diff of temp in Bhuj on 1st January → -7 + 4 = -3°C.

Hence, the second highest difference in temperature is on 1st November.

189. 5; Average = $\frac{12 + 9 + 15 + 2 + 5}{5} = \frac{43}{5} = 8.6$°C

190. 2; Reqd ratio = $\frac{9}{15} = 3:5$

191. 1; Number of Swift manufactured during 2007 to 2012 = (250 + 200 + 230 + 245 + 260 + 275) = 1460

Number of SX4 manufactured during 2007 to 2012 = (200 + 230 + 225 + 210 + 135 + 155) = 1155

Number of Ertiga manufactured during 2007 to 2012 = (128 + 150 + 142 + 170 + 180 + 230) = 1000

Number of Zen manufactured during 2007

to 2012 = (140 + 155 + 160 + 175 + 185 + 220) = 1035

Number of Echo manufactured during 2007 to 2012 = (115 + 120 + 135 + 125 + 130 + 120) = 745

Thus, Swift is manufactured in maximum number.

192. 1; Production of Swift in 2007 = 250 and in 2012 = 275

$\therefore$ Percentage increase = $\frac{275-250}{250}\times 100$

$= 10\%$

193. 2; The table shows that the production; of Zen increases continuously over the years.

194. 4; Production of Echo in 2011 = 130

Production of SX4 in 2010 = 210

Reqd% = $\frac{130\times 100}{210} = 61.90\%$

195. 3; Production of Zen in 2008 = 155 and that in 2010 = 175

$\therefore$ Percentage increase = $\frac{175-155}{155}\times 100$

$= \frac{20}{155}\times 100 = 12.9\%$

196. 1; Reqd % = $\frac{221}{4933} \times 100 = 4.48 = 4.5\%$

197. 2; Difference = $\frac{1}{5}$ {1542 - 1382} + (1545 - 1384) + (1550 - 1275) + (1570 - 1300) + (1580 - 1290)}

$= \frac{1}{5}$ {160 + 161 + 275 + 270 + 290}

$= \frac{1}{5} \times 1156 = 231.2 \approx 231$

198. 4; Reqd ratio = $\frac{5825}{5625} = \frac{233}{225} = 233 : 225$

199. 5; Total number of employees in all the departments of all the organisations together = 4933 + 4751 + 6631 + 7787 + 3867 + 221 = 28190

200. 3; Reqd % = $\frac{960}{5703} \times 100 = 16.83 \approx 17\%$

201. 2; Reqd ratio = $\frac{980}{1120} = \frac{7}{8} = 7 : 8$

202. 5; Average number of students in College A

$= \frac{5610}{6} = 935$

Average number of students in College C

$= \frac{5490}{6} = 915$

$\therefore$ Reqd difference = 935 - 915 = 20

203. 4; Total number of students in College B = 5810

Total number of students in College D = 5598

$\therefore$ Reqd difference = 5810 - 5598 = 212

204. 5; Average number of students in College E

$= \frac{5880}{6} = 980$

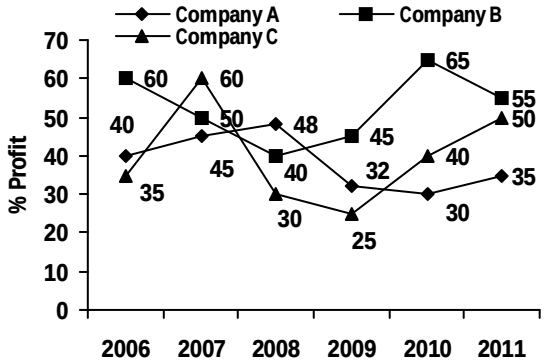
205. 1; $\therefore$ Reqd%

$= \left[\frac{\text{Number of students in College C in 2010}}{\text{Total number of students in 2010}}\times 100\right]\%$

$= \left[\frac{980}{4910}\times 100\right]\% = 19.95\% \approx 20\%$

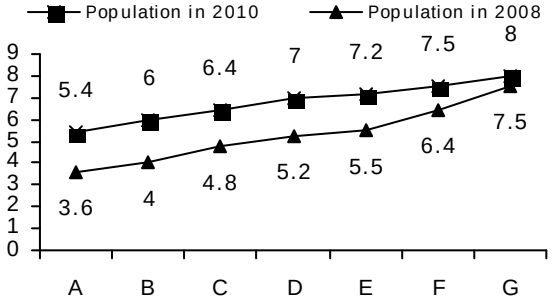
DATA INTERPRETATION LINE GRAPH

Directions (Q. 1-5): Following line-graph shows the percentage profit earned by three companies A, B and C in the period of 2006 to 2011.



1. If the expenditure of Company A in the year 2008 is ?55.5 lakh then what is its income in that year?
   (1) ` 78.841akh (2) ` 82.141akh (3) ` 84.61akh (4) ` 85.51akh (5) ` 87.21akh
2. What is the percentage rise in the percentage profit of Company B from 2008 to 2009?
   (1) 5% (2) 10% (3) 12.5% (4) 25% (5) None of these
3. If the total expenditure of Company A in the year 2006 and Company C in the year 2010 together is ` 94 lakh then what is the sum of the total income of A in 2006 and C in 2010?
   (1) ` 67.141akh (2) ` 131.61akh (3) ` 65.81akh (4) ` 134.28 lakh (5) None of these
4. If the income of Company A in year 2006 and expenditure of Company B in year 2007 are equal and ?91 lakh each then what is the difference between the income of B in 2007 and the expenditure of A in the year 2006?
   (1) ` 67.2 lakh (2) ` 69.8 lakh (3) ` 70.41 lakh (4) ` 71.5 lakh (5) None
5. If the expenditure of Company B in the year 2006 and the income of C in the year 2009 are equal then what is the ratio of the income of B in the year 2006 to the expenditure of C in the year 2009?
   (1) 2:1 (2) 1:2 (3) 12:5 (4) 5:12 (5) None of these

Directions (Q. 6 - 10): Following line-graph shows the population of seven cities (in lakh) and the table shows the percentage of literate population in these cities.



	% Literate 2008	% Literate 2010
A	57.8%	62.3%
B	63.1%	68.6%
C	59.2%	66.4%
D	64.5%	73.2%
E	67.7%	71.0%
F	65.8%	74.5%
G	68.9%	73.3%

6. What is the percentage rise in the population of City C from 2008 to 2010?
(1) 27.5% (2) 33.3% (3) 36.8% (4) 37.5% (5) 39%

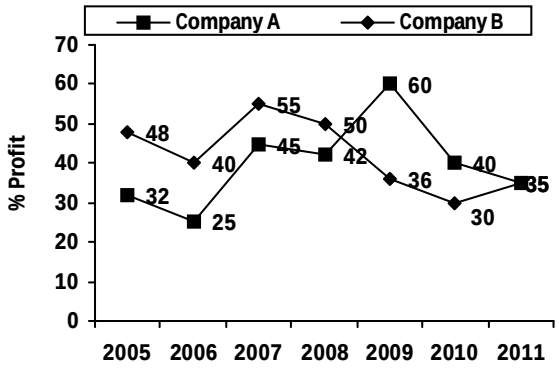
7. What is the total literate population of City A in the year 2008 and 2010 together (in lakh)?
(1) 5.126 (2) 5.248 (3) 5.312 (4) 5.445 (5) 5.560

8. What is the difference between the total illiterate population of City G and City F in the year 2008? (in lakh)
(1) 0.1437 (2) 0.1487 (3) 0.1527 (4) 0.1567 (5) 0.1687

9. The literate population of City E in the year 2010 is approximately what percentage more than its literate population in 2008?
(1) 27.5% (2) 32% (3) 34.8% (4) 36% (5) 37.3%

10. What is the difference between the Literate population and illiterate population of City D in the year 2008? (in lakh)
(1) 1.302 (2) 1.406 (3) 1.508 (4) 1.603 (5) 1.704

Directions (Q. 11-15): Following line-graph shows the percentage profit earned by two companies A and B during the period of 2005 to 2011.



11. If the income of Company A in 2007 and that of B in 2009 are ` 52.49 lakh and ?61.2 lakh respectively, what is the total expenditure of Company A in 2007 and that of B in 2009?
(1) ` 78.4 1akh (2) ` 79.6 1akh (3) ` 80.4 1akh (4) ` 81.2 1akh (5) ` 82.5 1akh

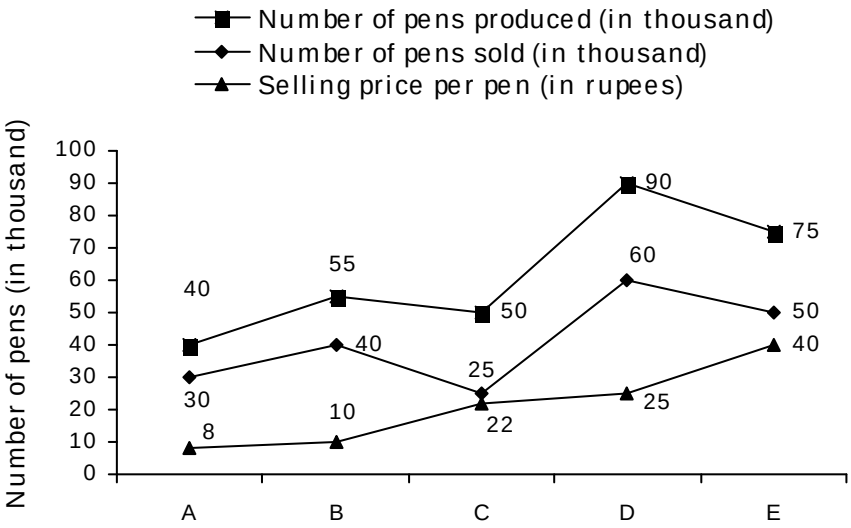
12. If the expenditure of Company A in 2005 and the income of B in 2006 are ` 48.5 lakh and ` 75.04 lakh respectively, what is the difference between the income of A in 2005 and the expenditure of B in 2006?
(1) ` 9.86 lakh (2) ` 9.92 lakh (3) ` 10.04 lakh (4) ` 10.24 lakh (5) `10.421akh

13. If the total income of Company B in 2006 and that of Company A in 2010 together is ?133 lakh, what is the sum of the expenditure of B, in 2006 and the expenditure of A in the year 2010?
(1) ` 95 1akh (2) ` 1.33 1akh (3) ` 186.2 1akh (4) ` 93.1 1akh (5) None of these

14. If the expenditure of Company A in 2006 is the same as the income of B in 2008, what would be the ratio of the expenditure of B in 2008 to the income of A in 2006?
(1) 4:7 (2) 4:9 (3) 7:15 (4) 8:15 (5) 4:15

15. If the expenditure of A in 2009 and the expenditure of B in 2005 are equal, the income of B in 2005 is approximately what percentage of the income of A in the year 2009?
(1) 87.5% (2) 92.5% (3) 94.5% (4) 96.5% (5) 108%

Directions (Q. 16-20): Following line graph shows the number of pens produced by a pen manufacturing company, the number of pens sold by it and the price of one pen of different types.



16. The average number of pens sold by the company i s what percentage of the average number of pens produced by it in all the five types together? (Answer in approximate value)
(1) 56% (2) 62% (3) 66% (4) 70% (5) 75%

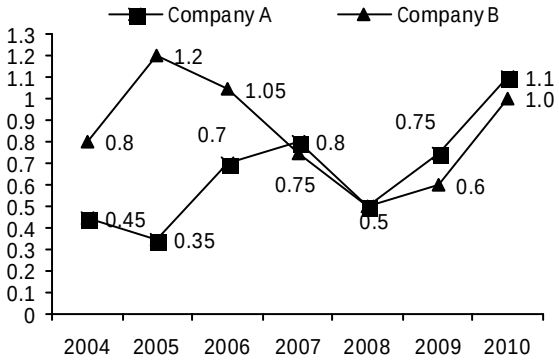
17. If the cost of manufacturing of Type A pens is ` 4.50 per pen, what is the net profit earned by the company by selling all pens of type A?
(1) ` 95 thousand (2) ` 1.05 lakh (3) ` 1.20 lakh (4) ` 1.25 lakh (5) None of these

18. What is the net amount received by the company by selling all the pens of all types?
(1) ` 46.91akh (2) ` 47.21akh (3) ` 48.81akh (4) ` 49.4 lakh (5) None of these

19. If the manufacturing cost of Type C and that of Type D pens is equal and it is ` 15 per pen, what is the net profit earned by the company by selling all pens of Type C and Type D?
(1) ` 6.81akh (2) ` 71akh (3) ` 7.21akh (4) ` 7.51akh (5) ` 7.751akh

20. The profit earned by selling all pens of Type B is what percentage of the total profit earned by selling all pens of Type E if the per unit cost of Type B pens is `5.5 and that of Type E pens is ` 25?
(1) 18% (2) 22% (3) 24% (4) 28% (5) 32%

Directions (Q. 21-25): The following graph shows the ratio of imports to exports of two companies A and B in different years.



21. The ratio of imports to exports of Company B in year 2006 is what percentage of the ratio of imports to exports of Company A in year 2009?
(1) 40% (2) 30% (3) 120% (4) 140% (5) 130%

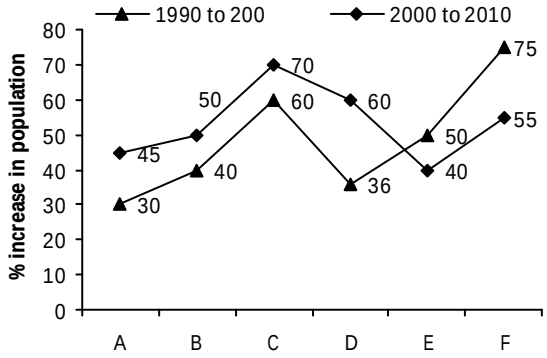
22. If imports of Company A in year 2008 was 78 lakh, what will be the exports of Company B in the same year?
(1) 78 lakh (2) 156 lakh (3) 39 lakh (4) 117 lakh (5) None of these

23. If the sum of exports of Company A in year 2007 and Company B in year 2004 is 180 lakh, what will be the sum of imports of Company A in year 2007 and Company B in year2004?
(1) 144 lakh (2) 180 lakh (3) 225 lakh (4) 90 lakh (5) None of these

24. If exports of A and imports of B in year 2009 are equal and they are 120 lakh each, what will be the difference between exports of B and imports of A in year 2009?
(1) 18 1akh (2) 40 lakh (3) 80 lakh (4) 110 lakh (5) 145 lakh

25. If the imports of Company A in year 2008 and exports of Company B in year 2005 are 80 lakh and 60 lakh respectively, the imports of Company B in year 2005 are what percentage of exports of Company A in year 2008?
(1) 45% (2) 90% (3) 75% (4) 222.22% (5) 111.11%

Directions (Q. 26-30): The following graph shows the percentage growth in population of six cities from 1990 to 2000 and 2000 to 2010.



26. If the population of City F in year 1990 was 12 lakh, what will be its population in year 2010?
(1) 31.65 lakh (2) 32.55 lakh (3) 33.4 lakh (4) 34.64 lakh (5) None of these

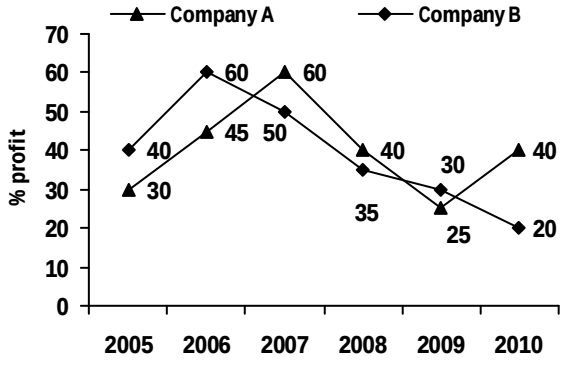
27. The population of City D in year 2000 was what per cent of its population in year 2010?
(1) 57.8% (2) 60% (3) 62.5% (4) 96% (5) 160%

28. In year 1990 the population of City A and City B are equal and the population of City A in year 2010 is 37.7 lakh. What is the population of City B in year 2010?
(1) 38.4 lakh (2) 42 lakh (3) 43.5 lakh (4) 44 lakh (5) 46.4 lakh

29. If the population of City C in year 2010 and that of City D in year 2000 are equal and they are 27.2 lakh each the population of City C in year 1990 is what percentage of population of City D in the same year?
(1) 50% (2) 75% (3) 80% (4) 120% (5) 200%

30. The population of City E in year 1990 was what fraction of its population in 2010?
(1) 8:19 (2) 10:19 (3) 8:21 (4) 10:21 (5) 15:19

Directions (Q. 31-35): In the following line-graph, the percentage profit earned by two companies A and B during the period 2005 to 2010 is given.



31. What is the percentage increase in the per cent profit of Company A from the year 2006 to 2007?

(1) 15% (2) 25% (3) $33\frac{1}{3}\%$ (4) $52\frac{1}{2}\%$ (5) ;None of these

32. If the incomes of Company A and B are equal in the year 2007, what is the ratio of the expenditure of A to that of B?

(1) 6:5 (2) 5:4 (3) 4:3 (4) 3:2 (5) None of these

33. If the income of Company A in 2009 and the expenditure of Company B in 2010 are equal and that are ` 90 lakh each, what is the difference between the income of B in 2010 and the expenditure of A in 2009?

(1) `18 lakh (2) ` 36 lakh (3) ` 45 lakh (4) ` 41 lakh (5) None of these

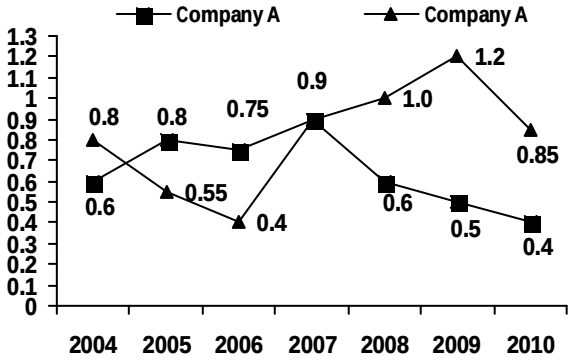
34. If the income of Company A in the year 2010 and the expenditure of Company B in the year 2005 are ` 98 lakh and ` 85 lakh respectively, what is the sum of the expenditure of A in 2010 and the income of B in the year 2005?

(1) `189 1akh (2) `183 1akh (3) `155 1akh (4) ` 217 lakh (5) None of these

35. The expenditure of Company B in the year 2006 is what percentage of its income in that year?

(1) 60% (2) 160% (3) 62.5% (4) 40% (5) 80%

Directions (Q. 36-40): Following line-graph shows the ratio of imports to exports of two countries A and B over the years.



36. If the value of imports of Country A in the year 2008 is ` 39.72 crore, what is the value of exports of Country Ain that year?

(1) 64.6 crore (2) 66.2 crore (3) 68.5 crore (4) 69.8crore (5) 72crore

37. If the exports of Country A in the year 2009 and the exports of Country B in the year2007 are equal and they are 96.4 crore each, what is the difference between the imports of B in the year 2007 and the import of A in the year 2009?

(1) ` 32.28 crore (2) ` 34.86 crore (3) ` 36.64 crore (4) ` 38.56 crore (5) ` 40.5 crore

38. If the total imports of Country A in the year 2006 and the total imports of B in the year 2004 are ` 63.6 crore and ` 62.8 crore respectively, what is the sum of exports of A in 2006 and exports of B in 2004?

(1) `161.1 crore (2) `162.2 crore (3) `163.3 crore (4) `164.4 crore (5) `165.5 crore

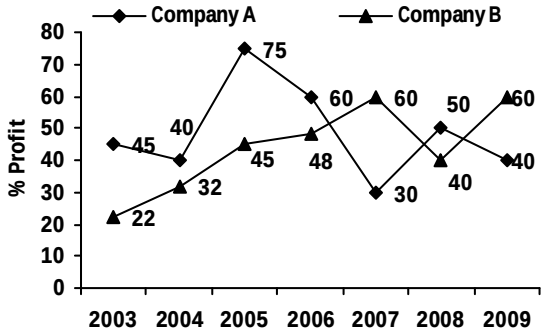
39. The ratio of imports to exports of Country B in the year 2005 is what percentage of the ratio of imports to exports of Country A in 2010?

(1) 112.5% (2) 137.5% (3) 150% (4) 72.72% (5) 87.5%

40. If, for Country A, in the year 2005, the import is increased by 25% and the export is decreased by 50%, what Will be the new ratio of import to export of Country A in 2005?

(1) 1.25 (2) 2 (3) 2.5 (4) 0.6 (5) 0.5

Directions (Q. 41-45): Following line-graph shows the percentage profit earned by two different companies A and B over the years.



41. In which of the following years the percentage of expenditure with respect to income is 62.5% for Company B?
(1) 2004 (2) 2005 (3) 2006 (4) 2007 (5) None of these

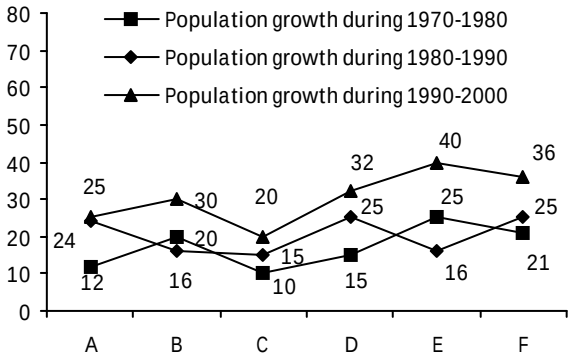
42. If the sum of expenditure of Company A in 2008 and that of Company B in 2004 is `175 lakh, what will be the sum of the income of A in the year 2008 and the income of B in 2004?
(1) `125 1akh (2) `245 lakh (3) `122.5 1akh (4) `250 1akh (5) None of these

43. If the expenditure of Ain 2009 is equal to the expenditure of B in the year 2004, the income of B in the year 2004 is what percentage of the income of A in the year 2009?
(1) 62.5% (2) 71.42% (3) 87.5% (4) 140% (5) 160%

44. If the expenditure of Ain the year 2005 and the income of B in the year 2003 are equal and it is `116 lakh each what is the difference between the income of Ain 2005 and the expenditure of B in 2003?
(1) `82.8 1akh (2) `84.6 1akh (3) `86.4 lakh (4) `88.2 lakh (5) `80.7 lakh

45. If the income of A in 2009 and the expenditure of B in 2005 are `112 lakh and `56 lakh respectively, what is the ratio of the expenditure of A in 2009 to the income of B in 2005?
(1) 3 : 5 (2) 5 : 7 (3) 7 : 9 (4) 1 : 3 (5) 1 : 2

Directions (Q. 46-50): Following line-graph shows the percentage growth of population of six cities (A, B, C, D, E and F) in three decades.



46. If the population of City C was 8.5 lakh in the year 1970, what is the population of City C in the year 2000?
(1) 11.256 lakh (2) 12.134 lakh (3) 12.903 lakh (4) 13.196 lakh (5) 13.427 lakh

47. If the population of City D is 2087250 in the the year 2000, what was its population in the year 1970?
(1) 11 lakh (2) 11.4 lakh (3) 12.2 lakh (4) 12.6 lakh (5) 13 lakh

48. If, in the year 2000, the populations of City A and B are 1388800 and 1302912 respectively, the population of City B in the year 1970 was what percentage of the population of City A in the year 1970?
(1) 72% (2) 75% (3) 90% (4) 96% (5) 108%

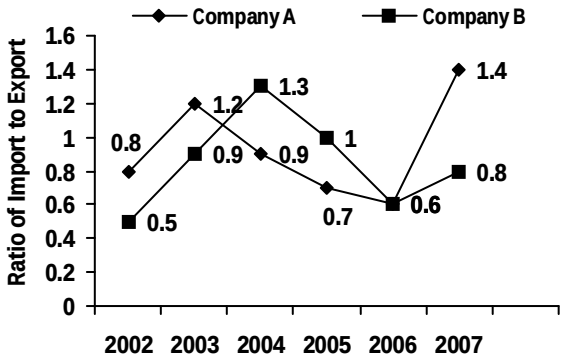
49. If the population of City E and City F in the year 1970 was 12.5 lakh and 10 lakh respectively, what is the difference between the population of City E and the population of City F in the year 2000?

(1) 3.615 lakh (2) 3.904 lakh (3) 4.264 lakh (4) 4.805 lakh (5) None of these

50. If the population of City C and that of City D were equal in the year 1970, what is the ratio of the population of City C to that of City D in 1990?

(1) 22:25 (2) 26:31 (3) 25:28 (4) 3:4 (5) 7:9

Directions (Q. 51-55): Following line-graphs show the ratio of imports to exports by two companies (A and B) during the period 2002-2007.



51. In how many years were the imports less than or equal to the exports for Company B?

(1) 4 (2) 2 (3) 3 (4) 5 (5) None of these

52. The import-to-export ratio of Company B in the year 2002 is what percentage of the import-to-export ratio of A in the year 2002?

(1) 60% (2) 160% (3) 162.5% (4) 62.5% (5) None of these

53. If the import of Company A in the year 2006 is 12 lakh, what is the total export of Company B in the same year?

(1) 7.2 lakh (2) 20 lakh (3) 12 lakh (4) 10 lakh (5) None of these

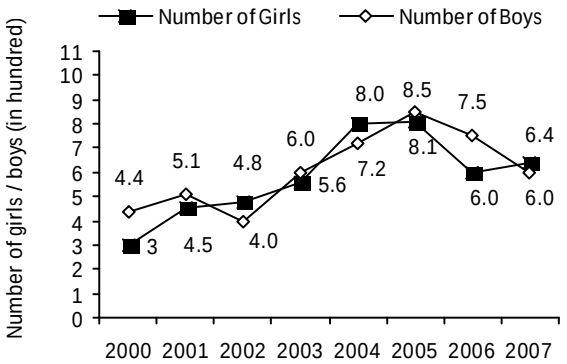
54. If the of exports of Company A and B are equal in the year 2003 and 40 lakh each, the total import of Company B is what percentage of the total import of Company A in that year?

(1) 133.33% (2) 75% (3) 90% (4) 33.33% (5) 25%

55. If the import of Company B in the year 2007 is 78 lakh, what is the difference between the total export and total import of Company B in that year?

(1) 15.6 lakh (2) 16.4 lakh (3) 19.5 lakh (4) 21.2 lakh (5) None of these

Directions (Q. 56-60): Following line-graph shows the number of boys and the number of girls admitted in a college in different years, Answer the questions given below based on this graph.



56. What is the difference between the total number of boys and that of girls admitted in all eight years together?

(1) 228 (2) 230 (3) 232 (4) 234 (5) 236

57. The number of girls admitted in the year 2000 and 2001 together is what percentage of the number of boys admitted in the year 2004 and 2007 together? (Answer in approximate value)

(1) 52.4% (2) 54.3% (3) 56.8% (4) 58% (5) 62.4%

58. What is the approximate percentage increase in the number of girls admitted in the year 2003 and 2004?

(1) 42.8% (2) 38.6% (3) 36.48% (4) 35% (5) 32%

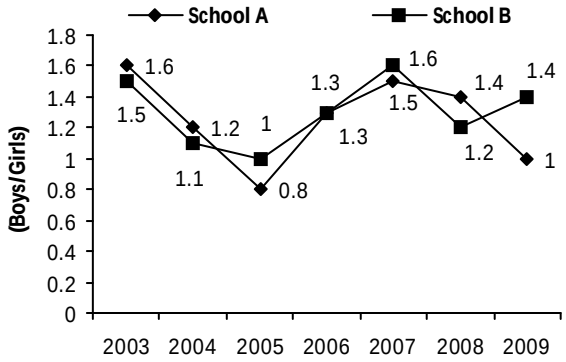
59. In which of the following years is the percentage rise in the number of boys the maximum compared to its previous year?

(1) 2001 (2) 2003 (3) 2004 (4) 2005 (5) None of these

60. The number of girls admitted in the year 2007 is what percentage more than the average number of girls admitted during the entire period of eight years ?

(1) 8.26% (2) 10.34% (3) 12.24% (4) 16% (5) 17.5%

Directions (Q. 61-65): Following line graph shows the ratio of the number of boys to the number of girls passed from two different schools A and B over the period 2003 to 2009.



61. If in year 2003 the number of boys passed from School A is 128, what is the difference between the number of boys passed and the number of girls passed from School A in 2003?

(1) 40 (2) 48 (3) 64 (4) 80 (5) None of these

62. In year 2007 the number of girls passed from School B is approximately what percentage of the number of boys passed in that year?

(1) 160% (2) 80% (3) 62.5% (4) 60% (5) None of these

63. In which year the difference between the number of boys passed and the number of girls passed is highest for School B?

(1) 2003 (2) 2005 (3) 2007 (4) 2009 (5) None of these

64. If the number of girls passed from School A in year 2005 and the number of girls passed from School B in 2006 are equal, the number of boys passed from School B in year 2006 is what percentage of the number of boys passed from School A in 2005?

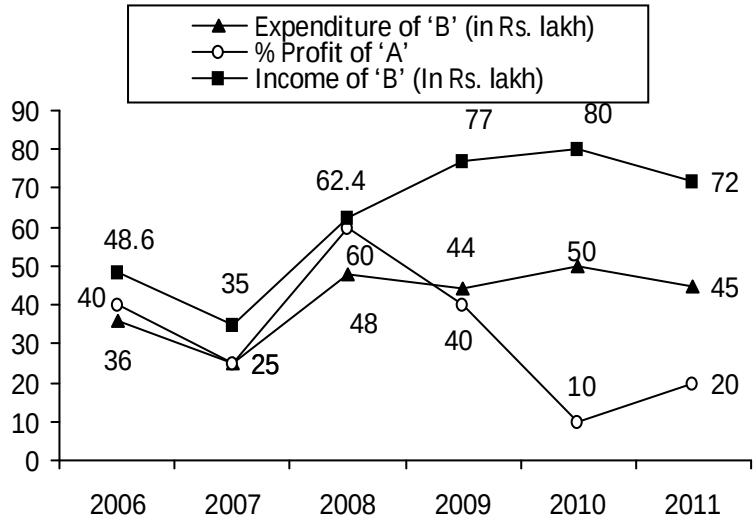
(1) 50% (2) 78.5% (3) 120% (4) 162.5% (5) None of these

65. If the number of girls passed from School B in year 2003 is 70, which is equal to the number of girls passed from School A in year 2006, the difference between the number of boys passed from B in 2003 and the number of boys passed from A in 2006, is what percentage of the total number

of girls passed from A in 2006 and B in 2003?

(1) 10% (2) 20% (3) 80% (4) 120% (5) 140%

Directions (Q. 66-70): Following line graph shows the per cent profit of Company A, income of Company B and expenditure of Company B from 1990 to 1995.



66. What is the difference between per cent profit of Company A and Company B in the year 2006?

(1) 5% (2) 7% (3) 11% (4) 12% (5) 15%

67. If the income of Company A in year 2007 was ` 32.5 lakh, what was the sum of the net profit of Company A and Company Bin 2007?

(1) `12.8 1akh (2) `13.2 1akh (3) `15 1akh (4) `16.5 lakh (5) None of these

68. In which of the following years was the per cent profit of Company B maximum?

(1) 2007 (2) 2008 (3) 2009 (4) 2010 (5) 2011

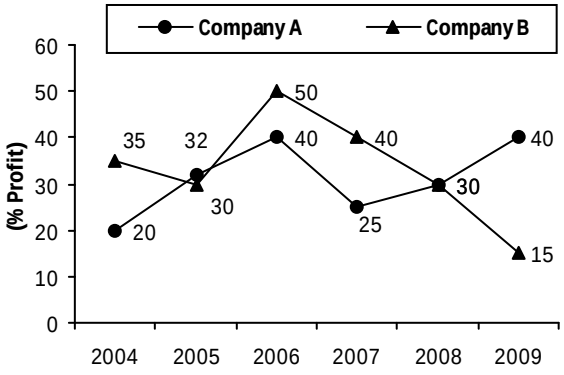
69. If the expenditure of Company A in year 2010 was `45 lakh the net profit of Company A is what per cent of net profit of company B in 2010?

(1) 15% (2) 25% (3) 40% (4) 75% (5) 80%

70. If the income of Company A in year 2011 was ` 90 lakh the net profit of Company B is what per cent more than the net profit of Company A?

(1) 30% (2) 60% (3) 75% (4) 80% (5) 90%

Directions (Q: 71-75): Following line graph shows the percentage profit earned by two companies A and B during the period 2004 to 2009. Answer the following questions based on this graph.



71. If the expenditure of Company B in the year 2004 was `17 lakh, what was its income in that year?
(1) `22.95 lakh (2) `23.151akh (3) `24.5 lakh (4) `25.65 lakh (5) `27.50 lakh

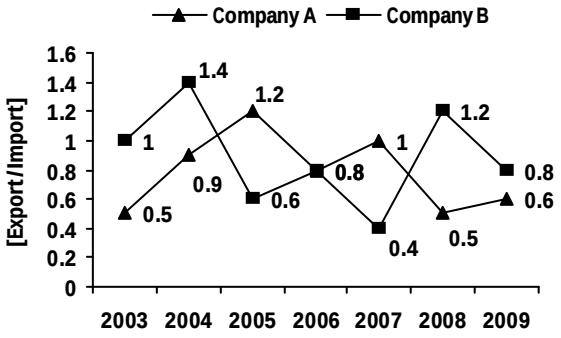
72. If the income of Company A in me year 2008 is `26 lakh, what is the expenditure of Company B in that year?
(1) `20 lakh (2) `33.81akh (3) `22.5 lakh (4) `21.6 lakh (5)Can't be determined

73. If the sum of expenditure of Company B in the year 2005 and 2008 together is `48 lakh, what is the total income of Company B in these two years together?
(1) `62.4 lakh (2) `36.2 lakh (3) `641akh (4) `65.5 lakh (5) None of these

74. In which year is the ratio of income to expenditure of Company A the maximum?
(1) 2004 (2) 2008 (3) 2006 (4) 2009 (5) None of these

75. If the expenditure of Company A in the year 2004 and Company B in die year 2009 are the same and the income of Company B in die year 2009 is `77 lakh, what is the income of Company A in the year 2004?
(1) `55 1akh (2) `66 lakh (3) `56 lakh (4) `64 lakh (5) None of these

Directions (Q. 76-80): Following line graph shows the ratio of import to export of two different Companies A and B during the period 2003 to 2009.



76. If the total import of Company B in year 2005 is 67.2 lakh, what is the total export of Company B in year 2005?
(1) 112 lakh (2) 96 lakh (3) 44.8 lakh (4) 40.32 lakh (5) None of these

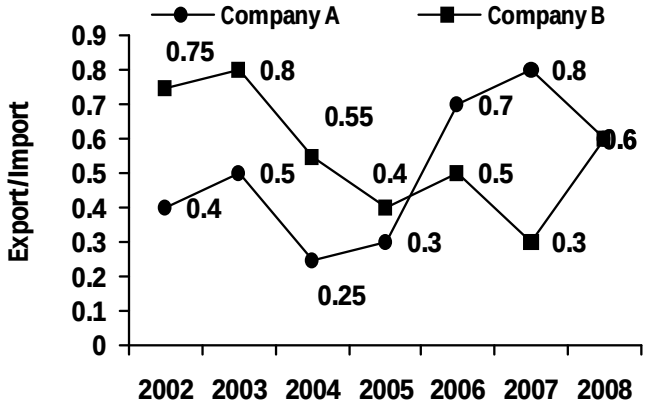
77. If the total export of Company A in year 2006 is 84 lakh, what will be the total import of Company B in year 2006?
(1) 105 lakh (2) 84 lakh (3) 67.2 lakh
(4) Can't be determined (5) None of these

78. If in year 2Q08 the export of Company A and import of Company B are 116 lakh and 117 lakh respectively, what will be the sum of imports of Company A and exports of Company B in 2008?
(1) 151.5 lakh (2) 152.5 lakh (3) 153.5 lakh (4) 154.5 lakh (5) 155.5 lakh

79. If in year 2005 the import of Company A is decreased by 25% and export is decreased by 50%, what will be the new ratio of import to export of Company A in 2005?
(1) 0.55 (2) 0.9 (3) 1.2 (4) 1.8 (5) 2.25

80. If the import of Company A in year 2005 and the export of Company B in year 2007 are 102.6 lakh and 112.5 lakh respectively, the export of A in 2005 is what percentage of the import of Company B in year 2007?
(1) 190% (2) 148% (3) 108% (4) 68.32% (5) 52.63%

Directions (Q. 81-85) : Following line graph shows the ratio of exports to imports of two companies A and B over the period 2002 to 2008.



81. If the import of Company A in 2004 was 96.8 lakh, what was the export of Company A in that year?
(1) 24.2 lakh (2) 36 lakh (3) 48.4 lakh (4) 64 lakh (5) None of these

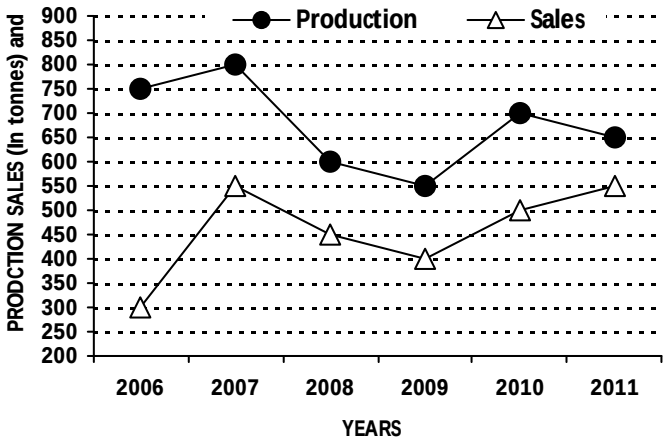
82. The ratio of export to import of Company B in year 2004 was what percentage of the ratio of export to import of Company A in year 2002?
(1) 72.72% (2) 97.5% (3) 115% (4) 137.5% (5) 150%

83. If the import of Company A in year 2007 and export of Company B in year 2008 are 86 lakh and 51 lakh respectively, what is the sum of export of Company A in 2007 and import of Company B in 2008?
(l) 1.536crore (2) 1.538crore (3) 1.540crore (4) 1.542 crore (5) 1.546 crore

84. If in year 2005 the export of Company B is increased by 125% and its import is decreased by 60%, what will the new ratio of export to import of Company B in 2005?
(1) 5 : 4 (2) 3 : 2 (3) 7 : 4 (4) 2 : 1 (5) 9 : 4

85. If the export of Company A in year 2005 and that of B in year 2002 were 23.4 lakh and 72 lakh respectively, then the import of Company A in year 2005 is what percentage of the import of Company B in year 2002?
(1) 81.25% (2) 83.5% (3) 85.75 (4) 87.5% (5) 123%

Directions (Q.86-90) Study the following information and answer the questions that follow :
THE GRAPH GIVEN BELOW REPRESENTS THE PRODUCTION (IN TONNES) AND SALES (IN TONNES) OF COMPANY A FROM 2006-2011.



The table given below represents the respective ratio of the production (in tonnes) of Company A to the production (in tonnes) of Company B. and the respective ratio of the sales (in tonnes) of Company A to the sales (in tonnes) of Company B.

Year	Production	Sales
2006	5 : 4	2 : 3
2007	8 : 7	11 : 12
2008	3 : 4	9 : 14
2009	11 : 12	4 : 5
2010	14 : 13	10 : 9
2011	13 : 14	1 : 1

86. What is the approximate percentage increase in the production of Company A (in tonnes) from the year 2009 to the production of Company A (in tonnes) in the year 2010 ?
(1) 18% (2) 38% (3) 23% (4) 27% (5) 32%

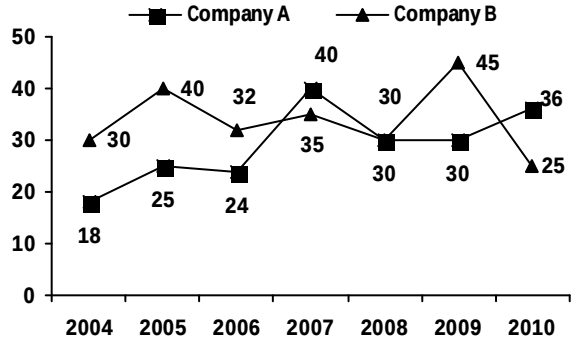
87 The sales of Company A in the year 2009 was approximately what percent of the production of Company A in the same year?
(1) 65% (2) 73% (3) 79% (4) 83% (5) 69%

88. What is the average production of Company B (in tonnes) from the year 2006 to the year-2011 ?
(1) 574 (2) 649 (3) 675 (4) 593 (5) 618

89 What is the respective ratio of the total production (in tonnes) of Company A to the total sales (in tonnes) of Company A?
(1) 81 : 64 (2) 64 : 55 (3) 71 : 81 (4) 71 : 55 (5) 81 : 55

90 What is the respective ratio of production of Company B (in tonnes) in the year 2006 to production of Company B (in tonnes) in the year 2008 ?
(1) 2 : 5 (2) 4 : 5 (3) 3 : 4 (4) 3 : 5 (5) 1 : 4

Directions (Q. 91-95): The following line graph shows the percentage profit of two companies over the years. Study it carefully and answer the questions that follow.



91. If the total income of Company A in the year 2006 was ` 55.8 crore then what was the expenditure of Company A in the same year?
(1) `42.5 crore (2) `45 crore (3) `47.5 crore (4) `50 crore (5) None of these

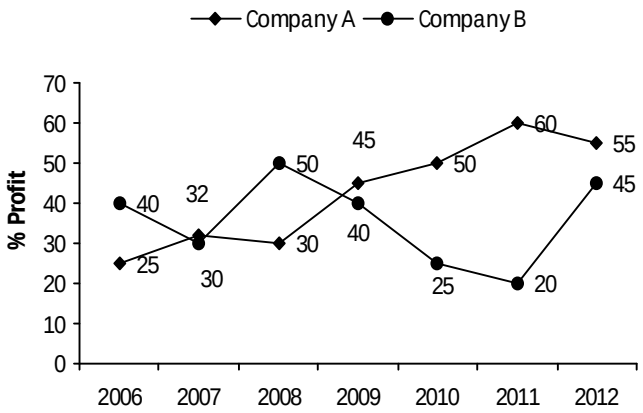
92. In which of the following years is the ratio of income to expenditure the maximum for Company B?
(1) 2004 (2) 2005 (3) 2008 (4) 2009 (5) 2010

93. If the total expenditure of Company A in 2009 and Company B in 2004 together was 7148 crore, what was the total income of Company A in 2009 and Company B in 2004 together?
(1) 7184.6 crore (2) 7188 crore (3) 7190.8 crore (4) 7192.4 crore (5) 7196 crore

94. If the expenditure of Company B in the year 2009 and the income of Company A in the year 2005 are equal and it is ` 56 crore each, what is the sum of the income of B in 2009 and the expenditure of A in 2005?
(1) 7124.8 crore (2) 7126 crore (3) 7127.5 crore (4) 7132 crore (5) 7134.8 crore

95. If the total income of Company A and Company B in the year 2008 is ` 78 crore what is the total expenditure of Company B in the year 2008?
(1) 30 crore (2) 39 crore (3) 60 crore (4) 7.8 crore (5) Data inadequate

Directions (Q. 96-100) : The following graph shows the net percentage profit of two companies, A and B for the period 2006 to 2012.



96. If the income of Company A in year 2007 is Rs 85.8 lakh, then what will be its expenditure (in Rs) in that year?
(1) 56 lakh (2) 65 lakh (3) 72.8 lakh (4) 97.64 lakh (5) 113.2561akh

97. If in year 2012 the expenditure of Company A was Rs 90.6 lakh, what was its income (in Rs) in that year?
(l) 139.181akh (2) 148 lakh (3) 138.2 lakh (4) 140.43 lakh (5) 144.64 lakh

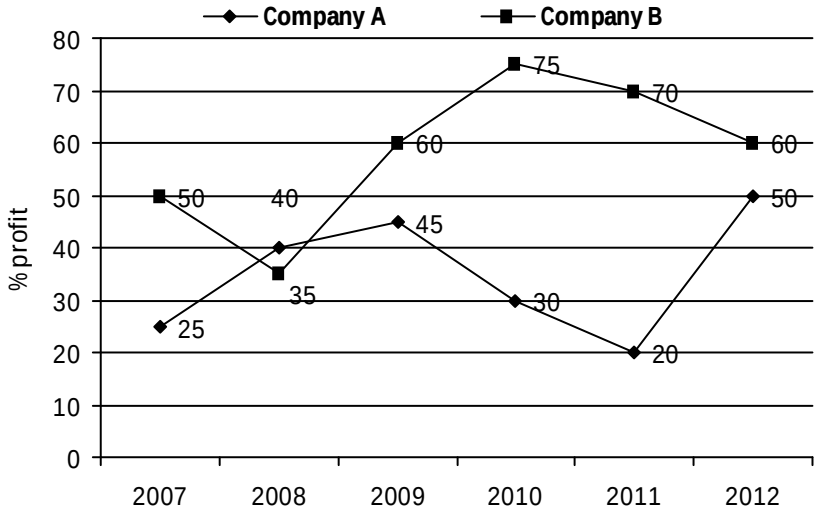
98. In which of the following years is the percentage increase in the profit of Company A the' highest over the preceding year?
(1) 2007 (2) 2009 (3) 2010 (4) 2011 (5) None of these

99. In which of the following year's is the difference between the income and the expenditure of Company B the maximum?
(1) 2006 (2) 2008 (3) 2011 (4) 2012 (5) None of these

100. If in the year 2008, the expenditure of Company A and the income of Company are Rs 84 lakh each, what is the difference (in Rs) between the income of Company A and the expenditure of Company B in that year?
(1) 48.6 lakh (2) 50.4 lakh (3) 51 lakh (4) 53.2 lakh (5) 57.6 lakh

Directions (Q. 101-105) : Following line graph shows the percentage profit gained by two companies A and B over the years 2007 to 2012.

$$\%\,\text{profit} = \frac{\text{Profit}}{\text{Expenditure}} \times 100$$



101. If the income of Company B in year 2008 was Rs 91.8 lakh, what was its expenditure (in Rs) in that year?
(1) 64 lakh (2) 68 lakh (3) 70 lakh (4) 72 lakh (5) 76 lakh

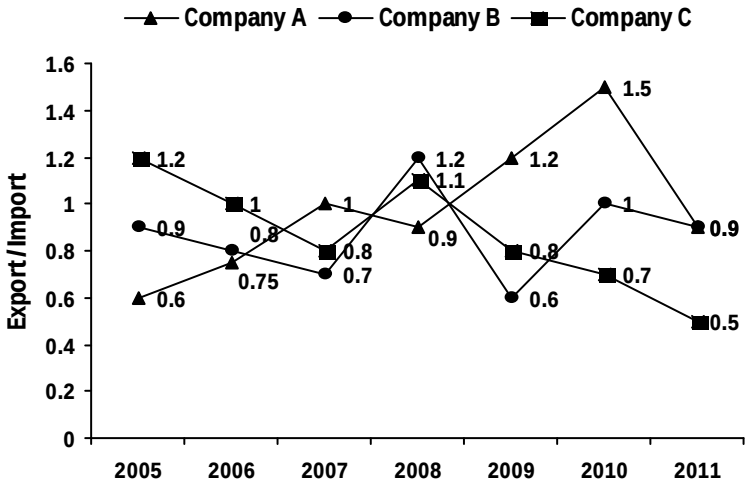
102. If the expenditure of Company A in the year 2010 and 2011 was in the ratio 6:5, what was the ratio of its incomes?
(1) 7 : 3 (2) 9 : 5 (3) 11 : 9 (4) 13 : 10 (5) None of these

103. If the expenditure of Company B in the year 2009 was Rs 40 lakh, what was its income (in Rs) in the year 2012?
(1) 60 lakh (2) 52 lakh (3) 70 lakh (4) 66.6 lakh (5) Data inadequate

104. The income of Company A in the year 2011 and the expenditure of Company B in the year 2008 was the same, that is Rs 90 lakh. What will be the ratio of the income of Company B in 2008 to the expenditure of Company A in the year 2011?
(1) 8 : 3 (2) 9 : 5 (3) 7 : 4 (4) 9 : 7 (5) 1 : 1

105. In which of the following years is the percentage of expenditure with respect to income 80% for Company A?
(1) 2007 (2) 2008 (3) 2009 (4) 2010 (5) 2011

Directions (Q. 106–110) : The following line graph shows the ratio of export to import of three companies A, B and C for the period 2005 to 2011.



106. If the export of Company A in year 2005 and that of Company B in year 2006 are 51 lakh and 54 lakh respectively, what is the difference between the import of A in 2005 and that of B in 2006?
(1) 12.5 lakh (2) 13 lakh (3) 15 lakh (4) 17.51akh (5) 18 lakh

107. If the import of Company A in year 2010 and the export of Company C in year 2011 are 64 lakh and 48 lakh respectively, what is the ratio of the export of A in 2010 to the import of Company C in 2011?
(1) 4 : 3 (2) 3 : 2 (3) 2 : 1 (4) 6 : 5 (5) 1 : 1

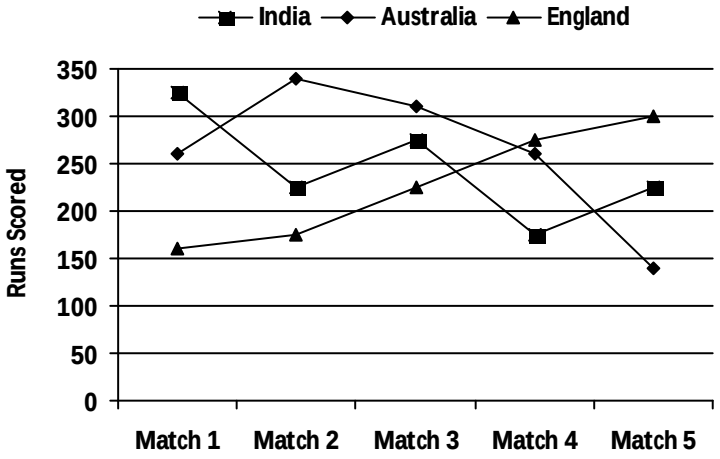
108. If the import of Company A in 2009 and the import of Company B in 2006 are equal and they are 55 lakh each, then the export of Company A in 2009 is approximately what per cent of the export of Company B in 2006?
(1) 66.66% (2) 78% (3) 112% (4) 140% (5) 150%

109. If the export of Company B in 2007 and the export of Company C in 2010 are 58.8 lakh and 56.7 lakh respectively, what is the difference between the import of Company B in 2007 and the import of Company C in 2010?
(1) 3 lakh (2) 4 lakh (3) 4.4 lakh (4) 6.2 lakh (5) 7.5 lakh

110. If in the year 2006 the export of Company A is increased by 200% and the import is increased by 50%, what will be the new ratio of export to import of Company A in 2006?
(1) 4 : 3 (2) 3 : 1 (3) 3 : 2 (4) 9 : 5 (5) 5 : 3

Directions (Q. 111-115) : Study the following graph carefully to answer the questions that follow:

Runs scored by three different teams in five different cricket matches.



111. The total runs scored by India and Australia in Match 4 together is approximately what percentage of the total runs scored by England in all the five matches together?
(1) 42 (2) 18 (3) 36 (4) 24 (5) 28

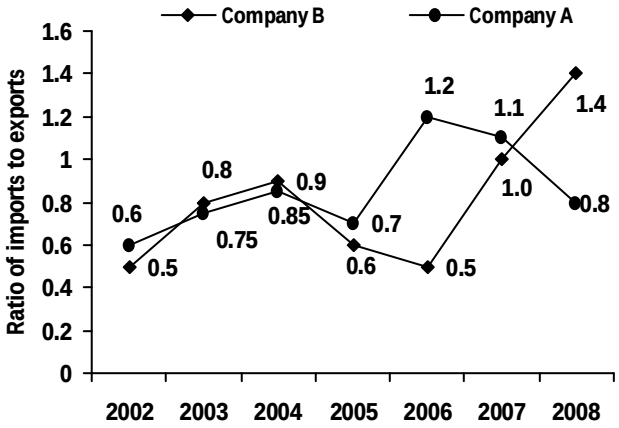
112. In which match is the difference between the runs scored by Australia and England the second lowest?
(1) Match 1 (2) Match 2 (3) Match 3 (4) Match 4 (5) Match 5

113. In which match are the total runs scored by India and England together the third highest/ lowest?
(1) Match 1 (2) Match 2 (3) Match 3 (4) Match 4 (5) Match 5

114. What is the ratio of the runs scored by India in Match 5, Australia in Match 1 and England in Match 2?
(1) 11 : 13 : 7 (2) 11 : 7 : 13 (3) 11 : 3 : 9 (4) 11 : 13 : 9 (5) None of these

115. What is the average runs scored by all the three teams in Match 3 together?
(1) 280 (2) 270 (3) 275 (4) 285 (5) None of these

Directions (Q. 116-120) : Study the given graph carefully and answer the following questions. The graph shows the ratio of imports to exports of two Companies A and B over the years.



116. If the total imports of Company A in the year 2005 was Rs 53.9 lakh, what was its total exports (in Rs) in that year?
(1) 37.73 lakh (2) 47.8 lakh (3) 68.3 lakh (4) 77 lakh (5) None of these

117. The ratio of imports to exports of Company B in the year 2004 was what percentage more than that of Company A in the year 2008?
(1) 10% (2) 12.5% (3) 20% (4) 25% (5) None of these

118. If in the year 2003 the imports of Company A increased by $33\frac{1}{3}$% and exports decreased by 20%, then what would be the new ratio of imports to exports of Company A in that year?

(1) 0.8 (2) 0.6 (3) 1.2 (4) 1.25 (5) None of these

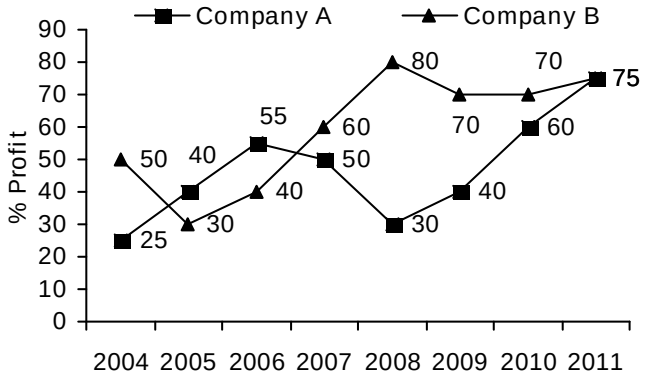
119. If the imports of Company A in the year 2008 and exports of B in the year 2004 were Rs 36 lakh and Rs 60 lakh respectively, then the imports of Company B in the year 2004 would be what percentage of the exports of Company A in the year 2008?

(1) 125% (2) 120% (3) 97.5% (4) 83.33% (5) 75%

120. In which of the following years was the value of exports less than the value of imports in the case of Company B?

(1) 2002 (2) 2006 (3) 2004 (4) 2007 (5) 2008

Directions (Q. 121-125) : The following graph shows the percentage profit of two companies A and B over the years. Study the graph carefully and answer following Questions.



121. If the income of Company B in, the year 2010 is Rs 136 lakh, then what is its total profit (in Rs) in the year 2010?

(1) 48 lakh (2) 56 lakh (3) 64 lakh (4) 72 lakh (5) 80 lakh

122. If the sum of the incomes of Company A in the year 2005 and the year 2009 together is Rs 171.5 lakh, then what is the total profit of Company A in the years 2005 and 2009 together?

(1) 42.5 lakh (2) 45 lakh (3) 47.5 lakh (4) 49 lakh (5) 52.5 lakh

123. If the income of Company A in the year 2011 was equal to the expenditure of Company B in the year 2004, then what was the ratio of the expenditure of Company A in 2011 to the income of Company B in 2004?

(1) 7 : 6 (2) 25 : 42 (3) 16 : 25 (4) 16 : 42 (5) None of these

124. If the expenditure of Company A in the year 2005 was equal to the income of Company B in the year 2008 and it was Rs 90 lakh, then the profit of Company A in the year 2005 is what per cent of the profit of Company B in the year 2008?

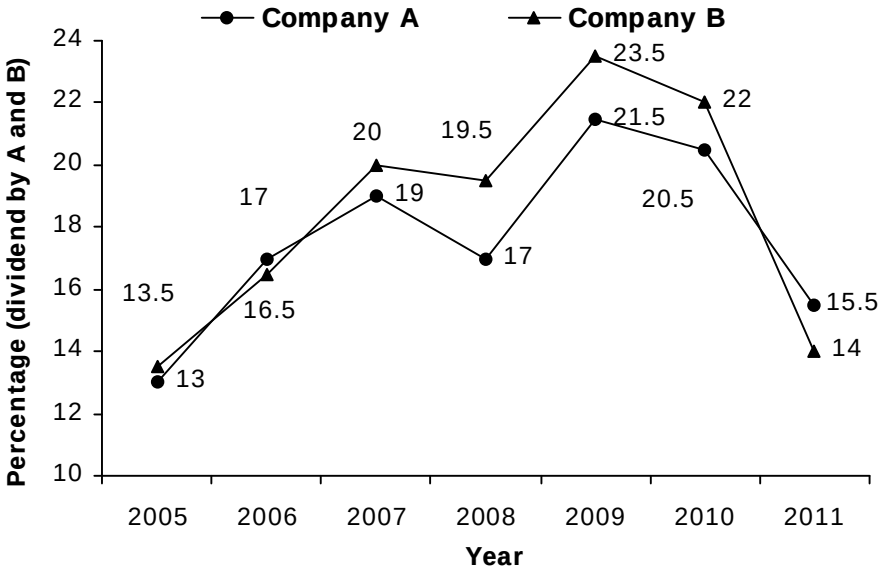
(1) 90% (2) 11.11% (3) 80% (4) 40% (5) 120%

125. For Company A, in which year is the per cent increase in profit over that of previous year the highest?

(1) 2005 (2) 2006 (3) 2009 (4) 2010 (5) 2011

Directions (Q. 126-130) : Study the following graph carefully to answer the questions given below:

Percentage annual dividend offered by two companies A and B over the years



126. Manav invested a total amount of ` 40000 in year 2005 for one year in two different companies together and got a total dividend of ` 5299. What was the amount invested in Company B?
(1) ` 20200 (2) ` 19800 (3) ` 31400
(4) Can't be determined (5) None of these

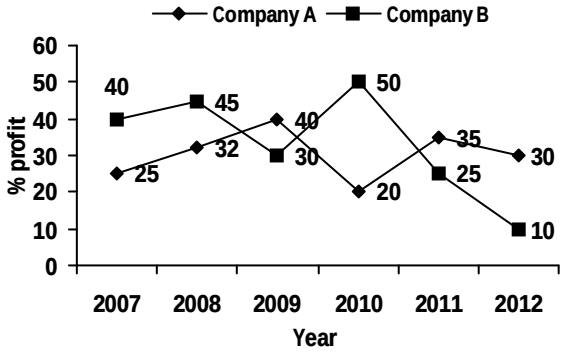
127. Priya invested `50000 in Company A in year 2009. After one year she transferred the entire amount with dividend to Company B for one year. What amount including dividend would she receive?
(1) ` 60750 (2) ` 61750 (3) ` 42750
(4) Can't be determined (5) None of these,

128. An amount of ` 3 7000 was invested in Company B in year 2007. After one year the same amount was reinvested for one year. What was the total dividend received at the end of two years?
(1) ` 17430 (2) ` 37312 (3) ` 14430 (4) ` 5305 (5) None of these

129. Rahul invested two different amounts in Company A and B in 2011 in the ratio of 7 : 9. What will be the ratio of dividends received from Company A and B?
(1) 31 : 36 (2) 36 : 31 (3) 35 : 32
(4) Can't be determined (5) None of these

130. Sukriti invested ` 75000 in Company A in the year 2010. How much more or less dividend would have she received had the amount been invested in Company B?
(1) ` 45221ess (2) ` 1011 less (3) `1 015 less (4) ` 1125 more (5) None of these

Directions (Q. 131-135) : The following graph shows the percentage profit earned by two companies A and B during 2007-2012.



131. If the expenditure of Company A in the year 2009 was ` 77.5 lakh what was its income (in `) in that year?
(1) 96 lakh (2) 102.4 lakh (3) 108.5 lakh (4) 112.5 lakh (5) None of these

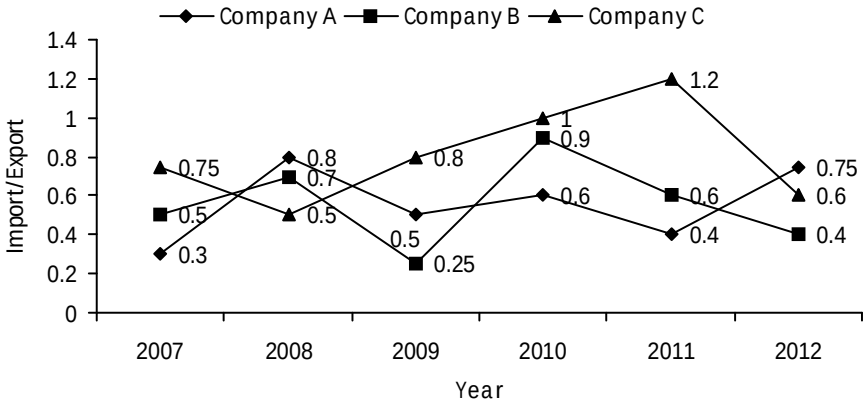
132. If the income of Company B in the year 2012 was ` 125.4 lakh what was its expenditure (in `) in that year?
(1) 94 lakh (2) 102 lakh (3) 108 lakh (4) 114 lakh (5) 117.5 lakh

133. If the expenditure of Company A in the year 2008 and the income of Company B in the year 2011 were equal to 85 lakh what was the difference between the profit of Company A in the year 2008 and the profit of Company B in the year 2011?
(1) 10.2 lakh (2) 11.4 lakh (3) 12.8 lakh (4) 15 lakh (5) 17.5 lakh

134. If the incomes of two Companies in the year 2010 were equal what was the ratio of their expenditures?
(1) 5 : 4 (2) 5 : 3 (3) 5 : 2 (4) 5 : 1 (5) None of these

135. If the income of Company A in the year 2010 and the expenditure of Company B in the year 2012 were equal and they were ` 171 lakh each, what was the difference between the income of Company B in the year 2012 and the expenditure of Company A in the year 2010?
(1) 41.2 lakh (2) 42.3 lakh (3) 43.4 lakh (4) 44.5 lakh (5) 45.6 lakh

Directions (Q. 136-140) : Following line graph shows the ratio of import to export of three companies over the period of 2007-2012.



136. If the import of Company A in the year 2007 was ` 23.58 lakh what was its export (in `) in that year?
(1) 70.74 lakh (2) 48.24 lakh (3) 70.74 lakh (4) 78.60 lakh (5) 81.5 lakh

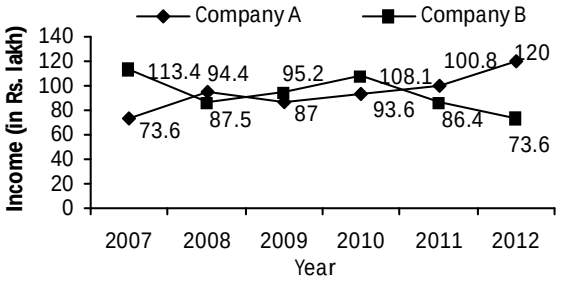
137. The ratio of import to export of Company A in the year 2012 is approximately what per cent of the ratio of import to export of Company C in the year 2011 ?
(1) 47.5% (2) 55% (3) 62.5% (4) 11.2% (5) 160%

138. If the export of Company A in the year 2012 and the import of Company C in the year 2009 were equal to ` 64 lakh each then the import of Company A in the year 2012 was approximately what per cent of the export of Company C in the year 2009?
(1) 20% (2) 40% (3) 60% (4) 80% (5) 100%

139. If the import of Company A and Company B in the year 2009 were ` 36 lakh and ` 27 lakh respectively what was the ratio of their exports in that year?
(1) 4:3 (2) 2:3 (3) 8:9 (4) 4:9 (5) 1:2

140. If the imports of Company C in year 2008 and 2012 were equal then the export of Company C in year 2008 was approximately what per cent of its export in year 2012?
(1) 40% (2) 60% (3) 80% (4) 100% (5) 120%

Directions (Q. 141-145): The following line-graph shows the income of two companies A and B over the period 2007 to 2012. Answer the given questions based on this graph.



141. If the percentage profit of Company A in the year 2007 was 15% what was its expenditure (in `) in that year?

(1) 60 lakh (2) 64 lakh (3) 68 lakh (4) 72 lakh (5) None of these

142. If the percentage profit of Company A in the year 2010 and that of Company B in the year 2011 was equal to 20%, what was the difference (in `) between the expenditure of Company A in the year 2010 and the expenditure of Company B in the year 2011 ?

(1) 4 lakh (2) 4.8 lakh (3) 5.4 lakh (4) 6 lakh (5) 6.5 lakh

143. If the expenditure of Company A and Company B were ` 75 land and ` 85 lakh respectively in the year 2009, what was the difference between their percentage profit in that year?

(1) 2% (2) 3% (3) 4% (4) 5% (5) 6%

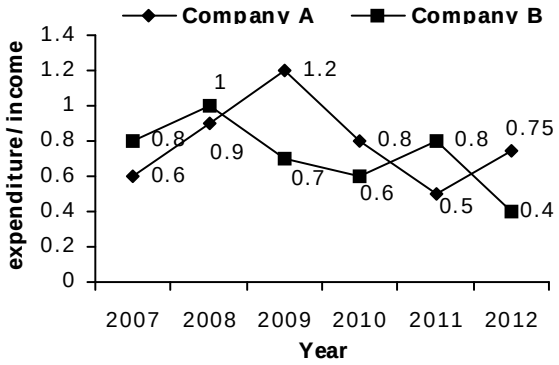
144. The income of Company A in the year 2010 was approximately what per cent of its income in the year 2012?

(1) 72% (2) 75% (3) 78% (4) 80% (5) 84%

145. If the percentage profit of Company A in the year 2011 and that of Company B in the year 2009 were equal to 12% each, what was the ratio of the expenditure of Company A in the year 2011 to the expenditure of Company B in the year 2009?

(1) 9 : 8 (2) 8 : 5 (3) 9 : 7 (4) 9 : 5 (5) None of these

Directions (Q. 146-150) : Following line-graph shows the ratio of expenditure to income of two companies A and B over the period of 2007 to 2012. Answer the given question based on this graph.



146. The ratio of expenditure to income of Company A in the year 2012 is-approximately what per cent of its ratio of expenditure to income in the year 2009?

(1) 60.5% (2) 62.5% (3) 72.5% (4) 52.25% (5) None of these

147. If the expenditure and income of Company B in the year 2009 are increased by 100% and 110% respectively, what will be its new ratio of expenditure to income in that year?

(1) 1 : 2 (2) 2 : 3 (3) 3 : 4 (4) 4 : 7 (5) 5 : 7

148. If the expenditure of Company B in the year 2009 was `14.7 lakh, what was its percentage profit that year? (Answer in approximate value)

(1) 32% (2) 37% (3) 40% (4) 43% (5) 44%

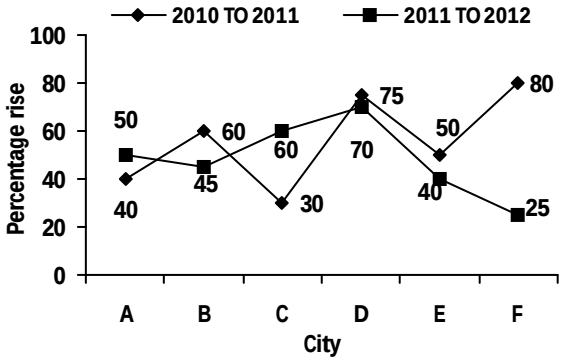
149. If the income of Company A in the year 2010 and the expenditure of Company B in the year 2007 were `18.5 lakh and `12.4 lakh respectively, what was the difference between their net profits?

(1) ` 60000 (2) ` 65000 (3) ` 70000 (4) ` 75000 (5) ` 80000

150. If the income of Company A in the year 2012 and the expenditure of Company B in the year 2011 were equal to ` 24 lakh then the profit of Company A in the year 2012 is approximately what per cent of the profit of Company B in the year 2011?

(1) 60% (2) 75% (3) 80% (4) 100% (5) 120%

Directions (Q. 151-155): The following graph shows the percentage rise in population of six different cities from 2010 to 2011 and 2011 to 2012.



151. If the population of City C was 4.5 lakh in the year 2010, what was its population in the year 2012?

(1) 5.85 lakh (2) 6.48 lakh (3) 7.42 lakh (4) 8.24 lakh (5) 9.36 lakh

152. The population of City D in the year 2010 was approximately what per cent of its population in the year 2011?

(1) 51% (2) 54% (3) 57% (4) 60% (5) 63%

153. If the rise in the population of City A from 2010 to 2012 was 2.75 lakh, what was its population in the year 2010?

(1) 2.4 lakh (2) 2.5 lakh (3) 2.8 lakh (4) 3 lakh (5) 3.2 lakh

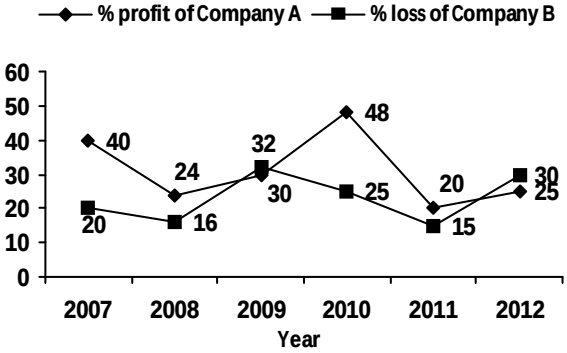
154. If the population of City E in the year 2010 was 3.2 lakh, what was its population in the year 2012?

(1) 5.48 lakh (2) 5.96 lakh (3) 6.24 lakh (4) 6.72 lakh (5) 7.12 lakh

155. In the year 2010, the population of cities B and F were equal, and the population of City F in the year 2012 was 5.4 lakh. What was the population of City B in the year 2012?

(1) 5.248 lakh (2) 5.568 lakh (3) 5.842 lakh (4) 6.214 lakh (5) 6.412 lakh

Directions (Q. 156-160): The following line graph shows the percentage profit of company A and the percentage loss of company B over the years. Answer the following questions based on this information.



156. If the expenditures of Company A and B are equal in the year 2008, and they are ?75 lakh each, what is the difference between the income of Company A and the income of Company B in that year?

(1) `24 1akh (2) `30 1akh (3) `36 1akh (4) `40 lakh (5) `44 1akh

157. If the incomes of Company A in the year 2007 and 2011 are equal and they are `84 lakh each, what is the difference between its expenditures in the year 2011 and 2007?

(1) `10 1akh (2) `12 1akh (3) `14 lakh (4) `16 1akh (5) `18 lakh

158. If the expenditure of Company A in the year 2012 and Company B in the year 2010 are equal, what is the ratio of the income of Company B in year 2010 to the income of Company A in year 2010?

(1) 1:1 (2) 2:5 (3) 3:5 (4) 4:5 (5) None of these

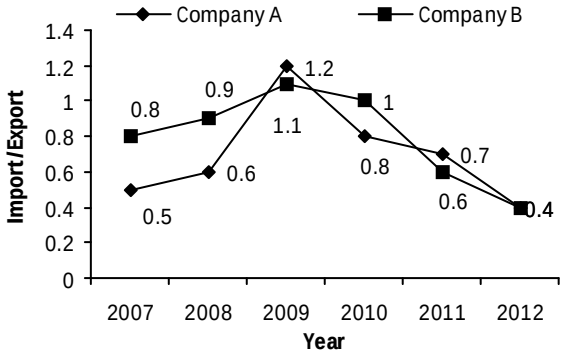
159. What is the percentage increase in the percentage profit of Company A from year 2008 to 2009?

(1) 6% (2) 20% (3) 24% (4) 25% (5) 27%

160. If the expenditure of Company A in the year 2008 and 2010 are `55 lakh and `35 lakh respectively then what is the profit of Company A in the year 2008 and 2010 together?

(1) ` 24 1akh (2) ` 28 1akh (3) ` 30 1akh (4) ` 32 lakh (5) ` 36 1akh

Directions (Q. 161-165): The following graph shows the ratio of imports to exports by two companies over the years.



161. If the export of Company A in the year 2008 was `105 lakh and the import of Company B in the year 2007 was `72 lakh, the import of Company A in the year 2008 is approximately what per cent of the export of Company B in the year 2007?

(1) 60% (2) 70% (3) 80% (4) 90% (5) None of these

162. If the import of Company A is increased by 50% and the export is decreased by 20% in the year 2010, what will be the new ratio of import to export of Company A in that year?

(1) 5:4 (2) 4:3 (3) 3:2 (4) 2:1 (5) None of these

163. If the import of Company A in the year 2010 and the export of Company B in the year 2011 are equal, what will be the ratio of the export of Company A in the year 2010 to the import of Company B in the year 2011 ?

(1) 2:5 (2) 3:5 (3) 4:5 (4) 6:5 (5) None of these

164. The ratio of import to export of Company A in the year 2011 is what per cent of the ratio of import to export of Company B in the year 2012?

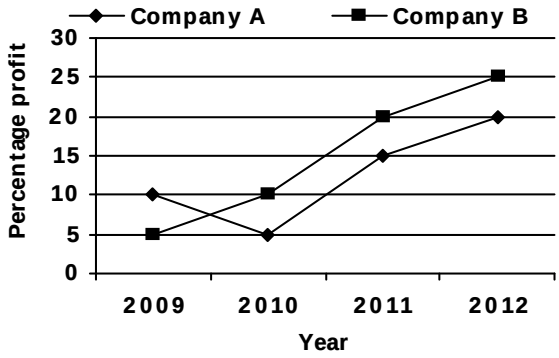
(1) 75% (2) 125% (3) 175% (4) 225% (5) None of these

165. If the import of Company A in the year 2010 and the import of Company B in the year 2008 are equal and they are `108 lakh each then the export of Company A in the year 2010 is what per cent of the export of Company B in the year 2008?

(1) 88.88% (2) 112.5% (3) 120% (4) 127.5% (5) 150%

Directions (Q. 166-170) : Study the graph carefully to answer the questions that follow:

$$\text{Profit\%} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$



166. If the income of Company A in the year 2009 is `440 crore, what is the expenditure (in `) of Company A in that year?

(1) 330cr (2) 450cr (3) 400cr (4) 225 cr (5) None of these

167. In which year is the ratio of expenditure to income of Company A the highest?

(1) 2009 (2) 2011 (3) 2010 (4) 2012 (5) Can't be determined

168. If the sum of income of Company A in the year 2009 and that of Company B in the year 2010 is `880 crore, find the sum of expenditures of Company A in the year 2009 and Company B in the year 2010.

(1) 775cr (2) 830cr (3) 800cr (4) 625 cr (5) Can't be determined

169. If the income of Company A in the year 2009 and the expenditure of Company B in the year 2012 are equal and the income of Company B in the year 2012 is ` 250 crore, then the expenditure of Company A in the year 2009 is approximately what per cent of the expenditure of Company B in the year 2012?

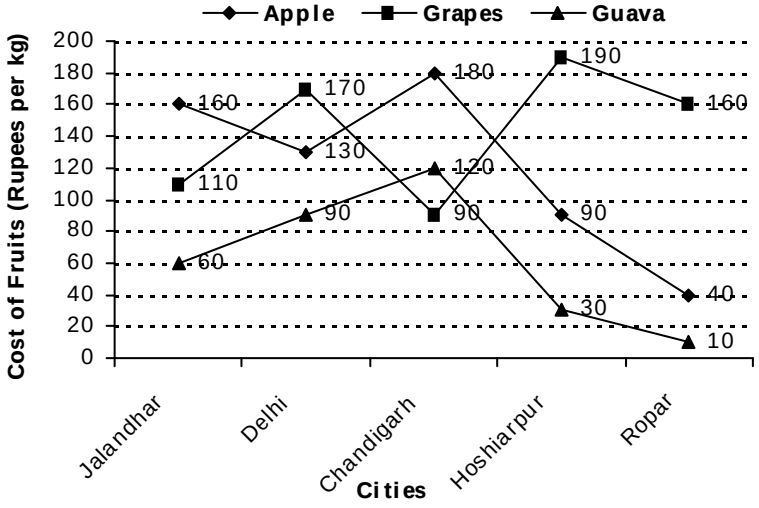
(1) 98% (2) 89% (3) 75% (4) 91% (5) None of these

170. If the ratio of expenditure of Company A in the year 2009 to that of Company B is 5 : 11, what is the ratio of their incomes in that year?

(1) 3:5 (2) 2:3 (3) 2:5 (4) 5 :2 (5) None of these

Directions (Q. 171-175) : Study the following graph carefully to answer the questions that follow:

Cost of three different fruits (in rupees per kg) in five different cities



171. In which city is the difference between the cost of one kg of apple and the cost of one kg of guava the second lowest?

(1) Jalandhar (2) Delhi (3) Chandigarh (4) Hoshiarpur (5) Ropar

172. The cost of one kg of guava in Jalandhar is approximately what per cent of the cost of two kg of grapes in Chandigarh?

(1) 66 (2) 24 (3) 28 (4) 34 (5) 58

173. What total amount will Ram pay to the shopkeeper for purchasing 3 kg of apples and 2 kg of guavas in Delhi?

(1) `530 (2) `450 (3) `570 (4) `620 (5) `490

174. Ravinder had to purchase 45 kg of grapes from Hoshiarpur. The shopkeeper gave him a discount of 4% per kg. What amount did he pay to the shopkeeper after the discount?

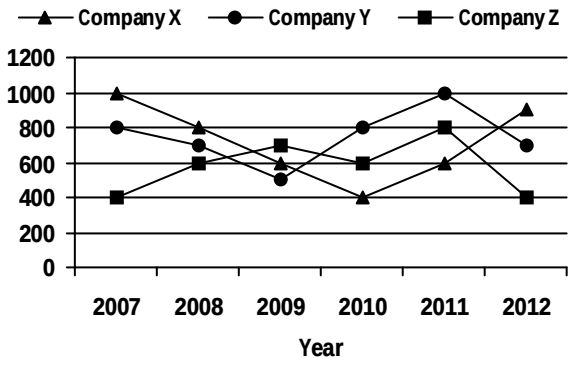
(1) `8208 (2) `8104 (3) `8340 (4) `8550 (5) `8410

175. What is the ratio of the cost of one kg of apples from Ropar to the cost of one kg of grapes from Chandigarh?

(1) 3 : 2 (2) 2 : 3 (3) $2^2 : 3^2$ (4) $4^2 : 9^2$ (5) $9^2 : 4^2$

Directions (Q. 176-180): Study the following graph care-fully to answer these questions:

Quantity of rice (in thousand tonnes) exported by three companies over the years



176. What is the percentage increase in export of Company Y from 2009 to 2012?

(1) 55% (2) 40% (3) 60% (4) 50% (5) None of these

177. What is the ratio of the total export of all the three companies from 2008 to 2012?

(1) 1:6 (2) 6:7 (3) 4:1 (4) 4:4 (5) None of these

178. The percentage decrease in export from previous years was the maximum during which of the following years for Company X?

(1) 2008 (2) 2010 (3) 2009 (4) 2011 (5) None of these

179. What are the average exports of Company Y in all the years (in thousand tonnes)?

(1) 650 (2) 850 (3) 750 (4) 800 (5) None of these

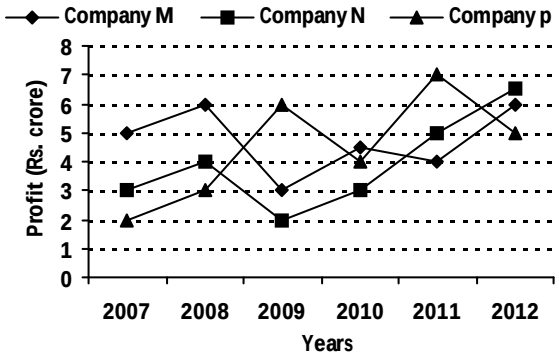
180. Total export of Company Z in all the years is approximately what per cent of the total export of Company Y in all the years?

(1) 66% (2) 82% (3) 78% (4) 76% (5) None of these

Directions: (Q. 181-185): Study the following information and answer the questions that follow:

The graph given below represents the profit (in lakh) of three companies M, N and P.

Profit = Income - Expenditure



181. In 2010, the profit of Company M is what percentage of the total profit of Company P and Company N together?

(1) 64.28% (2) 65.71% (3) 66.28% (4) 63.11% (5) 62.58%

182. If the expenditures of Company M and Company P in the year 2011 are `75 crore and `68 crore respectively, what is the ratio of the income of Company M to that of Company P?

(1) 74 : 71 (2) 81 : 79 (3) 82 : 75 (4) 79 : 75 (5) 79 : 71

183. What is the average income of all three companies in the year 2012, if the expenditure is 50%, 60% and 80% more than the profits of Company M, N and P respectively?

(1) `16.4 crore (2) `15.3 crore (3) `17.5 crore (4) `14.3 crore (5) `14.7 crore

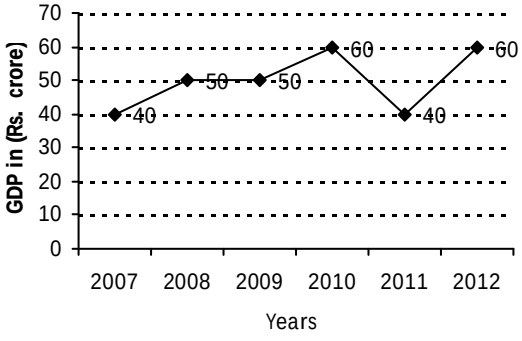
184. What is percentage increase in the profit of Company N from 2009 to 2012?

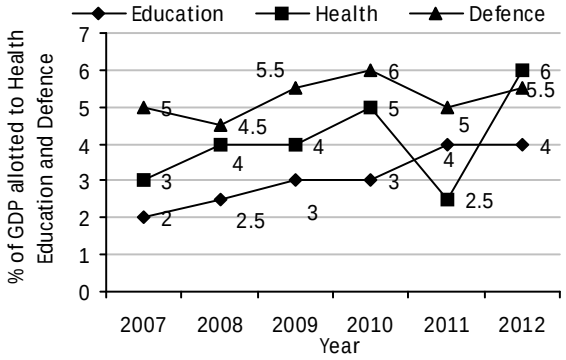
(1) 230% (2) 240% (3) 225% (4) 220% (5) 215%

185. In the year 2010, the income of Company P is `40 crore. If the income of Company M is 20% more than that of Company P in that year, what is the expenditure of Company M in the year 2010?

(1) `45.5 crore (2) `46.5 crore (3) `47.9 crore (4) `41.5 crore (5) `43.5 crore

Directions (Q. 186-191): Study the following line graph carefully and answer the questions given below. ?





186. In 2010, what is the ratio of amount spent on Defence to Education to Health?

(1) 3:5:6 (2) 4:5:6 (3) 3:4:6 (4) 4:3:6 (5) 3:2:5

187. The GDP growth from 2007 to 2008 is what per cent of the GD Pgrowth from 2011 to 2012?

(1) 42% (2) 44% (3) 46% (4) 48% (5) 50%

188. What is the total amount (in `) allotted to Defence during 2007-12?

(1) 17.5cr (2) 15.9cr (3) 16.8cr (4) 18.8cr (5) 19.4cr

189. In which of the following years is the total amount allotted to Education, Health and Defence the maximum?

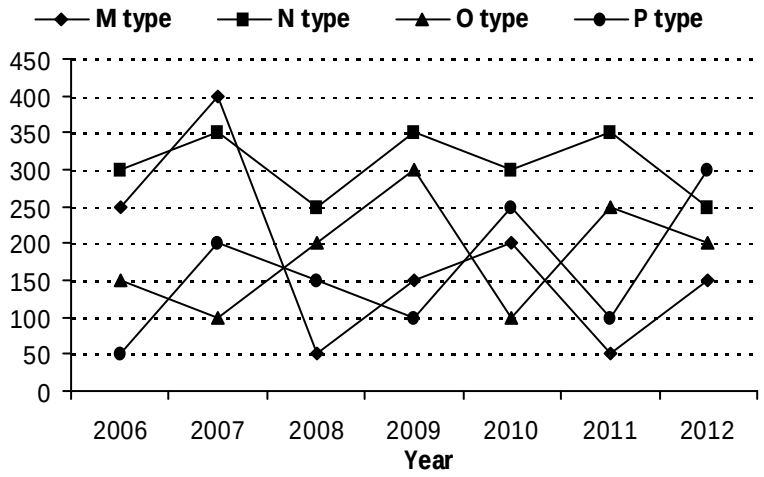
(1) 2012 (2) 2011 (3) 2010 (4) 2009 (5) 2008

190. What is the difference between the amount allotted to Education in 2009 and that in 2010?
(1) 34 lakh (2) 27 lakh (3) 32 lakh (4) 30 lakh (5) 28 lakh

191. Has the amount allotted to Education in 2010 remained the same in 2011 or increased or decreased? If it has increased or decreased, then by what per cent?
(1) Increased by 35.5% (2) Decreased by 33.3% (3) Increased by 37.7%
(4) Decreased by 31.1% (5) None of these

Directions (Q. 192-196): Answer the following questions based on the given graph:

In the line graph the prices (in `) of four types of tile M, N, O, P respectively are given for different years.



192. Which type of tiles shows the maximum percentage increase in the price over the given period?
(1) M (2) N (3) O (4) P (5) Both O and P

193. Which type of tiles shows the maximum average price over the years?
(1) M (2) N (3) O (4) P (5) Both M and N

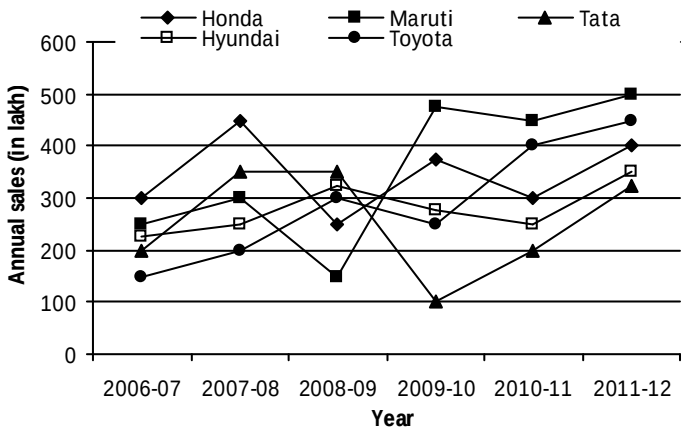
194. In which year is the average price of all four types of tiles the minimum?
(1) 2006 (2) 2008 (3) 2010 (4) 2011 (5) 2012

195. Total price of all four types of tiles in 2012 is what per cent more or less than the total price of all four types of tiles in 2009?
(1) 1% (2) 2% (3) 0% (4) 4% (5) 6%

196. What is the ratio of the price of tiles O in 2008 to that of tiles Pin 2010?
(1) 2:1 (2) 4:3 (3) 3:4 (4) 5:2 (5) 4:5

Directions (Q. 197-201): Study the line graph and answer the questions given below:

The graph shows sales of four-wheelers of different companies in India for FY 2006-07 to 2011-12.



197. What is the percentage increase in annual sales of all companies put together from FY 2006-07 to 2011 -12?
(1) 68% (2) 78.51% (3) 80% (4) 82.22% (5) 14.91%

198. Which company recorded the highest percentage increase in sale from FY 2006-07 to 2011 -12?
(1) Honda (2) Hyundai (3) Maruti (4) Toyota (5) Tata

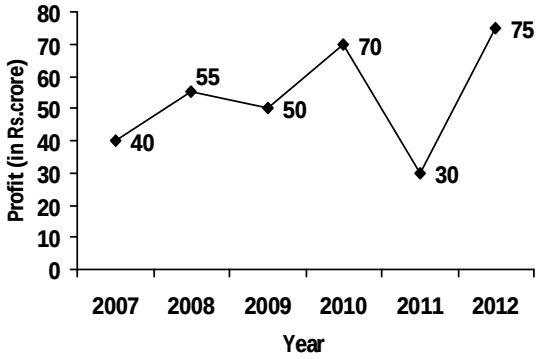
199. In which FY is the average sales of all the companies the minimum?
(1) 2007-08 (2) 2006-07 (3) 2010-11 (4) 2011-12 (5) 2008-09

200. The total sale of Hyundai and Maruti is what per cent more or less than the total sale of Tata and Honda in FY 2006-07?
(1) 4% less (2) 5% more (3) 5% less (4) 4% more (5) 2% less

201. The total sale of Honda is what per cent more than the total sale of Toyota for FY 2009-10?
(1) 71% (2) 70% (3) 49% (4) 50% (5) 25%

Directions (Q. 202-206): Study the following graph carefully to answer the questions that follow:

Profit earned by a company over the years



202. If the income of the company in the year 2010 was `120 crore, what was the percentage profit of the company in the year 2010?
(1) 100% (2) 120% (3) 133% (4) 125% (5) 140%

203. If the expenditure of the company in the year 2011 was `85 crore, what was the ratio of income to expenditure of the company in that year?
(1) 23:17 (2) 5:4 (3) 11:8 (4) 21:16 (5) None of these

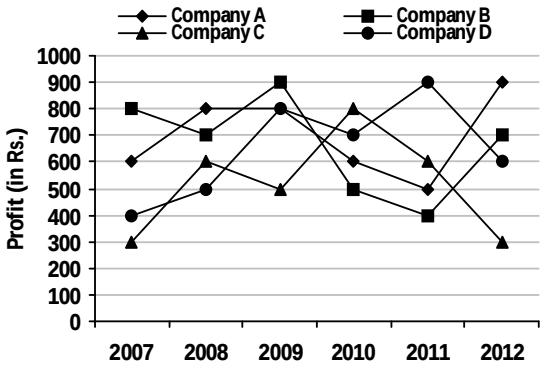
204. What is the approximate average profit (in ` crore) earned by the company over the years?
(1) 50 (2) 48 (3) 53 (4) 57 (5) 61

205. If the income of the company in the year 2007 was `950000000, what was the expenditure (in ?) of the company in that year?
(1) 50000000 (2) 550000000 (3) 40000000 (4) 350000000 (5) None of these

206. What is the percentage increase in the profit of the company in the year 2010 from the previous year?
(1) 43% (2) 46% (3) 50% (4) 40% (5) None of these

Directions (Q. 207-211): Study the following graph carefully to answer the given questions. The graph shows the profit of companies A, B, C and D in various years

$$\% \text{ Profit} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$



207. If the income of Company A in the year 2009 is `25000 and that in the year 2012 is `32000 then what is the average expenditure for the year 2009 and 2012?
(1) `29540 (2) `22790 (3) `27650 (4) `31320 (5) `19460

208. What is the ratio of the percentage profit of Company C in the year 2010 to that of Company B in the year 2012 if the income is `45000 and `65000 of Company C in the year 2010 and Company B in the year 2012 respectively?
(1) 8:7 (2) 5:3 (3) 13:12 (4) 2:7 (5) 2:3

209. If in the year 2009 incomes of both the companies A and B are the same ie `10000, what was the ratio of their expenditures in that year?
(1) 103:22 (2) 42:47 (3) 13:77 (4) 92:91 (5) 5:3

210. What is the percentage increase in profit of Company C in the year 2008 from the previous year?
(1) 12% (2) 105% (3) 92% (4) 89% (5) 100%

211. What is the ratio of the income of Company A to that of Company D in the year 2011, if their expenditures are `15000 and `22000 respectively?
(1) 155:229 (2) 3:5 (3) 16:19 (4) 239:331 (5) 65:189

SHORT ANSWER

1. (2)	2. (3)	3. (2)	4. (4)	5. (1)	6. (2)	7. (4)	8. (1)
9. (5)	10. (3)	11. (4)	12. (5)	13. (1)	14. (4)	15. (2)	16. (3)
17. (2)	18. (1)	19. (5)	20. (3)	21. (4)	22. (5)	23. (1)	24. (4)
25. (1)	26. (2)	27. (3)	28. (2)	29. (1)	30. (4)	31. (3)	32. (5)
33. (2)	34. (1)	35. (3)	36. (2)	37. (4)	38. (3)	39. (2)	40. (2)
41. (3)	42. (2)	43. (3)	44. (4)	45. (2)	46. (3)	47. (1)	48. (3)
49. (4)	50. (1)	51. (4)	52. (4)	53. (5)	54. (2)	55. (3)	56. (2)
57. (3)	58. (1)	59. (2)	60. (2)	61. (2)	62. (3)	63. (5)	64. (4)
65. (1)	66. (1)	67. (4)	68. (3)	69. (1)	70. (4)	71. (1)	72. (5)
73. (1)	74. (3)	75. (2)	76. (1)	77. (3)	78. (5)	79. (4)	80. (1)
81. (1)	82. (4)	83. (2)	84. (5)	85. (1)	86. (4)	87. (2)	88. (3)
89. (5)	90. (3)	91. (2)	92. (5)	93. (4)	94. (2)	95. (5)	96. (2)
97. (4)	98. (2)	99. (5)	100. (4)	101. (2)	102. (4)	103. (5)	104. (2)
105. (1)	106. (4)	107. (5)	108. (5)	109. (1)	110. (3)	111. (3)	112. (3)
113. (1)	114. (4)	115. (2)	116. (4)	117. (2)	118. (4)	119. (2)	120. (5)
121. (2)	122. (4)	123. (4)	124. (1)	125. (1)	126. (2)	127. (5)	128. (3)
129. (1)	130. (4)	131. (3)	132. (4)	133. (1)	134. (1)	135. (5)	136. (4)
137. (3)	138. (3)	139. (2)	140. (5)	141. (2)	142. (4)	143. (3)	144. (3)
145. (5)	146. (2)	147. (2)	148. (4)	149. (1)	150. (4)	151. (5)	152. (3)
153. (2)	154. (4)	155. (2)	156. (2)	157. (1)	158. (3)	159. (4)	160. (3)
161. (2)	162. (3)	163. (5)	164. (3)	165. (2)	166. (3)	167. (3)	168. (3)
169. (4)	170. (3)	171. (2)	172. (4)	173. (3)	174. (1)	175. (3)	176. (2)
177. (1)	178. (2)	179. (3)	180. (3)	181. (1)	182. (4)	183. (2)	184. (3)
185. (5)	186. (1)	187. (5)	188. (2)	189. (1)	190. (4)	191. (2)	192. (4)
193. (2)	194. (2)	195. (3)	196. (5)	197. (3)	198. (4)	199. (2)	200. (3)
201. (4)	202. (5)	203. (1)	204. (3)	205. (2)	206. (4)	207. (3)	208. (2)
209. (4)	210. (5)	211. (1)					

DETAIL - EXPLANATIONS

1. 2; % $\text{Profit}_{2008} = 48\%$,
Expenditure = 55.5 lakh

$\therefore$ Income = $55.5 \times \frac{(100+48)}{100}$

$= 55.5 \times 1.48 = 82.14$

2. 3; % $\text{Profit}_{2008} = 40\%$
% $\text{Profit}_{2009} = 45\%$

$\therefore$ % Rise = $\frac{(45-40)}{40} \times 100 = \frac{500}{40} = 12.5\%$

3. 2; % profit of A in 2006 and % profit of C in 2010 are equal and are 40%,

$\therefore$ Total income = $94 \times \frac{(100+40)}{100}$

$= 94 \times 1.4 = 131.6$ lakh

4. 4; % $\text{Profit}_A = 40\%$
and Income_A = 91 lakh

$\therefore \text{Expenditure}_A = 91 \times \frac{100}{140} = 65$ lakh

% $\text{Profit}_B = 50\%$, Expenditure_B = 91 lakh

$\therefore \text{Income}_B = 91 \times \frac{150}{100} = 136.6$ lakh

$\therefore$ Diff = 136.5 - 65 = 71.5 lakh

5. 1; Let $\text{Expenditure}_B = \text{Income}_c = x$

$\therefore \text{Income}_B = x \times \frac{(100+60)}{100} = \frac{8x}{5}$

$\text{Expenditure}_c = x \times \frac{100}{100+25} = \frac{4x}{5}$

$\therefore$ Reqd ratio = $\frac{\text{Income}_B}{\text{Expenditure}_c}$

$= \frac{8x}{5} \times \frac{5}{4x} = \frac{2}{1}$

6. 2; P_{2008} = 4.8 lakh, P_{2010} = 6.4 lakh
$\therefore$ Reqd % = $\frac{6.4-4.8}{4.8} \times 100 = \frac{160}{4.8} \approx 33.33\%$

7. 4; $\text{Literate}_{A\text{-}2008} = 3.6 \times \frac{57.8}{100} = 2.0808$ lakh

$\text{Literate}_{A\text{-}2010} = 5.4 \times \frac{62.3}{100} = 3.3642$ lakh

$\therefore$ Total = 2.0808 + 3.3642 = 5.445 lakh

8. 1; Illiterate_G - Illiterate_F

$= 7.5 \times \frac{(100-68.9)}{100} - 6.4 \times \frac{(100-65.8)}{100}$

Diff = 2.3325 - 2.1888 = 0.1437 lakh

9. 5; $E_{2008} = 5.5 \times \frac{67.7}{100} = 3.7235$ lakh

$E_{2010} = 7.2 \times \frac{71}{100} = 5.112$ lakh

$\therefore$ Reqd % = $\frac{5.112-3.7235}{3.7235} \times 100$

$= 37.29 \approx 37.3\%$

10. 3; Total population = 5.2 lakh
Percentage of literates = 64.5%
$\therefore$ Percentage of illiterates
= 100 - 64.5 = 35.5%
$\therefore$ Diff = 64.5 - 35.5 = 29%

$\therefore$ Reqd answer = $5.2 \times \frac{29}{100} = 1.508$ lakh

11. 4; $E_A = 52.49 \times \frac{100}{145} = 36.2$ lakh

$E_B = 61.2 \times \frac{100}{136} = 45$ lakh

$\therefore$ Total expenditure = 36.2 + 45 = 81.2 lakh

12. 5; $I_A = 48.5 \times \frac{132}{100} = 64.02$ lakh

$E_B = 75.04 \times \frac{100}{140} = 53.61$akh

Diff = 64.02 - 53.6 = 10.42 lakh

13. 1; Since their profit % is same, ie 40%, total expenditure = $133 \times \frac{100}{140} = 95$ 1akh

14. 4; $\%P_A = 25\%$ and $\%P_B = 50\%$
Let $E_{A\text{-}20O6} = I_{B\text{-}2008} = x$

$\therefore I_A = x \times \frac{125}{100} = \frac{5x}{4}$, $\therefore E_B = x \times \frac{100}{150} = \frac{2x}{3}$

$\therefore \frac{E_B}{I_A} = \frac{2x}{3} \times \frac{4}{5x} = \frac{8}{15}$

15. 2; $\%P_{A\text{-}2009} = 60\%$ and $\%P_{B\text{-}2005} = 48\%$
Let $E_A = E_B = x$

$\therefore I_A = x \times \frac{160}{100}$ and $I_B = x \times \frac{148}{100}$

$\therefore$ Reqd % = $\frac{\frac{148x}{100}}{\frac{160x}{100}} \times 100 = \frac{14800}{160} = 92.5\%$

16. 3; Total pens produced
= 40 + 55 + 50 + 90 + 75 = 310 thousand

Avg production = $\frac{310}{5}$ = 62 thousand

Total pens sold = 30 + 40 + 25 + 60 + 50
= 205 thousand

Avg of pens sold = $\frac{205}{5}$ = 41 thousand

Reqd % = $\frac{41}{62} \times 100 = 66.129 = 66\%$

17. 2; Cost price per pen = 4.50
Selling price per pen = 8
$\therefore$ Profit per pen = 3.50
Total number of pens sold = 30000
$\therefore$ Net profit = 30000 × 3.50 = 1.05 lakh

18. 1; Total amount = 8 × 30000 + 10 × 40000 + 22 × 25000 + 25 × 60000 + 40 × 50000
= 240000 + 400000 + 550000 + 1500000
= 46.9 lakh

19. 5; Manufacturing cost of type C = 15
Selling price of type C = 22
$\therefore$ Profit per pen = 7
$\therefore$ Total profit of type C = 25000 × 7
= 175000
Similarly,
Total profit of type D = 60000 × 10 = 600000
$\therefore$ Total profit = 7.75 lakh

20. 3; $Profit_B$ = 40000 (10 - 5.5) = , 180000
$Profit_E$ = 50000 (40 - 25) = 750000

Reqd % = $\frac{180000}{750000} \times 100 = 24\%$

21. 4; $\frac{I_B}{E_B} = 1.05$,

$\frac{I_A}{E_A} = 0.75$

$\therefore$ Reqd% = $\frac{1.05}{0.75} \times 100$

$= \frac{105}{0.75} = 140\%$

22. 5; Exports of B in year 2008 cannot be determined.

23. 1; $\because$ The ratio of imports to exports is the same for Company A in year 2007 and Company B in year 2004 the sum of their imports will be
$(I_A + I_B) = 0.8 \times (E_A + E_B) = 0.8 \times 180 = 144$ lakh

24. 4; $\frac{I_A}{E_A} = 0.75$

$\therefore I_A = 0.75 \times E_A$
= 0.75 × 120 = 90 lakh

$\frac{I_B}{E_B} = 0.6$

$\therefore E_B = \frac{I_B}{0.6} = \frac{120}{0.6}$
= 200 lakh
$\therefore$ Diff = 200 - 90 = 110 lakh

25. 1; $\frac{I_A}{E_A} = 0.5$

$E_A = \frac{I_A}{0.5} = \frac{80}{0.5} = 160$ lakh

$\frac{I_B}{E_B} = 1.2$

$\therefore I_B = 1.2 \times 60 = 72$ lakh

$\therefore$ Reqd% = $\frac{72}{160} \times 100 = 45\%$

26. 2; P_{1990} = 12 lakh,

$P_{2010} = 12 \times \frac{(100+75)}{100} \times \frac{(100+55)}{100}$

P_{2010} = 12 × 1.75 × 1.55 = 32.55 lakh

27. 3; Suppose the population in year 2000 was x.
$\therefore$ Its population in year 2010

$= x \times \frac{160}{100} = \frac{8x}{5}$

$\therefore$ Reqd % = $\frac{x}{(8x/5)} \times 100$

$= x \times \frac{5}{8x} \times 100 = 62.5\%$

28. 2; A_{2010} = 37.7 lakh

$A_{1990} = 37.7 \times \frac{100}{145} \times \frac{100}{130} = 20$ lakh

$\therefore B_{1990} = A_{1990} = 20$ lakh

$\therefore B_{2010} = 20 \times \frac{140}{100} \times \frac{150}{100} = 42$ lakh

29. 1; $C_{2010}0 = 27.2 = D_{2000}$

$\therefore C_{1990} = 27.2 \times \frac{100}{170} \times \frac{100}{160} = 10$ lakh

$D_{1990} = 27.2 \times \frac{100}{136} = 20$ lakh

$\therefore$ Reqd% = $\frac{10}{20} \times 100 = 50\%$

30. 4; Let the population of E in 1990 be 100.

$\therefore E_{2010} = 100 \times \frac{150}{100} \times \frac{140}{100} = 210$

$\therefore$ Reqd fraction = $\frac{100}{210} = \frac{10}{21}$

31. 3; Reqd % = $\frac{60-45}{45} \times 100 = \frac{1500}{45} = \frac{100}{3}$

$= 33\frac{1}{3}\%$

32. 5; Let the incomes of A and B each be x in the year 2007.

$\therefore E_A = \frac{x \times 100}{100+60} = \frac{5x}{8}$, $E_B = \frac{x \times 100}{100+50} = \frac{2x}{3}$

$\therefore$ Ratio = $\frac{5x}{8} \times \frac{3}{2x} = \frac{15}{16}$

33. 2; I_A = 90 lakh

$\therefore E_A = \frac{90 \times 100}{100+25} = 72$ lakh

E_B = 90 lakh

$\therefore I_B = 90 \times \frac{100+20}{100} = 108$ lakh

$\therefore$ Diff = 108 - 72 = 36 lakh

34. 1; I_A = 98

$\therefore E_A = 98 \times \frac{100}{100+40} = 70$ 1akh

$E_B = 85$

$\therefore I_B = 85 \times \frac{100+40}{100} = 119$ lakh

$\therefore$ Sum = 70 + 119 = 189 lakh

35. 3; Let the expenditure of B be x.

$\therefore$ Income = $x \times \frac{160}{100} = \frac{8x}{5}$

$\therefore$ Reqd % = $\frac{x}{8x/5} \times 100$

$= \frac{100x \times 5}{8x} = 62.5\%$

36. 2; $\because \frac{I}{E} = 0.6$

$= \frac{39.72}{0.6} = 66.2$ crore

37. 4; $\frac{I_A}{E_A} = 0.5$

$\therefore I_A = 0.5 \times 96.4 = 48.2$ crore

$\frac{I_B}{E_B} = 0.9$

$\therefore I_B = 0.9 \times 96.4 = 86.76$ crore

$\therefore$ Diff = 86.76 - 48.2 = 38.56 crore

38. 3; $\frac{I_A}{E_A} = 0.75$,

$E_A = \frac{I_A}{0.75} = 84.8$ crore

$\frac{I_B}{E_B} = 0.8$,

$E_B = \frac{I_B}{0.8} = \frac{62.8}{0.8} = 78.5$ crore

$\therefore$ Sum = 84.8 + 78.5 = 163.3 crore

39. 2; $\frac{I_B}{E_B} = 0.55$, $\frac{I_A}{E_A} = \frac{0.4}{}$

$\therefore$ Reqd % = $\frac{0.55}{0.4} \times 100 = 137.5\%$

40. 2; $\frac{I}{E} = 0.8 = \frac{4}{5}$

$I_1 = 4 + \frac{25}{100} \times 4 = 5$

$E_1 = 5 \pm 5 \times \frac{50}{100} = 2.5$

$\therefore$ Ratio = $\frac{I_1}{E_1} = \frac{5}{2.5} = 2.0$

41. 3; In 2006, let the expenditure be x. So, its income will be $x \times \frac{100+60}{100} = \frac{8x}{5}$

$\therefore$ Reqd% = $\frac{x}{(8x/5)} \times 100 = x \times \frac{5}{8x} \times 100$

= $\frac{500}{8}$ = 62.5%

42. 2; Since, percentage profit is same for A in 2008 and B in 2004,

$\therefore$ Sum of income = 175 × $\frac{140}{100}$ = 245 lakh

43. 3; Let $E_A = E_B = x$

$\because$ $\%P_A = 60\%$ and $\%P_B = 40\%$

$\therefore I_A = x \times \frac{160}{100} = \frac{8x}{5}$, $I_B = x \times \frac{140}{100} = \frac{7x}{5}$

$\therefore$ Reqd % = $\frac{7x}{5} \times \frac{5}{8x} \times 100 = 87.5\%$

44. 4; $E_A = I_B$ = 116 1akh
$\%P_A = 45\%$, $\%P_B = 45\%$

$\therefore I_A = 116 \times \frac{145}{100} = 168.2$ lakh

$E_B = 116 \times \frac{100}{145} = 80$ lakh

$\therefore$ Diff = 168.2 - 80 = 88.2 lakh

45. 2; I_A = 112 lakh $\%P_A = 60\%$

$\therefore E_A = 112 \times \frac{100}{160} = 70$ lakh

E_B = 56 lakh, $\%P_B = 75\%$ $\therefore I_B = 56 \times \frac{175}{100} = 98$ lakh

$\therefore$ Ratio = $\frac{70}{98} = \frac{5}{7}$

46. 3; Population = $8.5 \times \frac{110}{100} \times \frac{115}{100} \times \frac{120}{100}$

= 12.903 lakh

47. 1; Population$_{1970}$

$= 2087250 \times \frac{100}{115} \times \frac{100}{125} \times \frac{100}{132} = 11$ lakh

48. 3; Population-A_{1970}

$= 1388800 \times \frac{100}{112} \times \frac{100}{124} \times \frac{100}{125} = 8$ lakh

Population-B_{1970}

$= 1302912 \times \frac{100}{120} \times \frac{100}{116} \times \frac{100}{130} = 7.2$ lakh

Reqd percentage = $\frac{7.2}{8}$ × 100 = 90%

49. 4; $E_{2000} = 12.5 \times \frac{125}{100} \times \frac{116}{100} \times \frac{140}{100}$

= 25.375 lakh

$F_{2000} = 10 \times \frac{121}{100} \times \frac{125}{100} \times \frac{136}{100} = 20.57$ lakh

$\therefore$ Difference = 25.375 – 20.57 = 4.805 lakh

50. 1; Let the population of City C and City D be x at the beginning of 1970.

$\therefore$ Population-$C_{1990} = x \times \frac{110}{100} \times \frac{115}{100}$

Population-$D_{1990} = x \times \frac{115}{100} \times \frac{125}{100}$

$\therefore$ Ratio = $\frac{110}{125} = \frac{22}{25}$

51. 4; 2002, 2003, 2005, 2006, 2007.

52. 4; $(1 : E)_B = 0.5$ and $(I : E)_A = 0.8$

$\therefore$ Reqd% = $\frac{0.5}{0.8}$ × 100 = 62.5%

53. 5; Data given are not sufficient.

54. 2; I_A = 40 + 1.2 = 48 lakh
$I_B = 0.9 \times 40$ = 36 lakh

$\therefore$ Reqd% = $\frac{36}{48}$ × 100 = 75%

55. 3; $\frac{I_B}{E_B} = 0.8$,

$\therefore E_B = \frac{I_B}{0.8} = \frac{78}{0.8} = 97.5$ lakh

$\therefore$ Difference = 97.5 - 78 = 19.5 lakh

56. 2; Difference = 4870 - 4640 = 230

57. 3; Number of girls = 300+ 450 = 750
Number of boys = 720 + 600 = 1320

$\therefore$ Regd % $= \frac{750}{1320} \times 100 = 56.8\%$

58. 1; Girls 2003 = 560
Girls 2004 = 800

$\therefore \% = \frac{800 - 560}{560} \times 100 = \frac{24000}{560} = 42.8\%$

59. 2; % rise = $\frac{600 - 400}{400} \times 100 = 50\%$

60. 2; Girls 2007 = 640

Girls avg during whole period = $\frac{4640}{8} = 580$

$\therefore$ Reqd % = $\frac{(640 - 580)}{580} \times 100 \approx 10.34\%$

61. 2; $\frac{B}{G} = 1.6$

$\therefore G = \frac{B}{1.6} = \frac{128}{1.6} = 80$

$\therefore$ Diff = 128 - 80 = 48

62. 3: Reqd % = $\frac{1}{1.6} \times 100 = 62.5\%$

63. 5; Data is not sufficient to find the exact difference.

64. 4; Let $G_A = G_B = x$

$\because \frac{B_A}{G_A} = 0.8$

$\therefore B_A = 0.8x$

$\therefore \frac{B_B}{G_B} = 1.3$

$\therefore B_B = 1.3x$

$\therefore$ Reqd % = $\frac{1.3x}{0.8x} \times 100 = 162.5$

65. 1; $\frac{B_B}{G_B} = 1.5$

$\therefore B_B = 1.5 \times 70 = 105$, $B_A = 1.3 \times 70 = 91$
$B_B - B_A = 105 - 91 = 14$
and $G_A + G_B = 70 + 70 = 140$

$\therefore$ Reqd % = $\frac{14}{140} \times 100 = 10\%$

66. 1; $P_A = 40\%$, $P_B = \frac{48.6 - 36}{36} \times 100$
= 35%
$\therefore$ Difference = 40 - 35 = 5%

67. 4: $I_A = 32.5$, $\%P_A = 25\%$

$\therefore E_A = \frac{32.5}{100 + 25} = 26$ lakh

$\therefore P_A = 32.5 - 26 = 6.5$ lakh
$P_B = 35 - 25 = 10$ lakh
Net profit of A and B = 10 + 6.5 = 16.5 lakh

68. 3; 2009; % profit = $\frac{77 - 44}{44} \times 100 = 75\%$

69. 1; $E_A = 45$ 1akh

$\therefore I_A = 45 \times \frac{110}{100} = 49.5$ lakh

$\therefore P_A = 4.5$ lakh and $P_B = 80 - 50 = 30$ lakh

$\therefore \% = \frac{4.5}{30} \times 100 = 15\%$

70. 4; $I_A = 90$ lakh, $E_A = 90 \times \frac{100}{120} = 75$ lakh

$\therefore P_A = 15$ lakh, $P_B = 72 - 45 = 27$ lakh

$\therefore$ Reqd % = $\frac{27 - 15}{15} \times 100 = \frac{1200}{15} = 80\%$

71. 1; Income = $17 \times \frac{(100 + 35)}{100}$
= 17 × 1.35 = 22.95 lakh

72. 5; Data is not sufficient.

73. 1; As the per cent profit of B is same in both the years, the total income is

$48 \times \frac{100 + 30}{100} = 62.4$ lakh

74. 3; The ratio of income to expenditure is maximum when the percentage profit is maxi-mum. Hence in year 2006.

75. 2; $\text{Income}_{B\text{-}2009} = 77$

$\therefore \text{Expenditure}_{B\text{-}2009} = \frac{77 \times 100}{100 + 40} = 55$ lakh

$\therefore \text{Expenditure}_{A\text{-}2004} = 55$ lakh

$\therefore$ Hence $\text{income}_{A\text{-}2004} = \frac{55 \times (100 + 20)}{100}$
= 66 lakh

76. 1; $\frac{I}{E} = 0.6$

$\therefore E = \frac{I}{0.6} = \frac{67.2}{0.6} = 112$ lakh

77. 3; Import of B can't be determined because no relationship between Company A and B is given.

78. 5; $\frac{I_A}{E_A} = 0.5$

$\therefore I_A = 0.5 \times E_A = 0.5 \times 116 = 58$ lakh

$\frac{I_B}{E_B} = 1.2$

$\Rightarrow \frac{1170}{12} = E_B$

$\Rightarrow E_B = 97.5$ lakh

$\therefore$ Sum = 58 + 97.5 = 155.5 lakh

79. 4; $\frac{I}{E} = 1.2$

$\Rightarrow I_1 = I - \frac{25I}{100} = \frac{25I}{100} = \frac{75I}{100}$

$\Rightarrow E_1 = E - E \times \frac{50}{100} = \frac{50E}{100}$

$\therefore \frac{I_I}{E_I} = \frac{75I}{100} \times \frac{100}{50E} = \frac{3}{2} \times \frac{I}{E} = \frac{3}{2} \times 1.2 = 1.8$

80. 1; $\frac{I_A}{E_A} = 1.2$

$\therefore E_A = \frac{I_A}{1.2} = \frac{102.6}{1.2} = 85.5$ lakh $\Rightarrow \frac{I_B}{E_B} = 0.4$

$\therefore I_B = 0.4 \times E_B = 0.4 \times 112.5 = 45$ lakh

$\therefore$ Reqd % = $\frac{85.5}{45} \times 100 = 190\%$

81. 1; $\frac{E_A}{I_A} = 0.25$

$\therefore E_4 = 0.25 \times 96.8 = 24.2$ lakh

82. 4; $\frac{E_{B2004}}{I_{A2004}} = 0.55$

$\frac{E_{B2002}}{I_{A2002}} = 0.4$

$\therefore$ Reqd % = $\frac{0.55}{0.4} \times 100 = 137.5\%$

83. 2; $\frac{E_A}{I_A} = 0.8$

$\therefore E_A = 0.8 \times I_A = 0.8 \times 86 = 68.8$ lakh

$\frac{E_B}{I_B} = 0.6$

$\therefore I_B = \frac{E_B}{0.6} = \frac{51}{0.6} = 85$ lakh

$\therefore$ Sum = 85 + 68.8 = 153.8 lakh

84. 5; $\frac{E}{I} = 0.4$

Let the new export be E_1 and import be I_1 Then,

$E_1 = \frac{E + E \times 125}{100} = \frac{225E}{100}$

$I_1 = \frac{I - I \times 60}{100} = \frac{401}{100}$

New ratio = $\frac{E_1}{I_1} = \frac{225E}{100} \times \frac{100}{401}$

$= \frac{225}{40} \times \frac{E}{I} = \frac{225}{40} \times 0.4 = 9:4$

85. 1; $\frac{E_A}{I_A} = 0.3$

$\therefore I_A = \frac{E_A}{0.3} = \frac{23.4}{0.3} = 78$ lakh

$\frac{E_B}{I_B} = 0.75$

$\therefore I_B = \frac{E_B}{0.75} = \frac{72}{0.75} = 96$ lakh

$\therefore$ Reqd % = $\frac{78}{96} \times 100 = 81.25\%$

86. 4; Production of Company A in year 2009 = 550
Production of Company A in year 2010 = 700

Reqd % = $\frac{700 - 550}{550} \times 100 = \frac{150}{550} \times 100$

$= \frac{300}{100} = 27.27 \approx 27\%$

87. 2; Sales of Company A in year 2009 = 400
Production of Company A in year 2009 = 550

Reqd % = $= \frac{400}{550} \times 100 = \frac{800}{11} = 72.72 \approx 73\%$

88. 3; Average production of Company B

$= \frac{600 + 700 + 800 + 600 + 650 + 700}{6}$

$= \frac{4050}{6} = 675$

89. 5; Reqd ratio

$= \frac{\text{Total Production of Company A}}{\text{Total Sales of Company A}}$

$= \frac{4050}{2750} = \frac{81}{55} = 81:55$

90. 3; Production of Company B in the year 2006.
= 150 × 4 = 600
Production of Company B **in** the year 2008
= 200 × 4 = 800

$$\text{Ratio} = \frac{600}{800} = 3:4$$

91. 2; $\text{Income} = \text{Expenditure} \times \frac{100 + \%\text{Profit}}{100}$

$\text{Expenditure} = \text{Income} \times \frac{100}{100 + \%\text{Profit}} = 55.8 \times \frac{100}{100 + 24}$

$\text{Expenditure} = 55.8 \times \frac{100}{124} = 45 \text{ crore}$

92. 5; For $\frac{\text{Income}}{\text{Expenditure}}$ to be the minimum the % profit should be the minimum.

Hence, in the year 2010, $\frac{\text{Income}}{\text{Expenditure}}$ is the minimum.

93. 4; Since % profit is the same, the total income will be =

$\text{total expenditure} \times \frac{100 + \%P}{100}$

$\therefore$ Total Income = $148 \times \frac{130}{100} = 192.4$ crore

94. 2; Profit of Company A in the year 2005 = 25%
Income of company A in the year 2005 = 56 crore
Profit of company B in the 2009 year = 45%
Expenditure of Company B in the year 2009 = 56 crore

$\therefore \quad E_A = 56 \times \frac{100}{100 + 25} = 44.8$

$\therefore \quad I_B = 56 \times \frac{100 + 45}{100} = 81.2$

$\therefore$ Total = 44.8 + 81.2 = 126 crore

95. 5; Data are not sufficient.
We can find the total expenditure of A and B together in the year 2008 but we can't find their individual exenditures.

96. 2; Income of Company A in 2007

$I = E \times \frac{(100 + P)}{100}$

$\text{or } E = \frac{100 \times I}{(100 + P)} = \frac{85.8 \times 100}{(100 + 32)}$

$= \frac{8580}{132} = 65 \text{ lakh}$

97. 4; Company A's income in 2012

$= \text{Expenditure} \times \frac{(\%\text{ Profit} + 100)}{100}$

$\therefore 1 = 90.6 \times \frac{155}{100} = 140.43$ lakh

98. 2; Company B's percentage profits in different years are as follows

% Profit in 2007 → $\frac{32 - 25}{25} \times 100 = 28\%$

% Profit in 2009 → $\frac{45 - 30}{30} \times 100 = 50\%$

% Profit in 2010 → $\frac{50 - 45}{45} \times 100 = $ II.II% 45

% Profit in 2011 → $\frac{60 - 50}{50} \times 100 = 20\%$

99. 5; We can't find the exact value of the net profit from the given data.

100. 4; $E_A = I_a = 84$ lakhs
Percentage profit of Company A = 30%
Percentage profit of Company B = 50%

$I_A = E_A \times \frac{100 + P_A}{100} = 84 \times \frac{130}{100} = 109.2 \text{ lakh}$

$E_B = I_B \times \frac{100}{(100 + P_B)} = 84 \times \frac{100}{150} = 56 \text{ lakh}$

$\therefore$ Difference = 109.2 - 56 = 53.2 lakhs

101. 2; % profit = 35%

$\text{Expenditure} = \text{Income} \times \frac{100}{100 + \%P}$

Thus, $91.8 \times \frac{100}{135}$ = 68 lakh

102. 4; $\frac{E_1}{E_2} = \frac{6}{5}$ So, $E_1 = 6$, $E_2 = 5$

Now,

$I_1 = E_1 \times \frac{100 + 30}{100} = E_1 \times 1.3$

$I_2 = E_2 \times 1.2$

$\frac{I_1}{I_2} = \frac{E_1}{E_2} \times \frac{1.3}{1.2} = \frac{6 \times 1.3}{5 \times 1.2} = \frac{78}{60}$

$I_1 : I_2 = \frac{13}{10} = 13:10$

103. 5

104. 2; $\%\ P_A = 20\%$

$\text{Expenditure}_A = \frac{I}{1.2} = \frac{90}{1.2} = 75 \text{ lakhs}$

$\%\ P_B = 35\%$
$\text{Income}_B = 90 \times 1.35 = 135$ lakhs

$\text{Ratio} = \frac{135}{75} = \frac{9}{5}$

105. 1; Let the expenditure be x.

$\text{Income} = x \times \frac{100 + 25}{100} = 1.25x$

$\therefore \% = \frac{x}{1.25x} \times 100 = \frac{100}{1.25} = 80\%$

106. 4: $\frac{\text{Export of Company A}}{\text{Import of Company A}} = .6$

$\therefore$ Import of Company A $= \frac{51}{0.6} = 85$ lakh

$\frac{\text{Export of Company B}}{\text{Import of Company B}} = 0.8$

$\therefore$ Import of Company B $= \frac{54}{0.8} = 67.5$ lakh

$\therefore$ Difference = 85 – 67.5 = 17.5 lakh

107. 5; $\frac{\text{Export of Company A}}{\text{Import of Company A}} = 1.5$

$\therefore$ Export of Company A = 64 × 1.5 = 96 lakh

$\frac{\text{Export of Company C}}{\text{Import of Company C}} = 0.5$

$\therefore$ Import of Company C $= \frac{48}{0.5} = 96$ lakh

$\therefore$ Ratio = $\frac{96}{96} = \frac{1}{1}$

108. 5; $\frac{\text{Export of Company A}}{\text{Import of Company A}} = 1.2$

$\therefore$ Export of Company A = 1.2 × 55 = 66 lakh

$\frac{\text{Export of Company B}}{\text{Import of Company B}} = 0.8$

$\therefore$ Export of Company B = 55 × 0.8 = 44 lakh

$\therefore$ Reqd % = $\frac{66}{44}$ × 100 = 150%

109. 1; $\frac{\text{Export of Company B}}{\text{Import of Company B}} = 0.7$

$\therefore$ Import of Company B $= \frac{58.8}{0.7} = 84$ lakh

$\frac{\text{Export of Company C}}{\text{Import of Company C}} = 0.7$

$\therefore$ Import of Company C $= \frac{56.7}{0.7} = 81$ lakh

110. 3; $\frac{E}{I} = 0.75 = \frac{3}{4}$

$E_1 = E + \frac{200 \times E}{100} = 3E$

$I_1 = I + I + \frac{501}{100} = \frac{31}{2}$

$\therefore \frac{E_1}{I_1} = \frac{3E}{1} \times \frac{2}{31} = 2 \times \frac{E}{I} = 2 \times \frac{3}{4} = \frac{3}{2}$

111. 3; Total runs socred by India and Australia in Match 4 together = 220 + 190 = 410
Total runs scored by England in all the five matches togeather
= 160 + 180 + 230 + 270 + 300 = 1140

$\therefore$ Reqd % = $\frac{410}{114} \times 100 = 35.96 \approx 36\%$

112. 3; Difference between Australia and England in
Match 1 → 260 - 160 = 100
Match 2 → 330 - 180 = 150
Match 3 → 310 - 230 = 80
Match 4 → 270 - 220 = 50
Match 5 → 300 - 150 = 150
The second lowest difference of runs scored was in Match 3.

113. 1; Total runs scored by India and England in
Match 1 → 160 + 320 = 480
Match 2 → 180 + 240 = 420
Match 3 → 230 + 270 = 500
Match 4 → 270 + 190 = 460
Match 5 → 300 + 220 = 520
Hence the third highest / lowest was scored in Match 1.

114. 4; India scored in Match 5 = 220
England scored in Match 2 = 180
Australia scored in Match 1 = 260
$\therefore$ Ratio of India : Australia : England
220 : 260 : 180
11 : 13 : 9

115. 2; Average

$= \frac{230 + 370 + 310}{3} = \frac{810}{3} = 270$

116. 4; $\frac{I_A}{E_A} = 0.7$

or, $E_A = \frac{I_A}{0.7} = \frac{53.9}{0.7} = 77$ lakh

117. 2; $\frac{I_A}{E_A} = 0.8$

$\frac{I_B}{E_B} = 0.9$

$\therefore$ Reqd % = $\frac{0.9 - 0.8}{0.8} \times 100 = \frac{100}{8} = 12.5\%$

118. 4; $\frac{I_A}{E_A} = 0.75$ ...(I)

$I_A = I_A + \frac{I_A \times \frac{100}{3}}{100}$

$= I_A + \frac{I_A}{3} = \frac{41_A}{3}$

$E_{A1} = E_A - E_A \times \frac{20}{100} = \frac{80E_A}{100} = \frac{4}{5}E_A$

New ratio = $\frac{I_{A1}}{E_{A1}} = \frac{41_A}{3} \times \frac{5}{4E_A} = \frac{5}{3} \times \frac{I_A}{E_A}$

$= \frac{5}{3} \times 0.75 = 1.25$

119. 2; $\frac{I_A}{E_A} = 0.8$

$E_A = \frac{I_A}{0.8} = \frac{36}{0.8} = 45$ lakh

$I_B = E_B \times 0.9 = 60 \times 0.9 = 54$ lakh

$\therefore$ Reqd% = $\frac{54}{45} \times 100 = 120\%$

120. 5; $E_B < I_B$

$\therefore \frac{I_B}{E_B} > 1.0$

In year 2008 $\frac{I_B}{E_B} = 1.4$ ie > 1.0

121. 2; Expenditure$_B$

= Income$_B$ $\times \frac{100}{100 + \%\text{profit}}$

$= 136 \times \frac{100}{170} =$ Rs 80 lakh

$\therefore$ Profit of Company B

= 136 - 80 = `56 lakh

122. 4; Income of Company A in 2005 + Income of Company A in 2009

= `171.50 lakh

Expenditure of Company A in 2005 + Expenditure of Company A in 2009)

$= \frac{171.5 \times 100}{140} =$ Rs 122.5 lakh

{% profit is the same in year 2005 and 2009}

Total profit = 171.50 - 122.50 = `49 lakh

123. 4; % P_A = 75% and % P_B = 50%

$I_A = E_B$

$\therefore E_A = I_A \times \frac{100}{175}$ and $I_B = E_B \times \frac{150}{100}$

$\frac{E_A}{I_B} = \frac{100 \times I_A}{175} \times \frac{100}{150 \times E_B} = \frac{100 \times 100}{175 \times 150} = \frac{16}{42}$

= 16 : 42

124. 1; $E_A = I_B =$ Rs 90 lakh

$I_A = 90 \times \frac{140}{100} =$ Rs 126 lakh

$P_A = 90 \times \frac{40}{100} =$ Rs 36 lakh

I_B = 90 lakh

$\therefore E_B = \frac{90 \times 100}{180} =$ Rs 50 lakh

P_B = 90 - 50 = Rs 40 lakh

Reqd % = $\frac{36}{40} \times 100 = 90\%$

125. 1; $2005 \rightarrow \frac{40 - 25}{25} \times 100 = 60\%$

$2006 \rightarrow \frac{55 - 40}{40} \times 100 = 37.5\%$

$2009 \rightarrow \frac{40 - 30}{30} \times 100 = 33.33\%$

$2010 \rightarrow \frac{60 - 40}{40} \times 100 = 50\%$

$2011 \rightarrow \frac{75 - 60}{60} \times 100 = 25\%$

126. 2; Let Manav invest Rs x in Company B. Therefore, in Company A his investment would be Rs (40000 - x).

13.5% of x + 13% of (40000 – x) = 5299

or, $\frac{13.5}{100}x + \frac{13}{100} \times 40000 - \frac{13}{100}x = 5299$

or, $\frac{(13.5x - 13x)}{100} + 5200 = 5299$

or, $\frac{0.5x}{100} = 5299 - 5200 = 99$

$\therefore x = \frac{9900}{0.5} = \frac{99000}{5} =$ Rs19800

Therefore, Manav's investment in Company B is Rs 19800.

127. 5; Priya's amount in 2010 becomes

$50000 \times \frac{121.5}{100} = 60750$

Priya's amount in 2010 (when she invests Rs 60750 in Company B)

$= 60750 \times \frac{122}{100} =$ Rs 74115

128. 3; Total dividend

$= 37000 \times \frac{120}{100} + 19 \times \frac{37000}{100}$

$= 37000 \times \frac{39}{100} = 370 \times 39 =$ Rs14430

129. 1; Reqd ratio = $\frac{15.5 \times 7}{14 \times 9} = \frac{15.5}{18} = 31 : 36$

130. 4; Sukriti would have gained (22 - 20.5%) = 1.5% of investments. Therefore, she would have received 75000 $\times \frac{1.5}{100} =$ Rs1125

Hence, sukriti would have got Rs 1125 more.

131. 3; $I_{A2009} = Ex + \frac{Ex \times \%Profit}{100}$

$= 77.5 + \frac{77.5 \times 40}{100} = 77.5 + 31 = 108.5$ lakh

132. 4; $Ex_{B2012} = \frac{In \times 100}{\%P + 100} = \frac{125.4 \times 100}{10 + 100}$

$= \frac{12540}{100} = 114$ lakh

133. 1; Profit of Company A in the year 2008

$= 85 \times \frac{32}{100} = 27.2$ lakh

Profit of Company B in the year 2011

$= 85 - \frac{85 \times 100}{125} = 85 - 68 = 17$ lakh

$\therefore$ Difference = 27.2 - 17 = 10.2 lakh

134. 1; Let each of their incomes be I.

Expenditure of Company A in the year 2010

$= \frac{I \times 100}{\%P + 100} = \frac{I \times 100}{20 + 100} = \frac{1001}{120} = \frac{101}{12}$

Expenditure of Company B in the year 2010

$= \frac{1001}{150} = \frac{101}{15}$

$\therefore$ Ratio = $\frac{E_A}{E_B} = \frac{15}{12} = \frac{5}{4} = 5 : 4$

135. 5; Expenditure of Company A in the year 2010

$= \frac{171 \times 100}{120} = 142.5$ lakh

Income of Company B in the year 2012

$= 171 \times \frac{110}{100} = 188.1$ lakh

$\therefore$ Difference = 188.1 – 142.5 = 45.6 lakh

136. 4; $\frac{I_A}{E_A} = 0.3$

$\therefore E_A = \frac{23.58}{0.3} = 78.6$ lakh

137. 3; $\frac{I_{A2012}}{E_{A2012}} = 0.75$ $\quad$ $\frac{I_{C2011}}{E_{C2011}} = 1.2$

$\therefore$ Reqd % = $\frac{0.75}{1.2} \times 100 = \frac{75}{1.2} = 62.5\%$

138. 3; $\frac{I_{A2012}}{E_{A2012}} = 0.75$

I_{A2012} = 0.75 × 64 = 48 lakh

Again,

$\frac{I_{C2009}}{E_{C2009}} = 0.8$

$E_{C2009} = \frac{64}{0.8} = 80$ lakh

$\therefore$ Reqd % = $\frac{48}{80} \times 100 = 60\%$

139. 2; $\frac{I_{A2009}}{E_{A2009}} = 0.5$

$E_{A2012} = \frac{36}{0.5} = 72$ lakh

$\frac{I_{B2009}}{E_{B2009}} = 0.25$

Again,

$\therefore E_{B2009} = \frac{27}{0.25} = 108$ lakh

$\therefore$ Ratio = $\frac{72}{108} = \frac{2}{3} = 2 : 3$

140. 5; Let the import of Company C in 2008 and 2012 be x each.

$Export_{2008} = \frac{x}{0.5} = 2x$

$Export_{2012} = \frac{x}{0.6} = \frac{5x}{3}$

$\therefore$ Reqd % = $\frac{2x}{\left(\frac{5x}{3}\right)} \times 100 = \frac{6x}{5x} \times 100 = 120\%$

141. 2; Exp of Company A in the year 2007

$= \frac{73.6 \times 100}{100 + 15} = \frac{7360}{115} = 64$ lakh

142. 4; $Ex_{A2010} = \frac{93.6 \times 100}{120} = 78$ lakh

$Ex_{B2011} = \frac{86.4 \times 100}{120} = 72$ lakh

$\therefore$ Difference = 78 - 72 = 6 lakh

143. 3; Percentage profit of Company A

$= \frac{87 - 75}{75} \times 100 = 16\%$

Percentage profit of Company B

$= \frac{95.2 - 85}{85} \times 100 = 12\%$

$\therefore$ Difference = 16 - 12 = 4%

144. 3; Reqd % = $\frac{93.6 \times 100}{120} = 78\%$

145. 5; $Ex_{A2011} = \frac{100.8 \times 100}{112} = 90$ lakh

$Ex_{B2009} = \frac{95.2\times100}{112}$ = 85 lakh

$\therefore$ Ratio = $\frac{90}{85}=\frac{18}{17}=18:17$

146. 2; The ratio of expenditure to income of Company A in the year 2009 = 1.2

And the ratio of expenditure to income of Company A in the year 2012 = 0.75

$\therefore$ Reqd % = $\frac{0.75\times100}{1.2}=62.5\%$

147. 2; $\frac{E}{I}=\frac{7}{10}$ ...(i)

Let the new expenditure be E_1

Then, $E_1 = E + E\times\frac{100}{100} = 2E$

Now, let the new income be I_1, Then,

$I_1 = I + I\times\frac{110}{100}=\frac{211}{10}$

$\therefore$ New ratio =

$\frac{E_1}{I_1}=\frac{2E}{\left(\frac{211}{10}\right)}=2E\times\frac{10}{211}=\frac{20}{21}\times\frac{E}{I}=\frac{20}{21}\times\frac{7}{10}=\frac{2}{3}$

= 2 : 3

148. 4; Ratio of Company B = $\frac{E}{I}$ = 0.7

$\therefore$ 1 = $\frac{14.7}{0.7}$ = 21 lakh

$\therefore$ Profit = 21 - 14.7 = 6.3 lakh

$\therefore$ % profit = $\frac{6.3}{14.7}\times100=42.857\%\approx43\%$

149. 1; $\frac{E_{A2010}}{I_{A2010}}=0.8$

$\therefore E_{A2010}$ = 0.8 × 18.5 = 14.8 lakh

P_A= 18.5 - 14.8 = 3.7 lakh

Now, $\frac{E_{B2007}}{I_{B2007}}=0.8$

$\therefore I_B=\frac{E_B}{0.8}=\frac{12.4}{0.8}=15.5$ lakh

$\therefore P_B$ = 15.5 - 12.4 = 3.1 lakh

$\therefore$ Difference = 3.7 - 3.1 = 0.6 lakh = `60000

150. 4; $\frac{E_{A2012}}{I_{A2012}}=0.75$

$\therefore E_{A2012}$ = 0.75 × 24 = 18 lakh

$\therefore$ Profit of Company A_{2012}

= 24 - 18 = 6 lakh

$\frac{E_{B2011}}{I_{B2011}}=0.8$ $I_{B2011}=\frac{24}{0.8}=30$ lakh

Profit of Company B_{2011} = 30 - 24 = 6 lakh

$\therefore$ Reqd % = $\frac{6\times100}{6}=100\%$

151. 5; Population of City C_{2012}

$=4.5\times\frac{130}{100}\times\frac{160}{100}=9.36$ lakh

152. 3; Let the population of City D in the year 2010 be x.

Then population of City D in the year 2011

= x × $\frac{175}{100}$

$\therefore$ Required % = $\frac{x}{1}\times\frac{100}{175x}\times100$

$=57.14\approx57\%$

153. 2; Let the population of City A in the year 2010 be x.

$\therefore$ Then, its population in the year 2012

$=x\times\frac{140}{100}\times\frac{150}{100}=2.1x$

$\therefore$ Difference = 2.1x - x = l.1x

$\because$ 1.1x = 2.75 lakh

$\therefore x=\frac{2.75}{1.1}=2.5$ lakh

154. 4; Population of E in the year 2012

$=3.2\times\frac{150}{100}\times\frac{140}{100}=6.72$ lakh

155. 2; Population of City F in the year 2010

$=5.4\times\frac{100}{180}\times\frac{100}{125}=2.4$ lakh

[Population of B_{2010} = Population of F_{2010}]

$\therefore$ Population of $B_{2012}=2.4\times\frac{160}{100}\times\frac{145}{100}$

= 5.568 lakh

156. 2; $I_A = 75 + 75 \times \frac{124}{100} = 93$ lakh

$I_B = 75 - 75 \times \frac{16}{100} = 63$ lakh

$\therefore$ Difference = 93 - 63 = 30 lakh

157. 1; $Ex_{2007} = 84 \times \frac{100}{140} = 60$ lakh

$Ex_{2011} = 84 \times \frac{100}{120} = 70$ lakh

$\therefore$ Difference = 70 - 60 = 10 lakh

158. 3; Let their expenditures be x each.

$Income_A = x \times \frac{125}{100} = \frac{5x}{4}$

$Income_B = x \times \frac{75}{100} = \frac{3x}{4}$

$\therefore$ Ratio $= \frac{3x}{4} \times \frac{4}{5x} = 3 : 5$

159. 4; Reqd % $= \frac{30-24}{24} \times 100 = \frac{600}{24} = 25\%$

160. 3; Profit of $A_{2008} = 55 \times \frac{24}{100} = 13.2$ lakh

Profit of $A_{2010} = 35 \times \frac{48}{100} = 16.8$ lakh

$\therefore$ Total profit = 13.2 + 16.8 = 30 lakh

161. 2; $\frac{I_{A2008}}{E_{A2008}} = 0.6$

and $\frac{I_{B2007}}{E_{B2007}} = 0.8$

Now, $E_{B2007} = \frac{75}{0.8} = 90$ lakh

Again, $I_{A2008} = 0.6 \times 105 = 63$ lakh

$\therefore$ Reqd % $= \frac{63 \times 100}{90} = 70\%$

162. 3. Initially, $\frac{I_{A2010}}{E_{A2010}} = 0.8 = \frac{4}{5}$

Now $I_1 = I_A + I_A \times \frac{50}{100} = \frac{31_A}{2}$

$E_1 = E_A - \frac{20E_A}{100} = \frac{4E_A}{5}$

New ratio $= \frac{3I_A}{2} \times \frac{5}{4E_A}$

$= \frac{15}{8} \times \frac{I_A}{E_A} = \frac{15}{8} \times \frac{4}{5} = \frac{3}{2} = 3 : 2$

163. 5; $\frac{I_{A2010}}{E_{A2010}} = 0.8 = \frac{4}{5}$ ...(i)

$\frac{I_{B2011}}{E_{B2011}} = 0.6 = \frac{3}{5}$ ...(ii)

Now, from eqn (i), we have

$E_{A2010} = \frac{5}{4} I_{A2010}$

Again, from eqn (ii), we have

$I_{B2011} = \frac{3}{5} E_{B2011}$

$\therefore \frac{E_{A2010}}{I_{B2011}} = \frac{5I_A}{4} \times \frac{5}{3E_B} = \frac{25}{12} = 25 : 12$

[$\because$ $I_A = E_B$]

164. 3; $\therefore$ Reqd % $= \frac{0.7}{0.4}$ x 100 = 175%

165. 2; $\frac{I_{A2010}}{E_{A2010}} = 0.8$

$\therefore E_{A2010} = \frac{I_{A2010}}{0.8} = \frac{108}{0.8} = 135$ lakh

Now, $\frac{I_{B2008}}{E_{B2008}} = 0.9$

$E_{B2008} = \frac{I_{B2008}}{0.9} = \frac{108}{0.9} = 120$ lakh

$\therefore$ Reqd % $= \frac{135 \times 100}{120} = 112.5\%$

166. 3; Income of $A_{2009} = 440$ cr

$\therefore$ Expenditure of $A_{2009} = \frac{100}{110} \times 440$

= 400 cr

167. 3; The ratio of expenditure to income is the highest when profit is the lowest. Thus, in the year 2010 the profit of Company A is the lowest.

168. 3; Income of A_{2009} + Income of $B_{2009} = 880$ crore
Expenditure of A_{2009} + Expenditure of B_{2010}

$= \frac{100}{110} \times 880 = 800$ crore

169. 4; Income of B_{2012} = 250
Income of A_{2009} = Expenditure of A_{2009}

$= 250 \times \frac{100}{125} = 200$ cr

Then, expenditure of

$A_{2009} = 200 \times \frac{100}{110} = \frac{2000}{11}$ cr

$\therefore$ Expenditure of B_{2012} = 200 cr

Reqd% = $\frac{2000}{11 \times 200} \times 100 = \frac{1000}{11}\%$

$= 90.90\% \approx 91\%$

170. 3; Ratio of expenditure of Company A to Company B in the year 2009 = 5 : 11
Ratio of income of Company A to Company B in the year 2009

$= 5 \times \frac{110}{100} : 11 \times \frac{125}{100}$

$= \frac{11}{2} : \frac{55}{4}$

Reqd ratio = 2 : 5

171. 2; Cost of one kg apple in Jalandhar = `160
Cost of one kg guava in Jalandhar = `60
Difference = 160 - 60 = `100
Similarly, in Delhi → `(130 - 90) = `40
In Chandigarh → `(180 - 120) = `60
In Hoshiarpur → `(90 - 30) = `60
In Ropar → `(40 - 20) = `20
Hence, the second lowest difference between price of one kg apple and one kg guava is in Delhi.

172. 4; Cost of one kg of guava in Jalandhar = `60
Cost of two kg of grapes in Chandigarh = ` 90 × 2 = `180

Reqd% = $\frac{60}{180} \times 100 = \frac{1}{3} \times 100 = 33.33 \approx$ 34%

173. 3: Total amount = 3 × 130 + 90 × 2 = 390 + 180 = ` 570

174. 1; Cost of 45 kg grapes in Hoshiarpur = 45 × 190 = ` 8550
After 4% discount, cost price of grapes

= 8550 - $\frac{8550 \times 4}{100}$ = 8550 - 342 = ` 8208

Hence, Ravindcr had to pay ` 8208.

175. 3; Reqd ratio = $\frac{40}{90} = \frac{4}{9} = 2^2 : 3^2$

176. 2; Reqd % = $\frac{700 - 500}{500} \times 100$

$= \frac{200}{500} \times 100 = 40\%$

177. 1; Total export of all three companies in the year 2008 = 600 + 700 + 800 = 2100
Total export of all three companies in the year 2010 = 400 + 600 + 800 = 1800
$\therefore$ Reqd ratio = 2100 : 1800 = 7 : 6

178. 2; For Company X in the year

$2008 \rightarrow \frac{200}{1000} \times 100 = 20\%$ (decrease)

$2009 \rightarrow \frac{200}{800} \times 100 = 25\%$ (decrease)

$2010 \rightarrow \frac{200}{600} \times 100 = 33\frac{1}{3}\%$ (decrease)

$2011 \rightarrow \frac{200}{400} \times 100 = 50\%$ (increase)

$2012 \rightarrow \frac{300}{600} \times 100 = 50\%$ (increase)

179. 3; Average

$= \frac{800 + 700 + 500 + 800 + 1000 + 700}{6}$

= 750 thousand tonnes

180. 3; Reqd % = $\frac{3500 \times 100}{4500} = 77.77\% \approx 78\%$

181. 1; In 2010, profit of Company M
= 4.5 crore
Profit of Company (P + N) = (4 + 3) = 7 crore

$\therefore$ Reqd% = $\frac{4.5}{7} \times 100 = 4.5 = 64.28\%$

182. 4; Expenditure of Company M in the year 2011 is 75 crore.
Profit of Company M in year 2011 is 4 crore.
$\therefore$ Income of Company M in year 2011 is 75 + 4 = 79 crore
Now, expenditure of Company P in the year 2011 is 68 crore.
Profit of Company P in the year 2011 is 7 crore.
Income of Company P in the year 2011 is (68 + 7) = 75 crore
$\therefore$ Reqd ratio = 79 : 75

183. 2; In the year 2012 profit of Company M
= 6 crore

$\therefore$ Expenditure = $6\left(1+\frac{50}{100}\right)=9$ crore

Income = (9 + 6) = 15 crore

Profit of Company N in the year 2012 = 6.5 crore

$\therefore$ Expenditure $=6.5\left(1+\frac{60}{100}\right)$

$= 6.5 \times \frac{8}{5} = 1.3 \times 8 = 10.4$ crore

Hence, Income = (6.5 + 10.4) = 16.9 crore

Again, Profit of Company P in the year 2012 = 5 crore

$\therefore$ Expenditure = $5\left(1+\frac{80}{100}\right)=5\times\frac{9}{8}=$ 9 crore

Hence, Income = (9 + 5) = 14 crore

Now, average income of all three companies

$=\frac{1}{3}(15+16.9+14)=\frac{45.9}{3}=15.3$ crore

184. 3; Profit of Company N in the year 2009 = 2 crore

Profit of Company N in the year 2012. = 6.5 crore

Increase = (6.5 - 2) = 4.5 crore

% increase = $\frac{4.5}{2} \times 100 = 225\%$

185. 5; Income of Company P in the year 2010 = 40 crore

Income of Company M in the year 2010

$=40\left(1+\frac{20}{100}\right)=48$ crore

Now, profit of Company M in the year 2010 = 4.5 crore

$\therefore$ Expenditure of Company M in the year 2010 - (48 - 4.5) crore = 43.5 crore

186. 1; In 2010 total GDP = ` 60 crore

Expenditure on Education = $60\times\frac{3}{100}$

= 1.8 crore

Expenditure on Health = $60 \times \frac{5}{100} = 3$ crore

Expenditure on Defence = $60 \times \frac{6}{100} = 3.6$ crore

Reqd ratio = 1.8 : 3 : 3.6 = 3 : 5 : 6

187. 5; GDP growth during 2011-12 → 60 - 40 = 20 crore

GDP growth during 2007-08 → 50 - 40 = 10 crore

$\therefore$ Required percentage = $\frac{10}{20} \times 100 = 50\%$

188. 2; Total amount allotted to Defence during 2007-12 $=(40+\frac{5}{100}+50\times\frac{4.5}{100}+50\times\frac{5.5}{100}$

$+60\times\frac{6}{100}+40\times\frac{5}{100}+60\times\frac{5.5}{100})$ crore

= (2 + 2.25 + 2.75 + 3.6 + 2 + 3.3) crore = 15.9 crore

189. 1; Total amount allotted to Education, Health and Defence in the year 2007

$= 40 \times \frac{(2+3+5)}{100}$crore $= 40\times\frac{10}{100}$ crore

In 2008 = $50 \times \frac{(2.5+4+4.5)}{100}$ crore

$= 50 \times \frac{11}{100} = 5.5$crore

In 2009 = $50 \times \frac{(3+4+5.5)}{100}$ crore

$= 50 \times \frac{12.5}{100} = 6.25$ crore

In 2010 = $60 \times \frac{(3+5+6)}{100}$ crore

$= 60 \times \frac{14}{100} = 8.4$crore

In 2011 = $40 \times \frac{(2.5+4+5)}{100}$ crore

$= 40 \times \frac{11.5}{100} = 4.6$ crore

In 2012 = $60 \times \frac{(4+5.5+6)}{100}$ crore

$= 60 \times \frac{15.5}{100} = 9.3$ crore

In 2012, the allotted amount is the maximum.

190. 4; Amount allotted during 2009 to

Education = $65 \times \frac{3}{100}$ crore = 1.5 crore

In 2010 = $60 \times \frac{3}{100}$ crore = 1.8 crore

$\therefore$ Difference = (1.8 - 1.5) crore = 0.3 crore = 30 lakh

191. 2; In 2010, amount allotted to Education

$= 60 \times \frac{3}{100} = 1.8$crore

In 2012, amount allotted to Education

$= 40 \times \frac{3}{100} = 1.2$ crore

$\therefore$ Percentage decrease = $\frac{0.6}{1.8} \times 100 = 33.3\%$

192. 4; The graph shows that the price of M and N type tiles sdecreases over the period.

Now, for O type tiles the percentage increase from 2006 to 2012 is

$$\frac{200-150}{150} \times 100 = \frac{50}{150} \times 100 = 33\frac{1}{3}\%$$

For P type tiles the percentage increase from 2006 to 2012 is

$$\frac{300-50}{50} \times 100 = \frac{250}{50} \times 100 = 500\%$$

193. 2; Average price of M during 2006 to 2012

$= \frac{1}{7}$ (250 + 400 + 50 + 150 + 200 + 50 + 150) = $\frac{1250}{7}$ = `178.57

Average price of N during 2006 to 2012

$= \frac{1}{7}$ {300 + 350 + 250 + 350 + 300 + 350 + 250) = $\frac{2150}{7}$ = `307.14

Average price of O during 2006 to 2012

$= \frac{1}{7}$ (150 + 100 + 200 + 300 + 100 + 250 + 200) = $\frac{1300}{7}$ = `185.714

Average price of P during 2006 to 2012

$= \frac{1}{7}$ (50 + 200 + 150 + 100 t 250 + 100 + 300) = `164.28

Thus, N type of tiles' show the maximum average price during 2006 to 2012.

194. 2; Average price of all tiles in 2006

$= \frac{1}{4}$ (50 + 150 + 250 + 300) = `187.5

Average price of all tiles in 2007

$= \frac{1}{4}$ (100 + 200 + 350 + 400) = `262.5

Average price of all tiles in 2008

$= \frac{1}{4}$ (50 + 150 + 200 + 250) = `162.5

Average price of all tiles in 2009

$= \frac{1}{4}$ (100 + 150 + 300 + 350) = `225

Average price of all tiles in 2010

$= \frac{1}{4}$ (100 + 200 + 250 + 300) = `212.5

Average price of all tiles in 2011

$= \frac{1}{4}$ (50 + 100 + 250 + 350) = `187.5

Average price of all tiles in 2012

$= \frac{1}{4}$ (150 + 200 + 250 + 300) = `225

$\therefore$ In 2008, the average price of all four types of tiles is the minmum.

195. 3; Total price of alLfour types of tiles in 2012 is (150 + 200 + 250 + 300) = `900

Total price of all four types of tiles in 2009 is (100 + 150 + 300 + 350) = `900

Both are equal, so the required percentage is 0%.

196. 5; Reqd ratio

$$= \frac{\text{Price of O type tiles in 2008}}{\text{Price of P type tiles in 2009}}$$

$$= \frac{200}{250} = \frac{4}{5} = 4:5$$

197. 3; Annual sales of all companies in FY 2006-07 = 150 + 200 + 225 + 250 + 300 = 1125 lakh

Annual sales of all companies in FY 2011-12 = (325 + 350 + 400 + 450 + 500) = `2025 lakh

$\therefore$ Percentage increase

$$= \frac{2025-1125}{1125} \times 100 = 80\%$$

198. 4; Honda → Sales in FY 2006-07 = 300 lakh and in FY 2011-12 = 400 lakh

% increase in sales = $\frac{400-300}{300} \times 100$

= 33.33%

Maruti → Sales in the FY 2006-07
= 250 lakh and in FY 2011-12 = 500 lakh

$$\%\text{. increase in sales} = \frac{500-250}{250} \times 100$$

= 100%

Tata → Sales in FY 2006-07 = 200 lakh and in FY 2011-12 = 325 lakh

$$\%\text{ increase in sales} = \frac{325-200}{200} \times 100$$

= 62.5%

Hyundai → Sales in FY 2006-07 = 225 lakh and in FY 2011-12 = 350 lakh

$$\%\text{ increase in sales} = \frac{350-225}{225} \times 100$$

= 55.55%

Toyota → Sales in FY 2006-07 = 150 lakh and in FY 2011-12 = 450 lakh

$$\%\text{ increase in sales} = \frac{450-150}{150} \times 100 =$$

200%

Hence, Toyota recorded highest percentage increase in sales.

199. 2; Average sales of all companies

In FY 2006-07 $= \frac{1}{5} \times (150 + 200 + 225 +$ 250 + 300) = 235

In FY 2007-08 $= \frac{1}{5} \times (200 + 250 + 300 +$ 350 + 450) = 310

In FY 2008-09 $= \frac{1}{5} \times (150 + 250 + 300 +$ 325 + 350) = 275

In FY 2009-10 $= \frac{1}{5} \times (100 + 250 + 275 +$ 375 + 475) = 295

In FY 2010-11 $= \frac{1}{5} \times (200 + 250 + 300 +$ 400 + 450) = 320

In FY 2011-12 $= \frac{1}{5} \times (325 + 350 + 400 +$ 450 + 500) = 405

∴ Average minimum sales is in FY 2006-07.

200. 3; Total sales of Hyundai and Maruti in FY 2006-07 = (225 + 250) = 475 lakh
Total sales of Tata and Honda in FY 2006-07 = (200 + 300) = 500

$$\text{Reqd\%} = \frac{500-475}{500} \times 100 = \frac{25}{500} \times 100$$

= 5%.less. Hence, total sale of Maruti and Hyundai is 5% less than the total sales of Tata and Honda.

201. 4; Total sale of Honda in 2009-10 = 375
Total sale of Toyota in 2009-10 = 250.

$$\therefore \text{ Reqd \%} = \frac{375-250}{250} \times 100 = 50\%$$

202. 5; Expenditure = 120 - 70 = 50 crore

$$\therefore \text{ Profit \%} = \frac{70}{50} \times 100 = 140\%$$

203. 1; Income in 2011 = 85 + 30 = 115
∴ Reqd ratio = 115 : 85 = 23 : 17

204. 3;

$$\text{Average profit} = \frac{40+55+50+70+30+75}{6}$$

$$= \frac{320}{6} \approx 53 \text{ crore}$$

205. 2; Expenditure = Income - Profit
= 950000000 - 400000000 = `550000000

206. 4; % increase from previous year

$$= \frac{20}{50} \times 100 = 40\%$$

207. 3; Expenditure of Company A in the year 2009 = 25000 - 800 = 24200
Expenditure of Company A in the year 2012 = `32000 - 900 = `31100
∴ Average expenditure in both years

$$= \text{`}24200 + \text{`}31100 \times \frac{1}{2} = \text{`}27650$$

208. 2; Income of Company C in the year 2010 = `45000
Profit = `800
∴ Expenditure = `45000 - 800 = `44200

$$\%\text{ profit} = \frac{45000-44200}{44200} \times 100 = 1.80$$

Income of Company B in the year 2012 = `65000
Profit = ` 700
∴ Expenditure = `65000 - `700 = `64300

$$\%\text{ profit} = \frac{65000-64300}{64300} \times 100 = 1.08$$

∴ Reqd ratio = 1.80 : 1.08 = 5 : 3

209. 4; In 2009 profit of Company A = `800
Profit of Company B = `900
Income of Company A = `10000
Expenditure of Company A = 10000 - 800 = `9200
Expenditure of Company B = 10000 - 900 = 9100
∴ Reqd ratio = 9200 : 9100 = 92 : 91

210. 5; Profit of Company C in the year 2007 = `300
Profit of Company C in the year 2008 = `600
∴ % increase in profit

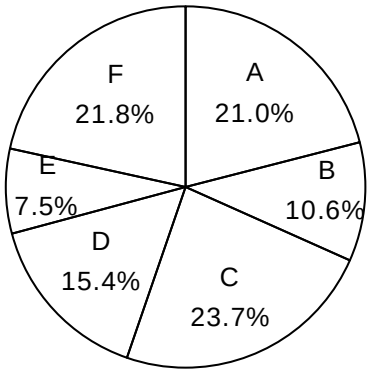
$$= \frac{(600-300)}{300} \times 100 = 100\%$$

211. 1; Income of Company A in the year 2011 = `(15000 + 500) = `15500
Income of Company D in the year 2011 = `(22000 + 900) = `22900
∴ Reqd ratio = 15500 : 22900 = 155 : 229

DATA INTERPRETATION
PIE CHART

Directions (Q. 1-5): The following pie-chart shows the percentage distribution of total population of six cities, and the table shows the percentage of males among them.

(Total population of City F = 1526000).

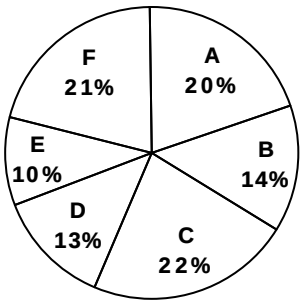


City	% Male
A	51.10%
B	53.20%
C	52.90%
D	53.80%
E	47.90%
F	49.20%

1. What is the total number of females in City A?
   (1) 718830 (2) 751170 (3) 724085 (4) 745915 (5) 739026
2. What is the difference between the male and the female population of City B?
   (1) 47448 (2) 47484 (3) 47488 (4) 47848 (5) 47844
3. The female population of City F is approximately what percentage of the female population of City E?
   (1) 174.8% (2) 224.5% (3) 257.5% (4) 283.5% (5) 296%
4. What is the total number of males in all six cities together?
   (1) 3573240 (2) 3605756 (3) 3614028 (4) 3625284 (5) None of these
5. The total number of females in all six cities together is what percentage of the total population of all six cities together? (Answer in approximate value)
   (1) 42.5% (2) 45% (3) 48.5% (4) 51% (5) 52.5%

Directions (Q. 6-10): Study the following information carefully and answer the given questions.

There are six companies, namely A, B, C, D, E and F, which produce two models (M_1 and M_2) of an item. The given pie-chart shows the percentage distribution of total production by the given six companies and the table shows the ratio of production of M_1 to that of M_2 and the percentage of profit earned on these items. (Total production cost of the six companies is `3.2 crore.)



Company	Ratio of production		%Profit earned	
	M_1	M_2	$\%P_{M1}$	$\%P_{M2}$
A	13	7	25%	32%
B	9	5	28%	30%
C	6	5	20%	24%
D	6	7	35%	25%
E	2	3	24%	21%
F	11	10	30%	20%

6. What is the total profit earned by Company A on model M_1 (in ` crore)?
(1) 0.124 (2) 0.112 (3) 0.104 (4) 0.140 (5) 0.122

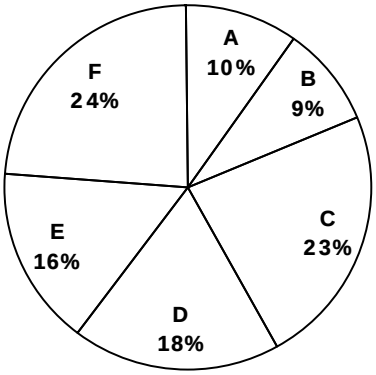
7. What is the total profit earned by Company B and Company C together on model M_2 (in ` crore)
(1) 0.1248 (2) 0.1284 (3) 0.1288 (4) 0.1244 (5) None of these

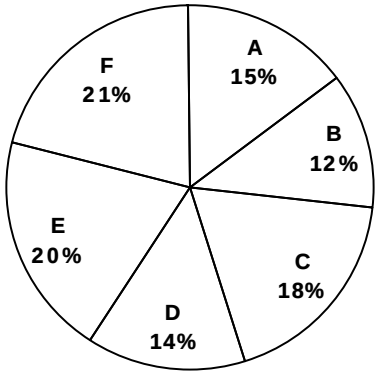
8. What is the ratio of the cost of production of model M_1 of Company D to that of model M_2 of Company F?
(1) 4 : 5 (2) 3 : 5 (3) 5 : 7 (4) 4 : 7 (5) 1 : 2

9. What is the difference beween the profit earned by Company C on model M_1 and the profit earned by Company E on model M_2? (in ` crore)
(1) 0.72768 (2) 0.74268 (3) 0.73428 (4) 0.77258 (5) None of these

10. The percentage profit earned by Company B on model M_1 is what percentage of the percentage profit earned by Company D on model M_2.
(1) 112% (2) 89.28% (3) 61% (4) 44.64% (5) Data inadequate

Directions (Q. 11-15): In the following charts, the first pie-chart shows the percentage distribution of total students studying in different schools and the second pie-chart shows the percentage distribution of total girl students studying in these schools. (Total number of students in all the schools together is 30000 and the ratio of boys to girls is 3:2.)





11. What is the difference between the total number of boys and the total number of girls studying in School D?
(1) 2020 (2) 2040 (3) 2066 (4) 2680 (5) 3720

12. The number of girls studying in School C is what percentage of the number of boys studying in School E?
(1) 60% (2) 70% (3) 75% (4) 80% (5) 90%

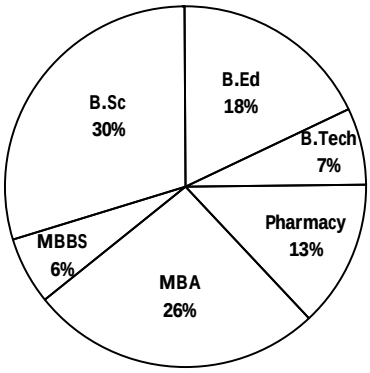
13. What is the average of the number of boys studying in school A, Band C?
(1) 2150 (2) 2200 (3) 2350 (4) 2400 (5) 2450

14. The number of girls in School F is what percentage more than the number of girls in School A?
(1) 25% (2) 30% (3) 40% (4) 50% (5) 60%

15. Total number of boys in School F is approximately what percentage more than the total number of boys in School
(1) 21.4% (2) 25.8% (3) 27.5% (4) 32% (5) 34.6%

Directions (Q. Nos. 16-20) ***Study the following Pie-chart carefully to answer these questions.***

Total Students = 6500 Percentage distribution of Students in different courses



16. What is the value of **half** of the difference between the number of students in MBA and MBBS?
(1) 800 (2) 1600 (3) 1300 (4) 650 (5) None of these

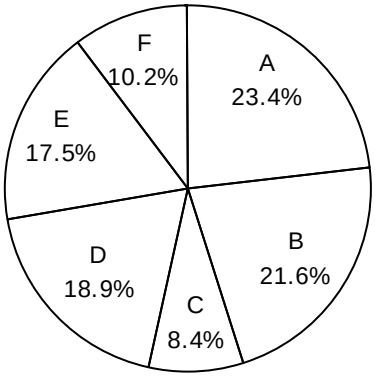
17. What percentage **(approximately)** of students are in MBA as compared to students in B.Ed.?
(1) 49 (2) 53 (3) 59 (4) 41 (5) 44

18. What is the total number of students in B.Ed., Pharmacy and MBBS together?
(1) 2465 (2) 2565 (3) 2405 (4) 2504 (5) None of these

19. What is the respective ratio between the number of students in Pharmacy and the number of students in B.Tech?
(1) 11 : 13 (2) 13 : 6 (3) 13 : 7 (4) 6 : 13 (5) None of these

20. Number of students in B.Sc. is **approximately** what percentage of the number of students in B.Ed.?
(1) 167 (2) 162 (3) 157 (4) 153 (5) 150

Directions (Q. 21 - 25): Following pie-chart shows the percentage distribution of the total population of six different cities and the table shows the percentage of adult population in them. (Population of City A = 1287000)



City	% Adult
A	77%
B	68%
C	73%
D	75%
E	69%
F	72%

21. What is the total adult population of City C?
(1) 337260 (2) 337262 (3) 337264 (4) 337266 (5) None of these

22. The total population of City A is approximately what percentage of the total population of City D?
(1) 117.5% (2) 123.8% (3) 125% (4) 127.6% (5) 129.2%

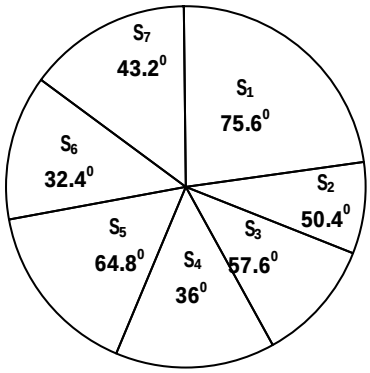
23. What is the total non - adult population of City F?
(1) 153010 (2) 154040 (3) 155300 (4) 1561020 (5) 157080

24. The total adult population of City B and C together is approximately what percentage of the total population of all six cities together?
(1) 16% (2) 21% (3) 25% (4) 27% (5) 30%

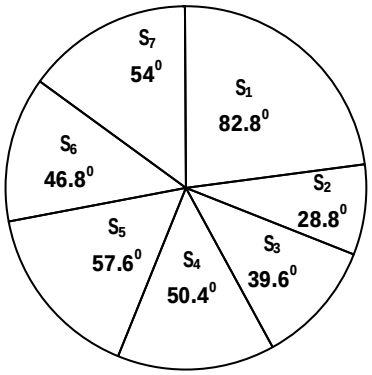
25. The total population of City D is approximately what percentage more than the total population of City E?

(1) 8% (2) 10% (3) 12% (4) 14% (5) 16%

Directions (Q. 26-30): Following pie-charts show the distribution of the total number of students selected in an entrance exam from seven different schools in 2010 and 2011. (The total number of students selected from School S_7 in 2010 and 2011 are 180 and 270 respectively.)



2010



2011

26. The total number of students selected from all seven schools together in the year 2011 is approximately what per cent of the total number of students selected from all seven schools in 2010?

(1) 83.33% (2) 120% (3) 71.42% (4) 140% (5) None of these

27. What is the per cent rise in the number of students selected from School S_4 from 2010 to 2011 ?

(1) 60% (2) 63% (3) 68% (4) 72% (5) 75%

28. The total number of students selected from School S_5 and S_7 together in the year 2010 is approximately what per cent of the number of students selected from School S_2 in the year 2011?

(1) 178.5% (2) 247.5% (3) 287.5% (4) 312.5% (5) 342.5%

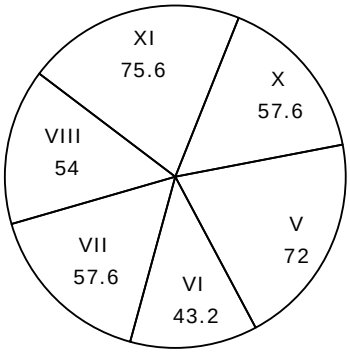
29. What is the difference between the average number of students selected from school S_1, S_2 and S_3 in the year 2010 and the average number of students selected from schools S_5, S_6 and S_7 in the year 2011 ?

(1) 9 (2) 12 (3) 15 (4) 18 (5) 21

30. In which of the following schools is the per cent rise or fall in the number of students selected from 2010 to 2011 the maximum?

(1) S_2 (2) S_3 (3) S_4 (4) S_5 (5) S_6

Directions (Q. 31 - 35): In the following pie - chart, the distribution of students of a school is given. The table gives the ratio of boys to girls among them. Total students studying in six different classes of the school is 1200.



Class	Boy : Girls
V	3 : 2
VI	3 : 1
VII	5 : 3
VIII	8 : 7
IX	4 : 3
X	1 : 1

31. What is the average of the number of girls studying in all six classes?
(1) 82 (2) 84 (3) 86 (4) 88 (5) None of these

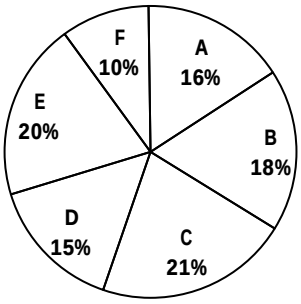
32. What is the difference between the total number of boys and the total number of girls in all six classes together?
(1) 208 (2) 210 (3) 212 (4) 214 (5) 216

33. In the given pairs of classes, which two classes have equal number of boys in them?
(1) V - VII (2) VII – X (3) VIII – X (4) IX – X (5) None of these

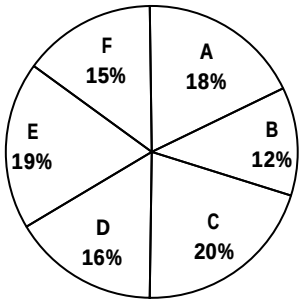
34. The difference between the number of boys and the number of girls in class V is what percentage of the difference between the number of boys and the number of girls in class VII??
(1) 60% (2) 80% (3) 100% (4) 120% (5) 150%

35. The total number of boys in class VI is what percentage more than the total number of girls in class X?
(1) 8.5% (2) 12.5% (3) 15% (4) 17.5% (5) None of these

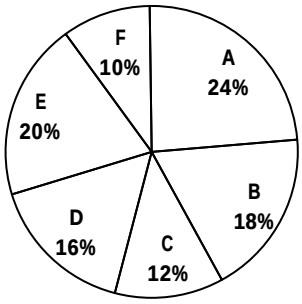
Directions (Q. 36-40): Following pie-chart shows the percentage distribution of total population of different cities and the distribution of literate males and females among them. Total population of all six cities together is 1.5 crore and the ratio of males to females among them is 8 : 7. Total literate males and females in all cities are 40 and 25 lakh respectively.



Total Population = 1.5 crore



Literate Males = 40 lakh



Literate Females = 25 lakh

36. What is the total illiterate population of City A?
(1) 10.8 lakh (2) 14.2 lakh (3) 16.8 lakh (4) 18 lakh (5) None of these

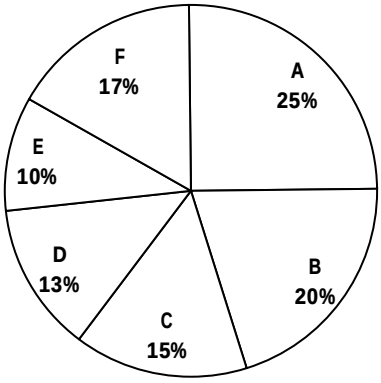
37. Total literate males of City E are what percentage of total literate females of City F?
(1) 32.89% (2) 118% (3) 196% (4) 240% (5) 304%

38. Total literate population of City E is what percentage of its total population?
(1) 25.33% (2) 16.66% (3) 26% (4) 42% (5) 64%

39. What is the difference between total illiterate population and total literate population of City C?
(1) 8.5 lakh (2) 9.5 lakh (3) 10.5 lakh (4) 11 lakh (5) 20.5 lakh

40. Total number of literate males of City D is what percentage more than the total number of literate female of City D?
(1) 60% (2) 38.46% (3) 61.538% (4) 120% (5) 160%

Directions (Q. 41-45): Following pie-chart shows the percentage distribution of items produced (I_1 and I_2) by six companies. The cost of total production (of both items) of all companies together is ` 24 crore. The given table shows the ratio of items I_1 and I_2 produced and percentage profit earned on these items.



	Ratio of production		Per cent profit earned	
	I_1	I_2	P_1	P_2
A	14	11	20	30
B	2	3	28	25
C	8	7	24	20
D	5	8	24	30
E	7	3	25	35
F	9	8	32	15

41. What is the total cost of production of item I_2 produced by companies E and F together?
(1) ` 2.12 crore (2) ` 2.44 crore (3) ` 2.64 crore (4) ` 2.86 crore (5) ` 2.96 crore

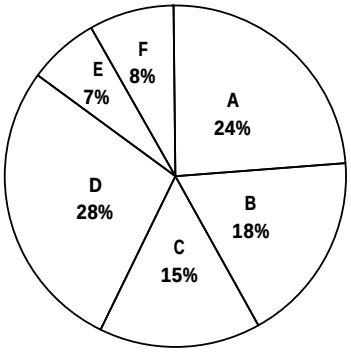
42. What is the difference between the cost of production of item I, by Company B and the cost of production of item I_2 by C?
(1) ` 21 lakh (2) ` 24 lakh (3) ` 27 lakh (4) ` 29.5 lakh (5) ` 32 lakh

43. What is the amount of profit earned by Company A on both items I_1 and I_2 together?
(1) ` 1.216 crore (2) ` 1.32 crore (3) ` 1.364 crore (4) ` 1.464 crore (5) ` 1.56 crore

44. What is the amount of profit earned on item I_2 by Company B and D together?
(1) ` 1.648 crore (2) ` 1.296 crore (3) ` 324 crore (4) ` 1.48 crore (5) ` 1.502 crore

45. What is the ratio of the profit earned by Company A to that earned by Company E on item I_1?
(1) 8 : 5 (2) 8 : 3 (3) 5 : 3 (4) 3 : 2 (5) None of these

Directions (Q. 46-50): Following pie-chart shows the percentage distribution of total items (I_1 and I_2) produced by six companies (A, B, C, D, E and F) and the table shows the ratio of I_1 to I_2 and percentage sale of I_1 and I_2.



Company	I_1	I_2	% Sold I_1	% Sold I_2
A	5	3	65%	62%
B	5	4	56%	78%
C	2	3	72%	66%
D	3	4	75%	60%
E	4	3	64%	55%
F	3	2	50%	48%

Total items (I_1 and I_2) = 16 lakh

46. What is the difference between the total items produced by Company A and B together and the total items produced by Company D?
(1) 3.84 lakh (2) 3.06 lakh (3) 2.96 lakh (4) 2.24 lakh (5) 1.78 lakh

47. What is the difference between the total number of I_1 items and the total number of I_2 items produced by Company F?
(1) 24800 (2) 25600 (3) 26300 (4) 27500 (5) 28300

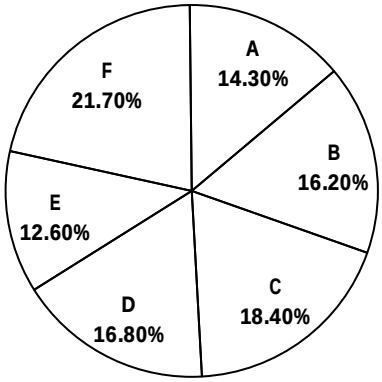
48. What is the average number of I_1 items sold by all six companies together?
(1) 89480 (2) 89580 (3) 89680 (4) 89780 (5) None of these

49. What is the difference between the number of I_1 items sold and the number of I_2 items sold by Company E?
(1) 14560 (2) 14480 (3) 14610 (4) 14340 (5) 14220

50. The number of I, items sold by Company A is what percentage of the number of I_1 items sold by Company F?

(1) 40.625% (2) 120% (3) 184.64% (4) 296.5% (5) None of these

Directions (Q. 51-55): Total number of cars sold by a company in six cities is 90000. Given pie-chart shows the percentage distribution of these cars sold in these cities. The table shows the proportion of three models among those cars sold.



[Total = 90000]

Model / City	M_1	M_2	M_3
A	7	7	4
B	2	5	2
C	3	3	4
D	4	3	2
E	2	2	1
F	3	2	5

51. What is the total number of M_2 cars sold in all cities together?

(1) 31155 (2) 31255 (3) 31355 (4) 31455 (5) 31555

52. What is the difference between M_1 cars sold in City D and City E?

(1) 2184 (2) 2204 (3) 2244 (4) 2284 (5) 2294

53. The number of M_1 cars sold in City D is approximately what percentage of the total number of M_3 cars sold in City A?

(1) 145% (2) 42.55% (3) 185% (4) 83.0% (5) 235%

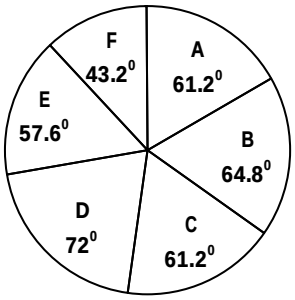
54. Total number of cars sold in City F is approximately what percentage more than the total number of cars sold in City B?

(1) 5.5% (2) 13% (3) 21% (4) 27.5% (5) 34%

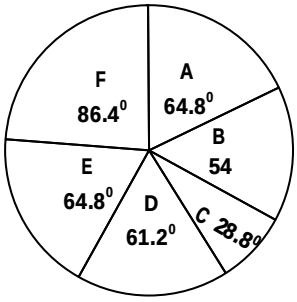
55. What is the ratio of the total number of cars sold in City C to the total number of M_2 cars sold in City D?

(1) 19:5 (2) 23:7 (3) 27:8 (4) 33:10 (5) 47:10

Directions (Q. 56-60): Following pie-charts show the distribution of items of six different types produced by a company in two years 2008 and 2009. Total number of items produced by the company in the year 2008 and 2009 are 48600 and 62500 respectively.



2008



2009

56. What is the total number of items of type C produced in the year 2008 and 2009 together?

(1) 12482 (2) 13262 (3) 14786 (4) 15200 (5) None of these

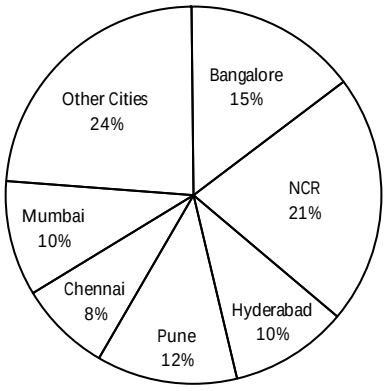
57. The number of type B items produced in 2008 is what percentage of the number of type B items produced in the year 2009? (approximate value)
(1) 78% (2) 84% (3) 87% (4) 90% (5) 93%

58. What is the ratio of the number of type D items produced in 2008 to the number of type F items produced in 2009?
(1) 13:17 (2) 83:116 (3) 81:125 (4) 103:147 (5) None of these

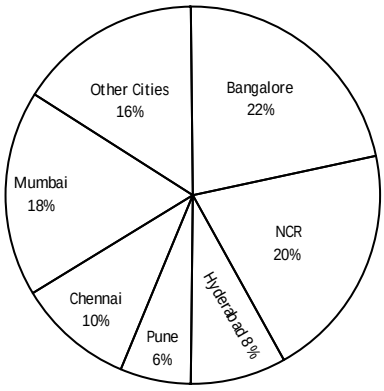
59. What is the total number of type A, B and C items produced by the company in the year 2008 and 2009 together?
(1) 48542 (2) 50897 (3) 51164 (4) 52324 (5) 54160

60. The number of type E items produced in the year 2009 is what per cent more than the number of type C items produced in 2009?
(1) 84% (2) 72% (3) 75% (4) 60% (5) None of these

Direction (Q. 61-65): Following pie-charts show the percentage distribution of job vacancies in IT industries in the year 2000 and 2010. In the year 2000, the total number of vacancies was 5.4 lakh, and in the year 2010, it was 8.6 lakh.



Year - 2000



Year - 2010

61. What is the difference between the number of vacancies available in Bangalore in the year 2010 and 2000?
(1) 108200 (2) 113120 (3) 118400 (4) 96400 (5) None of these

62. What is the average number of vacancies available in Hyderabad in the year 2000 and 2010?
(1) 41080 (2) 42740 (3) 58610 (4) 61400 (5) 62800

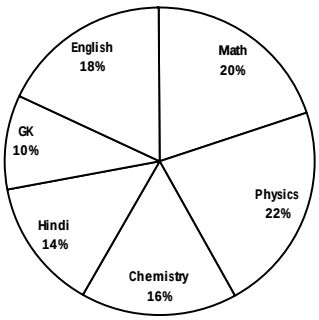
63. What is the total number of vacancies available in Chennai in 2000 and in Mumbai in the year 2010?
(1) 2.16 lakh (2) 2.04 lakh (3) 1.98 lakh (4) 1.92 lakh (5) None of these

64. If the number of vacancies in Pune is 48000 in the year 2010 and the percentage distribution is the same as given in the chart, what is number of vacancies available in NCR in 2010?
(1) 1.2 lakh (2) 1.32 lakh (3) 1.48 lakh (4) 1.60 lakh (5) 1.72 lakh

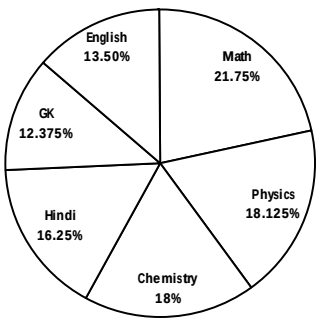
65. What is the percentage rise in vacancies available in Hyderabad from year 2000 to 2010? (Give approximate value only).
(1) 21.8% (2) 23.2% (3) 24.5% (4) 26.2% (5) 27.41%

Directions (Q. 66-70): Following pie-chart shows the percentage distribution of total marks scored by a student in Unit Test-I and Unit Test-2. In Unit Test-1 he got 750 marks and in Unit Test-2 he got 800.

Test - 1



Test - 2



66. What is the total marks scored by the student in Physics, Chemistry and Maths together in Unit Test–2?
(1) 461 (2) 463 (3) 465 (4) 467 (5) 469

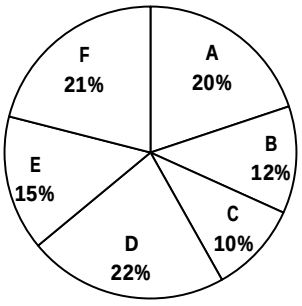
67. What is the difference of marks scored by him in Chemistry in Test–2 and that in English in Test-1?
(1) 36 (2) 15 (3) 9 (4) Nil (5) None of these

68. What is the percentage rise in marks scored by him in GK, from Unit Test–1 to Unit Test–2?
(1) 25% (2) 28% (3) 32% (4) 36% (5) 39%

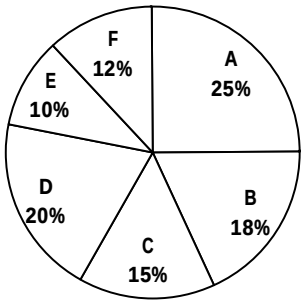
69. The marks scored by the student in Maths in Unit Test–2 is what percentage of the marks scored by him in the same subject in Unit Test–1 ?
(1) 86.2% (2) 92.5% (3) 96% (4) 116% (5) 124%

70. The marks scored by the student in Physics in both tests together is what percentage more than the marks scored by him in Hindi in both tests together? (Answer in approximate value)
(1) 27% (2) 30% (3) 32% (4) 35% (5) 37%

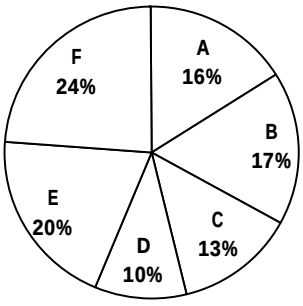
Directions (Q. 71-75): The total number of employees of a company is 8000, in which the ratio of Male to Female is 3 : 5 and Graduate to Non-graduate is 3 : 2. Following pie-chart shows the percentage distribution of these employees among different departments.



[Total = 8000]



Male



Graduate

71. What is the number of employees working in Department F who are non-graduate?
(1) 528 (2) 526 (3) 524 (4) 522 (5) 520

72. What is the total number of male graduate employees working in Department D?
(1) 600 (2) 1160 (3) 480 (4) 1280 (5) None of these

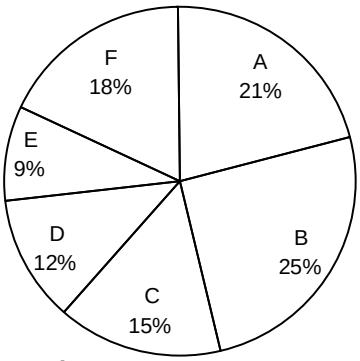
73. What is the difference between the total number of female employees and the total number of male employees working in the company?
(1) 1000 (2) 2000 (3) 3000 (4) 4000 (5) 5000

74. The number of graduate employees working in Department C is what percentage of the number of non-graduate employees working in Department E?

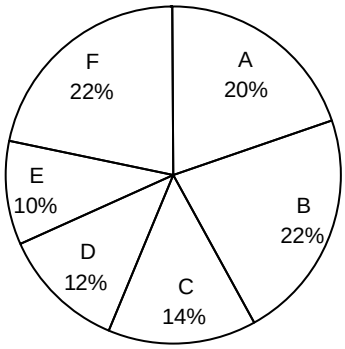
(1) 260% (2) 180% (3) 160% (4) 120% (5) 60%

75. The total number of female employees in Department F is what percentage more than the total number of male employees working in Department A?
(1) 72% (2) 74% (3) 76% (4) 78% (5) 80%

Directions (Q. 76-80) : In the given pie-charts the per cent distribution of sportspersons on the basis of their country is shown. Total persons who participated in the event is 2200 and the ratio of Male to Female among them is 15:7.



Total sports persons = 2200



Male players = 1500

76. The total Female participants from Country D is what percentage of the total Male participants from Country A?
(1) 21% (2) 24% (3) 28% (4) 31% (5) 36%

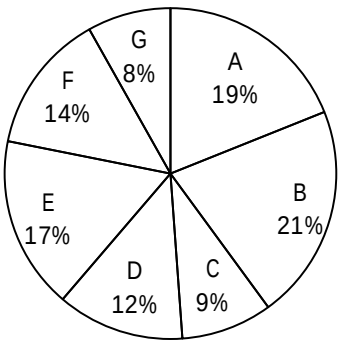
77. The total Male participants from Country B and F together is what percentage of the total participants of the event?
(1) 20% (2) 25% (3) 30% (4) 40% (5) 45%

78. The Male participants from Country E is what percentage more than the Female participants from Country C?
(1) 18% (2) 21% (3) 24% (4) 25% (5) 27%

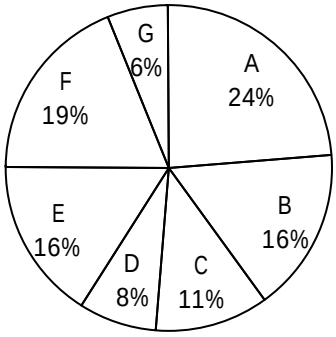
79. What is the ratio of the total participants from country D to the total female participants from Country E?
(1) 7 : 2 (2) 9 : 4 (3) 13 : 4 (4) 11 : 5 (5) None of these

80. If 20 additional female participants from Country C joined the event, the total number of female participants from Country C is what percentage of total participants from Country C?
(1) 30% (2) 40% (3) 50% (4) 60% (5) None of these

Directions (Q. 81-85) : Following pie-chart shows the percentage distribution of doctors from different cities and the second pie-chart shows the percentage distribution of female doctors among them. Total doctors in all seven cities together is 4800 and the ratio of male to female among them is 5 : 3.



Doctor



Female doctors

81. In how many cities is the number of female doctors more than the average number of female doctors, taking all cities together?
(1) Two (2) Three (3) Four (4) Five (5) Six

82. In which of the following cities is the number of male doctor less than the number of female doctors?
(1) A (2) B (3) D (4) E (5) F

83. In City A the number of female doctors is what percentage of the number of male doctors?
(1) 64% (2) 80% (3) 90% (4) 96% (5) 111%

84. What is the difference between the average number of male doctors of cities A, B and C together and the average number of male doctors of City E, F and G together?
(1) 88 (2) 96 (3) 100 (4) 108 (5) 112

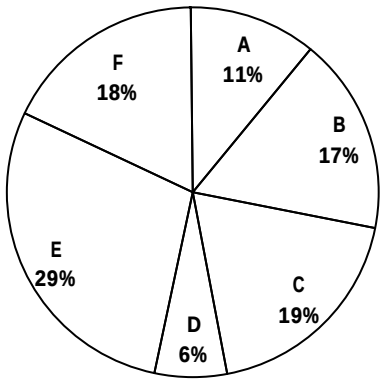
85. The total number of female doctors in City D is what per cent of the total number of male doctors in City B?
(1) 16% (2) 20% (3) 24% (4) 36% (5) 48%

Directions (Q. 86-90) : Study the following pie-chart and answer the following questions.

Percentagewise distribution of teachers in six different universities.

Total number of teachers = 6400

Percentage of Teachers



86. The number of teachers in University B is approximately what per cent of the total number of teachers in University D and University E together?.
(1) 55 (2) 59 (3) 49 (4) 45 (5) 65

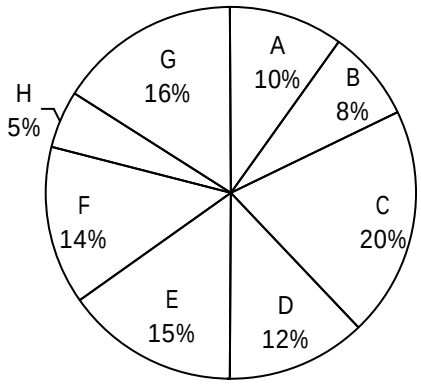
87. If twenty five per cent of the teachers in University C are females, what is the number of male teachers in University C?
(1) 922 (2) 911 (3) 924 (4) 912 (5) None of these

88. The difference between the total number of teachers in University A, University B and University C together and the total number of teachers in University D, University E and University F together is exactly equal to the number of teachers of which University?
(1) University A (2) University B (3) University C (4) University D (5) University F

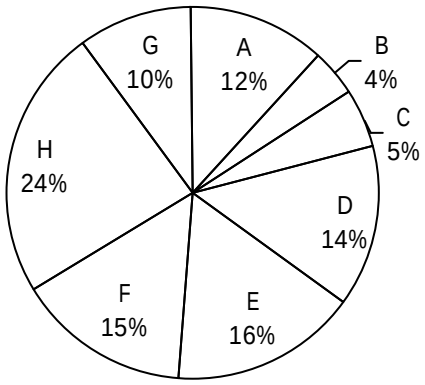
89. If one-thirtysixth of the teachers from University F are professors and the salary of each professor is ` 96000, what will be the total salary of all the professors together from University F?
(1) ` 307.2 lakh (2) ` 32.64 lakh (3) ` 3.072 lakh (4) ` 3.264 lakh (5) None of these

90. What is the average number of teachers in University A, University C, University D and University F together?
(1) 854 (2) 3546 (3) 3456 (4) 874 (5) None of these

Directions (Q. 91-95) : The following pie-chart shows the distribution of the number of cars of different models produced by a Company in 2005 and 2010.



Total cars in the year 2005 = 32000



Total cars in the year 2010 = 60000

91. What is the central angle made by cars of Model D, E and F in the year 2005?
(1) 147.6° (2) 158.2° (3) 164° (4) 167.5° (5) 172.5°

92. What is the percentage increase in number of Model A cars produced by the company from 2005 to year 2010?
(1) 75% (2) 90% (3) 112.5% (4) 125% (5) 137.5%

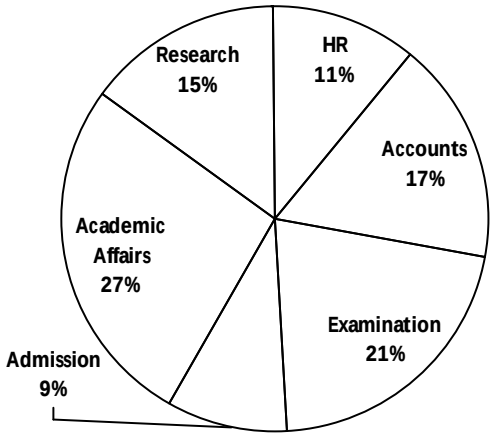
93. What is the ratio of the number of cars of model F in the year 2005 to the number of cars of model H in the year 2010?
(1) 16:35 (2) 10:27 (3) 15:38 (4) 16:45 (5) None of these

94. The number of cars of Model D in the year 2010 is what percentage of the number of Model C cars in the year 2005?
(1) 122.5% (2) 131.25% (3) 142.75% (4) 150% (5) 152.25%

95. The number of cars of Model G in the year 2010 is what percentage more than the number of same-model cars in 2005? (approximate value)
(1) 12% (2) 17% (3) 24% (4) 28% (5) 35%

Directions (Q. 96-100) : Study the following pie-chart and answer the following questions.

Total number of Employees = 12600 Percentagewise distribution of Employees



96. The number of employees in the department of Academic Affairs is approximately what per cent more than the number of employees in Examination department?
(1) 39 (2) 29 (3) 12 (4) 139 (5) 112

97. If 30 per cent of the number of employees of Research department is females, then what is the number of male employees in the Research department?
(1) 1343 (2) 1232 (3) 1323 (4) 1242 (5) None of these

98. The number of employees in Examination department is approximately what percentage of the total number of employees in the department of HR and Academic Affairs together?
(1) 69 (2) 65 (3) 61 (4) 55 (5) 51

99. What is the average number of employees in Accounts, Admission and Research department together?

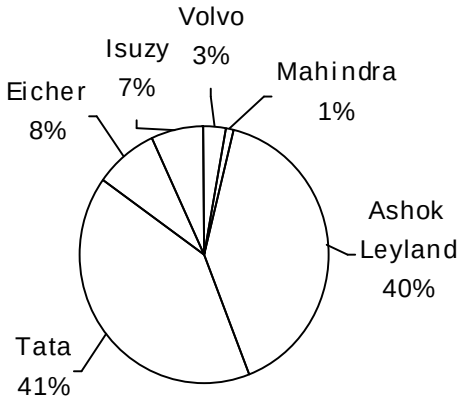
(1) 1722 (2) 1742 (3) 1786 (4) 1784 (5) None of these

100. What is the difference between the total number of employees in the department of HR and Admission together and the total number of employees in Accounts and Examination department together?

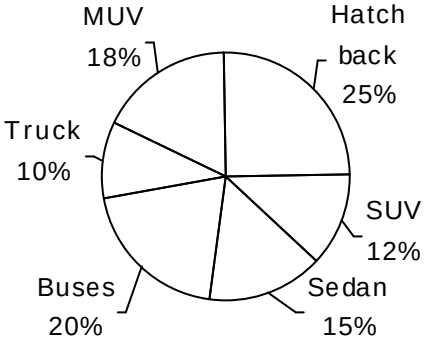
(1) 2268 (2) 2464 (3) 2286 (4) 2644 (5) None of these

Directions (Q. 101-105) : Study the graph below to answer the questions that follow.

% market share of sales of buses by six different companies in FY 2011-12



% of different models sold of Tata automobile company



101. The number of buses sold by Ashok Leyland is 40 thousand in FY 2010-11 and the percentage growth in sales of buses is $12\frac{1}{2}\%$ in FY 2011-12. How many units have been sold by Eicher in FY 2011 -12?

(1) 12000 units (2) 11000 units (3) 10000 units (4) 9000 units (5) None of these

102. What is the approximate percentage of buses sold by Isuzy with respect to that of SUVs sold by Tata in the FY 2011-12, if the number of units sold by Volvo is 3375?

(1) 28.5% (2) 31.5% (3) 35.5% (4) 32.5% (5) None of these

103. What is the ratio of the number of Eichers sold to the number of SUV sold by Tata in the year 2011-12?

(1) 2 : 5 (2) 1 : 2 (3) 1 : 3 (4) 3 : 7 (5) None of these

104. What is the approximate percentage of Volvos sold to that of MUVs sold by Tata in 2011-12?

(1) 10% (2) 7% (3) 8%

(4) Can't be determined (5) None of these

105. Referring to the data of question number 91, what is the average number of units of Volvo, Isuzy,

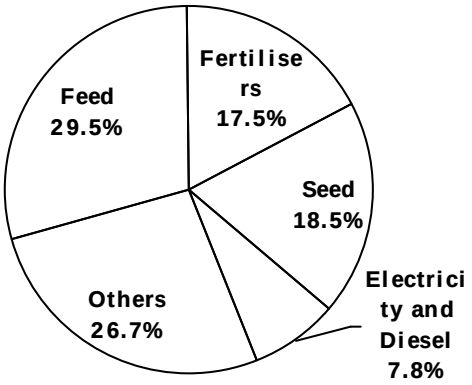
Eicher and Mahindra sold in FY 2011-12?

(1) 4035 (2) 2334.5 (3) 2137.5 (4) 5343.8 (5) None of these

Directions (Q. 106-100) : Study the given pie-charts carefully and answer the questions given below:

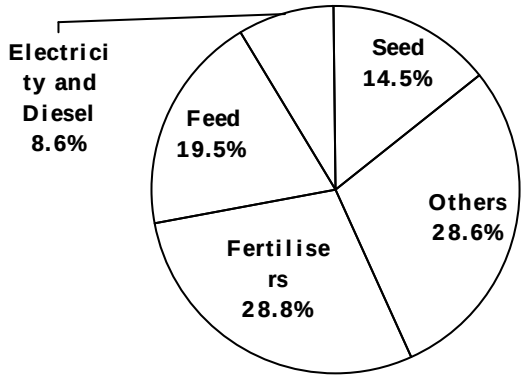
The pie-charts show the major expenses in agriculture under different heads in year 2000-01 and 2010-11

Total expenditure = ` 15432 crore



Year 2000-04

Total expenditure = ` 35349 crore



Year 2010-11

106. The total expenditure on electricity and diesel in year 2010-11 exceeded similar expenditure in year 2000-01 by approximately

(l) ` 1840crore (2) ` 1852crore (3) ` 7162 crore (4) ` 4544 crore (5) ` 6519 crore

107. The actual expenditure on fertilisers in year 2010-11exceeded the expenditure on the same in year 2000-01 by approximately

(1) 4 times (2) 3 times (3) 6 times (4) 5 times (5) 7 times

108. The expenditure on fertilisers and feed in year 2000-01 amounted to approximately

(1) ` 7253 crore (2) ` 8000crore (3) ` 7200crore (4) ` 3542 crore (5) None of these

109. The expenditure on feed in year 2010-11, as compared to that in year 2000-01, was approximately

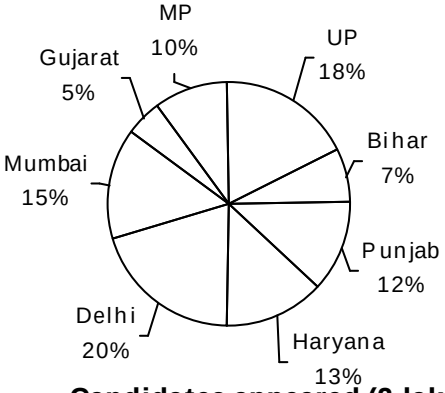
(1) 47% (less) (2) 53% (more) (3) 51% (more) (4) 53% (less) (5) 51% (less)

110. In terms of actual expenditure on electricity and diesel, the increase in year 2010-11, as compared to 2000-01, was roughly

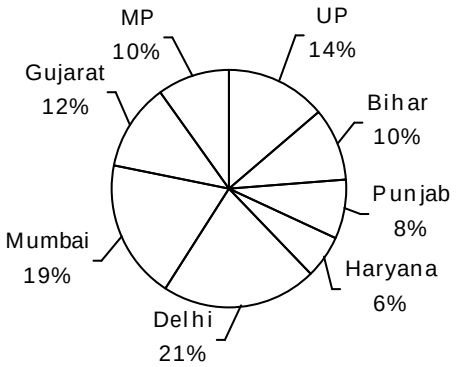
(1) 1.91 times (2) 1.53 times (3) 1.73 times (4) 1.83 times (5) 1.94 times

Directions (Q. 111-115) : Study the following pie-charts below and answer the questions that follow:

Classification of candidates from different states who appeared and qualified in a competitive exam



Candidates appeared (2 lakh)



Candidates qualified (16500)

111. What is the difference between the number of candidates who qualified from Gujarat, MP and UP together and that of those who qualified from Bihar and Punjab together?
(1) 3045 (2) 2603 (3) 2970 (4) 2556 (5) None of these

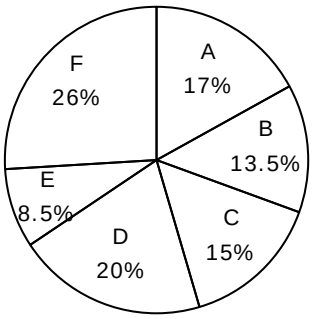
112. What is the percentage of the number of candidates who qualified from UP and Bihar together with respect to those who appeared from Delhi and Haryana?
(1) 9.6% (2) 6.8% (3) 6% (4) 8% (5) None of these

113. What is the ratio of the number of candidates who appeared from UP to that of those who qualified from Mumbai, Delhi, Haryana and Punjab together?
(1) 3:2 (2) 57:79 (3) 5:9 (4) 125:108 (5) None of these

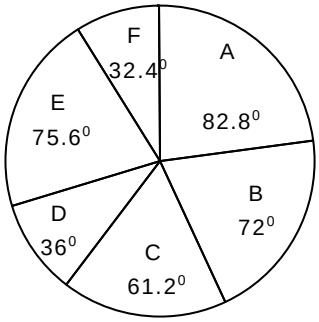
114. Which of the following states has the least percentage of number of candidates who qualified with respect to appeared from that state?
(1) Haryana (2) Delhi (3) Mumbai (4) Gujarat (5) UP

115. In which of the following states the percentage of the number of qualified candidates with respect to the number of appeared candidates is the maximum?
(1) Haryana (2) Gujarat (3) Mumbai (4) Delhi (5) UP

Directions (Q. 116-120): A total of 39 thousand students appeared in an entrance examination from six cities, in which the number of boys was 12000. The following pie-charts show the distribution of the total students and of the total boys from these cities who appeared in the exam.



TOTAL STUDENTS = 39000



BOYS = 12000

116. What is the total number of girls who appeared from City A?
(1) 3210 (2) 3440 (3) 3650 (4) 3870 (5) 3900

117. What is the difference between the total number of boys and the total number of girls who appeared in the exam from City E?
(1) 1725 (2) 1750 (3) 1775 (4) 1800 (5) 1825

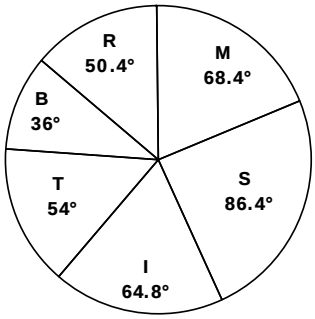
118. The total number of girls who appeared from City C is approximately what per cent of the total number of students who appeared from City D?
(1) 45% (2) 49% (3) 54% (4) 57% (5) 60%

119. What is the difference between the total number of boys who appeared from City A and that from City B?
(1) 320 (2) 330 (3) 340 (4) 350 (5) 360

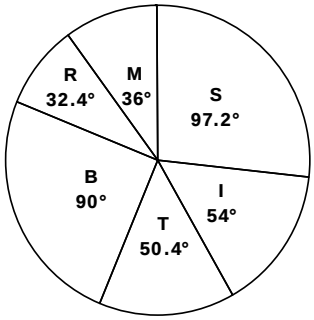
120. The number of girls who appeared from City F is approximately what per cent of the total number of girls who appeared from all six cities together?
(1) 31.5% (2) 32.5% (3) 33.5% (4) 34.5% (5) 35.5%

Directions (Q. 121-125) : The following pie-chart shows the distribution of expenditure of three companies A, B and C

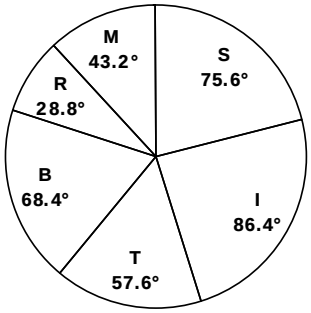
S → Salary, I → Infrastructure, T → Transportation, B → Bonus, R → Raw material, M → Miscellaneous and the total expenditures of Company A, B and C are ` 80 lakh, ` 90 lakh and ` 75 lakh respectively.



Expenditure of Company A
(` 80 lakh)



Expenditure of Company B
(` 90 lakh)



Expenditure of Company C (` 75 lakh)

121. What is the difference between (in `) the expenditure of Company A on salary and the expenditure of Company Bon raw material?
(1) 9.6 lakh (2) 11.1 lakh (3) 12.4 lakh (4) 13.4 lakh (5) 15.1 lakh

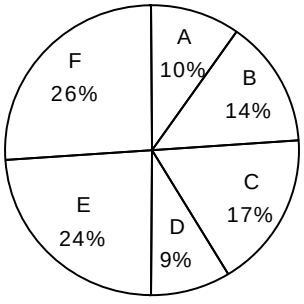
122. The expenditure of Company C on salary is approximately what percentage of the expenditure of Company A on transportation?
(1) 76.2% (2) 96% (3) 112.5% (4) 125% (5) 131%

123. What is the average expenditure (in `) of the three companies on infrastructure?
(1) 12.2 lakh (2) 15.3 lakh (3) 16.4 lakh (4) 17.5 lakh (5) None of these

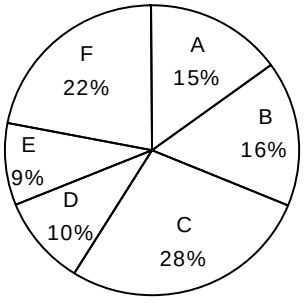
124. What is the ratio of the expenditure of Company A on infrastructure to the expenditure of Company B on transportation?
(1) 5 : 4 (2) 6 : 5 (3) 7 : 6 (4) 8 : 7 (5) 9 : 8

125. The expenditure of Company C on infrastructure is what percentage more or less than the expenditure of Company A on bonus?
(1) 80% (2) 100% (3) 120% (4) 125% (5) 150%

Directions (Q.126-130): The following pie-charts show the percentage distribution of the total number of readers of a newspaper in the year 2008 and 2012, among six different states.



Year 2008



Year 2012

126. If the number of readers from State D in the year 2008 and 2012 were 38700 and 57000 respectively, what is the difference between the total number of readers from State F in the year 2012 and that in 2008?

(1) 12400 (2) 13600 (3) 14200 (4) 15700 (5) 16800

127. If the ratio of the number of readers from State A in the year 2008 to that in 2012 was 2 : 5, what will be the ratio of the total number of readers from all six states together in year 2008 to that in 2012?

(1) 2 : 5 (2) 3 : 5 (3) 4 : 5 (4) 9 : 25 (5) 4 : 9

128. If the number of readers from State C in the year 2008 and that from State E in the year 2012 were 73100 and 51300 respectively, then what is the total number of readers from State B in the year 2008 and 2012 together?

(1) 1.324 lakh (2) 1.468 lakh (3) 1.514 lakh (4) 1.642 lakh (5) 1.728 lakh

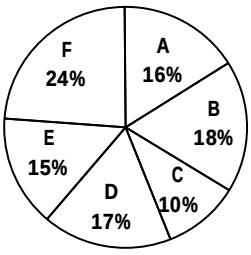
129. The percentage share of readers from State A in the year 2012 is approximately what per cent of the percentage share of readers from State E in the year 2008?

(1) 47.5% (2) 52.5% (3) 57.5% (4) 62.5% (5) None of these

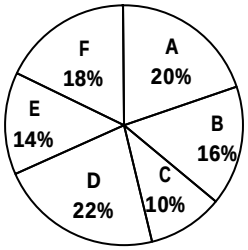
130. If the total number of readers from all six states together in year 2008 and 2012 were 4.3 lakh and 5.7 lakh respectively, what is the difference between the total number of readers from State B and State C together in the year 2008 and 2012?

(1) 1.175 lakh (2) 1.415 lakh (3) 1.625 lakh (4) 1.596 lakh (5) None of these

Directions (Q. 131-135): The following pie-charts show the percentage distribution of the total students passed from six different colleges. The second pie-chart shows the percentage distribution of the total girls passed from six different colleges. The total number of passed students is 7.5 thousand and 40% of them are girls.



Total students = 7500



Total girls = 3000

131. In which of the following colleges is the ratio of the number of passed boys to the number of passed girls 1:1?

(1) A (2) B (3) C (4) D (5) E

132. In which of the following colleges is the number of passed girl students more than the number of passed boy students?

(1) B (2) C (3) D (4) E (5) F

133. In which of the following colleges the difference between the number of passed boy students and the number of passed girl students is the maximum?

(1) B (2) C (3) D (4) E (5) F

134. The boy students who passed from College E is approximately what per cent of the girl students passed from College C?

(1) 165% (2) 185% (3) 205% (4) 235% (5) 275%

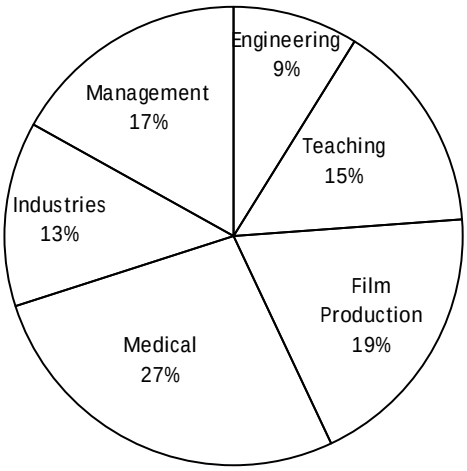
135. The number of boys who passed from College B is approximately what per cent more or less than the number of girls who passed from the same college?

(1) 67% (2) 72% (3) 81% (4) 87% (5) 92%

Directions (Q.136-140) : Study the following pie-chart and answer the following questions.

Percentage distribution of employees in six different professions

Total number of employees = 26800



136. What is the difference between the total number of employees in teaching and medical profession together and the number of employees in management profession?
(1) 6770 (2) 7700 (3) 6700 (4) 7770 (5) 7670

137. In management profession three-fourths of the number of employees are females. What is the number of male employees in management profession?
(1) 1239 (2) 1143 (3) 1156 (4) 1289 (5) 1139

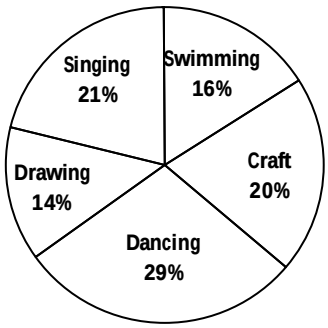
138. 25% of employees from film production profession went on a strike. What is the number of employees from film production who did not participate in the strike?
(1) 3271 (2) 3819 (3) 3948 (4) 1273 (5) 1246

139. What is the total number of employees in engineering profession and industries together?
(1) 5698 (2) 5884 (3) 5687 (4) 5896 (5) 5487

140. In teaching profession if three-fifths of the teachers are not permanent, what is the number of permanent teachers in the teaching profession?
(1) 1608 (2) 1640 (3) 1764 (4) 1704 (5) 1686

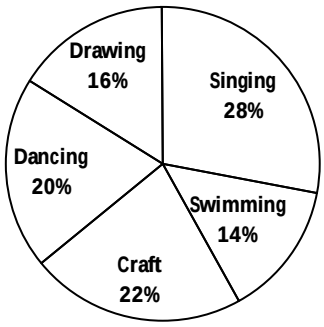
Directions (Q. 141-145): Study the charts carefully to answer the following questions:

Percentage of students enrolled in different activities in a school



Total students = 3000

Percentage break-up of girls enrolled in these activities



Total Girls = 1750

141. What is the ratio of the number of girls to boys enrolled in Swimming?
(1) 49:47 (2) 97:49 (3) 51:31 (4) 31:51 (5) None of these

142. The number of girls enrolled in Dancing form what per cent of the total number of students in the school (round off two digits after decimal)?
(1) 12.95% (2) 11.67% (3) 16.75% (4) 19.65% (5) None of these

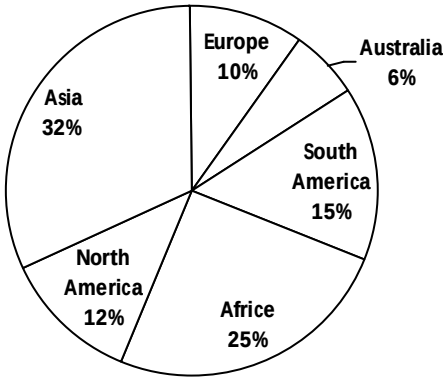
143. What is the total number of girls enrolled in Swimming and Drawing together?
(1) 625 (2) 550 (3) 490 (4) 525 (5) 455

144. How many boys are enrolled in Singing and Craft together?
(1) 610 (2) 590 (3) 640 (4) 720 (5) 355

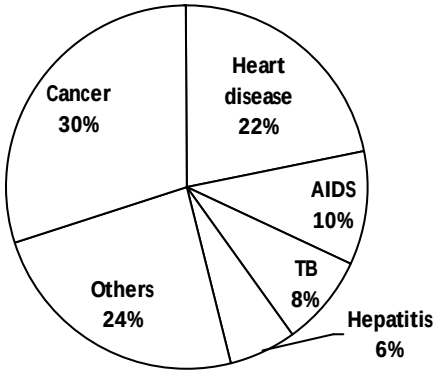
145. What is the approximate percentage of boys in the school?
(1) 42% (2) 56% (3) 49% (4) 58% (5) None of these

Directions (Q. 146-150): Study the following pie-charts carefully and answer the questions given below.

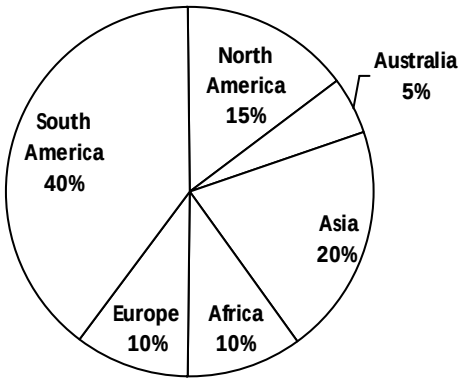
Total population of the world = 700 crore
Number of patients = 10% of the total population



Percentage of all the patients in various continents



Percentage of patients of various diseases in the world



Percentage of cancer patients in various continents

146. The number of cancer patients in Australia is what per cent of the total number of patients of heart disease in the world?
(1) 6.81% (2) 7.85% (3) 5.49% (4) 6.01% (5) 7.98%

147. If the number of cancer patients in South America decreases by 25%, what is the percentage decrease in total number of cancer patients in the world?
(1) 4% (2) 8% (3) 3% (4) 6% (5) 5%

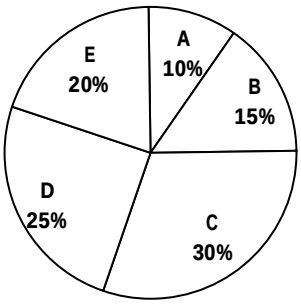
148. What is the ratio of the total number of patients in Africa to the total number of cancer patients in Asia and North America together?
(1) 350 : 347 (2) 360 : 347 (3) 350 : 334 (4) 352 : 250 (5) None of these

149. If the total number of patients increases by 10% every year in Europe then what is the difference between the total number of patients in Europe after 2 years and the total number of cancer patients in South America now?
(1) 8 lakh (2) 9 lakh (3) 6 lakh (4) 7 lakh (5) 5 lakh

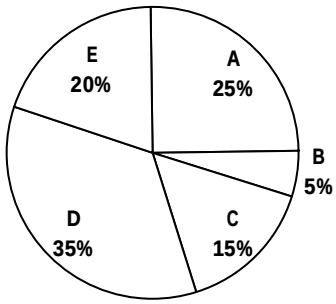
150. If the number of hepatitis patients increases by 6% and that of heart disease ones by 22%, what will be their new ratio?
(1) 1110 : 2396 (2) 1245 : 4925 (3) 1113 : 4697 (4) 1346 : 3411 (5) 1496 : 2541

Directions (Q. 151-155): Study the following pie-charts carefully and answer the questions given below:

Disturibution of candidates studying Arts and Commerce in five different Institutions A, B, C, D and E



Total number of students studying Arts = 5000



Total number of students studying Commerce = 6000

151. How many students study Arts and Commerce in Institute D and E together?
(1) 4525 (2) 5550 (3) 6550 (4) 5525 (5) 6750

152. What is the ratio of the number of students studying Arts in Institute D to that studying Commerce in Institute C?
(1) 3:5 (2) 5:3 (3) 17:25 (4) 25:17 (5) 25:18

153. The total number of students studying both Commerce and Arts in Institute B and E together is what per cent of the total number of students studying Arts?
(1) 71% (2) 61% (3) 72% (4) 51% (5) None of these

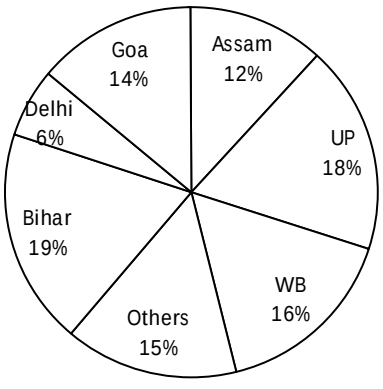
154. The number of students studying Arts in Institute A is approximately what per cent of the total number of students studying Commerce in Institute B?
(1) 167% (2) 143% (3) 198% (4) 189% (5) 193%

155. What is the ratio of the total number of students studying Arts in Institute C to that studying Commerce in Institute A and E together?
(1) 9:5 (2) 8:9 (3) 5:9 (4) 4:9 (5) 2:3

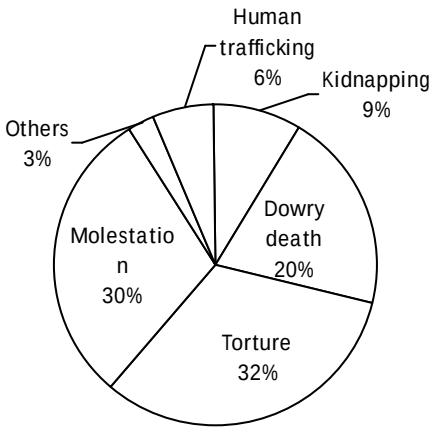
Directions (Q. 156-160): Study the following pie-charts carefully and answer the given questions.

The following pie-charts show the crimes against women in the year 2012

Total number of cases registered as crimes against women in2012 = 101akh



Statewise % crimes against women in 2012



Incidence of crimes committed against women in 2012

Note: The proportion of the nature of crimes remains the same for each state.

156. During 2012, the number of registered cases in WB and Goa together exceeded the number of cases in Assam and Others together by (in numbers)
(1) 32000 (2) 30000 (3) 31000 (4) 37000 (5) None of these

157. Approximately how many cases of Dowry deaths were registered per day in Goa in the year 2012?
(1) 77 (2) 72 (3) 78 (4) 79 (5) 70

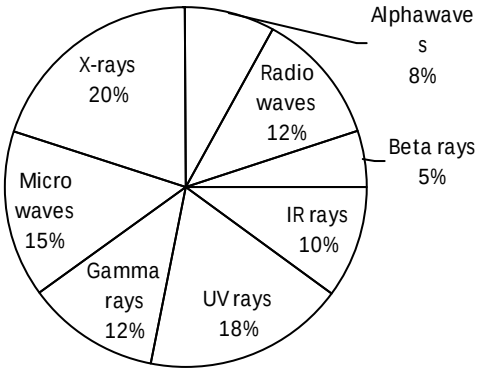
158. The number of cases of Human trafficking registered in UP exceeded that in WB by
(1) 1652 (2) 1700 (3) 1400 (4) 1200 (5) None of these

159. Which of the following crimes against women in Bihar is less than 5800?
(1) Others (2) Kidnapping (3) Dowry death (4) Torture (5) None of these

160. During 2012, the number of cases of Torture and Others together exceeded the number of cases of Molestation by
(1) 49000 (2) 30000 (3) 35000 (4) 45000 (5) None of these

Directions (Q. 161-165): Study the following pie-chart and answer the questions given below:

Constituents of sun rays received in 1 minute



Total sun rays received in 1 minute = 3600 units

161. If the human body can withstand a maximum 8750 units of IR rays when exposed to the sun continuously, then what is the maximum time that any one can stand in the sun without crossing the threshold limit of IR rays?
(1) 24.3 minutes (2) 45 minutes (3) 20 minutes (4) 15 minutes (5) 30 minutes

162. The amount of UV rays received in 5 minutes is how many times the amount of IR rays received in 2 minutes?
(1) 4 (2) 2.1 (3) 4.5 (4) 3.6 (5) 5.2

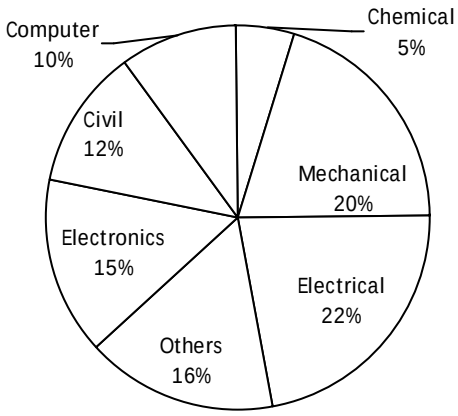
163. If presently the ozone layer in the atmosphere reflects 55% of the sun rays then what would be the amount of Gamma rays received in one minute, if the ozone layer were to disappear completely?
(1) 342 (2) 432 (3) 531 (4) 135 (5) 351

164. The amount of microwaves received in 4 minutes is how much more/less than the amount of Alpha rays received in 3 minutes?
(1) 1435 (2) 1142 (3) 1378 (4) 1296 (5) 1526

165. How many minutes of exposure to the sun in a day would be enough to ensure that the body receives enough amount of vitamin D, given that the body requires 40 units of vitamin D every day and that 30 units of Beta rays generate 1 unit of vitamin D?
(1) $4\frac{2}{3}$ (2) $3\frac{1}{3}$ (3) $5\frac{1}{3}$ (4) $6\frac{2}{3}$ (5) $7\frac{1}{3}$

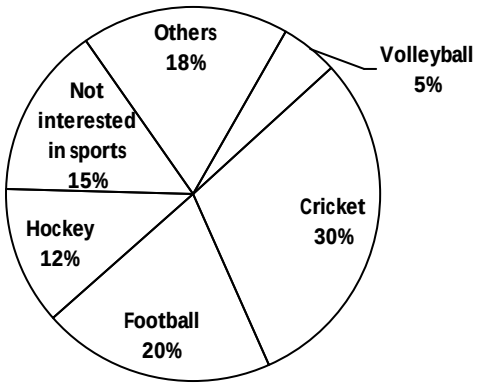
Directions (Q, 166-170): Study the pie-charts given below and answer the following questions.

Percentage of students studying in various branches of an Engineering college



Total students = 2500

Percentage of students interested in various sports of the Engineering college



Total students = 2500

166. If 10% of Civil students, 20% of Mechanical students and 12% of Electrical students are not interested in sports then what is the average number of students of these branches who are interested in sports? (Calculate approximate value)
(1) 362 (2) 378 (3) 315 (4) 385 (5) 316

167. What is the ratio of the number of students who play Volleyball to the number of students who study in Mechanical branch?
(1) 2 : 3 (2) 1 : 4 (3) 4 : 1 (4) 3 : 2 (5) 5 : 6

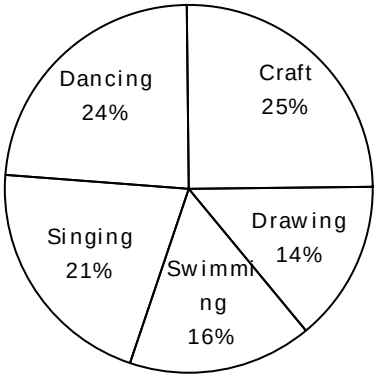
168. If 20% students of Electronics branch fail, and out of these 60% are not interested in sports, then the number of failed Electronics students who are not interested in sports is what per cent of the total number of students who are not interested in sports?
(1) 14% (2) 18% (3) 16% (4) 22% (5) 12%

169. If 50% Mechanical students and 40% Electrical students are interested in Football then what is their ratio?
(1) 25 : 22 (2) 21 : 19 (3) 22 : 37 (4) 23 : 47 (5) 17 : 11

170. The percentage of students who are interested in other games are same (20%) in all branches. What is the difference between the number of students of Electrical and Mechanical branches who are interested in other games?
(1) 12 (2) 18 (3) 10 (4) 16 (5) 15

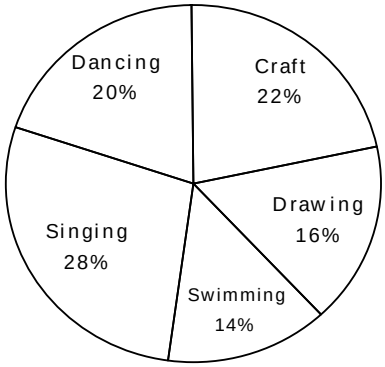
Directions (Q. 171-175): Study the following pie-charts carefully to answer the given questions.

Percentage of students enrolled for different activities in School N



total number of students = 3000

Percentage break-up of the girls enrolled for these activities



Total number of girl students = 1750

171. The number of girls enrolled for Dancing forms what per cent of the total number of students in School N?(Rounded off to two digits after decimal)
(1) 12.35 (2) 14.12 (3) 11.67 (4) 10.08 (5) None of these

172. How many boys are enrolled for Singing and Craft together?
(1) 505 (2) 610 (3) 485 (4) 420 (5) 705

173. What is the ratio of the number of girls to the number of boys enrolled for Swimming?
(1) 47 : 49 (2) 23 : 29 (3) 29 : 23 (4) 49 : 47 (5) None of these

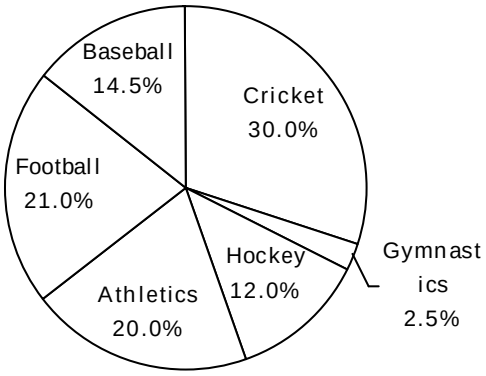
174. What is the total number of girls enrolled for Swimming and Drawing together?
(1) 480 (2) 525 (3) 505 (4) 495 (5) None of these

175. What is the approximate percentage of boys in the school?
(1) 34 (2) 56 (3) 28 (4) 50 (5) 42

Directions (Q. 176-180): Study the information carefully and answer the questions that follow:

The following pie-chart shows the percentage of employees of Bank X who are interested in different sports activities.

Total number of employees = 65000



176. The number of employees interested in Athletics is approximately what per cent of the number of employees interested in Baseball?
(1) 138% (2) 128% (3) 148% (4) 127% (5) None of these

177. What is the difference between the number of employees interested in Cricket and the total number of employees interested in Baseball, Hockey and Gymnastics together?
(1) 6500 (2) 650 (3) 6565 (4) 6050 (5) 1300

178. What is the ratio of employees interested in Gymnastics to the number of employees interested in Baseball?
(1) 5 : 39 (2) 29 : 5 (3) 25 : 29 (4) 14 : 29 (5) 5 : 29

179. The number of employees interested in Hockey is approximately what per cent of the employees interested in Football, Atheletics and Baseball together?
(1) 32% (2) 42% (3) 22% (4) 52% (5) 18%

180. The number of employees interested in Gymnastics is what percentage of the number of employees interested in Hockey? (Calculate approximate percentage)
(1) 21% (2) 31% (3) $16\frac{2}{3}\%$ (4) $33\frac{1}{3}\%$ (5) 50%

SHORT ANSWER

1. (1)	2. (3)	3. (4)	4. (2)	5. (3)	6. (3)	7. (1)	8. (2)
9. (5)	10. (1)	11. (2)	12. (5)	13. (4)	14. (3)	15. (2)	16. (4)
17. (5)	18. (3)	19. (3)	20. (1)	21. (1)	22. (2)	23. (5)	24. (2)
25. (1)	26. (2)	27. (3)	28. (4)	29. (1)	30. (5)	31. (1)	32. (5)
33. (3)	34. (3)	35. (2)	36. (1)	37. (5)	38. (4)	39. (2)	40. (1)
41. (3)	42. (2)	43. (4)	44. (2)	45. (1)	46. (4)	47. (2)	48. (3)
49. (1)	50. (5)	51. (5)	52. (1)	53. (5)	54. (5)	55. (2)	56. (2)
57. (5)	58. (3)	59. (2)	60. (5)	61. (1)	62. (4)	63. (3)	64. (4)
65. (5)	66. (2)	67. (3)	68. (3)	69. (4)	70. (3)	71. (1)	72. (5)
73. (2)	74. (1)	75. (3)	76. (3)	77. (3)	78. (4)	79. (5)	80. (2)
81. (3)	82. (5)	83. (3)	84. (3)	85. (2)	86. (3)	87. (4)	88. (4)
89. (5)	90. (5)	91. (1)	92. (4)	93. (5)	94. (2)	95. (2)	96. (2)
97. (3)	98. (4)	99. (1)	100. (1)	101. (4)	102. (1)	103. (5)	104. (3)
105. (4)	106. (1)	107. (2)	108. (1)	109. (3)	110. (2)	111. (3)	112. (3)
113. (5)	114. (1)	115. (2)	116. (4)	117. (1)	118. (2)	119. (5)	120. (3)
121. (2)	122. (5)	123. (2)	124. (4)	125. (4)	126. (2)	127. (2)	128. (3)
129. (4)	130. (1)	131. (1)	132. (3)	133. (5)	134. (4)	135. (3)	136. (3)
137. (5)	138. (2)	139. (4)	140. (1)	141. (1)	142. (2)	143. (4)	144. (5)
145. (1)	146. (1)	147. (3)	148. (5)	149. (4)	150. (3)	151. (2)	152. (5)
153. (5)	154. (1)	155. (3)	156. (2)	157. (1)	158. (4)	159. (1)	160. (5)
161. (1)	162. (3)	163. (2)	164. (4)	165. (4)	166. (4)	167. (2)	168. (5)
169. (1)	170. (3)	171. (3)	172. (1)	173. (4)	174. (2)	175. (5)	176. (1)
177. (2)	178. (5)	179. (3)	180. (1)				

DETAIL EXPLANATIONS

1. 1; Total number of people in all six cities

$= \frac{1526000 \times 100}{21.8} = 7000000$

$\therefore$ Total population of City A

$= 7000000 \times \frac{21}{100} = 1470000$

$\text{Male}_A = 1470000 \times \frac{51.1}{100} = 751170$

$\therefore \text{Female}_A = 1470000 - 751170 = 718830$

2. 3; $\text{Total}_B = 7000000 \times \frac{10.6}{100} = 742000$

$\because$ Males are 53.2%, so females
$= 100 - 53.2 = 46.8\%$
$\therefore$ Diff $= 53.2 - 46.8 = 6.4\%$

$\therefore$ Reqd answer $= 742000 \times \frac{6.4}{100} = 47488$

3. 4; $\text{Female}_E = 7000000 \times \frac{21.8}{100} \times \frac{(100 - 49.2)}{100}$

$= 700 \times 21.8 \times 50.8 = 775208$

$\text{Female}_F = 7000000 \times \frac{7.5}{100} \times \frac{(100 - 47.9)}{100}$

$= 700 \times 7.5 \times 52.1 = 273525$

$\therefore$ Reqd % $= \frac{775208}{273525} \times 100 = 283.4 \approx 283.5\%$

4. 2; Total males $= \frac{7000000}{100 \times 100}$ {21 × 51.1 + 10.6 × 53.2 + 23.7 × 52.9 + 15.4 × 53.8 + 7.5 × 47.9 + 21.8 × 49.2}
$= 700$ {1073.1 + 563.92 + 1253.73 + 828.52 + 359.25 + 1072.56}
$= 700 \times 5151.08 = 3605756$

5. 3; Total population in all six cities = 7000000
Total females in all six cities
$= 7000000 - 3605756 = 3394244$

$\therefore$ Reqd % $= \frac{3394244}{7000000} \times 100 = 48.489 \approx 48.5\%$

6. 3; $A_{M1} = 3.2 \times \frac{20}{100} \times \frac{13}{20} \times \frac{25}{100} = 0.104$ crore

7. 1; $B_{M2} = 3.2 \times \frac{14}{100} \times \frac{5}{14} \times \frac{30}{100} = 0.048$

$C_{M2} = 3.2 \times \frac{22}{100} \times \frac{5}{11} \times \frac{24}{100} = 0.0768$

$\therefore$ Sum = 0.048 + 0.0768 = 0.1248

8. 2; $(\text{Production-}M_1) = 3.2 \times \frac{13}{100} \times \frac{6}{13} = \frac{3.2 \times 6}{100}$

$(\text{Production-}M_2) = 3.2 \times \frac{21}{100} \times \frac{10}{21} = \frac{3.2 \times 10}{100}$

$\therefore$ Ratio $= \frac{6}{10} = \frac{3}{5}$

9. 5; $C_{M1} = 3.2 \times \frac{22}{100} \times \frac{6}{11} \times \frac{20}{100} = 0.0768$ crore

$E_{M2} = 3.2 \times \frac{10}{100} \times \frac{3}{5} \times \frac{21}{100} = 0.04032$ crore

$\therefore$ Diff = 0.0768 - 0.04032 = 0.03648

10. 1; $\%P_{BM1} = 28\%, \%P_{DM2} = 25\%$

$\therefore$ Reqd % $= \frac{28}{25} \times 100 = 112\%$

11. 2; Total students = 30000 × 0.18 = 5400
Girls = 12000 × 0.14 = 1680
$\therefore$ Boys = 5400 – 1680 = 3720
$\therefore$ Diff = 3720 – 1680 = 2040

12. 5; $\text{Total}_c = 30000 \times 0.23 = 6900$
$\text{Girls}_c = 12000 \times 0.18 = 2160$
$\text{Total}_E = 30000 \times 0.16 = 4800$
$\text{Girls}_E = 12000 \times 0.20 = 2400$
$\therefore \text{Boys}_E = 4800 - 2400 = 2400$

$\therefore$ Reqd % $= \frac{2160}{2400} \times 100 = 90\%$

13. 4; $\text{Boys}_A = (30000 \times 0.10) - (12000 \times 0.15)$
$= 3000 - 1800 = 1200$
$\text{Boys}_B = (30000 \times 0.09) - (12000 \times 0.12)$
$= 2700 - 1440 = 1260$
$\text{Boys}_c = (30000 \times 0.23) - (12000 \times 0.18)$
$= 6900 - 2160 = 4740$
$\therefore$ Avg $= (1200 + 1260 + 4740) \div 3$
$= 7200 \div 3 = 2400$

14. 3; $\text{Girls}_F = 12000 \times 0.21 = 2520$
$\text{Girls}_A = 12000 \times .15 = 1800$

$\therefore$ rise% $= \frac{(2520 - 1800)}{1800} \times 100 = \frac{72000}{1800} = 40\%$

15. 2; $\text{Boys}_F = (30000 \times 0.24) - (12000 \times 0.21)$
$= 7200 - 2520 = 4680$
$\text{Boys}_D = (30000 \times 0.18) - (12000 \times 0.14)$
$= 5400 - 1680 = 3720$

$\therefore$ Reqd % = $\frac{(4680-3720)}{3720}\times 100$

$=\frac{96000}{3720}=25.8\%$

16. 4; Diff = 26% – 6% = 20%
Half = 10%
$\therefore$ 10% of 6500 = 650

17. 5; $\frac{26-18}{18}\times 100=44.44\%$

18. 3; 18% + 13% + 6% = 37
37% of 6500 = 2405

19. 3; Reqd ratio = 13 : 7

20. 1; Reqd % = $\frac{30}{18}\times 100=166.67\%$

21. 1; Population of A = 1287000
$\therefore$ Total population of all six cities

$=\frac{1287000}{23.4}\times 100=5500000$

$\therefore \text{Adult}_c=5500000\times\frac{8.4}{100}\times\frac{73}{100}=337262$

22. 2; A = 55 × $\frac{23.4}{100}$ lakh, D = 55 × $\frac{18.9}{100}$ lakh

$\therefore$ Reqd % = $\frac{23.4}{18.9}\times 100=123.8$

23. 5; In City F, adult population is 72%. So, population of non-adults is 28%.

$\therefore$ Reqd answer = $5500000\times\frac{10.2}{100}\times\frac{28}{100}$

= 157080

24. 2; $\text{Adult}_{(B+C)}=\frac{5500000}{100\times 100}\{21.6\times 68+8.4\times 73\}$

= 550 × (1468.8 + 613.2) = 550 × 2082
Total population of all six cities = 5500000

$\therefore$ Reqd % = $\frac{550\times 2082}{5500000}\times 100$

$=20.82\%\approx 21\%$

25. 1; Total population of D = 18.9% of 55 lakh
Total population of E = 17.5% of 55 lakh

$\therefore$ Reqd% = $\frac{(18.9-17.5)\times 100}{17.5}=\frac{140}{17.5}=8\%$

26. 2; $\text{Total}_{2010}=180\times\frac{360^\circ}{43.2^\circ}=1500$

$\text{Total}_{2011}=270\times\frac{360^\circ}{54^\circ}=1800$

$\therefore$ Reqd % = $\frac{1800}{1500}\times 100=120\%$

27. 3; $S_4(2010)=1500\times\frac{36^\circ}{360^\circ}=150$

$S_4(2011)=1800\times\frac{50.4^\circ}{360^\circ}=252$

$\therefore$ % rise = $\frac{252-150}{150}\times 100=\frac{10200}{150}=68\%$

28. 4; $S_5+S_7=1500\times\frac{(64.8^\circ+43.2^\circ)}{360^\circ}$

$=\frac{1500\times 108^\circ}{360^\circ}=450$

$S_2=1800\times\frac{28.8^\circ}{360^\circ}=144$

$\therefore$ Reqd% = $\frac{450}{144}\times 100=312.5\%$

29. 1; Avg of S_1, S_2 and S_3

$=\frac{(75.6+50.4+57.6)}{360\times 3}\times 1500=255$

Avg of S_5, S_6 and S_7

$=\frac{(57.6+46.8+54)}{360\times 3}\times 1800=264$

$\therefore$ Diff = 264 – 255 = 9

30. 5; $S_1=\frac{(414-315)}{315}\times 100=31.42\%$

$S_2=\frac{(210-144)}{210}\times 100=31.42\%$

$S_3=\frac{(240-198)}{240}\times 100=17.5\%$

$S_4=\frac{(252-150)}{150}\times 100=68\%$

$S_5=\frac{(288-270)}{270}\times 100=6.66\%$

$S_6=\frac{(234-135)}{135}\times 100=73.33\%$

$S_7=\frac{(270-180)}{180}\times 100=50\%$

31. 1; Total students in class V

$= \frac{72}{360} \times 1200 = 240$

$\therefore$ Girls $= \frac{240}{5} \times 2 = 96$

Total students in class VI

$= \frac{43.2}{360} \times 1200 = 144$

$\therefore$ Girls $= \frac{144}{4} \times 1 = 36$

Similarly, $VII_{girls} = 72$, $VIII_{girls} = 84$,
$IX_{girls} = 108$, $X_{girls} = 96$

$\therefore$ Avg $= \frac{96+36+72+84+108+96}{6} = \frac{492}{6}$

$= 82$

32. 5; Total girls = 492
Total boys = 1200 – 492 = 708
Diff = 708 – 492 = 216

33. 3; $Total_{VIII} = \frac{54}{360} \times 1200 = 180$

$\therefore$ Boys $= \frac{180}{15} \times 8 = 96$

$Total_X = \frac{57.6}{360} \times 1200 = 192$

$\therefore$ Boys $= \frac{192}{2} \times 1 = 96$

34. 3; $Total_V = \frac{72}{360} \times 1200 = 240$

$\therefore$ $Boys_V = \frac{240}{5} \times 3 = 144$, $Girls_V = 96$

$\therefore$ Diff = 48

$Total_{VII} = \frac{57.6}{360} \times 1200 = 192$

$\therefore$ $Boys_{VII} = \frac{192}{8} \times 5 = 120$, $Girls_{VII} = 72$

$\therefore$ Diff = 48

$\therefore$ Reqd% $= \frac{48}{48} \times 100 = 100\%$

35. 2; $Boys_{VI} = 108$
$Girls_X = 96$

$\therefore$ Reqd% $= \frac{108-96}{96} \times 100 = \frac{1200}{96} = 12.5\%$

36. 1; Total population of A

$= 1.5 \times \frac{16}{100} = 0.24$ crore $= 2400000$

Total literate males of A

$40 \times \frac{18}{100} = 7.2$ lakh $= 720000$

Total literate females of A

$= 25 \times \frac{24}{100} = 6$ lakh $= 600000$

$\therefore$ Total illiterate population
$= 2400000 - (720000 + 600000) = 1080000$

37. 5; (E) Literate males $= 40 \times \frac{19}{100} = 7.61$akh

(F) Literate females $= 25 \times \frac{10}{100} = 2.5$ lakh

$\therefore$ Reqd% $= \frac{7.6}{2.5} \times 100 = 304\%$

38. 4; Total population of E

$= 1.5 \times \frac{20}{100} = 0.30$ crore $= 30$ lakh

Total literate males of E

$= 40 \times \frac{19}{100} = 7.6$ lakh

Total literate females of E

$= 25 \times \frac{20}{100} = 5$ 1akh

$\therefore$ Total literate = 7.6 + 5 = 12.6 lakh

$\therefore$ Reqd% $= \frac{12.6}{30} \times 100 = 42\%$

39. 2; TotaL $= 1.5 \times \frac{21}{100} = 0.315$ crore $= 31.5$ lakh

Literate males $= 40 \times \frac{20}{100} = 8$ lakh

Literate females $= 25 \times \frac{12}{100} = 3$ lakh

$\therefore$ Total literate = 8 + 3 = 11 lakh
$\therefore$ Total illiterate = 31.5 – 11 = 20.5 lakh
$\therefore$ Difference = 20.5 – 11 = 9.5 lakh

40. 1; Literate males = $40 \times \frac{16}{100}$ = 6.4 lakh

Literate females = $25 \times \frac{16}{100}$ = 4 lakh

Reqd % = $\frac{(6.4-4)}{4} \times 100 = 60\%$

41. 3; Production cost

$= 24\left[\frac{10}{100} \times \frac{3}{10} + \frac{17}{100} \times \frac{8}{17}\right]$

= 24[0.03 + 0.08] = 24 × 0.11 = 2.64 crore

42. 2; $B_{I_1} = 24 \times \frac{20}{100} \times \frac{2}{5} = 1.92$ crore

$C_{I_2} = 24 \times \frac{15}{100} \times \frac{7}{15} = 1.68$ crore

∴ Diff = 1.92 – 1.68 = 0.24 crore = 24 lakh

43. 4; $\text{Profit}_{(I_1+I_2)} = 24 \times \frac{25}{100}\left[\frac{14}{25} \times \frac{20}{100} + \frac{11}{25} \times \frac{30}{100}\right]$

Profit = $24 \times \frac{25}{100} \times \frac{1}{250}[28 + 33] = 1.464$ crore

44. 2; $\text{Profit}_B = 24 \times \frac{20}{100} \times \frac{3}{5} \times \frac{25}{100} = 0.72$ crore

$\text{Profit}_D = 24 \times \frac{13}{100} \times \frac{8}{13} \times \frac{30}{100} = 0.576$ crore

∴ $\text{Profit}_{(B+D)}$ = 0.72 + 0.576 = 1.296 crore

45. 1; $\text{Profit}_A = 24 \times \frac{25}{100} \times \frac{14}{25} \times \frac{20}{100}$

$\text{Profit}_E = 24 \times \frac{10}{100} \times \frac{7}{10} \times \frac{25}{100}$

∴ Ratio = $\frac{14 \times 20}{7 \times 25} = \frac{8}{5}$

46. 4; Total items = $\frac{[(24+18)-28]}{100} \times 16$

= 2.24 lakh

47. 2; $\text{Total}_F = 16 \times \frac{8}{100}$ = 1.28 lakh,

$I_1 = \frac{1.28}{5} \times 3 = 0.768$

$I_2 = \frac{1.28}{5} \times 2 = 0.512$

∴ Diff = 0.768 – 0.512 = 0.256 lakh = 25600

48. 3; I_1 sold by A = $16 \times \frac{24}{100} \times \frac{5}{8} \times \frac{65}{100}$ = 1.56 lakh

Similarly,

Total I_1 = 1.56 + 0.896 + 0.6912 + 1.44 + 0.4096 + 0.384 = 5.3808 lakh

∴ Average = $\frac{5.3808}{6}$ = 0.8968 lakh = 89680

49. 1; $I_1 = 16 \times \frac{7}{100} \times \frac{4}{7} \times \frac{64}{100} = 0.4096$,

Similary, I_2 = 0.2640

∴ Diff = 0.4096 - 0.2640 = 0.1456 lakh

= 14560 lakhs

50. 5; I_1 sold by A = 156000,

I_1 sold by F = 38400

∴ Reqd % = $\frac{156000}{38400} \times 100 = 406.25\%$

51. 5; Total number

$= \frac{90000}{100}\left[\frac{14.3 \times 7}{18} + \frac{16.2 \times 5}{9} + \frac{18.4 \times 3}{10} + \frac{16.8 \times 3}{9} + \frac{12.6 \times 2}{5} + \frac{21.7 \times 2}{10}\right]$

= 5005 + 8100 + 4968 + 5040 + 4536 + 3906 = 31555

52. 1; $T_D = 90000 \times \frac{16.8}{100} \times \frac{4}{9} = 6720$,

$T_E = 90000 \times \frac{12.6}{100} \times \frac{2}{5} = 4536$

∴ Diff = 6720 - 4536 = 2184

53. 5; $M_{1\text{-}D} = 90000 \times \frac{16.8}{100} \times \frac{4}{9} = 6720$,

$M_{3\text{-}A} = 90000 \times \frac{14.3}{100} \times \frac{4}{18} = 2860$

∴ Reqd % = $\frac{6720}{2860} \times 100 = 234.96 = 235\%$

54. 5; $\text{Total}_F = \frac{90000}{100} \times 21.7 = 19530$,

$\text{Total}_B = \frac{90000}{100} \times 16.2 = 14580$

∴ Reqd % = $\frac{(19530-14580)}{14580} \times 100$

$= \frac{495000}{14580} = 33.95 = 34$

55. 2; $\text{Total}_C = \frac{90000}{100} \times 18.4 = 16560$

$M_{2\text{-}D} = \frac{90000}{100} \times 16.8 \times \frac{3}{9} = 5040$

$\therefore$ Ratio $= \frac{16560}{5040} = \frac{23}{7}$

56. 2; $\frac{61.2}{360} \times 48600 + \frac{28.8}{360} \times 62500$

$= 8262 + 5000 = 13262$

57. 5; $B_{2008} = \frac{64.8}{360} \times 48600 = 8748,$

$B_{2009} = \frac{54}{360} \times 62500 = 9375$

$\therefore \% = \frac{8748}{9375} \times 100 = 93.31\% \approx 93\%$

58. 3

59. 2; Sum = 8262 + 8748 + 8262 + 11250 + 9375 + 5000 = 50897

60. 5; $E_{2009} = \frac{64.8}{360} \times 62500 = 11250,$

$C_{2009} = \frac{28.8}{360} \times 62500 = 5000$

Percentage $= \frac{11250 - 5000}{5000} \times 100 = \frac{625000}{5000}$

$= 125\%$

61. 1; Difference $= 8.6 \times \frac{22}{100} - 5.4 \times \frac{15}{100}$

$= 1.892 - 0.81 = 1.082$ lakh

62. 4; $H_{2000} = 5.4 \times \frac{10}{100} = 0.54$ lakh,

$H_{2010} = 8.6 \times \frac{8}{100} = 0.688$

Avg $= \frac{0.54 + 0.688}{2} = \frac{1.228}{2}$ lakh $= 61400$

63. 3; Sum $= 5.4 \times \frac{8}{100} + 8.6 \times \frac{18}{100}$

$= 0.432 + 1.548 = 1.98$ lakh

64. 4; Total number of vacancies in 2010

$= \frac{48000 \times 100}{6} = 800000$

$\therefore$ vacancies in NCR = 20% of 800000

$= 160000$

65. 5; $H_{20OO} = 5.4 \times \frac{10}{100} = 0.54$ lakh

$H_{2010} = 8.6 \times \frac{8}{100} = 0.688$ lakh

$\therefore$ % rise $= \frac{(0.688 - 0.54)}{0.54} \times 100 \approx 27.4\%$

66. 2; Total Marks in Unit Test – 2 in (Physics + Chemistry + Math)

$= \frac{800}{100}(18.125 + 18 + 21.75)$

$= 8 \times 57.875 = 463$

67. 3; Chemistry = 800 × $\frac{80}{100}$ = 144

English = 750 × $\frac{18}{100} = 135$

$\therefore$ Difference = 144 – 135 = 9

68. 3; $GK_1 = 750 \times \frac{10}{100} = 75$

$GK_2 = 800 \times \frac{12.375}{100} = 99$

$\therefore$ % Rise $= \frac{99 - 75}{75} \times 100 = \frac{2400}{75} = 32\%$

69. 4; $\text{Math}_1 = 750 \times \frac{20}{100} = 150$

$\text{Math}_2 = 800 \times \frac{21.75}{100} = 174$

$\therefore$ Reqd% $= \frac{174}{150} \times 100 = 116\%$

70. 3; Physics (Test-1 + Test-2)

$= 750 \times \frac{22}{100} + 800 \times \frac{18.125}{100} = 165 + 145 = 310$

Hindi (Testl + Test2)

$= 750 \times \frac{14}{100} + 800 \times \frac{16.25}{100} = 105 + 130 = 235$

$\therefore$ Reqd % $= \frac{310 - 235}{235} \times 100 = \frac{7500}{235}$

$= 31.91\% \approx 32\%$

71. 1; Total = 8000

Graduate : Non-graduate = 3 : 2

$\therefore$ Graduate = 4800 and Non-graduate = 3200

$\therefore$ Graduate F = 4800 × $\frac{24}{100} = 1152$

$\therefore$ Non-graduate F = $8000 \times \frac{21}{100} - 1152$

$\therefore$ 1680 – 1152 = 528

72. 5; No relationship between the number of males and the number of graduates is given. Hence, (5).

73. 2; Total = 8000 $\Rightarrow$ Male : Female = 3 : 5

$\therefore$ Males = $\frac{8000}{8} \times 3 = 3000$

And Females = 8000 - 3000 = 5000

$\therefore$ Difference = 5000 - 3000 = 2000

74. 1; $G_C = 4800 \times \frac{13}{100} = 624$

$NG_E = 8000 \times \frac{15}{100} - 4800 \times \frac{20}{100}$

= 1200 - 960 = 240

$\therefore$ Reqd % = $\frac{624}{240} \times 100 = 260\%$

75. 3; $Male_A = 3000 \times \frac{25}{100} = 750$

$Female_F = 8000 \times \frac{21}{100} - 3000 \times 12$

= 1680 - 360 = 1320

$\therefore$ Reqd % = $\frac{(1320 - 750)}{750} \times 100$

$\therefore \frac{57000}{750} = 76\%$

76. 3; $Male_A = \frac{20}{100} \times 1500 = 300$

$Female_D = 2200 \times \frac{12}{100} - 1500 \times \frac{12}{100}$

= 264 - 180 = 84

$\therefore$ Reqd% = $\frac{84}{300} \times 100 = 28\%$

77. 3; $Male_F = 1500 \times \frac{22}{100} = 330$

$Male_P = \frac{22}{100} \times 1500 = 330$

$\therefore$ $Male_B + Male_F = 660$

$\therefore$ Reqd% = $\frac{660}{2200} \times 100 = 30\%$

78. 4; $Male_E = 1500 \times \frac{10}{100} = 150$

$Female_c = 2200 \times \frac{15}{100} - 1500 \times \frac{14}{100}$

= 330 - 210 = 120

$\therefore$ Reqd% = $\frac{150 - 120}{120} \times 100$

$= \frac{30}{120} \times 100 = 25\%$

79. 5; $E_{Female} = 2200 \times \frac{9}{100} - 1500 \times \frac{10}{100}$

= 198 – 150 = 48

$D_{Total} = 2200 \times \frac{12}{100} = 264$

$\therefore$ Ratio = $\frac{264}{48} = \frac{11}{2} = 11 : 2$

80. 2; $C_{Total} = 330$ $C_{Female} = 120$

Now, $C_{1\ Total} = 330 + 20 = 350$

$C_{1\ Female} = 120 + 20 = 140$

$\therefore$ Reqd% = $\frac{140}{350} \times 100 = 40\%$

81. 3; Average of female doctors = $\frac{1800}{7}$

= 257.1 $\approx$ 257

In City A, female doctors = 432

In City B, female doctors = 288

In City C, female doctors.= 198

In City D, female doctors = 144

In City E, female doctors = 288

In City F, female doctors = 342

In City G, female doctors = 108

There are four cities in which the number of female doctors is more than the average number of female doctors.

These Cities are A, B, E and F.

82. 5; Total doctors in F = $4800 \times \frac{14}{100} = 672$

Female doctors in F = $1800 \times \frac{19}{100} = 342$

$\therefore$ Male doctors = 672 – 342 = 330

83. 3; Total number of doctors in city A

$= 4800 \times \frac{19}{100} = 912$

$Female_A = 1800 \times \frac{24}{100} = 432$

$Male_A$ = 912 – 432 = 480

Reqd % = $\frac{432}{480} \times 100 = 90\%$

84. 3; Number of male doctors in City A

$= 4800 \times \frac{19}{100} - 1800 \times \frac{24}{100} = 912 - 432 = 480$

Similarly,

Number of male doctors in City B

$= 21 \times \frac{4800}{100} - 1800 \times \frac{16}{100} = 1008 - 288 = 720$

And the number of male doctors in City C

$= 9 \times \frac{4800}{100} - 1800 \times \frac{11}{100} = 432 - 198 = 234$

Total number of male doctors in cities A, B and C together = 480 + 720 + 234 = 1434
Total number of male doctors in cities E, F and G together = 528 + 330 + 276 = 1134

$\therefore$ Average of (A, B, C) = $\frac{1434}{3}$ = 478

$\therefore$ Average of (E, F, G)

$= \frac{528 + 330 + 276}{3} = 378$

$\therefore$ Difference = 478 – 378 = 100

85. 2; $D_{Total} = 4800 \times \frac{12}{100} = 576$

$D_{Female} = 1800 \times \frac{8}{100} = 144$

$B_{Total} = 4800 \times \frac{21}{100} = 1008$

$Female_B = 1800 \times \frac{16}{100} = 288$

$Male_B$ = 720

$\therefore$ Reqd % = $\frac{144}{720} \times 100 = 20\%$

86. 3; Number of teachers in University B

$= \frac{17 \times 6400}{100} = 1088$

Number of teachers in University D

$= \frac{6 \times 6400}{100} = 384$

Number of teachers in University E

$= \frac{29 \times 6400}{100} = 1856$

$\therefore$ Required percentage = $\frac{1088}{1856 + 384} \times 100$

$= \frac{108800}{2240} = 48.57 \approx 49\%$

87. 4; Number of teachers in University C

$= \frac{19 \times 6400}{100} = 1216$

Number of female teachers in University C $= 1216 \times \frac{25}{100} = 1216 \times \frac{1}{4} = 304$

Number of male teachers in University C = 1216 – 304 = 912

88. 4; Number of teachers in University A

$= \frac{11 \times 6400}{100} = 704$

Number of teachers in University B

$= \frac{17 \times 6400}{100} = 1088$

Number of teachers- in University C

$= \frac{19 \times 6400}{100} = 1216$

Number of teachers in University D

$= \frac{6 \times 6400}{100} = 384$

Number of teachers in University E

$= \frac{29 \times 6400}{100} = 1856$

Number of teachers in University F

$= \frac{18 \times 6400}{100} = 1152$

$\therefore$ Difference = 3392 - 3008 = 384

Quicker Method:

Difference = (D + E + F)% – (A + B + C)%}
= (53 – 47) = 6%
6% of 6400 = 384
Hence, University of D is equal to 6%.

89. 5; Number of teachers in University F

$= \frac{18 \times 6400}{100} = 1152$

Number of professors in University F

$= 1152 \times \frac{1}{36} = 32$

$\therefore$ Total salary of professors in University F = 32 × 96000 = 30.72 lakh

90. 5; Average $= \frac{704 + 1216 + 384 + 1152}{4} = \frac{3456}{4}$ = 864

91. 1; Central angle = (12 + 15 + 14) × $\frac{360}{100}$

= 41 × 3.6 = 147.6°

92. 4; Car $A_{2005} = \frac{10}{100}$ × 32000 = 3200

Car $A_{2010} = \frac{20}{100}$ × 60000 = 7200

$\therefore$ % rise = $\frac{7200 - 3200}{3200}$ × 100 = 125%

93. 5; Ratio = $\frac{0.14 \times 32000}{0.24 \times 60000} = \frac{14}{45} = 14 : 45$

94. 2; Car D_{2010} = 0.14 × 60000 = 8400
Car C_{2005} = 0.20 × 32000 = 6400

$\therefore$ Reqd% = $\frac{8400}{6400} \times 100 = 131.25$

95. 2; Reqd % = $\frac{6000-5120}{5120} \times 100 = 17.1875 \approx 17\%$

96. 2; Number of employees in Academic affairs

$= \frac{27 \times 12600}{100} = 3402$

Number of employees in Examination department $= \frac{21 \times 12600}{100} = 2646$

$\therefore$ Reqd % = $\frac{3402-2646}{2646} \times 100$

$= \frac{756}{2646} \times 100 = 28.57 \approx 29\%$

97. 3; Number of employees in Research department $= \frac{15 \times 12600}{100} = 1890$

$\therefore$ female employees in Research department $= \frac{1890 \times 30}{100} = 567$

Hence number of male employees in Research department = 1890 - 567 = 1323

98. 4; Number of employees in examination department $= \frac{21 \times 12600}{100} = 2646$

Number of employees in the HR department $= \frac{11 \times 12600}{100} = 1386$

Number of employees in Academic Affairs

$= \frac{27 \times 12600}{100} = 3402$

$\therefore$ Total number of employees in both the departments Academic Affairs and HR together = 3402 + 1386 = 4788

$\therefore$ Reqd % = $\frac{2646}{4748} \times 100 = 55.26 \approx 55$

99. 1; Number of employees in Accounts Department = $\frac{17 \times 12600}{100}$ = 1890

$\therefore$ Average = $\frac{2142+1134+1890}{3} = \frac{5166}{3} = 1722$

100. 1; Difference = (38% of 12600 – 20% of 12600)

= 18% of 12600 = $\frac{18 \times 12600}{100} = 2268$

101. 4; Sales of Ashok Leyland in FY 2010-11 = 40 thousand FY 2011-12 = 40 × 1.125 = 45000

Sales of Eicher = 45000 × $\frac{8}{40}$ = 9000 units

102. 1; Number of buses by Isuzy

= 3375 × $\frac{7}{3}$ = 7875 units

Number of buses sold by Tata

= 3375 × $\frac{41}{3}$ = 46125

SUVs sold by Tata = 46125 × $\frac{12}{20}$ = 27675

$\therefore$ Reqd % = $\frac{7875}{27675} \times 100 = 28.45 \approx 28.5\%$

103. 5; Let total vehicles sold by all companies = 100

Vehicles sold by Eicher = 8

Vehicles sold by Tata = 41

SUVs sold by Tata = $\frac{41 \times 12}{20} = 24.6$

$\therefore$ Ratio = $\frac{8}{24.6} = \frac{40}{123} = 40:123$

104. 3; Let total buses sold = 100

Number of Volvos sold = 3

Number of MUVs sold by Tata $= \frac{41}{20} \times 18 = \frac{369}{10}$

Reqd % = $\frac{3 \times 10}{369} \times 100$ = 8.13% ≈ 8% (approx)

105. 4; Average of Volvo, Isuzy, Eicher and Mahindra = $\frac{3+7+8+1}{4} = \frac{19}{4}\%$

Now, sales of Ashok Leyland in FY 2010-11 = 40 thousand

FY 2011-12 = 40 × 1.125 = 45000

$\frac{19}{4}\% = \frac{45000}{40} \times \frac{19}{4} = 5343.8$

106. 1; Expenditure on electricity and diesel in the year 2000-01 = 7.8% of 15432 = Rs 1203.696 crore

And expenditure on electricity and diesel in the year 2010-11 = 8.6% of 35349 = Rs 3040.014 crore

Exceeding amount = 3040.014 - 1203.696 = 1836.318 crore ≈ 1840 crore

107. 2; Expenses on fertilisers in the year 2000-01 = 17.5% of 15432 = 2700.6 crore = 2701 crore

Now, the expenses on fertilisers in the year 2010-11 = 28.8% of 35349 = 10180.512 crore

$\therefore$ Difference = (10180.512 ≈ 10181)

= 10181 - 2701 = 7480 crore

Number of times = $\frac{7480}{2701} = 2.76 \approx 3$ times

108. 1; Expenses on Fertilisers in 2000-01 = 2700.6 crore

And that on Feed in 2000-01

= 29.5% of 15432 = 4552.44 crore

Total = 2700.6 + 4552.44 = 7253.04 crore ≈ 7253 crore

109. 3; Expenses on Feed in 2000-01 = Rs 4552.44 crore

And the expenses on Feed in 2010-11

= 19.5% of 35349 = Rs 6893 crore

$$\% \text{ increase} = \frac{6893-4552}{4552}\times 100$$

$$= 51.427 \approx 51\%$$

110. 2; Expenses on Electricity and Diesel in 2000-01 = 7.8% of 15432 = 1203.696 crore And in the year 2010-11 expenses = 8.6% of 35349

= 3040.014 crore

∴ Difference of expenses on the same = 3040.014 - 1203.696 = 1836.318 crore

$$\text{Number of times of increase} = \frac{1836.318}{1203.696}$$

= 1.525 ≈ 1.53 times

111. 3; Reqd difference in the number of qualified candidate

= 36% of 16500 - 18% of 16500

$$= 18\% \text{ of } 16500 = \frac{18\times 16500}{100} = 2970$$

112. 3; The number of qualified candidates (UP + Bihar) → 24% of 16500 = 3960

No. of candidates appeared (Delhi and Haryana) → 33% of 2 lakh = 66000

$$\text{Reqd}\% = \frac{3960}{66000}\times 100 = 6\%$$

113. 5; The number of candidates appeared from UP → 18% of 2 lakh = 36000

The number of candidates qualified from Mumbai, Delhi, Haryana and Punjab = 54% of 16500 = 8910

$$\therefore \text{Ratio} = \frac{36000}{8910} = \frac{400}{99} = 400:99$$

114. 1; Haryana

115. 2; Gujarat

116. 4; Total number of students from City A

$$= 39000\times\frac{17}{100} = 6630$$

Total number of boys from City A

$$= \frac{12000}{360}\times 82.8 = 2760$$

∴ Girls = 6630 - 2760 = 3870

117. 1; Total number of students from City E

$$= 39000\times\frac{8.5}{100} = 3315$$

Number of boys from City E

$$= 12000\times\frac{75.6}{360} = 2520$$

Number of girls = 3315 - 2520 = 795

∴ Difference = 2520 - 795 = 1725

118. 2; Total number of students from City C

$$= 39000\times\frac{15}{100} = 5850$$

Total number of boys from City C

$$= 12000\times\frac{61.2}{360} = 2040$$

∴ Number of girls from City C

= 5850 – 2040 = 3810

Total number of students from City D

$$= 39000\times\frac{20}{100} = 7800$$

$$\therefore \text{Reqd } \% = \frac{3810\times 100}{7800} = 48.84 \approx 49\%$$

119. 5; Difference = $12000\times\frac{(82.8-72)}{360}$

$$= \frac{12000\times 10.8}{360} = 360$$

120. 3; Total number of students from City F

$$= 39000\times\frac{26}{100} = 10140$$

Number of boys from City F = $12000\times\frac{32.4}{360} = 1080$

Number of girls from City F = 10140 - 1080 = 9060

Total number of giris = 39000 - 12000 = 27000

$$\therefore \text{Reqd } \% = \frac{9060}{27000}\times 100 = 33.55\%$$

121. 2; ∴ Difference = $80\times\frac{86.4}{360} - \frac{90\times 32.4}{360}$

= 19.2 - 8.1 = 11.1 lakh

122. 5; Expenditure of Company C on Salary

$$= 75\times\frac{75.6}{360} = 15.75 \text{ lakh}$$

Expenditure of Company A on transportation

$$= 80\times\frac{54}{360} = 12 \text{ lakh}$$

$\therefore$ Reqd % $= \frac{15.75 \times 100}{12} = 131.25\% \approx 131\%$

123. 2; Average $= \frac{1}{3}\left\{80 \times \frac{64.8}{360} + 90 \times \frac{54}{360} + 75 \times \frac{86.4}{360}\right\}$

$= \frac{1}{3}\{14.4 + 13.5 + 18\} = \frac{45.9}{3} = 15.3$ lakh

124. 4; Expenditure of Company A on infrastructure $= 80 \times \frac{64.8}{360} = 14.4$ lakh

Expenditure of Company B on transportation

$= 90 \times \frac{50.4}{360} = 12.6$ lakh

$\therefore$ Ratio $= \frac{14.4}{12.6} = \frac{8}{7} = 8:7$

125. 4; Expenditure of Company C on infrastructure $= 75 \times \frac{86.4}{360} = 18$ lakh

Expenditure of Company A on bonus

$= 80 \times \frac{36}{360} = 8$ lakh

$\therefore$ Reqd % $= \frac{(18-8)}{8} \times 100 = \frac{1000}{8} = 125\%$

126. 2; Number of readers from State F in the year 2012

$= 57000 \times \frac{100}{10} \times \frac{22}{100}$

= 1.254 lakh.

Number of readers from State F in the year 2008

$= 38700 \times \frac{100}{9} \times \frac{26}{100} = 1.118$ lakh

$\therefore$ Difference = 1.254 - 1.118 = 0.136 lakh

= 13600

127. 2; Let the number of readers from State A be 2x and 5x respectively in the year 2008 and 2012

$\therefore$ Total number of readers from State A in the year 2008

$= 2x \times \frac{100}{10} = 20x$

Total number of readers from State A in the year 2012 = $5x \times \frac{100}{15} = \frac{100x}{3}$

$\therefore$ Ratio = $20x \times \frac{3}{100x} = 3:5$

128. 3; The number of readers from State B in the year 2008

$= 73100 \times \frac{100}{17} \times \frac{14}{100} = 60200$

The number of readers from State B in the year 2012 = $51300 \times \frac{100}{9} \times \frac{16}{100} = 91200$

$\therefore$ Total = 60200 + 91200 = 151400

129. 4; Reqd % = $\frac{15}{24} \times 100 = 62.5\%$

130. 1; The number of readers from State B in the year 2008 = $430000 \times \frac{14}{100} = 60200$

The number of readers from State C in the year 2008 = $430000 \times \frac{17}{100} = 73100$

$\therefore$ Total number of readers from State B and C = 133300

The number of readers from state B in the year 2012 = $570000 \times \frac{16}{100} = 91200$

The number of readers from state C in the year 2012 = $570000 \times \frac{28}{100} = 159600$

The number of readers from State B and C in the year 2012 = 91200 + 159600 = 250800

$\therefore$ Difference = 250800 - 133300 = 117500

131. 1; Total number of students in College A

$= 7500 \times \frac{16}{100} = 1200$

Number of girl students in College A

$= 3000 \times \frac{20}{100} = 600$

$\therefore$ Number of boy students in College A

= 1200 - 600 = 600

$\therefore$ Reqd ratio = 1 : 1

132. 3; Total number of students in College D

$= 7500 \times \frac{17}{100} = 1275$

Number of girl students in College D

$= 3000 \times \frac{22}{100} = 660$

Number of boy students in College D

= 1275 - 660 = 615

133. 5: Total number of students in College F

$= 7500 \times \frac{24}{100} = 1800$

Number of girl students in College F

$= 3000 \times \frac{18}{100} = 540$

Number of boy students in College F

= 1800 - 540 = 1260

$\therefore$ Difference = 1260 - 540 = 720,

which is maximum.

134. 4; Number of boy students in College E

$= 7500 \times \frac{15}{100} - 3000 \times \frac{14}{100} = 1125 - 420 = 705$

Number of girl students in College C

$= 3000 \times \frac{10}{100} = 300$

$\therefore$ Reqd % $= \frac{705}{300} \times 100 = 235\%$

135. 3; Total number of students in College B

$= 7500 \times \frac{18}{100} = 1350$

Number of girl students in College B

$= 3000 \times \frac{16}{100} = 480$

Number of boy students in College B

= 1350 - 480 = 870

$\therefore$ Reqd % $= \frac{(870 - 480)}{480} \times 100 = \frac{39000}{480}$

$= 81.25\% \approx 81\%$

136. 3; Number of employees in Teaching profession = $26800 \times \frac{15}{100} = 4020$

Number of employees in Medical profession

$= 26800 \times \frac{27}{100} = 7236$

Total number of employees = 4020 + 7336 =11256

Number of employees in Management profession = $26800 \times \frac{17}{100}$ = ` 4556

$\therefore$ Reqd difference = 11256 – 4556 = 6700

Quicker Method:

Reqd difference = (15 + 27 – 17)% of 26800 = 25% of 26800 = 6700

137. 5; Total number of employees in Management profession = $26800 \times \frac{17}{100} = 4556$

Number of female employees in Management profession = $4556 \times \frac{3}{4} = 3417$

$\therefore$ Required number of male employees in Management profession

= 4556 – 3417 = 1139

138. 2; Total number of employees from Film Production = $26800 \times \frac{19}{100} = 5092$

Now, number of employees from Film Production who went on strike

$= 5092 \times \frac{25}{100} = 1273$

$\therefore$ Number of employees who have not participated in strike = 5092 – 1273 = 3819

Quicker Method:

Required number of employees who have not participated in strike

$= 26800 \times \frac{19}{100} \times \frac{75}{100} = 3819$

139. 4; Required number of employees who participated in both Engineering and Industries professions = $26800 \times \frac{(9+13)}{100}$

= 268 × 22 = 5896

140. 1; Total number of teachers

$= 26800 \times \frac{15}{100} = 4020$

Number of teachers who are not permanent

$= 4020 \times \frac{3}{5} = 804 \times 3 = 2412$

$\therefore$ Number of teachers who are permanent $= 4020 - 2412 = 1608$

141. 1; The number of girls enrolled in Swimming

$= 1750 \times \frac{14}{100} = 245$

The number of boys enrolled in Swimming

$= \left(\frac{3000 \times 16}{100} - 245\right) = 480 - 245 = 235$

Ratio of girls to boys in Swimming

$= 245 : 235 = 49 : 47$

142. 2; The number of girls enrolled in Dancing

$= \frac{1750 \times 20}{100} = 350$

Reqd % $= \frac{350}{3000} \times 100 = 11.66\% \approx 11.67\%$

143. 4; The number of girls enrolled in Swimming

$= \frac{1750 \times 14}{100} = 245$

The number of girls enrolled in Drawing

$= \frac{1750 \times 16}{100} = 280$

$\therefore$ Total number of girls $= 245 + 280 = 525$

144. 5; The number of boys enrolled in Singing

$= \frac{3000 \times 21}{100} - \frac{1750 \times 28}{100}$

$= 630 - 490 = 140$

The number of boys enrolled in Craft

$= \left(\frac{3000 \times 20}{100} - \frac{1750 \times 22}{100}\right)$

$= 600 - 385 = 215$

Total number of boys $= 140 + 215 = 355$

145. 1; Number of boys $= 3000 - 1750 - 1250$

Read % $= \frac{1250}{3000} \times 100 = 41.66 \approx 42\%$

146. 1; Total population = 7000000000

Total number of patients in the world

$= 7000000000 \times \frac{10}{100} = 700000000 = 70$ crore Now, cancer patients in the world

$= 70 \times \frac{30}{100} = 21$ crore

$\therefore$ Cancer Patients in Australia $= 21 \times \frac{5}{100}$

$= 1.05 = 1$ crore 5 lakh

Total number of patients of heart disease in the world $= 70 \times \frac{22}{100} = 15.40$ crore

$= 15$ crore 40 lakh

$\therefore$ Reqd $\% = \frac{10500000}{154000000} \times 100 = 6.81\%$

147. 3; Cancer patients in South America

$= 70 \times \frac{30}{100} \times \frac{40}{100} = 8.4$ crore

After decrease of 25%, number of patients in South America $= 84000000 \times \frac{75}{100}$

$= 63000000 = 6.3$ crore

$\therefore$ Decrease $= 84000000 - 63000000$

$= 21000000 = 2.1$ crore

$\therefore$ Percentage decrease in the number of total cancer patients in the world

$= \frac{21000000}{700000000} \times 100 = 3\%$

148. 5; Total number of patients in Africa

$= 70 \times \frac{25}{100} = 17.5$ crore $= 175000000$

Total number of cancer patients in the world

$= 70 \times \frac{30}{100} = 21$ crore

Now, total number of cancer patients in Asia and North America

$= 21 \times \frac{35}{100} = 73500000$

$\therefore$ Ratio $= \frac{175000000}{73500000} = \frac{1750}{735} = \frac{350}{147}$

$= 350 : 147$

149. 4; Total number of patients in Europe

$= 70 \times \frac{10}{100} = 7$ crore

After 2 years, the number of patients in Europe $= 7 \text{ crore} \left(1 + \frac{10}{100}\right)^2$

$= 7 \times \frac{11}{100} \times \frac{11}{10} = 8.47$crore

= 8 crore 47 lakh

Number of cancer patients in South America = $21 \times \frac{40}{100}$ = 84000000

= 8.4 crore

∴ Difference = 8.47 – 8.40 = 7 lakh

150. 3; Number of patients of hepatitis

= $70 \times \frac{6}{100}$ = 42000000 = 4.2 crore

After increase, 6% of the number of patients of hepatitis = $42000000\left(1+\frac{6}{100}\right)$

= $42000000 \times \frac{106}{100}$ = 44520000

= 4.452 crore

Number of patients of heart disease

= $70 \times \frac{22}{100}$ = 154000000 = 15.4 crore

After 22% increase, the number of patients of heart disease

= $154000000 \times \frac{122}{100}$ = 187880000

∴ Reqd ratio = 4452 : 18788 = 1113 : 4697

151. 2; Total number of students studying Arts and Commerce in Institute D and E together

= 45% of 5000 + 55% of 6000

$= \frac{45}{100} \times 5000 + \frac{55 \times 6000}{100}$

= 2250 + 3300 = 5550

152. 5; Reqd ratio = $\frac{25\% \text{ of } 5000}{15\% \text{ of } 6000}$

$= \frac{125}{90} = \frac{25}{18} = 25:18$

153. 5; Total number of students studying both Commerce and Arts in Institute B and E together = 25% of 6000 + 35% of 5000

= 1500 + 1750 = 3250

Read % = $\frac{3250}{5000} \times 100 = 65\%$

154. 1; Total number of students studying Arts in Institute A = 10% of 5000 = 500

Total number of students studying Commerce in Institute B

= 5% of 6000 = 300

Reqd % = $\frac{500}{300} \times 100 = 166.66\% \approx 167\%$

155. 3; Number of students studying Arts in Institute C = 30% of 5000

Number of students studying Commerce in Institute A and E together

= 45% of 6000 = 2700

Reqd ratio = $\frac{1500}{2700}$ = 5 : 9

156. 2; Cases registered in WB = $10 \times \frac{16}{100}$

= 1.6 lakh

Cases registered in Goa = $10 \times \frac{14}{100}$

= 1.4 lakh

∴ Total number of cases in (WB + Goa)

= 1.6 + 1.4 = 3 lakh

Now, the number of cases registered in Assam

= $10 \times \frac{12}{100}$ = 1.2 lakh

Number of cases registered in Others

= $10 \times \frac{15}{100}$ = 1.5 lakh

∴ Total number of cases = 1.2 + 1.5

= 2.7 lakh

Exceeded number of cases = 3 – 27

= 0.3 lakh = 30000

157. 1; Total number of cases registered in Goa in 2012 = $10 \times \frac{14}{100}$ = 1.4 lakh

Number of cases of Dowry death registered in Goa = $1.4 \times \frac{20}{100}$

= 0.28 lakh = 28000

Number of cases registered per day in Goa = $\frac{28000}{366}$ $76.502 \approx 77$

(Since 2012 is a leap year, there would be 366 day.)

158. 4; Number of Human trafficking cases in UP = 10 × 18% × 6% = 0.108 = 10800

Number of cases of Human trafficking in WB = 10 × 16% × 6% = 0.096 lakh = 9600

∴ Excess = 10800 – 9600 = 1200

159. 1; Total number of crimes registered in

Bihar in 2012 = $10 \times \frac{19}{100}$ = 1.9 lakh

Now, number of cases registered for Dowry deaths = $1.9 \times \frac{20}{100}$ = 0.38 lakh = 38000

Number of registered cases of Torture

$= \frac{1.9 \times 32}{100}$ = 0.608 lakh = 60800

Number of registered cases of Molestation

$= \frac{1.9 \times 30}{100}$ = 0.57 lakh = 57000

Number of registered cases of Others

$= \frac{1.9 \times 3}{100}$ = 0.057 = 5700

Number of registered cases of Human trafficking = $\frac{1.9 \times 6}{100}$ = 0.114 lakh = 11400

160. 5; In 2012, the number of cases of Torture

= $10 \times \frac{32}{100}$ = 3.2 lakh

In 2012, the number of cases of Others

= $10 \times \frac{30}{100}$ = 0.3 lakh = 30000

$\therefore$ Total cases in (Torture + Others)
= 3.2 + 30000 = 3.5 lakh

Again, number of cases of Molestation

= $10 \times \frac{30}{100}$ = 3 lakh

$\therefore$ Exceeding number = 3.5 – 3
= 0.5 lakh = 50000

161. 1; Total IR rays received in 1 minute

$= 3600 \times \frac{10}{100} = 360$ units

Time taken to receive 8750 units of IR

= $\frac{8750}{360}$ minutes = 24.3 minutes

162. 3; Amount of UV rays in 5 minutes

= $3600 \times \frac{18}{100} \times 5$ = 3240 units

Amount of IR rays received in 2 minutes

$= 3600 \times \frac{10}{100} \times 2$ = 720 units

Amount of UV rays in 5 minutes of sun rays is $\left(\frac{3240}{720}\right)$ = 4.5 times the amount of IR rays received in 2 minutes.

163. 2; The amount of Gamma rays received when the ozone layer cover completely disappears = 100%

The amount of Gamma rays received in one minute if the ozone layer were to completely disappear = $3600 \times \frac{12}{100}$ units

= 432 units

164. 4; Amount of Microwaves received in 4 minutes = $3600 \times \frac{15}{100} \times 4$ = 2160 units

Amount of Alpha rays received in 3 minutes = $3600 \times \frac{8}{100} \times 3$ = 864 units

$\therefore$ Amount of Microwavers received in 4 minutes is (2160 - 864) units = 1296 units more than the amount of Alpha rays received in 3 minutes

165. 4; Given that the body requires 40 units of vitamin D every day.

To generate 1 unit of vitamin D, requirement of Beta rays = 30

To generate 40 units of vitamin D, requirement of Beta rays
= (30 × 40) = 1200 units

Now, in l minute $3600 \times \frac{5}{100}$ = 180 units Beta rays are received.

$\because$ 180 units Beta rays are received in 1 minute

$\therefore$ 1200 units Beta rays are received in

$\frac{1}{180} \times 1200 = \frac{120}{18} = 6\frac{2}{3}$ minutes

166. 4; Number of Civil students not interested in sports = $2500 \times \frac{12}{100} \times \frac{10}{100} = 30$

Now, number of Civil students interested in Sports

$= 2500 \times \frac{12}{100} - 30 = 300 - 30 = 270$

Number of Mechanical students not interested in sports

$= 2500 \times \frac{20}{100} \times \frac{20}{100} = 100$

$\therefore$ Number of Mechanical students interested in sports

$= 2500 \times \frac{20}{100} - 100 = 400$

Again, number of Electrical students interested in sports

$= 2500 \times \frac{22}{100} - 2500 \times \frac{22}{100} \times \frac{12}{100} = 484$

$\therefore$ Average number of students of these branches who are interested in sports

$= \frac{270 + 400 + 484}{3} = \frac{1154}{3} = 384.66 \approx 385$

167. 2; $\therefore$ Reqd ratio $= 2500 \times \frac{5}{100} : 2500 \times \frac{20}{100}$

$= 125 : 500 = 1 : 4$

168. 5; Number of failed students of Electronics branch $= 2500 \times \frac{15}{100} \times \frac{20}{100} = 75$

Now, failed Electronic students who are not interested in sports $= 75 \times \frac{60}{100} = 45$

Total number of students of all branches who are not interested in sports

$= 2500 \times \frac{15}{100} = 375$

$\therefore$ Reqd % $= \frac{45 \times 100}{375} = 12\%$

169. 1; Number of Mechanical students interested in Football $= 2500 \times \frac{20}{100} \times \frac{50}{100} = 250$

Number of Electrical students interested in Football $= 2500 \times \frac{22}{100} \times \frac{40}{100} = 220$

$\therefore$ Reqd ratio = 25 : 22

170. 3; Students of Mechanical branch interested in other games $= 2500 \times \frac{20}{100} \times \frac{20}{100} = 100$

Student of Electrical branch interested in other games $= 2500 \times \frac{22}{100} \times \frac{20}{100} = 110$

$\therefore$ Difference = (110 – 100) = 10

171. 3: Reqd % $= \frac{1750 \times 20}{3000} \times 100\%$

$= \frac{350}{3000} \times 100 = \frac{35}{3} = 11.67\%$

172. 1; Number of boys enrolled in Singing and Craft together

$= 3000 \times \frac{46}{100} - 1750 \times \frac{50}{100}$

= 1380 – 875 = 505

173. 4; Reqd ratio

$= \frac{14\% \text{ of } 1750}{16\% \text{ of } 3000 - 14\% \text{ of } 1750}$

$= \frac{245}{480 - 245} = \frac{245}{235} = \frac{49}{47} = 49 : 47$

174. 2: Total number of girls in Swimming and Drawing together $= 1750 \times \frac{30}{100} = 525$

175. 5; Reqd % of boys

$= \frac{(3000 - 1750)}{3000} \times 100\% = \frac{1250}{3000} \times 100\%$

$= 41.67 \approx 42\%$

176. 1; Number of employees interested in Athletics $= \frac{65000 \times 20}{100} = 13000$

Number of employees interested in Baseball

$= \frac{65000 \times 14.5}{100} = 9425$

$\therefore$ Reqd % $= \frac{13000}{9425} \times 100 = 137.93 \approx 138\%$

177. 2; Reqd difference

$= \frac{65000}{100}\{30 - (14.5 + 12 + 2.5)\}$

$= \frac{65000}{100} \times (30 - 29) = 650$

178. 5; Reqd ratio $= \frac{2.5}{14.5} = \frac{25}{145} = 5 : 29$

179. 3; Number of employees interested in Hockey

$= \frac{65000 \times 12}{100} = 7800$

Number of employees interested in Football, Athletics and Baseball

together $= \frac{65000}{100}(21 + 20 + 14.5)$

$= 650 \times 55.5 = 36075$

$\therefore$ Reqd % $= \frac{7800}{36075} \times 100 = 21.62 \approx 22\%$

180. 1; Number of employees interested in Gymnastics $= \frac{65000 \times 2.5}{100} = 1625$

Number of employees interested in Hockey

$= \frac{65000 \times 12}{100} = 7800$

$\therefore$ Reqd % $= \frac{1625}{7800} \times 100 = 20.83\% \approx 21\%$

DATA SUFFICIENCY

NATIONALISED BANKS & IBPS SO/MT/SO

Directions (1-5): Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read the question and both the statements and—

Give answer (1) if the data in **statement I alone are sufficient** to answer the question, while the data in **statement II alone are not sufficient** to ansswer the question.

Give answer (2) if the data in **statement II alone are sufficient** to answer the question, while the data in **statement I alone** are not **sufficient** to answer the question.

Give answer (3) if the data **either** in **statement I alone or in statement II** alone are **sufficient** to answer the question.

Give answer (4) if the data even in both the **statements I and II together** are not **sufficient** to answer the question.

Give answer (5) if the data in both the **statements I and II together are necessary** to answer the question.

Canara Bank PO Exam. 09.02.2003)

1. Train 'A' running at a certain speed crosses another train 'B' running at a certain speed in the opposite direction in 12 seconds. What is the length of train 'B' ?
I. The length of both the trains together is 450 metres.
II. Train 'A' is slower than train 'B'.

2. Area of a rectangle is equal to the area of a right angled triangle. What is the length of the rectangle ?
I. The base of the triangle is 40 cms.
II. The height of the triangle is 50 cms.

3. What was the total compound interest on a sum after three years ?
I. The interest after one year was Rs. 100/ and the sum was Rs. 1,000/-.
II. The difference between simple and compound interest on a sum of Rs. 1,000/- at the end of two years was Rs. 10/-.

4. What is the two digit number where the digit at the unit place is smaller ?
I. The difference between the two digits is 5.
II. The sum of the two digits is 7.

5. What is the speed of the boat in still water ?
I. It takes 2 hours to cover distance between A and B downstreams.
II. It takes 4 hours to cover distance between A and B upstreams.

Directions (6-10) : Each of the questions below consists of a question and two statements numbered I and II given below it. You have have to decide whether the data provided in the statements are sufficient to answer the question. Read the question and both the statements and.

Give answer (1) if the data in **statement I alone are sufficient** to answer the question, while the data in **statement II alone are not sufficient** to answser the question.

Give answer (2) if the data in **statement II alone are sufficient** to answer the question, while the data in **statement I alone** are not **sufficient** to answer the question.

Give answer (3) if the data **either** in **statement I** alone or in **statement II** alone are **sufficient** to answer the question.

Give answer (4) if the data in **both the statements I and II** together are not **sufficient** to answer the question.

Give answer (5) if the data in **both the statements I and II together are necessary** to answer the question.

(Syndicate Bank PO Exam. 10.10.2004)

6. What is the annual income of Mr. Krishnamurthy form April 2001 to March 2002 ?
I. The annual income of Mr. Krishnamurthy is 70% more than that of his boss.
II. In April 2001 the income of Krishnamurthy was Rs. 12000 and his income increases by 10% per month.

7. What is the area of circular field ?
I. The circumference of the field is *x* metres.
II. The diameter of the field is *y* metres.

8. What is the speed of boat in still water ?
I. The boat covers a distance of 35 kms in 5 hours in downstream.
II. The boat takes 7 hours to cover the same distance in upstream.

9. How many students are there in the class ?
I. The ratio between the number of boys and girls is 5 : 6 respectively.
II. The difference between the numbers of boys and girls is 7.

10. What is the amount of earned profit ?
I. On selling the article for Rs. 1740, 20% profit is earned.
II. The cost price ofthe article is Rs. 1400.

Directions (11-15) : In each of the following questions, a question is followed by information given in three statements. You have to study the

question alongwith the statements and decide, the information given in which of the statement(s) is necessary to answer the question.

(Corporation Bank PO Exam. 29.07.2006)

11. In how many days can 16 men and 8 women together complete the piece of work ?
I. 8 men complete the piece of work in 10 days.
II. 16 women complete the piece of work in 10 days.
III. 5 women take 32 days to complete the piece of work.
(1) Only I and II
(2) Only I and III
(3) Only II and III
(4) Only I and either II or III
(5) Any two of the three

12. What is the speed of the train ?
I. Train crosses a pole in 10 seconds
II. Length of the train is 240 metres.
III. Train crosses a platform of equal length in 20 seconds.
(1) Only I and II
(2) Only II and III
(3) All I, II and III
(4) Any two of the three
(5) II and either I or III

13. What is the area of the square ?
I. Measure of diagonal of the square is given.
II. Measure of one side of square is given.
III. Perimeter of the square is given.
(1) Only II
(2) Only III
(3) Only I and III
(4) Only II and III
(5) Any one of the three

14. What is the two digit number ?
I. The number obtained by interchanging the digits of the number is greater than the original number by 18.
II. Sum of the two digits of the number is 14.
III. Difference between the two digits of the number is 2.
(1) Any two of the three
(2) Only I and III
(3) II and either I or III
(4) All the three
(5) III and either I or II

15. What is the rate of interest p.c.p.a. ?
I. Simple interest earned per annum is Rs. 5,300
II. The difference between the compound and simple interest on an amount is Rs. 1,060 at the end of 2 years.
III. An amount doubles itself in 5 years with simple interest.
(1) All the three
(2) Only III
(3) Either II or III
(4) Only III or I and II
(5) Question cannot be answered even with the information in all three statements

Directions (16-20): Each of these questions consists of a question followed by information in three statements. You have to study the question and the statenents and decide that information in which of the statements is/are necessary to answer the question.

(Indian Overseas Bank PO Exam. 15.06.2008)

16. What is the capacity of the cylindrical tank ?
I. Radius of the base is half of its height ?
II. Area of the base is 616 square metres.
III. Height of the cylinder is 28 metres.
(1) I and II Only
(2) II and III Only
(3) I and III Only
(4) All 1, II and III
(5) Any two of the three

17. What is the speed of the train ?
I. The train crosses a signal pole in 18 seconds.
II. The train crosses a platform of equal length in 36 seconds.
III. Length of the train is 300 metres.
(1) I and III Only
(2) II and III Only
(3) I and II Only
(4) III and either I or II only
(5) Any two of the three

18. What is the staff strength of company 'X' ?
I. Male and female employees are in the ratio of 2 : 3 respectively.
II. Of the officer employees 80% are males.
III. Total number of officers is 132.
(1) I and III only
(2) II and either III or I only
(3) All I, II and III
(4) Any two of the three
(5) Question cannot be answered even with the information in all three statements

19. What is the two digit number ?
I. Number obtained by interchanging the digits is more than the original number by 9.
II. Sum of the digits is 7.
III. Difference between the digits is 1.
(1) 1 and III only
(2) I and II only

(3) II and III only
(4) All I, II and III
(5) Question cannot be answered even with the information in all three statements

20. How many articles were sold ?
I. Total profit earned was Rs. 1,596
II. Cost-price per article was Rs. 632
III. Selling price per article was Rs. 765
(1) II and III only
(2) I and II only
(3) All I, II and III
(4) Any two of the three
(5) Question cannot be answered even with the information in all three statements

Directions (21-25): Each question below is followed by two statements A and B. You have to detemine whether the data given in the statement is sufficient for answering the question. You should use the data and your knowledge of Mathematics to choose between the possible answers.

Give answer (1) if the statement A alone is sufficient to answer the question, but the statement B alone is not sufficient.

Give answer (2) if the statement B alone is sufficient to answer the question, but the statement A alone is not sufficient.

Give answer (3) if both the statements A and B together are needed to answer the question.

Give answer (4) if either the statement A alone or statement B alone is sufficient to answer the question.

Give answer (5) if you cannot get the answer from the statements A and B together, but need even more data.

(Andhra Bank PO Exam. 14.09.2008)

21. Is A an odd number ?
A. A multiplied by an odd number is equal to an odd number.
B. A is not divisible by 2.

22. The ages of Pradumn and Gunit are in the ratio of 7 : 5 . What is the age of Pradumn.
A. The ages of Pradumn and Nandini are in the ratio of 3 : 1.
B. After 7 years the ratio of Pradumn's and Aviral's ages will be 4 : 3.

23. What is the salary of B, in a group of A, B, C and D whose average salary is Rs. 62,880 ?
A. Total of the salary of A and C is exact multiple of 8.
B. Average of the salary of A, C and D is Rs. 61,665.

24. What is the three digit number ?
A. The three digit number is divisible by 9.
B. The first and the third digit is 6.

25. What is the profit earned by selling a printer for Rs. 3,000 ?
A. The cost price of 6 such printers is equal to selling price of 5 such printers.
B. 20% profit is earned by selling each printer.

Directions (26-30): Each of the questions below consists of a question and two statements numbered I and II are given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read both the statements and

Give answer (1) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.

Give answer (2) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.

Give answer (3) if the data either in Statement I alone or in Statement II alone are sufficient to answer the question.

Give answer (4) if the data in both the Statements I and II are not sufficient to answer the question.

Give answer (5) if the data in both the Statements I and II together are necessary to answer the question.

(Bank Of Baroda Specialist Officer Exam. 05.10.2008)

26. What is the ratio of the number of freshers to the number of seniors in a college ?
I. The ratio of males and females in the college is 2 : 3.
II. There are 1125 female freshers in the college.

27. What is Nidhi's age ?
I. Nidhi is 3 times younger to Rani.
II. Surekha is twice the age of Rani and the sum of their ages is 72 years.

28. What is the ratio of the total number of girls to the total number of boys in the school ?
I. The ratio of the total number of boys to the total number of girls, last year was 4 : 5.
II. There are 3500 students in the school out of which 60% are boys.

29. What is Mr. Mehta's present income ?
I. Mr. Mehta's income increases by 10% every year.
II. His income will increase by Rs. 2500 this year.

30. What is the speed of the bus ?
I. The bus covers a distance of 80 kms. in 5 hrs.

II. The bus covers a distance of 160 kms. in 10 hrs.

Directions (31-35): Each question below is followed by two statements A and B. You are to determine whether the data given in the statement is sufficient for answering the question. You should use the data and your knowledge of Mathematics to choose between the possible answers.

Give answer (1) if the statement A alone is sufficient to answer the question, but the statement B alone is not sufficient.

Give answer (2) if the statement B alone is sufficient to answer the question, but the statement A alone is not sufficient.

Give answer (3) if both statements A and B together are needed to answer the question.

Give answer (4) if neither the statement A alone or statement B alone is sufficient to answer the question.

Give answer (5) if you cannot get the answer from the statement A and B together, but need even more data.

(Oriental Bank of Commerce PO Exam. 21.12.2008)

31. Is X an even number ?
- **A.** X is divisible by 2.
- **B.** X + an odd number is an odd number.

32. What is the three digit number ?
- **A.** One-third of that number is less by 27 of the half of that number.
- **B.** One-fifth of that number is 20% of that number.

33. The ages of Tanish and Shivay are in the ratio of 6 : 5. What is the age of Shivay ?
- **A.** The ages of Tanish and Danish are in the ratio of 3 : 2.
- **B.** After 6 years the ratio of Danish's and Shivay's ages will be 6 : 7.

34. In how many days 15 men can complete a piece of work ?
- **A.** If 16 children can complete the same piece of work in 48 days.
- **B.** If 9 men can complete the same piece of work in 27 days.

35. What is the salary of C, in a group of A, B, C, D, E and F whose average salary is Rs. 60,000 ?
- **A.** Total of the salary of A and E is Rs. 64,500
- **B.** Total of the salary of B and F is Rs. 52,600

Directions (36 - 40): Each question below is followed by two statements A and B. You are to determine whether the data given in the statement is sufficient for answering the question. You should use the data and your knowledge of Mathematics to choose between the possible answers.

Give answer (1) if the statement A alone is sufficient to answer the question, but the statement B alone is not sufficient.

Give answer (2) if the statement B alone is sufficient to answer the question, but the statement A alone is not sufficient.

Give answer (3) if both statements A and B together are needed to answer the question.

Give answer (4) if either the statement A alone or statement B alone is sufficient to answer the question.

Give answer (5) if you cannot get the answer from the statement A and B together, but need even more data.

(Canara Bank PO Exam. 15.03.2009)

36. How many women can complete a piece of work in 15 days ?
- **A.** If 12 women can complete the same piece of work in 20 days.
- **B.** If 10 men can complete the same piece of work in 12 days.

37. What is the rate of interest p.c.p.a. on an amount of Rs. 12,000 deposited in a Bank ?
- **A.** The difference between the simple interest and compound interest is Rs. 172.8.
- **B.** The simple interest for two years is Rs. 2,880.

38. What is the three digit number ?
- **A.** The three digit number is exact multiple of 13.
- **B.** The first and third digit are 7.

39. What is the age of C, in a group of A, B, C, D and E whose average age is 45 years ?
- **A.** Average of the ages of A and B is 53 years.
- **B.** Average of the ages of D and E is 47 years.

40. What is the profit earned by selling a Laptop of Rs. 26,250 ?
- **A.** The cost price of 5 such Laptops is equal to selling price of 4 such Laptops.
- **B.** 25% profits is earned by selling each Laptop.

Directions (41-42) : Each question below is followed by two statements A and B. You are to determine whether the data given in the statement is sufficient for answering the question. You should use the data and your knowledge of Mathematics to choose between the possible answers.

Give answer (1) if the Statement alone is sufficient to answer the question, but the Statement B alone is not sufficient.

Give answer (2) if the Statement B alone is sufficient to answer the luestion, but the Statement A alone is not sufficient.

Give answer (3) if both statements A and B together are needed to answer the question.

Give answer (4) if either the statement A alone or statement B alone is sufficient to answer the question.

Give answer (5) if you cannot get the answer from the statements A and B together, but needed even more data.

(Indian Overseas Bank PO Exam. 05.04.2009)

41. What is the speed of the car ?
A. The car covers a distance of 135 kms in 3 hours.
B. The car covers a distance of 270 kms. in 6 hours.

42. What is the value of the two digit number ?
A. The product of the digits is 72 and the difference between the digits is 1.
B. The digit at the unit's place is greater than the other.

43. What is the ratio of the total number of girls to the total number of boys in a college ?
A. There are 2500 students in the college out of which 40% are boys.
B. The ratio of the total number of boys to the total number of girls in the last year was 5 : 6.

44. What is the number ?
A. 75% of that number is three- fourth of that number.
B. One-third of that number is less by 27 of the half of that number.

45. What is Deepali's age ?
A. Deepali is two times younger than Nisha.
B. Supriya is twice the age of Nisha.

Directions (46-50) : In each of these questions, a question is given followed by information in three statements. You have to find out the data in which statement(s) is sufficient to answer the question and mark your answer accordingly.

(United Bank of India PO Exam. 21.06.2009)

46. What is the average age of the six members A, B, C, D, E and F in the family ?
I. Total age of D and E is 14 years.
II. Average age of A, B, C and F is 50 years.
III. Average age of A, B, D and E is 40 years.
(1) Only I and II
(2) Only I and III
(3) Only II and III
(4) All I, II and III
(5) None of these

47. What is the area of the right angled triangle ?
I. Base of the triangle is X cms.
II. Height of the triangle is Y cms.
III. Hypotenuse of the triangle is Z cms.
(1) Only I and II
(2) Only II
(3) Only II and III
(4) Any two of three
(5) None of these

48. In how many days will B alone complete the work ?
I. A and B together can complete the work in 8 days.
II. B and C together can complete the work in 10 days.
III. A and C together can complete the work in 12 days.
(1) Only I and II
(2) Only II and III
(3) All I, II and III
(4) Question cannot be answerd even with the information in all the three statements
(5) None of these

49. What is the rate of interest p.c.p.a. ?
I. An amount doubles itself at simple interest in 10 years.
II. Difference between the compound interest and simple interest on an amount of Rs 15,000 in two years is Rs 150.
III. The compound interest accured in 8 years is more than the amount (principal).
(1) Only I
(2) Only II
(3) Only II and III
(4) Only I and III
(5) Only either I or II

50. What are the marks scored by Abhijit in English ?
I. Marks scored by Abhijit in Maths are more than his marks in Science by 20.
II. Total marks scored by Abhijit in Maths, Science and and English are 198.
III. Marks scored by Abhijit in Science are more than his marks in English by 12.
(1) Any two of the three
(2) Only II and III
(3) All I, II and III
(4) Question cannot be answered even with the information in all the three statements
(5) None of these

Directions (51-55): Each of the questions below consists of a question and two statements numbered I and II are given below it. You have to decide whether the data provided in the statements are sufficient to answer question. Read both the statements and

Give answer (1) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.

Give answer (2) if the data in statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.

Give answer (3) if the data in Statement I alone or in Statement II alone sufficient to answer the question.

Give answer (4) if the data in both the Statements 1 and II are not sufficient to answer the question.

Give answer (5) if the data in both the Statements I and II together are necessary to answer the question.

CAUTION: Do not mark your anawer unless you consider **both** the statements carefully.

(Andhra Bank PO Exam. 05.07.2009)

51. What is Nikita's share in the profit of Rs. 50,000 earned in the business run by her in partnership with Sharmila ?
I. Nikita invested an amount 150% of the amount invested by Sharmila.
II. Amount invested by Sharmila is two-third of the amount invested by Nikita.

52. What is the two digit number ?
I. Sum of the digits is 6.
II. Digit in the ten's place is double the digit in the unit's place.

53. What is the rate of interest p.c.p.a. ?
I. The amount doubles itself in 10 years.
II. The simple interest accrued in 5 years is Rs. 5,000.

54. In how many days can B alone complete the work ?
I. B and C together can complete the work in 8 days.
II. A and B together can complete the work in 12 days.

55. What is the mother's present age ?
I. Present ages of mother and child are in the ratio of 5 : 1 respectively.
II. Four years hence the ratio of the ages of mother and child will be 17 : 5 respectively.

Directions (56-60) : In each of the following questions, a question followed by two statements numbered I and II are given. You have to read both the statements and then

Give answer (1) if the data given in statement I alone are sufficient to answer the question whereas the data given in statement II alone are not sufficient to answer the question.

Give answer (2) if the data given in statement II alone are sufficient to answer the question whereas the data given in statement I alone are not sufficient to answer the question.

Give answer (3) if the data in either statement I alone or in statement II alone are sufficient to answer the question

Give answer (4) if the data in both the statements I and II are not sufficient to answer the question

Give answer (5) if the data given in both the statements I and II are necessary to answer the question

(Bank Of India Banking Officer Exam. 24.01.2010)

56. What is the two digit number ?
I. The difference between the two digits of the number is 0.
II. The sum of the two digits of the number is 18.

57. What is Supriya's present age ?
I. Supriya is three years older than Priya.
II. The ratio between Priya's and Reshma's age is 3 : 4 respectively.

58. What is the speed of the boat in still water ?
I. The boat travels at the speed of 4 km/hr upstream
II. The boat travels at the speed of 6 km/hr downstream.

59. How many marks did Nigam get in Biology ?
I. Nigam got 42 marks in English which were half the marks he got in Biology
II. Nigam's marks in Biology were 14 per cent of the total marks he got in all the subjects together

60. What is the rate of interest p.c.p.a. ?
I. An amount of Rs. 6200 fetches simple interest of Rs. 1736 in two years.
II. An amount of Rs. 4500 fetches compound interest of Rs. 1348.2 in two years.

Directions (61-65): Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read both the statements and

Give answer (1) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.

Give answer (2) If the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.

Give answer (3) if the data In Statement I alone or in Statement II alone are sufficient to answer the question.

Give answer (4) if the data in both the Statements I and II are not sufficient to answer the question.

Give answer (5) if the data in both the Statements I and II together are necessary to answer the question

(Allahabad Bank PO Exam.21.02.2010)

61. What is the perimeter of a semicircle ?
I. The radius of the semicircle is equal to half the side of a square.
II. The area of the square is 196 sq.cm.

62. What is the exact average of *n*, 35, 39, 42, *p* and *w* ?
I. *n* is six more than *w*.
II. *w* is four less than *p*.

63. What was the percent profit/loss made/ incurred by selling an article for Rs.24,000 ?
I. The ratio between the selling price and the cost price of the article is 5 : 3 respectively.
II. The difference between the cost price and the selling price is Rs.9,600.

64. What will be the difference between two two-digit numbers ?
I. The square of the first number is 9 times the second number.
II. The ratio between the first number and the second number is 3 : 4 respectively.

65. What is the ratio between two numbers *x* and *y* ?
I. 40% of *x* is 20% of 50
II. 30% of *y* is 25% of 72.

Directions (66-70): Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement are sufilcienit to answer the question. Read both the statements and

Give answer (1) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.

Give answer (2) If the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.

Give answer (3) if the data in Statement I alone or in Statement II alone are sufficient to answer the question.

Give answer (4) if the data in both the Statements I and II are not sufficient to answer the question.

Give answer (5) if the data in both the Statements I and II together are necessary to answer the question.

(Syndicate Bank PO Exam. 29.08.2010)

66. In how many days will five women complete a work ?
I. Two women, five boys and three men together complete the work in six days.
II. Eight women complete the same work in twelve days.

67. What will be Shyam's age after 6 years ?
I. The ratio between Ram's and Shyam's present age is 4 : 3 respectively.
II. Ram is seven years older than Shyam.

68. What is the average speed of a car ?
I. Average speed of the car is five times average speed of a truck whereas the average speed of a bus is 45 km/hr.
II. Average speed of the truck is half the average speed of a train whereas the average speed of a bus is 45 km/hr.

69. What is the simple interest accrued on a principal of Rs. 5,000 in six years ?
I. The rate of the simple interest for the first three years is 5 p.c.p.a. ?
II. The rate of simple interest for the next three years is 8 p.c.p.a.?

70. What is the total cost of 4 kgs of apples and 3 kgs of mangoes together ?
I. Cost of 2 kgs of apples is Rs. 170 and one kg of mangoes is Rs. 50.
II. The total cost of 5 kgs of apples and 4 kgs of mangoes is Rs. 410.

Directions (71-75): Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read both the statements and

Give answer (1) if the data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.

Give answer (2) if the data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.

Give answer (3) if the data either in statement I alone or in statement II alone are sufficient to answer the question.

Give answer (4) if the data given in both the statements I and II together are not sufficient to answer the question, and

Give answer (5) if the data in both the statements I and II together are necessary to answer the question.

(Punjab National Bank Specialist Officer Exam. 24.10.2010)

71. What is the circumference of the circle ?
I. The diameter of the circle is 21 cm.
II. The area of the circle is 346.5 sq. cm.

72. What is the average age of the boys in the class ?
I. The average age of girls in the class is 14.
II. The number of boys in the class is twice the number of the girls in the class.

73. How many pieces of 80 cm. each can be cut from the reel of cloth ?
I. The length of the reel of cloth is 900 cm.
II. After cutting all the pieces 20 cms. of the cloth from the reel remains.

74. What is the selling price of the wrist watch ?
I. The cost price of the wrist watch is Rs 6,400.
II. 31.25 per cent profit is earned after selling the wrist watch.

75. What is the two-digit number ?
I. The sum of the two digits of the number is 6.
II. The number is divisible by 7.

Directions (76-80): Each of the questions below consists of a questions and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read both the statements and —

Give answer (1) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.

Give answer (2) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.

Give answer (3) if the data in Statement I alone or in Statement II alone are sufficient to answer the question

Give answer (4) if the data in both the Statements I and II are not sufficient to answer the question.

Give answer (5) if the data in both the Statements I and II together are necessary to answer the questions.

(Bank Of India PO Exam. 31.10.2010)

76. In how many days will five women complete a work ?
I. Two women, five boys and three men together complete the work in eight days.
II. Six women complete the same work in sixteen days.

77. What is Sonu's monthly income ?
I. Rita's monthly income is Rs 5000 more than Sonu's monthly income.
II. Sonu's monthly income is Rs. 8,000 more than Sohan's monthly income.

78. What will be Pravin's age after 4 years ?
I. The ratio between Pravin's and Shekhar's present age is 2 : 3 respectively.
II. Shekhar is six years older than Pravin.

79. What is the area of a circle ?
I. The radius of the circle is equal to length of a rectangle.
II. The breadth of the rectangle is equal to 22 cms.

80. What is the average speed of a car ?
I. Average speed of the car is four times average speed of a truck which covers 220 kms in 11 hours.
II. Average speed of the truck is half the average speed of a train whereas the average speed of a jeep is 40 km/hr.

Directions (81-85): Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide Whether the data provided in the statements are sufficient to answer the question. Read both the statements and

Give answer (1) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.

Give answer (2) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.

Give answer (3) if the data in Statement I alone or in Statement II alone are sufficient to answer the question.

Give answer (4) if the data in both the Statements I and II are not sufficient to answer the question.

Give answer (5) if the data in both the Statements I and II together are necessary to answer the question.

(Oriental Bank Of Commerce PO Exam. 26.12.2010 (1st Sitting)

81. What is the area of the circle ?
I. The breadth of a rectangle is three-fourth of the radius of the circle.
II. The radius of the circle is equal to the side of a square of area 144 sq. cm.

82. What is the cost of three tables and two chairs ?
I. Cost of four chairs is twice the cost of three tables.
II. Cost of two tables is equal to cost of one cot, i.e. Rs 500 ?

83. What is Jyoti's annual income ?

I. Jyoti's monthly income is Rs 8,500 more than Amit's monthly income.
II. Rohit's monthly income is Rs 3.5 thousand which is half of Amit's monthly income.

84. What will be Suraj's age after eight years ?
I. The ratio between Kamya's and Suraj's present age is 4 : 7 respectively.
II. Kamya is 15 years younger than Suraj.

85. What is the minimum passing percentage in a test ?
I. Raju scored 162 marks in a test and failed by 104 marks.
II. The maximum marks of the test are 538 more marks obtained by Raju.

Directions (86-90): Each of the questions given below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements sufficient to answer the question. Read both the statements and

Give answer (1) if the data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.

Give answer (2) if the data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.

Give answer (3) if the data in statement I alone or in statement II alone are sufficient to answer the question.

Give answer (4) if the data in both the statements I and II are not sufficient to answer the question.

Give answer (5) if the data in both the statements I and II together are necessary to answer the question.

(Indian Bank PO Exam. 02.01.2011)

86. What is the salary of C, in a group of A, B, C, D and E whose average salary is Rs 48,250 ?
I. C's salary is 1.5 times B's salary.
II. Average salary of A and B is Rs 23,500.

87. What is the per cent profit earned by selling a car for Rs 6,40,000 ?
I. The amount of profit earned on selling the car was Rs 3,20,000.
II. The selling price of the car was twice the cost price.

88. What is the rate of interest p.c.p.a. ?
I. An amount of Rs 14,350 gives a simple interest of Rs 11,480 in four years.
II. The amount doubles itself in 5 years with simple interest.

89. What is the two digit number ?
I. The difference between the two digits of the number is 9.
II. The product of the two digits of the number is 0.

90. What is the perimeter of the rectangle ?
I. The area of the rectangle is 252 sq. m.
II. The ratio of length to breadth of the rectangle is 9 : 7 respectively.

Directions (91-95): Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read both the statements and

Give answer (1) if the data in Statement **I alone** are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.

Give answer (2) if the data in Statement **II alone** are sufficient to answer the question, while the data in statement 1 alone are not sufficient to answer the question.

Give answer (3) if the data in **either** Statement I alone or in statement II alone are sufficient to answer the question.

Give answer (4) if the data in both the Statements I and II are **not** sufficient to answer the question.

Give answer (5) if the data in **both** the Statements I and II together are necessary to answer the question.

(IDBI Bank Officer Exam. 16.09.2012)

91. What is the minimum passing percentage in a test ?
I. Raman scored 25% marks in the test and Sunil scored 288 marks which is 128 more than Raman.
II. Raman scored 64 marks less than the minimum passing marks.

92. What is the value of $x^2 + y + z$?
I. $4x + 3y + 5z = 60$ and $2x = y$, $2y = z$
II. $3x + 3y + 2z = 34$, $2x + 5y + 6z = 72$

93. Whose body weight is second highest among the five boys Arun, Vinay, Suraj, Raju and Pratap ?
I. Average weight of Arun, Suraj and Vinay is 68 kg and average weight of Raju and Pratap is 72 kg. Also Suraj is 78 kg, Raju is 68 kg and Binay is 46 kg.
II. Average weight of Arun, Suraj, Vinay and Raju is 68 kg and also Suraj is 78 kg, Raju is 68 kg and Vinay is 46 kg. All of them have different weight.

94. What is the respective ratio between the length of a rectangle and side of a square ?
I. Area of the square is 576 sq. cm. and the area of the rectangle is 600 sq.cm.
II. Breadth of the rectangle is half the side of the square.

95. What is the smaller angle of a parallelogram ?
I. Ratio between the angles of a triangle is 3 : 5 : 4 and the larger angle of the parallelogram is 34° more than the largest angle of the triangle.
II. Larger angle of the parallelogram is 38° more than its smaller angle.

Directions (96-100) : Each of the following questions consists of a question followed by three statements I, II and III. You have to study the question and the statements and decide which of the statement (s) is/are necessary to answer the question.

(IBPS Bank PO/MT CWE-III 26.10.2013)

96. What is the speed of boat in still water ?
I. The boat covers 12 km in 2 hours in downstream.
II. The boat covers same distance in 4 hours in upstream.
III. The speed of stream is one third of that of boat in still water.
(1) Both I and II
(2) I and either II or III
(3) All I, II and III
(4) Question cannot be answered even with the information in all three statements.
(5) None of these

97. What is the speed of train ?
I. The length of train is 240 metre.
II. The train crosses a pole in 24 seconds.
III. The train crosses a platform in 48 seconds.
(1) Both I and III
(2) Both I and II
(3) Both II and III
(4) Any two of three
(5) None of these

98. What is the age of class teacher ?
I. There are 11 students in the class.
II. The average age of students and the teacher is 14 years.
III. The average age of the teacher and students is 3 years more than that of students.
(1) Both I and III
(2) Both I and II
(3) II and either I or III
(4) All I, II and III
(5) None of these

99. Sri Gupta borrowed a sum at compound interest. What is the amount returned in 2 years ?
I. The rate of interest is 5% per annum.
II. The simple interest incurred on the sum in 1 year is Rs. 600.
III. The borrowed sum is ten times the amount earned as simple interest in two years.
(1) Only I
(2) Only III
(3) Both II and III
(4) Either I or III
(5) All I, II and III

100. What is the area of the given right angled triangle ?
I. The length of hypotenuse is 5 cm.
II. The perimeter of triangle is four times of its base.
III. One of the angles of triangle is 60°
(1) Only II
(2) Only III
(3) Either II or III
(4) Both I and III
(5) Question cannot be answer even with the information in all three statements

Directions (101 - 105): In each of the following questions, a question and three statements following it have been given. You are required to study the question and statements and decide that the information given in which statement(s) is necessary to answer the question ?

(Corporation Bank Specialist Officer (Marketing) Exam 22.12.2014)

101. What is the principal ?
I. The simple interest accrued on that sum at the rate of 12% per annum in 2 years is Rs. 360 less than the compound interest on the same sum at 12% per annum in 2 years.
II. The sum doubles itself in 10 years at 10% per annum rate of simple interest.
III. The compound interest on the sum in 2 years at the rate of 12% per annum is Rs 6360.
(1) I or II (2) I and III
(3) I and II (4) II and III
(5) None of these

102. What will be the area of a 2 metre wide boundary around a rectangular field ?
I. The breadth of the field is one-fourth of its perimeter. Area of the field is 144 metre.
II. The ratio of the length and breadth is respectively 3 : 2.
III. Area of the field is 216 sq. metre.

(1) I, II and III
(2) only I or II and III
(3) I and II
(4) Anyone of the three statements
(5) None of these

103. What are the marks obtained by Arnab in History ?
I. The average marks obtained by Arnab in History, Geography and Civics are 65.
II. Marks obtained by Arnab in Geography is 6 more than that obtained in History.
III. Marks obtained in Geography is as much more than that in Civics as the marks obtained in Civics is more than that in History.
(1) I and II (2) II and III
(3) I and III (4) I, II and III
(5) None of these

104. What is the speed of train ?
I. Train crosses a signal post in 15 seconds.
II. Train crosses a 250 metre long platform in 27 seconds.
III. Train crosses another train running in the same direction on a parallel track in 32 seconds.
(1) I and II
(2) I and III
(3) II and III
(4) Any two of the three
(5) None of these

105. What percent of discount is offered ?
I. The profit earned on a article sold for Rs. 252 after discount is Rs. 52.
II. If no discount were allowed, there would have been a profit of Rs. 80.
III. If there were no discount, there would have been a gain of 40%.
(1) I and II
(2) II and either I or III
(3) I and III
(4) I and either II or III
(5) None of these

SBI PO EXAMS

Directions (1-5) : In each of these questions, a question is followed by information given in three statements. You have to study the question alongwith the information given in the statements and decide the information in which of the statement(s) is/are necessary and **sufficient** to answer the question.

Directions (1-5): Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the state-ments are sufficient to answer the question. Read the question and both the statements and —

(RBI Grade-B Officer Exam. 2008)

Give answer (1) if the data in **statement I alone are sufficient** to answer the question, while the data in **statement II alone are not sufficient** to answer the question.

Give answer (2) if the data in **statement II alone are sufficient** to answer the question, while the data in **statement I alone** are not **sufficient** to answer the question.

Give answer (3) if the data **either** in **statement I alone or in statement II** alone are **sufficient** to answer the question.

(SBI Banks PO Exam. 11.02.2001)

1. In how many days can the work be completed by A, B and C together ?
(I) A and B together can complete the work in 6 days.
(II) B and C together can complete the work in $3\frac{3}{4}$ days.
(III) A and C together can complete the work in $3\frac{1}{3}$ days.
(1) Only I
(2) Only II
(3) Only III
(4) Any one of the three
(5) All the three statements are necessary to answer the question.

2. What is the cost of painting the two adjacent walls of a hall which has no windows or doors ?
(I) The area of the hall is 24 sq. metres.
(II)
(III) Area of one wall is 30 square metres.
(1) Only I
(2) Only II
(3) Only III
(4) Either I or III
(5) Data inadequate

3. What is the total compound interest earned at the end of the three years ?
(I) Simple interest earned on that amount at the same rate and period is Rs. 4,500
(II) The rate of interest is 10 percent per annum
(III) Compound interest for three years is more than the simple interest for that period by Rs. 465.
(1) Only I and II
(2) Only II and III
(3) Only I and III

(4) Any two of the three
(5) Either II or III only

4. What is the percent profit earned by the shopkeeper on selling the articles in his shop ?
(I) Labelled price of the articles sold was 130% of the cost price.
(II) Cost price of each article was Rs. 550.
(III) A discount of 10% on labelled price was offered.
(1) Only I
(2) Both I and II
(3) Only III
(4) All the three statements are necessary to answer the question
(5) Questions cannot be answered even with the information given in all the three statements.

5. What is the average salary of 15 employees ?
(I) Average salary of 7 Clerical cadre (out of the 15 employees) employees is Rs. 8,500.
(II) Average salary of 5 Officer cadre (out of the 15 employees) employees is Rs. 10,000.
(III) Average salary of the 3 substaff employees (out of the 15 employees) is Rs. 2,500.
(1) None
(2) Only I
(3) Only II
(4) Only III
(5) Question cannot be answered even with information given in all three statements

Directions (6-10) : Each of the questions consists of a question followed by three statements. You have to study the questions and the statements and decide which of the statement(s) is/are **necessary** to answer the question.

(SBI Associate Banks PO Exam. 21.07.2002)

6. What is the area of the hall ?
I. Material cost of flooring per sq metre is 250
II. Labour cost of flooring the hall is Rs. 3,500
III. Total cost of flooring the hall is Rs. 14,500
(1) I and II only
(2) II and III only
(3) All I, II and III
(4) Any two of the three
(5) None of these

7. What was the percentage of discount offered ?
I. Profit earned by selling the article for Rs. 252 after giving discount was Rs. 52.
II. Had there been no discount the profit earned would have been Rs. 80.
III. Had there been no discount the profit earned would have been 40%.
(1) I and II only
(2) II and either I or III only
(3) I and III only
(4) I and either II or III only
(5) None of these

8. What is the speed of the train ?
I. The train crosses a signal pole in 13 sec.
II. The train crosses a platform of length 250 mts. in 27 seconds.
III. The train crosses another train running in the same direction in 32 seconds.
(1) I and II only
(2) I and III only
(3) II and III only
(4) Any two of the three
(5) None of these

9. What is the population of State A ?
I. After increasing the population of State A by 15% it becomes 1.61 lakhs
II. Ratio of population of State A to that of. State B is 7 : 8 respectively
III. Population of State B is 1.6 lakhs
(1) I only
(2) II and III only
(3) I and II only
(4) Either only I or II and III
(5) All I, II and III

10. How many workers are required for completing the construction work in 10 days ?
I. 20% of the work can be completed by 8 workers in 8 days
II. 20 workers can complete the work in 16 days
III. One eighth of the work can be completed by 8 workers in 5 days
(1) I and II only
(2) II and III only
(3) I only
(4) III only
(5) Any one of three

Directions (11-16) : Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are **sufficient** to answer the question. Read both the statements and—

Give answer (1) if the data in **statement I alone are sufficient** to answer the question, while the data in **statement II alone** are **not sufficient** to answer the question.

Give answer (2) if the data in **statement II alone** are **sufficient** to answer the question, while the data in **statement I alone** are **not sufficient** to answer the question.

Give answer (3) if the data either in **statement I alone** or in **statement II alone** are **sufficient** to answer the question.

Give answer (4) if the data even in **both statements I and II together** are **not sufficient** to answer the question.

Give answer (5) if the data in **both statements I and II together** are **necessary** to answer the question.

(SBI Banks PO Exam. 18.05.2003)

11. By selling a product with 20% profit, how much profit was earned ?
I. The difference between, cost and selling price is Rs. 40/-.
II. The selling price is 120 per cent of the cost price.

12. A train crosses another train running in the opposite direction in *x* seconds. What is the speed of the train ?
I. Both the trains have the same length and are running at the same speed.
II. One train crosses a pole in 5 seconds.

13. What is the two digit number ?
I. The difference between the two digits is 9.
II. The sum of the digits is equal to the difference between the two digits.

14. A spherical ball of given radius *x* cm is melted and made into a right circular cylinder. What is the height of the cylinder ?
I. The volume of the cylinder is equal to the volume of the ball.
II. The area of the base of the cylinder is given.

15. Area of a square is equal to the area of a circle. What is the circumference of the circle ?
I. The diagonal of the square is *x* inches.
II. The side of the square is *y* inches.

16. A, B and C are positive integers. Is their product an even number ?
I. A is an even number.
II. The product of A and B is an even number and that of A and C is also an even number.

Directions (17-21) : In each of the following questions, a question is followed by information given in three statements. You have to study the question alongwith the statements and decide the information given in which of the statement(s) is **necessary and sufficient** to answer the question.

(SBI PO Exam. 09.01.2005)

17. What is the average weight of girls in the class ?
I. Average weight of all the 60 students is 42 kg.
II. Average weight of boys is 43 kg.
III. Total weight of all girls together is 1144 kg.
(1) Any two of three
(2) All I, II & III
(3) I & II only
(4) II & III only
(5) Question cannot be answer even with information in all three statements.

18. What is the selling price of the T.V. set if no discount is offers
I. Profit earned was 20%.
II. Had 10% discount been offered on selling price the profit have been Rs. 1,200.
III. Cost price is Rs. 15,000.
(1) Any two of the three
(2) Only I & II
(3) Only I & III
(4) Only II & III
(5) None of these

19. What is the speed of the train ?
I. Length of the platform is 150% of the length of the train.
II. The train crosses the platform in 25 seconds.
III. The train crosses the signal pole in 19 seconds.
(1) All I, II & III
(2) I and either II or III
(3) Only II & III
(4) Any two of the three
(5) Question cannot be answered even with the information in all three statements.

20. How many children are there in the class ?
I. 20% children are there in the class speak Hindi.
II. 44 children can speak languages other than Hindi.
III. There are 30 boys in the class.
(1) All I, II & III
(2) Any two of the three
(3) II and either I or III
(4) I and II only
(5) None of these

21. What is the volume of the cylindrical tank ?
I. Area of the base is X square metres.
II. Height of the tank is Y metres.
III. Diameter of the base is equal to height of the tank.
(1) Only I & II

(2) Only II
(3) Only I & III
(4) All I, II & III.
(5) Any two of the three

Directions (22-26) : Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are **sufficient** to answer the questions. Read both the statements and -

Give answer **(1)** if the data in **statement I** alone are sufficient to answer the question, while the data in statement **II** alone are not sufficient to answer the question.

Give answer (2) if the data in **statemeat II** alone are sufficient to answer the question, while the data in statement **I** alone are not sufficient to answer the question.

Give answer (3) if the data **either** in statement I alone or in statement II alone are sufficient to answer the question.

Give answer (4) if the data even in both the statements I and II together are not sufficient to answer the question.

Give answer (5) if the data in both the statements I and II together are neccessary to answer the question.

(SBI Associate Banks PO Exam. 07.01.2007)

22. By selling a product for Rs. 100/-how much profit was earned ?
I. 20% profit would have been earned if it had been sold for Rs. 90/-
II. The profit was one-third of the purchase price.

23. A train crosses another train running in the opposite direction in *x* seconds. What is the speed of the train ?
I. Both the trains are running at the same speed.
II. The first train is *y* cm. long.

24. The difference between the two digits of a number is 6. What is the number ?
I. The digit at the unit place is bigger than the other digit.
II. The sum of the two digits is 12.

25. X, Y and Z are integers. Is X an odd number ?
I. An odd number is obtained when X is divided by 5.
II. (X + Y) is an odd number.

26. What is the capacity of a cylindrical tank ?
I. Radius of the base is half of its height which is 28 metres.
II. Area of the base is 616 sq. metres and its height is 28 metres.

Directions : (27-31) : Each question below is followed by two statements A and B. You are to determine whether the data given in the statement is **sufficient** for answering the question. You should use the data and your knowledge of Mathematics to choose the best possible answers.

Give answer (1) if the statement A alone is sufficient to answer the question, but the statement B alone is not sufficient.

Give answer (2) if the statement B alone is sufficient to answer the question, but the statement A alone is not sufficient.

Give answer (3) if both statements A and B together are needed to answer the question.

Give answer (4) either the statement A alone or statement B alone is sufficient to answer the question

Give answer (5) if you cannot get the answer from the statements A and B together, but need even more data.

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

27. Triangle ABC has angle BAC equal to 90°. What is the measure of the angle ABC ?
A. The angle ACB is 35°.
B. The angle CBA is 55°.

28. X, Y and Z are three consecutive even numbers (not necessarily in this order). What is the sum of these numbers ?
A. The difference between X and Z is 4.
B. One-third of Y is 14.

29. What is the salary of P, in a group of P, Q, R, S, T and U, whose average salary is Rs. 35,000 ?
A. Total of the salary of Q and S is Rs. 54000.
B. Total of the salary of T and U is Rs. 58000.

30. What is the rate of p.c.p.a. on an amount of Rs. 6,000 deposited in a Bank ?
A. The simple interest for four years is Rs. 36000.
B. The difference between the simple interest and compound interest is Rs. 894.0375.

31. What is the number ?
A. 20% of that number is one-fifth of that number.
B. $\frac{5}{6}$ th of that number is less by 15 of that number.

Directions (32 - 36) : Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements

are sufficient to answer the question. Read both the statements and—

(SBI Management Executive Exam. 23.02.2014)

Give answer (1) if the data in statement **I alone** are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.

Give answer (2) if the data in statement **II alone** are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.

Give answer (3) if the data **either** in statement I alone or in statement II alone are sufficient to answer the question.

Give answer (4) if the data given in both the statements I and II together are not sufficient to answer the question, and

Give answer (5) if the data in both the statements I and II together are necessary to answer the question.

32. What will be the amount at the end of 2 years, if the interest is compounded yearly.
I. The simple interest on the same sum for a period of 2 years is Rs. 400 at the same rate of 5% per annum.
II. The difference between the simple interest and the compound interest for 2 years at the rate of 5% per annum is Rs. 100.

33. Find the average of five consecutive odd numbers.
I. The difference of fifth number and the first number is 7.
II. The sum of the first two numbers is 5 more than the fifth number.

34. Find the ratio of the area of the bigger circle and smaller circle.



I. The radius of the smaller circle is 24 cm.
II. The difference between the radii of bigger and the smaller circles is 3 cm.

35. What is the length of the train ?
I. The train crosses a signal post in 9 seconds.
II. If the train with speed *x* kmph crosses another train 100m long coming from the opposite direction at 60 kmph in 15 seconds.

36. Find the radius of the semi-circle.
I. The area of semi-circle is equal to the area of the rectangle.
II. The breadth of rectangle is 5 cm less than its length and its perimeter is 50 cm.

RBI GRADE-B OFFICER EXAMS

Directions (1-5): Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the question. Read the question and both the statements and —

(RBI Grade-B Officer Exam. 2008)

Give answer (1) if the data in **statement I alone are sufficient** to answer the question, while the data in **statement II alone are not sufficient** to answer the question.

Give answer (2) if the data in **statement II alone are sufficient** to answer the question, while the data in **statement I alone** are not **sufficient** to answer the question.

Give answer (3) if the data **either** in **statement I alone or in statement II** alone are **sufficient** to answer the question.

Give answer (4) if the data even in both the **statements I and II together** are not **sufficient** to answer the question.

Give answer (5) if the data in both the **statements I and II together are necessary** to answer the question.

1. Train 'A' running at a certain speed crosses another train 'B' running at a certain speed in the opposite direction in 12 seconds. What is the length of train 'B' ?
I. The length of both the trains together is 450 metres.
II. Train 'A' is slower than train 'B'.

2. Area of a rectangle is equal to the area of a right angled triangle. What is the length of the rectangle ?
I. The base of the triangle is 40 cms.
II. The height of the triangle is 50 cms.

3. What was the total compound interest on a sum after three years ?
I. The interest after one year was Rs. 100/- and the sum was Rs. 1,000/-.
II. The difference between simple and compound interest on a sum of Rs. 1,000/- at the end of two years was Rs. 10/-.

4. What Is the two digit number where the digit at the unit place is smaller ?
I. The difference between the two digits is 5.
II. The sum of the two digits is 7.

5. What is the speed of the boat in still water ?
I. It takes 2 hours to cover distance between A and B downstreams.

II. It takes 4 hours to cover distance between A and B upstreams.

Directions (6-10): Each of the questions given below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements is sufficient to answer the question. Read both the statements.

Give answer (1) if the data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question.

Give answer (2) if the data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.

Give answer (3) if the data in statement I alone or in statement II alone is Sufficient to answer the question.

Give answer (4) if the data in both the statements I and II is not stifficient to answer the question.

Give answer (5) if the data in both the statements I and II together is necessary to answer the questions.

(RBI Grade-B Officer Exam.06.02.2011)

6. What is the two digit positive number ?
I. Sum of the two digits of the number is 8.
II. Sum of the two digits is 3 more than the higher digit.

7. The symbol ∞ represents one of the the following operations: addition, subtraction, multiplication or division. What is the value of $6 \infty 4$?
I. $0 \infty 5 = 5$
II. $5 \infty 0 = 5$

8. At what time did Poonam finish the job ?
I. Poonam started working with out a break on the job at exactly 9 a.m. and by noon she had utilised exactly half the time that it took her to finish the job.
II. Poonam took exactly 6 hours to finish the job.

9. Last year an employee received an annual salary of Rs 6,18,000 which was paid in equal pay cheques throughout the year. What was the salary received in each of the paycheques ?
I. The employee received a total of 24 paycheques durning the year ?
II. The employee received a paycheque twice a month each month during the year.

10. Is the sum of integers *x* and *y* greater than 85 ?
I. The product of *x* and y is greater than 85.
II. One of the variables is greater than 83.

INSURANCE EXAMS

Directions (1-5) : In each of the following questions, a question is asked followed by three statements. You have to the questions and all the three statements given and decide whether information provided in the statement(s) are **redundant** and **can be expensed with** while answering the questions ?

(LIC Assistant Administrative Officer (AAO) Exam, 24.04.2005)

1. What is the area of the given rectangle ?
(A) Perimeter of the rectangle is 60 cms.
(B) Breadth of the rectangle is 12 cms,
(C) Sum of two adjacent sides is 30 cms.
(1) A and B only
(2) A only
(3) B only
(4) C only
(5) A or C only

2. Who is the tallest among M, P, Q and R ?
(A) P is taller than Q but not as R.
(B) R is taller than M.
(C) M is taller than P but not as tall as R.
(1) C only
(2) B only
(3) B or C only
(4) A or B only
(5) A or C only

3. What will be the ratio between ages of Samir and Anil after five years?
(A) Samir's present age is more than Anil's present age by 4 years.
(B) Anil's present age is 20 years.
(C) Anil and Samir's present ages are in the ratio of 3 : 4.
(1) A or B or C only
(2) B only
(3) C only
(4) A or C only
(5) B or C only

4. Mr. X borrowd a sum of money on compound interest. What will be the amount to be repaid if he is repaying entire amount at the end of two years ?
(A) The rate of interest is 5 percent p.a.
(B) Simple Interest fetched on the same amount in one year is Rs. 600/-.
(C) The amount borrowed is 10 times the simple interest in two years.
(1) C only
(2) A only
(3) A or B only
(4) A or C only

(5) All A, B and C are required to answer the question.

5. A boat will take how much time to cross the river against the stream of the river ?
(A) In still water the speed of the boat is 15 km/hour.
(B) The width of the river is 8 km.
(C) The speed of the stream is 2 km/hr.
(1) Only A
(2) Only B
(3) Only C
(4) All A, B, C are required to answer the question
(5) It is not possible to answer the question with the help of statements all three A, B and C.

Directions (6-10) : Each of the questions consists of a question followed by three statements. You have to study the questions and the statements and decide which of the statement(s) is/are necessary to answer the question.

(LIC Assistant Administrative Officer (AAO) Exam. 2006)

6. What is the area of the hall ?
I. Material cost of flooring per sq metre is 250
II. Labour cost of flooring the hall is Rs. 3,500
III. Total cost of flooring the hall is Rs. 14,500
(1) I and II only
(2) II and III only
(3) All I, II and III
(4) Any two of the three
(5) None of these

7. What was the percentage of discount offered ?
I. Profit earned by selling the article for Rs. 252 after giving discount was Rs. 52.
II. Had there been no discount the profit earned would have been Rs. 80.
III. Had there been no discount the profit earned would have been 40%.
(1) I and II only
(2) II and either I or III only
(3) I and III only
(4) I and either II or III only
(5) None of these

8. What is the speed of the train ?
I. The train crosses a signal pole in 13 sec.
II. The train crosses a platform of length 250 mts. in 27 seconds.
III. The train crosses another train running in the same direction in 32 seconds.
(1) I and II only
(2) I and III only
(3) II and III only
(4) Any two of the three
(5) None of these

9. What is the population of State 'A' ?
I. After increasing the population of State A by 15% it becomes 1.61 lakhs
II. Ratio of population of State A to that of State B is 7 : 8 respectively
III. Population of State B is 1.6 lakhs
(1) 1 only
(2) II and III only
(3) I and II only
(4) Either only I or II and III
(5) All I, II and III

10. How many workers are required for completing the construction work in 10 days ?
I. 20% of the work can be completed by 8 workers in 8 days
II. 20 workers can complete the work in 16 days
III. One eighth of the work can be completed by 8 workers in 5 days
(1) I and II only
(2) II and III only
(3) I only
(4) III only
(5) Any one of three

Directions (11-15): In each of the following questions, a question and three statements I, II and III are given. You have to decide whether the data given in the statements are sufficient to answer the question or not.

(LIC Assistant Administrative Officer Exam. 2008)

11. How much profit did the company earn in the year 2002 ?
I. The company earned 40% more profit in the year 2003 than that in the year 2001.
II. The company earned a total profit of Rs. 20 crores in the years 2001 and 2002 taken together,
III. In the year 2003, the company earned 80 per cent, profit of that in 2002.
(1) Any two of I, II and III are sufficient
(2) Either I and II or II and III are sufficient
(3) I and II or III are sufficient
(4) All I, II and III are necessary to answer the question

12. How many students secured at least 60 per cent marks in Mathematics in a class of 240 students ?
I. 20 per cent of the students in the class secured 80 per cent and above marks in Mathematics.

II. 80 students have secured more than 50 per cent but less than 60 per cent marks in Mathematics.
III. The number of students who secured marks between 60 and 79 per cent was equal to the number of students who secured less than 50 per cent marks in Mathematics.
(1) All I, II and III are necessary to answer the question
(2) Only I and III are sufficient
(3) Only II and III are sufficient
(4) Only I and II are sufficient
(5) The question cannot be answered even with all I, II and III

13. In an institute employing managers, observers and assistants what is the monthly salary of an assistant ?
I. Each observer gets Rs. 12000 per month more than an assistant.
II. An observer and an assistant together get Rs. 32000 per month.
III. The total salary per month of a manager and an observer is Rs. 57000.
(1) Either I or II and III are sufficient
(2) Only I and II are sufficient
(3) Any two of I, II and III are sufficient
(4) All I, II and III are necessary to answer the question
(5) The question cannot be answered even with all I, II and III

14. What is the circumference of semicircle in cm ?
I. The area of semi-circle is half of the area of parallelogram.
II. The length of parallelogram is 1.5 times the radius of the semicircle
III. The difference between the length and breadth of parallelogram is 8 cm.
(1) Only II and III are sufficient
(2) Only I and III are sufficient
(3) Only I and II are sufficient
(4) All I, II and III are necessary to answer the question
(5) The question cannot be answered even with all I, II and III

15. In a partnership business of A, B and C, what profit did B get at the end of 2 years ?
I. A and B started the business by investing in ratio 3 : 5
II. C joined after 6 months with Rs. 4 lakhs
III. After two years, A's share in profit was Rs. 60000.
(1) Only I and III are sufficient
(2) Only II and III are sufficient
(3) Either I or II and III are sufficient
(4) All I, II and III are necessary to answer the question
(5) The question cannot be answered even with all I, II and III

Directions (16-20): Each question below is followed by two statements A and B. You are to determine whether the data given in the statement is sufficient for answering the question. You should use the data and your knowledge of Mathematics to choose between the possible answers.

(United India Insurance AO Exam. 26.05.2013

Give answer (1) if the statement A alone is sufficient to answer the question, but the statement B alone is not sufficient.

Give answer (2) if the statement B alone is sufficient to answer the question, but the statement A alone is not sufficient.

Give answer (3) if both statements A and B together are needed to answer the question.

Give answer (4) if either the statement A alone or statement B alone is sufficient to answer the question.

Give answer (5) if you cannot get the answer from the statements A and B together, but need even more data.

16. Is A an odd number ?
A. A multiplied by an odd nmber is equal to an odd number.
B. A is not divisible by 2.

17. The ages of Pradumn and C are in the ratio of 7 : 5 . What is the age of Pradumn.
A. The ages of Pradumn and Nandini are in the ratio of 3 :1.
B. After 7 years the ratio of Pradumn's and Aviral's ages will be 4 : 3.

18. What is the salary of B, in a group of A, B, C and D whose average salary is Rs. 62,880 ?
A. Total of the salary of A and C is exact multiple of 8.
B. Average of the salary of A , C and D is Rs. 61,665.

19. What is the three digit nunmber ?
A. The three digit number is divisible by 9.
B. The first and the third digit is 6.

20. What is the profit earned by selling a printer for Rs. 3,000 ?
A. The cost price of 6 such printers is equal to selling price of 5 such printers.
B. 20% profit is earned by selling each printer.

SHORT ANSWERS

NATIONALISED BANKS & IBPS PO/MT/SO

1. (4) **2.** (4)
3. (3) **4.** (5)
5. (4) **6.** (2)
7. (3) **8.** (5)
9. (5) **10.** (1)
11. (4) **12.** (5)
13. (5) **14.** (3)
15. (2) **16.** (5)
17. (4) **18.** (5)
19. (2) **20.** (3)
21. (4) **22.** (5)
23. (2) **24.** (3)
25. (4) **26.** (4)
27. (5) **28.** (2)
29. (5) **30.** (3)
31. (4) **32.** (1)
33. (3) **34.** (2)
35. (5) **36.** (1)
37. (2) **38.** (3)
39. (3) **40.** (4)
41. (4) **42.** (3)
43. (1) **44.** (2)
45. (5) **46.** (1)
47. (4) **48.** (3)
49. (5) **50.** (3)
51. (3) **52.** (5)
53. (1) **54.** (4)
55. (5) **56.** (2)
57. (4) **58.** (5)
59. (1) **60.** (3)
61. (5) **62.** (4)
63. (3) **64.** (4)
65. (5) **66.** (2)
67. (5) **68.** (4)
69. (5) **70.** (1)
71. (3) **72.** (4)
73. (1) **74.** (5)
75. (5) **76.** (2)
77. (4) **78.** (5)
79. (4) **80.** (1)
81. (2) **82.** (5)
83. (5) **84.** (5)
85. (5) **86.** (4)
87. (3) **88.** (3)
89. (5) **90.** (5)
91. (5) **92.** (1)
93. (5) **94.** (5)
95. (3) **96.** (2)
97. (2) **98.** (4)
99. (4) **100.** (4)
101. (1) **102.** (2)
103. (4) **104.** (1)
105. (4)

SBI PO EXAMS

1. (5) **2.** (5)
3. (5) **4.** (2)
5. (1) **6.** (3)
7. (4) **8.** (1)
9. (4) **10.** (5)
11. (1) **12.** (4)
13. (1) **14.** (2)
15. (3) **16.** (3)
17. (2) **18.** (1)
19. (5) **20.** (4)
21. (5) **22.** (3)
23. (4) **24.** (5)
25. (1) **26.** (3)
27. (4) **28.** (3)
29. (5) **30.** (1)
31. (2) **32.** (3)
33. (4) **34**. (5)
35. (5) **36.** (5)

RBI GRADE-B OFFICER EXAMS

1. (4) **2.** (4)
3. (3) **4.** (5)
5. (4) **6.** (4)
7. (5) **8.** (1)
9. (3) **10.** (5)

INSURANCE EXAMS

1. (1) **2.** (4)
3. (3) **4.** (5)
5. (4) **6.** (3)
7. (4) **8.** (1)
9. (4) **10.** (5)
11. (4) **12.** (5)
13. (2) **14.** (5)
15. (1) **16.** (4)
17. (5) **18.** (2)
19. (3) **20.** (4)

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (4) Time taken in crossing each other

$$= \frac{\text{Total length of trains}}{\text{Relative speed}}$$

The information given in both statements is not sufficient as length of train A and individual speed of each train are required.

2. (4) Area of rectangle = Area of triangle.
From the information given in both the statements, we can find area of triangle or area of rectangle. For finding length, breadth is required, which is not known.

3. (3) From the statement I,

$r = \frac{100 \times 100}{1000} = 10\%$

Thus we have,
P = Rs. 1000, r = 10%, t = 3years
Hence, C.I. can be determined
From the statement II.

$S.I = \frac{1000 \times r \times 2}{100} = 20r$

$C.I = 1000\left[\left(1+\frac{r}{100}\right)^2 - 1\right]$

$\therefore$ C.I - S.I

$= 1000\left[\frac{200r + r^2}{10000}\right] - 20r$

$\Rightarrow 2000r + r^2 - 200r = 100$

$\Rightarrow r = 10$

Hence, C.I. can be determined

4. (5) Let the unit's digit be x and ten's digit be y and $x < y$.
$\therefore$ Number = $10y + x$
From statement I,
$y - x = 5$ ....(i)
From statement II,
$y + x = 7$ ....(ii)
From (i) and (ii), x, y can be calculated and two digit number can be found.

5. (4) Let the distance between A and B be z km.
Again, let speed of boat in still water be x kmph and that of stream be y kmph.
$\therefore$ Rate downstream = $(x + y)$ kmph
Rate upstream = $(x - y)$ kmph
From statement I,

$\frac{z}{x+y} = 2$ ...(i)

From statement II

$\frac{z}{x-y} = 4$ ....(ii)

6. (2) From statement I,
If the. annual income of boss be Rs. x, then Mr. Krishnamurthy's annual income
= 170% of x
But we donot know the value of. x. Hence this statement is not sufficient from
From statement II,
Initial income = Rs. 12000
% Increase = 10% per month
Time = 12 months
$\therefore$ Income in May = Rs. 12000 + 10% of 12000
= Rs. (12000 + 1200) = Rs. 13200
Similarly, the income for other months can be calculated.

7. (3) From statement I,
If the radius of the circular field be r metres, then $2\pi r = x$
This relation gives us the value of radius.
$\therefore$ Area = $\pi \times (\text{Radius})^2$
Hence, statement I alone is sufficient to answer the question
From statement II,
Diameter = y

$\therefore \text{Area} = \left(\frac{\text{Diameter}}{2}\right)^2$

Hence, statement II alone is also sufficient to answer the question.

8. (5) Let the speed of boat in still water be x kmph and that of current be y kmph
Rate upstream = $(x - y)$ kmph
Rate downstream = $(x + y)$ kmph
From statement I,

$x + y = \frac{35}{5}$

$\Rightarrow x + y = 7$ kmph ....(i)

From statement II,

$x - y = \frac{35}{7}$

$\Rightarrow x - y = 5$ kmph ...(ii)

From combined statements I and II, we can get the required answer.

9. (5) From statement I,
Let the number of boys and girls be $5x$ and $6x$ respectively.
From statement II,
$6x - 5x = 7$
$\Rightarrow x = 7$
$\therefore 5x = 35$ and
$6x = 42$
Clearly, both statement are required to answer the question.

10. (1) From statement I,
SP = Rs. 1740
% Profit = 20

$\therefore CP = \frac{100 \times 1740}{120} = \text{Rs } 1450$

$\therefore$ Profit = Rs. (1740 - 1450)
= Rs. 290
Information in statement II is not required.
Hence, statement I alone is sufficient to answer the question.

11. (4) We need equivalence between one day's work of man and woman.

From statement I, we can get 1 man's 1 day's work.
From statement II or III, we can get 1 woman's 1 day's work.
Hence, we can establish relation between man's and woman's work and get the required answer.

12. (5) When a train crosses a pole, then
Speed of train

$$= \frac{\text{Length of train}}{\text{Time taken}}$$

When a train crosses a platform, then Speed of train

$$= \frac{\text{Length of platform and train}}{\text{Time}}$$

Clearly, statement II and either I or III supplement the required data to determine the speed of train.

13. (5) Area of the square = $(\text{side})^2$

$$= \frac{1}{2}(\text{diagonal})^2$$

Again, Perimeter = 4 × side
Clearly, from any one of the three statements we can determine area of the square.

14. (3) Let the number be $10x + y$.
From statement I,
$(10y + x) - (10x + y) = 18$
$\Rightarrow 9(y - x) = 18$
$\Rightarrow y - x = 2$ ...(i)
From statement II,
$x + y = 14$ ...(ii)
From statement III,
$y - x = 2$ ...(ili)
Clearly, statement II and either I or III can supplement the data to know x and y and hence the number.

15. (2) From statement I,
We do not know the principal.
From statement II,
Data are incomplete,
i.e. principal is unknown.
From statement III,
We get the required data, i.e.
If principal be Rs. x.
Interest = Rs. x
Time = 5 years,

$$\text{Rate} = \frac{\text{Interest} \times 100}{\text{Principe} \times \text{time}}$$

16. (5) From statements I,
Volume of the cylindrical tank = $\pi r^2 h$
Where r = radius of the base and h = height of the tank.
From statement II,
$\pi r^2 = 616$
From statement III,
$h = 28$ metre
Hence, we can determine the capacity of the tank by any two of the statements

17. (4) Let the length of the train be x metre
From statements I and II,

$$\text{Speed of train} = \frac{x}{18} = \frac{2x}{36}$$

It gives no result.
From statements I and III,

$$\text{Speed of train} = \frac{300}{18}$$

$$= \frac{50}{3} \text{ metre/second}$$

From statements II and III

$$\text{Speed of train} = \frac{300 + 300}{36}$$

$$= \frac{50}{3} \text{ metre/second.}$$

18. (5) Question cannot be answered even with the information in all three statements.

19. (2) Let the two digit number be $10x + y$.
From statement I,
$\Rightarrow 10y + x = 10x + y + 9$
$\Rightarrow 9(y - x) = 9$
$\Rightarrow y - x = 1$ ...(i)
From statement II,
$x + y = 7$ ...(ii)
From statement III,
$x - y = 1$ or $y - x = 1$ ...(iii)
From statements I and II,
The number obtained = 34
From statements II and III,
Numbers = 34 or 43
Hence, we get no unique answer.

20. (3) From statements II and III
Profit on each article = 765 - 632
From statements I, II and III
Number of articles sold

$$= \frac{1596}{133} = 12$$

21. (4) From statement A
3 × 5 = 15 ; 5 × 9 = 45 (An odd number)
It is also obvious from statement B.

22. (5) The answer is not possible with the help of even both the statements. We need more

information like sum or average of their ages or ratio of their after some time or before sometime etc.

23. (2) A + B + C + D
= Rs. (4 × 62880)
From statement B,
A + C + D = Rs.(3 × 61665)
∴ B's salary = (A + B + C + D)'s salary - (A + C + D)'s salary

24. (3) From statement A,
The three digit number is divisible by 9.
From statement B,
Number = 6*6
A number is divisible by 9 if sum of its digits is divisible by 9.
Clearly, * = 6
because 666 ÷ 9 = 74

25. (4) From statement A,
Let CP of 1 printer = Re. 1
∴ CP of 5 printers = Rs. 5
and SP of 5 printers = Rs. 6

$$\therefore \text{Gain \%} = \frac{1}{5} \times 100 = 20\%$$

$$\therefore \text{CP} = \frac{100}{120} \times 3000 = \text{Rs.}2500$$

∴ Gain = Rs. (3000 - 2500) = Rs. 500
From statement B, we can also find the answer.

26. (4) The given data are inadequate.

27. (5) From statement II,
If the age of Rani = x years,
then Surekha's age = $2x$ years
$\therefore x + 2x = 72$
$\Rightarrow 3x = 72$ years

$$\Rightarrow x = \frac{72}{3} = 24 \text{ years}$$

∴ Rani's age = 24 years
As per the given information in statement I, Nidhi's age can be determined.

28. (2) Statement I is superfluous.
From statement II,
Number of boys in the school

$$= 3500 \times \frac{60}{100} = 2100$$

Number of girls in the school = 3500 - 2100 = 1400.
∴ Required ratio = 2100 : 1400 = 3 : 2

29. (5) Let Mr. Mehta's present income be Rs. x.
From statements I and II,
10% of x = 2500

$$\Rightarrow x \times \frac{10}{100} = 2500$$

$\Rightarrow x = 2500 \times 10 =$ Rs. 25000

30. (3) From statement I,
Speed of the bus

$$= \frac{\text{Distance covered}}{\text{Time Taken}}$$

$$= \frac{80}{5} = 16 \text{ kmph}$$

As per the information in statement II, the speed ofthe bus can also be determined.

31. (4) From statement A,
A number divisible by 2 is an even number.
From statement B,
2 + 3 = 5 (odd number)
4 + 9 = 13 (odd number)

32. (1) From statement A,
Let the number be x.

$$\Rightarrow \frac{x}{2} - \frac{x}{3} = 27 \Rightarrow \frac{3x - 2x}{6} = 27$$

$\Rightarrow x = 27 \times 6 = 162$
Statement B is superfluous.

33. (3) From both statements,
T : S = 6 : 5
T : D = 3 : 2 = 6 : 4
∴ T : S : D = 6 : 5 : 4
Let the present ages of Danish and Shivay be $4x$ and $5x$ years respectively.

$$\therefore \frac{4x + 6}{5x + 6} = \frac{6}{7}$$

$\Rightarrow 30x + 36 = 28x + 42$
$\Rightarrow 2x = 6 \Rightarrow x = 3$
Hence, the age of Shivay can be determined.

34. (2) From statement B,

Men	Days
9	27
15	x

$\Rightarrow 15 : 9 = 27 : x$
$\Rightarrow 15 \times x = 9 \times 27$

$$\Rightarrow x = \frac{9 \times 27}{15} = \frac{81}{5} \text{ days}$$

35. (5) From statement A,
A + E = 64500
From statement B,
B + F = 52600
∴ C + D = (A + B + C + D + E + F) - (A + B + E + F)
Hence, the salary of C cannot be determined.

36. (1) From statement 'A

Days	Women
20	12
15	x

$\therefore 15 : 20 = 12 : x$

$\Rightarrow 15 \times x = 12 \times 20$

$\Rightarrow x = \frac{12 \times 20}{15} = 16$ women

Statement B is superfluous.

37. (2) From statement B,

$SI = \frac{\text{Principal} \times \text{time} \times \text{Rate}}{100}$

$\Rightarrow 2800 = \frac{12000 \times 2 \times \text{Rate}}{100}$

$\Rightarrow \text{Rate} = \frac{2880 \times 100}{12000 \times 2}$

= 12% per annum

38. (3) From both the statements,

Number = 7 * 7

$\because$ 7 * 7 is exactly divisible by 13.

$\therefore$ * = 6

$\therefore$ Number = 767

39. (3) From both the statements,

A + B + C + D + E = 5 × 45

= 225 years (i)

A + B = 106 years (ii)

D + E = 94 years (iii)

By Equation (i) - (ii) - (Iii), the age of C can be determined.

40. (4) From statement A,

CP of 5 laptops

= Rs. (4 × 26250)

= Rs. 105000

SP of 5 laptops

= Rs. (5 × 26250)

= Rs. 131250

$\therefore$ Gain / laptop

$= \frac{\text{Rs.}(131250 - 105000)}{5}$

$= \text{Rs.} \frac{26250}{5} = \text{Rs. } 5250$

From statement B,

CP of each laptop

$= \frac{100}{125} \times 26250 = \text{Rs. } 21000$

$\therefore$ Gain = Rs. (26250 - 21000)

= Rs. 5250

41. (4) From statement A,

$\text{Speed of car} = \frac{\text{Distance covered}}{\text{Time taken}}$

$= \frac{135}{3} = 45$ kmph

From statement B,

$\text{Speed of car} = \frac{270}{6} = 45$ kmph

42. (3) From statements A and B,

Let the number be $10y + x$

where $x > y$

$xy = 72$ ....(i)

$x - y = 1$ ...(ii)

$\therefore (x + y)^2 = (x - y)^2 + 4xy$

$= 1 + 4 \times 72$

$= 1 + 288 = 289$

$\therefore x + y = \sqrt{289} = 17$ .... (iii)

From equations (ii) and (iii),

$x = 9$ $y = 8$

$\therefore$ Number = 89

43. (1) From statement A,

Number of boys

$= 2500 \times \frac{40}{100} = 1000$

Number of girls

= 2500 - 1000 = 1500

$\therefore$ Required ratio

= 1500 : 1000 = 3 :2

Statement B is superfluous.

44. (2) Statement A is superfluous.

From statement B,

Let the number be x.

$\frac{x}{2} - \frac{x}{3} = 27$

$\Rightarrow \frac{3x - 2x}{6} = 27$

$\Rightarrow x = 27 \times 6 = 162$

45. (5) Both the statements together are not sufficient to answer the question.

46. (1) From statements (I) and (II),

D + E = 14

and A+B + C + F = 4 × 50 = 200

$\therefore \frac{A + B + C + D + E + F}{6}$

$= \frac{14 + 200}{6} = \frac{214}{6} = 35\frac{2}{3}$ years

47. (4) Area of the right angled triangle

$= \frac{1}{2} \times$ base × height

48. (3) From all three statements,

(A + B)'s 1 day's work = $\frac{1}{8}$ ....(i)

(B + C)'s 1 day's work = $\frac{1}{10}$ ....(ii)

(A + C)'s 1 day's work = $\frac{1}{12}$ ...(iii)

Adding all three equations,
2(A + B + C)'s 1 day's work

$= \frac{1}{8}+\frac{1}{10}+\frac{1}{12}$

$= \frac{15+12+10}{120} = \frac{37}{120}$

$\therefore$ (A + B + C)'s 1 days work = $\frac{37}{240}$ ....(iv)

By equation (iv) - (iii),

B'sl day's work = $\frac{37}{240} - \frac{1}{12}$

$= \frac{37-20}{240} = \frac{17}{240}$

$\therefore$ B will complete the work in $\frac{240}{17}$ days.

49. (5) From statement (I),

$x = \frac{x \times 10 \times r}{100}$

$\Rightarrow r$ = 10% / annum
From statement (II),

Principal = Difference$\left(\frac{100}{\text{Rate}}\right)^2$

50. (3) From all three statements,
M + Sc. + E = 198
Let Abhijit get x marks in English.
$\therefore$ Marks in science = x + 12 and
marks in Maths = x + 32
$\therefore x + x + 12 + x + 32 = 198$
$\Rightarrow 3x + 44 = 198$
$\Rightarrow 3x = 198 - 44 = 154$

$\Rightarrow x = \frac{154}{3}$

51. (3) The data in statement I alone or in statement II alone are sufficient. We can find the ratio of profit sharing as ratio of amounts of investment can be determined

52. (5) Let the number be $10x + y$.
From statement I,
$x + y = 6$
Many such combinations are possible.
From statement II
$x = 2y$
Taking both the statement together,
$\therefore 3y = 6 \Rightarrow y = 2$ and hence $x = 4$
$\therefore$ Number = 42

53. (1) From statement I alone,
Principal = Rs. x, R = ?, S.I. = x
T = 10 years.

$\therefore R = \frac{\text{S.I} \times 100}{\text{Principal} \times \text{Time}}$

$= \frac{x \times 100}{x \times 10} = 10\%$

54. (4) Statement II gives no conclusion.
From statement II,

(A + B)'s 1 day's work = $\frac{1}{12}$

A's 1 day's work is not known.

55. (5) From both the statements,

$\frac{5x+4}{x+4} = \frac{17}{5}$

$\Rightarrow 25x + 20 = 17x + 68$
$\Rightarrow 8x = 48 \Rightarrow x = 6$
$\therefore$ Mother's present age = 5 × 8 = 40 years

56. (2) From statement I,
$x - y = 0$.
From statement II,
$x + y = 18$
It is possible only when $x = y = 9$.

57. (4) Data are inadequate.

58. (5) From both statements,
Speed of boat in still water

$= \frac{1}{2}(4 + 6) = 5$ kmph

59. (1) From statement I
Nigam's marks in Biology = 2 × 42 = 84

60. (3) From statement I,

Rate = $\frac{\text{S.I} \times 100}{\text{Time} \times \text{Principal}}$

$= \frac{1736 \times 100}{6200 \times 2} = 14\%$ Per annum

By using C.I.

$= P\left[\left(1+\frac{R}{100}\right)^T - 1\right]$

we get the required rate of interest.

61. (5) From statements I and II,
Side of the square

$= \sqrt{196}$ = 14cm.

$\therefore$ Circumference of the semi-circle
$= (\pi \div 2) \times$ radius $= 7(\pi \div 2)$ cm

62. (4) Data are inadequate.

63. (3) From statement I,

Cost price of the article = $\frac{3}{5} \times 24000$

= Rs. 14400
Gain = Rs. (24000 - 14400) = Rs 9600

$\therefore$ Gain per cent $= \frac{9600}{14400} \times 100 = 66\frac{2}{3}\%$

From statement II,
Cost price = Rs. (24000 - 9600) = Rs. 14400

Hence, profit per cent can be determined.

64. (4) From statement I,
$x^2 = 9y$
Required answer is not possible From statement II,
$4x - 3x = ?$

65. (5) From statement I,

$x \times \frac{40}{100} = \frac{50 \times 20}{100}$

$\Rightarrow x = 25$

From statement II,

$y \times \frac{30}{100} = \frac{72 \times 25}{100}$

$\Rightarrow y = 60$

$\therefore x : y = 25 : 60 = 5 : 12$

66. (2) From statement II,
$M_1D_1 = M_2D_2$
$\Rightarrow 8 \times 12 = 5 \times D_2$

$\Rightarrow D_2 = \frac{8 \times 12}{5} = \frac{96}{5} = 19\frac{1}{5}$ days

67. (5) From statement II,
If the present age of Shyam be x years then Ram's present age = $(x + 7)$ years
From statement I,

$\frac{x+7}{x} = \frac{4}{3}$

$\Rightarrow 4x = 3x + 21$

$\Rightarrow x = 21$

$\therefore$ Shyam's age after 6 years
= 21 + 6 = 27 years

68. (4) Data from both the statements are inadequate.

69. (5) From statements I and II,
Simple interest

$= \text{Rs.} \left(\frac{5000 \times 3 \times 5}{100} + \frac{5000 \times 3 \times 8}{100} \right)$

= Rs. (750 + 1200) = Rs. 1950

70. (1) From statement I,
Required CP.
= Rs. (4 × 85 + 3 × 50)
= Rs. (340 + 150)
= Rs. 490

71. (3) From statement I,
Circumference of circle
$= \pi \times$ diameter $= 21\pi$ cm
From statement II,
$\pi r^2 = 346.5$

$\Rightarrow \frac{22}{7} \times r^2 = 346.5$

$\Rightarrow r^2 = \frac{346.5 \times 7}{22} = 110.25$

$\Rightarrow r = \sqrt{110.25} = 10.5$

$\therefore$ Circumference $= 2\pi r$
$= 21\pi$ cm

72. (4) Data in both the statements are inadequate

73. (1) From statement I,
Required number of pieces

$= \frac{900}{80} \approx 11$

Data in statement II are inadequate.

74. (5) From statement I and II, Selling price of wrist watch

$= \text{Rs.} \left(6400 \times \frac{131.25}{100} \right)$

= Rs. 8400

75. (5) From statement I,
Numbers = 15, 51, 24, 42, 33, 60
Number divisible by 7 = 42

76. (2) From statement I, no equivalence relation among a woman, a boy and a man can be established.
From statement II,
$M_1D_1 = M_2D_2$
$\Rightarrow 6 \times 16 = 5 \times D_2$

$\Rightarrow D_2 = \frac{6 \times 16}{5} = \frac{96}{5}$

$= 19\frac{1}{5}$ days

77. (4) From both statements,
Rita = Sonu + 5000
Sonu = Sohan + 8000
Two equations and can't established the relation.
Hence, answer can't be determined.

78. (5) From statement I,
Pravin's present age = $2x$ years
Shekhar's present age = $3x$ years
From statements I and II,
$3x = 2x + 6$
$\Rightarrow x = 6$
$\therefore$ Pravin's age after 4 years

$= 2x + 4$
$= 2 \times 6 + 4 = 16$ years

79. (4) From both the statements answer can't be determined.
Radius = Length of rectangle
Breadth of rectangle = 22cm.
We get no information about the length of rectangle.

80. (1) From statement I,
Average speed of truck

$$= \frac{\text{Distance covered}}{\text{Time taken}}$$

$$= \frac{220}{11} = 20 \text{ kmph}$$

$\therefore$ Average speed of car = (4 × 20) kmph
= 80 kmph
From statement II,
we get no result.

81. (2) From statement I, we will not get radius of circle.
From statement II,
Radius of circle = Side of square

$= \sqrt{\text{Area}} = \sqrt{144} = 12$ cm
Area of circle = πr^2

$$= \frac{22}{7} \times 12 \times 12 = \frac{3168}{7} \text{ sq.cm.}$$

82. (5) From statement II,
$\because$ CP of 2 tables = Rs. 500

$$\therefore \text{CP of 3 tables} = \frac{500}{2} \times 3 = \text{Rs. } 750$$

From statement I,
$\because$ CP of 4 chairs = Rs. (2 × 750)
$\therefore$ CP of 2 chairs = Rs. 750
$\therefore$ 3 tables + 2 chairs
= 750 + 750 = Rs. 1500

83. (5) From statement II,
Amit's monthly income
= 2 × 3.5 = Rs. 7 thousand
From statement I,
Jyoti's monthly income
= Rs. (7000 + 8500) = Rs. 15500
$\therefore$ Jyoti's annual income
= 12 × 15500 = Rs. 186000

84. (5) From both statements,
$7x - 4x = 15$
$\Rightarrow 3x = 15 \Rightarrow x = 5$
$\therefore$ Suraj's present age
$= 7x = 7 \times 5 = 35$ years
$\therefore$ Suraj's age after 8 years
= 35 + 8 = 43 years

85. (5) From both statements,
Maximum marks of the exam
= 538 + 162 = 700
If the minimum passing percentage be x, then

$$\frac{700 \times x}{100} = 162 + 104 = 266$$

$$\Rightarrow x = \frac{266 \times 100}{700} = 38\%$$

86. (4) We do not have the average salary of D and E.
From both statements, A + B + C + D + E
$= 5 \times 48250$ ...(i)
$C = 1.5\ B$ ...(ii)
$A + B = 2 \times 23500$ ...(iii)
Clearly, C's salary cannot be determined.

87. (3) From statement I,
CP. = Rs. (640000 - 320000)
= Rs. 3,20000

$$\therefore \text{Profit per cent} = \frac{320000}{320000} \times 100 = 100\%$$

From statement II,
If the CP. be Rs. x then
S.P. = Rs. $2x$

$$\text{Gain percent} = \frac{x}{x} \times 100 = 100\%$$

88. (3) From statement I,

$$\text{Rate} = \frac{\text{S.I.} \times 100}{\text{principal} \times \text{Time}}$$

$$= \frac{11480 \times 100}{14350 \times 4} = 20\% \text{ per annum}$$

From statement II,
If principal be Rs. x, then amount = Rs. $2x$.
$\therefore$ S.I. = Rs. x, Time = 5 years

$$\text{Rate} = \frac{\text{S.I.} \times 100}{\text{Principal} \times \text{Time}}$$

$$= \frac{x \times 100}{x \times 5} = 20\% \text{ per annum}$$

89. (5) From statement II,
Unit digit = 0
From statement I,
ten's digit = 9
$\therefore$ Number = 90

90. (5) From statements I and II,
If the length of rectangle be $9x$ metre and its breadth be $7x$ metre, then
$9x \times 7x = 252$

$$\Rightarrow x^2 = \frac{252}{9 \times 7} = 4$$

$$\therefore x = \sqrt{4} = 2$$

$\therefore$ Perimeter of rectangle
= 2 (length + breadth)
= 2 (9x + 7x)
= 32x = 32 × 2 = 64 metre

91. (5) From statements I and II,
If the total marks be x, then

$$x \times \frac{25}{100} = 288\text{-}128 = 160$$

$\Rightarrow x = 640$
If the minimum pass percentage be y, then

$$640 \times \frac{y}{100} = 160 + 64 = 224$$

$$\Rightarrow y = \frac{224 \times 100}{640} = 35$$

92. (1) From statement I,
$4x + 3y + 5z = 60$
Again, $2x = y$
$2y = z \Rightarrow z = 4x$
$\therefore 4x + 6x + 20x = 60$
$\Rightarrow 30x = 60 \Rightarrow x = 2$
$\therefore y = 4; z = 8$
$\therefore x^2 + y + z = 4 + 4 + 8 = 16$
From statement II,
We have three unknown quantities but two equations.

93. (5) From statement I,
Arun + Vinay + Suraj + Raju + Pratap
= 68 × 3 + 72 × 2
= 204 + 144 = 348 kg
$\therefore$ Arun + Pratap = 348 - 78 - 68 - 46
= 156 kg
From statement II,
Arun + Suraj + Vinay + Raju
= 68 × 4 = 272 kg
$\therefore$ Pratap = 348 - 272 = 76 kg
Arun = 156 - 76 = 80 kg
Clearly, weight of Suraj is the second biggest.

94. (5) From statements I and II,
Side of square $\Rightarrow \sqrt{576} = 24$ cm
Area of rectangle = 600
$\Rightarrow l \times b = 600$
$\Rightarrow l \times 12 = 600$
$\Rightarrow l = 50$ cm
$\therefore$ Required ratio = 50 : 24 = 25 : 12

95. (3) From statement I,
$3x + 5x + 4x = 180$
$\Rightarrow 12x = 180$
$\Rightarrow x = 15$
Larger angle of parallelogram
= $5x + 34 = 75 + 34 = 109°$
$\therefore$ Second angle = 180 - 109 = 71°
From statement II,
$x + x + 38 = 180°$
$\Rightarrow 2x = 180° - 38° = 142°$
$\Rightarrow x = 71°$

96. (2) From statements I and II,
Rate downstream

$$= u + v = \frac{12}{2} = 6 \text{ kmph}$$

$$\text{Rate upstream} = u - v = \frac{12}{4} = 3 \text{ kmph}$$

$$u = \frac{1}{2}(u + v) + (u - v)$$

$$= \frac{1}{2}(6 + 3) = \frac{9}{2} \text{ kmph}$$

From statements I and III,
$u + v = 6$ kmph

$$\Rightarrow u + \frac{u}{3} = 6$$

$$\Rightarrow 4u = 18$$

$$\Rightarrow u = \frac{18}{4} = \frac{9}{2} \text{ kmph}$$

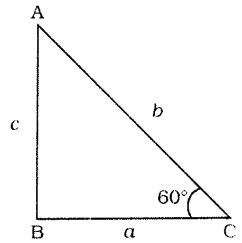
97. (2) From statements I and II,

$$\text{Speed of train} = \frac{240}{24} = 10 \text{ m/sec}$$

98. (4) From all three statements,
Total age of 11 students + 1 teacher
= 14 × 12 = 168 years
Average age of 11 students + teacher
= Average age of 11 students + 3
$\Rightarrow$ Average age of 11 students = 14 - 3 = 11 years
Their total age = 11 × 11 = 121 years
Teacher's age = 168 - 121 = 47 years

99. (3) From statements II and III,
SI for two years = Rs. 1200
Principal = 10 × 1200 = Rs. 12000
Thus we can find C.I. and amount.

100. (4) From statements I and III.



$$\frac{AB}{AC} = \cos 60° \Rightarrow BC = \frac{5}{2}$$

$a = \frac{5}{2}$, $b = 5$ and $\theta = 60°$

$\therefore$ We get area by A = $\frac{1}{2}ab\sin C$

101. (I) From statement I,

Difference = $\frac{P \times R^2}{10000}$

$\Rightarrow 360 = \frac{P \times 12 \times 12}{10000}$

$\Rightarrow P = \frac{360 \times 10000}{12 \times 12}$

Rs. 25000

From statement II,

Interest = 2 P - P = Rs, P

= Principal

Principal = $\frac{S.I \times 100}{Time \times Rate}$

From statement III,

$C.I = P\left[\left(1+\frac{Rate}{100}\right)^{Time} - 1\right]$

$\Rightarrow 6360 = P\left[\left(1+\frac{12}{100}\right)^{2} - 1\right]$

$\Rightarrow 6360 = P\left[\left(\frac{28}{25}\right)^{2} - 1\right]$

$\Rightarrow 6360 = P\left[\left(\frac{784-625}{628}\right)\right] = \frac{P \times 159}{625}$

$\Rightarrow P = \frac{6360 \times 625}{159}$ = Rs 25000

Note : You need not calculate during exam. You are required to examine whether requirements of a formula are satisfied or not.

102. (2) From statement I,

If length = x metre and breadth = y metre, then

$y = \frac{1 \times 2}{4}(x + y)$

$\Rightarrow 2y = x + y \Rightarrow y = x$

$\therefore x^2 = 144$

$\Rightarrow x = \sqrt{144} = 12$ metre

Area of field = $(16)^2 - (12)^2 = 256 - 144$

= 112 sq. metre

From statement II and III,

Length $3x$ metre

Breadth = $2x$ metre

$\therefore 3x \times 2x = 216$

$\Rightarrow x^2 = \frac{316}{3 \times 2} = 36$

$\Rightarrow x = 6$

$\therefore$ Length =18 metre, Breadth = 12 metre

Area of the field with boundary

= (18 + 2 × 2) (12 + 2 × 2)

= 22 × 16 = 352 sq. metre

$\therefore$ Area of boundary = 352 - 216 = 136 sq. m.

103. (4) From statement I,

H + G + C = 65 × 3 = 195 ...(i)

From statement II,

G = H + 6

From statement III,

G - C = C - H

$\Rightarrow$ 2C = G + H

$\Rightarrow$ 2C = H + 6 + H = 2H + 6

$\Rightarrow$ C = H + 3

$\therefore$ H + G + C = 195

$\Rightarrow$ H + H + 6 + H + 3=195

$\Rightarrow$ 3H = 195 - 9 = 186

$\Rightarrow H = \frac{186}{3} = 62$

104. (1) From statements I and II,

If the length of the train be x metre, then

Speed of train = $\frac{x}{15} = \frac{x+250}{27}$

$\Rightarrow \frac{x}{5} = \frac{x+250}{9}$

$\Rightarrow 9x = 5x + 1250$

$\Rightarrow 4x = 1250$

$\Rightarrow x = \frac{1250}{4}$ = 312.5 m/sec.

105. (4) From statement I and II,

CP. of article =252 - 52 = Rs. 200

Discount = 280 - 252 = Rs. 28

Marked price = Rs. 280

$\therefore 280 \times \frac{x}{100} = 28$

$\Rightarrow x$ = 10%

Statement I and III will also give same result.

SBI PO EXAMS

1. (5) The question has 3 variables, i.e., the rate of work done per day by each A, B and C. We need to have 3 equations to solve the problem. This is possible only by using

all the 3 statements given. Let the rate of work done per day by A, B and C be denoted by x, y and z respectively.

$\therefore x + y = \frac{1}{6}$

$y + z = \frac{4}{15}$

$x + z = \frac{3}{10}$

Adding above equations we get,

$2(x + y + z) = \frac{1}{6} + \frac{4}{15} + \frac{3}{10}$

$2(x + y + z) = \frac{5 + 8 + 9}{30}$

$2(x + y + z) = \frac{22}{30}$

$x + y + z = \frac{11}{30}$

Therefore, (A + B + C) together can complete the work in $\frac{30}{11}$

$= 2\frac{8}{11}$ days.

2. (5) The rate of painting has not been given in any of the statements. Further, we need to kow the exact dimensions of the hall for which any two of the above statements will be required.

3. (5) Simple interest is the interest earned only on the Principal sum. But Compound interest is equal to the sum of the Simple interest and the interest earned on this Simple interest during this period. So, we should know the amount of the Simple interest and the rate of Interest to calculate the additional interest or the difference between them or, we should know the difference in the two Interests and the Simple Interest. Only then we can find the amount of Compound interest during that period. Therefore, either statement II or III can be dispensed with.

When statements I and II are used,

$C.I = S.I\left[\frac{r^2 + 300r + 30000}{30000}\right]$

$= 4500\left[\frac{100 + 3000 + 30000}{30000}\right]$

= Rs. 4965

When statements I and II are used,

C.I. = S.I. + Difference in two interests.

= 4500 + 465 = Rs. 4965.

4. (2) Statement II can be dispensed with as only percent profit is to be found.

Let the cost price be x

$\therefore$ Labelled price = 1.30 x.

Selling price after 10% discount on labelled price = $0.9 \times 1.30\,x = 1.17x$

$\%$ profit = $\frac{1.17x - x}{x} \times 100$

= 17%

Thus, we see that the value of the cost price is not required.

5. (1) In order to find the average salary of all the 15 employees,

we need to know the total salary of all the 15 employees. And for that the average salary and the number in each of the three categories of employees is required.

Total salary of 15 employees

= (7 × 8500) (5 × 10000) + (3 × 2500)

= 59500 + 50000 + 7500 = 117000

$\therefore$ Average salary of 15 employees

$= \frac{117000}{15}$ = Rs 7800

6. (3) Let the area of the hall be x sq. m.

Then total material cost

= Rs. 250 x

Labour cost = 3500

$\therefore 250x + 3000 = 14{,}500$ (total cost)

By this equation, we can find area = x

Hence, all three are required.

7. (4) From statement I,

CP. of article = 252 - 52

= Rs. 200

From statement II.

Marked price = 200 + 800 = Rs. 280

$\therefore$ Discount = Rs. (280 - 252) = Rs. 28

$\therefore$ Discount % = $\frac{28 \times 100}{280}$ = 10%

From statement III, Profit % without discount = 40%

$\therefore$ Profit = 80

By this discount % can be calculated

8. (1) When a train crosses a signal post it travels its own length.

When it crosses a platform its crosses its length and the length of platform.

Let the length of train = x m.

From statements (I) and (II) we can write,

$\frac{x}{13} = \frac{x + 250}{27}$

Now x can be determined Speed of train

$= \frac{x}{13}$ m/s

9. (4) From statement I.
Population of State A

$= \frac{100}{115} \times$ l.61 lakh

From statements (II) and (III)
Population of State A

$= \frac{7}{8} \times$ 1.6 lakh

10. (5) The no. of workers can be determined by any of the three pieces of information.

11. (1) From statement I, the difference between selling price and cost price gives profit. The data in statement II alone are not sufficient to answer the question.

12. (4) From statement I.
Let the speed of each train be u m/s and the length of each train be y metres.

$\therefore \frac{2y}{2u} = x$

or, $u = \frac{y}{x}$

From statement II,

$\frac{y}{u} = 5$

Obviously, the data even in both statements I and II together are not sufficient to answer the question as in that case the length of train is required.

13. (1) We have the digit 0, 1,2, 3, 4, 5, 6, 7, 8 and 9.
From statement I.
9 - 0 = 9
Hence, the number is 90.
From statement II,
9 + 0 = 9, 9 - 0 = 9
2 + 0 = 2, 2 - 0 = 2 etc.
Hence, many numbers are possible.

14. (2) When a spherical ball is melted and made into a right circular cylinder the volume remains same,
i.e., Volume of the cylinder = Volume of the spherical ball.
Again, volume of the cylinder = Area of the base × height
From statement II,
Area of the cylinder is given.
Hence, its height can be obtained by equating volume of the cylinda to $\frac{4}{3}\pi r^2$

15. (3) From statement I,
The diagonal of the square = x inches.

$\therefore$ Area of the square = $\frac{1}{2}x^2$

Now, $\frac{1}{2}x^2 = \pi r^2$

where r is the radius of the circle From this relation, r can be determined and hence the circumference.
From statement II,
Area of the square = y^2
Now, $y^2 = \pi r^2$ gives us the value of radius and hence the circumference.

16. (3) From statement 1,
A is an even number.
Therefore, the product of A and any other positive number will be an even number.
From statement II,
A × B = even number
A × C = even number
Obviously, at least one of these three is an even number. Hence their product will be an even number.

17. (2) From statement I,
Total weight of 60 students
= 60 × 42 = 2520kg
From statement III,
Total weight of all the girls = 1144 kg
$\therefore$ Total weight of all the boys = 2520 - 1144 = 1376 kg.
From statement II,

Number of boys = $\frac{1376}{32} = 32$

$\therefore$ Number of girls = 60 - 32 = 25
From statement III,

Average weight of girls = $\frac{1144}{28} \approx 40.86$ kg

18. (1) From statement I and III,
S.P. can be obtained.
From statements II and III,
Let the SP be Rs. x.
After 10% discount,

SP = 90% of $x = \frac{9}{10}x$

$\therefore \frac{9}{10}x - 15000 = 1200$

$\Rightarrow 9x = 150000 + 12000$

$\Rightarrow x = \frac{162000}{9}$ = Rs. 18000

From statements I and II,
Let the CP = Rs. x

$\therefore$ S.P. = $\frac{120}{100}x$ = Rs. $\frac{6x}{5}$

Now, after 10% discount on

Rs. $\frac{6x}{5}$, New S.P. = $\frac{90}{100} \times \frac{6x}{5}$

Rs. $\frac{54x}{50}$

$\therefore \frac{54x}{50} - x = 1200$

$\Rightarrow 54x - 50x = 50 \times 1200$

$\Rightarrow x = \frac{50 \times 1200}{4}$ = Rs. 15000

Hence S.P. can be calculated as S.P. Rs. $\frac{6x}{5}$

19. (5) From statement I,
Let the length of the train be x metre.
$\therefore$ Length of platform $3x$

= $\frac{3x}{2}$ metre

From statement II,

Speed of train = $\frac{\frac{3x}{2} + x}{25}$

$= \frac{5x}{50} = \frac{x}{10}$ ...(i)

But x is not known. Hence, we proceed.
From statement III, x

Speed of train = $\frac{x}{19}$ ...(ii)

Clearly, we reach at no unique conclusion.

20. (4) From statement I,
80% children speak languages other than Hindi.
From statement II,
$\therefore$ Total number of children

= $\frac{44 \times 100}{80}$ = 55

21. (5) From statements I and II,
Volume of tank = XY cubic metres
From statements II and III,

Radius of base = $\frac{Y}{2}$

$\therefore$ Volume = π × (radius)^2 × height
From statements I and III,
Area of base = X sq. metres.
Hence, radius can be determined.
Height = 2 × radius.
Clearly, after knowing height and area of base, volume can be determined.

22. (3) S.P. of the product = Rs. 100
From statement I,

CP. = $\frac{100}{120} \times 90$ = Rs. 75

$\therefore$ Actual profit =100 - 75 = Rs. 25
From statement II,
Let the CP. be Rs. x

Then profit = $\frac{x}{3}$

$\therefore x + \frac{x}{3} = 100$

$\Rightarrow 4x = 300$

$\Rightarrow x$ = Rs. 75

$\therefore$ Profit = Rs. 25

23. (4) Time taken in seconds to cross

= $\frac{\text{Sum of length of both trains in metres}}{\text{Relative speed in m/ sec.}}$

$\therefore x = \frac{y + z}{u}$

where z m is the length of other train.
To know u, z is required.

24. (5) Let the digits at unit's and ten's places be x and y respectively.
When $x > y$ then $x - y = 6$
Then $x - y = 6$
and $x + y = 12$ gives
$x = 9$ and $y = 3$
Again, where $y > x$
$y - x = 6$
Hence, $y - x = 6$
and $x + y = 12$ gives
$y = 9$ and $x = 3$
Hence, the numbers are 93 or 39.
Obviously, when statement I is not taken into consideration, two numbers are possible.

25. (1) From statements I,

$\frac{X}{5}$ = odd number. It is possible only when X is odd.
From statement II,
X + Y = an odd number
But, X = 2 and Y = 5 gives
X + Y = 7, an odd number

Hence only statement 1 is sufficient to answer the question.

26. (3) From statement I,
Capacity of tank
$= \pi \times 14 \times 14 \times 28 = 17248\ m^3$.
From Statement II,
Capacity of tank $= 616 \times 28 = 17248\ m^3$.
Obviously, either statement I or statement II is required.

27. (4) In ΔABC, $\angle BAC = 90°$
$\therefore \angle BAC + \angle ABC + \angle ACB = 90°$
From statement A,
$\angle ACB = 35°$
$\therefore \angle ABC = 90° - 35° = 55°$
From the statement B,
$\angle CBA = 55°$

28. (3) From the statement A,
X - Z = 4 or Z - X = 4
From the statement B,

$\frac{Y}{3} = 14$

$\Rightarrow Y = 14 \times 3 = 42$
Taking both statements together,
we have
X is the first number and Z is the last number or vice versa as the difference between the two consecutive even numbers is 2.
$\therefore$ X = 40, Y = 42, Z = 44
Or, Z = 40 , Y = 42, X = 44

29. (5) P + Q + R + S + T + U
= Rs. (6 × 35000)
= Rs. 210000 .....(i)
From the statements A and B,
Q + S = 54000 .....(ii)
T + U = 58000 .....(iii)
From all these equations, we can find
P + R i.e. total salary of P and R by equation (i) - (ii) - (iii)

30. (1) Principal = Rs. 6000
From the statement A,
S.I. = Rs. 3600
T = 4 years

$$\text{Rate} = \frac{\text{S.I} \times 100}{\text{Principal} \times \text{Time}}$$

will give the result.
Statement B alone will give no result.

31. (2) Statement A will give no result as it is superfluous.
From the statement B,
If the number be *x*,

$\Rightarrow \frac{5}{6}x = x - 15$

$\Rightarrow x - \frac{5}{6}x = 15\ x$

$\Rightarrow \frac{x}{6} = 15 \Rightarrow x = 90$

32. (3) From statement I,

$$P = \frac{\text{S.I.} \times 100}{R \times T}$$

$= \frac{400 \times 100}{5 \times 2} = \text{Rs } 4000$

Using,

$A = \left(1 + \frac{R}{100}\right)^T$ Amount can be determined.

From statement II,

$$\text{Difference} = \frac{PR^2}{(100)^2}$$

We can find principal and hence amount.

33. (4) From both the statements, we get no result.

34. (5) From both statements,
$r_1 = 24$ cm
$r_2 - r_1 = 3$ cm
$\Rightarrow r_2 = r_1 + 3 = 24 + 3 = 27$ cm
$\therefore$ Required ratio

$= \frac{\pi r_2^2}{\pi r_1^2} = \frac{r_2^2}{r_1^2} = \frac{27^2}{24^2}$

$= \frac{84}{64} = 21 : 16$

35. (5) From both the statements,
If the length of the train be *y* metre,

Speed of train $= \frac{y}{9}$ m / sec

$= \frac{5y}{9 \times 18}$ kmph

Again $\left(\frac{5y}{9 \times 18} + 60\right) \times \frac{15}{60 \times 60} = \frac{y + 100}{1000}$

$\Rightarrow \left(\frac{5y}{162} + 60\right) \times \frac{15}{36} = \frac{y + 100}{10}$ ....(i)

It is to be noted that when a train crosses a pole, the distance covered = length of the train.
When it crosses another train, Relative speed × time = sum of lengths of both trains.
Hence, from equation (i), we can determine length of train.

36. (5) From statement II,
length = x cm, breadth = $(x - 5)$ cm
$\therefore 2(x + x - 5) = 50$
$\Rightarrow 2x - 5 = 25$
$\Rightarrow 2x = 25 + 5 = 30$
$\Rightarrow x = 15$ cm = length
$\therefore$ Breadth = 15 - 5 = 10 cm
From statement I,

$$\frac{\pi r^2}{2} = 15 \times 10$$

$$\Rightarrow \pi r^2 = 300$$

$$\Rightarrow r = \sqrt{\frac{300}{\pi}}$$

This gives us value of radius.

RBI GRADE-B OFFICER EXAMS

1. (4) Time taken in crossing each other

$$= \frac{\text{Total length of trains}}{\text{Relative speed}}$$

The information given in both statements is not sufficient length of train A and individual speed of each train are required.

2. (4) Area of rectangle = Area of triangle.
From the information given in both the statements, we can find area of triangle or area of rectangle. For finding lenght breadth is required, which is not known.

3. (3) From the statement I,

$$r = \frac{100 \times 100}{1000} = 10\%$$

Thus we have,
P = Rs. 1000, r = 10%,
t = 3 years
Hence, C.I. can be determined
From the statement II.

$$\text{S.I} = \frac{100 \times r \times 2}{100} = 20r$$

$$\text{CI} = 1000\left[\left(1 + \frac{r}{100}\right)^2 - 1\right]$$

$\therefore$ C.I. - S.I.

$$= 1000\left[\frac{200r + r^2}{10000}\right] - 20r$$

$\Rightarrow 2000r + r^2 - 200r = IO0$
$\Rightarrow r = 10\%$
Hence, C.I. can be determined

4. (5) Let the unit's digit be x and ten's digit be y and $x < y$.
$\therefore$ Number = $10y + x$
From statement I,
$y - x = 5$ ...(i)
From statement II,
$y + x = 7$ ...(ii)
From (i) and (ii), x, y can be calculated and two digit number can be found.

5. (4) Let the distance between A ard B be z km.
Again, let speed of boat in still water be x kmph and that a stream be y kmph.
$\therefore$ Rate downstream = $(x + y)$ kmph
Rate upstream = $(x - y)$ kmph
From statement I,

$$= \frac{z}{x + y} = 2$$

From statement II,

$$\frac{z}{x - y} = 4$$

6. (4) From both the statements,
Number = 53 or 35

7. (1) From statements I,
0 + 5 = 5
From statement II,
5 + 0 = 5 and 5 - 0 = 5
$\therefore 6 \infty 4 = 6 + 4 = 10$

8. (1) From statement I,
Total time = 6 hours
$\therefore$ The work was completed at 3 p.m.

9. (3) From statements I,
Salary recieved in each paycheques

$$= \text{Rs } \frac{618000}{24} = \text{Rs } 27750$$

From statements II,
Salary received in each paycheque

$$= \text{Rs.}\left(\frac{618000}{24}\right) = \text{Rs } 25750$$

We get answer also from statement II.

10. (5) From both the statements,
if $x \geq 84$, then $y \geq 2$
$\therefore x + y > 85$ ...(ii)

INSURANCE EXAMS

1. (1) Area of a rectangle = $l \times b$
Where l = Length of the rectangle
b = Breadth of the rectangle
From the statement A
Perimeter of the rectangle
$= 2(l + b) = 60$
$l + b = 30$ — (1)
From the statement B
b = 12 cm —- (2)

l = 30 -12 = 18 cm.
Area = 18 × 12
So for calculating the area of a rectangle the statements A as well as B are necessary.

2. (4) **From the statement A**
R > P > Q
From the statement B,
R > M
From the statements A and B it is clear that the R is the tallest among M, P, Q, R

3. (3) Let x is Samir's present age and y is the Anil's present age
From the statement A
$x = y + 4$
From the statement B
y = 20 years
From the statement C,
$y : x = 3 : 4$
$\Rightarrow x : y = 4 : 3$
Therefore, the only statement C is sufficient to answer the question.

4. (5) From given statement,
Sum of money borrowed by x
= unknown
C.I = ? T = 2
To calculate C.I. the formula

$$= P\left(1+\frac{r}{100}\right)^{t} - 1$$

Where P $\rightarrow$ sum of money
$r \rightarrow$ rate of interest annually
$t \rightarrow$ time
From the statement A,
Annual rate = 5%
From the statement B

$$\text{S.I.} = \frac{P \times r \times t}{100}$$

$$600 = \frac{P \times 5 \times 2}{100}$$

$$P = \frac{600 \times 100}{10}$$

= Rs.6000
Now from the statement A, B and C, we have all data required to calculate the C.I. So statements A, B and C are necessary to solve the question statement.

5. (4) In still water the velocity of boat is
= 15 km/hour
Velocity of the stream = 2km/ hour
Relative velocity of the boat
=15 - 2 km/hour = 13 km/hour
Distance = 8 km

$$\text{Time} = \frac{\text{Distance}}{\text{Velocity}} = \frac{8}{13}$$

Therefore, to answer the question all data given in statements A, B and C are required.

6. (3) Let the area of the hall be x sq. m.
Then total material cost
= Rs. 250x
Labour cost = Rs 3500
$\therefore$ 250x + 3500 = 14,500 (total cost)
By this equation, we can find area = x
Hence, all three are required.

7. (4) From statement I,
CP. of article = 252 - 52
= Rs. 200
From statement II.
Marked price = 200 + 80 = Rs. 280
$\therefore$ Discount = Rs. (280 - 252) = Rs. 28

$$\therefore \text{Discount \%} = \frac{28 \times 100}{280} = 10\%$$

From statement III,
Profit % without discount = 40%
$\therefore$ Profit = 80
By this discount % can be calculated

8. (1) When a train crosses a signal post it travels its own length. When it crosses a platform its crosses its length and the length of platform.
Let the length of train = x m.
From statements (I) and (II) we can write,

$$\frac{x}{13} = \frac{x+250}{27}$$

Now x can be determined

$$\text{Speed of train} = \frac{x}{13}\text{ m/s}$$

9. (4) From statement I,
Population of State A

$$= \frac{100}{115} \times 1.61 \text{ lakh}$$

From statements (II) and (III) Population of State A

$$= \frac{7}{8} \times 1.6 \text{ lakh}$$

10. (5) The no. of workers can be determined by any of the three pieces of information.

11. (4) Taking all statements together,
Let the profit earned by company in 2001 = Rs. x and in 2002 = Rs. y
$\therefore$ Profit earned in 2003 = 1.4x,
$x + y$ = Rs. 20 crore ....(i)
From statement III,

$1.4x = y \times \frac{80}{100}$

$\Rightarrow x = \frac{4}{5} \times \frac{1}{1.4} y = \frac{4}{7} y$ ...(ii)

From equations (i) and (ii), we can get the required profit.

12. (5) The equation cannot be answered even with all I, II and III.

13. (2) From statement I,
Observer = Assistant + Rs. 12000 ...(i)
From statement II,
Observer + Assistant = Rs. 32000 ...(ii)
From statement III,
Manager + Observer = Rs. 57000 ...(iii)
We can find the salary of an assistant from equations (i) and (ii) only.

14. (5) The question cannot be answered even with all I, II, and III.

15. (1) From statements I and III we can get the share of B in the profit.

16. (4) From statement A,
$3 \times 5 = 15$; $5 \times 9 = 45$ (An odd number)
It is also obvious from statement B.

17. (5) The answer is not possible with the help of even both the statements. We need more information like sum or average of their ages or ratio of their after some time or before sometime etc.

18. (2) A + B + C + D = Rs. (4 × 62880)
From statement B,
A + C + D = Rs.(3 × 61665)
∴ B's salary = (A + B + C + D)'s salary - (A + C + D)'s salary

19. (3) From statement A,
The three digit number is divisible by 9.
From statement B,
Number = 6 * 6
A number is divisible by 9 if sum of its digits is divisible by 9.
Clearly, * = 6
because 666 * 9 = 74

20. (4) From statement A,
Let CP of 1 printer = Re. 1
∴ CP of 5 printers = Rs. 5
and SP of 5 printers = Rs. 6

$\therefore \text{Gain\%} = \frac{1}{5} \times 100 = 20\%$

$\therefore$ CP = $\frac{100}{120} \times 3000 = \text{Rs.}2500$

∴ Gain = Rs. (3000 - 2500) = Rs. 500
From statement B, we can also find the answer.

MODEL EXERCISES

Directions (1-5): Each of these has a question followed by two statements numbered I and II. Mark answer as

(1) if statement I alone is sufficient but statement II alone is not sufficient to answer the question.

(2) if statement II alone is sufficient but statement I alone is not sufficient to answer the question.

(3) if both statements I and II together are required to answer the question but neither statement alone is sufficient.

(4) if statements I and II together are not sufficient to answer the question.

1. If a certain village is losing 12% of its water supply each day because of a burst water pipe, then what is the loss in rupees per day ?

I. The daily water supply to the village is 700 m gallon.

II. The cost to the village for every 24000 gallons of water lost is Rs 25.

2. Dipesh had a lawyer's appointment on a certain day. Was the appointment on a Tuesday ?

I. Exactly 60 hours before the appointment, it was Sunday.

II. The appointment was between 1.00 pm and 9.00 pm.

3. What is the weighted average of marks obtained by Ashokan ?

I. Math, Science and English have weights 5, 4 and 3 respectively.

II. Simple arithmetic mean of Math and Science is 140, which is twice his average of Science and English.

4. Total amount of Rs 38,500 was distributed among Anuj, Sonam and Deepak. How much does each get ?

I. Anuj gets $\frac{2}{9}$ of what the other two together get.

II. Sonam gets $\frac{3}{11}$ of what the other two together get.

5. Nitu and Arun start walking towards each other simultaneously. What is the distance between them when they start ?

I. 30 minutes after crossing each other they were 1200 m apart.

II. After crossing each other, Nitu reaches the starting point of Arun in twice as much time as Arun takes to reach the starting point of Nitu

Directions (6-10): Two quantities A and B are given in each of the following questions. Mark answer as

(1) if Quantity A is greater than Quantity B.

(2) if Quantity B is greater than Quantity A.

(3) if Quantities A and B are equal.

(4) if comparison cannot be made.

6. A. The number of ways you can divide 6 different candies to 4 different people.

B. The number of ways you can divide 7 different candies to 3 different people.

7. Arun and Rakesh bought the same brand of car. Arun got a 10% discount from the dealer after which 7% sales tax was added. For Rakesh 7% tax was added and only then a 10% discount was given.

A. The amount Arun paid for the car.

B. The amount Rakesh paid for the car.

8. The average height in the class didn't change after Shaan and John left and Aarti and Geeta came.

A. The height of Shaan.

B. The height of Aarti.

9. Travis received an allowance. With a fifth of the money, he bought a ticket to a movie, and with a quarter of what is left he bought a slice of pizza.

A. The price of a movie ticket.

B. The price of a slice of pizza.

10. Mukesh and Vijay each have some stamps while Sharon has none. They decided to give Sharon half of the number that Mukesh has and half of the number that Vijay has.

A. The average number of stamps that Mukesh and Vijay have before giving Sharon any stamps.

B. The number of stamps that Sharon gets from Mukesh and Vijay.

SHORT ANSWERS

1. (3) **2.** (3)
3. (4) **4.** (3)
5. (4) **6.** (2)
7. (3) **8.** (4)
9. (3) **10.** (3)

EXPLANATIONS

1. (3) Loss of water supply = 12%
Therefore, using statement I,
Loss of water supply
= 700 million gallon × 12% = 84000000 gallon
Now, using statement II, we can find the loss in rupees
Thus, both the statements are required to answer the question.

2. (3) From statement I, it is not evident that appointment was on Tuesday or not. For example, if we assume that the appointment was at 10 am. then before 60 *h*. it is not Sunday but when we assume that the appointment was at 2 pm, then before 60 *h,* it is Sunday. So, statement I alone is not sufficient to answer the question
But with the help of statement II, we can say that if the appointment was between 1:00 pm and 9:00 am, then before 60 *h*, it is certainly Sunday.

3. (4) From statement II,

$$\frac{M+S}{2} = 140 = \frac{1}{2}\left(\frac{S+E}{2}\right)$$

$\Rightarrow$ M + S = 280 $\Rightarrow$ S + E = 560
From these equations, we can't find the values of his marks in Math, Science and English.
In statement I, only weights of subjects are given. Thus, both the statements are not sufficient to answer the question.

4. (3) Let the amounts got by Anuj, Sonam and Deepak be *x*, *y* and
{38500 - (*x* + *y*)}, respectively.
From statement I,

$$x = \frac{2}{9}\left[y + \{38500 - (x - y)\}\right]$$

$$= \frac{2}{9}\left[y + 38500 - x - y\right]$$

$$\Rightarrow x = \frac{2}{9}(38500 - x)$$

$\Rightarrow 9x = 77000 - 2x$
$\Rightarrow 11x = 77000$
$\Rightarrow x = 7000$
From statement II,

$$y = \frac{3}{11}[x + 38500 - x - y]$$

Similarly as above, we can get the value of *y* from this equation. Hence, amount got by each can be determined to find which, both of the statements are required.

5. (4) Answer can't be given using both the statements also

6. (2) A Number of ways to divide 6 different candies to 4 different people

$$= \frac{6!}{2!4!} = \frac{6\times5}{2} = 5$$

B. Number of ways to divide 7 different candies to 3 different people

$$= {}^7P_3 = \frac{7!}{4!3!} = \frac{7\times6\times5}{3\times2} = 35$$

$\therefore$ A < B

7. (3) Let marked price of the car be Rs. *x.*
A. Amount given by Arun for the car
= 107% of 90% of *x*
= 107 × 0.9 × *x* = 0.963 *x*
B. Amount given by Rakesh for the car
= 90% of 107% of *x* = 0.963 *x*
Thus, A = B

8. (4) From the given statement,
Total height of Shaan and John = Total height of Aarti and Geeta But we can't find the separate heights of all the four students.
Thus, comparison between A and B can't be made.

9. (3) Let allowance got by Travels be Rs. *x*.
Then, price of the ticket to the *x* movie

$$= \text{Rs. } \frac{x}{5}$$

Then, price of the slice of pizza

$$= \frac{1}{4}\left(x - \frac{x}{5}\right) = \text{Rs } \frac{x}{5}$$

A. Price of the movie ticket = Rs. $\frac{x}{5}$

B. Price of the slice of pizza = Rs. $\frac{x}{5}$

$\therefore$ A = B

10. (3) Let Mukesh and Vijay have *x* and *y* number of stamps, respectively.
A. Average number of stamps

Mukesh and Vijay = $\frac{x+y}{2}$

B. Number of stamps that Sharon gets from Mukesh and Vijay

$$\frac{x}{2} + \frac{y}{2} = \frac{x+y}{2}$$

$\therefore$ A = B

DISCOUNT

NATIONALISED BANKS & IBPS PO/MT/SO

1. Deepa bought a calculator with 30% discount on the listed price. Had she not got the discount, she would have paid Rs. 82.50 extra. At what price did she buy the calculator ?
(1) Rs. 192.50
(2) Rs. 275
(3) Rs. 117.85
(4) Cannot be determined
(5) None of these

Canara Bank PO Ezam. 09.02.2003)

2. Naresh purchased a TV set for Rs. 11,250 after getting discount of 10% on the labelled price. He spent Rs. 150 on transport and Rs. 800 on installation. At what price should it be sold so that the proflt earned would have been 15% if no discount was offered ?
(1) Rs. 12937.50 (2) Rs. 14,030
(3) Rs. 13,450 (4) Rs. 15467.50
(5) None of these

(United Bank of India PO Exam. 21.06.2009)

3. Manoj sold an article for Rs. 15,000. Had he offered a discount of 10% on the selling price he would have earned a profit of 8%. What is the cost price ?
(1) Rs. 12,500 (2) Rs. 13,500
(3) Rs. 12,250 (4) Rs. 13,250
(5) None of these

Andhra Bank PO Exaxn. 05.07.2009)

4. A man buys a single apple for Rs. 25. If he were to buy a dozen apples, he would have to pay a total amount of Rs. 250. What would be the **approximate** percent discount he would get on buying a dozen apples ?
(1) 32 (2) 20
(3) 12 (4) 8
(5) 17

(Bank Of India Banking Officer Exam. 24.01.2010)

5. A discount of 15% is given on the marked price of an article. The shopkeeper charges sales tax of 6% on the discounted price. If the selling price be Rs. 1081.20, what is the marked price of the article ?
(1) Rs. 1185.20 (2) Rs. 1250.20
(3) Rs. 302 (4) Rs. 1200
(5) None of these

(Indian Overseas Bank PO Online Exam. 01.09.2013)

SBI PO EXAMS

1. A shopkeeper sells 25 articles at 45/- per article after giving 10% discount and gains 50% proflt. If the discount is not given what is the proflt gained ?
(1) 60% (2) $60\frac{2}{3}\%$
(3) $66\frac{2}{3}\%$ (4) 66%
(5) None of these

(SBI Associate Banks PO Exam. 16.07.2000)

2. Deepa bought a calculator with 30% discount on the listed price. Had she not got the discount, she would have paid Rs. 82.50 extra. At what price did she buy the calculator ?
(1) Rs. 192.50
(2) Rs. 275
(3) Rs. 117.85
(4) Cannot be determined
(5) None of these

(SBI Associate Banks PO Exam. 07.01.2007)

3. While selling a watch, a shopkeeper gives a discount of 15%. If he gives a discount of 20%, he earns Rs. 51 less as proflt. What is the original price of the watch ?
(1) Rs. 920 (2) Rs. 985
(3) Rs. 1125 (4) Rs. 1020
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

RBI GRADE-B OFFICE EXAMS

1. A shopkeeper sold a T.V. set for Rs. 17,940, with a discount of 8% and gained 19.6% If no discount is allowed, what will be his gain per cent ?
(1) 25%
(2) 26.4%
(3) 24.8%
(4) Cannot be determined
(5) None of these

(RBI Grade-B Officer Exam.17.11.2002)

INSURANCE EXAMS

1. A company decided to sell Rs. 50.000, T.V. set for Rs. 46,000/- as a world cup offer. What is the percentage discount offered by the company ?

(1) 7.5 (2) 7
(3) 6.5 (4) 8.5
(5) 8%

(United India Insurance Co. AAO Exam. 21.04.2002)

2. The labelled price of a cupboard is Rs. 6,500/-. The shopkeeper sold it by giving 5% discount on the labelled price and earned a profit of 15%. What **approximately** is the cost price of the cupboard ?
(1) Rs. 5,800/- (2) Rs. 5,600/-
(3) Rs. 5,000/- (4) Rs. 6,000/-
(5) Rs. 5,369/-

(LIC Assistant Adminiatrative Officer (AAO) Exam. 24.04.2005)

3. On a Rs. 10,000 payment order, a person has choice between 3 successive discounts of 10%, 10% and 30%, and 3 successive discounts of 40%, 5% and 5%. By choosing the better one he can save (in Rupees) :
(1) 200 (2) 255
(3) 400 (4) 433

(United India Insurance Co. (AAO) Exam. 11.03.2007)

4. A book is listed at Rs. 150, with a discount of 20 per cent. What additional discount must be offered to bring the net price to Rs. 108 ?
(1) 8 per cent (2) 10 per cent
(3) $12\frac{1}{2}$ per cent (4) 15 per cent

(LIC Assitant Administrative Offlcer (AAO) Exam. 07.06.2009)

5. A shopkeeper marks his goods at such a price that after allowing a discount of 12.5% on the marked price, he still earns a profit of 10%. The marked price of an article which costs him Rs. 4,900 is
(1) Rs. 5,390 (2) Rs. 6,160
(3) Rs. 5,490 (4) Rs. 6,260

(New India Assurance AO Exam. 25.10.2009)

6. A T-shirt marked at Rs. 400 is sold for Rs. 360. The rate of discount is :
(1) 12 (2) 10
(3) 15 (4) 17

(NICL (GIC) AÓ (FInance) Exam. 08.09.2013 (Paper-I)

7. The difference between a discount of 40% on Rs. 500 and two successive discounts of 36% and 4% on the same amount is
(1) Rs 0 (2) Rs 2
(3) Rs 1.93 (4) Rs 7.20

(NICL (GIC) AO Exam. 08.09.2013 (Paper-I)

8. A dealer offers a discount of 10% on the marked price of an article and stili makes a profit of 20%. If its marked price is Rs. 800, then the cost price is:
(1) Rs. 600 (2) Rs. 700
(3) Rs. 800 (4) Rs. 900

(NICL (GIC) Administrative Offlcer Exam. 15.12.2013)

9. A seller allows a discount of 5% on a watch. If he allows a discount of 7%, he earns Rs 15 less in the profit What is the marked price ?
(1) Rs. 697.50
(2) Rs. 712.50
(3) Rs. 750
(4) Rs. 817.50

(NICL (GIC) AO (Finance) Offlcer Exam. 15.12.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (1) **2.** (2)
3. (1) **4.** (5)
5. (4)

SBI PO EXAMS

1. (3) **2.** (1)
3. (4)

RBI GRADE-B OFFICER EXAMS

1. (5)

INSURANCE EXAMS

1. (5) **2.** (5)
3. (2) **4.** (2)
5. (2) **6.** (2)
7. (4) **8.** (1)
9. (3)

EXPLANAT10NS
NATIONALISED BANKS IBPS & PO/MT/SO

1. (1) Let the listed price be Rs. x.
Discount = 30% of x

$= \frac{30x}{100} = \text{Rs.}\frac{3x}{10}$

According to the question,

$\frac{3x}{10} = 82.5$

$\Rightarrow x = \frac{82.5\times10}{3} = \text{Rs. } 275$

$\therefore$ Required cost price of calculator
= 70% of 275

$= \text{Rs } \frac{70\times275}{100} = \text{Rs. } 192.50$

2. (2) Total cost of the TV set
= Rs. (11250 + 150 + 800)
= Rs. 12200
$\therefore$ Required SP

$= \text{Rs. } \left(12200\times\frac{115}{100}\right)$

= Rs. 14030

3. (1) Let the cost price be Rs. x.
$\therefore$ 90% of 15000 = 108% of x

$\Rightarrow x = \frac{90\times15000}{108}$

$\therefore x$ = Rs. 12500

4. (5) C.P in first case
= Rs. (25 × 12) = Rs. 300
$\therefore$ Discount = Rs. 50.
If the discount be x %, then

$\frac{300\times x}{100} = 50$

$\Rightarrow x = \frac{50\times100}{300} = 16.67$

5. (4) If the marked price of that article be Rs. x, then
Price after discount of 15%

$= \text{Rs. } \frac{85x}{100} = \text{Rs. } \frac{17x}{20}$

Actual S.P. = Rs. $\left(\frac{17x}{20}\times\frac{106}{100}\right)$

$\therefore \frac{17x}{20}\times\frac{106}{100} = 1081.20$

$\Rightarrow x = \frac{1081.20\times20\times100}{17\times106} = \text{Rs. } 1200$

SBI PO EXAMS

1. (3) Total marked Price of article
= 25 × 45 = Rs. 1125/-
Selling Price (Giving 10% discount)

$= \frac{90}{100}$ of 1125 = Rs 1012.5

CP = $\frac{1012.50}{150}$ × 100 = Rs 675

Now the selling price is Rs. 1125 then profit
= 1125 - 675 = Rs 450

% profit = $\frac{450}{675}\times 100 = 66\frac{2}{3}\%$

2. (1) Let the listed price be Rs. x.
$\therefore$ Discount = 30% of x

$= \frac{30x}{100} = \text{Rs. } \frac{3x}{10}$

According to the question,

$\frac{3x}{10} = 82.5$

$\Rightarrow x = \frac{82.5\times10}{3} = \text{Rs. } 275$

$\therefore$ Required cost price of calculator
= 70% of 275

= Rs. $\frac{70\times275}{100}$ = Rs. 192.50

3. (4) Let the marked price be Rs. x
$\therefore$ First S.P. = 85% of Rs. x
= Rs. 0.85x
Second S.P. = 80% of Rs. x
= Rs. 0.8x
According to the question,
0.85x - 0.8x = 51

x = $\frac{51}{0.05}$ = Rs. 1020

RBI GRADE-B OFFICER EXAMS

1. (5) MP = $\frac{17940}{92}$ × 100 = Rs 19500

CP = $\frac{17940}{119.6}$ × 100 = Rs 15000
If no discount is given,
Profit = 19500 - 15000 = Rs 4500

$\therefore$ Profit% = $\frac{4500}{15000}$ × 100 = 30%

INSURANCE EXAMS

1. (5) Required percentage discount

= $\frac{(50000-46000)}{50000}$ × 100

= $\frac{4}{50}$ × 100 = 8%

2. (5) According to question,

Selling Price = $\frac{6500\times95}{100}$ = Rs. 6175

$\therefore$ Cost Price = $\frac{6175}{115}$ × 100

= Rs. 5269.96 $\approx$ Rs. 5369

3. (2) Selling price in the first case
= 70% of 90% of 90% of Rs. 10000

= $\frac{70}{100}\times\frac{90}{100}\times\frac{90}{100}\times$ 10000

= Rs. 5670
Selling price in the second case
= 95% of 95% of 60% of Rs. 10000

= $\frac{95}{100}\times\frac{95}{100}\times\frac{60}{100}\times$ 10000

= Rs. 5415
$\therefore$ Saving = Rs. (5670 - 5415) = Rs. 255

4. (2) After a discount of 20% cost of the book
= 80% of Rs. 150

= $\frac{150\times80}{100}$ = Rs. 120
Second discount
= 120 - 108 = Rs. 12

$\therefore$ additional discount% = $\frac{12}{120}$ × 100 = 10%

5. (2) Tricky approach
SP of the article

= 4900 × $\frac{110}{100}$ = Rs. 5390

$\therefore$ Marked price = $\frac{100}{87.5}$ × 5390

= Rs. 6160

6. (2) Discount% = $\frac{400-360}{400}$ × 100 = 10%

7. (4) Single equivalent discount for successive discounts of 36% and 4%

= $\left(36+4-\frac{36\times4}{100}\right)\%$

= 38.56%
$\therefore$ Difference
= 500 × (40 - 38.56)%

= $\frac{500\times1.44}{100}$ = Rs. 7.20

8. (1) Marked Price = Rs. 800

S.P = $\frac{800\times90}{100}$ = Rs. 720

$\therefore$ CP = $\frac{100}{120}$ × 720 = Rs. 600

9. (3) If the marked price be Rs. x then,
(7- 5)% of x = 15

$\Rightarrow$ $\frac{x\times2}{100}$ = 15

$\Rightarrow x$ = $\frac{1500}{2}$ = Rs. 750

MODEL EXERCISES

1. What price should a shopkeeper mark on an article costing him Rs 153 to gain 20% after allowing a discount of 15% ?
(1) Rs 162 (2) Rs 184
(3) Rs 216 (4) Rs 224
(5) None of these

2. Sunder purchased an office bag with a price tag of Rs 600 in a sale where 25% discount was being offered on the tag price. He was given a further discount of 10% on the amount arrived at after giving usual 25% discount. What was the final amount paid by Sunder ?
(1) Rs 210 (2) Rs 540
(3) Rs 405 (4) Rs 450
(5) None of these

3. A bicycle originally costs Rs 100 and was discounted 10%. After three months, it was sold after being discounted 15%. Howmuch was the bicycle sold for ?
(1) Rs 55.5 (2) Rs 95.25
(3) Rs 76.5 (4) None of these
(5) All of these

4. A shopkeeper sold a TV set for Rs 17940, with a discount of 8% and gained 19.6%. If no discount is allowed, then what will be his gain per cent ?
(1) 25% (2) 26.4 %
(3) 24.8% (4) 30%
(5) None of these

5. A cash payment that wlll settle a bill for 250 chairs at Rs 50 per chair less 20% and 15% with a further discount of 5% on cash payment is :
(1) Rs 8075 (2) Rs 7025
(3) Rs 8500 (4) Rs 7125
(5) None of these

6. A shopkeeper marks the prices of his goods at 25% higher than the original price. After that, he allows a discount of 12% discount. What profit or loss did he get ?
(1) 15% profit (2) 10% profit
(3) 10% loss (4) 15% loss
(5) None of these

7. A shopkeepers announce the same price of Rs 700 for a shirt. The first offers successive discounts of 30% and 6% while the second offers successive discounts of 20% and 16%. The shopkeeper that offers better discount is more of
(1) Rs 22.40 (2) Rs 16.80
(3) Rs 9.80 (4) Rs 36.40
(5) None of these

8. A tradesman gives 4% discount on the marked price and 1 article free with every 15 articles bought and still gains 35%. The marked price is more than the cost price by-
(1) 40% (2) 39%
(3) 20% (4) 50%
(5) None of these

9. What is the maximum percentage discount that a merchant can offer on her marked price so that she ends up selling at no profit or loss, if she had initially marked her goods up by 50% ?
(1) 16.67% (2) 20%
(3) 50% (4) 33.33%
(5) None of these

10. An article is listed at Rs 65. A customer bought this article for Rs 56.16 with two successive discounts of which one is 10%. The other discount of this discount scheme that was allowed by the shopkeeper is
(1) 4% (2) 3%
(3) 6% (4) 2.5%
(5) None of these

SHORT ANSWERS

1. (3) **2.** (3)
3. (3) **4.** (4)
5. (1) **6.** (2)
7. (3) **8.** (4)
9. (4) **10.** (1)

EXPLANATIONS

1. (3) According to question,
CP = Rs 153, gain 20%

$\therefore$ SP = $153 \times \frac{120}{100}$ = Rs 183.60

Now, MP = $\frac{183.60}{85} \times 100$ = Rs 216

2. (3) Final amount after giving successive discounts of 25% and 10%
= 600 × 0.75 × 0.9 = Rs 405

3. (3) According to question,

SP of bicycle $100 \times \frac{90}{100} \times \frac{85}{100}$ = Rs 76.50

4. (4) SP = 17940, Discount = 8%

$\therefore$ MP = $\frac{17940}{0.92}$ = Rs 19500

Gain = 19.6% (given)

$\therefore$ CP = $\frac{17940}{1.196}$ = Rs 15000

New SP without discount = Rs 19500
Gain = (19500- 15000) = Rs 4500

$\therefore$ Gain percent = $\frac{4500}{15000} \times 100 = 30\%$

5. (1) By question, originai price of 250 chairs
= 250 × 50 = Rs 12500
Price after discount

= $12500 \times \frac{80}{100} \times \frac{85}{100} \times \frac{95}{100}$ = Rs 8075

6. (2) By question, Profit per cent or loss per cent.

= + 25 - 12 - $\frac{25 \times 12}{100}$ = +10%

As the sign is +ve so, there is profit of 10%.

7. (3) According to the question,
Selling price of first shopkeeper.

= $700 \times \frac{70}{100} \times \frac{94}{100}$ = Rs 460.60

Selling price of second shopkeeper

= $700 \times \frac{80}{100} \times \frac{84}{100}$ = Rs 470.40

Required difference
= 470.40 - 460.60 = Rs 9.80

8. (4) According to the question,

Discount on articles = $\frac{1}{16} \times 100 = 6.25\%$

Overall discount = -4-6.25 + $\frac{4 \times 6.25}{100}$ = - 10%

Let cost price = Rs 100, then
selling price = Rs 135
So, 90% of marked price = 135

Marked price = $\frac{135 \times 100}{90}$ = Rs 150

Marked price is increased by

= $\frac{150 - 100}{100} \times 100$ = 50%

9. (4) Let cost price = Rs 100
Marked price = Rs150
$\therefore$ Discount per cent

= $\frac{50}{150} \times 100 = 33.33\%$

10. (1) Let the other discount be x%

$65 \times \frac{90}{100} \times \left(\frac{100 - x}{100}\right) = 56.16$

$\Rightarrow 100 - x = \frac{56.16 \times 100 \times 100}{65 \times 90}$

$\Rightarrow 100 - x = 96$

$\Rightarrow x = 4\%$

EQUATIONS AND INEQUATIONS

NATIONALISED BANKS & IBPS SO/MT/SO

Directions (1-5): For the two given equations I and II.

Give answer (1) if p is greater than q.
Give answer (2) if p is smaller than q.
Give answer (3) if p is equal to q.
Give answer (4) if p is either equal to or greater than q.
Give answer (S) if p is either equal to or smaller than q.

(Canara Bank PO Exam. 09.02.2003)

1. I. $p^2 + 5p + 6 = 0$
II. $q^2 + 3q + 2 = 0$

2. I. $p^2 = 4$
II. $q^2 + 4q = -4$

3. I. $p^2 + p = 56$
II. $q^2 - 17q + 72 = 0$

4. I. $3p + 2q - 58 = 0$
II. $4q + 4p = 92$

5. I. $3p^2 + 17p + 10 = 0$
II. $10q^2 + 9q + 2 = 0$

Directions (6-10): In each of the following questions two equations are given. You have to solve the equations and

Give answer (1) if $x < y$
Give answer (2) if $x \leq y$
Give answer (3) if $x = y$
Give answer (4) if $x \geq y$
Give answer (5) if $x > y$

(Syndicate Bank PO Exam. 10.10.2004)

6. I. $4x^2 - 8x + 3 = 0$
II. $2y^2 - 7y + 6 = 0$

7. I. $x^2 + x - 6 = 0$
II. $2y^2 - 13y + 21 = 0$

8. I. $x^2 - x - 6 = 0$
II. $2y^2 + 13y + 21 = 0$

9. I. $x^2 = 4$
II. $y^2 + 6y + 9 = 0$

10. I. $2x + 3y = 4$
II. $3x + 2y = 11$

Directions (11-15): In each of the following questions two equations I and II are given. You have to solve both the equations and -

Give answer if
(1) $x > y$ (2) $x < y$
(3) $x < y$ (4) $x > y$
(5) $x = y$ or relationship betweei x and y cannot be established

(Union Bank of India PO Exam. 7.11.2005

11. I. $4x + 2y = 51$
II. $15y + 13x = 221$

12. I. $8x^2 + 3x = 38$
II. $6y^2 + 34 = 29y$

13. I. $x^2 + 91 = 20x$
II. $10y^2 - 29y + 21 = 0$

14. I. $6x^2 + 13x + 5 = 0$
II. $9y^2 + 22y + 8 = 0$

15. I. $(x + y)^2 = 784$
II. $92551 = 92567 - y$

16. If $4x + 5y = 83$ and $\frac{3x}{2y} = \frac{21}{22}$ then what is the value of $y - x$?
(1) 3 (2) 4
(3) 7 (4) 11
(5) None of these

(Indian Overseas Bank PO Exam. 15.06.2008)

Directions (17-21): In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer (1) if $x > y$
Give answer (2) if $x \geq y$
Give answer (3) if $x < y$
Give answer (4) if $x \leq y$
Give answer (5) if $x = y$
or the relationship cannot be established

(UCO Bank PO Exam. 22.03.2009)

17. I. $x^2 - 14x + 48 = 0$
II. $y^2 + 6 = 5y$

18. I. $x^2 + 9x + 20 = 0$
II. $y^2 + 7y + 12 = 0$

19. I. $x^2 = 529$
II. $y = \sqrt{529}$

20. I. $x^2 + 13x = -42$
II. $y^2 + 16y + 63 = 0$

21. I. $2x + 3y = 14$
II. $4x + 2y = 16$

22. If $3y + 9x = 54$ and $\frac{28x}{13y} = \frac{140}{39}$ then what is the value of $y - x$?
(1) -1 (2) -2
(3) 2 (4) 1
(5) None of these

(Indian Overseas Bank PO Exam. 05.04.2009)

Directions (23-27) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer (1) If $x > y$
Give answer (2) If $x \geq y$
Give answer (3) If $x < y$
Give answer (4) If $x \leq y$
Give answer (5) If $x = y$ or the relationship cannot be established.

(Corporation Bank PO

Exam. 22.11.2009)

23. I. $x^2 - l = o$
II. $y^2 + 4y + 3 = 0$

24. I. $x^2 - 7x + 12 = 0$
II. $y^2 - 12y + 32 = 0$

25. I. $x^3 - 371 = 629$
II. $y^3 - 543 = 788$

26. I. $5x + 2y = 31$
II. $3x + 7y = 36$

27. I. $2x^2 + 11x + 12 = 0$
II. $5y^2 + 27y + 10 = 0$

Directions (28-32) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer (1) If $x > y$
Give answer (2) If $x \geq$ y
Give answer (3) If $x <$ y
Give answer (4) If $x \leq y$
Give answer (5) If x = y or the relationship cannot be established.

(Indian Bank PO Exam. 17.10.2010)

28. I. $2x^2 + 11x + 14 = 0$
II. $4y^2 + 12y + 9 = 0$

29. I. $x^2 - 4 = 0$
II. $y^2 + 6y + 9 = 0$

30. I. $x^2 - 7x + 12 = 0$
II. $y^2 + y - 12 = 0$

31. I. $x^2 = 729$
II. $y = \sqrt{729}$

32. I. $x^4 - 227 = 398$
II. $y^2 + 321 = 346$

Directions (33-37) : In the following questions two equations numred I and II are given. You have to solve both the equations and

Give answer (1) if $x > y$
Give answer (2) if $x \geq y$
Give answer (3) if $x < y$
Give answer (4) if $x \leq y$
Give answer (5) if $x = y$ or the f.ationship cannot be established.

(Punjab & Sind Bank PO Exam. 16.05.2010)

33. 1. $x^2 - x - 12 = 0$
II. $y^2 + 5y + 6 = 0$

34. I. $x^2 - 8x + 15 = 0$
II. $y^2 - 3y + 2 = 0$

35. I. $x^2 - 32 = 112$
II. $y - \sqrt{l69} = 0$

36. I. $x - \sqrt{121} = 0$
II. $y^2 - 121 = 0$

37. I. $x^2 - 16 = 0$
II. $y^2 - 9y + 20 = 0$

Directions (38-42) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer (1) if $x > y$
Give answer (2) if $x \geq y$
Give answer (3) if $x < y$
Give answer (4) if $x \leq y$
Give answer (5) if $x = y$ or the relationship cannot be established

(Bank Of Baroda PO Exam. 30.05.2010)

38. I. $3x^2 + 8x + 4 = 0$
II. $4y^2 - 19y + 12 = 0$

39. I. $x^2 + x - 20 = 0$
II. $y^2 - y - 30 = 0$

40. I. $x^2 - 365 = 364$
II. $y - \sqrt{324} = \sqrt{81}$

41. I. $\frac{4}{\sqrt{x}} + \frac{7}{\sqrt{x}} = \sqrt{x}$
II. $y^2 - \frac{(11)^{\frac{5}{2}}}{\sqrt{y}} = 0$

42. I. $225x^2 - 4 = 0$
II. $\sqrt{225y} + 2 = 0$

Directions (43-47) : In the following questions two equations numbered I and II are given. You have to solve both the equations and —

Give answer (1) If $x > y$
Give answer (2) If $x \geq y$
Give answer (3) If $x < y$
Give answer (4) If $x \leq y$
Give answer (5) If $x = y$ or the relationship cannot be established

(Central Bank Of India PO Exam. 25.07.2010)

43. I. $5x^2 - 18x + 9 = 0$
II. $20y^2 - 13y + 2 = 0$

44. I. $x^3 - 878 = 453$
II. $y^2 - 82 = 39$

45. I. $\frac{3}{\sqrt{x}} + \frac{4}{\sqrt{x}} = \sqrt{x}$
II. $y^3 - \frac{(7)^{\frac{7}{2}}}{\sqrt{y}} = 0$

46. I. $9x - 15.45 = 54.55 + 4x$
II. $\sqrt{y+155} - \sqrt{36} = \sqrt{49}$

47. I. $x^2 + 11x + 30 = 0$
II. $y^2 + 7y + 12 = 0$

Directions (48-52): In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer (1) if $x > y$

Give answer (2) if $x \geq y$
Give answer (3) if $x < y$
Give answer (4) if $x \leq y$
Give answer (5) if $x = y$ or the relationship cannot be established

(Syndicate Bank PO Exam. 29.08.2010)

48. I. $\sqrt{x} - \frac{\sqrt{6}}{\sqrt{x}} = 0$

II. $y^3 - 6^{\left(\frac{3}{2}\right)} = 0$

49. I. $3x - 2y = 10$
II. $5x - 6y = 6$

50. I. $x^2 + x - 12 = 0$
II. $y^2 - 5y + 6 = 0$

51. I. $x^2 + 9x + 18 = 0$
II. $y^2 - 13y + 40 = 0$

52. I. $\sqrt{x+6} = \sqrt{121} - \sqrt{36}$
II. $y^2 + 112 = 473$

Directions (53-57): In the following questions two equations numbered I and II are given. You have to solve both the equations and —

Give answer (1) if $x > y$
Give answer (2) if $x \geq y$
Give answer (3) if $x < y$
Give answer (4) if $x \leq y$
Give answer (5) if $x = y$ or the relationship cannot be established.

(Bank Of India PO Exam. 31.10.2010)

53. I. $x^2 - 1200 = 244$
II. $y + 122 = 159$

54. I. $14x - 25 = 59 - 7x$

II. $\sqrt{y+222} - \sqrt{36} = \sqrt{81}$

55. I. $144x^2 - 16 = 9$

II. $12y + \sqrt{4} = \sqrt{49}$

56. I. $x^2 - 9x + 20 = 0$
II. $y^2 - 13y + 42 = 0$

57. I. $\frac{\sqrt{x}}{5} + \frac{3\sqrt{x}}{10} = \frac{1}{\sqrt{x}}$

II. $\frac{10}{\sqrt{y}} - \frac{2}{\sqrt{y}} = 4\sqrt{y}$

58. If $2x + 3y = 78$ and $3x + 2y = 72$, what is the value of $x + y$?
(1) 36
(2) 32
(3) 30
(4) Cannot be determined
(5) None of these

(PNB Management Trainee Exam. 28.11.2010)

Directions (59-63) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer if

(1) $x > y$
(2) $x \geq y$
(3) $x < y$
(4) $x \leq y$
(5) $x = y$ or the relationship cannot be established

(Bank Of Maharashtra Exam. 19.12.2010)

59. I. $20x^2 - x - 12 = 0$
II. $20y^2 + 27y + 9 = 0$

60. I. $x^2 - 218 = 106$
II. $y^2 - 37y + 342 = 0$

61. I. $\frac{7}{\sqrt{x}} + \frac{5}{\sqrt{x}} = \sqrt{x}$

II. $y^2 - \frac{(12)^{\frac{5}{2}}}{\sqrt{y}} = 0$

62. I. $\sqrt{361}\,x + \sqrt{16} = 0$

II. $\sqrt{441}\,y + 4 = 0$

63. I. $\frac{15}{\sqrt{x}} - \frac{2}{\sqrt{x}} = 6\sqrt{x}$

II. $\frac{\sqrt{y}}{4} + \frac{7\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

Directions (64-68) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer If

(1) $x > y$
(2) $x \geq y$
(3) $x < y$
(4) $x \leq y$
(5) $x = y$ or the relationship cannot be established

(Oriental Bank Of Commerce PO Exam. 26.12.2010 (1st Sitting)

64. I. $\sqrt{x+18} = \sqrt{144} - \sqrt{49}$
II. $y^2 + 409 = 473$

65. I. $x^2 - 7x + 12 = 0$
II. $y^2 - 9y + 20 = 0$

66. I. $y^2 - x^2 = 32$
II. $y - x = 2$

67. I. $\sqrt{x} - \frac{\sqrt{5}}{\sqrt{x}} = 0$

II. $y^3 - 5^{(3/2)} = 0$

68. I. $3x + 5y = 28$
II. $8x - 3y = 42$

Directions (69-73) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer if

(1) $x > y$

(2) $x \geq y$

(3) $x < y$

(4) $x \leq y$

(5) $x = y$ or the relationship cannot be established

(Union Bank Of India PO Exam. 09.01.2001)

69. I. $\sqrt{289x} + \sqrt{25} = 0$

II. $\sqrt{676y} + 10 = 0$

70. I. $8x^2 - 78x + 169 = 0$

II. $20y^2 - 117y + 169 = 0$

71. I. $\frac{15}{\sqrt{x}} + \frac{9}{\sqrt{x}} = 11\sqrt{x}$

II. $\frac{\sqrt{y}}{4} + \frac{5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

72. I. $\frac{8}{\sqrt{x}} + \frac{6}{\sqrt{x}} = \sqrt{x}$

II. $y^3 - \frac{(14)^{\frac{7}{2}}}{\sqrt{y}} = 0$

73. I. $x^2 - 208 = 233$

II. $y^2 - 47 + 371 = 0$

Directions (74-78) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer if

(1) $x > y$

(2) $x \geq y$

(3) $x < y$

(4) $x \leq y$

(5) $x = y$ or the relationship cannot be established

(Corporation Bank PO Exam. 16.01.2011)

74. I. $x^2 - 11x + 24 = 0$

II. $2y^2 - 9y + 9 = 0$

75. I. $x^3 \times 13 = x^2 \times 247$

II. $y^{1/3} \times 14 = 294 \div y^{2/3}$

76. I. $\frac{12 \times 4}{x^{\frac{4}{7}}} - \frac{3 \times 4}{x^{\frac{4}{7}}} = x^{\frac{10}{7}}$

II. $y^3 + 783 = 999$

77. I. $\sqrt{500}\,x + \sqrt{402} = 0$

II. $\sqrt{360}\,y + (200)^{1/2} = 0$

78. I. $(17)^2 + 144 \div 18 = x$

II. $(26)^2 - 18 \times 21 = y$

Directions (79-83) : In each of these questions, two equations are given. You have to solve these equations and find out the values of x and y and

Give answer If

(1) $x < y$

(2) $x > y$

(3) $x \leq y$

(4) $x \geq y$

(5) $x = y$

(Punjab & Sind Bank PO Exam. 23.01.2011)

79. I. $16x^2 + 20x + 6 = 0$

II. $10y^2 + 38y + 24 = 0$

80. I, $18x^2 + 18x + 4 = 0$

II. $12y^2 + 29y + 14 = 0$

81. I. $8x^2 + 6x = 5$

II. $12y^2 - 22y + 8 = 0$

82. I. $17x^2 + 48x = 9$

II. $13y^2 = 32y - 12$

83. I. $4x + 7y = 209$

II. $12x - 14y = -38$

Directions (84-88) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer (1) if $x > y$

Give answer (2) if $x \geq y$

Give answer (3) if $x < y$

Give answer (4) if $x \leq y$

Give answer (5) if $x = y$ or the relationship cannot be established

(UCO Bank PO Exam. 30.01.2011

84. I. $\frac{25}{x^2} - \frac{12}{x} + \frac{9}{x^2} = \frac{4}{x^2}$

II. $9.84 - 2.64 = 0.95 + y^2$

85. I. $\sqrt{901}\,x + \sqrt{1295} = 0$

II. $(257)^{1/4}y + (217)^{1/3} = 0$

86. I. $\frac{(3)^5 + (7)^3}{3} = x^3$

II. $7y^3 = -(15 \times 2) + 17y^3$

87. I. $(x^{1/4} \div 16)^2 = 144 \div x^{3/2}$

II. $y^{1/3} \times y^{2/3} \times 3104 = 16 \times y^2$

88. I. $3x^2 - 19x + 28 = 0$

II. $5y^2 - 18y + 16 = 0$

Directions (89-93) : In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give answer If

(1) $x > y$

(2) $x \geq y$

(3) $x < y$

(4) $x \le y$

(5) $x = y$ or the relationship cannot be established

(Bank Of Baroda PO Exam. 13.03.2011)

89. I. $\sqrt{1225}\,x + \sqrt{4900} = 0$

II. $(81)^{\frac{1}{4}}y + (343)^{\frac{1}{3}} = 0$

90. I. $\frac{18}{x^2} + \frac{6}{x} - \frac{12}{x^2} = \frac{8}{x^2}$

II. $y^2 + 9.68 + 5.64 = 16.95$

91. I. $\frac{(2)^5 + (11)^3}{6} = x^3$

II. $4y^3 = -(589 \div 4) + 5y^3$

92. I. $12x^2 + 11x + 12 = 10x^2 + 22x$

II. $13y^2 - 18y + 3 = 9y^2 - 10y$

93. I. $\left(x^{\frac{7}{5}} \div 9\right) = 169 \div x^{\frac{3}{5}}$

II. $y^{\frac{1}{4}} \times y^{\frac{1}{4}} \times 7 = 273 \div y^{\frac{1}{2}}$

Directions (94-98) : In the following questions two equations numbred I and II are given. You have to solved both the equations and

Give answer If

(1) $x > y$

(2) $x \ge y$

(3) $x < y$

(4) $x \le y$

(5) $x = y$ or the relationship cannot be established

(Allahabad Bank PO Exam. 17.04.2011)

94. $\sqrt{25x^2} - 125 = 0$

II. $\sqrt{361}\,y + 95 = 0$

95. I. $\frac{5}{7} - \frac{5}{21} = \frac{\sqrt{x}}{42}$

II. $\frac{\sqrt{y}}{4} + \frac{\sqrt{y}}{16} = \frac{250}{\sqrt{y}}$

96. $(625)^{\frac{1}{4}}x + \sqrt{1225} = 155$

II. $\sqrt{196}\,y + 13 = 279$

97. I. $5x^2 - 18x + 9 = 0$

II. $3y^2 + 5y - 2 = 0$

98. I. $\frac{13}{\sqrt{x}} + \frac{9}{\sqrt{x}} = \sqrt{x}$

II. $y^4 - (13 \times 2)^{\frac{9}{2}}$

Directions (99-103): In the following questions, two equations numbred I and II are given. You have to to solve both the equations and

—

Give answer If

(1) $x > y$

(2) $x \ge y$

(3) $x < y$

(4) $x \le y$

(5) $x = y$ or the relation- ship cannot be established

(Indian Overseas Bank PO Exam. 22.05.2011)

99. I. $x^2 - 19x + 84 = 0$

II. $y^2 - 25y + 156 = 0$

100. I. $x^3 - 468 = 1729$

II. $y^2 - 1733 + 1564 = 0$

101. I. $\frac{9}{\sqrt{x}} + \frac{19}{\sqrt{x}} = \sqrt{x}$

II. $y^5 - \frac{(2 \times 14)^{\frac{11}{2}}}{\sqrt{y}} = 0$

102. I. $\sqrt{784}\,x + 1234 = 1486$

II. $\sqrt{1089}\,y + 2081 = 2345$

103. I. $\frac{12}{\sqrt{x}} - \frac{23}{\sqrt{x}} = 5\sqrt{x}$

II. $\frac{\sqrt{y}}{12} - \frac{5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

Directions (104-108): In the following questions two equations numbered I and II are given. You have to solve both the equations and

(IDBI Bank Officer Exam. 16.09.2012)

Give answer if

(1) $x > y$

(2) $x \ge y$

(3) $x < y$

(4) $x \le y$

(5) $x = y$ or the relationship cannot be established

104. I. $\frac{15}{\sqrt{x}} - \frac{9}{\sqrt{x}} = (x)^{\frac{1}{2}}$

II. $y^{10} - (36)^5 = 0$

105. I. $5x + 2y = 96$

II. $3(7x + 5y) = 489$

106. I. $(441)^{\frac{1}{2}}x^2 - 111 = (15)^2$

II. $\sqrt{121}\,y^2 + (6)^3 = 260$

107. I. $17x = (13)^2 + \sqrt{196} + (5)^2 + 4x$

II. $9y - 345 = 4y - 260$

108. I. $3x^2 - 13x + 14 = 0$

II. $y^2 - 7y + 12 = 0$

Directions (109-113): In each of the following questions two equations are given. Solve

these equations and give answer :

(1) if $x \geq y$, i.e., x is greater than or equal to y
(2) if $x > y$, i.e., x is greater than y
(3) if $x \leq y$, i.e., x is less than or equal to y
(4) if $x < y$, i.e., x less than y
(5) $x = y$ or no relation can be established between x and y

(IBPS Specialist Officer CWE 17.03.2013)

109. I. $x^2 + 5x + 6 = 0$
II. $y^2 + 7y + 12 = 0$

100. I. $x^2 + 20 = 9x$
II. $y^2 + 42 = 13y$

111. I. $2x + 3y = 14$
II. $4x + 2y = 16$

112. I. $x = \sqrt{625}$
II. $y = \sqrt{676}$

113. I. $x^2 + 4x + 4 = 0$
II. $y^2 - 8y + 16 = 0$

Directions (114-118) : In each of the following questions, two equations are given. You have to solve these equations and

(IBPS Bank PO/MT CWE-III 26.10.2013)

Give answer If
(1) $x < y$
(2) $x > y$
(3) $x = y$
(4) $x \geq y$
(5) $x \leq y$

114. I. $x^2 - 24x + 144 = 0$
II. $y^2 - 26y + 169 = 0$

115. I. $2x^2 + 3x - 20 = 0$
II. $2y^2 + 19y + 44 = 0$

116. I. $6x^2 + 77x + 121 = 0$
II. $y^2 + 9y - 22 = 0$

117. I. $x^2 - 6x = 7$
II. $2y^2 + 13y + 15 = 0$

118. I. $10x^2 - 7x^2 + 1 = 0$
II. $35y^2 - 12y + 1 = 0$

Directions (119- 123): In each of the following questions equation I and equation II have been given. You have to solve both of these equations and

(Corporation Bank Specialist Officer (Marketing) Exam. 22.12.2014)

Give answer If
(1) $x < y$
(2) $x > y$
(3) $x \leq y$
(4) $x \geq y$
(5) $x = y$ or no relation between two can be established.

119. 1. $4x^2 - 32x + 63 = 0$
II. $2y^2 - 11y + 15 = 0$

120. I. $x^3 = \left(\sqrt[3]{216}\right)^3$
II. $6y^2 = 150$

121. I. $12x^2 + 17x + 6 = 0$
II. $6y^2 + 5y + 1 = 0$

122. I. $20x^2 + 9x + 1 = 0$
II. $30y^2 + 11y + 1 = 0$

123. I. $x^2 + 17x + 72 = 0$
II. $y^2 + 19y + 90 = 0$

SBI PO EXAMS

Directions (1-3): In each question below one or more equation (s) is/are provided. On the basis of these you have to find out relation between p and q.

Give answer (1) if $p = q$
Give answer (2) if $p > q$.
Give answer (3) if $q > p$
Give answer (4) if $p \geq q$ and
Give answer (5) if $q \geq p$

(SBI Banks PO Exam. 20.08.2000)

1. I. $4q^2 + 8q = 4q + 8$
II. $p^2 + 9p = 2p - 12$

2. I. $2p^2 + 40 = 18p$
II. $q^2 = 13q - 42$

3. I. $6q^2 + \frac{1}{2} = \frac{7}{2}q$
II. $12p^2 + 2 = 10p$

Directions (4-8): In each of the questions a pair of equations is given. You have to find the values of x and y and give answer.

(l) if $x < y$ (2) if $x \leq y$
(3) if $x = y$ (4) if $x > y$
(5) if $x \geq y$

(SBI Associate Banks PO Exam. 21.07.2002)

4. I. $2x^2 - 7x + 6 = 0$
II. $4y^2 = 9$

5. I. $4x^2 - 4x - 3 = 0$
II. $4y^2 + 12y + 5 = 0$

6. I. $4x^2 = 49$
II. $9y^2 - 66y + 121 = 0$

7. I. $x^2 + 9x + 14 = 0$
II. $y^2 + y - 2 = 0$

8. I. $9x^2 - 18x + 5 = 0$
II. $2y^2 - 9y + 10 = 0$

Directions (9-13) : For the two given equations I and II.

Give answer (1) if p is greater than q
Give answer (2) if p is smaller than q
Give answer (3) if p is equal to q.

Give answer (4) if p is either equa to or greater than q.

Give answer (5) if p is either equa to or smaller than q.

(SBI Banks PO Exam. 18.05.2003)

9. I. $6p^2 + 5p + 1 = 0$
II. $20p^2 + 9q = -1$

10. I. $3p^2 + 2p - 1 = 0$
II. $2q^2 + 7q + 6 = 0$

11. I. $3p^2 + 15p = -18$
II. $q^2 + 7q + 12 = 0$

12. I. $p = \frac{\sqrt{4}}{\sqrt{9}}$
II. $9q^2 - 12q + 4 = 0$

13. I. $p^2 + 13p + 42 = 0$
II. $q^2 = 36$

Directions (14-18) : In each of these questions two equations numbered I & II are given. You have to solve both the equations and give answer

(1) if $a < b$
(2) if $a > b$
(3) if relationship between a & b cannot be established
(4) if $a \geq b$
(5) if $a \leq b$

(SBI PO Exam. 09.01.2005)

14. I. $a^2 + 5a + 6 = 0$
II. $b^2 + 3b + 2 = 0$

15. I. $2a^2 + 3a + 1 = 0$
II. $12b^2 + 7b + 1 = 0$

16. I. $a^2 = 4$
II. $a^2 = 9$

17. I. $6a^2 - 25a + 25 = 0$
II. $15b^2 - 16b + 4 = 0$

18. I. $4a^2 - 20a + 21 = 0$
II. $2b^2 - 5b + 3 = 0$

Directions (19-23): In each question below one or two equation(s) is/ are provided. On the basis of these you have to find out relation between p and q.

Give answer (1) if $p = q$
Give answer (2) if $p > q$
Give answer (3) if $p < q$
Give answer (4) if $p \geq q$ and
Give answer (5) if $q \geq p$

(SBI PO Exam. 26.11.2006) j

19. I. $p^2 + 24 = 10p$
II. $2q^2 + 18 = 12q$

20. $pq + 30 = 6p + 5q$

21. I. $q^2 + q = 2$
II. $p^2 + 7p + 10 = 0$

22. I. $p^2 + 16 = 8p$
II. $4q^2 + 64 = 32q$

23. I. $2p^2 + 12p + 16 = 0$
II. $2q^2 + 14q + 24 = 0$

Directions (24-28): In each of the following questions two equations are given. You have to solve them and

Give answer (1) if $p < q$
Give answer (2) if $p > q$
Give answer (3) if $p \leq q$
Give answer (4) if $p \geq q$
Give answer (5) if $p = q$

(SBI Associate Banks PO Exam. 07.01.2007)

24. I. $p^2 - 7p = -12$
II. $q^2 - 3q + 2 = 0$

25. I. $12p^2 - 7p = -1$
II. $6q^2 - 7q + 2 = 0$

26. I. $p^2 + 12p + 35 = 0$
II. $2q^2 + 22q + 56 = 0$

27. I. $p^2 - 8p + 15 = 0$
II. $q^2 - 5q = -6$

28. I. $2p^2 + 20p + 50 = 0$
II. $q^2 = 25$

RBI GRADE-B OFFICER EXAMS

1. Which value of x does satisfy the inequality $2x^2 + x - 3 < 0$?

(1) $-\frac{3}{2} < x < 1$ (2) $-1 < x < \frac{2}{3}$

(3) $x > 1$ (4) $x < \frac{-2}{5}$

(5) None of these

(RBI Grade-B Officer Exam.17.11.2002

Directions : (2 - 6) In each of the following questions, one or two equations (s) is/are given. On their basis you have to determine relation betweer x and y and then,

(RBI Grade-B Officer Exam.17.11.2002)

Give answer (1) if $x < y$
Give answer (2) if $x > y$
Give answer (3) if $x < y$
Give answer (4) if $x > y$
Give answer (5) if $x = y$

2. I. $x^2 + 3x + 2 = 0$
II. $2y^2 = 5y$

3. I. $2x^2 + 5x + 2 = 0$
II. $4y^2 = 1$

4. I. $y^2 + 2y - 3 = 0$
II. $2x^2 - 7x + 6 = 0$

5. I. $x^2 + 2x - 8 = 0$
II. $y^2 - 2 = 7$

6. I. $x^2 - 5x + 6 = 0$
II. $y^2 + y - 6 = 0$

Directions (7-11) : For the two given equations I and II.

Give answer (1) if p is greater than q.
Give answer (2) if p is smaller than q.
Give answer (3) if p is equal to q.
Give answer (4) if p is either equal to or greater than q.
Give answer (5) if p is either equal to or smaller than q.

(RBI Orade-B Officer Exam. 2007)

7. I. $p^2 + 5p + 6 = 0$
II. $q^2 + 3q + 2 = 0$

8. I. $p^2 = 4$
II. $q^2 + 4q = -4$

9. I. $p^2 + p = 56$
II $q^2 - 17q + 72 = 0$

10. I. $3p + 2q - 58 = 0$
II. $4q + 4p = 92$

11. I. $3p^2 + 17p + 10 = 0$
II. $10q^2 + 9q + 2 = 0$

Directions (12-16) : In each of these questions, two equations are given. You have to solve these equations and find out the values of x and y and :

Give answer
(1) if $x < y$ (2) if $x > y$
(3) if $x \leq y$ (4) if $x \geq y$
(5) if $x = y$

(RBI Grade-B Office Exam. 2008)

12. **I.** $8x^2 + 6x = 5$
II. $12y^2 - 22y + 8 = 0$

13. **I.** $4x + 7y = 209$
II. $12x - 14y = -38$

14. **I.** $17x^2 + 48x = 9$
II. $13y^2 = 32y - 12$

15. **I.** $18x^2 + 18x + 4 = 0$
II. $12y^2 + 29y + 14 = 0$

16. **I.** $16x^2 + 20x + 6 = 0$
II. $10y^2 + 38y + 24 = 0$

Directions (17-21): In the following questions two equations numbered I and II are given. You have to solve both the equations and

Give Answer If
(1) $x > y$
(2) $x \geq y$
(3) $x < y$
(4) $x \leq y$
(5) $x = y$ or the relationship can not be established

(RBI Grade-B Officer Exam.06.02.2011)

17. I. $x^2 + 5x + 6 = 0$
II. $y^2 + 3y + 2 = 0$

18. I. $x^2 - 10x + 24 = 0$
II. $y^2 - 9y + 20 = 0$

19. I. $(x)^2 = 961$
II. $y = \sqrt{961}$

20. I. $x^2 - 72 = x$
II. $y^2 = 64$

21. I. $x^2 - 463 = 321$
II. $y^2 - 421 = 308$

Directions (22-26): In the following questions three equations numbered I, II and III are given. You have to solve all the equations either together or separately, or two together and one separately, or by any other method and—

(RBI Grade 'B' Officer's Exam. 18.12.2011)

Give answer If
(1) $x < y = z$
(2) $x < y < z$
(3) $x < y > z$
(4) $x = y > z$
(5) $x = y = z$ or if none of the above relationship is established

22. I. $7x + 6y + 4z = 122$
II. $4x + 5y + 3z = 88$
III. $9x + 2y + z = 78$

23. I. $7x + 6y = 110$
II. $4x + 3y = 59$
III. $x + z = 15$

24. I. $x = \sqrt{\left[(36)^{\frac{1}{2}} \times (1296)^{\frac{1}{4}}\right]}$
II. $2y + 3z = 33$
III. $6y + 5z = 71$

25. I. $8x + 7y = 135$
II. $5x + 6y = 99$
III. $9y + 8z = 121$

26. I. $(x + y)^3 = 1331$
II. $x - y + z = 0$
III. $xy = 28$

INSURANCE EXAMS

Directions (1-5) : In each of the questions a pair of equations is given. You have to find the values of x and y and give answer.

(I) if $x < y$ (2) if $x \leq y$
(3) if $x = y$ (4) if $x > y$
(5) if $x \geq y$

(LIC Assistant Administrative Officer (AAO) Exam. 2006)

1. I. $2x^2 - 7x + 6 = 0$
II. $4y^2 = 9$

2. I. $4x^2 - 4x - 3 = 0$
II. $4y^2 + 12y + 5 = 0$

3. I. $4x^2 = 49$
II. $9y^2 - 66y + 121 = 0$

4. I. $x^2 + 9x + 14 = 0$
II. $y^2 + y - 2 = 0$

5. I. $9x^2 - 18x + 5 = 0$
II. $2y^2 - 9y + 10 = 0$

SHORT ANSWERS

NATIONALISED BANKS & IBPS PO/MT/SO

1. (5) **2.** (4)
3. (2) **4.** (1)
5. (2) **6.** (2)
7. (1) **8.** (5)
9. (5) **10.** (5)
11. (1) **12.** (2)
13. (1) **14.** (5)
15. (3) **16.** (2)
17. (1) **18.** (4)
19. (5) **20.** (2)
21. (3) **22.** (2)
23. (2) **24.** (4)
25. (3) **26.** (1)
27. (5) **28.** (3)
29. (1) **30.** (2)
31. (4) **32.** (5)
33. (2) **34.** (1)
35. (3) **36.** (1)
37. (4) **38.** (3)
39. (4) **40.** (4)
41. (5) **42.** (5)
43. (1) **44.** (2)
45. (5) **46.** (5)
47. (3) **48.** (5)
49. (1) **50.** (4)
51. (3) **52.** (2)
53. (5) **54.** (1)
55. (5) **56.** (3)
57. (5) **58.** (3)
59. (2) **60.** (4)
61. (5) **62.** (3)
63. (1) **64.** (5)
65. (4) **66.** (3)
67. (5) **68.** (1)
69. (3) **70.** (2)
71. (1) **72.** (5)
73. (5) **74.** (2)
75. (3) **76.** (5)
77. (3) **78.** (3)
79. (2) **80.** (2)
81. (3) **82.** (1)
83. (5) **84.** (2)
85. (1) **86.** (1)
87. (3) **88.** (1)
89. (1) **90.** (3)
91. (1) **92.** (2)
93. (5) **94.** (1)
95. (3) **96.** (1)
97. (1) **98.** (3)
99. (4) **100.** (2)
101. (5) **102.** (1)
I03. (1) **104.** (5)
105. (1) **106.** (1)
107. (3) **108.** (3)
109. (1) **110.** (4)
111. (4) **112.** (5)
113. (4) **114.** (1)
115. (4) **116.** (5)
117. (2) **118.** (4)
119. (2) **120.** (2)
121. (1) **122.** (3)
123. (4)

SBI PO EXAMS

1. (3) **2.** (3)
3. (4) **4.** (5)
5. (5) **6.** (1)
7. (2) **8.** (1)
9. (2) **10.** (1)
11. (4) **12.** (3)
13. (5) **14.** (5)
15. (1) **16.** (3)
17. (2) **18.** (4)
19. (2) **20.** (3)
21. (5) **22.** (1)
23. (4) **24.** (2)
25. (1) **26.** (3)
27. (4) **28.** (3)

RBI GRADE-B OFFICER EXAMS

1. (1) **2.** (1)
3. (3) **4.** (2)
5. (1) **6.** (4)
7. (5) **8.** (4)
9. (2) **10.** (1)
11. (2) **12.** (3)
13. (5) **14.** (1)
15. (4) **16.** (2)
17. (4) **18.** (2)
19. (5) **20.** (2)
21. (5) **22.** (1)
23. (3) **24.** (2)
25. (4) **26.** (5)

INSURANCE EXAMS

1. (5) **2.** (5)
3. (1) **4.** (2)
5. (1)

EXPLANATIONS

NATIONALISED BANKS & IBPS SO/MT/SO

1. (5) I. $\Rightarrow p^2 + 3p + 2p + 6 = 0$
$\Rightarrow p(p + 3) + 2(p + 3) = 0$
$\Rightarrow (p + 3)(p + 2) = 0$
$\Rightarrow p = -2$ or -3

II. $\Rightarrow q^3 + q + 2q + 2 = 0$
$\Rightarrow q(q + 1) + 2(q + 1) = 0$
$\Rightarrow (q + 1)(q + 2) = 0$
$\Rightarrow q = -1$ or -2
Obviously $p \leq q$

2. (4) I. $\Rightarrow p = \pm 2$
II. $\Rightarrow q^2 + 2q + 2q + 4 = 0$
$\Rightarrow q(q + 2) + 2(q + 2) = 0$
$\Rightarrow (q + 2)(q + 2) = 0$
$\Rightarrow q = -2$
Obviously, $p \geq q$

3. (2) I. $\Rightarrow p^5 + p - 56 = 0$
$\Rightarrow p^6 + 8p - 7p - 56 = 0$
$\Rightarrow p(p + 8) - 7(p + 8) = 0$
$\Rightarrow (p + 8)(p - 7) = 0$
$\Rightarrow p = 7$ or -8
II. $\Rightarrow q^2 - 8q - 9q + 72 = 0$
$\Rightarrow q(q - 8) - 9(q - 8) = 0$
$\Rightarrow (q - 8)(q - 9) = 0$
$\Rightarrow q = 8$ or 9
Obviously, $p < q$

4. (1) We have,
$3p + 2q = 58$ ...(i)
$4p + 4q = 92$
$\Rightarrow 2p + 2q = 46$ ...(ii)
By equation (i) - (ii) we get
$p = 12$
From equation (i), $3 \times 12 + 2q = 58$
$\Rightarrow 2q = 58 - 36 = 22$
$\Rightarrow q = 11$
Hence, $p > q$

5. (2) I. $\Rightarrow 3p^2 + 15p + 2p + 10 = 0$
$\Rightarrow 3p(p + 5) + 2(p + 5) = 0$
$\Rightarrow (p + 5)(3p + 2) = 0$
$\Rightarrow p = -5$ or $-\frac{2}{3}$
II. $\Rightarrow 10q^2 + 5q + 4q + 2 = 0$
$\Rightarrow 5q(2q + 1) + 2(2q + 1) = 0$
$\Rightarrow (2q + 1)(5q + 2) = 0$
$\Rightarrow q = -\frac{1}{2}$ or $-\frac{2}{5}$
Obviously, $p < q$

6. (2) I. $4x^2 - 8x + 3 = 0$
$\Rightarrow 4x - 6x - 2x + 3 = 0$
$\Rightarrow 2x(2x - 3) - 1(2x - 3) = 0$
$\Rightarrow (2x - 1)(2x - 3) = 0$
$\therefore x = \frac{1}{2}$ or $\frac{2}{3}$
II. $2y^2 - 7y + 6 = 0$
$\Rightarrow 2y^2 - 4y - 3y + 6 = 0$
$\Rightarrow 2y(y - 2) - 3(y - 2) = 0$
$\Rightarrow (2y - 3)(y - 2) = 0$
$\therefore y = 2$ or $\frac{2}{3}$
Hence $x \leq y$

7. (1) I. $x^2 + x - 6 = 0$
$\Rightarrow x^2 + 3x - 2x - 6 = 0$
$\Rightarrow x(x + 3) - 2(x + 3) = 0$
$\Rightarrow (x + 3)(x - 2) = 0$
$\therefore x = -3$ or 2
II. $2y^2 - 13y + 21 = 0$
$\Rightarrow 2y^2 - 7y - 6y + 21 = 0$
$\Rightarrow y(2y - 7) - 3(2y - 7) = 0$
$\Rightarrow (2y - 7)(y - 3) = 0$
$\therefore y = \frac{7}{2}$ or 3
Hence $x < y$

8. (5)1. $x^2 - x - 6 = 0$
$\Rightarrow x^2 - 3x + 2x - 6 = 0$
$\Rightarrow x(x - 3) + 2(x - 3) = 0$
$\Rightarrow (x + 2)(x - 3) = 0$
$\Rightarrow x = 2$ or 3
II. $2y^2 + 13y + 21 = 0$
$\Rightarrow 2y^2 + 7y + 6y + 21 = 0$
$\Rightarrow y(2y + 7) + 3(2y + 7) = 0$
$\Rightarrow (y + 3)(2y + 7) = 0$
$\therefore y = -3$ or $\frac{-7}{2}$
Clearly, $x > y$

9. (5) I. $x^2 = 4$
$\Rightarrow x = \pm 2$
II. $y^2 + 6y + 9 = 0$
$\Rightarrow (y + 3)^2 = 0$
$\Rightarrow y + 3 = 0$
$\Rightarrow y = -3$
Clearly, $x > y$

10. (5) I. $2x + 3y = 4$
II. $3x + 2y = 11$
Multiplying equation I by 3 and equation II by 2 and subtracting equation II from equation I
$6x + 9y - 6x - 4y = 10 - 22$
$5y = -10$
$\therefore y = -2$
Putting the value of $y = -2$ in equation I,
$2x + 3x - 2 = 4$
$\Rightarrow 2x = 6 + 4$
$\Rightarrow x = \frac{10}{2} = 5$

Clearly, $x > y$

11. (1) I. $4x + 2y = 51$

II. $13x + 15y = 221$

Multiplying equation I by 15 an II by 2 and by I - II,

$60x + 30y - 26x - 30y = 765 - 442$

$34x = 323$

$$x = \frac{323}{34} = 9.5$$

From equation I,

$4 \times 9.5 + 2y = 51$

$\Rightarrow 38 + 2y = 51$

$\Rightarrow 2y = 51 - 38$

$$\Rightarrow y = \frac{13}{2} = 6.5$$

Clearly, $x > y$

12. (2) I. $8x^2 + 3x - 38 = 0$

$\Rightarrow 8x^2 + 19x - 16x - 38 = 0$

$\Rightarrow x(8x + 19) - 2(8x + 19) = 0$

$\Rightarrow (8x + 19)(x - 2) = 0$

$$\Rightarrow x = 2 \text{ or } -\frac{19}{8}$$

II. $6y^2 - 29y + 34 = 0$

$\Rightarrow 6y^2 - 17y - 12y + 34 = 0$

$\Rightarrow y(6y - 17) - 2(6y - 17) = 0$

$\Rightarrow (y - 2)(6y - 17) = 0$

$$\Rightarrow y = 2 \text{ or } \frac{17}{6}$$

Clearly, $x \leq y$.

13. (1) I. $x^2 - 20x + 91 = 0$

$\Rightarrow x^2 - 13x - 7x + 91 = 0$

$\Rightarrow x(x - 13) - 7(x - 13) = 0$

$\Rightarrow (x - 7)(x - 13) = 0$

$\Rightarrow x = 7 \text{ or } 13$

II. $10y^2 - 29y + 21 = 0$

$\Rightarrow 10y^2 - 15y - 14y + 21 = 0$

$\Rightarrow 5y(2y - 3) - 7(2y - 3) = 0$

$\Rightarrow (2y - 3)(5y - 7) = 0$

$$\Rightarrow y = \frac{3}{2} \text{ or } \frac{7}{5}$$

Clearly, $x > y$.

14. (5) I. $6x^2 + 13x + 5 = 0$

$\Rightarrow 6x^2 + 10x + 3x + 5 = 0$

$\Rightarrow 2x(3x + 5) + 1(3x + 5) = 0$

$\Rightarrow (3x + 5)(2x + 1) = 0$

$$\Rightarrow x = \frac{-5}{3} \text{ or, } -\frac{1}{2}$$

II. $9y^2 + 22y + 8 = 0$

$\Rightarrow 9y^2 + 18y + 4y + 8 = 0$

$\Rightarrow 9y(y + 2) + 4(y + 2) = 0$

$\Rightarrow (y + 2)(9y + 4) = 0$

$$\Rightarrow y = -2 \text{ or } \frac{-4}{9}$$

Relationship can't be established.

15. (3) II. $y = 92567 - 92551 = 16$

I. $(x + y)^2 = 784$

$\Rightarrow x + y = 28$

$\Rightarrow x = 28 - 16 = 12$

Clearly, $x < y$

16. (2) $4x + 5y = 83$ ...(i)

$$\frac{3x}{2y} = \frac{21}{22} \Rightarrow \frac{x}{y} = \frac{21}{22} \times \frac{2}{3} = \frac{7}{11}$$

$$\Rightarrow x = \frac{7}{11}y \quad \text{...(ii)}$$

From equation (i),

$$4 \times \frac{7}{11}y + 5y = 83$$

$\Rightarrow 28y + 55y = 913$

$\Rightarrow 83y = 913$

$$\Rightarrow y = \frac{913}{83} = 11$$

From equation (ii),

$$x = \frac{7}{11} \times 11 = 7$$

$\therefore y - x = 11 - 7 = 4$

17. (1) I. $x^2 - 14x + 48 = 0$

$\Rightarrow x^2 - 8x - 6x + 48 = 0$

$\Rightarrow x(x - 8) - 6(x - 8) = 0$

$\Rightarrow (x - 6)(x - 8) = 0$

$\therefore x = 6 \text{ or } 8$

II. $y^2 - 5y + 6 = 0$

$\Rightarrow y^2 - 3y - 2y + 6 = 0$

$\Rightarrow y(y - 3) - 2(y - 3) = 0$

$\Rightarrow (y - 2)(y - 3) = 0$

$\therefore y = 2 \text{ or } 3$

Clearly, $x > y$

18. (4) I. $x^2 + 9x + 20 = 0$

$\Rightarrow x^2 + 5x + 4x + 20 = 0$

$\Rightarrow x(x + 5) + 4(x + 5) = 0$

$\Rightarrow (x + 4)(x + 5) = 0$

$\therefore x = -4 \text{ or } -5$

II. $y^2 + 7y + 12 = 0$

$\Rightarrow y^2 + 4y + 3y + 12 = 0$

$\Rightarrow y(y + 4) + 3(y + 4) = 0$

$\Rightarrow (y + 3)(y + 4) = 0$

$\therefore y = -3 \text{ or } -4$

Clearly, $x \leq y$

19. (5) I. $x^2 = 529$

$\therefore x = \sqrt{529} = \pm 23$

II. $y = \sqrt{529} = \pm 23$

Clearly, $x = y$

20. (2) I. $x^2 + 13x + 42 = 0$

$\Rightarrow x^2 + 7x + 6x + 42 = 0$

$\Rightarrow x(x + 7) + 6(x + 7) = 0$

$\Rightarrow (x + 6)(x + 7) = 0$

$\therefore x = -6$ or -7

II. $y^2 + 16y + 63 = 0$

$\Rightarrow y^2 + 9y + 7y + 63 = 0$

$\Rightarrow y(y + 9) + 7(y + 9) = 0$

$\Rightarrow (y + 9)(y + 7) = 0$

$\therefore y = -9$ or -7

Clearly, $x \geq y$

21. (3) I. $2x + 3y = 14$

II. $4x + 2y = 16$

By equation 1 × 2 - equation II, we have

$4x + 6y - 4x - 2y = 28 - 16$

$\Rightarrow 4y = 12$

$\Rightarrow y = \frac{12}{4} = 3$

From equation I,

$2x + 3 \times 3 = 14$

$\Rightarrow 2x = 5$

$\Rightarrow x = \frac{5}{2}$

Clearly, $x < y$

22. (2) $3y + 9x = 54$

$\Rightarrow 3(y + 3x) = 54$

$\Rightarrow 3x + y = \frac{54}{3} = 18$

and,

$\Rightarrow \frac{28x}{13y} = \frac{140}{39}$

$\Rightarrow \frac{x}{y} = \frac{140}{39} \times \frac{13}{28} \Rightarrow \frac{x}{y} = \frac{5}{3}$

$\Rightarrow 5y = 3x$

$\Rightarrow 5y - 3x = 0$ ...(ii)

Adding equations (i) and (ii), we get

$3x + y + 5y - 3x = 18$

$\Rightarrow 6y = 18 \Rightarrow y = \frac{18}{6} = 3$

From equation (ii),

$5 \times 3 - 3x = 0$

$\Rightarrow 3x = 15$

$\Rightarrow x = \frac{15}{3} = 5$

$\therefore y - x = 3 - 5 = -2$

23. (2) I. $x^2 - 1 = 0$

$\Rightarrow (x + 1)(x - 1) = 0$

$\Rightarrow x = -1$ or 1

II. $y^2 + 4y + 3 = 0$

$\Rightarrow y^2 + 3y + y + 3 = 0$

$\Rightarrow y(y + 3) + 1(y + 3) = 0$

$\Rightarrow (y + 1)(y + 3) = 0$

$\Rightarrow y = -1$ or -3

Clearly, $x \geq y$

24. (4) I. $x^2 - 7x + 12 = 0$

$\Rightarrow x^2 - 4x - 3x + 12 = 0$

$\Rightarrow x(x - 4) - 3(x - 4) = 0$

$\Rightarrow (x - 3)(x - 4) = 0$

$\Rightarrow x = 3$ or 4

II. $y^2 - 12y + 32 = 0$

$\Rightarrow y^2 - 8y - 4y + 32 = 0$

$\Rightarrow y(y - 8) - 4(y - 8) = 0$

$\Rightarrow (y - 4)(y - 8) = 0$

$\Rightarrow y = 4$ or 8

Clearly, $x \leq y$

25. (3)1. $x^3 - 371 = 629$

$\Rightarrow x^3 = 371 + 629 = 1000$

$\Rightarrow x = \sqrt[3]{1000} = 10$

II. $y^3 = 543 + 788 = 1331$

$\Rightarrow y = \sqrt[3]{1331} = 11$

Clearly, $x < y$

26. (1) By equation 1 × 3 - equation II × 5, we have,

$15x + 6y - 15x - 35y = 93 - 180$

$\Rightarrow -29y = -87 \Rightarrow y = \frac{87}{29} = 3$

From equation I,

$5x + 2 \times 3 = 31$

$\Rightarrow 5x = 31 - 6 = 25$

$\Rightarrow x = 5$

Clearly, $x > y$

27. (5)1. $2x^2 + 11x + 12 = 0$

$\Rightarrow 2x^2 + 8x + 3x + 12 = 0$

$\Rightarrow 2x(x + 4) + 3(x + 4) = 0$

$\Rightarrow (x + 4)(2x + 3) = 0$

$\Rightarrow x = -4$ or $-\frac{3}{2}$

II. $5y^2 + 27y + 10 = 0$

$\Rightarrow 5y^2 + 25y + 2y + 10 = 0$

$\Rightarrow 5y(y + 5) + 2(y + 5) = 0$

$\Rightarrow (y + 5)(5y + 2) = 0$

$\Rightarrow y = -5$ or $-\frac{2}{5}$

Relationship can't be established.

28. (3) I. $2x^2 + 7x + 4x + 14 = 0$

$\Rightarrow x(2x + 7) + 2(2x + 7) = 0$

$\Rightarrow (x + 2)(2x + 7) = 0$

$\therefore x = -2 \text{ or } -\frac{7}{2}$

II. $4y^2 + 2.2y.3 + 9 = 0$

$\Rightarrow (2y + 3)^2 = 0$

$\Rightarrow 2y + 3 = 0$

$\Rightarrow y = -\frac{3}{2}$

Clearly, $x < y$

29. (1) I. $(x - 2)(x + 2) = 0$

$x = 2 \text{ or } -2$

II. $y^2 + 2.y.3 + 9 = 0$

$\Rightarrow (y + 3)^2 = 0$

$\Rightarrow y + 3 = 0$

$y = -3$

Clearly, $x > y$

30. (2) I. $x^2 - 4x - 3x + 12 = 0$

$\Rightarrow x(x - 4) - 3(x - 4) = 0$

$\Rightarrow (x - 3)(x - 4) = 0$

$\therefore x = 3 \text{ or } 4$

II. $y^2 + 4y - 3y - 12 = 0$

$\Rightarrow y(y + 4) - 3(y + 4) = 0$

$\Rightarrow (y - 3)(y + 4) = 0$

$\therefore y = 3 \text{ or } -4$

Clearly, $x \geq y$

31. (4) I. $x^2 = 729$

$\therefore x = \pm 27$

II. $y = \sqrt{729} = +27$

Clearly, $x \leq y$

32. (5) I. $x^4 = 227 + 398 = 625 = 5^4$

$\Rightarrow x = 5$

II. $y^2 = 346 - 321 = 25 = 5^2$

$\Rightarrow y = 5$

Clearly, $x = y$

33. (2) I. $x^2 - x - 12 = 0$

$\Rightarrow x^2 - 4x + 3x - 12 = 0$

$\Rightarrow x(x - 4) + 3(x - 4) = 0$

$\Rightarrow (x - 4)(x + 3) = 0$

$\therefore x = 4 \text{ or } -3$

II. $y^2 + 5y + 6 = 0$

$\Rightarrow y^2 + 3y + 2y + 6 = 0$

$\Rightarrow y(y + 3) + 2(y + 3) = 0$

$\Rightarrow (y + 3)(y + 2) = 0$

$\therefore y = -3 \text{ or } -2$

Clearly, $x \geq y$

34. (1) I. $x^2 - 8x + 15 = 0$

$\Rightarrow x^2 - 5x - 3x + 15 = 0$

$\Rightarrow x(x - 5) - 3(x - 5) = 0$

$\Rightarrow (x - 3)(x - 5) = 0$

$\therefore x = 3 \text{ or } 5$

II. $y^2 - 3y + 2 = 0$

$\Rightarrow y^2 - 2y - y + 2 = 0$

$\Rightarrow y(y - 2) - 1(y - 2) = 0$

$\Rightarrow (y - 1)(y - 2) = 0$

$\therefore y = 1 \text{ or } 2$

Clearly, $x > y$

35. (3) I. $x^2 = 32 + 112 = 144$

$\therefore x = \pm 12$

II. $y = \sqrt{169} = +13$

Clearly $x < y$

36. (1) I. $x = \sqrt{121} = +11$

II. $y^2 = 121$

$\therefore y = \pm 11$

Clearly $x \geq y$

37. (4) I. $x^2 = 16$

$\Rightarrow x = \pm 4$

II. $y^2 - 9y + 20 = 0$

$\Rightarrow y^2 - 4y - 5y + 20 = 0$

$\Rightarrow y(y - 4) - 5(y - 4) = 0$

$\Rightarrow (y - 5)(y - 4) = 0$

$\therefore y = 5 \text{ or } 4$

Clearly, $x \leq y$

38. (3) I. $3x^2 + 8x + 4 = 0$

$\Rightarrow 3x^2 + 6x + 2x + 4 = 0$

$\Rightarrow 3x(x + 2) + 2(x + 2) = 0$

$\Rightarrow (x + 2)(3x + 2) = 0$

$\therefore x = -2 \text{ or } -\frac{2}{3}$

II. $4y^2 - 19y + 12 = 0$

$\Rightarrow 4y^2 - 16y - 3y + 12 = 0$

$\Rightarrow 4y(y - 4) - 3(y - 4) = 0$

$\Rightarrow (y - 4)(4y - 3) = 0$

$\therefore y = 4 \text{ or } \frac{3}{4}$

Clearly, $x < y$

39. (4) I. $x^2 + x - 20 = 0$

$\Rightarrow x^2 + 5x - 4x - 20 = 0$

$\Rightarrow x(x + 5) - 4(x + 5) = 0$

$\Rightarrow (x + 5)(x - 4) = 0$

$\therefore x = -5 \text{ or } 4$

II. $y^2 - y - 30 = 0$

$\Rightarrow y^2 - 6y + 5y - 30 = 0$

$\Rightarrow y(y - 6) + 5(y - 6) = 0$

$\Rightarrow (y - 6)(y + 5) = 0$

$\therefore y = 6 \text{ or } -5$

Clearly, $x \leq y$

40. (4) I. $x^2 = 365 + 364 = 729$
$\therefore x = \pm 27$
II. $y - \sqrt{324} = \sqrt{81}$
$\Rightarrow y - 18 = 9$
$\Rightarrow y = 27$
Clearly, $x \leq y$

41. (5) I. $4 + 7 = \sqrt{x} \times \sqrt{x}$
$\Rightarrow x = 11$

II. $y^2 - \frac{11^{\frac{5}{2}}}{\sqrt{y}} = 0$

$\Rightarrow y^{2+\frac{1}{2}} = 11^{\frac{5}{2}}$

$\Rightarrow y^{\frac{5}{2}} = 11^{\frac{5}{2}}$

$\Rightarrow y = 11$
Clearly, $x = y$

42. (3) I. $225x^2 = 4$

$\Rightarrow x^2 = \frac{4}{225} \Rightarrow x = \pm \frac{2}{15}$

II. $\sqrt{225y} + 2 = 0$

$\Rightarrow \sqrt{225y} = -2$
Squaring on both sides, $225y = 4$

$\Rightarrow y = \frac{4}{225}$
Clearly $x < y$

43. (1) I. $5x^2 - 18x + 9 = 0$
$\Rightarrow 5x^2 - 15x - 3x + 9 = 0$
$\Rightarrow 5x(x - 3) - 3(x - 3) = 0$
$\Rightarrow (x - 3)(5x - 3) = 0$

$\therefore x = 3$ or $\frac{3}{5}$
II. $20y^2 - 13y + 2 = 0$
$\Rightarrow 20y^2 - 8y - 5y + 2 = 0$
$\Rightarrow 4y(5y - 2) - 1(5y - 2) = 0$
$\Rightarrow (4y - 1)(5y - 2) = 0$

$\therefore y = \frac{1}{4}$ or $\frac{2}{5}$
Clearly, $x > y$

44. (2) I. $x^3 = 878 + 453 = 1331$
$\therefore x = \sqrt[3]{1331} = 11$
II. $y^2 = 82 + 39 = 121$
$\therefore y = \pm 11$
Clearly $x \geq y$

45. (5) I. $\frac{3}{\sqrt{x}} + \frac{4}{\sqrt{x}} = \sqrt{x}$
$\Rightarrow 3 + 4 = x \Rightarrow x = 7$

II. $y^3 - \frac{(7)^{\frac{7}{2}}}{\sqrt{y}} = 0$

$\Rightarrow y^{3+\frac{1}{2}} - (7)^{\frac{7}{2}} = 0$

$\Rightarrow y^{\frac{7}{2}} = 7^{\frac{7}{2}} \Rightarrow y = 7$
Clearly $x = y$

46. (5) I. $9x - 4x = 54.55 + 15.45$
$\Rightarrow 5x = 70$
$\Rightarrow x = 14$
II. $\sqrt{y + 155} = 7 + 6 = 13$
$\Rightarrow y + 155 = 169$
$\Rightarrow y = 169 - 155 = 14$
Clearly $x = y$

47. (3) I. $x^2 + 11x + 30 = 0$
$\Rightarrow x^2 + 6x + 5x + 30 = 0$
$\Rightarrow x(x + 6) + 5(x + 6) = 0$
$\Rightarrow (x + 5)(x + 6) = 0$
$\therefore x = -5$ or -6
II. $y^2 + 7y + 12 = 0$
$\Rightarrow y^2 + 4y + 3y + 12 = 0$
$\Rightarrow y(y + 4) + 3(y + 4) = 0$
$\Rightarrow (y + 3)(y + 4) = 0$
$\therefore y = -3$ or -4
Clearly, $x < y$

48. (5) I. $\sqrt{x} - \frac{\sqrt{6}}{\sqrt{x}} = 0$

$\Rightarrow x - \sqrt{6} = 0 \Rightarrow \sqrt{6}$

II. $y^3 = 6^{\frac{3}{2}} = \left(\sqrt{6}\right)^3$

$\Rightarrow y = \sqrt{6}$
Clearly $x = y$

49. (1) By equation 1 × 3 - equation II
$9x - 6y - 5x + 6y = 30 - 6$
$\Rightarrow 4x = 24 \Rightarrow x = 6$
From equation I,
$3 \times 6 - 2y = 10$
$\Rightarrow 2y = 18 - 10 = 8$
$\Rightarrow y = 4$
Clearlr $x > y$

50. (4) I. $x^2 + x - 12 = 0$
$\Rightarrow x^2 + 4x - 3x - 12 = 0$
$\Rightarrow x(x + 4) - 3(x + 4) = 0$

$\Rightarrow (x - 3)(x + 4) = 0$
$\Rightarrow x = 3$ or -4]
II. $y^2 - 5y + 6 = 0$
$\Rightarrow y^2 - 3y - 2y + 6 = 0$
$\Rightarrow y(y - 3) - 2(y - 3) = 0$
$\Rightarrow (y - 2)(y - 3) = 0$
$\Rightarrow y = 2$ or 3
Clearly $x \leq y$

51. (3) I. $x^2 + 9x + 18 = 0$
$\Rightarrow x^2 + 6x + 3x + 18 = 0$
$\Rightarrow x(x + 6) + 3(x + 6) = 0$
$\Rightarrow (x + 3)(x + 6) = 0$
$\Rightarrow x = -3$ or -6
II. $y^2 - 13y + 40 = 0$
$\Rightarrow y^2 - 8y - 5y + 40 = 0$
$\Rightarrow y(y - 8) - 5(y - 8) = 0$
$\Rightarrow (y - 5)(y - 8) = 0$
$\Rightarrow y = 5$ or 8

52. (2) I. $\sqrt{x+6} = 11 - 6 = 5$
$\Rightarrow x + 6 = 25$
$\Rightarrow x = 25 - 6 = 19$
II. $y^2 = 473 - 112 = 361$
$\Rightarrow y = \pm 19$
Cleary $x \geq y$

53. (5) I. $x^2 - 1200 + 244 = 1444$
$\therefore x = \pm 38$
II. $y = 159 - 122 = 37$
Clearly, $x > y$ or $x < y$

54. (1) I. $14x + 7x = 59 + 25$
$\Rightarrow 21x = 84$

$$\Rightarrow x = \frac{84}{21} = 4$$

II. $\sqrt{y+222} = \sqrt{36} + \sqrt{81}$

$$\Rightarrow \sqrt{y+222} = 6 + 9 = 15$$

$\therefore y + 222 = 225$
$\Rightarrow y = 225 - 222 = 3$
Clearly, $x > y$

55. (4) I. $144x^2 = 16 + 9 = 25$

$$\Rightarrow x^2 = \frac{25}{144}$$

$$\Rightarrow x = \pm \frac{5}{12}$$

II. $12y = \sqrt{49} - \sqrt{4} = +5$

$$\Rightarrow y = \frac{5}{12}$$

Clearly $x \leq y$

56. (3) I. $x^2 - 9x + 20 = 0$
$\Rightarrow x^2 - 5x - 4x + 20 = 0$
$\Rightarrow x(x - 5) - 4(x - 5) = 0$
$\Rightarrow (x - 5)(x - 4) = 0$
$\therefore x = 5$ or 4
II. $y^2 - 7y - 6y + 42 = 0$
$\Rightarrow y(y - 7) - 6(y - 7) = 0$
$\Rightarrow (y - 6)(y - 7) = 0$
$\therefore y = 6$ or 7
Clearly, $x < y$

57. (5) I. $\dfrac{2\sqrt{x} + 3\sqrt{x}}{10} = \dfrac{1}{\sqrt{x}}$

$\Rightarrow 5\sqrt{x} \times \sqrt{x} = 10$
$\Rightarrow 5x = 10$
$\Rightarrow x = 2$

II. $\dfrac{10 - 2}{\sqrt{y}} = 4\sqrt{y}$

$\Rightarrow 4y = 8$

$$\Rightarrow y = \frac{8}{4} = 2$$

Clearly $x = y$

58. (3) $2x + 3y = 78$
$3x + 2y = 72$
On adding,
$5x + 5y = 150 \Rightarrow 5(x + y) = 150$

$$\Rightarrow x + y = \frac{150}{5} = 30$$

59. (2) I. $20x^2 - x - 12 = 0$
$\Rightarrow 20x^2 - 16x + 15x - 12 = 0$
$\Rightarrow 4x(5x - 4) + 3(5x - 4) = 0$
$\Rightarrow (5x - 4)(4x + 3) = 5$
$\Rightarrow 5x - 4 = 0$ or $4x + 3 = 0$

$$\Rightarrow x = \frac{4}{5} \text{ or } \frac{-3}{4}$$

II. $20y^2 + 27y + 9 = 0$
$20y^2 + 15y + 12y + 9 = 0$
$\Rightarrow 5y(4y + 3) + 3(4y + 3) = 0$
$\Rightarrow (5y + 3)(4y + 3) = 0$

$$\Rightarrow y = \frac{-3}{5} \text{ or } \frac{-3}{4}$$

Clearly, $x \geq y$

60. (4) I. $x^2 = 106 + 218 = 324$
$\therefore x = \sqrt{324} = \pm 18$
II. $y^2 - 37y + 342 = 0$
$\Rightarrow y^2 - 18y - 19y + 342 = 0$
$\Rightarrow y(y - 18) - 19(y - 18) = 0$
$\Rightarrow (y - 19)(y - 18) = 0$

$\Rightarrow y = 19$ or 18

Clearly $x \leq y$

61. (5) I. $\frac{7}{\sqrt{x}} + \frac{5}{\sqrt{x}} = \sqrt{x}$

$\Rightarrow 7 + 5 = \sqrt{x} \times \sqrt{x}$

$\Rightarrow x = 12$

II. $\frac{(12)^{\frac{5}{2}}}{\sqrt{y}} = 0$

$\Rightarrow y^{2+\frac{1}{2}} - (12)^{\frac{5}{2}} = 0$

$\Rightarrow y^{5/2} = 12^{5/2}$

$\Rightarrow y = 12$

62. (3) I. $19x + 4 = 0$

$\Rightarrow 19x = -4$

$\Rightarrow x = \frac{-4}{19}$

II. $21y + 4 = 0$

$\Rightarrow y = \frac{-4}{21}$

Clearly $x < y$

63. (1) I. $\frac{15}{\sqrt{x}} - \frac{2}{\sqrt{x}} = 6\sqrt{x}$

$\Rightarrow 15 - 2 = 6x$

$\Rightarrow 13 = 6x$

$\Rightarrow x = \frac{13}{6}$

II. $\frac{\sqrt{y}}{4} + \frac{7\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

$\Rightarrow \frac{3\sqrt{y} + 7\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

$\Rightarrow \frac{10\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

$\Rightarrow 10y = 12$

$\Rightarrow y = \frac{12}{10} = \frac{6}{5}$

Clearly $x > y$

64. (5) 1. $\sqrt{x+18} = \sqrt{144} - \sqrt{49}$

$\Rightarrow \sqrt{x+18} = \pm(12 - 7) = \pm 5$

$\Rightarrow x + 18 = 25$

$\Rightarrow x = 25 - 18 = 7$

II. $y^2 = 473 - 409 = 64$

$\Rightarrow y = \sqrt{64} = \pm 8$

Relationship can't be established.

65. (4) I. $x^2 - 7x + 12 = 0$

$\Rightarrow x^2 - 4x - 3x + 12 = 0$

$\Rightarrow x(x - 4) - 3(x - 4) = 0$

$\Rightarrow (x - 3) - (x - 4) = 0$

$\Rightarrow x = 3$ or 4

II. $y^2 - 9y + 20 = 0$

$\Rightarrow y^2 - 5y - 4y + 20 = 0$

$\Rightarrow y(y - 5) - 4(y - 5) = 0$

$\Rightarrow (y - 4)(y - 5) = 0$

$\therefore y = 4$ or 5

Clearly, $x \leq y$

66. (3) Dividing equation I by II,

$\frac{(y - x)(y + x)}{y - x} = \frac{32}{2}$

$\Rightarrow y + x = 16$ ....(i)

and $y - x = 2$ ....(ii)

Adding both equations,

$2y = 18 \Rightarrow y = 9$

From equation (i),

$x = 16 - 9 = 7$

Clearly, $x < y$

67. (5) I. $\sqrt{x} - \frac{\sqrt{5}}{\sqrt{x}} = 0$

$\Rightarrow \sqrt{x} \times \sqrt{x} - \sqrt{5} = 0$

$\Rightarrow x = \sqrt{5}$

II. $y^3 = 5^{3/2}$

$\Rightarrow y^3 = \left(\sqrt{5}\right)^3$

$\Rightarrow y = \sqrt{5}$

Clearly $x = y$

68. (1) By equation 1 × 3 + equation II × 5,

$9x + 15y = 84$

$40x - 15y = 210$

$49x = 294$

$\Rightarrow x = \frac{294}{49} = 6$

From equation I,

$3 \times 6 + 5y = 28$

$\Rightarrow 5y = 28 - 18 = 10$

$\Rightarrow y = \frac{10}{5} = 2$

Clearly, $x > y$

69. (3) I. $\sqrt{289x} = -\sqrt{25}$

Squaring both sides,

$289x = 25$

$\Rightarrow x = \frac{25}{289}$

II. $\sqrt{676y} = -10$

Squaring both sides,
$676y = 100$

$\Rightarrow y = \frac{100}{676}$

Clearly, $x < y$

70. (2) I. $8x^2 - 78x + 169 = 0$
$\Rightarrow 8x^2 - 26x - 52x + 169 = 0$
$\Rightarrow 2x(4x - 13) - 13(4x - 13) = 0$
$\Rightarrow (2x - 13)(4x - 13) = 0$

$\therefore x = \frac{13}{2}$ or $\frac{13}{4}$

II. $20y^2 - 117y + 169 = 0$
$\Rightarrow 20y^2 - 52y - 65y + 169 = 0$
$\Rightarrow 4y(5y - 13) - 13(5y - 13) = 0$
$\Rightarrow (4y - 13)(5y - 13) = 0$

$\therefore y = \frac{13}{4}$ or $-\frac{13}{5}$

Clearly, $x \geq y$

71. (1) I. $\frac{15 + 9}{\sqrt{x}} = 11\sqrt{x}$

$\Rightarrow 11\sqrt{x} \times \sqrt{x} = 24$

$\Rightarrow 11x = 24 \Rightarrow x = \frac{24}{11}$

II. $\frac{\sqrt{y}}{4} + \frac{5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

$\Rightarrow \frac{3\sqrt{y} + 5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

$\Rightarrow \frac{8\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

$\Rightarrow 8\sqrt{y} \times \sqrt{y} = 12$

$\Rightarrow y = \frac{12}{8} = \frac{3}{2}$

Clearly, $x > y$

72. (5) I. $\frac{8}{\sqrt{x}} + \frac{6}{\sqrt{x}} = \sqrt{x}$

$\Rightarrow \frac{8 + 6}{\sqrt{x}} = \sqrt{x} \Rightarrow x = 14$

II. $y^3 - \frac{(14)^{\frac{7}{2}}}{\sqrt{y}} = 0$

$\Rightarrow y^3 - \frac{(14)^{\frac{7}{2}}}{\sqrt{y}} = 0$

$\Rightarrow y^3 \times \sqrt{y} = (14)^{\frac{7}{2}}$

$\Rightarrow y^{\frac{7}{2}} = (14)^{\frac{7}{2}} \Rightarrow y = 14$

Clearly $x = y$

73. (5) I. $x^2 = 208 + 233 = 441$

$\therefore x = \sqrt{441} = \pm 21$

II. $y^2 - 47 + 371 = 0$
$\Rightarrow y^2 + 324 = 0$

$\therefore y = \sqrt{-324}$ = An imaginary number.

Relationship can't be established.

74. (2) I. $x^2 - 11x + 24 = 0$
$\Rightarrow x^2 - 8x - 3x + 24 = 0$
$\Rightarrow x(x - 8) - 3(x - 8) = 0$
$\Rightarrow (x - 3)(x - 8) = 0$
$\therefore x = 3$ or 8

II. $2y^2 - 9y + 9 = 0$
$\Rightarrow 2y^2 - 3y - 6y + 9 = 0$
$\Rightarrow y(2y - 3) - 3(2y - 3) = 0$
$\Rightarrow (2y - 3)(y - 3) = 0$

$\therefore y = \frac{3}{2}$ or 3

Clearly, $x \geq y$

75. (3) I. $x^3 \times 13 = x^2 \times 247$

$\Rightarrow \frac{x^3}{x^2} = \frac{247}{13} \Rightarrow x = 19$

II. $y^{\frac{1}{3}} \times 14 = \frac{294}{y^{\frac{2}{3}}}$

$\Rightarrow y^{\frac{1}{3}} \times y^{\frac{2}{3}} = \frac{294}{14}$

$\Rightarrow y^{\frac{1}{3} + \frac{2}{3}} = 21 \Rightarrow y = 21$

Clearly, $x < y$

76. (5) I. $\frac{48}{x^{\frac{4}{7}}} - \frac{12}{x^{\frac{4}{7}}} = x^{\frac{10}{7}}$

$\Rightarrow \frac{48 - 12}{x^{\frac{4}{7}}} = x^{\frac{10}{7}} \Rightarrow 36 = x^{\frac{10}{7} + \frac{4}{7}}$

$\Rightarrow 36 = x^2 \Rightarrow x = \sqrt{36} = \pm 6$

II. $y^3 = 999 - 783 = 216$

$\therefore y = \sqrt[3]{216} = 6$

Relationship can't be established.

77. (3) I. $\sqrt{500}\,x = -\sqrt{402}$

$\Rightarrow x = \sqrt{\frac{402}{500}} \approx -\sqrt{\frac{400}{500}} \approx -0.9$

II. $\sqrt{360}\,y = -\sqrt{200}$

$y = -\sqrt{\frac{200}{360}} = -0.74$

Clearly $x < y$

78. (3) I. $x = 17^2 + 144 \times \frac{1}{18}$

$= 289 + 8 = 297$

II. $y = 26^2 - 18 \times 21$

$= 676 - 378 = 298$

Clearly, $x < y$

79. (2) I. $16x^2 + 20x + 6 = 0$

$\Rightarrow 8x^2 + 10x + 3 = 0$

$\Rightarrow 8x^2 + 6x + 4x + 3 = 0$

$\Rightarrow 2x\,(4x + 3) + 1\,(4x + 3) = 0$

$\Rightarrow (2x + 1)\,(4x + 3) = 0$

$\therefore x = -\frac{1}{2}$ or $-\frac{3}{4}$

II. $10y^2 + 38y + 24 = 0$

$\Rightarrow 5y^2 + 19y + 12 = 0$

$\Rightarrow 5y^2 + 15y + 4y + 12 = 0$

$\Rightarrow 5y\,(y + 3) + 4\,(y + 3) = 0$

$\Rightarrow (y + 3)\,(5y + 4) = 0$ 4

$\therefore y = -3$ or $-\frac{4}{5}$

Clearly, $x > y$

80. (4) I. $18x^2 + 18x + 4 = 0$

$\Rightarrow 9x^2 + 9x + 2 = 0$

$\Rightarrow 9x^2 + 6x + 3x + 2 = 0$

$\Rightarrow 3x\,(3x + 2) + 1\,(3x + 2) = 0$

$\Rightarrow (3x + 1)\,(3x + 2) = 0$

$\therefore x = -\frac{1}{3}$ or $-\frac{2}{3}$

II. $12y^2 + 29y + 14 = 0$

$\Rightarrow 12y^2 + 21y + 8y + 14 = 0$

$\Rightarrow 3y\,(4y + 7) + 2\,(4y + 7) = 0$

$\Rightarrow (3y + 2)\,(4y + 7) = 0$

$\therefore y = -\frac{2}{3}$ or $-\frac{7}{4}$

Clearly $x \geq y$

81. (3) I. $8x^2 + 6x - 5 = 0$

$\Rightarrow 8x^2 + 10x - 4x - 5 = 0$

$\Rightarrow 2x\,(4x + 5) - 1\,(4x + 5) = 0$

$\Rightarrow (2x - 1)\,(4x + 5) = 0$

$\therefore x = \frac{1}{2}$ or $-\frac{5}{4}$

II. $12y^2 - 22y + 8 = 0$

$\Rightarrow 6y^2 - 11y + 4 = 0$

$\Rightarrow 6y^2 - 8y - 3y + 4 = 0$

$\Rightarrow 2y\,(3y - 4) - 1\,(3y - 4) = 0$

$\Rightarrow (3y - 4)\,(2y - 1) = 0$

$\Rightarrow y = \frac{4}{3}$ or $\frac{1}{2}$

Clearly, $x \leq y$

82. (1) I. $17x^2 + 48x - 9 = 0$

$\Rightarrow 17x^2 + 51x - 3x - 9 = 0$

$\Rightarrow 17x\,(x + 3) - 3\,(x + 3) = 0$

$\Rightarrow (x + 3)(17x - 3) = 0$

$\therefore x = -3$ or $\frac{3}{17}$

II. $13y^2 - 32y + 12 = 0$

$\Rightarrow 13y^2 - 26y - 6y + 12 = 0$

$\Rightarrow 13y\,(y - 2) - 6\,(y - 2) = 0$

$\Rightarrow (y - 2)(13y - 6) = 0$

$\therefore y = 2$ or $\frac{6}{13}$

Clearly, $x < y$

83. (5) By equation I × 2 + equation II,

$8x + 14y = 418$

$12x - 14y = -38$

$20x = 380$

$\therefore x = \frac{380}{20} = 19$

From equation I,

$4 \times 19 + 7y = 209$

$\Rightarrow 7y = 209 - 76 = 133$

$\therefore y = \frac{133}{7} = 19$

Clearly $x = y$

84. (2) I. $\frac{25}{x^2} + \frac{9}{x^2} - \frac{4}{x^2} = \frac{12}{x}$

$\Rightarrow \frac{25 + 9 - 4}{x^2} = \frac{12}{x}$

$\Rightarrow \frac{30}{x} = 12$

$\Rightarrow 12x = 30$

$\Rightarrow x = \frac{30}{12} = \frac{5}{2} = 2.5$

II. $9.84 - 2.64 = 0.95 + y^2$

$\Rightarrow 7.2 - 0.95 = y^2$

$\Rightarrow 6.25 = y^2$

$\Rightarrow y = \sqrt{6.25} = \pm 2.5$

Clearly $x \geq y$

85. (1) I. $\sqrt{901}\,x = -\sqrt{1295}$

$\Rightarrow \sqrt{901}\,x \approx -\sqrt{1296}$

$\Rightarrow 30x \approx -36$

$\Rightarrow x \approx \frac{-36}{30} \approx \frac{-6}{5} \approx -1.2$

II. $(256)^{\frac{1}{4}}y \approx -(216)^{\frac{1}{3}}$

$\Rightarrow 4y \approx -6$

$\Rightarrow y \approx \frac{-6}{4} \approx -1.5$

Clearly $x > y$

86. (1) I. $\frac{243+343}{3} = x^3$

$\Rightarrow \frac{586}{3} = x^3$

II. $7y^3 = -30 + 17y^3$

$\Rightarrow 10y^3 = 30$

$\Rightarrow y^3 = 3$

Clearly $x > y$

87. (3) I. $\left(\frac{x^{\frac{1}{4}}}{16}\right)^2 = \frac{144}{x^{\frac{3}{2}}}$

$\Rightarrow \frac{x^{\frac{1}{2}}}{256} = \frac{144}{x^{\frac{3}{2}}}$

$\Rightarrow x^{\frac{1}{2}} \times x^{\frac{3}{2}} = 256 \times 144$

$\Rightarrow x^2 = 256 \times 144$

$\therefore x = \sqrt{256 \times 144}$

$= \pm 16 \times 12 = \pm 192$

II. $y^{\frac{1}{3}} \times y^{\frac{2}{3}} \times 3104 = 16 \times y^2$

$\Rightarrow y \times 3104 = 16 \times y^2$

$\Rightarrow 3104 = 16y$

$\Rightarrow y = \frac{3104}{16} = 194$

Clearly $x < y$

88. (1) I. $3x^2 - 19x + 28 = 0$

$\Rightarrow 3x^2 - 12x - 7x + 28 = 0$

$\Rightarrow 3x(x - 4) - 7(x - 4) = 0$

$\Rightarrow (x - 4)(3x - 7) = 0$

$\therefore x = 4$ or $\frac{7}{3}$

II. $5y^2 - 18y + 16 = 0$

$\Rightarrow 5y^2 - 10y - 8y + 16 = 0$

$\Rightarrow 5y(y - 2) - 8(y - 2) = 0$

$\Rightarrow (5y - 8)(y - 2) = 0$

$\therefore y = \frac{8}{5}$ or 2

Clearly $x > y$

89. (1) I. $\sqrt{1225}\,x + \sqrt{4900} = 0$

$\Rightarrow 35x + 70 = 0$

$\Rightarrow 35x = -70 \Rightarrow x = \frac{-70}{35} = -2$

II. $\left(3^4\right)^{\frac{1}{4}}y + \left(7^3\right)^{\frac{1}{3}} = 0$

$\Rightarrow 3y + 7 = 0$ $\quad [(a^m)^n = a^{mn}]$

$\Rightarrow y = \frac{-7}{3}$

Clearly, $x > y$

90. (3) I. $\frac{18}{x^2} - \frac{12}{x^2} - \frac{8}{x^2} = \frac{-6}{x}$

$\Rightarrow \frac{18-12-8}{x^2} = \frac{-6}{x}$

$\Rightarrow \frac{-2}{x^2} = \frac{-6}{x} \Rightarrow \frac{2}{x} = 6 \Rightarrow 6x = 2$

$\Rightarrow x = \frac{2}{6} = \frac{1}{3} \Rightarrow x^3 = \frac{1}{27} = 0.037$

$\Rightarrow y^3 = 16.95 - 9.68 - 5.64 = 1.63$

Clearly $x < y$

91. (1) I. $x^3 = \frac{32+1331}{6} = \frac{1363}{6} \approx 227$

II. $4y^3 - 5y^3 = -\frac{589}{4}$

$\Rightarrow -y^3 = -\frac{589}{4}$

$\Rightarrow y^3 = \frac{589}{4} = 147$

Clearly $x > y$

92. (2) I. $12x^2 + 11x + 12 - 10x^2 - 22x = 0$

$\Rightarrow 2x^2 - 11x + 12 = 0$

$\Rightarrow 2x^2 - 8x - 3x + 12 = 0$

$\Rightarrow 2x(x - 4) - 3(x - 4) = 0$

$\Rightarrow (x - 4)(2x - 3) = 0$

$\Rightarrow x = 4 \text{ or } \frac{3}{2}$

II. $13y^2 - 18y + 3 - 9y^2 + 10y = 0$
$\Rightarrow 4y^2 - 8y + 3 = 0$
$\Rightarrow 4y^2 - 6y - 2y + 3 = 0$
$\Rightarrow 2y(2y - 3) - 1(2y - 3) = 0$
$\Rightarrow (2y - 1)(2y - 3) = 0$

$\Rightarrow y = \frac{1}{2} \text{ or } \frac{3}{2}$

Clearly $x \geq y$

93. (5) I. $\frac{x^{\frac{7}{5}}}{9} = \frac{169}{x^{\frac{3}{5}}}$

$\Rightarrow x^{\frac{7}{5}} \times x^{\frac{3}{5}} = 9 \times 169$

$\Rightarrow x^{\frac{7}{5}+\frac{3}{5}} = 9 \times 169$
$\Rightarrow x^2 = 9 \times 169$
$\Rightarrow x = 3 \times 13 = 39$

II. $y^{\frac{1}{4}} \times y^{\frac{1}{4}} \times 7 = \frac{273}{y^{\frac{1}{2}}}$

$\Rightarrow y^{\frac{1}{4}+\frac{1}{4}+\frac{1}{2}} = \frac{273}{7}$

$\Rightarrow y = 39$

Clearly $x = y$

94. (1) I. $\sqrt{25x^2} - 125 = 0$

$\Rightarrow \sqrt{25x^2} = 125$
$\Rightarrow 25x^2 = 125 \times 125$

$\Rightarrow x^2 = \frac{125 \times 125}{25} = 625$

$\therefore x = \sqrt{625} = 25$

II. $\sqrt{361}\, y + 95 = 0$
$\Rightarrow 19y = -95$
$\Rightarrow y = -5$

Clearly $x > y$

95. (3) I. $\frac{5}{7} - \frac{5}{21} = \frac{\sqrt{x}}{42}$

$\Rightarrow \frac{15-5}{21} = \frac{\sqrt{x}}{42}$

$= \sqrt{x} = \frac{10}{21} \times 42 = 20$

$\therefore x = 20 \times 20 = 400$

II. $\frac{\sqrt{y}}{4} + \frac{\sqrt{y}}{16} = \frac{250}{\sqrt{y}}$

$\Rightarrow \frac{4\sqrt{y} + \sqrt{y}}{16} = \frac{250}{\sqrt{y}}$

$\Rightarrow 5\sqrt{y} \times \sqrt{y} = 250 \times 16$
$\Rightarrow 5y = 250 \times 16$

$\Rightarrow y = \frac{250 \times 16}{5} = 800$

clearly $x < y$

96. (1) I. $(625)^{\frac{1}{4}} x + \sqrt{1225} = 155$

$\Rightarrow \left(5^4\right)^{\frac{1}{4}} x + 35 = 155$
$\Rightarrow 5x = 155 - 35$
$\Rightarrow 5x = 120$

$\Rightarrow x = \frac{120}{5} = 24$

II. $\sqrt{196}\, y + 13 = 279$
$\Rightarrow 14y = 279 - 13 = 266$

$\Rightarrow y = \frac{266}{14} = 19$

Clearly $x > y$

97. (1) I. $5x^2 - 18x + 9 = 0$
$\Rightarrow 5x^2 - 15x - 3x + 9 = 0$
$\Rightarrow 5x(x - 3) - 3(x - 3) = 0$
$\Rightarrow (5x - 3)(x - 3) = 0$

$\Rightarrow x = \frac{3}{5}$ or, 3

II. $3y^2 + 5y - 2 = 0$
$\Rightarrow 3y^2 + 6y - y - 2 = 0$
$\Rightarrow 3y(y + 2) - 1(y + 2) = 0$
$\Rightarrow (3y - 1)(y + 2) = 0$ 1

$\Rightarrow y = \frac{1}{3}$ or, -2

Clearly $x > y$

98. (3) I. $\frac{13}{\sqrt{x}} + \frac{9}{\sqrt{x}} = \sqrt{x}$

$\Rightarrow 13 + 9 \Rightarrow \sqrt{x} \times \sqrt{x} = x$
$\Rightarrow x = 22$

II. $y^4 = \frac{(26)^{\frac{9}{2}}}{\sqrt{y}}$

$\Rightarrow y^4 \times y^{\frac{1}{2}} = (26)^{\frac{9}{2}}$

$\Rightarrow (y)^{\frac{9}{2}} = (26)^{\frac{9}{2}}$

$\Rightarrow y = 26$

Clearly $x < y$

99. (4) I. $x^2 - 19x + 84 = 0$

$\Rightarrow x^2 - 12x - 7x + 84 = 0$

$\Rightarrow x(x - 12) - 7(x - 12) = 0$

$\Rightarrow (x - 7)(x - 12) = 0$

$\therefore x = 7$ or 12

II. $y^2 - 25y + 156 = 0$

$\Rightarrow y^2 - 12y - 13y + 156 = 0$

$\Rightarrow y(y - 12) - 13(y - 12) = 0$

$\Rightarrow (y - 13)(y - 12) = 0$

$\therefore y = 13$ or 12

Clearly $x \leq y$

100. (2) I. $x^3 = 1729 + 468 = 2197$

$\therefore x = \sqrt[3]{2197} = 13$

II. $y^2 = 1733 - 1564 = 169$

$\therefore y = \sqrt{169} = \pm 13$

Clearly $x \geq y$

101. (5) I. $9 + 19 = \sqrt{x} \times \sqrt{x}$

$\Rightarrow x = 28$

II. $y^5 \times y^{\frac{1}{2}} = (28)^{\frac{11}{2}}$

$\Rightarrow y^{\frac{11}{2}} = (28)^{\frac{11}{2}}$

$\Rightarrow y = 28$

Clearly $x = y$

102. (1) I. $\sqrt{784}\, x = 1486 - 1234 = 252$

$\Rightarrow x = \frac{252}{28} = 9$

$\Rightarrow y = \frac{264}{\sqrt{1089}} = \frac{264}{33} = 8$

103. (1) I. $12 - 23 = 5\sqrt{x} \times \sqrt{x}$

$\Rightarrow -11 = 5x$

$\Rightarrow x = \frac{-11}{5}$

II. $\frac{\sqrt{y} - 5\sqrt{y}}{12} = \frac{1}{\sqrt{y}}$

$\Rightarrow -4\sqrt{y} \times \sqrt{y} = 12$

$\Rightarrow -4y = 12$

$\Rightarrow y = -3$

Clearly $x > y$

104. (5) I. $\frac{15}{\sqrt{x}} - \frac{9}{\sqrt{x}} = \sqrt{x}$

$\Rightarrow \frac{6}{\sqrt{x}} = \sqrt{x} \Rightarrow x = 6$

II. $y^{10} = (36)^5 = (6^2)^5 = 6^{10}$

$\Rightarrow y = 6$

Clearly $x = y$

105. (1) I. $5x + 2y = 96$

II. $3(7x + 5y) = 489$

$\Rightarrow 7x + 5y = 489 \div 3 = 163$

By equation I × 5 - equation II × 2,

$25x + 10y = 480$

$\underline{14x + 10y = 326}$

$11x = 154$

$\Rightarrow x = \frac{154}{11}$

$\Rightarrow x = 14$

From equation I,

$14 \times 5 + 2y = 96$

$\Rightarrow 2y = 96 - 70 = 26$

$\Rightarrow y = 13$

Clearly, $x > y$

106. (1) I. $(441)^{\frac{1}{2}} \cdot x^2 - 111 = 225$

$\Rightarrow 21x^2 = 225 + 111 = 336$

$\Rightarrow x^2 = \frac{336}{21} = 16$

$\Rightarrow x = 4$

II. $\sqrt{121}\, y^2 + (6)^3 = 260$

$\Rightarrow 11y^2 + 216 = 260$

$\Rightarrow 11y^2 = 260 - 216 = 44$

$\Rightarrow y^2 = \frac{44}{11} = 4$

$\Rightarrow y = 2$

Clearly $x > y$

107. (3) I. $17x = 169 + 14 + 25 + 4x$

$\Rightarrow 13x = 208$

$\Rightarrow x = \frac{208}{13} = 16$

II. $9y - 345 = 4y - 260$

$\Rightarrow 9y - 4y = 345 - 260$

$\Rightarrow 5y = 85$

$\Rightarrow y = 17$

Clearly, $x < y$

108. (3) I. $3x^2 - 13x + 14 = 0$

$\Rightarrow 3x^2 - 7x - 6x + 14 = 0$

$\Rightarrow x(3x - 7) - 2(3x - 7) = 0$

$\Rightarrow (x - 2)(3x - 7) = 0$

$x = 2$ or $\frac{7}{3}$

II. $y^2 - 7y + 12 = 0$
$\Rightarrow y^2 - 4y - 3y + 12 = 0$
$\Rightarrow y(y - 4) - 3(y - 4) = 0$
$\Rightarrow (y - 3)(y - 4) = 0$
$\Rightarrow y = 3$ or 4
Clearly $x < y$

109. (1) I. $x^2 + 5x + 6 = 0$
$\Rightarrow x^2 + 2x + 3x + 6 = 0$
$\Rightarrow x(x + 2) + 3(x + 2) = 0$
$\Rightarrow (x + 3)(x + 2) = 0$
$\Rightarrow x = -3$ or -2
II. $y^2 + 7y + 12 = 0$
$\Rightarrow y^2 + 4y + 3y + 12 = 0$
$\Rightarrow y(y + 4) + 3(y + 4) = 0$
$\Rightarrow (y + 3)(y + 4) = 0$
$\Rightarrow y = -3$ or -4
Clearly $x \geq y$

110. (4) I. $x^2 - 9x + 20 = 0$
$\Rightarrow x^2 - 5x - 4x + 20 = 0$
$\Rightarrow x(x - 5) - 4(x - 5) = 0$
$\Rightarrow (x - 4)(x - 5) = 0$
$\Rightarrow x = 4$ or 5
II. $y^2 - 13y + 42 = 0$
$\Rightarrow y^2 - 7y - 6y + 42 = 0$
$\Rightarrow y(y - 7) - 6(y - 7) = 0$
$\Rightarrow (y - 6)(y - 7) = 0$
$\Rightarrow y = 6$ or 7
Clearly $x < y$

111. (4) $2x + 3y = 14$ I
$4x + 2y = 16$ II
By equation (I) × 2 - equation II,
$4x + 6y - 4x - 2y = 28 - 16$
$\Rightarrow 4y = 12 \Rightarrow y = 3$
From equation I,
$2x + 3 \times 3 = 14$

$$\Rightarrow 2x = 14 - 9 = 5 \Rightarrow x = \frac{5}{2}$$

Clearly $x < y$

112. (5) I. $x = \sqrt{625} = \pm 25$

II. $y = \sqrt{676} = \pm 26$

113. (4) I. $x^2 + 4x + 4 = 0$
$(x + 2)^2 = 0 \Rightarrow x = -2$
II. $y^2 - 8y + 16 = 0$
$\Rightarrow (y - 4)^2 = 0$
$\Rightarrow y = 4$
Clearly $x < y$

114. (1) I. $x^2 - 24x + 144 = 0$
$\Rightarrow x^2 - 12x - 12x + 144 = 0$
$\Rightarrow x(x - 12) - 12(x - 12) = 0$
$\Rightarrow (x - 12)(x - 12) = 0$
$\Rightarrow x = 12$
II. $y^2 - 26y + 169 = 0$
$\Rightarrow (y - 13)^2 = 0$
$\Rightarrow y - 13 = 0$
$\Rightarrow y = 13$
Clearly, $x < y$

115. (4) I. $2x^2 + 3x - 20 = 0$
$\Rightarrow 2x^2 + 8x - 5x - 20 = 0$
$\Rightarrow 2x(x + 4) - 5(x + 4) = 0$
$\Rightarrow (x + 4)(2x - 5) = 0$

$$\Rightarrow x = -4 \text{ or } \frac{5}{2}$$

II. $2y^2 + 19y + 44 = 0$
$\Rightarrow 2y^2 + 11y + 8y + 44 = 0$
$\Rightarrow y(2y + 11) + 4(2y + 11) = 0$
$\Rightarrow (y + 4)(2y + 11) = 0$

$$\Rightarrow y = -4 \text{ or } -\frac{11}{2}$$

Clearly, $x \geq y$

116. (5) I. $6x^2 + 77x + 121 = 0$
$\Rightarrow 6x^2 + 66x + 11x + 121 = 0$
$\Rightarrow 6x(x + 11) + 11(x + 11) = 0$
$\Rightarrow (x + 11)(6x + 11) = 0$

$$\Rightarrow x = -11 \text{ or } -\frac{11}{6}$$

II. $y^2 + 11y - 2y - 22 = 0$
$\Rightarrow y(y + 11) - 2(y + 11) = 0$
$\Rightarrow (y - 2)(y + 11) = 0$.
$\Rightarrow y = 2$ or -11
Clearly $x \leq y$

117. (2) I. $x^2 - 6x - 7 = 0$
$\Rightarrow x^2 - 7x + x - 7 = 0$
$\Rightarrow x(x - 7) + 1(x - 7) = 0$
$\Rightarrow (x + 1)(x - 7) = 0$
$\Rightarrow x = -1$ or 7
II. $2y^2 + 13y + 15 = 0$
$\Rightarrow 2y^2 + 10y + 3y + 15 = 0$
$\Rightarrow 2y(y + 5) + 3(y + 5) = 0$
$\Rightarrow (y + 5)(2y + 3) = 0$

$$\Rightarrow y = -5 \text{ or } -\frac{3}{2}$$

Clearly $x > y$

118. (4) I. $10x^2 - 7x + 1 = 0$
$\Rightarrow 10x^2 - 5x - 2x + 1 = 0$
$\Rightarrow 5x(2x - 1) - 1(2x - 1) = 0$
$\Rightarrow (5x - 1)(2x - 1) = 0$

$$\Rightarrow x = \frac{1}{5} \text{ or } \frac{1}{2}$$

II. $35y^2 - 12y + 1 = 0$
$\Rightarrow 35y^2 - 7y - 5y + 1 = 0$
$\Rightarrow 7y(5y - 1) - 1(5y - 1) = 0$
$\Rightarrow (7y - 1)(5y - 1) = 0$

$\Rightarrow x = \frac{1}{7}$ or $\frac{1}{5}$

Clearly $x \geq y$

119. (2) I. $4x^2 - 32x + 63 = 0$
$\Rightarrow 4x^2 - 14x - 18x + 63 = 0$
$\Rightarrow 2x(2x - 7) - 9(2x - 7) = 0$
$\Rightarrow (2x - 7)(2x - 9) = 0$

$\Rightarrow x = \frac{7}{2}$ or $\frac{9}{2}$

II. $2y^2 - 11y + 15 = 0$
$\Rightarrow 2y^2 - 6y - 5y + 15 = 0$
$\Rightarrow 2y(y - 3) - 5(y - 3) = 0$
$\Rightarrow (y - 3)(2y - 5) = 0$

$\Rightarrow y = 3$ or $\frac{5}{2}$

Clearly, $x > y$

120. (2) I. $x^3 = (216)^{\frac{1}{3}\times 3} = 216$

$\Rightarrow x = \sqrt[3]{216} = 6$

II. $6y^2 = 150$

$\Rightarrow y^2 = \frac{150}{6} = 25$

$\Rightarrow y = \sqrt{25} = \pm 5$

Clearly, $x > y$

121. (1) I. $12x^2 + 17x + 6 = 0$
$\Rightarrow 12x^2 + 9x + 8x + 6 = 0$
$\Rightarrow 3x(4x + 3) + 2(4x + 3) = 0$
$\Rightarrow (4x + 3)(3x + 2) = 0$

$\Rightarrow x = -\frac{3}{4}$ or $-\frac{2}{3}$

II. $6y^2 + 5y + 1 = 0$
$\Rightarrow 6y^2 + 2y + 3y + 1 = 0$
$\Rightarrow 2y(3y + 1) + 1(3y + 1) = 0$
$\Rightarrow (3y + 1)(2y + 1) = 0$

$\Rightarrow y = -\frac{1}{3}$ or $-\frac{1}{2}$

Clearly, $x < y$

122. (3) I. $20x^2 + 9x + 1 = 0$
$\Rightarrow 20x^2 + 5x + 4x + 1 = 0$
$\Rightarrow 5x(4x + 1) + 1(4x + 1) = 0$
$\Rightarrow (4x + 1)(5x + 1) = 0$

$\Rightarrow x = -\frac{1}{4}$ or $-\frac{1}{5}$

II. $30y^2 + 11y + 1 = 0$
$\Rightarrow 30y^2 + 6y + 5y + 1 = 0$
$\Rightarrow 6y(5y + 1) + 1(5y + 1) = 0$
$\Rightarrow (5y + 1)(6y + 1) = 0$

$\Rightarrow y = -\frac{1}{5}$ or $-\frac{1}{6}$

Clearly, $x \leq y$

123. (4) I. $x^2 + 17x + 72 = 0$
$\Rightarrow x^2 + 8x + 9x + 72 = 0$
$\Rightarrow x(x + 8) + 9(x + 8) = 0$
$\Rightarrow (x + 9)(x + 8) = 0$
$\Rightarrow x = -9$ or -8
II. $y^2 + 19y + 90 = 0$
$\Rightarrow y^2 + 10y + 9y + 90 = 0$
$\Rightarrow y(y + 10) + 9(y + 10) = 0$
$\Rightarrow (y + 9)(y + 10) = 0$
$\Rightarrow y = -9$ or -10
Clearly, $x > y$

SBI PO EXAMS

1. (3) I. $4q^2 + 8q = 4q + 8$
$4q^2 + 4q - 8 = 0$
$q^2 + q - 2 = 0$
$q^2 + 2q - q - 2 = 0$
$q(q + 2) - 1(q + 2) = 0$
$(q + 2)(q - 1) = 0$
$q = -2, 1$
II. $p^2 + 9p = 2p - 12$
$p^2 + 7p + 12 = 0$
$p^2 + 3p + 4p + 12 = 0$
$p(p + 3) + 4(p + 3) = 0$
$(p + 3)(p + 4) = 0$
$p = -3, -4$
$\therefore q > p$

2. (3) I. $2p^2 + 40 = 18p$
$p^2 + 20 = 9p$
$p^2 - 9p + 20 = 0$
$p^2 - 4p - 5p + 20 = 0$
$p(p - 4) - 5(p - 4) = 0$
$(p - 4)(p - 5) = 0$
$p = 4, 5.$
II. $q^2 = 13q - 42$
$q^2 - 13q + 42 = 0$
$q^2 - 6q - 7q + 42 = 0$
$q(q - 6) - 7(q - 6) = 0$
$(q - 6)(q - 7) = 0$
$q = 6, 7$
$q > p$

3. (4) I. $6q^2 + \frac{1}{2} = \frac{7}{2}q$

$\frac{12q^2 + 1}{2} = \frac{7}{2}q$

$12q^2 + 1 = 7q$
$12q^2 - 7q + 1 = 0$
$12q^2 - 3q - 4q + 1 = 0$
$3q\,(4q - 1) - 1\,(4q - 1) = 0$
$(4q - 1)\,(3q - 1) = 0$

$q = \frac{1}{4}, \frac{1}{3}$

II. $12p^2 + 2 = 10p$
$6p^2 + 1 = 5p$
$6p^2 - 5p + 1 = 0$
$6p^2 - 3p - 2p + 1 = 0$
$3p\,(2p - 1) - (2p - 1) = 0$
$(2p - 1)\,(3p - 1) = 0$

$p = \frac{1}{2}, \frac{1}{3}$

$\therefore p \geq q$

4. (5) Equation I
$2x^2 - 7x + 6 = 0$
$\Rightarrow 2x^2 - 4x - 3x + 6 = 0$
$\Rightarrow (x - 2)\,(2x - 3) = 0$

$\Rightarrow x = 2$ or $x = \frac{3}{2}$

Equation II. $4y^2 = 9$

$\Rightarrow y^2 = \sqrt{\frac{9}{4}} = \pm\frac{3}{2}$

Obviously, $y \leq x$

5. (5) Equation I
$4x^2 - 4x - 3 = 0$
$\Rightarrow 4x^2 - 6x + 2x - 3 = 0$
$\Rightarrow 2x\,(2x - 3) + 1\,(2x - 3) = 0$
$\Rightarrow (2x - 3)\,(2x + 1) = 0$

$\Rightarrow x = \frac{3}{2}$ or, $x = -\frac{1}{2}$

Equation II. $4y^2 + 12y + 5 = 0$
$\Rightarrow 4y^2 + 10y + 2y + 5 = 0$
$\Rightarrow 2y\,(2y + 5) + 1\,(2y + 5) = 0$
$\Rightarrow (2y + 5)\,(2y + 1) = 0$

$\Rightarrow y = -\frac{5}{2}$ or, $y = -\frac{1}{2}$

Obviously, $x \geq y$

6. (1) Equation I
$4x^2 = 49$

$\Rightarrow x^2 = \frac{49}{4}$

$\Rightarrow x = \pm\frac{7}{2}$

Equation II
$9y^2 - 66y + 121$
$\Rightarrow 9y^2 - 33y - 33y + 121 = 0$
$\Rightarrow 3y\,(3y - 11) - 11\,(3y - 11) = 0$
$\Rightarrow (3y - 11)\,(3y - 11) = 0$

$\Rightarrow y = \frac{11}{3}$

Obviously, $x < y$

7. (2) Equation I
$x^2 + 9x + 14 = 0$
$\Rightarrow x^2 + 2x + 7x + 14 = 0$
$\Rightarrow x\,(x + 2) + 7\,(x + 2) = 0$
$\Rightarrow (x + 2)\,(x + 7) = 0$
$\Rightarrow x = -2$ or, $x = -7$
Equation II
$y^2 + y - 2 = 0$
$\Rightarrow y^2 + 2y - y - 2 = 0$
$\Rightarrow y\,(y + 2) - 1\,(y + 2) = 0$
$\Rightarrow (y - 1)(y + 2) = 0$
$\Rightarrow y = 1$ or, $y = -2$
Clearly, $x \leq y$

8. (1) Equation I
$9x^2 - 18x + 5 = 0$
$\Rightarrow 9x^2 - 15x - 3x + 5 = 0$
$\Rightarrow 3x\,(3x - 5) - (3x - 5) = 0$
$\Rightarrow (3x - 5)\,(3x - 1) = 0$

$\Rightarrow x = \frac{5}{3}$ or, $x = \frac{1}{3}$

Equation II
$2y^2 - 9y + 10 = 0$
$\Rightarrow 2y^2 - 5y - 4y + 10 = 0$
$\Rightarrow y\,(2y - 5) - 2\,(2y - 5) = 0$
$\Rightarrow (2y - 5)\,(y - 2) = 0$

$\Rightarrow y = \frac{5}{2}$ or $y = 2$

Clearly, $x < y$

9. (2) I. $6p^2 + 3p + 2p + 1 = 0$
or, $3p\,(2p + 1) + 1\,(2p + 1) = 0$
or, $(2p + 1)\,(3p + 1) = 0$
Now, $2p + 1 = 0$

or, $p = -\frac{1}{2}$

or, $3p + 1 = 0$

or, $p = -\frac{1}{3}$

$\therefore p = -\frac{1}{2}$ or $-\frac{1}{3}$

II. $20q^2 + 9q + 1 = 0$
or, $20q + 5q + 4q + 1 = 0$
or, $5q\,(4q + 1) + 1\,(4q + 1) = 0$
or, $(4q + 1)\,(5q + 1) = 0$

or, $q = -\frac{1}{4}$ or $-\frac{1}{5}$

Obviously, $p < q$

10. (1) I. $3p^2 + 3p - p - 1 = 0$
or, $3p(p + 1) - 1(p + 1) = 0$
or, $(p + 1)(3p - 1) = 0$

or, $p = -1$ or $\frac{1}{3}$

II. $2q^2 + 4q + 3q + 6 = 0$
or, $2q(q + 2) + 3(q + 2) = 0$
or, $(q + 2)(2q + 3) = 0$

or, $q = -2$ or $-\frac{3}{2}$

Obviously, $p > q$.

11. (4) I. $3p^2 + 15p + 18 = 0$
or, $3p^2 + 9p + 6p + 18 = 0$
or, $3p(p + 3) + 6(p + 3) = 0$
or, $(p + 3)(3p + 6) = 0$

or, $p = -3$ or $-\frac{6}{3} = -2$

II. $q^2 + 4q + 3q + 12 = 0$
or, $q(q + 4) + 3(q + 4) = 0$
or, $(q + 3)(q + 4) = 0$
or, $q = -3$ or -4
Obviously $p \geq q$

12. (3) I. $p = \frac{\sqrt{4}}{\sqrt{9}} = \frac{2}{3}$

II. $(3q - 2)^2 = 0$
or, $(3q - 2) = 0$

$\Rightarrow q = \frac{2}{3}$

Obviously, $p = q$

13. (5) $p^2 + 7p + 6p + 42 = 0$
or, $p(p + 7) + 6(p + 7) = 0$
or, $(p + 6)(p + 7) = 0$
or, $p = -6$ or -7
II. $q^2 = 36$
or, $q = 6$ or -6
Obviously, $p \leq q$

14. (5) I. $a^2 + 5a + 6 = 0$
$\Rightarrow a^2 + 2a + 3a + 6 = 0$
$\Rightarrow a(a + 2) + 3(a + 2) = 0$
$\Rightarrow (a + 2)(a + 3) = 0$
$\Rightarrow a = -2$ or -3
II. $b^2 + 3b + 2 = 0$
$\Rightarrow b^2 + 2b + b + 2 = 0$
$\Rightarrow b(b + 2) + 1(b + 2) = 0$
$\Rightarrow (b + 1)(b + 2) = 0$
$\Rightarrow b = -1$ or -2
Clearly, $a \leq b$

15. (1) I. $2a^2 + 3a + 1 = 0$
$\Rightarrow 2a^2 + 2a + a + 1 = 0$
$\Rightarrow 2a(a + 1) + 1(a + 1) = 0$
$\Rightarrow (a + 1)(2a + 1) = 0$

$\Rightarrow a = -1$ or $-\frac{1}{2}$

II. $12b^2 + 7b + 1 = 0$
$\Rightarrow 12b^2 + 3b + 4b + 1 = 0$
$\Rightarrow 3b(4b + 1) + 1(4b + 1) = 0$
$\Rightarrow (3b + 1)(4b + 1) = 0$

$\Rightarrow b = -\frac{1}{3}$ or $-\frac{1}{4}$

Clearly, $a < b$

16. (3) I. $a^2 = 4 \Rightarrow a = \pm 2$
II. $b^2 = 9 \Rightarrow b = +3$

17. (2) I. $6a^2 - 25a + 25 = 0$
$\Rightarrow 6a^2 - 15a - 10a + 25 = 0$
$\Rightarrow 3a(2a - 5) - 5(2a - 5) = 0$
$\Rightarrow (3a - 5)(2a - 5) = 0$

$\Rightarrow a = \frac{5}{3}$ or $\frac{5}{2}$

II. $15b^2 - 16b + 4 = 0$
$\Rightarrow 15b^2 - 10b - 6b + 4 = 0$
$\Rightarrow 5b(3b - 2) - 2(3b - 2) = 0$
$\Rightarrow (5b - 2)(3b - 2) = 0$

$\Rightarrow b = \frac{2}{5}$ or $\frac{2}{3}$

Clearly, $a > b$

18. (4) I. $4a^2 - 20a + 21 = 0$
$\Rightarrow 4a^2 - 14a - 6a + 21 = 0$
$\Rightarrow 2a(2a - 7) - 3(2a - 7) = 0$
$\Rightarrow (2a - 7)(2a - 3) = 0$

$\Rightarrow a = \frac{7}{2}$ or $\frac{3}{2}$

II. $2b^2 - 5b + 3 = 0$
$\Rightarrow 2b^2 - 3b - 2b + 3 = 0$
$\Rightarrow b(2b - 3) - 1(2b - 3) = 0$
$\Rightarrow (b - 1)(2b - 3) = 0$

$\Rightarrow b = 1$ or $\frac{3}{2}$

Clearly $a \geq b$

19. (2) I. $p^2 + 24 = 10p$
$p^2 - 10p + 24 = 0$
$p^2 - 6p - 4p + 24 = 0$
$p(p - 6) - 4(p - 6) = 0$
$(p - 6)(p - 4) = 0$
$p = 4, 6$
II. $2q^2 + 18 = 12q$

$q^2 + 9 = 6q$
$q^2 - 6q + 9 = 0$
$(q - 3)^2 = 0$
$q = 3$
Thus, $p > q$

20. (3) I. $pq + 30 = 6p + 5q$
$6p + 5q - pq = 30$

$$\frac{p}{5} + \frac{q}{6} - \frac{pq}{30} = 1$$

$p = 5$ and $q = 6$
$\therefore q > p$

21. (5) I. $q^2 + q = 2$
$q^2 + q - 2 = 0$
$q^2 - q + 2q - 2 = 0$
$q(q - 1) + 2(q - 1) = 0$
$(q - 1)(q + 2) = 0$
$q = -2, 1$
II. $p^2 + 7p + 10 = 0$
$p^2 + 5p + 2p + 10 = 0$
$p(p + 5) + 2(p + 5) = 0$
$(p + 5)(p + 2) = 0$
$p = -5, -2$
Thus, $q \geq p$

22. (1) I. $p^2 + 16 = 8p$
$p^2 - 8p + 16 = 0$
$(p - 4)^2 = 0$
$p = 4$
II. $4q^2 + 64 = 32q$
$q^2 + 16 = 8q$
$q^2 - 8q + 16 = 0$
$(q - 4)^2 = 0$
$\Rightarrow q = 4$
Thus, $p = q$

23. (4) I. $2p^2 + 12p + 16 = 0$
$p^2 + 6p + 8 = 0$
$p^2 + 2p + 4p + 8 = 0$
$p(p + 2) + 4(p + 2) = 0$
$(p + 2)(p + 4) = 0$
$p = -2, -4$
II. $2q^2 + 14q + 24 = 0$
$q^2 + 7q + 12 = 0$
$q^2 + 3q + 4q + 12 = 0$
$q(q + 3) + 4(q + 3) = 0$
$(q + 3)(q + 4) = 0$
$q = -3, -4$
Thus, $p \geq q$

24. (2) (I) $p^2 - 7p = -12$
$\Rightarrow p^2 - 7p + 12 = 0$
$\Rightarrow p^2 - 4p - 3p + 12 = 0$
$\Rightarrow p(p - 4) - 3(p - 4) = 0$
$\Rightarrow (p - 4)(p - 3) = 0$
$\Rightarrow p = 3$ or 4
(II) $q^2 - 3q + 2 = 0$
$\Rightarrow q^2 - 2q - q + 2 = 0$
$\Rightarrow q(q - 2) - 1(q - 2) = 0$
$\Rightarrow (q - 2)(q - 1) = 0$
$\Rightarrow q = 1$ or 2
Obviously $p > q$

25. (1) (I) $12p^2 - 7p = -1$
$\Rightarrow 12p^2 - 7p + 1 = 0$
$\Rightarrow 12p^2 - 4p - 3p + 1 = 0$
$\Rightarrow 4p(3p - 1) - 1(3p - 1) = 0$
$\Rightarrow (3p - 1)(4p - 1) = 0$

$$\Rightarrow p = \frac{1}{4} \text{ or } \frac{1}{3}$$

(II) $6q^2 - 7q + 2 = 0$
$\Rightarrow 6q^2 - 4q - 3q + 2 = 0$
$\Rightarrow 2q(3q - 2) - 1(3q - 2) = 0$
$\Rightarrow (3q - 2)(2q - 1) = 0$

$$\Rightarrow q = \frac{2}{3} \text{ or } \frac{1}{2}$$

Obviously, $p < q$

26. (3) (I) $p^2 + 12p + 35 = 0$
$\Rightarrow p^2 + 7p + 5p + 35 = 0$
$\Rightarrow p(p + 7) + 5(p + 7) = 0$
$\Rightarrow (p + 7)(p + 5) = 0$
$\Rightarrow p = -5$ or -7
(II) $2q^2 + 22q + 56 = 0$
$\Rightarrow 2q^2 + 14q + 8q + 56 = 0$
$\Rightarrow 2q(q + 7) + 8(q + 7) = 0$
$\Rightarrow (q + 7)(2q + 8) = 0$
$\Rightarrow q = -7$ or -4
Obviously, $p \leq q$

27. (4) (I) $p^2 - 8p + 15 = 0$
$\Rightarrow p^2 - 3p - 5p + 15 = 0$
$\Rightarrow p(p - 3) - 5(p - 3) = 0$
$\Rightarrow (p - 3)(p - 5) = 0$
$\Rightarrow p = 3$ or 5
(II) $q^2 - 5q = -6$
$\Rightarrow q^2 - 5q + 6 = 0$
$\Rightarrow q^2 - 3q - 2q + 6 = 0$
$\Rightarrow q(q - 3) - 2(q - 3) - 0$
$\Rightarrow (q - 3)(q - 2) = 0$
$\Rightarrow q = 3$ or 2
Obviously, $p \geq q$

28. (3) (I) $2p^2 + 20p + 50 = 0$
$\Rightarrow p^2 + 10p + 25 = 0$
$\Rightarrow (p + 5)^2 = 0$
$\Rightarrow p + 5 = 0$
$\Rightarrow p = -5$
(II) $q^2 = 25$
$\Rightarrow q = \pm 5$
Obviously, $p \leq q$

RBI GRADE-B OFFICER EXAMS

1. (1) $2x^2 + x - 3 < 0$
$\Rightarrow 2x^2 - 2x + 2x - 3 < 0$
$\Rightarrow 2x(x - 1) + 3(x - 1) < 0$
$\Rightarrow (2x + 3)(x - 1) < 0$
$-\frac{3}{2} < x < 1$

2. (1) $\Rightarrow x^2 + 2x + x + 2 = 0$
$\Rightarrow x(x + 2) + 1(x + 2) = 0$
$\Rightarrow (x + 1)(x + 2) = 0$
$\therefore x = -1$ or -2
II. $\Rightarrow 2y^2 - 5y = 0$
$\Rightarrow y(2y - 5) = 0$
$\Rightarrow y = 0$ or $\frac{5}{2}$
Obviously, $x < y$

3. (3) I. $2x^2 + 5x + 2 = 0$
$\Rightarrow 2x^2 + 4x + x + 2 = 0$
$\Rightarrow 2x(x + 2)(x + 2) = 0$
$\Rightarrow x = -\frac{1}{2}$ or, -2
II. $4y^2 = 1$
$\Rightarrow y = \sqrt{\frac{1}{4}} = \pm\frac{1}{2}$
Clearly, $x \leq y$

4. (2) I. $\Rightarrow y^2 + 3y - y - 3 = 0$
$\Rightarrow y(y + 3) - 1(y + 3) = 0$
$\Rightarrow (y + 3)(y - 1) = 0.$
$\therefore y = -3$ or 1
II. $\Rightarrow 2x^2 - 4x - 3x + 6 = 0$
$\Rightarrow 2x(x - 2) - 3(x - 2) = 0$
$\Rightarrow (2x - 3)(x - 2)$
$\therefore x = \frac{3}{2}$ or 2
Obviously, $x > y$

5. (1) I. $\Rightarrow x^2 + 4x - 2x - 8 = 0$
$\Rightarrow x(x + 4) - 2(x + 4) = 0$
$\Rightarrow (x - 2)(x + 4) = 0$
$\therefore x = 2$ or -4
II. $\Rightarrow y^2 - 9 = 0$
$\Rightarrow (y + 3)(y - 3) = 0$
$\therefore y = 3$ or -3
Obviously, $x < y$.

6. (4) I. $\Rightarrow x^2 - 3x - 2x + 6 = 0$
$\Rightarrow x(x - 3) - 2(x - 3) = 0$
$\Rightarrow (x - 2)(x - 3) = 0$
$\therefore x = 2$ or 3
II. $\Rightarrow y^2 + 3y - 2y - 6 = 0$
$\Rightarrow y(y + 3) - 2(y + 3) = 0$
$\Rightarrow (y - 2)(y + 3) = 0$
$\therefore y = 2$ or -3.
Obviously, $x > y$

7. (5) I. $\Rightarrow p^2 + 3p + 2p + 6 = 0$
$\Rightarrow p(p + 3) + 2(p + 3) = 0$
$\Rightarrow (p + 3)(p + 2) = 0$
$\Rightarrow p = -2$ or -3
II. $\Rightarrow q^2 + q + 2q + 2 = 0$
$\Rightarrow q(q + 1) + 2(q + 1) = 0$
$\Rightarrow (q + 1)(q + 2) = 0$
$\Rightarrow q = -1$ or -2
Obviously $p \leq q$

8. (4) I. $\Rightarrow p = \pm 2$
II. $\Rightarrow q^2 + 2q + 2q + 4 = 0$
$\Rightarrow q(q + 2) + 2(q + 2) = 0$
$\Rightarrow (q + 2)(q + 2) = 0$
$\Rightarrow q = -2$
Obviously, $p \geq q$

9. (2) I. $\Rightarrow p^2 + p - 56 = 0$
$\Rightarrow p^2 + 8p - 7p - 56 = 0$
$\Rightarrow p(p + 8) - 7(p + 8) = 0$
$\Rightarrow (p + 8)(p - 7) = 0$
$\Rightarrow p = 7$ or -8
II. $\Rightarrow q^2 + 17q + 72 = 0$
$\Rightarrow q - 8q - 9q + 72 = 0$
$\Rightarrow q(q - 8) - 9(q - 8) = 0$
$\Rightarrow (q - 8)(q - 9) = 0$
$\Rightarrow q = 8$ or 9
Obviously, $p < q$

10. (1) We have,
$3p + 2q = 58$ ...(i)
$4p + 4q = 92$
$\Rightarrow 2p + 2q = 46$ ...(ii)
By (i) - (ii) we get $p = 12$
From (i), $3 \times 12 + 2q = 58$
$\Rightarrow 2q = 58 - 36 = 22$
$\Rightarrow q = 11$
Hence, $p > q$

11. (2) I. $\Rightarrow 3p^2 + 15p + 2p + 10 = 0$
$\Rightarrow 3p(p + 5) + 2(p + 5) = 0$
$\Rightarrow (p + 5)(3p + 2) = 0$
$\Rightarrow p = -5$ or $-\frac{2}{3}$
II. $\Rightarrow 10q^2 + 5q + 4q + 2 = 0$
$\Rightarrow 5q(2q + 1) + 2(2q + 1) = 0$
$\Rightarrow (2q + 1)(5q + 2) = 0$

12. (3) I. $8x^2 + 6x - 5 = 0$

$\Rightarrow 8x^2 - 4x + 10x - 5 = 0$
$\Rightarrow 4x(2x - 1) + 5(2x - 1) = 0$
$\Rightarrow (4x + 5)(2x - 1) = 0$

$\Rightarrow x = -\frac{5}{4}$ or $\frac{1}{2}$

II. $12y^2 - 22y + 8 = 0$
$\Rightarrow 6y^2 - 11y + 4 = 0$
$\Rightarrow 6y^2 - 8y - 3y + 4 = 0$
$\Rightarrow 2y(3y - 4) - 1(3y - 4) = 0$
$\Rightarrow (2y - 1)(3y - 4) = 0$

$\Rightarrow y = \frac{1}{2}$ or $\frac{4}{3}$

Clearly, $x \leq y$

13. (5) Equation (i) × 2 + Equation (ii),
$8x + 14y + 12x - 14y = 418 - 38$
$\Rightarrow 20x = 380$

$\Rightarrow x = \frac{380}{20} = 19$

From (i),
$4 \times 19 - 7y = 209$
$\Rightarrow 7y = 76 - 209$
$\Rightarrow 7y = -133$
$\Rightarrow y = -19$
Clearly, $x > y$

14. (1) I. $17x^2 + 48x - 9 = 0$
$\Rightarrow 17x^2 + 51x - 3x - 9 = 0$
$\Rightarrow 17x(x + 3) - 3(x + 3) = 0$
$\Rightarrow (17x - 3)(x + 3) = 0$

$\Rightarrow x = \frac{3}{17}$ or -3

II. $13y^2 - 32y + 12 = 0$
$\Rightarrow 13y^2 - 26y - 6y + 12 = 0$
$\Rightarrow 13y(y - 2) - 6(y - 2) = 0$
$\Rightarrow (13y - 6)(y - 2) = 0$

$\Rightarrow y = \frac{6}{13}$ or 2

Clearly, $x < y$

15. (4) I. $18x^2 + 18x + 4 = 0$
$\Rightarrow 9x^2 + 9x + 2 = 0$
$\Rightarrow 9x^2 + 6x + 3x + 2 = 0$
$\Rightarrow 3x(3x + 2) + 1(3x + 2) = 0$
$\Rightarrow (3x + 1)(3x + 2) = 0$

$\Rightarrow x = -\frac{1}{3}$ or $-\frac{2}{3}$

II. $12y^2 + 29y + 14 = 0$
$\Rightarrow 12y^2 + 21y + 8y + 14 = 0$
$\Rightarrow 3y(4y + 7) + 2(4y + 7) = 0$
$\Rightarrow (3y + 2)(4y + 7) = 0$

$\Rightarrow y = -\frac{2}{3}$ or $-\frac{7}{4}$

Clearly, $x \geq y$

16. (2) I. $16x^2 + 20x + 6 = 0$
$\Rightarrow 8x^2 + 10x + 3 = 0$
$\Rightarrow 8x^2 + 6x + 4x + 3 = 0$
$\Rightarrow 8x^2 + 6x + 4x + 3 = 0$
$\Rightarrow 2x(4x + 3) + 1(4x + 3) = 0$

$\Rightarrow x = -\frac{1}{2}$ or $-\frac{3}{4}$

II. $10y^2 + 38y + 24 = 0$
$\Rightarrow 5y^2 + 19y + 12 = 0$
$\Rightarrow 5y^2 + 15y + 4y + 12 = 0$
$\Rightarrow 5y(y + 3) + 4(y + 3) = 0$

$\Rightarrow y = -\frac{4}{5}$ or -3

Clearly, $x > y$

17. (4) I. $x^2 + 5x + 6 = 0$
$\Rightarrow x^2 + 2x + 3x + 6 = 0$
$\Rightarrow x(x + 2) + 3(x + 2) = 0$
$\Rightarrow (x + 3)(x + 2) = 0$
$\therefore x = -3$ or -2
II. $y^2 + 3y + 2 = 0$
$\Rightarrow y^2 + 2y + y + 2 = 0$
$\Rightarrow y(y + 2) + 1(y + 2) = 0$
$\Rightarrow (y + 1)(y + 2) = 0$
$\therefore y = -1$ or -2
Clearly, $x \leq y$

18. (2) I. $x^2 - 10x + 24 = 0$
$\Rightarrow x^2 - 6x - 4x + 24 = 0$
$\Rightarrow x(x - 6) - 4(x - 6) = 0$
$\Rightarrow (x - 4)(x - 6) = 0$
$\therefore x = 4$ or 6
II. $y^2 - 9y + 20 = 0$
$\Rightarrow y^2 - 5y - 4y + 20 = 0$
$\Rightarrow y(y - 5) - 4(y - 5) = 0$
$\Rightarrow (y - 4)(y - 5) = 0$
$\therefore y = 4$ or 5
$\therefore x \geq y$

19. (5) I. $x^2 = \sqrt{961} = \pm 31$

II. $y = \sqrt{961} = \pm 31$

20. (2) I. $x^2 - x - 72 = 0$
$\Rightarrow x^2 - 9x + 8x - 72 = 0$
$\Rightarrow x(x - 9) + 8(x - 9) = 0$
$\Rightarrow (x + 8)(x - 9) = 0$
$\therefore x = -8$ or 9
II. $y^2 = 64$

$\Rightarrow y = \sqrt{64} = \pm 8$

Clearly, $x \geq y$

21. (5) I. $x^2 = 463 + 321 = 784$

$\therefore x = \sqrt{784} = \pm 28$

II. $y^2 = 308 + 421 = 729$

$\therefore y = \sqrt{729} = \pm 27$

22. (1)

$7x + 6y + 4z = 122$ ... (i)

$4x + 5y + 3z = 88$ ... (ii)

$9x + 2y + z = 78$ ... (iii)

By equation (iii) × 3 - equation (ii),

$27x + 6y + 3z = 234$

$4x + 5y + 3z = 88$

$23x + y = 146$ ...(iv)

By equation (iii) × 4 - equation (i),

$36x + 8y + 4z = 312$

$7x + 6y + 4z = 122$

$29x + 2y = 190$ ...(v)

By equation (iv) × 2 - equation (v)

$46x + 2y = 292$

$29x + 2y = 190$

$17x = 102$

$\Rightarrow x = 6$

From equation (iv),

$23 \times 6 + y = 146$

$\Rightarrow y = 146 - 138 = 8$

From equation (iii),

$9 \times 6 + 2 \times 8 + z = 78$

$\Rightarrow 54 + 16 + z = 78$

$\Rightarrow z = 78 - 70 = 8$

Clearly, $x < y = z$

23. (3) By equation II × 2 - equation (I)

$8x + 6y = 118$

$7x + 6y = 110$

$x = 8$

From equation (I),

$7 \times 8 + 6y = 110$

$\Rightarrow 6y = 110 - 56 = 54$

$\Rightarrow y = 9$

From equation (III),

$8 + z = 15$

$\Rightarrow z = 7$

Clearly, $x < y > z$

24. (2) I. $x = \sqrt{(36)^{\frac{1}{2}} \times (1296)^{\frac{1}{4}}}$

$= \sqrt{6 \times 6} = \pm 6$

By equation II × 3 - equation I

$6y + 9z = 99$

$6y + 5z = 71$

$4z = 28$

$\Rightarrow z = 7$

From equation II,

$2y + 3 \times 7 = 33$

$\Rightarrow 2y = 33 - 21 = 12$

$\Rightarrow y = 6$

$x < y < z$

25. (4) By equation I × 5 - II × 8

$40x + 35y = 675$

$40x + 48y = 792$

$-13y = -117$

$\Rightarrow y = 9$

From equation I,

$8x + 7 \times 9 = 135$

$\Rightarrow 8x = 135 - 63 = 72$

$\Rightarrow x = 9$

From equation III,

$9 \times 9 + 8z = 121$

$\Rightarrow 8z = 121 - 81 = 40$

$\Rightarrow z = 5$

Clearly, $x = y > z$

26. (5) I. $(x + y)^3 = 1331$

$\Rightarrow x + y = 11$

$\Rightarrow y = 11 - x$

From equation III,

$x(11 - x) = 28$

$\Rightarrow 11x - x^2 = 28$

$\Rightarrow x^2 - 11x + 28 = 0$

$\Rightarrow x^2 - 7x - 4x + 28 = 0$

$\Rightarrow x(x - 7) - 4(x - 7) = 0$

$\Rightarrow (x - 7)(x - 4) = 0$

$\Rightarrow x = 7, 4$

From equation I

$y = 4, 7$

From equation II

$7 - 4 + z = 0 \Rightarrow z = -3$

or, $4 - 7 + z = 0 \Rightarrow z = 3$

INSURANCE EXAMS

1. (5) Equation I

$2x^2 - 7x + 6 = 0$

$\Rightarrow 2x^2 - 4x - 3x + 6 = 0$

$\Rightarrow 2x(x - 2) - 3(x - 2) = 0$

$\Rightarrow (x - 2)(2x - 3) = 0$

$\Rightarrow x = 2$ or $x = \frac{3}{2}$

Equation II. $4y^2 = 9$

$\Rightarrow y^2 = \frac{9}{4}$

$\Rightarrow y = \pm \frac{3}{2}$

Obviously, $y \leq x$

2. (5) Equation I

$4x^2 - 4x - 3 = 0$

$\Rightarrow 4x^2 - 6x + 2x - 3 = 0$

$\Rightarrow 2x(2x - 3) + 1(2x - 3) = 0$

$\Rightarrow (2x - 3)(2x + 1) = 0$

$\Rightarrow x = \frac{3}{2}$ or, $x = -\frac{1}{2}$

Equation II. $4y^2 + 12y + 5 = 0$

$\Rightarrow 4y^2 + 10y + 2y + 5 = 0$

$\Rightarrow 2y(2y + 5) + 1(2y + 5) = 0$

$\Rightarrow (2y + 5)(2y + 1) = 0$

$\Rightarrow y = -\frac{5}{2}$ or, $y = -\frac{1}{2}$

Obviously, $x \geq y$

3. (1) Equation I $4x^2 = 49$

$\Rightarrow x^2 = \frac{49}{4}$

$\Rightarrow x = \pm \frac{7}{2}$

Equation II

$9y^2 - 66y + 121$

$\Rightarrow 9y^2 - 33y - 33y + 121 = 0$

$\Rightarrow 3y(3y - 11) - 11(3y - 11) = 0$

$\Rightarrow (3y - 11)(3y - 11) = 0$

$\Rightarrow y = \frac{11}{3}$

Obviously, $x < y$

4. (2) Equation I

$x^2 + 9x + 14 = 0$

$\Rightarrow x^2 + 2x + 7x + 14 = 0$

$\Rightarrow x(x + 2) + 7(x + 2) = 0$

$\Rightarrow (x + 2)(x + 7) = 0$

$\Rightarrow x = -2$ or, $x = -7$

Equation II

$y^2 + y - 2 = 0$

$\Rightarrow y^2 + 2y - y - 2 = 0$

$\Rightarrow y(y + 2) - 1(y + 2) = 0$

$\Rightarrow (y - 1)(y + 2) = 0$

$\Rightarrow y = 1$ or, $y = -2$

Clearly, $y \geq x$

5. (1) Equation I

$9x^2 - 18x + 5 = 0$

$\Rightarrow 9x^2 - 15x - 3x + 5 = 0$

$\Rightarrow 3x(3x - 5) - 1(3x - 5) = 0$

$\Rightarrow (3x - 5)(3x - 1) = 0$

$\Rightarrow x = \frac{5}{3}$ or, $x = \frac{1}{3}$

Equation II

$2y^2 - 9y + 10 = 0$

$\Rightarrow 2y^2 - 5y - 4y + 10 = 0$

$\Rightarrow y(2y - 5) - 2(2y - 5) = 0$

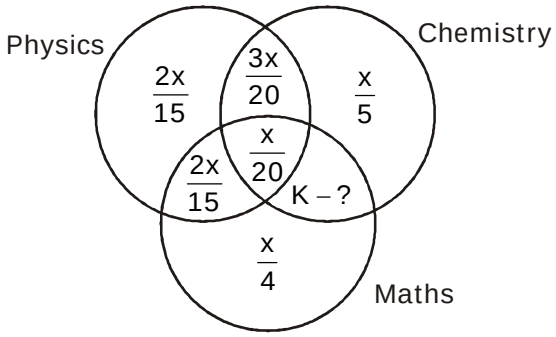
$\Rightarrow (2y - 5)(y - 2) = 0$

$\Rightarrow y = \frac{5}{2}$ or $y = 2$

Clearly, $y > x$

MATHEMATICAL PUZZLE -122

Directions (Q. 1 - 5): A total of x students appeared in a class test consisting of three papers, viz Physics, Chemistry and Maths. The following Venn diagram shows the number of students who passed these three papers. None of the students failed in all the three papers together. Answer the given questions based on this diagram.



1. How many students are there who passed in Chemistry and Maths but failed in Physics?

   (1) $\frac{x}{6}$ (2) $\frac{x}{12}$ (3) $\frac{x}{20}$ (4) $\frac{x}{5}$ (5) None of these

2. What is the difference between the number of students who passed in Chemistry and the number of students who passed in Physics? (The number of students who passed in all the three papers is 15.)

   (1) 5 (2) 10 (3) 15 (4) 20 (5) None of these

3. How many students are there who passed in exactly one paper, if the number of students who passed in exactly two papers is 110.

   (1) 160 (2) 165 (3) 170 (4) 175 (5) 180

4. The number of students who passed in at least two papers is what percentage of the total number of students?

   (1) $33\frac{1}{2}\%$ (2) $41\frac{2}{3}\%$ (3) 44% (4) $47\frac{1}{2}\%$ (5) None of these

5. The number of students who passed in only Maths is what percentage more than the number of students who passed in all the three papers?

   (1) 20% (2) 80% (3) 120% (4) 200% (5) 400%

Q.6-10. (Study the following information carefully to answer the questions that follow.:)

There are two Trains, Train-A and Train-B. Both trains have four different types of Coaches viz. General Coaches. 'Sleeper Coaches, First Class Coaches and AC Coaches. In Train A there are total 700 passengers. Train B has thirty percent more passengers than Train A. Twenty percent of the passengers of Train A are in "General Coaches. One-fourth of the total number of passengers of Train A are in AC coaches. Twenty three percent of the passengers of Train A are in Sleeper Class Coaches. Remaining passengers of Train-A are in first class coaches. Total number of passengers in AC coaches in both the trains together is 480. Thirty percent of the number of passengers of Train B is in Sleeper Class Coaches. Ten percent of the total passengers of Train B are in first class coaches. Remaining passengers of Train-B are in general class coaches.

6. What is the respective ratio between the number of passengers in first class Coaches of Train A to the number of passengers in Sleeper Class coaches of Train B ?

   (1) 13 : 7 (2) 7:13 (3) 32:39 (4) Data Inadequate (5) None of these

7. What is the total number of passengers in the General Coach of Train A and the AC Coach of Train B together?
(1) 449 (2) 459 (3) 435 (4) 445 (5) None of these

8. What is the difference between the number of passengers in the AC Coach of Train A and the total number of passengers in Sleeper class coaches and First class coaches together of Train B ?
(1) 199 (2) 178 (3) 187 (4) 179 (5) None of these

9. Total number of passengers in General Class Coaches in both the Trains together is approximately what percentage of total number of passengers in Train-B ?
(1) 35 (2) 42 (3) 46 (4) 38 (5) 31

10. If cost of per ticket of First class coach Rs 450/-. What will be the total amount generated from First class coaches of Train-A?
(1) Rs 1,00,080/- (2) Rs 1,08,000/- (3) Rs 1,00,800/- (4) Rs 10,800/- (5) None of these

Directions (Q. 11-15): Read the following information carefully and answer the questions.

From a class a total of 200 students appeared in an examination consisting three papers P_1, P_2 and P_3. 56% of students passed in paper P_1, 63% passed in P_2 and 56.5% passed in P_3. 11% students passed only in paper P_1 and P_2, 8% passed only in paper P_1 and P_3, and 22% students passed in all three papers.

No student failed in all three papers.

11. How many students passed in paper P_2 and P_3 but failed in P_1?
(1) 22 (2) 24 (3) 25 (4) 28 (5) 32

12. What is the ratio of the number of students who passed only in paper P_2 to the number of students who passed in P_3 only?
(1) 3:2 (2) 4:3 (3) 5:4 (4) 6:5 (5) 7:6

13. The number of students who passed in paper P_1 only is what percentage of the number of students who passed in paper P_1 and P_3 but failed in paper P_2?
(1) 187.5% (2) 157.5% (3) 112.5% (4) 97.5% (5) 53.33%

14. The number of students who passed in at most one paper is what percentage of the total number of students in the class?
(1) 43.5% (2) 44.5% (3) 45.5% (4) 46.5% (5) 47.5%.

15. What is the difference between the number of students who passed in paper P_3 and the number of students who passed only in paper P_3?
(1) 81 (2) 83 (3) 85 (4) 87 (5) 89

Directions (Q. 16-20): Study the following information carefully and answer the given questions.

Among 400 cricket players, 45% played in IPL 1 and 6.25% played only in IPL 1. Again, 57.5% players played in IPL 2 and 11.25% players played only in IPL 2. Again, 72.5% players played in IPL 3 and 27.5% players played only in IPL 3. Twenty per cent players played in all three IPL tournaments..

16. How many players are there who played in IPL 1 and IPL 2 but not in IPL 3?
(1) 30 (2) 35 (3) 40 (4) 65 (5) 45

17. What is the percentage of players who played in IPL 2 and IPL 3 but not in IPL 1?
(1) 12.5% (2) 16.25% (3) 22.75% (4) 24% (5) None of these

18. What is the percentage of the players who played in at least two IPL tournaments?
(1) 30% (2) 35% (3) 45% (4) 50% (5) 55%

19. The number of players who played only in either IPL 1 or IPL 2 is what percentage of the players who played in all three IPL?
(1) 72% (2) 75% (3) 82.5% (4) 85% (5) None of these

20. The number of players who played in at most one IPL tournament is what percentage more/less than the number of players who played in at least one IPL?
(1) 55% more (2) 55% less (3) 40% more (4) 40% less (5) None of these

Directions (Q. 21-25): Study the following information carefully and answer the given questions.

From a group of x sportsmen 50% participated in Olympic games, 53% participated in Asiad and 42% participated in Commonwealth Games (CWG). Ten percent participated in Olympics and Asiad but not in CWG, 14% participated in Olympics CWG but not in Asiad, and 5% participated in Asiad and CWG but not in Olympic games. Based on the information given, answer the following questions.

21. If x is equal to 3 00, how many sportsmen participated in all three events?
(1) 20 (2) 24 (3) 30 (4) 36 (5) 40

22. If the number of persons who participated in only Olympic games is 81, what is the total number of sportsmen?
(1) 360 (2) 420 (3) 450 (4) 480 (5) 510

23. If the number of sportsmen who participated in all three events is 48, what is the number of sportsmen, who participated in Asiad?
(1) 272 (2) 296 (3) 310 (4) 318 (5) 330

24. If the number of sportsmen who participated in exactly two events is 58, what is the number of persons who participated in all three events?
(1) 16 (2) 20 (3) 24 (4) 12 (5) 28

25. If x is equal to 300, what is the ratio of the number of sportsmen who participated only in Olympic games to the number of persons who participated only in CWG?
1) 3 : 2 (2) 4 : 3 (3) 5 : 4 (4) 6 : 5 (5) 7 : 6

Directions (Q. 26-30): Study the following information carefully and answer the given questions.

Three companies — A, B, and C produce a particular item in two different types —I and II. Total number of items of both types produced by all three companies is 62000 and total items I and II produced by company A is 15200. The ratio of the numbers of type I to type II items produced by A is 9 : 10. Type I items produced by Company B is 175% of type 1 items produced by A. Total items (both I and II) produced by B is 150% of total items produced by A. The number of type I items produced by C is 20% more than the number of type II items produced by A.

26. What is the number of type II items produced by B?
(1) 9600 (2) 10200 (3) 14400 (4) 12600 (5) None

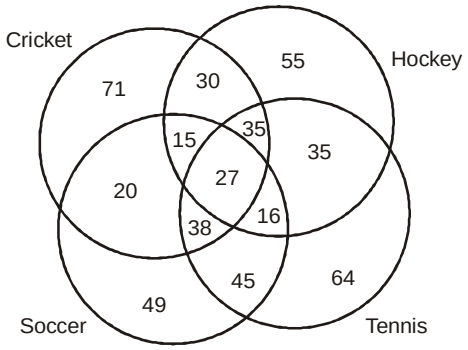
27. What is the ratio of the number of type I items to the number of type II items produced by Company C?
(1) 2:3 (2) 3:4 (3) 4:5 (4) 5:6 (5) None of these

28. What is the average number of type I items produced by all three companies?
(1) 9650 (2) 9800 (3) 9960 (4) 10200 (5) .None of these

29. The number of type II items produced by C is what percentage of the total number of items produced by C?
(1) 80% (2) 75% (3) 60% (4) 50% (5) 40%

30. What is the difference between the total number of type II items and the total number of type I items produced by all three companies together?
(1) 2750 (2) 2800 (3) 3000 (4) 3150 (5) None of these

Directions (Q. 31-35): Following Venn-diagram shows the result of a survey conducted on people about their interest in different sports. Answer the following questions based on this diagram.



Total number of people = 500

31. How many people are there who like exactly three types of sports out of the given four?
(1) 98 (2) 104 (3) 108 (4) 112 (5) 115

32. The number of people who like exactly two types of sports is what percentage of the total number of people surveyed?
(1) 40% (2) 32% (3) 30% (4) 26% (5) 24%

33. What is the difference between the number of people who like either only Cricket or only Soccer and the number of people who like either only Hockey or only Tennis?
(1) 1 (2) 3 (3) 5 (4) 7 (5) 9

34. What is the ratio of the number of people who like Cricket to the number of people who like only Tennis?
(1) 40:13 (2) 59:16 (3) 63:23 (4) 64:25 (5) 70:29

35. The number of people who like exactly one type of sports is what percentage of the total number of people surveyed?
(1) 42.5% (2) 45.4% (3) 45.8% (4) 51.1% (5) 54%

Q.36-40. (Study the following information and answer the questions that follow :)

The premises of a bank are to be renovated. The renovation is in terms of flooring. Certain areas are to be floored either with marble or wood. All rooms/halls and pantry are rectangular. The area to be renovated comprises of a hall for customer transaction measuring 23 m by 29 m, branch manager's room measuring 13 m by 17 m, a pantry measuring 14 m by 13 m, a record keeping cum server room measuring 21m by 13 m and locker area measuring 29 m by 21 m. The total area of the bank is 2000 square meters. The cost of wooden flooring is Rs 170/- per square meter and the cost of marble flooring is Rs 190/- per square meter. The locker area, record keeping cum server room and pantry are to be floored with marble. The branch manager's room and the hall for customer transaction are to be floored with wood. No other area is to be renovated in terms of flooring.

36. What is the respective ratio of the total cost of wooden flooring to the total cost of marble flooring ?
(1) 1879 : 2527 (2) 1887 : 2386 (3) 1887 : 2527 (4) 1829 : 2527 (5) 1887 : 2351

37. If the four walls and ceiling of the branch managers room (the height of the room is 12 meters) are to be painted at the cost of Rs 190/- per square meter, how much will be the total cost of renovation of the branch manager's room including the cost of flooring ?
(1) Rs 1,36,800/- (2) Rs 2,16,660- (3) Rs 1,78,790/- (4) Rs 2,11,940- (5) None of these

38. If the remaining area of the bank is to be carpeted at the rate of Rs 110/- per square meter, how much will be the increment in the total cost of renovation of bank premises ?
(1) Rs 5,820/- (2) Rs 4,848/- (3) Rs 3,689/- (4) Rs 6690/- (5) None of these

39. What is the percentage area of the bank that is not to be renovated ?
(1) 2.2% (2) 2.4% (3) 4.2% (4) 4.4% (5) None of these

40. What is the total cost of renovation of the hall for customer transaction and the locker area ?
(1) Rs 2,29,100/- (2) Rs 2,30,206/- (3) Rs 2,16,920/- (4) Rs 2,42,440/- (5) None of these

Directions (Q. 41-45) : In a society, there are total 200 families. Out of that 47.5% people have taken subscription of AXN channel, 53.5% have taken HBO and 54% Star Movies channel. 6% of them have taken subscription of AXN and HBO but not Star Movies. 10% of the people have taken Star Movies and HBO but not AXN and 15% people have taken the subscription of all the three channels.

Answer the following questions based on the given information.

41. How many families are there who have taken the subscription of Star Movies and AXN but not HBO?
(1) 12 (2) 20 (3) 18 (4) 30 (5) 48

42. How many families are there who have taken the subscription of either only AXN or only HBO?
(1) 75 (2) 80 (3) 90 (4) 85 (5) None of these

43. What percentage of families have taken the subscription of Star Movies only?
(1) 15% (2) 54% (3) 39% (4) 20% (5) 22.5%

44. How many families are there who have taken the subscription of exactly two channels out of the given three?
(1) 50 (2) 80 (3) 120 (4) 30 (5) None of these

45. The number of families who have taken the subscription of at least two channels from the given three options is what percentage of the number of families who have taken the subscription of exactly one channel from the given three options?
(1) 45% (2) 60% (3) 72% (4) 75% (5) None of these

Directions (Q. 46-50) ; Study the following information carefully and answer the given questions.

In a class test, total 250 students appeared. Out of that 16% students passed only in Physics in which B : G is 3 : 2. Again, 24% passed only in Chemistry paper in which B: G is 7 : 5. Also, 30% students passed only in Maths and B : G is 7 : 8. Besides 8% students passed in Physics and Chemistry but failed in Maths and among them B : G is 2 :3. Again, 6% students passed in all the three papers and among them B : G = 2 : 3. Also, 9.6% students passed in Physics and Maths but failed in Chemistry and among them B : G is 3 : 1. None of them failed in all the three subjects and the difference between the total number of boys and the total number of girls who appeared in the exam is 14.

Answer the following questions based on the above information. (B : G is the ratio of the number of boys to the number of girls.)

46. What is the ratio of the number of boys to the number of girls who passed in Chemistry and Math but failed in Physics?
(1) 3:5 (2) 4:5 (3) 4:7 (4) 5:7 (5) 5:8

47. How many girls are there who passed exactly in one subject?
(1) 72 (2) 75 (3) 78 (4) 81 (5) 85

48. What is the total number of boys who passed exactly in two subjects?
(1) 28 (2) 32 (3) 36 (4) 40 (5) 42

49. The number of girls who passed in Physics and Chemistry but failed in Maths is what percentage of the number of girls who passed in Chemistry and Maths but failed in Physics?
(1) 75% (2) 80% (3) 120% (4) 150% (5) 200%

50. The number of boys who passed only in Chemistry is what percentage of the number of girls who passed either only in Maths or only in Physics?
(1) 72.5% (2) 65% (3) 62.5% (4) 60% (5) 47.5%

Directions (Q. 51-55) : In a company total 500 persons work there. Out of that, 53.6% employees have bank Account in SBI, 44% in ICICI and 34% in HDFC. 5% of total employees have account in SBI and HDFC but not in ICICI, and 180% of this number have account in SBI and ICICI but not in HDFC. 3.6% employees have account in all three banks and 10.4% employees have account in ICICI and HDFC but not in SBI.

Answer the following questions based on these information.

51. How many employees are there who have account in ICICI Bank only?
(1) 75 (2) 105 (3) 180 (4) 220 (5) None of these

52. What percentage of employees hold account in HDFC only?

(1) 9% (2) 12% (3) 15% (4) 24% (5) 36%

53. How many people hold account in exactly one bank?
(1) 270 (2) 320 (3) 350 (4) 360 (5) 370

54. The number of employees who hold account in all three banks is what percentage of the number of employees who have account in SBI only?
(1) 6% (2) 8% (3) 10% (4) 12% (5) 18%

55. The number of employees who hold account in at least two banks is what percentage of the number of employees who hold account in almost two banks?
(Answer in approximation.)
(1) 24% (2) 29% (3) 34% (4) 39% (5) 45%

Directions (Q. 56-60) :Study the information carefully to answer the questions that follow.

A company produced five different products, viz mobile phone, pen drive, calculator, television and washing machine; The total number of all the five products is 1650.24% of the total number of products is mobile phones. One-sixth of the total number of products is pen drives. 14% of the total number of products is calculators. Remaining products are either television or washing machine. The number of washing machines is 50 more than the number of televisions produced.

56. What is the ratio of the number of washing machines to the number of calculators produced by the company?
(1) 17 : 11 (2) 19 : 11 (3) 11 : 17 (4) 19 : 13 (5) None of these

57. If 24 per cent of the pen drives are defective, what is the number of pen drives which are not defective?
(1) 209 (2) 215 (3) 219 (4) 225 (5) None of these

58. The number of televisions produced is approximately what per cent of the total number of calculators and washing machines produced together?
(1) 63 (2) 55 (3) 59 (4) 51 (5) 67

59. What is the difference between the total number of televisions and mobile phones together and the number of calculators produced?
(1) 534 (2) 524 (3) 511 (4) 523 (5) None of these

60. What is the total number of pen drives, calculators and washing machines produced by the company?
(1) 907 (2) 917 (3) 925 (4) 905 (5) None of these

Directions (Q. 61-65) : Study the following information carefully and answer the questions given below

In a survey three questions (Q_1, Q_2 and Q_3) were asked from 600 people. 20% of them answered only question Q_1 and double of them answered question Q_3. $\frac{1}{4}$ of the total people surveyed answered only Question Q_2. $\frac{1}{24}$ of the total people surveyed answered all three questions and this number is 20 less than the number of people who answered question Q_1 and Q_3 but not Q_2. The number of people who answered question Q_2 and Q_3 but not question Q_1 is 25% of the number of people who answered question Q_1 and Q_2 but didn't answer question Q_3. The number of people who didn't answer any of the three questions is 10.

61. How many people are there who answered question Q_2 and Q_3 but didn't answer question Q_1?
(1) 20 (2) 40 (3) 80 (4) 25 (5) None of these

62. What is the number of people who answered exactly one question from the given three questions?
(1) 400 (2) 420 (3) 440 (4) 460 (5) 480

63. The number of people who answered question Q_1 is what percentage of the total number of people surveyed?
(1) 32% (2) 35% (3) 40% (4) 45% (5) 48%

64. What is the difference between the number of people who answered at least one question and the number of people who answered at most one question?

(1) 120 (2) 160 (3) 430 (4) 210 (5) 80

65. What is the ratio of the number of people who answered question Q_1 to the number of people who answered only question Q_3?
(1) 7 : 5 (2) 9 : 5 (3) 7 : 3 (4) 7 : 4 (5) 9 : 4

Directions (Q. 66-70) : Study the following information carefully answer the questions given below:

In an examination (consisting of two papers Physics and Chemistry) total 300 students appeared. Out of that the ratio of boys to girls is 3 : 2. The number of boys who passed only in Physics is 25% of the total number of boys and this number is $\frac{3}{2}$ of the number of girls who passed only in Chemistry. The number of girls who passed in both the papers is $13\frac{1}{3}\%$ of the total number of students and the number of boys who passed in both the papers is 180% of the number of girls who passed in both the papers. None of the candidate failed in both the papers.

66. How many girls are there who passed only in Physics paper?
(1) 35 (2) 40 (3) 45 (4) 50 (5) 60 .

67 The number of boys who passed only in Chemistry is what percentage of the total number of students who appeared in the examination?
(1) 21% (2) 36% (3) 48% (4) 72% (5) 84%

68. How many students passed in Physics?
(1) 192 (2) 197 (3) 201 (4) 203 (5) 207

69. What is the ratio of the number of boys who passed in Chemistry to the number of girls who passed only in Physics?
(1) 23 : 8 (2) 25 : 11 (3) 27 : 10 (4) 29 : 15 (5) 31 : 16

70. How many students are there who passed at most in one subject?
(1) 172 (2) 178 (3) 181 (4) 188 (5) 192

Directions (Q. 71-75) : Study the information carefully to answer the questions that follow.

In a school there are total 120 staff members and 800 students. 65 per cent of the number of staff members are teachers and the remaining staff members are administrative officials. Out of the total number of students 45 per cent are girls. Twenty per cent of the number of girls can speak only Hindi. The remaining girls can speak both Hindi and English. Three-fourths of the number of boys can speak only Hindi. The remaining boys can speak both Hindi and English. Two-thirds of the number of teachers are males. Five-fourteenths of the number of administrative officials are females.

71. What is the difference between the number of boys (student) who can speak both Hindi and English and the number of girls (student) who can speak both Hindi and English?
(1) 164 (2) 178 (3) 188 (4) 174 (5) None of these

72. The total number of girls (student) is what percentage of the total number of staff members in the school?
(1) 350 (2) 300 (3) 400 (4) 450 (5) None of these

73. What is the difference between the total number of female administrative officials, female teachers and the number of male administrative officials?
(1) 16 (2) 12 (3) 18 (4) 14 (5) None of these

74. What is the ratio of the total number of teachers to the number of boys (student) who can speak Hindi only?
(1) 11 : 56 (2) 13 : 54 (3) 13 : 56 (4) 11 : 54 (5) None of these

75. What is the total number of male administrative officials, female teachers and girls (student) who can speak Hindi only?
(1) 125 (2) 115 (3) 127 (4) 117 (5) None of these

Directions (Q. 76-80) : Study the following information carefully and answer the questions given below

From a school 500 students appeared in all in an examination consisting of three papers P_1, P_2 and P_3. 52.6% students passed in P_1, 57% in P_2 and 50.4% in P_3. 18% of students passed only in P_1 and P_2,

10.2% passed only in P_1 and P_3 and 8.4% passed in all three papers. The number of girls passed in Paper P_1 only is 60% of the number of boys who passed only in Paper P_1. The ratio of the number of boys to the number of girls who passed only in Paper P_2 is 7 : 6. The number of girls who passed in Paper P_3 only is 6% of the total number of students who appeared in the examination.

Answer the following questions based on the above information.

76. How many students are there who passed in Paper P_2 and P_3, but failed in P_1?
(1) 60 (2) 75 (3) 80 (4) 90 (5) 120

77. What is the difference between the number of boys and the number of girls who passed in Paper P_1 only?
(1) 10 (2) 20 (3) 30 (4) 40 (5) 50

78. What percentage of students passed only in Paper P_2?
(1) 14.2% (2) 15% (3) 15.6% (4) 16% (5) 16.4%

79. What is the ratio of the number of students who passed only in Paper P_1 and P_2 to the number of boys who passed only in Paper P_3?
(1) 5:4 (2) 7:4 (3) 5:3 (4) 9:5 (5) 8:5

80. What is the percentage of students who failed in any two papers?
(1) 48.4% (2) 52.6% (3) 54% (4) 57.5% (5) 60%

Directions (Q. 81-85): Study the following information carefully to give answer to the following questions.

In a school there are total 800 students. Among them, 55.625% like Cricket, 32.5% like Football and 43.125% like Hockey. 5% students like all three games. 9.375% students like only Cricket and Football, and 5.625% students like only Football and Hockey. Answer the following questions based on this information.

81. What percentage of students like only Cricket?
(1) 12.5% (2) 23.125% (3) 30% (4) 32.5% (5) 35%

82. How many students are there who like Cricket and Hockey but do not like Football?
(1) 40 (2) 50 (3) 60 (4) 70 (5) 75

83. The number of students who like only Football is approximately, what per cent of the number of students who like Football?
(1) 32% (2) 34% (3) 36% (4) 38% (5) 42%

84. The number of students who like only Hockey is approximately what per cent of the number of students who like all three games?
(1) 425% (2) 450% (3) 475% (4) 500% (5) 525%

85. What is the percentage of students who like at least two games among the given three types of games?
(1) 22.25% (2) 24.25% (3) 26.25% (4) 28.25% (5) 32.5%

Directions (Q. 86-90): Study the information carefully and answer the questions given below.

There are 500 students in a class who appeared in an exam comprising Paper P_1 and P_2. The ratio of the number of boys to girls is 7 : 3. Out of the total boys, 44% passed in Paper P_2 only, which is 82 more than the total number of girls who passed in Paper P_2 only. 28% of girls passed in Paper P_1 only, which is one-third of the total number of boys who passed in both the papers. No one failed in both the papers.

Answer the following questions based on this information.

86. The total number of boys who passed in Paper P_1 only is what per cent of the total number of students who appeared in the exam?
(1) 12% (2) 14% (3) 16% (4) 18% (5) 20%

87. What percentage of girls passed in both the papers?
(1) 16% (2) 20% (3) 24% (4) 28% (5) 32%

88. What is the difference between the total number of boys who passed in Paper P_1 and the total number of girls who passed in Paper P_2?

(1) 82 (2) 84 (3) 86 (4) 88 (5) 90

89. What percentage of the students passed in both the papers?

(1) 32.4% (2) 34.6% (3) 36.8% (4) 38.2% (5) 40%

90. The total number of boys who passed in Paper P_2 is what percentage of the total number of students who appeared in the exam?

(1) 36% (2) 40% (3) 48% (4) 51% (5) 56%

Direction (Q. 91-95) : Study the following information carefully and answer the questions given below:

An organisation consists of 1500 employees. The ratio of males to females is 17 : 13. All the employees work at five different levels I, II, III, IV and V. 28% females are at level I. 18% males work at level II. 20% males work at level V. The ratio of females to males at level II is 2 : 3. 25% of the total number of employees are at level III. Females working at level V are 60% of the males working at level 1.18% of the females are at level IV. The remaining females are at level III. 20% of the males work at level I, and the remaining males work at level IV.

91. What is the total number of males who work at level I and III together?

(1) 495 (2) 498 (3) 447 (4) 398 (5) None of these

92. The number of females working at the level II is approximately what per cent of the total number of employees in the organisation?

(1) 12% (2) 11% (3) 15% (4) 7% (5) 21%

93. What is the number of females who work at level V?

(1) 115 (2) 105 (3) 125 (4) 315 (5) 102

94. What is the total number of females who work at level I and V together?

(1) 384 (2) 184 (3) 484 (4) 284 (5) 584

95. Find the ratio of the number of males working at level I to the number of females working at level III.

(1) 107:174 (2) 147:710 (3) 170:147 (4) 129:117 (5) None of these

Directions (Q. 96-100) : Study the following information carefully to answer these questions.

An institute having 450 employees has sent all its employees for training in one or more areas of HR, Computer skills and Financial skills. The employees are classified into two categories - officers and clerks, who are in the ratio of 4 : 5.10% of the officers take training only in Computer skills, 16% of the clerks take training only in HR and this is equal to the number of officers taking training only in Financial skills and equal to 50% of the number of officers taking training in HR and Financial skills both. 6% of the total employees take training in all the three, of which $\frac{2}{3}$ are officers. 10% of the total employees take training in HR and Computer skills both, which is five times the number of clerks taking training in Computer skills and Financial skills. 10% of the clerks take training in HR and Computer skills both. The number of officers taking training only in HR is 25% of the number of clerks taking training only in HR. 20% of the total number of employees take training only in Computer skills. The number of clerks taking training in HR and Financial skills both is 20% of the total number of clerks.

96. How many employees take training in Financial skills but not in HR?

(1) 174 (2) 172 (3) 134 (4) 162 (5) None of these

97. How many officers take training in Computer skills?

(1) 78 (2) 68 (3) 98 (4) 88 (5) 91

98. How many clerks take training in HR but not in Computer skills?

(1) 51 (2) 4 (3) 63 (4) 81 (5) 91

99. How many employees take training in Computer and HR skills both only?

(1) 10 (2) 25 (3) 18 (4) 16 (5) None of these

100. How many clerks take training in HR, Computer and Financial skills only?
(1) 6 (2) 9 (3) 11 (4) 74 (5) None of these

Directions (Q. 101-105): Study the following information and answer the questions that follow.

The premises of an institute are to be renovated. Only the floor is to be renovated either with marble or with wood. All rooms, halls and pantry are rectangular. The area to be renovated comprises a hall measuring 33m by 39m. The director's room measures 13m by 12m and the pantry measures 14m by 12m. A record keeping-cum-server room measures 23m by 13m and the accounts room measures 12m by 23m. The total area of the institute is 2500 square metres. The cost of wooden flooring is ` 170 per square metre and the cost of marble flooring is ` 190 per square metre. The accounts room, the record keeping-cum-server room, and the pantry are to be floored with marble. The director's room and the hall are to be floored with wood.

101. What is the ratio of the total cost of wooden flooring to the total cost of marble flooring?
(1) 1443 : 735 (2) 8177 : 4655 (3) 1443 : 4655 (4) 24531 : 14117 (5) 9177 : 4655

102. If four walls and ceiling of the room (the height of the room is 15 metres) are to be painted at the cost of `190 per square metre, how much will be the total cost of renovation of the director's room, including the cost of flooring?
(1) `198660 (2) ` 178680 (3) ` 198880 (4) ` 22876 (5) ` 188680

103. If the remaining area of the institute is to be carpeted at the rate of ` 210 per square metre, by how much will the cost of renovation of institute premises increase?
(1) ` 75000 (2) ` 72840 (3) ` 65940 (4) ` 75940 (5) ` 64940

104. What is the percentage area of the institute that is not to be renovated?
(1) 16.44% (2) 13.56% (3) 14.55% (4) 12.56% (5) 11.44%

105. What is the total cost of renovation of the hall and the accounts room?
(1) ` 287700 (2) ` 277230 (3) ` 266600 (4) ` 298870 (5) ` 271230

Directions (Q. 106-108) : Study the information carefully to answer these questions.

In a team there are 240 members (males and females). Two-thirds of them are males. Fifteen per cent of males are graduates. Remaining males are non-graduates. Three-fourths of the females are graduates. Remaining females are non-graduates.

106. What is the difference between the number of females who are non-graduates and the number of males who are graduates?
(1) 2 (2) 24 (3) 4 (4) 116 (5) 36

107. What is the sum of the number of females who are graduates and the number of males who are non-graduates?
(1) 184 (2) 96 (3) 156 (4) 84 (5) 196

108. What is the ratio of the total number of males to the number of females who are non-graduates?
(1) 6:1 (2) 8:1 (3) 8:3 (4) 5:2 (5) 7:2

Directions (Q. 109-112): Study the table carefully to answer the questions that follow:

A candidate is contesting an election from a constituency which has seven electoral zones -A, B, C, D, E, F and G. The following table shows the voter population, the estimated voter turnout per 1000 persons and the time required to campaign in each zone.

	A	B	C	D	E	F	G
Voter population	20000	24000	35000	42000	30000	21000	28000
Estimated voter turnout per 1000 persons	875	725	800	700	600	500	650
Number of days required for campaigning	2	3	4	5	3	3	3

Also note that:

(1) if the candidate starts campaigning in a zone, he has to meet the entire voter population in that zone.

(2) on any day, the candidate is allowed to campaign in only one electoral zone.

(3) if the candidate has a limited time to campaign, the candidate campaigns according to the given estimation in such a way that he maximises the voter turnout.

109. What is the maximum possible voter population that the candidate can meet over a total campaigning period of 20 days?

(1) 176000 (2) 179000 (3) 189500 (4) 200000 (5) None of these

110. If the candidate can campaign for only 15 days, in which of the following electoral zones would he not campaign?

(1) A, B and F (2) B, F and E (3) C and F (4) D and F (5) None of these

111. If the candidate has a permission to campaign only in four zones, which zone would he select?

(1) A, B, C, D (2) C, D, E, G (3) B, C, D, E (4) A, B, E, F (5) None of these

112. Which of the following statements is/are not true?

I. Maximum voter turnout (estimated) is from Zone C.

II. The voter turnout is more than 60% in five zones.

III. The voter turnout in Zone G is 60% less than that in Zone C

(1) I and II (2) Only II (3) I and III (4) I, II and III (5) None of these

Directions (Q. 113-117): Study the following information carefully and answer the questions that follow:

There are two universities - University U_1 and University U_2. Both universities have four different departments, viz Mathematics Department, Geography Department, Physics Department and Chemistry Department. In University U_1 there are total 1400 staff. University U_2 has forty per cent more staff than University U_1 Twentyfive per cent of the staff of University U_1 are in Mathematics Department. One-fifth of the total number of staff of University U_1 are in Chemistry Department. Thirtyfive per cent of the staff of University U_1 are in Geography Department, Remaining staff of University U_1 are in Physics Department, The total number of staff in Geography Department is 686. Fortyfive per cent of the staff of University U_2 are in Chemistry Department. Fifteen per cent of the total staff of University U_2 are in Mathematics Department. The remaining staff of University U_2 are in Physics Department.

113. If the monthly salary of a staff member in Mathematics Department is ` 9500, what will be the total amount paid by University U_2,to the entire staff in Mathematics Department?

(1) ` 27.93 lakh (2) ` 2.793 lakh (3) ` 20.7 lakh (4) ` 27.05 lakh (5) ` 33.25 lakh

114. The total number of staff in Geography Departments of both the universities together is what per cent of the total number of staff in University U_1?

(1) 39% (2) 49% (3) 65% (4) 35% (5) 51%

115. What is the difference between the total number of staff in Chemistry Department of University U_2 and the total number of staff in Physics and Chemistry Departments together of University U_1?

(1) 222 (2) 252 (3) 322 (4) 232 (5) 482

116. What is the ratio of the number of staff in Geography Department of University U_2 to the total number of staff in Mathematics Department of University U_1?

(1) 25:14 (2) 4:35 (3) 7:5 (4) 7:6 (5) 14:25

117. The total number of staff in Chemistry Department of University U_1 is approximately what per cent of the total number of staff of University U_2?

(1) 14% (2) 18% (3) 15% (4) 28% (5) None of these

Directions (Q. 118-122): Study the following information carefully to answer that follow.

A bank has five different types of accounts, viz Savings Account, Recurring Account, NRI Account, Current Account and Senior Citizenship Account. The total number of account holders is 2050. 24% of the total accounts are Savings Accounts. One-fifth of the total number of accounts is Current Account.

16% of the total accounts are NRI Accounts. Remaining accounts are either Senior Citizenship Accounts or Recurring Accounts. The number of Recurring Accounts is 182 more than the number of Senior Citizenship Accounts.

118. What is the ratio of the total number of Current Accounts to the total number of Senior Citizenship and Recurring Accounts together?
(1) 2 : 1 (2) 1 : 2 (3) 3 : 4 (4) 7 : 6 (5) None of these

119. If 20% of Current Accounts are non-operative, what is the number of Current Accounts which are operative?
(1) 382 (2) 164 (3) 328 (4) 428 (5) 82

120. The number of NRI accounts is approximately what per cent of the total number of Savings Accounts and Current Accounts together?
(1) 63% (2) 26% (3) 46% (4) 56% (5) 36%

121. What is the total number of Senior Citizenship, NRI and Current Accounts together?
(1) 1027 (2) 1157- (3) 1057 (4) 957 (5) 1257

122. What is the difference between the total number of Senior Citizenship and Savings Accounts together and the number of Recurring Accounts?
(1) 310 (2) 410 (3) 210 (4) 390 (5) 610

SHORT ANSWER

1. (2)	2. (1)	3. (4)	4. (2)	5. (5)	6. (3)	7. (4)	8. (5)
9. (2)	10. (3)	11. (3)	12. (3)	13. (1)	14. (4)	15. (3)	16. (3)
17. (2)	18. (5)	19. (5)	20. (2)	21. (2)	22. (3)	23. (4)	24. (1)
25. (4)	26. (2)	27. (1)	28. (2)	29. (3)	30. (5)	31. (2)	32. (4)
33. (1)	34. (2)	35. (3)	36. (3)	37. (5)	38. (5)	39. (2)	40. (1)
41. (3)	42. (2)	43. (4)	44. (1)	45. (5)	46. (1)	47. (4)	48. (2)
49. (3)	50. (3)	51. (2)	52. (3)	53. (4)	54. (3)	55. (2)	56. (2)
57. (1)	58. (2)	59. (5)	60. (4)	61. (1)	62. (2)	63. (4)	64. (2)
65. (2)	66. (4)	67. (1)	68. (5)	69. (3)	70. (4)	71. (2)	72. (2)
73. (4)	74. (5)	75. (1)	76. (2)	77. (2)	78. (3)	79. (3)	80. (1)
81. (5)	82. (2)	83. (4)	84. (5)	85. (3)	86. (2)	87. (3)	88. (4)
89. (1)	90. (5)	91. (4)	92. (4)	93. (5)	94. (4)	95. (3)	96. (4)
97. (4)	98. (4)	99. (3)	100. (2)	101. (4)	102. (1)	103. (3)	104. (4)
105. (5)	106. (3)	107. (5)	108. (2)	109. (2)	110. (4)	111. (2)	112. (3)
113. (1)	114. (2)	115. (3)	116. (5)	117. (1)	118. (2)	119. (3)	120. (5)
121. (3)	122. (1)						

DETAIL - EXPLANATIONS

1. 2; $\frac{2x}{15}+\frac{3x}{20}+\frac{x}{5}+\frac{2x}{15}+\frac{x}{20}+\frac{x}{4}+K=x$

$\therefore \frac{55x}{60}+K=x,$

$\therefore K=x-\frac{55x}{60}=\frac{5x}{60}=\frac{x}{12}$

2. 1; Chemistry = $\frac{x}{5}+\frac{3x}{20}+\frac{x}{20}+\frac{x}{12}=\frac{29x}{60}$

Physics = $\frac{2x}{15}+\frac{3x}{20}+\frac{x}{20}+\frac{2x}{15}=\frac{7x}{15}$

$\therefore$ Diff = $\frac{29x}{60}-\frac{7x}{15}=\frac{29x-28x}{60}=\frac{x}{60}$

$\because \frac{x}{20}=15 \qquad \therefore x=300$

So'diff = $\frac{x}{60}=\frac{300}{60}=5$

3. 4; Students who passed in exactly one paper

$=\frac{2x}{15}+\frac{x}{5}+\frac{x}{4}=\frac{7x}{12}$

Students who passed in exactly two papers

$=\frac{3x}{20}+\frac{2x}{15}+\frac{x}{12}=\frac{11x}{30}$

$\because \frac{11x}{30}=110 \qquad \therefore x=300$

So, $\frac{7x}{12}=175$

4. 2; Students who passed in at least two papers

$=\frac{3x}{20}+\frac{2x}{15}+\frac{x}{12}+\frac{x}{20}=\frac{5x}{12}$

Total number of students = x

$\therefore$ Reqd% = $\frac{5x/12}{x}\times 100=\frac{125}{3}=41\frac{2}{3}\%$

5. 5; Students who passed in only Maths = $\frac{x}{4}$.

Students who passed in all three papers

$=\frac{x}{20}$

$\therefore$ Reqd % = $\frac{\left(\frac{x}{4}-\frac{x}{20}\right)}{\frac{x}{20}}\times 100=\frac{4x}{20}\times\frac{20}{x}\times 100$

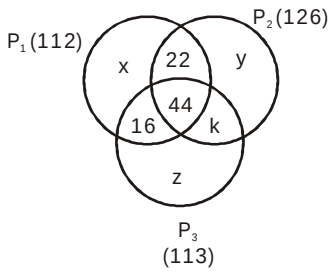
$=400\%$

(6 - 10):

	Train A (700)	Train B (910)
General coaches	140	241
Sleeper coaches	161	273
First class	224	91
AC coaches	175	205

6. 3 7. 4 8. 5 9. 2 10. 3

(11 - 15):



P_1 = 56% of 200 = 112
P_2 = 63% of 200 = 126
P_3 = 56.5% of 200 = 113
$P_1 + P_2$ = 11% of 200 = 22
$P_1 + P_3$ = 8% of 200 = 16
$P_I + P_2 + P_3$ = 22% of 200 = 44
x + 22 + 16 + 44 = 112
or, x = 112 - 82 = 30
22 + 44 + y + k = 126
or, y + k = 60 ... (1)
z + k + 16 + 44 = 113
or, z + k = 53 ... (2)
30 + 22 + y + 16 + 44 + k + z = 200
or, y + z + k = 88 ... (3)
From eqn (1), (2) and (3),
k = 25, y = 35, z = 28

11. 3
12. 3; Only P_2 = 35 and only P_3 = 28

$\therefore$ Ratio $= \frac{35}{28} = \frac{5}{4}$

13. 1; Only $P_1 = 30$ Only $P_1 + P_3 = 16$

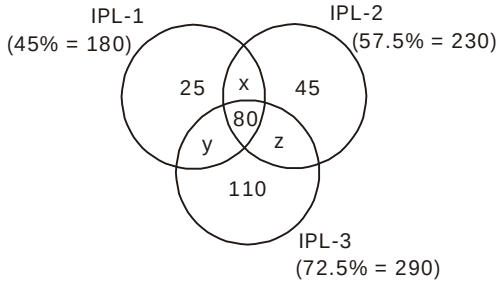
$\therefore$ Reqd % $= \frac{30}{16} \times 100 = 187.5\%$

14. 4; At most one paper = 30 + 35 + 28 = 93

$\therefore$ Reqd % $= \frac{93}{200} \times 100 = 46.5\%$

15. 3; $P_3 = 16 + 44 + 25 + 28$
$= 113$
Only $P_3 = 28$
$\therefore$ Difference = 113 - 28 = 85

(16 - 20):



$x + y + 80 + 25 = 180$
or $x + y = 75$ ... (I)
$y + z + 80 + 110 = 290$
or $y + z = 100$ ... (II)
$x + z + 80 + 45 = 230$
or $x + z = 105$ ... (III)
From eqn (I), (II) and (III),
$x = 40, y = 35, z = 65$

16. 3; 40

17. 2; Reqd percentage $= \frac{65}{400} \times 100 = 16.25\%$

18. 5; At least two IPL = 40 + 35 + 65 + 80 = 220

Reqd % $= \frac{220}{400} \times 100 = 55\%$

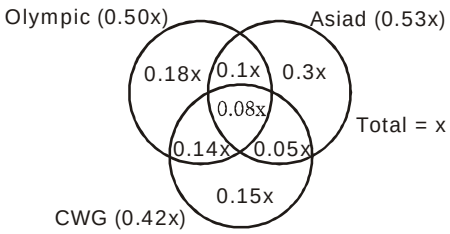
19. 5; Only IPL 1 = 25, Only IPL 2 = 45
$\therefore$ Total = 70

Reqd % $= \frac{70}{80} \times 100 = 87.5\%$

20. 2; At least on IPL = 400
At most one IPL = 25 + 45 + 110 = 180

$\therefore$ Reqd % $= \frac{400-180}{400} \times 100 = \frac{22000}{400} = 55\%$

(21-25):



21. 2 22. 3 23. 4 24. 1 25. 4

26. 2;

27. 1; Ratio $= \frac{9600}{14400} = \frac{2}{3}$ ie 2 : 3

28. 2; Average number of type I items produced

$= \frac{7200 + 12600 + 9600}{3}$

$= \frac{29400}{3} = 9800$

29. 3; Type II produced by C = 14400
$Total_c = 24000$

Reqd % $= \frac{14400}{24000} \times 100 = 60\%$ of no. of items by C

30. 5; Type I = 7200 +12600 + 9600 = 29400
Type II = 8000 + 10200 + 14400 = 32600
$\therefore$ Diff = 32600 - 29400 = 3200

31. 2; 15 + 35 + 38 + 16 = 104

32. 4; People who like exactly two types of sports
= 30 + 20 + 45 + 35 = 130
Total number of people surveyed = 500

$\therefore$ % $= \frac{130}{500} \times 100 = 26\%$

33. 1; Difference = (71 + 49) - (55 + 64)
= 120 - 119 = 1

34. 2; Reqd ratio $= \frac{236}{64} = \frac{59}{64}$ ie 59 : 16

35. 3; People who like exactly one sports
= 71 + 55 + 49 + 64 = 239
Total number of people = 500

$\therefore \% = \frac{239}{500} \times 100 = 47.8\%$

(36-40):

Area of hall 23 × 29 = 667 m^2
Area of branch manager room

$\rightarrow 13 \times 17 = 221m^2$

Area of pantry = $14 \times 13 = 182m^2$

Area of record keeping $\Rightarrow 21 \times 13 = 273m^2$

Area of locker = $29 \times 21 = 609m^2$

Area of flooring area = $1952\ m^2$

Cost of wooden flooring = Rs 170 per sq. m

Cost of marble flooring = Rs. 190 per sq. m

36. 3; Total flooring area with marble
= locker area + record keeping + pantry
= 182 + 273 + 609 - 1064 sqm
Cost of flooring = 1064 × 190
Total flooring area with wooden
= Branch Manager room + Hall
= 221 + 667 = 888 sqm
Cost of flooring = 888 × 170
Ratio = 888 × 170 : 1064 × 190
= 888 × 17 : 1064 × 19
= 15096 : 20216
= 1887 : 2527

37. 5; Cost of flooring of branch manager room = 221 × 170 = Rs 37570
Cost Of pamting = [2(17 × 12 + 13 × 12) + 13 × 17} × 190
= [2(204 + 156) + 221] × 190 = (2 × 360 + 221) × 190
= (720 + 221) × 190 = 941 × 190 = Rs 178790
Total cost = 178790 + 37570 = Rs 216360

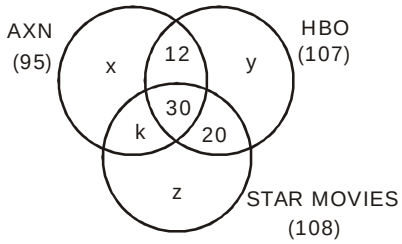
38. 5; total area of bank = 2000 sq m
Total flooring area = 1952 sq fn
Remaining area = 2000 - 1952 = 48 sq m
$\therefore$ Cost of carpeting = 48 × 110 = Rs 5280

39. 2; Area not to be renovated = 48 sq m

$$\therefore \text{Reqd\%} = \frac{48}{2000} \times 100 = 2.4\%$$

40. 1; Cost of renovation of hall + locker area
= 667 × 170 + 609 × 190
= 113390 + 115710 = Rs 229100

(41-45)



y = 107 - 12 - 30 - 20 = 45
x + k + 12 + 30 = 95
$\therefore$ x + k = 53 ...(I)
z + k + 30 + 20 = 108
z + k = 58 ...(II)
x+12 + y + k + 30 + 20 + z = 200
$\therefore$ x + z + k = 93 ...(III)
From eqn (I), (II) and (III)
x = 35, z = 40, k = 18

41. 3

42. 2; Only AXN = 35
Only HBO = 45
$\therefore$ Sum = 80

43. 4; Only Star Movies = 40

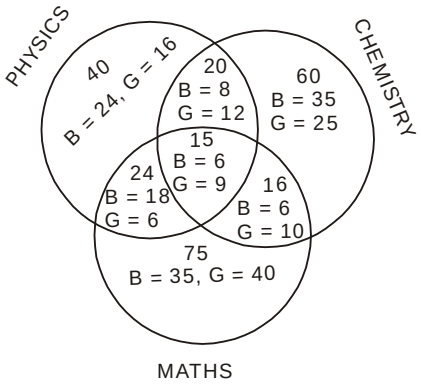
$$\therefore \text{Reqd\%} = \frac{40}{200} \times 100 = 20\%$$

44. 1

45. 5; At least two channels = 80
Exactly one channel = 120

$$\therefore \text{Reqd\%} = \frac{80}{120} \times 100 = \frac{200}{3} = 66\frac{2}{3}\%$$

(46-50) :



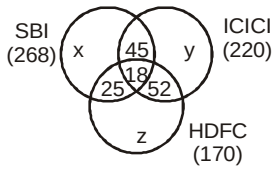
Total = 250
B = 32, G = 118

46. 1 47. 4 48. 2 49. 3

50. 3; Boys who passed only in Chemistry = 35
Girls who passed only in Maths = 40 And girls who passed only in Physics =16

$$\therefore \text{Reqd \%} = \frac{35}{56} \times 100 = 62.5\%$$

(51-55) :



Total number of persons = 500
Number of persons having account in SBI

$$= 53.6 \times \frac{500}{100} = 268$$

Number of persons having account in ICICI

$= 44 \times \frac{500}{100} = 220$

Number of persons having account in

HDFC = $34 \times \frac{500}{100} = 170$

∴ Persons having account in SBI and HDFC

$= 5 \times \frac{500}{100} = 25$

∴ Persons having account in SBI and ICICI

$= 180 \times \frac{25}{100} = 45$

∴ Persons having account in all the three banks

$= 3.6 \times \frac{500}{100} = 18$

∴ Persons having account in HDFC and ICICI

$= 10.4 \times \frac{500}{100} = 52$

x + y + z = 500 - (45 + 18 + 25 + 52)
= 500 - 140 = 360
x = 268 - (45 + 25 + 18) = 180
y = 220-(45 + 18 + 52) = 105
z = 170 - (25 +18 + 52) = 75

51. 2 52. 3 53. 4 54. 3

55. 2; No. of employees having account in at least two banks = 45 + 25 + 18 + 52 = 140
No. of employees having account in atmost two banks = 500 - 18 = 482

∴ Reqd % = $\frac{140}{482} \times 100 = 29\%$

(56-60) :

No of Mobile phones $\Rightarrow \frac{1650 \times 24}{100} = 396$

No of pen drives = $1650 \times \frac{1}{6} = 275$

No of calculators = $1650 \times \frac{14}{100} = 231$

No of Televisions + washing machines = 748
No of washing machines = T + 50
T + W = 748
T + T + W = 748
2T = 748 – 50 = 698

$\therefore T = \frac{698}{2} = 349$

washing machines 349 + 50 = 399

56. 2; Ratio = $\frac{399}{231} = \frac{133}{77} = \frac{19}{11}$ = 19 : 11

57. 1; Number of pen drives which are not defective

$= 275 - 275 \times \frac{24}{100} = 275 - 66 = 209$

58. 2; Required %

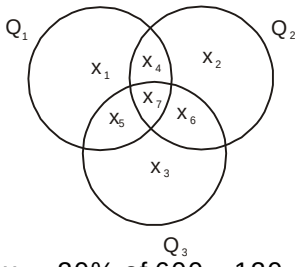
$= \frac{349}{231 + 399} \times 100 = \frac{349}{640} \times 100$

= 54.53 = 55%

59. 5; Difference = 349 + 396 - 231
= 745 - 231 = 514

60. 4; Total number of pen drives, calculators and washing machines
= 275 + 231 + 399 = 905

(61-65) :



x_1 = 20% of 600 = 120

$x_2 = \frac{1}{4} \times 600 = 150$

$x_7 = \frac{1}{24} \times 600 = 25$

x_5 = 25 + 20 = 45

x_6 = 25% of x4 = $\frac{x_4}{4}$

or $x_3 + x_5 + x_6 + x_7 = 2 \times x_1 = 240$
or $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 600 - 10 = 590$
or $510 - x_6 + x_4 + x_6 = 590$
∴ x_4 = 590 – 510 = 80

$x_6 = \frac{80}{4} = 20$ and $x_3 = 240 - (45 + 25 + 20) = 150$

61. 1

62. 2

63. 4; Total number of people who answered Q_1 = 270

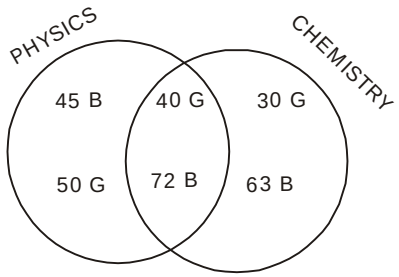
∴ Reqd % = $\frac{270}{600} \times 100 = 45\%$

64. 2; Number of people who answered at most one question
= 600 - 10 = 590
Number of people who most one question
= 120 + 150 + 150 + 10 = 430

Difference = 590 - 430 = 160

65. 2; Ratio = $\frac{270}{150} = \frac{9}{5} = 9:5$

(66-70) :



Total = 300
Bays : Girls = 3 : 2
Boys = 180, Girls = 120

66. 4

67. 1; Reqd % = $\frac{63}{300} \times 100 = 21\%$

68. 5; Total students who passed in Physics = 45 + 50 + 40 + 72 = 207

69. 3; Ratio = $\frac{72+63}{50} = \frac{135}{50} = \frac{27}{10} = 27:10$

70. 4; Students who passed at most in one subject = 45 + 50 + 30 + 63 = 188

(71-75) :

Staff member = 120, students = 800

Number of teacher = $\frac{120 \times 65}{100}$ = 78

Administrative officer = 120 – 78 = 42

Number of girl students = $\frac{800 \times 45}{100}$ = 360

Number of boy students = 440
Number of girls who speak Hindi

= $\frac{360 \times 20}{100}$ = 72

Number of girls who speak both Hindi and English = 360 – 72 = 288
Number of boys who speak only Hindi

$\Rightarrow 440 \times \frac{3}{4} = 330$

∴ Number of boys who speak Hindi and English = 440 – 330 = 110

∴ Number of male teachers = $78 \times \frac{2}{3}$ = 52

71. 2; Difference = 288 - 110 = 178

72. 2; Required percentage

$= \frac{360}{120} \times 100 = 300\%$

73. 4; Female administrative officers

$= \frac{5}{14} \times 42 = 15$

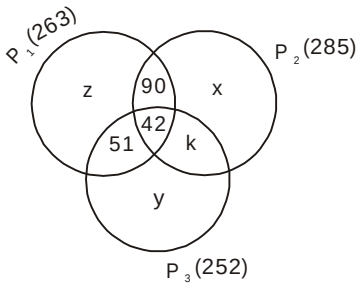
Male teachers = $= \frac{2}{3} \times 78 = 52$

Female teachers = 78 - 52 = 26
Male administrative officers
= 42 - 15 = 27
∴ Difference = 26 + 15 - 27 = 14

74. 5; Ratio = 78 : 330
= 39 : 165 = 13 : 55

75. 1; 27 + 26 + 72 = 125

(76-80) :



z = 263 – (90 + 51 + 42)
263 – 183 = 80
x + k = 285 – (90 + 42) = 153 .....(i)
y + k = 252 – (51 + 42) = 159 .....(ii)
x + y + k = 500 – (90 + 42 + 51 + 80) = 273 ...(iii)
Solving (i), (ii) and (iii)
x = 78, y = 84, k = 75
Let the number of boys who passed in p_1 only be B The number of girls who passed in p_1 only will be $\frac{3B}{5}$

∴ B + $\frac{3B}{5}$ = 80

∴ B = 50

50 number of girls = $\frac{3B}{5}$ = 30

The number of students who passed only in pz is 78 where B : G = 7 : 6 i.e
B + G = 78 ...(i)
And 6B = 79 solving these two egrs, B = 42
G = 36
Number of girls who passed only in

p_3 = 6% of 500 $\Rightarrow \frac{6}{100} \times 500 = 30$

So the number of boys who passed only in p_3 is 84 – 30 = 54

76. 2 .

77. 2; The number of boys who passed in Paper P_1 = 50
Number of girls who passed in Paper P_2 = 30
$\therefore$ Difference = 50 - 30 = 20

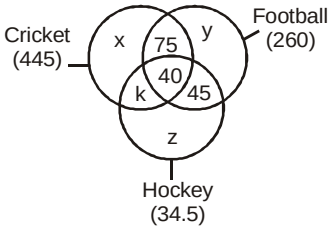
78. 3; $\text{Reqd}\% = \frac{78}{500} \times 100 = 15.6\%$

79. 3; Ratio = $\frac{90}{54} = \frac{5}{3}$ = 5 : 3

80. 1; Required number = 80 + 78 + 84 = 242

$\text{Reqd}\% = \frac{242}{500} \times 100 = 48.4\%$

(81-85):



Cricket = 55.625% of 800 = 445
Football = 32.5% of 800 = 260
Hockey = 43.125% of 800 = 345
All three games = 5% of 800 = 40
Cricket + Football = 9.375% of 800 = 75
Football + Hockey = 5.625% of 800 = 45
y = 260 - (75 + 40 + 45) = 100
x + k = 445 - (75 + 40) = 330
... (i)
z + k = 345 - (40 + 45) = 260
... (ii)
x + z + k = 800 - (75 + 40 + 45 + 100) = 540
... (iii)
eqn (i) + (ii) - eqn (iii)
(x + z + 2k) - (x + z + k) = (330 + 260) - 540
Solving this, k = 590 - 540 = 50
x = 330 - 50 = 280
z = 260 - 50 = 210

81. 5; Reqd % = $\frac{280}{800} \times 100 = 35\%$

82. 2

83. 4; Only Football = 100, Football = 260,

Reqd % = $\frac{100}{260} \times 100 = 38.46 \approx 38\%$

84. 5; Only Hockey = 210, all three = 40

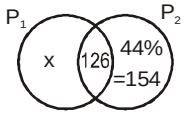
Reqd % = $\frac{210}{40} \times 100 = 525\%$

85. 3; The number of students who like at least two games = 75 + 50 + 45 + 40 = 210

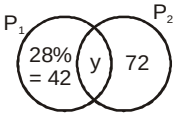
Reqd % = $\frac{210}{800} \times 100 = 26.25\%$

(86-90):

Ratio of Boys to Girls = 7 : 3
Boys = 350
Girls = 150



Boys = 350
$\therefore$ x + 126 + 154 = 350
$\therefore$ x = 350 – 280 = 70



Girls = 150
$\therefore$ y + 72 + 42 = 150
or, y = 150 – 114 = 36

86. 2; Reqd % = $\frac{70}{500} \times 100 = 14\%$

87. 3; Reqd % = $\frac{36}{150} \times 100 = 24\%$

88. 4; Difference = 196 - 108 = 88

89. 1; Total number of students passed in both the papers = 126 + 36 =162

$\therefore$ Reqd % = $\frac{162}{500} \times 100 = 32.4\%$

90. 5; Reqd % = $\frac{280}{500} \times 100 = 56\%$

(91-95):

Level	I	II	III	IV	V	Total
Male	170	153	228	129	170	850
Female	182	102	147	117	102	650

91. 4; Total number of males working at level I and III together = 170 + 228 = 398

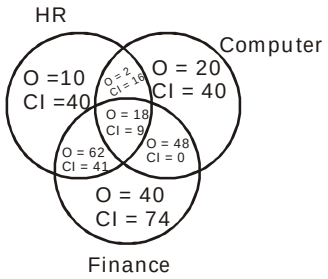
92. 4; Reqd % = $\frac{102}{1500} \times 100 = 6.8\% \approx 7\%$

93. 5

94. 4; Total number of females working a level I and V together = 182 + 102 = 284

95. 3; Reqd ratio = $\frac{170}{147}$ = 170 : 147

(96 - 100):



Total employees = 450

Number of officers = 200

Number of clerks = 250

96. 4; Required number of employees = 74 + 40 + 48 = 162

97. 4; Required number of officers = 20 + 48 + 2+ 18 = 88

98. 4; Required number of clerks = 40 + 41 = 81

99. 3; Required number = 16 + 2 = 18

100. 2; Required number = 9

(101-105):

Area of hall = 33 × 39 = 1287m^2

Area of Director's room = 13 × 12 = 156m^2

Area of record keeping-cum-server room = 23 × 13

= 299m^2

Area of pantry = 14 × 12 = 168 m^2

Area of accounts room = 12 × 23 = 276m^2

∴ Total area to be floored = 2186m^2

Cost of wooden flooring = `170 per sq m

Cost of marble flooring = `190 per sq m

101. 4; Total flooring area with wood = 1287 + 156 = 1443 sqm

Cost of flooring area with wood

= 1443 × 170

= `245310

Total flooring area with marble = accounts room + record keeping cum-server room + pantry

Cost of flooring area with marble = 743 × 190 = `141170

Reqd ratio = $\frac{245310}{141170} = \frac{24531}{14117}$

= 24531 : 14117

102. 1; Area of wall = 2(15 × 13 + 15 × 12) = 750

Area of director's room = 13 × 12 = 156

Cost of painting = 190 × (750 + 156) = `172140

Cost of flooring = 170 × 156 = ` 26520

∴ Total cost = 172410 + 26520 = ` 198660

103. 3; Total area of the institute = 2500 sq m

∴ Remaining area = 2500 - 2186 = 314 sq m

Cost of renovation of the remaining area

= 314 × 210 = ` 65940

104. 4; Area not to be renovated = 314 sq m

∴ Reqd % = $\frac{3.14}{2500} \times 100 = 12.56$

105. 5; Cost of renovation of the hall = 1287 × 170 = 21879.0

Cost of renovating the accounts room

= 276 × 190 = 52440

Total cost = 218790 + 52440 = ` 271230

(106-108):

Total number of members (males + females) = 240

Number of males = 240 × $\frac{2}{3}$ = 160

Number of males who are graduates

= 160 × $\frac{15}{100}$ = 240

Number of males who are non graduates

= 160 – 24 = 136

Number of females = 240 – 160 = 80

Now the number of females who are graduates = 80 × $\frac{3}{4}$ = 60

Number of females who are non graduates

= 80 – 60 = 20

106.3; ∴ Difference = 24 - 20 = 4

107.5; Reqd sum = 60 + 136 = 196

108. 2; Reqd ratio = $\frac{160}{20}$ = 8 : 1

(109-112):

In Zone A, the estimated voter turnout is 17500 and the minimum number of days required is 2.

In Zone B, the estimated voter turnout is

17400 and the minimum number of days required for campaigning is 3.

In Zone C, estimated voter turnout is 28000 and the minimum number of days required for it campaigning 4.

In Zone D, the estimated voter turn out is 29400 and the minimum number of days required for campaigning is 5.

In Zone E, the estimated voter turn out is 18000 and the minimum number of days for campaigning is 3.

In Zone F, the estimated voter turnout is 10500 and the minimum number of days required is 3.

In Zone G, the estimated voter turn out 18200 and the minimum number of days campaigning is 3.

109. 2; Total number of days required for campaigning in all the zones = 2 + 3 + 4+ 5 + 3 + 3 + 3 = 23. In 20 days the candiate can campaign in at most 6 zones. The zone in which he would not campaign is zone F, which requires 3 days to campaign and has the least estimated voter turnout'.

$\therefore$ The maximum number of voter population that the candidate can meet in 20 days

= 20000 + 24000 + 35000 + 42000 + 30000 + 28000 = 179000

110. 4; Since the candidate has to ensure that the total voter turnout is the maximum, he will not campaign in Zone D and F. Here the estimated voter turnout is less and the time taken to campaign is more for both the zones.

111. 2; He would choose only those zones in which the voter turnout is the maximum. From the above table, the sum of the voter turnout in the zones C, D, E, and G is the maximum

112. 3; From the above table, only statement II is true.

(113 - 117):

	University U_1 (1400)	University U_2 (1960)
Physics	280	588
Chemistry	280	882
Mathematics	350	294
Geography	490	196

113. 1; Total amount paid by University U_2 to the staff of Mathematics Department

= 294 × 9500 = 2793000 = ` 27.93 lakh

114. 2; Total number of staff in Geography Deptt in University U_1 and U_2 together = 686

$\therefore$ Reqd% $= \frac{686}{1400} \times 100 = 49\%$

115. 3; Total number of staff in Chemistry Department of University U_2 = 882

Total number of staff in Physics and Chemistry Departments in University U_1 and U_2 together = 280 + 280 = 560

$\therefore$ Difference = 882 - 560 = 322

116. 5; Reqd ratio = $\frac{196}{350} = \frac{14}{25} = 14 : 25$

117. 1; Reqd % $= \frac{280}{1960} \times 100 = 14.28 \approx 14\%$

(118-122):

Number of Savings Aecounts

$= \frac{24 \times 2050}{100} = 492$

Number of Current Accounts = 2050 x $\frac{1}{5}$

= 410

Number of NRI Accounts = $\frac{16 \times 2050}{100} = 328$

Number of Senior Citizenship and Recurring Accounts = 820.

Number of Recurring Accounts = No. of Senior Citizenship Aecounts + 182

Senior Citizenship + Recurring Accounts = 820

or Senior Citizenship + Senior Citizenship + 182 = 820

or, 2 Senior Citizenship = 820 - 182 = 638

$\therefore$ Senior Citizenship $= \frac{638}{2} = 319$

$\therefore$ Recurring Account = 319 + 182 = 501

118. 2; Reqd ratio = $\frac{410}{820} = 1 : 2$

119. 3; Number of non - operative accounts

$= \frac{410 \times 20}{100} = 82$

$\therefore$ Number of accounts which are operative = 410 - 82 = 328

120. 5; Reqd % = $\frac{328}{902} \times 100 = 36.36 \approx 36\%$

121. 3; Total number of Senior Citizenship, NRI and Current Accounts = 328 + 319 + 410 = 1057

122. 1; Difference = 319 + 492 - 501

= 811 - 501 = 310

MENSURATION

NATIONALISED BANKS & IBPS SO/MT/SO

1. The area of a rectangular field is 460 square metres. If the length is 15 per cent more than the breadth, what is breadth of the rectangular field ?
(1) 15 metres
(2) 26 metres
(3) 34.5 metres
(4) Cannot be determined
(5) None of these

(Canara Bank PO Exam. 09.02.2003)

2. What will be the cost of gardening 1 metre broad boundary around a rectangular plot having perimeter of 340 metres at the rate of Rs. 10 per square metre ?
(1) Rs. 3,400/-
(2) Rs. 1,700/-
(3) Rs. 3,440/-
(4) Cannot be determined
(5) None of these

(Cneara Bank PO Exam. 09.02.2003)

3. What would be the cost of building a 7 metres wide garden around a circular field with diameter equal to 280 metres if the cost per sq. metre for building the garden is Rs. 21 ?
(1) Rs. 1,56,242
(2) Rs. 2,48,521
(3) Rs. 1,11,624
(4) Rs. 2,06,118 .
(5) None of these

(Corporation Bank PO Exam. 29.07.2006)

4. The ratio of length and breadth of a rectangular plot is 8 : 5 respectively. If the breadth is 60 metre less than the length, what is the perimeter of the rectangular plot ?
(1) 260 metres
(2) 1600 metres
(3) 500 metres
(4) Cannot be determined
(5) None of these

(Bank Of Maharashtra PO Exam. 25.05.2008)

5. What will be the area (in square metres) of 1.5 metre wide garden developed around all the four sides of a rectangular field having area equal to 300 square metres and breadth equal to three-fourth of the length ?
(1) 96
(2) 105
(3) 114
(4) Cannot be determined
(5) None of these

(Indian Overseas Bank PO Exam. 15.06.2008)

6. The circumference of a circle is equal to the side of a square whose area measures 407044 sq. cms. What is the area of the circle ?
(1) 22583.2 sq. cms.
(2) 32378.5 sq. cms.
(3) 41263.5 sq. cms.
(4) 39483.4 sq. cms.
(5) Cannot be determined

(PNB Agriculture Officer Exam. 04.01.2009)

7. What is the area of a circle whose circumference is 1047.2 metres ?
(1) 87231.76 sq.metres
(2) 85142.28 sq.metres
(3) 79943.82 sq.metres
(4) 78621.47 sq.metres
(5) 69843.23 sq.metres

(UCO Bank PO Exam. 22.03.2009)

8. An order was placed for supply of carpet of breadth 3 metres, the length of carpet was 1.44 times of breadth. Subsequently the breadth and length were increased by 25 and 40 per cent respectively. At the rate of Rs. 45 per square metre, what would be the increase in the cost of the carpet ?
(1) Rs. 1020.6 (2) Rs. 398.8
(3) Rs. 437.4 (4) Rs. 583.2
(5) None of these

(Indian Overseas Bank PO Exam. 05.04.2009)

9. The length of a rectangular plot is thrice its breadth. If the area of the rectangular plot is 7803 sq. metre, what is the breadth of the rectangular plot ?
(1) 51 metres (2) 153 metres
(3) 104 metres (4) 88 metres
(5) None of these

(Indian Overseas Bank PO Exam. 05.04.2009)

10. The cost of fencing a circular plot at the rate of Rs. 15 per metre is Rs. 3,300. What will be the cost of flooring the plot at the rate of Rs. 100 per square metre ?
(1) Rs. 3,85,000
(2) Rs. 2,20,000
(3) Rs. 3,50,000
(4) Cannot be determined
(5) None of these

(United Bank of India PO Exam. 21.06.2009)

11. If the area of a circle is 616 cm.^{2}, what is its circumference ?
(1) 76 cm. (2) 84 cm.
(3) 96 cm. (4) 80 cm.
(5) None of these

(Andhra Bank PO Exam. 05.07.2009)

12. The length of a rectangular floor is twice its breadth. If Rs. 256 is required to paint the floor at the rate of Rs. 2 per square metres, then what would be the length of floor ?
(1) 16 metres (2) 8 metres
(3) 12 metres (4) 32 metres
(5) 20 metres

(Corporation Bank PO Exam. 22.11.2009)

13. The circumference of two circles is 132 metres and 176 metres respectively. What is the difference between the area of the larger circle and the smaller circle ?
(1) 1048 sq. metres
(2) 1076 sq. metres
(3) 1078 sq. metres
(4) 1090 sq. metres
(5) None of these

(Indian Bank PO Exam. 17.10.2010)

14. What would be the cost of building a fence around a circular field with area equal to 32378.5 sq. metres, if the price per metre for building the fence was Rs. 154 ?
(1) Rs. 84,683
(2) Rs. 86,495
(3) Rs. 79,326
(4) Rs. 98,252
(5) None of these

(Bank Of India Banking Officer Exam. 24.01.2010)

15. The perimeter of a square is equal to twice the perimeter of a rectangle of length 8 cms. and breadth 7 cms. What is the circumference of a semicircle whose diameter is equal to the side of the square ? (Rounded off to the two decimal places)
(1) 38.57 cms.
(2) 23.57 cms.
(3) 42.46 cms.
(4) 47.47 cms.
(5) None of these

(Punjab & Sind Bank PO Exam. 16.05.2010)

16. The perimeter of a square is double the perimeter of a rectangle. The area of the rectangle is 240 sq. cm. What is the area of the square ?
(1) 100 sq. cm.
(2) 36 sq. cm.
(3) 81 sq. cm.
(4) Cannot be determined
(5) None of these

(Punjab & Sind Bank PO Exam. 16.05.2010)

17. The total area of a circle and a square is equal to 5450 sq.cm. The diameter of the circle is 70 cms. What is the sum of the circumference of the circle and the perimeter of the square ?
(1) 360 cm
(2) 380 cm
(3) 270 cm
(4) Cannot be determined
(5) None of these

(Bank Of Baroda PO Exam. 30.05.2010)

18. The area of a square is 1024 sq. cm. What is the respective ratio between the length and the breadth of a rectangle whose length is twice the side of the square and breadth is 12 cm. less than the side of the square ?
(1) 5 : 18 (2) 16 : 7
(3) 14 : 5 (4) 32 : 5
(5) None of these

(Central Bank Of India PO Exam. 25.07.2010)

19. There are two circles of different raddi. The area of a square is 196 sq.cm, whose side is half the radius of the larger circle. The radius of the smaller circle is three-seventh that of the larger circle. What is the circumference of the smaller circle ?
(1) 12π cm (2) 16π cm
(3) 24π cm (4) 32π cm
(5) None of these

(Syndicate Bank PO Exam. 29.08.2010)

20. The cost of building a fence around a circular field is Rs 7,700 at the rate of Rs 14 per foot. What is the area of the circular field ?
(1) 24062.5 sq.ft.
(2) 23864.4 sq.ft.
(3) 24644.5 sq.ft.
(4) Cannot be determined
(5) None of these

(United Bank Of India PO Exam. 14.11.2010)

21. The total area of a circle and a square is equal to 2611 sq.cm. The diameter of the circle is 42 cms. What is the sum of the circumference of the circle and the perimeter of the square ?
(1) 272 cms.
(2) 380 cms.
(3) 280 cms.

(4) Cannot be determined
(5) None of these

(Bank Of Maharashtra Exam. 19.12.2010)

22. The area of a square is 196 sq. cm. whose side is **half** the radius of a circle. The circumference of the circle is equal to breadth of a rectangle. If perimeter of the rectangle is 712 cm, what is the length of the rectangle ?
(1) 196 cm (2) 186 cm
(3) 180 cm (4) 190 cm
(5) None of these

(Oriental Bank Of Commerce PO Exam. 26.12.2010 (1st Sitting)

23. What would be the cost of building a fence around a square plot with area 462.25 sq. ft. at the rate of Rs 34 per foot ?
(1) Rs 2,924
(2) Rs 2,682
(3) Rs 2,846
(4) Cannot be determined
(5) None of these

(Indian Bank PO Exam. 02.01.2011)

24. The length of a rectangle is twice the diameter of a circle. The circumference of the circle is equal to the area of a square of side 22 cm. What is the breadth of the rectangle if its perimeter is 688 cm ?
(1) 24 cm
(2) 26 cm
(3) 52 cm
(4) Cannot be determined
(5) None of these

(Union Bank Of India PO Exam. 09.01.2001)

25. The circumference of two circles is 88 metres and 220 metres respectively. What is the difference between the area of the larger circle and the smaller circle ?
(1) 3422 sq. metre
(2) 3242 sq. metre
(3) 3244 sq. metre
(4) 3424 sq. metre
(5) None of these

(Corporation Bank PO Exam. 16.01.201l)

26. What is the area of a circle whose radius is equal to the side of a square whose perimeter is 112 metres ?
(1) 176 sq. m. (2) 2504 sq. m.
(3) 284 sq. m. (4) 1956 sc m.
(5) None of these

(Punjab & Sind Bank PO Exam. 23.01.2011)

27. The circumference of a semicircle of area 1925 sq. cm is equal to the breadth of a rectangle. If the length of the rectangle is equal to the perimeter of a square of side 48 cm. What is the perimeter of the rectangle ?
(1) 734 cm
(2) 754 cm
(3) 745 cm
(4) Cannot be determined
(5) None of these

(UCO Bank PO Exam. 30.01.2011)

28. The total area of a circle and a rectangle is equal to 1166 sq.cm. The diameter of the circle is 28 cms. What is the sum of the circumference of the circle and the perimeter of the rectangle if the length of the rectangle is 25 cm ?
(1) 186 cm
(2) 182 cm
(3) 184 cm
(4) Cannot be determined
(5) None of these

(Bank Of Baroda PO Exam. 13.03.2011)

29. The sum of the circumference of a circle and the perimeter of a square is equal to 272 cm. The diameter of the circle is 56 cm. What is the sum of the area of the circle and the area of the square ?
(1) 2464 sq. cm.
(2) 2644 sq. cm.
(3) 3040 sq. cm.
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 17.04.2011)

30. The length of a rectangle is three-fifth of the side of a square. The radius of a circle is equal to side of the square. The circumference of the circle is 132 cm. What is the area of the rectangle if the breadth of the rectangle is 8 cm ?
(1) 112.4 sq. cm.
(2) 104.2 sq. cm.
(3) 100.8 sq. cm.
(4) Cannot be determined
(5) None of these

(Indian Overseas Bank PO Exam. 22.05.2011)

31. Smallest side of a right angled triangle is 8 cm less than the side of a square of perimeter 56 cm. Second largest side of the right angled triangle is 4 cm less than the length of rectangle of area 96 sq. cm. and breadth 8 cm. What is the largest side of the right angled triangle ?

(1) 20 cm (2) 12 cm
(3) 10 cm (4) 15 cm
(5) None of these

(Indian Overseas Bank PO Exam. 22.05.2011)

32. The area of a square is 1444 square metre. The breadth of a rectangle is $\frac{1}{4}$th of the side of the square and the length of the rectangle is thrice the breadth. What is the difference between the area of the square and the area of the rectangle ?
(1) 1152.38 sq.mtr.
(2) 1169.33 sq.mtr.
(3) 1181.21 sq.mtr.
(4) 1173.25 sq.mtr.
(5) None of these

(IBPS Bank PO/MT CWE 17.06.2012)

33. Circumference of a circle-A is $1\frac{4}{7}$ times perimeter of a square. Area of the square is 784 sq cm. What is the area of another circle-B whose diameter is half the radius of the circle-A ?
(1) 38.5 sq.cm
(2) 156 sq.cm
(3) 35.8 sq.cm
(4) 616 sq.cm
(5) None of these

(IDBI Bank Officer Exam. 16.09.2012)

34. A horse is tethered to a peg with a 14 metre long rope at the corner of a 40 metre long and 24 metre wide rectangular grassfield. What area of the field will the horse graze ?
(1) 154 m^2 (2) 308 m^2
(3) 240 m^2 (4) 480 m^2
(5) None of these

(IBPS Specialist Officer CWE 17.03.2013)

35. The area of circle is seven times the numerical value of its circumference. What is the circumference of the circle ?
(1) 616 units
(2) 132 units
(3) 88 units
(4) Cannot be determined
(5) None of these

(IBPS Specialist Officer CWE 17.03.2013)

36. Length of the floor of a rectangular auditorium is 6 metre more than the radius of a circle with a circumference of 572 m. The perimeter of the floor of the rectangular auditorium is 356 m. What will be cost of flooring the auditorium (only the floor of the auditorium), if the cost of flooring is Rs. 12/ m^2 ?
(1) Rs. 87,954
(2) Rs. 91,236
(3) Rs. 94,284
(4) Rs. 75,490
(5) None of these

(IBPS RRBs Office Assistant CWE Exam. 09.09.2012)

37. The circumference of a circular playground is 308 metre. There is 7 metre wide path around the ground. The area of the path is
(1) 2130 sq. metre
(2) 2410 sq. metre
(3) 2510 sq. metre
(4) 2310 sq. metre
(5) None of these

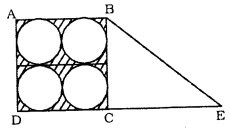
(Indian Overseas Bank PO Online Exam. 01.09.2013)

38. The edge of an ice cube is 14 cm. The volume of the largest cylindrical ice cube that can be formed out of it is
(1) 2200 cu.cm
(2) 2000 cu.cm
(3) 2156 cu.cm
(4) 2400 cu.cm
(5) None of these

(IBPS Bank PO/MT CWE-III 26.10.2013)

Directions (39-40) : Study the following diagram to answer the questions.

(IBPS Bank PO/MT CWE-III 26.10.2013)



39. If the diameter of each circle is 14cm and DC = CE, the area of $\triangle$BDE is
(1) 784 sq.cm (2) 748 sq.cm
(3) 874 sq.cm (4) 441 sq.cm
(5) None of these

40. The area of the shaded region of square ABCD is
(1) 186 sq. cm (2) 168 sq. cm
(3) 188 sq. cm (4) 198 sq. cm
(5) None of these

SBI PO EXAMS

1. If the height of a triangle is decreased 20% and its base is increased by 20% what will be the effect on its area?
(1) No change (2) 8% increase
(3) 12% decrease (4) 16% increase

(5) None of these

(SBI Associate Banks PO Exam. 14.02.1999)

2. A circular ground whose diameter is 35 metres, has a 1.4 metre broad garden around it. What is the area of the garden in square metres ?
(1) 160.16
(2) 6.16
(3) 1122.66
(4) Data inadequate
(5) None of these

(SBI Associate Banks PO Exam. 14.02.1999)

3. 2 metres broad pathway is to be constructed around a rectangular plot. The area of the plot is 96 sq.m. The cost of construction is Rs. 50 per sq. metre. Then find the total cost of production ?
(1) Rs. 4,800 (2) Rs. 4,000
(3) Rs. 2400 (4) Data inadequate
(5) None of these

(SBI Associate Banks PO Exam. 16.07.2000)

4. If length of the rectangle is increased by 50% and breadth is decreased by 20%. Then what is the percentage change in the area ?
(1) by decrease 20%
(2) 20% increase
(3) 80% increase
(4) 30% decrease
(5) None of these

(SBI Associate Banks PO Exam. 16.07.2000)

5. The length and breadth of the floor of a room are 20 feet and 10 feet respectively. Square tiles of 2 feet length of three different colours are to be laid on the floor. Black tiles are laid in the first row on all sides. If white titles are laid in the one-third of the remaining and blue tiles in the rest, how many blue tiles will be there ?
(1) 16 (2) 32
(3) 48 (4) 24
(5) None of these

(SBI Banks PO Exam. 20.08.2000)

6. The area of a rectangular field is 460 square metres. If the length is 15 per cent more than the breadth, what is breadth of the rectangular field ?
(1) 15 metres
(2) 26 metres
(3) 34.5 metres
(4) Cannot be determined
(5) None of these

(SBI Associate Banks PO Exam. 07.01.2007)

7. What will be the cost of gardening 1 metre broad boundary around a rectangular plot having perimeter of 340 metres at the rate of Rs. 10 per square metre ?
(1) Rs. 3,400/-
(2) Rs. 1,700/-
(3) Rs. 3,440/-
(4) Cannot be determined
(5) None of these

(SBI Associate Banks PO Exam. 07.01.2007)

8. The ratio of the length and the breadth of a rectangular plot is 6 : 5 respectively. If the breadth of the plot is 34 metres less than the length, what is the perimeter of the rectangular plot ?
(1) 374 metres (2) 408 metres
(3) 814 metres (4) 748 metres
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

9. The diameter of a circle is equal to the perimeter of a square whose area is 3136 sq. cms. What is the circumference of the circle ?
(1) 352 cms.
(2) 704 cms.
(3) 39424 cms.
(4) Cannot be determined
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

10. If the length of a rectangular field is increased by 20% and the breadth is reduced by 20%, the area of the rectangle will be 192 m^2. What is the area of the original rectangle ?
(1) 184 m^2 (2) 196 m^2
(3) 204 m^2 (4) 225 m^2
(5) None of these

(SBI Associate Banks PO Exam. 07.08.2011)

11. Inside a square plot a circular garden is developed which exactly fits in the square plot and the diameter of the garden is equal to the side of the square plot which is 28 metres. What is the area of the space left out in the square plot after developing the garden ?
(1) 98 m^2 (2) 146 m^2
(3) 84 m^2 (4) 168 m^2
(5) None of these

(SBI Associate Banks PO Exam. 07.08.2007)

RBI GRADE-B OFFICER EXAMS

1. The length of rectangular plot is thrice its breadth. If the area of the rectangular plot is 6075 sq. metres, what is the length ?
(1) 145 metres (2) 130 metres
(3) 75 metres (4) 45 metres
(5) None of these
(RBI Grade-B Officcer Exam 2007)

2. A triangle has two of its angle in the ratio of 1 : 2. If the measure of one of its angles is 30 degrees, what is the measure of the largest angle of the triangle in degrees ?
(1) 100
(2) 90
(3) 135
(4) Cannot be determined
(5) None of these
(RBI Grade-B Officer B Exam.06.02.2011)

INSURANCE EXAMS

1. A wall 3m × 2.7m × 0.2m of concrete weight 320 kg per cubic meter. What will be the total weight of the wall in kg.?
(1) 5184.0 (2) 51.84
(3) 518.4 (4) 51840
(United India Insurance Co. AAO Exam. 21.04.2002)

2. The area of a square is 2.25 cm^2. What is its perimeter ?
(1) 9.0 cms (2) 6.0 cms
(3) 1.5 cms (4) 4.5 cms

3. The length of a rectangle is 20**%** more than its breadth. What will be the ratio of the area of this rectangle to the area of a square whose side is equal to the breadth of the rectangle ?
(1) 5 : 6
(2) 6 : 5
(3) 2 : 1
(4) Data inadequate
(5) None of these
LIC Assistant Administrative Officer (AAO) Exam. 24.04.2005)

4. A well with 14 m inside diameter is dug 10 m deep. Earth taken out of it has been evenly spread all around it to a width of 21 m to form an embankment. The height (in metres) of the embankment is :
(1) $\frac{1}{2}$ (2) $\frac{2}{3}$
(3) $\frac{3}{4}$ (4) $\frac{3}{5}$
(United India Insurance Co. (AAO) Exam. 11.03.2007)

5. A rectangular carpet has an area of 120 sq. metres and a perimeter of 46 metres. The length of its diagonal (in metres) is :
(1) 11 (2) 13
(3) 15 (4) 17
(United India Insurance Co. (AAO) Exam. 11.03.2007)

6. The area of a square of side 8 cm is equal to a rectangle. Which of the following statement/s is/ are definitely true about the rectangle ?
(1) The length of rectangle is 16 times its breadth
(2) The length of rectangle is 32 times its breadth
(3) The breadth of rectangle is $\frac{1}{6}$ of its length
(4) The breadth of rectangle is $\frac{1}{9}$ of its length
(5) None of these
(LIC Assistant Administrative Officer Exam. 2008)

7. An equilateral triangle and a regular hexagon have equal perimeters. If the area of the triangle is 2 cm^2, then the area of the hexagon is
(1) 2 cm^2 (2) 3 cm^2
(3) 4 cm^2 (4) 6 cm^2
(New India Assurance AO Exam. 25.10.2009)

8. A solid is in the form of a right circular cylinder with hemispherical ends. The total length of the solid is 35 cm. The diameter of the cylinder is 1/4 of its height. The surface area of the solid is
(Take $\pi = \frac{22}{7}$)
(1) 462 cm^2 (2) 693 cm^2
(3) 750 cm^2 (4) 770 cm^2
(New India Assurance AO Exam. 25.10.2009)

9. A solid metalic sphere of radius r is converted into a solid right circular cylinder of radius R. If the height of the cylinder is twice the radius of the sphere, then
(1) R = r (2) R = $r\sqrt{\frac{2}{3}}$
(3) R = $\sqrt{\frac{2r}{3}}$ (4) R = $\frac{2r}{3}$

(New India Assurance AO Exam. 25.10.2009)

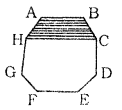
10. The radius of a circle is twice the side of a square of area 196 sq.cm. Length of a rectangle is twice the diameter of the circle. What is the perimeter of the rectangle if its breadth is half the length of the rectangle ?
(1) 244 cm
(2) 168 cm
(3) 336 cm
(4) Cannot be determined
(5) None of these

(United India Insurance AO Exam. 27.03.2011)

11. The smallest side of a right angled triangle is 6 cm. and second largest side is 8 cm. Side of a square is thrice the largest side of the triangle. What is the diagonal of the square ?
(1) $30\sqrt{2}$ cm
(2) $60\sqrt{2}$ cm
(3) 30 cm.
(4) Cannot be determined
(5) None of these

(United India Insurance AO Exam. 27.03.2011)

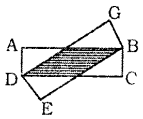
12. In the figure, ABCDEFGH is a regular octagon. What fraction of its area is shaded ?



(1) $\frac{1}{3}$ (2) $\frac{1}{4}$
(3) $\frac{1}{5}$ (4) $\frac{3}{8}$

(New India Insurance AAO Exam. 22.05.2011)

13. In the figure there are two rectangles ABCD and DEBG, each of length 7 cm and width 3 cm. The area of shaded region, in cm^2, is approximately



(1) 12 (2) 10
(3) 8 (4) 4

(New India Insurance AAO Exam. 22.05.2011)

14. If the area of a square inscribed in a circle is 15 cm^2, then the area of the square inscribed in a semicircle of the same circle, in cm^2, is
(1) 5 (2) 6
(3) 7.5 (4) $\sqrt{50}$

(New India Insurance AAO Exam. 22.05.2011)

15. Diameter of a cylindrical jar is increased by 25%. By what percent must the height be decreased so that there is no change in its volume ?
(1) 18% (2)25%
(3) 32% (4) 36%

(New India Insurance AAO Exam. 22.05.2011)

16. The area of a rectangle gets reduced by 9 sq. metre if its length is reduced by 5 m and breadth is increased by 3 m. If we increase the length by 2 m and breadth by 3 m, the area is increased by 67sq. metre. The length of the rectangle is:
(1) 9 m (2) 15.6 m
(3) 17 m (4) 18.5 m

(General Insurance Corporation AAO Exam. 11.12.2011)

17. Height of a cylindrical jar is decreased by 36%. By what percent must the radius be increased so that there is no change in its volume ?
(1) 25 (2) 35
(3) 36 (4) 40

(Oriental Insurance Company AAO Exam. 08.04.2012)

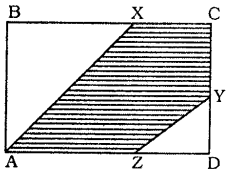
18. Suface area of a cuboid is 22 cm^2 and the sum of the lengths of all its edges is 24 cm. Length of each diagonal of the cuboid (in cm) is
(1) $\sqrt{11}$ (2) $\sqrt{13}$
(3) $\sqrt{14}$ (4) $\sqrt{15}$

(Oriental Insurance Company AAO Exam. 08.04.2012)

19. An equilateral triangle and a regular hexagon have equal perimeters. The ratio of the area of the triangle and that of the hexagon is
(1) 1 : 1 (2) 2 : 3
(3) 3 : 2 (4) 3 : 4

(Oriental Insurance Company AAO Exam. 08.04.2012)

20. In the figure, ABCD is a parallelogram with area 120 cm^2, and BX : XC = 3 : 2, CY : YD = 2 : 1 and AZ: ZD = 3 : 1. Area (in cm^2) of pentagon AXCYZ is



(1) 47 (2) 63
(3) 73 (4) 79

(United India Insurance AAO Exam. 03.06.2012)

21. A and B travel around a circular path at uniform speeds in opposite directions, starting from diametrically opposite points, at the same time. They meet each other first after B has travelled 100 metres and meet again 60 metres before A completed one round. The circumference of the park is
(1) 240 m (2) 300 m
(3) 320 m (4) 480 m

(United India Insurance AAO Exam. 03.06.2012)

22. The perimetres of a square and a regular hexagon are equal. The ratio of the area of the hexagon to the area of the square is
(1) $2\sqrt{3}$: 3 (2) $\sqrt{3}$: 1
(3) $3\sqrt{3}$: 2 (4) $\sqrt{2}$: 3

(United India Insurance AAO Exam. 03.06.2012)

23. The surface of water in a swimming pool forms a rectangle with length 40 m and breadth 15 m. The depth of water increases uniformly from 1.2 m to 2.4 m at the other end. The volume (in m^3) of water in the pool is
(1) 500 (2) 540
(3) 720 (4) 1080

(United India Insurance AAO Exam. 03.06.2012)

24. The circumference of a park is 750 m. A and B start walking from the same point in the same direction at 6.75 kmph and 4.75 kmph. In what time will they meet each other again ?
(1) 3 hours (2) 2.5 hours
(3) 3.5 hours (4) 4 hours
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 12.05.2013)

25. A hollow speherical silver ball has an external diameter 4 cm and internal diameter 2 cm thick. Then the volume of the silver used in the ball is :
(1) $\frac{28}{3}\pi$ (2) 1
(3) $\frac{4}{3}\pi$ (4) 7

(NICL (GIC) AO (Finance) Exam. 08.09.2013 (Paper-I)

26. The areas of three consesutive faces of a cuboid are 12 cm^2, 20 cm^2 and 15 cm^2 , then the volume (in cm^3) of the cuboid is
(1) 3600 (2) 100
(3) 80 (4) 60

(NICL (GIC) AO Exam. 08.09.2013 (Paper-I)

27. The wheel of a motor car makes 1000 revolutions in moving 440 m. The diameter (in metre) of the wheel is
(1) 0.44 (2) 0.14
(3) 0.24 (4) 0.34

(NICL (GIC) AO Exam. 08.09.2013 (Paper-I)

28. The perimeters of a circular and a square field are equal. What is the diameter of the circular field if the area of the square field 484 $metre^2$?
(1) 14 metre (2) 21 metre
(3) 28 metre (4) None of these

(NICL (GIC) Administratis Officer Exam. 15.12.2013

29. For a sphere of radius 10 cm the numerical value of the surface area is how many per cent of the numerical value of its volume ?
(1) 24% (2) 26.5%
(3) 30% (4) 45%

(NICL (GIC) AO (Finance Exam. 15.12.2013)

30. A circle and a rectangle have the same perimeter. The sides of the rectangle are 18 cm and 26 cm. What is the area of the circle ?
(1) 88 cm^2 (2) 154 cm^2
(3) 616 cm^2 (4) 1250 cm^2

(NICL (GIC) AO (Finance Exam. 15.12.2013)

31. A rectangular courtyard 3.78 m long and 5.25 m broad is to be paved exactly with square tiles all of the same size. The minimum number of such tiles is
(1) 430 (2) 440
(3) 450 (4) 460

(NICL (GIC) AO (Finance Exam. 15.12.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1.	(5)	**2.**	(3)
3.	(5)	**4.**	(5)
5.	(3)	**6.**	(2)
7.	(1)	**8.**	(3)
9.	(1)	**10.**	(1)
11.	(5)	**12.**	(1)
13.	(3)	**14.**	(4)
15.	(1)	**16.**	(4)
17.	(2)	**18.**	(5)
19.	(3)	**20.**	(1)
21.	(1)	**22.**	(3)
23.	(1)	**24.**	(2)
25.	(5)	**26.**	(5)
27.	(5)	**28.**	(2)
29.	(3)	**30.**	(3)
31.	(3)	**32.**	(4)
33.	(5)	**34.**	(1)
35.	(3)	**36.**	(3)
37.	(4)	**38.**	(3)
39.	(1)	**40.**	(2)

SBI PO EXAMS

1.	(2)	**2.**	(1)
3.	(4)	**4.**	(2)
5.	(1)	**6.**	(5)
7.	(3)	**8.**	(4)
9.	(2)	**10.**	(5)
11.	(4)		

RBI GRADE-B OFFICER EXAMS

1.	(5)	**2.**	(4)

INSURANCE EXAMS

1.	(3)	**2.**	(2)
3.	(2)	**4.**	(2)
5.	(4)	**6.**	(1)
7.	(2)	**8.**	(4)
9.	(2)	**10.**	(3)
11.	(1)	**12.**	(4)
13.	(1)	**14.**	(2)
15.	(4)	**16.**	(3)
17.	(1)	**18.**	(3)
19.	(2)	**20.**	(4)
21.	(4)	**22.**	(1)
23.	(4)	**24.**	(1)
25.	(1)	**26.**	(4)
27.	(2)	**28.**	(3)
29.	(3)	**30.**	(3)
31.	(3)		

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (5) Let the breadth of rectangular field be x metres.

$\therefore$ Length = $x \times \frac{115}{100} = \frac{23x}{20}$ metres

Now, Length × Breadth = Area

$\Rightarrow \frac{23x}{20} \times x = 460$

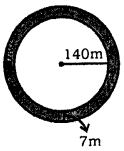
$\Rightarrow x = \frac{460 \times 20}{23}$

$\Rightarrow x^2 = 20 \times 20$

$\Rightarrow x = \sqrt{20 \times 20} = 20$ metres

2. (3) Let the length of rectangular plot of x and breadth be y.
Perimeter = $2(x + y) = 340$ metres
Area of boundary = $(x + 2) + (y + 2) - xy$
$= xy + 2x + 2y + 4 - xy$
$= 2x + 2y + 4$
$= 2(x + y) + 4$
$= 340 + 34 = 344$
$\therefore$ Cost of gardening = 344 × 10 = Rs. 3440

3. (5)



Radius of the field = 140 m
Width of garden = 7 m
$\therefore$ Area of garden
$= \pi(147^2 - 140^2)$

$= \frac{22}{7}(147 + 140)(147 - 140)$

$= 22 \times 287 = 6314\ m^2$
Required cost = Rs. (21 × 6314)
= Rs. 132594

4. (5) Let the length and breadth of the rectangular plot be $8x$ and $5x$ metres respectively,
According to the question,
$8x - 5x = 60$
$\Rightarrow 3x = 60$

$\therefore x = \frac{60}{3} = 20$

Perimeter of the plot = $2(8x + 5x)$
$= 26x$ metre = 26 × 20 = 520 metre

5. (3) Let the length of the rectangular field = x metre

$\therefore$ Its breadth = $\frac{3x}{4}$ metre

According to the question,

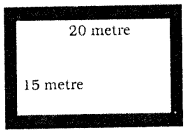
$x \times \frac{3x}{4} = 300$

$\Rightarrow x^2 = 300 \times \frac{4}{3} = 400$

$\Rightarrow x = \sqrt{400} = 20$

$\therefore$ length = 20 metre

and breadth = $\frac{3}{4} \times 20 = 15$ metre



Length of the field with garden
= 20 + 1.5 × 2 = 23 metre
Breadth of the field with garden
= 15 + 1.5 × 2 = 18 metre
$\therefore$ Its area = 23 × 18 = 414 square metre
$\therefore$ Area of the garden = 414 - 300
= 114 square metre

6. (2) Side of the square = $\sqrt{\text{Area}}$

= $\sqrt{407044}$ = 638 cm

$\therefore$ Circumference of the circle = 638 cm

$\Rightarrow 2\pi r = 638$

$\Rightarrow 2 \times \frac{22}{7} \times r = 638$

$\Rightarrow r = \frac{638 \times 7}{2 \times 22} = \frac{29 \times 7}{2}$ cm

$\therefore$ Area of the circle

$= \frac{22}{7} \times \frac{29 \times 7 \times 29 \times 7}{2 \times 2} = 32378.5$ sq. cm.

7. (1) $2\pi r = 1047.2$

$\Rightarrow 2 \times \frac{22}{7} \times r = 1047.2$

$\Rightarrow r = \frac{1047.2 \times 7}{2 \times 22} = 166.6$ sq.metre

Area of the circle = πr^2

$= \frac{22}{7} \times 166.6 \times 166.6 = 87231.76$ sq. metre

8. (3) Initial area of the carpet
= 3 × (3× l.44) sq. metre = 12.96 sq. metre
After corresponding changes in dimensions,
Area of the carpet =

$\left(3 \times \frac{125}{100}\right) \times \left(3 \times 1.44 \times \frac{140}{100}\right)$

= 22.68 sq. metre
$\therefore$ Increase in area = (22.68 -12.96) sq. metre
= 9.72 sq. metre
$\therefore$ Increase in the cost = Rs. (9.72 × 45)
= Rs. 437.4

9. (1) Let the breadth of the rectangular plot be x metre.
$\therefore$ Length = $3x$ metre
According to the question, $3x \times x = 7803$

$\Rightarrow x^2 = \frac{7803}{3} = 2601$

$\therefore x = \sqrt{2601} = 51$ metre

10. (1) Total cost of fencing = Rs. 3300
Rate = Rs. 15 /metre

$\therefore$ Circumference of circular plot = $\frac{3300}{15}$

= 220 metre
$\therefore 2\pi r = 220$

$\Rightarrow \frac{2 \times 22}{7} \times r = 220$

$\Rightarrow r = \frac{220 \times 7}{2 \times 22} = 35$ metre

$\therefore$ Area of the circular plot = πr^2

$= \frac{22}{7} \times 35 \times 35 = 3850$ sq.metre

$\therefore$ Cost of flooring =3850 × 100 = Rs. 385000

11. (5) $\pi r^2 = 616 \Rightarrow \frac{22}{7} r^2 = 616$

$\Rightarrow r^2 = \frac{616 \times 7}{22} = 196$

$\therefore r = \sqrt{196} = 14$ cm .
$\therefore$ Circumfernce of the circle = $2\pi r$

$= 2 \times \frac{22}{7} \times 14 = 88$ cm

12. (1) Rate of painting = Rs. 2/ sq.metre
$\therefore$ Area of the rectangular floor

$= \frac{256}{2} = 128$ sq.m.

Let the breadth of floor be x metre
$\therefore$ length = $2x$ metre
$\therefore 2x \times x = 128$

$\Rightarrow x^2 = \frac{128}{2} = 64$

$\therefore x = \sqrt{64} = 8$

$\therefore$ Length of floor = $2x = 2 \times 8 = 16$ metre

13. (3) $2\pi r_1 = 132$

$\Rightarrow 2 \times \frac{22}{7} \times r_1 = 132$

$\Rightarrow r_1 = \frac{132 \times 7}{2 \times 22} = 21$ metre

Similarly,

$2\pi r_2 = 176$

$\Rightarrow r_2 = \frac{176 \times 7}{2 \times 22} = 28$ metre

$\therefore$ Difference between their area

$= \pi\left(r_2^2 - r_1^2\right) = \pi(r_2 + r_1)(r_2 - r_1)$

$= \frac{22}{7} \times 49 \times 7 = 1078$ Sq. metre.

14. (4) $\pi r^2 = 32378.5$

$\Rightarrow r^2 = \frac{32378.5 \times 7}{22}$

$\therefore r = \sqrt{\frac{32378.5 \times 7}{22}} = 101.5$ metres

$\therefore$ Circumference of the circle

$= 2\pi r = 2 \times \frac{22}{7} \times 101.5 = 638$ metre

$\therefore$ Expenditure on fencing = Rs. (154 × 638)
= Rs. 98252

15. (1) Perimeter of square
= 2 × Perimeter of rectangle
= 2 × 2 (8 + 7) = 60 cm.

$\therefore$ Side of square = $\frac{60}{4} = 15$ cm.

$\therefore$ Diameter of semi-circle = 15 cm.

$\therefore$ Cicumference of semi-cricle

$= \frac{\pi d}{2} + d = \frac{22}{7 \times 2} \times 15 + 15 = 38.57$ cm.

16. (4) Data are inadequate.

17. (2) Tricky approach

If the side of the square be x cm then,

$\pi \times 35 \times 35 + x^2 = 5450$

$\Rightarrow \frac{22}{7} \times 35 \times 35 + x^2 = 5450$

$\Rightarrow x^2 = 5450 - 3850 = 1600$

$\therefore x = 40$ cm.

$\therefore$ Required sum = $\pi \times d + 4x$

$= \left(\frac{22}{7} \times 70 + 4 \times 40\right)$ cm = 380 cm

18. (5) Side of the square = $\sqrt{1024} = 32$ cm.

$\therefore$ Length of rectangle = 2 × 32 = 64 cm.

Breadth of rectangle
= 32 - 12 = 20 cm.

$\therefore$ Required ratio = 64 : 20 = 16 : 5

19. (3) Radius of larger circle

$= 2 \times \sqrt{196} = 28$ cm

Radius of smaller circle = $\left(\frac{3}{7} \times 28\right)$ cm

= 12 cm

Circumference of smaller circle

$= 2\pi r = 2\pi \times 12 = 24\pi$ cm

20. (1) Circumference of circular plot

$= \frac{7700}{14} = 550$ feet

$\therefore 2\pi r = 550$

$\Rightarrow r = \frac{550}{2\pi} = \frac{550 \times 7}{2 \times 22} = 87.5$ feet.

$\therefore$ Area = $\frac{22}{7} \times 87.5 \times 87.5 = 24062.5$ sq.feet.

21. (1) If the side of square be x cm
then,

$\pi r^2 + x^2 = 2611$

$\Rightarrow \frac{22}{7} \times 21 \times 21 + x^2 = 2611$

$\Rightarrow 1386 + x^2 = 2611$

$\Rightarrow x^2 = 2611 - 1386 = 1225$

$\therefore x = \sqrt{1225} = 35$ cm

$\therefore$ Circumference of circie + Perimeter of square = $2\pi r + 4x$

$= 2 \times \frac{22}{7} \times 21 + 4 \times 35$

$= 132 + 140 = 272$ cm

22. (3) Side of square

$= \sqrt{\text{Area}} = \sqrt{196} = 14$ cm

$\therefore$ Radius of circle = 28 cm

$\therefore$ Circumfence of circle

$= 2 \times \frac{22}{7} \times 28 = 176$ cm

If the lenght of rectangle be x cm
then,

$2(x + 176) = 712$

$\Rightarrow x + 176 = \frac{712}{2} = 356$

$\therefore x = 356 - 176 = 180$ cm

23. (1) Side of the square plot = $\sqrt{\text{Area}}$

$= \sqrt{462.25} = 21.5$ feet

$\therefore$ Perimeter = 4 × Side

= 4 × 21.5 = 86 feet

$\therefore$ Cost of building a fence

= Rs. (34 × 86) = Rs. 2924

24. (2) Area of the square = 22 × 22 = 484 sq.cm

$\therefore$ Circumference of circle = 484 cm

$\Rightarrow \pi \times$ Dimater = 484

$\Rightarrow \frac{22}{7} \times$ Dimater = 484

$\therefore$ Dimater = $\frac{484}{22} \times 7 = 154$ cm

$\therefore$ Lenght of rectangle = 2 × 154 cm

= 308 cm

$\therefore$ 2(lenght + breadht) = Perimeter of rectangle

$\Rightarrow 2(308 + x) = 668$ [Breadht = x (let)]

$\Rightarrow 308 + x = \frac{668}{2} = 334$

$\Rightarrow x = 334 - 308 = 26$ cm

25. (5) $2\pi r_1 = 88$

$\Rightarrow 2 \times \frac{22}{7} \times r_1 = 88$

$\Rightarrow r_1 = \frac{88 \times 7}{2 \times 22} = 14$ metre

$2 \pi r = 220$

$\Rightarrow 2 \times \frac{22}{7} \times r_2 = 220$

$\Rightarrow r_2 = \frac{220 \times 7}{2 \times 22} = 35$ metre

Required diffrence

$\Rightarrow \pi \left(r_2^2 - r_1^2\right) = \frac{22}{7}\left(r_2 + r_1\right)\left(r_2 - r_1\right)$

$\Rightarrow \frac{22}{7}(35 + 14)(35 - 14)$

$\Rightarrow \frac{22}{7} \times 49 \times 21 = 3234$ sq.metre

26. (5) Side of a square = $\frac{\text{Perimeter}}{4}$

$= \frac{112}{4} = 28$ metre

$\therefore$ Radius of circle = 28 metre

$\therefore$ Area of circle = πr^2

$= \frac{22}{7} \times 28 \times 28 = 2464$ sq. metre

27. (5) $\frac{\pi r^2}{2} = 1925$

$\Rightarrow \pi r^2 = 1925 \times 2$

$\Rightarrow \frac{22}{7} \times r^2 = 1925 \times 2$

$\Rightarrow r^2 = \frac{1925 \times 2 \times 7}{22} = 1225$

$\therefore r = \sqrt{1225} = 35$

$\therefore$ Breadth of rectangle

$= \pi r + 2r$

$= (\pi + 2) r = \left(\frac{22}{7} + 2\right) \times 35 = 180$ cm

Lenght of rectangle = Perimeter of square

= 4 × 48 = 192 cm

$\therefore$ Perimeter of recangle

= 2 (lenght + breadht)

= 2 (192 + 180) = 744 cm

28. (2) Area of circle = πr^2

$= \frac{22}{7} \times 14 \times 14 = 616$ sq. cm.

$\therefore$ Area of rectangle = 1166 - 616

= 550 sq.cm.

Breadth of rectangle = $\frac{550}{25} = 22$cm.

$\therefore$ Circumference of circle = $\pi \times$ diameter

$= \frac{22}{7} \times 28 = 88$ cm.

Perimeter of rectangle

= 2 (length + breadth)

= 2 (25 + 22) = 94 cm.

Required sum = 88 + 94 = 182 cm.

29. (3) Circumference of circle

= π × diameter

$= \frac{22}{7} \times 56 = 176$ cm

$\therefore$ Perimeter of square

= 272 - 176 = 96 cm

$\therefore$ Side of square = $\frac{96}{4} = 24$ cm

$\therefore$ Area of square = 24 × 24 = 576 sq.cm.

Area of circle = πr^2

$= \frac{22}{7} \times 28 \times 28 = 2464$ sq.cm

$\therefore$ Required sum = (576 + 2464) sq. cm.
= 3040 sq. cm.

30. (3) $2\pi r = 132$

$\Rightarrow 2 \times \frac{22}{7} \times r = 132$

$\Rightarrow r = \frac{132 \times 7}{2 \times 22} = 21$ cm

$\therefore$ Side of square = 21 cm

$\therefore$ Length of the rectangle = $\frac{3 \times 21}{5}$ cm

$\therefore$ Area of the rectangle = $\frac{3 \times 21}{5} \times 8$

= 100.8 sq.cm.

31. (3) Side of a square = $\frac{\text{Perimeter}}{4}$

$= \frac{56}{4} = 14$ cm

$\therefore$ Smallest side of the right angled triangle
= 14 - 8 = 6 cm.

Lenght of rectangle = $\frac{\text{Area}}{\text{Breadht}}$

$= \frac{96}{8} = 12$ cm

$\therefore$ Second side of the triangle
= 12 - 4 = 8 cm

$\therefore$ Hypotenuse of the right angled tringle

$= \sqrt{6^2 + 8^2}$

$= \sqrt{36 + 64} = \sqrt{100} = 10$ cm

32. (4) Side of square = $\sqrt{1444} = 38$ metre

Breadht of rectangle = $\frac{1}{4} \times 38 = \frac{19}{2}$ metre

Lenght of rectangle = $\frac{57}{2}$ metre

Required difference = $1444 - \frac{19}{2} \times \frac{57}{2}$

= 1444 - 270.75 = 1173.25 sq. metre

33. (5) Side of square = $\sqrt{784} = 28$ cm

$\therefore$ Circumference of circle-A

$= 4 \times 28 \times \frac{11}{7}$

$\Rightarrow 2\pi r = 16 \times 11$

$\Rightarrow 2 \times \frac{22}{7} \times r = 16 \times 11$

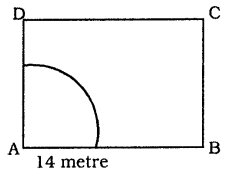
$\Rightarrow r = \frac{16 \times 11 \times 7}{2 \times 22} = 28$

Radius of circle-B = $\frac{28}{4} = 7$ cm

$\therefore$ Area = πr^2

$= \frac{22}{7} \times 7 \times 7 = 154$ sq. cm.

34. (1)



Required area = $\frac{1}{4} \times \pi R^2$

$= \frac{1}{4} \times \frac{22}{7} \times 14 \times 14$

= 154 sq. metre

35. (3) $\pi R^2 = 7 \times 2\pi R$

$\Rightarrow$ R = 14

$\therefore$ Circumference of circle

$= 2\pi R = 2 \times \frac{22}{7} \times 14$

= 88 units

36. (3) Circumference of circle = $2\pi r$

$\Rightarrow 572 = 2 \times \frac{22}{7} \times r$

$\Rightarrow r = \frac{572 \times 7}{2 \times 22} = 91$

$\therefore$ Length of floor = 91 + 6 = 97 metre = l

$\therefore 2(l + b) = 356$

$\Rightarrow 2(97 + b) = 356$

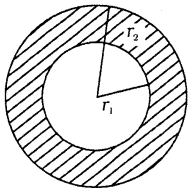
$\Rightarrow 97 + b = 178$

$\Rightarrow b = 178 - 97 = 81$ metre

$\therefore$ Area of floor = 97 × 81 = 7857 sq. metre

$\therefore$ Required cost = 12 × 7857 = Rs. 94284

37. (4) $2\pi r_1 = 308$ where r_1 = radius of circular playground



$\Rightarrow 2 \times \frac{22}{7} \times r_1 = 308$

$\Rightarrow r_1 = \frac{308 \times 7}{2 \times 22} = 49$ metre

$\therefore$ Area of the path = $\pi\left(r_2^2 - r_1^2\right)$

$= \frac{22}{7}(56 + 49)(56 - 49)$

$= 22 \times 105 = 2310$ sq. m.

38. (3) Radius of cylinder = $\frac{14}{2} = 7$ cm

Height of cylinder = 14 cm

$\therefore$ Required volume = $\pi r^2 h$

$= \frac{22}{7} \times 7 \times 7 \times 14 = 2156$ cu.cm.

39. (1) In Δ BDE,
DE = 28 + 28 = 56 cm
BC = 28 cm

Area of ΔBDE = $\frac{1}{2} \times$ DE $\times$ BC

$= \frac{1}{2} \times 56 \times 28 = 784$ sq.cm

40. (2) Area of square = 28 × 28 = 784 sq. cm.
Area of four circles = $4\pi r^2$

$= 4 \times \frac{22}{7} \times 7 \times 7 = 616$ sq.cm

Area of shaded region
= 784 - 616 = 168 sq.cm

SBI PO EXAMS

1. (2) Let height and base of triangle be h and b respectively

Area of triangle = $\frac{1}{2}\ hb$

Accordig this condition area of triangle

$= \frac{1}{2}(h$ - 20% of $h)$ $(b$ + 20% of $b)$

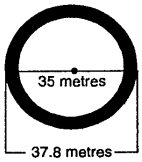
$= \frac{1}{2}\left(h - \frac{h}{5}\right)\left(b + \frac{b}{5}\right)$

Change in area = $\frac{12hb}{25} - \frac{hb}{2} = \frac{-hb}{25}$

change in area of triangle = $\frac{\frac{-hb}{25}}{\frac{hb}{2}} \times 100$

= -8 %
Hence the area of triangle decreased by 8%.

2. (1)



Required area
= π [$(18.9)^2 - (17.5)^2$]

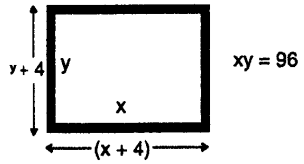
$= \frac{22}{7}$ [(18.9 + 17.5) (18.9 - 17.5)]

$= \frac{22}{7}$ [36.4 × 1.4]

$= \frac{22}{7} \times 36.4 \times 1.4$

= 22 × 36.4 × 0.2 = 160.16 sq. metres.

3. (4)



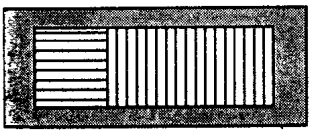
$(x + 4)(y + 4) = xy + 4(x + y) + 16$
$= 96 + 4(x + y) + 16$
New Area = $112 + 4(x + y)$
$\therefore$ We should have either x or y to complete the question.

4. (2) % change = $x + y + \frac{xy}{100}$

$= +50 - 20 + \frac{50 \times -20}{100}$

= 30 - 10 = 20% (increase)

5. (1) Area of the floor = 20 × 10
= 200 sq. ft.



Area covered by black tiles
= 2 (20 + 20 + 6 + 6) = 104 sq.ft.
Remaing in space = 200 - 104 = 96 sq.ft.

Area covered by white tiles = $\frac{1}{3} \times 96$

= 32 sq.feet
Area covered by blue tiles = 96 - 32
= 64 sq. feet
Area of 1 tile = $(2)^2$ = 4 sq.ft.

No. of blue tile = $\frac{64}{4}$ = 16

6. (5) Let the breadth of rectangular field be x metres.

$\therefore$ Length = $x \times \frac{115}{100} = \frac{23x}{20}$ metres

Now, Length x Breadth = Area

$\Rightarrow \frac{23x}{20} \times x = 460$

$\Rightarrow x^2 = \frac{460 \times 20}{23}$

$\Rightarrow x^2 = 20 \times 20$

$\Rightarrow x = \sqrt{20 \times 20} = 20$ metres

7. (3) Total area of rectangular plot to be gardened = 340 × 1 + 4 × 1 = 344 sq.m.
$\therefore$ Total cost = 344 × 10 = Rs. 3440

8. (4) Let the length and breadth of the rectangular plot be $6x$ and $5x$ metres respectively.
According to the question,
$6x - 5x = 34$
$\Rightarrow x = 34$
$\therefore$ Perimeter of the plot = 2 (length + breadth)
= 2 ($6x + 5x$) = $22x$ metre
= 22 × 34 = 748 metre

9. (2) Area of the square = 3136 cm^2
$\therefore$ Side of the square = $\sqrt{3136}$ = 56 cm.
$\therefore$ Its perimeter = 4 × 56 = 224 cm
$\therefore$ Diameter of circle = 224 cm
$\therefore$ Circumference of circle = π × diameter

= $\frac{22}{7} \times 224 = 704$ cm

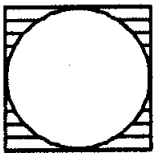
10. (5) Let the length and breadth of the original rectangle be x and y metres respectively.
$\therefore$ Area of the original rectangle
= xy square metre
According to the question,

$\therefore x \times \frac{120}{100} \times y \times \frac{80}{100} = 192$

$\Rightarrow x \times 1.2 \times y \times 0.8 = 192$

$\Rightarrow xy = \frac{192}{1.2 \times 0.8} = 200$ square metre

11. (4)



Radius of the circular garden

= $\frac{28}{2}$ = 14 metre

$\therefore$ Its area = πr^2

= $\frac{22}{7} \times 14 \times 14 = 616$ sq. metre

Area of the square plot = 28 × 28
= 784 sq. metre
$\therefore$ Area of the shaded region
= 784 - 616 = 168 sq. metre

RBI GRADE-B OFFICER EXAMS

1. (5) Let the breadth of rectangular plot = x m.
$\therefore$ Length = $3x$ m
According to the question,
$3x \times x = 6075$

$\Rightarrow x^2 = \frac{6075}{3} = 2025$

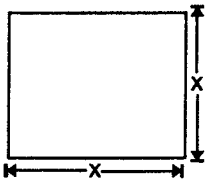
$\Rightarrow x = \sqrt{2025} = 45$

2. (4) If the angle be 30°, then according to ratio, Second angle = 60°
$\therefore$ Third angle = 180° - 60° - 30° = 90°
If the given ratio be of two other angles, then
$x + 2x = 180° - 30° = 150°$
$\Rightarrow 3x = 150° \Rightarrow x = 50°$
$\therefore$ Second angle = 2 × 50 = 100°
Hence, the largest angle = 90° or 100°

INSURANCE EXAMS

1. (3) Required answer
= (3m × 2.7m × 0.2m) × 320 kg.
= 518.4 kg.

2. (2)



According to the question,
$x \times x = 2.25$ cm^2

$\therefore x = \sqrt{2.25}$ cm = 1.5 cm
$\therefore$ Perimeter = $4x$ = 4 × 1.5 cm = 6 cm

3. (2) Let the breadth of the rectangle be x
$\therefore$ According to question,
Length = 1.20 x
$\therefore$ Area of rectangle = 1.20 $x \times x$ = 1.20 x^2
Area of square = $x \times x = x^2$

$\therefore$ Required ratio = $\frac{1.20x^2}{x^2}$

$= \frac{12}{10} = \frac{6}{5}$ = 6 : 5

4. (2) Volume of earth dug out

$= \left(\frac{22}{7} \times 7 \times 7 \times 10\right)$ m^3 = 1540 m^3

Area of embankment $= \frac{22}{7}\left[(28)^2 - (7)^2\right]$ m^2

$= \frac{22}{7} \times (28 + 7)(28 - 7)$ m^2

$= \frac{22}{7} \times 35 \times 21$ m^2 = 2310 m^2

$\therefore$ Height of embankment = $\frac{\text{Volume}}{\text{Area}}$

$= \frac{1540}{2310}$ m = $\frac{2}{3}$ m

5. (4) Let the length of carpet
= a m and breadth = b m.
Then,
2 (a + b) = 46
$\Rightarrow a + b = 23$
Again, ab = 120
$\therefore$ Diagonal = $\sqrt{a^2 + b^2}$

$= \sqrt{(a+b)^2 - 2ab}$

$= \sqrt{(23)^2 - 240}$

$= \sqrt{289}$ = 17 m

6. (1) From statement (1),
If breadth of rectangle be x cm,
length = 16x
$\therefore 16x \times x = (8)^2 \Rightarrow x = 2$
$\therefore$ Length = 32 cm.
From statement (2),
If breadth = $\sqrt{2}$ cm
Length = $32\sqrt{2}$ cm
Hence, area of rectangle = area of square.
Clearly, these two statements are definitely true.

7. (2) <u>Tricky approach</u>

$\frac{\sqrt{3}}{4} \times$ side^2 = Area of equilateral triangle

$= \frac{\sqrt{3}}{4} \times a^2 = 2$

$\Rightarrow a^2 = \frac{8}{\sqrt{3}}$

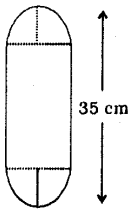
Now, side of a regular hexagon

$= \frac{3}{6} \times$ side $= \frac{a}{2}$

$\therefore$ Area of regular hexagon

$= \frac{3\sqrt{3}}{2} \times \frac{a^2}{4} = \frac{3\sqrt{3}}{2} \times \frac{8}{\sqrt{3} \times 4}$ = 3 cm^2

8. (4)



Let radius of the cylinder be r cm and height = h cm.

Then, $r = \frac{h}{8}$ given

and $h = 35 - 2r = 35 - 2 \times \frac{h}{8}$

$\Rightarrow h + \frac{h}{4} = 35$

$\Rightarrow \frac{5h}{4} = 35$

$\Rightarrow \frac{35 \times 4}{5} = 28$ cm.

$\therefore r = \frac{h}{8} = \frac{28}{8} = \frac{7}{2}$ cm

$\therefore$ Radius of the hemisphere = $\frac{7}{2}$ cm

Total surface of area the solid
= (2π rh + 2 × $2\pi r^2$) sq. cm.
= $2\pi r$ (h + 2r) sq. cm.

$= 2 \times \frac{22}{7} \times \frac{7}{2} \times \left(28 + 2 \times \frac{7}{2}\right)$ sq. cm.

$= 2 \times \frac{22}{7} \times \frac{7}{2} \times 35$ sq. cm. = 770 sq. cm.

9. (2) Tricky approach
In both cases, the volume will be the same.

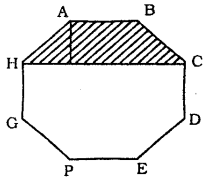
$\therefore \frac{4}{3}\pi r^3 = \pi R^2.2r \Rightarrow \frac{2}{3}r^3 = R^2$

$\therefore R = r\sqrt{\frac{2}{3}}$

10. (3) Side of the square
$= \sqrt{196} = 14$ cm
Radius of circle = 2 × 14 = 28 cm
Length of rectangle = 2 × 2 × 28 = 112 cm
Breadth = $\frac{112}{2} = 56$ cm
Perimeter = 2 (112 + 56) = (2 × 168) cm
= 336 cm

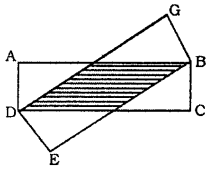
11. (1) For a right angled triangle,
Hypotenuse
$= \sqrt{6^2+8^2} = \sqrt{36+64}$
$= \sqrt{100} = 10$ cm = Largest side
$\therefore$ Side of square = 3 × 10 = 30 cm
Dignonal of square
$= \sqrt{2} \times 30 = 30\sqrt{2}$ cm

12. (4)



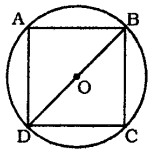
Area of octagon = $8 \times \frac{\sqrt{3}}{4}x^2$
$= 2\sqrt{3}x^2$ where, x = AB
Area of ABCH
$= \frac{1}{2}(2x + x) \times$ AI
$= \frac{1}{2} \times 3x \times \sqrt{x^2 - \frac{x^2}{4}}$
$= \frac{1}{2} \times 3x \times \frac{\sqrt{3}x}{2} = \frac{3\sqrt{3}}{4}x^2$
Required shaded portion
$= \frac{3\sqrt{3}}{4}x^2 \div 2\sqrt{3}x^2 = \frac{3}{8}$

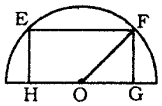
13. (1)



Required area = 7 × 3 - 3 × 3 = 12 sq.cm.

14. (2)

I. 

II. 

DB = $\sqrt{2}$ × side
$= \sqrt{2} \times \sqrt{15} = \sqrt{30}$
$\therefore$ Radius of Circle = $\frac{\sqrt{30}}{2}$ cm
In second figure,
OF = $\frac{\sqrt{30}}{2}$
If FG = x, then
OF = $\sqrt{x^2 + \frac{x^2}{4}} = \sqrt{\frac{5x^2}{4}}$
$\Rightarrow \frac{\sqrt{30}}{2} = \sqrt{\frac{5x^2}{4}}$
$= \frac{30}{4} + \frac{5x^2}{4} \Rightarrow x^2 = 6$ sq.cm

15. (4) Original volume of the cylindrical jar
$= \pi r^2 h$
$\therefore \pi\left(\frac{125r}{100}\right)^2 \times h_1 = \pi r^2 h$
$\Rightarrow \frac{25}{16}h_1 = h$
$\Rightarrow \frac{h_1}{h} = \frac{16}{25}$
$\therefore$ Required percentage = $\frac{h - h_1}{h} \times 100$
$= \frac{25-16}{25} \times 100 = 36\%$

16. (3) Let teh length of the rectangle be x metre and breatdh be y metre.
$\therefore xy - (x - 5)(y + 3) = 9$
$\Rightarrow xy - (xy - 5y + 3x - 15) = 9$
$\Rightarrow 5y - 3x + 15 = 9$
$\Rightarrow 3x - 5y - 6 = 0$ ....(i)

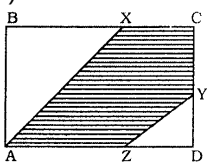
Again, $(x + 3)(y + 2) - xy = 67$
$3y + 2x + 6 = 67$
$\Rightarrow 2x + 3y - 61 = 0$ ....(ii)
By equation (i) × 3 + (ii) × 5
$9x - 15y - 18 = 0$
$10x + 15y - 305 = 0$
$19x = 323$
$\Rightarrow x = 17$ metre

17. (1) Volume of the cylindrical jar
$= \pi r^2 h$
New volume on reducing height
$= \pi r^2 \times \frac{64}{100} h = \frac{16}{25} \pi r^2 h$
$\therefore$ If $r_1^2 = \frac{25}{16} R^2$ the volume will not change.
$\therefore r_1 = \frac{5}{4} R = 1 + \frac{1}{4}$ i.e. increase of 25%

18. (3) Let the length, breadth and height of the cuboid be a, b and c cm respectively.
$2(ab + bc + ca) = 22$
and, $4(a + b + c) = 24$
$\Rightarrow a + b + c = 6$
$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$
$\Rightarrow 36 = a^2 + b^2 + c^2 + 22$
$\Rightarrow a^2 + b^2 + c^2 = 14$
$\Rightarrow \sqrt{a^2 + b^2 + c^2} = \sqrt{14}$
= Diagonal of cuboid

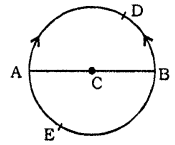
19. (2) $3x = 6y$ where x = side of triangle;
y = side of regular hexagon.
$\Rightarrow x = 2y$
$\Rightarrow y = \frac{x}{2}$
$\therefore$ Area of triangle $= \frac{\sqrt{3}}{4} x^2$
Area of hexagon $= 6 \times \frac{\sqrt{3}}{4} y^2$
$= 6 \times \frac{\sqrt{3}}{4} \times \frac{x^2}{4} = \frac{3\sqrt{3}}{8} x^2$
$\therefore$ Required ratio
$= \frac{\sqrt{3}}{4} x^2 : \frac{3\sqrt{3}}{8} x^2 = 2 : 3$

20. (4)



BC = BX + XC = $3x + 2x = 5x$ cm
CD = CY + YD = $2y + y = 3y$ cm
$\therefore 5x \times 3y = 120$
$\Rightarrow xy = 8 (= 2 \times 4)$
BC = 20 cm
CD = 6 cm
$BX = \frac{3}{5} \times 20 = 12$ cm
$YD = \frac{1}{3} \times 6 = 2$ cm
$ZD = \frac{1}{4} \times 20 = 5$ cm
$\therefore$ Area of the shaded region
$= 120 - \Delta ABX - \Delta ZYD$
$= 120 - \frac{1}{2} \times 12 \times 6 - \frac{1}{2} \times 2 \times 5$
$= 120 - 36 - 5 = 79$ sq.cm.

21. (4)



Distance covered in first meeting = πr
Distance covered in second meeting = $3\pi r$
BD = 100 metre
AD = 60 metre
If $2\pi r$ = 480 metre then,
A's speed : B's speed = 7 : 5

22. (1) Side of the square = x units and side of regular hexagon = y units.
$\therefore 4x = 6y \Rightarrow x = \frac{3}{2} y$
Area of square $= \frac{9}{4} y^2$
Area of hexagon $= \frac{3\sqrt{3}}{2} \times y^2$
$\therefore$ Required ratio
$= \frac{3\sqrt{3}}{2} \times y^2 : \frac{9}{4} y^2 = 2\sqrt{3} : 3$

23. (4) Volume of water in the pool
$= 40 \times 15 \times \left(\frac{1.2 + 2.4}{2}\right)$
= 1080 cubic metre

24. (1) Time taken by A in completing one revolution

$= \dfrac{750}{\dfrac{6.75\times5}{18}}$ sec $= \left(\dfrac{750\times18}{6.75\times5}\right)$sec

Time taken by B $= \left(\dfrac{750\times18}{4.75\times5}\right)$sec

Required time $= \dfrac{\text{LCM of numerators}}{\text{HCF of denominators}}$

$= \dfrac{750\times1800}{125}$ = 1800 × 6 seconds

$= \dfrac{1800\times6}{3600}$ = 3 hours

25. (1) Volume of the silver used in the ball

$= \dfrac{4}{3}\pi\left(r_2^3 - r_1^3\right)$

$= \dfrac{4}{3}\pi\left(2^3 - 1^3\right)$

$= \dfrac{28\pi}{3}$cu.cm

26. (4) Let the sides of the cuboid be *a*, *b* and c.
$\therefore ab$ = 12 sq. cm.
bc = 20 sq.cm.
ac = 15 sq.cm.
Volume of cuboid = *abc*

$= \sqrt{a^2b^2c^2} = \sqrt{12\times20\times15}$

$= \sqrt{3600}$ = 60 cu.cm.

27. (2) Distance covered in 1 revolution
$= \pi \times$ diameter $= \pi d$
$\therefore 1000 \times \pi d = 440$

$\Rightarrow 1000 \times \dfrac{22}{7} \times d = 440$

$\Rightarrow d = \dfrac{440\times7}{1000\times22}$ = 0.14 metre.

28. (3) Side of square field = $\sqrt{484}$ = 22 metre
Circumference of circular field =
Perimeter of square field

$\Rightarrow 2\pi r = 4 \times 22$

$\Rightarrow 2 \times \dfrac{22}{7} \times r = 4 \times 22$

$\Rightarrow r$ = 14 metre
$\therefore$ Diameter = 28 metre

29. (3) Surface area of the sphere = $4\pi r^2$
$= 4\pi \times 10^2 = 400\pi$ sq. cm.

Volume o sphere = $\dfrac{4}{3}\pi r^3$

$= \dfrac{4}{3}\pi \times 10^3 = \dfrac{4000\pi}{3}$ cu. cm.

$\therefore$ Required percentage $= \dfrac{400\pi}{\dfrac{4000\pi}{3}} \times 100$

= 30%

30. (3) Perimeter of rectangle = 2(18 + 26)
= 2 × 44 = 88 cm
$\therefore$ Circumference of circle = 88 cm

$\Rightarrow 2 \times \dfrac{22}{7} \times r = 88$

$\Rightarrow r = \dfrac{88\times7}{2\times22}$ =14 cm

$\therefore$ Area of circle = πr^2

$= \dfrac{22}{7} \times 14 \times 14 = 616$ sq. cm.

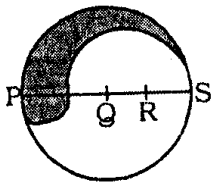
31. (3) The number of tiles will be minimum if size of each marble is maximum.
Size of each tile = HCF of 3.78 and 5.25 metre = 0.21 metre

```
378) 525 (1
     378
    ----
     147) 378 (2
          294
         -----
           84) 147 (1
               84
              ----
               63) 84 (1
                   63
                  ----
                   21) 63 (3
                       63
                      ----
                       ×
```

$\therefore$ Number of tiles $= \dfrac{3.78\times5.25}{0.21\times0.21} = 450$

MODEL EXERCISES

1. *PQRS* is a diameter of a circle whose radius is *r*. The lengths of *PQ, QR* and *RS* are equal. Semicircles are drawn on *PQ* and *QS* to create the shaded figure below
The perimeter of the shaded figure is —



(1) $2\pi r$ (2) $\frac{4\pi r}{3}$

(3) $\frac{5\pi r}{3}$ (4) $\frac{3\pi r}{2}$

(5) None of these

2. The Length of an edge of a hollow cube open at one face is $\sqrt{3}$ m. What is the length of the largest pole that it can accommodate ?

(1) $\sqrt{3}$ m (2) $3\sqrt{3}$ m

(3) 3 m (4) $\frac{3}{\sqrt{3}}$ m

(5) None of these

3. A tank 30 m long 20 m wide and 12 m deep is dug in a field 500 m long and 30 m wide. By how much will the level of the field rise if the earth dug out of the tank is evenly spread over the field ?

(1) 0.33 m (2) 0.5 m
(3) 0.25 m (4) 0.4 m
(5) None of these

4. A water tank in the form of a cuboid has its base 20 m long, 7 m wide and 10 m deep. Initially, the tank is full but later when water is taken out of it, the level of water in the tank reduces by 2 m. The volume of the water left in the tank is —

(1) 1120 m^3 (2) 400 m^3
(3) 280 m^3 (4) 140 m^3
(5) None of these

5. The floor of a rectangular room is 15 m long and l2 m wide.The Room is surrounded by a verandah of width 2 m on its sides. The area of the verandah is —

(1) 124 m^2 (2) 120 m^2
(3) 108 m^2 (4) 58 m^2
(5) None of these

6. A cylindrical vessel of radius 4 cm contains water. A solid sphere of radius 3 cm is lowered into the water until it is completely immersed. The water level in the vessel will rise by —

(1) $\frac{9}{2}$ cm (2) $\frac{9}{4}$ cm

(3) $\frac{4}{9}$ cm (4) $\frac{2}{9}$ cm

(5) None of these

7. A cylinder is circumscribed about a hemisphere and a cone is inscribed in the cylinder so as to have its vertex at the centre of one end and other end as its base. The volume of the cylinder, hemisphere and the cone are respectively in the ratio :

(1) 2 : 3 : 2 (2) 3 : 2 : 1
(3) 3 : 1 : 2 (4) 1 : 2 : 3
(5) None of these

8. A conical cavity is drilled in a circular cylinder of height 15 cm and base radius 8 cm. The height and the base radius of the cone are also same. Then, the whole surface of the remaining solid is —

(1) 440π cm^2 (2) 240π cm^2
(3) 640π cm^2 (4) 960π cm^2
(5) None of these

9. A conical flask has base radius '*a*' cm and height '*h*' cm. It is completely filled with milk. The milk is poured into a cylindrical thermos flask whose base radius is *'p'* cm. What will be the height of the solution level in the flask ?

(1) $\frac{a^2h}{3p^2}cm$ (2) $\frac{3hp^2}{a^2}$ cm

(3) $\frac{p^2}{3h^2}$ cm (4) $\frac{3a^2}{hp^2}$ cm

(5) None of these

10. The height of a room is 40% of its semiperimeter. It cost Rs 260 to paper the walls of the room with paper 50 cm wide at the rate of Rs 2 per m allowing an area of 15 m^2 for doors and windows. The height of the room is —

(1) 2.6 m (2) 3.9 m
(3) 4 m (4) 4.2 m
(5) None of these

11. The area of a figure formed by a square of side 8 cm and an isosceles triangle with base as one side of the square and perimeter as 18 cm is —

(1) 80 cm^2 (2) 84 cm^2
(3) 104 cm^2 (4) 76 cm^2
(5) None of these

12. A toy is in the form of a cone mounted on a hemisphere of radius 3.5 cm. The total height of the toy is 15.5 cm. Find the total surface area. $\left(\text{use}\pi = \frac{22}{7}\right)$

(1) 137.5 cm^2 (2) 214.5 cm^2
(3) 154 cm^2 (4) 291.5 cm^2
(5) None of these

13. A solid cylinder and a solid cone have equal base and equal height. If the radius and the height be in the ratio of 4 : 3, the ratio of the total surface area of the cylinder to that of the cone is in the ratio of —

(1) 10 : 9 (2) 11 : 9
(3) 12 : 9 (4) 14 : 9
(5) None of these

14. The volume of a cube is numerically equal to the sum of its edges. What is its total surface area in square units ?

(1) 66 (2) 183
(3) 36 (4) 72
(5) None of these

15. An ice-cream company makes a popular brand of ice-cream in rectangular shaped bar 6 cm long, 5 cm wide and 2 cm thick. To cut costs, the company has decided to reduce the volume of the bar by 20% the thickness will remain the same, but the length and wide will be decreased by the same percentage amount. The new length *L* will satisfy. —

(1) $5.5 < L < 6$ (2) $5 < L < 5.5$
(3) $4.5 < L < 5$ (4) $4 < L < 4.5$
(5) None of these

16. From a square piece of paper having each side equal to 10 cm, the largest possible circle is cut out. The ratio of the area of the circle to the area of the original square is closest to —

(1) $\frac{4}{5}$ (2) $\frac{3}{5}$
(3) $\frac{5}{6}$ (4) $\frac{6}{7}$
(5) None of these

17. The perimeter of a rectangular field is 480 m and the ratio between the length and the breadth is 5 : 3. The area is —

(1) 15500 m^2 (2) 13500 m^2
(3) 1550 m^2 (4) 1350 m^2
(5) None of these

18. The length breadth and height of a cuboid are in the ratio 1 : 2 : 3. The length, breadth and height of the cuboid are increased by 100%, 200% and 200% respectively. Then, the increase in the volume of the cuboid will be —

(1) 5 times (2) 6 times
(3) 12 times (4) 17 times
(5) None of these

19. A sphere of radius 3 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 6 cm. If the sphere is submerged completely then the surface of the water is raised by-

(1) $\frac{1}{4}$ cm (2) $\frac{1}{2}$ cm
(3) 1 cm (4) 2 cm
(5) None of these

20. If the length of the diagonal of a square and that of the side of another square are both 10 cm, the ratio of the area of the first square to that of the second is —

(1) 1 : 2 (2) 1 : 3
(3) 1 : 4 (4) 2 : 3
(5) None of these

21. If the circumference and the area of a circle are numerically equal, then what is the numerical value of the diameter ?

(1) 1 (2) 2
(3) 4 (4) π
(5) None of these

22. A circular park of 20 m diameter has a circular path just inside the boundary of width 1 m. The area of the path is —

(1) 15π (2) 17π
(3) 19π (4) 25π
(5) None of these

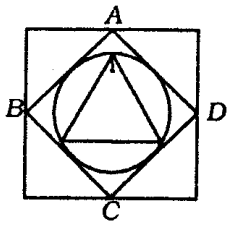
23. An edge of a cube measures 10 cm. If the largest possible cone is cut out of this cube, then the volume of the cone is —

(1) 260 cm^3 (2) 260.9 cm^3
(3) 261.9 cm^3 (4) 262.7 cm^3
(5) None of these

24. A square and an equilateral triangle have the same perimeter. If the diagonal of the square is $12\sqrt{2}$ cm, then the area of the triangle is —

(1) $24\sqrt{3}$ cm^2 (2) $24\sqrt{2}$ cm^2
(3) $64\sqrt{3}$ cm^2 (4) $32\sqrt{3}$ cm^2
(5) None of these

25. What is the area of the inner equilateral triangle, if the side of the outermost square is *'a'* ?
(ABCD is a square) —



(1) $\frac{3\sqrt{3}a^2}{32}$ (2) $\frac{3\sqrt{3}a^2}{64}$ (3) $\frac{5\sqrt{3}a^2}{32}$ (4) None of these

(5) $\frac{5\sqrt{3}a^3}{33}$

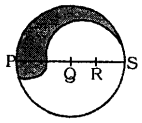
SHORT ANSWERS

1. (1) **2.** (3)
3. (2) **4.** (1)
5. (1) **6.** (2)
7. (2) **8.** (1)
9. (1) **10.** (3)
11. (4) **12.** (2)
13. (4) **14.** (4)
15. (2) **16.** (1)
17. (2) **18.** (4)
19. (3) **20.** (1)
21. (3) **22.** (3)
23. (3) **24.** (3)
25. (4)

EXPLANATIONS

1. (1) According to the question,
Diameter PS = $2r$

$\therefore$ PQ = QR = RS = $\frac{2r}{3}$



Perimeter of shaded portion

$$= \frac{1}{2}\left(2\pi r \times \frac{r}{3}\right) + \frac{1}{2}\left(2\pi r \times \frac{2r}{3}\right) + \frac{1}{2}(2\pi r)$$

$$= \frac{\pi r}{3} + \frac{2\pi r}{3} + \pi r$$

$$= \frac{\pi r + 2\pi r + 3\pi r}{3}$$

$$= \frac{6\pi r}{3} = 2\pi r$$

2. (3) According to the question.
Largest pole can be accommodated along the diagonal of cube.

$\therefore$ Diagonal of cube = $\sqrt{3}$ × side

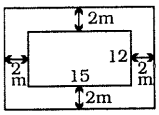
= $\sqrt{3} \times \sqrt{3}$ = 3 m

3. (2) Given that,
Volume of the earth taken out
= 30 × 20 × 12 = 7200 m^3
Area of the remaining portion (leaving the area of dug out portion)
= 470 × 30 + 30 × 10
= 14100 + 300 = 14400m^2
Let hbe the height to which the field is raised by spreading the earth dugout Then, 14400 × h = 7200

$h = \frac{7200}{14400}$ = 0.5 m

4. (1) Given that,
Volume of the tank = l × b × h
= 20 × 7 × (10 - 2) = 1120 cm^3

5. (1) Given that
Area of outer rectangle = 19 × 16 = 304 m^2



Area of inner rectangle = 15 × 12 = 180 m^2
$\therefore$ Required area = (304 - 180) = 124 m^3

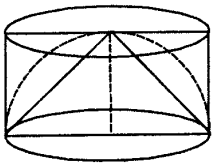
6. (2) According to question, volume of the sphere = Volume of the water displaced
Let the required height to which the water rises be h.

Then, $\pi r_1^2 h$

$\Rightarrow 16\,h = \frac{4}{3} \times 27$

$\Rightarrow h = \frac{39}{16} = \frac{9}{4}$ cm

7. (2) According to the information given in the question and the figure it is clear that Radius of the hemisphere = radius of cone = height of cone = height of cylinder.
let it be r, then ratio of volume of cylinder, hemisphere and cone

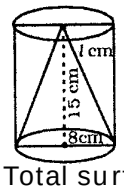


$= \pi r^3 : \frac{2}{3}\pi r^3 : \frac{1}{3}\pi r^2 h$

$= 1 : \frac{2}{3} : \frac{1}{3} = 3 : 2 : 1$

8. (1) According to question,

$l = \sqrt{15^2 + 8^2}$ = 17 cm



Total surface area of the remaining solid
= $2\pi rl + \pi r^2 + \pi rl$
= $2\pi \times 8 \times 15 + \pi(8)^2 + \pi \times 8 \times 17$
= $240\pi + 64\pi + 136\pi$
= 440π cm^2

9. (1) According to question,
Volume of conical flask = volume of cylindrical flask up to the required height (x)

$$\frac{1}{3}\pi a^2 h = \pi p^2 \times x = x = \frac{ha^2}{3p^2}$$

10. (3) Let the length, breadth and height of the room be l, b and h respectively.
Then, $h = 0.4\ (l + b)$
Area of the four walls = $2\ (l + b)\ h$
$= 2\ (l + b) \times 0.4\ (l + b) = 0.8\ (l + b)^2$
Required area where paper has to be pasted

$$= \frac{4}{5}(l + b)^2 - 15$$

Now, area of paper = area of wall Length × breadth $= \frac{4}{5}(l + b)^2 - 15$

$$\text{Length} = \frac{\frac{4}{5}(l+b)^2 - 15}{\frac{1}{2}}$$

$$\text{Given, } 2\left[\frac{4}{5}(l+b)^2 - 15\right] \times 2 = 260$$

$$\Rightarrow \frac{16}{5}(l + b)^2 - 60 = 260$$

$$\Rightarrow \frac{16}{5}(l + b)^2 = 320$$

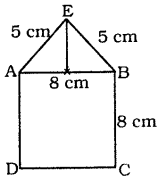
$$\Rightarrow (l + b)^2 = 100$$

$$\Rightarrow (l + b)^2 = 10$$

$$\therefore h = 0.4 \times 10 = 4 \text{ m}$$

11. (4) By question,
AE + AB + EB = 18 cm
AE + EB= 10 cm
AE = EB = 5 cm

$$EF = \sqrt{(5)^2 - (4)^2} = 3 \text{ cm}$$

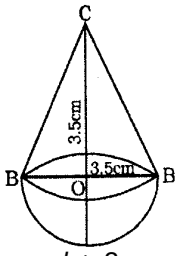


Now, area of the required figure = area of square + area of isosceles triangle

$$= 8 \times 8 + \frac{1}{2} \times 8 \times 3$$

$$= 64 + 12 = 76 \text{ cm}^2$$

12. (2) According to the question,
Total surface area of the figure = Total surface area of cone + Total surface area of hemisphere



$$= \pi rl + 2\pi r$$

$$= \pi r \times \sqrt{r^2 + h^2} + 2\pi r^2$$

$$= \frac{22}{7} \times 3.5 \sqrt{(3.5)^2 + (12)^2} + 2 \times \frac{22}{7} \times (3.5)^2$$

$$= 11 \sqrt{12.25 + 144} + \frac{44}{7} \times 12.25$$

$$= 11 \times 12.5 + 44 \times 1.75$$

$$= 137.5 + 77 = 214.5 \text{ cm}^2$$

13. (4) Let the radius be $4x$ and height be $3x$, then required ratio

$$= \frac{2\pi rh + 2\pi r^2}{\pi rl + \pi r^2} = \frac{2(h + r)}{\sqrt{r^2 + h^2} + r}$$

(where l is he slant height and

$$l = \sqrt{h^2 + r^2}$$

$$= \frac{2(7x)}{\sqrt{16x^2 + 9x^2} + 4x}$$

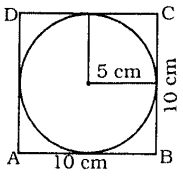
$$= \frac{14x}{5x + 4x} = \frac{14x}{9x} = 14 : 9$$

14. (4) Let x be the side, (edge of the cube)
Given, $a^3 = 12a$
$\therefore\ a^2 = 12$
Total surface area = $6a^2$
$= 6 \times 12 = 72$

15. (2) According to question, volume of ice-cream = $6 \times 5 \times 2 = 60 \text{ cm}^2$
Also, $L \times b \times 2 = 48$
$\Rightarrow$ $L \times b = 24$
Now, $L = 6 - 6 \times 10\% = 5.4$
$b = 5 - 5 \times 10\% = 4.5$
$L \times b = 5.4 \times 4.5 = 24.3$
Clearly, $5 < L < 5.5$

16. (1) By question, area of square = $(10)^2 = 100 \text{cm}^2$

$$\text{Area of circle} = \frac{22}{7} \times (5)^2 = \frac{22 \times 25}{7}$$



$\therefore$ Required ratio = $\frac{22\times25}{7\times100} = \frac{22}{28} = \frac{11}{14}$

= 0.785 which is closed to $\frac{4}{5}$

17. (2) Let the length and breadth be $5x$ and $3x$ respectively,
then 2 (length + breadth) = 480
$\therefore$ 2 ($5x + 3x$) = 480
$\Rightarrow 8x = 240$
$\Rightarrow x = 30$m
$\therefore$ Sides are 150 m and 90 m
Hence, area = 150 × 90 = 13500 m^2

18. (4) Let the length, breadh and height of the cube be x, $2x$ and $3x$ respectively.
Then, volume = $x \times 2x \times 3x = 6x^2$
New length, breadth and height
= $2x$, $6x$ and $9x$ respectively.
Now volume = $108x^3$
Hence, increase in volume

= $\left(\frac{108-6}{6}\right)x^3$ = 17 times of original volume.

19. (3) Let the rise in the level of water be h cm.
Then, volume of the sphere = volume of the water displaced

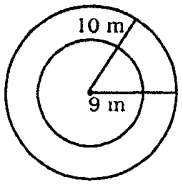
$\Rightarrow \frac{4}{3}\pi(3)^2 = \pi(6)^2 \times h$

$\Rightarrow h = \frac{\frac{4}{3}\times27}{36}$ = 1 cm

20. (1) Let the side of the 1st suare be x cm.
Then, for first square $x^2 + x^2 = 100$
$x^2 = 50$ cm^2
Hence, area of 1st square is 50 cm^2.
Area of IInd square =100 cm^2.
$\therefore$ Required ratio = 50 : 100 = 1 : 2

21. (3) Let the radius of the circle be r cm.
Then, $2\pi r = \pi r^2$
$\Rightarrow r$ = 2 cm
Diameter = 2 × 2 = 4 cm

22. (3) According to the question,
Area of circular path



= $\pi(10^2 - 9^2) = 19\pi$

23. (3) According to the question,
Height of the cone = 10 cm
Radius = 5 cm
$\therefore$ Required volume

= $\frac{1}{3}\times \pi \times r^2 \times h$

= $\frac{1}{3}\times\frac{22}{7}\times 25 \times 10$

= 261.9 cm^3

24. (3) Let the side of the square be x.

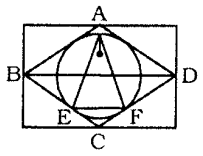
Then, $2x^2 = \left(12\sqrt{2}\right)^2$

$\Rightarrow x$ = 12
Now, perimeter of equilateral triangle
= 12 × 4 = 48 cm
Side of equilateral triangle

= $\frac{48}{3}$ = 16 cm
Area of equilateral triangle

= $\frac{\sqrt{3}}{4}\times(16)^2$ = 64 73cm^2

25. (4) According to the question,
BD || EF and BD = 2EF



$\therefore$ EF = $\frac{1}{2}\times a$

$\therefore$ Area of equilateral triangle

= $\frac{\sqrt{3}}{4}\left(\frac{1}{2}a\right)^2$

= $\frac{\sqrt{3}}{4}\times\frac{1}{4}a^2 = \frac{\sqrt{3}}{16}a^2$

MISCELLANEOUS

NATIONALISED BANKS & IBPS SO/MT/SO

1. if $\frac{(a-b)^2-(a+b)^2}{-4a} = \frac{x}{y}$

On simplifying the above mentioned equation, what will be the equation ?

(1) $xy = b$ (2) $bx = y$
(3) $by = x$ (4) $ab = x$
(5) $ay = x$

(Corporation Bank PO Exam. 22.11.2009)

2. Angle 'A' of the quadrilateral ABCD is 26° less than angle B. Angle B is twice angle C and angle C is 10° more than the angle D. What would be the measure of angle A ?

(1) 104° (2) 126°
(3) 56° (4) 132°
(5) 106°

(Corporation Bank PO Exam. 22.11.2009)

3. In every 30 minutes the time of a watch increases by 3 minutes. After setting the correct time at 5 a.m., what time will the watch show after 6 hours ?

(1) 10 : 54 a.m. (2) 11 : 30 a.m.
(3) 11: 36 a.m. (4) 11 : 42 a.m.
(5) 11: 38 p.m.

(Corporation Bank PO Exam. 22.11.2009)

Directions (4-6) : In the following questions three squares A, B and C are given which contain four figures/ numbers related to each other. The figures/ numbers in the first two squares are given as examples and bear a certain relationship to each other within the square. This relationship can be established/calculated vertically, horizontally or diagonally. Depending on this relationship between the figures/numbers in the first two squares, find out what should come in place of the question mark in the third square - C.

(United Bank Of India PO Exam. 14.11.2010)

4.

32 | 72 92 | 52	16.5 | 56.5 76.5 | 36.5	? | 13 33 | -7
A	B	C

(1) 27 (2) -13
(3) 6 (4) 13
(5) -27

5.

9 | 540 45 | 180	5 | 300 25 | 100	2.4 | 144 12 | ?
A	B	C

(1) 48 (2) 36
(3) 24 (4) 52

6.

6 | 12 48 | 24	4 | 8 32 | 16	25 | ? 20 | 10
A	B	C

(1) 4 (2) 5
(3) 5.5 (4) 6.5
(5) 9

7. Rachita enters a shop to buy icecreams, cookies and pastries. She has to buy atleast 9 units of each. She buys more cookies than ice-creams and more pastries than cookies. She picks up a total of 32 items. How many cookies does she buy ?

(1) Either 12 or 13
(2) Either 11 or 12
(3) Either 10 or 11
(4) Either 9 or 11
(5) Either 9 or 10

(IBPS Bank PO/MT CWE 17.06.2012)

8. The fare of a bus is Rs *x* for the first five kilometres and Rs13 per kilometres thereafter. If a passenger pays Rs 2,402 for a journey of 187 kilometres, what is the value of X ?

(1) Rs 29 (2) Rs 39
(3) Rs 36 (4) Rs 31
(5) None of these

(IBPS Bank PO/MT CWE 17.06.2012)

9. Varun got a monthly increment of 12 pecent of Puja's monthly salary. Puja's monthly salary is Rs 7800. Varun's monthly salary before increment was Rs 6400. What amount will he earn in four months after his increment ?

(1) Rs 29344 (2) Rs 29434
(3) Rs 28434 (4) Rs 28344
(5) None of these

(IDBI Bank Officer Exam.16.09 2012)

10. In a school some sweats were to be distributed among 420 children on the occasion of Teacher's day. But 140 children remained absent on that particular day and hence each child got one sweat extra. How many sweats each child would have got originally ?

(1) Cannot be determined
(2) 2
(3) 5
(4) 4
(5) 1

(IBPS Specialist Officer CWE 17.03.2013)

11. The cost of 14 keyboards and 8 mouse-pads is Rs. 26,240. What is the cost of 35 keyboards and 20 mouse-pads ?
(1) Rs. 65,600
(2) Rs. 58,800
(3) Rs. 76,550
(4) Cannot be determined
(5) None of these

(IBPS RRBs Office Assistant CWE Exam. 09.09.2012)

12. A boat covers 20 km in 4 hours along the current and 9 km in 3 hours against the current. What is the speed of the current ?
(1) 2 kmph (2) 1 kmph
(3) 1.5 kmph (4) 1.75 kmph
(5) None of these

(Indian Overseas Bank PO Online Exam. 01.09.2013)

13. A pump can fill a tank with water in 2 hours. Because of a leak, it took $2\frac{1}{3}$ hours to fill the tank, The leak can drain all the water of the tank in
(1) $4\frac{1}{3}$ hours (2) 7 hours
(3) 8 hours (4) 14 hours
(5) None of these

(IBPS Bank PO/MT CWE-III 26.10.2013)

14. Pipe A can fill a tank In 30 minutes while pipe B can fill it in 45 minutes. An other pipe C can empty a full tank in 60 minutes. If all three pipes are opened simultaneously, the empty tank will be filled in
(1) $\frac{2}{7}$ hour (2) $\frac{3}{7}$ hour
(3) $\frac{4}{7}$ hour (4) $\frac{5}{7}$ hour
(5) None of these

(Corporation Bank Specialist Officer (Marketing) Exam. 22.12.2014)

SBI PO EXAMS

1. If $2x + y = 15$, $2y + z = 25$ and $2z + x = 26$, what is the value of z ?
(1) 4 (2) 7
(3) 9 (4) 12
(5) None of these

(SBI Associate Banks PO Exam. 14.02.1999)

2. The top of a 15 metre high tower makes an angle of elevation of 60° with the bottom of an electric pole and an angle of elevation of 30° with the top of the pole. What is the height of the electric pole ?
(1) 5 metres (2) 8 metres
(3) 10 metres (4) 12 metres
(5) None of these

(SBI Associate Banks PO Exam. 14.02.1999)

3. Which of the following values of P satisfy the inequality P (P - 3) < 4P - 12 ?
(1) P > 14 or P < 13
(2) $24 \leq P < 71$
(3) P > 13; P < 51
(4) 23 < P < 41
(5) P = 4, P = +3

(SBI Associate Banks PO Exam. 14.02.1999)

4. Four of the following five parts numbered (1), (2), (3), (4) and (5) in the following equation are exactly equal. Which of the parts is not equal to the other four ? The number of that part is the answer.
(1) $xy^2 - x^2y + 2x^2y^2 = xy^2 (1 - 2x) + x^2y$
(2) $(2x - 1) = xy^2$
(3) $[y (1 + x) - (1 - y)] = xy [(x + y)^2 + y$
(4) $(1 + x) - x^2y (1- y) = xy$
(5) $(1 - y) - x(1 + x)$

5. Four of the following five parts numbered (1), (2), (3), (4) and (5) are equal. Number of the part which is not equal to the other four parts is the answer.
(1) $136 \times 12 \div 17 + 4 =$
(2) $152 \times 12 \div 19 + 48 \div 12 =$
(3) $268 \div 67 \times 4^2 + 6^2 =$
(4) $234 \div 13 \times 4 + 84 \div 3 =$
(5) $378 \div 42 \times 9 + 3^2 + 4$

(SBI Associate Banks PO Exam. 16.07.2000)

6. In the following equation of equations four out of the five parts numbered (1), (2), (3), (4) and (5) are exactly equal. Which one part is not equal to the other four? The number of that part is the answer.
(1) $120 \times 12 - 22 \times 20 =$
(2) 10% of 500 + $\frac{2}{5}$ of 1200 =
(3) $80 \times 40 - 20 \times 110 =$
(4) $8640 \div 60 + 53.5 \times 16 =$
(5) 5314 - 3029- 1285 =

(SBI Banks PO Exam. 20.08.2000)

7. In the following equation four out of the five parts numbered (1), (2), (3), (4) and (5) are

exactly equal. Which part is not equal to the other four? The number of that part is the answer.
(1) $5^3 + 3^3 + 48 =$
(2) $5^2 \times 3^3 - 475 =$
(3) $3^5 - 44 =$
(4) $4^3 + 2 \times 17 \times 4 =$
(5) $(6)^3 - (2)^4 =$

(SBI Banks PO Exam. 20.08.2000)

8. If $3x + 7 = x^2 + M = 7x + 5$,
what is the value of M ?
(1) 2
(2) $8\frac{1}{4}$
(3) $8\frac{1}{2}$
(4) Cannot be determined
(5) None of these

(SBI Banks PO Exam. 20.08.2000)

9. If $2x + 3y = 34$ and $\frac{x+y}{y} = \frac{13}{8}$, then $5y + 7x$ = ?
(1) 82 (2) 175
(3) 178 (4) 185
(5) None of these

(SBI Banks PO Exam. 11.02.2001)

10. Taps A, B and C are connected to a water tank and the rate of flow of water is 42 litres/hr., 56 litres/ hr. and 48 litres/hr. respectively. Taps A and B fill the tank while tap C empties the tank. If all the three taps are opened simultaneously, the tank gets completely filled up in 16 hours. What is the capacity of the tank ?
(1) 960 litres (2) 2346 litres
(3) 1600 litres (4) 800 litres
(5) None of these

(SBI Banks PO Exam. 11.02.2001)

11. Four of the following five parts numbered (1), (2), (3), (4) and (5) are equal to other four parts is the answer.
(1) $(2x + 3y)^2 =$
(2) $(2y - x)^2 + y(16x + 5y) =$
(3) $4x(x + 2y) + y(4x + 9y) =$
(4) $(2x + 2y)^2 + y(4x + 5y) =$
(5) $(2x - y)^2 + 8y(2x + y)$

(SBI Banks PO Exam. 11.02.2001)

12. A boat running down stream covers a distance of 16 km in 2 hours while for covering the same distance upstream it takes 4 hours. What is the speed of the boat in still water ?
(1) 4 kmph
(2) 6 kmph
(3) 8 kmph
(4) Data inadequate
(5) None of these

(SBI Associate Banks PO Exam. 21.07.2002)

13. If $2x + 3y + z = 55$; $x + z - y = 4$ and $y - x + z = 12$ then what is the value of y ?
(1) 7 (2) 8
(3) 12 (4) 9
(5) None of these

(SBI Associate Banks PO Exam. 21.07.2002)

14. If $3y + 2x = 47$ and $11x = 7y$ then what is value of $y - x$?
(1) 4 (2) 6
(3) 7 (4) 5
(5) None of these

(SBI Associate Banks PO Exam. 21.07.2002)

15. Four of the following five parts numbered (1), (2), (3), (4) and (5) are exactly equal. The number of the part which is not equal to the other four is the answer
(1) $136 \times 12 \div 9 \times 3 =$
(2) $17 \times 64 \div 6 \times 3 =$
(3) $36 \times 17 \div 9 \times 8 =$
(4) $56 \times 8 \div 14 \times 17 =$
(5) $76 \times 6 \div 19 \times 17$

SBI Associate Banks PO Exam. 21.07.2002

16. For which of the following values of x the inequality
$x(x + 3) < 10$ is satisfied ?
(1) $x > 2, x < -5$
(2) $-5 < x < 2$
(3) $-2 < x < 5$
(4) $x < -2, x > 5$
(5) None of these

(SBI Associate Banks PO Exam. 21.07.2002)

17. A boat takes 6 hours to travel from place M to N downstream and back from N to M upstream. If the speed of the boat in still water is 4 km./hr., what is the distance between the two places ?
(1) 8 kms.
(2) 12 kms.
(3) 6 kms.
(4) Data inadequate
(5) None of these

(SBI PO Exam. 28.11.2006)

18. The speed of a boat when travelling downstream is 32 kmph whereas when travelling upstream it is 28 kmph. What is the speed of the boat in still water ?

(1) 27 Kmph
(2) 29 Kmph
(3) 31 Kmph
(4) Cannot be determined
(5) None of these

(SBI & Rural Business PO Exam. 18.04.2010)

INSURANCE EXAMS

1. A boat running down stream covers a distance of 16 km in 2 hours while for covering the same distance upstream it takes 4 hours. What is the speed of the boat in still water ?
(1) 4 kmph (2) 6 kmph
(3) 8 kmph (4) Data inadequate
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 2006)

2. River is running at 2 kmph. It took a man twice as long to row up as to row down the river. The rate (in km ph) of the man in still water is :
(1) 8 (2) 10
(3) 4 (4) 6

(United India Insurance Co. (AAO) Exam. 11.03.2007)

3. Three pipes A, B and C can fill a cistern in 6 hours. After working at it together for 2 hours, C is closed and A and B can fill the remaining part in 7 hours. The number of hours taken by C alone to fill the cistern is :
(1) 12 (2) 14
(3) 16 (4) 18

(United India Insurance Co. (AAO) Exam. 11.03.2007)

4. The angles of a convex hexagon in degrees are integers and in arithmetic progression. L and M denote the largest of these 6 angles. Then the maximum value that M can take is :
(1) 125 degree (2) 150 degree
(3) 175 degree (4) 179 degree

(General Insurance Corporation AAO Exam. 11.12.2011)

5. The inequality $2x^2 + 9x + 4 < 0$ is satisfied for which of the the following values of'x'?

(1) $-4 < x - \frac{1}{2}$ (2) $\frac{1}{2} < x < 4$
(3) $1 < x < 2$ (4) $-2 < x < 1$
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 24.04.2005)

6. If $2x + 3y + z = 55$; $x + z - y = 4$and $y - x + z = 12$ then what is the value of y ?
(1) 7 (2) 8
(3) 12 (4) 9
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 2006)

7. If $3y + 2x = 47$ and $11x = 7y$ then what is value of $y - x$?
(1) 4 (2) 6
(3) 7 (4) 5
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 2006)

8. Two straight lines can divide a circular disk into a maximum of 4 parts. Likewise into how many parts can four straight lines divide a circular disk ?
(1) 8 (2) 9
(3) 10 (4) 11

(General Insurance Corporation AAO Exam. 11.12.2011)

9. A person can row a boat d km upstream and the same distance downstream $5\frac{1}{4}$ hours. Also he can row the boat $2d$ km upstream in 7 hours. He will row the same distance downstream in

(1) $3\frac{1}{2}$ hours (2) $3\frac{1}{4}$ hours

(3) $1\frac{1}{4}$ hours (4) 4 hours

(Oriental Insurance Company AAO Exam. 08.04.2012)

10. The rate at which a river flows is one-third the speed of a boat in still water. If that boat travels down the river for 2 hours and then back up river for 2 hours, it will be 16 km short of its starting point. The speed (km/hour) of the boat in still water is
(1) 4 (2) 6
(3) 8 (4) 12

(United India Insurance AAO Exam. 03.06.2012)

11. If $a^x = b^y = c^z$ and $\frac{b}{a} = \frac{c}{b}$ then $\frac{2z}{x+z} = ?$

(1) $\frac{y}{x}$ (2) $\frac{x}{y}$

(3) $\frac{x}{z}$ (4) $\frac{z}{x}$

(United India Insurance AAO Exam. 03.06.2012)

12. If $f(x) = 2x^3 + 3x^2 + 5$ then $f(2) = ?$
(1) 31 (2) 32
(3) 33 (4) 35
(United India Insurance AAO Exam. 03.06.2012)

13. For what value(s) of k, the roots of the equation $9x^2 + 2Kx + 4 = 0$ will be equal ?
(1) 6 (2) -6
(3) ±6 (4) ±5
(United India Insurance AAO Exam, 03.06.2012)

14. A 10 metre, long ladder is placed against a wall. It is inclined at an angle of 30° to the ground. The distance of the foot of the ladder from the wall is
(1) 7.32 m (2) 8.26 m
(3) 8.66 m (4) 8.16 m
(NICL (GIC) Administrative Officer Exam. 15.12.2013)

15. How many different triangles are there for which the lengths of the sides are 3, 8 and n, where n is an integer and $3 < n < 8$?
(1) Two (2) Three
(3) Four (4) Five
(NICL (GIC) AO (Finance) Exam. 15.12.2013)

16. The value of k for which $x - 1$ is a factor of $4x^3 + 3x^2 - 4x + k$, is
(1) 3 (2) 1
(3) -2 (5) -3
(NICL (GIC) AO (Finance) Exam. 18.12.2013)

17. Each inch on ruler A is marked in equal $\frac{1}{8}$- inch units, and each inch on ruler B is marked in equal $\frac{1}{12}$- inch units. When ruler A is used, a side of triangle measures 12 of the $\frac{1}{8}$- inch units. When ruler B is used, how many $\frac{1}{12}$- inch units will the same side measure ?
(1) 8 (2) 12
(3) 18 (4) 20
(NICL (GIC) AO (Finance) Exam. 15.12.2013)

18. Some students planned a picnic. The budget for food was Rs 500. But, 5 of them failed to go and thus the cost of food for each member increased by Rs 5. How many students attended the picnic ?
(1) 15 (2) 20
(3) 25 (4) 30
(NICL (GIC) AO (Finance) Exam. 15.12.2013)

SHORT ANSWERS

NATIONALISED BANKS & IBPS PO/MT/SO

I.	(3)	**2.**	(5)
3.	(3)	**4.**	(5)
5.	(1)	**6.**	(2)
7.	(3)	**8.**	(3)
9.	(1)	**10.**	(2)
11.	(1)	**12.**	(2)
13.	(4)	**14.**	(2)

SBI PO EXAMS

1.	(5)	**2.**	(3)
3.	(5)	**4.**	(2)
5.	(5)	**6.**	(2)
7.	(3)	**8.**	(2)
9.	(5)	**10.**	(4)
11.	(2)	**12.**	(2)
13.	(5)	**14.**	(1)
15.	(5)	**16.**	(5)

INSURANCE EXAMS

1.	(2)	**2.**	(4)
3.	(2)	**4.**	(4)
5.	(1)	**6.**	(5)
7.	(1)	**8.**	(1)
9.	(1)	**10.**	(4)
11.	(1)	**12.**	(3)
13.	(3)	**14.**	(3)
15.	(1)	**16.**	(4)
17.	(1)	**18.**	(2)

EXPLANATIONS

NATIONALISED BANKS & IBPS PO/MT/SO

1. (3) $\frac{(a-b)^2-(a+b)^2}{-4a}=\frac{x}{y}$

$\Rightarrow \frac{-4ab}{-4a}=\frac{x}{y} \Rightarrow \frac{x}{y}=b \Rightarrow x=by$

2. (5) Let $\angle A = x^\circ$

$\therefore \angle B = x + 26$

$\angle C = \frac{x+26}{2} = \frac{x}{2} + 13$

$\angle D = \frac{x}{2} + 3$

$\therefore x + x + 26 + \frac{x}{2} + 13 + \frac{x}{2} + 3 = 360^\circ$

$\Rightarrow 3x = 360 - 42 = 318^\circ$

$\Rightarrow x = \frac{318}{3} = 106^\circ$

3. (3) Time gained in 6 hours
$= 12 \times 3 = 36$ minutes
$\therefore$ Required time $= 11 : 36$ a.m.

4. (5) Square A $\Rightarrow 92 - 32 = 60$;
$72 - 52 = 20$
Square B $\Rightarrow 76.5 - 16.5 = 60$;
$56.5 - 36.5 = 20$
Square C $\Rightarrow ? = 33 - 60 = -27$

5. (1) Square A $\Rightarrow 9 \times 5 = 45$;
$180 \times 3 = 540$
Square B $\Rightarrow 5 \times 5 = 25$;
$100 \times 3 = 300$
Square C $\Rightarrow 2.4 \times 5 = 12$

$? = \frac{144}{3} = 48$

6. (2) Square A $\Rightarrow 6 \times 8 = 48$
$12 \times 2 = 24$
Square B $\Rightarrow 4 \times 8 = 32$;
$8 \times 2 = 16$
Square C $\Rightarrow 2.5 \times 8 = 20$

$? = \frac{10}{2} = \underline{5}$

7. (3) Look :
Ice-creams + Cookies + Pastries = 32
$\Rightarrow 9 + 11 + 12 = 32$
or, $9 + 10 + 13 = 32$
$10 + 11 + 12 \neq 32$

8. (3) According to the question,
$x + 182 \times 13 = 2402$
$\Rightarrow x + 2366 = 2402$
$\Rightarrow x = 2402 - 2366$
= Rs. 36

9. (1) Monthly increase in salary of Varun

$= \frac{7800 \times 12}{100} =$ Rs. 936

$\therefore$ New monthly salary of Varun
$= 6400 + 936 =$ Rs. 7336
$\therefore$ Required income $= 4 \times 7336$
= Rs. 29344

10. (2) If the number of all sweats be x, then

$\frac{x}{280} - \frac{x}{420} = 1$

$\Rightarrow \frac{3x-2x}{840} = 1$

$\Rightarrow x = 840$

$\therefore$ Required answer $= \frac{840}{420} = 2$

11. (1) CP. of 1 key board = Rs. x.
CP. of 1 mouse pad = Rs. y.
$\therefore 14x + 8y = 26240$
Dividing both sides by 2,
$7x + 4y = 13120$
Multiplying both sides by 5,
$35x + 20y = 65600$

12. (2) Rate downstream

$= \frac{20}{4} = 5$ kmph

Rate upstream $= \frac{9}{3} = 3$ kmph

$\therefore$ Speed of current

$= \frac{1}{2}(5 - 3) = 1$ kmph

13. (4) Part of tank emptied in 1 hour by the leak

$= \frac{1}{2} - \frac{3}{7} = \frac{1}{14}$

The leak will empty the tank in 14 hours.

14. (2) Part of tank filled by all three pipes in 1 minute

$= \frac{1}{30} + \frac{1}{45} - \frac{1}{60} = \frac{6+4-3}{180} = \frac{7}{180}$

$\therefore$ Time taken $= \frac{180}{7}$ minutes

$= \frac{180}{7\times 60} = \frac{3}{7}$ hours

SBI PO EXAMS

1. (5) According to question,

$y = \frac{25-z}{2}$

$x = 26 - 2z$

$\therefore 2(26 - 2z) + \frac{25-z}{2} = 15$

$\Rightarrow 52 - 4z + \frac{25-z}{2} = 15$

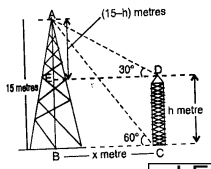
$\Rightarrow 104 - 8z + 25 - z = 30$

$\Rightarrow -9z = 30 - 129$

$\Rightarrow -9z = -99$

$\therefore z = 11$

2. (3)



Now, according to question,
AB = Height of Tower = 15 metres
CD = Height of Electric Pole
= h metres
$\angle ADE = 30°$
$\angle ACB = 60°$
$\angle ACB = 60°$
AE = (15 - h) metres
Distance between B and C
= x metres
We know that

$\text{Tan}\,\theta = \frac{\text{Perpendicular}}{\text{Base}}$

$\therefore \tan 30° = \frac{15-h}{x}$

or, $\frac{1}{\sqrt{3}} = \frac{15-h}{x}$

or, $\sqrt{3} = \frac{15}{x}$

$\therefore x = \sqrt{3}(15 - h)$ ....(i)

Now, $\tan 60° = \frac{15}{x}$

$\therefore x = \frac{15}{\sqrt{3}}$ ....(ii)

From equation (i) and (ii)

$\sqrt{3}(15 - h) = \frac{15}{\sqrt{3}}$

or, $15 - h = \frac{15}{3}$

or, $h = 15 - 5 = 10$ metres.

3. (5) $P(P - 3) = 4P - 12$
$P^2 - 3P = 4P - 12$
$P^2 - 7P + 12 = 0$

$\because P = \frac{-(-7) \pm \sqrt{49 - 4\times 12}}{2}$

$= \frac{7 \pm \sqrt{1}}{2} = \frac{7+1}{2}, \frac{7-1}{2} = 4, 3$

4. (2) **Option (1)**
$\Rightarrow xy^2 - x^2y + 2x^2y^2$
Option (2)
$\Rightarrow xy^2(1 - 2x) + x^2y(2y - 1)$
$\Rightarrow xy^2 - 2x^2y^2 + 2x^2y^2 - x^2y$
$= xy^2 - x^2y$
Option (3)
$xy^2(1 + x) - x^2y(1 - y)$
$= xy^2 + x^2y^2 - x^2y + x^2y^2$

$= xy^2 - x^2y + 2x^2y^2$

5. (5) (1) = 136 × 12 ÷ 17 + 4 = 100
(2) = 152 × 12 ÷ 19 + 48 ÷ 12

$= \frac{152 \times 12}{19} + \frac{48}{12} = 100$

(3) = 268 ÷ 67 × 4^2 + 6^2

$= \frac{268 \times 16}{67} + 36 = 100$

(4) = 234 ÷ 13 × 4 + 84 ÷ 3

$= \frac{234 \times 4}{13} + \frac{84}{3} = 100$

(5) = 378 ÷ 42 × 9 + 3^2 + 4

$= \frac{378 \times 9}{42} + 13 = 94$

6. (2) (1) 120 × 12 - 22 × 20
= 1440 - 440 = 1000

(2) 10% of 5000 + $\frac{2}{5}$ of 1200

$= \frac{10}{100} \times 5000 + \frac{2}{5} \times 1200$

= 500 + 480 = 980
(3) 80 × 40 - 20 × 110
= 3200 - 2200 = 1000
(4) 8640 ÷ 60 + 53.5 × 16
= 144 + 856 = 1000
(5) 5314 - 3029 - 1285
= 2285 - 1285 = 1000

7. (3) (1) $5^3 + 3^3 + 48$
= 125 + 27 + 48 = 200
(2) $5^2 \times 3^3 - 475$
= 25 × 27 - 475
= 675 - 475 = 200
(3) $3^5 - 44 = 243 - 44 = 199$
(4) $4^3 + 2 \times 17 \times 4$
= 64 + 136 = 200
(5) $(6)^3 - (2)^4 = 216 - 16 = 200$

8. (2) $3x + 7 = x^2 + M = 7x + 5$
We can write .
$7x + 5 = 3x + 7$
or, $4x = 2$

or, $x = \frac{1}{2}$

Now, $3x + 7 = x^2 + M$
or, $M = 3x + 7 - x^2$

$= 3 \times \frac{1}{2} + 7 - \left(\frac{1}{2}\right)^2$

$= 1\frac{1}{2} + 7 - \frac{1}{4} = 1\frac{1}{4} + 7 = 8\frac{1}{4}$

9. (5) $\frac{x+y}{y} = \frac{13}{8}$

$8x + 8y = 13y$
$8x - 5y = 0$ ....(i)
Also, $2x + 3y = 34$
Multiplying this equation by 4 we can write,
$8x + 12y = 136$ ...(ii)
Subtracting (i) from (ii) we get,
$17y = 102$
$y = 6$
Substituting this value in
$2x + 3y = 34$ we get,
$2x + (3 \times 6) = 34$
or, $2x = 16$
$\therefore x = 8$
$\therefore 5y + 7x = 5 \times 6 + 7 \times 8$,
= 30 + 56
= 86

10. (4) Net amount of water filled in the tank in 1 hour when all three taps are opened simultaneously
= 42 + 56 - 48 litres
= 50 litres.
The tank gets completely filled in 16 hours.
$\therefore$ Capacity of the tank
= 16 × 50 = 800 litres.

11. (2) (1) $(2x + 3y)^2 = 4x^2 + 9y^2 + 12xy$
(2) $(2y - x)^2 + y(16x + 5y)$
$= 4y^2 + x^2 - 4xy + 16xy + 5y^2$
$= x^2 + 9y^2 + 12xy$
(3) $4x(x + 2y) + y(4x + 9y)$
$= 4x^2 + 8xy + 4xy + 9y^2$
$= 4x^2 + 9y^2 + 12xy$
(4) $(2x + 2y)^2 + y(4x + 5y)$
$= 4x^2 + 4y^2 + 8xy + 4xy + 5y^2$
$= 4x^2 + 9y^2 + 12xy$
(5) $(2x - y)^2 + 8y(2x + y)$
$= 4x^2 + y^2 - 4xy + 16xy + 8y^2$
$= 4x^2 + 9y^2 + 12xy$
Thus, we see that except (2), other four expressions are equal.

12. (2) Rate upstream

$= \frac{16}{2}$ kmph = 8 kmph

Rate downs tream $= \frac{16}{4}$

= 4 kmph
$\therefore$ Rate In still water

$= \frac{1}{2}$ (8 + 4) kmph = 6 kmph

13. (5) Given,
$2x + 3 + z = 55$ ....(i)

$x + z - y = 4$ ...(ii)
$y - x + z = 12$ ...(iii)
From (ii) and (iii)
$2z = 16 \Rightarrow z = 8$
From (ii), $x + 8 - y = 4$
$\Rightarrow x = 4 - 8 + y = y - 4$
Putting this value in (i) we get
$2(y - 4) + 3y + 8 = 55$
$\Rightarrow 2y - 8 + 3y + 8 = 55$
$\Rightarrow 5y = 55$
$\Rightarrow y = 11$

14. (1) Given $3y + 2x = 47$
$7y - 11x = 0$ ....(ii)

From (ii), $x = \frac{7}{11}y$

$\therefore$ Equation (i) reduces to

$3y + 2 \times \frac{7}{11}y = 47$

$\Rightarrow \frac{33y + 14y}{11} = 47$

$\Rightarrow y = \frac{11 \times 47}{47} = 11$

From (ii), $x = \frac{7 \times 11}{11} = 7$

$\therefore$ Required difference
$= y - x$
$= 11 - 7 = 4$

15. (5)
Option (1) : $136 \times 12 \div 9 \times 3$

$= 136 \times 12 \times \frac{1}{9} \times 3 = 544$

Option (2) : $17 \times 64 \div 6 \times 3$

$= 17 \times 64 \times \frac{1}{6} \times 3 = 544$

Option (3) : $36 \times 17 \div 9 \times 8$

$= 56 \times 8 \times \frac{1}{4} \times 18 = 544$

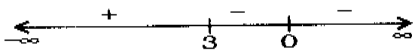
Option (4) : $56 \times 8 \div 14 \times 17$

$= 56 \times 8 \times \frac{1}{4} \times 18 = 544$

Option (5) : $76 \times 6 \div 19 \times 17$

$= 76 \times 6 \times \frac{1}{19} \times 17 = 408$

16. (5) Given inequality :
$x(x + 3) < 0$
Corresponding equation:
$x(x + 3) = 0$
$\Rightarrow x = 0$
or $x = -3$



The given inequality holds for
$-3 < x < 0$

17. (4) Total time = 6 hours.
Speed of the boat in still water = 4 km/hr.
Let the distance between M and N be D. and the speed of the stream be x.

$D\left[\frac{1}{4+x} + \frac{1}{4-x}\right] = 6$

$D\left[\frac{4 - x + 4 + x}{(4+x)(4-x)}\right] = 6$

$D\left[\frac{8}{4^2 - x^2}\right] = 6$

$\frac{8D}{16 - x^2} = 6$

$D = \frac{6}{8}(16 - x^2) = \frac{3}{4}(16 - x^2)$

Since the speed of the stream (x) is not given, the distance D cannot be determined.

18. (5) Tricky Approach
Speed of the boat in still water

$= \frac{1}{2}$ (Rate down stream + Rate upstream)

$= \frac{1}{2}(32 + 28) = 30$ kmph.

INSURANCE EXAMS

1. (2) Rate upstream

$= \frac{16}{2}$ kmph $= 8$ kmph

Rate downs tream $= \frac{16}{4}$

$= 4$ kmph
$\therefore$ Rate in still water

$= \frac{1}{2}(8 + 4)$ kmph $= 6$ kmph

2. (4) Let rate upstream be x kmph.
Then, rate downstream
$= 2x$ kmph
$\therefore$ Rate of current

$\frac{1}{2}(2x - x) = \frac{x}{2}$ kmph

$\therefore \frac{x}{2} = 2 \Rightarrow x = 4$

$\therefore$ Rate upstream = 4 kmph
Rate downstream = 8 kmph
$\therefore$ Rate in still water

$= \frac{1}{2}(8 + 4) = 6$ kmph

3. (2) Part of the tank filled by three pipes (A + B + C) in 1 hour

$= \frac{1}{6}$

$\therefore$ Part filled in 2 hours $= \frac{2}{6} = \frac{1}{3}$

Remaining part $= 1 - \frac{1}{3} = \frac{2}{3}$

Part filled by (A + B) in 7 hours $= \frac{2}{3}$

$\therefore$ (A + B)'s 1 hour's work $= \frac{2}{21}$

$\therefore$ Part filled by C in 1 hour
= Part filled by (A + B + C) in 1 hour - Part filled by (A + B) in 1 hour

$= \frac{1}{6} - \frac{2}{21} = \frac{14-8}{84} = \frac{6}{84} = \frac{1}{14}$

C alone can fill the tank in 14 hours

4. (4) In convex polygon, each angle < 180°

5. (1) $2x^2 + 9x + 4 = 0$

$$x = \frac{-(9) \pm \sqrt{(9)^2 - 4 \times 2 \times 4}}{2 \times 2}$$

$$= \frac{-(9) \pm \sqrt{81-32}}{4} = \frac{-(9) \pm \sqrt{49}}{4}$$

$$= \frac{-(9) \pm 7}{4} = \frac{1}{2}, -4$$

Therefore, the required answer is

$-4 < x < -\frac{1}{2}$

6. (5) Given,
$2x + 3 + z = 55$ ....(i)
$x + z - y = 4$ ...(ii)
$y - x + z = 12$ ...(iii)
From (ii) and (iii)
$2z = 16 \Rightarrow z = 8$
From (ii), $x + 8 - y = 4$
$\Rightarrow x = 4 - 8 + y = y - 4$
Putting this value in (i) we get
$2(y - 4) + 3y + 8 = 55$
$\Rightarrow 2y - 8 + 3y + 8 = 55$
$\Rightarrow 5y = 55$
$\Rightarrow y = 11$

7. (1) Given $3y + 2x = 47$ ...(i)
$7y - 11x = 0$ ....(ii)

From(ii), $x = \frac{7}{11}y$

$\therefore$ Equation (i) reduces to

$3y + 2 \times \frac{7}{11}y = 47$

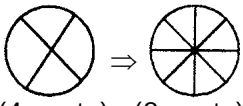
$\Rightarrow \frac{33y + 14y}{11} = 47$

$\Rightarrow y = \frac{11 \times 47}{47} = 11$

From(ii), $x = \frac{7 \times 11}{11} = 7$

$\therefore$ Required difference
$= y - x$
$= 11 - 7 = 4$

8. (1)



$\Rightarrow$ $\Rightarrow$ 8 parts
(4 parts) (8 parts)

9. (1) Let the speed of boat in still water be x kmph and that of current by y kmph. According to the question,

$\therefore \frac{d}{x+y} + \frac{d}{x-y} = \frac{21}{4}$ ...(i)

and,

$\frac{2d}{x-y} = 7 \Rightarrow \frac{d}{x-y} = \frac{7}{2}$ ...(ii)

By equation (ii) - (i),

$\frac{d}{x+y} = \frac{21}{4} - \frac{7}{2} = \frac{21-14}{4} = \frac{7}{4}$

$\Rightarrow \frac{2d}{x+y} = \frac{7}{2} = 3\frac{1}{2}$ hours

10. (4) Speed of boat in still water = x kmph.

Speed of current $= \frac{x}{3}$ kmph

$\therefore$ Speed downstream

$= x + \frac{x}{3} = \frac{4x}{3}$ kmph.

Speed upstream

$= x - \frac{x}{3} = \frac{2x}{3}$ kmph

$\therefore \frac{4x}{3} \times 2 - \frac{2x}{3} \times 2 = 16$

$\Rightarrow \frac{4x}{3} = 16$

$\Rightarrow x = \frac{16 \times 3}{4} = 12$ kmph

11. (1) $\frac{b}{a} = \frac{c}{a} = b^2 = ca$

Again, $a^x = b^y = c^z = k$

$\Rightarrow a = k^{\frac{1}{2}}, b = k^{\frac{1}{y}}, c = k^{\frac{1}{2}}$

$\therefore b^2 = ac$

$\Rightarrow k^{\frac{2}{y}} = k^{\frac{1}{x}} \cdot k^{\frac{1}{z}}$

$\Rightarrow \frac{2}{y} = \frac{1}{x} + \frac{1}{z} = \frac{z+x}{zx}$

$\Rightarrow \frac{2z}{x+z} = \frac{y}{x}$

12. (3) $f(2) = 2 \times 2^3 + 3 \times 2^2 + 5$
$= 16 + 12 + 5 = 33$

13. (3) Sum of roots $= 2\alpha = \frac{-2k}{9}$

$\Rightarrow \alpha = \frac{-k}{9}$

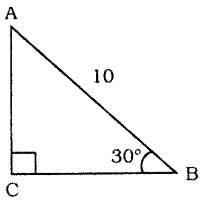
Product of roots $= \alpha^2 = \frac{4}{9}$

$\therefore \left(\frac{-k}{9}\right)^2 = \frac{4}{9}$

$\Rightarrow \frac{k^2}{81} = \frac{4}{9} \Rightarrow k^2 = 36$

$\Rightarrow k = \pm 6$

14. (3)



AB is ladder.

$\angle ABC = 30°$

AC is a wall.

$\cos 30° = \frac{BC}{AB}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{BC}{10}$

$\Rightarrow BC = 10 \times \frac{\sqrt{3}}{2}$

$= 5\sqrt{3}$ metre

$= 5 \times 1.732 = 8.66$ metre

15. (1) The sum of two sides of a triangle is greater than the third side.

When $n = 6, 7$

$3 + 6 > 8$

$3 + 7 > 8$

16. (4) $(x - 1)$ is a factor of $4x^3 + 3x^2 - 4x + k$.

On putting $x = 1$

$4(1)^3 + 3(1)^2 - 4(1) + k = 0$

$\Rightarrow 4 + 3 - 4 + k = 0$

$\Rightarrow k = -3$

17. (1) $\frac{1}{8}$ of A $\equiv \frac{1}{12}$ of B

$\therefore 12$ of A $= \frac{1}{12} \times 12 \times 8 = 8$

18. (2) Number of students who went on picnic $= x$

Students who planned $= x + 5$

$\therefore \frac{500}{x} - \frac{500}{x+5} = 5$

$\Rightarrow 500\left(\frac{1}{x} - \frac{1}{x-5}\right) = 5$

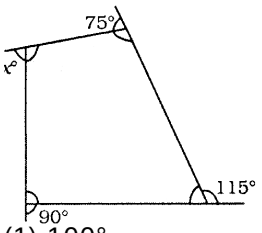
$\Rightarrow 500\left(\frac{x+5-x}{x(x+5)}\right) = 5$

$\Rightarrow x(x + 5) = 500$

$\Rightarrow x = 20$

MODEL EXERCISES

1. The sides of a quadrilateral are extended to make the angles as shown below.
What is the value of x ?



(1) 100° (2) 90°
(3) 80° (4) 75°
(5) None of these

2. What is the area of the region in the cartesian plane whose points (x, y) satisfy $|x| + |y| + |x + y) \leq 2$?
(1) 2.5 (2) 3
(3) 2 (4) 4
(5) None of these

3. From a horizontal distance of 50 m, the angles of elevation of the top and the bottom of a vertical cliff face are 45° and 30° respectively. The height of the cliff face (in m) is

(1) $\frac{50}{\sqrt{3}}$ (2) $\frac{50}{\sqrt{2}}$

(3) $\frac{50}{2\sqrt{3}}$ (4) $50\left(1-\frac{1}{\sqrt{3}}\right)$

(5) None of these

4. If $G = H + \sqrt{\frac{4}{L}}$ then L equals

(1) $\frac{4}{(G-H)^2}$ (2) $4(G - H)^2$

(3) $\frac{4}{(G^2-H^2)}$ (4) $4(G^2 - H)^2$

(5) None of these

5. If $\frac{1}{x} = \frac{1}{y} + \frac{1}{z}$, then z equals

(1) $\frac{xy}{(x-y)}$ (2) $x - y$

(3) $\frac{xy}{(y-x)}$ (4) $\frac{(x-y)}{xy}$

(5) None of these

6. If $(x - 3)(2x + 1) = 0$, then possible value of $2x + 1$ are —
(1) Only 0 (2) 0 and 3
(3) $-\frac{1}{2}$ and 3 (4) 0 and 7
(5) None of these

7. If $x + \frac{1}{x} = 3$, then the value of $x + \frac{1}{x^2}$ is —
(1) 9 (2) 10
(3) 27 (4) 7
(5) None of these

8. If $\log_8 x + \log_8 \frac{1}{6} = \frac{1}{3}$, then the value of x is —
(1) 18 (2) 24
(3) 16 (4) 12
(5) None of these

9 If $x + \frac{1}{x} = 5$, then the value of $x^3 + \frac{1}{x^3}$ is —
(1) 125 (2) 110
(3) 45 (4) 75
(5) None of these

10. Of the following quadratic equations, which is the one whose roots are 2 and -15 ?
(1) $x^2 - 2x + 15 = 0$
(2) $x^2 + 15x - 2 = 0$
(3) $x^2 + 13x - 30 = 0$
(4) $x^2 - 30 = 0$
(5) None of these

11. The locus of a point equidistant from the two fixed points is
(1) any straight line bisecting the segment joining the fixed points
(2) any straight line perpendicular to the segment joining the fixed points
(3) the straight line which is perpendicular bisector of the segment joining the fixed points
(4) any straight line perpendicu-lar to the line joining the fixed points
(5) None of these

12. Any cyclic parallelogram having unequal adjacent sides is necessarily a
(1) square (2) rectangle
(3) rhombus (4) trapezium
(5) None of these

13. The lengths of three sides of a triangle are known. In which of the cases given below, it is impossible to construct a triangle ?
(1) 15 cm, 12 cm, 10 cm
(2) 3.6 cm, 4.3 cm, 5.7 cm
(3) 17 cm, 12 cm, 6 cm
(4) 2.3 cm, 4.4 cm, 6.8 cm

(5) None of these

14. Two non-intersecting circles, one lying inside another, are of diameters a and b. The minimum distance between their circumferences is c. The distance between their centres is

(1) $a - b - c$ (2) $a + b - c$

(3) $\frac{1}{2}(a - b - c)$ (4) $\frac{1}{2}(a - b)$ - c

(5) None of these

15. A tin of oil was four fifths full. When six bottles of oil were taken out and four bottles of oil were poured into it, was three fourths full. How many bottles of oil were contained by the tin ?

(1) 10 (2) 20

(3) 30 (4) 40

(5) None of these

16. A number consists of two digits. If the digits in the unit's place and the ten's place are 7 and x respectively, the number is —

(1) $x + 7$ (2) 10 ($x + 7$)

(3) $70 + x$ (4) $10x + 7$

(5) None of these

17. The sum of the digits of a three-digit number is 16. If the ten's digit of the number is 3 times the unit's digit and the unit's digit is one-fourth of the hundredth digit, then what is the number ?

(1) 446 (2) 561

(3) 682 (4) 862

(5) None of these

18. If $x^{11} = y°$ and $x = 2y$, then y is equal to —

(1) $\frac{1}{2}$ (2) 1

(3) -1 (4) -2

(5) None of these

19. Pipes A and B can fill a tank in 5 and 6 h respectively. Pipe C can empty empty it in 12 h. The tank is half full. All the three pipes are in operation simultaneously. After how much time the tank will be full ?

(1) $3\frac{9}{17}$ h (2) 11 h

(3) $2\frac{8}{11}$ h (4) $1\frac{13}{17}$ h

(5) None of these

20. In climbing a 21 m long round pole, a monkey climbs 6 m in the first minute and slips 3 m in the next minute. What time (in minutes) the monkey would take to reach the top of the pole ?

(1) 11 (2) 14

(3) $14\frac{2}{3}$ (4) 9

(5) None of these

21. The solution of the equation $\sqrt{25 - x^2}$ $= x - 1$ are

(1) $x = 3$ and $x = 4$

(2) $x = 5$ and $x = 1$

(3) $x = -3$ and $x = 4$

(4) $x = 4$ and $x \neq -3$

(5) None of these

22. Which one of the following is a factor of $x^3 - 19x + 30$?

(1) $x - 2$ (2) $x + 2$

(3) $x - 1$ (4) $x + 1$

(5) None of these

23. The value of $\left(\frac{1}{x^2}\right) + \left(\frac{1}{y^2}\right)$, where $x = 2 + \sqrt{3}$ and $y = 2 - \sqrt{3}$,is

(1) 14 (2) 12

(3) 10 (4) 16

(5) None of these

24. If $x : y : z :: 1 : 3 : 5$, then the value of $\frac{\sqrt{x^2 + 7y^2 + 9z^2}}{x}$ is

(1) 7 (2) 17

(3) 13 (4) 1

(5) None of these

25. If $(x + 1)$ is factor of $2x^3 - ax^2 - (2a - 3)x + 2$, then the value of '$a$' is

(1) 3 (2) 2

(3) $\frac{3}{2}$ (4) $\frac{1}{2}$

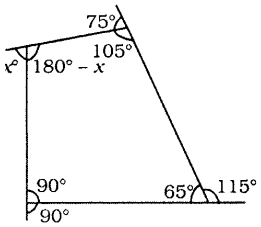
(5) None of these

SHORT ANSWERS

1.	(3)	**2.**	(3)
3.	(4)	**4.**	(1)
5.	(3)	**6.**	(4)
7.	(4)	**8.**	(4)
9.	(2)	**10.**	(3)
11.	(3)	**12.**	(2)
13.	(4)	**14.**	(4)
15.	(4)	**16.**	(4)
17.	(4)	**18.**	(1)
19.	(4)	**20.**	(1)
21.	(4)	**22.**	(1)
23.	(1)	**24.**	(2)
25.	(1)		~

EXPLANATIONS

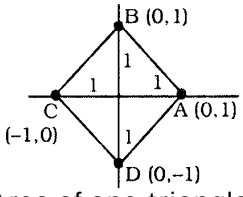
1. (3) We know that sum of all the interior angles of the polygon is 360°.



$\therefore (180° - x) + 105° + 65° + 90° = 360°.$

$\Rightarrow x = 80°$

2. (3) The region given by the inequality is shown by the graph

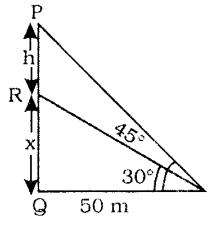


Area of one triangle

$= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$

Total area $= 4 \times \frac{1}{2} = 2$

3. (4) Let the height of cliff PR is h m.



$\therefore \tan 30° = \frac{x}{50}$

$\frac{1}{\sqrt{3}} = \frac{x}{50} \Rightarrow x = \frac{50}{\sqrt{3}}$

And $\tan 45° = \frac{(x+h)}{50}$

$\Rightarrow x + h = 50$

$\therefore h = 50 - \frac{50}{\sqrt{3}} = 50\left(1 - \frac{1}{\sqrt{3}}\right)$

4. (1) $G = H + \sqrt{\frac{4}{L}}$

$\Rightarrow (G - H) = \frac{4}{L}$

$\Rightarrow L = \frac{4}{(G-H)^2}$

5. (3) $\frac{1}{x} = \frac{1}{y} + \frac{1}{z}$

$\Rightarrow \frac{1}{z} = \frac{1}{x} - \frac{1}{y} = \frac{y-x}{xy}$

$\Rightarrow z = \frac{xy}{(y-x)}$

6. (4)$(x - 3)(2x + 1) = 0$
Then, $(x - 3) = 0$
$\Rightarrow x = 3$
and $(2x + 1) = 0$
If $x = 3$
Then, $(2x + 1) = 7$
$\therefore$ Possible values of $(2x + 1)$ is 0 and 7.

7. (4) $\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2$

$\Rightarrow (9 - 2) = x^2 + \frac{1}{x^2}$

$\therefore x^2 + \frac{1}{x^2} = 7$

8. (4) $\log_8 x + \log_8 \frac{1}{6} = \frac{1}{3}$

$\Rightarrow \log_8\left(x \times \frac{1}{6}\right) = \frac{1}{3}$

$\Rightarrow \log_6\left(\frac{x}{6}\right) = \frac{1}{3}$

$\Rightarrow (8)^{\frac{1}{3}} = \frac{x}{6}$

[$\because$ if $\log_a y = x$, then $(a)^x = y$]

$\Rightarrow (2^3)^{\frac{1}{3}} = \frac{x}{6}$

$\Rightarrow x = 12$

9. (2)

$$\left(x + \frac{1}{x}\right)^3 = \left(x^3 + \frac{1}{x^3}\right) + 3\left(x + \frac{1}{x}\right)$$

$$(5)^3 = \left(x^3 + \frac{1}{x^3}\right) + 15$$

$$\therefore x^3 + \frac{1}{x^3} = 125 - 15 = 110$$

10. (3) Sum of roots = -13
Product of roots = - 30
$\therefore$ Equation
x^2 - x (sum of roots) + product of roots = 0
$\Rightarrow x^2 + 13x - 30 = 0$

11. (3) Locus is the straight line which is perpendicular bisector of the segment joining the fixed points.

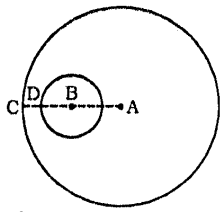
12. (2) It is a rectangle.

13. (4) To construct a triangle sum of any two sides is greater than the third side. Hence, option (d) is the correct answer.

14. (4) According to question,

$AC = \frac{a}{2}$

$BD = \frac{b}{2}$ and $CD = c$



Then, AB = AC - BC

$= \frac{a}{2}$ - (BD + CD)

$= \frac{a}{2} - \frac{b}{2} - c$

$= \frac{1}{2}(a - b) - c$

15. (4) Let the full capacity of the tin be x.

Then, $\frac{4}{5}x - 6 + 4 = \frac{3}{4}x$

$\Rightarrow \frac{4}{5}x - \frac{3}{4}x = 2$

$\Rightarrow x = 40$

16. (4) Unit place = 7
Tens place = 10
Hence, the number is $10x + 7$.

17. (4) Let a, b and c be the digits at the hundredth, ten's and unit place respectively.

$\therefore a + b + c = 16$ ...(i)

$b = 3c$ ...(ii)

and $c = \frac{1}{4}a$ ...(iii)

From Eqs. (ii) and (iii), $b = \frac{3}{4}a$ ...(iv)

From Eqs. (i), (iii) and (iv), we get

$a + \frac{3}{4}a + \frac{1}{4}a = 16$

$\Rightarrow a = 8$, $b = 6$ and $c = 2$

Therefore, the three digit number is 862.

18. (1) If $x^{11} = y° = 1$

$\Rightarrow x^{11} = 1 \Rightarrow x = 1$

Given, $x = 2y$

$2y = 1 \Rightarrow y = \frac{1}{2}$

19. (4) In one hour work done by all the pipes together

$= \frac{1}{5} + \frac{1}{6} - \frac{1}{12}$

$= \frac{17}{60}$ tank

In other work, tank is filled by all the pipes working together in $\frac{60}{17}$ h.

$\therefore \frac{1}{2}$ tank will be filled by all the pipes working together in $\frac{60}{17} \times \frac{1}{2}$

i.e. $1\frac{13}{17}h$

20. (1) Net distance travelled by monkey in 2 min = 3 m
$\therefore$ In 10 min, distance travelled (climbed) = 15 m.
As, it is given in the question that in first minute it climbs 6 m and in other minute it slips down 3 m.
And this happens alternately.
Hence, money will travel remaining 6 m in next minute.
$\therefore$ Total time taken = (10 + 1) = 11 min

21. (4) $\sqrt{25-x^2} = x - 1$

$\Rightarrow 25 - x^2 = (x-1)^2$

$\Rightarrow 25 - x^2 = x^2 + 1 - 2x$

$\Rightarrow 2x^2 - 2x - 24 = 0$

$\Rightarrow x^2 - x - 12 = 0$

$\Rightarrow (x-4)(x-3)$

$\Rightarrow x = 4, x = -3$

$\Rightarrow x = 4$

$(\because x = -3)$ is not satisfy the give equation.

22. (1) $x^3 - 19x + 30$

$(x - 2)$ is the fatctor of the above expression.

Because for $x = 2$

$(2)^3 - 19 \times (2) + 30 = 0$

23. (1) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{\left(2+\sqrt{3}\right)^2} + \frac{1}{\left(2-\sqrt{3}\right)^2}$

$= \frac{1}{\left(7+4\sqrt{3}\right)} + \frac{1}{\left(7-4\sqrt{3}\right)}$

$= \frac{14}{49-16\times 3} = 14$

24. (2) Let $\frac{x}{1} = \frac{y}{3} = \frac{z}{5} = K$

$\Rightarrow x = K, y = 3K, z = 5K$

$\therefore \frac{\sqrt{x^2+7y^2+9z^2}}{x}$

$= \frac{\sqrt{K^2+7\times 9K^2+9\times 25K^2}}{K}$

$= \sqrt{289} = 17$

25. (1) Let $f(x) = 2x^3 - ax^2 - (2a - 3)x + 2 = 0$

If $(x + 1)$ is a factor of the avobe expression, then $f(-1) = 0$

$\therefore f(-) = 2(-1)^3 - a(-1)^2 - (2a - 3) \times -1 + 2 = 0$

$\Rightarrow -2 - a + 2a - 3 + 2 = 0$

$\Rightarrow a - 3 = 0$

$\Rightarrow a = 3$

NUMBER SERIES

NATIONALISED BANKS & IBPS SO/MT/SO

Directions (1-5): In the following number series, a wrong number is given. Find out that wrong number.

(Canara Bank PO Exam. 09.02.2003)

1. 2 11 38 197 1172 8227 65806
(1) 11 (2) 38
(3) 197 (4) 1172
(5) 8227

2. 16 19 21 30 46 71 107
(1) 19 (2) 21
(3) 30 (4) 46
(5) 71

3. 7 9 16 25 41 68 107 173
(1) 107 (2) 16
(3) 41 (4) 68
(5) 25

4. 4 2 3.5 7.5 26.25 118.125
(1) 118.125 (2) 26.25
(3) 3.5 (4) 2
(5) 7.5

5. 16 4 2 1.5 1.75 1.875
(1) 1.875 (2) 1.75
(3) 1.5 (4) 2
(5) 4

Directions (6-10): What will come in place of the question mark (?) in the following number series ?

(Syndicate Bank PO Exam. 10.10.2004)

6. 3 10 32 100 ?
(1) 345 (2) 460
(3) 308 (4) 440
(5) None of these

7. 5 3 4 ? 38
(1) 8.5 (2) 6
(3) 7.5 (4) 8
(5) None of these

8. 5 6 ? 57 244
(1) 21 (2) 16
(3) 17 (4) 15
(5) None of these

9. 3 10 21 ? 51
(1) 34 (2) 32
(3) 33 (4) 37
(5) None of these

10. 5 11 ? 55 117
(1) 21 (2) 27
(3) 23 (4) 25
(5) None of these

Directions (11-15): In each of the following questions a number series is given. After the series a number is given followed by (a), (b), (c), (d) and (e). You have to complete the series starting with the number given, following the sequence of the original series and answer the questions that follow the series.

(Union Bank of India PO Exam. 27.11.2005)

11. 12 30 120 460 1368 2730
16 (a) (b) (c) (d) (e)
What will come in place of (d) ?
(1) 1384 (2) 2642
(3) 2808 (4) 1988
(5) None of these

12. 154 462 231 693 346.5 1039.5
276 (a) (b) (c) (d) (e)
What will come in place of (e) ?
(1) 1746 (2) 621
(3) 1242 (4) 983
(5) None of these

13. 7 91 1001 7007 35035 105
14.5 (a) (b) (c) (d) (e)
What will come in place of (c) ?
(1) 21132.5 (2) 14514.5
(3) 20020.5 (4) 13864.5
(5) None of these

14. 582 574 601 537 662 446
204 (a) (b) (c) (d) (e)
What will come in place of (d) ?
(1) 284 (2) 68
(3) 174 (4) 331
(5) None of these

15. 85 43 44 67.5 137 345
125 (a) (b) (c) (d) (e)
What will come in place of (c) ?
(1) 86 (2) 107.5
(3) 112.5 (4) 97.5
(5) None of these

Directions (16-22) : What will come in place of the question mark (?) in the following number series ?

(Corporation Bank Po Exam. 29.07.2006)

16. 1 ? 27 64 125
(1) 8 (2) 4
(3) 6 (4) 9
(5) None of these

17. 25 16 ? 4 1
(1) 3 (2) 6
(3) 12 (4) 18
(5) None of these

18. 1 6 36 240 1960 ?
(1) 19660 (2) 3680
(3) 36800 (4) 19600
(5) None of these

19. 12 14 17 13 8 14 21 13 4 ?

(1) 14 (2) 13
(3) 15 (4) 2
(5) None of these

20. 2 5 7 12 19 31 50 ?
(1) 53 (2) 81
(3) 69 (4) 74
(5) None of these

21. 15 12 17 10 ? 8 21 6
(1) 3 (2) 7
(3) 21 (4) 19
(5) None of these

22. 4 6 12 30 90 315 ?
(1) 945 (2) 1102
(3) 1260 (4) 1417.5
(5) None of these

Directions (16-22) : What should come in place of the question mark (?) in the following number series ?

(Bank Of Maharashtra PO Exam. 29.07.2006)

23. 1548 516 129 43 ?
(1) 11 (2) 10.75
(3) 9.5 (4) 12
(5) None of these

24. 949 189.8 ? 22.776 11.388 6.8328
(1) 48.24 (2) 53.86
(3) 74.26 (4) 56.94
(5) None of these

25. 121 144 190 259 ? 466
(1) 351 (2) 349
(3) 374 (4) 328
(5) None of these

26. 14 43.5 264 ? 76188
(1) 3168 (2) 3176
(3) 1587 (4) 1590
(5) None of these

27. 41 164 2624 ? 6045696
(1) 104244 (2) 94644
(3) 94464 (4) 102444
(5) None of these

Directions (28-32): What should come In place of question mark (?) in the following number series ?

(Indian Overseas Bank PO Exam. 15.06.2008)

28. 12 12 18 45 180 1170 ?
(1) 12285 (2) 10530
(3) 11700 (4) 12870
(5) 7605

29. 444 467 513 582 674 789 ?
(1) 950 (2) 904
(3) 927 (4) 881
(5) 973

30. 1 16 81 256 625 1296 ?
(1) 4096 (2) 2401
(3) 1764 (4) 3136
(5) 6561

31. 23 25 53 163 657 3291 ?
(1) 16461 (2) 13169
(3) 9877 (4) 23045
(5) 19753

32. 13 13 65 585 7605 129285 ?
(1) 2456415 (2) 2235675
(3) 2980565 (4) 2714985
(5) 2197845

Directions (33-37) : What should come in place of question mark (?) in the following number series ?

(Andhra Bank PO Exam. 14.09.2008)

33. 40280625 732375 16275 465 18.6 1.24 ?
(1) 0.248 (2) 0.336
(3) 0.424 (4) 0.512
(5) 0.639

34. 14 12 21 59 231 1149 ?
(1) 6987 (2) 6787
(3) 6887 (4) 6687
(5) 6587

35. 1728 2744 4096 5832 8000 10648 ?
(1) 12167 (2) 13824
(3) 15625 (4) 9261
(5) 17576

36. 120 15 105 17.5 87.5 ?
(1) 18.5 (2) 19.5
(3) 21.875 (4) 17.5
(5) 90

37. 3 6 21 28 55 66 ? 120
(1) 103 (2) 104
(3) 108 (4) 106
(5) 105

Directions (38-42) : In each of the following questions a number se-ries is given which has only one **wrong** number. You have to find out the **wrong** number.

(Bank Of Baroda Specialist Officer Exam. 05.10.2008)

38. 7.25 47.5 87.5 157.5 247.5 357.5 487.5
(1) 357.5 (2) 87.5
(3) 157.5 (4) 7.5
(5) 47.5

39. 13 16 21 27 39 52 69
(1) 21 (2) 39
(3) 27 (4) 52
(5) 16

40. 1500 1581 1664 1749 1833 1925 2016
(1) 1581 (2) 1664
(3) 1833 (4) 1925 (5) 1749

41. 66 91 120 153 190 233 276

(1) 120 (2) 233
(3) 153 (4) 276
(5) 190

42. 1331 2197 3375 4914 6859 9261 12167
(1) 4914 (2) 6859
(3) 9261 (4) 2197
(5) 12167

Directions (43-47): What should come in place of the question mark (?) in the following number series ?

(Oriental Bank of Commerce PO Exam. 21.12.2008)

43. 20 24 33 49 74 110 ?
(1) 133 (2) 147
(3) 159 (4) 163
(5) 171

44. 529 841 961 1369 1681 1849 ?
(1) 2809 (2) 2601
(3) 3249 (4) 3481
(5) 2209

45. 16 24 48 120 360 1260 ?
(1) 3780 (2) 4725
(3) 5355 (4) 5040
(5) 4410

46. 8 31 122 485 1936 7739 ?
(1) 30950 (2) 46430
(3) 34650 (4) 42850
(5) 38540

47. 499 622 868 1237 1729 2344 ?
(1) 3205 (2) 3082
(3) 2959 (4) 3462
(5) 2876

Directions (48-52) : In the following number series only one number is **wrong.** Find out the **wrong** number.

(PNB Agriculture Officer Exam. 04.01.2009)

48. 1 4 27 256 3125 46658
(1) 46658 (2) 4
(3) 27 (4) 3125
(5) None of these

49. 18000 3600 720 142.2 28.8 5.76
(1) 28.8 (2) 3600
(3) 5.76 (4) 142.2
(5) None of these

50. 12 237 406 527 604 657
(1) 237 (2) 406
(3) 527 (4) 657
(5) None of these

51. 3 35 226 1160 4660 13998
(1) 13998 (2) 4660
(3) 226 (4) 1160
(5) None of these

52. 18 119 708 3534 14136 42405
(1) 708 (2) 3534
(3) 14136 (4) 42405
(5) None of these

Directions (53-57): What should come in place of question mark (?) in the following number series ?

(Canara Bank PO Exam. 15.03.2009)

53. 5 9 18 34 59 95 ?
(1) 272 (2) 168
(3) 116 (4) 148
(5) 144

54. 1200 480 192 76.8 30.72 12.288 ?
(1) 4.9152 (2) 5.8192
(3) 6.7112 (4) 7.6132
(5) 8.5172

55. 963 927 855 747 603 423 ?
(1) 209 (2) 208
(3) 207 (4) 206
(5) 205

56. 841 961 1089 1225 1369 1521 ?
(1) 1581 (2) 1681
(3) 1781 (4) 1881
(5) 1981

57. 18 20 44 138 560 2810 ?
(1) 16818 (2) 16836
(3) 16854 (4) 16872
(5) 16890

Directions (58-62) : In the following number series only one number is **wrong.** Find out the **wrong** number.

(UCO Bank PO Exam. 22.03.2009)

58. 4 6 18 49 201 1011
(1) 1011 (2) 201
(3) 18 (4) 49
(5) None of these

59. 48 72 108 162 243 366
(1) 72 (2) 108
(3) 162 (4) 243
(5) None of these

60. 2 54 300 1220 3674 7350
(1) 3674 (2) 1220
(3) 300 (4) 54
(5) None of these

61. 8 27 64 125 218 343
(1) 27 (2) 218
(3) 125 (4) 343
(5) None of these

62. 19 68 102 129 145 154
(1) 154 (2) 129
(3) 145 (4) 102
(5) None of these

Directions (63-67): What should come in place of the question mark (?) in the following number series ?

(Indian Overseas Bank PO Exam. 05.04.2009)

63. 0 5 18 43 84 145 ?
(1) 220 (2) 240
(3) 260 (4) 280
(5) None of these

64. 10 17 48 165 688 3475 ?
(1) 27584 (2) 25670
(3) 21369 (4) 20892
(5) None of these

65. 1 3 24 360 8640 302400 ?
(1) 14525100 (2) 154152000
(3) 14515200 (4) 15425100
(5) None of these

66. 12 14 32 102 416 2090 ?
(1) 15522 (2) 12552
(3) 13525 (4) 17552
(5) None of these

67. 10 15 15 12.5 9.375 6.5625 ?
(1) 4.375 (2) 3.2375
(3) 4.6275 (4) 3.575
(5) None of these

Directions (68-72) : What will come in place of the question mark (?) in each of the following series ?

68. 17 52 158 477 ? 4310
(1) 1433 (2) 1432
(3) 1435 (4) 1434
(5) None of these

69. 3 22 ? 673 2696 8093
(1) 133 (2) 155
(3) 156 (4) 134
(5) None of these

70. 6 13 38 ? 532 2675
(1) 129 (2) 123
(3) 172 (4) 164
(5) None of these

71. 286 142 ? 34 16 7
(1) 66 (2) 72
(3) 64 (4) 74
(5) None of these

72. 17 9 ? 16.5 35 90
(1) 5 (2) 15
(3) 10 (4) 20
(5) None of these

Directions (73-77): What will come in place of the question mark (?) in each of the following rtamber series ?

(Andhra Bank PO Exam 05.07.2009)

73. 2 8 26 ? 242
(1) 78 (2) 72
(3) 82 (4) 84
(5) None of these

74. 3 4 12 ? 196
(1) 45 (2) 40
(3) 41 (4) 49
(5) None of these

75. 9 17 ? 65 129
(1) 32 (2) 24
(3) 35 (4) 33
(5) None of these

76. 7 13 ? 49 97
(1) 27 (2) 25
(3) 23 (4) 29
(5) None of these

77. 5 3 6 ? 64.75
(1) 15 (2) 15.5
(3) 17.5 (4) 17.25
(5) None of these

Directions (78-82) : What will come in place of the question mark (?) in each of the following number series ?

(PNB Specialist Officer's Exam. 16.08.2009)

78. 16 8 12 30 ?
(1) 75 (2) 105
(3) 95 (4) 115
(5) None of these

(United Bank of India PO Exam. 21.06.2009)

79. 5 6 14 45 ?
(1) 138 (2) 154
(3) 118 (4) 184
(5) None of these

80. 7 12 32 105 ?
(1) 428 (2) 214
(3) 218 (4) 416
(5) None of these

81. 11 23 47 95 ?
(1) 189 (2) 193
(3) 181 (4) 195
(5) None of these

82. 9 17 33 65 ?
(1) 113 (2) 131
(3) 129 (4) 118
(5) None of these

Directions (83-84) : In the following number series only one number is **wrong.** Find out the **wrong** number.

(Corporation Bank PO Exam. 22.11.2009)

83. 8 11 17 47 128 371 1100
(1)11 (2)47
(3) 17 (4) 371
(5) 128

84. 1 5 13 31 61 125 253
(1) 1 (2) 5
(3)31 (4)61
(5) 125

Directions (85-89) : In the following number series a **wrong** number is given. Find out the **wrong** number.

(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

85. 150 290 560 1120 2140 4230 8400
(1) 2140 (2) 560
(3) 1120 (4) 4230
(5) 290

86. 10 8 13 35 135 671 4007
(1) 8 (2) 671
(3) 135 (4) 13
(5) 35

87. 80 42 24 13.5 8.75 6.375 5.1875
(1) 8.75 (2) 13.5
(3) 24 (4) 6.375
(5) 42

88. 125 75 45 25 16.2 9.72 5.832
(1) 25 (2) 45
(3) 9.72 (4) 16.2
(5) 75

89. 29 37 21 43 13 53 5
(1) 37 (2) 53
(3) 13 (4) 21
(5) 43

Directions (90-94): In the following number series only one number is **wrong.** Find out the **wrong** number.

(Indian Bank PO Exam. 17.10.2010)

90. 13 25 40 57 79 103 130
(1) 25 (2) 40
(3) 57 (4) 79
(5) None of these

91. 850 600 550 500 475 462.5 456.25
(1) 600 (2) 550
(3) 500 (4) 462.5
(5) None of these

92. 2 10 18 54 162 486 1458
(1) 18 (2) 54
(3) 162 (4) 10
(5) None of these

93. 8 12 24 46 72 108 152
(1) 12 (2) 24
(3) 46 (4) 72
(5) None of these

94. 142 119 100 83 65 59 52
(1) 65 (2) 100
(3) 59 (4) 119
(5) None of these

Directions (95-99) : What should come in place of the question mark in the following number series ?

(Bank Of India Banking Officer Exam. 24.01.2010)

95. 5 54 90 115 131 140 ?
(1) 149 (2) 146
(3) 142 (4) 152
(5) None of these

96. 7 4 5 9 ? 52.5 160.5
(1) 32 (2) 16
(3) 14 (4) 20
(5) None of these

97. 6 42 ? 1260 5040 15120 30240
(1) 546 (2) 424
(3) 252 (4) 328
(5) None of these

98. 4 10 40 190 940 ? 23440
(1) 4690 (2) 2930
(3) 5140 (4) 3680
(5) None of these

99. 2 9 30 ? 436 2195 13182
(1) 216 (2) 105
(3) 178 (4) 324
(5) None of these

Directions (100-104): In each question below, a number series is given in which one number is **wrong.** Find out the **wrong** number.

(Allahabad Bank PO Exam. 21.02.2010)

100. 484 240 120 57 26.5 11.25 3.625
(1) 240 (2) 120
(3) 57 (4) 26.5
(5) 11.25

101. 3 5 13 43 176 891 5352
(1) 5 (2) 13
(3) 43 (4) 176
(5) 891

102. 6 7 16 41 90 154 292
(1) 7 (2) 16
(3) 41 (4) 90
(5) 154

103. 5 7 16 57 244 1245 7506
(1) 7 (2) 16
(3) 57 (4) 244
(5) 1245

104. 4 2.5 3.5 6.5 15.5 41.25 126.75
(1) 2.5 (2) 3.5
(3) 6.5 (4) 15.5
(5) 41.25

Directions (105-109) : What should come in place of the question mark (?) in the following number series.

(Corporation Bank PO Exam. 09.05.2010)

105. 325 314 292 259 215 ?
(1) 126 (2) 116
(3) 130 (4) 160
(5) None of these

106. 45 46 70 141 ? 1061.5
(1) 353 (2) 353.5
(3) 352.5 (4) 352
(5) None of these

107. 620 632 608 644 596 ?
(1) 536 (2) 556
(3) 656 (4) 646
(5) None of these

108. 15 25 40 65 ? 195
(1) 115 (2) 90
(3) 105 (4) 120
(5) None of these

109. 120 320 ? 2070 5195 13007.5
(1) 800 (2) 920
(3) 850 (4) 900
(5) None of these

Directions (110-114): In the following number series only one number is **wrong.** Find out the wrong number.

(Punjab & Sind Bank PO Exam. 16.05.2010)

110. 32 34 37 46 62 87 123
(1) 34 (2) 37
(3) 62 (4) 87
(5) 46

111. 7 18 40 106 183 282 403
(1) 18 (2) 282
(3) 40 (4) 106
(5) 183

112. 850 843 829 808 788 745 703
(1) 843 (2) 829
(3) 808 (4) 788
(5) 745

113. 33 321 465 537 573 590 600
(1) 321 (2) 465
(3) 573 (4) 537
(5) 590

114. 37 47 52 67 87 112 142
(1) 47 (2) 52
(3) 67 (4) 87
(5) 112

Directions (115-119) : What will come in place of the question mark (?) in the following number series ?

(Bank Of Baroda PO Exam. 30.05.2010)

115. 13 16 22 33 51 (?)
(1) 89 (2) 78
(3) 102 (4) 69
(5) None of these

116. 39 52 78 117 169 (?)
(1) 246 (2) 182
(3) 234 (4) 256
(5) None of these

117. 62 87 187 412 812 (?)
(1) 1012 (2) 1437
(3) 1337 (4) 1457
(5) None of these

118. 7 8 24 105 361 (?)
(1) 986 (2) 617
(3) 486 (4) 1657
(5) None of these

119. 656 432 320 264 236 (?)
(1) 222 (2) 229
(3) 232 (4) 223
(5) None of these

Directions (120-124): What will come in place of the question mark (?) in the following number series ?

(Central Bank Of India PO Exam. 25.07.2010)

120. 7 20 46 98 202 (?)
(1) 420 (2) 410
(3) 310 (4) 320
(5) None of these

121. 210 209 213 186 202 (?)
(1) 138 (2) 77
(3) 177 (4) 327
(5) None of these

122. 27 38 71 126 203 (?)
(1) 212 (2) 202
(3) 301 (4) 312
(5) None of these

123. 435 354 282 219 165 (?)
(1) 103 (2) 112
(3) 120 (4) 130
(5) None of these

124. 4 200 369 513 634 (?)
(1) 788 (2) 715
(3) 734 (4) 755
(5) None of these

Directions (125-129) : What will come in place of the question mark (?) in the following number series ?

(Syndicate Bank PO Exam. 29.08.2010)

125. 495 485 465 425 345 ?
(1) 195 (2) 165
(3) 185 (4) 175
(5) None of these

126. 16 22 33 49 70 ?
(1) 95 (2) 96
(3) 85 (4) 91
(5) None of these

127. 32 36 52 88 152 ?
(1) 266 (2) 232
(3) 242 (4) 256
(5) None of theses

128. 17 289 425 493 527 ?
(1) 534 (2) 542
(3) 544 (4) 594
(5) None of these

129. 13 27 55 97 153 ?
(1) 243 (2) 265
(3) 215 (4) 223
(5) None of these

Directions (130-134) : What should come in place of the question mark (?) in the following number series ?

(Punjab National Bank Specialist Officer Exam. 24.10.2010)

130. 50 60 75 97.5 ? 184.275 267.19875
(1) 120.50 (2) 130.50
(3) 131.625 (4) 124.25
(5) None of these

131. 12 15 36 ? 480 2415 14508
(1) 115 (2) 109
(3) 117 (4) 121
(5) None of these

132. 1 2 6 21 88 445 ?
(1) 2230 (2) 2676
(3) 2580 (4) 2670
(5) None of these

133. 20 21 25 34 50 ? 111
(1) 70 (2) 65
(3) 60 (4) 75
(5) None of these

134. 600 125 30 ? 7.2 6.44 6.288
(1) 6 (2) 10
(3) 15 (4) 12
(5) None of these

Directions (135-139): What will come in the place of the question mark (?) in the following number series ?

(Bank Of India PO Exam. 31.10.2010)

135. 11 15 31 67 131 (?)
(1) 233 (2) 221
(3) 243 (4) 231
(5) None of these

136. 483 471 435 375 291 (?)
(1) 183 (2) 184
(3) 185 (4) 186
(5) None of these

137. 5 7 13 25 45 (?)
(1) 67 (2) 75
(3) 65 (4) 55
(5) None of these

138. 4 11 25 53 109 (?)
(1) 221 (2) 234
(3) 212 (4) 222
(5) None of these

139. 15 21 33 51 75 (?)
(1) 113 (2) 103
(3) 105 (4) 115
(5) None of these

Directions (140-144): In the following number series only one number is **wrong.** Find out the **wrong** number.

(United Bank Of India PO Exam. 14.11.2010)

140. 5 348 564 689 716 780 788
(1) 716 (2) 788
(3) 348 (4) 689
(5) 780

141. 4444 2224 1114 556 281.5 142.75 73.375
(1) 2224 (2) 281.5
(3) 1114 (4) 556
(5) 142.75

142. 4.5 16 25 33 38.5 42 43.5
(1) 33 (2) 38.5
(3) 42 (4) 43.5
(5) 25

143. 6 49 305 1545 6196 18603 37218
(1) 6196 (2) 49
(3) 305 (4) 1545
(5) 18603

144. 8 5 6.5 11 26 68 207.5
(1) 68 (2) 6.5
(3) 11 (4) 26
(5) 207.5

Directions (145-149) : What should come in place of the question mark (?) in the following number series ?

(PNB Management Trainee Exam. 28.11.2010)

145. 586 587 586 581 570 ? 522
(1) 545 (2) 543
(3) 551 (4) 557
(5) None of these

146. 64 54 69 49 74 44 ?
(1) 89 (2) 69
(3) 59 (4) 99
(5) None of these

147. 4000 2008 1012 ? 265 140.5 78.25
(1) 506 (2) 514
(3) 520 (4) 512
(5) None of these

148. 5 5 15 75 ? 4725 51975
(1) 520 (2) 450
(3) 525 (4) 300
(5) None of these

149. 52 26 26 39 78 ? 585
(1) 195 (2) 156
(3)234 (4)117
(5) None of these

Directions (150-154) .-What will come in place of question mark (?) in the following number series ?

(Bank Of Maharashtra Exam. 19.12.2010)

150. 10 14 25 55 140 (?)
(1) 386 (2) 398
(3) 388 (4) 396
(5) None of these

151. 119 131 155 191 239 (?)

(1) 289 (2) 290
(3) 279 (4) 280
(5) None of these

152. 11 57 149 333 701 (?)
(1) 1447 (2) 1347
(3) 1368 (4) 1437
(5) None of these

153. 697 553 453 389 353 (?)
(1) 328 (2) 337
(3) 362 (4) 338
(5) None of these

154. 336 224 168 140 126 (?)
(1) 119 (2) 118
(3) 116 (4) 121
(5) None of these ,

Directions (155-159): What will come in place of the question mark (?) in the following number series ?

(Oriental Bank Of Commerce PO Exam. 26.12.2010 (1st Sitting)

155. 9 15 27 51 99 ?
(1) 165 (2) 195
(3) 180 (4) 190
(5) None of these

156. 13 21 36 58 87 ?
(1) 122 (2) 128
(3) 133 (4) 123
(5) None of these

157. 7 9 19 45 95 ?
(1) 150 (2) 160
(3) 145 (4) 177
(5) None of these

158. 14 15 23 32 96 ?
(1) 121 (2) 124
(3) 152 (4) 111
(5) None of these

159. 20 24 36 56 84 ?
(1) 116 (2) 124
(3) 120 (4) 128
(5) None of these

Directions (160-164) : What should come in place of the question mark (?) In the following number series ?

(Indian Bank PO Exam. 02.01.2011)

160. 3 732 1244 1587 1803 1928 ?
(1) 2144 (2) 1992
(3) 1955 (4) 2053
(5) None of these

161. 16 24 ? 210 945 5197.5 33783.75
(1) 40 (2) 36
(3) 58 (4) 60
(5) None of these

162. 45030 9000 1795 355 68 ? 1.32
(1) 11.6 (2) 12.2
(3) 10.4 (4) 9.8
(5) None of these

163. 5 12 36 123 ? 2555 15342
(1) 508 (2) 381
(3) 504 (4) 635
(5) None of these

164. 8 11 17 ? 65 165.5 498.5
(1) 27.5 (2) 32
(3) 28 (4) 30.5
(5) None of these

Directions (165-169) :What will come in place of the question mark (?) in the following number series ?

(Union Bank Of India PO Exam. 09.01.2001)

165. 117 389 525 593 627 (?)
(1) 654 (2) 640
(3) 634 (4) 630
(5) None of these

166. 7 11 23 51 103 (?)
(1) 186 (2) 188
(3) 185 (4) 187
(5) None of these

167. 18 27 49 84 132 (?)
(1) 190 (2) 183
(3) 180 (4) 193
(5) None of these

168. 33 43 65 99 145 (?)
(1) 201 (2) 203
(3) 205 (4) 211
(5) None of these

169. 655 439 314 250 223 (?)
(1) 205 (2) 210
(3) 195 (4) 190
(5) None of these

Directions (170-174): What will come in place of the question mark (?) in the following number series ?

(Corporation Bank PO Exam. 16.01.2011)

170. 15 21 39 77 143 (?)
(1) 243 (2) 240
(3) 253 (4) 245
(5) None of these

171. 33 39 57 87 129 (?)
(1) 183 (2) 177
(3) 189 (4) 199
(5) None of these

172. 15 19 83 119 631 (?)
(1) 731 (2) 693
(3) 712 (4) 683
(5) None of these

173. 19 26 40 68 124 (?)
(1) 246 (2) 238
(3) 236 (4) 256
(5) None of these

174. 43 69 58 84 73 (?)
(1) 62 (2) 98
(3) 109 (4) 63
(5) None of these

Directions (175-179): What should come in place of the question mark (?) in the following number series ?

(Punjab & Sind Bank PO Exam. 23.01.2011)

175. 15 18 16 19 17 20 ?
(1) 23 (2) 22
(3) 16 (4) 18
(5) None of these

176. 1050 420 168 67.2 26.88 10.752 ?
(1) 4.3008 (2) 6.5038
(3) 4.4015 (4) 5.6002
(5) None of these

177. 0 6 24 60 120 210 ?
(1) 343 (2) 280
(3) 335 (4) 295
(5) None of these

178. 32 49 83 151 287 559 ?
(1) 1118 (2) 979
(3) 1103 (4) 1120
(5) None of these

179. 462 552 650 756 870 992 ?
(1) 1040 (2) 1122
(3) 1132 (4) 1050
(5) None of these

Directions (180-184): What will come in place of the question mark (?) in the following number series ?

(UCO Bank PO Exam. 30.01.2011)

180. 28 39 63 102 158 (?)
(1) 232 (2) 242
(3) 233 (4) 244
(5) None of these

181. 7 16 141 190 919 (?)
(1) 1029 (2) 1019
(3) 1020 (4) 1030
(5) None of these

182. 12 17 32 57 92 (?)
(1) 198 (2) 195
(3) 137 (4) 205
(5) None of these

183. 19 25 45 87 159 (?)
(1) 254 (2) 279
(3) 284 (4) 269
(5) None of these

184. 83 124 206 370 698 (?)
(1) 1344 (2) 1324
(3) 1364 (4) 1334
(5) None of these

Directions (185-189): What will come in place of the question mark (?) in the following number series.

(Bank Of Baroda PO Exam.13.03.2011)

185. 1 7 49 343 (?)
(1) 16807 (2) 1227
(3) 2058 (4) 2401
(5) None of these

186. 13 20 39 78 145 (?)
(1) 234 (2) 244
(3) 236 (4) 248
(5) None of these

187. 12 35 81 173 357 (?)
(1) 725 (2) 715
(3) 726 (4) 736
(5) None of these

188. 3 100 297 594 991 (?)
(1) 1489 (2) 1479
(3) 1478 (4) 1498
(5) None of these

189. 112 119 140 175 224 (?)
(1) 277 (2) 276
(3) 287 (4) 266
(5) None of these

Directions (190-194): What will come in place of the question mark (?) in the following number series ?

(Allahabad Bank PO Exam.17.04.2011)

190. 958 833 733 658 608 (?)
(1) 577 (2) 583
(3) 567 (4) 573
(5) None of these

191. 11 10 18 51 200 (?)
(1) 885 (2) 1025
(3) 865 (4) 995
(5) None of these

192. 25 48 94 186 370 (?)
(1) 738 (2) 744
(3) 746 (4) 724
(5) None of these

193. 14 24 43 71 108 (?)
(1) 194 (2) 154
(3) 145 (4) 155
(5) None of these

194. 144 173 140 169 136 (?)
(1) 157 (2) 148
(3) 164 (4) 132
(5) None of these

Directions (195-199): What will come in place of the question mark (?) in the following number series ?

(Indian Overseas Bank PO Exam. 22.05.2011)

195. 8 10 18 44 124 (?)
(1) 344 (2) 366

(3) 354 (4) 356
(5) None of these

196. 13 25 61 121 205 (?)
(1) 323 (2) 326
(3) 324 (4) 313
(5) None of these

197. 656 352 200 124 86 (?)
(1) 67 (2) 59
(3) 62 (4) 57
(5) None of these

198. 454 472 445 463 436 (?)
(1) 436 (2) 456
(3) 454 (4) 434
(5) None of these

199. 12 18 36 102 360 (?)
(1) 1364 (2) 1386
(3) 1384 (4) 1376
(5) None of these

Directions (200-204): In the following number series only one number is **wrong.** Find out the wrong number.

(IBPS Bank PO/MT CWE Exam. 18.09.2011)

200. 7 12 40 222 1742 17390 208608
(1) 222 (2) 12
(3)40 (4) 1742
(5) 208608

201. 6 91 584 2935 11756 35277 70558
(1) 6 (2) 70558
(3) 584 (4) 2935
(5) 35277

202. 9050 5675 3478 2147 1418 1077 950
(1) 950 (2) 1418
(3) 5675 (4) 2147
(5) 1077

203. 1 4 25 256 3125 46656 823543
(1) 4 (2) 823543
(3) 46656 (4) 25
(5) 256

204. 8424 4212 2106 1051 526.5 263.25 131.625
(1) 526.5 (2) 1051
(3) 4212 (4) 8424
(5) 263.25

Directions (205-209): In each of these questions a number series is given. In each series **only one** number is wrong. Find out the **wrong** number.

(IBPS Bank PO/MT CWE 17.06.2012)

205. 5531 5506 5425 5304 5135 4910 4621
(1) 5531 (2) 5425
(3) 4621 (4) 5135
(5) 5506

206. 6 7 9 13 26 37 69
(1) 7 (2) 26
(3) 69 (4) 37
(5) 9

207. 1 3 10 36 152 760 4632
(1) 3 (2) 36
(3) 4632 (4) 760
(5) 152

208. 4 5 13 40 105 229 445
(1) 4 (2) 13
(3) 105 (4) 445
(5) 229

209. 157.5 45 15 6 3 2 1
(1) 1 (2) 2
(3) 6 (4) 157.5
(5) 45

Directions (210-215): What will come in place of the question mark (?) in the following number series ?

(IDBI Bank Officer Exam. 16.09.2012)

210. 123 277 459 669 907 ?
(1) 1179 (2) 1173
(3) 1167 (4) 1169
(5) None of these

211. 456.5 407 368.5 341 324.5 ?
(1) 321 (2) 319
(3) 317 (4) 323
(5) None of these

212. 23 42.2 80.6 157.4 311 ?
(1) 618.2 (2) 623.2
(3) 624.2 (4) 616.6
(5) None of these

213. 36 154 232 278 300 ?
(1) 304 (2) 313
(3) 308 (4) 307
(5) None of these

214. 24 536 487 703 678 ?
(1) 768 (2) 748
(3) 764 (4) 742
(5) None of these

215. 224 576 752 840 884 ?
(1) 960 (2) 890
(3) 906 (4) 908
(5) None of these

Directions (216-220) : What should come in place of the question mark (?) in the following series ?

(IBPS RRBs Office Assistant CWE Exam. 09.09.2012)

216. 5 6 16 57 ? 1245
(1) 244 (2) 148
(3) 296 (4) 271
(5) None of these

217. 12 ? 168 504 1260 2520

(1) 96 (2) 59
(3) 61 (4) 48
(5) None of these

218. 4 9 29 ? 599 3599
(1) 117 (2) 347
(3) 258 (4)174
(5) None of these

219. 177 170 159 146 ? 110
(1) 132 (2) 106
(3) 129 (4) 127
(5) None of these

220. 2 3 11 38 102 ?
(1) 402 (2) 182
(3) 227 (4) 168
(5) None of these

Directions (221-225): What will come in place of the question mark (?) in the following number series ?

(Indian Overseas Bank PO Online Exam. 01.09.2013)

221. 21 10.5 ? 15.75 31.5 78.75
(1) 10.5 (2) 11.5
(3) 12.5 (4) 10.25
(5) None of these

222. 6 19 58 ? 214 331
(1) 113 (2) 123
(3) 133 (4) 143
(5) None of these

223. ? 16 28 58 114 204
(1) 7 (2) 9
(3) 14 (4) 6
(5) 10

224. 13. 76 14.91 17.21 20.66 ? 31.01
(1) 25.66 (2) 24.36
(3) 24.26 (4) 25.26
(5) 25.36

225. 15 ? 24 33 97 122
(1) 20 (2) 19
(3) 17 (4) 18
(5) 16

Directions (226-230) : In each of the following number series, a number is **wrong.** Find out that wrong number.

(Corporation Bank Specialist Officer (Marketing) Exam 22.12.2014)

226. 2 6 15 30 45 43.5 22.5
(1) 6 (2) 30
(3) 45 (4) 15
(5) 43.5

227. 950 661 436 269 146 65 16
(1) 436 (2) 65
(3) 269 (4) 661
(5) 146

228. 6.5 11.8 22.4 38.3 59.5 87.3 117.8
(1) 22.4 (2) 59.5
(3) 11.8 (4) 38.3
(5) 87.3

229. 1 2 4 9 23 69 186
(1) 2 (2) 9
(3) 23 (4) 4
(5) 69

230. 250 239 216 181 136 75 4
(1) 239 (2) 181
(3) 75 (4) 216
(5) 136

SBI PO EXAMS

Directions (1-5): One number is wrong in each of the number series given in each of the following questions. You have to identify that number and assuming that a new series starts with that number following the same logic as in the given series, which of the numbers given in (1), (2), (3), (4) and (5) given below each series will be the **third** number in the new series ?

(SBI Associate Banks PO Exam. 14.02.1999)

1. 3 5 12 38 154 914 4634
(1) 1636 (2) 1222
(3) 1834 (4) 3312
(5) 1488

2. 3 4 10 34 136 685 4116
(1) 22 (2) 276
(3) 72 (4) 1374
(5) 12

3. 214 18 162 62 143 90 106
(1) 34 (2) 110
(3) 10 (4) 91
(5)38

4. 160 80 120 180 1050 4725 25987.5
(1) 60 (2) 90
(3) 3564 (4) 787.5
(5) 135

5. 2 3 7 13 25 47 78
(1) 11 (2) 13
(3) 15 (4) 18
(5) 20

Directions (6-8): In each of the following questions, a number **series** is given. After the series, below it, a number alongwith (a), (b), (c), (d) and (e) is given. You have to complete the series following the same sequence as that of the given series. Then answer the question that follows.

(SBI Associate Banks PO Exam. 16.07.2000)

6. 2 3 10 29 172 885
1 (a) (b) (c) (d) (e)
What will come in place of (b) ?

(1) 11 (2) 7
(3) 9 (4) 8
(5) None of these

7. 5 7 10 36 136 690
2 (a) (b) (c) (d) (e)
What will come in place of (e) ?
(1) 310 (2) 330
(3) 110 (4) 64
(5) None of these

8. 8 4 6 15 52.5 236.25
4 (a) (b) (c) (d) (e)
Which of the following will come in place of (d) ?
(1) 36.25 (2) 33.25
(3) 26.75 (4) 32.75
(5) None of these

Directions (9-10) : In each of the following questions, a number series is established if the positions of two out of the five marked numbers are interchanged. The position of the first unmarked number remains the same and it is the beginning of the series. The earlier of the two marked numbers whose positions are interchanged is the answer. For example, if an interchange of number marked ' 1' and the number marked '4' is required to establish the series, your answer is T. If it is not necessary to interchange the position of the numbers to establish the series, give 5 as your answer. Remember that when the series is established, the numbers change from left to right (i.e. from the unmarked number to the last marked number) in a specific order.

(SBI Banks PO Exam. 20.08.2000)

9. 40 14 60 24 80 19
(1) (2) (3) (4) (5)

10. 120 15 105 21.875 87.5 17.5
(1) (2) (3) (4) (5)

Directions (11-15) : In each of the following number-series only one number is **wrong.** If the wrong number is corrected, the series gets established following a certain logic. Below the series a number is given followed by (a), (b), (c), (d), (e) and (f). You have to complete the series following the same logic as in the given series after correcting the **wrong** number, now answer the following questions giving the correct values for the letter in the questions.

(SBI Banks PO Exam. 11.02.2001)

11. 2 3 2 15 76 254 1434
3 (a) (b) (c) (d) (e) (f)
What will come in place of (c) ?
(1) 18 (2) 22
(3) 24 (4) 21
(5) None of these

12. 1 2 8 33 148 740 4626
2 (a) (b) (c) (d) (e) (f)
What will come in place of (d) ?
(1) 156 (2) 164
(3) 168 (4) 152
(5) None of these

13. 2 4.5 11 30 93 312 1136
1 (a) (b) (c) (d) (e) (i)
What will come in place of (b) ?
(1) 6 (2) 81
(3) 16.75 (4) 18.75
(5) None of these

14. 2 14 18 46 82 176 338
4 (a) (b) (c) (d) (e) (i)
What will come in place of (e) ?
(1) 238 (2) 338
(3) 218 (4) 318
(5) None of these

15. 1 3 7 11 21 43 85
4 (a) (b) (c) (d) (e) (f)
What will come in place of (f) ?
(1) 275 (2) 279
(3) 277 (4) 273
(5) None of these

Directions (16-20) : Find out the wrong number in the following given sequence.

(SBI Associate Banks PO Exam. 21.07.2002

16. 7 4 6 9 20 52.5 160.5
(1) 6 (2) 4
(3) 20 (4) 9
(5) 52.5

17. 4 6 12 30 75 315 1260
(1) 315 (2) 75
(3) 12 (4) 6
(5) 30

18. 3 4 13 38 87 166 289
(1) 38 (2) 13
(3) 87 (4) 166
(5) 4

19. 4 5 9 29 111 556 3325
(1) 5 (2) 9
(3) 29 (4)111
(5) 556

20. 2 6 16 38 84 176 368
(1) 6 (2) 16
(3) 38 (4) 84
(5) 176

Directions (21-26): In each of thefollowing number series, a **wrong** number is given. Find out the **wrong** number.

(SBI Banks PO Exam. 18.05.2003)

21. 2 3 6 18 109 194 209952
(1) 3 (2) 6
(3) 18 (4) 109
(5) 1944

22. 1 3 6 11 20 39 70

(1) 3 (2) 39
(3) 11 (4) 20
(5) 6

23. 2 13 27 113 561 3369 23581
(1) 13 (2) 27
(3) 113 (4) 561
(5) 3369

24. 50 51 47 56 42 65 29
(1) 51 (2) 47
(3) 56 (4) 42
(5) 65

25. 3 9 23 99 479 2881 20159
(1) 9 (2) 23
(3) 99 (4) 479
(5) 2881

26. 2 4 5 8 13 21 34
(1) 4 (2) 5
(3) 8 (4) 13
(5) 21

Directions (27-31) : In each of the following questions a number series is given. After the **series** a number is given followed by (a), (b), (c), (d) and (e). You have to complete the series starting with the given number, following the sequence of original series and answer the questions that follow the series.

(SBI PO Exam. 09.01.2005)

27. 3 19 103 439 1381 2887
5 (a) (b) (c) (d) (e)
What will come in place of (b) ?
(1) 139 (2) 163
(3) 161 (4) 157
(5) None of these

28. 4 13 40 135 552 2765
2 (a) (b) (c) (d) (e)
What will come in place of (c) ?
(1) 123 (2) 133
(3) 127 (4) 131
(5) None of these

29. 5 12 4 10 3 8
6 (a) (b) (C) (d) (e)
What will come in place of (d) ?
(1) 3 (2) 5
(3) 4 (4) 7
(5) None of these

30. 3 13 37 87 191 401
1 (a) (b) (c) (d) (e)
What will come in place of (d) ?
(1) 169 (2) 161
(3) 171 (4) 159
(5) None of these

31. 8 4 6 15 52.5 236.25
12 (a) (b) (c) (d) (e)
What will come in place of (c) ?
(1) 18.25 (2) 19
(3) 22.5 (4) 20.75
(5) None of these

Directions (32-36): In each of the following questions a number series is given. After the **series,** a number is given followed by (a), (b), (c), (d) and (e). You have to complete the series starting with the number given following the sequence of the given series. Then answer the question given below it.

(SBI PO Exam. 26.11.2006)

32. 9 19.5 41 84.5
12 (a) (b) (c) (d) (e)
Which of the following numbers will come in place of (c) ?
(1) 111.5 (2) 118.5
(3) 108.25 (4) 106.75
(5) None of these

33. 4 5 22 201
7 (a) (b) (c) (d) (e)
Which of the following number will come in place of (d) ?
(1) 4948 (2) 4840
(3) 4048 (4) 4984
(5) None of these

34. 5 5.25 11.5 36.75
(a) (b) (c) (d) (e)
Which of the following number will come in place of (c) ?
(1) 34.75 (2) 24.75
(3) 24.5 (4) 34.5
(5) None of these

35. 38 19 28.5 71.25
18 (a) (b) (c) (d) (e)
Which of the following number will come in place of (d) ?
(1) 118.75 (2) 118.25
(3) 108.25 (4) 118.125
(5) None of these

36. 25 146 65 114
39 (a) (b) (c) (d) (e)
Which of the following number will come in place of (e) ?
(1) 122 (2)119
(3) 112 (4) 94
(5) None of these

Directions (37-41) : In each of these questions a number series is given. Only one number is **wrong** in each series. You have to find out the **wrong** number.

(SBI Associate Banks PO Exam. 07,01.2007)

37. 10 15 24 35 54 75 100
(1) 35 (2) 75
(3) 24 (4) 15
(5) 54

38. 1 3 4 7 11 18 27 47
(1) 4 (2) 11
(3) 18 (4) 7
(5) 27

39. 3 2 3 6 12 37.5 115.5
(1) 37.5 (2) 3
(3) 6 (4) 2
(5) 12

40. 2 8 32 148 765 4626 32431
(1) 765 (2) 148
(3) 8 (4) 32
(5) 4626

41. 2 3 11 38 102 229 443
(1) 11 (2) 229
(3) 102 (4) 38
(5) 3

Directions (42-46): What should come in place of the question mark(?) in the following number series ?

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

42. 7413 7422 7440 ? 7503 7548
(1) 7464 (2) 7456
(3) 7466 (4) 7477
(5) None of these

43. 4 16 36 64 100 ?
(1) 120 (2) 180
(3) 136 (4) 144
(5) None of these

44. 12 33 96 ? 852 2553
(1) 285 (2) 288
(3) 250 (4) 384
(5) None of these

45. 70000 14000 2800 ? 112 22.4
(1) 640 (2) 420
(3) 560 (4) 540
(5) None of these

46. 102 99 104 97 106 ?
(1) 96 (2) 95
(3) 100 (4) 94
(5) None of these

Directions (47-51) : What will ome in place of the question mark (?) in the following number series which as only one number wrong by a margin of + 1 or - 1 ? The first and last number in the series are correct ?

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

47. 93 95 99 ? 110 121 134
(1) 104 (2) 96
(3) 82 (4) 103
(5) None of these

48. 8 12 18 26 40.5 60.75 136.6875
(1) 104.125 (2) 121.125
(3) 96.125 (4) 83.125
(5) None of these

49. 4 7 11 18 28 ? 76 12
(1) 59 (2) 38
(3) 46 (4) 53
(5) None of these

50. 3 10 ? 172 886 5346 3747 299832
(1) 39 (2) 27
(3) 24 (4) 34
(5) None of these

51. 15 22 57 183 ? 748 3751 22542
(1) 709 (2) 698
(3) 748 (4) 800
(5) None of these

Directions (52-56) : In each o these questions a number series is given. In each series **only one** number is **wrong.** Find out the **wrong** number.

(SBI Associate Banks PO Exam. 07.08.2011)

52. 3601 3602 1803 604 154 36 12
(1) 3602 (2) 1803
(3) 604 (4) 154
(5) 36

53. 4 12 42 196 1005 6066 42511
(1) 12 (2) 42
(3) 1005 (4) 196
(5) 6066

54. 2 8 12 20 30 42 56
(1) 8 (2) 42
(3) 30 (4) 20
(5) 12

55. 32 16 24 65 210 945 5197.5
(1) 945 (2) 16
(3) 24 (4) 210
(5) 65

56. 7 13 25 49 97 194 385
(1) 13 (2) 49
(3) 97 (4) 194
(5) 25

Directions (57-61): In each of the following questions, a number series is given. After the series a number is given followed by (a), (b), (c), (d) and (e). You have to complete the 'series starting with the number given, following the sequence of the original series and answer the questions that bllow the series.

(SBI Management Executive Exam. 23.02.2014)

57. 37 19 20 31.5 65 165
21 (a) (b) (c) (d) (e)
What will come in the place of (e) ?
(1) 105 (2) 41
(3) 110 (4) 108
(5) 116

58. 5 6 16 57 244 1245
9 (a) (b) (c) (d) (e)
What will come in the place of (d) ?
(1) 366 (2) 364
(3) 368 (4) 378
(5) 382

59. 7 5 11 49 335 3005
13 (a) (b) (c) (d) (e)
What will come in the place of (b) ?
(1) 31 (2) 27
(3) 29 (4) 28
(5) 30

60. 12 47 152 467 1412 4247
33 (a) (b) (c) (d) (e)
What will come in the place of (d) ?
(1) 3131 (2) 1133
(3) 3311 (4) 3113
(5) 3123

61. 54 50 84 188 496 1456
42 (a) (b) (c) (d) (e)
What will come the in the place of (d) ?
(1) 304 (2) 286
(3) 293 (4) 281
(5) 301

RBI GRADE-B OFFICER EXAMS

Directions (1-5) : In each of the following questions a number series is given. After the series a number is given followed by (a), (b) (c), (d) and (e). You have to complete the series starting with the number given, following the sequence of the original series and answer the questions that follow the series.

(RBI Grade-B Officer Exam.17.11.2002)

1. 5 6 16 57 244 1245
2 (a) (b) (c) (d) (e)
What will come in place of (d) ?
(1) 46 (2) 39
(3) 156 (4) 172
(5) None of these

2. 3 5 9 17 33 65
7 (a) (b) (c) (d) (e)
What will come in place of (d)
(1) 95 (2) 51
(3) 99 (4) 49
(5) None of these

3. 7 4 5 9 20 52.5
3 (a) (b) (c) (d) (e)
What will come in place of (c) ?
(1) 4.5 (2) 2
(3) 6 (4) 7
(5) None of these

4. 3 10 32 111 460 2315
2 (a) (b) (c) (d) (e)
What will come in place of (b) ?
(1) 29 (2) 30
(3) 26 (4) 28
(5) None of these

5. 5 8 6 10 7 12
7 (a) (b) (c) (d) (e)
What will come in place of (c) ?
(1) 14 (2) 16
(3) 9 (4) 11
(5) None of these

Directions (6-10) : What should come in place of the question mark (?) in the following number series ?

(RBI Grade-B Officer Exam. 2007)

6. 104 109 99 114 94 9
(1) 69 (2) 124
(3) 120 (4) 78
(5) None of these

7. 980 392 156.8 ? 25.088 10.0352
(1) 65.04 (2) 60.28
(3) 62.72 (4) 63.85
(5) None of these

8. 14 16 35 109 441 ?
(1) 2651 (2) 2205
(3) 2315 (4) 2211
(5) None of these

9. 1331 2197 4913 6859 ? 24389
(1) 12167 (2) 13824
(3) 9261 (4) 15625
(5) None of these

10. 3600 725 150 35 12 ?
(1) 8 (2) 7.4
(3) 10.5 (4) 10
(5) None of these

Directions (11-15) : What should come in place of quesbon mark (?) in the following number series ?

(RBI Grade-B Officer Exam. 2008)

11. 13 14 30 93 376 1885 ?
(1) 10818 (2) 10316
(3) 11316 (4) 11318
(5) None of these

12. 4 6 9 13.5 20.25 30.375
(1) 40.25 (2) 45.5625
(3) 42.7525 (4) 48.5625
(5) None of these

13. 400 240 144 86.4 51.84 31.104 ?
(1) 19.2466 (2) 17.2244
(3) 16.8824 (4) 18.6625
(5) None of these

14. 9 4.5 4.5 6.75 13.5 33.75 ?
(1) 101.25 (2) 103.75
(3) 99.75 (4) 105.50
(5) None of these

15. 705 728 774 843 935 1050 ?

(1) 1190 (2) 1180
(3) 1185 (4) 1187
(5) None of these

Directions (16-20) : In each of these questions a number series is given. Below the series one number is given followed by (a), (b), (c), (d) and (e) You have to complete this series following the same logic as in the original series and answer the question that tollows.

(RBI Grade-B Officer Exam.11.10.2009)

16. 5 9 25 91 414 2282 5
3 (a) (b) (c) (d) (e)
What will come in place of (c) ?
(1) 63.25 (2) 63.75
(3) 64.25 (4) 64.75
(5) None of these

17. 15 9 8 12 36 170
19 (a) (b) (C) (d) (e)
What will come in place of (b) ?
(1) 18 (2) 16
(3) 22 (4) 24
(5) None of these

18. 7 6 10 27 104 515
9 (a) (b) (c) (d) (e)
What will come in place of (d) ?
(1) 152 (2) 156
(3)108 (4)112
(5) None of these

19. 6 16 57 244 1245 7506
4 (a) (b) (c) (d) (e)
What will come in place of (d) ?
(1) 985 (2) 980
(3) 1004 (4) 1015
(5) None of these

20. 8 9 20 63 256 1285
5 (a) (b) (c) (d) (e)
What will come in place of (e)
(1) 945 (2) 895
(3) 925 (4) 845
(5) None of these

Directions (21-25): In the following number series only one number is **wrong.** Find out the **wrong** number.

(RBI Grade-B Officer Exam.06.02.2011)

21. 4 3 4.5 8.5 20 53 162.5
(1) 3 (2) 4.5
(3) 8.5 (4) 20
(5) 53

22. 12000 2395 472 89.8 12.96 - 2.408 -5.4816
(1) -5.4816 (2) 472
(3) 12.96 (4) - 2.408
(5) 2395

23. 1 8 28 99 412 2075 12460
(1) 28 (2) 99
(3) 412 (4) 2075
(5) 12460

24. 144 215 540 1890 8505 46777.5 304053.75
(1) 215 (2) 540
(3) 1890 (4) 8505
(5) 46777.5

25. 2222 1879 1663 1538 1474 1447 1440
(1) 1879 (2) 1538
(3) 1474 (4) 1447
(5) 1440

Directions (26 - 30) : What will come in place of the question mark (?) in the following number series ?

(RBI Grade 'B' Officer's Exam. 18.12.2011)

26. 9 31 73 141 (?)
(1) 164 (2) 280
(3) 239 (4) 241
(5) None of these

27. 35 256 451 620 763 (?)
(1) 680 (2) 893
(3) 633 (4) 880
(5) None of these

28. 130 139 155 180 216 (?)
(1) 260 (2) 290
(3) 265 (4) 996
(5) None of these

29. 2890 (?) 1162 874 730 658
(1) 1684 (2) 1738
(3) 1784 (4) 1672
(5) None of these

30. 14 1004 1202 1251.5 1268 (?)
(1) 1267.5 (2) 1276.25
(3) 1324.5 (4) 1367.25
(5) None of these

Directions (31-35) : What will come in place of the question mark (?) in the following number series ?

(RBI Officer Grade 'B' Online Exam. 25.08.2013

31. 224 576 752 840 884 ?
(1) 960 (2) 890
(3) 906 (4) 908
(5) None of these

32. 55 66.15 88.45 121.9 166.5 ?
(1) 212.25 (2) 322.25
(3) 224.25 (4) 222.25
(5) None of these

33. 36 49 75 88 114 (?)
(1) 130 (2) 140
(3) 132 (4) 128
(5) 127

INSURANCE EXAMS

1. What will come in place of the question mark (?) in the following series ?
3 7 18 26 ? 53 64 96
(1) 34 (2) 37
(3) 32 (4) 38

(United India Insurance Co. AAO Exam. 21.04.2002)

2. What will come in place of the question mark (?) in the following series ?
1.7 3.2 2.7 4.2 3.7 ? 4.7 6.2
(1) 6.2 (2) 5.5
(3) 5.2 (4) 4.3

(United India Insurance Co. AAO Exam. 21.04.2002)

Directions (3-7) : In each of the following questions, a number series in given. **Only** one number is **wrong** in this series. Find out that **wrong** number, and taking this wrong number as the first term of the second series formed following the same logic, find out the fourth term of the second series.

(LIC Assistant Administrative Officer (AAO) Exam. 24.04.2008)

3. 8 4 4 6 12 28 90
(1) 18 (2) 42
(3) 21 (4) 24
(5) None of these

4. 17 17.25 18.25 20.75 24.5 30.75
(1) 23.25 (2) 24.25
(3) 24,5 (4) 24,75
(5) None of these

5. 438 487 447 476 460 469
(1) 485 (2) 425
(3) 475 (4) 496
(5) None of these

6. 2 7 18 45 99 209 431
(1) 172 (2) 171
(3) 174 (4) 175
(5) None of these

7. 6 8 10 42 146 770 4578
(1) 868 (2) 8872
(3) 858 (4) 882
(5) None of these

Directions (8-12) : Find out the wrong number in the following given sequence.

(LIC Assistant Administrative Officer (AAO) Exam. 2006)

8. 7 4 6 9 20 52.5 160.5
(1) 6 (2) 4
(3) 20 (4) 9
(5) 52.5

9. 4 6 12 30 75 315 1260
(1) 315 (2) 75
(3) 12 (4) 6
(5) 30

10. 3 4 13 38 87 166 289
(1) 38 (2) 13
(3) 87 (4) 166
(5) 4

11. 4 5 9 29 111 556 3325
(1) 5 (2) 9
(3) 29 (4) 111
(5) 556

12. 2 6 16 38 84 176 368
(1) 6 (2) 16
(3) 38 (4) 84
(5) 176

Directions (13 - 17) : What should come in place of the question mark (?) in the following number series ?

(New India Assurance AO Exam. 25.10.2009)

13. 3 52 88 113 129 ?
(1) 128 (2) 142
(3) 133 (4) 145
(5) None of these

14. 2 3 8 ? 112 565
(1) 36 (2) 14
(3) 27 (4) 45
(5) None of these

15. 6 4 8 23 ? 385.25
(1) 84.5 (2) 73
(3) 78.5 (4) 82
(5) None of these

18. 8 84 216 512 ? 1728
(1) 729 (2) 1331
(3) 684 (4) 1000
(5) None of these

17. 5 11 32 108 444 ?
(1) 1780 (2) 2230
(3) 1784 (4) 2225
(5) None of these

18. If $S = 1^2 - 2^2 + 3^2 - 4^2 + ...+ 199^2 - 200^2$, then the value of S is
(1) 19900 (2) 20100
(3) -20100 (4) -19900

(New India Assurance AO Exam. 25.10.2009)

19. Theexpression $\frac{3}{4}+\frac{5}{36}+\frac{7}{144}+....\frac{17}{5184}+\frac{19}{8100}$ is equal to
(1) 0.9 (2) 0.95
(3) 0.99 (4) 1.91

Directions (20- 24) : What will come in place of the question mark (?) in the following number series ?

(United India Insurance AO Exam. 27.03.2011)

20. 8 14 32 70 136 (?)
(1) 248 (2) 247
(3) 237 (4) 238
(5) None of these

21. 25 41 89 169 281 (?)
(1) 425 (2) 415
(3) 409 (4) 419
(5) None of these

22. 461 474 465 478 469 (?)
(1) 460 (2) 482
(3) 456 (4) 478
(5) None of these

23. 980 516 284 168 110 (?)
(1)73 (2)71
(3) 83 (4) 91
(5) None of these

24. 4 4 10 34 94 (?)
(1) 230 (2) 214
(3) 220 (4) 209
(5) None of these

25. The sum 1 + 3 - 5 + 7 + 9 - 11 +13 + 15- 17 +.....+ 61 + 63 - 65 is equal to
(1) 319 (2) 330
(3) 341 (4) 451

(New India Insurance AAO Exam. 22.05.2011)

26. If $x = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{63}$ then value of $\frac{1}{x}$ is closest to
(1) 1.1 (2) 1
(3) 0.9 (4) 0.8

(Ntw India Insurance AAO Exam. 22.05.2011)

27. If $\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)....\left(1-\frac{1}{2011^2}\right) = \frac{x}{2\times 2011}$ then the value of x is
(1) 1 (2) 2010
(3) 2011 (4) 2012

(United India Insurance AAO Exam. 03.06.2012)

Directions (28 - 32) : Find the wrong number in the following number series .

(LIC Assistant Administrative Officer (AAO) Exam. 12.05.2013)

28. 1050 510 242 106 46 16 3
(1) 3 (2) 106
(3) 242 (4) 510
(5) None of these

29. 550 546 537 521 494 460 411
(1) 494 (2) 546
(3) 521 (4) 460
(5) None of these

30. 8 21 47 86 140 203 281
(1) 47 (2) 86
(3) 140 (4) 203
(5) None of these

31. 4 24 161 965 4795 19176 57525
(1) 161 (2) 965
(3) 57525 (4) 19176
(5) None of these

32. 1 2 8 24 120 720 5040
(1)120 (2) 24
(3) 8 (4) 720
(5) None of these

Directions (33-38) : What should come in place of the question mark (?) in the following number series ?

(United India Insurance AO Exam. 26.05.2013

33. 1548 516 129 43 ?
(1) 11 (2) 10.75
(3) 9.5 (4) 12
(5) None of these

34. 949 189.8 ? 22.776 11.388 6.8328
(1) 48.24 (2) 53.86
(3) 74.26 (4) 56.94
(5) None of these

35. 121 144 190 259 ? 466
(1) 351 (2) 349
(3) 374 (4) 328
(5) None of these

36. 14 43.5 264 ? 76188
(1) 3168 (2) 3176
(3) 1587 (4) 1590
(5) None of these

37. 41 164 2624 ? 6045696
(1) 104244 (2) 94644
(3) 94464 (4) 102444
(5) None of these

38. Find the missing number in the series :
2, 5, 9, ?, 20, 27
(1) 14 (2)16
(3)18 (4)24

(NICL (GIC) Administrative Officer Exam. 15.1.2.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (4) **2.** (1) **3.** (4) **4.** (3) **5.** (2) **6.** (3) **7.** (5) **8.** (2) **9.** (1) **10.** (4) **11.** (3) **12.** (5) **13.** (2) **14.** (1) **15.** (4) **16.** (1) **17.** (5) **18.** (1) **19.** (1) **20.** (2) **21.** (4) **22.** (3) **23.** (2) **24.** (4) **25.** (1) **26.** (5) **27.** (3) **28.** (1) **29.** (3) **30.** (2) **31.** (5) **32.** (4) **33.** (1) **34.** (3) **35.** (2) **36.** (3) **37.** (5) **38.** (5) **39.** (3) **40.** (3) **41.** (2) **42.** (1) **43.** (3) **44.** (5) **45.** (4) **46.** (1) **47.** (2) **48.** (1) **49.** (4) **50.** (5) **51.** (3) **52.** (2) **53.** (5) **54.** (1) **55.** (3) **56.** (2) **57.** (4) **58.** (3) **59.** (5) **60.** (1) **61.** (2) **62.** (4) **63.** (5) **64.** (4) **65.** (3) **66.** (2) **67.** (1) **68.** (3) **69.** (4) **70.** (1) **71.** (5) **72.** (3) **73.** (5) **74.** (1) **75.** (4) **76.** (2) **77.** (3) **78.** (2) **79.** (4) **80.** (1) **81.** (5) **82.** (3) **83.** (3) **84.** (3) **85.** (3) **86.** (2) **87.** (3) **88.** (1) **89.** (5) **90.** (3) **91.** (1) **92.** (4) **93.** (3) **94.** (1) **95.** (5) **96.** (4) **97.** (3) **98.** (1) **99.** (2) **100.** (2) **101.** (4) **102.** (5) **103.** (1) **104.** (3) **105.** (4) **106.** (2) **107.** (3) **108.** (5) **109.** (5) **110.** (1) **111.** (3) **112.** (4) **113.** (5) **114.** (1) **115.** (2) **116.** (3) **117.** (2) **118.** (1) **119.** (1) **120.** (2) **121.** (2) **122.** (5) **123.** (3) **124.** (3) **125.** (3) **126.** (2) **127.** (5) **128.** (3) **129.** (4) **130.** (3) **131.** (3) **132.** (2) **133.** (4) **134.** (5) **135.** (4) **136.** (1) **137.** (2) **138.** (1) **139.** (3) **140.** (1) **141.** (4) **142.** (5) **143.** (3) **144.** (3) **145.** (3) **146.** (5) **147.** (2) **148.** (3) **149.** (1) **150.** (3) **151.** (5) **152.** (4) **153.** (2) **154.** (1) **155.** (2) **156.** (4) **157.** (4) **158.** (1) **159.** (3) **160.** (2) **161.** (4) **162.** (1) **163.** (1) **164.** (4) **165.** (5) **166.** (4) **167.** (4) **168.** (2) **169.** (5) **170.** (4) **171.** (1) **172.** (1) **173.** (3) **174.** (5) **175.** (4) **176.** (1) **177.** (5) **178.** (3) **179.** (2) **180.** (3) **181.** (5) **182.** (3) **183.** (4) **184.** (5) **185.** (4) **186.** (4) **187.** (1) **188.** (5) **189.** (3) **190.** (2) **191.** (4) **192.** (1) **193.** (2) **194.** (5) **195.** (2) **196.** (4) **197.** (1) **198.** (3) **199.** (2) **200.** (4) **201.** (3) **202.** (5) **203.** (4) **204.** (2) **205.** (1) **206.** (2) **207.** (4) **208.** (3) **209.** (1) **210.** (2) **211.** (2) **212.** (1) **213.** (5) **214.** (4)

215. (3) **216.** (1)
217. (4) **218.** (5)
219. (3) **220.** (3)
221. (1) **222.** (2)
223. (3) **224.** (4)
225. (5) **226.** (5)
227. (3) **228.** (5)
229. (5) **230.** (5)

SBI PO EXAMS

1. (3) **2.** (3)
3. (4) **4.** (5)
5. (1) **6.** (4)
7. (2) **8.** (5)
9. (3) **10.** (3)
11. (4) **12.** (5)
13. (5) **14.** (1)
15. (3) **16.** (1)
17. (2) **18.** (4)
19. (3) **20.** (5)
21. (4) **22.** (2)
23. (1) **24.** (4)
25. (3) **26.** (1)
27. (2) **28.** (1)
29. (3) **30.** (4)
31. (3) **32.** (3)
33. (1) **34.** (2)
35. (4) **36.** (3)
37. (1) **38.** (5)
39. (4) **40.** (4)
41. (2) **42.** (5)
43. (4) **44.** (1)
45. (3) **46.** (2)
47. (4) **48.** (5)
49. (5) **50.** (1)
51. (3) **52.** (4)
53. (2) **54.** (1)
55. (3) **56.** (4)
57. (1) **58.** (2)
59. (3) **60.** (4)
61. (5)

RBI GRADE-B OFFICER EXAMS

1. (4) **2.** (5)
3. (3) **4.** (2)
5. (1) **6.** (5)
7. (3) **8.** (4)
9. (1) **10.** (2)
11. (3) **12.** (2)
13. (4) **14.** (1)
15. (5) **16.** (4)
17. (2) **18.** (1)
19. (5) **20.** (3)
21. (3) **22.** (2)
23. (5) **24.** (1)
25. (5) **26.** (4)
27. (4) **28.** (3)
29. (2) **30.** (2)
31. (3) **32.** (4)
33. (5)

INSURANCE EXAMS

1. (2) **2.** (3)
3. (3) **4.** (2)
5. (1) **6.** (5)
7. (4) **8.** (1)
9. (2) **10.** (4)
11. (3) **12.** (5)
13. (5) **14.** (3)
15. (1) **16.** (4)
17. (2) **18.** (3)
19. (3) **20.** (4)
21. (1) **22.** (2)
23. (5) **24.** (5)
25. (1) **26.** (1)
27. (4) **28.** (2)
29. (1) **30.** (3)
31. (2) **32.** (3)
33. (2) **34.** (4)
35. (1) **36.** (5)
37. (3) **38.** (1)

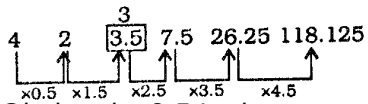
EXPLANATIONS
NATIONALISED BANKS
& IBPS PO/MT/SO

1. (4) The series is based on the following pattern:
$11 = 2 \times 3 + 5$
$38 = 11 \times 4 - 6$
$197 = 38 \times 5 + 7$
$1172 \neq 197 \times 6 - 8$
$\therefore$ **1172** is wong and it should be replaced by $197 \times 6 - 8 = 1174$

2. (1) The series is based on the following pattern:
$107 - 71 = 36 = 6^2$
$71 - 46 = 25 = 5^2$
$46 - 30 = 16 = 4^2$
$30 - 21 = 9 = 3^2$
$21 - 19 = 2 \neq 2^2$
$\therefore$ **19** I should be replaced by 17 for which $21 - 17 = 2^2$

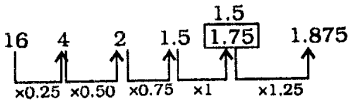
3. (4) The series is based on the following pattern:
$16 = 9 + 7$
$25 = 16 + 9$
$41 = 25 + 16$
68 $\neq 41 + 25$

4. (3) The series is based on the following pattern:



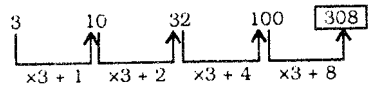
Obviously, 3.5 Is the wrong number which should be replaced by 3.

5. (2) The series is based on the following pattern:



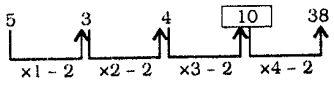
Obviously, 1.75 is the wrong number which should be replaced by 1.5.

6. (3) The given series is based on the following pattern:



Hence, 308 will come in place of question mark.

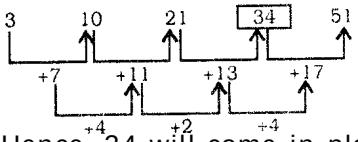
7. (5) The given series is based on the following pattern:



Hence, 10 will come in place of question mark.

8. (2) The given series is based on the following pattern:
$5 \times 1 + (1)^2 = 6$
$6 \times 2 + (2)^2 =$ **16**
$16 \times 3 + (3)^2 = 57$
$57 \times 4 + (4)^2 = 244$
Hence, 16 will come in place of question mark.

9. (1) The given series is based on the following patterns.

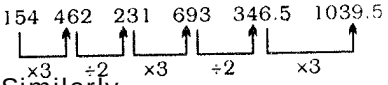


Hence, 34 will come in place of question mark.

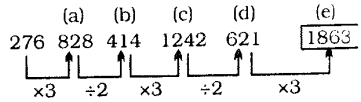
10. (4) The given series is based on the following pattern:
$5 \times 2 + 1 = 11$
$11 \times 2 + 3 =$ **25**
$25 \times 2 + 5 = 55$
$55 \times 2 + 7 = 117$

11. (3) The given series is based on the following pattern:
$30 = 12 \times 6 - 7 \times 6$
$120 = 30 \times 5 - 6 \times 5$
$460 = 120 \times 4 - 5 \times 4$
$1368 = 460 \times 3 - 4 \times 3$
$2730 = 1368 \times 2 - 3 \times 2$
Similarly,
(a) $= 16 \times 6 - 7 \times 6 = 96 - 42 = 54$
(b) $= 54 \times 5 - 6 \times 5 = 240$
(c) $= 240 \times 4 - 5 \times 4 = 940$
(d) $= 940 \times 3 - 4 \times 3 =$ **2808**
Hence, 2808 will come in place of (d).

12. (5) The given series is based on the following pattern:

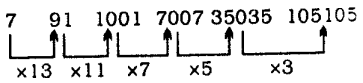


Similarly,

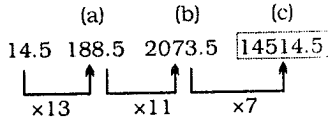


Hence, 1863 will come in place of (e).

13. (2) The given series is based on the following pattern:

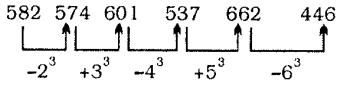


Similarly,

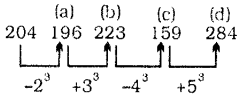


Hence, 14514.5 will come in place of (c).

14. (1) The given series is based on the following pattern :

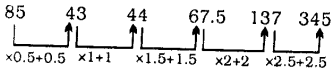


Similarly,

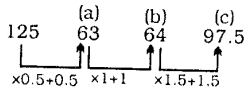


Hence, 284 will come in place of(d).

15. (4) The given series is based on the following pattern:



Similarly,



Hence, **97.5** will come in place of (c).

16. (1) The given series is based on the following pattern :

$1 = 1^3$ $? = 2^3 =$ **8**

$27 = 3^3$ $64 = 4^3$

$125 = 5^3$

Hence, 8 will come in place of the question mark.

17. (5) The given series is based on the following pattern :

$25 = 5^2$ $16 = 4^2$

$? = 3^2 =$ **9** $4 = 2^2$ and

$1 = 1^2$

Hence, 9 will come in place of the question mark,

18. (1) The given series Is based on the following pattern:

$1 \times 2 + 2 \times 2 = 6$

$6 \times 4 + 4 \times 3 = 36$

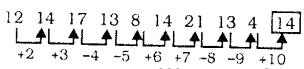
$36 \times 6 + 6 \times 4 = 240$

$240 \times 8 + 8 \times 5 = 1960$

$1960 \times 10 + 10 \times 6 =$ **19660**

Hence, 19660 will come in place of the quesdon mark.

19. (1) The given series is based on the following pattern :



Hence, 14 will come in place ol the question mark.

20. (2) The given series is based on the following pattern :

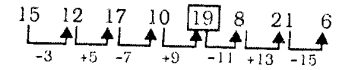
$2 + 5 = 7$ $7 + 5 = 12$

$12 + 7 = 19$ $19 + 12 = 31$

$31 + 19 = 50$ $50 + 31 =$ **81**

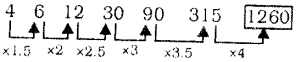
Hence, 81 will come in place of the question mark.

21. (4) The given series is based on the following pattern:



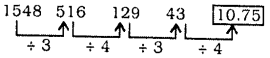
Hence, 19 will come in place of the question mark.

22. (3) The given series is based on the following pattern :



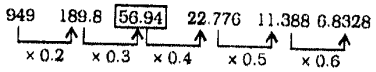
Hence, 1260 will come in place of the question mark.

23. (2) The given number series is based on the following pattern :



Hence, 10.75 will replace the quesdon mark.

24. (4) The given number series is based on the following pattern :



Hence, 56.94 will replace the question mark.

25. (1) The given number series is based on the following pattern :

$121 + 23 \times 1 = 144$

$144 + 23 \times 2 = 190$

$190 + 23 \times 3 = 259$

$\therefore ? = 259 + 23 \times 4$

$= 259 + 92 = 351$

Hence, **351** will replace the question mark.

26. (5) The given number series is based on the following pattern :

$14 \times 3 + 1.5 = 43.5$

$43.5 \times 6 + 1.5 \times 2 = 264$

$264 \times 12 + 1.5 \times 4 =$ **3174**

$3174 \times 24 + 1.5 \times 8 = 76188$

Hence, 3174 will replace the question mark.

27. (3) The given number series is based on the following pattern :

$41 \times 2^2 = 164$

$164 \times 4^2 = 2624$

$2624 \times 6^2 =$ **94464**

$94464 \times 8^2 = 6045696$

Hence, 94464 will replace the question mark.

28. (1) The given number series is based on the following pattern :

$12 \times 1 = 12$

$12 \times 1.5 = 18$

$18 \times (1 + 1.5) = 18 \times 2.5 = 45$
$45 \times (1.5 + 2.5) = 45 \times 4 = 180$
$180 \times (4 + 2.5) = 180 \times 6.5 = 1170$
$\therefore ? = 1170 \times (4 + 6.5) =$ **12285**
Hence, 12285 will replace the quesdon mark.

29. (3) The given number series is based on the following pattern :
$467 - 444 = 23 = 23 \times 1$
$513 - 467 = 46 = 23 \times 2$
$582 - 513 = 69 = 23 \times 3$
$674 - 582 = 92 = 23 \times 4$
$789 - 674 = 115 = 23 \times 5$
$\therefore ? = 789 + 23 \times 6$
$= 789 + 138 =$ **927**
Hence, 927 will replace the question mark.

30. (2) The given number series is based on the following pattern :
$1 = l^4$; $16 = 2^4$;
$81 = 3^4$; $256 = 4^4$;
$625 = 5^4$; $1296 = 6^4$;
$\therefore ? = 7^4 = 7 \times 7 \times 7 \times 7$
$=$ **2401**
Hence, 2401 will replace the question mark.

31. (5) The given number series is based on the following pattern :
$23 \times 1 + 2 = 25$
$25 \times 2 + 3 = 53$
$53 \times 3 + 4 = 163$
$163 \times 4 + 5 = 657$
$657 \times 5 + 6 = 3291$
$\therefore ? = 3291 \times 6 + 7$
$= 19746 + 7 =$ **19753**
Hence, 19753 will replace the question mark.

32. (4) The given number series is based on the following pattern :
$13 \times 1 = 13$
$13 \times 5 = 65$
$65 \times 9 = 585$
$585 \times 13 = 7605$
$7605 \times 17 = 129285$
$\therefore ? = 129285 \times 21 =$ **2714985**
Hence, 2714985 will replace the question mark.

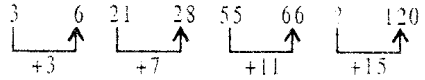
33. (1) The given number series is based on the following pattern :
$40280625 \div 55 = 732375$
$732375 \div 45 = 16275$
$16275 \div 35 = 465$
$465 \div 25 = 18.6$
$18.6 \div 15 = 1.24$
$\therefore ? = 1.24 \div 5 =$ **0.248**
Hence, 0.248 will replace the question mark.

34. (3) The given number series is based on the following pattern :
$14 \times 1 - 2 = 14 - 2 = 12$
$12 \times 2 - 3 = 24 - 3 = 21$
$21 \times 3 - 4 = 63 - 4 = 59$
$59 \times 4 - 5 = 236 - 5 = 231$
$231 \times 5 - 6 = 1155 - 6 = 1149$
$\therefore ? = 1149 \times 6 - 7$
$= 6894 - 7 =$ **6887**
Hence, 6887 will replace the question mark.

35. (2) The given number series is based on the following pattern :
$12 \times 12 \times 12 = 1728$
$14 \times 14 \times 14 = 2744$
$16 \times 16 \times 16 = 4096$
$18 \times 18 \times 18 = 5832$
$20 \times 20 \times 20 = 8000$
$22 \times 22 \times 22 = 10648$
$\therefore ? = 24 \times 24 \times 24 =$ **13824**
Hence, 13824 will replace the question mark.

36. (3) The given number series is based on the following pattern :
$120 \div 8 = 7$
$15 \times 7 = 105$
$105 \div 6 = 17.5$
$17.5 \times 5 = 87.5$
$\therefore = 87.5 \div 4 =$ **21.875**
Hence, 21.875 will replace the question mark.

37. (5) The given number series is based on the following pattern :



Hence, 105 will replace the question mark.

38. (5) The given number series is based on the following pattern :
$487.5 - 357.5 = 130$
$357.5 - 247.5 = 110$
$247.5 - 157.5 = 90$
$157.5 - 87.5 = 70$
$87.5 - 47.5 =$ **40**
$87.5 - 37.5 = 50$
$37.5 - 7.5 = 30$
Clearly, **47.5** is the wrong number. It should be replaced by 37.5.

39. (3) The given number series is based on the following pattern :
$13 + 3 = 16$
$16 + 5 = 21$
$21 + 7 = 28 \neq$ **27**
$28 + 11 = 39$
$39 + 13 = 52$
$52 + 17 = 69$

Clearly, 27 is the wrong num-ber. It should be replaced by 28.

40. (3) The given number series is based on the following pattern :
1500 + 81 = 1581
1581 + 83 = 1664
1664 + 85 = 1749
1749 + 87 = 1836 $\neq$ **1833**
1836 + 89 = 1925
1925 + 91 = 2016
Clearly, 1833 is the wrong number. It should be replaced by 1836.

41. (2) The given number series is based on the following pattern :
66 + 25 = 91
91 + 29 = 120
120 + 33 = 153
153 + 37 = 190
190 + 41 = 231 $\neq$ **233**
231 + 45 = 276
Clearly, 233 is the wrong number. It should be replaced by 231.

42. (1) The given number series is based on the following pattern :
11 × 11 × 11 = 1331
13 × 13 × 13 = 2197
15 × 15 × 15 = 3375
17 × 17 × 17 = 4913 $\neq$ **4914**
19 × 19 × 19 = 6859
Clearly, 4914 is the wrong number. It should be replaced by 4913.

43. (3) The given number series is based on the following pattern :
$20 + 2^2 = 24$
$24 + 3^2 = 33$
$33 + 4^2 = 49$
$49 + 5^2 = 74$
$74 + 6^2 = 110$
$\therefore ? = 110 + 7^2$
= 110 + 49 = **159**

44. (5) The given number series is based on the following pattern :
529 = 23 × 23
841 = 29 × 29
961 = 31 × 31
1369 = 37 × 37
1681 = 41 × 41
1849 = 43 × 43
$\therefore$? = 47 × 47 = **2209**
Here, the numbers are formed by squaring the prime numbers greater than 23.

45. (4) The given number series is based on the following pattern :
16 × 1.5 = 24
24 × 2 = 48
48 × 2.5 = 120
120 × 3 = 360
360 × 3.5 = 1260
$\therefore$? = 1260 × 4 = **5040**

46. (1) The given number series is based on the following pattern :
8 × 4 - 1= 32 - 1 = 31
31 × 4 - 2 = 124 - 2 = 122
122 × 4 - 3 = 488 - 3 = 485
485 × 4 - 4 = 1940 - 4 = 1936
1936 × 4 - 5 = 7744 - 5 = 7739
$\therefore$? = 7739 × 4 - 6
= 30956 - 6 = **30950**

47. (2) The given number series is based on the following pattern :
499 + 1 × 123 = 622
622 + 2 × 123 = 868
868 + 3 × 123 = 1237
1237 + 4 × 123 = 1729
1729 + 5 × 123 = 2344
$\therefore$? = 2344 + 6 × 123
= 2344 + 738 = **3082**

48. (1) The given number series is based on the following pattern
$1^1 = 1$; $2^2 = 4$
$3^3 = 27$; $4^4 = 256$
$5^5 = 3125$; 6^6 = **46656**
Hence 46658 is the wrong number.

49. (4) The given number series is based on the following pattern
18000 $\div$ 5 = 3600
3600 $\div$ 5 = 720
720 $\div$ 5 = 144 $\neq$ **142.2**
144 $\div$ 5 = 28.3
28.8 $\div$ 5 = 5.76
Hence 142.2 is the wrong number.

5O. (5) The given number series is based on the following pattern :
$12 + 15^2 = 12 + 225 = 237$
$237 + 13^2 = 237 + 169 = 406$
$406 + 11^2 = 406 + 121 = 527$
527 + 81 = 608
$608 + 7^2 = 608 + 49 = 657$
Hence 604 is the wrong number.

51. (3) The given number series is based on the following pattern :
3 × 7 + 2 × 7 = 21 + 14 = 35
35 × 6 + 3 × 6 = 210 + 18
= 228 $\neq$ **226**
228 × 5 + 4 × 5 = 1140 + 20 = 1160
1160 × 4 + 5 × 4 = 4640 + 20 = 4660
4660 × 3 + 6 × 3 = 13980 + 18 = 13998
Hence 226 is the wrong number

52. (2) The given number series i based on the following pattern :

$18 \times 7 - 7 = 126 - 7 = 119$
$119 \times 6 - 6 = 714 - 708$
$708 \times 5 - 5 = 3540 - 5 = 3535 \neq$ **3534**
$3535 \times 4 - 4 = 14140 - 4 = 14136$
Hence 3534 is the wrong number.

53. (5) $5 + 2^2 = 5 + 4 = 9$
$9 + 3^2 = 9 + 9 = 18$
$18 + 4^2 = 18 + 16 = 34$
$34 + 5^2 = 34 + 25 = 59$
$59 + 6^2 = 59 + 36 = 95$
$\therefore ? = 95 + 7^2 = 95 + 49 =$ **144**

54. (1) $1200 \div 2.5 = 480$
$480 \div 2.5 = 192$
$192 \div 2.5 = 76.8$
$76.8 \div 2.5 = 30.72$
$30.72 \div 2.5 = 12.288$
$\therefore ? = 12.288 \div 2.5 =$ **4.9152**

55. (3) $963 - 1 \times 36 = 963 - 36 = 927$
$927 - 2 \times 36 = 927 - 72 = 855$
$855 - 3 \times 36 = 855 - 108 = 747$
$747 - 4 \times 36 = 747 - 144 = 603$
$603 - 5 \times 36 = 603 - 180 = 423$
$\therefore ? = 423 - 6 \times 36 = 423 - 216 =$ **207**

56. (2) $29 \times 29 = 841$
$31 \times 31 = 961$
$33 \times 33 = 1089$
$35 \times 35 = 1225$
$37 \times 37 = 1369$
$39 \times 39 = 1521$
$\therefore ? = 41 \times 41 =$ **1681**

57. (4) $18 \times 1 + 2 = 18 + 2 = 20$
$20 \times 2 + 4 = 40 + 4 = 44$
$44 \times 3 + 6 = 132 + 6 = 138$
$138 \times 4 + 8 = 552 + 8 = 560$
$560 \times 5 + 10 = 2800 + 10 = 2810$
$\therefore ? = 2810 \times 6 + 12 = 16860 + 12 =$ **16872**

58. (3) $4 \times 1 + 2 = 4 + 2 = 6$
$6 \times 2 + 3 = 12 + 3 = 15 \neq$ **18**
$15 \times 3 + 4 = 45 + 4 = 49$
$49 \times 4 + 5 = 196 + 5 = 201$
$201 \times 5 + 6 = 1005 + 6 = 1011$

59. (5) $48 \times \frac{3}{2} = 72$; $72 \times \frac{3}{2} = 108$

$108 \times \frac{3}{2} = 162$: $162 \times \frac{3}{2} = 243$

$243 \times \frac{3}{2} = 364.5 \neq$ **366**

60. (1) $2 \times 6 + 7 \times 6 = 12 + 42 = 54$
$54 \times 5 + 6 \times 5 = 270 + 30 = 300$
$300 \times 4 + 5 \times 4 = 1200 + 20 = 1220$
$1220 \times 3 + 4 \times 3 = 3660 + 12$
$= 3672 \neq$ **3674**
$3672 \times 2 + 3 \times 2 = 7344 + 6$
$= 7350$

61. (2) $2^3 = 8 : 3^3 = 27$
$4^3 = 64 : 5^3 = 125$
$6^3 = 216 \neq$ **218**
$7^3 = 343$

62. (4) $19 + 7^2 = 19 + 49 = 68$
$68 + 6^2 = 68 + 36 = 104 \neq$ **102**
$104 + 5^2 = 104 + 25 = 129$
$129 + 4^2 = 129 + 16 = 145$
$145 + 3^2 = 145 + 9 = 154$

63. (5)
$0 + 5 = 5$
$5 + 13 = 18$
$18 + 25 = 43$
$43 + 41 = 84$
$84 + 61 = 145$
$\therefore ? = 145 + 85 =$ **230**

64. (4) $10 \times 1 + 1 \times 7 = 10 + 7 = 17$
$17 \times 2 + 2 \times 7 = 34 + 14 = 48$
$48 \times 3 + 3 \times 7 = 144 + 21 = 165$
$165 \times 4 + 4 \times 7 = 660 + 28 = 688$
$688 \times 5 + 5 \times 7 = 3440 + 35 = 3475$
$\therefore ? = 3475 \times 6 + 6 \times 7$
$= 20850 + 42 =$ **20892**

65. (3) $1 \times 3 = 3$
$3 \times 8 = 24$
$24 \times 15 = 360$
$360 \times 24 = 8640$
$8640 \times 35 = 302400$
$\therefore ? = 302400 \times 48$
$=$ **14515200**

66. (2) $12 \times 1 + 2 \times 1 = 12 + 2 = 14$
$14 \times 2 + 2 \times 2 = 28 + 4 = 32$
$32 \times 3 + 2 \times 3 = 96 + 6 = 102$
$102 \times 4 + 2 \times 4 = 408 + 8 = 416$
$416 \times 5 + 2 \times 5 = 2080 + 10$
$= 2090$
$\therefore ? = 2090 \times 6 + 2 \times 6$
$= 12540 + 12 =$ **12552**

67. (1) $10 \times \frac{3}{2} = 15$

$15 \times \frac{4}{4} = 15$

$15 \times \frac{5}{6} = 12.5$

$12.5 \times \frac{6}{8} = 9.375$

$9.375 \times \frac{7}{10} = 6.5625$

$\therefore ? = 6.5625 \times \frac{8}{12} =$ **4.375**

68. (3) The pattern of the number series is :
$17 \times 3 + 1 = 51 + 1 = 52$
$52 \times 3 + 2 = 156 + 2 = 158$
$158 \times 3 + 3 = 474 + 3 = 477$
$477 \times 3 + 4 = 1431 + 4 =$ **1435**

69. (4) The pattern of the number series is :
$3 \times 7 + 1 = 21 + 1 = 22$
$22 \times 6 + 2 = 132 + 2 =$ **134**
$134 \times 5 + 3 = 670 + 3 = 673$
$673 \times 4 + 4 = 2692 + 4 = 2696$

70. (1) The pattern of the number series is :
$6 \times 1 + 1 \times 7 = 6 + 7 = 13$
$13 \times 2 + 2 \times 6 = 26 + 12 = 38$
$38 \times 3 + 3 \times 5 = 114 + 15 =$ **129**
$129 \times 4 + 4 \times 4 = 516 + 16 = 532$

71. (5) The pattern of the number series is :
$\frac{286}{2} - 1 = 143 - 1 = 142$
$\frac{142}{2} - 1 = 71 - 1 =$ **70**
$\frac{70}{2} - 1 = 35 - 1 = 34$
$\frac{34}{2} - 1 = 17 - 1 = 16$

72. (3) The pattern of the number series is :
$17 \times 0.5 + 0.5 = 9$
$9 \times 1 + 1 =$ **10**
$10 \times 1.5 + 1.5 = 16.5$
$16.5 \times 2 + 2 = 35$

73. (5) The pattern is :
$2 \times 3 + 2 = 6 + 2 = 8$
$8 \times 3 + 2 = 24 + 2 = 26$
$26 \times 3 + 2 = 78 + 2 =$ **80**
$80 \times 3 + 2 = 240 + 2 = 242$

74. (1) The pattern is :
$3 \times 1 + 1^2 = 3 + 1 = 4$
$4 \times 2 + 2^2 = 8 + 4 = 12$
$12 \times 3 + 3^2 = 36 + 9 =$ **45**
$45 \times 4 + 4^2 = 180 + 16 = 196$

75. (4) The pattern is :
$9 \times 2 - 1 = 18 - 1 = 17$
$17 \times 2 - 1 = 34 - 1 =$ **33**
$33 \times 2 - 1 = 66 - 1 = 65$
$65 \times 2 - 1 = 130 - 1 = 129$

76. (2) The pattern is :
$7 \times 2 - 1 = 14 - 1 = 13$
$13 \times 2 - 1 = 26 - 1 =$ **25**
$25 \times 2 - 1 = 50 - 1 = 49$
$49 \times 2 - 1 = 98 - 1 = 97$

77. (3) The pattern is :
$5 \times 0.5 + 0.5 = 2.5 + 0.5 = 3$
$3 \times 1.5 + 1.5 = 4.5 + 1.5 = 6$
$6 \times 2.5 + 2.5 = 15 + 2.5 =$ **17.5**
$17.5 \times 3.5 + 3.5 = 61.25 + 3.5 = 64.75$

78. (2) The pattern is :
$16 \times 0.5 = 8 \quad 8 \times 1.5 = 12$
$12 \times 2.5 = 30 \quad 30 \times 3.5 =$ **105**

79. (4) The pattern is :
$5 \times 1 + 1 = 6$
$6 \times 2 + 2 = 14$
$14 \times 3 + 3 = 45$
$45 \times 4 + 4 = 184$

80. (1) The pattern is :
$7 \times 1 + 1 \times 5 = 12$
$12 \times 2 + 2 \times 4 = 32$
$32 \times 3 + 3 \times 3 = 105$
$105 \times 4 + 4 \times 2 =$ **428**

81. (5) The pattern is :
$11 \times 2 + 1 = 23$
$23 \times 2 + 1 = 47$
$47 \times 2 + 1 = 95$
$95 \times 2 + 1 =$ **191**

82. (3) The pattern is :
$9 \times 2 - 1 = 17$
$17 \times 2 - 1 = 33$
$33 \times 2 - 1 = 65$
$65 \times 2 - 1 =$ **129**

83. (3) The pattern of the number series is :
$8 + 3 = 11$
$11 + 3^2 = 11 + 9 = 20 \neq$ **17**
$20 + 3^3 = 20 + 27 = 47$
$47 + 3^4 = 47 + 81 = 128$
$128 + 3^5 = 128 + 243 = 371$

84. (3) The pattern of the number series is :
$1 + 2^2 = 1 + 4 = 5$
$5 + 2^3 = 5 + 8 = 13$
$13 + 2^4 = 13 + 16 = 29 \neq$ **31**
$29 + 2^5 = 29 + 32 = 61$
$61 + 2^6 = 61 + 64 = 125$

85. (3) The pattern is :
$150 \times 2 - 1 \times 10$
$= 300 - 10 = 290$
$290 \times 2 - 2 \times 10$
$= 580 - 20 = 560$
$560 \times 2 - 3 \times 10 = 1120 - 30$
$= 1090 \neq$ **1120**
$1090 \times 2 - 4 \times 10 = 2180 - 40 = 2140$
$2140 \times 2 - 5 \times 10 = 4280 - 50 = 4230$

86. (2) The pattern is : $10 \times 1 - 2 = 8$
$8 \times 2 - 3 = 13$
$13 \times 3 - 4 = 35$
$35 \times 4 - 5 = 135$
$135 \times 5 - 6 = 675 - 6$
$= 669 \neq$ **671**
$669 \times 6 - 7 = 4014 - 7 = 4007$

87. (3) The pattern is :
$(80 \div 2) + 2 = 40 + 2 = 42$

$(42 \div 2) + 2 = 21 + 2 = 23 \neq$ **24**
$(23 \div 2) + 2 = 11.5 + 2 = 13.5$
$(13.5 \div 2) + 2 = 6.75 + 2 = 8.75$
$(8.75 \div 2) + 2 = 4.375 + 2 = 6.375$

88. (1) The pattern is :

$125 \times \frac{3}{5} = 75$

$75 \times \frac{3}{5} = 45$

$45 \times \frac{3}{5} = 27 \neq$ **25**

$27 \times \frac{3}{5} = 16.2$

$16.2 \times \frac{3}{5} = 9.72$

89. (5) The pattern is :
$29 + 1 \times 8 = 37$
$37 - 2 \times 8 = 37 - 16 = 21$
$21 + 3 \times 8 = 21 + 24 = 45 \neq$ **43**
$45 - 4 \times 8 = 45 - 32 = 13$
$13 + 5 \times 8 = 13 + 40 = 53$
$53 - 6 \times 8 = 53 - 48 = 5$

90. (3) The pattern is:
$13 + 12 = 25$; $25 + 15 = 40$
$40 + 18 = 58 \neq$ **57**
$58 + 21 = 79$

91. (1) The pattern is :
$850 - 200 = 650 \neq$ **600**
$650 - 100 = 550$
$550 - 50 = 500$
$500 - 25 = 475$
$475 - 12.5 = 462.5$

92. (4) The pattern is:
$2 \times 3 = 6 \neq$ **10**
$6 \times 3 = 18$; $18 \times 3 = 54$
$54 \times 3 = 162$

93. (3) The pattern is:
$8 + 4 \times 1 = 12$; $12 + 4 \times 3 = 24$
$24 + 4 \times 5 = 44 \neq$ **46**
$44 + 4 \times 7 = 72$
$72 + 4 \times 9 = 108$

94. (1) The pattern is :
$142 - 23 = 119$; $119 - 19 = 100$
$100 - 17 = 83$
$83 - 13 = 70 \neq$ **65**
$70 - 11 = 59$
$59 - 7 = 52$

95. (5) The pattern is :
$5 + 7^2 = 54$
$54 + 6^2 = 90$
$90 + 5^2 = 115$
$115 + 4^2 = 131$
$131 + 3^2 = 140$
$140 + 2^2 = 140 + 4 =$ **144**

96. (4) The pattern is :
$7 \times 0.5 + 0.5 = 3.5 + 0.5 = 4$
$4 \times 1 + 1 = 4 + 1 = 5$
$5 \times 1.5 + 1.5 = 7.5 + 1.5 = 9$
$9 \times 2 + 2 = 18 + 2 =$ **20**

97. (3) The pattern is :
$6 \times 7 = 42$
$42 \times 6 =$ **252**
$252 \times 5 = 1260$

98. (1) The pattern is:
$4 \times 5 - 10 = 10$
$10 \times 5 - 10 = 40$
$40 \times 5 - 10 = 190$
$190 \times 5 - 10 = 940$
$940 \times 5 - 10 = 4700 - 10$
$=$ **4690**

99. (2) The pattern is :
$2 \times 1 + 1 \times 7 = 9$
$9 \times 2 + 2 \times 6 = 30$
$30 \times 3 + 3 \times 5 =$ **105**
$105 \times 4 + 4 \times 4 = 436$
$436 \times 5 + 5 \times 3 = 2195$

100. (2) The pattern of the numbe series is :
$(484 \div 2) - 2 = 242 - 2 = 240$
$(240 \div 2) - 2 = 120 - 2 = 118 = 120$
$(118 \div 2) - 2 = 59 - 2 = 57$
$(57 \div 2) - 2 = 28.5 - 2 = 26\ 5$

101. (4) The pattern of the number series is :
$3 \times 1 + 2 = 5$
$5 \times 2 + 3 = 13$
$13 \times 3 + 4 = 43$
$43 \times 4 + 5 = 177 \neq$ **176**
$177 \times 5 + 6 = 891$

102. (5) The Pattern of the number series is :
$6 + l^2 = 6 + 1 = 7$
$7 + 3^2 = 7 + 9 = 16$
$16 + 5^2 = 16 + 25 = 41$
$41 + 7^2 = 41 + 49 = 90$
$90 + 9^2 = 90 + 81 = 177 \neq$ **154**
$171 + 11^2 = 171 + 121 = 292$

103. (1) The pattern of the number series is :
$5 \times 1 + 1^2 = 6 \neq$ **7**
$6 \times 2 + 2^2 = 16$
$16 \times 3 + 3^2 = 57$
$57 \times 4 + 4^2 = 228 + 16 = 244$
$244 \times 5 + 5^2 = 1220 + 25 = 1245$

104. (3) The pattern of the number series is :
$4 \times 0.5 + 0.5 = 2 + 0.5 = 2.5$
$2.5 \times 1 + 1 = 3.5$
$3.5 \times 1.5 + 1.5 = 6.75 = \neq$ **65**
$6.75 \times 2 + 2 = 15.5$
$15.5 \times 2.5 + 2.5 = 38.75 + 25 = 41.25$
$41.25 \times 3 + 3 = 123.75 + 3 = 126.75$

105. (4) The pattern of the number series is :
325- 1 × 11 = 314
314 - 2 × 11 = 292
292 - 3 × 11 = 259
259 - 4 × 11 = 215
215 - 5 × 11 = $\neq$ **160**

106. (2) The pattern of the number series is :
45 × 1 + 1 = 46
46 × 1.5 + 1 = 70
70 × 2 + 1 = 141
141 × 2.5 + 1
= 352.5 + 1 = **353.5**

107. (3) The pattern of the number series is :
620 + 1 × 12 = 632
632 - 2 × 12 = 608
608 + 3 × 12 = 644
644 - 4 + 12 = 596
596 + 5 × 12 = |656 |

108. (5) The pattern of the number series is :
15 × 2 - 1 × 5 = 25
25 × 2 - 2 × 5 = 40
40 × 2 - 3 × 5 = 65
65 × 2 - 4 × 5 = **110**
110 × 2 - 5 × 5 = 195

109. (5) The pattern of the number series is :
120 × 2.5 + 20 = 320
320 × 2.5 + 20 = **820**
820 × 2.5 + 20 = 2070
2070 × 2.5 + 20 = 5195

110. (1) The pattern of the number series is :
32 + 1^2 = 32 + 1 = 33 $\neq$ **34**
33 + 2^2 = 33 + 4 = 37
37 + 3^2 = 37 + 9 = 46
46 + 4^2 = 46 + 16 = 62
62 + 5^2 = 62 + 25 = 87

111. (3) The pattern of the number series is :
7 + 1 × 11 = 7 + 11 = 18
18 + 3 × 11 = 18 + 33 = 51 $\neq$ **40**
51 + 5 × 11 = 51 + 55 = 106
106 + 7 × 11 = 106 + 77 = 183
183 + 9 × 11 = 183 + 99 = 282

112. (4) The pattern of the number series is :
850 - 1 × 7 = 843
843 - 2 × 7 = 829
829 - 3 × 7 = 808
808 - 4 × 7 = 780 $\neq$ **788**
780 - 5 × 7 = 745
745 - 6 × 7 = 703

113. (5) The pattern of the number series is :
33 + 288 = 321
321 + 144 = 465
465 + 72 = 537
537 + 36 = 573
573 + 18 = 591 $\neq$ **590**
591 + 9 = 600

114. (1) The pattern of the number series is :
37 + 1 × 5 = 42 $\neq$ **47**
42 + 2 × 5 = 52
52 + 3 × 5 = 67
67 + 4 × 5 = 87
87 + 5 × 5 = 112
112 + 6 × 5 = 142

115. (2) The pattern of the number series is :
13 + 3 = 16
16 + (3 + 3) = 22
22 + (6 + 5) = 33
33 + (11 + 7) = 51
51 + (18 + 9) = **78**

116. (3) The pattern of the number series is :
39 + 1 × 13 = 52
52 + 2 × 13 = 78
78 + 3 × 13 = 117
117 + 4 × 13 = 169
169 + 5 × 13 = **234**

117. (2) The pattern of the number series is :
62 + 5^2 = 62 + 25 = 87
87 + 10^2 = 87 + 100 = 187
187 + 15^2 = 187 + 225 = 412
412 + 20^2 = 412 + 400 = 812
812 + $(25)^2$ = 812 + 625 = **1437**

118. (1) The pattern of the number series is :
7 + 1^2 = 8
8 + 4^2 = 24
24 + 9^2 = 105
105 + 16^2 = 361
361 + 25^2 = **986**

119. (1) The pattern of the number series is :
656 - 224 = 432
432 - 112 = 320
320 - 56 = 264
264 - 28 = 236
236 - 14 = **222**

120. (2) The pattern of the number series is :
7 × 2 + 6 = 20
20 × 2 + 6 = 46
46 × 2 + 6 = 98
98 × 2 + 6 = 202
202 × 2 + 6 = 404 + 6 = **410**

121. (2) The pattern of the number series is :
210 - 1^3 = 209
209 + 2^2 = 213
213 - 3^3 = 186
186 + 4^2 = 202
202 - 5^3 = 202 - 125 = **77**

122. (5) The pattern of the number series is :
27 + 11 = 38
38 + 33 = 71
71 + 55 = 126
126 + 77 = 203
203 + 99 = **302**

123. (3) The pattern of the number series is :
435 - 9 × 9 = 354
354 - 9 × 8 = 282
282 - 9 × 7 = 219
219 - 9 × 6 = 165
165 - 9 × 5 = **120**

124. (3) The paatem of the number series is :
$4 + 14^2 = 4 + 196 = 200$
$200 + 13^2 = 200 + 169 = 369$
$369 + 12^2 = 369 + 144 = 513$
$513 + 11^2 = 513 + 121 = 634$
$634 + 10^2 = 634 + 100 =$ **734**

125. (3) The pattern of the number series is :
495 - 1 × 10 = 485
485 - 2 × 10 = 465
465 - 4 × 10 = 425
425 - 8 × 10 = 345
345 - 16 × 10 = **185**

126. (2) The pattern of the number series is :
16 + 6 = 22
22 + 11 = 33
33 + 16 = 49
49 + 21 = 70
70 + 26 = **96**

127. (5) The pattern of the number series is :
$32 + 2^2 = 36$
$36 + 4^2 = 52$
$52 + 6^2 = 88$
$88 + 8^2 = 152$
$152 + 10^2 = 252$

128. (3) The pattern of the number series is :
17 + 272 = 289
289 + 136 = 425
425 + 68 = 493
493 + 34 = 527
527 + 17 = **544**

129. (4) The pattern of the numbe series is :
13 + 1 × 14 = 27
27 + 2 × 14 = 55
55 + 3 × 14 = 97
97 + 4 × 14 = 153
153 + 5 × 14 = **223**

130. (3) The pattern of the number series is :
50 × 1.2 = 60
60 × 1.25 = 75
75 × 1.3 = 97.5
97.5 × 1.35 = **131.625**
131.625 × 1.4 = 184.275

131. (3) The pattern of the number series is :
12 × 1 + 3 × 1 = 15
15 × 2 + 3 × 2 = 36
36 × 3 + 3 × 3 = **117**
117 × 4 + 3 × 4 = 480
480 × 5 + 3 × 5 = 2415

132. (2) The pattern of the number series is :
1 × 1 + 1 = 2
2 × 2 + 2 = 6
6 × 3 + 3 = 21
21 × 4 + 4 = 88
88 × 5 + 5 = 445
445 × 6 + 6 = **2676**

133. (4) The pattern of the number series is :
$20 + 1^2 = 21$
$21 + 2^2 = 25$
$25 + 3^2 = 34$
$34 + 4^2 = 50$
$50 + 5^2 =$ **75**

134. (5) The pattern of the number series is :
$\frac{600}{5} + 5 = 125$
$\frac{125}{5} + 5 = 30$
$\frac{30}{5} + 5 =$ **11**
$\frac{11}{5} + 5 = 7.2$

135. (4) The pattern of the number series is :
$11 + 2^2 = 11 + 4 = 15$
$15 + 4^2 = 15 + 16 = 31$
$31 + 6^2 = 31 + 36 = 67$
$67 + 8^2 = 67 + 64 = 131$
$131 + 10^2 = 131 + 100 =$ **231**

136. (1) The pattern of the number series is :
483 - 1 × 12 = 483 - 12 = 471
471 - 3 × 12 = 471 - 36 = 435
435 - 5 × 12 = 435 - 60 = 375
375 - 7 × 12 = 375 - 84 = 291
291 - 9 × 12 = 291 - 108 = **183**

137. (2) The pattern of the number series is :
5 + 1 × 2 = 7
7 + 2 × 3 = 13
13 + 3 × 4 = 25
25 + 4 × 5 = 45
45 + 5 × 6 = **75**

138. (1) The pattern of the number series is :
4 +1 × 7 = 11
11 + 2 × 7 = 25
25 + 4 × 7 = 53
53 + 8 × 7 = 109
109 + 16 × 7 = 109 + 112 = **221**

139. (3) The pattern of the number series is :
15 + 6 × 1 = 21
21 + 6 × 2 = 33
33 + 6 × 3 = 51
51 + 6 × 4 = 75
75 + 6 × 5 = 105

140. (1) The pattern of the number series is :

$5 + 7^3 = 5 + 343 = 348$
$348 + 6^3 = 348 + 216 = 564$
$564 + 5^3 = 564 + 125 = 689$
$689 + 4^3 = 689 + 64 = 753$, not **716**
$753 + 3^3 = 753 + 27 = 780$

141. (4) The pattern of the number series is :

$\frac{4444}{2} + 2 = 2224$

$\frac{2224}{2} + 2 = 1114$

$\frac{1114}{2} + 2 = 559$ not **556**

$\frac{559}{2} + 2 = 281.5$

142. (5) The pattern of the number series is :
$4.5 + 11.5 = 16$
$16 + 9.5 = 25.5$, not **25**
$25.5 + 7.5 = 33$
$33 + 5.5 = 38.5$

143. (3) The pattern of the number series is :
$6 \times 7 + 1 \times 7 = 49$
$49 \times 6 + 2 \times 6 = 306$, not **305**
$306 \times 5 + 3 \times 5 = 1545$
$1545 \times 4 + 4 \times 4 = 6196$
$6196 \times 3 + 5 \times 3 = 18603$

144. (3) The pattern of the number series is :
$8 \times 0.5 + 1 = 5$
$5 \times 1 + 1.5 = 6.5$
$6.5 \times 1.5 + 2 = 9.75 + 2 = 11.75$, not **11**
$11.75 \times 2 + 2.5 = 23.5 + 2.5 = 26$
$26 \times 2.5 + 3 = 68$

145. (3) The pattern of the number series is :
$586 + 1 = 587$
$587 + (1 - 2) = 587 - 1 = 586$
$586 + (-1 - 4) = 586 - 5 = 581$
$581 + (-5 - 6) = 581 - 11 = 570$
$570 + (-11 - 8) = 570 - 19 = 551$
$551 + (-19 _ 10) = 551 - 29 = 522$

146. (5) The pattern of the number series is :
$64 - 10 = 54$
$54 + 15 = 69$
$69 - 20 = 49$
$49 + 25 = 74$
$74 - 30 = 44$
$44 + 35 =$ **79**

147. (2) The pattern of the number series is :
$(4000 \div 2) + 8 = 2008$
$(2008 \div 2) + 8 = 1012$
$(1012 \div 2) + 8 =$ **514**
$(514 \div 2) + 8 = 265$

148. (3) The pattern of the number series is :
$5 \times 1 = 5$
$5 \times 3 = 15$
$15 \times 5 = 75$
$75 \times 7 =$ **525**
$525 \times 9 = 4725$

149. (1) The pattern of the number series is :

$52 \times \frac{1}{2} = 26$

$26 \times 1 = 26$

$26 \times \frac{3}{2} = 39$

$39 \times 2 = 78$

$78 \times \frac{5}{2} =$ **195**

150. (3) The pattern of the number series is :
$14 - 10 = 4$
$25 - 14 = 11 = 4 \times 3 - 1$
$55 - 25 = 30 = 11 \times 3 - 3$
$140 - 55 = 85 = 30 \times 3 - 5$
$\therefore ? = 140 + 85 \times 3 - 7$
$= 140 + 248 =$ **388**

151. (5) The pattern of the number series is :
$119 + 1 \times 12 = 131$
$131 + 2 \times 12 = 155$
$155 + 3 \times 12 = 191$
$191 + 4 \times 12 = 239$
$239 + 5 \times 12 =$ **299**

152. (4) The pattern of the number series is :
$11 + 1 \times 46 = 11 + 46 = 57$
$57 + 2 \times 46 = 57 + 92 = 149$
$149 + 2 \times 92 = 149 + 184 = 333$
$333 + 2 \times 184 = 333 + 368 = 701$
$701 + 2 \times 368 = 701 + 736 =$ **1437**

153. (2) The pattern of the number series is :
$697 - 553 = 144 = 12^2$
$553 - 453 = 100 = 10^2$
$453 - 389 = 64 = 8^2$
$389 - 353 = 36 = 6^2$
$\therefore ? = 353 - 4^2 = 353 - 16 =$ **337**

154. (1) The pattern of the number series is :
$336 - 224 = 112$
$224 - 168 = 56$
$168 - 140 = 28$
$140 - 126 = 14$
$\therefore ? = 126 - 7 = 119$

155. (2) The pattern of the number series is :
$9 \times 2 - 3 = 18 - 3 = 15$
$15 \times 2 - 3 = 30 - 3 = 27$
$27 \times 2 - 3 = 54 - 3 = 51$
$51 \times 2 - 3 = 102 - 3 = 99$
$99 \times 2 - 3 = 198 - 3 =$ **195**

156. (4) The pattern of the number series is :
$13 + 8 = 21$
$21 + 8 + 7 = 21 + 15 = 36$

36 + 15 + 7 = 36 + 22 = 58
58 + 22 + 7 = 58 + 29 = 87
87 + 29 + 7 = 87 + 36 = **123**

157. (4) The pattern of the number series is :
7 + 2 + 0 = 9
9 + (2 + 8) = 19
19 + (10 + 16) = 45
45 + (26 + 24) = 95
95 + (50 + 32) = **177**

158. (1) The pattern of the number series is :
$14 + I^2 = 15$
$15 + 2^3 = 23$
$23 + 3^2 = 32$
$32 + 4^3 = 96$
$96 + 5^2 = 96 + 25 =$ **121**

159. (3) The pattern of the number series is :
20 + 1 × 4 = 20 + 4 = 24
24 + 3 × 4 = 24 + 12 = 36
36 + 5 × 4 = 36 + 20 = 56
56 + 7 × 4 = 56 + 28 = 84
84 + 9 × 4 = 84 + 36 = **120**

160. (2) The pattern of the number series is :
$732 - 3 = 729 = 9^3$
$1244 - 732 = 512 = 8^3$
$1587 - 1244 = 343 = 7^3$
$1803 - 1587 = 216 = 6^3$
$1928 - 1803 = 125 = 5^3$
$\therefore ? = 1928 + 4^3 = 1928 + 64 =$ **1992**

161. (4) The pattern of the number series is :
16 × 1.5 = 24
24 × 2.5 = **60**
60 × 3.5 = 210
210 × 4.5 = 945

162. (1) The pattern of the number series is :
(45030 ÷ 5) - 6 = 9000
(9000 ÷ 5) - 5 = 1795
(1795 ÷ 51 - 4 = 355
(355 ÷ 5) - 3 = 68
(68 ÷ 5) - 2 = 13.6 - 2 = **11.6**

163. (1) The pattern of the number series is :
5 × 1 + 1 × 7 = 12
12 × 2 + 2 × 6 = 36
36 × 3 + 3 × 5 = 123
123 × 4 + 4 × 4 = 492 + 16 = **508**
508 × 5 + 5 × 3 = 2540 + 15 = 2555

164. (4) The pattern of the number series is :
8 × 0.5 + 7 = 4 + 7=11
11 × 1 + 6 = 17
17 × 1.5 + 5 = 25.5 + 5 = **30.5**
30.5 × 2 + 4 = 61 + 4 = 65

165. (5) The pattern of the number series is :
389 - 117 = 272
525 - 389 = 136
593 - 525 = 68
627 - 593 = 34
$\therefore ?$ = 627 + 17 = **644**

166. (4) The pattern of the number series is :
7 + 1 × 4 = 11
11 + (1 + 2) 4 = 11 + 3 × 4 = 23
23 + (3 + 4) 4 = 23 + 7 × 4 = 51
51 + (7 + 6) 4 = 51 + 13 × 4 = 103
103 + (13 + 8) 4 = 103 + 21 × 4 = **187**

167. (4) The pattern of the number series is :
18 + 9 = 27
27 + (9 + 13) = 49
49 + (9 + 26) = 84
84 + (9 + 39) = 132

168. (2) The pattern of the number series is :
33 + 10 = 43
43 + (10 + 12) = 65
65 + (10 + 24) = 99
99 + (10 + 36) = 145
145 + (10 + 48) = **203**

169. (5) The pattern of the number series is :
$655 - 439 = 216 = 6^3$
$439 - 314 = 125 = 5^3$
$314 - 250 = 64 = 4^3$
$250 - 223 = 27 - 3^3$
$\therefore ? = 223 - 2^3 = 223 - 8$
= **215**

170. (4) The pattern of the number series is :
15 + 6 = 21
21 + 18 (= 6 + 12) = 39
39 + 38 (= 18 + 20) = 77
77 + 66 (= 38 + 28) = 143
143 + 102 (= 66 + 36) = **245**

171. (1) The pattern of the number series is :
33 + 6 = 39
39 + 18 (= 6 + 12) = 57
57 + 30 (= 18 + 12) = 87
87 + 42 (= 30 + 12) = 129
129 + 54 (= 42 + 12) = **183**

172. (1) The pattern of the number series is :
$19 - 15 = 4 = 2^2$
$83 - 19 = 64 = 4^3$
$119 - 83 = 36 = 6^2$
$631 - 119 = 512 = 8^3$
$\therefore ? = 631 + 10^2 = 631 + 100 =$ **731**

173. (3) The pattern of the number series is :
19 + 1 × 7 = 19 + 7 = 26
26 + 2 × 7 = 26 + 14 = 40
40 + 4 × 7 = 40 + 28 = 68
68 + 8 × 7 = 68 + 56 = 124
124 + 16 × 7 = 124 + 112
= **236**

174. (5) The pattern of the number series is :
69 - 43 = 26
58 - 69 = - 11
84 - 58 = 26
73 - 84 = -11

$\therefore ? = 73 + 26 =$ **99**

175. (4) The pattern of the numbe series is :
$15 + 3 = 18$
$18 - 2 = 16$
$16 + 3 = 19$
$19 - 2 = 17$
$17 + 3 = 20$
$20 - 2 =$ **18**

176. (1) The pattern of the number series is :

$1050 \times \frac{2}{5} = 420$

$420 \times \frac{2}{5} = 168$

$168 \times \frac{2}{5} = 67.2$

$\therefore 10.752 \times \frac{2}{5} =$ **4.3008**

177. (5) The pattern of the number series is :
$0 + 1 \times 6 = 6$
$6 + 2 \times 9 = 24$
$24 + 3 \times 12 = 60$
$60 + 4 \times 15 = 120$
$120 + 5 \times 18 = 210$
$210 + 6 \times 21 = 210 + 126 =$ **336**

178. (3) The pattern of the number series is :
$32 + 1 \times 17 = 32 + 17 = 49$
$49 + 2 \times 17 = 49 + 34 = 83$
$83 + 4 \times 17 = 83 + 68 = 151$
$151 + 8 \times 17 = 151 + 136 = 287$
$287 + 16 \times 17 = 287 + 272 = 559$
$559 + 32 \times 17 = 559 + 544 =$ **1103**

179. (2) The pattern of the number series is :
$552 - 462 = 90$
$650 - 552 = 98$
$756 - 650 = 106$
$870 - 756 = 114$
$992 - 870 = 122$
$\therefore ? = 992 + 130 =$ **1122**

180. (3) The pattern of the number se ries is :
$28 + 11 = 39$
$39 + 24 (= 11 + 13) = 63$
$63 + 39 (= 24 + 15) = 102$
$102 + 56 (= 39 + 17) = 158$
$158 + 75 (= 56 + 19) =$ **233**

181. (5) The pattern of the number series is :
$7 + 3^2 = 7 + 9 = 16$
$16 + 5^3 = 16 + 125 = 141$
$141 + 7^2 = 141 + 49 = 190$
$190 + 9^3 = 190 + 729 = 919$
$919 + 11^2 = 919 + 121 =$ **1040**

182. (3) The pattern of the number series is :
$12 + 5 \times 1 = 17$
$17 + 5 \times 3 = 32$
$32 + 5 \times 5 = 57$
$57 + 5 \times 7 = 92$
$92 + 5 \times 9 = 137$

183. (4) The pattern of the number series is :
$19 + 2 \times 3 = 19 + 6 = 25$
$25 + 4 \times 5 = 25 + 20 = 45$
$45 + 6 \times 7 = 45 + 42 = 87$
$87 + 8 \times 9 = 87 + 72 = 159$
$159 + 10 \times 11 = 159 + 110 =$ **269**

184. (5) The pattern of the number series is :
$83 + 41 \times 1 = 124$
$124 + 41 \times 2 = 124 + 82 = 206$
$206 + 41 \times 4 = 206 + 164 = 370$
$370 + 41 \times 8 = 370 + 328 = 698$
$698 + 41 \times 16 = 698 + 656 =$ **1354**

185. (4) The pattern of the number series is :
$1 \times 7 = 7$
$7 \times 7 = 49$
$49 \times 7 = 343$
$343 \times 7 =$ **2401**

186. (4) The pattern of the number series is :
$13 + 7 = 20$
$20 + 19 (= 7 + 12) = 39$
$39 + 39 (=19 + 20) = 78$
$78 + 67 (= 39 + 28) = 145$
$145 + 103 (= 67 + 36) =$ **248**

187. (1) The pattern of the number series is :
$12 + 1 \times 23 = 35$
$35 + 2 \times 23 = 35 + 46 = 81$
$81 + 2 \times 46 = 81 + 92 = 173$
$173 + 2 \times 92 = 173 + 184 = 357$
$357 + 2 \times 184 = 357 + 368 =$ **725**

188. (5) The pattern of the number series is :
$3 + 97 = 100$
$100 + 197 = 297$
$297 + 297 = 594$
$594 + 397 = 991$
$991 + 497 =$ **1488**

189. (3) The pattern of the number series is :
$112 + 1 \times 7 = 119$
$119 + 3 \times 7 = 119 + 21 = 140$
$140 + 5 \times 7 = 140 + 35 = 175$
$175 + 7 \times 7 = 175 + 49 = 224$
$224 + 9 \times 7 = 224 + 63 =$ **287**

190. (2) The pattern of the number series is :
$958 - 833 = 125$
$833 - 733 = 100$
$733 - 658 = 75$
$658 - 608 = 50$
$\therefore ? = 608 - 25 =$ **583**

191. (4) The pattern of the number series is :
$11 \times 1 - 1 = 10$
$10 \times 2 - 2 = 18$
$18 \times 3 - 3 = 51$

51 × 4 - 4 = 200
200 × 5 - 5 = **995**

192. (1) The pattern of the number series is :
25 × 2 - 2 = 50 - 2 = 48
48 × 2 - 2 = 96 - 2 = 94
94 × 2 - 2 = 188 - 2 = 186
186 × 2 - 2 = 372 - 2 = 370
370 × 2 - 2 = 740 - 2 = **738**

193. (2) The pattern of the number series is :
14 + 10 = 24
24 + 19 (=10 + 9) = 43
43 + 28 (= 19 + 9) = 71
71 + 37 (= 28 + 9) = 108
108 + 46 (=37 + 9) = **154**

194. (5) The pattern of the number series is :
144 + 29 = 173
173 - 33 = 140
140 + 29 = 169
169 - 33 = 136
136 + 29 = **165**

195. (2) The pattern of the number series is :
8 + 2 = 10
10 + 8 (= 2 × 3 + 2) = 18
18 + 26 (= 3 × 8 + 2) = 44
44 + 80 (=3 × 26 + 2) = 124
124 + 242 (= 3 × 80 + 2) = **366**

196. (4) The pattern of the number series is :
13 + 1 × 12 = 13 + 12 = 25
25 + 3 × 12 = 25 + 36 = 61
61 + 5 × 12 = 61 + 60 = 121
121 + 7 × 12 = 121 + 84 = 205
205 + 9 × 12 = 205 + 108
= **313**

197. (1) The pattern of the number series is :
$\frac{656}{2}$ + 24 = 328 + 24 = 352
$\frac{352}{2}$ + 24 = 176 + 24 = 200
$\frac{200}{2}$ + 24 = 100 + 24 = 124
$\frac{124}{2}$ + 24 = 62 + 24 = 86
$\frac{86}{2}$ + 24 = 43 + 24 = **67**

198. (3) The pattern of the number series is :
454 + 18 = 472
472 - 27 = 445
445 + 18 = 463
463 - 27 = 436
436 + 18 = **454**

199. (2) The pattern of the number series is :
12 × 4 - 30 = 48 - 30 = 18
18 × 4 - 36 = 72 - 36 = 36
36 × 4 - 42 = 144 - 42 = 102
102 × 4 - 48 = 408 - 48 = 360
360 × 4 - 54 = 1440 - 54 = **1386**

200. (4) The pattern of the number series is :
7 × 2 - 2 = 12
12 × 4 - (2 + 6) = 48 - 8 = 40
40 × 6 - (8 + 10) = 240 - 18 = 222
222 × 8 - (18 + 14) = 1776 - 32
= 1744 $\neq$ **1742**
1744 × 10 - (32 + 18) = 17440 - 50 = 17390

201. (3) The pattern of the number series is :
6 × 7 + 7^2 = 42 + 49 = 91
91 × 6 + 6^2 = 546 + 36 = 582
582 × 5 + 5^2 = 2910 + 25 = 2935
2935 × 4 + 4^2 = 11740 + 16 = 11756
11756 × 3 + 3^2 = 35268 + 9 = 35277

202. (5) The pattern of the number series is :
9050 - 15^3 = 9050 - 3375 = 5675
5675- 13^3 = 5675 - 2197 = 3478
3478 - 11^3 = 3478 - 1331 = 2147
2147 - 9^3 = 2147 - 729 = 1418
1418 - 7^3 = 1418 - 343 = 1075 $\neq$ **1077**

203. (4) The pattern of the number series is :
1 = 1
2^2 = 4
3^3 = 27 $\neq$ **25**
4^4 = 256
5^5 = 3125
6^6 = 46656

204. (2) The pattern of the number series is :
8424 ÷ 2 = 4212
4212 ÷ 2 = 2106
2106 ÷ 2 = 1053 $\neq$ **1051**
1053 ÷ 2 = 526.5
526.5 ÷ 2 = 263.25

205. (1) The pattern is :
5531 - 5506 = 25 = 5^2
5555 - 5506 = 49 = 7^2
5506 - 5425 = 81 = 9^2
5425 - 5304 = 121 = 11^2
5304 - 5135 = 169 = 13^2
5135 - 4910 = 225 = 15^2
4910 - 4621 = 289 = 17^2
Clearly, 5531 is wrong which should be substituted by 5555.

206. (2) The pattern is :
6 + 1 = 7
7 + 1 × 2 = 9
9 + 2 × 2 = 13
13 + 8 = 21 $\neq$ **26**
21 + 16 = 37
37 + 32 = 69

207. (4) The pattern is :
1 × 1 + 2 = 3

$3 \times 2 + 4 = 10$
$10 \times 3 + 6 = 36$
$36 \times 4 + 8 = 152$
$152 \times 5 + 10 = 770 \neq$ **760**
$770 \times 6 + 12 = 4632$

208. (3) The pattern is :
$4 + 1^3 = 5$
$5 + 2^3 = 13$
$13 + 3^3 = 40$
$40 + 4^3 = 104 \neq$ **105**
$104 + 5^3 = 229$
$229 + 6^3 = 445$

209. (1) The pattern is :
$157.5 \div 3.5 = 45$
$45 \div 3 = 15$
$15 \div 2.5 = 6$
$6 \div 2 = 3$
$3 \div 1.5 = 2$
$2 \div 1 = 2 \neq$ **1**

210. (2) The pattern is :
$123 + 11 \times 14 = 123 + 154 = 277$
$277 + 13 \times 14 = 277 + 182 = 459$
$459 + 15 \times 14 = 459 + 210 = 669$
$669 + 17 \times 14 = 669 + 238 = 907$
$907 + 19 \times 14 = 907 + 266$
$=$ **1173**

211. (2) The pattern is :
$456.5 - 407 = 49.5$
$407 - 368.5 = 38.5$
$368.5 - 341 = 27.5$
$341 - 324.5 = 16.5$
$\therefore ? = 324.5 - 5.5 =$ **319**

212. (1) The pattern is :
$23 + 1 \times 19.2 = 42.2$
$42.2 + 2 \times 19.2 = 80.6$
$80.6 + 4 \times 19.2 = 157.4$
$157.4 + 8 \times 19.2 = 311$
$311 + 16 \times 19.2 = 311 + 307.2$
$=$ **618.2**

213. (5) The pattern is :
$154 - 36 = 118$
$232 - 154 = 78$
$278 - 232 = 46$
$300 - 278 = 22$
$\therefore ? - 300 = 6$
$\Rightarrow ? =$ **306**

214. (4) The pattern is ;
$24 + 8^3 = 24 + 512 = 536$
$536 - 7^2 = 536 - 49 = 487$
$487 + 6^3 = 487 + 216 = 703$
$703 - 5^2 = 703 - 25 = 678$
$678 + 4^3 = 678 + 64 = 742$

215. (3) The pattern is :
$576 - 224 = 352$
$752 - 576 = 176$
$840 - 752 = 88$
$884 - 840 = 44$
$\therefore ? = 884 + 22 =$ **906**

216. (1) The pattern is :
$5 \times 1 + 1^2 = 5 + 1 = 6$
$6 \times 2 + 2^2 = 12 + 4 = 16$
$16 \times 3 + 3^2 = 48 + 9 = 57$
$57 \times 4 + 4^2 = 228 + 16 =$ **244**

217. (4) The pattern is :
$12 \times 4 =$ **48**
$48 \times 3.5 = 168$
$168 \times 3 = 504$
$504 \times 2.5 = 1260$
$1260 \times 2 = 2520$

218. (5) The pattern is :
$4 \times 2 + 1 = 8 + 1 = 9$
$9 \times 3 + 2 = 27 + 2 = 29$
$29 \times 4 + 3 = 116 + 3 =$ **119**
$119 \times 5 + 4 = 595 + 4 = 599$
$599 \times 6 + 5 = 3594 + 5 = 3599$

219. (3) The pattern is :
$177 - 7 = 170$
$170 - 11 = 159$
$159 - 13 = 146$
$146 - 17 =$ **129**
$129 - 19 = 110$
Note : Consecutive prime numbers have been subtracted.

220. (3) The pattern is :
$2 + 1^3 = 2 + 1 = 3$
$3 + 2^3 = 3 + 8 = 11$
$11 + 3^3 = 11 + 27 = 38$
$38 + 4^3 = 38 + 64 = 102$
$102 + 5^3 = 102 + 125 =$ **227**

221. (1) The pattern of the number series is :
$21 \times 0.5 = 10.5$
$10.5 \times 1 =$ **10.5**
$10.5 \times 1.5 = 15.75$
$15.75 \times 2 = 31.50$
$31.50 \times 2.5 = 78.75$

222. (2) The pattern of the number series is :
$6 + 1 \times 13 = 6 + 13 = 19$
$19 + 3 \times 13 = 19 + 39 = 58$
$58 + 5 \times 13 = 58 + 65 =$ **123**
$123 + 7 \times 13 = 123 + 91 = 214$
$214 + 9 \times 13 = 214 + 117 = 331$

223. (3) The pattern of the number series is :
14 $+ 1 \times 2 =$ **16**
$16 + 3 \times 4 = 16 + 12 = 28$
$28 + 5 \times 6 = 28 + 30 = 58$
$58 + 7 \times 8 = 58 + 56 = 114$
$114 + 9 \times 10 = 114 + 90 = 204$

224. (4) The pattern of the number series is :
$13.76 + 1 \times 1.15 = 14.91$
$14.91 + 2 \times 1.15 = 14 + 2.30 = 17.21$

17.21 + 3 × 1.15 = 17.21 + 3.45 = 20.66
20.66 + 4 × 1.15 = 20.66 + 4.60 = **25.26**
25.26 + 5 × 1.15 = 25.26 + 5.75 = 31.01

225. (5) The pattern of the number series is :
$15 + 1^2 = \mathbf{16}$
$16 + 2^3 = 16 + 8 = 24$
$24 + 3^2 = 24 + 9 = 33$
$33 + 4^3 = 33 + 64 = 97$
$97 + 5^2 = 97 + 25 = 122$

226. (5) The pattern is :
$2 \times 3 = 6$
$6 \times 2.5 = 15$
$15 \times 2 = 30$
$30 \times 1.5 = 45$
$45 \times 1 = 45 \neq \mathbf{43.5}$
$45 \times 0.5 = 22.5$

227. (3) The pattern is :
$950 - 661 = 289 = 17^2$
$661 - 436 = 225 = 15^2$
$436 - \mathbf{269} = 167 \neq 13^2$
$\therefore 436 - 267 = 169 = 13^2$
$267 - 146 = 121 = 11^2$
$146 - 65 = 81 = 9^2$

228. (5) The pattern is :
$6.5 + 5.3 = 11.8$
$11.8 + 2 \times 5.3 = 11.8 + 10.6 = 22.4$
$22.4 + 3 \times 5.3 = 22.4 + 15.9 = 38.3$
$38.3 + 4 \times 5.3 = 38.3 + 21.2 = 59.5$
$59.5 + 5 \times 5.3 = 59.5 + 26.5 = 86$
$\neq \mathbf{87.3}$
$86 + 6 \times 5.3 = 86 + 31.8 = 117.8$

229. (5) The pattern is :
$1 \times 3 - 1 = 2$
$2 \times 3 - 2 = 4$
$4 \times 3 - 3 = 9$
$9 \times 3 - 4 = 23$
$23 \times 3 - 5 = 69 - 5 = 64 \neq \mathbf{69}$
$64 \times 3 - 6 = 192 - 6 = 186$

230. (5) The pattern is :
$250 - 11 = 239$
$239 - (11 \times 2 + 1) = 239 - 23 = 216$
$216 - (11 \times 3 + 2) = 216 - 35 = 181$
$181 - (11 \times 4 + 3)$
$= 181 - 47 = 134 \neq \mathbf{136}$
$134 - (11 \times 5 + 4)$
$= 134 - 59 = 75$
$75 - (11 \times 6 + 58)$
$= 75 - 71 = 4$

SBI PO EXAMS

1. (3) The series is based on following pattern:
$3 \times 1 + 2 = 5$
$5 \times 2 + 2 = 12$
$12 \times 3 + 2 = 38$
$38 \times 4 + 2 = 154$
$154 \times 5 + 2 = \mathbf{772}$
$772 \times 6 + 2 = 4634$
Therefore, the number 914 is wrong.
$\therefore$ According to question, the new series is as follows:
$914 \times 1 + 2 = 916$
$916 \times 2 + 2 = \mathbf{1834}$
$1834 \times 3 + 2 = 5504$
Therefore, the required number is 1834.

2. (3) The series is based on following pattern :
$3 \times 1 + 1 = 4$
$4 \times 2 + 2 = 10$
$10 \times 3 + 3 = \mathbf{33}$
$33 \times 4 + 4 = 136$
$136 \times 5 + 5 = 685$
$685 \times 6 + 6 = 4116$
Therefore, the number 34 is wrong.
$\therefore$ According to question, the new series starts from the number 34 in the same pattern.
$34 \times 1 + 1 = 35$
$35 \times 2 + 2 = \mathbf{72}$
Hence, the number 72 is required answer.

3. (4) The series is based on following pattern :
$214 - (14)^2 = 18$
$18 + (12)^2 = 162$
$162 - (10)^2 = 62$
$62 + (8)^2 = \mathbf{126}$
$126 - (6)^2 = 90$
$90 + (4)^2 = 106$
Therefore the number 143 is wrong.
$\therefore$ According to question, the new series starts from the number 143 in
$143 - (14)^2 = -53$
$-53 + (12)^2 = \mathbf{91}$
Hence, the number 91 is required answer.

4. (5) The series in based on following pattern:
$160 \times 0.5 = 80$
$80 \times 1.5 = 120$
$120 \times 2.5 = \mathbf{300}$
$300 \times 3.5 = 1050$
$1050 \times 4.5 = 4725$
$4725 \times 5.5 = 25987.5$
Therefore, the number 180 is wrong.
$\therefore$ According to question, the new series starts from the number 180 in the same pattern:
$180 \times 0.5 = 90$
$90 \times 1.5 = \mathbf{135}$
Hence, the number 135 is required answer.

5. (1) The series is based on following pattern:
$2 + I^2 - 0 = 3$
$3 + 2^2 - 1 = \mathbf{6}$
$6 + 3^2 - 2 = 13$
$13 + 4^2 - 3 = 26$
$26 + 5^2 - 4 = 47$

$47 + 6^2 - 5 = 78$

Therefore, the number 7 is wrong. According to question, the new series starts from the number 7 in the same pattern.

$7 + l^1 - 0 = 8$

$8 + 2^2 - 1 =$ **11**

Hence, the number 11 is required answer.

6. (4) The series is based on following pattern :

$2 \times 1 + l^2 = 3$

$3 \times 2 + 2^2 = 10$

$10 \times 3 + 3^2 = 39$

$39 \times 4 + 4^2 = 172$

$172 \times 5 + 5^2 = 885$

Similarly, the new series is as follows :

$1 \times 1 + 12 = 2$.....(a)

$2 \times 2 + 2^2 = 8$.....(b)

$8 \times 3 + 3^2 = 33$ .....(c)

Therefore, the number 8 will come in place of (b).

7. (2) The series is based on the following pattern:

$5 \times 1 + 2 = 7$

$7 \times 2 - 4 = 10$

$10 \times 3 + 6 = 36$

$36 \times 4 - 8 = 136$

$136 \times 5 + 10 = 690$

Similarly, the new series is as follows:

$2 \times 1 + 2 = 4$ ....(a)

$4 \times 2 - 4 = 4$ ....(b)

$4 \times 3 + 6 = 18$ ....(c)

$18 \times 4 - 8 = 64$ ....(d)

$64 \times 51 + 10 = 330$ .....(e)

Therefore, the number 330 will come in palce of (e).

8. (5) The series is based on following pattern:

$8 \times 0.5 = 4$

$4 \times 1.5 = 6$

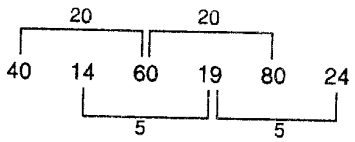
$6 \times 2.5 = 15$

$15 \times 3.5 = 52.6$

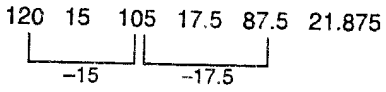
$52.5 \times 4.5 = 236.25$

Therefore, the number 236.25 will come in place of (d).

9. (3) Interchanging (3) and (5)



10. (3) Interchanging (3) and (5)



11. (4) The series is based on following pattern

$2 \times 1 + (1)^2 = 3$

$3 \times 2 - (2)^2 = 2$

$2 \times 3 + (3)^2 = 15$

$15 \times 4 - (4)^2 = 44$

$44 \times 5 + (5)^2 = 245$

$245 \times 6 - (6)^2 = 1434$

Similarly,

$3 \times 1 + (1)^2 = 4$ ....(a)

$4 \times 2 - (2)^2 = 4$ ....(b)

$4 \times 3 + (3)^2 =$ **21** ....(c)

$21 \times 4 - (4)^2 = 68$ ....(d)

Therefore, the 21 will come in place of (c).

12. (5) The series is based on following pattern

$1 \times 1 + (1)^2 = 2$

$2 \times 2 + (2)^2 = 8$

$8 \times 3 + (3)^2 = 33$

$33 \times 4 + (4)^2 = 148$

$148 \times 5 + (5)^2 = 765$

$765 \times 6 + (6)^2 = 4626$

Similarly,

$2 \times 1 + (1)^2 = 3$ ....(a)

$3 \times 2 + (2)^2 = 10$ ....(b)

$10 \times 3 + (3)^2 = 39$ .....(c)

$39 \times 4 + (4)^2 =$ **172** ....(d)

Therefore, the number 172 will come in place of (d).

13. (5) The series is based on following pattern :

$2 \times 2 + 0.5 = 4.5$

$4.5 \times 2 + (0.5) \times 4 = 11$

$11 \times 2 + 2 \times 4 = 30$

$30 \times 2 + 8 \times 4 = 92$

$92 \times 2 + 32 \times 4 = 312$

$312 \times 2 + 128 \times 4 = 1136$

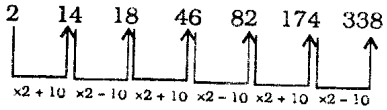
Similarly,

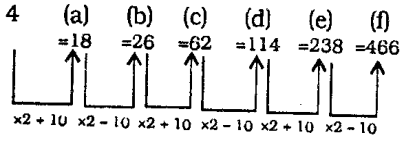
$1 \times 2 + 0.5 = 2.5$ ....(a)

$2.5 \times 2 + (0.5) \times 4 =$ **7** ....(b)

Therefore, the number 7 wil come in place of (b)

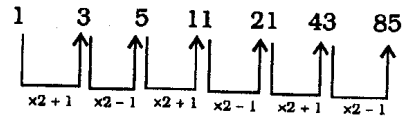
14. (1)

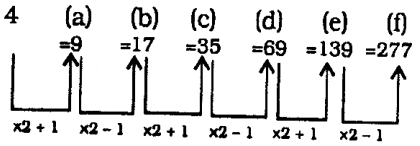




In the given series 176 should be replace br 174.238 will come in place of (e)

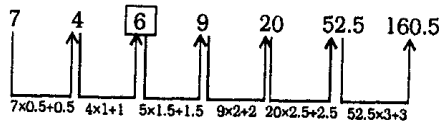
15. (3)





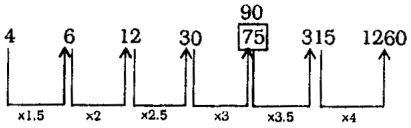
In the given series 7 should be replaced by 5. and 277 should come in place of (f).

16. (1) The given number series is based on the following pattern



Hence the wrong number is 6

17. (2) The given number series is based on the following pattern:



Hence, the wrong number is 75

18. (4)The given number series is basei on the following pattern

$4 - 3 = 1^2$

$13 - 4 = 9 = 32$

$38 - 13 = 25 = 5^2$

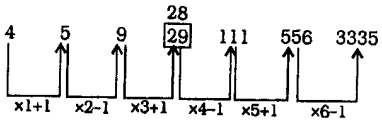
$87 - 38 = 49 = 7^2$

$168 - 87 = 81 = 9^2$

$289 - 168 = 121 = 11^2$

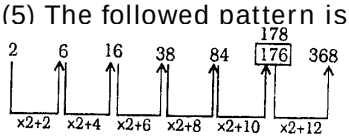
Obviously, 166 is the wrong number.

19. (3) The number series follows the rule as mentioned below:



Hence 29 is the wrong number.

20. (5) The followed pattern is:



Hence the wrong number is 176

21. (4) The given series is based on tne following pattern

$2 \times 3 = 6$

$6 \times 3 = 18$

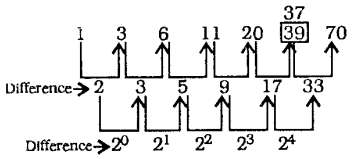
$18 \times 6 \neq 109$ but 108

$108 \times 18 = 1944$

$1944 \times 108 = 209952$

Obviously, 109 is the wrong number and it should be replaced with 108.

22. (2) The given series is based on the following pattern :



Obviously, 39 is the wrong number and it should be replaced with 37.

23. (1) The given series is based on the following pattern :

$2 \times 2 + 7 = 11$ (not 13)

$11 \times 3 - 6 = 27$

$27 \times 4 + 5 = 113$

$113 \times 5 - 4 = 561$

Obviously the number 13 is wrong and it should be replaced with 11.

24. (4) The given series is based on the following pattern.

$50 + (1^2) = 51$

$51 - (2^2) = 47$

$47 + (3^2) = 56$

$56 - (4^2) = 40$ (not 42)

$40 + (5^2) = 65$

Obviously, the number 42 is wrong and it should be replaced with 40.

25. (3) The given series is based on the following pattern :

$3 \times 2 + 3 = 9$

$9 \times 3 - 4 = 23$

$23 \times 4 + 5 = 97$ (not 99)

$97 \times 5 - 6 = 479$

Obviously, the number 99 is wrong and it should be replaced with 97.

26. (1) The given series is based on the following pattern:

$2 + 3 = 5$

$5 + 3 = 8$

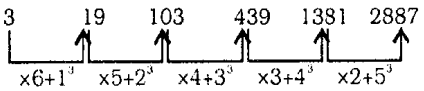
$8 + 5 = 13$

$13 + 8 = 21$

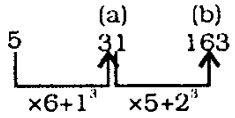
$21 + 13 = 34$

Obviously, the number 4 is wrong and it should be replaced with 3.

27. (2) The given series is based on the following pattern :



Similarly,



Hence, 163 will come in place of (b).

28. (1) The given series is based on the following pattern

$13 = 4 \times 1 + 1 \times 9$

$40 = 13 \times 2 + 2 \times 7$

$135 = 40 \times 3 + 3 \times 5$

$552 = 135 \times 4 + 4 \times 3$

$2765 = 552 \times 5 + 5 \times 1$

Similarly,

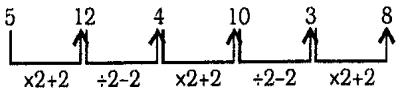
(a) $= 2 \times 1 + 1 \times 9 = 11$

(b) $= 11 \times 2 + 2 \times 7 = 36$

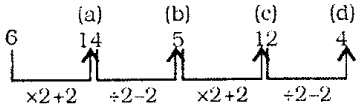
(c) $= 36 \times 3 + 3 \times 5 = 123$

Hence, 123 will come in place of (c).

29. (3) The given series is based on the following pattern:

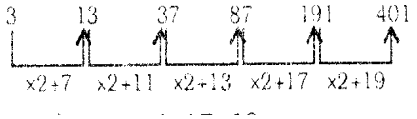


Similarly,



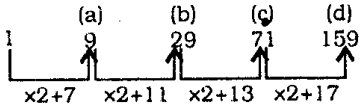
Hence, 4 will come in place of (d).

30. (4) The given series is based on the following pattern :



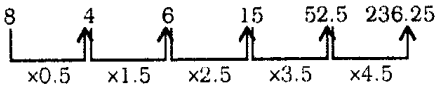
7, 11, 13, 17, 19, ... are consecutive prime numbers)

Similarly,

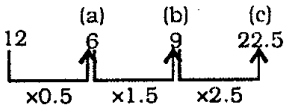


Hence, 159 will come in place of (d).

31. (3) The given series is based on the following pattern :



Similarly,



Hence, 22.5 will come in place of (c).

32. (3) The given series is based on the following pattern :

$9 \times 2 + 1.5 = 19.5$

$19.5 \times 2 + 2 = 41$

$41 \times 2 + 2.5 = 84.5$

Therefore, the new series is as follows :

$12 \times 2 + 1.5 = 25.5$ ....(a)

$25.5 \times 2 + 2 = 53$ ....(b)

$53 \times 2 + 2.5 =$ **108.5** ....(c)

$108.5 \times 2 + 3 = 220$ ....(d)

$220 \times 2 + 3.5 = 443.5$ ....(e)

Therefore, the number 108.5 will come in place of (C) in the new series.

33. (1) The series is based on following pattern:

$4 \times 1 + 1 = 5$

$\downarrow +3$

$5 \times 4 + 2 = 22$

$\downarrow +5$

$22 \times 9 + 3 = 201$

Similarly the new series is as follows :

$7 \times 1 + 1 = 8$ ....(a)

$8 \times 4 + 2 = 4$ ....(b)

$34 \times 9 + 3 = 309$ ....(c)

$309 \times 16 + 4 =$ **4948** ....(d)

Therefore, the number 4948 will come in place of (d) in the new series.

34. (2) The series is based on following pattern :

$5 \times 1 + 0.25 \times 1 = 5.25$

$\downarrow +3$

$5.25 \times 2 + 0.25 \times 4 = 11.5$

$\downarrow +5$

$11.5 \times 3 + 0.25 \times 9 = 36.75$

Similarly, the new series is as follows.

$3 \times 1 + 0.25 \times 1 = 3.25$ ....(a)

$3.25 \times 2 + 0.25 \times 4 = 7.5$ ....(b)

$7.5 \times 3 + 0.25 \times 9 =$ **24.75** ....(c)

Therefore, the number 24.75 will come in place of (c) in the new series.

35. (4) The series is based on following pattern :

$38 \times 0.5 = 19$

$19 \times 1.5 = 28.5$

$28.5 \times 2.5 = 71.25$

Similarly, the new series is as follows :

$18 \times 0.5 = 9$ ....(a)

$9 \times 1.5 = 13.5$ .....(b)

$13.5 \times 2.5 = 33.75$ ....(c)

$33.75 \times 3.5 =$ **118.125.....(d)**

Therefore, the number 118.125 will come

in place of (d) in the new series.

36. (3) The series is based on following pattern:
$25 + (11)^2 \Rightarrow 25 + 121 = 146$
$146 - (9)^2 \Rightarrow 146 - 81 = 65$
$65 + (7)^2 \Rightarrow 65 + 49 = 114$
Similarly, the new series is as follows :
$39 + (11)^2 \Rightarrow 39 + 121$
$= 190$.....(a)
$160 - (9)^2 \Rightarrow 160 - 81$
$= 79$ .....(b)
$79 + (7)^2 \Rightarrow 79 + 49$
$= 128$ .....(c)
$128 + (5)^2 \Rightarrow 128 - 25$
$= 103$ .....(d)
$103 + (3)^2 \Rightarrow 103 + 9$
$= 111$ ....(e)
Therefore, the number 112 will come in place of (e) in new series.

37. (1) The given series is based on following pattern
$15 - 10 = 5$
$24 - 15 = 9$
$37 - 24 = 13$
$54 - 37 = 17$
$75 - 54 = 21$
$100 - 75 = 25$
Obviously, 35 is wrong number.

38. (5) Here the middle number = difference of succeeding number and preceding number.
i.e., $4 - 1 = 3$
$7 - 3 = 4$
$11 - 4 = 7$
$18 - 7 = 11$
$27 - 11 = 16$
Here the sequence gets disturbed
$\therefore 29 - 11 = 18$
$47 - 18 = 29$
Hence, 27 is the wrong number.

39. (5) The sequence is based on following pattern:
$3 \times 0.5 + 0.5 = 2$
$2 \times 1 + 1 = 3$
$3 \times 1.5 + 1.5 = 6$
$6 \times 2 + 2 = 14$
$14 \times 2.5 + 2.5 = 37.5$
$37.5 \times 3 + 3 = 115.5$
Obviously, 12 is the wrong number.

40. (4) $32431 = 7 \times 4626 + 7^2$
$4626 = 6 \times 765 + 6^2$
$765 = 5 \times 148 + 5^2$
$148 = 4 \times 32 + 4^2$
But $148 = 4 \times 33 + 4^2$
$33 = 3 \times 8 + 3^2$
$8 = 2 \times 2 + 2^2$
Obviously 32 is the wrong number.

41. (2) The sequence is based on following pattern:
$3 - 2 = 1^3$
$11 - 3 = 8 = 2^3$
$38 - 11 = 27 = 3^3$
$102 - 38 = 64 = 4^3$
But,
$\mathbf{229 - 102 = 127 \neq 5^3}$
$\mathbf{227 - 102 = 125 = 5^3}$
$443 - 227 = 216 = 6^3$
Obviously 229 is the wrong number.

42. (5) The given number series is based on the following pattern :
$7413 + 9 \times 1 = 7422$
$7422 + 9 \times 2 = 7440$
$7440 + 9 \times 3 = 7467$
$7467 + 9 \times 4 = 7503$
Hence, 7467 will replace the question mark.

43. (4) The given number series is based on the following pattern :
$4 = 2^2$; $16 = 4^2$;
$36 = 6^2$; $64 = 8^2$;
$100 = 10^2$.
$\therefore ? = 12^2 =$ **144**
Hence, 144 will replace the question mark.

44. (1) The given number series is based on the following pattern:
$12 \times 3 - 3 = 33$
$33 \times 3 - 3 = 96$
$96 \times 3 - 3 =$ **285**
$285 \times 3 - 3 = 852$
Hence, 285 will replace the question mark.

45. (3) The given number series is based on the following pattern :
$70000 \div 5 = 14000$
$14000 \div 5 = 2800$
$2800 \div 5 =$ **560**
$560 \div 5 = 112$
$112 \div 5 = 22.4$
Hence, 560 will replace the question mark.

46. (2) The given number series is based on the following pattern :
$102 - 3 = 99$
$99 + 5 = 104$
$104 - 7 = 97$
$97 + 9 = 106$
$106 - 11 =$ **95**
Hence, 95 will replace the question mark.

47. (4) The given number series is based on the following pattern
93 + 2 (prime number) = 95
$95 + 3 = 98 \neq 99$
$98 + 5 =$ **103**

103 + 7 = 110
110 + 11 = 121
121 + 13 = 134
Hence, 103 will replace the question mark

48. (5) The given number series is based on the following pattern:
8 × 1.5 = 12
12 × 1.5 = 18
18 × 1.5 = 27 ≠ 26
27 × 1.5 = 40.5
40.5 × 1.5 = 60.75
∴ ? = 60.75 × 1.5 = **91.125**
Hence, 91.125 will replace the question mark.

49. (5) The given number series is based on the following pattern : 4 + 7 = 11
11 + 7 = 18
18 + 11 = 29 ≠ 28
∴ ? = 29 + 18 = **47**
Hence, 47 will replace the question mark.

50. (1) The given number series is based on the following pattern:
3 × 2 + 2^2 = 10
10 × 3 + 3^2 = **39**
39 × 4 + 4^2 = 172
172 × 5 + 5^2 = 885 ≠ 886
885 × 6 + 6^2 = 5346
Hence, 39 will replace the question mark.

51. (3) The given number series is based on the following pattern :
15 × 1 + 1 × 7 = 22
22 × 2 + 2 × 6 = 56 ≠ 57
56 × 3 + 3 × 5 = 183
183 × 4 + 4 × 4 = **748**
748 × 5 + 5 × 3 = 3755
3755 × 6 + 6 × 2 = 22542
Hence, 748 will replace the question mark.

52. (4) The pattern of the number series is :
3601 ÷ 1 + 1 = 3602
3602 ÷ 2 + 2- 1801 + 2 = 1803
1803 ÷ 3 + 3 - 601 + 3 - 604
604 ÷ 4 + 4 = 151 + 4 = 155 ≠ **154**
155 ÷ 5 + 5 = 31 + 5 = 36
36 ÷ 6 + 6 = 6 + 6 = 12

53. (2) The pattern of the number series is :
4 × 2 + 2^2 = 8 + 4 = 12
12 × 3 + 3^2 = 36 + 9 = 45 ≠ **42**
45 × 4 + 4^2 = 180 + 16 = 196
196 × 5 + 5^2 = 980 + 25 = 1005
1005 × 6 + 6^2 = 6030 + 36 = 6066

54. (1) The pattern of the number series is :
2 + 4 = 6 ≠ **8**
6 + 6 = 12
12 + 8 = 20
20 + 10 = 30
30 + 12 = 42

55. (5) The pattern of the number series is :

$32 \times \frac{1}{2} = 16$

$16 \times \frac{3}{2} = 24$

$24 \times \frac{5}{2} = 60 \neq$ **65**

$60 \times \frac{7}{2} = 210$

$210 \times \frac{9}{2} = 945$

$945 \times \frac{11}{2} = 5197.5$

56. (4) The pattern of the number series is :
7 × 2 - 1 = 14 - 1 = 13
13 × 2 - 1 = 26 - 1 = 25
25 × 2 - 1 = 50 - 1 = 49
49 × 2 - 1 = 98 - 1 = 97
97 × 2 - 1 = 194 - 1 = 193 ≠ **194**
193 × 2 - 1 = 386 - 1 = 385

57. (1) The pattern of the given series Is :
37 × 0.5 + 0.5 = 18.5 + 0.5 = 19
19 × 1 + 1 = 19 + 1 - 20
20 × 1.5 + 1.5 = 30 + 1.5 - 31.5
31.5 × 2 + 2 = 63 + 2 = 65
65 × 2.5 + 2.5 = 162.5 + 2.5 - 165
Similarly,
21 × 0.5 + 0.5 = 10.5 + 0.5 = 11(a)
11 × 1 + 1 = 11 + 1 = 12 (b)
12 × 1.5 + 1.5 = 18 + 1.5 = 19.5 (c)
19.5 × 2 + 2 = 39 + 2 = 41 (d)
41 × 2.5 + 2.5 = 102.5 + 2.5 = **105** (e)

58. (2) The pattern of the given series is :
5 × 1 + 1^2 = 5 + 1 = 6
6 × 2 + 2^2 = 12 + 4 = 16
16 × 3 + 3^2 = 48 + 9 = 57
57 × 4 + 4^2 = 228 + 16 = 244
244 × 5 + 5^2 = 1220 + 25 = 1245
Similarly,
9 × 1 + 1^2 = 9 + 1 = 10 (a)
11 × 2 + 2^2 = 22 + 4 = 26 (b)
26 × 3 + 3^2 = 78 + 9 = 87 (c)
87 × 4 + 4^2 = 348 + 16 = **364** (d)

59. (3) The pattern of the given series is :
7 × 1 - 2 = 7 - 2 = 5
5 × 3 - 4 = 15 - 4 = 11
11 × 5 - 6 = 55 - 6 = 49
49 × 7 - 8 = 343 - 8 = 335
335 × 9 - 10 = 3015 - 10 = 3005
Similarly,

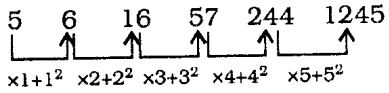
13 × 1 - 2 = 13 - 2 = 11 (a)
11× 3 - 4 = 33 - 4= **29** (b)

60. (4) The pattern of the given series is :
12 × 3 + 11 = 36 + 11 = 47
47 × 3 + 11 = 141 + 11 = 152
152 × 3 + 11 = 456 + 11 = 467
467 × 3 + 11 = 1401 + 11 = 1412
1412 × 3 + 11 = 4236 + 11 = 4247
Similarly,
33 × 3 + 11 = 99 + 11 = 110 (a)
110 × 3 + 11 = 330 + 11 = 341 (b)
341 × 3 + 11 = 1023 + 11 = 1034(c)
1034 × 3 + 11 = 3102 + 11
= **3113** (d)

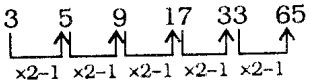
61. (5) The pattern of the given series is :
68 × 1 - 8 = 60
60 × 1.5 + 14 = 90 + 14 = 104
104 × 2 - 20 = 208 - 20 = 188
188 × 2.5 + 26 = 470 + 26 = 496
496 × 3 - 32 = 1488 - 32 = 1456
Similarly,
42 × 1 - 8 = 42 - 8 = 34 (a)
34 × 1.5 + 14 = 51 + 14 = 65 (b)
65 × 2 - 20 = 130 - 20 = 110 (c)
110 × 2.5 + 26 = 275 + 26 = **301** (d)

RBI GRADE-B OFFICER EXAMS

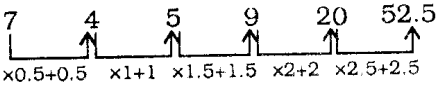
I. (4)The given series is based on the following pattern:



2. (5) The given series is based on the following pattern:

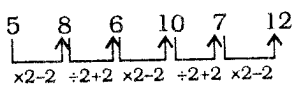


3. (3) The given series is based on the following pattern:

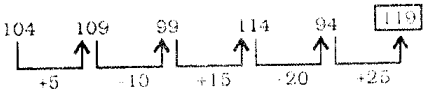


4. (2) 30 (According to question)

5. (1) The given series is based on the following pattern:

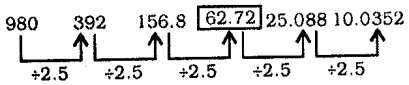


6. (5) The given series is based on the following pattern:



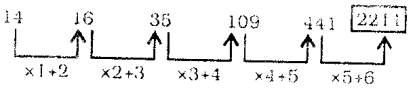
Hence, 119 will come in place of the question mark.

7. (3) The given series is based on the following pattern :



Hence, 62.72 will come ir place of the question mark.

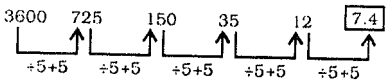
8. (4) The given series is based on the following pattern :



Hence, 2211 will come in place of the question mark.

9. (1) The given series is based on the following pattern:
Numbers are cubes of consecutive prime numbers. i.e.
11^3 = 1331
13^3 = 2197
17^3 = 4913
19^3 = 6859
23^3 = **I2167**
29^3 = 24389
Hence, 12167 will come in place of the question mark.

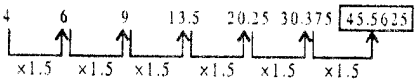
10. (2) The given series is based on the following pattern



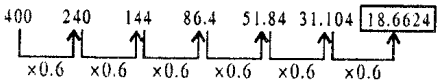
Hence, 7.4 will come in place of the question mark.

11. (3) The given number series is based on the following pattern :
13 × 1 + 1 = 14
14 × 2 + 2 = 30
30 × 3 + 3 = 93
93 × 4 + 4 = 376
376 × 5 + 5 = 1885
∴ ? = 1885 × 6 + 6 = **11316**
Hence, number 11316 will replace the question mark.

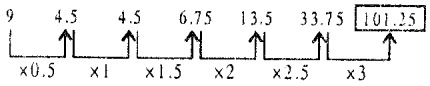
12. (2)



13. (4)



14. (1)



15. (5)
705 + 1 × 23 = 728
728 + 2 × 23 = 774
774 + 3 × 23 = 843
843 + 4 × 23 = 935
935 + 5 × 23 = 1050
$\therefore$? = 1050 + 6 × 23 = 1050 + 138 = 1188

16. (4) The pattern of the given series is :
5 × 1.5 + 1.5 = 7.5 + 1.5 = 9
9 × 2.5 + 2.5 = 22.5 + 2.5 = 25
25 × 3.5 + 3.5 = 87.5 + 3.5 = 91
91 × 4.5 + 4.5 = 409.5 + 4.5 = 414
Similarly,
(a) $\Rightarrow$ 3 × 1.5 + 1.5 = 4.5 + 1.5 = 6
(b) $\Rightarrow$ 6 × 2.5 + 2.5 = 15 + 2.5 = 17.5
(c) $\Rightarrow$ 17.5 × 3.5 + 3.5 = 61.25 + 3.5 = **64.75**

17. (2) The pattern of the given se ries is :
15 × 1 - 1 × 6 = 15 - 6 = 9
9 × 2 - 2 × 5 = 18 - 10 = 8
8 × 3 - 3 × 4 = 24 - 12 = 12
12 × 4 - 4 × 3 = 48 - 12 = 36
36 × 5 - 5 × 2 = 180 - 10 = 170
Similarly,
(a) $\Rightarrow$ 19 × 1 - 1 × 6 = 19 - 6 = 13
(b) $\Rightarrow$ 13 × 2 - 2 × 5 = 26 - 10 = **16**

18. (1) The pattern of the given series is :
7 × 1 - 1 = 6
6 × 2 - 2 = 10
10 × 3 - 3 = 27
27 × 4 - 4 = 104
104 × 5 - 5 = 515
Similarly,
(a) $\Rightarrow$ 9 × 1 - 1 = 8
(b) $\Rightarrow$ 8 × 2 - 2 =14
(c) $\Rightarrow$ 14 × 3 - 3 = 39
(d) $\Rightarrow$ 39 × 4 - 4 = **152**

19. (5) The pattern of the given series is :
6 × 2 + 2^2 = 12 + 4 = 16
16 × 3 + 3^2 = 48 + 9 = 57
57 × 4 + 4^2 = 228 + 16 = 244
Similarly,
(a) $\Rightarrow$ 4 × 2 + 2^2 = 8 + 4 = 12
(b) $\Rightarrow$ 12 × 3 + 3^2 = 36 + 9 = 45
(c) $\Rightarrow$ 45 × 4 + 4^2 = 180 + 16 = 196
(d) $\Rightarrow$ 196 × 5 + 5^2 = 980 + 25 = **1005**

20. (3) The pattern of the given series is :
8 × 1 + 1 = 9
9 × 2 + 2 = 20
20 × 3 + 3 = 63
63 × 4 + 4 = 256
Similarly,
(a) $\Rightarrow$ 5 × 1 + l = 6
(b) $\Rightarrow$ 6 × 2 + 2 = 14
(c) $\Rightarrow$ 14 × 3 + 3 = 45
(d) $\Rightarrow$ 45 × 4 + 4 = 184
(e) $\Rightarrow$ 184 × 5 + 5 = **925**

21. (3) The pattern of the number series is :
4 × 0.5 + 1 = 2 + 1 = 3
3 × 1 + 1.5 = 3 + 1.5 = 4.5
4.5 × 1.5 + 2 = 6.75 + 2
= 8.75 $\neq$ **8.5**
8.75 × 2 + 2.5 = 17.5 + 2.5 = 20
20 × 2.5 + 3 = 50 + 3 = 53

22. (2) The pattern of the number series is :
12000 $\div$ 5 - 5 = 2400 - 5 = 2395
2395 $\div$ 5 - 5 = 479 - 5
= 474 $\neq$ **472**
474 $\div$ 5 - 5 = 94.8 - 5 = 89.8
89.8 $\div$ 5 - 5= 17.96 - 5 = 12.96

23. (5) The pattern of the number series is :
l × 1 + 7 × l = l + 7 = 8
8 × 2 + 6 × 2 = 16 + 12 = 28
28 × 3 + 5 × 3 = 84 + 15 = 99
99 × 4 + 4 × 4 = 396 + 16 = 412
412 × 5 + 3 × 5 = 2060 + 15 = 2075
2075 × 6 + 2 × 6 = 12450 + 12
= 12462 $\neq$ **12460**

24. (1) The pattern of the number series is :
144 × 1.5 = 216 $\neq$ **215**
216 × 2.5 = 540
540 × 3.5 = 1890
1890 × 4.5 = 8505
8505 × 5.5 = 46777.5

25. (5) The pattern of the number series is :
2222 - 7^3 = 2222 - 343 = 1879
1879 - 6^3 = 1879 - 216 = 1663
1663 - 5^3 = 1663 - 125 = 1538
1538 - 4^3 = 1538 - 64 = 1474
1474 - 3^3 = 1474 - 27 = 1447
1447 - 2^3 = 1447 - 8
= 1439 $\neq$ 440

26. (4) The pattern is :
2^3 + 1^2 = 9
3^3 + 2^2 = 31
4^3 + 3^2 = 73
5^3 + 4^2 = 141
6^3 + 5^2 = **241**

27. (4) The pattern is :

35 + 221 = 256
256 + (221 - 26) = 451
451 + 169 (=195 - 26) = 620
620 + 143 (=169 - 26) = 763
763 + 117 = **880**

28. (3) The pattern is :
130 + 3^2 = 139
139 + 4^2 = 155
155 + 5^2 = 180
180 + 6^2 = 216
216 + 7^2 = **265**

29. (2) The pattern is :
658 + 72 = 730
730 + 144 = 874
874 + 288 = 1162
1162 + 576= **1738**

30. (2) The pattern is :
14 + 990 = 1004

$1004 + \frac{990}{5} = 1202$

$1202 + \frac{198}{4} = 1251.5$

$1251.5 + 16.5 \left(= \frac{49.5}{3}\right) = 1268$

1268 + 8.25 = **1276.25**

31. (3) The pattern is :
576 - 224 = 352
752 - 576 = 176
840 - 752 = 88
884 - 840 = 44
∴ ? = 884 + 22 = **906**

32. (4) The pattern is :
55 + 11.15 = 66.15
66.15 + 2 × 11.15 = 88.45
88.45 + 3 ×11.15 = 121.9
121.9 + 4 × 11.15 = 166.5
166.5 + 5 × 11.15
= 166.5 + 55.75 = **222.25**

33. (5) The pattern is
36 + 13 = 49
49 + 2 × 13 = 75
75 + 13 = 88
88 + 2 × 13 = 114
114 + 13 = **127**

INSURANCE EXAMS

1. (2) The series is based on following pattern :
3 + 4 × (2)° = 7
7 + 11 = 18
18 + 4 × $(2)^1$ = 26
26 + 11 = **37**
37 + 4 × $(2)^2$ = 53
53 + 11 = 64
64 + 4 × $(2)^3$ = 96
Therefore, the number 37 will come in place of question mark (?) in the series.

2. (3) The series is based on following pattern :
1.7 + 1.5 = 3.2
3.2 - 0.5 = 2.7
2.7 + 1.5 = 4.2
4.2 - 0.5 = 3.7
3.7 + 1.5 = **5.2**
5.2 - 0.5 = 4.7
4.7 + 1.5 = 6.2
Therefore, the number 5.2 will come in place of question mark (?) in the series.

3. (3) The original series is based on following pattern:

$8 \times \frac{1}{2} = 4$

4 × 1 = 4
4 × 1.5 = 6
6 × 2 = 12
12 × 2.5 = 30
30 × 3 = 90
Therefore, the number 28 is wrong. Hence, the new series is as follows:

$28 \times \frac{1}{2} = 14$ ....2nd term
14 × 1 = 14 ....3rd term
14 × 1.5 = **21 - 4th term**
21 × 2 = 42
Therefore, the fourth term of new series is 21.

4. (2) The original series is based on following pattern:
17 + 0.25 × $(1)^2$ = 17.25
17.25 + 0.25 × $(2)^2$ = 18.25
18.25 + 0.25 × $(3)^2$
= **20.50**
20.50 + 0.25 × $(4)^2$ = 24.50
24.50 + 0.25 × $(5)^2$ = 30.75
Therefore, the number 20.75 is wrong. Hence, the new series is as follows:
20.75 + 0.25 × l^2 = 21.00 .... 2nd term
21.00 + 0.25 × $(2)^2$ = 22.00 .... 3rd term
22.00 + 0.25 × $(3)^2$ = 24.25 ....4th term
Therefore, the fourth term of the new series is 24.25.

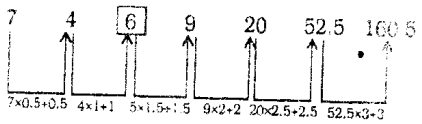
5. (1) The original series is based on following pattern:
438 + $(7)^2$ = 487
487 - $(6)^2$ = **451**
451 + $(5)^2$ = 476
476 + $(4)^2$ = 460
460 + $(3)^2$ = 469

Therefore, the number 447 is wrong. Hence the new series is as follows:
$447 + (7)^2 = 496$ .....2nd term
$496 - (6)^2 = 460$ .....3rd term
$460 + (5)^2 =$ **485 - 4th term**
$485 - (4)^2 = 469$
Therefore, the fourth term of the new series is 485.

6. (5) The original series is based on following pattern:
$2 \times 2 + 3 = 7$
$7 \times 2 + 5 =$ **19**
$19 \times 2 + 7 = 45$
$45 \times 2 + 9 = 99$
$99 \times 2 + 11 = 209$
$209 \times 2 + 13 = 431$
Therefore, the number 18 is wrong.
Hence, the new series is as follows:
$18 \times 2 + 3 = 39$ — 2nd term
$39 \times 2 + 5 = 83$ — 3rd term
$83 \times 2 + 7$ = **173 - 4th term**
$173 \times 2 + 9 = 355$
Therefore, the fourth term of the new series is 173,

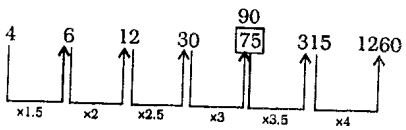
7. (4) The original series is based on following pattern:
$6 \times 1 + 1 \times 2 = 8$
$8 \times 2 - 2 \times 3 = 10$
$10 \times 3 + 3 \times 4 = 42$
$42 \times 4 - 4 \times 5 = 148$
$148 \times 5 + 5 \times 6 = 770$
$770 \times 6 - 6 \times 7 = 4578$
Therefore, the number 146 is wrong.
Hence, the new series is as follows:
$146 \times 1 + 1 \times 2 = 148$
= 2nd term
$148 \times 2 - 2 \times 3$
= 290 -- 3rd term
$290 \times 3 + 3 \times 4$
= **882 - 4th term**
Therefore, the fourth term of the new series is 882.

8. (1) The given number series is based on the following pattern



Hence the wrong number is 6.

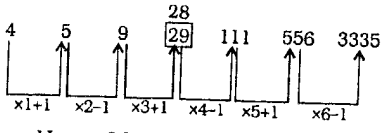
9. (2) The given number series is based on the following pattern :



Hence, the wrong number is 75.

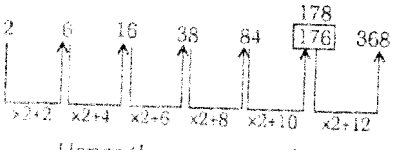
10. (4) The given number series is based on the following pattern:
$4 - 3 = 1^2$
$13 - 4 = 9 = 3^2$
$38 - 13 = 25 = 5^2$
$87 - 38 = 49 = 7^2$
$168 - 87 = 81 = 9^2$
$289 - 168 = 121 = 11^2$
Obviously, 166 is the wrong number.

11. (3) The number series follows the rule as mentioned below:



Hence 29 is the wrong number.

12. (5) The followed pattern is :



Hence the wrong number is 176.

13. (5) The pattern of the number series is :
$3 + 7^2 = 3 + 49 = 52$
$52 + 6^2 = 52 + 36 = 88$
$88 + 5^2 = 88 + 25 = 113$
$113 + 4^2 = 113 + 16 = 129$
$129 + 3^2 = 129 + 9 =$ **138**

14. (3) The pattern of the number series is :
$2 \times 1 + 1 = 52$
$3 \times 2 + 2 = 8$
$8 \times 3 + 3 =$ **27**
$27 \times 4 + 4 = 112$
$112 \times 5 + 5 = 565$

15. (1) The pattern of the number series is :
$6 \times 0.5 + 1 = 4$
$4 \times 1.5 + 2 = 8$
$8 \times 2.5 + 3 = 23$
$23 \times 3.5 + 4 =$ **84.5**
$84.5 \times 4.5 + 5 = 385.25$

16. (4) The pattern of the number series is :
$2^3 = 8$; $4^3 = 64$
$6^3 = 216$; $8^3 = 512$
$10^3 =$ **1000** ; $12^3 = 1728$

17. (2) The pattern of the number series is :
$5 \times 1 + 1 \times 6 = 11$
$11 \times 2 + 2 \times 5 = 32$
$32 \times 3 + 3 \times 4 = 108$
$108 \times 4 + 4 \times 3 = 444$
$444 \times 5 + 5 \times 2 =$ **2230**

18. (3) $S = (1^2 - 2^2) + (3^2 - 4^2) + (5^2 - 6^2) +$ .....to 100 terms

= -3 - 7 - 11 - 15 - .... to 100 terms
= - (3 + 7 + 11 + 15 + ... to 100 terms)

$= \frac{100}{2}[2 \times 3 + (100 - 1)4]$

$\left[S_n = \frac{n}{2}\left[2a + (n-1)d\right]\right]$

= - 50 × 402 = - 20100

19. (3) Tricky approach

$\frac{3}{4} + \frac{5}{36} + \frac{7}{44} ..... + \frac{17}{5184} + \frac{19}{8100}$

$= \left(1 - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{9}\right) + \left(\frac{1}{9} + \frac{1}{16}\right) ..... + \left(\frac{1}{81} - \frac{1}{100}\right)$

$= 1 - \frac{1}{100} = \frac{99}{100} = 0.99$

20. (4) The pattern is :
8 + 6 = 14
14 + 18 (= 6 + 12) = 32
32 + 38 (= 18 + 20) = 70
70 + 66 (= 38 + 28) = 136
136 + 102 (= 66 + 36)
= **238**

21. (1) The pattern is :
25 + 1 × 16 = 41
41 + 3 × 16 = 41 + 48 = 89
89 + 5 × 16 = 89 + 80 = 169
169 + 7 × 16 = 169 + 112 = 281
281 + 9 × 16 = 281 + 144 = **425**

22. (2) The pattern is :
461 + 13 = 474
474 - 9 = 465
465 + 13 = 478
478 - 9 = 469
469 + 13 = **482**

23. (5) The pattern is :
(980 ÷ 2) + 26 = 516
(516 ÷ 2) + 26 = 284
(284 ÷ 2) + 26 = 168
(168 ÷ 2)+ 26 = 110
(110 ÷ 2) + 26 = **81**

24. (5) The pattern is :
4 + 0 = 4
4 + 6= 10
10 + 24 (= 6 + 18) = 34
34 + 60 (= 6 + 54) = 94
94 + 168 (= 6 + 162) = **262**

25. (1) Expression =
(l + 7 + 13 + 19 + ... + 61) + (3 - 5 + 9 - 11 + ... + 63 - 65)
= (1 + 7 +13 + ... + 61) - 2 ×11
First Part = 1 + 7 + 13 +....+ 61
$t_n = a + (n-1)d$

$\Rightarrow 61 = l + (n-1)d$
$\Rightarrow 61 - 1 = (n-1)6$
$\Rightarrow (n-1)6 = 60$
$\Rightarrow n - 1 = 10$
$\Rightarrow n = 11$

$S_n = \frac{n}{2}[a + l] = \frac{11}{2}(1 + 61) = 341$

$\therefore$ Expression = 341 - 22 = 319

26. (1) $x = \frac{1}{1\times 2} + \frac{1}{2\times 3} + \frac{1}{3\times 4} + ... + \frac{1}{7\times 8} + \frac{1}{7\times 9}$

$= 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + ... + \frac{1}{6} - \frac{1}{7} + \frac{1}{7} - \frac{1}{8} + \frac{1}{7\times 9}$

$= 1 - \frac{1}{8} + \frac{1}{63}$

$= \frac{504 - 63 + 8}{8\times 63} = \frac{449}{504}$

$\therefore \frac{1}{x} = \frac{504}{449} \approx 1.1$

27. (4)

$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) .... \left(1 - \frac{1}{2011^2}\right) = \frac{x}{2\times 2011}$

$\Rightarrow \left(1 - \frac{1}{2}\right)\left(1 + \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 + \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 + \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\left(1 + \frac{1}{5}\right) .... \left(1 - \frac{1}{2011}\right) \times \left(1 + \frac{1}{2011}\right) = \frac{x}{2\times 2011}$

$\Rightarrow \frac{1}{2} \times \frac{3}{2} \times \frac{2}{3} \times \frac{4}{3} \times \frac{3}{4} \times \frac{5}{4} \times \frac{4}{5} \times \frac{6}{5} .... \frac{2010}{2011} \times \frac{2012}{2011} = \frac{x}{2\times 2011}$

$\Rightarrow \frac{1}{2} \times \frac{2012}{2011} = \frac{x}{2\times 2011}$

$\Rightarrow x = 2012$

28. (2) The pattern is :

$\frac{1050 - 30}{2} = 510$

$\frac{510-26}{2}$ = 242

$\frac{242-22}{2}$ = 100 $\neq$ **106**

$\frac{110-18}{2}$ = 46

$\frac{46-14}{2}$ = 16.

29. (1) The pattern is
550 - 2^2 = 550 - 4 = 546
546 - 3^2 = 546 - 9 = 537
537 - 42 = 537 - 16 = 521
521 - 5^2 = 521 - 25
= 496 $\neq$ **494**
496 - 6^2 = 496 - 36 = 460

30. (3) The pattern is ;
8 + 1 × 13 = 21
21 + 2 ×13 = 21 +26 = 47
47 + 3 × 13 = 47 + 39 = 86
86 + 4 × 13 = 86 + 52
= 138 $\neq$ **140**
138 + 5 × 13 = 138 + 65 = 203
203 + 6 × 13 = 203 + 78 = 281

31 (2) The pattern is ;
4 × 8 - 8 = 32 - 8 = 24
24 × 7 - 7 = 168 - 7 = 161
161 × 6 - 6 = 966 - 6
= 960 $\neq$ **965**
960 × 5 - 5 = 4800 - 5 = 4795

32. (3) The pattern is :
1 × 2 = 2
2 × 3 = 6 $\neq$ **8**
6 × 4 = 24
24 × 5 = 120
120 × 6 = 720

33. (2) The given number series is based on the following pattern :
1548 ÷ 3 = 516
516 ÷ 4 = 129
129 ÷ 3 = 43
43 ÷ 4 = **10.75**
Hence, 10.75 will replace the question mark.

34. (4) The given number series is ' based on the following pattern :
949 × 0.2 = 189.8
189.8 × 0.3 = **56.94**
56.94 × 0.4 = 22.776
22.776 × 0.5 = 11.388
11.388 × 0.6 = 6.8328
Hence, 56.94 will replace the question mark.

35. (1) The given number series is based on the following pattern :
121 + 23 × 1 = 144
144 + 23 × 2 = 190
190 + 23 × 3 = 259
$\therefore$? = 259 + 23 × 4 = 259 + 92 = **351**
Hence, 351 will replace the question mark.

36. (5) The given number series is based on the following pattern :
14 × 3 + 1.5 = 43.5
43.5 × 6+ 1.5 × 2 = 264
264 × 12 + 1.5 × 4 = **3174**
3174 × 24 + 1.5 × 8 = 76188
Hence, 3174 will replace the question mark.

37. (3) The given number series is based on the following pattern :
41 × 2^2 = 164
164 × 4^2 = 2624
2624 × 6^2 = **94464**
94464 × 8^2 = 6045696
Hence 94464 will replace the question mark.

38. (1) The pattern is :
2 + 3 = 5
5 + 4 = 9
6 + 5 = **14**
14 + 6 = 20
20 + 7 = 27

MODEL EXERCISES

1. The interior angles of a polygon are in AP, the smallest angle is 120° and the common difference is 5. Then, the number of sides of the polygon are —
(1) 16 (2) 9
(3) 8 (4) 12
(5) None of these

2. A man arranges to pay off a debt of Rs 3600 in 40 annual instalments which form an AP. When 30 of the instalments are paid, he dies leaving one-third of the debt unpaid. Find the value of the first instalment.
(1) 55 (2) 53
(3) 51 (4) 49
(5) None of these

3. Find $1^3 + 2^3 + 3^3 + ... + 15^3$
(1) 11025 (2) 13400
(3) 900 (4) 14400
(5) None of these

4. The value of
$(1^3 + 2^3 + 3^3 + ... + 15^3)$ -
$(1 + 2 + 3 + ... + 15)$ is —
(1) 14280 (2) 14400
(3) 12280 (4) 13280
(5) None of these

5. What is the next number in the series given below ?
53, 48, 50, 50, 47
(1) 51 (2) 46
(3) 53 (4) 52
(5) None of these

6. In a GP, the first term is 5 and the common ratio is 2. The eighth term is —
(1) 640 (2) 1280
(3) 256 (4) 160
(5) None of these

7. If the arithmetic mean of two numbers is 5 and geometric mean is 4, then the numbers are —
(1) 4, 6 (2) 4, 7
(3) 3, 8 (4) 2, 8
(5) None of these

8. What is the next number in the series given below ?
2, 5, 9, 14, 20
(1) 25 (2) 26
(3) 27 (4) 28
(5) None of these

9. The sum of 40 terms of an AP whose first term is 4 and common difference is 4, will be —
(1) 3200 (2) 1600
(3) 200 (4) 2800
(5) None of these

10. Let S_n denote the sum of the first 'n' terms of an AP

$S_{2n} = 3S_n$. Then, the ratio $\frac{S_{3n}}{S_n}$ is equal to
(1) 4 (2) 6
(3) 8 (4) 10
(5) None of these

11. The missing number in the series
8, 24, 12, 36, 18, 54 is —
(1) 27 (2) 108
(3) 68 (4) 72
(5) None of these

12. The sum of the 6th and 15th elements of an arithmetic progression is equal to the sum of 7th, 10th and 12th elements of the same progression. Which element of the series should necessarily be equal to zero ?
(1) 10th (2) 8th
(3) 1st (4) 9th
(5) None of these

13. If p, q, r, s are in harmonic progression and $p > s$, then —
(1) $\frac{1}{ps} < \frac{1}{qr}$ (2) $q + r = p + s$
(3) $\frac{1}{q} + \frac{1}{p} = \frac{1}{r} + \frac{1}{s}$ (4) None of these

(MAT Exam. Sept. 2003)

14. What is the eighth term of the sequence
1, 4, 9, 16, 25 ..... ?
(1) 8 (2) 64
(3) 128 (4) 200
(5) None of these

15. In a geometric progression, the sum of the first and the last term is 66 and the product of the second and the last but one term is 128. Determine the first term of the series.
(1) 64 (2) 64 or 2
(3) 2 or 32 (4) 32
(5) None of these

16. A sequence is generated by the rule that the xth term is $x^2 + 1$ for each positive integer x. In this sequence, for any value $x > 1$, the value of $(x + 1)$th term less the value of xth term is —
(1) $2x^2 > + 1$ (2) $x^2 + 1$
(3) $2x + 1$ (4) $x + 2$
(5) None of these

17. Four different integers form an increasing AP. If one of these numbers is equal to the sum of the squares of the other three numbers, then the numbers are —

(1) -2, -1, 0, 1 (2) 0, 1, 2, 3
(3) -1, 0, 1, 2 (4) 1, 2, 3, 4
(5) None of these

18. How many terms are there in an AP whose first and fifth terms are -14 and 2 respectively and the sum of terms is 40 ?
(1) 15 (2) 10
(3) 5 (4) 20
(5) None of these

19. The first three numbers in a series are -3, 0, 3, the 10th number in the series will be —
(1) 18 (2) 21
(3) 24 (4) 27
(5) None of these

SHORT ANSWERS

1.	(2)	**2.**	(3)
3.	(4)	**4.**	(1)
5.	(4)	**6.**	(1)
7.	(4)	**8.**	(3)
9.	(1)	**10.**	(2)
11.	(1)	**12.**	(2)
13.	(4)	**14.**	(1)
15.	(4)	**16.**	(2)
17.	(2)	**18.**	(3)
19.	(3)		
1.	(4)	**2.**	(1)
3.	(4)	**4.**	(3)
5.	(2)	**6.**	(3)
7.	(5)	**8.**	(2)
9.	(1)	**10.**	(4)
11.	(3)	**12.**	(5)
13.	(2)	**14.**	(1)
15.	(4)	**16.**	(1)
17.	(5)	**18.**	(1)
19.	(1)		

EXPLANATIONS

1. (2) Let the polygon has n sides.
Given, the smallest interior angle is 120°, hence the greatest exterior angle will be (180° -120°) = 60°
We know sum of exterior angles of a polygon = 360°
60 + 55 + 50 + ... = 360
{Common difference = -5}

$$\therefore \frac{n}{2}[2a + (n - 1)\,d] = 360$$

$$\frac{n}{2}[120 + (n - 1) \times -5] = 360$$

$\Rightarrow n^2 - 25n + 144 = 0$
$\Rightarrow n = 9, 16$
Number of sides cannot be 16.
Hence, $n = 9$

2. (3) According to question,
Sum of 40 instalments S_{40}
$= 3600 = 20\,(2a + 39d)$
$\Rightarrow 2a + 39d = 180$ ...(i)
Sum of 30 instalments
$S_{30} = 2400 = 15\,(2a + 29d)$
$\Rightarrow 2a + 29d = 160$ ...(ii)
Solving Eqs. (i) and (ii), we get
$a = 51$ and $d = 2$
$\therefore$ The value of first instalment
= Rs 51

3. (4) According to question, we have,

$$1^3 + 2^3 + 3^3 + ... + n^3 = \left[\frac{n \times (n+1)}{2}\right]^2$$

Here, n = number of terms = 15

$$\therefore \left[\frac{n(n+1)}{2}\right]^2 = \left[\frac{15 \times 16}{2}\right]^2$$

$= (120)^2 = 14400$

4. (1) According to question,
$(1^3 + 2^3 + 3^3 + ... + 15^3)$ -
$(1 + 2 + 3 + ... + 15)$

$$= \left[\frac{n(n+1)}{2}\right]^2 - \left[\frac{n(n+1)}{2}\right]$$

$$= \left[\frac{15 \times 16}{2}\right]^2 - \left[\frac{15 \times 16}{2}\right]$$

$= (120)^2 - (120)$
$= 120 \times 119 = 14280$

5. (4) According to question,
53, 48, 50, 50, 47....
The above series can be splitted into two series one in ascending order and other in descending order 53, 50, 47 and other is 48, 50, 52.
Hence, 52 will be the next number.

6. (1) According to question,
nth term of a GP = a^{n-1}
$\therefore$ 8th term = $5 \times (2)^{8-1} = 5 \times (2)^7$
$= 5 \times 128 = 640$

7. (4) Let the two numbers be x and y.
Then, AM,

$$\frac{x + y}{2} = 5$$

$\Rightarrow x + y = 10$

and GM, $\sqrt{xy} = 4$ ...(i)

$\Rightarrow xy = 16$
$\Rightarrow (x - y)^2 = (x + y)^2 - 4xy$
$100 - 64 = 36$
$x - y = 6$ ...(ii)
Or
Solving Eqs. (i) and (ii),
$x = 8$ and $y = 2$

8. (3) According to question,
2 + 3 =5; 5 + 4 = 9;
9 + 5 = 14; 14 + 6 = 20;
20 + 7 = 27
Hence, the next number of the series will be 27.

9. (1) According to question,

$$S_{40} = \frac{n}{2}[2a + (n - 1)d]$$

$= 20\,[4 + 39 \times 4]$
$= 20 \times 160 = 3200$

10. (2) Let a be the first term and d be the common difference.

Then, $S_n = \frac{n}{2}(2a + (n - 1)d]$

$S_2 = \frac{2n}{2}[2a + (2n - 1)d]$

and $S_{3n} = \frac{3n}{2}[2a + (3n - 1)d]\ ]$

Given, $S_{2n} = 3S_n$

$\therefore \frac{2n}{2}[2a + (2n - 1)d] =$

$2\frac{n}{2}[2a + (n - 1)d]$

$\Rightarrow 4a + (4n - 2)d = 6a + (3n - 3)d$

$\Rightarrow d(4n - 2 - 3n + 3) = 2a$

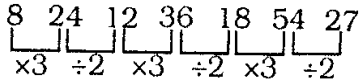
$\Rightarrow d = \frac{2a}{n+1}$

$\therefore S_n = \frac{2an^2}{n+1}$

and $S_3n = \frac{12an^2}{n+1}$

$\therefore \frac{S_n}{S_{3n}} = \frac{2an^2}{n+1} \times \frac{n+1}{12an^2} = \frac{1}{6} = \frac{S_{3n}}{S_n} = 6$

11. (1) According to question,
8, 24, 12, 36, 18, 54



Hence, 27 will come in the blank space.

12. (2) Let the first term and common term of the AP be a and d respectively.
Then, $(a + 5d) + (a + 14d) =$
$(a + 6d) + (a + 9d) + (a + 11d)$
$\Rightarrow 2a + 19d = 3a + 26d$
$\Rightarrow a + 7d = 0$
$\therefore$ 8th term is 0.

13. (4) According to question,
If p, q, r, s are in HP.

$\Rightarrow \frac{1}{p}, \frac{1}{q}, \frac{1}{r}, \frac{1}{s}$ are in AP

$\Rightarrow \frac{1}{q} - \frac{1}{p} = \frac{1}{s} - \frac{1}{r}$

$\Rightarrow \frac{1}{q} + \frac{1}{r} = \frac{1}{s} + \frac{1}{p}$

Hence the none of these be answer

14. (2) According to question,
1, 4, 9, 16, 25
$(1)^2\ (2)^2\ (3)^2\ (4)^2\ (5)^2$
Each term of the progression is the square of a natural number.
Hence, the eighth term of the sequence will be $(8)^2 = 64$

15. (2) Let the last term be n,
then $a + ar^{n-1} = 66$
and $ar.\ ar^{n-2} = 128$
$a^2r^{n-1'} = 128$
From Eqs. (i) and (ii),
$a(66 - a) = 128$
$\Rightarrow a^2 - 66a + 128 = 0$
$\Rightarrow a = 64, 2$

16. (3) According to question,
$(x + 1)^{th}$ term $-x^{th}$ term
$= (x + 1)^2 + 1 - (x^2 + 1)$
$= x^2 + 2x + 1 + 1 - x^2 - 1$
$= 2x + 1$

17. (3) By hit and trial or common sense, we have,
$2 = (-1)^2 + (0)^2 + (1)^2$
Hence the numbers are -1, 0, 1, 2

18. (2) According to question,
$T_5 = a + (n - 1).d$
$2 = -14 + 4d$

$d = \frac{16}{4} = 4$

$\therefore S_n = \frac{n}{2}[2a + (n - 1) \times d]$

$40 = \frac{n}{2}[-28 + (n - 1) \times 4]$

$\Rightarrow 80 = -28n + 4n^2 - 4n$
$\Rightarrow 4n^2 - 32n - 80 = 0$
$n^2 - 8n - 20 = 0$
$\Rightarrow (n - 10)(n + 2) = 0$
$\therefore n = 10\ (\because n \neq -2)$

19. (3) According to question,
$a = -3.\ d = 3$
$\therefore T_{10} = a + (10 - 1).\ d$
$T_{10} = -3 + 9 \times 3 = 24$

PARTNERSHIP

NATIONALISED BANKS & IBPS PO/MT/SO

1. A, B and C started a business with investment in the ratio 5 : 6 : 8 respectively. After one year C withdrew 50% of his capital and A increased his capital by 60% of his investment. After two years in what ratio should the earned profit be distributed among A, B and C respectively ?
(1) 2 : 3 : 3
(2) 4 : 3 : 2
(3) 13 : 12 : 12
(4) Cannot be determined
(5) None of these

(Syndicate Bank PO Bxam. 10.10.2004}

2. Tanvi started a business investing Rs. 45,000. After 8 months Anisha joined her with a capital of Rs. 52,000. At the end of the year the total profit was Rs. 56,165. What is the share of profits of Anisha ?
(1) Rs. 21,450 (2) Rs. 24,440
(3) Rs. 27,635 (4) Rs. 31,765
(5) None of these

(Orientai Bank of Commerce PO Exam. 21.12.2008)

3. Karan starts a business by investing Rs. 60,000. six months later Shirish joins him by investing Rs. 1,00,000. At the end of one year from commencement of the business, they eam a profit of Rs. 1,51,800. What is the Shirish's share in the profit ?
(1) Rs. 55200 (2) Rs. 82800
(3) Rs. 62500 (4) Rs. 96600
(5) None of these

(UCO Bank PO Exam. 22.03.2009)

4. Prakash, Sunil and Anil started a business jointly investing Rs. 11 lakhs, Rs. 16.5 lakhs and Rs. 8.25 lakhs respectively. The profit earned by them in the business at the end of three years was Rs. 19.5 lakhs. What will be the 50% of Anil's share in the profit ?
(1) Rs. 4.5 lakhs (2) Rs.2.25 lakhs
(3) Rs. 2.5 lakhs (4) Rs.3.75 lakhs
(5) None of these

(United Bank of India PO Exam. 21.06.2009)

5. Sonu invested 10% more than Mona. Mona invested 10% less than Raghu. If the total sum of their investment is Rs. 5,780. How much amount did Raghu invest ?
(1) Rs. 2010 (2) Rs. 2000
(3) Rs. 2100 (4) Rs. 2210
(5) None of these

(Indiain Bank PO Exam. 17.10.2010)

6. Gina invests Rs 48,000 to start a business. Four months later Shrayon joins her by investing Rs 62,000 and another two months later Deepika joins them both by investing Rs 80,000. At the end of one year the business eams a profit of Rs 20,661. What is Deepika's share in the profit ?
(1) Rs 7,668 (2) Rs 6,603
(3) Rs 7,240 (4) Rs 6,390
(5) None of these

(United Bank Of India PO Exam. 14.11.2010)

7. Mrudul invested an amount of Rs 29,500 in order to start a business. Shalaka joined her 4 months later by investing an amount of Rs 33,500. If the business earned a profit of Rs 1,20,575 at the end of two years, what was Mrudul's share of the profit ?
(1) Rs 60,725 (2) Rs 61,950
(3) Rs 59,250 (4) Rs 58,625
(5) None of these

(Indian Bank PO Exam. 02.01.2011)

8. Rs 73,689, are divided between A and B in the ratio 4 : 7. What is the difference between thrice the share of A and twice the share of B ?
(1) Rs 36,699 (2) Rs 46,893
(3) Rs 20,097 (4) Rs 26,796
(5) Rs 13,398

(IBPS Bank PO/MT CWE 17.06.2012)

9. Three men A, B and C start a business together. They invest Rs. 30000, Rs. 24000 and Rs. 42000 respectively in the beginning. After 4 months. B took out Rs. 6000 and C took out Rs. 10000. They get a profit of Rs. 11960 at the end of the year. B's share in the profit is
(1) Rs. 2700 (2) Rs. 2803
(3) Rs. 2900 (4) Rs. 2785
(5) None of these

(IBPS Bank PO/MT CWE-III 26.10.2013)

10. Geeta, Shweta and Sita started business with investment of Rs 4200, Rs. 8400 and Rs. 5400 respectively. In the end of the year, there was a profit of Rs 24000. Sita invested 32% of her share in profit in the business. The remaining amount of profit with Sita is
(1) Rs. 4986 (2) Rs. 4689
(3) Rs. 4886 (4) Rs. 4896
(5) None of these

(Corporation Bank Speciallist Officer (Marketing) I Exam. 22.12.2014)

SBI PO EXAMS

1. A man has divided his total money in his will in such a way that half of it goes to his wife, $\frac{2}{3}$rd of the remaining among his three sons equally and the rest among his four daughter equally. If each daughter gets Rs. 20,000. how much money will each son get ?
(1) Rs. 50,333.33
(2) Rs. 48,233.33
(3) Rs. 53,333.33
(4) Data inadequate
(5) None of these
(SBI Banks PO Exam. 20.08.2000)

2. Sunetra invested an amount of Rs. 50,000 to start a software business. After six months Nikhil joined her with an amount of Rs 80,000. At the end of 3 years they earned a profit of Rs. 24,500. What is Sunetra's share in in the profit ?
(1) Rs, 14,000 (2) Rs. 9423
(3) Rs. 10,250 (4) Rs. 12,500
(5) None of these
(SBI Associate Banks PO Exam. 21.07.2002)

3. Mr. 'X' invested certain amounts in two different schemes 'A' & 'B' Scheme 'A' offers simple interest at 12 p.c.p.a. and Scheme 'B' of offers compound interest at 10 p.c.p.a. Interest accrued on the amount invested in Scheme A in 2 years was Rs. 3,600 and the total amount invested was Rs 35,000. What was interest accrued on the amount invested in profits of Anisha ?
(1) Rs. 4,800
(2) Rs. 4,200
(3) Rs. 4,000
(4) Cannot be determined
(5) None of these
(SBI PO Ezam. 09.01.2005)

4. Ninad, Vikas and Manav enter into a partnership. Ninad invests some amount at the beginning. Vikas invests doublé the amount after 6 months and Manav invests thrice the amount invested by Ninad after 8 months. They eam a proflt of Rs. 45,000 at the end of the year. What is Manav's share in the profit ?
(1) Rs. 25,000 (2) Rs. 15,000
(3) Rs. 12,000 (4) Rs. 9,000
(5) None of these
(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

RBI GRADE-B OFFICER EXAMS

1. Raksha invested an amount of Rs. 60,000 to start a software business. After six months, Kamal joined her with an amount of Rs. 90,000. After one year from the commencement of the business, Raksha put in an additional amount of Rs. 20,000. At the end of three years, they earned a proflt of Rs. 71,20,000. What is Raksha's share in the proflt ?
(1) Rs. 35,20,000
(2) Rs. 26,40,000
(3) Rs. 27,00,000
(4) Rs. 38,40,000
(5) None of these
(RBI Grade-B Offìcer Exam. 2008)

ISURANCE EXAMS

1. A and B started a partnership business investing some amount in the ratio of 3 : 5. C joined them after six months with an amount equal to that of B; proflt at the end of one year should be distributed in what proportion among A, B and C ?
(1) 6 : 10 : 5 (2) 3 : 5 : 5
(3) 3 : 5 : 2 (4) Data inadequate
(5) None of these.
(LIC Assistant Administrative Offìcer (AAO) Exam. 24.04.2005)

2. Sunetra invested an amount of Rs. 50,000 to start a software business. After six months Nikhil joined her with an amount of Rs. 80,000. At the end of 3 years they eamed a proflt of Rs. 24,500. What is Sunetra's share in the proflt ?
(1) Rs, 14,000 (2) Rs. 9423
(3) Rs. 10, 250 (4) Rs. 12,500
(5) None of these
(LIC Assistant Administrative Officer (AAO) Exam. 2006)

3. A, B and C started a business with their investment in the ratio 1 : 3 : 5. After 4 months, A invested the same amount as before and B as well as C withdrew half of their investments. The ratio of their profits at the end of the year was :
(1) 5 : 6 : 10 (2) 6 : 5 : 10
(3) 10 : 5 : 6 (4) 4 : 3 : 5
(United India Insurance Co. (AAO) Exam. 11.03.2007)

4. Dheeraj distributed his total assets to his wife, three sons, two daughters and five grandchildren in such a way that each grandchild got one-eighth of each son or

one-tenth of each daughter.His wife got 40 per cent of the total share of his sons and daughters together. If each daughter received asset of worth Rs. 1.25 lakh, what was the total worth of the assets received by his wife and the three grandchildren together ?

(1) Rs. 32,500
(2) Rs. 2,57,500
(3) Rs. 2,82,500
(4) Cannot be determined
(5) None of these

(LIC Assistant Administrative Officer Exam. 2008)

5. X, Y and Z were sharing profits in the ratio 4 : 3 : 2. Y retired from the firm and X and Z decide to share profits in the ratio 3 : 2. Calculate the gaining ratio.

(1) 7 : 8 (2) 5 : 9
(3) 4 : 7 (4) 5 : 8

(LIC Assistant Administrative Officer (AAO) Exam. 07.06.2009)

6. A, B and C enter into a partnership. A invests some money at the beginning, B invests doublé the amount after 6 months and C invests thrice the amount after 8 months. If the annual proflt be Rs. 27000 C's share is:

(1) Rs. 9000 (2) Rs. 11250
(3) Rs. 10,800 (4) Rs. 8625

(General Insurance Corporation AAO Exam. 11.12.2011)

7. Tanvi started a business investing Rs. 45,000. After 8 months Anisha joined her with a capital of Rs. 52,000. At the end of the year the total profit was Rs. 56,165. What is the share of profits of Anisha ?

(1) Rs. 21,450 (2) Rs. 24,440
(3) Rs. 27,635 (4) Rs. 31,765
(5) None of these

(United India Insurance AO Exam. 26.05.2013

8. A, B, C started a shop by investing Rs. 27000, Rs. 81000 and Rs. 72000 respectively. At the end of one year, B's share in total profit was Rs. 36000. What was the total profit ?

(1) Rs. 108000 (2) Rs. 116000
(3) Rs. 80000 (4) Rs. 92000

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (3) **2.** (5)
3. (5) **4.** (2)
5. (2) **6.** (4)
8. (5) **9.** (2)
10. (4)

SBI PO EXAMS

1. (3) **2.** (5)
3. (2) **4.** (2)

RBI GRADE-B OFFICER EXAMS

1. (1)

INSURANCE EXAMS

1. (4) **2.** (5)
3. (1) **4.** (2)
5. (1) **6.** (1)
7. (5) **8.** (3)

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (3) Total equivalent capital of A
= $5x \times 12 + 8x \times 12$ = Rs. $156x$
Total equivalent capital of B for 1 month
= $6x \times 24$ = Rs. $144x$
Total equivalent capital of C
= $8x \times 12 + 4x \times 12$ = Rs. $144x$
$\therefore$ Required ratio = A : B : C
= $156x : 144x : 144x$
= 13 : 12 : 12

2. (5) Ratio of the earned profit = Ratio of the equivalent capitate of Tanvi and Anisha
= $45000 \times 12 : 52000 \times 4$
= $45 \times 3 : 52$
= 135 : 52
Sum of ratios = 135 + 52 = 187
$\therefore$ Anisha's share

= Rs. $\left(\frac{52}{187} \times 56165\right)$ = Rs. 156180.07

3. (5) Ratio of the proflt = Ratio of the equivalent capitals of Karan and Shirish
= $60000 \times 12 : 100000 \times 6$
= 720000 : 600000 = 6 : 5
$\therefore$ Shirish's share in the profit

= Rs. $\left(\frac{5}{11} \times 151800\right)$ = Rs. 69000

4. (2) Ratio of the equivalent capitals of Prakash, Sunil and Anil
= $11 \times 36 : 16.5 \times 36 : 8.25 \times 36$
=4 : 6 : 3
Anil's share in the proflt

= Rs. $\left[\frac{3}{(4+6+3)} \times 19.5\right]$ lakh

= Rs. 4.5 lakh
50% of Rs. 4.5 lakh
= Rs. 2.25 lakh

5. (2) Let Mona's investment = Rs.100
$\therefore$ Sonu's investment = Rs. 110 and Raghu's investment = $\frac{100}{90} \times 100$ = Rs. $\frac{1000}{9}$
$\therefore$ Ratio of Mona's, Sonu's and Raghu's investments

= 100 : 110 : $\frac{1000}{9}$ = 90 : 99 : 100

Sum of ratios = 90 + 99 + 100 = 289
$\therefore$ Raghu's investment

= Rs. $\left(\frac{100}{289} \times 5870\right)$ = Rs. 2000

6. (4) Ratio of the proflts of Gina, Shrayon and Deepika
= $48000 \times 12 : 62000 \times 8 : 80000 \times 6$
= $48 \times 12 : 62 \times 8 : 80 \times 6$
= 36 : 31 : 30
Sum of ratios
= 36 + 31 +30 = 97
$\therefore$ Deepika's share

= Rs. $\left(\frac{30}{97} \times 20661\right)$ = Rs. 6390

7. (2) Ratio of the equivalent capitals of Mrudul and Shalaka
= $29500 \times 24 : 33500 \times 20$ = 354 : 335
Sum of the ratios = 354 + 335 = 689
$\therefore$ Mrudul's share

Rs $\left(\frac{354}{689} \times 120575\right)$

= Rs. 61950

8. (5) Required difference

= $\left(\frac{7}{11} \times 2 - \frac{4}{11} \times 3\right)$

= $\frac{2}{11}$ × 73689 = Rs. 13398

9. (5) Ratio of equivalent capitals
= $30000 \times 12 : (24000 \times 4 + 18000 \times 8)$

: (42000 × 4 + 32000 × 8)
= 360000 : 240000 : 424000
= 90 : 60 : 106
= 45 : 30 : 53
Sum of ratios = 45 + 30 + 53 = 128
∴ B's share

$= \frac{30}{128} \times 11960 = \text{Rs. } 2803.125$

10. (4) Ratio of share of proflt
= Geeta : Shweta : Sita
= 4200 : 8400 : 5400
= 7 : 14 : 9
Sita's share

$= \frac{9}{(7+14+9)} \times 24000$

$= \frac{9}{30} \times 24000 = \text{Rs. } 7200$

Remaining amount after investing 32%
= 7200 × (100 - 32)%

$= \frac{7200 \times 68}{100} = \text{Rs.}4896$

SBI PO EXAMS

1. (3) Let the total money be Rs. *x*

Half of the total money goes to wife = $\frac{x}{2}$

$\frac{2}{3}$ of the remaining $\frac{x}{2} = \frac{x}{3}$ is divided equally among three sons.
∴ The share of each son

$= \frac{1}{3} \times \frac{x}{3} = \frac{x}{9}$

$\frac{x}{2} - \frac{x}{3} = \frac{x}{6}$ is divided equally among four daughters.
∴ The share of each daughter

$\frac{x}{24} = 20000$

x = 480000
The share of each son

$= \frac{x}{9} = \frac{480000}{9}$

= Rs.53333.33

2. (5) The ratio of capital invested by each partner i.e. Sunetra and Nikhil for in order to get the same proflt is :
36 × 50000 : 30 × 80000
= 3 : 4
∴ Share of Sunetra out of

Rs. 24500 = Rs. $\frac{3}{7} \times 24500$

= Rs. 10,500

3. (2) Let the amount invested in scheme A be Rs. *x.*

$\therefore \frac{x \times 12 \times 2}{100} = 3600$

$\Rightarrow x = \frac{3600 \times 100}{24} = \text{Rs. } 15000$

Total investment = Rs. 35000
∴ Amount invested in scheme B
= Rs. (35000 - 15000) = Rs. 20000

$\therefore \text{C.I} = P\left[\left(1+\frac{R}{100}\right)^{T} - 1\right]$

$= 20000\left[\left(1+\frac{10}{100}\right)^{2} - 1\right]$

= 20000 (1.21 - 1)
= 20000 × 0.21
= Rs. 4200

4. (2) Let Ninad invest Rs.*x*
∴ Vikas's investment = Rs.2*x*
Manav's investment = Rs. 3*x*
Ratio of the equivalent capital of Ninad, Vikas and Manav respectively
= *x* ×12 : 2*x* × 6 : 4 × 3*x*
= 1 : 1 : 1
Clearly, the proflt will be shared equally
∴ Manav's share

= Rs. $\frac{1}{3} \times 45000 = \text{Rs. } 15000$

RBI GRADE-B OFFICER EXAMS

1. (1) Equivalent capital of Raksha for 3 year
= Rs. (60000 × 1 + 80000 × 2)
= Rs. (60000 + 160000) = Rs. 220000
Equivalent capital of Kamal for 3 year

$= \text{Rs. } 90000 \times 2\frac{1}{2}$

$= \text{Rs. } \left(90000 \times \frac{5}{2}\right) = \text{Rs } 225000$

Ratio of capitals = 220000 : 225000
= 44 : 45
Sum of ratios = 44 + 45 = 89
Total profit = Rs. 7120000

$\therefore$ Raksha's share

= Rs $\left(\frac{44}{89}\times 7120000\right)$

= Rs. 3520000

INSURANCE EXAMS

1. (4) Let the amount invested by A and B is $3x$ and $4x$ respectively after 6 month C joined amount equal to B.
Then, Ratio of A, B and C in profit
= $3x \times 12 : 5x \times 12 : 5x \times 6 = 6 : 10 : 5$

2. (5) The ratio of capital invested by each partner i.e. Sunetra and Nikhil for 1 month in order to get the same profit is :
$36 \times 50000 : 30 \times 80000 = 3 : 4$
$\therefore$ Share of Sunetra out of

Rs. 24500 = Rs. $\frac{3}{7}\times 24500$

= Rs. 10,500

3. (1) Let their initial investments be Rs. x, Rs. $3x$ and Rs. $5x$ respectively.
Then, A : B : C

$= (x \times 4 + 2x \times 8) : (3x \times 4 + \frac{3x}{2} \times 8)$

$: (5x \times 4 + \frac{5x}{2} \times 8)$

$= (4x + 16x) : (12x + 12x) : (20x + 20x)$
$= 20x : 24x : 40x = 5 : 6 : 10$

4. (2) Share of 1 grandchild

= $\frac{1}{10}\times 1.25$ = Rs. 0.125 lakh

$\therefore$ Each son gets = 8×0.125 = Rs. 1 lakh
$\therefore$ Share of 3 sons = Rs. 3 lakhs
Share of 2 daughters = 2×1.25 = Rs. 2.5 lakhs
Total share of sons and daughters = Rs. 5.5 lakhs
$\therefore$ Wife's share = $\frac{2}{5}\times 5.5$

= Rs. 2.2 lakhs
Share of three grandchildren
= 3×0.125 = Rs. 0.375 lakh
$\therefore$ Required answer
= Rs. (2.2 + 0.375) lakh
= Rs. 2.575 lakhs
= Rs. 257500

5. (1) Gaining ratio

$= \left(\frac{3}{5}-\frac{4}{9}\right):\left(\frac{2}{5}-\frac{2}{9}\right)$

$= \frac{27-20}{45}:\frac{18-10}{45} = 7 : 8$

6. (1) Ratio of profit
$= x \times 12 : 2x \times 6 : 3x \times 4$
$= 1 : 1 : 1$

$\therefore$ C's share = $\frac{1}{3}\times 27000$

= Rs. 9000

7. (5) Ratio of the earned profit = Ratio of the equivalent capitals of Tanvi and Anisha
$= 45000 \times 12 : 52000 \times 4$
$= 45 \times 3 : 52 = 135 : 52$
Sum of ratios = 135 + 52 = 187
$\therefore$ Anisha's share

= Rs. $\left(\frac{52}{187}\times 56165\right)$ = Rs. 15618.074

8. (3) Ratio of their profit
= 27000 : 81000 : 72000 = 3 : 9 : 8
If total profit be Rs. x,

then B's share = $\frac{9x}{20}$ = 36000

$\Rightarrow x = \frac{36000\times 20}{9}$ = Rs. 80000

MODEL EXERCISES

1. Rs 770 nave been divided among A B and C such that A receives $\frac{2}{9}$th of what B and C together receive. Then, A's share is
(1) Rs 140 (2) Rs 154
(3) Rs 165 (4) Rs 170
(5) None of these

2. A sum of Rs 370 is to be divided among A, B and C such that
$\frac{\text{A's share}}{\text{B's share}} = \frac{\text{B's share}}{\text{C's share}} = \frac{3}{4}$.
Then, A's share is
(1) Rs 240 (2) Rs 120
(3) Rs 100 (4) Rs 90
(5) None of these

3. A and B enter into a partnership with Rs 50000 and Rs 60000 respectively. C joins them after *x* months contribuing Rs 70000 and B leaves *x* months before the end of the year. If they share the profit in the ratio of 20 : 18 : 21, then the value of *x* is
(1) 3 (2) 9
(3) 6 (4) 8
(5) None of these

4. Ajay, Aman, Suman and Geeta rented a house and agreed to share the rent as follows (Ajay : Aman = 8 : 15, Aman : Suman = 5 : 8 and Suman : Geeta = 4 : 5. The part of rent paid by Suman will be
(1) 24/77 (2) 13/66
(3) 12/55 (4) 13/77
(5) None of these

5. Three partners invested capital in the rado 2 : 7 : 9. The time period for which each of them invested was in the ratio of the reciprocal of the amount invested. Find the share of the partner who brought in the highest capital, if profit is Rs 1080.
(1) Rs 120 (2) Rs 360
(3) Rs 540 (4) Rs 420
(5) None of these

6. A man left one-half of the capital for his wife. One-third to his son and the remainder to his daughter and her share was worth Rs 45,000. Find the total capital.
(1) Rs 245000 (2) Rs 260000
(3) Rs 270000 (4) Rs 275000
(5) None of these

7. X and Y entered into partnership with Rs 700 and Rs 600 respectively. After 3 months, X withdrew 2/7 of his stock but after 3 months, he puts back 3/5 of what he had withdrawn. The profit at the end of the year is Rs 726. How much of this should X receive ?
(1) Rs 336 (2) Rs 366
(3) Rs 633 (4) Rs 663
(5) None of these

8. A, B and C are three partners. They altogether invested Rs 14000 in business. At the end of the year, A got Rs 337.50, B Rs 1125 and C Rs 637.50 as proflt. The difference between the investments of B and A was
(1) Rs 2200 (2) Rs 3200
(3) Rs 4200 (4) Rs 5250
(5) None of these

10. Anu is a working partner and Bimla is a sleeping partner in a business. Anu puts in Rs 5000 and Bimla puts in Rs 6000. Anu receives 12.5% of the proflt for managing the business and the rest is divided in proportion to their capital. What does each get out of a proflt of Rs 880 ?
(1) Rs 400 and Rs 480
(2) Rs 460 and Rs 420
(3) Rs 450 and Rs 430
(4) Rs 470 and Rs 410
(5) None of these

10. X and Y are partners in a business. X contributed 1/3 of the capital for 9 months and Y received 2/5 of the profits. For how long was Y's money used in the business ?
(1) 4 months (2) 3 months
(3) 2 months (4) 5 months
(5) None of these

11. Two companies A and B quote for a tender. On the tender opening day, A realizes that the two quotes are in the ratio 7 : 4 and hence decreases its price during negotiations to make it Rs 1 lakh lower than B's quoted price. B then realizes that the final quotes of the two were in the ratio 3 : 4. By how much did A decrease its price in order to win the bid ?
(1) Rs 7 lakhs (2) Rs 4 lakhs
(3) Rs 9 lakhs (4) None of these
(5) None of these

SHORT ANSWERS

1. (1) **2**. (4)
3. (1) **4**. (1)
5. (2) **6**. (2)
7. (3) **8**. (2)
9. (4) **10**. (2)
11. (2) **12**. (2)

EXPLANATIONS

1. (1) According to question,
A + B + C = 770 ...(i)

$A=\frac{2}{9}(B + C)$ ...(ii)

From Eqs. (i) and (ii), we get

$A+\frac{9}{2}A = 770$

$\Rightarrow 11A = 770 \times 2$

$\Rightarrow A = 140$

Thus, A's share is Rs 140

2. (4) According to question,

$\frac{A}{B}=\frac{3\times3}{4\times3}=\frac{9}{12}:\frac{B}{C}=\frac{3\times4}{4\times4}:\frac{12}{16}$

$\Rightarrow$ A : B : C = 9 : 12 : 16

A's share = $\frac{9}{37}\times 370$ = Rs 90

3. (1) According to question,
Investment of A = 50000 ×12 = Rs 600000
Investment of B = 60000 × (12 - x)
Investment of C = 70000 × (12 - x)

$\therefore \frac{600000}{60000\times(12-x)}=\frac{20}{18}$

$\Rightarrow 180 = 240 - 20x$

$\Rightarrow x = 3$

4. (1) According to question,
Ajay : Aman = 8 : 15
Aman : Suman = 5 : 8
Suman : Geeta = 4 : 5
$\therefore$ Ajay : Aman : Suman : Geeta
= 8 : 15 : 24 : 30
$\therefore$ Part of rent paid by Suman

$=\frac{24}{8+15+24+30}=\frac{24}{77}$

5. (2) According to question,
Ratio of capital = 2 : 7 : 9

Ratio of time = $\frac{1}{2}:\frac{1}{7}:\frac{1}{9}$

$\therefore$ Ratio of investment

$= 2\times\frac{1}{2}: 7\times\frac{1}{7}: 9\times\frac{1}{9}$

= 1 : 1 : 1

$\therefore$ Share of each partner

$=\frac{1}{3}\times 1080$ = Rs 360

6. (3) Let that man left Rs x.

$\therefore$ Share of daughter = $x - \frac{x}{2}-\frac{x}{3}$

$\Rightarrow 45000 = \frac{x}{6}$

$\Rightarrow x$ = Rs 270000

7. (2) X's investment = (700 × 3)

$+\left(700\times\frac{5}{7}\times3\right)+\left(500+200\times\frac{3}{5}\right)\times 6$

= Rs 7320
Y's investment = 600 ×12 = Rs 7200
X's share from profit

$=\frac{7320}{(7320+7200)}\times 726$ = Rs 366

8. (4) By question,
Ratio of investments of A, B and C
= 337.50 : 1125 : 637.50
= 27 : 90 : 51 : = 9 : 30 : 17
Let the investments of A, B and C be 9x, 30x and 17x respectively
$\therefore 9x + 30x + 17x = 14000$
$\Rightarrow x = 250$
So, required difference
$= 30x - 9x = 21x$ = Rs 5250

9. (2) By question, 2.5% of profit

$=\frac{12.5}{100}\times 880$ = Rs 110
Remaining = Rs 770 is divided in the ratio
= 5000 : 6000 = 5 : 6 Proflt of Anu

$=\frac{5}{11}\times 770 + 110$ = Rs 460 Proflt of Bimla

$=\frac{6}{11}\times 770$ = Rs 420

10. (2) By question,

Ratio of capital of X and Y = $\frac{1}{3}:\frac{2}{3}$ = 1 : 2

Ratio of their proflt = $\frac{3}{5}:\frac{2}{5}$ = 3 : 2

Let Y's money was used for n months.
$\therefore (1 \times 9) : (2 \times n) = 3 : 2$
$\Rightarrow n$ = 3 months

12. (2) A quotes Rs. 7x initially.
B quotes Rs. 4x initially.
The new ratio changes to 3 : 4.
Change is decrease of Rs. 4 lakhs.

PERMUTATION, COMBINATION & PROBABILITY

1. There are 12 students in a class. If a group of 5 students has to be selected from the class in such a way that one particular student is always included, in how many different ways can the selection be made?

   (1) 210 (2) 220 (3) 320 (4) 330 (5) 420

2. If the letters of the word PREVIOUS be arranged at random, what is the probability that all the vowels come together?

   (1) $\frac{1}{8}$ (2) $\frac{7}{8}$ (3) $\frac{1}{14}$ (4) $\frac{7}{14}$ (5) $\frac{1}{16}$

3. Letters of the word "SERIES" are arranged in such a way that all the vowels always come together. What is the total number of ways of making such an arrangement?

   (1) 720 (2) 180 (3) 144 (4) 72 (5) 36

4. From a pack of 52 cards, two cards are drawn at random. What is the probability of both of them being queens?

   (1) $\frac{200}{221}$ (2) $\frac{20}{221}$ (3) $\frac{201}{221}$ (4) $\frac{1}{221}$ (5) $\frac{32}{221}$

5. The ratio of the number of girls to the number of boys is the 5 : 2 in a class of 28 students. A group of three students is to to be selected at random amongst them. What is the probability that the selected group of students contain one boy and two girls?

   (1) $\frac{14}{71}$ (2) $\frac{14}{117}$ (3) $\frac{20}{71}$ (4) $\frac{20}{117}$ (5) None of these

6. 18 persons are sitting around a circular table. In how many ways can they be seated if six particular persons are to always sit together?

   (1) 18! × 6! (2) 17! × 5! (3) 13! × 6! (4) 12! × 6! (5) 12! × 5!

7. In how many ways can the letters of the word "PRODUCT" be arranged so that all the vowels never come together?

   (1) 5040 (2) 720 (3) 4320 (4) 1440 (5) 3600

8. There are seven boys and six girls. They sit in a row randomly. What is the probability that all the girls do not sit together?

   (1) $\frac{1}{429}$ (2) $\frac{428}{429}$ (3) $\frac{2}{429}$ (4) $\frac{427}{429}$ (5) $\frac{1}{1716}$

9. There are seven boys and four girls in a row. In how many ways can they be seated in the row so that all the girls do not sit together?

   (1) 7! × 4! (2) 11! - 7! × 4! (3) 11! - 7! (4) 11! - 4! (5) 11! - 8! × 4!

10. A bag contains four red, six black and and seven green balls. Three balls are drawn randomly. What is the probability that the balls drawn contain exactly two red balls?

    (1) $\frac{39}{340}$ (2) $\frac{301}{340}$ (3) $\frac{55}{408}$ (4) $\frac{353}{408}$ (5) $\frac{54}{91}$

11. In how many ways can 6 boys and 5 girls be seated in a row so that they sit alternately?

    (1) 43200 (2) 86400 (3) 21600 (4) 840 (5) 720

12. When two dice are thrown, what is the probability that the difference of the numbers shown by them will be 3?

    (1) $\frac{2}{9}$ (2) $\frac{1}{36}$ (3) $\frac{1}{6}$ (4) $\frac{1}{18}$ (5) None of these

13. In how many different ways can the letters of the word “PRODUCTION’ be arranged so that the vowels always come together?

(1) 15120 (2) 30240 (3) 60480 (4) 120960 (5) $\frac{11!}{2!}$

14. From a group of 8 men and 5 women, in how many different ways can six of them be included so that at least one woman is chosen?

(1) 1715 (2) 1688 (3) 896 (4) 996 (5) 1714

15. In how many ways can you choose a group of four persons out of three girls and seven boys such that the group has only one girl?

(1) 105 (2) 210 (3) 120 (4) 224 (5) 240

16. A card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is either a Diamond or a Jack?

(1) $\frac{1}{13}$ (2) $\frac{2}{13}$ (3) $\frac{3}{13}$ (4) $\frac{4}{13}$ (5) $\frac{5}{13}$

17. In a party, every person shakes hands with every other person. If there occur a total of 351 handshakes in the party, how many persons were present at the party?

(1) 25 (2) 26 (3) 27 (4) 28 (5) 29

18. A bag contains 10 white, 8 black and 6 blue balls. If two balls are drawn at random, find the probability of both the balls being black?

(1) $\frac{1}{3}$ (2) $\frac{1}{12}$ (3) $\frac{7}{69}$ (4) $\frac{9}{69}$ (5) None of these

19. In how many different ways can the letters of the word “SCHOOL” be arranged?

(1) 720 (2) 360 (3) 180 (4) 120 (5) 60

20. A card is drawn from a pack of 52 cards. What is the probability that it is either a Queen or a Jack?

(1) $\frac{1}{13}$ (2) $\frac{1}{26}$ (3) $\frac{2}{13}$ (4) $\frac{3}{13}$ (5) None of these

21. From a pack of 52 cards, two cards are drawn at random. What is the probability that one is a Queen and the other is an Ace?

(1) $\frac{1}{221}$ (2) $\frac{32}{221}$ (3) $\frac{20}{663}$ (4) $\frac{8}{663}$ (5) None of these

22. If a total of nine students appear in an examination, in how many ways can the result be announced?

(1) 1024 (2) 512 (3) 256 (4) 2048 (5) 128

23. How many numbers between 300 and 800 can be made by using digits 2,4,5,6 and 0?

(1) 36 (2) 72 (3) 144 (4) 108 (5) None of these

24. If two dice are thrown simultaneously, what is the probability that the sum of the numbers appeared is less than seven?

(1) $\frac{7}{12}$ (2) $\frac{5}{12}$ (3) $\frac{7}{36}$ (4) $\frac{5}{18}$ (5) $\frac{5}{36}$

25. From a group of 8 boys and 6 girls, in how many ways can a group of 5 be chosen so as to include at least one girl?

(1) 240 (2) 1260 (3) 1720 (4) 1840 (5) 1946

26. A bag contains 5 red and 8 black balls. A ball is drawn out of it and replaced in the bag. Then a ball is drawn out again. What is the probability that both balls are red?

(1) $\frac{5}{13}$ (2) $\frac{8}{13}$ (3) $\frac{25}{169}$ (4) $\frac{64}{169}$ (5) None of these

27. How many different arrangements can be made from the letters of the word "WEDNESDAY" such that all vowels come together?
(1) 7560 (2) 15120 (3) 2520 (4) 5040 (5) None of these

28. Seven persons are sitting around a round table. What is the probability that three particular persons are sitting together?
(1) $\frac{1}{7}$ (2) $\frac{2}{7}$ (3) $\frac{1}{5}$ (4) $\frac{2}{5}$ (5) $\frac{1}{6}$

29. In a party, every member shakes hands with every other member. If the total number of handshakes is 561, how many members are there in the party?
(1) 36 (2) 35 (3) 34 (4) 33 (5) 32

30. In a throw of two dice, what is the probablity that the sum of numbers appeared is greater than or equal to eight?
(1) $\frac{5}{36}$ (2) $\frac{5}{18}$ (3) $\frac{5}{12}$ (4) $\frac{1}{6}$ (5) None of these

31. A bag contains 4 Red, 8 Black and 12 White balls. Three balls are drawn randomly. What is the probability that the balls drawn are of different colours?
(1) $\frac{12}{253}$ (2) $\frac{16}{253}$ (3) $\frac{24}{253}$ (4) $\frac{48}{253}$ (5) $\frac{205}{253}$

32. How many different words can be formed with the letters of the word 'HINDUSTAN' so that all the vowels come together?
(1) 30240 (2) 15120 (3) 7560 (4) 362880 (5) 181440

33. In a group of 12 persons, 5 are females. In how many different ways can they sit in a row so that no two females sit together?
1) 3386.88×10^3 2) 3386.88×10^2 3) 3386.88×10^4 4) 3386.88 5) None of these

34. 12 persons are to be seated around a round table. What is the probability that 5 particular persons sit together?
(1) $\frac{2}{33}$ (2) $\frac{1}{66}$ (3) $\frac{4}{65}$ (4) $\frac{3}{65}$ (5) $\frac{26}{65}$

35. In how many different ways can the letters of the word THERAPY' be arranged so that the vowels never come together ?
(1) 720 (2) 1440 (3) 5040 (4) 3600 (5) 4800

36. A bag contains 13 white and 7 black balls. Two balls are drawn at random. What is the probability that they are of the same colour ?
(1) $\frac{41}{190}$ (2) $\frac{21}{190}$ (3) $\frac{59}{190}$ (4) $\frac{99}{190}$ (5) $\frac{77}{190}$

37. How many numbers between 300 and 1000 can be made with the digits 1, 2, 3, 4, 5, 6, 7 and 0?
(1) 280 (2) 210 (3) 5040 (4) 300 (5) None of these

38. 15 persons are seated around a table. What is the probability that four particular persons always sit together?
(1) $\frac{1}{546}$ (2) $\frac{1}{91}$ (3) $\frac{4}{455}$ (4) $\frac{1}{2184}$ (5) None of these

39. What numbers between 500 and 1900 can be made with the digits 3, 4, 5, 6, 7 and 0 (Repetition of digits is not allowed) ?
(1) 60 (2) 90 (3) 120 (4) 150 (5) None of these

40. From a pack of 52 cards, 3 cards are drawn. What is the probability that it has all three kings?

(1) $\frac{1}{52}$ (2) $\frac{16}{5525}$ (3) $\frac{1}{5525}$ (4) $\frac{48}{5525}$ (5) None of these

41. In how many different ways can the letters of the word PRODUCTION be arranged in such a way that all the vowels always come together?

(1) 362880 (2) 1814400 (3) 10080 (4) 60480 (5) 120960

42. Six boys and seven girls are made to sit in a row. What is the probability that all the boys do not sit together?

(1) $\frac{1}{429}$ (2) $\frac{2}{429}$ (3) $\frac{428}{429}$ (4) $\frac{427}{429}$ (5) None of these

43. In how many ways can six different things be divided equally among three persons?

(1) 3375 (2) 1125 (3) 375 (4) 90 (5) 22

44. A box contains 8 Red, 4 White and 3 Blackballs. Three balls are drawn randomly from the box. What is the probability that the three balls drawn randomly are of different colours?

(1) $\frac{3}{91}$ (2) $\frac{3}{455}$ (3) $\frac{1}{5}$ (4) $\frac{91}{455}$ (5) $\frac{96}{455}$

45. From a group of five males and six females, in how many ways can four be chosen to include exactly one female?

(1) 210 (2) 180 (3) 120 (4) 80 (5) 60

46. A bag contains 6 red, 7 blue and 8 green balls. Three balls are drawn randomly. What is the probability that the balls drawn contain exactly two blue balls?

(1) $\frac{147}{665}$ (2) $\frac{518}{665}$ (3) $\frac{54}{455}$ (4) $\frac{44}{455}$ (5) $\frac{401}{455}$

47. In how many different ways can the letters of the word 'REPLACE' be arranged?

(1) 2630 (2) 5040 (3) 1680 (4) 2580 (5) None of these

48. How many words can be formed from the letters of the word REGULAR?

(1) 1260 (2) 5040 (3) 2520 (4) 720 (5) 2880

49. A bag contains 3 black and 6 white balls. Two draws of one ball each are made without replacement. What is the probability that one ball is black and the other is white?

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{6}$ (4) $\frac{2}{9}$ (5) $\frac{7}{9}$

50. In how many ways can 6 Americans, 5 Indians and 3 Russians be seated in a row so that all the persons of the same nationality sit together?

(1) 1555200 (2) 777600 (3) 1036800 (4) 3110400 (5) 518400

51. In a box there are 5 red and 6 white balls. Aball is drawn out of it and replaced in the box. Then a ball is drawn again. What is the probability that the first ball was red and the second white?

(1) $\frac{1}{30}$ (2) $\frac{30}{121}$ (3) $\frac{1}{121}$ (4) $\frac{5}{121}$ (5) $\frac{1}{11}$

52. From a group of 5 girls and 8 boys, in how many ways can 8 person be chosen so as to include at least one girl?

(1) 984 (2) 1024 (3) 1132 (4) 1242 (5) 1286

53. In a bag there are 9 white and 7 black balls. Two draws of, one ball each are made without replacement. What is the probability that one ball is black and the other is white?

(1) $\frac{19}{40}$ (2) $\frac{21}{40}$ (3) $\frac{19}{56}$ (4) $\frac{37}{56}$ (5) $\frac{24}{95}$

54. If the letters of the word NISHANT are arranged at random, what is the probability that the letters I and A will come together?

(1) $\frac{2}{7}$ (2) $\frac{1}{7}$ (3) $\frac{5}{7}$ (4) $\frac{6}{7}$ (5) None of these

55. How many different numbers of 8 digits can be formed from the digits 0, 1,2,3,4, 5,6,7 that are divisible by 5? (The digits should not be repeated.)
(1) 4320 (2) 5040 (3) 9360 (4) 18720 (5) 4680

56. In how many ways can 6 boys and 5 girls be seated in a row so that they are alternate to one another?
(1) 86400 (2) 518400 (3) 14400 (4) 2880 (5) 30

57. An unbiased coin is tossed 9 times. Find the chances that exactly five tails will appear.

(1) $\frac{21}{256}$ (2) $\frac{63}{256}$ (3) $\frac{15}{256}$ (4) $\frac{63}{128}$ (5) $\frac{21}{128}$

58. A bag contains 9 red, 11 blue and 12 green balls (all the balls are distinct in some manner) . Find out in how many different ways can be three balls selected so that all the three balls are of different colours.
(1) 1138 (2) 1182 (3) 1178 (4) 1188 (5) None of these

59. Seven boys and seven girls are to be seated around a circular table in such a way that no two girls are together. In how many different ways can they be arranged?
(1) 3682880 (2) 3688200 (3) 3628800 (4) 3682200 (5) None of these

60. In how many different ways can we select a queen or a spade from a full pack of cards?
(1) 13 (2) 14 (3) 16 (4) 12 (5) None of these

61. A box contains 3 red, 4 black and 5 green balls. If 3 balls are drawn at random, then what is the probability that they are of the same colour?

(1) $\frac{3}{41}$ (2) $\frac{38}{41}$ (3) $\frac{3}{44}$ (4) $\frac{41}{44}$ (5) None of these

62. In how many different ways can the letters of the word DISPLAY be arranged so that the vowels always come together?
(1) 1440 (2) 720 (3) 5040 (4) 2520 (5) None of these

63. A bag contains 3 red, 4 black and 5 green balls. 13 balls are drawn randomly. What is the probability that the balls drawn are certainly green?

(1) $\frac{5}{41}$ (2) $\frac{5}{44}$ (3) $\frac{7}{41}$ (4) $\frac{7}{44}$ (5) None of these

64. In how many ways can the letters of the word VOLUME be arranged so,, that the vowels always come together?
(1) 720 (2) 360 (3) 180 (4) 144 (5) 24

65. In a party every person shakes hands with every other person. If there were a total of 990 handshakes in the party, find the number of persons who were present in the parry.
(1) 75 (2) 60 (3) 55 (4) 48 (5) 45

66. A and B sit in a round table conference with 15 other persons. If the 17 persons are seated all random, what is the probability that there are exactly 5 persons between A and B?

(1) $\frac{1}{8}$ (2) $\frac{2}{17}$ (3) $\frac{1}{9}$ (4) $\frac{2}{15}$ (5) $\frac{1}{7}$

67. In how many different ways can the letters of the word TRIANGLE be arranged?
(1) 5040 (2) 40320 (3) 43020 (4) 720 (5) 4240

68. A bag contains 6 red and 5 white balls. Three balls are drawn at random. Find the probability that one ball is red and two balls are white.

(1) $\frac{10}{33}$ (2) $\frac{12}{33}$ (3) $\frac{15}{33}$ (4) $\frac{55}{230}$ (5) None of these

Directions (Q. 69-70): Study the given information carefully and answer the questions that follow.

69. A box contains 6 red, 3 yellow and 2 green balls. If two balls are picked at random, what is the probability that both of them are red?

(1) $\frac{3}{11}$ (2) $\frac{1}{2}$ (3) $\frac{4}{15}$ (4) $\frac{1}{7}$ (5) None of these

70. If two balls are picked at random from the box given in the above question, what is the probability that at least one of them is yellow?

(1) $\frac{22}{91}$ (2) $\frac{27}{55}$ (3) $\frac{1}{456}$ (4) $\frac{7}{11}$ (5) None of these

71. A bag contains 7 blue balls and 5 yellow balls. If two balls are selected at random, what is the probability that none is yellow?

(1) $\frac{5}{33}$ (2) $\frac{5}{22}$ (3) $\frac{7}{22}$ (4) $\frac{7}{33}$ (5) $\frac{7}{66}$

72. A die is thrown twice. What is the probability of getting a sum 7 from both the throws?

(1) $\frac{5}{18}$ (2) $\frac{1}{18}$ (3) $\frac{1}{9}$ (4) $\frac{1}{6}$ (5) $\frac{5}{36}$

73. The letters of the word PUMPKIN are arranged in such a way that all the vowels come together. Find out the total number of ways of making such an arrangement.

(1) 360 (2) 5040 (3) 720 (4) 1440 (5) None of these

74. What is the probability that a card drawn at random from a well-shuffled pack of 52 cards is either a king or a diamond?

(1) $\frac{11}{52}$ (2) $\frac{4}{13}$ (3) $\frac{3}{13}$ (4) $\frac{13}{52}$ (5) None of these

75. 5 persons are chosen at random from a group of 4 men, 3 women and 5 children. The probability that exactly 3 of them are children is

(1) $\frac{36}{48}$ (2) $\frac{35}{132}$ (3) $\frac{34}{139}$ (4) $\frac{35}{221}$ (5) $\frac{37}{135}$

76. A box contains 4 white balls, 3 black balls and 9 red balls. In how many ways can 4 balls be drawn from the box, if at least one white ball is to be included in a draw?

(1) 1325 (2) 1421 (3) 325 (4) 428 (5) 912

77. The probability that A can solve a problem is $\frac{2}{3}$, and that B can solve it is $\frac{3}{4}$. If both of them attempt the problem, what is the probability that the problem will be solved?

(1) $\frac{6}{12}$ (2) $\frac{5}{12}$ (3) $\frac{11}{12}$ (4) $\frac{10}{12}$ (5) $\frac{9}{12}$

78. Out of 9 consonants and 5 vowels, how many words of 3 consonants and 2 vowels can be formed?

(1) 100800 (2) 100700 (3) 200800 (4) 200700 (5) 152535

79. From a group of 8 men and 7 women, 6 persons are to be selected to form a committee so that at least 4 men are there on the committee. In how many different ways can this be done?

(1) 2010 (2) 1950 (3) 1720 (4) 1890 (5) 2375

80. If two dice are thrown simultaneously, what is the probability that the sum of the numbers appearing on the top of the dice is less than 5?

(1) $\frac{2}{5}$ (2) $\frac{1}{3}$ (3) $\frac{7}{9}$ (4) $\frac{8}{17}$ (5) None of these

81. In how many ways can 3 boys and 3 girls sit in a row so that neither any two boys nor any two girls sit together?
(1) 720 (2) 36 (3) 72 (4) 12 (5) 35

82. What is the probability that a card drawn from a pack of 52 cards will be either a diamond or a king?
(1) $\frac{2}{13}$ (2) $\frac{4}{13}$ (3) $\frac{1}{13}$ (4) $\frac{1}{52}$ (5) $\frac{2}{37}$

83. Out of 15 students studying in a class, 7 are from Maharastra, 5 from Karnataka and 3 from Goa. 4 students are to be selected at random. What are the chances that at least one is from Karnataka?
(1) $\frac{12}{13}$ (2) $\frac{11}{13}$ (3) $\frac{100}{15}$ (4) $\frac{14}{27}$ (5) $\frac{37}{47}$

84. To fill 12 vacancies there are 25 candidates, of which 5 are from local institutes. 3 vacancies are reserved for the local - institute candidates and the rest are open to all. Find the number of ways in which the selection can be made!
(1) 16796 (2) 167960 (3) 1679600 (4) 176900 (5) 196700

Directions (Q. 85-86): A box contains 9 red, 11 green and 12 yellow balls. Find out in how many ways can 3 balls be selected so that

85. All 3 balls are of the same colour.
(1) 459 (2) 468 (3) 469 (4) 569 (5) None of these

86. All three balls are of different colours.
(1) 1178 (2) 1188 (3) 1172 (4) 1168 (5) None of these

87. In how many different ways can the letters of the word TRANSPIRATION be arranged so that the vowels always come together?
(1) 2429500 (2) 2721600 (3) 1627800 (4) 2826200 (5) 1360800

88. In a shelf, there are 8 Hindi, 7 English and 6 Urdu books. One book is picked up randomly. What is the probability that it is neither in Hindi nor in Urdu?
(1) $\frac{7}{21}$ (2) $\frac{13}{21}$ (3) $\frac{17}{21}$ (4) $\frac{19}{21}$ (5) $\frac{23}{47}$

89. If the letters of the word MOHAN are arranged at random, what is the probability that all the vowels will come together?
(1) $\frac{2}{5}$ (2) $\frac{3}{5}$ (3) $\frac{4}{5}$ (4) $\frac{3}{4}$ (5) $\frac{1}{2}$

90. In how many different ways can 5 girls and 4 boys be seated in a row so that they are alternate to one another?
(1) 2440 (2) 2660 (3) 2880 (4) 3220 (5) None of these

91. In how many different ways can the letters of the word GENERAL be arranged?
(1) 3280 (2) 2640 (3) 2830 (4) 2520 (5) 2250

92. In how many different ways can the letters of the word COMMENCEMENT be arranged so that the vowels always come together?
(1) 60480 (2) 120960 (3) 122840 (4) 50480 (5) 64080

93. What is the probability of getting a sum 9 from two throws of a dice?
(1) $\frac{1}{8}$ (2) $\frac{1}{7}$ (3) $\frac{2}{5}$ (4) $\frac{1}{9}$ (5) $\frac{2}{7}$

94. A box contains 6 green, 4 blue and 7 red pens. 4 pens are drawn at random. What is the probability that they all are not of the same colour?
(1) $\frac{12}{13}$ (2) $\frac{379}{780}$ (3) $\frac{5341}{6873}$ (4) $\frac{189}{581}$ (5) $\frac{2329}{2380}$

SHORT ANSWER

1. (4)	2. (3)	3. (5)	4. (4)	5. (4)	6. (4)	7. (5)	8. (4)
9. (5)	10. (1)	11. (2)	12. (3)	13. (3)	14. (2)	15. (1)	16. (4)
17. (3)	18. (3)	19. (2)	20. (3)	21. (4)	22. (2)	23. (1)	24. (2)
25. (5)	26. (3)	27. (1)	28. (3)	29. (3)	30. (3)	31. (4)	32. (2)
33. (3)	34. (2)	35. (4)	36. (4)	37. (2)	38. (2)	39. (1)	40. (3)
41. (4)	42. (4)	43. (4)	44. (5)	45. (5)	46. (1)	47. (5)	48. (3)
49. (1)	50. (4)	51. (2)	52. (5)	53. (2)	54. (1)	55. (3)	56. (1)
57. (2)	58. (4)	59. (3)	60. (3)	61. (3)	62. (1)	63. (4)	64. (4)
65. (5)	66. (1)	67. (2)	68. (2)	69. (1)	70. (2)	71. (3)	72. (4)
73. (3)	74. (2)	75. (2)	76. (1)	77. (3)	78. (1)	79. (4)	80. (5)
81. (3)	82. (2)	83. (2)	84. (3)	85. (3)	86. (2)	87. (5)	88. (1)
89. (1)	90. (3)	91. (4)	92. (1)	93. (4)	94. (5)		

DETAIL - EXPLANATIONS

1. 4; As one particular student is included in every selection, we have to choose (5-1) students from (12-1) students.

   $\therefore$ Total number of selections = $^{11}C_4$

   $$= \frac{11\times10\times9\times8}{4\times3\times2} = 330$$

2. 3; There are total eight letters

   $\therefore$ n(S) = $^8P_8$ = 8!

   As all vowels should come together, we assume them as one letter. Here E, I, O and U together are taken as one, so the number of letters is 4 + 1 = 5 and it can be arranged in $^5P_5$ = 5! ways and the vowels can be arranged in 4! ways among themselves

   $\therefore$ n(E) = 4! × 5!

   $$\therefore P(E) = \frac{4!5!}{8!}$$

   $$\therefore P(E) = \frac{4\times3\times2\times5!}{8\times7\times6\times5!}$$

   $$= \frac{4\times3\times2}{8\times7\times6} = \frac{1}{14}$$

3. 5; Out of the total six letters, three are vowels (E, I, E) and three are consonants (SRS).

   Here, E and S come twice. If we take all the vowels together, the number of letters is 3 + 1 = 4

   So, letters can be arranged in $\frac{4!}{2!}$ ways and

   three vowels can be arranged in $\frac{3!}{2!}$ ways

   $\therefore$ Total number of ways = $\frac{4!}{2!}\times\frac{3!}{2!}$

   = 12 × 3 = 36

4. 4; From 52 cards, 2 cards are drawn.

   $\therefore$ n(s) = $^{52}C_2$.

   In a pack, there are four queen cards from .which two queen cards have to be drawn.

   $\therefore$ n(E) = $^4C_2$

   $$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{^4C_2}{^{52}C_2} = \frac{6}{1326} = \frac{1}{221}$$

5. 4; Total = 28 $\therefore$ Boys = 20, Girls = 8

   n(S) = $^{28}C_3$ = 3276

   n(E) = $^{20}C_1$ × $^8C_2$ = 20 × 28 = 560

   $$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{560}{3276} = \frac{140}{819} = \frac{20}{117}$$

6. 4; We assume the six particular persons as one. So, total number of persons is 18 - 6 + 1 = 13 and they can sit around a circular table in 12! ways. Six particular persons can sit in 6! ways among themselves.

   $\therefore$ Total number of ways = 12! × 6!

7. 5; Here, the total letters are 7, out of which two are vowels.

   Total arrangements without restriction = 7!

   If we take the two vowels as one letter, number of letters = 5 + 1 = 6

   $\therefore$ Number of arrangements is 6! and the two vowels can be arranged in 2! ways among themselves.

   $\therefore$ Total arrangements = 6! × 2!

   $\therefore$ Reqd answer = 7! - 6! × 2! = 5040 - (720 × 2) = 5040 - 1440 = 3600

8. 4; Total number of persons is 7 + 6 = 13

   $\therefore$ n(S) = $^{13}P_{13}$ = 13!

   If we take all six girls as one unit, total number of persons = 7 + 1 = 8.

   They can be arranged in 8! ways and all the girls can be arranged in $^6P_6$ = 6! ways among themselves. So, when all the girls are together, total number of arrangements = 8! × 6! Probability of all girls sitting together

   $$= P(E) = \frac{8!\times6!}{13!} = \frac{2}{429}$$

   Probability of all girls not sitting together

   $$= 1 - \frac{2}{429} = \frac{427}{429}$$

9. 5; Without any restriction, 11 persons can sit in 11! ways. If we take all the girls together as one, the total number of persons will be 7 + 1 = 8, who can sit in 8! ways and the four girls can sit in 4! ways among themselves. So, when all the girls sit together, the number of ways = 8! × 4!

   When they don't sit together, the total number of ways = 11! - 8! × 4!

10. 1; Total balls = 4 + 6 + 7 = 17

$\therefore$ n(S) = $^{17}C_3$ = 680

Two red balls can be selected from four red balls in $^4C_2$ = 6 ways, and the third ball can be selected from the remaining 13 balls in $^{13}C_1$ = 13 ways.

$\therefore P(E) = \frac{13 \times 6}{680} = \frac{39}{340}$

11. 2; Number of arrangements = 6! × 5!

= 720 × 120 = 86400

12. 3; n(s) = 36 n(E) = 6

$\therefore P(E) = \frac{6}{36} = \frac{1}{6}$

$\because$ Possible combinations are {(4, 1) (1, 4) (5, 2) (2, 5) (6, 3) (3, 6)} = 6

13. 3; There are ten letters in which there are four vowel (O, O, U, I), and 'O' comes twice. If we consider all vowels as one letter the total number of letters will be 6 + 1 = 7 so the number of arrangements = 7!, and four vowels can be arranged in $\frac{4!}{2!}$ ways as 'O' comes twice.

$\therefore$ Total arrangements = $7! \times \frac{4!}{2!}$ = 60480

14. 2;

Man	Woman	Arrangements
5	1	$^8C_5 \times {}^5C_1 = 56 \times 5 = 280$
4	2	$^8C_4 \times {}^5C_2 = 70 \times 10 = 700$
3	3	$^8C_3 \times {}^5C_3 = 56 \times 10 = 560$
2	4	$^8C_2 \times {}^5C_4 = 28 \times 5 = 140$
1	5	$^8C_1 \times {}^5C_5 = 8 \times 1 = 8$

$\therefore$ Total = 280 + 700 + 560 + 140 + 8 = 1688

15. 1; One girl is chosen from 3 girls in $^3C_1$ ways and three boys can be chosen from 7 boys in $^7C_3$ ways. So, total number of ways

= $^3C_1 \times {}^7C_3 = 3 \times 35 = 105$

16. 4; Total cards = 52;

$\therefore$ n(S) = 52 in which there is a total of 13 Diamond cards and four Jack of cards.

So, for Diamond n(E) = 13 and for Jack $n(E_1) = 4$

$\therefore P(E) = \frac{13}{52} = \frac{1}{4}$,

$P(E_1) = \frac{4}{52} = \frac{1}{13}$

As one card is Jack of Diamond, so $n(E \cap E_1) = 1$

$\therefore P(E \cap E_1) = \frac{1}{52}$

$\therefore$ Probability of the card being either a Jack or a Diamond.

$= P(E \cup E_1)$

$= P(E) + P(E_1) - P(E \cap E_1)$

$\frac{1}{4} + \frac{1}{13} - \frac{1}{52}$

$= \frac{13 + 4 - 1}{52} = \frac{4}{13}$

17. 3; Let the number of handshakes in the party be $^nC_2$.

$\because {}^nC_2 = 351$

$\frac{n(n-1)}{2} = 351 \quad \therefore n(n-1) = 702 = 27 \times 26$

$\therefore$ n = 27

18. 3; Total number of balls = 24.

Two balls are drawn.

$\therefore$ n(S) = $^{24}C_2$ = 276

There are eight black balls, out of which two balls are drawn in $^8C_2$ ways.

$\therefore$ n(E) = 28 $\therefore P(E) = \frac{28}{276} = \frac{7}{69}$

19. 2; Total number of arrangements

= $\frac{6!}{2!}$ { $\because$ 'O' comes twice}

= 6 × 5 × 4 × 3 = 360

20. 3; There are total four Queen cards and four Jack cards in a pack of 52 cards.

$\therefore P(\text{Queen}) = \frac{4}{52} = \frac{1}{13}$, $P(\text{Jack}) = \frac{4}{52} = \frac{1}{13}$

$\therefore P(E) = \frac{1}{13} + \frac{1}{13} = \frac{2}{13}$

21. 4; n(S) = $^{52}C_2 = \frac{52!}{(52-2)!\,2!} = \frac{52!}{50!\,2!}$

= 26 × 51

There are total four Queens and four Aces in a pack of 52 cards.

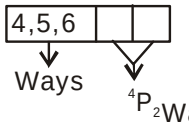
$\therefore$ n(E) = $^4C_1 \times {}^4C_1 = 4 \times 4 = 16$

$\therefore P(E) = \frac{16}{26 \times 51} = \frac{8}{663}$

22. 2; There are two possibilities for every students: either to pass or to fail.

$\therefore$ Total number of ways = $(2)^9 = 512$

23. 1,



$\therefore$ Reqd. numbers = $3 \times {}^4P_2 = 3 \times 12 = 36$

24. 2; Sum of the numbers can be 2, 3, 4, 5, and 6.

n (E) = 15, n (S) = 36 $\therefore$ P (E) = $\frac{15}{36} = \frac{5}{12}$

25. 5; Different possibilities are $\Rightarrow$

Girl	Boy
1	$4 = {}^6C_1 \times {}^8C_4 = 6 \times 70 = 420$
2	$3 = {}^6C_2 \times {}^8C_3 = 15 \times 56 = 840$
3	$2 = {}^6C_3 \times {}^8C_2 = 20 \times 28 = 560$
4	$1 = {}^6C_4 \times {}^8C_1 = 15 \times 8 = 120$
5	$0 = {}^6C_5 \times {}^8C_0 = 6 \times 1 = 6$

$\therefore$ Total No. of ways = 1946

26. 3; Probablity that both balls are red

$$= \frac{5}{13} \times \frac{5}{13} = \frac{25}{169}$$

27. 1; There are nine letters. Taking the three vowels (E, E, A) as a unit, total number of letters becomes 7.

D is repeated twice, and E is repeated twice among vowels.

$\therefore$ Total number of arrangements = $\frac{7!}{2!} \times \frac{3!}{2!}$

= 2520 × 3 = 7560

28. 3; n(s) = No. of ways in which seven persons can sit around a round table = 6!

As three persons can be taken as a unit, total persons = 5 and number of ways in which five persons can sit around the table = 4! and three persons can sit in 3! ways among themselves.

$\therefore$ n(E) = 4! × 3!

$\therefore$ P(E) = $\frac{4! \times 3!}{6!} = \frac{1}{5}$

29. 3; Number of handshakes = ${}^nC_2 = 561$

$\therefore \frac{n(n-1)}{2} = 561$

$\therefore$ n(n - 1) = 1122

or n × (n - l) = 34 × 33

$\therefore$ n = 34

30. 3; The sum can be 8, 9, 10, 11 and 12.

$\therefore$ n(E) = 5 + 4 + 3 + 2 + 1 = 15

and, n(s) = 36

$\therefore$ P(E) = $\frac{15}{36} = \frac{5}{12}$

31. 4; Total number of balls = 4 + 8 + 12 = 24

n(s) = $^{24}C_3 = 2024$

Number of ways to select one ball of each colour

= n(E) = $^4C_1 \times {}^8C_1 \times {}^{l2}C_1 = 4 \times 8 \times 12 = 384$

$\therefore$ P(E) = $\frac{384}{2024} = \frac{48}{253}$

32. 2; Total number of letters is nine, in which N comes twice and total vowels are three (I, U, A).

We consider the three vowels as one letter. So, the total number of letters is 6 + 1 = 7 in which N comes twice. These seven letters can be arranged in $\frac{7!}{2!}$ ways and the three vowels can be arranged in 3! ways.

$\therefore$ Total numbers of ways = $\frac{7!}{2!} \times 3! = 15120$

33. 3; (12 - 5) = 7 boys can sit in a row in 7! ways. Now the females can sit in 8 places (including both ends) in $^8p_5$ ways.

$\therefore$ Total number of ways = $^8P_5 \times 7! = \frac{8!}{3!} \times 7!$

= 6720 × 5040 = 3386.88×10^4

34. 2; 12 persons can sit at a round table in (12 - 1)! = 11! ways. If we take 5 particular persons always together then total number of persons = (12 - 5 + 1) = 8. So, they can sit in 7! ways and those five persons can sit in 5! ways among themselves.

$\therefore$ P(E) = $\frac{7! \times 5!}{11!} = \frac{1}{66}$

35. 4; Total number of letters is 7, and these letters can be arranged in 7! ways

= 1 × 2 × 3 × 4 × 5 × 6 × 7 = 5040 ways

There are seven letters in the word THERAPY including 2 vowels. (E, A) and five consonants.

Consider two vowels as one letter.

We have 6 letters which can be arranged in $^6P_6$ = 6 ways.

But vowels can be arranged in 2! ways.

Hence, the number of ways, all vowels will come together = 6! x 2!

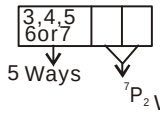
= 1 × 2 × 3 × 4 × 5 × 6 × 2 = 1440

Total number of ways in which vowels will never come together = 5040 - 1440 = 3600

36. 4; Total number of balls = 13 + 7 = 20
Number of Sample space = n(S) = $^{20}C_2 = 190$
Number of events = n(E)
$= {}^{13}C_2 + {}^{7}C_2 = 78 + 21 = 99$

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{99}{190}$

37. 2;

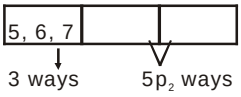


$= 5 \times \frac{7!}{5!} = 5 \times 42 = 210$

38. 2; n(S) = Total number of ways 15 persons sit at a round table =14!
Taking four persons as one total numbers = 12 persons
$\therefore$ Number of ways they can sit at the round table = 11!
Four persons sit among themselves in 4! ways.
So n(E) = 11!4!

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{11! \times 4!}{14!} = \frac{1}{91}$

39. 1; When the number is of three digits



$\therefore$ Numbers of three digits = $3 \times \frac{5!}{3!}$

$= 3 \times 5 \times 4 = 60$

When the number is of four digits



Only digit' 1' can be placed but we don't have that digit So, numbers of 4 digits = 0

40. 3; n(s) = $^{52}C_3 = 22100$
There are total four kings in the pack.
$\therefore$ n(E) = $^4C_3 = 4$

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{4}{22100} = \frac{1}{5525}$

41. 4; Total number of letters = 10, vowels = 4, and 'O' comes twice.
We can consider all the vowels as one unit.
So, total letters = 7, ie number of arrangements = 7!.
Now vowels can be arranged in 4! ways among them selves and 'O' comes twice.
$\therefore$ Total number of ways

$= \frac{7! \times 4!}{2!} = \frac{5040 \times 24}{2} = 60480$

42. 4; n(s) = $^{13}P_{13} = 13!$
We can consider all the six boys as one unit. So, total persons will be 7 + 1 and they can be seated in 8! ways and all the six boys can be seated in 6! ways among themselves. So, total number of ways in which all the six boys sit together = $\frac{8! \times 6!}{13!} = \frac{2}{429}$

$\therefore$ Probability of all the boys not sitting together

$= 1 - \frac{2}{429} = \frac{427}{429}$

43. 4; 1st person will get 2 things from 6 in $^6C_2$ ways, 2nd will get 2 things from the remaining 4 in $^4C_2$ ways and 3rd in $^2C_2$ ways.
Total number of ways = $^6C_2 \times {}^4C_2 \times {}^2C_2$
$= 15 \times 6 \times 1 = 90$ ways

44. 5; Total number of balls = 8 + 4 + 3 = 15
n(S) = $^{l5}C_3 = 455$
n(E) = $^8C_1 \times {}^4C_1 \times {}^3C_1 = 8 \times 4 \times 3 = 96$

$\therefore P(E) = \frac{96}{455}$

45. 5; Required number of ways
$^6C_1 \times {}^5C_3 = 6 \times 10 = 60$ ways

46. 1; Total balls = 6 + 7 + 8 = 21
n(s) = $^{21}C_3 = 1330$
Two blue balls can be selected from 7 blue balls in $^7C_2 = 21$ ways and the remaining one ball can be selected from the remaining 14 balls in $^{14}C_1 = 14$ ways
$\therefore$ n(E) = 21 × 14 = 294

$\therefore P(E) = \frac{294}{1330} = \frac{147}{665}$

47. 5; The number of letters in the word REPLACE = 7.
The number of arrangements = 7!
But 'E' comes twice.
So, number of arrangements

$= \frac{7!}{2!} = 3 \times 4 \times 5 \times 6 \times 7 = 2520$

48. 3; Total number of letters = 7, and 'R' comes twice.

$\therefore$ Number of words = $\frac{7!}{2} = 2520$

49. 1; Two balls out of 9 can be drawn in $^9C_2$ ways.

n(S) = $\frac{9!}{2!7!} = \frac{9 \times 8}{2} = 36$ ways

One black ball out of 3 balls can be drawn in $^3C_1 = 3$ ways. One white ball out of 6 white balls can be drawn in $^6C_1 = 6$ ways.
n(E) = 3 × 6 = 18 ways

$\therefore P(E) = \frac{18}{36} = \frac{1}{2}$

50. 4; Take the persons of the same nation as one unit. Total three units are there. Total number of arrangements = 3! 6! 5! 3! = 6 × 720 × 120 × 6 = 3110400

51. 2; The first was red and the second white.

$\frac{5 \times 6}{11 \times 11} = \frac{30}{121}$

52. 5; The different possibilities are:
girl = 1 + boy = 7
No. ofways = $^5C_1 \times {}^8C_7$
= 5 × 8 = 40
girl = 2 + boy = 6
No. ofways = $^5C_2 \times {}^8C_6$
= 10 × 28 = 280
girl = 3 + boy = 5
No. ofways = $^5C_3 \times {}^8C_5$
= 10 × 56 = 560
girl = 4 + boy = 4
No. of ways = $^5C_4 \times {}^8C_4$
= 5 × 70 = 350
girl = 5 + boy = 3
No. ofways = $^5C_5 \times {}^8C_3$
= 1 × 56 = 56
∴ Total number of ways = 40 + 280 + 560 + 350 + 56 = 1286

53. 2; Total balls = 7 + 9 = 16, from which two balls can be drawn in $^{16}C_2$ ways, ie n(s) = 120, One white ball can be drawn from 9 white balls in $^9C_1$ = 9 ways and one black ball can be drawn from 7 black balls in $^7C_1$ = 7 ways. n(E) = 9 × 7 = 63

$\therefore P(E) = \frac{63}{120} = \frac{21}{40}$

54. 1; The total letters are 7, so n(s) = $^7P_7$ = 7!
We consider A and I as a unit. So total letters = 5 + 1 = 6. So the number of arrangements = $^6P_6$ = 6! and A and I can be arranged in 2! ways among themselves, so total number of ways = 6! × 2!

$\therefore P(E) = \frac{6! \times 2!}{7!} = \frac{2}{7}$

55. 3; For divisibility by 5 the unit's place should have 0 or 5
(I) When 0 is in unit's place then the arrangement will be in 7! ways, ie 5040 ways.
(II) When 5 is in the unit's place then 0 should not be in the first place. So the toal numbers = 7! - 6! = 5040 - 720 = 4320
Total numbers = 5040 + 4320 = 9360

56. 1;

B	G	B	G	B	G	B	G	B	G	B

Number of ways = 61 × 5! = 720 × 120
= 86400

57. 2; Reqd probability = $^9C_5 \times \left(\frac{1}{2}\right)^5 \times \left(\frac{1}{2}\right)^4$

$= \frac{9!}{5! \times 4!} \times \frac{1}{32} \times \frac{1}{16} = \frac{63}{256}$

58. 4; All the 3 balls of different colours canbe selected in $^9C_1 \times {}^{11}C_1 \times {}^{12}C_1$
= 9 × 11 × 12 = 99 × 12 = 1188

59. 3; To arrange 7 boys and 7 girls around a circular table so that no two girls sit together, we need to arrange the boys and girls alternately.
∴ 7 boys in a circular table can be arranged in (7 - 1)! = 6! ways and the 7 girls will be arranged in a linear manner in 7! ways. Therefore, total number of ways = 6! × 7! = 3628800

60. 3; Number of queens = 4
∴ Number of ways to select one queen = $^4C_1$ = 4 ways
Number of spades =13
∴ Number of ways to select 1 spade = $^{13}C_1$ = 13 ways.
But, as we know, the set of spades and the set of queens have one card in common - the queen of spade. Therefore, it is included twice, and hence needs to be subtracted once while adding both the sets.
∴ Number of ways to select either a queen or a spade
$= n(Q \cup S) = n(Q) + n(S) - n(Q \cap S)$
= 4 + 13 - 1 = 16
Therefore, there are only 16 ways to select a queen or a spade.

61. 3; $n(S) = {}^{12}C_3 = \frac{\lfloor 12}{\lfloor 9 \times \lfloor 3}$

$= \frac{12 \times 11 \times 10}{3 \times 2} = 220$

All balls being of same colour can be arranged in $^3C_3 + {}^4C_3 + {}^5C_3$ ways.
∴ n(E) = $^3C_3 + {}^4C_3 + {}^5C_3$ = 1 + 4 + 10 = 15

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{15}{220} = \frac{3}{44}$

62. 1; There are seven letters, of which two are vowels (I, A). We consider these two letters as one {because they should always come

together}, so total number of letters will be 7 - 2 + 1 = 6

There is no repetition; hence the total number of arrangements = 6! × 2! = 1440

63. 4;

$$n(S) = {}^{12}C_3 = \frac{12!}{3! \times 9!} = \frac{12!}{3! \times 9!} = \frac{12 \times 11 \times 10}{3 \times 2} = 220$$

There is no green ball in the selection, ie 3 balls are selected from 3 red and 4 black balls in ${}^7C_3$ ways.

$$\therefore n(E) = \frac{7 \times 6 \times 5}{3 \times 2} = 35$$

$$\therefore P(E) = \frac{35}{220} = \frac{7}{44}$$

64. 4; We consider all three vowels as one.

So, the total number of letters will be 3 + 1 = 4, which can be arranged in 4! ways and three vowels can be arranged in 3! ways among themselves.

Total number of ways of arrangement

= 3! × 4! = 6 × 24 = 144

65. 5; The number of handshakes in the party is ${}^nc_2$ where n = number of persons.

$$\therefore \frac{n \times (n-1)}{2!} = 990$$

or, n(n - 1) = 1980

or, n^2 - n - 1980 = 0

or, n^2 - 45 + 44n - 1980 = 0

or, n(n - 45) + 44(n - 45) = 0

or, (n - 45) (n + 44) = 0

∴ n = 45, -44 (discard - ve value)

Hence, n = 45

66. 1; Let the first person A sit on any place.

Then there are 16 places for B to sit

So n(s) = 16. If there are exactly 5 persons between A and B then there can be only two possible seats for B, so n(E) = 2

$$P(E) = \frac{n(E)}{n(S)} = \frac{2}{16} = \frac{1}{8}$$

67. 2; The number of letters in the word TRIANGLE is 8. Hence (the number of ways) = 8!

= 1 × 2 × 3 × 4 × 5 × 6 × 7 × 8 = 40320

68. 2; Total number of balls = 6 + 5 = 11

Let S be the sample space. Then

$$n(S) = {}^{11}C_3 = \frac{9 \times 10 \times 11}{1 \times 2 \times 3} = 165$$

$$n(E) = {}^6C_1 = {}^6C_1 \times {}^5C_2 = 6 \times 10 = 60$$

$$p(E) = \frac{n(E)}{n(S)} = \frac{60}{165} = \frac{12}{33}$$

69. 1; Total number of balls = 6 + 3 + 2 = 11

Sample space n(S) = ${}^{11}C_2$

Now, the number of events n(E) = ${}^6C_2$

Required probability P(E) = $\frac{n(E)}{n(S)} = \frac{{}^6C_2}{{}^{11}C_2}$

$$= \frac{15}{55} = \frac{3}{11}$$

70. 2; At least one is yellow = 1 - P (None is yellow)

$$= 1 - \frac{{}^8C_2}{{}^{11}C_2} = 1 - \frac{28}{55} = \frac{27}{55}$$

71. 3; Total number of balls = 7 + 5 = 12

Sample space = n(S) = ${}^{12}C_2 = \frac{12!}{10!2!} = 66$

Number of events = n(E) = ${}^7C_2$

$$= \frac{7!}{5! \times 2!} = \frac{7 \times 6}{2} = 21$$

72. 4; Sample space n(S) = 6 × 6 = 36

For a sum of 7, events = (1, 6), (6, 1), (5, 2), (2, 5), (4, 3), (3, 4)

n(E) = 6

Probability P(E) = $\frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$

73. 3; There are two vowels I and U in the given words. They may be treated as a single letter. Thus, there are six letters in which there are two Ps. Hence, the number of arrangements

$$= \frac{6!}{2!} \times 2! = 360 \times 2 = 720$$

The two vowels can be arranged in 2! ways

74. 2; Let E and F be the event of getting diamond and a king respectively.

Then, $E \cap F$ is the event of getting a king of diamond.

∴ n(E) = 13, n(F) = 4 and $n(E \cap F) = 1$

So, P(E) = $\frac{13}{52} = \frac{1}{4}$, P(E) = $\frac{4}{52} = \frac{1}{13}$ and

$$P(E \cap F) = \frac{1}{52}$$

∴ P(a diamond or king)

$$= \frac{1}{4} + \frac{1}{13} - \frac{1}{52} = \frac{13+4-1}{52} = \frac{16}{52} = \frac{4}{13}$$

75. 2; n(S) = Number of ways of selecting 5 persons out of 12

$$= {}^{12}C_5 = \frac{12}{5!7!} = 792$$

n(E) = Number of ways of selecting 3 children out of 5, and 2 persons out of (4 + 3 =)

7 persons = ${}^5C_3 \times {}^7C_2 = \frac{\lfloor 5}{\lfloor 3 \lfloor 2} \times \frac{\lfloor 7}{\lfloor 5 \lfloor 2} = 210$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{210}{792} = \frac{35}{132}$$

76. 1; Reqd number of ways
= $({}^4C_1 \times {}^{12}C_3) + ({}^4C_2 \times {}^{12}C_2) + ({}^4C_3 \times {}^{12}C_1) + {}^4C_4$

$$= \left(4 \times \frac{12 \times 11 \times 10}{6}\right) + \left(\frac{4 \times 3}{2} \times \frac{12 \times 11}{2}\right) + (4 \times 12) + 1$$

= (4 × 22 × 10) + (6 × 6 × 11) + 48 + 1
= 880 + 396 + 48 + 1 = 1325

77. 3; Problem gets solved = Problem solved . by A but not by B + Problem solved by B but not by A + Problem solved by both = P(A and $\overline{B}$) + P($\overline{A}$ × and B) + P(A) P(B) Probability of

Problem solved by A = P(A) = $\frac{2}{3}$

Problem not solved byA = P($\overline{A}$) = $1 - \frac{2}{3} = \frac{1}{3}$

Problem solved by B = P(B) = $\frac{3}{4}$

Problem not solved by B = P($\overline{B}$) = $1 - \frac{3}{4} = \frac{1}{4}$

∴ Problem solved

$$= \left(\frac{2}{3} \times \frac{1}{4}\right) + \left(\frac{3}{4} \times \frac{1}{3}\right) + \left(\frac{2}{3} \times \frac{3}{4}\right)$$

$$= \frac{1}{6} + \frac{1}{4} + \frac{1}{2} = \frac{11}{12}$$

78. 1; Number of ways of selecting (3 consonants out of 9) and (2 vowels out of 5)

$$\Rightarrow {}^9C_3 \times {}^5C_2 = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} \times \frac{5 \times 4}{2 \times 1} = 840$$

Again, each group contains 5 letters.
Number of ways of arranging 5 letters among themselves = $\lfloor 5$ = 120
∴ Reqd number of ways = 120 × 840
= 100800

79. 4; Required number of ways

$$= \left({}^8C_4 \times {}^7C_2\right) + ({}^8C_5 \times {}^7C_1) + ({}^4C_6 \times {}^7C_0)$$

$$= \left(\frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2} \times \frac{7 \times 6}{1 \times 2}\right) + \left(\frac{8 \times 7 \times 6}{1 \times 2 \times 3} \times \frac{7}{1}\right) + \frac{8 \times 7}{2}$$

= (70 × 21) + (56 × 7) + 28 = 1470 + 392 + 28
= 1890

80. 5; Let the number of favourable events be n(E).
Then n(E) = (1, 1) (1, 2) (1, 3) (2, 2)
(2, 1) (3, 1)
∴ n(E) = 6
∴ Number of sample space = n(S) = 6 × 6

$$\therefore \text{Reqd probability} = P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

81. 3; Arrangement BGBGBG or GBGBGB
[B → Boys G → Girls]
In first arrangement, boys can sit in 3! ways and girls can sit in 3! ways.
∴ Total arrangements = 3 × 3 = 6 × 6 = 36 ways
In second arrangement, girls can sit in 3! ways and boys can sit in 3! ways
∴ Total arrangements = 3 × 3 = 6 × 6
= 36 ways
Hence, required number of ways
= (First + Second) = (36 + 36) = 72

82. 2; Here n(s) = 52
There are 13 cards of diamond (including one king) and there are 3 more kings. Let E be the event of getting a diamond, or a king:
Then, n(E) = (13 + 3) = 16

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{16}{52} = \frac{4}{13}$$

83. 2; We may have (1 Karnataka student and 3 non-Karnataka students) + (2 Karnataka students and 2 non-Karnataka students) + (3 Karnataka students and I non-Karnataka student) + (4 Karnataka students)
∴ Reqd number of ways = $({}^5C_1 \times {}^{10}C_3) + ({}^5C_2 \times {}^{10}C_2) + ({}^5C_3 \times {}^{10}C_1) + {}^5C_4$

$$= \left(5 \times \frac{10 \times 9 \times 8}{3 \times 2}\right) + \left(\frac{5 \times 4}{2} \times \frac{10 \times 9}{2}\right)$$

$$+ \left(\frac{5 \times 4}{2} \times 10\right) + 5 = 1155$$

Now, total number of ways = n(S) = ${}^{15}C_4$
= 1365

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{1155}{1365} = \frac{11}{13}$

84. 3; Total vacancies = 12
Number of candidates = 25
Number of local candidates = 5
Number of vacancies for local candidates = 3
Which can be filled in ${}^5C_3$ ways.
Remaining vacancies = 12 - 3 = 9
Remaining candidates = 25 - 5 = 20
These vacancies can be filled in ${}^{20}C_9$ ways.
$\therefore$ Reqd number of ways = ${}^5C_3 \times {}^{20}C_9$
= 1679600

85. 3; All the three are of the same colour means we choose 3 balls from 9 red or we select 3 balls from 11 green balls Or we select 3 balls from 12 yellow balls. So it can be done in
${}^9C_3 + {}^{11}C_3 + {}^{12}C_3 = 84 + 165 + 220 = 469$ ways

86. 2; All the three balls of different colours can be selected in
${}^9C_1 + {}^{11}C_1 + {}^{12}C_1 = 9 \times 11 \times 12 = 1188$ ways

87. 5; In the word TRANSPIRATION we treat the vowels AIAIO as one letter.
Thus we have TRNSPRTN (AIAIO)
Now we have to arrange 9 letters, out of which T, R, N appear twice and S, P appear once.
$\therefore$ Number of ways of arranging these letters

$= \frac{9!}{2! \times 2! \times 2!} = 45360$

Now AIAIO has 5 letters, in which A and I appear twice and O appears once.
$\therefore$ Number of ways of arranging these letters

$= \frac{5!}{2! \times 2!} = 30$

$\therefore$ Reqd number of ways = (45360 × 30)
= 1360800

88. 1; Total number of books = 8 + 7 + 6 = 21
Let E be the event that the picked book is neither in Hindi nor in Urdu or the event that the book picked is in English
n(E) = 7

$\therefore P(E) = \frac{7}{21}$

89. 1; Here, n(S) = 5!
n(E) = 4!2!

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{\lfloor 4 \rfloor 2}{\lfloor 5} = \frac{2}{5}$

90. 3;

G	B	G	B	G	B	G	B	G

$\therefore$ Number of ways = 5! × 4!
= 120 × 24 = 2880

91. 4; The word GENERAL contains 7 letters, and E comes twice.
$\therefore$ Reqd number of ways

$= \frac{7!}{2!} = 7 \times 6 \times 5 \times 4 \times 3 = 2520$

92. 1; In the word COMMENCEMENT we treat the vowels OEEE as one letter. Thus we have CMMNCMNT (OEEE). We have to arrange 9 letters out of which M occurs 3 times, C occurs 2 times, N occurs 2 times and rest are different letters. Now OEEE has 4 letters in which E occurs 3 times and O only once.
Number of ways of arranging these letters

$= \frac{9!}{3!2!2!} \times \frac{4!}{3!} = 60480$ ways

93. 4; In two throws of a dice n(S) = 6 × 6 = 36
Let E = Event of getting a sum 9
= {(3, 6), (4, 5), (5, 4), (6, 3)}

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{4}{36} = \frac{1}{9}$

94. 5; Let S be the sample space.
Then, n(S) = Number of ways of drawing 4 pens out of 17 pens

$= {}^{17}C_4 = \frac{17 \times 16 \times 15 \times 14}{4 \times 3 \times 2 \times 1} = 2380$

Let E = Event of drawing 4 pens of the same colour

$n(E) = {}^6C_4 + {}^4C_4 + {}^7C_4$

$= \frac{6 \times 5}{2} + 1 + \frac{7 \times 6 \times 5}{3 \times 2} = 15 + 1 + 35 = 51$

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{51}{2380}$

Event of drawing 4 pens such that they all are not of the same colour = 1 - P(E)

$= 1 - \frac{51}{2380} = \frac{2329}{2380}$

PROFIT AND LOSS

NATIONALISED BANKS & IBPS PO/MT/SO

1. Sri Ganesh bought 40 kgs of wheat at Rs. 12.50 per kg and 25 kgs of it at Rs. 15.10 per kg. He mixed them together. At what rate should he sell the mixture to earn 10% proflt ?
(1) Rs. 13.50 (2) Rs. 13.25
(3) Rs. 14.75 (4) Rs. 14.85
(5) None of these
(Syndicate Bank PO Exam. 10.10.2004)

2. The proflt earned after selling an article for Rs. 1,516 is the same as loss incurred after selling the article for Rs. 1,112. What is the cost price of the article ?
(1) Rs. 1,314 (2) Rs. 1,343
(3) Rs. 1,414 (4) Rs. 1,434
(5) None of these
(Orientai Bank of Commerce PO Exam. 21.12.2008)

3. The proflt earned after selling an article for Rs. 1,754 is the same as loss incurred after selling the article for Rs. 1,492. What is the cost price of the article ?
(1) Rs. 1,623 (2) Rs. 1,523
(3) Rs. 1,689 (4) Rs. 1,589
(5) None of these
(Canara Bank PO Exam. 15.03.2009)

4. Giridhar purchased 100 sarees @ Rs. 450 per piece. While selling he offered 10% discount on the labelled price and earned a proflt of 20%. What was the labelled price of each saree ?
(1) Rs. 540 (2) Rs. 650
(3) Rs. 590 (4) Rs. 600
(5) None of these
(PNB Specialist Officer's Exam. 16.08.2009)

5. Vinod makes a proflt of Rs. 110 if he sells a certain number of pencils he has at the price of Rs. 2.5 per pencil and incurs a loss of Rs. 55 if he sells the same number of pencils for Rs. 1.75 per pencil. How many pencils does Vinod have ?
(1) 220
(2) 240
(3) 200
(4) Cannot be determined
(5) None of these
(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

6. A shopkeeper bought 30 kg. of wheat at the rate of Rs.45 per kg. He sold forty percent of the total quantity at the rate of Rs. 50 per kg. **Approximately,** at what price per kg. should he sell the remaining quantity to make 25 per cent overall proflt ?
(1) Rs. 54 (2) Rs. 52
(3) Rs. 50 (4) Rs. 60
(5) Rs. 56
(Allahabad Bank PO Exam. 21.02.2010)

7. Meenal purchased a car for Rs. 2,50,000 and sold it for Rs. 3,48,000. What is the per cent profit she made on the car ?
(1) 40 (2) 39.2
(3) 38.4 (4) 38
(5) None of these
(Corporation Bank PO Exam. 09.05.2010)

8. Rajni purchased a mobile phone and a refrigerator for Rs. 12,000 and Rs. 10,000 respectively. She sold the refrigerator at a loss of 12 per cent and mobile phone at a proflt of 8 per cent. What is her overall loss/ profit ?
(1) Loss of Rs. 280
(2) Profit of Rs. 2,160
(3) Loss of Rs. 240
(4) Profit of Rs.2,060
(5) None of these
(Punjab & Sind Bank PO Exam. 16.05.2010)

9. A shopkeeper sells notebooks at the rate of Rs. 45 each and earns a commission of 4%. He also sells pencil box at the rate of Rs. 80 each and earns a commission of 20%. How much amount of commission will he earn in two weeks if he sells 10 notebooks and 6 pencil boxes a day ?
(1) Rs. 1,956 (2) Rs. 1,586
(3) Rs. 1,496 (4) Rs. 1,596
(5) None of these
(Central Bank Of India PO Exam. 25.07.2010)

10. A man sold a wristwatch for Rs. 2,400 at a loss of twenty five percent. At what rate should he have sold the wristwatch to earn a proflt of twenty five per cent ?
(1) Rs. 3,600
(2) Rs. 4,000
(3) Rs. 3,500
(4) Rs. 3,800
(5) None of these
(Bank Of India PO Exam. 31.10.2010)

11. What profit/loss per cent did Ravi earn if he purchased an item of Rs 5,600 and sold it at three-fourth of its cost price ?
(1) Loss of 20 per cent
(2) Gain of 25 per cent
(3) Neither gain nor loss
(4) Loss of 15 per cent
(5) None of these

(Orientai Bank Of Commerce PO Exam. 26.12.2010 (Ist Sitting)

12. Prasad sold his work tools for Rs 1,850 and earned a profit of 25 per cent. At what price did Prasad buy the work tools ?
(1) Rs 1,360 (2) Rs 1,300
(3) Rs 1,240 (4) Rs 1,480
(5) None of these

(Indian Bank PO Exam. 02.01.2011)

13. Rehaan purchased a bike for Rs 54,000. He sold it at a loss of 8 per cent. With that money he again purchased another bike and sold it at a proflt of 10 per cent. What is his overall loss/profit ?
(1) Loss of Rs 657
(2) Profit of Rs 567
(3) Loss of Rs 648
(4) Profit of Rs 648
(5) None of these

(Corporation Bank PO Exam. 16.01.2011)

14. Meera purchased an item for Rs 62,000 and sold it at loss of 25 percent. With that amount she purchased another item and sold it at a gain of 30 percent. What was her overall gain/loss ?
(1) Los of Rs 560
(2) Profit of Rs 1560
(3) Loss of Rs 1550
(4) Profit of Rs 1550
(5) None of these

(UCO Bank PO Exam. 30.01.2011)

15. Seema purchased an item for Rs 9,600 and sold it for a loss of 5 percent. From that money she purchased another item and sold it for a gain of 5 percent. What is her overall gain/loss ?
(1) Loss of Rs 36
(2) Profit of Rs 24
(3) Loss of Rs 54
(4) Profit of Rs 36
(5) None of these

(Bank Of Baroda PO Exam. 13.03.2011)

16. Kamya purchased an item of Rs 46,000 and sold it at a loss of 12 percent. With that amount she purchased another item and sold it at a gain of 12 percent. What was her overall gain/loss ?
(1) Loss of Rs 662.40
(2) Profit of Rs 662.40
(3) Loss of Rs 642.80
(4) Profit of Rs 642.80
(5) None of these

(Allahabad Bank PO Exam. 17.04.2011)

17. An article was purchased for Rs 78,350. Its price was marked up by 30%. It was sold at a discount of 20% on the marked up price. What was the profit percent on the cost price ?
(1) 4 (2) 7
(3) 5 (4) 3
(5) 6

(IBPS Bank PO/MT CWE 17.06.2012)

18. A man sells calculator at the rate of Rs 250 each which includes a profit of 14 per cent. What amount of profit will he eam in 19 days if he sells seven calculators per day ?
(1) Rs 4665 (2) Rs 4565
(3) Rs 4545 (4) Rs 4655
(5) None of these

(IDBI Bank Officer Exam. 16.09.2012)

19. Manish bought 25 kg of rice at Rs. 32 per kg and 15 kg of rice at Rs. 36 per kg. What profit did he get when he mixed the two varieties together and sold it at Rs. 40.20 per kg ?
(1) 25% (2) 40%
(3) 30% (4) 20%
(5) None of these

20. A shopkeeper sells two watches for Rs. 308 each. On one he gets 12% proflt and on the other 12% loss. His profit or loss in the entire transaction was
(1) $1\frac{11}{25}\%$ loss
(2) $1\frac{11}{25}\%$ gain
(3) $3\frac{2}{25}\%$ loss
(4) $3\frac{2}{25}\%$ gain
(5) None of these

(IBPS Bank PO/MT CWE-III 26.10.2013)

21. An article was sold for Rs. 5220 at a loss of 42% of the cost price. What will be the selling price of the article for a proflt of 42% ?
(1) Rs. 12580 (2) Rs. 17280

(3) Rs. 12780 (4) Rs. 15280
(5) None of these

(Corporation Bank Specialist Officer (Marketing) Exam. 22.12.2014)

SBI PO EXAMS

1. Mr. X, a businessman had the income in the year 1995, such that he earned a proflt of 20% on his investment in the business. In the year 1996 his investment was less by Rs. 5,000 but still had the same income (Income = Investment + Profit) as that in 1995. Thus the per cent profit earned in 1996 increased by 6%. What was his investment in 1995 ?
(1) Rs. 1,02,000
(2) Rs. 1,50,500
(3) Rs. 1,05,000
(4) Data inadequate
(5) None of these

(SBI Banks PO Exam. 11.02.2001)

2. Mr. Shivkumar started a businees investing Rs. 25,000 in 1996. In 1997 he invested an additional amount of Rs. 10,000 and Mr. Rakesh joined him with an amount of Rs. 35,000. In 1998 Mr. Shivkumar invested another additional amount of Rs. 10,000 and Mr. Suresh joined them with an amount of Rs. 35,000. What will be Rakesh's share in the profit of Rs. 1,50,000 earned at the end of three years from the start of the business in 1996 ?
(1) Rs. 70,000 (2) Rs. 50,000
(3) Rs. 45,000 (4) Rs. 75,000
(5) None of these

(SBI Banks PO Exam. 11.02.2001)

3. If on selling 12 notebook any seller makes a profit equal to the selling price of 4 notebooks. What is his per cent profit ?
(1) 50 (2) 25
(3) $16\frac{2}{3}$ (4) Data inadequate
(5) None of these

(SBI PO Exam. 26.11.2006)

4. The proflt after selling a pair trousers for Rs. 863 is the same as the loss incurred after selling the same pair of trousers for Rs.631. What is the cost price of the pair of trousers ?
(1) Rs. 750
(2) Rs. 800
(3) Rs. 763
(4) Cannot be determined
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

5. A trader sells 145 metres of cloths for Rs. 12,325 at the profit of Rs. 10 per metre of cloth. What is cost price of 1 metre of cloth ?
(1) Rs. 65 (2) Rs. 75
(3) Rs. 95 (4) Rs. 85
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

6. The profit earned after selling an article for Rs. 878 is the same as loss incurred after selling the article for Rs. 636. What is the cost price of the article ?
(1) Rs. 797 (2) Rs. 787
(3) Rs. 767 (4) Rs. 757
(5) None of these

(SBI PO Preliminary (Tire-I Exam. 27.07.20O8)

7. The value of machine depreciates at the rate of 12 per cent per annum. It was purchased three years ago. Its present value is Rs. 29644.032, what was the purchase price of the machine ?
(1) Rs. 48700 (2) Rs. 43500
(3) Rs. 38900 (4) Rs. 39000
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

8. Proflt earned by an organization is distributed among officers and clerks in the ratio of 5 : 3 respectively. If the number of officers is 45 and the number of clerks is 80 and the amount received by each officer is Rs 25,000, what was the total amount of proflt earned ?
(1) Rs 22 lakhs
(2) Rs 18.25 lakhs
(3) Rs 18 lakhs
(4) Rs 23.25 lakhs
(5) None of these

(SBI Associate Banks PO Exam. 07.08.2011)

9. A shopkeeper labelled the price of his articles so as to earn a profit of 30% on the cost price. He then sold the articles by offering a discount of 10% on the labelled price. What is the actual percent profit earned in the deal ?
(1) 18%
(2) 15%
(3) 20%
(4) Cannot be determined
(5) None of these

(SBI Associate Banks PO Exam. 07.08.2011)

INSURANCE EXAMS

1. If a shirt costing Rs. 385 is sold at a loss of 5% of the cost price, flnd the selling price.
(1) Rs. 364 (2) Rs. 364.74
(3) Rs. 365 (4) Rs. 365.75
(United India Insurance Co. AAO Exam. 21.04.2002)

2. If the cost price of 24 articles is equal to the selling price of 21 articles, what is the percentage gain or loss ?
(1) $14\frac{2}{7}$% gain
(2) $14\frac{2}{7}$% loss
(3) 12.5% loss
(4) 12.5% gain
(United India Insurance Co. AAO Exam. 21.04.2002)

3. A merchant has 1000 kg of sugar, part of which he sells at 8% proflt and the rest at 18% profit. He gains 14% on the whole. The quantity (in kg.) sold at 18% profit is :
(1) 560 (2) 600
(3) 400 (4) 640
(United India Insurance Co. (AAO) Exam. 11.03.2007)

4. A shopkeeper sold a T.V. set for Rs 17,940/-, with a discount of 8% and gained 19.6% If no discount is allowed, what will be his gain per cent ?
(1) 25%
(2) 26.4%
(3) 24.8%
(4) Cannot be determined
(5) None of these
(LIC Assistant Administrative Offlcer Exam. 2008)

5. The income of a broker remains unchanged though the rate of commission is increased from 4 per cent to 5 per cent. The percentage of slump in business is :
(1) 10 per cent
(2) 15 per cent
(3) 20 per cent
(4) 30 per cent
(LIC Assistant Administrative Offlcer (AAO) Exam. 07.06.2009)

6. A watch is sold at a proflt of 20%. If both the cost price and the selling price of the watch are decreased by Rs. 100, the profit would be 5% more. Originai cost price of the watch is
(1) Rs. 450 (2) Rs. 500
(3) Rs. 550 (4) Rs. 600
(New India Assurance AO Exam. 25.10.2009)

7. A vendor sells calculators at the rate of Rs 250 each and eams a commission of 20% on each. He also sells pens at the rate of Rs 50 each and earns a commission of 10% on each. How much amount of commission will he earn in three days if he sells 10 calcula-tors and 5 pens a day ?
(1) Rs 1,575 (2) Rs 1,445
(3) Rs 1,550 (4) Rs 1,450
(5) None of these
(United India Insurance AO Exam. 27.03.2011)

8. A person bought 864 articles and sold 800 of them for the price he paid for 864 articles. He sold the remaining articles at the same price per article as the other 800. The percentage gain on the entire transaction is
(1) 7.5% (2) 8%
(3) 8.5% (4) 9%
(New India Insurance AAO Exam. 22.05.2011)

9. The currencies in countries M and N are denoted by *m* and *n* respectively. The exchange rate in 1990 was l *m*, 0.6 *n* the price level in 2006 in M and N are 150 and 400 respectively with 1990 as a base of 100. The exchange rate in 2006 based solely on the purchasing power parity consideration is 1 *m* :
(1) 0.225 *n* (2) 0.625 *n*
(3) 1.6 *n* (4) 3.6 *n*
(General Insurance Corporation AAO Exam. 11.12.2011)

10. Prof. Chatterjee bought a car and got 15% of its original price as a dealer's discount. He then sold it at 20% profit on his purchase price. What percentage profit did he get on the original price ?
(1) 2% (2) 12%
(3) 5% (4) 17%
(General Insurance Corporation AAO Exam. 11.12.2011)

11. Cost of 4 fans and 3 blowers is Rs 16,500. Also cost of 2 fans, 2 tables and 2 blowers is Rs 12,000. Cost of one table is Rs 1,000. What is the cost of 3 fans and one blower ?
(1) Rs 8,000
(2) Rs 7,500
(3) Rs 8,500
(4) Cannot be determined
(5) None of these

(United India Insurance AO Exam. 27.03.2011)

12. By selling 18 chocolates, a vendor loses the selling price of 2 chocolates. Find his loss percent.

(1) 9% (2) 10%
(3) 11% (4) 12%

(United India Insurance AAO Exam. 03.06.2012)

13. The profit earned after selling an article for Rs. 1,516 is the same as loss incurred after selling the article for Rs. 1,112. What is the cost price of the article ?

(1) Rs. 1,314 (2) Rs. 1,343
(3) Rs. 1,414 (4) Rs. 1,434
(5) None of these

(United India Insurance AO Exam. 26.05.2013

14. If the cost price of 15 articles is equal to the selling price of 12 articles, find gain per cent.

(1) 20 (2) 25
(3) 18 . (4) 21

(NICL (GIC) AO Exam. 08.09.2013 (Paper-I)

15. By selling a table for Rs. 350 instead of Rs. 400, loss percent increases by 5%. The cost price of the table is:

(1) Rs. 435 (2) Rs. 417.50
(3) Rs. 1000 (4) Rs. 1050

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

16. Ram bought 1600 eggs at Rs. 3.75 per dozen. He sold 900 of them at 2 for Re 1 and the remaining at 5 for Rs. 2. His gain per cent is :

(1) 40% (2) 45%
(3) 42% (4) 46%

(NICL (GIC) AO (Finance) Exam. 15.12.2013)

17. If the manufacturer gains 10%, the Wholesale dealer 15% and the retailer 25%, then the cost of production of an article whose retail price is Rs 1265, is:

(1) Rs. 632.50 (2) Rs. 800
(3) Rs. 814 (4) Rs. 834.34

(NICL (GIC) AO (Finance) Exam. 15.12.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (4) **2.** (1)
3. (1) **4.** (4)
5. (1) **6.** (4)
7. (2) **8.** (3)
9. (5) **10.** (2)
11. (5) **12.** (4)
13 (4) **14.** (3)
15. (5) **16.** (1)
17. (1) **18.** (5)
19. (4) **20.** (1)
21. (3)

SBI PO EXAMS

1. (3) **2.** (2)
3. (1) **4.** (5)
5. (2) **6.** (4)
7. (2) **8.** (4)
9. (5)

INSURANCE EXAMS

1. (4) **2.** (1)
3. (2) **4.** (5)
5. (3) **6.** (2)
7. (1) **8.** (2)
9. (1) **10.** (1)
11. (1) **12.** (2)
13. (1) **14.** (2)
15. (3) **16.** (4)
17. (2)

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (4) Total weight of the mixture
= 40 + 25 = 65 kg
Total cost price of wheat
= Rs. (40 x 12.50 + 25 × 15.10)
= Rs. (500 + 377.50) = Rs. 877.50
Total selling price of wheat

$$= \text{Rs. } \frac{877.50 \times 110}{100}$$

= Rs. 965.25

$$\therefore \text{SP per kg} = \text{Rs. } \frac{965.25}{65}$$

= Rs. 14.85

2. (1) Let the cost price of the artice be Rs. x.
According to the question,
1516 - x = x - 1112
$\Rightarrow 2x = 1516 + 1112 = 2628$

$$\Rightarrow x = \frac{2628}{2} = \text{Rs. } 1314$$

3. (1) Let the CP of the article be Rs. x
According to the question,
1754 - x = x - 1492
$\Rightarrow 2x = 1754 + 1492 = 3246$

$$\Rightarrow x = \frac{3246}{2} = \text{Rs. } 1623$$

4. (4) Let the labelled price of each sari be Rs. x.
According to the question,

$$90\% \text{ of } x = \frac{120 \times 450}{100}$$

$$\Rightarrow \frac{90 \times x}{100} = \frac{120 \times 450}{100}$$

$$\Rightarrow x = \frac{120 \times 450}{90} = \text{Rs. } 600$$

5. (1) Let Vinod have x pencils.
$\therefore 2.5 \times x - 1.75 \times x = 110 + 55$
$\Rightarrow 0.75 \times x = 165$

$$\Rightarrow x = \frac{165}{0.75} = \text{Rs } 220$$

6. (4) Cost price of 30 kg of wheat
= 30 × 45
= Rs. 1350
Total SP for an overall profit of

$$25\% = \frac{1350 \times 125}{100} = \text{Rs } 1687.5$$

$$\text{Sp of } \left(\frac{30 \times 40}{100}\right) = 12 \text{ kg of wheat}$$

= 12 × 50 = Rs. 600
Expected SP of 18kg of remaining wheat
= 1687.5 - 600 = Rs. 1087.5
Required selling price per kg

$$= \frac{1087.5}{18} \approx \text{Rs. } 60$$

7. (2) <u>Tricky Approach</u>
Gain per cent

$$= \frac{348000 - 250000}{250000} \times 100$$

= 39.2%

8. (3) Total CP
= Rs. (12000 + 10000) = Rs. 22000

$$\text{Total S.P.} = \left(\frac{12000 \times 108}{100} + \frac{10000 \times 88}{100}\right)$$

= Rs. (12960 + 8800) = Rs. 21760
$\therefore$ Loss = Rs. (22000 - 21760) = Rs. 240

9. (5) Required amount of commission

$= \text{Rs.} \left(\frac{10\times45\times4}{100}+\frac{80\times20\times6}{100}\right)$

= Rs. (18 + 96) = Rs. 114

10. (2) CP. of wristwatch

$= \frac{100}{100-\text{loss}\%} \times \text{S.P}$

$= \text{Rs.} \left(\frac{100}{75}\times2400\right)$

= Rs. 3200

∴ Required S.P. of wristwatch

$= \text{Rs.} \left(\frac{125}{100}\times3200\right)$

= Rs. 4000

11. (5) S.P. = $5600 \times \frac{3}{4}$

= Rs. 4200

Loss = 5600 - 4200

= Rs. 1400

∴ Loss percentage

$= \frac{\text{Loss}}{\text{Cost price}} \times 100$

$= \frac{1400}{5600} \times 100 = 25\%$

12. (4) CP = $\frac{100}{100+\text{Profit per cent}} \times \text{S.P}$

$= \frac{100}{125} \times 1850 = \text{Rs. } 1480$

13. (4) First S.P. = $\frac{54000\times92}{100}$ = Rs 49680

Second SP = $49680 \times \frac{110}{100}$ = Rs 54648

∴ Profit = 54648 - 54000 = Rs 648

14. (3) Last selling price

$= 62000 \times \frac{75}{100} \times \frac{130}{100}$ = Rs. 60450

∴ Loss = Rs. (62000 - 60450) = Rs. 1550

15. (5) First S.P. = $\frac{9600\times95}{100}$ = 9120

Second S.P. = $\frac{9120\times105}{100}$ = Rs. 9576

∴ Loss = Rs. (9600 - 9576) = Rs. 24

16. (1) First S.P. = $\frac{46000\times88}{100}$ = Rs. 40480

Second S.P. = $\frac{40480\times112}{100}$ = Rs. 45337.6

∴ Loss = Rs. (46000 - 45337.6) = Rs. 662.4

17. (1) Gain per cent

$= \left(30-20-\frac{30\times20}{100}\right)\% = 4\%$

18. (5) Profit on one calculator

$= \text{Rs.} \left(250\times\frac{14}{114}\right)$

∴ Total profit = $\frac{7\times19\times250\times14}{114}$ = Rs. 4083.33

19. (4) CP. of 40 kg of mixture

= Rs [(25 × 32) + (15 × 36)]

= Rs (800 + 540) = Rs 1340

S.P. of 40 kg of mixture

= Rs (40 × 40.2) = Rs 1608

Profit = Rs (1608 - 1340) = Rs 268

Profit % = $\frac{268}{1340} \times 100 = 20\%$

20. (1) Loss per cent

$= \left(\frac{\text{Common gain or less}}{10}\right)^2 \%$

$= \left(\frac{12}{10}\right)^2 \% = \frac{36}{25}\% = 1\frac{11}{25}\%$

21. (3) CP. of article

$= 5220 \times \frac{100}{(100-42)} = \frac{5220\times100}{58}$ = Rs. 9000

∴ Required S.P. = $\frac{9000\times142}{100}$ = Rs 12780

SBI PO EXAMS

1. (3) Let the investment in 1995 be Rs. x

∴ Income in 1995 with 20% profit = 1.20x

Also the income in 1996 = 1.20x

Investment in 1996 = Rs. (x - 5000)

Profit in 1996 = (20 + 6)% = 26%

Income in 1996 with 26% profit

= 1.26 (x - 5000)

∴ 1.26 (x - 5000) = 1.20x

0.06 x = 1.26 × 5000

$x = \frac{6300}{0.06}$

x = Rs. 105000

2. (2) Shivkumar's equivalent capital

= (25000 × 3) + (10000 × 2) + (10000 × 1)

= 105000

Rakesh's equivalent capital

= 35000 × 2 = 70000
Suresh's equivalent capital
= 35000 × 1 = 35000
Total profit of Rs. 150000 will be distributed in the ratio of theii equivalent capitate.
Shivkumar : Rakesh : Suresh
= 105 : 70 : 35 = 3 : 2 : 1
Rakesh's share in the total profit

$= \frac{2}{3+2+1} \times 150000$ = Rs. 50000

3. (1) Profit = Selling price of 4 notebooks cost price = selling price of(12 - 8) = 4 notebooks.

$\therefore$ % proflt = $\frac{4}{8} \times 100 = 50$

4. (5) Let the cost price of the trouser be Rs. *x*.
According to the question,
863 - *x* = *x* - 631
$\Rightarrow$ 2*x* = 863 + 631 = 1494

$\Rightarrow x = \frac{1494}{2}$ = Rs. 747

5. (2) Cost price of 145m long cloth
= Rs. (12325 - 145 × 10) = Rs. 10875

$\therefore$ CP of 1 metre long cloth = $\frac{10875}{145}$ = Rs. 75

6. (4) let the CP. of the article be Rs. *x*.
According to the question,
878 - *x* = *x* - 636
$\Rightarrow$ 2*x* = 878 + 636 = 1514

$\Rightarrow x = \frac{1514}{2}$ = Rs. 757

7. (2) $29644.032 = P\left(1-\frac{12}{100}\right)^3$

$\Rightarrow 29644.032 = P \times (0.88)^3$

$\Rightarrow P = \frac{29644.032}{0.88 \times 0.88 \times 0.88}$ = Rs. 43500

8. (4) Profit received by each officer
= Rs. 25000
$\therefore$ Proflt received by each clerk 3

$= \frac{3}{5} \times 25000$ = Rs. 15000

$\therefore$ Total earned profit
= Rs. (45 × 25000 + 80 × 15000)
= Rs. (1125000 + 1200000) = Rs. 23.25 lakh

9. (5) Let the cost price of the article be Rs. 100.
$\therefore$ Marked price = Rs.130

$\therefore$ S.P. of the article = $\frac{130 \times 90}{100}$

= Rs. 117
$\therefore$ Actual gain per cent = 17

INSURANCE EXAMS

1. (4) Required answer

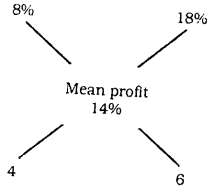
$= \frac{95}{100} \times 385$ = Rs. 365.75

2. (1) Percentage gain

$= \frac{(24-21)}{21} \times 100 = \frac{100}{7}\%$

$= 14\frac{2}{7}\%$

3. (2)
Profit on first part Profit on second part



Ratio of first and second parts
= 4 : 6 = 2 : 3
$\therefore$ Quantity of 2nd part

$= \left(\frac{3}{5} \times 1000\right)$ kg = 600 kg

4. (5) $\because$ If Rs. 92 is the S.P., the Marked Price is Rs. 100.
$\therefore$ If Rs. 17,940 is the S.P., the Marked Price

$= \frac{100}{92} \times 17940$ =Rs. 19,500

Now, Cost Price

$= 17940 \times \frac{100}{100+19.6}$

$= 17940 \times \frac{100}{119.6}$ = Rs. 15,000

If there is no discount,
S.P. = Rs. 19,500

$\therefore$ % Profit = $\frac{19,500-15000}{15000} \times 100$

$= \frac{4500}{15000} \times 100 = 30\%$

5. (3) Let the business value changes from Rs. *x* to Rs. *y*.
$\therefore$ 4% of *x* = 5% of *y*

$\Rightarrow \frac{4x}{100} = \frac{5y}{100}$

$\Rightarrow y = \frac{4}{5}x$

Change in business

$= \left(x - \frac{4}{5}x\right) = \frac{x}{5}$

$\therefore$ Percentage slump in business

$= \frac{x}{5} \times \frac{1}{x} \times 100 = 20\%$

6. (2) Let the CP of the watch be Rs. x

$\therefore$ First SP $= \frac{120x}{100} =$ Rs. $\frac{6x}{5}$

Case II,
CP = Rs. (x - 100)

SP = Rs. $\left(\frac{6x}{5} - 100\right)$

Profit $= \left(\frac{6x}{5} - x\right) =$ Rs. $\frac{x}{5}$

$\therefore \frac{\frac{x}{5}}{x - 100} \times 100 = 25$

$\Rightarrow 25x - 2500 = 20x$
$= 5x = 2500$

$x = \frac{2500}{5} =$ Rs. 500

7. (1) Commission on 1 calculator

$= \frac{250 \times 20}{100} =$ Rs. 50

Commission on 1 pen

$= \frac{50 \times 10}{100} =$ Rs. 5

$\therefore$ Required amount of commissior
= Rs. 3 × (10 × 50 + 5 × 5) = Rs. 1575

8. (2) Profit percent

$= \frac{64}{800} \times 100 = 8\%$

9. (1) $\frac{1m}{150} = \frac{0.6n}{400}$

$\Rightarrow 1m = \frac{0.6}{400} \times 150n = 0.225n$

10. (1) Gain percent

$= \left(20 - 15 - \frac{20 \times 15}{100}\right)\% = 2\%$

11. (1) Let C.P. of 1 fan = Rs. x
CP of 1 blower = Rs. y
CP of 1 table = Rs. 1000

$\therefore 4x + 3y = 16500$ (i)
$2x + 2y + 2 \times 1000 = 12000$
$\Rightarrow 2x + 2y = 12000 - 2000$
$= 10000$
$\Rightarrow x + y = 5000$ (ii)
Equation (ii) × 4 - (i),
$y = 20000 - 16500 = 3500$
$\therefore x = 5000 - 3500 = 1500$
$\therefore 3x + y = 3 \times 1500 + 3500$
= Rs. 8000

12. (2) If the S.P. of 1 chocolate be Re. 1, then,
S.P. of 18 chocolates = Rs. 18
C.P. = 18 + 2 = Rs. 20

$\therefore$ Loss percent $= \frac{2}{20} \times 100 = 10\%$

13. (1) Let the cost price of the artice be Rs. x.
According to the question,
$1516 - x = x - 1112$
$\Rightarrow 2x = 1516 + 1112 = 2628$

$\Rightarrow x = \frac{2628}{2} =$ Rs. 1314

14. (2) Gain percent

$= \frac{15 - 12}{12} \times 100 = 25\%$

15. (3) If the C.P. of table be Rs. x,
then

$\frac{400 - 350}{x} \times 100 = 5$

$\Rightarrow 50 \times 100 = 5x$
$\Rightarrow x =$ Rs. 1000

16. (4) $\because$ C.P. of 12 eggs = Rs. 3.75
$\therefore$ C.P. of 1600 eggs

$= \frac{3.75 \times 1600}{12} =$ Rs. 500

S.P. of 900 eggs $= \frac{1}{2} \times 900 =$ Rs. 450

S.P. of remaining 700 eggs

$= \frac{2}{5} \times 700 =$ Rs. 280

Total S.P. = 450 + 280 = Rs. 730
Gain = 730 - 500 = Rs. 230

Gain per cent $= \frac{230}{500} \times 100 = 46\%$

17. (2) C.P. of article

$= 1265 \times \frac{100}{110} \times \frac{100}{115} \times \frac{100}{125}$

= Rs. 800

MODEL EXERCISES

1. By selling 12 marbles for a rupee, a shopkeeper loses 20%. In order to gain 20% in the transaction, he should sell the marbles at the rate of how many marbles for a rupee ?
(1) 8 (2) 6
(3) 4 (4) 3
(5) None of these

2. In a certain store, the profit is 320% of the cost. If the cost increase by 25% but the selling price remains Constant, approximately what percentage of the selling price is the profit ?
(1) 30% (2) 70%
(3) 100% (4) 250%
(5) None of these

3. A horse and a carriage together cost Rs 8000. If by selling the horse at a profit of 10% and the carriage at a loss of 10% a total profit of 2.5% is made, then what is the cost price of the horse ?
(1) Rs 3000 (2) Rs 3500
(3) Rs 4000 (4) Rs 5000
(5) None of these

4. A piece of cloth costs Rs 35. If the length of the piece would have been 4 m longer and each metre costs Rs 1 less, the cost would have remained unchanged. How long is the piece ?
(1) 14 m (2) 10 m
(3) 12 m (4) 9 m
(5) None of these

5. A house costs C rupees. Later it was sold for a profit of 25%. What is the capital gains tax if it is 50% ofthe profit?
(1) $\frac{C}{24}$ (2) $\frac{C}{8}$
(3) $\frac{C}{4}$ (4) $\frac{C}{2}$
(5) None of these

6. If selling price is doubled, the profit triples. Find the profit per cent.
(1) $66\frac{2}{3}\%$ (2) 100%
(3) $105\frac{1}{3}\%$ (4) 120%
(5) None of these

7. Vishal goes to a shop to buy a radio costing Rs 2568. The rate of sales tax is 7%. He tells the shopkeeper to reduce the price of the radio to such an extent that he has to pay Rs 2568, inclusive of sales tax. Find the reduction needed in the price of the radio.
(1) Rs 179.76 (2) Rs 170
(3) Rs 168 (4) Rs 169
(5) None of these

8. *A* bought 4 bottles of beer and *B* bought one bottle of lager, lager per bottle costing twice that of the beer. C bought nothing but paid Rs 50, for his share of the drink which they mixed together and shared equally. If C's Rs 50 covered his share, then what is the cost of the lager ?
(1) Rs 50 (2) Rs 75
(3) Rs 30 (4) Rs 46
(5) Noncof these

9. Ravi sells an article at a gain of $12\frac{1}{2}\%$. if he had sold it at Rs 22.50 more, he would gained 25%. The cost price of the article is
(1) Rs 162 (2) Rs 140
(3) Rs 196 (4) Rs 180
(5) None of these

10. The sale price of an article including the sales tax is Rs 616. The rate of sales tax is 10%. If the shopkeeper has made a profit of 12%, then the cost price of the article is
(1) Rs 500 (2) Rs 515
(3) Rs 550 (4) Rs 600
(5) None of these

11. A reduction of 20% in the price of sugar enables a purchaser to obtain $2\frac{1}{2}$ kg more for Rs 160. Find the original price per kg of sugar.
(1) Rs 12 (2) Rs 15
(3) Rs 16 (4) Rs 18
(5) None of these

12. If the cost of 12 pencils is equal to the selling price of 10 pencils, the profit per cent in the transactions is
(1) $16\frac{2}{3}\%$ (2) 18%
(3) 20% (4) 25%
(5) None of these

13. Two motor cars were sold for Rs 9900 each, gaining 10% on one and losing 10% on the other. The gain or loss per cent in the whole transaction is
(1) neither loss no gain
(2) 1% gain

(3) $\frac{100}{99}$% profit
(4) 1% loss
(5) None of these

14. By selling 33 m of cloth, a shopkeeper gains the price of 11 m of cloth. His gain per cent is :
(1) 7% (2) 50%
(3) 20% (4) 22%
(5) None of these

15. On selling a pen at 5% loss and a book at 15% gain, Karim gains Rs 7. If he sells the pen at 5% gain and the book at 10% gain, the he gains Rs 13. The actual price of the book is
(1) Rs 100 (2) Rs 80
(3) Rs 10 (4) Rs 400
(5) None of these

16. My friend collects antique stamps. She purchased two, but found that she needed to raise money urgently So, she sold them for Rs 8000 each. On one she made 20% profit and on the others she lost 20%. How much did she gain or loss in the entire transaction ?
(1) No loss / profit
(2) Rs 667 loss
(3) Rs 667 profit
(4) Rs 658 profit
(5) None of these

17. A man sells an article at 5% profit. If he had bought it at 5% less and sold it for Rs 1 less, he would have gained 10%. The cost price of the article is
(1) Rs 200 (2) Rs 150
(3) Rs 250 (4) Rs 240
(5) None of these

18. A dairyman pays Rs 6.4 per litre of milk. He adds water and sells the mixture at Rs 8 per litre, thereby making 37.5% profit. The proportion of water to milk received by the customers is
(1) 1 : 15 (2) 1 : 10
(3) 1 : 20 (4) 1 : 20
(5) None of these

19. The cost price of 20 articles is the same as the selling price of *x* articles. If the profit is 25%, then the value of *x* is
(1) 25 (2) 18
(3) 16 (4) 15
(5) None of these

20. A person had deposited Rs 13200 in a bank which pays 14% interest. He withdraws the money and invests in Rs 100 stock at Rs 110 which pays a dividend of 15%. How much does he gain or loss ?
(1) Loss Rs 48 (2) Gain Rs 48
(3) Loss Rs 132 (4) Gain Rs 132
(5) None of these

21. A video magazine distributor made 3500 copies of the March issue of the magazine at a cost of Rs 350000. He gave 500 cassettes free to some key video libraries. He also allowed a 25% discount on the market price of the cassette and gave one extra cassette free with every 29 cassettes bought at a time. In this manner, he was able to sell all the 3500 cassettes that were produced. If the market price of a cassette was Rs150, then what is his gain or loss per cent for the March issue of video magazine ?
(1) 25% loss (2) 10% gain
(3) 40% gain (4) 6.8% loss
(5) None of these

22. A firm of ready made garments makes both men's and women's shirts. Its average profit is 6% of the sales. Its profit in men's shirts average 8% of the sales and women's shirts comprise 60% of the output. The average proflt per sales rupee in women's shirts is :
(1) 0.0466 (2) 0.0666
(3) 0.0166 (4) 0.0669
(5) None of these

23. A manufacturer of a certain item can sell all he can produce at the selling price of Rs 60 each. If costs him Rs 40 in materials and labour to produce each item and he has overhead expenses of Rs 3000 per week in order to operate that plant. The number of units he should produce and sell in order to make a profit of at least 1000 per week is —
(1) 250 (2) 300
(3) 400 (4) 200
(5) None of these

24. If the selling price of a product is increased by 162, then the business would make a proflt of 17% instead of a loss of 19%. What is the cost price of the product ?
(1) Rs 540 (2) Rs 450
(3) Rs 360 (4) Rs 600
(5) None of these

25. Sita buys a fridge at $\frac{15}{16}$ of its original value and sells it for 10% more than its value. Then, gain per cent is
(1) 15.55 (2) 11.67
(3) 16.67 (4) None of these
(5) 16.59

SHORT ANSWERS

1.	(1)	**2.**	(2)
3.	(4)	**4.**	(2)
5.	(2)	**6.**	(2)
7.	(3)	**8.**	(1)
9.	(4)	**10.**	(1)
11.	(3)	**12.**	(3)
13.	(4)	**14.**	(2)
15.	(2)	**16.**	(2)
17.	(1)	**18.**	(2)
19.	(3)	**20.**	(1)
21.	(4)	**22.**	(1)
23.	(4)	**24.**	(2)
25.	(4)		

EXPLANATIONS

1. (1) According to the question,
SP of 12 marbles = Rs 1, loss = 20%
CP of 12 marbles

$= \text{Rs } \frac{1}{0.8} = \text{Rs } 1.25$

Now, SP of 12 marbles at a gain of 20%
CP × 1.2 = 1.25 × 1.2 = Rs 1.5
It means in order to gain 20%, he should sell 12 marbles for Rs. 5 Or for Rs 1, he should sell $\frac{12}{1.5}$ = 8 marbles

2. (5) Let the original cost price be Rs 100.
Then, profit = Rs 320 and SP = Rs 420
New CP = Rs 125
$\therefore$ New profit = Rs 295
Required percentage

$= \frac{295}{420} \times 100 = 70.23\%$

3. (4) Let the CP of horse be Rs x.
then CP of carriage would be Rs (8000 -x)
SP of horse = 1.1x
SP of carriage = 0.9 (8000 - x)
$\therefore$ 1.1x × 0.9 (8000 - x)
= 8000 × 1.025 = 8200
1.1x + 7200 - 0.9x = 8200
$\Rightarrow$ 0.2x = 1000
$\Rightarrow x$ = Rs 5000

4. (2) Let the length of the piece be x *m*
Cost of price = Rs 35

Then, price per metre = Rs $\frac{35}{x}$

$\therefore (x + 4)\left(\frac{35}{x} - 1\right) = 35$

$\Rightarrow x$ = 10 m [from Eq. (i)]

5. (2) According to the question,
Cost price of house = Rs C

Profit = 25% of C = $\frac{C}{4}$

Capital gain tax = 5% of $\frac{C}{4}$ = Rs $\frac{C}{8}$

6. (2) Let CP be x and SP be y.
Then, profit = (y - x)
If SP = 2y, then profit = 3(y - x)
Given 2y - x = 3(y - x)
$\Rightarrow y = 2x$
$\therefore$ Profit is 100%

7. (3) Let reduced price of the radio be Rs x.
$\therefore x$ + 7% of x = 2568

$\Rightarrow x + \frac{7x}{100} = 2568$

$\Rightarrow x = \frac{256800}{107} \Rightarrow x = 2400$

$\therefore$ Reduction needed in the price of radio
= (2568 - 2400) = Rs 168

8. (1) According to the question,
Total amount spent by them = Rs 150
Let the cost of one beer be x, then the cost of one lager will be 2x.
Total money spent on beer and larger
= (4x + 2x)

$\therefore 6x = 150 \Rightarrow x = \frac{150}{6} = 25$

$\therefore$ The cost of the larger = 25 × 2 = 50

9. (4) Let the CP of the article be Rs 100.

SP atagain of $12\frac{1}{2}\%$

= 100 × 1.125 = Rs 112.50
SP at a gain of 25% = Rs 125
If SP is Rs 12.5 more than CP = 100
If SP Rs 22.50 more than CP

$= \frac{100}{12.5} \times 22.5 = \text{Rs } 180$

10. (1) Let the CP of the article be Rs x
Then, SP = x × 1.12 × 1.1
Given, x × 1.12 × 1.1 = 616

$x = \frac{616}{1.232} = \text{Rs } 500$

11. (3) According to question purchasing capacity = Rs 160
A reduction of 20% means, now a person gets $\frac{5}{2}$ kg for Rs 32 and this is the present price of that commodity.

$\therefore$ Present price per kg = $\frac{32}{5} \times 2$

= Rs 12.8

Let the originai price be Rs x, then new price is arrived after reduction 20% of it.

$\therefore x \times 0.8 = 12.8 \Rightarrow x$ = Rs 16

12. (3) Let CP of 12 pencils = SP of 10 pencils = Rs 1

Then, CP of 1 pencil = Rs $\frac{1}{12}$

and SP of 1 pencil = Rs $\frac{1}{10}$

$\therefore$ Profìt = $\frac{1}{10} - \frac{1}{12} = \frac{2}{120}$ = Rs $\frac{1}{60}$

Profit per cent =

$$\frac{\frac{1}{60}}{\frac{1}{12}} \times 100 = \frac{1}{60} \times 12 \times 100 = 20\%$$

13. (4) According to question, where SP is same and also gain % and loss % is same there is always a loss and such loss %

$$\left(\frac{(\text{Common gain or loss \%})}{10}\right)^2 = \left(\frac{10}{10}\right)^2$$

= 1% loss

14. (2) SP of 33 m of cloth
= CP of 33m + gain
SP of 33 m = CP of 33 m + SP of 11 m
$\Rightarrow$ SP of 22 m = CP of 33 m = 1 (say)

$\Rightarrow$ SP of l m = $\frac{1}{22}$ CP of l m $\frac{1}{33}$

Gain = $\frac{1}{22} - \frac{1}{33} = \frac{11}{22 \times 33}$

$\therefore$ Gain per cent = $\frac{\frac{11}{22 \times 33}}{\frac{1}{33}} \times 100$

= $\frac{11}{22 \times 33} \times 33 \times 100$ = 50%

15. (2) Let the CP of the pen and book be Rs x and Rs y respectively.
$\therefore 0.95x + 1.15y = (x + y) + 7$
$\Rightarrow 0.15\ y - 0.05\ x = 7$
and $1.05x + 1.1y = (x + y) + 13$
$\Rightarrow 0.05x + 0.1y = 13$
Solving Eqs. (i) and (ii), we get
y = Rs 80

16. (5) According to the question,
CP of stamp sold at a profit of 20%

= $\frac{80000}{12} = \frac{20000}{3}$ = 6666.66

CP of stamp sold at a loss cf 20%

= $\frac{8000}{0.8}$ = 10000

Total SP 6666.66 + 10000 = 16666.66
Total SP = 2 × 8000 = Rs16000
Loss = 16666.66 - 1600 = Rs 666.66

17. (1) Let the CP of the article be Rs 100.
Then, SP = Rs 105
Now CP = Rs 95
New SP = Rs 104.5
If he sells it for Rs 0.5 less than CP
= Rs 100
If he sells it for Rs 1 less than CP 100

= $\frac{100}{0.5} \times 1$ = Rs 200

18. (2) Let the quantity of milk purchase be x and quantity of water added be y.
Then, ratio of water to milk be $y : x$.
CP = $6.4x$
SP = $8\ (x + y)$
Profit per cent = 37.5%
$\therefore 8\ (x + y) = 6.4\ x \times 1.375$
$8x + 8y = 8.8x$
$\Rightarrow 8y = 0.8\ x$

$\Rightarrow \frac{x}{y} = \frac{80}{8} = \frac{10}{1}$

$\Rightarrow y : x = 1 : 10$

19. (3) According to question,
CP of 20 articles = SP of x articles = 1

$\therefore$ CP = of 1 articles = $\frac{1}{20}$,

SP of 1 articles = $\frac{1}{x}$

Profit per cent = $\frac{\frac{1}{x} - \frac{1}{20}}{\frac{1}{20}} = \frac{25}{100}$

$\Rightarrow \frac{20 - x}{x} = \frac{1}{4}$

$\Rightarrow$ 80 - $4x = x$

$\Rightarrow 5x = 80$

$\Rightarrow x = 16$

20. (1) According to the question,
Interest earned on Rs 13200 @ 14%
= Rs 1848
$\therefore$ Number of shares purchased

$= \frac{13200}{110} = 120$

Dividend earned by him on 120 shares Rs 15 per share = Rs 1800
$\therefore$ Net loss = Rs 48

21. (5) According to the question,
CP of 3500 cassettes = Rs 350000
The SP of one cassette after discount is Rs 112.50
SP of each set of 30 cassettes = (29 +1)
= Rs 29 × 112.50 = Rs 3262.50
$\therefore$ SP of 3500 cassettes including 500 free cassettes = Rs 326250
$\therefore$ Overall loss on Rs 350000 is Rs 23750

$\therefore$ Loss percent = $\frac{23750}{350000} \times 100 = 6.78\%$

22. (1) According to the question,
Women's shirts comprise 60% of the output.
$\therefore$ Men's shirts comprise 40% of the output.
$\therefore$ Average profit from men's shirts
= 8% of 40 = 3.2 out of 40
Overall average profit = 6 out of 100
Average profit from women's shirts = 2.8 out of 60,
i.e., 0.0466 out of each shirt

23. (4) Required number of items

$= \frac{(3000+1000)}{(60-40)} = \frac{4000}{20} = 200$

24. (2) According to question,
(17 + 19) = 36% of the cost price = Rs 162
100% of the cost price

$= \frac{162}{36} \times 100 = \text{Rs } 450$

25. (4) Let the original value of fridge be Rs = x.

Then, Cost price = $\frac{15}{16}x$

Selling price = $\frac{110}{100} \times x$

$\therefore$ Gain per cent = $\frac{\frac{110}{100}x - \frac{15}{16}x}{\frac{15}{16}} \times 100$

= 17.33%
Hence, the answer is none of these.

RATIO AND PROPORTION

NATIONALISED BANKS & IBPS PO/MT/SO

1. Seats for Maths, Physics and Biology are in the ratio of 5 : 7 : 8 respectively. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the respective ratio of increased seats ?
(1) 2 : 3 : 4
(2) 6 : 7 : 8
(3) 6 : 8 : 9
(4) Cannot be determined
(5) None of these

Canara Bank PO Exam. 09.02.2003)

2. Samira, Mahira and Kiara rented a set of DVDs at a rent of Rs. 578. If they used it for 8 hours, 12 hours and 14 hours respectively, what is Kiara's share of rent to be paid ?
(1) Rs. 238 (2) Rs. 204
(3) Rs. 192 (4) Rs. 215
(5) None of these

(Bank Of Maharashtra PO Exam. 25.05.2008)

3. A sum of money is to be divided among four persons in the ratio of 2 : 3 : 4 : 5. Out of the four, one person gets Rs. 200 more than the other and Rs. 100 less than another. What is the sum ?
(1) Rs. 2800
(2) Rs. 1400
(3) Rs. 4200
(4) Cannot be determined
(5) None of these

(Indian Overseas Bank PO Exam. 15.06.2008)

4. In a college the number of students studying Arts, Commerce and Science are in the ratio of 3 : 5 : 8 respectively. If the num- ber of students studying Arts, Commerce and Science is increased by 20%, 40% and 25% respectively, what will be the new ratio of students in Arts, Commerce and Science respectively ?
(1) 18 : 35 : 50 (2) 3 : 10 : 10
(3) 4 : 8 : 5 (4) 32 : 35 : 25
(5) None of these

(Indian Overseas Bank PO Exam. 15.06.2008)

5. 20 boys and 25 girls form a group of social workers. During their membership drive, the same number of boys and girls joined the group (e.g. if 7 boys joined, 7 girls joined). How many members does the group have now, if the ratio of boys to girls is 7 : 8 ?
(1) 75 (2) 65
(3) 70 (4) 60
(5) None of these

(Bank Of Baroda Specialist Officer Exam. 05.10.2008)

6. A sum of money is divided among A, B, C and D in the ratio of 3 : 4 :9 : 10 respectively. If the share of C is Rs. 2,580 more than the share of B, then what is the total amount of money of A and D together ?
(1) Rs. 5,676 (2) Rs. 6,192
(3) Rs. 6,708 (4) Rs. 7,224
(5) None of these

(Canara Bank PO Exam. 15.03.2009)

7. Production of company A is 120% of the production of company B and 80% of the production of company C. What is the ratio between the productions of companies A, B and C respectively ?
(1) 6 : 5 : 9 (2) 6 : 5 : 4
(3) 12 : 10 : 15 (4) 10 : 12 : 15
(5) None of these

(PNB Specialist Officer's Exam. 16.08.2009)

8. Number of students in Arts and Science faculties in an institute are in the ratio of 5 : 8 respectively. If 150 more students join Art faculty while 80 more students join Science faculty, the respective ratio becomes 3 : 4. Originally what was the total number of students in both faculties together ?
(1) 1200
(2) 1400
(3) 1150
(4) Cannot be determined
(5) None of these

(PNB Specialist Officer's Exam. 16.08.2009)

9. 75% of a number is equal to $\frac{5}{8}$ th of another number. What is the ratio between the first number and the second number respectively ?
(1) 5 : 4 (2) 6 : 5
(3) 4 : 5 (4) 5 : 6
(5) None of these

(PNB Specialist Officer's Exam. 16.08.2009)

10. In a test, a candidate secured 336 marks out of maximum marks 'x'. If the maximum marks '*x*' were converted into 400 marks, he would have secured 192 marks. What were the maximum marks of the test ?

(1) 700 (2) 750
(3) 500 (4) 650
(5) 800

(Corporation Bank PO Exam. 22.11.2009)

11. Which of the following represents $ah = 64$?
(1) $8 : a = 8 : b$ (2) $a : 16 = b : 4$
(3) $a : 8 = b : 8$ (4) $32 : a = b : 2$
(5) None of these

(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

12. The ratio of the number of students studying in schools A, B and C is 5 : 8 : 4 respectively. If the number of students studying in each of the schools is increased by 20%, 25% and 30% respectively, what will be the new respective ratio of the students in schools A, B and C ?
(1) 13 : 25 : 15
(2) 20 : 25 : 13
(3) 15 : 25 : 13
(4) Cannot be determined
(5) None of these

(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

13. When 30% of one number is subtracted from another number, the second number reduces to its own four-fifth. What is the ratio between the first and the second numbers respectively ?
(1) 4 : 7
(2) 3 : 2
(3) 2 : 5
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 21.02.2010)

14. The largest and the second largest angles of a triangle are in the ratio of 3 : 2 respectively. The smallest angle is 20% of the sum of the largest and the second largest angles. What is the sum of the smallest and the second largest angles ?
(1) 80° (2) 60°
(3) 100% (4) 90°
(5) None of these

(Bank Of Baroda PO Exam. 30.05.2010)

15. The ratio between the angles of a quadrilateral is 7 : 2 : 5 : 6 respectively. What is the sum of double the smallest angle and half the largest angle of the quadrilateral ?
(1) 162° (2) 198°
(3) 99° (4) 135°
(5) None of these

(Bank Of Baroda PO Exam. 30.05.2010)

16. The angles of a quadrilateral are in the ratio of 2 : 4 : 7 : 5. The smallest angle of the quadrilateral is equal to the smallest angle of a triangle. One of the angles of the triangle is twice the smallest angle of the triangle. What is the second largest angle of the triangle ?
(1) 80°
(2) 60°
(3) 120°
(4) Cannot be determined
(5) None of these

(Central Bank Of India PO Exam. 25.07.2010)

17. The ratio between the angles of a quadrilateral is 3 : 4 : 6 : 7. Half the second largest angle of the quadrilateral is equal to the smaller angle of a parallelogram. What is the value of adjacent angle of the parallelogram ?
(1) 136° (2) 126°
(3) 94° (4) 96°
(5) None of these

(Syndicate Bank PO Exam. 29.08.2010)

18. The ratio between the three angles of a quadrilateral is 1 : 4 : 5 respectively. The value of the fourth angle of the quadrilateral is 60°. What is the difference between the value of the largest and the smallest angles of the quadrilateral ?
(1) 120° (2) 90°
(3) 110° (4) 100°
(5) None of these

(Bank Of India PO Exam. 31.10.2010)

19. Mr. Pandit owned 950 gold coins all of which he distributed amongst his three daughters Lalita, Amita and Neeta. Lalita gave 25 gold coins to her husband, Amita donated 15 gold coins and Neeta made jewellery out of 30 gold coins. The new respective ratio of the coins left with them was 20 : 73 : 83. How many gold coins did Amita receive from Mr. Pandit ?
(1) 380 (2) 415
(3) 400 (4) 350
(5) None of these

(PNB Management Trainee Exam. 28.11.2010)

20. The largest and the second largest angles of a triangle are in the ratio of 13 : 12 respectively. The smallest angle is 20% of the sum of the largest and the second largest angles. What is the sum of the smallest and the second largest angles ?
(1) 120° (2) 108°
(3) 100° (4) 102°
(5) None of these

(Bank Of Maharashtra Exam. 19.12.2010)

21. Twenty five percent of Pranab's annual salary is equal to eighty percent of Surya's annual salary. Surya's monthly salary is forty percent of Dheeru's monthly salary. If Dhreeu's annual salary is Rs 6 lacs, what is Pranab's monthly salary ? (At some places annual income and in some place monthly income are given)
(1) Rs 7.68 lacs (2) Rs 56,000
(3) Rs 8.4 lacs (4) Rs 64,000
(5) None of these

(Bank Of Maharashtras Exam. 19.12.2010)

22. The ratio between the three angles of a quadrilateral is 1 : 6 : 2 respectively. The value of the fourth angle of the quadrilateral is 45°. What is the difference between the value of the largest and the smallest angles of the quadrilateral ?
(1) 165° (2) 140°
(3) 175° (4) 150°
(5) None of these

(Bank Of Maharashtra Exam. 19.12.2010)

23. The ratio between the angles of a quadrilateral is 3 : 4 : 6 : 5. Two-third of the largest angle of the quadrilateral is equal to the smaller angle of a parallelogram. What is the value of adjacent angle of the parallelogram ?
(1) 120° (2) 110°
(3) 100° (4) 130°
(5) None of these

(Oriental Bank Of Commerce P0 Exam. 26.12.2010 (1st Sitting)

24. Rohit has some 50 paisa coins, some 2 rupee coins, some 1 rupee coins and some 5 rupee coins. The value of all the coins is Rs 50. Number of 2 rupee coins is 5 more than that of the 5 rupee coins. 50 paisa coins are double in number than 1 rupee coins Value of 50 paisa coins and 1 rupee coins is Rs 26. How many 2 rupee coins does he have ?
(1) 4
(2) 2
(3) 7
(4) Cannot be determined
(5) None of these

(Union Bank Of India PO Exam. 09.01.2001)

25. The ratio between the adjacent angles of a parallelogram is 2 : 3 respectively. Half the smaller angle of the parallelogram is equal to the smallest angle of a quadrilateral. Largest angle of quadrilateral is four times its smallest angle. What is the sum of largest angle of quadrilateral and the smaller angle of parallelogram
(1) 252° (2) 226°
(3) 144° (4) 180°
(5) None of these

(Union Bank Of India PO Exam. 09.01.2001)

26. One of the angles of a triangle is two-third of sum of adjacent angles of parallelogram. Remaining angles of the triangle are in ratio 5 : 7 respectively. What is the value of second largest angle of the triangle ?
(1) 25°
(2) 40°
(3) 35°
(4) Cannot be determined
(5) None of these

(Corporation Bank PO Exam. 16.O1.2011)

27. The largest and the smallest angles of a triangle are in the ratio of 3 : 1 respectively. The second largest angle of the triangle is equal to 44°. What is the value of 150 per cent of the largest angle of the triangle ?
(1) 149 (2) 129
(3) 153 (4) 173
(5) None of these

(UCO Bank PO Exam. 30.01.2011)

28. One of the angles of a quadrilateral is thrice the smaller angle of a parallelogram. The respective ratio between the adjacent angles of the parallelogram is 4 : 5. Remaining three angles of the quadrilateral are in ratio 4 : 11 : 9 respectively. What is the sum of the largest and the smallest angles of the quadrilateral ?
(1) 255° (2) 260°
(3) 265° (4) 270°
(5) None of these

(UCO Bank PO Exam. 30.01.2011)

29. Smallest angle of a triangle is equal to two-third of the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is 3 : 4 : 5 : 6. Largest angle of the triangle is twice its smallest angle. What is the sum of second largest angle of the triangle and largest angle of the quadrilateral ?
(1) 160° (2) 180°
(3) 190° (4) 170°
(5) None of these

(Bank Of Baroda PO Exam. 13.03.2011)

30. The largest and the second largest angles of a triangle are in the ratio of 4 : 3 respectively. The smallest angle is half the largest angle. What is the difference between the smallest and the largest angles of the triangle ?
(1) 30° (2) 60°
(3) 40° (4) 20°
(5) None of these

(Allahabad Bank PO Exam. 17.04.2011)

31. The ratio between the three angles of a quadrilateral is 13 : 9 : 5 respectively. The value of the fourth angle of the quadrilateral is 36°. What is the difference between the **largest** and the **second smallest** angles of the quadrilateral ?
(1) 104° (2) 108°
(3) 72° (4) 96°
(5) None of these

(Allahabad Bank PO Exam. 17.04.2011)

32. The ratio between the adjacent angles of a parallelogram is 7 : 8 respectively. Also the ratio between the angles of quadrilateral is 5 : 6 : 7 : 12. What is the sum of the smaller angle of parallelogram and second largest angle of the quadrilateral ?
(1) 168° (2) 228°
(3) 156° (4) 224°
(5) None of these

(Indian Overseas Bank PO Exam. 22.05.2011)

33. The ages of Sulekha and Arunima are in the ratio of 9 : 8 respectively. After 5 years the ratio of their ages will be 10 : 9. What is the difference in years between their ages.
(1) 4 years (2) 5 years
(3) 6 years (4) 7 years
(5) None of these

(Andhra Bank PO Exam. 14.09.2008)

34. The ages of Sonal and Nitya are in the ratio of 9 : 5 respectively. After 8 years the ratio of their ages will be 13 : 9. What is the difference in years between their ages ?
(1) 4 years (2) 12 years
(3) 6 years (4) 14 years
(5) None of these

(Oriental Bank of Commerce PO Exam. 21.12.2008)

35. The ratio of the ages of a father and son is 17 : 7 respectively. 6 years ago the ratio of their ages was 3 : 1 respectively. What is the father's present age ?
(1) 64
(2) 51
(3) 48
(4) Cannot be detremined
(5) None of these

(UCO Bank PO Exam. 22.03.2009)

36. Ratio of Rani's and Komal's age is 3 : 5 respectively. Ratio of Komal's and Pooja's age is 2 : 3 respectively. If Rani is two-fifth of Pooja's age, what is Rani's age ?
(1) 10 years
(2) 15 years
(3) 24 years
(4) Cannot be determined
(5) None of these

(Indian Overseas Bank PO Exam. 05.04.2009)

37. Present ages of Amit and his father are in the ratio of 2 : 5 respectively. Four years hence the ratio of their ages becomes 5 : 11 respectively. What was father's age five years ago ?
(1) 40 years (2) 45 years
(3) 30 years (4) 35 years
(5) None of these

(Andhra Bank PO Exam. 05.07.2009)

38. Four years ago Shyam's age was $\frac{3}{4}$ times that of Ram. Four years hence, Shyam's age will be $\frac{5}{6}$ times that of Ram. What is the present age of Shyam ?
(1) 15 years (2) 20 years
(3) 16 years (4) 24 years
(5) 8 years

(Corporation Bank PO Exam. 22.11.2009)

39. The ratio of the ages of Tina and Rakesh is 9 : 10 respectively. Ten years ago the ratio of their ages was 4 : 5 respectively. What is the present age of Rakesh ?
(1) 25 years (2) 20 years
(3) 30 years (4) 24 years
(5) None of these

(Indian Bank PO Exam. 17.10.2010)

40. The present ages of Vishal and Shekhar are in the ratio of 14 : 17 respectively. Six years from now, thier ages will be in the ratio of 17 : 20 respectively. What is Shekhar's present age ?
(1) 17 years (2) 51 years
(3) 34 years (4) 28 years
(5) None of these

(Bank Of India Banking Officer Exam. 24.01.2010)

41. The ratio between the ages of a father and a son at present is 5 : 2 respectively. Four

years hence the ratio between the ages of the son and his mother will be 1 : 2 respectively. What is the ratio between the present ages of the father and the mother respectively ?
(1) 3 : 4
(2) 5 : 4
(3) 4 : 3
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 21.02.2010)

42. Radha's present age is three years less than twice her age 12 years ago. Also the respective ratio between Raj's present age and Radha's present age is 4 : 9. What will be Raj's age after 5 years ?
(1) 12 years
(2) 7 years
(3) 21 years
(4) Cannot be determined
(5) None of these

(Punjab & Sind Bank PO Exam. 16.05.2010)

43. The ratio of the present ages of Meena and Fiona is 16 : 13 respectively. Four years ago the respective ratio of their ages was 14 : 11. What will be Fiona's age four years from now ?
(1) 28 years (2) 32 years
(3) 26 years (4) 36 years
(5) None of these

(United Bank Of India PO Exam. 14.11.2010)

44. The respective ratio of the present ages of Swati and Trupti is 4 : 5. Six years hence the respective ratio of their ages will be 6 : 7. What is the difference between their ages ?
(1) 2 years
(2) 3 years
(3) 4 years
(4) Cannot be determined
(5) None of these

(PNB Management Trainee Exam. 28.11.2010)

45. The respective ratio between the present ages of Ram and Rakesh is 6 : 11. Four years ago the ratio of their ages was 1 : 2 respectively. What will be Rakesh's age after five years ?
(1) 45 years
(2) 29 years
(3) 49 years
(4) Cannot be determined
(5) None of these

(Corporation Bank PO Exam. 16.01.2011)

46. The respective ratio between the present ages of son, mother, father and grandfather is 2 : 7 : 8 : 12. The average age of son and mother is 27 years. What will be mother's age after 7 years ?
(1) 40 years (2) 41 years
(3) 48 years (4) 49 years
(5) None of these

(UCO Bank PO Exam. 30.01.2011)

47. The respective ratio between the present ages of Ram, Rohan and Raj is 3 : 4 : 5. If the average of their present ages is 28 years then what would be the sum of the ages of Ram and Rohan together after 5 years ?
(1) 45 years (2) 55 years
(3) 52 years (4) 59 years
(5) None of these

(Bank Of Baroda PO Exam. 13.03.2011)

48. The respective ratio between present age of Manoj and Wasim is 3 : 11. Wasim is 12 years younger than Rehana. Rehana's age after 7 years will be 85 years. What is the present age of Manoj's father who is 25 years older than Manoj ?
(1) 43 years (2) 67 years
(3) 45 years (4) 69 years
(5) None of these

(Indian Overseas Bank PO Exam. 22.05.2011)

49. The respective ratio between the present age of Aarti and Savita is 5 : *x*. Aarti is 9 years younger than Jahnavi. Jahnavi's age after 9 years will be 33 years. The difference between Savita's and Aarti's age is same as the presen age of Jahnavi. What will come in place of *x* ?
(1) 21
(2) 37
(3) 17
(4) Cannot be determined
(5) None of these

(IBPS Bank PO/MT CWI Exam.18.09.2011)

50. An amount of money is to be divided among P, Q and R in in the ratio of 3 : 5 : 7 respectively. If the amount received by R is Rs 4,000 more than the amount received by Q, what will be the be total amount received by P and Q together ?
(1) Rs. 8,000
(2) Rs. 12,000
(3) As. 16,000
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 21.02.2010)

51. Rita invested 25% more than Sunil. Sunil invested 30% less than Abhinav who invested Rs 6,000. What is the respective ratio between the amount that Rita invested and the total amount invested by all of them together ?
(1) 35 : 104 (2) 13 : 29
(3) 101 : 36 (4) 35 : 103
(5) None of these

(Punjab & Sind Bank PO Exam. 16.05.2013

52. When X is subtracted from the numbers 9, 15 and 27, the remainders are in continued proportion. What is the value of X ?
(1) 8 (2) 6
(3) 4 (4) 5
(5) None of these

(IBPS Bank PO/MT CWE 17.06.2012)

53. A certain amount was to be distributed among A, B and C in the ratio 2 : 3 : 4 respectively, but was erroneously distributed the ratio 7 : 2 : 5 respectively. As a result of this, B got Rs 40 less. What is the amount ?
(1) Rs 210 (2) Rs 270
(3) Rs 230 (4) Rs 280
(5) None of these

(IBPS Bank PO/MT CWE 17.06.2012)

54. A particular sum was divided among A, B and C in the ratio 2 : 6 : 7 respectively. If the amount received by A was Rs. 4,908, what was the difference between the amounts received by B and C ?
(1) Rs. 2,454 (2) Rs. 3,494
(3) Rs. 2,135 (4) Rs. 2,481
(5) None of these

(IBPS RRBs Office Assistant CWE Exam. 09.09.2012)

55. The average age of a man and his son is 30 years. The ratio of their ages four years ago was 10 : 3 respectively. What is the difference between the present ages of the man and his son ?
(1) 28 years (2) 16 years
(3) 26 years (4) 44 years
(5) None of these

(IBPS RRBs Office Assistant CWE Exam. 09.09.2012)

56. A sum of Rs. 221 is divided among X, Y and Z such that X gets Rs. 52 more than Y. Y gets Rs. 26 more than Z. The ratio of the shares of X, Y and Z respectively is :
(1) 9 : 5 : 3 (2) 9 : 3 : 5
(3) 5 : 9 : 3 (4) 10 : 6 : 5
(5) None of these

(Indian Overseas Bank PO Online Exam. 01.09.2013)

57. In two vessels A and B, there is mixture of milk and water. The ratio of milk and water in these vessels is 5 : 2 and 8 : 5 respectively. In what ratio these mixtures be mixed together so that the ratio of milk and water in the new mixture becomes 9 : 4 ?
(1) 7 : 2 (2) 2 : 7
(3) 3 : 5 (4) 5 : 3
(5) 7 : 9

(Indian Overseas Bank PO Online Exam. 01.09.2013)

58. If 50% of a certain number is equal to $\frac{3}{4}$ th of another number, what is the ratio between the numbers ?
(1) 3 : 2 (2) 2 : 5
(3) 5 : 2 (4) 3 : 4
(5) 4 : 3

(Corporation Bank Specialist Officer (Marketing) Exam. 22.12.2014)

59. The ratio of the present ages of Mahesh and Ajay is respectively 3 : 2. After 8 years, ratio of their ages will be 11 : 8. What will be the present age of Mahesh's son if his age is half of the present age of Ajay ?
(1) 12 years (2) 24 years
(3) 18 years (4) 9 years
(5) None of these

(Corporation Bank Specialist Officer (Marketing) Exam. 22.12.2014)

SBI PO EXAM

1. When a number is added to a second number, the sum is $333\frac{1}{3}$ percent of the second number. What is the ratio between the first number to the second number ?
(1) 3 : 7 (2) 7 : 4
(3) 7 : 3 (4) Data inadequate
(5) None of these

(SBI Banks PO Exam. 20.08.2000)

2. A sum of money is to be distributed among P, Q and R in the ratio 6 : 19 : 7. If R gives Rs. 200 from his share to Q, the ratio of P, Q and R becomes 3 : 10 : 3, what is the total sum ?
(1) Rs. 6400 (2) Rs. 12800
(3) Rs. 3200 (4) Data inadequate
(5) None of these

(SBI Banks PO Exam. 20.08.2000)

3. In a school the number of boys and that of the girls are in the respective ratio of 2 : 3. If the number of boys is increased by 20% and that of girls is increased by 10%, what will be the new ratio of number of boys to that of the girls ?
(1) 14 : 5 (2) 5 : 8
(3) 13 : 4 (4) Data inadequate
(5) None of these

(SBI Banks PO Exam. 11.02.2001)

4. Income of two companies A and B are in the ratio of 5 : 8. Had the income of company 'A' been more by Rs. 25 lakhs, the ratio of their incomes would have been 5 : 4 respectively. What is the income of company 'B' ?
(1) Rs. 80 lakhs
(2) Rs. 50 lakhs
(3) Rs. 40 lakhs
(4) Rs. 60 lakhs
(5) None of these

(SBI Banks PO Exam. 11.02.2001)

5. Ratio of the earnings of A and B is 4 : 7 respectively. If the earnings of A increase by 50% and the earnings of B decrease by 25% the new ratio of their earnings becomes 8 : 7 respectively. What are A's earnings ?
(1) Rs 26,000
(2) Rs 28, 000
(3) Rs 21,000
(4) Data inadequate
(5) None of these

(SBI Associate Banks PO Exam. 21.07.2002)

6. Salaries of A, B and C are in the ratio of 2 : 3 : 5 respectively. If their salaries were increased by 15%, 10% and 20% respectively. What will be the new respective ratio of their salaries ?
(1) 3 : 3 : 10
(2) 23 : 33 : 60
(3) 10 : 11 : 20
(4) Can't be determined
(5) None of these

(SBI Associate Banks PO Exam. 21.07.2002)

7. Present ages of Seema and Naresh are in the respective ratio of 5 : 7. Five years hence the ratio of their ages becomes 3 : 4 respectively. What is Naresh's present age in years ?
(1) 25
(2) 40
(3) 30
(4) Can not be determined
(5) None of these

(SBI PO Exam. 09.01.2008)

8. Seats for Maths, Physics and Biology are in the ratio of 5 : 7 : 8 respectively. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the respective ratio of increased seats ?
(1) 2 : 3 : 4
(2) 6 : 7 : 8
(3) 6 : 8 : 9
(4) Cannot be determined
(5) None of these

(SBI Associate Banks PO Exam. 07.01.2007)

9. A sum of money is divided among A, B, C and D in the ratio of 3 : 5 : 9 : 13 respectively. If the share of C is Rs. 2412 more than the share of A, then what is the total amount of money of B and D together ?
(1) Rs. 4422 (2) Rs. 7236
(3) Rs. 6030 (4) Rs. 4824
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

10. The ages of Khushi and Jagriti are in the ratio of 5 : 8 respectively. After 8 years the ratio of their ages will be 3 : 4. What is the difference in their ages ?
(1) 16 years (2) 8 years
(3) 10 years (4) 12 years
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

11. The respective ratio of the present ages of a mother and daughter is 7 : 1. Four years ago the respective ratio of their ages was 19 : l. What will be the mother's age four years from now ?
(1) 42 years (2) 38 years
(3) 46 years (4) 36 years
(5) None of these

(SBI & Rural Business PO Exam. 18.04.2010)

12. Mr. X invested a certain amount in Debit and Equity funds in the ratio of 4 : 5 respectively. At the end of one year, he earned a tolal dividend of 30% on his investment. After one year he reinvested the amount including dividend in the ratio of 6 : 7 in Debit and Equity Funds. If the amount reinvested in Equity Funds was f 94, 500, what was the original amount invested in Equity Funds ?
(1) Rs 75,000 (2) Rs 81,0007
(3) Rs 60,000 (4) Rs 65,0007

(5) None of these

(SBI Associate Banks PO Exam. 07.08.2011)

RBI GRADE-B OFFICER EXAMS

1. The ages of Samir and Tanuj are in the ratio of 8 : 15 years respectively. After 9 years the ratio of their ages will be 11 : 18. What is the difference in years between their ages ?
(1) 24 years (2) 20 years
(3) 33 years (4) 21 years
(5) None of these

(RBI Grade-B Officer Exam. 2007)

2. Ratio of earnings of A and B is 8 : 9 respectively. If the earnings of A increase by 50% and the earnings of B decrease by 25%, the new ratio of their earnings becomes 16 : 9 respectively. What are A's earnings ?
(1) Rs. 37,000
(2) Rs. 28,500
(3) Rs. 22,000
(4) Cannot be determined
(5) None of these

(RBI Grade-B Officer Exam. 2008)

3. In a college, the ratio of boys to girls is 31 : 23 respectively. When 75 more girls join the college, this ratio becomes 124 : 107. How many more girls should join the college to make the number of boys and girls equal ?
(1) 75 (2) 90
(3) 60 (4) 85
(5) None of these

(RBI Grade-B Officer Exam.06.02.2011)

4. The ratio of the salaries of A and B is 8 : 9. If A's salary is increased by 50% and B's salary is reduced by 25%, their ratio becomes 16 : 9. What is the salary of A ?
(1) Rs. 22000
(2) Rs. 28500
(3) Rs. 37000
(4) Cannot be determined
(5) None of these

(RBI Officer Grade 'B' Online Exam. 25.08.2013

INURANCE EXAMS

I. The number of employees shed through voluntary retirement in company A was 23,000 and that in company B was 6325. What is the ratio of employees retired voluntarily from company A to that retired from company B ?
(1) 40 : 11 (2) 10 : 11
(3) 11 : 10 (4) 11 : 40

(United India Insurance Co. AAO Exam. 21.04.2002

2. Ratio of the earnings of A and B is 4 : 7 respectively. If the earnings of A increase by 50% and the earnings of B decrease by 25% the new ratio of their earnings becomes 8 : 7 respectively. What are A's earnings ?
(1) Rs 26,000
(2) Rs 28, 000
(3) Rs 21,000
(4) Data inadequate
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 2006

3. Salaries of A, B and C are in the ratio of 2 : 3 : 5 respectively.If their salaries were increased by 15%, 10% and 20% respectively. What will be the new respective ratio of their salaries ?
(1) 3 : 3 : 10
(2) 23 : 33 : 60
(3) 10 : 11 : 20
(4) Can't be determined
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 2006

4. The present ages of A, B and C are in the ratio of 8 : 14 : 22 respectively. The present ages of B, C and D are in the ratio of 21 : 33 : 44 respectively. Which of the following represents the ratio of the present ages of A, B, C and D respectively ?
(1) 12 : 21 : 33 : 44
(2) 12 : 22 : 31 : 44
(3) 12 : 21 : 36 : 44
(4) Cannot be determined
(5) None of these

(LIC Assistant Administrative Officer Exam. 2008.

5. Three-fourth of a number is equal to 60% of another number. What is the difference between the numbers ?
(1) 18
(2) 32
(3) 24
(4) Cannot be determined
(5) None of these

(LIC Assistant Administrative Officer Exam. 2008)

6. The sum of three numbers is 136. If the ratio between first and second be 2 : 3 and that between second and third be 5 : 3, then the second number is:

(1) 40 (2) 48
(3) 52 (4) 60

LIC Assistant Administrative Officer (AAO) Exam. 07.06.2009)

7. The sides of a triangle are in the ratio $\frac{1}{2}$: $\frac{1}{3}$: $\frac{1}{4}$ and its perimeter is 104 cm. The length of the longest side (in cm.) is :
(1) 26 (2) 32
(3) 48 (4) 52

LIC Assistant Administrative Officer (AAO) Exam. 07.06.2009)

8. Cost of a diamond varies directly as the square of its weight. A diamond broke into four pieces with their weights in the ratio 1 : 2 : 3 : 4. If the loss in the total value of the diamond was Rs. 70,000, the price of the original diamond was
(1) Rs. 1,00,000
(2) Rs. 1,40,000
(3) Rs. 1,50,000
(4) Rs. 1,75,000

(New India Assurance AO Exam. 25.10.2009)

9. The angles of a quadrilateral are in ratio of 3 : 5 : 9 : 7. The second largest angle of the quadrilateral is equal to the largest angle of a triangle. One of the angles of the triangle is 25°. What is the value of second largest angle of the triangle ?
(1) 60° (2) 50°
(3) 40° (4) 20°
(5) None of these

(United India Insurance AO Exam. 27.03.2011)

10. Abhijit invested in three schemes A, B and C the amounts in the ratio of 2 : 3 : 4 respectively. If the schemes offered interest @ 20 p.c.p.a., 16 p.c.p.a. and 15 p.c.p.a. respectively, what will be the respective ratio of the amounts after one year ?
(1) 10 : 8 : 5
(2) 12 : 14 : 15
(3) 12 : 15 : 22
(4) Cannot be determined
(5) None of these

(LIC Assistant Administrative Officer Exam. 2008)

11. The price of a diamond is proportional to the square of its mass, which is measured in carats. A 6 carat diamond was broken into two parts and total price of the two pieces is $\frac{5}{8}$ of the price of original diamond. The masses (in carat) of two pieces are
(1) 3.5 and 2.5
(2) 5 and 1
(3) 4.5 and 1.5
(4) 3.2 and 2.8

(Oriental Insurance Company AAO Exam. 08.04.2012)

12. A shopeeper placed on display some shirts each with a marked price. He then posted a board "$\frac{1}{4}$ off on shirts". If the cost of a shirt was $\frac{2}{3}$ of the price at which it was actually sold, the ratio of cost and marked price of shirt was
(1) 1 : 2 (2) 1 : 3
(3) 2 : 3 (4) 3 : 4

(United India Insurance AAO Exam. 03.06.2012)

13. A camel pursues an elephant and takes 5 leaps for every 7 leaps of the elephant, but 5 leaps of elephant are equal to 3 leaps of camel. What is the ratio of speeds of camel and elephant ?
(1) 21 : 25 (2) 25 : 21
(3) 23 : 21 (4) 21 : 23

(United India Insurance AAO Exam. 03.06.2012)

14. The sum of the three consecutive odd numbers is 285. What is the ratio of the smallest and largest numbers respectively ?
(1) 97 : 95 (2) 93 : 95
(3) 95 : 93 (4) 93 : 97
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 12.05.2013)

15. The ages of Sonal and Nitya are in the ratio of 9 : 5 respectively. After 8 years the ratio of their ages will be 13 : 9. What is the difference in years between their ages ?
(1) 4 years (2) 12 years
(3) 6 years (4) 14 years
(5) None of these

(United India Insurance AO Exam. 26.05.2013

16. If $x : y = 4 : 5$, then $(3x + y) : (5x + 3y) =$
(1) 3 : 5 (2) 5 : 3
(3) 17 : 35 (4) 35 : 17

(NICL (GIC) AO Exam. 08.09.2013 (Paper-I)

17. The ratio between two numbers is 2 : 3. If each number is increased by 4, the ratio between them becomes 5 : 7, the difference between the numbers is
(1) 8 (2) 6
(3) 4 (4) 2
(NICL (GIC) AO Exam. 08.09.2013 (Paper-I)

18. The ratio between the rates of travelling of A and B is 2 : 3 and therefore A takes 10 minutes more than the time taken by B to reach a destination. If A had walked at double the speed, he would have covered the distance in
(1) 30 min (2) 25 min
(3) 20 min (4) 15 min
(NICL (GIC) Administrative Officer Exam. 15.12.2013)

19. The price of sugar is increased by 20%. If the expenditure is not allowed to increase, the ratio between the reduction in consumption and the original consumption is
(1) 1 : 3 (2) 1 : 4
(3) I : 6 (4) 1 : 5
(NICL (GIC) Administrative Officer Exam. 15.12.2013)

20. The difference between the present ages of Anil and Sudhir is 6 years. The ratio between their ages after 4 years will be 3 : 4. What can be the present age of Sudhir ?
(1) 15 years (2) 18 years
(3) 20 years (4) 24 years
(NICL (GIC) Administrative Officer Exam. 15.12.2013)

21. In a ratio which is equal to 5 : 8, if the antecedent is 40, then the consequent is:
(1) 25 (2) 64
(3) 48 (4) None of these
(NICL (GIC) Administrative Officer Exam. 15.12.2013)

22. The ratio of the number of students studying in schools A, B and C is 5 : 6 : 8. If the number of students in each of the schools is increased by 30%, 25% and 25% respectively. What will be the new ratio of the students in schools A, B and C ?
(1) 14 : 15 : 20 (2) 13 : 15 : 20
(3) 13 : 14 : 15 (4) 15 : 17 : 19
(NICL (GIC) AO (Finance) Exam. 15.12.2013)

23. The average score of a class of boys and girls in an examination is A. The ratio of boys and girls in the class is 3 : 1. If the average score of the boys is (A + 1), the average score of the girls is :
(1) (A - 1) (2) (A - 3)
(3) (A + 1) (4) (A + 3)
(NICL (GIC) AO (Finance) Exam. 15.12.2013)

24. Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:
(1) 1 : 3 (2) 3 : 2
(3) 3 : 4 (4) None of these
(NICL (GIC) AO (Finance) Exam. 15.12.2013)

25. If A : B = 1 : 3, B : C = 5 : 7, C : D = 9 : 7, then A : B : C : D = ?
(1) 15 : 45 : 63 : 49
(2) 15 : 49 : 45 : 63
(3) 45 : 15 : 63 : 49
(4) 49 : 15 : 45 : 63
(NICL (GIC) AO (Finance) Exam. 15.12.2013)

26. The cost of making an article is divided between materials, labour and overheads in the ratio of 3 : 4 : 1. If the materials cost Rs. 234, then the labour cost is:
(1) Rs. 176 (2) Rs. 312
(3) Rs. 78 (4) Rs. 390
(NICL (GIC) AO (Finance) Exam. 15.12.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (1) **2**. (1)
3. (2) **4**. (1)
5. (1) **6**. (3)
7. (3) **8**. (5)
9. (4) **10**. (1)
11. (4) **12**. (3)
13. (5) **14**. (4)
15. (4) **16**. (2)
17. (2) **18**. (1)
19. (1) **20**. (4)
21. (4) **22**. (3)
23. (3) **24**. (3)
25. (5) **26**. (3)
27. (3) **28**. (2)
29. (2) **30**. (3)
31. (4) **32**. (1)
33. (2) **34**. (5)
35. (2) **36**. (4)
37. (4) **38**. (3)
39. (2) **40**. (3)
41. (4) **42**. (5)
43. (5) **44**. (2)
45. (3) **46**. (4)
47. (4) **48**. (1)
49. (5) **50**. (3)
51. (4) **52**. (5)
53. (1) **54**. (1)
55. (1) **56.** (1)
57. (1) **58.** (1)
59. (1) **60**. (1)

SBI PO EXAMS

1. (3) **2.** (1)
3. (5) **4.** (3)
5. (4) **6.** (2)
7. (5) **8.** (1)
9. (2) **10**. (5)
11. (3) **12**. (1)

RBI GRADE-B OFFICER EXAMS

1. (4) **2.** (4)
3. (4) **4.** (4)

INSURANCE EXAMS

1. (1) **2**. (4)
3. (2) **4**. (1)
5. (4) **6**. (4)
7. (3) **8**. (1)
9. (2) **10.** (5)
11. (3) **12**. (1)
13. (2) **14.** (4)
15. (5) **16.** (3)
17. (1) **18.** (4)
19. (3) **20.** (3)
21. (2) **22.** (2)
23. (2) **24.** (4)
25. (1) **26.** (2)

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (1) Let the inital seats for Maths Physics and Biology be $5x$, $7x$ and $8x$ respectively.

Now, new seats for Maths = $\frac{5x \times 140}{100}$

For physics = $\frac{7x \times 150}{100}$

and for Biology = $\frac{8x \times 175}{100}$

$\therefore$ Required ratio

$= \frac{5x \times 140}{100} : \frac{7x \times 150}{100} : \frac{8x \times 175}{100}$

$= 5 \times 140 : 7 \times 150 : 8 \times 175$

$= 2 : 3 : 4$

2. (1) Ratio of rent's sharing
$= 8 : 12 : 14 = 4 : 6 : 7$
Total rent = Rs. 578

Share of Kiara = $\frac{7}{17} \times 578$ = Rs 238

3. (2) From the options,
$2x + 3x + 4x + 5x = 1400$
$\Rightarrow 14x = 1400 \Rightarrow x = 100$
$\therefore$ shares are : Rs. 200, Rs. 300.
Rs. 400 and Rs. 500.
Hence, total sum = 200 + 300 + 400 + 500
= Rs. 1400

4. (1) Let the number of students in Arts, Commerce and Science be $3x$, $5x$ and $8x$ respectively.
On increasing their respective numbers,

Required ratio = $3x \times \frac{120}{100} : 5x \times \frac{140}{100} : 8x \times \frac{125}{100}$

$= 360 : 700 : 1000$
$= 18 : 35 : 50$

5. (1) Let x boys and x girls joined the group.
According to the question,

$= \frac{20+x}{25+x} = \frac{7}{8}$

$\Rightarrow 160 + 8x = 175 + 7x$
$\Rightarrow 8x - 7x = 175 - 160$

$\Rightarrow x = 15$

$\therefore$ New number of members
$= 20 + x + 25 + x = 45 + 2x$
$= 45 + 2 \times 15 = 75$

6. (3) Let the amounts received by A, B, C and D be Rs. $3x$, $4x$, $9x$, and Rs. $10x$ respectively.
According to the question,
$9x - 4x = 2580$
$\Rightarrow 5x = 2580$

$$\Rightarrow x = \frac{2580}{5} = 516$$

Total amount of the money of A and D
$= 3x + 10x$
$= 13x = 13 \times 516 =$ Rs. 6708

7. (3) Let the production of company B = 100 units
$\therefore$ Production of company A = 120 units
Production of company C

$$= \frac{120 \times 100}{80} = 150 \text{ units}$$

$\therefore$ Required ratio = 120 : 100 : 150
= 12 : 10 : 15

8. (5) Let the original number of students in Arts and Science faculties be $5x$ and $8x$ respectively.
According to the question,

$$\frac{5x + 150}{8x + 80} = \frac{3}{4}$$

$\Rightarrow 24x + 240 = 20x + 600$
$\Rightarrow 4x = 360$

$$\Rightarrow x = \frac{360}{4} = 90$$

$\therefore$ Original number of students
$= 5x + 8x = 13x$
$= 13 \times 90 = 1170$

9. (4) Let the number be x and y respectively.

$$\therefore \frac{75x}{100} = \frac{5y}{8}$$

$$\Rightarrow \frac{x}{y} = \frac{5}{8} \times \frac{100}{75} = \frac{5}{6}$$

10. (1) $x : 336 = 400 : 192$
$\Rightarrow x \times 192 = 336 \times 400$

$$\Rightarrow x = \frac{336 \times 400}{192} = 700$$

11. (4) $\frac{32}{a} = \frac{b}{2} \Rightarrow ab = 64$
$\Rightarrow 32 : a = b : 2$

12. (3) Required ratio $= \frac{5 \times 120}{100} : \frac{8 \times 125}{100} : \frac{4 \times 130}{100}$
$= 5 \times 120 : 8 \times 125 : 4 \times 130$
$= 15 : 25 : 13$

13. (5) let the numbers be x and y respectively.
According to the question,

$$y - \frac{x \times 30}{100} = \frac{4}{5}y$$

$$\Rightarrow \frac{y}{5} = \frac{3x}{10}$$

$$\Rightarrow \frac{x}{y} = \frac{10}{3 \times 5} = 2 : 3$$

14. (4) Tricky approach
If the largest and the second largest angles be $3x°$ and $2x°$. respectively then, third angle = x
$\therefore x + 2x + 3x = 180°$
$\Rightarrow x = 30°$
$\therefore$ Required sum $= x + 2x = 3x = 90°$

15. (4) Tricky approach
$7x + 2x + 5x + 6x = 360°$
$\Rightarrow 20x = 360°$

$$\Rightarrow x = \frac{360}{20} = 18$$

$$\therefore \text{Required answer} = 2 \times 2x + \frac{7x}{2}$$

$$= \frac{15x}{2} = \frac{15 \times 18}{2} = 135°$$

16. (2) $2x + 4x + 7x + 5x = 360°$
$\Rightarrow 18x = 360°$

$$= x = \frac{360}{18} = 20°$$

$\therefore$ Smallest angle of the triangle
$= 2 \times 20° = 40°$
Second angle $= 2 \times 40° = 80°$
$\therefore$ Required angle $= 180° - 80° - 40°$
$= 60°$

17. (2) $3x + 4x + 6x + 7x = 360°$
$\Rightarrow 20x = 360° \Rightarrow x = 18°$

$$\therefore \text{smaller angle of the parallelogram} = \frac{6x}{2}$$

$= 3x = 54°$
$\therefore$ Adjacent angle of parallelogram
$= 180° - 54° = 126°$

18. (1) $x + 4x + 5x + 60 = 360°$
$\Rightarrow 10x = 300° \Rightarrow x° = 30$
$\therefore$ Required difference $= 5x - x = 4x$

= 4 × 30 = 120°

19. (1) $20x + 73x + 83x = 950 - 25 - 15 - 30$

$\Rightarrow 176x = 880 \Rightarrow x = \frac{880}{176} = 5$

$\therefore$ Number of coins got by Amita = $73x + 15$
$= 73 \times 5 + 15 = 380$

20. (4) Sum of three angles of a triangle =180°
Largest angle = $13x°$
Second largest angle = $12x°$

$\therefore$ Third angle = $(13x + 12x) \times \frac{1}{5} = 5x°$

$\therefore 13x + 12x + 5x = 180°$
$\Rightarrow 30x = 180°$

$\Rightarrow x = \frac{180}{30} = 6°$

$\therefore$ Required sum = $5x + 12x = 17x$
$= 17 \times 6 = 102°$

21. (4) Pranab $\times \frac{25}{100}$ = Surya $\times \frac{80}{100}$

$\Rightarrow \frac{\text{Pranab}}{\text{Surya}} = \frac{80}{25} = \frac{16}{5}$

Pranab : Surya = 16 : 5

$\frac{\text{Surya}}{\text{Dheeru}} = \frac{40}{100} = \frac{2}{5}$

$\Rightarrow$ Surya : Dheeru = 2 : 5
Pranab : Surya : Dheeru
= 16 × 2 : 5 × 2 : 5 × 5
= 32 : 10 : 25
Now,
$\because 25 \equiv 600000$

$\therefore 32 \equiv \frac{600000}{25} \times 32$

= Rs. 768000 = Pranab' annual income
$\therefore$ Pranab's monthly salary

$= \frac{768000}{12}$ = Rs. 64000

22. (3) Sum ofthe angles of quadrilateral = 360°
$\Rightarrow x + 6x + 2x + 45 = 360°$
$\Rightarrow 9x = 360 - 45 = 315°$

$\Rightarrow x = \frac{315}{9} = 35°$

$\therefore$ Required difference = $6x - x = 5x$
$= 5 \times 35 = 175°$

23. (3) Sum of the angles of a quadrilateral = 360°
$\therefore 3x + 4x + 6x + 5x = 360°$
$\Rightarrow 8x = 360° \Rightarrow x = 20°$

$\therefore$ The largest angle ofthe quadrilateral
$= 6 \times 20 = 120°$
$\therefore$ Smaller angle of parallelogram

$= 120 \times \frac{2}{3} = 80°$

$\therefore$ Its adjacent angle = 180 - 80 = 100°

24. (3) If the number of 2-rupee coins be x, then number of 5 rupee coins = x - 5
$\therefore 2x + 5(x - 5) = 50 - 26$
$\Rightarrow 2x + 5x - 25 = 24$
$\Rightarrow 7x = 24 + 25 = 49$

$\Rightarrow x = \frac{49}{7} = 7$

25. (5) If the adjacent angles of parallelogram be $2x°$ and $3x°$ respectively, then
$2x° + 3x° = 180°$
$\Rightarrow 5x° = 180°$
$\Rightarrow x° = 36°$
$\therefore$ Smaller angle of parallelogram
$= 2x = 72°$
$\therefore$ Smallest angle of the quadrilateral = 36°
$\therefore$ Its largest angle = 4 × 36 = 144°
$\therefore$ Required sum = 144 + 72 = 216°

26. (3) Sum of adjacent angles of a parallelogram = 180°
$\therefore$ One of the angles of triangle

$= \frac{2}{3} \times 180° = 120°$

Sum of three angles of a triangle = 180°
$\therefore 5x + 7x = 180 - 120$
$\Rightarrow 12x = 60 \quad \Rightarrow x = 5$
Second angle of triangle = 5 × 5 = 25°
Third angle of triangle = 7 × 5 = 35°
$\therefore$ The second largest angle of triangle = 35°

27. (3) Sum of angles of a triangle = 180°
$\Rightarrow 3x + x + 44 = 180°$
$\Rightarrow 4x = 180 - 44 = 136°$

$\Rightarrow x = \frac{136}{4} = 34°$

$\therefore$ Largest angle of triangle = 3 × 34 = 102°

$\therefore$ 150% of 102 = $\frac{102 \times 150}{100} = 153°$

28. (2) For the Parallelogram,
$4x° + 5x° = 180°$
$\Rightarrow 9x = 180°$

$\Rightarrow x = \frac{180}{9} = 20°$

$\therefore$ Smaller angel of parallelogram
= 4 × 20 = 80°
$\therefore$ One angle of the quadrilateral

$= 3 \times 80 = 240°$

$\therefore 4y + 11y + 9y = 360 - 240 = 120°$

$\Rightarrow 24y = 120° \quad \Rightarrow y = \frac{120}{24} = 5°$

$\therefore$ Its smallest angle $= 4 \times 5 = 20°$

$\therefore$ Required sum $= 240° + 20° = 260°$

29. (2) Sum of the angles of quadrilateral $= 360°$

$\therefore 3x + 4x + 5x + 6x = 360°$

$\Rightarrow 18x = 360°$

$\Rightarrow x = \frac{360}{18} = 20°$

$\therefore$ Smallest angle of quadrilateral $= 3 \times 20 = 60°$

Largest angle of quadrilateral $= 6 \times 20 = 120°$

$\therefore$ Smallest angle of triangle $= 60 \times \frac{2}{3} = 40°$

Largest angle of triangle $= 2 \times 40 = 80°$

$\therefore$ Third angle of triangle $= 180° - 40° - 80° = 60°$

$\therefore$ Required answer $= 60 + 120 = 180°$

30. (3) The smallest angle of triangle is half of the largest angle.

Ratio of three angles $= 4 : 3 : 2$

$\therefore 4x + 3x + 2x = 180$

$\therefore 9x = 180$

$\Rightarrow x = 20$

$\therefore$ Required difference $= 4x - 2x = 2x$
$= 2 \times 20 = 40°$

31. (4) Let the three angles of quadrilateral be $13x°$, $9x°$ and $5x°$ respectively.

$\therefore 13x + 9x + 5x = 360 - 36$

$\Rightarrow 27x = 324° \Rightarrow x = \frac{324}{27} = 12°$

$\therefore$ Required difference $= 13x - 5x = 8x$
$= 8 \times 12 = 96°$

32. (1) Let the adjacent angles be $7x°$ and $8x°$.

$\therefore 7x + 8x = 180°$

$\Rightarrow 15x = 180°$

$\Rightarrow x = 12°$

$\therefore$ Smaller angle $= 7 \times 12 = 84°$

Again, $5y + 6y + 7y + 12y = 360°$

$\Rightarrow 30y = 360°$

$\Rightarrow y = \frac{360°}{30} = 12°$

$\therefore$ Second largest angle of the quadrilaterals $7 \times 12 = 84°$

$\therefore$ Required sum $= 84 + 84 = 168°$

33. (2) Let the present ages of Sulekhaa and Arunima be $9x$ and $8x$ years respectively.

According to the question,

After 5 years,

$= \frac{9x+5}{8x+5} = \frac{10}{9}$

$\Rightarrow 81x + 45 = 80x + 50$

$\Rightarrow 81x - 80x = 50 - 45$

$\Rightarrow x = 5$

$\therefore$ Required difference $= 9x - 8x = x = 5$ years

34. (5) Let the present ages of Sonal and Nitya be $9x$ and $5x$ year respectively

According to the question,

$= \frac{9x+8}{5x+8} = \frac{13}{9}$

$\Rightarrow 81x + 72 = 65x + 104$

$\Rightarrow 81x - 65x = 104 - 72$

$\Rightarrow 16x = 32$

$\Rightarrow x = \frac{32}{16} = 2$

$\therefore$ Required difference $= 9x - 5x$
$= 4x = 4 \times 2 = 8$ years

35. (2) Let the present ages of father and son be $17x$ years and $7x$ years respectively.

According to the question,

$= \frac{17x-6}{7x-6} = \frac{3}{1}$

$\Rightarrow 21x - 18 = 17x - 6$

$\Rightarrow 21x - 17x = 18 - 6$

$\Rightarrow 4x = 12$

$\Rightarrow x = \frac{12}{4} = 3$

$\therefore$ Father's present age $= 17 \times 3 = 51$ years

36. (4) Rani : Komal $= 3 : 5 = 6 : 10$

Komal : Pooja $= 2 : 3 = 10 : 15$

$\therefore$ Rani : Komal : Pooja $= 6 : 10 : 15$

We have insufficient data to solve this question.

37. (4) Let the present ages of Amit and his father be $2x$ years and $5x$ years respectively.

$\therefore \frac{2x+4}{5x+4} = \frac{5}{11}$

$\Rightarrow 25x + 20 = 22x + 44$

$\Rightarrow 3x = 24 \Rightarrow x = \frac{24}{3} = 8$

$\therefore$ Father's age 5 years ago $= 5x - 5$
$= 5 \times 8 - 5 = 35$ years

38. (3) Four years ago,

Shyam : Ram $= 3 : 4$

After four years,

$\frac{3x+8}{4x+8} = \frac{5}{6}$

$\Rightarrow 20x + 40 = 18x + 48$

$\Rightarrow 2x = 48 - 40 = 8$

$\Rightarrow x = \frac{8}{2} = 4$

$\therefore$ Shyam's present age = $3x + 4$
$= 3 \times 4 + 4 = 16$ years

39. (2) Let the prestent ages of Tina and Rakesh be $9x$ and $10x$ years respectively.
10 years ago,

$\frac{9x-10}{10x-10} = \frac{4}{5}$

$\Rightarrow 45x - 50 = 40x - 40$

$\Rightarrow 5x = 10 \Rightarrow x = \frac{10}{5} = 2$

$\therefore$ Rakesh's present age = $10x$
$= 10 \times 2 = 20$ years.

40. (3) Let the present ages of Vishal and Shekhar be $14x$ and $17x$ years respectively.
After 6 years,

$\frac{14x+6}{17x+6} = \frac{17}{20}$

$\Rightarrow 280x + 120 = 289x + 102$

$\Rightarrow 9x = 18$

$\Rightarrow x = 2$

$\Rightarrow$ Shekhar's present age = $17x$
$= 17 \times 2 = 34$ years

41. (4) Let the present ages of father and son be $5x$ and $2x$ years respectively.
After 4 years,
Son's age = y years
and mother's age = $2y$ years

$\therefore y = 2x + 4 \Rightarrow x = \frac{y-4}{2}$

$\therefore$ Father's present age = $5\frac{(y-4)}{2}$ years

Mother's present age = $(2y - 4)$ years
Clearly, data are inadequate.

42. (5) Let Radha's present age = x years.
$\therefore x = 2(x - 12) - 3$
$\Rightarrow x = 2x - 24 - 3$
$\Rightarrow x = 27$

$\therefore$ Raj's present age = $\frac{4}{9} \times 27 = 12$ years

$\therefore$ Raj's age after 5 years = $12 + 5 = 17$ years

43. (5) Let the present ages of Meena and Fiona be $16x$ and $13x$ years respectively.
According to the question,

$\frac{16x-4}{13x-4} = \frac{14}{11}$

$\Rightarrow 176x - 44 = 182x - 56$

$\Rightarrow 182x - 176x = 56 - 44$

$\Rightarrow 6x = 12$

$\Rightarrow x = 2$

$\therefore$ Fiona's age after four years = $13x + 4$
$= 13 \times 2 + 4 = 30$ years

44. (2) Let Swati's present age = $4x$ years
Trupti's present age = $5x$ years

$\therefore \frac{4x+6}{5x+6} = \frac{6}{7}$

$\Rightarrow 30x + 36 = 28x + 42$

$\Rightarrow 2x = 42 - 36 = 6$

$\Rightarrow x = 3$ = difference of their ages

45. (3) Let Ram's present age be $6x$ years and that of Rakesh be $11x$ years.
Four years ago,

$\frac{6x-4}{11x-4} = \frac{1}{2}$

$\Rightarrow 12x - 8 = 11x - 4$

$\Rightarrow x = 8 - 4 = 4$

$\therefore$ Rakesh's age after five years = $11x + 5$
$= 11 \times 4 + 5 = 49$ years

46. (4) According to the question,

$\frac{2x+7x}{2} = 27$

$\Rightarrow 9x = 27 \times 2 = 54$

$= x = \frac{54}{9} = 6$

$\therefore$ Mother's age after 7 years = $7x + 7$
$= 7 \times 6 + 7 = 49$ years

47. (4) Let the present ages of Ram, Rohan and Raj be $3x$, $4x$ and $5x$ years respectively.
$\therefore 3x + 4x + 5x = 3 \times 28$
$\Rightarrow 12x = 84$

$\Rightarrow x = \frac{84}{12} = 7$

Sum of the ages of Ram and Rohan after 5 years $= 3x + 4x + 10 = 7x + 10$
$= 7 \times 7 + 10 = 59$ years

48. (1) Rehana's present age = $85 - 7 = 78$ years
Wasim's present age = $78 - 12 = 66$ years

$\therefore$ Manoj's present age = $\frac{3}{11} \times 66 = 18$ years

$\therefore$ Manoj's father's present age = $25 + 18$
$= 43$ years

49. (5) Jahnavi's present age = $33 - 9 = 24$ years
$\therefore$ Aarti's present age = $24 - 9 = 15$ years
Now, Aarti : Savita = $5 : x = 15 : 3x$
$\therefore$ Savita's present age = $3x$ years

$\therefore 3x - 15 = 24$

$\Rightarrow 3x = 24 + 15 = 39$

$\Rightarrow x = \frac{39}{3} = 13$

50. (3) Let the amount received by P, Q and R be Rs. $3x$, Rs. $5x$ and Rs. $7x$ respectively.

$\therefore 7x - 5x = 4000$

$\Rightarrow x = \frac{4000}{2} = 2000$

$\therefore$ Amount received by P and Q together $= 8x = 8 \times 2000 =$ Rs. 16000

51. (4) Abhinav's investment = Rs. 6000

Sunil's investment = $\frac{70 \times 6000}{100}$ = Rs. 4200

Rita's investment = $\frac{4200 \times 125}{100}$ = Rs. 5250

$\therefore$ Required ratio
$= 5250 : (6000 + 1200 + 5250)$
$= 5250 : 15450 = 35 : 103$

52. (5) $\frac{9-x}{15-x} = \frac{15-x}{27-x}$

$\Rightarrow 243 - 9x - 27x + x^2 = 225 - 30x + x^2$

$\Rightarrow 6x = 243 - 225 = 18$

$\Rightarrow x = 3$

53. (1) Let the amount be Rs. x.

$\therefore \left(\frac{3}{9} - \frac{2}{14}\right)x = 40$

$\Rightarrow \left(\frac{1}{3} - \frac{1}{7}\right)x = 40$

$\Rightarrow \left(\frac{7-3}{21}\right)x = 40$

$\Rightarrow x = \frac{40 \times 21}{4}$ = Rs. 210

54. (1) If the total amount be Rs. x, then

$\frac{2x}{15} = 4908$

$\Rightarrow x = \frac{4908 \times 15}{2}$ = Rs. 36810

$\therefore$ Required difference $= \frac{7-6}{15} \times 36810$

= Rs. 2454

55. (1) Four years ago,
Father's age = $10x$ years
Son's age = $3x$ years
$\therefore 10x + 3x + 8 = 60$

$\Rightarrow 13x = 60 - 8 = 52$

$\Rightarrow x = 4$

$\therefore$ Required difference $= 7x = 7 \times 4 = 28$ years

56. (1) $x = y + 52$

$z = y - 26$

$\therefore x + y + z = 221$

$\Rightarrow y + 52 + y + y - 26 = 221$

$\Rightarrow 3y = 221 - 26 = 195$

$\Rightarrow y = \frac{195}{3} = 65$

$\therefore x = 65 + 52 = 117$

$\therefore z = 65 - 26 = 39$

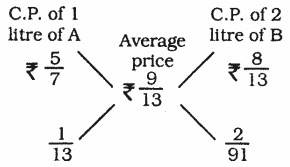
$\therefore x : y : z = 117 : 65 : 39 = 9 : 5 : 3$

57. (1) Let. CP. of milk per litre be Rs 1

Milk in 1 litre of A = $\frac{5}{7}$ litre

Milk in 1 litre of B = $\frac{8}{13}$ litre

Milk in 1 litre of mixture = $\frac{9}{13}$ litre



$\therefore$ Required ratio = $\frac{1}{13} : \frac{2}{91} = 7 : 2$

58. (1) First number = x
Second number = y

$\therefore x \times \frac{50}{100} = y \times \frac{3}{4}$

$\Rightarrow \frac{x}{2} = y \times \frac{3}{4}$

$\Rightarrow \frac{x}{y} = \frac{3}{4} \times 2 = \frac{3}{2}$

59. (1) Mahesh = $3x$ years
Ajay = $2x$ years
After 8 years,

$\frac{3x+8}{2x+8} = \frac{11}{8}$

$\Rightarrow 24x + 64 = 22x + 88$

$\Rightarrow 2x = 88 - 64 = 24$

$\Rightarrow x = 12$

$\therefore$ Ajay's age $= 2x = 2 \times 12 = 24$ years

$\therefore$ Age of Mahesh's son = $\frac{1}{2} \times 24 = 12$ years

SBI PO EXAMS

1. (3) Let the first number be x and the second number be y.

$x + y = 333\frac{1}{3}\%$ of y

$\Rightarrow x + y = \frac{1000}{3\times100}y$

$\Rightarrow x + y = \frac{10}{3}y$

$\Rightarrow 3x + 3y = 10y$

$\Rightarrow 3x = 7y$

$\Rightarrow \frac{x}{y} = \frac{7}{3} = 7 : 3$

2. (1) Let P's share be $6x$
Q's share be $19x$ and
R's share be $7x$.
Total sum = $6x + 19x + 7x = 32x$
$6x : 19x + 200 : 7x - 200 = 3 : 10 : 3$
We can write
$6x = 7x - 200$ or, $x = 200$
$\therefore$ Total sum = $32x$ = Rs. 6400

3. (5) Ratio of no. boys : girls = 2 : 3
Let the no. of boys = $2x$
Then the no. of girls = $3x$
No. of boys after 20% increase
$= 1.20 \times 2x = 2.4x$.
No. of girls after 10% increase
$= 1.10 \times 3x = 3.3x$

Required ratio = $\frac{2.4x}{3.3x} = \frac{8}{11} = 8 : 11$

4. (3) A : B = 5 : 8
Let A's income be $5x$ and B's income be $8x$.

or, $\frac{5x+25}{8x} = \frac{5}{4}$

or, $\frac{5(x+5)}{4\times2x} = \frac{5}{4}$ or, $\frac{x+5}{2x} = 1$

$\Rightarrow x + 5 = 2x$

$\Rightarrow x = 5$

B's income = $8x = 8 \times 5$ = Rs. 40 lakhs.

5. (4) Let the earnings of A and B be Rs. $4x$ and $7x$ respectively.
After 50% increase,
A's earnings = 150% of $4x$
After 25% decrease,
B's earning = 75% of $7x$
Ratio = 150% of $4x$: 75% of $7x$ = 8 : 7
But their total earnings are unknown. Hence A's earnings can't be known.

6. (2) Let the salaries of A, B and C be Rs. $2x$, Rs. $3x$ and Rs. $5x$ respectively.
After respective increase of 15% 10% and 20% their salaries will be

Rs. $\frac{115\times2x}{100}$, $\frac{110\times3x}{100}$ and $\frac{120\times5x}{100}$

$\therefore$ Required ratio

$= \frac{115\times2x}{100} : \frac{110\times3x}{100} : \frac{120\times5x}{100} = 23 : 33 : 60$

7. (5) Let the present ages of Seema and Naresh be $5x$ and $7x$ years respectively.
According to the question,

$\frac{5x+5}{7x+5} = \frac{3}{4}$

$\Rightarrow 21x + 15 = 20x + 20$

$\Rightarrow x = 20 - 15 = 5$

$\therefore$ Naresh's present age = $7x$ years
$= 7 \times 5 = 35$ years

8. (1) Let the inital seats for Maths, Physics and Biology be $5x$, $7x$ and $8x$ respectively.

Now, new. seats for Maths = $\frac{5x\times140}{100}$

For Physics = $\frac{7x\times150}{100}$

and for Biology = $\frac{8x\times175}{100}$

$\therefore$ Required ratio = $\frac{5x\times140}{100} : \frac{7x\times150}{100} : \frac{8x\times175}{100}$

$= 5 \times 140 : 7 \times 150 : 8 \times 175$
$= 2 : 3 : 4$

9. (2) Let the original sum be Rs.x.
Sum of the Ratios = 3 + 5 + 9 + 13 = 30

$\therefore$ C's share = Rs. $\frac{9x}{30}$ = Rs. $\frac{3x}{10}$

A's share = Rs. $\frac{3x}{30}$ = Rs. $\frac{x}{10}$

According to the question,

$\frac{3x}{10} - \frac{x}{10} = 2412$

$\Rightarrow \frac{2x}{10} = 2412$

$\Rightarrow x = 2412 \times 5$ = Rs. 12060

$\therefore$ Amount received by B and D together

Rs $\left(\frac{(5+13)}{30}\times12060\right)$ = Rs. 7236

10. (5) Let the present ages of Khushi and Jagriti be $5x$ and $8x$ years respectively.
After 8 years,

$$\frac{5x+8}{8x+8}=\frac{3}{4}$$

$\Rightarrow 24x + 24 = 20x + 32$

$\Rightarrow 4x = 32 - 24 = 8$

$$\Rightarrow x = \frac{8}{4} = 2$$

$\therefore$ Required difference = $(8x - 5x)$ years
$= 3x = 3 \times 2 = 6$ years

11. (3) Let the ages of the mother and daughter be $7x$ and x years respectively.

$$\therefore \text{Four years ago, } \frac{7x-4}{x-4}=\frac{19}{1}$$

$\Rightarrow 19x - 76 = 7x - 4$

$\Rightarrow 12x = 72 \Rightarrow x = 6$

$\therefore$ Mother's age after four years
$= 7x + 4 = 7 \times 6 + 4 = 46$ years

12. (1) Let the original amounts invested in Debit and Equity funds be Rs. $4x$ and $5x$ respectively.
Dividend at the end of the year

$$= \frac{9x\times 30}{100} = \text{Rs } \frac{27x}{10}$$

Total investment after one year

$$= \text{Rs } \left(9x+\frac{27x}{10}\right) = \text{Rs } \left(\frac{117x}{10}\right)$$

$$\therefore \frac{7}{13}\times\frac{117x}{10} = 94500$$

$$\Rightarrow x = \frac{94500\times 13\times 10}{7\times 117} = 15000$$

$\therefore$ The original amount invested in Equity funds = $5x = 5 \times 15000$ = Rs. 75000

RBI GRADE-B OFFICER EXAMS

1. (4) Let the present ages of Samir and Tanuja be $8x$ and $15x$ years respectively.
According to the question,

$$\text{After 9 years, } \frac{8x+9}{15x+9}=\frac{11}{18}$$

$\Rightarrow 165x + 99 = 144x + 162$

$\Rightarrow 165x - 144x = 162 - 99$

$\Rightarrow 21x = 63$

$$\Rightarrow x = \frac{63}{21} = 3$$

$\therefore$ Samir's present age = $8x = 24$ years
Tanuj's present age = $15x = 45$ years
$\therefore$ Required difference = 45 - 24 = 21 years

2. (4) Let the earnings of A and B be Rs. $8x$ and Rs. $9x$ respectively.
Now, after changes in their earnings,

$$\text{A's earning} = 8x \times \frac{150}{100} = \text{Rs. } 12x$$

$$\text{B's earning} = 9x \times \frac{75}{100} = \text{Rs. } \frac{27x}{4}$$

Clearly, we need sum of their earnings etc. to get the desired result.

3. (4) Original number of boys in the college $= 31x$
Original number of girls $= 23x$

$$= \frac{31x}{23x+75} = \frac{124}{107}$$

$\Rightarrow 2852x + 9300 = 3317x$

$\Rightarrow 3317x - 2852x = 9300$

$\Rightarrow 465x = 9300$

$$\Rightarrow x = \frac{9300}{465} = 20$$

$\therefore$ Number of boys $= 31 \times 20 = 620$
New number of girls $= 23 \times 20 + 75 = 535$
$\therefore$ Required answer $= 620 - 535 = 85$

4. (4) A's salary = Rs. $8x$ and B's salary = Rs. $9x$

$$\frac{8x\times 150\%}{9x\times 75\%} = \frac{16}{9}$$

$$\Rightarrow \frac{8x\times 150}{9x\times 75} = \frac{16}{9}$$

$$\Rightarrow \frac{\frac{12x}{27x}}{4} = \frac{16}{9}$$

$$\Rightarrow x = \frac{48}{27} = \frac{16}{9}$$

Hence, A's salary cannot be determined.

INSURANCE EXAMS

1. (1) Required answer

$$= \frac{23000}{6325} = \frac{40}{11}$$

2. (4) Let the earnings of A and B be Rs. $4x$ and $7x$ respectively.
After 50% increase,
A's earnings = 150% of $4x$
After 25% decrease,
B's earning = 75% of $7x$

Ratio = 150% of $4x$: 75% of $7x$ = 8 : 7
But their total earnings are unknown.
Hence A's earnings can't be known.

3. (2) Let the salaries of A, B and C be Rs. $2x$, Rs. $3x$ and Rs. $5x$ respectively.
After respective increase of 15% 10% and 20% their salaries will be

Rs $\frac{115\times 2x}{100}$, $\frac{110\times 3x}{100}$ and $\frac{120\times x}{100}$

$\therefore$ Required ratio = $\frac{115\times 2x}{100} : \frac{110\times 3x}{100} : \frac{120\times x}{100}$

= 23 : 33 : 60

4. (1) A : B : C = 8 : 14 : 22
= 12 : 21: 33
B : C : D = 12 : 21 : 33
$\therefore$ A : B : C : D = 12 : 21 : 33 : 44

5. (4) Let the numbers be x and y respectively.
Then,

$\frac{3}{4}x$ = 60% of y

$\Rightarrow \frac{3}{4}x = \frac{60}{100}y \Rightarrow \frac{3}{4}x = \frac{3}{5}y$

$\Rightarrow y = \frac{3}{4}\times\frac{5}{3}x = \frac{5}{4}x$

$\therefore\ y - x = \frac{5}{4}x - x = \frac{x}{4}$

Clearly, no unique answer is possible.

6. (4) Let the three numbers be a, b and c respectively.
Now,
$a: b$ = 2 : 3
b : c = 5 : 3
$\therefore a : b : c$ = 2 × 5 : 3 × 5 : 3 × 3
= 10 : 15 : 9
According to the question,
$a + b + c$ = 136
$\Rightarrow 10x + 15x + 9x = 136$

$\Rightarrow 34x = 136 \Rightarrow x = \frac{136}{34} = 4$

$\therefore b = 15x = 15 \times 4 = 60$

7. (3) $a : b : c = \frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ = 6 : 4 : 3

$\therefore$ The longest side = $\frac{6}{13}\times 104$ = 48 cm.

8. (1) <u>Tricky approach</u>
Let the weights of the four prices of diamond be x, $2x$, $3x$ and $4x$ units respectively.
Total weight of the original diamond
$= x + 2x + 3x + 4x = 10x$ units
Price of the original diamond
$= k(100x^2)$ where k is a constant.
Cost of four pieces of diamond
$= k\,(x^2 + 4x^2 + 9x^2 + 16x^2)$
$= k.30x^2$
Loss in the cost of diamond
$= k(100x^2 - 30x^2) = k.70x^2$
$\therefore k.70x^2$ = 70000
$\Rightarrow kx^2$ = Rs. 1000
$\therefore$ Cost of the original diamond
$= k\,(100x^2)$
= 1000 × 100 = Rs. 100000

9. (2) Sum of ratios
= 3 + 5 + 9 + 7 = 24
Second largest angle of quadrilateral

= $\frac{7}{24}\times 360 = 105°$

= Largest angle of triangle
$\therefore$ Third angle of triangle = 180° - 105° - 25°
= 50°= second largest angle of triangle

10. (5) Let Abhijit invested Rs. $2x$, Rs. $3x$ and Rs. $4x$ in three schemes A, B and C respectively.
$\therefore$ The required ratio

= $2x \times \frac{120}{100} : 3x \times \frac{116}{100} : 4x \times \frac{115}{100}$
= 2 × 120 : 3 × 116 : 4 × 115
= 60 : 87 : 115

11. (3) Actual price of diamond = $k \times 6^2$
= $36k$ where k is constant of proportionality.
New price of diamond
= $kx_1^2 + kx_2^2$

$\Rightarrow 36k \times \frac{5}{8} = k(x_1^2 + x_2^2)$

$\Rightarrow \frac{45}{2} = x_1^2 + x_2^2$

$\Rightarrow 22.5 = x_1^2 + x_2^2$

$\Rightarrow (4.5)^2 + (1.5)^2 = x_1^2 + x_2^2$

12. (1) S.P. of shirt = Rs. x

CP. = Rs. $\frac{2x}{3}$

$\therefore$ Marked price = $\frac{4}{3}\times x$ = Rs. $\frac{4x}{3}$

$\therefore$ Required ratio = $\frac{2x}{3} : \frac{4x}{3}$ = 1 : 2

13. (2) Required ratio = $\frac{5}{3} : \frac{7}{5}$

$= \frac{5}{3} \times 15 : \frac{7}{5} \times 15 = 25 : 21$

14. (4) $x + x + 2 + x + 4 = 285$

$\Rightarrow 3x = 285 - 6 = 279$

$\Rightarrow = \frac{279}{3} = 93$

$\therefore$ Required ratio = 93 : 97

15. (5) Let the present ages of Sonal and Nitya be $9x$ and $5x$ years respectively.
According to the question,

$= \frac{9x+8}{5x+8} = \frac{13}{9}$

$\Rightarrow 81x + 72 = 65x + 104$

$\Rightarrow 81x - 65x = 104 - 72$

$\Rightarrow 16x = 32$

$\Rightarrow x = \frac{32}{16} = 2$

$\therefore$ Required difference = $9x - 5x = 4x$
$= 4 \times 2 = 8$ years

16. (3) $\frac{x}{y} = \frac{4}{5}$

$\therefore \frac{3x+y}{5x+3y} = \frac{3\frac{x}{y}+1}{5\frac{x}{y}+3}$

$= \frac{\frac{3\times4}{5}+1}{5\times\frac{4}{5}+3} = \frac{\frac{12+5}{5}}{7} = \frac{17}{35}$

17. (1) Numbers = $2x$ and $3x$

$\therefore \frac{2x+4}{3x+4} = \frac{5}{7}$

$\Rightarrow 15x + 20 = 14x + 28$

$\Rightarrow x = 28 - 20 = 8$

= Difference between numbers.

18. (4) Time taken by B = x minutes.
Time taken by A = $(x + 10)$ minutes.

$\therefore \frac{2}{3} = \frac{x}{x+10}$

$\therefore 2x + 20 = 3x$

$\therefore x = 20$ minutes

$\therefore$ Time taken by A = 30 minutes.

$\therefore$ Time taken by A when he doubles his speed = $\frac{30}{2}$ =15 minutes

19. (3) In the beginning.
The rate of sugar = Rs. 100/kg
Expenditure = 1 kg.
Now,
Rate = Rs. 120 per kg

$\therefore$ In Rs. 100, sugar bought = $\frac{100}{120} = \frac{5}{6}$ kg

$\therefore$ Required ratio = $\left(1 - \frac{5}{6}\right) : 1$

$= \frac{1}{6} : 1 = 1 : 6$

20. (3) After four years,
Anil = $3x$ years
Sudhir = $4x$ years
In the present,
$4x - 4 - 3x + 4 = 6$
$\Rightarrow x = 6$
$\therefore$ Sudhir's present age = $4x - 4$
$= 4 \times 6 - 4 = 20$ years

21. (2) Ratio = $5 : 8 = 5 \times 8 : 8 \times 8 = 40 : 64$

22. (2) Number of students :
A = $5x$
B = $6x$
C = $8x$

$\therefore$ Required ratio = $\frac{5x\times130}{100} : \frac{6x\times125}{100} : \frac{8x\times125}{100}$
= 650 : 750 : 1000 = 13 : 15 : 20

23. (2) Average marks of girls = x

$\therefore A = \frac{3\times(A+1)+x}{3+1}$

$\Rightarrow 4A = 3A + 3 + x$

$\Rightarrow x = 4A - 3A - 3 = A - 3$

24. (4) Length of first train = x metre
Length of second train = y metre

Speed of first train = $\left(\frac{x}{27}\right)$ m/sec

Speed of second train = $\left(\frac{y}{17}\right)$ m/sec

Relative Speed = $\left(\frac{x}{27} + \frac{y}{17}\right)$ m/sec

$\therefore 23 = \frac{x+y}{\frac{x}{27}+\frac{y}{17}}$

$\Rightarrow \frac{x}{27}+\frac{y}{17}=\frac{x}{23}+\frac{y}{23}$

$\Rightarrow \frac{x}{23}-\frac{x}{27}=\frac{y}{17}-\frac{y}{23}$

$\Rightarrow \frac{27x-23x}{23\times 27}=\frac{23y-17y}{23\times 17}$

$\Rightarrow \frac{4x}{27}=\frac{6y}{17}$

$\Rightarrow \frac{x}{y}=\frac{6}{17}\times\frac{27}{4}=\frac{81}{34}$

25. (1) A : B = 1 : 3 = 15 : 45
B : C = 5 : 7 = 5 × 7 : 7 × 9
= 45 : 63
C : D = 9 : 7 =9 × 7 : 7 × 7 = 63 : 49
$\therefore$ A : B : C : D = 15 : 45 : 63 : 49

26. (2) Materials : labour : Overheads
= 3 : 4 : 1
Sum of ratios = 3 + 4 +1 = 8
If the total cost be Rs. x, then

$\frac{3x}{8}$ = 234

$\Rightarrow x = \frac{234\times 8}{3}$ = Rs. 624

$\therefore$ Labour cost = $\frac{4}{8}\times 624$

= Rs. 312

MODEL EXERCISES

1. M The total emoluments of A and B are equal. However, A gets 65% of his basic salary as allowances and B gets 80% of his basic salary as allowances. What is the ratio ofthe basic salaries of A and B ?
(1) 16 : 13 (2) 5 : 7
(3) 12 : 11 (4) 7 : 9
(5) None of these

2. A contractor employed 25 labourers on a job. He was paid 275 for the work. After retaining 20% of this sum, he distributed the remaining amount amongst the labourers. If the number of men to women labourers was in the ratio 2 : 3 and their wages in the ratio 5 : 4, what wages did a woman labourer get ?
(1) 10 (2) 8
(3) 12 (4) 15
(5) None of these

3. The dimensions of a rectangular room when increased by 4 m are in the ratio of 4 : 3 and when decreased by 4 m are in the ratio of 2 : 1. The dimensions of the room are
(1) 6 m and 4 m
(2) 12 m and 8 m
(3) 16 m and 12 m
(4) 24 m and 16 m
(5) None of these

4. If 1066 are divided among A, B, C and D such that A : B = 3 : 4, B : C = 5 : 6 and C : D = 7 : 5, who will get the maximum ?
(1) B (2) A
(3) C (4) D
(5) None of these

5. Two numbers are such as the square of one is 224 less than 8 times the square of the other. If the numbers be in the ratio of 3 : 4 their values are
(1) 12, 16 (2) 6, 8
(3) 9, 12 (4) 6, 9
(5) All of these

6. If a : b = 2 : 5, then the value of $(2a + 2b) : (7a + 5b)$ is
(1) $\frac{19}{39}$ (2) $\frac{19}{13}$
(3) $\frac{31}{19}$ (4) $\frac{19}{31}$
(5) None of these

7. P = $\frac{x^2+36}{x^2-49}$ and Q = $\frac{x+6}{x-7}$ then the value of $\frac{P}{Q}$ is
(1) $\frac{x-6}{x-7}$ (2) $\frac{x-6}{x+7}$
(3) $\frac{x-7}{x+6}$ (4) $\frac{x+6}{x-7}$
(5) None of these

8. If the ratio of boys to girls in a class is B and the ratio of girls to boys is G, then 3(B + G) is
(1) equal to 3 (2) less than 3
(3) more than 3 (4) less than $\frac{1}{3}$
(5) None of these

9. The monthly incomes of two persons are in the ratio of 4 : 5 and their monthly expenditures are in the ratio of 7 : 9. If each saves Rs 50 a month, then what are their monthly incomes ?
(1) Rs 100, Rs 125
(2) Rs 300, Rs 375
(3) Rs 300, Rs 375
(4) Rs 400, Rs 500
(5) None of these

10. A bag contains Rs 216 in the form of one rupee, 50 paise and 25 paise coins in the ratio of 2 : 3 : 4. The number of 50 paise coins is
(1) 96 (2) 144
(3) 114 (4) 141
(5) None of these

11. Half the girls and one-third of the boys of a college reside in the hostel. What fractional part of the student body is hostel dwellers if the total number of girls in the college is 100 and is $\frac{1}{4}$ of the total strength ?
(1) $\frac{2}{5}$ (2) $\frac{5}{12}$
(3) $\frac{1}{5}$ (4) None of these
(5) All of these

12. Given that 24 carat gold is pure gold, 18 carat gold is $\frac{3}{4}$ gold and 20 carat gold is $\frac{5}{6}$ gold, the ratio of the pure gold in 18 carat gold to the pure gold in 20 carat gold is
(1) 5 : 8 (2) 9 : 10
(3) 15 : 24 (4) 8 : 5
(5) None of these

13. If $\frac{y}{z-x}=\frac{y+x}{z}=\frac{x}{y}$, then find *x : y : z.*
(1) 1 : 2 : 3 (2) 3 : 2 : 1
(3) 4 : 2 : 3 (4) 2 : 4 : 7
(5) None of these

14. The ratio of the rate of flow of water in pipes varies inversely as the square of the radius of the pipes. What is the ratio of the rates of flow in 2 pipes of diameter 2 cm and 4 cm ?
(1) 1 : 2 (2) 2 : 1
(3) 1 : 8 (4) 4 : 1
(5) None of these

15. Salaries of A, B and C were in the ratio 3 : 5 : 7, respectively. If their salaries were increased by 50%, 60% and 50% respectively, what will be the new ratio of the their respective salaries ?
(1) 4 : 5 : 7 (2) 3 : 6 : 7
(3) 4 : 15 : 18 (4) 9 : 16 : 21
(5) None of these

16. A sink contains exactly 12 L of water. If water is drained from the sink untill it holds exactly 6 L of water less than the quantity drained away, how many litres of water were drained away ?
(1) 2 L (2) 6 L
(3) 3 L (4) 9 L
(5) None of these

17. Railway fares of 1st, 2nd and 3rd classes between two stations we are in the ratio of 8 : 6 : 3. The fares of 1st and 2nd class were subsequently reduced by 1/6 and 1/12 respectively. If during a year, the ratio between the passengers of 1st, 2nd and 3rd classes was 9 : 12 : 26 and total amount collected by the sale of tickets was Rs 1088, then find the collection from the passengers of 1st class.
(1) Rs 260 (2) Rs 280
(3) Rs 300 (4) Rs 320
(5) None of these

18. At Narmada Sarovar Bachao (NSB; demonstration, supporters of Ms Patkar outnumbered the police by 9 : 1. The police arrested 135 NSB supporters averaging 5 for even 3 policemen. How many supporters of NSB were there in the demonstration ?
(1) 1215 (2) 665
(3) 405 (4) None of these
(5) All of these

19. In a proportion the multiplication of 1st and 4th terms is 10 and that of 2nd and 3rd terms is 2.5*x*, then the value of *x* is
(1) 16 (2) 36
(3) 75 (4) 90
(5) None of these

20. A vessel contains liquid P and Q in the ratio 5 : 3. If 16 L of the mixture are removed and the same quantity of liquid Q is added the ratio become 3 : 5. What quantity does the vessel hold ?
(1) 35 L (2) 45 L
(3) 40 L (4) 50 L
(5) None of these

21. A bottle is full of dettol. One-third of it is taken out and then an equal amount of water is poured into the bottle to fill it. This operationis done four times. Find the final ratio of dettol and water in the bottle.
(1) 13 : 55 (2) 20 : 74
(3) 16 : 65 (4) 10 : 48
(5) None of these

22. Divide Rs 671 among A, B and C such that if their shares be increased by Rs 3, Rs 7 and Rs 9 respectively, the remainder shall be in the ratio 1 : 2 : 3.
(1) Rs 110, Rs 220 and Rs 336
(2) Rs 112. Rs 223 and Rs 336
(3) Rs 105, Rs 223 and Rs 330
(4) None of these
(5) All of these

23. The ratio of the ages of the father and the son at present is 7 : 1. After 4 years, the ratio will become 4 : 1. What is the sum of the present ages of the father and the son ?
(1) 29 years (2) 35 years
(3) 32 years (4) 36 years
(5) None of these

24. The ratio between the number of passengers travelling by I and II classes between the two railway stations is 1 : 50, whereas the ratio of I and II classes fares between the same stations is 3 : 1. If on a particular day, Rs 1325 were collected from the passengers travelling between these stations, then what was the amount collected from the II class passengers ?
(1) Rs 750 (2) Rs 1000
(3) RS 850 (4) Rs 1250
(5) None of these

25. Mira's expenditure and savings are in the ratio 3 : 2. Her income increases by 10%. Her expenditure also increases by 12%. By how much per cent do her savings increase ?
(1) 7% (2) 9%
(3) 10% (4) 13%
(5) None of these

SHORT ANSWERS

1.	(1)	**2.**	(2)
3.	(2)	**4.**	(3)
5.	(2)	**6.**	(1)
7.	(1)	**8.**	(3)
9.	(4)	**10.**	(2)
11.	(4)	**12.**	(2)
13.	(3)	**14.**	(4)
15.	(4)	**16.**	(4)
17.	(4)	**18.**	(4)
19.	(1)	**20.**	(3)
21.	(3)	**22.**	(2)
23.	(3)	**24.**	(4)
25.	(1)		

EXPLANATIONS

1. (1) Let the basic salary of A be Rs x and that of B be Rs y, then

$$A \times \frac{65}{100} = B \times \frac{80}{100}$$

$\Rightarrow A : B = 16 : 13$

2. (2) According to the question,

$$\text{Number of men} = \frac{2}{5} \times 25 = 10$$

$$\text{Number of women} = \frac{3}{5} \times 25 = 15$$

Amount distributed among men and women = 275 × 80% = Rs 220

Let the wages paid to a man by $5x$ and to a woman be $4x$, then

$10 \times 5x + 15 \times 4x = 220$

$\Rightarrow 50x + 60x = 220$

$\Rightarrow x = 2$

$\therefore$ Wages received by a woman = 2 × 4 = Rs 8

3. (2) According to the question,

$$\therefore \frac{L+4}{B+4} = \frac{4}{3} \Rightarrow 3L + 12 = 4B + 16$$

$\Rightarrow 3L - 4B = 4$ ...(i)

and $\frac{L-4}{B-4} = \frac{2}{1}$

$\Rightarrow L - 4 = 2B - 8$

$\Rightarrow L - 2B = -4$ ...(ii)

Solving Eqs. (i) and (ii), we get

L = 12 m and B = 8 m

4. (3) According to question,

$$\frac{A}{B} = \frac{3\times5}{4\times5} = \frac{15}{20};$$

$$\frac{B}{C} = \frac{5\times4}{6\times4} = \frac{20}{24}$$

$$\frac{B}{C} = \frac{20\times7}{24\times7} = \frac{140}{168};$$

$$\frac{C}{D} = \frac{7\times24}{5\times24} = \frac{168}{120}$$

$$\therefore \frac{A}{B} = \frac{15\times7}{20\times7} = \frac{105}{140}$$

Now, $\frac{A}{B} = \frac{105}{140}, \frac{B}{C} = \frac{140}{168}, \frac{C}{D} = \frac{168}{120}$

$\therefore A : B : C : D = 105 : 140 : 168 : 120$

Hence, C gets the maximum share.

5. (2) Let the numbers be $3x$ and $4x$. Then,

$16x^2 = 8\times(9x)^2 - 224$

$\Rightarrow 16x^2 = 72 \times x^2 - 224$

$\Rightarrow 56x^2 = 224$

$\Rightarrow x^2 = 4$

$\Rightarrow x = 2$

Hence, numbers are 6, 8.

6. (1) According to the question,

$$\frac{a}{b} = \frac{2}{5}; \ \frac{2a+3b}{7a+5b} = \frac{\frac{2a}{b}+3}{\frac{7a}{b}+5}$$

$$= \frac{2\times\frac{2}{5}+3}{7\times\frac{2}{5}+5} = \frac{\frac{19}{5}}{\frac{39}{5}} = \frac{19}{5}\times\frac{5}{39} = \frac{19}{39}$$

7. (1) According to the question,

$$P = \frac{x^2-36}{x^2-49} = \frac{(x-6)(x+6)}{(x-7)(x+7)}$$

$$Q = \frac{x+6}{x+7}$$

$$\therefore \frac{P}{Q} = \frac{\frac{(x-6)(x+6)}{(x-7)(x+7)}}{\frac{(x+6)}{(x+7)}} = \frac{x-6}{x-7}$$

8. (3) Let the number of boys be x and number of girls be y.

Then, $\frac{x}{y} = B$ and $\frac{y}{x} = G$

$$\therefore 3(B + G) = 3\left(\frac{x}{y}+\frac{y}{x}\right)$$

$$= \frac{3(x^2+y^2)}{xy} > 3$$

9. (4) Let the income of two persons be $4x$ and $5x$ and their expenses be $7y$ and $9y$.
Then, $4x - 7y = 50$ ...(i)
and $5x - 9y = 50$ ...(ii)
Solving Eqs. (i) and (ii), we get
$x = 100$ and $y = 50$
$\therefore$ The income of persons are Rs 400 and Rs 500.

10. (2) By question amount of 50 paise coins

$= \frac{3}{4} \times 216 = \text{Rs } 72$

$\therefore$ Number of 50 paise coins = $72 \times 2 = 144$
(As Rs 1 has 2 fifty paise coins)

11. (4) By question total students = 400
No. of girls = 100
$\therefore$ No. of boys = 300
$\therefore$ No. of hostel dwellers

$= \frac{1}{2} \times 100 + \frac{1}{3} \times 300 = 150$

Required fraction = $\frac{150}{400} = \frac{15}{40}$

12. (2) According to the question

18 carat gold = $\frac{3}{4}$ pure gold = $\frac{3}{4} \times 24 = 18$

20 carat gold = $\frac{5}{6}$ pure gold = $\frac{5}{6} \times 24 = 20$

$\therefore$ Required ratio = 18 : 20 = 9 : 10

13. (3) According to the question,

$\frac{y}{x-z} = \frac{y+x}{z}$

$\Rightarrow yz = xy - x^2 - yz - xz$ ...(i)

Also, $\frac{x}{y} = \frac{y}{x-z}$

$\Rightarrow x^2 - xz = y^2$ ...(ii)
Using Eqs. (i) and (ii), we get
$yz = xy - yz + y^2$
$\Rightarrow 2yz = xy + y^2$
$\therefore 2z = x + y$ ...(iii)
Only option (3) satisfies the Eq. (iii).

14. (4) Radius of pipes are 1 cm and 2 cm.

$\therefore$ Required ratio = $\frac{1}{(1)^2} : \frac{1}{(2)^2} = 4 : 1$

15. (4) According to the question,

New ratio = $3 \times \frac{150}{100} : 5 \times \frac{160}{100} : 7 \times \frac{150}{100}$

$= \frac{9}{2} : 8 : \frac{21}{2} = 9 : 16 : 21$

16. (4) Since, Water (W) + Drained (D) =12
$\Rightarrow$ W + D = 12
Now, W = D - 6
$\Rightarrow$ D - 6 + D = 12
$\Rightarrow$ D = 9L

17. (4) New ratio of fares (1st, 2nd and 3rd)

$= 8 \times \frac{5}{6} : 6 \times \frac{11}{12} : 3 \times 1$

= 80 : 66 : 36 = 40 : 33 : 18
Ratio of passengers = 9 : 12 : 26
$\Rightarrow$ Ratio of amount collected
= 40 × 9 : 12 × 33 : 26 × 18
= 90 : 99 : 117
Amount collected from 1st class fares

$= \frac{99}{306} \times 1088 = \text{Rs } 320$

18. (4) The number of police involved in the security operation $= \frac{3}{5} \times 135 = 3 \times 27 = 81$

Thus, the required number of supporters
= 81 × 9 = 729

19. (1) According to the question in a proportion, multiplication of 1st and 4th terms = Multiplication of 2nd and 3rd terms
$40 = 2.5x$
$\Rightarrow x = 16$

20. (3) Let the quantity of liquid P and Q be $5x$ and $3x$ L respectively.

Quantity of P removed = $\frac{5}{5+3} \times 16 = 10$ L

Quantity of Q removed = $\frac{3}{5+3} \times 16 = 6$L

Now, $\frac{5x-10}{3x-6+16} = \frac{3}{5}$

$\Rightarrow 25x - 50 = 9x + 30$
$\Rightarrow 16x = 80$
$\Rightarrow x = 5$
$\therefore$ Quantity that vessel hold = 8 × 5 = 40 L

21. (3) Amount of dettol water

First operation = $\frac{2}{3}$

Second operations = $\frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$

Third operation = $\frac{4}{9} \times \frac{2}{3} = \frac{8}{27}$

Fourth operation = $\frac{8}{27} \times \frac{2}{3} = \frac{16}{81}$

Amount of water after fourth operation

$= 1 - \frac{16}{81} = \frac{65}{81}$

$\therefore$ Required ratio = $\frac{16}{81} : \frac{65}{81}$ = 16 : 65

22. (2) Let the shares of A, B and C be (x - 3), (2x - 7) and (3x - 9) respectively.
Then,
(x - 3) + (2x - 7) + (3x - 9) = 671
$\Rightarrow$ 6x = 690
$\therefore$ x = Rs 115
A = Rs 112
B = Rs 223
and C = Rs 336

23. (3) Let the present age of father and son be 7x and x years respectively.
After 4 years,
age of father = (7x + 4) years
age of son = (x + 4) years

Given, $\frac{7x+4}{x+4} = \frac{4}{1}$

$\Rightarrow$ 7x + 4 = 4x + 16
$\Rightarrow$ 3x =12
$\therefore$ x = 4
$\therefore$ 7x + x = 28 + 4 = 32 years

24. (4) Let the number of passengers travelling by Class I and Class II be x and 50x respectively.
Then amount collected from Class I and Class II will be Rs 3x and Rs 50x respectively.
Given, 3x + 50x = 1325
$\Rightarrow$ 53x = 1325
$\Rightarrow$ x = 25
$\therefore$ Amount collected from Class II
= 50 × 25 = 1250

25. (1) Let expenditure be Rs 60 and savings be Rs 40.
Total income = Rs 100
New income = Rs 110
New expenditure = Rs 67.2
New saving = 110 - 67.2
= Rs 42.2
$\therefore$ Percentage increase in saving

$\frac{2.8}{40} \times 100 = 7\%$

SIMPLE INTEREST

NATIONALISED BANKS & IBPS SO/MT/SO

1. A certain amount earns simple Interest of Rs. 1,750/- after 7 years. Had the interest been 2% more, how mueh more interest would it have earned ?
(1) Rs. 35/-
(2) Rs. 350/-
(3) Rs. 245/-
(4) Cannot be determined
(5) None of these

(Canara Bank PO Exam. 09.02.2003)

2. Vishwas borrowed a total amount of Rs. 30,000 part of it on simple interest, rate of 12 p.c.p.a. and remaining on simple interest rate of 10 p.c.p.a. If at the end of 2 years he paid in all Rs. 36,480 to settle the loan amount, what was the amount borrowed at 12 p.c.p.a.?
(1) Rs. 16000 (2) Rs. 18000
(3) Rs. 17500 (4) Rs. 12000
(5) None of these

(Indian Overseas Bank PO Exam. 15.06.2008)

3. The simple interest accrued on an amount of Rs. 2,500 at the end of six years is Rs. 1,875. What would be the simple interest accrued on an amount of Rs. 6,875 at the same rate and same period ?
(1) Rs. 4,556.5 (2) Rs. 5,025.25
(3) Rs. 4,895.25 (4) Rs. 5,245.5
(5) None of these

(PNB Agriculture Officer Exam. 04.01.2009)

4. What amount a man would have received on a principal of Rs. 4,000 after two years at simple interest at the rate of 5 per cent per annum ?
(1) Rs 4,161 (2) Rs 5,200
(3) Rs 4,400 (4) Rs 4,100
(5) None of these

(Corporation Bank PO Exam. 22.11.2009)

5. Shamita took a loan at simple interest rate of 6 p.c.p.a in the year and it increased by 1.5 p.c.p.a every year. If she pays Rs. 8,190 as interest at the end of 3 years, what was her loan amount ?
(1) Rs. 36000
(2) Rs. 35400
(3) Rs. 36800
(4) Cannot be determined
(5) None of these

(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

6. Arun invested a sum of money at a certain rate of simple interest for a period of four years. Had he invested the same sum for a period of six years the total interest earned by him would have been fifty percent more than the earlier interest amount. What was the rate of interest per cent per annum ?
(1) 4
(2) 8
(3) 5
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 21.02.2010)

7. The simple interest accrued on a sum of certain principal is Rs 1,200 in four years at the rate of 8 p.c.p.a. What would be the simple interest accrued on thrice of that principal at the rate of 6 p.c.p.a. in 3 years ?
(1) Rs 2,025 (2) Rs 3,025
(3) Rs 2,250 (4) Rs 2,150
(5) None of these

(Oriental Bank Of Commerce PO Exam. 26.12.2010 (Ist Sitting)

8. What total amount would Mithilesh get at the end of three years if he invests an amount of Rs 11,200 in a scheme which offers simple interest at the rate of 8.5 p. c. p. a. for three years ?
(1) Rs 14,056 (2) Rs 14,348
(3) Rs 13,852 (4) Rs 15,064
(5) None of these

(Indian Bank PO Exam. 02.01.2011)

9. Ravi borrowed some money at the rate of 4 p.c.p.a. for the first three years, at the rate of 8 p.c.p.a. for the next two years and at the rate of 9 p.c.p.a. for the period beyond 5 years. If he pays a total simple interest of Rs 19,550 at the end of 7 years, how much money did he borrow ?
(1) Rs 39,500 (2) Rs 42,500
(3) Rs 41,900 (4) Rs 43,000
(5) None of these

(Punjab & Sind Bank PO Exam. 23.01.2011)

10. A sum of Rs. 3200 becomes Rs 3456 in two years at a certain rate of simple interest. What is the rate of interest per annum ?
(1) 5.5% (2) 6%
(3) 4% (4) 4.5%
(5) None of these

(Indian Overseas Bank PO Online Exam. 01.09.2013)

11. A sum of Rs. 2200 is invested at two different rates of interest. The difference between the interests got after 4 years is Rs. 202.40. What is the difference between the rates of interest ?
(1) 3.3% (2) 2.3%
(3) 3.5% (4) 2.5%
(5) None of these

(Indian Overseas Bank PO Online Exam. 01.09.2013)

12. A sum of Rs. 16800 is divided into two parts. One part is lent at the simple interest of 6% per annum and the other at 8% per annum. After 2 years total sum received is Rs. 19000. The sum lent at 6% of simple interest is
(1) Rs. 12200 (2) Rs. 12000
(3) Rs. 11000 (4) Rs. 10000
(5) None of these

(IBPS Bank PO/MT CWE-III 26.10.2013)

13. What will be the difference between the interest accrued on a sum of Rs. 4500 at 12% per annum for 2 years and that on a sum of Rs. 5600 at 9% per annum for 2 years ?
(1) Rs 75 (2) Rs 72
(3) Rs 69 (4) Rs 76
(5) None of these

(Corporation Bank Specialist Officer (Marketing) Exam. 22.12.2014)

SBI PO EXAMS

1. Mr. A lends 40% of sum at 15% p.a. 50% of rest sum at 10% p.a. and the rest at 18% p.a. rate of interest. What would be the rate of interest if the interest is calculated on the whole sum ?
(1) 13.4% p.a. (2) 14.33% p.a.
(3) 14.4% p.a. (4) 13.33% p.a.
(5) None of these

SBI Associate Banks PO Exam16.07.2000

2. A sum of Rs. 5000/- amounts to Rs. 6,050/- in two years.What is the rate of interest ?
(1) 15% p.a. (2) 13% p.a.
(3) 11% p.a. (4) 21% p.a.
(5) None of these

SBI Associate Banks PO Exam. 16.07.2000

3. Srinivasan invests two equal amounts in two banks giving 10% and 12% rate of interest respectively. At the end of year the interest earned is Rs. 1650/-.
Find the sum invested in each.
(1) Rs. 8,500/- (2) Rs. 15,000/-
(3) Rs. 7,500/- (4) Rs. 17,000/-
(5) None of these

SBI Associate Banks PO Exam. 16.07.2000

4. Rs. 800 becomes Rs. 956 in 3 years at certain simple rate of interest. If the rate of interest is increased by 4%, what amount will Rs. 800 become in 3 years ?
(1) Rs. 1020.8 (2) Rs. 1025
(3) Rs. 1052 (4) Data inadequate
(5) None of these

SBI PO Exam. 26.11.2006

5. A certain amount earns simple interest of Rs. 1,750/- after 7 years. Had the interest been 2% more, how much more interest would it have earned ?
(1) Rs. 35/-
(2) Rs. 350/-
(3) Rs. 245/-
(4) Cannot be determined
(5) None of these

SBI Associate Banks PO Exam. 07.01.2007

8. Veena obtained an amount of Rs. 8,376 as simple interest on a certain amount at 8 p.c.p.a. after 6 years. What is the amount invested by Veena ?
(1) Rs.17,180 (2) Rs. 18,110
(3) Rs. 16,660 (4) Rs. 17,450
(5) None of these

SBI PO Preliminary (Tire-I) Exam. 27.04.2008

INSURANCE EXAMS

1. A sum of money at simple interest amounts to Rs. 14,160 in 3 years. If the rate of interest is increased to 25%, the same sum amounts to Rs. 14,700 in the same time. The rate of interest is
(1) 5% (2) $5\frac{1}{2}\%$
(3) 6% (4) 7%

(New India Insurance AAO Exam. 22.05.2011)

2. Simple interest on a certain sum at a certain annual rate of interest is 16% of the sum. If the numbers representing rate percent and time in years be equal, then the rate of interest is
(1) 4% (2) 6%
(3) 4.5% (4) 6.5%
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 12.05.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (4) **2.** (4)
3. (5) **4.** (3)
5. (5) **6.** (4)
7. (1) **8.** (1)
9. (2) **10.** (3)
11. (2) **12.** (1)
13. (2)

SBI PO EXAMS

1. (3) **2**. (5)
3. (3) **4**. (3)
5. (4) **6**. (4)

INSURANCE EXAMS

1. (3) **2**. (1)

EXPLANAT10NS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (4) Let the principal be Rs. x and rate of interest be r%

Case I : $\frac{x \times r \times 7}{100} = 1750$

$\Rightarrow xr = \frac{1750 \times 100}{7}$

= Rs. 25000
Case II :

S.I. = $\frac{x \times (r+2) \times 7}{100}$

Which cannot be determined with the help of given information.

2. (4) Let the sum borrowed at the rate of 12 p.c.p.a. be Rs. x.
$\therefore$ Sum borrowed at 10 p.c.p.a.
= Rs. (30000 - x)
Simple interest
= Rs. (36480 - 30000)
= Rs. 6480
According to the question,

$\frac{x \times 2 \times 12}{100} + \frac{(30000 - x) \times 2 \times 10}{100} = 6480$

$24x + 600000 - 20x = 648000$
$\Rightarrow 4x = 648000 - 600000$
$\Rightarrow 4x = 48000$

$\Rightarrow x = \frac{48000}{4} = 12000$

3. (5) $\because$ SI on Rs. 2500 for 6 years at a certain rate = Rs. 1875
$\Rightarrow$ SI on Rs. 6875 for the same period at the same rate

$= \frac{1875 \times 6875}{2500}$

= Rs. 5156.25

4. (3) S.I.

$= \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$

$= \text{Rs}\left(\frac{4000 \times 2 \times 5}{100}\right) = \text{Rs } 400$

$\therefore$ Amount received
= Rs. (4000 + 400) = Rs. 4400

5. (5) Let the loan amount be Rs. x

$\therefore 8190 = \frac{x \times 6 \times 1}{100} + \frac{x \times 7.5 \times 1}{100} + \frac{x \times 9 \times 1}{100}$

$\Rightarrow \frac{22.5x}{100} = 8190$

$\Rightarrow x = \frac{8190 \times 100}{22.5} = \text{Rs. } 36400$

6. (4) According to the question,

$\frac{P \times R \times 6}{100} = \frac{P \times R \times 4}{100} \times \frac{150}{100}$

This relation gives no result.

7. (1) Principal = $\frac{SI \times 100}{\text{Time} \times \text{Rate}}$

$= \frac{1200 \times 100}{4 \times 8} = \text{Rs. } 3750$

New principal = 3 × 3750
= Rs. 11250

$\therefore \text{S.I} = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$

$= \frac{11250 \times 3 \times 6}{100} = \text{Rs } 2025$

8. (1)

$\therefore \text{S.I} = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$

$= \frac{11200 \times 3 \times 8.5}{100} = \text{Rs. } 2856$

$\therefore$ Required amount
= Rs. (11200 + 2856)
= Rs. 14056

9. (2) If the amount borrowed be Rs. x, then

$\frac{x\times4\times3}{100}+\frac{x\times8\times2}{100}+\frac{x\times9\times2}{100}$ = 19550

$\Rightarrow \frac{12x}{100}+\frac{16x}{100}+\frac{18x}{100}$ = 19550

$\Rightarrow$ 12x + 16x + 18x = 1955000

$\Rightarrow$ 46x = 1955000

$\Rightarrow x = \frac{1955000}{46}$

= Rs. 42500

10. (3) Interest = 3456 - 3200 = Rs. 256

Rate = $\frac{\text{S.I.}\times100}{\text{Principal}\times\text{Time}}$

= $\frac{256\times100}{3200\times2}$ = 4% per annum

11. (2) If the rates of interest be r_1 and r_2% per annum, then

$\frac{2200\times4(r_1-r_2)}{100}$ = 202.40

$\Rightarrow r_1 - r_2 = \frac{202.40\times100}{2200\times4}$

= 2.3% per annum

12. (1) Sum lent at 6% rate of interest = Rs. x

S.I. = 19000 - 16800

= Rs. 2200

$\therefore \frac{x\times6\times2}{100}+\frac{(16800-x)\times8\times2}{100}$ = Rs. 2200

$\Rightarrow$ 12x + 16800 × 16 - 16x = 220000

$\Rightarrow$ 4x = 268800 - 220000

$\Rightarrow$ 4x = 48800

$\Rightarrow x$ = Rs. 12200

13. (2) S.I.

= $\frac{\text{Principal}\times\text{Time}\times\text{Rate}}{100}$

$\therefore$ Diffrence

= $\frac{4500\times2\times12}{100}-\frac{5600\times2\times9}{100}$

= 1080 - 1008

= Rs. 72

SBI PO EXAMS

1. (3) Let the amount be Rs. x

Investment is done as given below.

Amount left = $x - \frac{40}{100}x = \frac{60x}{100}$

$\frac{40}{100}x$ at 15% p.a

$\frac{50}{100}$ of $\frac{60x}{100} = \frac{30x}{100}$ at 10% p.a

Rest amount

= $x - \frac{40x}{100} - \frac{30x}{100} = \frac{30x}{100}$ at 18% p.a

Interest eamed by each at end of 1 year

By 1st $\Rightarrow \frac{15}{100}\times\frac{40x}{100} = \frac{60}{1000}x$

By 2nd $\Rightarrow \frac{10}{100}\times\frac{30x}{100} = \frac{30}{1000}x$

By 3rd $\Rightarrow \frac{18}{100}\times\frac{30x}{100} = \frac{54}{1000}x$

Total interest = $\frac{144}{1000}x$

$\therefore$ Rate% = $\frac{\frac{144x}{1000}}{x}\times100$ = 14.4%

2. (5) S.I = 6050 - 5000 = Rs 1050

R% = $\frac{S.I\times100}{P\times T}$ = $\frac{1050\times100}{5000\times2}$ = 10.5%

3. (3) Let sum be Rs.x

According to the question,

$\frac{x\times10\times1}{100}+\frac{x\times12\times1}{100}$ = 1650

22x = 1650 × 100

x = $\frac{1650\times100}{22}$ = Rs 7500

4. (3) Increase is interest in 3 years due to increase in rate by 4%

= $\frac{800\times3\times4}{100}$ = Rs. 96

Total amount at the end of 3 years

= Rs. 956 + Rs. 96 = Rs. 1052

5. (4) Let the principal be Rs. x and rate of interest be r%

Case I : $\frac{x\times r\times7}{100}$ = 1750

$\Rightarrow xr = \frac{1750\times100}{7}$

= Rs. 25000

Case II :

S.I. = $\frac{x \times (r+2) \times 7}{100}$

Which cannot be determined with the help of given information.

6. (4) Let the amount invested by Veena be Rs. *x*.

$\therefore$ Principal = $\frac{\text{SI} \times 100}{\text{Time} \times \text{Rate}}$

= $\frac{8376 \times 100}{6 \times 8}$ = Rs. 17450

INSURANCE EXAMS

1. (3) P + $\frac{\text{P}r \times 3}{100}$ = 14160 .....(i)

P + P × $\frac{5r}{400} \times 3$ = 14700 ....(ii)

$\Rightarrow$ P + $\frac{5}{4}$ (14160 - P) = 14700

$\Rightarrow$ 4P + 70800 - 5P = 58800

$\Rightarrow$ P = 70800 - 58800 = 12000

$\therefore$ From equation (i),

12000 + $\frac{12000 \times r \times 3}{100}$ =14160

$\Rightarrow$ 120 × 3 × *r* = 14160 - 12000 = 2160

$\Rightarrow$ *r* = $\frac{2160}{120 \times 3}$ = 6%

2. (1) Let the sum be Rs. *x*.

S.I. = Rs. $\frac{16x}{100}$

Rate = Time = R

$\therefore$ Rate = $\frac{\text{S.I.} \times 100}{\text{Principal} \times \text{Time}}$

$\Rightarrow$ R = $\frac{16x \times 100}{100x \times R}$

$\Rightarrow$ R^2 = 16

$\Rightarrow$ R = 4% Per annum

MODEL EXERCISES

1. I derive an annual income of Rs 688.25 from Rs 10000 invested partly at 8% per annum and partly at 5% per annum simple interest. How much of my money is invested at 5% ?
(1) Rs 3725 (2) Rs 4225
(3) Rs 4800 (4) Rs 5000
(5) None of these

2. Two equal sums of money were invested, one at 4% and the other at $4\frac{1}{2}$%. At the end of 7 yr, the simple interest received from the latter exceeded that received from the former by Rs 31.50. Each sum was
(1) Rs 1000 (2) Rs 500
(3) Rs 750 (4) Rs 900
(5) None of these

3. What total amount would Mohan get at the end of three years if he invests an amount of Rs11,200 in a scheme which offers simple interest at the rate of 8.5 per cent per annum for three years ?
(1) Rs 14,348 (2) Rs 14,056
(3) Rs 13,852 (4) Rs 15,064
(5) None of these

4. Ravi gave Rs 1200 on loan. Some amount he gave at 4% per annum simple interest and remaining at 5% per annum simple interest. After two years, he got Rs 110 as interest. Then, the amounts given at 4% and 5% per annum simple interest are, respectively
(1) Rs 500, Rs 700
(2) Rs 400, Rs 800
(3) Rs 800, Rs 300
(4) Rs 1100, Rs 1100
(5) None of these

5. Two equal sums of money are lent at the same time at 8% and 7% per annum simple interest. The former is recovered 6 months earlier than the latter and the amount in each case is Rs 2560. The sum and the time for which the sums of money are lent out are :
(1) Rs 2000, 3.5 year and 4 year
(2) Rs 1500, 3.5 year and 4 year
(3) Rs 2000, 4 year and 5.5 year
(4) Rs 3000, 4 year and 4.5 year
(5) None of these

6. Sumit lent some money to Mohit at 5% per annum simple interest. Mohit lent the entire amount to Birju on the same day at $8\frac{1}{2}$ % per annum. In this transaction after a year, Mohit earned a proflt of Rs 350. Find the sum of money lent by Sumit to Mohit.
(1) Rs 10000 (2) Rs 9000
(3) Rs 10200 (4) Rs 8000
(5) None of these

7. A computer is available for Rs 39000 cash or Rs 17000 as cash down payment followed by five monthly instalments of Rs 4800 each. What is the rate of interest per annum under the instalment plant ?
(1) 35.71% (2) 37.71%
(3) 36.71% (4) 38.71%
(5) None of these

8. The ratio of the amount for two years under CI annually and for one year under SI is 6 : 5. when the ratio of interest is same, then the value of the rate of interest is
(1) 12.5% (2) 18%
(3) 20% (4) 16.66%
(5) None of these

9. Consider the following statements If a sum of money is lent at simple interest, then the
(I) money gets doubled in 5 yr, if the rate of interest is $16\frac{2}{3}$ %.
(II) money gets doubled in 5 yr, if the rate of interest is 20%.
(III) money becomes four times in 10 yr, if it gets doubled in 5yr.
(1) I and III are correct
(2) II alone is correct
(3) III alone is correct
(4) II and III are correct
(5) None of these

10. Divide Rs 1586 in three parts in such a way that their amounts at the end of 2, 3 and 4 yr respectively, at 5% per annum simple interest be equal
(1) Rs 552, Rs 528, Rs 506
(2) Rs 560, Rs 520, Rs 506
(3) Rs 556, Rs 524, Rs 506
(4) Rs 548, Rs 528, Rs 510
(5) None of these

11. Anu owes Biresh Rs 1120 payable 2 year hence, Biresh owes Anu Rs 1081.50 payable 6 months. If they decide to settle their accounts forthwith by payment of ready money and the rate of interest be 6% per annum, they who should pay and how much ?
(1) Anu, Rs 70 (2) Biresh, Rs 50
(3) Anu, Rs 50 (4) Biresh, Rs 70

(5) None of these

12. The present worth of bill due 7 months, hence is Rs 1200. If the bill were due at the end of $2\frac{1}{2}$ yr, its present worth would be Rs 1016. The rate per cent of bill is:
(1) 16% (2) 8%
(3) 10% (4) 18%
(5) None of these

13. The simple interest and the true discount on a certain sum of a given and at a given rate are Rs 25 and Rs 20 respectively. The sum is
(1) Rs 500 (2) Rs 200
(3) Rs 250 (4) Rs 100
(5) None of these

14. A trader owes a merchant Rs 10028 due 1 yr, hence the trader wants to settle the account after 3 months. If the rate of interest is 12% per annum, how much cash should he pay ?
(1) Rs 9025 (2) Rs 9200
(3) Rs 9600 (4) Rs 9500
(5) None of these

15. A person invested some amount at the rate of 12% simple interest and a certain amount at the rate of 10% simple interest. He received yearly interest of Rs 130. But if he had interchanged the amounts invested, he would have received Rs 4 more as interest. How much did he invest at 12% simple interest ?
(1) Rs 700 (2) Rs 500
(3) Rs 800 (4) Rs 400
(5) None of these

16. The effective annual rate of interest corresponding to a nominal rate of 8% per annum payable half yearly is
(1) 8% (2) 8.01%
(3) 8.13% (4) 8.16%
(5) None of these

17. A sum was put at simple interest at a certain rate for 3 yr. Had it been put at 1% higher rate, it would have fetched Rs 5100 more. The sum is
(1) Rs 170000 (2) Rs 150000
(3) Rs 125000 (4) Rs 120000
(5) None of these

18. Prem invested a certain sum of money in a simple interest bond whose value grew to Rs 300 at the end of 3 yr and to Rs 400 at the end of another 5 yr. What was the rate of interest in which he invested his sum ?
(1) 12% (2) 12.5%
(3) 6.67% (4) 8.33%
(5) None of these

19. If Rs 1000 be invested at interest rate of 5% and the interest be added to the principal after 10 year, then the number of years in which it will amount to Rs 2000 is
(1) $16\frac{2}{3}$ year (2) $16\frac{1}{4}$ year
(3) 16 year (4) 11 year
(5) None of these

20. A sum of Rs 7700 is to be divided among three brothers Sunil, Sumant and Surat in such a way that simple interest on each part at 5% per annum after 1, 2 and 3 yr respectively remains equal. The share of Sunil is more than that of Surat by :
(1) Rs 2800 (2) Rs 2500
(3) Rs 3000 (4) Rs 2700
(5) None of these

21. What will be the ratio of simple interest earned by a certain amount at the same rate of interest for 6 years and 9 years ?
(1) 1:3
(2) 1 : 4
(3) 2 : 3
(4) Data inadequate
(5) None of these

22. An automobile flnancier claims to be lending money at simple interest but he includes the interest every six months for calculating the principal. If he is charging an interest of 10%, the effective rate of interest becomes
(1) 10% (2) 10.25%
(3) 10.5% (4) 10.35%
(5) None of these

23. Sultan took a loan from the bank at 8% per annum, and was supposed to pay a sum of Rs 2240 at the end of 4 years. If the same sum is cleared off in four equal annual installments at the same rate, the amount of annual installment will be
(1) Rs 500 (2) Rs 550
(3) Rs 600 (4) Rs 1000
(5) None of these

SHORT ANSWERS

1. (1) **2.** (4)
3. (2) **4.** (1)
5. (1) **6.** (1)
7. (4) **8.** (3)
9. (2) **10.** (1)
11. (2) **12.** (3)
13. (4) **14.** (2)
15. (2) **16.** (4)
17. (1) **18.** (4)
19. (1) **20.** (1)
21. (3) **22.** (2)
23. (1)

EXPLANATIONS

1. (1) Let money invested at 5% be Rs x. According to question,

$$\frac{x\times1\times5}{100}+\frac{(10000-x)\times1\times8}{100}=688.25$$

$\Rightarrow 5x - 8x - 80000 = 68825$

$\Rightarrow 3x = 11175$

$\Rightarrow x =$ Rs 3725

2. (4) Let each sum be Rs x, then

$$\frac{x\times4\frac{1}{2}\times7}{100}-\frac{x\times4\times7}{100}=31.50$$

$$\Rightarrow \frac{7x}{100}\times\frac{1}{2}=\frac{63}{2}$$

$\Rightarrow x =$ Rs 900

3. (2) S.I

$$\frac{\text{Principal}\times\text{Time}\times\text{Rate}}{100}$$

$$=\frac{11200\times3\times8.5}{100}=\text{Rs } 2856$$

$\therefore$ Required Amount = 11200 + 2856
= Rs 14056

4. (1) Let the amount given at 4% per annum be Rs x.
$\therefore$ Amount given at 5% per annum
= Rs (1200 - x)

$$\therefore \frac{x\times4\times2}{100}+\frac{(1200-x)\times5\times2}{100}=110$$

$$\Rightarrow \frac{-2x+12000}{100}=100$$

$\Rightarrow x =$ Rs 500
Aslo, (1200 - x) = 1200 - 500 = Rs 700

5. (1) Let each amount be Rs x and time be t yr.

Then $x+\left(t-\frac{1}{2}\right)\times x\times\frac{8}{100}$

$$= x+\frac{x\times t\times7}{100}$$

$$\Rightarrow \frac{2xt}{25}-\frac{x}{25}=\frac{7xt}{100}$$

$$\Rightarrow \frac{xt}{100}=\frac{x}{25}$$

$\Rightarrow t = 4$ yr

$\therefore$ First money was lent for $4-\frac{1}{2}=3.5$ yr.

$$\therefore \text{Amount} = x+\frac{3.5\times x\times8}{100}=2560$$

$x =$ Rs 2000

6. (1) According to question,

$8\frac{1}{2}\% - 5\% = 350$

$$\therefore 100\% = \frac{350}{3.5}\times100 = \text{Rs } 10000$$

7. (4) According to question cash price,
CP = Rs 39000
Cash down payment, DP = Rs 17000
Balance due, after lst instalment,
BD = Rs 22000
P = value of instalment = Rs 4800
n = no. of instalments = 5
R = rate of interest

$$\therefore \left(1+\frac{nR}{1200}\right)\text{BD} = \left\{1+\frac{(n-1)R}{2400}\right\}nP$$

$$\left(1+\frac{5R}{1200}\right)22000 = \left\{1+\frac{4R}{2400}\right\}5\times4800$$

$$\Rightarrow 11+\frac{55R}{1200}=12+\frac{24R}{1200}$$

$$\Rightarrow \frac{55R}{1200}-\frac{24R}{1200}=1 \Rightarrow \frac{31R}{1200}=1$$

$$\Rightarrow \text{R} = \frac{1200}{31}=38.71\%$$

8. (3) According to question,

$$\frac{P\left(1+\frac{r}{100}\right)^2}{\left(P+\frac{Pr}{100}\right)} = \frac{6}{5}$$

$$= \left(1+\frac{r}{100}\right) = \frac{6}{5} \Rightarrow r = 20\%$$

9. (2) Let the sum of money lent be Rs P.
Then,
I. SI received after 5 yr

$$= \frac{P\times 50\times 5}{3\times 100} = \frac{5}{6}P$$

Amount after 5 yr

$$= P + \frac{5}{6}P = \frac{11P}{6}$$

Therefore, I is not correct.
II. SI after 5 yr = P + P = 2P
Therefore, II is correct.
III. Amount after 10 yr. at the rate of 20%

$$= P + \frac{P\times 20\times 10}{100} = P + 2P = 3P$$

Therefore, III is not correct.
Hence, (2) is the correct option.

10. (1) Let the three parts be Rs x,
Rs y, and Rs z.
According to the question,

$$x + \frac{x\times 2\times 5}{100} = y + \frac{y\times 3\times 5}{100} = z + \frac{z\times 4\times 5}{100}$$

$$\Rightarrow 1.1x = 1.15y = 1.2z$$

$$\Rightarrow \frac{x}{y} = \frac{1.15}{1.1} = \frac{23}{22}$$

and $\frac{y}{z} = \frac{1.2}{1.15} = \frac{24}{23}$

$$\Rightarrow x : y : z = 276 : 264 : 253$$

$$\Rightarrow x = \frac{276}{793}\times 1586 = \text{Rs } 552,$$

$$\Rightarrow y = \frac{264}{793}\times 1586 = 528$$

and $z = \frac{253}{793}\times 1586 = \text{Rs } 506$

Hence, the required three parts are Rs 552, Rs 528 and Rs 506.

11. (2) According to question, Present worth of money for Anu

$$1120 - P = \frac{P\times 6\times 2}{100} \Rightarrow P = \text{Rs } 1000$$

Present worth of money for Biresh

$$1081.50 - P = \frac{P\times 6\times 1}{2\times 100}$$

$$108150 - 100P = 3P$$

$$P = \text{Rs } 1050$$

$\therefore$ Biresh should pay Rs 50.

12. (3) Let the rate of interest be r%
Then,

$$A = 1200 + \frac{1200\times r\times 7}{12\times 100}$$

$$\Rightarrow A = 1200 + 7r$$

Again $1200 + 7r =$

$$1016 + \frac{1200\times r\times 25}{100}$$

$$\Rightarrow 1200 + 7r = 1016 + 25.4r$$

$$\Rightarrow 18.4r = 184$$

$$\therefore r = \frac{184}{18.4} = 10\%$$

13. (4) Let that amount be Rs x and rate is r%
Then, for simple interest

$$\frac{x\times r}{100} = 25$$

$$\Rightarrow x \times r = 2500$$

For true discount

$$\frac{(x-20)\times r}{100} = 20$$

$$\Rightarrow \frac{x\times r - 20r}{100} = 20$$

$$\Rightarrow xr - 20r = 2000$$

From Eqs. (i) and (ii), we get

$$2500 - 20r = 2000$$

$$\Rightarrow r = 25\%$$

From Eq. (i)

$$x \times 25 = 2500$$

$$\Rightarrow x = 100$$

14. (5) Let the present value of money be x.

Then, $\frac{x\times 12}{100} + x = 10028$

$$\Rightarrow 0.12x + x = 10028$$

$$\Rightarrow x = \frac{10028}{1.12}$$

This amount after 3 months

$= \frac{\frac{10028}{112}\times 12\times 3}{12\times 100} + 10028$

$= \frac{10028\times 3}{1.12\times 100} + \frac{10028}{1.12}$

$= \frac{10028\times 103}{112}$ = Rs 9222.17

15. (2) Amount invested at 12% = Rs x .
Amount invested at 10% = Rs y

$130 = \frac{x\times 12\times 1}{100} + \frac{y\times 10\times 1}{100}$

$\Rightarrow 13000 = 12x + 10y$

and $134 = \frac{x\times 10\times 1}{100} + \frac{y\times 12\times 1}{100}$

$\Rightarrow 13400 = 10x + 12y$

Solving Eqs. (i) and (ii), we get x = Rs 500
So, amount invested at 12% is Rs 500.

16. (4) Rate of 8% per annum payable half yearly. So, effective rate Effective annual rate

$= 4 + 4 + \frac{4\times 4}{100} = 8.16\%$

17. (1) Simple interest for 1 yr = $\frac{5100}{3}$

= Rs 1700
1% of sum 1700

$\therefore$ Sum = $\frac{1700\times 100}{1}$

= Rs 170000

18. (4) Let principle be P and rate of interest is r. Then,

$\frac{P\times r\times 3}{100} + P = 300$

and $\frac{P\times r\times 8}{100} + P = 400$

Substracting Eq. (i) from Eq. (ii), we get

$\frac{P\times r\times 5}{100} = 100$

$\therefore P \times r = 2000$
From Eq. (i),

$\frac{2000\times 3}{100} + P = 300 \Rightarrow P$ = Rs 240

$\therefore 240 \times r = 2000$

$\Rightarrow r = 8.33\%$

19. (1) According to question,

SI for 10 years = $\frac{1000\times 5\times 10}{100}$ = Rs 500

Now,P = Rs 1500, A = Rs 2000
$\therefore$ SI = Rs 500

$\therefore 500 = \frac{1500\times 5\times T}{100}$

$\Rightarrow T = \frac{500\times 100}{1500\times 5} = 6\frac{2}{3}$ years

$\therefore$ Total time = $16\frac{2}{3}$ years

20. (1) Let Sunil, Sumant and Surat get x, y and z amount respectively.

$\frac{x\times 5\times 1}{100} = \frac{y\times 5\times 2}{100} = \frac{z\times 5\times 3}{100}$

$\Rightarrow x = 2y = 3z$

$\Rightarrow x : y : z = 6 : 3 : 2$

$\therefore$ Required amount $\frac{6-2}{6+3+2}\times 7700$

= Rs 2800

21. (3) According to question, here, rate of interest is same and principal is also same.
$\therefore$ Required ratio = 6 : 9 = 2 : 3

22. (2) According to question, S.I. on Rs. 100 at 10% for six months =

$\frac{100\times 10\times 6}{100\times 12}$

= Rs. 5
Principal for next six months = Rs. 105

$\therefore$ SI = $\frac{105\times 10\times 6}{100\times 12}$

= Rs. 5.25
$\therefore$ Total interest = Rs. 10.25

23. (1) Let the annual payment be Rs. x. Since the first instalment is to be paid after 1 year, the sum of amounts of Rs. x for 3 years, of Rs. x for 2 years, of Rs. x for 1 year and Rs. x, gives the amount of debt.
Now, amount of Rs. x for three years

= Rs $\left[x + \frac{x\times 8\times 3}{100}\right]$ = Rs. $\left(\frac{31x}{25}\right)$

Amount of Rs. x for 2 years

= Rs $\left(x + \frac{x\times 8\times 2}{100}\right)$ = Rs. $\left(\frac{29x}{25}\right)$

Amount of Rs. x for 1 year

$= \text{Rs}\left(x + \frac{x \times 8 \times 1}{100}\right) = \text{Rs.}\frac{27x}{25}$

$\therefore \left(\frac{31x}{25} + \frac{29x}{25} + \frac{27x}{25} + x\right) = 2240$

$\Rightarrow \frac{31x + 29x + 27x + 25x}{25} = 2240$

$\Rightarrow \frac{112x}{25} = 2240$

$\Rightarrow x = \frac{2240 \times 25}{112} = \text{Rs. } 500$

TIME AND DISTANCE

NATIONALISED BANKS & IBPS PO/MT/SO

1. A motor starts with the speed of 70 kmph with its speed increasing every two hours by 10 kmph. In how many hours will it cover 345 kms ?

(1) $2\frac{1}{4}$ hours

(2) $4\frac{1}{2}$ hours

(3) 4 hours 5 minutes
(4) Cannot be determined
(5) None of these

(Canara Bank PO Exam. 09.02.2003)

2. A train running at the speed of 20 metres/second crosses a pole in 24 seconds less than the time it requires to cross a platform thrice its length at the same speed. What is the length of the train ?
(1) 270 metres
(2) 340 metres
(3) 180 metres
(4) Cannot be determined
(5) None of these

(Union Bank of India PO Exam. 27.11.2005)

3. A man takes 6 hours 35 minutes in walking to a certain place and riding back. He would have taken 2 hours less by riding both ways. What would be the time he would take to walk both ways ?
(1) 4 hours 35 minutes
(2) 8 hours 35 minutes
(3) 10 hours
(4) 8 hours 25 minutes
(5) None of these

(Bank Of Maharashtra PO Exam. 25.05.2008)

4. A train travelling at the speed of 60 kmph crosses a platform in 20 seconds. What is the length of the train ?
(1) 333 metres
(2) 300 metres
(3) 336 metres
(4) Cannot be determined
(5) None of these

(Indian Overseas Bank PO Exam. 15.06.2008)

5. A boat running downstreams covers a distance of 30 kms in 2 hours. While coming back the boat takes 6 hours to cover the same distance. If the speed of the current is half that of the boat, what is the speed of that boat in kmph ?
(1) 15
(2) 5
(3) 10
(4) Cannot be determined
(5) None of these

(Indian Overseas Bank PO Exam. 15.06.2008)

6. Samir drove at the speed of 45 kmph. from home to a resort. Returning over the same route, he got stuck in traffic and took an hour longer, also he could drive only at the speed of 40 kmph. How many kilometres did he drive each way ?
(1) 250 kms. (2) 300 kms.
(3) 310 kms. (4) 275 kms.
(5) None of these

(Bank Of Baroda Specialist Officer Exam. 05.10.2008)

7. A 240 metres long train crosses a platform twice its lenght in 40 seconds. What is the speed of the train ?
(1) 6 metres/sec.
(2) 28 metres/sec.
(3) 18 metres/sec.
(4) 16 metres/sec.
(5) None of these

(PNB Agriculture Officer Exam. 04.01.2009)

8. A boat running at the speed of 34 kmph downstream covers a distance of 4.8 kms, in 8 minutes. The same boat while running upstream at same speed covers the same distance in 9 minutes. What is the speed of the current ?
(1) 2.4 kmph (2) 3 kmph
(3) 2 kmph (4) 3.2 kmph
(5) None of these

(PNB Specialist Officer's Exam. 16.08.2009)

9. A bus started its journey from Ramgarh and reached Devgarh in 44 minutes with its average speed of 50 km/hour. If the average speed of the bus is increased by 5 km/hour, how much time will it take to cover the same distance ?
(1) 40 minutes (2) 38 minutes
(3) 36 minutes (4) 31 minutes
(5) 49 minutes

(Corporation Bank PO Exam. 22.11.2009)

10. A man walked at a speed of 4 km/hr from point A to B and came back from point B to

A at the speed of 6 km/hr. What would be the ratio between the time taken by man in walking from point A to B to point B to A respectively ?
(1) 5 : 3 (2) 2 : 3
(3) 2 : 1 (4) 4 : 3
(5) 3:2

(Corporation Bank P0 Exam. 22.11.2009)

11. A bus covered a certain distance from village A to village B at the speed of 60 km/hr. However on its return journey it got stuck in traffic and covered the same distance at the speed of 40 km/hr. and took 2 hours more to reach its destination. What is the distance covered between village A and B ?
(1) 240 km.
(2) 260 km.
(3) 200 km.
(4) Cannot be determined
(5) None of these

(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

12. A train speeds past a pole in 20 seconds and speeds past a platform 100 metres in length in 30 seconds. What is the length of the train ?
(1) 100 metre (2) 150 metre
(3) 180 metre (4) 200 metre
(5) None of these

(Indian Bank Rural Marketing Officer Exam. 03.01.2010

13. A 320 metre long train takes 80 seconds more to cross a platform twice its length than it takes to cross a pole at the same speed. What is the speed of the train in metre/second ?
(1) 16
(2) 10
(3) 6
(4) Cannot be determined
(5) None of these

(Bank Of India Banking Officer Exam. 24.01.2010)

14. A 180-metre long train crosses another 270-metre long train running in the opposite direction in 10.8 seconds. If the speed of the first train is 60 kmph., what is the speed of the second train in kmph ?
(1) 80
(2) 90
(3) 150
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 21.02.2010)

15. The respective ratio between the speeds of a car, a train and a bus is 5 : 9 : 4. The average speed of the car, the bus and the train is 72 kmph together. What is the average speed of the car and the train together ?
(1) 82 kmph
(2) 78 kmph
(3) 84 kmph
(4) Cannot be determined
(5) None of these

(Punjab & Slnd Bank PO Exam. 16.05.2010)

16. The ratio between the speed of a train and a car is 16 : 15 respectively. Also, a bus covered a distance of 480 kms. in 8 hours. The speed of the bus is three-fourth the speed of the train. How much distance will the car cover in 6 hours ?
(1) 450 km
(2) 480 km
(3) 360 km
(4) Cannot be determined
(5) None of these

(Bank Of Baroda PO Exam. 30.05.2010)

17. A man crosses a stationary train in 12 minutes. The same train crosses a man in 54 seconds. What is the respective ratio between the speed of the train and the man ?
(1) 40 : 7
(2) 400 : 3
(3) 40 : 3
(4) Cannot be determined
(5) None of these

(Bank Of Baroda PO Exam. 30.05.2010)

18. Train-A crosses a stationary train -B in 50 seconds and a pole in 20 seconds with the same speed. The length of the train-A is 240 metres. What is the length of the stationary Train-B ?
(1) 360 metres
(2) 260 metres
(3) 300 metres
(4) Cannot be determined
(5) None of these

(Central Bank Of India PO Exam. 25.07.2010)

19. The respective ratio between the speeds of a car, a jeep and tractor is 3 : 5 : 2. The speed of the jeep is 250 percent of the speed of the tractor which covers 360 km in 12 hours. What is the average speed of car and jeep together ?
(1) 60 km/hr.
(2) 75 km/hr.

(3) 40 km/hr.
(4) Cannot be determined
(5) None of these

(Central Bank Of India PO Exam. 25.07.2010)

20. A man crosses a stationary bus in 18 seconds. The same bus crosses a pole in 4 seconds. What is the respective ratio between the speed of the bus and the speed of the man ?
(1) 9 : 2
(2) 9 : 4
(3) 18 : 5
(4) Cannot be determined
(5) None of these

(Syndicate Bank PO Exam. 29.08.2010)

21. The speed of a car is 1.5 times the speed of a bus. If the car travels at the speed of 60 km/ hr., what will be the difference in the time taken by the bus and the time taken by the car to cover 720 km ?
(1) 5 hours (2) 6 hours
(3) 4 hours (4) 8 hours
(5) None of these

(Punjab National Bank Specialist Officer Exam. 24.10.2010)

22. A 320 metre long train crosses a platform thrice its length in 40 seconds. What is the speed of the train in km/hour ?
(1) 120.6
(2) 115.2
(3) 108.4
(4) Cannot be determined
(5) None of these

(United Bank Of India PO Exam. 14.11.2010)

23. A bike covers a certain distance at the speed of 64 km/hr. in 8 hours. If the bike was to cover the same distance is approximately 6 hours, at what **approximate** speed should the bike travel ?
(1) 80 kmph (2) 85 kmph
(3) 90 kmph (4) 75 kmph
(5) 70 kmph

(PNB Management Trainee Exam. 28.11.2010)

24. The ratio between the speed of a train and a car is 18 : 13 respectively. Also, a bus covered a distance of 480 kms. in 12 hours. The speed of the bus is five-ninth the speed of the train. How much distance will the car cover in 5 hours ?
(1) 250 kms.
(2) 280 kms.
(3) 260 kms.
(4) Cannot be determined
(5) None of these

(Bank Of Maharashtra Exam. 19.12.2010)

25. A 300 metre long train moving with an average speed of 126 km/hr. crosses a platform in 24 seconds. A man crosses the Same platform in 5 minutes. What is the speed of man in meter/second ?
(1) 1.8 metre/second
(2) 1.2 metre/second
(3) 1.5 metre/second
(4) Cannot be determined
(5) None of these

(Bank Of Maharashtra Exam. 19.12.2010)

26. Train-A crosses a stationary train B in 35 seconds and a pole in 14 seconds with the same speed. The length of the train-A is 280 metres. What is the length of the stationary train-B ?
(1) 360 metres
(2) 480 metres
(3) 400 metres
(4) Cannot be determined
(5) None of these

(Bank Of Maharashtra Exam. 19.12.2010)

27. A car covers a distance of 540 km in 9 hours. Speed of a train is **double** the speed of the car. Two-third of the speed of the train is equal of a bike. How much distance will the bike cover in 5 hours ?
(1) 450 km (2) 360 km
(3) 400 km (4) 500 km
(5) None of these

(Oriental Bank Of Commerce PO Exam. 26.12.2010 (1st Sitting)

28. Train-A crosses a pole in 25 seconds and another Train-B crosses a pole in 1 minute and 15 seconds. Length of train-A is half length of train-B. What is the respective ratio between the speeds of Train-A and Train-B ?
(1) 3 : 2
(2) 3 : 4
(3) 4 : 3
(4) Cannot be determined
(5) None of these

(Union Bank Of India PO Exam. 09.01.2001)

29. The average speed of a car is $1\frac{4}{5}$ times the average speed of a bus. A tractor covers 575

km in 23 hours. How much distance will the car cover in 4 hours if the speed of the bus is twice the speed of the tractor ? (1) 340 km (2) 480 km
(3) 360 km (4) 450 km
(5) None of these

(Corporation Bank PO Exam. 16.01.2011)

30. A car covers the first 39 km. of its journey in 45 minutes and covers the remaining 25 km. in 35 minutes. What is the average speed of the car ?
(1) 40 kmph (2) 64 kmph
(3) 49 kmph (4) 48 kmph
(5) None of these

(Punjab & Sind Bank PO Exam. 23.01.2011)

31. An aeroplane flies with an average speed of 756 km/hr. A helicopter takes 48 hours to cover **twice** the distance covered by aeroplane in 9 hours. How much distance will the helicopter cover in 18 hours ? **(assuming that flights are non-stop and moving with uniform speed)**
(1) 5014 km (2) 5140 km
(3) 5130 km (4) 5103 km
(5) None of these

(UCO Bank PO Exam. 30.01.2011)

32. A 320 metre long train moving with an average speed of 120 km/hr crosses a platform in 24 seconds. A man crosses the same platform in 4 minutes. What is the speed of man in metre/second ?
(1) 2.4 (2) 1.5
(3) 1.6 (4) 2.0
(5) None of these

(Bank Of Baroda PO Exam. 13.03.2011)

33. The ratio between the speed of a bus and train is 15 : 27 respectively. Also, a car covered a distance of 720 km. in 9 hours. The speed of the bus is three-fourth of the speed of the car. How much distance will the train cover in 7 hours ?
(1) 760 km.
(2) 756 km.
(3) 740 km.
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 17.04.2011)

34. A 280 metre long train moving with an average speed of 108 km/hr crosses a platform in 12 seconds. A man crosses the same platform in 10 seconds. What is the speed of the man in metre/ second ?
(1) 5 m/s
(2) 8 m/s
(3) 12 m/s
(4) Cannot be determined
(5) None of these

(Allahabad Bank PO Exam. 17.04.2011)

35. The average speed of a train is $1\frac{3}{7}$ times the average speed of a car. The car covers a distance of 588 km in 6 hours. How much distance will the train cover in 13 hours ?
(1) 1750 km (2) 1760 km
(3) 1720 km (4) 1850 km
(5) None of these

(Indian Overseas Bank PO Exam. 22.05.2011]

36. Amit, Sucheta and Neeti start running around a circular track and complete one round in 18, 24 and 32 seconds respectively. In how many seconds will the three meet again at the starting point. if they all have started running at the same time ?
(1) 196
(2) 288
(3) 324
(4) Cannot be determined
(5) None of these

(Bank Of India Banking Officer Exam. 24.01.2010

37. Philip, Tom and Brad start jogging around a circular field and complete a single round in 18, 22 and 30 seconds respectively In how much time will they meet again at the starting point ?
(1) 8 mins. 15 sees.
(2) 21 mins.
(3) 16 mins. 30 sees.
(4) 12 mins.
(5) None of these

(Indian Bank PO Exam. 02.01.2011)

38. A 320 metre long train crosses a pole in 50 seconds. What is the speed of train ?
(1) 4.6 m/sec
(2) 7.2 m/sec
(3) 5.4 m/sec
(4) 6.4 m/sec
(5) Cannot be determined

(IBPS Specialist Officer CWE 17.03.2013)

39. A car covers a distance of 525 km. in a certain time at a speed of 66 km./hr. How much distance would a truck cover at an average speed which is 24 km hr. less than that of the speed of the car in time which is 7 hours more than that taken by the car ?

(1) 336 km. (2) 682 km.
(3) 598 km. (4) 630 km.
(5) None of these

(IBPS RRBs Office Assistant CWE Exam. 09.09.2012

40. A person travels from P to Q at a speed of 40 kmph and returns to Q by increasing his speed by 50%. What is his average speed for both the trips ?
(1) 36 kmph (2) 45 kmph
(3) 48 kmph (4) 50 kmph
(5) None of these

(IBPS Bank PO/MT CWE-III 26.10.2013)

41. A 320 metre long train crosses a pole in 16 seconds. It stops five times of duration 18 minutes each. What time will it take in covering a distance of 576 km ?
(1) 9 hours (2) $9\frac{1}{4}$ hours
(3) $9\frac{1}{2}$ hours (4) $8\frac{1}{2}$ hours
(5) None of these

(Corporation Bank Specialist Officer (Marketing) Exam. 22.12.2014)

SBI PO EXAMS

1. A 260 metre long train crosses a 120 meter long wall in 19 seconds. What is the speed of the train ?
(1) 27 km/hr (2) 49 km/hr
(3) 72 km/hr (4) 70 km/hr
(5) None of these

(SBI Associate Banks PO Exam. 14.02.1999)

2. Two cars starts at the same time from A and B which is 120 km apart. If the two cars tarvel in opposite direction they meet after one hour and if they travel in same direction (from A towards B) then A meet B after 6 hours. What is the speed of car starting from A ?
(1) 70 kmph (2) 120 kmph.
(3) 60 kmph (4) Data inadequate
(5) None of these

(SBI Associate Banks PO Exam. 14.02.1999)

3. Find the speed of train in km/hr whose length is 200 m and crosses a platform of length 240 m in 22 seconds ?
(1) 60 (2) 48
(3) 53 (4) Data inadequate
(5) None of these

(SBI Associate Banks PO Exam. 14.02.1999)

4. A man starts going for morning walk every day. The distance walked by him on the first day was 2 kms. Everyday he walks half of the distance walked on the previous day. What can be the **maximum** total distance walked by him in his life time ?
(1) 4 kms. (2) 120 kms.
(3) 18 kms. (4) Data inadequate
(5) None of these

(SBI Banks PO Exam. 11.02.2001)

5. A car covers its journey at the speed of 80 km/hour in 10 hours. If the same distance is to be covered in 4 hours, by how much the speed of car will have to increase ?
(1) 8 km/hr (2) 10 km/hr
(3) 12 km/hr (4) 16 km/hr
(5) None of these

(SBI PO Exam. 26.11.2006)

6. A motor starts with the speed of 70 kmph with its speed increasing every two hours by 10 kmph. In how many hours will it cover 345 kms ?
(1) $2\frac{1}{4}$ hours
(2) $4\frac{1}{2}$ hours
(3) 4 hours 5 minutes
(4) Cannot be determined
(5) None of these

(SBI Associate Banks PO Exam. 07.01.2007)

7. Raman drove from home to a neighbouring town at the speed of 50 kms./hr and on his returning journey, he drove at the speed of 45 kms./hr. and also took an hour longer to reach home. What distance did he cover each way ?
(1) 450 kms. (2) 225 kms.
(3) 900 kms. (4) 500 kms.
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

RBI GRADE-B OFFICER EXAMS

1. Nitin rides a bicycle at the speed of 15 kms/hr., but stops for 10 minutes to take rest every 20 kms. How much time will he take to cover a distance of 150 kms. ?
(1) 12 hours 30 minutes
(2) 11 hours 10 minutes
(3) 10 hours 20 minutes
(4) 12 hours 10 minutes
(5) None of these

(RBI Grade-B Officer Exam. 2008)

2. A 280 metres long train, travelling at a uniform speed, crosses a platform in 60 seconds and a man standing on the platform in 20 seconds. What is the length of the platform ?
(1) 640 metres
(2) 420 metres
(3) 280 metres
(4) Cannot be determined
(5) None of these

(RBI Grade-B Officer Exam.06.02.2011)

3. Train A crosses a pole and platform in 18 secnds and 39 seconds respectively. The length of platform is 157.5 metre. What will be the length of train B if it is equal to the sum of half of the length of train A and twice the length of the platform ?
(1) 382.5 metre
(2) 328.5 metre
(3) 238.5 metre
(4) 315 metre
(5) None of these

(RBI Officer Grade 'B' Online Exam. 25.08.2013

INSURANCE EXAMS

1. A and B started towards each other at the same time from places 114 kms. apart and met in 8 hours. If 'A's rate was 6 kms. an hour, what was 'B's ?
(1) 14.25 km/hr. (2) 7 km./hr
(3) $7\frac{1}{8}$ km./hr (4) $8\frac{1}{2}$ km/hr.
(5) $8\frac{1}{4}$ km/hr.

(United India Insurance Co. AAO Exam. 21.04.2002)

2. Sunil left for city '*x*' from city '*y*' at 5.50a.m. He travelled at the speed of 80 kms/hr. for 2 hours 15 minutes. After that the speed was reduced to 60 kms/hour. If the distance between two cities is 350 kms., at what time did Sunil reach the city '*Y*?
(1) 10.05 a.m. (2) 10.35 a.m.
(3) 9.50 a.m. (4) 9.55 a.m.
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 24.04.2005)

3. Two trains A and B of equal length of 200 metres running in opposite direction cross each other in 16 seconds. What is the speed of train A ?
(1) 90 kms/hr.
(2) 40 kms/hr.
(3) 80 kms/hr.
(4) Data inadequate
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 24.04.2005)

4. An employee may claim Rs. 7.00 for each km when he travels by taxi and Rs. 6.00 for each km if he drives his own car. If in one week he claimed Rs. 675 for travelling 90 km, how many kms did he travel by taxi ?
(1) 135 (2) 155
(3) 162 (4) 170

(United India Insurance Co. (AAO) Exam. 11.03.2007)

5. A train B speeding with 120 kmph crosses another train C running in the same direction, in 2 minutes. If the lengths of the trains B and C be 100 m and 200 m respectively, what is the speed (in kmph) of the train C ?
(1) 111 (2) 123
(3) 127 (4) 129

(United India Insurance Co. (AAO) Exam. 11.03.2007)

6. A train crosses a 300 metre long platform in 38 seconds while it crosses a signal pole in 18 seconds. What is the speed of the train in kmph ?
(1) Cannot be determined
(2) 72
(3) 48
(4) 54
(5) None of these

(LIC Assistant Administrative Officer Exam. 2008)

7. A man goes uphill with an average speed of 24 kmph. and comes down with an average speed of 36 kmph. The distance travelled in both the cases being the same. The average speed (in kmph) for the entire journey is :
(1) 30 (2) 28.8
(3) 32.6 (4) 30.8

(LIC Assistant Administrative Officer (AAO) Exam. 07.06.2009)

8. A train overtakes two persons who are walking in the same direction in which the train is going at the rate of 2 kmph and 4 kmph, and passes them completely in 9 and 10 seconds respectively. The length of the train (in metres) is :
(1) 72 (2) 54
(3) 50 (4) 45

(LIC Assistant Administrative

Officer (AAO) Exam. 07.06.2009)

9. Train-A crosses a pole in 33 seconds and another Train-B crosses a pole in 55 seconds. Length of Train-A is three-fourth of Train-B. What is the respective ratio between the speed of Train-A and Train-B ?
(1) 5 : 11
(2) 5 : 4
(3) 11 : 3
(4) Cannot be determined
(5) None of these

(United India Insurance AO Exam. 27.03.2011)

10. There is a hill behind a person's house. He walks up the top of the hill at a speed of $1\frac{1}{2}$ km/hour, but walks down it at $4\frac{1}{2}$ km/hour. If it takes him 6 hours for the entire journey, the distance, in km, from his house to the top of the hill is

(1) $5\frac{1}{4}$ (2) 6

(3) $6\frac{3}{4}$ (4) 9

(New India Insurance AAO Exam. 22.05.2011)

Directions (11-13): The cost of fuel running the engine of an army tank is proportional to the square of the speed and Rs. 64 per hour for a speed of 16 kmph. Other costs amount to Rs. 400 per hour. The tank has to make a journey of 400 km at a constant speed.

(General Insurance Corporation AAO Exam. 11.12.2011)

11. The most economical speed for this journey is :
(1) 20 kmph (2) 30 kmph
(3) 35 kmph (4) 40 kmph

12. The total cost for the journey at this most economical speed is :
(1) Rs. 6000 (2) Rs. 8000
(3) Rs. 10000 (4) Rs. 11000

13. Ravi can walk a certain distance in 40 days when he rests 9 hours a day. How long will he take to walk twice the distance, twice as fast and rest twice as long each day ?
(1) 40 days (2) 80 days
(3) 50 days (4) 100 days

(New India Assurance AO Exam. 25.10.2009)

14. A can walk a certain distance in 20 days when he rests 8 hours a day. How long will he take to walk twice the distance, twice as fast and rests twice as long each day ?
(1) 40 days (2) 60 days
(3) 80 days (4) 100 days

(Oriental Insurance Company AAO Exam. 08.04.2012)

15. A person leaves a place A to place B at 6 a.m. and reaches place B at 10 a.m. Another person leaves B at 7.30 a.m. and reaches A at 11 a.m. They will meet each other at
(1) 7.40 a.m. (2) 8.20 a.m.
(3) 8.40 a.m. (4) 9.20 a.m.

(Oriental Insurance Company AAO Exam. 08.04.2012)

16. A can walk a certain distance in 10 days when he rests 9 hours a day. How long will he take to walk twice the distance, twice as fast and rest twice as long each day ?
(1) 4 days (2) 10 days
(3) 15 days (4) 25 days

(United India Insurance AAO Exam. 03.06.2012)

17. A 222 metre long train crosses a pole in 6 seconds. The same train crosses a man running in the same direction in 10 seconds. What will be the speed of man ?
(1) 15 m/sec (2) 17 m/sec
(3) 18 m/sec (4) 14 m/sec
(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 12.05.2013)

18. How long (time in seconds) will a 450 metre long train take to cross a girl walking with a speed of 5 km/hr in the direction of the moving train. The speed of the train is 65 km/hr ?
(1) 27 (2) 28
(3) 30 (4) 29

(NICL (GIC) AO (Finance Exam. 08.09.2013 (Paper-I)

19. An express train travelled at an average speed of 100 km/hr stopping for 3 minutes after 75 km. A local train travelled at a speed of 50 km/hr, stopping for 1 minute after every 25 km. If the trains began travelling at the same time, how many kms did the local train travel in the time it took the express train to travel 600 km ?
(1) 307.5 km (2) 900 km
(3) 1000 km (4) 1200 km

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

20. If a student walks from his house to school at 5 km/hr, he is late by 30 minutes. However, if he walks at 6 km/hr, he is late

by 5 minutes only. The distance of his school from his house is.

(1) 2.5 km (2) 3.6 km
(3) 5.5 km (4) 12.5 km

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

21. A plane left 30 minutes later than the scheduled time and in order to reach the destination 1500 km away in time, it had to increase the speed by 250 km/hr from the usual speed. Its usual speed is:

(1) 720 km/hr (2) 730 km/hr
(3) 740 km/hr (4) 750 km/hr

(NICL (GIC) AO (Finance) Exam. 5.12.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (2) **2.** (5)
3. (2) **4.** (4)
5. (3) **6.** (5)
7. (3) **8.** (3)
9. (1) **10.** (5)
11. (1) **12.** (4)
13. (5) **14.** (2)
15. (3) **16.** (1)
17. (3) **18.** (1)
19. (1) **20.** (1)
21. (2) **22.** (2)
23. (2) **24.** (3)
25. (1) **26.** (5)
27. (3) **28.** (1)
29. (3) **30.** (4)
31. (4) **32.** (4)
33. (2) **34.** (2)
35. (5) **36.** (2)
37. (3) **38.** (4)
39. (4) **40.** (3)
41. (3)

SBI PO EXAMS

1. (3) **2.** (1)
3. (5) **4.** (1)
5. (5) **6.** (2)
7. (1)

RBI GRADE-B OFFICER EXAMS

1. (2) **2.** (5)
3. (1)

INSURANCE EXAMS

1. (5) **2.** (5)
3. (5) **4.** (1)
5. (1) **6.** (4)
7. (2) **8.** (3)
9. (2) **10.** (3)
11. (4) **12.** (2)
13. (4) **14.** (1)
15. (3) **16.** (4)
17. (1) **18.** (1)
19. (1) **20.** (4)
21. (4)

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (2) Initial speed of motor = 70 kmph.
Distance covered in first 2 hours = 2 × 70 = 140 kms.
For next two hours speed of motor = 80 kmph.
Distance covered in next 2 hours = 2 × 80 = 160 kms.
Distance covered in first 4 hours
= 140 + 160 = 300 Remaining distance = 345 - 300 = 45 km
This distance will be covered at the speed of 90 kmph.

$\therefore$ Time taken = $\frac{45}{90} = \frac{1}{2}$ hour

$\therefore$ Total time

$= 4 + \frac{1}{2} = 4\frac{1}{2}$ hours

2. (5) Let the length of train be x metres.

Time taken in crossing the pole = $\frac{x}{20}$

Time taken in crossing the bridge

$= \frac{4x}{20}$

As given, $\frac{4x}{2} - \frac{x}{20} = 24$

$\Rightarrow \frac{3x}{20} = 24 \Rightarrow \frac{24 \times 30}{2} = 160$ m

3. (2) Time taken in walking one way + riding other way
= 6 hours 35 minutes ... (i)
Time taken in riding both ways
= 4 hours 35 minutes ...(ii)
By equation (i) × 2 - (ii),
2 × Time taken in walking one way
= 13 hours 10 minutes - 4 hours 35 minutes
= 8 hours 35 minutes

4. (4) When a train crosses a platform, the distance covered
= Length of platform and the train.

$$\text{Speed} = \frac{\text{Length of (platform + train)}}{\text{Time taken}}$$

Thus we have inadequate data.

5. (3) Let the speed of boat in still water = x kmph

$\therefore$ Speed of current = $\frac{x}{2}$ kmph

When boat runs downstream,

$$\frac{30}{x + \frac{x}{2}} = 2$$

$\Rightarrow \frac{30 \times 2}{3x} = 2 \Rightarrow 6x = 60$

$\Rightarrow x = \frac{60}{6} = 10$ kmph

6. (5) Let the required distance be x km.
According to the question,

$\frac{x}{40} - \frac{x}{45} = 1$

$\Rightarrow \frac{9x - 8x}{360} = 1$

$\Rightarrow \frac{x}{360} = 1$

$\Rightarrow x = 360$ km.

7. (3) Length of the train = 240 metre
$\therefore$ Length of the platform = 480 metre
We know that when a train crosses a platform, it covers a distance equal to the sum of its length and that of platform.
$\therefore$ Speed of train

$= \left(\frac{240 + 480}{40}\right)$ m/sec

$\frac{720}{40} = 18$ m/sec

8. (3) Let the speed of boat in still water be x kmph and that of current be y kmph.

$\therefore x + y = \frac{4.8}{\frac{8}{60}} = \frac{4.8 \times 60}{8}$

$\Rightarrow x + y = 36$ ....(i)

and, $x - y = \frac{4.8}{\frac{9}{60}} = \frac{4.8 \times 60}{9}$

$\Rightarrow x - y = 32$ ....(ii)
By equation (i)-(ii),
$x + y - x + y$
$= 36 - 32 = 4$

$\Rightarrow 2y = 4 \Rightarrow y = \frac{4}{2} = 2$ kmph

9. (1) Distance between Ramgarh and Devgarh

$= \frac{50 \times 44}{60} = \frac{110}{3}$ km
New speed = 55 kmph

$= \frac{55}{60}$ km/minute

$\therefore$ Required time $= \frac{\text{Distance}}{\text{Speed}}$

$= \frac{110}{3} \times \frac{60}{55} = 40$ minutes

10. (5) Required ratio = 6 : 4 = 3 : 2
as time taken is inversely proportional to the speed.

11. (1) Let the distance between villages A and B be x km.

$\therefore \frac{x}{40} - \frac{x}{60} = 2 \Rightarrow \frac{3x - 2x}{120} = 2$

$\Rightarrow x = 2 \times 120 = 240$ km

12. (4) Let the length of train be x metre.

$\therefore$ Speed of train $= \frac{x}{20} = \frac{x + 100}{30}$

$\Rightarrow 3x = 2x + 200$
$\Rightarrow x = 200$ m = length of train

13. (5) Let the speed of the train be x m/sec.

$\therefore \frac{320 \times 3}{x} - \frac{320}{x} = 80$

$\Rightarrow \frac{320 \times 2}{x} = 80$

$\Rightarrow x = \frac{320 \times 2}{80} = 8$ m/sec = Speed of train

14. (2) Time taken by trains in crossing each other $= \frac{\text{Sum of lengths of trains}}{\text{Relative speed}}$

60 kmph $= \frac{60 \times 5}{18} = \frac{50}{3}$ m/sec.
If the speed of other train be x m/sec. then,

$10.8 = \frac{180 + 270}{\frac{50}{3} + x}$

$\Rightarrow 180 + 10.8x = 450$
$\Rightarrow 10.8x = 450 - 180 = 270$

$\Rightarrow x = \frac{270}{10.8} = 25$ metre/sec.

$= 25 \times \frac{18}{5}$ kmph = 90 kmph

15. (3) $5x + 9x + 4x = 72 \times 3$
$\Rightarrow 18x = 72 \times 3$

$\therefore x = \frac{72 \times 3}{18} = 12$ kmph

$\therefore$ Average speed of car and train $= \frac{5x + 9x}{2}$

$= 7x = 7 \times 12 = 84$ kmph

16. (1) Tricky approach

Speed of the bus = $\frac{480}{8}$ = 60 kmph

$\therefore$ Speed of the train = $60 \times \frac{4}{3}$ = 80 kmph

$\therefore$ Speed of the car = $\frac{15}{16} \times 80$ = 75 kmph

$\therefore$ Required distance = Speed × time
= 75 × 6 = 450 km.

17. (3) Tricky approach
If the length of the train be *l* metre, then

Required ratio = $\frac{l}{54} : \frac{l}{12 \times 60}$ = 40 : 3

18. (1) If the length of train-B be *x* metre, then

Speed of train = $\frac{240 + x}{50} = \frac{240}{20}$

$\Rightarrow \frac{240 + x}{50} = 12$

$\Rightarrow 240 + x = 600$

$\Rightarrow x = 360$ metre

19. (1) Speed of tractor = $\frac{360}{12}$ = 30 kmph

Speed of jeep = $30 \times \frac{250}{100}$ = 75 kmph

$\therefore$ Speed of car = $\frac{3}{5} \times 75$ = 45 kmph

$\therefore$ Average speed of car and jeep together

= $\left(\frac{75 + 45}{2}\right)$ kmph = 60 kmph

20. (1) If the length of the bus be *x* metre, then

Required ratio = $\frac{x}{4} : \frac{x}{18}$ = 9 : 2

21. (2) Speed of bus = $\frac{60}{1.5}$ = 40 kmph

$\therefore$ Required difference = $\left(\frac{720}{40} - \frac{720}{60}\right)$ hours

= (18 - 12) hours = 6 hours

22. (2) Speed of train

= $\frac{\text{Length of (train + platform)}}{\text{Time taken}}$

= $\left(\frac{320 + 3 \times 320}{40}\right)$ metre/second

= 32 metre/second

$\left(\frac{32 \times 18}{5}\right)$ kmph = 115.2 kmph

23. (2) Distance = 64 × 8 = 512km.

$\therefore$ Required speed = $\frac{\text{Distance}}{\text{Time}}$

= $\frac{512}{6}$ = 85.33 $\approx$ 85 kmph.

24. (3) Speed of bus

= $\frac{\text{Distance covered}}{\text{Time taken}} = \frac{480}{12}$ = 40 kmph

$\therefore$ Speed of train = $\frac{9}{5} \times 40$ = 72 kmph

$\because 18 \equiv 72$

$\therefore 1 \equiv \frac{72}{18} = 4$

$\therefore$ 4 × 13 = 52 kmph = speed of car
Distance covered by car in 5 hours
= 5 × 52 = 260 km

25. (1) Let the length of the platform be *x* metre.
Speed of train =126 kmph

= $\left(\frac{126 \times 5}{18}\right)$ m/sec. = 35 m/sec.

$\therefore$ Speed of train = $\frac{x + 300}{24}$

$\Rightarrow 35 = \frac{x + 300}{24}$

$\Rightarrow 35 \times 24 = x + 300$

$\Rightarrow 840 = x + 300$

$\Rightarrow x = 840 - 300 = 540$ metre

$\therefore$ Speed of man = $\frac{\text{Distance}}{\text{Time}}$

= $\frac{540}{5 \times 60}$ =1.8 m/sec.

26. (5) Let the length of train B be *x* metre.
$\therefore$ Speed of train

= $\frac{\text{Sum of lengths of trains}}{\text{Time taken}}$

= $\frac{\text{Length of train - A}}{\text{Time taken}}$

$\Rightarrow \frac{280 + x}{35} = \frac{280}{14}$

$\Rightarrow \frac{280 + x}{35} = 20$

$\Rightarrow 280 + x = 700$

$\Rightarrow x = 700 - 280 = 420$ metre

27. (3) Speed of car = $\frac{\text{Distance}}{\text{Time}}$

$= \frac{540}{9} = 60$ kmph

Speed of train = 2 × 60
= 120 kmph

$\therefore$ Speed of bike

$= \frac{2}{3} \times 120 = 80$ kmph

$\therefore$ Distance covered by bike in 5 hours
= 5 × 80 = 400 km

28. (1) If the length of train A be x metre, then length of train B = $2x$ metre.
When a train crosses a pole, it covers a distance equal to its own length.

$\therefore$ Required ratio = $\frac{x}{25} : \frac{2x}{75}$

$= \frac{1}{25} \times 75 : \frac{2}{75} \times 75 = 3 : 2$

29. (3) Speed of tractor = $\frac{\text{Distance}}{\text{Time}}$

$= \frac{575}{23} = 25$ kmph

$\therefore$ Speed of bus = 50 kmph

$\therefore$ Speed of car

$= \frac{9}{5} \times 50 = 90$ kmph

$\therefore$ Distance covered by car in 4 hours
= 4 × 90 = 360 km

30. (4) Average speed

$= \frac{\text{Total distance}}{\text{Total time}}$

$= \frac{(39+25)\text{km}}{(45+35)\text{minute}}$

$= \frac{64}{\frac{80}{60}}$ kmph

$= \left(\frac{64 \times 60}{80}\right)$

= 48 kmph

31. (4) Distance covered by aeroplane in 9 hours
= Speed × Time = 9 × 756 = 6804 km

$\therefore$ Speed of helicopter = $\frac{2 \times 6804}{48}$

= 283.5 kmph

$\therefore$ Distance covered by helicopter in 18 hours = (283.5 × 18) km = 5103 km

32. (4) Speed of train = 120 kmph

$= \frac{120 \times 5}{18} = \frac{100}{3}$ m/sec

$\therefore \frac{100}{3} = \frac{(\text{Length of train and platform})}{\text{Time taken in crossirg}}$

$\Rightarrow \frac{100}{3} = \frac{320 + x}{24}$

Where x = length of platform

$\Rightarrow \frac{24 \times 100}{3} = 320 + x$

$\Rightarrow 800 = 320 + x$

$\Rightarrow x = 800 - 320 = 480$ m

$\therefore$ Man's speed

$= \frac{480}{4 \times 60} = 2$ m/sec

33. (2) Speed of car

$= \frac{\text{Distance covered}}{\text{Time taken}}$

$= \frac{720}{9} = 80$ kmph

$\therefore$ Speed of bus = $\frac{3}{4} \times 80 = 60$ kmph

$\therefore$ Speed of train = $\frac{27}{15} \times 60 = 108$ kmph

$\therefore$ Distance covered by train in 7 hours
= 7 × 108 = 756 km.

34. (2) Speed of train =108 kmph

$= \frac{108 \times 5}{18} = 30$ m/second

If the length of platfrom be x metre, then

$\frac{x + 280}{12} = 30$

$\Rightarrow x + 280 = 30 \times 12 = 360$

$\Rightarrow x = 360 - 280 = 80$ metre

$\therefore$ Man'speed = $\frac{\text{Distance}}{\text{Time}}$

$= \frac{80}{10} = 8$ m/second

35. (5) Average speed of car

$= \frac{\text{Distance}}{\text{Time}} = 98$ kmph

Average speed of train = $\frac{98 \times 10}{7} = 140$ kmph

Distance covered by train in 13 hours
= Speed × Time = 140 × 13 = 1820 km

36. (2) Required time = LCM of 18, 24 and 32 seconds. = 288 seconds

37. (3) Required time = LCM of 18, 22 and 30 seconds

$= 990 \text{ seconds} = \frac{990}{60} \text{ minutes}$

= 16 minutes 30 seconds

38. (4) Speed of train $= \frac{320}{50} = 6.4$ m/sec.

39. (4) Time taken by car $= \frac{528}{66} = 8$ hours

Case II,
Time =15 hours
Speed = 66 - 24 = 42 kmph
$\therefore$ Distance covered by truck
= Speed × Time
= 15 × 42 = 630 km

40. (3) Speed of return $= \frac{40 \times 150}{100}$

= 60 kmph

Average speed $= \frac{2 \times 40 \times 60}{40 + 60}$

$= \frac{4800}{100} = 48$ kmph

41. (3) Speed of train

$= \frac{\text{Length of train}}{\text{Time taken in crossing}}$

$= \frac{320}{16} = 20$m/sec.

$= 20 \times \frac{18}{5} = 72$ kmph

Total period of stoppage = 5 × 18

$= 90 \text{ minutes} = \frac{3}{2}$ hours

Total time taken in covering a distance of 576 km.

$= \left(\frac{576}{72} + \frac{3}{2}\right)$ hours

$= 8 + \frac{3}{2} = 9\frac{1}{2}$ hours

SBI PO EXAMS

1. (3) We know that

$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \left(\frac{260 + 120}{19}\right)$ m/ sec.

$= \frac{380}{19}$ m/sec. = 20 m/sec.

$= \frac{20 \times 18}{5}$ km/hr = 72 km/hr

2. (1) Let speed of car starting at A = x km/hr
and speed of car starting at B = y km/hr
$\therefore x + y = 120$ .... eq (i)
$6x - 6y = 120$ ....eq (ii)
$\therefore x = 70$ km/hr. $y = 50$ km/hr.

3. (5) Speed of train

$= \frac{\text{length of (train + platform)}}{\text{time taken to cross}}$

$= \frac{440}{22} = 20$ m/s

$= 20 \times \frac{18}{5} = 72$ km/hr

4. (1) The distance walked on the first day = 2 kms.
The distance walked on subsequent days is half the distance walked on the previous days.

Total distance walked = $2 + 1 + \frac{1}{2} + \frac{1}{2} + \ldots$

This is a Geometric series Whose
First term, $a = 2$

Common ratio, $r = \frac{1}{2}$

Maximum total distance walked by the person in his life-time means the number of terms in the series would be infinite. Hence, the series would be an infinite Geometric series. Sum of an infinite Geometric series is given by

$S = \frac{a}{1-r}$ $\qquad$ $S = \frac{2}{1 - \frac{1}{2}}$

$S = \frac{2}{\frac{1}{2}}$, $\qquad$ S = 4 kms.

5. (5) Initial speed = 80 km/hr
Total distance = 80 × 10 = 800 km.

New speed $= \frac{800}{4} = 200$ *km / hr.*

Increase in speed
= 200 - 80 = 120 km/hr.

6. (2) Initial speed of motor = 70 kmph.
Distance covered in first 2 hours = 2 × 70

= 140 kms.
For next two hours speed of motor
= 80 kmph.
$\therefore$ Distance covered in next 2 hours
= 2 × 80 = 160 kms.
Distance covered in first 4 hours
= 140 + 160 = 300 km
Remaining distance = 345 - 300 = 45 km
This distance will be covered at the speed of 90 kmph.

$\therefore$ Time taken = $\frac{45}{90} = \frac{1}{2}$ hour

$\therefore$ Total time = $4 + \frac{1}{2} = 4\frac{1}{2}$ hours

7. (1) Let the required distance be = x km.
According to the question,

$\frac{x}{45} - \frac{x}{50} = 1$

$\Rightarrow \frac{10x - 9x}{450} = 1$

$\Rightarrow x$ = 450 km.

RBI GRADE-B OFFICER EXAMS

1. (2)Nitin will take rest sever, times i.e. 70 minutes in covering 150 km.
Time taken to cover 150 km at 15 kmph

= $\frac{150}{15}$ = 10 hours

$\therefore$ Total time
= 10 hours + 70 minutes
= 11 hours 10 minutes

2. (5) When a train crosses a platform it covers a distance equal to the sum of lengths of platform and train.
Let the length of platform be x metre.

$\therefore$ Speed of train = $\frac{x + 280}{60} = \frac{280}{20} = 14$

$\Rightarrow x + 280 = 60 \times 14 = 840$
$\therefore x$ = 840 - 280 = 560 metre

3. (1) Length of train A = x metre

$\therefore \frac{x}{18} = \frac{x + 157.5}{39}$
$\Rightarrow 13x = 6x + 157.5 \times 6$
$\Rightarrow 7x = 945$

$\Rightarrow x = \frac{945}{7}$ = 135 metre

$\therefore$ Length of train B = $\left(\frac{135}{2} + 2 \times 157.5\right)$
= (67.5 + 315) metre = 382.5 metre

INSURANCE EXAMS

1. (5) According to the question, Distance covered by A in 8 hrs.
= 6 × 8 = 48 km
$\therefore$ Distance covered by B in 8 hrs.
= (114 - 48) km = 66 km

$\therefore$ Required speed = $\frac{66}{8}$ kmph = $8\frac{1}{4}$ kmph

2. (5) Distance between two cities = 350 km
Distance travelled by Sunil with the speed of 80 km/hr in $2\frac{1}{2}$ hr or 2.25 hr
= 80 × 2.25 km = 180 km
Remaining distance
= 350 - 180 = 170 km
Speed = 60 km/hr
Time taken to reach in city

$x = \frac{170}{60}$ = 2.83 hr

Total time = 2.25 + 2.83
= 5.08 hr = 305 minute
= 6 hr. 5 minute
Sunil started journey at 5:50 a.m. & finished the journey = 11:55 a.m.

3. (5) Distance = 200 metres
time =16 seconds

Speed = $\frac{\text{distance}}{\text{time}} = \frac{200}{16}$ = 12.5m/sec.

= $\frac{12.5 \times 1000}{3600}$ km/hr = 3.47 Km/hr.

4. (1) Let the employee travelled x kms by taxi.
$\therefore$ Distance covered by him by his own car
= (90 - x) km.
According to the question,
$x \times 7 + (90 - x) \times 6 = 675$
$7x + 540 - 6x = 675$
$x = 675 - 540 = 135$
$\therefore$ Required distance = 135 km.

5. (1) Let the speed of train C be x kmph.
Speed of train B relative to C
= (120 - x) kmph

= $\left[(120 - x) \times \frac{5}{18}\right]$ m/sec

= $\left(\frac{600 - 5x}{18}\right)$

Distance covered = 100 + 200 = 300m

$$\therefore \frac{300}{\left(\frac{600-5x}{18}\right)} = 120$$

$$\Rightarrow 300 = \frac{120(600-5x)}{18}$$

$\Rightarrow 10 \times 9 = 2\,(600 - 5x)$

$\Rightarrow 90 = 1200 - 10x$

$\Rightarrow 10x = 1200 - 90$

$$\Rightarrow x = \frac{1110}{10} = 111$$

Hence, the speed of train C is 111 kmph.

6. (4) When a train crosses a platform it covers a distance equal to the sum of lengths of platform and the train itself. If the length of train be x metres, then

Speed of train

$$= \frac{x+300}{38} \text{ m/sec} \qquad \ldots(i)$$

When the train crosses a signal post it covers its own length.

$\therefore$ Speed of train x

$$= \frac{x}{18} \text{ m/sec} \qquad \ldots(ii)$$

From equations (i) and (ii)

$$\frac{x+300}{38} = \frac{x}{18}$$

$\Rightarrow 38x - 18x = 300 \times 18$

$$\Rightarrow x = \frac{300 \times 18}{20}$$

$= 270$ metres

$$\therefore \text{Speed of train} = \frac{270}{18}$$

$= 15$m/sec.

$$= 15 \times \frac{18}{5} = 54 \text{ kmph}$$

7. (2) Average speed $= \dfrac{2xy}{x+y}$

(when the same distances are covered)

$$= \left(\frac{2 \times 24 \times 36}{24+36}\right) \text{kmph}$$

$$= \frac{2 \times 24 \times 36}{60} = 28.8 \text{ kmph}$$

8. (3) 2 kmph $= \left(\dfrac{2 \times 5}{18}\right)$ metre/sec. $= \dfrac{5}{9}$ metre/sec.

and 4 kmph $= \dfrac{4 \times 5}{18}$ metre/sec. $= \dfrac{10}{9}$ metre/sec.

Let the length of the train be x metre and its speed be y metre/ sec. Then,

$$\frac{x}{y - \frac{5}{9}} = 9$$

$\Rightarrow 9y - 5 = x$

$\therefore 9y - x = 5$ ..... (i)

$$\text{and} = \frac{x}{y - \frac{10}{9}} = 10$$

$\Rightarrow 10\,(9y - 10) = 9x$

$\Rightarrow 90y - 9x = 100$ ..... (ii)

By equation (i) × 10 - equation (ii), we have

$$\begin{array}{r} 90y - 10x = 50 \\ 90y - 9x = 100 \\ \hline -x = -50 \end{array} \quad (- \; + \; -)$$

$\Rightarrow x = 50$ m

9. (2) Let the length of train B = x metre

$$\therefore \text{Length of train A} = \frac{3x}{4} \text{ metre}$$

$$\therefore \text{Required ratio} = \frac{3x}{4 \times 33} : \frac{x}{55} = 5 : 4$$

10. (3) If the distance be x km, then

$$\frac{x}{\frac{3}{2}} + \frac{x}{\frac{9}{2}} = 6$$

$$\Rightarrow \frac{2x}{3} + \frac{2x}{9} = 6$$

$$\Rightarrow \frac{6x + 2x}{9} = 6$$

$\Rightarrow 8x = 9 \times 6$

$$\Rightarrow x = \frac{54}{8} = \frac{27}{4} = 6\frac{3}{4} \text{ km.}$$

11. (4) $C = kx^2$

When $x = 16$ kmph

C = Rs. 64

$\therefore 64 = k \times 16^2$

$\Rightarrow k = \frac{1}{4}$

$\therefore C = \frac{1}{4} \times x^2$

Total expenditure per hour = $\frac{1}{4}x^2 + 400$

When speed = 40 kmph, the expenditure will be minimum.

12. (2) Total expenditure

$= \frac{1}{4} \times 40 \times 40 \times 10 + 400 \times 10 =$ Rs. 8000

13. (4) Tricky approach

Working hours per day = 24 - 9 = 15 hours

Total time = 15 × 40 = 600 hours

Decrease in walking hours per day after increase in hours of rest = 24 - (2 × 9)

= 6 hours

Required time = $\frac{600}{6}$ = 100 days

14. (1) Working hours per day

= 24 - 8 = 16 hours

Total working time = 16 × 20 = 320 hours

In case II,

Distance is doubled and speed too.

Working hours per day = 24 - 16 = 8 hours

$\therefore$ Required time = $\frac{320}{8}$ = 40 days

15. (3) If AB = x km, then Speed of person starting from A = $\frac{x}{4}$ kmph

Speed of person starting from B

$= \frac{x}{\frac{7}{2}} = \frac{2x}{7}$ kmph

Relative speed

$= \frac{x}{4} + \frac{2x}{7} = \frac{7x + 8x}{28} = \frac{15x}{28}$ kmph

Distance covered by first person till 7:30 a.m. = $\frac{x}{4} \times \frac{3}{2} = \frac{3x}{8}$ km

Remaining distance = $x - \frac{3x}{8} = \frac{5x}{8}$ km.

Time of meeting = $\frac{\text{Remaining distance}}{\text{Relative speed}}$

$= \frac{\frac{5x}{8}}{\frac{15x}{28}} = \frac{5x}{8} \times \frac{28}{15x} = \frac{7}{6}$ hours

$= \frac{7}{6} \times 60 = 70$ minutes

$\therefore$ Requisite time = 8:40 a.m.

16. (4) Working hours per day = 24 - 9 = 15

Total working hours = 15 × 10 = 150

Distance is doubled and speed too. Both cancel each other.

New working hours = 24 - 18 = 6 hours

$\therefore$ Required time = $\frac{150}{6}$ = 25 days

17. (1) Speed of train = $\frac{222}{6}$ = 37m/sec.

If the speed of man be x m/sec, then

$\frac{220}{37 - x} = 10$

$\Rightarrow$ 220 = 370 - 10x

$\Rightarrow 10x$ = 370 - 220 = 150

$\Rightarrow x$ = 15 m/sec

18. (1) Relative speed

= 65 - 5 = 60 kmph

$= \frac{60 \times 5}{18}$ m/sec = $\frac{50}{3}$ m/sec

$\therefore$ Required time = $\frac{450}{\frac{50}{3}}$

$= \frac{450 \times 3}{50}$ = 27 seconds

19. (1) Effective time taken by express train in covering 75 km

$= \left(\frac{60}{100} \times 75 + 3\right)$ minutes = 48 minutes

Time taken in covering 600 km

= (48 × 7 + 45) minutes

= 336 + 45 = 381 minutes

Effective time taken by local train in covering 25 km = 31 minutes

$\therefore$ 31)381(12

31

71

62

9

Distance covered in 9 minutes

$= \frac{50}{60} \times 9 = 7.5$ km

Distance covered in (31 × 120 minutes

= 12 × 25 = 300 km

Total distance = 300 + 7.5 = 307.5 km

20. (4) If the required distance be x km, then

$$\frac{x}{5} - \frac{x}{6} = \frac{30-5}{60}$$

$$\Rightarrow \frac{6x-5x}{30} = \frac{25}{60} = \frac{5}{12}$$

$$\Rightarrow x = \frac{30\times 5}{12} = 12.5 \text{ km}$$

21. (4) Usual speed of plane = x kmph

$\therefore$ New speed = $(x + 250)$ kmph

$$\therefore \frac{1500}{x} - \frac{1500}{x+250} = \frac{30}{60}$$

$$\Rightarrow 1500\left(\frac{x+250-1}{x(x+250)}\right) = \frac{1}{2}$$

$\Rightarrow x(x + 250) = 1500 \times 500 = 750000$

$\Rightarrow x\,(x + 250) = 750\,(750 + 250)$

$\Rightarrow x = 750$ kmph

MODEL EXERCISES

1. A train running between two stations A and B arrives at its destination 10 min late when its speed is 50 km/h and 50 min late when its speed is 30 km/h. What is the distance between the stations A and B ?
(1) 40 km (2) 50 km
(3) 60 km (4) 70 km
(5) None of these

2. A student rides on a bicycles at 8 km/h and reaches his school 2.5 min late. The next day he increases his speed to 10 km/h and reaches the school 5 min early. How far is the school from his house ?
(1) 1.25 km (2) 8 km
(3) 5 km (4) 10 km
(5) None of these

3. A small aeroplane can travel at 320 km/h in still air. The wind is blowing at a constant speed of 40 km/h. The total time for a journey against the wind is 135 min. What will be the time in minutes for the return journey with the wind? (Ignore take off and landing time for the aeroplane)
(1) 94.5 (2) 105
(3) 108.125 (4) 120
(5) None of these

4. Two cyclists start on a circular track from a given point but in opposite directions with speeds of 7 m/s and 8 m/s respectively. If the circumference of the circle is 300 m after what time will they meet at the starting point ?
(1) 20 s (2) 100 s
(3) 300 s (4) 200 s
(5) None of these

5. During a journey of 80 km a train covers first 60 km with a speed of 40 km/h and completes the remaining distance with a speed of 20 km/h. What is the average speed of the train during the whole journey ?
(1) 30 km/h (2) 32 km/h
(3) 36 km/h (4) 40 km/h
(5) None of these

6. If a man travels at 30 km/h. he reaches his destination late by 10 min but if he travels at 42 km/h, then he reaches 10 min earlier. Therefore, the distance travelled by him is
(1) 36 km (2) 35 km
(3) 40 km (4) 45 km
(5) None of these

7. Two cars start together in the same direction from the same place. The first goes with a uniform speed of 10 km/h. The second goes at a speed of 8 km/h in the first hour and increase the speed by $\frac{1}{2}$ km each succeeding hour. After how many hours will the second car overtaken the first if both go non stop ?
(1) 9 h (2) 5 h
(3) 7 h (4) 8 h
(5) None of these

8. Two rockets approach each other, one at 42000 mile/h and the other at 18000 mile/h. They start 3256 miles a part. How far are they a part (in miles) 1 min before impact ?
(1) 1628 (2) 1000
(3) 826 (4) 1200
(5) None of these

9. Dinesh travels 760 km to his home, partly by train and partly by car. He takes 8 h if he travels 160 km by train and the rest by car. He takes 12 min more if he travels 240 km by train and the rest by car. The speeds of the train and the car respectively are
(1) 80 km/h, 100 km/h
(2) 100 km/h, 80 km/h
(3) 120 km/h, 100 km/h
(4) 100 km/h, 120 km/h
(5) None of these

10. In a flight of 3000 km an aircraft was slowed down by bad weather. If average speed for the trip was reduced by 100 km/h and the time increased by one hour, find the original duration of the flight.
(1) 5 h (2) 6 h
(3) 4 h (4) None of these
(5) None of these

11. The diameter of a cycle wheel is 70 cm. A cyclist takes 30 h to reach a destination at the speed of 22 km/h. How many revolutions will the whell make during this journey ?
(1) 3 million (2) 3 lakh
(3) 4 lakh (4) 5 lakh
(5) None of these

12. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 min more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the speed of the car is
(1) 4 : 3 (2) 3 : 4
(3) 3 : 2 (4) 2 : 3
(5) None of these

13. Points A and B are 70 km apart on a highway. One car starts from A and the another one from B at the same time. If they travel in the same direction, they meet in 7 h. But if they travel towards each other, they meet in one hour. The speeds of the two cars are
(1) 45 and 25 km/h
(2) 70 and 10 km/h
(3) 40 and 30 km/h
(4) 60 and 40 km/h
(5) None of these

14. A journey of 192 km between two cities takes two hours less by a fast train than by a slow train. If the average speed of the slow train is 16 km/h less than that of the fast train, then the average speed of the fast train is
(1) 36 km/h (2) 64 km/h
(3) 7 km/h (4) 48 km/h
(5) None of these

15. A train 100 m long passes a bridge at the rate of 72 km/h in 25 s. The length of the bridge is
(1) 150 m (2) 400 m
(3) 300 m (4) 200 m
(5) None of these

16. A train 100 m in length travels at 60 km/h. How much time does the train take in passing a man walking at 6 km/h against the train
(1) 6 s (2) 12 s
(3) 10 s (4) 18 s
(5) None of these

17. A car driver travels from the plains to the hill station, which are 200 km a part at an average speed of 40 km/h. In the return trip he covers the same distance at an average speed of 20 km/h the car. The average speed of the car over the entire distance of 400 km is
(1) 16.56 km/h
(2) 17.89 km/h
(3) 26.67 km/h
(4) 35 km/h
(5) None of these

18. A man in a train notices that he can count 21 telephone posts in one minute. If they are known to be 50 m apart, then at what speed is the train travelling ?
(1) 57 km/h (2) 60 km/h
(3) 63 km/h (4) 55 km/h
(5) None of these

19. A circular running path is 726 m in circumstance. Two men start from the same point and walk in opposite directions at the rate of 3.75 km/h and 4.5 km/h respectively. When will they meet for the first time ?
(1) 5.5 min (2) 6.0 min
(3) 5.28 min (4) 4.9 min
(5) None of these

20. A gun is fired at a distance of 3.32 km from Babar. He hears the sound 10 s later. Find the speed of the sound.
(1) 301 m/s (2) 302 m/s
(3) 332 m/s (4) 340 m/s
(5) None of these

21. The jogging track in a sports complex is 726 m in circumference. Suresh and his wife start from the same point and walk in opposite direction at 4.5 km/h and 3.75 km/h respectively. They will meet for the first time in
(1) 5.5 min (2) 6.0 min
(3) 5.28 min (4) 4.9 min
(5) None of these

22. There are 20 poles with a constant distance between each pole. A car takes 24 s to reach the 12 th pole. How much more time will it take to reach the last pole ?
(1) 25.25 s (2) 17.45 s
(3) 35.75 s (4) 41.45 S
(5) None of these

23. A train can travel 20% faster than a car. Both start from the point A at the same time and reach point B, 75 km away from A at the same time. On the way however the train lost about 12.5 min while stopping at the stations. The speed of the car is
(1) 50 km/h (2) 55 km/h
(3) 60 km/h (4) 65 km/h
(5) None of these

24. Two trains of equal length are running on parallel lines in the same direction at 46 km/h and 36 km/h. The faster train passes the slower train in 36 s. The length of each train is
(1) 50 m (2) 80 m
(3) 72 m (4) 82 m
(5) None of these

25. An aeroplane first flew with a speed of 440 km/h and covered a certain distance. It still had to cover 770 km less than what it had already covered but it flew with a speed of 660 km/h. The average speed for the entire flight was 550 km/h. Find the total distance covered.
(1) 3250 km (2) 2750 km
(3) 4400 km (4) 1375 km
(5) None of these

SHORT ANSWERS

1.	(2)	**2.**	(3)
3.	(2)	**4.**	(3)
5.	(1)	**6.**	(2)
7.	(1)	**8.**	(2)
9.	(1)	**10.**	(1)
11.	(2)	**12.**	(2)
13.	(3)	**14.**	(4)
15.	(2)	**16.**	(1)
17.	(3)	**18.**	(2)
19.	(3)	**20.**	(3)
21.	(1)	**22.**	(4)
23.	(3)	**24.**	(1)
25.	(2)		

EXPLANATIONS

1. (2) Let the distance between two stations be x km.

Then, $\frac{x}{50} - \frac{1}{6} = \frac{x}{30} - \frac{5}{6}$

$\Rightarrow \frac{x}{30} - \frac{x}{50} = \frac{2}{3}$

$\Rightarrow \frac{5x - 3x}{150} = \frac{2}{3}$

$\Rightarrow 6x = 300$

$\Rightarrow x = 50$ km

2. (3) Let the distance be x km.

Then, $\frac{x}{8} - \frac{2.5}{60} = \frac{x}{10} + \frac{5}{60}$

$\frac{x}{8} - \frac{x}{10} = \frac{5}{60} + \frac{2.5}{60}$

$\Rightarrow \frac{2x}{80} = \frac{7.5}{60}$

$x = \frac{7.5 \times 80}{2 \times 60} = 5$ km

3. (2) According to the question,
Speed of aeroplane against the wind
= (320 - 40) = 280 km/h
Let the distance be x km.

Then, $\frac{135}{60} = \frac{x}{280}$

$\Rightarrow x = 630$ km
Speed of Aeroplane with the wind
= (320 + 40)
= 360 km/h
$\therefore$ Time taken by aeroplane with the wind

$= \frac{630}{360} \times 60$ min = 105 min

4. (3) According to the question,
Relative speed = (8 - 7) =1 m/s
Distance = 300 m
$\therefore$ Time taken to cover this distance

$= \frac{300}{1} = 300$ s

5. (1) According to the question,
The clock gains 15 m in 24 h. Then, in 16 h it will gain 10 min
Hence, the time shown by this clock will be 4 : 10 am.

6. (2) Let the distance be x km.
Then, correct time at a speed of 30 km/h

$= \frac{x}{30} - \frac{10}{60}$

Correct time at a speed of 42 km/h

$= \frac{x}{42} + \frac{10}{60}$

$\Rightarrow \frac{x}{30} - \frac{10}{60} = \frac{x}{42} + \frac{10}{60}$

$\Rightarrow \frac{x}{30} - \frac{x}{42} = \frac{2}{6}$

$\Rightarrow \frac{x(12)}{1260} = \frac{2}{6}$

$\Rightarrow x = \frac{1260 \times 2}{72}$

$\Rightarrow x = 35$ km

7. (1) Let second car overtake the first car after n h.
$\therefore$ Distance covered by first car = Distance covered by second car

$\Rightarrow 10n = 8 + \left(8 + \frac{1}{2}\right) + \left(8 + \frac{1}{2}\right) + \left(8 + \frac{n-1}{2}\right)$

$\Rightarrow 10n = 8n + \frac{1}{2}[1 + 2 + ... + (n - 1)]$

$\Rightarrow 10n = 8n + \frac{1}{2}\frac{n(n-1)}{2}$

$\Rightarrow 2n = \frac{1}{4}(n^2 - n)$

$\Rightarrow n = 9$
as $n \neq 0$
Hence the answer is 9 h.

8. (2) Relative speed of rockets
= (42000 + 18000)
= 60000 mile/h

It means both of them together cover a distance of 60000 miles between then in 60 min or 1000 miles 1 min.
Hence, they should be 1000 miles apart, 1 min before impact.

9. (1) Let the speed of the train be x km/h and that of car be y km/h.

Then, $\frac{160}{x} + \frac{600}{y} = 8$ ...(i)

$\frac{240}{x} + \frac{520}{y} = \frac{41}{5}$ ...(ii)

Solving Eqs. (i) and (ii), we get
$x = 80$ and $y = 100$

10. (1) Let the original time taken by the plane x h.
The, original speed

$= \frac{3000}{x}$ km/h

$\therefore \frac{3000}{x+1} = \frac{3000}{x} - 100$

$\Rightarrow x = 5$ h

11. (2) According to the question, Circumference of the wheel

$= 2\pi r = 2 \times \frac{22}{7} \times 35 = 220$ m

$\therefore$ Number of revolution made by the wheel during the journey
($\because$ 22 km = 2200000 cm)
= 300000 = 3 lakh

12. (2) Let the speed of the train be x km/h and car be y km/h.

Then, $\frac{120}{x} + \frac{400}{y} = 8$ ...(i)

and $\frac{200}{x} + \frac{400}{y} = \frac{25}{3}$ ...(ii)

From Eqs. (i) $120y + 480x = 8xy$

From Eqs. (ii) $200y + 400x = \frac{25}{3}xy$

$\therefore \frac{120y + 480x}{8}$

$= \frac{3(200y + 400x)}{25}$

$\Rightarrow 25y + 60x = 24y + 48x$

$\Rightarrow 12x = 9y$

$\Rightarrow \frac{x}{y} = \frac{3}{4}$

$\Rightarrow x : y = 3 : 4$

13. (3) Let the speed of the cars be x km/h and y km/h.
Relative speed in same direction
$= (x - y)$ km/h
Relative speed in opposite direction
$= (x + y)$ km/h

Now, $\frac{70}{x+y} = 1$

$\Rightarrow x + y = 70$ ...(i)

and $\frac{70}{x-y} = 7$

$\Rightarrow x - y = 10$

Solving Eqs. (i) and (ii),
$x = 40$ km/h and $y = 30$ km/h

14. (4) Let the speed of the faster train be x km/h.
$\therefore$ Speed of the slower train = $(x - 16)$ km/h

$\Rightarrow \frac{192}{x} = \frac{192}{x-16} - 2$

$\Rightarrow x = 48$ km/h

15. (2) Let the length be x m.

Then, $(x + 100) = 72 \times 25 \times \frac{5}{18} = 500$

$\therefore x = 500 - 100 = 400$ m

16. (1) According to the question,

Relative $= (60 + 6) = 66$ km/h $= \frac{66 \times 5}{18}$ m/s

Distance = 110 m

Time $= \frac{110 \times 18}{66 \times 5} = 6$ s

17. (3) According to the question,

Average speed $= \frac{2xy}{x+y}$ where x and y are two different speeds covering same distance

$\therefore$ Average speed $= \frac{2 \times 40 \times 20}{(20+40)} = 26.67$ km/h

18. (2) $\because$ Distance between 21 posts
$= (21/01) \times 50 = 1000$ m
$\therefore$ Speed of train = 1 km/min = 60 km/h

19. (3) $\because$ Relative speed
$= (4.5 + 3.75) = 8.25$ km/h

$\therefore$ Distance = 726 m $= \frac{726}{1000} = 0.726$ km

$\therefore$ Required time $= \frac{0.726}{8.25} \times 60 = 5.28$ min

20. (3) $\because$ Distance = 3.32 × 100
Time = 10 s

Speed = $\frac{3320}{10}$ = 332 m/s

21. (1) $\because$ Relative speed = (500 + 700)
= 1200 m/min

$\therefore$ Time taken = $\frac{4200}{1200}$ = 3.5 min

22. (4) Let the distance between each pole be x m.
Then, distance up to 12th pole = 11x m

$\therefore$ Speed = $\frac{11x}{22}$ s

$\therefore$ Time to cover total distance up to 20th pole = $\frac{19x \times 24}{11x}$ = 41.45 s

23. (3) Let the speed of the car be x km/h.

$\therefore$ Speed of train = $x \times 1.2x = \frac{6x}{5}$ km/h

$\therefore \frac{5}{6x} \times 75 + \frac{25}{2 \times 60} = \frac{75}{x}$

$= \frac{75 \times 2 \times 60}{6 \times 25} = 60$ km/h

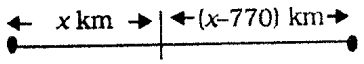
24. (1) Let the length of each train be xm.
Then, total distance covered = (x + x)
= 2x m
Relative speed = (46 - 36)

= 10 km/h = $\frac{10 \times 5}{18}$ m/s

Given (time) 36 = $\frac{2x \times 18}{50}$

$\Rightarrow x$ = 50 m

25. (2) Let the plane covers x km with 440 km/h and (x - 770) km at a speed of 660 km/h. Hence, it covers a total distance of (2x - 770) km at a speed of 500 km/h



Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$500 = \frac{2x - 770}{\frac{x}{440} + \frac{x - 770}{660}}$

$\Rightarrow \frac{2x - 770}{500} = \frac{x}{440} + \frac{x - 770}{660}$

$\Rightarrow x = 1760$

$\therefore$ Total distance covered
= 2x - 770 = 2 × 770 - 770 = 2750 km

TIME AND WORK

NATIONALISED BANKS & IBPS PO/MT/SO

1. 12 men can complete a piece of work in 36 days. 18 women can complete the same piece of work in 60 days. 8 men and 20 women work together for 20 days. If only women were to complete the remaining piece of work in 4 days, how many women would be required ?
(1) 70 (2) 28
(3) 66 (4) 40
(5) None of these

(Union Bank of India PO Exam. 27.11.2005)

2. 9 children can complete a piece of work in 360 days. 18 men can complete the same piece of work in 72 days and 12 women can complete the piece of work in 162 days. In how many days can 4 men, 12 women and 10 children together complete the piece of work ?
(1) 124 (2) 81
(3) 68 (4) 96
(5) None of these

(Corporation Bank PO Exam. 29.07.2006)

3. Fifty six men can complete a piece of work in 24 days. In how many days can 42 men complete the same piece of work ?
(1) 18 (2) 32
(3) 98 (4) 48
(5) None of these

(Bank Of Maharashtra PO Exam. 25.05.2008)

4. 42 women can do a piece of work in 18 days. How many women would be required do the same work in 21 days ?
(1) 36 (2) 24
(3) 30 (4) 44
(5) None of these

(Andhra Bank PO Exam. 14.09.2008)

5. 30 men can do a piece of work in 16 days. How many men would be required to do the same work in 20 days ?
(1) 12 (2) 36
(3) 48 (4) 24
(5) None of these

(Oriental Bank of Commerce PO Exam. 21.12.2008)

6. 8 men can complete a piece of work in 4 days. 12 women can complete the same piece of work in 4 days whereas 8 children can complete the same piece of work in 8 days. 2 men, 8 children and 3 women work together for 2 days. If only women were to finish the remaining work in 2 days, how many total women would be required ?
(1) 12 (2) 18
(3) 24 (4) 20
(5) None of these

(PNB Agriculture Officer Exam. 04.01.2009)

7. 18 children can do a piece of work in 12 days. How many children would be required to do the same work in 8 days ?
(1) 12 (2) 18
(3) 24 (4) 27
(5) None of these

(Canara Bank PO Exam. 15.03.2009)

8. 8 men and 4 women together can complete a piece of work in 6 days. Work done by a man in one day is double the work done by a woman in one day. If 8 men and 4 women started working and after 2 days, 4 men left and 4 new women joined, in how many more days will the work be completed ?
(1) 5 days (2) 8 days
(3) 6 days (4) 4 days
(5) 9 days

(United Bank of India PO Exam. 21.06.2009)

9. B and C together can complete a work in 8 days, A and B together can complete the same work in 12 days and A and C together can complete the same work in 16 days. In how many days can A, B and C together complete the same work ?

(1) $3\frac{9}{13}$ (2) $7\frac{5}{13}$

(3) $7\frac{5}{12}$ (4) $3\frac{5}{12}$

(5) None of these

(Andhra Bank PO Exam. 05.07.2009)

10. 10 men can complete a piece of work in 8 days. 20 women can complete the same piece of work in 6 days. In how many days 16 men 18 women together can complete the same piece of work ?

(1) $2\frac{5}{7}$ (2) $3\frac{6}{7}$

(3) $3\frac{3}{7}$ (4) $2\frac{3}{7}$

(5) None of these

(PNB Specialist Officer Exam. 16.08.2O09)

11. Ayesha can complete a piece of work in 16 days. Amita can complete the same piece of work in 8 days. If both of them work together in how many days can complete the same piece of work ?

(1) 6 days (2) $4\frac{2}{5}$ days

(3) $5\frac{1}{3}$ days (4) 12 days

(5) None of these

(Indian Bank Rural Marketing Officer Exam. 03.01.2010)

12. 6 men can complete a piece work in 12 days. 8 women can complete the same piece of work in 18 days whereas 18 children can complete the piece of work in 10 days. 4 men, 12 women and 20 children work together for 2 days. If only men were to complete the remaining work in 1 day how many men would be required totally ?

(1) 36
(2) 24
(3) 18
(4) Cannot be determined
(5) None of these

(Bank Of India Banking Officer Exam. 24.01.2010)

13. 3 men can complete a piece of work in 6 days. 5 women can complete the same work in 18 days. In how many days will 4 men and 10 women together complete the same work ?

(1) 3 days (2) 5 days
(3) 2 days (4) 4 days
(5) None of these

(Bank Of Baroda PO Exam. 30.05.2010)

14. 8 men can complete a piece of work in 20 days. 8 women can complete the same work in 32 days. In how many days will 5 men and 8 women together complete the same work ?

(1) 16 days (2) 12 days
(3) 14 days (4) 10 days
(5) None of these

(Central Bank Of India PO Exam. 25.07.2010)

15. 2 men can complete a piece of work in 6 days. 2 women can complete the same piece of work in 9 days, whereas 3 children can complete the same piece of work in 8 days. 3 women and 4 children worked together for 1 day. If only men were to finish the remaining work in 1 day, how many total men would be required ?

(1) 4
(2) 8
(3) 6
(4) Cannot be determined
(5)None of these

(United Bank Of India PO Exam. 14.11.2010)

16. Vikas gets Rs 350 for every day that he works. If he earns Rs 9,800 in a month of 31 days, for how many days did he work ?

(1) 25 days (2) 30 days
(3) 24 days (4) 28 days
(5) None of these

(PNB Management Trainee Exam. 28.11.2010)

17. A water tank has three taps A, B and C. Tap A, when opened, can fill the water tank alone in 4 hours. Tap B, when opened, can fill the water tank alone in 6 hours and tap C, when opened, can empty the water tank alone in 3 hours. If taps A, B and C are opened simultaneously how long will it take to fill the tank completely ?

(1) 10 hours (2) 8 hours
(3) 18 hours (4) 12 hours
(5) None of these

(Indian Bank PO Exam. 02.01.2011)

18. 4 girls can do a piece of work in 8 days, 3 boys can do the same piece of work in 9 days, 7 men do the same piece of work in 2 days and 5 women can do the same piece of work in 4 days. Who is least efficient ?

(1) Boys (2) Girls
(3) Women (4) Men
(5) Boys and Men both

(Union Bank Of India PO Exam. 09.01.2001)

19. Two men alone or three women alone can complete a piece of work in 4 days. In how many days can 1 woman and one man together complete the same piece of work ?

(1) 6 days

(2) $\frac{24}{5}$ days

(3) $\frac{12}{1.75}$ days

(4) Cannot be determined
(5) None of these

(Corporation Bank PO Exam. 16.01.2011)

20. Four examiners can examine a certain number of answer papers in 10 days by working for 5 hours a day. For how many hours in a day would 2 examiners have to work in order to examine twice the number of answerpapers in 20 days ?

(1) 8 hours (2) $7\frac{1}{2}$ hours

(3) 10 hours (4) $8\frac{1}{2}$ hours

(5) None of these

(Punjab & Sind Bank PO Exam. 23.01.2011)

21. 6 women and 6 men together can complete a piece of work in 6 days. In how many days can 15 men alone complete the piece of work if 9 women alone can complete the work in 10 days ?

(1) 6
(2) 5
(3) 7.2
(4) Cannot be determined
(5) None of these

(Punjab & Sind Bank PO Exam. 23.01.2011)

22. A and B together can complete a task in 20 days. B and C together can complete the same task in 30 days. A and C together can complete the same task in 40 days. What is the respective ratio of the number of days taken by A when completing the same task alone to the number of days taken by C when completing the same task alone ?

(1) 2 : 5 (2) 2 : 7
(3) 3 : 7 (4) 1 : 5
(5) 3 : 5

(IBPS Bank PO/MT CWE 17.06.2012)

SBI PO EXAMS

1. Three men, four women and six children can complete a work in 7 days. A woman does double the work a man does and a child does half the work a man does. How many women alone can complete this work in 7 days ?

(1) 8
(2) 7
(3) 12
(4) Cannot be determined
(5) None of these

(SBI Banks PO Exam. 18.05.2003)

2. 12 men take 36 days to do a work while 12 women complete $\frac{3}{4}$th of the same work in 36 days. In how many days 10 men and 8 women together will complete the same work ?

(1) 6 (2) 27
(3)12 (4) Data inadequate
(5) None of these

(SBI PO Exam. 26.11.2006)

3. 'A' can complete a piece of work in 12 days. 'A' and 'B' together can complete the same piece of work in 8 days. In how many days can 'B' alone complete the same piece of work ?

(1) 15 days (2) 18 days
(3) 24 days (4) 28 days
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.04.2008)

4. 12 men can do a piece of work in 10 days. How many men would be required to do the same work in 8 days ?

(1) 14 (2) 18
(3) 16 (4) 12
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

5. A and B together complete a piece of work in T days. If A alone completes the work in T +3 days and B alone completes the piece of work in T +12 days, what is T ?

(1) 3 days
(2) 12 days
(3) 9 days
(4) Cannot be determined
(5) None of these

(SBI PO Preliminary (Tire-I) Exam. 27.07.2008)

6. 4 men can complete a piece of work in 2 days. 4 women can complete the same piece of work in 4 days whereas 5 children can complete the same piece of work in 4 days. If 2 men, 4 women and 10 children work together, in how many days can the work be completed ?

(1) 1 day (2) 3 days
(3) 2 days (4) 4 days
(5) None of these

(SBI & Rural Business PO Exam. 18.04.2010)

7. Amit and Sujit together can complete an assignment of data entry in 5 days. Sujit's speed is 80% of Amit's speed and the total

key depressions in the assignment are 5,76,000. What is Amit's speed in key depressions per hour if they work for 8 hours a day ?

(1) 4800 (2) 6400
(3) 8000 (4) 7200
(5) None of these

(SBI Associate Banks PO Exam. 07.08.2011)

RBI GRADE-B OFFICER EXAMS

1. 9 Men working for 7 hours a day can complete a piece of work in 15 days. In how many days can 6 men working for 9 hours a day, complete the same piece of work ?

(1) $17\frac{1}{2}$ days (2) $16\frac{3}{4}$ days

(3) 16 days (4) $15\frac{3}{4}$ days

(5) None of these

(RBI Grade-B Officer Exam. 2008)

2. If 12 boys or 15 girls can do a work in 48 days, in what time will 24 boys and 6 girls do twice the work ?

(1) 42 days (2) 40 days
(3) 45 days (4) 30 days
(5) None of these

(RBI Officer Grade 'B' Online Exam. 25.08.2013

INSURANCE EXAMS

1. If it takes A four days to dig a certain ditch, whereas B can dig it in 8 days and A, B, C together can dig it in $2\frac{2}{7}$ days, how long C alone would take to dig it ?

(1) 8 (2) 4
(3) 6 (4) 12
(5) 16

(United India Insurance Co. AAO Exam. 21.04.2002)

2. P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If P and Q work together, working 8 hours a day, in how many days can they complete the work ?

(1) $6\frac{6}{11}$ (2) $6\frac{5}{11}$

(3) $5\frac{6}{11}$ (4) $5\frac{5}{11}$

(5) None of these

(LIC Assistant Administrative Officer (AAO) Exams. 24.04.2005)

3. A man's basic pay for a 40 hours week is Rs 200. Overtime is paid at 25% above the basic rate. In a certain week, he worked overtime and his total was Rs 300. He therefore, worked for a total of (in hours):

(1) 52 (2) 56
(3) 58 (4) 62

(United India Insurance Co. AAO Exam. 11.03.2007)

4. If 9 men working $7\frac{1}{2}$ hours a day can finish a piece of work in 20 days will be taken how many days will be taken by 12 men, working 6 hours a day to finish the work ? It is being given that 2 men of latter type work as much as 3 men of the former type ?

(1) $9\frac{1}{2}$ (2) 11

(3) $12\frac{1}{2}$ (4) 13

(United India Insurance (AAO) Exam. 11.03.2007)

5. Seven men, five women and eight children were given an assignment of distributing 2000 books to students in a school over a period of three days. All of then distributed books on the first day. On the second day two women and three children remained absent and on the third day three men and five children remained absent. If the ratio of the number of books distributed in a day by a man, a woman and a child was 5 : 4 : 2 respective, a total of **approximately** how many books were distributed on the second day ?

(1) 1000 (2) 800
(3) 650 (4) 900
(5) Cannot be determined

(LIC Assistant Administrator Officer Exam 2008)

6. 15 men take 21 days of 8 hours each to do a piece of work. How many days of 6 hours each would 21 women take, if 3 wonen do as much work as 2 men ?

(1) 18 (2) 20
(3) 25 (4) 30

(LIC Assistant Administrative Officer (AAO) Exam. 07.06.20O9)

7. A is thrice as good a workman as B and so takes 60 days less than B for doing a job.

The time in which they can do the job together is

(1) 30 days (2) 60 days

(3) $22\frac{1}{2}$ days (4) 45 days

(New India Assurance AO Exam. 25.10.2009)

8. A certain number of persons can complete a work in 100 days. If there be 10 persons less, it would have taken 10 days more for the work to be completed. The number of persons in the beginning was

(1) 90 (2) 105

(3) 110 (4) 120

(New India Insurance AA0 Exam. 22.05.2011)

9. A is twice as good a workman as B and together they finish a piece of work in 14 days. The number of days taken by A alone to finish the work is :

(1) 11 (2) 21

(3) 28 (4) 42

(General Insurance Corporation AAO Exam. 11.12.2011)

10. M and N can do a work in 10 days and 15 days respectively. If M starts on the work and both work alternately day after day, in how many days will the work be completed ?

(1) 10 (2) 12

(3) 8 (4) 9

(5) None of these

(LIC Assistant Administrative Officer (AAO) Exam. 12.05.2013)

11. 30 men can do a piece of work in 16 days. How many men would be required to do the same work in 20 days ?

(1) 12 (2) 36

(3) 48 (4) 24

(5) None of these

(United India Insurance AO Exam. 26.05.2013

12. A work can be completed by P and Q in 12 days, Q and R in 15 days, R and P in 20 days. In how many days P alone can finish the work ?

(1) 10 (2) 20

(3) 30 (4) 60

(NICL (GIC) AO Exam. 08.09.2013 (Paper-I)

13. '*x*' number of men can finish a piece of work in 30 days. If there were 6 men more, the work could be finished in 10 days less. The original number of men is

(1) 6 (2) 10

(3) 12 (4) 15

(NICL (GIC) AO Exam. 08.09.2013 (Paper-I)

14. A man, a woman and a boy together complete a piece of work in 3 days. If a man alone can do it in 6 days and a boy alone in 18 days, how long will a woman take to complete the work ?

(1) 9 days (2) 21 days

(3) 24 days (4) 27 days

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

15. 8 men can finish a piece of work in 40 days. If 2 more men join with them, then the work will be completed in

(1) 30 days (2) 32 days

(3) 36 days (4) 25 days

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

16. 10 women can complete a work in 7 days and 10 children take 14 days to complete the work. How many days will 5 women and 10 children take to complete the work ?

(1) 3

(2) 5

(3) 7

(4) Cannot be determined

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

17. 9 children can complete a piece of work in 360 days, 18 men can complete the same piece of work in 72 days and 12 women can complete it in 162 days. In how many days can 4 men, 12 women and 10 children together complete the piece of work ?

(1) 68 days (2) 81 days

(3) 96 days (4) 124 days

(NICL (GIC) Administrative Officer Exam. 15.12.2013)

SHORT ANSWERS
NATIONALISED BANKS & IBPS PO/MT/SO

1.	(1)	**2.**	(2)
3.	(2)	**4.**	(1)
5.	(4)	**6.**	(1)
7.	(4)	**8.**	(1)
9.	(2)	**10.**	(5)
11.	(3)	**12.**	(1)
13.	(1)	**14.**	(1)
15.	(2)	**16.**	(4)
17.	(4)	**18.**	(2)
19.	(2)	**20.**	(3)
21.	(5)	**22.**	(4)

SBI PO EXAMS

1.	(2)	**2.**	(2)
3.	(3)	**4.**	(5)
5.	(5)	**6.**	(1)
7.	(3)		

RBI GRADE-B OFFICER EXAMS

1.	(5)	**2.**	(4)
3.	(2)	**4.**	(3)
5.	(3)	**6.**	(4)
7.	(3)	**8.**	(3)
9.	(2)	**10.**	(2)
11.	(4)	**12.**	(3)
13.	(3)	**14.**	(1)
15.	(2)	**16.**	(3)
17.	(2)		

EXPLANATIONS
NATIONALISED BANKS & IBPS PO/MT/SO

1. (1) $\because$ 12 men can complete the work in 36 days.
$\therefore$ 12 × 36 men can complete the work in 1 day.
Again,
$\because$ 18 women can complete the work in 60 days.
$\therefore$ 18 × 60 women can complete the work in 1 day.
12 × 36 men = 18 × 60 women
$\Rightarrow$ 2 men = 5 women
Now, 8 men + 20 women
= (4 × 5 + 20) women = 40 women
$\because$ 18 women complete the work in 60 days.
$\therefore$ 40 womens' 20 days' work

$= \frac{40\times 20}{18\times 60} = \frac{20}{27}$

$\therefore$ Remaining work = $1 - \frac{20}{27} = \frac{7}{27}$

$\therefore$ 18 × 60 women do 1 work in 1 day.

$\therefore$ 1 woman does = $\frac{1}{80\times 60}$ Work in 1 day

$\therefore$ 1 woman does in 4 days

$= \frac{4}{180\times 60} = \frac{1}{18\times 15}$ Work

$\therefore \frac{1}{18\times 15}$ work is done in 4 days by 1 woman

$\therefore \frac{7}{27}$ work is done in 4 days by $= \frac{18\times 15\times 7}{27}$

= 70 days

2. (2) Clearly,
9 × 360 children = 18 × 72 men
= 12 × 162 women
$\Rightarrow$ 45 children =18 men = 27 women
$\Rightarrow$ 5 children = 2 men = 3 women
Now, 4 men +12 women +10 children
= 4 men + 8 men + 4 men = 16 men
$\therefore$ 18 men can complete the work in 72 days.
$\therefore$ 16 men can complete the work

in= $\frac{18\times 72}{16}$ = 81 days

3. (2) $\therefore$ 56 men can complete 1 work in 24 days.
$\therefore$ 1 man complete the work in 24 × 56 days.
$\therefore$ 42 men will do the same work in

$= \frac{24\times 56}{42}$ = 32 days

<u>QuickerMethod</u>

$\therefore M_1 D_1 = M_2 D_2$
$\therefore 56 \times 24 = 42 \times D_2$

$\Rightarrow D_2 = \frac{56\times 24}{42}$ = 32 days

4. (1) Days Women

$\begin{matrix}18\\21\end{matrix}\uparrow \qquad \begin{matrix}42\\x\end{matrix}\downarrow\downarrow$

$\therefore$ 21 : 18 : ; 42 : x

$\Rightarrow x = \frac{42\times 18}{21} = 36$

5. (4) Days Men

$\begin{matrix}16\\20\end{matrix}\uparrow \qquad \begin{matrix}30\\x\end{matrix}\downarrow$

$\therefore$ 20 : 16 = 30 : x

$\Rightarrow 20 \times x = 16 \times 30$

$\Rightarrow x = \frac{30\times16}{20}$ = 24 men

6. (1) As per given information,
8 × 4 men = 12 × 4 women
= 8 × 8 children
$\Rightarrow$ 32 men = 48 women = 64 children
$\Rightarrow$ 2 men = 3 women = 4 children
2 men + 8 children + 3 women
= (3 + 6 + 3) women =12 women

12 women's 2 day's work = $\frac{1}{2}$

Remaining work = $\frac{1}{2}$

$\therefore$ Required number of women = 12

7. (4) Days Children

$12 \uparrow \quad 18$
$8 \quad\quad x \downarrow$

$\therefore 8 : 12 = 18 : x$
$\Rightarrow 8x = 12 \times 18$

$\therefore x = \frac{12\times18}{8} = 27$

Note : you can use the formula :
$M_1 D_1 = M_2 D_2$ and solved it.

8. (1) According to the question,
1 man = 2 women
$\therefore$ 8 men + 4 women
= (16 + 4) women = 20 women
4 men + 8 women = 16 women
20 women's 2 days' work

= $\frac{2}{6} = \frac{1}{3}$ Part

Remaining work = $1 - \frac{1}{3} = \frac{2}{3}$

$\because$ 20 women complete 1 work in 6 days.

$\because$ 16 women will do $\frac{2}{3}$ work in

= $\frac{20\times6}{16} \times \frac{2}{3} = 5$ days

9. (2) (B+C)'s 1 day's work = $\frac{1}{8}$ ...(i)

(A+B)'s 1 day's work = $\frac{1}{12}$ ...(ii)

(A+C)'s 1 day's work = $\frac{1}{16}$ ...(iii)

On adding all these three equations,
2 (A + B + C)'s 1 day's work

$\Rightarrow \frac{1}{8}+\frac{1}{12}+\frac{1}{16} = \frac{6+4+3}{48} = \frac{13}{48}$

$\Rightarrow$ (A + B + C)'s 1 day's work = $\frac{13}{96}$

$\therefore$ A, B and C together can complete the work in $\frac{96}{13} = 7\frac{5}{13}$ days

10. (5) $\because$ 10 men complete the work in 8 days.
$\therefore$ 80 men will complete the work in 1 day.
$\because$ 20 women complete the work in 6 days.
$\therefore$ 120 women complete the work I in 1 day.
$\therefore$ 80 men = 120 women
$\therefore$ 2 men = 3 women
$\therefore$ 16 men + 18 women = 16 men

+ 18 × $\frac{2}{3}$ men = 28 men

$\because$ 10 men can do the work in 8 days
$\therefore$ 28 men can do the work in

$\frac{10\times8}{28} = \frac{20}{7} = 2\frac{6}{7}$ days.

11. (3) Ayesha's and Amita's 1 day's work

= $\frac{1}{16}+\frac{1}{8} = \frac{1+2}{16} = \frac{3}{16}$

$\therefore$ Ayesha and Amita together can complete the work in $\frac{16}{3} = 5\frac{1}{3}$ days

12. (1) 72 men = 144 women = 180 children
2 men = 4 women = 5 children 4 men +12 women + 20 children
= 4 men+ 6 men+ 8 men = 18 men
When 6 men do 1 work in 12 days
18 men will do the same in 4 days.
$\therefore$ In 2 days half work will done.

$\therefore$ Remaining work = $\frac{1}{2}$

$\therefore \frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$

$\Rightarrow \frac{1}{2} \times 6 \times 12 = I \times M_2$

$\Rightarrow M_2$ = 36 = Number of men

13. (1) <u>Tricky approach</u>
18 men = 5 × 18 = 90 women
$\therefore$ 1 man = 5 women
$\therefore$ 4 men +10 women = 30 women
$\therefore M_1D_1 = M_2D_2$
$\Rightarrow 5 \times 18 = 30\ D_2$

$\Rightarrow D_2 = \frac{5\times18}{30} = 3$ days

14. (1) 8 × 20 men = 8 × 32 women
$\Rightarrow$ 5 men = 8 women
5 men + 8 women =16 women
$M_1D_1 = M_2D_2$
$\Rightarrow 8 \times 32 = 16 \times D_2$

$\Rightarrow D_2 = \frac{8\times32}{16} = 16$ days

15. (2) 2 × 6 men = 18 women = 24 children
2 men = 3 women = 4 children
3 women + 4 children = 4 men

part of work done in 1 day by 4 men = $\frac{1}{3}$

$\therefore$ Remaining $\frac{2}{3}$ work will be finished by 8 men in 1 day. Hence, 8 men would be required.

16. (4) Number of days = $\frac{9800}{350} = 28$

17. (4) Part of the tank filled in 1 hour when all the taps are opened

$= \frac{1}{4} + \frac{1}{6} - \frac{1}{3}$

$= \frac{3+2-4}{12} = \frac{1}{12}$

Hence, the tank will be filled in 12 hours.

18. (2) 4 × 8 girls = 3 × 9 boys
= 7 × 2 men = 5 × 4 women
$\Rightarrow$ 32 girls = 27 boys = 14 men
= 20 women

19. (2) 2 men = 3 women
1 man + 1 woman

$= \left(\frac{3}{2}+1\right)$ women = $\frac{5}{2}$ women

$\therefore M_1D_1 = M_2D_2$

$\Rightarrow 3 \times 4 = \frac{5}{2} \times D_2$

$\Rightarrow D_2 = \frac{3\times4\times2}{5} = \frac{24}{5}$ days

20. (3)

Examiners work		Days		Hours/day	
4	$\uparrow$	1	$\downarrow$	10	$\uparrow$	5	$\downarrow$
2		2		20		x	

$M_1D_1H_1W_2 = M_2D_2H_2W_1$
$\Rightarrow 2 \times 20 \times x = 4 \times 10 \times 5 \times 2$

$\Rightarrow x = \frac{4\times10\times5\times2}{2\times20} = 10$ hours

21. (5) $\because$ 9 women can complete the work in 10 days.
$\therefore$ 6 women can complete the work in $\frac{10\times9}{6}$ = 15 days.

Part of work done by 6 women = $\frac{6}{15} = \frac{2}{5}$

$\therefore$ Part of work done by 6 men = $1 - \frac{2}{5} = \frac{3}{5}$

$\because$ $\frac{3}{5}$ part of work done by 6 men in 1 days.

$\therefore$ 1 part of work done by 6 men in $\frac{6}{3} \times 5$ = 10 days.

$\therefore$ 15 men can complete the work in $\frac{10\times6}{15}$ = 4 days

22. (4) (A + B)'s 1 day's work = $\frac{1}{20}$

(B + C)'s 1 day's work = $\frac{1}{30}$

(C + A)'s 1 day's work = $\frac{1}{40}$

On adding,
2(A + B + C)'s 1 day's work

$= \frac{1}{20} + \frac{1}{30} + \frac{1}{40}$

$= \frac{6+4+3}{120} = \frac{13}{120}$

$\therefore$ (A + B + C)'s 1 day's work

$= \frac{13}{240}$

$\therefore$ A's 1 day's work

$= \frac{13}{240} - \frac{1}{30} = \frac{13-8}{240}$

$= \frac{5}{240} = \frac{1}{48}$

C's 1 day's work

$= \frac{13}{240} - \frac{1}{20} = \frac{13-12}{240} = \frac{1}{240}$

$\therefore$ Required ratio = 48 : 240 = 1 : 5

SBI PO EXAMS

1. (2) 2 men = 1 woman

$\therefore$ 1 man = $\frac{1}{2}$ woman

$\therefore$ 3 men = $\frac{3}{2}$ women

Again, 2 children = 1 man

= $\frac{1}{2}$ woman

$\therefore$ 1 child = $\frac{1}{4}$ woman

$\therefore$ 6 children = $\frac{6}{4} = \frac{3}{2}$ women

Now, three men, four women and six children

= $\frac{3}{2} + 4 + \frac{3}{2} = 7$ women

Hence, 7 women complete the work in 7 days.

2. (2) In 36 days 12 men can do 1 complete work.

In 36 days 12 women can do $\frac{3}{4}$th of the work

Since time and the no. of persons is the same is both cases,

1 woman's daily work = $\frac{3}{4}$th of 1 man's daily work

8 women's daily work = $\frac{3}{4} \times 8 = 6$ men's daily work

(10 men + 8 womens daily work) = (10 men + 6 men)

= 16 men's daily work.

12 men can do the work is 36 days

$\therefore$ 16 men can do the work is

$36 \times \frac{12}{16} = 27$ days.

3. (3) A's 1 day's work = $\frac{1}{12}$

(A + B)'s 1 day's work = $\frac{1}{8}$

$\therefore$ B's 1 day's work

= $\frac{1}{8} - \frac{1}{12} = \frac{3-2}{24} = \frac{1}{24}$

$\therefore$ B alone can do the work in 24 days.

4. (5) Days Men

$10 \uparrow$ $12 \downarrow$

8 x

Where x = number of men

$\therefore 8 : 10 :: 12 : x$

$\Rightarrow 8 \times x = 10 \times 12$

$= x = \frac{10 \times 12}{8} = 15$ men

5. (5) (A + B)'s 1 day's work = $\frac{1}{T}$

A's 1 day's work = $\frac{1}{T+3}$

B's 1 day's work = $\frac{1}{T+12}$

$\therefore \frac{1}{(T+3)} + \frac{1}{(T+12)} = \frac{1}{T}$

$\Rightarrow \frac{T+12+T+3}{(T+3)(T+12)} = \frac{1}{T}$

$\Rightarrow \frac{2T+15}{T^2+15T+36} = \frac{1}{T}$

$\Rightarrow 2T^2 + 15T = T^2 + 15T + 36$

$\Rightarrow T^2 = 36$

$\Rightarrow T = 6$ days

6. (1) <u>Tricky Approach</u>

4×2 men = 4×4 women = 20 children

$\Rightarrow$ 2 men = 4 women = 5 children

$\therefore$ 2 men + 4 women +10 children

= 20 children

$\therefore M_1D_1 = M_2D_2$

$\Rightarrow 5 \times 4 = 20 \times D_2$

$\Rightarrow D_2 = 1$ day

7. (3) Let Amit's speed be x key depressions per day.

$\therefore$ Sujit's speed = $\frac{4x}{5}$ key depressions per day

$\therefore 5x + 5 \times \frac{4x}{5} = 576000$

$\Rightarrow 9x = 576000$

$\Rightarrow x = \frac{576000}{9} = 64000$

$\therefore$ Amit's speed per hour

= $\frac{64000}{8} = 8000$

RBI GRADE-B OFFICER EXAMS

1. (1) Men Hours/day Days

$$\begin{matrix} 9 \\ 6 \end{matrix}\uparrow \qquad \begin{matrix} 7 \\ 9 \end{matrix}\uparrow \qquad \begin{matrix} 15 \\ x \end{matrix}\downarrow$$

Where x = number of days

$$\therefore \left.\begin{matrix} 6:9 \\ 9:7 \end{matrix}\right\} ::15:x$$

$\Rightarrow 6 \times 9 \times x = 9 \times 7 \times 15$

$\Rightarrow \frac{9\times7\times15}{6\times9} = \frac{35}{2} = 17\frac{1}{2}$ days

2. (2) $\because$ 12 boys = 15 girls

$\therefore$ 24 boys = 30 girls

$\therefore$ 24 boys + 6 girls

= 36 girls

$\therefore \frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$

$\Rightarrow \frac{15\times48}{1} = \frac{36\times D_2}{2}$

$\Rightarrow D_2 = \frac{15\times48}{18} = 40$ days

INSURANCE EXAMS

1. (5) According to the question,

A's one day work = $\frac{1}{4}$

B's one day work = $\frac{1}{8}$

(A + B)'s one day work

$= \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$

Similarly,

(A + B + C)'s one day work = $\frac{7}{16}$

C's one day work

$= \frac{7}{16} - \frac{3}{8} = \frac{1}{16}$

Therefore, C can complete the whole work in 16 days.

2. (4) $\because$ P can complete a work in 12 days working 8 hours a day

$\therefore$ P can do a work in 12 days working

1 hour a day = $\frac{1}{8}$

$\therefore$ P's 1 day work working 1 hour a day

$= \frac{1}{8\times12}$

Similarly, Q's 1 day work working 1 hour a day = $\frac{1}{8\times10}$

$\therefore$ (A + B)'s 1 day work working 1 hour a day

$= \frac{1}{8\times12} + \frac{1}{8\times10}$

$= \frac{5+6}{60\times8} = \frac{11}{60\times8}$

$\therefore$ (A + B)'s 1 day work working 8 hours a day = $\frac{11\times8}{60\times8} = \frac{11}{60}$

$\therefore$ Required answer

$= \frac{60}{11} = 5\frac{5}{11}$ days

3. (2) Let the man worked overtime for x hours.

Basic pay of 40 hours = Rs. 200.

$\therefore$ Basic pay of 1 hour = $\frac{200}{40}$ = Rs. 5

According to the question,

$200 + x \times \frac{5\times125}{100} = 300$

$\Rightarrow \frac{25}{4}x = 300 - 200$

$\Rightarrow x = \frac{100\times4}{25} = 16$

$\therefore$ Total hours of work = 40 + 16 = 56

4. (3) More men, less days (Indirect) Less hrs/day, more days (Indirect)

Less efficiency, more days (Indirect)

$$\left.\begin{matrix} \text{Men} & 12:9 \\ \text{Hrs / Day} & 6:\frac{15}{2} \\ \text{Efficiency} & \frac{1}{2}:\frac{1}{3} \end{matrix}\right\}$$

$\therefore 12 \times 6 \times \frac{1}{2} \times x = 9 \times \frac{15}{2} \times \frac{1}{3} \times 20$

$\Rightarrow x = \frac{9\times15\times2\times20}{2\times3\times12\times6}$

$= \frac{25}{2} = 12\frac{1}{2}$

$\therefore$ Required number of days = $12\frac{1}{2}$

5. (3) Let the number of books distributed by a man, a woman and a child per day be $5x$,

$4x$ and $2x$ respectively
Books distributed on the first day
$= 7 \times 5x + 5 \times 4x + 8 \times 2x$
$= 35x + 20x + 16x = 71x$
Book distributed on the second day
$= 7 \times 5x + 3 \times 4x + 5 \times 2x$
$= 35x + 12x + 10x = 57x$
Books distributed on the third day
$= 4 \times 5x + 5 \times 4x + 3 \times 2x = 46x$
$\therefore 71x + 57x + 46x = 2000$
$\Rightarrow 174x = 2000$

$\therefore 57x = \frac{2000}{174} \times 57 = 655.172 \approx 650$

6. (4) 3 women = 2 men
$\Rightarrow$ 21 women = 14 men

men		working hours/day		days	
15	↑	8	↑	21	↓
14		6		x	

$\therefore \left.\begin{matrix} 14:15 \\ 6:8 \end{matrix}\right\} :: 21 : x$

$\therefore 14 \times 6 \times x = 15 \times 8 \times 21$

$\Rightarrow x = \frac{15 \times 8 \times 21}{14 \times 6} = 30$ days

7. (3) <u>Tricky approach</u>

A is three times as good a workman as B.
$\therefore$ If A completes a work in 30 days, B will complete it in 90 days.
$\therefore$ (A + B)' 1 day's work

$= \frac{1}{30} + \frac{1}{90} = \frac{3+1}{90} = \frac{2}{45}$

$\therefore$ Hence, both will complete the work in $\frac{45}{2}$

$= 22\frac{1}{2}$ day.

8. (3)

Men		Days	
x	↑	100	↓
$x-10$		110	

$\Rightarrow \frac{x-10}{x} = \frac{100}{110}$
$\Rightarrow 11x - 110 = 10x$
$\Rightarrow x = 110$ men

9. (2) If time taken by B be $2x$ days, then time taken by A = x days.

$\therefore \frac{1}{x} + \frac{1}{2x} = \frac{1}{14}$

$\Rightarrow \frac{2+1}{2x} = \frac{1}{14} \Rightarrow \frac{3}{x} = \frac{1}{7}$

$\Rightarrow x = 3 \times 7 = 21$ days

10. (2) Work done in 1st two days = $\frac{1}{10} + \frac{1}{15}$

$= \frac{3+2}{30} = \frac{1}{6}$

$\therefore$ Number of days = 12

11. (4)

Days		Men	
16	↑	30	↓
20		x	

$\therefore 20 : 16 = 30 : x$
$\Rightarrow 20 \times x = 16 \times 30$

$\Rightarrow x = \frac{30 \times 16}{20} = 24$

12. (3) (P + Q)'s 1 day's work = $\frac{1}{12}$

(Q + R)'s 1 day's work = $\frac{1}{15}$

(R + P)'s 1 day's work = $\frac{1}{20}$

On adding,
2 (P + Q + R)'s 1 day's work

$= \frac{1}{12} + \frac{1}{15} + \frac{1}{20} = \frac{5+4+3}{60} = \frac{1}{5}$

(P + Q + R)'s 1 day's work

$= \frac{1}{10}$

$\therefore$ P's 1 day's work = $\frac{1}{10} - \frac{1}{15}$

$= \frac{3-2}{30} = \frac{1}{30}$

Hence, P alone will complete the work in 30 days.

13. (3) $M_1D_1 = M_2D_2$
$\Rightarrow x \times 30 = (x + 6) \times 20$
$\Rightarrow 3x = 2x + 12$
$x = 12$ men

14. (1) (1 man + 1 woman + 1 child)'s 1 day's work = $\frac{1}{3}$

$\therefore$ Woman's 1 day's work = $\frac{1}{3} - \frac{1}{6} - \frac{1}{18}$

$= \frac{6-3-1}{18} = \frac{1}{9}$

$\therefore$ The woman alone will complete the work in 9 days.

15. (2) $M_1D_1 = M_2D_2$

$\Rightarrow 8 \times 40 = 10 \times D_2$

$\Rightarrow D_2 = \frac{8\times40}{10} = 32$ days

16. (3) 10 women complete the work in 7 days.
10 children complete the same work in 14 day's

$\therefore$ 1 woman = 2 children

$\therefore$ 5 women +10 children = 20 children

$\therefore$ By $M_1D_1 = M_2D_2$

$10 \times 14 = 20 \times D_2$

$\Rightarrow D_2 = \frac{10\times14}{20} = 7$ days

17. (2) According to the question,

9×360 children = 18×72 men

$= 12 \times 162$ women

$\Rightarrow$ 5 children = 2 men = 3 women

$\therefore$ 4 men + 12 women +10 children

= 10 children + 20 children + 10 children

= 40 children

$\therefore M_1D_1 = M_2D_2$

$\Rightarrow 9 \times 360 = 40 \times D_2$

$\Rightarrow D_2 = \frac{9\times360}{40} = 81$ days

MODEL EXERCISES

1. A certain job was assigned to a group of men to did it in 20 days. But 12 men did not turn up for the job and the remaining men did the job in 32 days. The original number of men in the group was
(1) 32 (2) 34
(3) 36 (4) 40
(5) None of these

2. A mother and a daughter working together can complete a work in 4 days. But if the mother works alone she can complete the work in 6 days. Both of them worked for one day and then the mother had to leave. How long will the daughter take to complete the remaining work ?
(1) 7 days (2) 8 days
(3) 9 days (4) 10 days
(5) None of these

3. Ramesh is twice as good workman as Sunil and finished a piece of work in 3 h less than Sunil. In how many hours they together could finish that piece of work ?
(1) $2\frac{1}{3}$ (2) 2
(3) $1\frac{2}{3}$ (4) 4
(5) None of these

4. A can do a work in 18 days, B in 9 days and C in 6 days. A and B start working together and after 2 days C joins them. What is the total number of days taken to finish the work ?
(1) 4.33 (2) 4.0
(3) 4.66 (4) 5.33
(5) None of these

5. In a house, there are six 40 W lamps which are on for 5 h a day and three 80 W fans which are on for 10 h a days. If electricity costs 2 per kilowatt hours what is the monthly electricity bill ?
(1) Rs 216 (2) Rs 280
(3) Rs 315 (4) Rs 400
(5) None of these

6. A worker is paid Rs 56 for 35 h week. Up to 40 h, he is paid at the normal rate and on overtime, 1.5 times the normal. How many hours did he work to get Rs 88 ?
(1) 48 (2) 50
(3) 58 (4) 55
(5) None of these

7. $\frac{1}{3}$rd of the contents of a container evaporated on the 1st day. $\frac{3}{4}$th of the remaining evaporated on the second day. What part of the contents of the container is left at the end of the second day ?
(1) $\frac{1}{4}$ (2) $\frac{1}{2}$
(3) $\frac{1}{18}$ (4) $\frac{1}{6}$
(5) None of these

8. Two men and 7 children complete a certain piece of work in 4 days while 4 men and 4 children complete the same work in only 3 days. The number of days required by 1 man to complete the work is
(1) 60 days (2) 15 days
(3) 6 days (4) 51 days
(5) None of these

9. Mr. Ram is an tour and he has Rs 360 for his expenses. If he exceeds his tour by 4 days he must cut down daily expenses by Rs 3. The number of days of Mr. Ram's tour programme is
(1) 20 days (2) 24 days
(3) 40 days (4) 42 days
(5) None of these

10. A worker makes a basket in $\frac{2}{3}$ of an hours. If he works for $7\frac{1}{2}$ h, then how many baskets will he make ?
(1) $10\frac{3}{4}$ (2) $11\frac{1}{4}$
(3) $12\frac{1}{2}$ (4) 13
(5) None of these

11. If 5 men take an hour to dig a pitch, then how long should 12 men take to dig a ditch of the same type ?
(1) 25 min (2) 30min
(3) 28 min (4) 20 min
(5) None of these

12. A and B can finish a job in 10 days while B and C can do it in 18 days. A started the job worked for 5 days, then B worked for 10 days and the remaining job was finished

by C in 15 days. In how many days could C alone have finished the whole job ?
(1) 30 days (2) 15 days
(3) 45 days (4) 24 days
(5) None of these

13. A can do a piece of work in 10 days, while B alone can do it in 15 days. They work together for 5 days and the rest of the work is done by C in 2 days. If they get Rs 450 for the whole work, how should they divide the money ?
(1) Rs 225, Rs 150, Rs 75
(2) Rs 250, Rs 100, Rs 100
(3) Rs 200, Rs 150, Rs 100
(4) Rs 175, Rs 175, Rs 100
(5) None of these

14. A can build up a wall in 8 days while B can break it in 3 days. A has worked for 4 days and then B joined to work with A for another 2 days only. In how many days will A alone build up the remaining part of the wall ?

(1) $13\frac{1}{3}$ days (2) $6\frac{1}{3}$ days

(3) $7\frac{2}{3}$ days (4) 7 days

(5) None of these

15. 4 men and 10 women were put on a work. They completed 1/3 of the work in 4 days. After this 2 men and 2 women were increased. They completed 2/9 more of the work in 2 days. If the remaining work is to be completed in 3 days, then how many more women must be increased ?
(1) 32 (2) 8
(3) 50 (4) 55
(5) None of these

16. C is twice efficient as A. B takes thrice as many days as C. A takes 12 days to finish the work alone If they work in pairs (i.e., AB, BC, CA) starting with AB on the first day, BC on the second day and AC on the third day and so on, then how many days are required to finish the work ?

(1) $6\frac{1}{5}$ days (2) 4.5 days

(3) $5\frac{1}{9}$ days (4) 8 days

(5) None of these

17. Two men undertake to do a piece of work for Rs 600. One alone could do it in 6 days and the other in 8 days. With the assistance of a boy, they finish it in 3 days. The boy's share should be .
(1) Rs 75 (2) Rs 225
(3) Rs 300 (4) Rs 100
(5) None of these

18. 15 men can complete a work in 210 days. They started the work but at the end of 10 days 15 additional men, with double efficiency, were inducted. How many days in all did they take to finish the work ?

(1) $72\frac{1}{2}$ days (2) $84\frac{3}{4}$ days

(3) $76\frac{2}{3}$ days (4) 70 days

(5) None of these

19. Two men A and B working together complete a piece of work which it would have taken them 30 and 40 days respectively to complete if they worked separately. If they received a payment of Rs 2100. B's share is
(1) Rs 900 (2) Rs 800
(3) Rs 1200 (4) Rs 1300
(5) None of these

20. Five painters can paint a wall 100 m long in 10 days of 8 h each. How many days of 6 h each will it take for 8 men to paint a wall 30 m long ?

(1) $2\frac{1}{2}$ (2) 1

(3) 3 (4) $1\frac{3}{4}$

(5) None of these

21. There are two types of workers-category I and category II. A category I worker can finish a piece of work in 2 m per 3 h and a category II worker can finish in m h. If one worker of category I and two workers of category II are employed, the work can be finished in how many hours ?
(1) 7 m/2 (2) 9 m/2
(3) 2 m/7 (4) 2 m/9
(5) None of these

22. Two coal loading machines each working 12 h per day for 8 days handle 9000 tonnes of coal with an efficiency of 90%; while 3 other coal loading machines at an efficiency of 80% are set to handle 12000 tonnes of coal in 6 days. Find how many hours per day each should work ?
(1) 20 h/day (2) 18 h/day
(3) 16 h/day (4) 14 h/day
(5) None of these

23. If 36 men can dig a trench 200 m long, 3 m wide and 2 m deep in 6 days working 10 h a day, in how many days, working 8 h a day will 10 men dig a trench 100 m long, 4 m wide and 3 m deep ?

(1) 15 days (2) 27 days
(3) 20 days (4) 54 days
(5) None of these

24. Bhavika alone would take 8 h more to complete the job than when Bhavika and Rita worked together. If Rita worked alone, she would take $4\frac{1}{2}$ h more to complete the job than when Bhavika and Rita worked together. What time would they take if both Bhavika and Rita worked together ?

(1) 5 days (2) 4 days
(3) 3 days (4) 6 days
(5) None of these

25. Ten men can finish a piece of work in 10 days, whereas it takes 12 women undertake the work, how many days will they take to complete it ?

(1) $4\frac{1}{2}$ days (2) 4 days
(3) 5 days (4) 6 days
(5) None of these

SHORT ANSWERS

1. (1) **2.** (3)
3. (2) **4.** (2)
5. (1) **6.** (2)
7. (4) **8.** (4)
9. (1) **10.** (2)
11. (1) **12.** (3)
13. (1) **14.** (3)
15. (2) **16.** (3)
17. (1) **18.** (3)
19. (1) **20.** (1)
21. (3) **22.** (3)
23. (2) **24.** (4)
25. (3)

EXPLANATIONS

1. (1) Let originally there were x men in the group.
$\therefore (x - 12)$ men did the job in 32 days
$\Rightarrow 20x = 32(x - 12)$
$\Rightarrow x = 32$

2. (3) According to the question,

(M + D)'s one day's work = $\frac{1}{4}$

Remaining work = $\frac{3}{4}$

M's one day's work = $\frac{1}{6}$

D's one day's work = $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$

$\therefore \frac{1}{12}$ th of the work is done by the daughter in 1 day

$\frac{3}{4}$ th of the work will be done by daughter in $\frac{12\times3}{4}$ = 9 days

3. (2) According to the question,
Let Sunil finishes the job in x h.

Then, Ramesh will finish the job in $\frac{x}{2}$ h.

Given, $x - \frac{x}{2} = 3$

$\Rightarrow x = 6$
$\therefore$ Sunil finishes a job in 6 h and Ramesh does it in 3 h.
Work done by both of them in hour

$\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ h

$\therefore$ They together could finish a piece of work in 2 h

4. (2) Let (A + B)'s 2 days work

$= 2\left(\frac{1}{18}+\frac{1}{9}\right) = \frac{2}{6} = \frac{1}{3}$

Remaining work = 1 - $\frac{1}{3} = \frac{2}{3}$

(A + B + C)'s one day's work

$= \left(\frac{1}{18}+\frac{1}{9}=\frac{1}{6}\right) = \frac{1}{3}$

$\frac{1}{3}$ work is completed working all of them together in 1 day.

Hence, $\frac{2}{3}$ work will be completed in 2 days.

$\therefore$ Total number of days taken to finish the work is 4 days.

5. (1) By question,
Electricity consumption per day
= 6 × 40 × 5 + 3 × 80 × 10
= 1200 + 2400 = 3600 watt
= 3.6 kW
$\therefore$ Consumption in a month
= 3.6 × 30 =108 kW
$\therefore$ Monthly electricity bill
= 108 × 2 = Rs 216

6. (2) Let the worker worked for (40 + x) h.
Then,

$\frac{56}{35} \times 40 + \frac{x\times1.5\times56}{35} = 88$

$\Rightarrow 2.4x = 24$

$\Rightarrow x = 10$

$\therefore$ The worker worked for (40 + 10) = 50 h

7. (4) According to question, after one day contents left

$= \left(1-\frac{1}{3}\right) = \frac{2}{3}$

Second day contents evapored

$= \frac{2}{3}\times\frac{3}{4} = \frac{6}{12}$

Contents left after second day

$= \frac{2}{3} - \frac{6}{12} = \frac{8-6}{12} = \frac{2}{12} = \frac{1}{6}$

8. (2) According to question,

(2M + 7C)'s 1 day work = $\frac{1}{4}$

It means that 1 work will be finished by (8M + 28C)

Again, (4M + 4C)'s 1 day's work = $\frac{1}{3}$

or 1 work will be completed by 12M + 12C

$\therefore$ 8M + 28C = 12M + 12C

$\Rightarrow$ M = 4C

$\therefore$ 4M + 4C = 5M

Since, 5 M complete a work in 3 days. Then, 1 M will complete it in 15 days.

9. (1) Let Ram undertakes a tour of x days.

Then, expenses for each day = $\frac{360}{x}$

$\Rightarrow \frac{360}{x+4} = \frac{360}{x} - 3$

$\Rightarrow 360\left(\frac{1}{x} - \frac{1}{x+4}\right) = 3$

$\Rightarrow x^2 + 4x - 480 = 0$

$\Rightarrow x = -24$ or $x = 20$

Since, $x \neq -24$

Hence, $x = 20$ days

10. (2) By question,

In $\frac{2}{3}$ h he makes 1 basket.

$\therefore$ In $\frac{15}{2}$ h he will make $\frac{1}{2/3} \times \frac{15}{2}$ baskets

$= \frac{45}{4} = 11\frac{1}{4}$ baskets

11. (1) According to question,

Required time = $\frac{5}{12}$ h

$= \frac{5}{12} \times 60$ min = 25 min

12. (3) Let complete the work in x days.

Work done by (A + B) in 1 day = $\frac{1}{10}$ work

Work done by (B + C) in 1 day = $\frac{1}{18}$ work

A's 5 day's work + B's 10 day's work + C's 15 day's work = 1

(A + B)'s 5 day's work + (B + C)'s 5 day's work + C's 10 day's work = 1

$\frac{5}{10} + \frac{5}{18} - \frac{10}{x} = 1$

$\Rightarrow x = 45$

13. (1) According to question,

Work done by A and B in 5 days

$= \left(\frac{1}{10} + \frac{1}{15}\right) \times 5 = \frac{5}{6}$

Rest Work = 1- $\frac{5}{6} = \frac{1}{6}$

$\therefore$ C alone can do the work in 6 × 2 =12 days

Ratio of their work = $\frac{5}{10} : \frac{5}{15} : \frac{2}{12}$

=3 : 2 : 1

Share of money = Rs 225, Rs 150 and Rs 75

14. (3) According to question, work done by A in 4 days = $\frac{4}{8} = \frac{1}{2}$

Net work done by (A + B) in 1 day

$\left(\frac{1}{8} - \frac{1}{3}\right) = \frac{-5}{24}$

Work done by (A+B) in 2 days

$= \frac{-5}{24} \times 2 = \frac{-5}{12}$

$\therefore$ Work done in 6 days = $\frac{1}{2} + \left(-\frac{5}{12}\right) = \frac{1}{12}$

$\therefore$ Remaining $\frac{11}{12}$ of the wall is built by A in

$\frac{8 \times 11}{12} = \frac{88}{12} = \frac{22}{3} = 7\frac{1}{3}$ days

15. (2) Let one man takes x days to complete the work and one woman takes y days to complete the work independently.

Then, $\frac{4 \times 4}{x} + \frac{10 \times 4}{y} = \frac{1}{3}$

and $\frac{6 \times 2}{x} + \frac{12 \times 2}{y} = \frac{2}{9}$

Solving above equations, we get

$x = 108$, $y = 216$

Let z women be added to complete the work in 3 days.

Then, $\frac{6 \times 3}{108} + \frac{3(12+z)}{216}$

$= 1 - \left(\frac{1}{3} + \frac{2}{9}\right) = \frac{4}{9}$

$\Rightarrow 36 + 36 + 3z = \frac{216+4}{9} = 96$

$\Rightarrow 3z = 96 - 72 = 24 \Rightarrow z = 8$

16. (3) According to question,

	A		B		C
Efficiency	3	:	2	:	6
No. of days	2	:	3	:	1

$\Rightarrow$ Number of days taken by A = 12,
Number of days taken by B =18
and Number of days taken by C = 6

1 day's work of (A + B) = $\frac{5}{36}$

1 day's work of (B + C) = $\frac{8}{36}$

1 day's work of (C + A) = $\frac{9}{36}$

In 5 days total work done

$= \frac{5}{36} + \frac{8}{36} + \frac{9}{36} + \frac{5}{36} + \frac{8}{36} = \frac{35}{36}$

Now, the rest of the work

$\left(ie, \frac{1}{36}\right)$ is done by AC

Number of days taken by AC for the rest of the work $= \frac{\frac{1}{36}}{\frac{9}{36}} = \frac{1}{9}$

Therefore, total time taken to complete the work $= 5 + \frac{1}{9} = 5\frac{1}{9}$ days

17. (1) According to the question,
Ratio of efficiencies of the three persons

$= \frac{24}{6} : \frac{24}{8} : \frac{24}{3} - \left(\frac{24}{6} + \frac{24}{8}\right) = 4 : 3 : 1$

$\therefore$ Boy's share $= \frac{1}{(4+3+1)} \times 600$

$= \frac{1}{8} \times 600 =$ Rs 75

18. (3) Total time taken to finish the work

$= 10 + \frac{15(210-10)}{(15+15\times 2)} = 10 + \frac{15\times 200}{45}$

$= 10 + \frac{200}{3} = 76\frac{2}{3}$ days

19. (1) A and B can complete the work in 30 and 40 days respectively.
So, ratio of their wages

$= \frac{1}{30} : \frac{1}{40} = 40 : 30$

$\therefore$ Share of B $= \frac{30}{40+30} \times 2100 =$ Rs 900

20. (1) By question,

$\therefore \frac{M_1D_1T_1}{W_1} = \frac{M_2D_2T_2}{W_2}$

$\therefore \frac{5\times 10\times 8}{100} = \frac{8\times D_2\times 6}{30}$

$\Rightarrow D_2 = \frac{5\times 10\times 8\times 30}{100\times 8\times 6} = 2\frac{1}{2}$ days

21. (3) Let one worker of category I can finish the work in $\frac{2m}{3}$ h.

Two workers of category II can finish the work in $\frac{m}{2}$ h.

so, required time $= \frac{\frac{2m}{3}\times\frac{m}{2}}{\frac{2m}{3}+\frac{m}{2}}$

$= \frac{2m^2}{7m} = \frac{2}{7}$ m h

22. (3) According to question
$\because M_1 \times D_1 \times W_1 = M_2 \times D_2 \times T_2 \times W_1$
$\therefore 2 \times 12 \times 8 \times 12000 \times 0.9$
$= 3 \times 6 \times x \times 9000 \times 0.8$

$\Rightarrow x = \frac{2\times 12\times 8\times 12000\times 0.9}{3\times 6\times 9000\times 0.8}$

= 16 h / day

23. (2) According to question
$\because M_1 \times D_1 \times T_1 \times W_2 = M_2 \times D_2 \times T_2 \times W_1$
$36 \times 6 \times 10 \times 1200 = 10 \times D_2 \times 8 \times 1200$

$\therefore D_2 = \frac{36\times 6\times 10\times 1200}{10\times 9\times 1200} = 27$ days

24. (4) Let Bhavika and Rita working together can finish a job in *x* days.
Then, in 1 day they will complete

$= \frac{1}{x}$ work

Bhavika working alone do in 1 day

$= \frac{1}{(x+8)}$ work

Rita working alone will do in 1 day

$= \frac{1}{\left(x+\frac{9}{2}\right)}$ work

$\therefore \frac{1}{(x+8)} + \frac{1}{(2x+9)} = \frac{1}{x}$

$\Rightarrow \frac{(2x+9)+2(x+8)}{(x+8)(2x+9)} = \frac{1}{x}$

$\Rightarrow \frac{4x+25}{(x+8)(2x+9)} = \frac{1}{x}$

$\Rightarrow x = 6$ days

25. (3) Work done by 1 man in 1 day

$= \frac{1}{100}$

Work done by 15 men and 6 women

$= \frac{15}{100} + \frac{1}{120}$

$= \frac{3}{20} + \frac{1}{20}$

$= \frac{4}{20} = \frac{1}{5}$ work

$\therefore$ 15 men and women will take 5 days to complete the work.